# SWAFL Desktop

## Complete Technical Manual

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Soil Water Atmosphere Field Loop workbench

Version 1.1.0

Step by step use, every control and trigger,
every layer and simulation, and the equations
behind every output

**Naziru Halilu**

Coupled Water Fluxes, Pore Scale Dynamics, Precision Irrigation and Mulching Feedback Loops

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# About this manual

This manual documents **SWAFL Desktop**, the Soil Water Atmosphere Field Loop decision-support workbench that accompanies the repository *Coupled Water Fluxes, Pore-Scale Dynamics, Precision Irrigation and Mulching Feedback Loops*. It is written for three readers: a farmer or adviser who wants to use the screens, a student who wants to know what each picture means, and a scientist who wants to know exactly which equation produced each number.

**How the manual is organised.** This is the manual of version 1.1.0. It follows the manual of version 1.0 chapter by chapter, with every picture taken again from version 1.1.0 and every numbered arrow placed on the item it names in the new interface, and it adds the chapters and screens of the new release. The manual of version 1.0 stays in the repository (`docs/SWAFL_Manual.pdf`).

1. Chapters 1 to 3 install the program and explain the window and every control.
2. Chapters 4 to 5 explain the clock (dates) and the two computing engines that sit under every layer.
3. Chapters 6 to 10 document each *layer* of the Layers panel: what it designs or shows, how the picture is generated, and the governing equations.
4. Chapters 11 to 13 document the three irrigation design layers (surface, drip, sprinkler) in full, including the irrigation schedule and the “When to irrigate” message.
5. Chapters 14 to 16 document the dashboard, step-by-step workflows, exports and troubleshooting.
6. Chapter 17 states the validity limits.
7. Chapters 18 to 24 document the additions of version 1.1: running a design and finding the most efficient inputs, weather data (your own file or the online weather of a place), the real field on the map (satellite indices, image analysis of the satellite photo, water sources and the choice of the irrigation system), the crop library and the crop water requirement, the seasonal data, the reports, and every change from version 1.0.
8. Chapter 25 is the screenshot atlas. It shows every screen of the workbench (all layers, every section of the three design layers, the menus, the toolbox, the Properties panel, the Dashboard and the dialogs of version 1.1) from the program itself, and prints under each screenshot every equation that the screen draws on, with its number, source chapter and provenance. A reverse index gives, for every numbered equation, the screens that use it.
9. Chapters 26 to 39 document the Daisy workbench: the overview of its groups of tabs, the governing equations and guiding principles (the last sections of Chapter 26), every screen with numbered arrows (Input data with online weather, Experiments and scenarios, the 27 best combinations, Results and analysis with the figures of real Daisy runs, the guided path with Back and Next and its hand-over to the layers of the SWAFL project, Model development, Documentation), the 107 reference topics, the tested parameters, the factors and responses of the experiments, the Daisy library as it comes with the program, and the weather of the bundled stations.
10. The appendices give symbols, control value lists, data sets, the provenance ledger, the references and the reference of all numbered equations.

**The map of the manual.** Every chapter carries an icon in the colour of its part, in the table of contents and on its first page:

| Part | Colour | Chapters | Icons |
|---|---|---|---|
| Start: install, window, controls, clock, engines | blue | 1 to 5 | ⚙ ▦ ☰ ◷ ⚛ |
| Layers of the field | green | 6 to 10 | ☂ ⊞ ☷ ☘ ❦ ♻ |
| Irrigation design | teal | 11 to 13 | ☔ ✚ ❄ |
| Outputs: dashboard, workflows, exports, limits, runs, reports | orange | 14 to 18, 23 | ▤ ✔ ✎ ⚠ ⚑ |
| Data: weather, the real field, crops, seasons | brown | 19 to 22 | ☀ ⊞ ✿ ◷ |
| Reference: changes, atlas, appendices | grey | 24, 25, appendices | ★ ▥ ▦ |
| The Daisy workbench | purple | 26 to 39 | ⚘ ☀ ❖ ⚖ ▤ ⚑ ⚙ ✎ ☰ ⚗ ❦ ☁ ⚠ |

**The manual inside the program.** This manual comes with SWAFL and opens from the program itself, at the page you need: **F1** or **Help > Manual** opens it whole; the button **Manual p. N** on the bar of every tab of the Daisy workbench and above every layer of the SWAFL project opens it at the section of that screen; and the tab **The SWAFL manual** of the Daisy workbench (also in the Layers panel, Daisy model, Documentation) lists all its sections with their pages, with a search. On the desktop it opens in a window of SWAFL at that page; in the web version, in a tab of the browser. The page of every section is read from this PDF each time the manual is built (tools/make_manual_index.py writes app/data/manual_index.json), so the buttons always point to the right page.

**A note on honesty of the equations.** SWAFL mixes three kinds of computation, and the manual labels them so that nothing is over-claimed. Every numbered equation carries its label directly beneath it, and the provenance ledger in the appendices lists them all:

| Label | Meaning |
|---|---|
| **[Literature]** | A published relation implemented as written, cited to its source. |
| **[Data-driven]** | Obtained by interpolating or scaling the 10,190 Daisy simulations (Appendix C of the repository), not computed from a closed-form law. |
| **[Engineered]** | A parameterisation chosen for the workbench (for drawing, for speed, or because no closed form exists). Stated explicitly with its constants so that it can be audited, but not presented as a physical law. |

**Where the simulation model is named.** The program itself shows results only; the simulation model behind them (Daisy, 10,190 runs on ten soils) is described in this manual.

**Units.** Unless stated, depths are in mm, lengths in m, suction in cm of water (1 cm $\approx$ 0.0980665 kPa), flows in L $s^{-1}$ or $m^3$ $min^{-1}$ as marked on the screen, times in min or h, and areas in ha.

# Chapter 1

# ⚙ Installing and starting SWAFL

## 1.1 What you receive

SWAFL Desktop is a self-contained Electron program. The simulation engine is one HTML page (`app/engine/engine.html`) with a few script files beside it, which the program serves to itself on a local port, so **no internet connection is needed once it is installed**. The internet is used only if you ask for the online weather of a location (Chapter 19). Four ways of running it are provided.

| Route | Who it suits | Steps |
|---|---|---|
| **A. Build and install on Windows** | Most users | Unzip `SWAFL_Desktop.zip`, double-click `BUILD_AND_INSTALL_SWAFL.bat`. |
| **B. Windows program** | Users who received `SWAFL.exe` | Double-click `SWAFL.exe`. |
| **C. From source** | Developers | `npm install`, then `npm start` (desktop) or `npm run web` (browser). |
| **D. Online, nothing to install** | Anyone with a browser | Open the online workbench (link below). |

## 1.2 Daisy comes with SWAFL: nothing to choose

The Daisy program (version 7.1.14, the folder `daisy` next to the program, about 47 MB with its library) is part of SWAFL. You never choose a file. When a run starts, SWAFL finds Daisy by itself, in this order:

1. The folder `daisy` that comes with SWAFL (next to the program, or in the resources of the installed program).
2. Its own copy in the folder of the user (`%LOCALAPPDATA%\SWAFL\Daisy-7.1.14`): SWAFL makes this copy by itself a few seconds after it starts, so Daisy is still found when the program runs from a temporary folder (the portable SWAFL.exe) or is moved.
3. A Daisy installed on the computer: any folder of Program Files whose name has *daisy* in it.
4. The Daisy installer, if you have downloaded it (daisy-7.1.14-Windows-python3.13.exe, in Downloads, on the Desktop or in OneDrive): SWAFL runs it once, silently. Windows asks for permission once, because the installer needs it; after that Daisy is found as in 3.

Only if all four fail does a message say that Daisy is missing, with the places looked at; unzip the whole SWAFL zip again into a folder of your own. The Scenarios tab shows where Daisy was found (*Daisy is ready: …*). *Use another Daisy program* (a small link there) is for experts who want another version; if you choose the installer there, it is run, not refused.

## 1.3 What is inside SWAFL.exe: one file, nothing to install

`SWAFL.exe` is a portable program. You double click it: it needs no installation, no administrator rights, no Node.js, no Python and no network, because everything the program uses is packed inside the one file. It is made from four parts.

| Part inside the file | What it is |
|---|---|
| The Electron runtime | The window, the menus and the engine of the screen (the same code that runs in a browser) |
| The SWAFL application | The shell, the two simulation engines, the design modules, the library of 48 crops, the soil classes and the built in weather record, all written in JavaScript |
| The Daisy 7.1.14 program with its library | About 47 MB, run from its own folder inside the package; the installer that the Daisy site offers is never needed |
| The manual | This manual as a PDF, opened with F1 or from a Manual button at the page of the layer in use |

When the file starts, it unpacks itself to a temporary folder of your account and opens the window from that copy. A few seconds later SWAFL keeps its own copy of Daisy in the folder of the user (see the list above), so Daisy is still found if the temporary folder is cleared or the file is moved. Your projects, runs and reports are saved to files that you choose, never into the package. Only the optional online services (online weather, satellite images, soil and terrain of a field) need a connection; every other number is computed on your computer, so a single file on a computer that is not connected still gives you the field, the designs, Daisy, the decision layers and this manual.

## 1.4 Route A: build and install, step by step

1. Install **Node.js** (the LTS version) from nodejs.org if it is not already installed. Close and reopen any terminal afterwards.
2. Unzip `SWAFL_Desktop.zip` to a short path outside cloud-synchronised folders, for example `C:\SWAFL_Desktop`. The zip contains a folder `SWAFL_Desktop`; the folder you work in is the one that contains `package.json`.
3. Double-click **`BUILD_AND_INSTALL_SWAFL.bat`**. It installs the packages, builds the program and puts **SWAFL** on the Desktop. The first build downloads Electron and takes a few minutes.
4. Start **SWAFL** from the Desktop. The title bar reads *SWAFL 1.1.0*. If Windows SmartScreen shows “Windows protected your PC”, click **More info** and then **Run anyway** (the program is not code-signed).

`START_SWAFL.bat` starts the program from the source folder without building it, and `START_SWAFL_WEB.bat` opens the same program in your browser from your own computer. On a Mac the same two files are `START_SWAFL.command` and `START_SWAFL_WEB.command`. None of them needs Node.js or npm to be installed: each looks for Node.js on the computer (also in the usual install folders, which is where “npm is not recognised” came from) and otherwise downloads a portable copy of about 30 to 40 MB into the folder `tools/node`, which changes nothing else. On a Mac, if a file will not open, type `chmod +x *.command tools/*.sh` in Terminal in that folder and open the file once with a right click and Open. The Daisy program that comes with SWAFL is a Windows program; on a Mac everything else works, and Daisy runs when SWAFL is pointed to a Daisy built for the Mac.

## 1.5 Route D: the online workbench

The same program runs online in any current browser, with nothing to install and no server or Node.js:

SWAFL 1.1.0 online workbench

https://halilunaziru73-creator.github.io/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/Software/v1.1.0/SWAFL_Desktop/app/index.html?v=1.1.0

## 1.6 Route C: from source

```
git clone https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops.git
cd Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/Software/v1.1.0/SWAFL_Desktop
npm install
npm start          # desktop window
npm run web        # the same program in a tab of your browser
npm run get-app    # build SWAFL.exe, copy it to the Desktop and start it
```

## 1.7 If the build stops

**The command is not found.** The script is called `get-app`, with a hyphen, and `npm run get_app` and `npm run getapp` do the same. If npm answers *Missing script: "get_app"*, the folder holds an older `package.json`: unzip the new zip into a new folder and run the command there.

**The message "Cannot create symbolic link : A required privilege is not held by the client".** It appears during the step *packaging* of the build, after the line that downloads `winCodeSign`, and it names the files `libcrypto.dylib` and `libssl.dylib`. The build tool unpacks a code-signing package that contains two symbolic links, and Windows allows a normal user to create symbolic links only when Developer Mode is on or the terminal runs as Administrator. The program does not need code signing, so there are three ways out, in this order:

1. Run `npm run get-app` again. This version of the script notices the error, deletes the unfinished `dist` folder and builds again with the signing step switched off, so the build finishes without any change of settings. The program that results works in the same way; only the file properties of `SWAFL.exe` carry no icon and version signature.
2. Turn on **Developer Mode** (Settings, System, For developers), delete the folder `%LOCALAPPDATA%\electron-builder\Cache\winCodeSign`, and run the command again.
3. Open the terminal with **Run as administrator**, go to the folder that holds `package.json`, and run the command again.

**A folder in OneDrive.** The same build is slower and sometimes fails when the folder is inside a cloud-synchronised folder (for example `C:\Users\name\OneDrive\...`), because the synchroniser locks files while the build writes them. Unzip to a short path such as `C:\SWAFL_Desktop` and build there.

**Other messages.** *Running scripts is disabled* (PowerShell): use a Command Prompt, or run `cmd /c npm install`. *npm is not recognised*: install Node.js (LTS) and open a new terminal. *ENOENT, package.json*: the zip makes a folder inside a folder, so go into the inner one. *Electron download fails*: check the connection or VPN, delete `node_modules` and run `npm install` again.

**Daisy.** The Daisy 7.1.14 program for Windows is inside the zip (the folder `daisy`), is built into the program by the build and starts by itself when a Daisy run is asked for. Do not download or choose the Daisy installer (`daisy-7.1.14-Windows-python3.13.exe`): it is not the program, it asks for administrator rights, and choosing it as the program gives the message *spawn ... EACCES* and runs that end with *Daisy gave no results (code -1)*.

## 1.8 Requirements

- Windows 10 or 11 (64-bit) for the built program; any current Chrome, Edge or Firefox for the browser route.

- A screen of at least 1280 × 720 px. The layout is designed for 1600 × 900; panels can be hidden (View menu) or the drawing shown alone (*Focus view*, F9).
- About 400 MB of free memory. The 20-year daily simulation is computed in the program; no server is involved.

## 1.9 First-run check

After start-up, verify three things: (i) the status bar at the bottom reads *Ready* and ends with *SWAFL 1.1.0*; (ii) the Results tab of the dashboard shows *On* followed by today's date with numbers below; (iii) clicking **Surface irrigation design** in the Layers panel opens the design with its inputs and a *Run the design* box. If the drawing area stays blank, see Chapter 16.

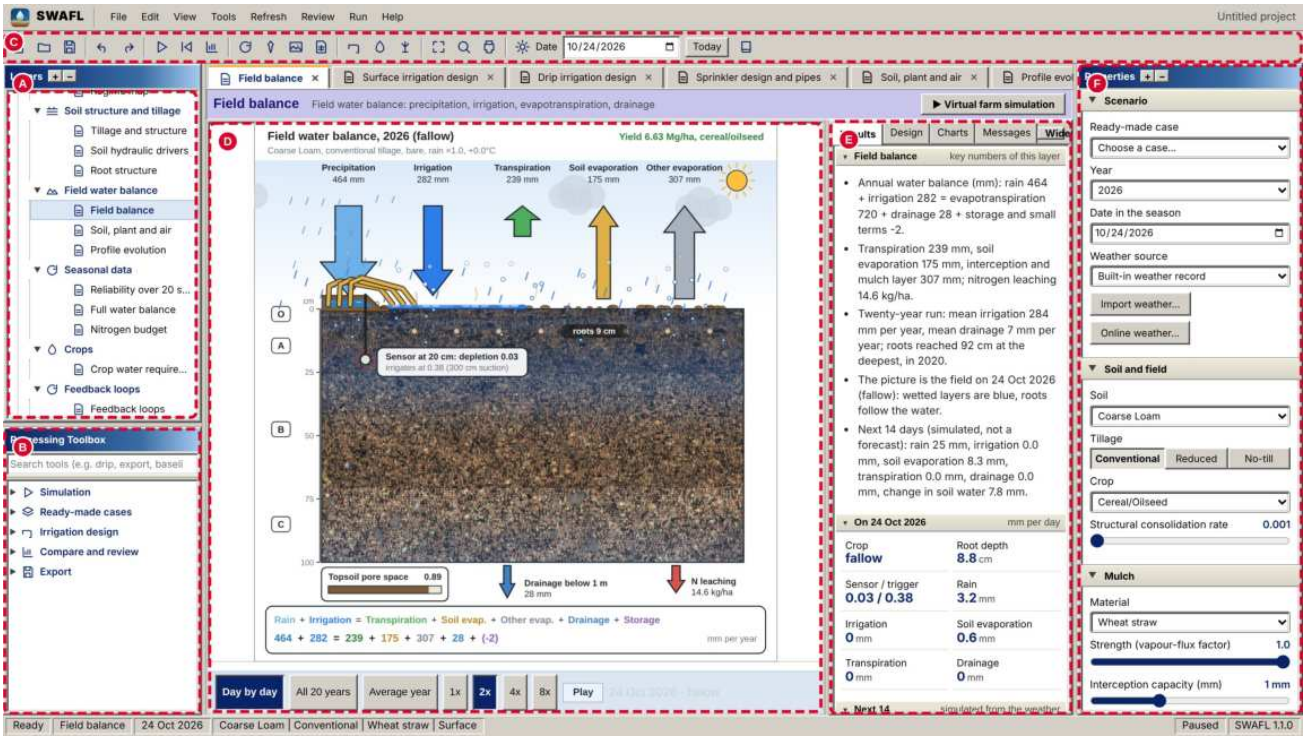



Figure 1.1: The workbench on first start: the Field balance layer for today's date. Left: Layers and Processing Toolbox. Centre: the drawing. Right of the drawing: the dashboard. Far right: Properties.

# Chapter 2

# The window: a guided tour

## 2.1 Anatomy

The window has eight regions, named here as they are named in the program.

1. **Menu bar**, File, Edit, View, Tools, Refresh, Review, Run, Help.
2. **Toolbar**, icon buttons for New, Open, Save, Undo, Redo, Play/Pause, Restart, 20-year simulation, the three design layers, fit-to-window, search, the report, and the **Date** control with a **Today** button.
3. **Layers** panel (top left), the project tree. Each layer is one drawing of the model; each design layer lists its sections.
4. **Processing Toolbox** (bottom left), a searchable list of ready-to-run actions.
5. **Document tabs and view bar** (above the drawing), one tab per open layer; the view bar names the layer and gives the buttons of its sections.
6. **Drawing** (centre), the active layer, with the playback bar and the legend under it.
7. **Dashboard** (right of the drawing), four tabs: *Results, Design, Charts, Messages*. Every block can be folded and unfolded by clicking its header.
8. **Properties** (far right), every model input, in collapsible groups.

The **status bar** shows, from left to right: the state (*Ready*), the active layer, the working date, the soil, tillage, mulch and irrigation method, *Paused* or *Running*, and the version (*SWAFL 1.1.0*).

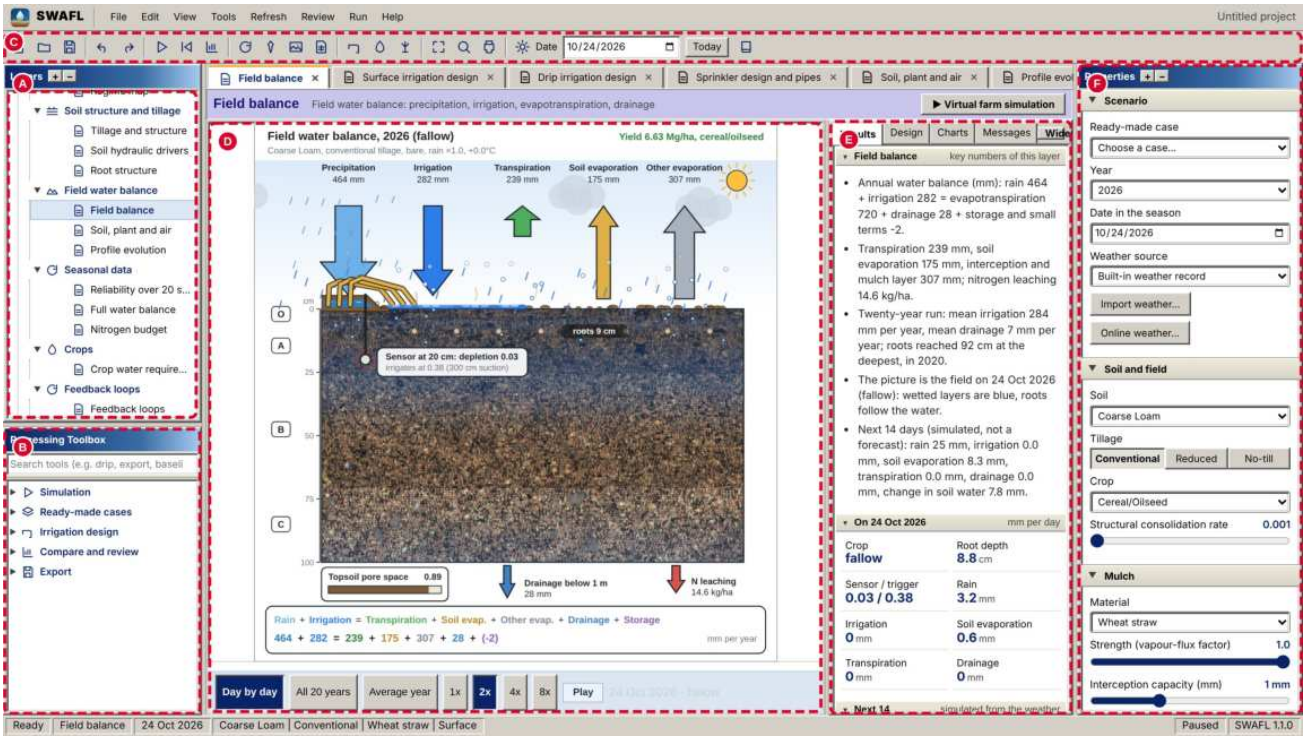



Figure 2.1: The window with the Field balance layer open.

## 2.2 Menu bar reference

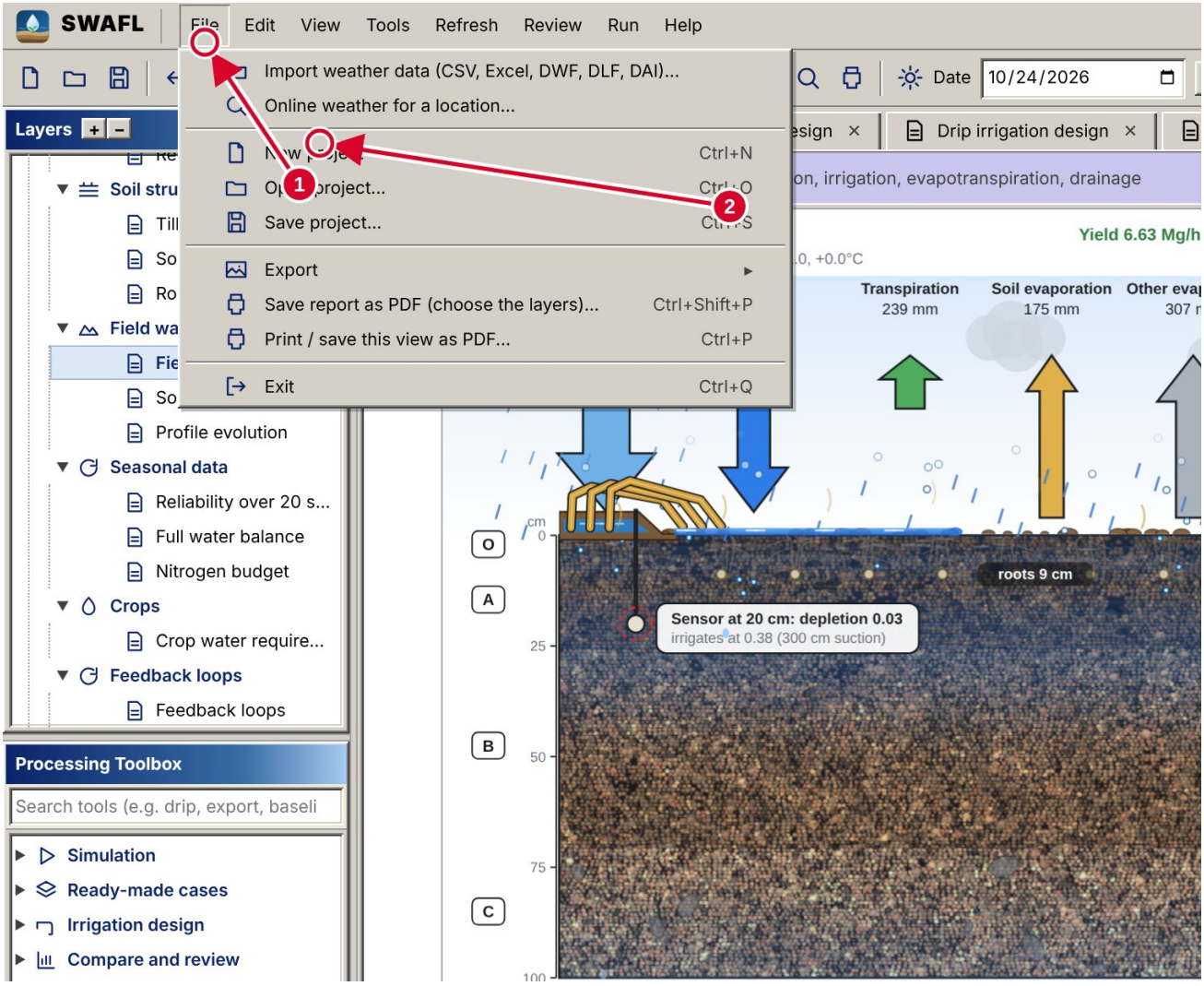



Figure 2.2: The File menu.

| Menu | Item | Shortcut | Action |
|---|---|---|---|
| File | Import weather data (CSV, Excel, DWF, DLF, DAI)… | | Reads a daily weather file; every layer, the dashboard and the crop water requirement then use it (Chapter 19). |
| | Online weather for a location… | | Gets twenty years of daily weather and a 16-day forecast for a place (Chapter 19). |
| | New project | Ctrl+N | Resets every control to the defaults (Appendix B). |
| | Open project… / Save project… | Ctrl+O / Ctrl+S | Loads or saves a `.swafl` file: JSON with the version, the open layer, every control, the year and the date. |

| Menu | Item | Shortcut | Action |
|---|---|---|---|
| | Export ▸ | Ctrl+E | Picture of the window (PNG); yearly results (CSV); design results (CSV). |
| | Save report as PDF (choose the layers)… | Ctrl+Shift+P | Asks which layers to include (all are ticked to start with) and writes the report (Chapter 23). |
| | Print / save this view as PDF… | Ctrl+P | Prints the open layer through the system print dialog. |
| | Exit | Ctrl+Q | Closes the program. |
| Edit | Undo / Redo | Ctrl+Z / Ctrl+Y | Steps through the history of control changes. |
| | Reset all settings; Copy settings as text | | Defaults for the controls only; a text list of every control value on the clipboard. |
| | Pin results as baseline / Clear baseline | Ctrl+B | Freezes the 20-year averages for comparison (Chapter 14). |
| | Find a tool… | Ctrl+F | Moves the cursor to the Toolbox search box. |
| View | Design layouts ▸ | | Jumps to the layout drawing of the surface, drip or sprinkler design. |
| | Field sketch to design… | | The field sketch (Chapter 18). |
| | Layers and Processing Toolbox / Properties / Dashboard | | Show or hide each dock. |
| | Fit drawing to window / Drawing at full width (scroll) / Focus view (hide panels) | F9 | Three ways of showing the drawing. |
| | Expand all layers / Collapse all layers; Full screen | F11 | Tree display; full screen. |
| Tools | Field sketch to design…; Compare the irrigation systems on this field; Find the most efficient design inputs | | Chapter 18. |
| | Import weather data…; Online weather for a location… | | Chapter 19. |
| | Ready-made cases ▸; Irrigation design ▸; Irrigation method ▸; Mulch material ▸ | | Set the control directly. |
| | Run 20-year season simulation; Find a tool… | | Chapter 5. |
| Refresh | Refresh display (F6); Reload simulation engine; Clear message log | | |
| Review | Results dashboard; Design results; Yearly charts; Messages; Pin / Clear baseline; Go to top of the drawing | | Switches the dashboard tab. |
| Run | Run / pause the season (F5); Restart from day 1 (Shift+F5); 20-year season simulation; Optimise surface layout; Ready-made cases ▸ | | |

| Menu | Item | Shortcut | Action |
|---|---|---|---|
| Help | Manual (PDF) (F1); How to use SWAFL; Troubleshooting (running the app); Keyboard shortcuts; SWAFL on GitHub; Full documentation on GitHub; Report a problem; About SWAFL | | |

The eight menus are pictured, opened, in the screenshot atlas (Chapter 25).

## 2.3 The command palette (Ctrl+K)

Press **Ctrl+K** anywhere to open a search box over *every* layer, section, tool, setting and date. Type part of a name (for example `drip`, `trigger`, `mulch`, `2031`) and press **Enter** to run the first match, or use the arrow keys to choose.

## 2.4 Layers panel

The tree has the project root *SWAFL project* and eight groups, in the order in which a field is usually designed and studied. Click a group to expand it, click a layer to open it as a tab, and click the small triangle beside a layer to list its sections; a section jumps the drawing to that part.

| Group | Layers |
|---|---|
| **Irrigation design** | Surface irrigation design; Drip irrigation design; Sprinkler design and pipes, each with its sections |
| **Irrigation control** | Sensor and trigger; Trigger law |
| **Mulch and evaporation** | Mulch against bare soil; Mulch control; Evaporation; Fate of saved water; Regime map |
| **Soil structure and tillage** | Tillage and structure; Soil hydraulic drivers; Root structure |
| **Deterministic phenotyping** | Deterministic phenotyping |
| **Field water balance** | Field balance; Soil, plant and air; Profile evolution |
| **Seasonal data** | Reliability over 20 seasons; Full water balance; Nitrogen budget |
| **Crops** | Crop water requirement |
| **Feedback loops** | Feedback loops |

In version 1.0 every layer carried a short code (S2, V1, N4 ...). The codes have gone from the program; this manual uses the layer names. Open layers stay as tabs above the drawing; closing a tab does not lose any setting, because the controls belong to the project, not to the layer.

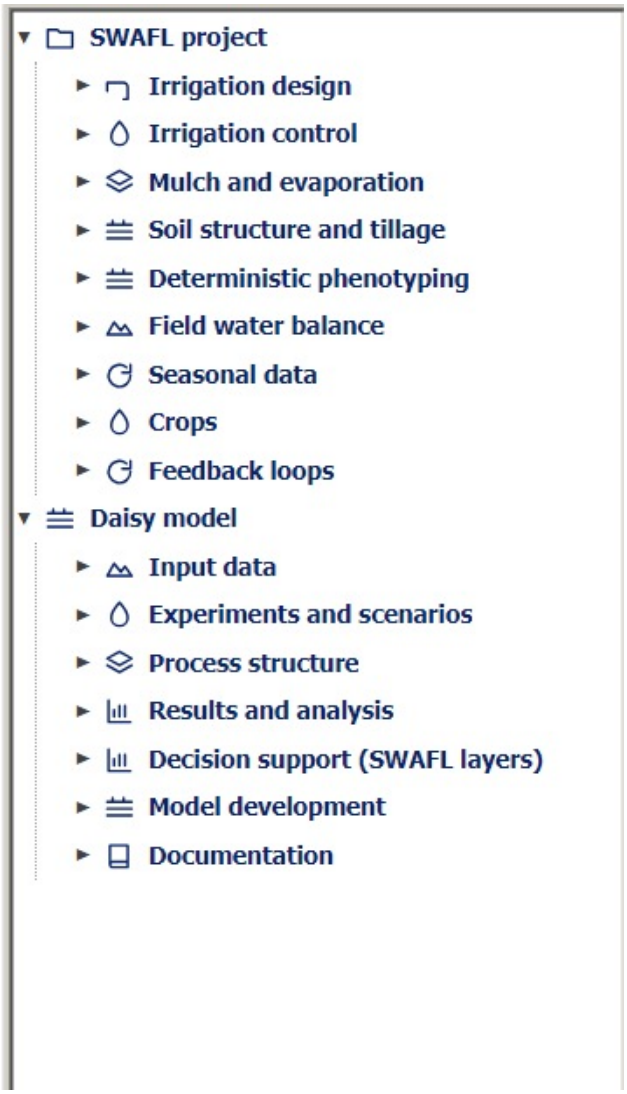



Figure 2.3: The Layers panel of version 1.1.0 as it opens: the SWAFL project with its nine groups (Irrigation design, Irrigation control, Mulch and evaporation, Soil structure and tillage, Deterministic phenotyping, Field water balance, Seasonal data, Crops, Feedback loops) and the Daisy model with its seven groups (Input data, Experiments and scenarios, Process structure, Results and analysis, Decision support (SWAFL layers), Model development, Documentation). A click on the arrow opens a group.

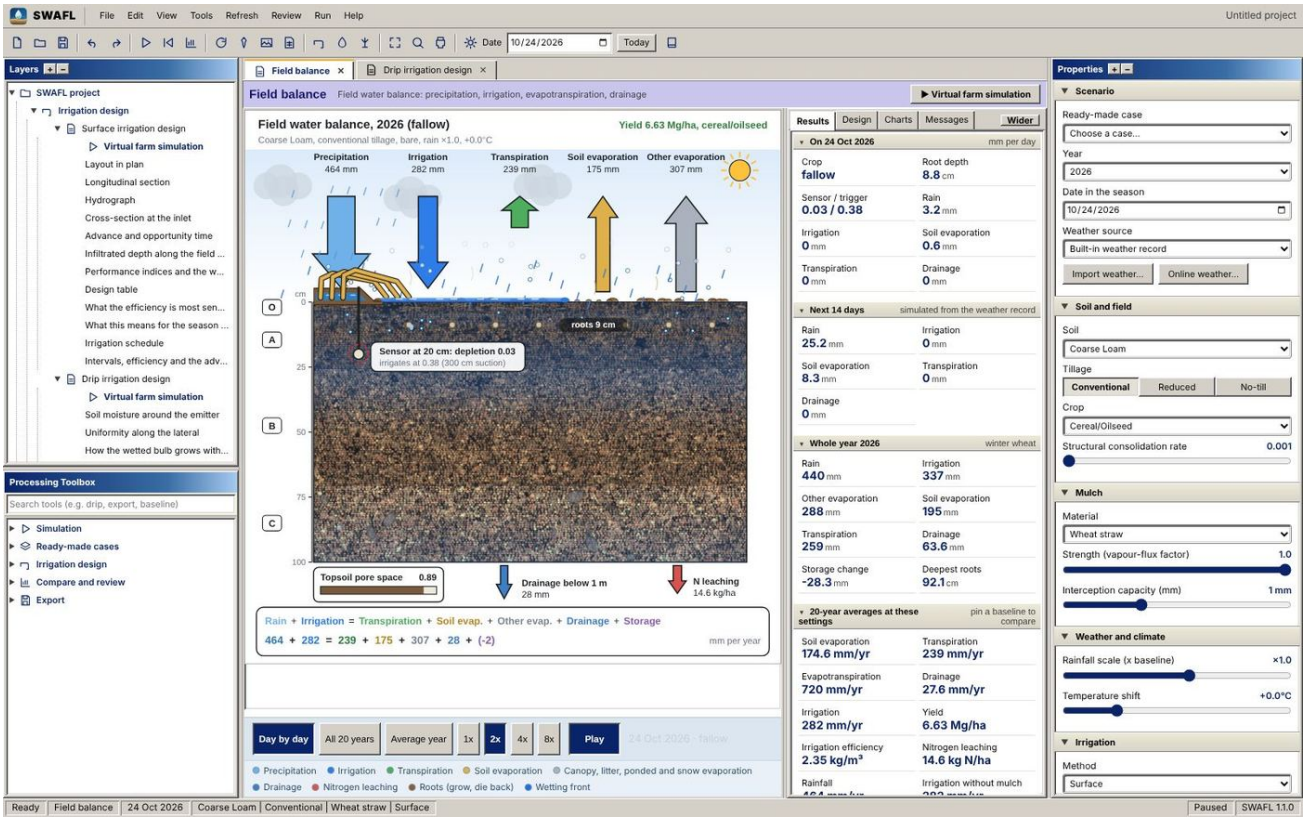



Figure 2.4: The window with a layer open: the Layers panel and the Processing Toolbox on the left, the drawing in the middle, the Results dock and Properties on the right.

## 2.5 The view bar

Above the drawing, the view bar names the open layer. For the design layers it carries one button per section (*Whole design, Layout design, Simulation, ...*).

## 2.6 Processing Toolbox

Double-click (or select and press Enter) to run an action. Typing in the search box filters all tools by name and by hidden keywords.

| Group | Tools |
|---|---|
| Simulation | Play/pause the season; Restart from day 1; Run 20-year season simulation; Reset all settings |
| Ready-made cases | The seven cases of Section 3.8 |
| Irrigation design | Open the surface, drip or sprinkler design; Optimise surface layout; set the irrigation method |
| Compare and review | Pin and clear the baseline; show the Results, Design or Charts tab |
| Export | PNG of the window; yearly results CSV; design results CSV; save project; print / PDF |

## 2.7 Resizing and remembering the layout

Drag the thin splitters between the docks to resize them. The program remembers the panel widths between sessions. The *Wider* button above the dashboard widens it; **View ▸ Drawing at full width** lets the drawing keep its design width and scroll; **Focus view** (F9) hides the panels.

## 2.8 Reading the drawings

- **Hot spots.** Many drawings are clickable. Hovering shows a tooltip; clicking applies a setting (for example, clicking a year bar jumps to that year, clicking a soil on a chart selects that soil).
- **Value boxes.** Every input of the three design layers is a value box. Click it: a small window opens in which you type the exact value, choose its unit (for example m, km or ft; L/s, $m^3$/h or L/min; ha, $m^2$ or acre; kPa, bar, m of water or psi) or pick a value from the list of the allowed range. A value outside the range is brought back to its nearest limit.
- **Nothing runs while you type.** Every layer, not only the three designs, has a **Run simulation** bar and shows its figures only after the button (or the Play button of the toolbar) is pressed for the present controls (Chapter 18); when a control changes, the bar asks to run again.
- **Colours.** Water blues, soil browns, crop greens; text is dark on a white sheet; labels sit in white boxes drawn after the pictures, so no arrow covers a label.

# Chapter 3

# ☰ Every control, value by value

All model inputs live in the **Properties** panel (far right). Each group can be collapsed; **+** and **−** in the panel header expand or collapse all. The sliders are *indices* into lists of discrete values, because every value in the lists corresponds to a set of simulations (Chapter 5). The number shown beside a slider is the value the index stands for. The inputs of the three irrigation designs are not in Properties: they are the value boxes inside each design layer (Chapters 11 to 13 and 18).

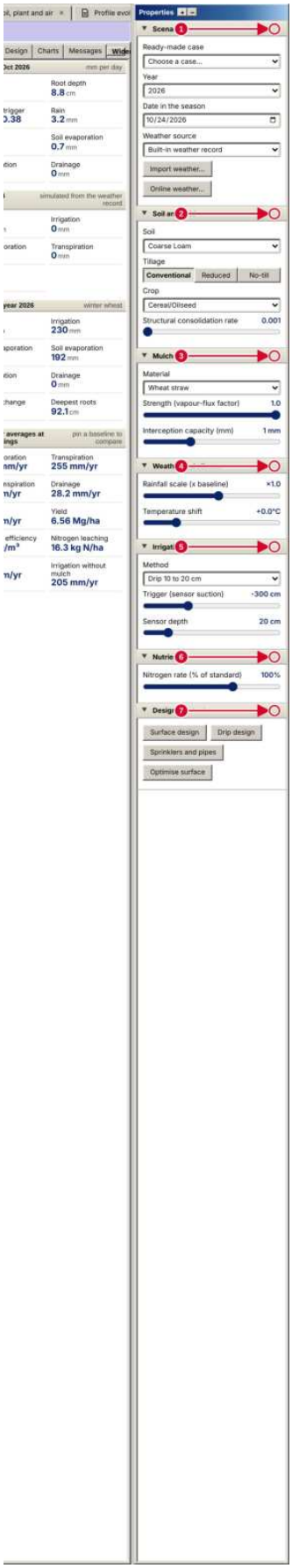



Figure 3.1: The Properties panel, with every group open (the arrows point to the groups; Chapter 25).

## 3.1 Scenario

| Control | Values | Default | What it does |
|---|---|---|---|
| **Ready-made case** | seven cases (Section 3.8) | n/a | Replaces many controls at once, as one undoable step. |
| **Year** | 1980 … 2100 | the current year | The calendar year of the working date. Years 1980 to 1999 use the recorded weather of that year; later years use the typical year of the record, or your own weather when it covers the year (Chapter 4 and Chapter 19). |
| **Date in the season** | any calendar date | today | The working date: the day drawn, the “On …” block of the dashboard and the first day of every irrigation schedule. |
| **Weather source** | Built-in weather record; Imported weather file; Online weather of a location | built-in | Which daily weather every layer uses. The second and third choices become active only after you have imported a file or fetched the weather of a place (Chapter 19). |
| **Import weather… / Online weather…** | buttons | | Open the two weather dialogs. |

## 3.2 Soil and field

| Control | Values | Default | Effect |
|---|---|---|---|
| **Soil** | Sand, Loamy Sand, Sandy Loam, Coarse Sand, Coarse Loam, Loam, Silt Loam, Sandy Clay Loam, Clay Loam, Silty Clay Loam | Coarse Loam | Selects the soil’s texture, its van Genuchten parameters $(\alpha, n)$ (Table 5.3), the baseline run, plant-available water per cm (`AWC`) and the trigger anchor. Re-computes every layer and cancels the last run of every design. |
| **Tillage** | Conventional / Reduced / No-till | Conventional | Sets the initial pore index $p_0$ (1.00, 0.72, 0.42) and therefore the starting structure of the topsoil (Section 8.1). |
| **Crop** | the simulated cereal/oilseed rotation and the 48 crops of the library, in the groups *Staple food crops* (9), *Cash and oil crops* (8), *Vegetables* (4) and *Trees, vines and palms* (27) | Cereal/Oilseed | Chooses the crop calendar, the crop coefficients, the root depth, the depletion fraction and the yield response (Chapter 21). The cereal rotation is the simulated crop; the library crops are transferred from it with their FAO-56 crop coefficients (Section 21.4). |

| Control | Values | Default | Effect |
|---|---|---|---|
| **Structural consolidation rate** | 0.001, 0.0025, 0.005, 0.01, 0.02 $d^{-1}$ | 0.001 | The rate $\tau_c$ at which conventionally tilled pores close (Eq. 8.1). In the simulations it changes the results of the five finer soils (loam to silty clay loam, up to about 6 mm a year of drainage and 14 kg N/ha of leaching); in the five coarser soils every rate gives the same result. |

## 3.3 Mulch

| Control | Values | Default | Effect |
|---|---|---|---|
| **Material** | Rice husk, Wheat straw, Wood chips, Sawdust, Grass clippings, Black plastic film | Wheat straw | Sets both sliders below to typical values for the material and the drawn layer thickness (Table 3.1). |
| **Strength (vapour-flux factor)** $\varphi$ | 0.1, 0.2, ... 1.0 | **1.0 (no mulch)** | The fraction of the bare-soil vapour flux that passes the mulch. 1.0 means bare; 0.1 means the flux is cut by 90 %. |
| **Interception capacity** | 0.5, 1, 2, 4 mm | 1 mm | Water the mulch can hold and lose before it reaches the soil. |

Table 3.1. Mulch materials as defined in the engine.

| Material | $\varphi$ | capacity (mm) | drawn thickness (relative) |
|---|---|---|---|
| Rice husk | 0.5 | 2 | 1.0 |
| Wheat straw | 0.4 | 4 | 1.35 |
| Wood chips | 0.3 | 2 | 1.7 |
| Sawdust | 0.6 | 1 | 0.9 |
| Grass clippings | 0.7 | 4 | 1.2 |
| Black plastic film | 0.1 | 0.5 | 0.25 |

The material only *sets* the sliders. Moving a slider afterwards keeps the material for drawing but uses the slider value for the calculation. Any $\varphi \geq 0.95$ is treated as “no mulch” in the drawings.

## 3.4 Weather and climate

| Control | Values | Default | Effect |
|---|---|---|---|
| **Rainfall scale** | $×$0.5, 0.6, ... 1.4 | $×$1.0 | Multiplies the baseline annual rainfall and selects the matching runs. When weather is imported, it is set to the step nearest to the ratio of the imported to the recorded annual rain (Eq. 19.4). |

| Control | Values | Default | Effect |
|---|---|---|---|
| **Temperature shift** | −1.0, −0.5, 0, +0.5, … +3.5 °C | 0 °C | Selects the warming runs and changes the evaporative demand (Eq. 5.26). When weather is imported, it is set to the step whose demand is nearest the imported reference evapotranspiration (Eq. 19.5). |

## 3.5 Irrigation

| Control | Values | Default | Effect |
|---|---|---|---|
| **Method** | Surface; Overhead; Drip 10 to 20 cm; Drip 20 to 30 cm; Drip 40 to 50 cm; Drip 50 to 60 cm | Surface | Selects the irrigation run set and, in the day-by-day engine, where each dose is placed (Section 5.2). |
| **Trigger (sensor suction)** | −50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm | −300 cm | Irrigation starts when the soil at the sensor depth is as dry as this suction. A *less negative* number means wetter, more frequent irrigation. |
| **Sensor depth** | 5, 10, 15, … 100 cm (20 values) | 20 cm | Depth of the simulated tensiometer that is compared with the trigger. |

**Sand and loamy sand.** In the simulations of these two soils the sensor-triggered runs at the reference trigger of −300 cm at 20 cm never irrigate, because the coarse soil drains past the sensor before it dries to that suction. A ratio to zero is undefined, so for these soils the irrigation of another trigger, sensor depth or method is added to the reference amount as the difference of the two simulated amounts (Eq. 5.12a). At −50 cm the sand then receives 400 mm a year instead of the reference 278 mm.

**Trigger mode (Trigger law layer).** The slider above applies one suction to every soil (*common* mode). The **Trigger law** layer offers an *own* mode, in which each soil gets the suction that corresponds to a soil-independent depletion fraction $f^*$ (default 0.45): see Chapter 6.

## 3.6 Nutrients

| Control | Values | Default | Effect |
|---|---|---|---|
| **Nitrogen rate** | 10, 25, 40, 55, 70, 85, **100**, 115, 130, 150 % of the standard rate | 100 % | Selects the nitrogen runs; changes yield and nitrogen leaching and, slightly, the maximum rooting depth (Eq. 5.18). |

## 3.7 Ready-made cases

| # | Case | Changes |
|---|---|---|
| 0 | Default: Coarse Loam, bare, normal climate | all defaults |
| 1 | Dry year on Sand, no mulch | Sand; rain $×$0.6; +1 °C |
| 2 | Wheat straw mulch on Loam, drip 10 to 20 cm | Loam; straw; drip 10 to 20 |
| 3 | Wet year on Silty Clay Loam, no-till | Silty Clay Loam; no-till; rain $×$1.3 |
| 4 | Hot year (+3 °C), tomato on Sandy Loam | Sandy Loam; tomato; +3 °C |
| 5 | Plastic film, deep drip 40 to 50 cm, Clay Loam | Clay Loam; plastic; drip 40 to 50 |
| 6 | Low nitrogen (40 %), cowpea on Silt Loam | Silt Loam; cowpea; N 40 % |

Every case first resets the controls to their defaults and then applies its changes, so cases never accumulate.

## 3.8 Design shortcuts

The last group of Properties opens the three designs (*Surface design, Drip design, Sprinklers and pipes*) and runs **Optimise surface**, which searches the surface design for the inflow and cut-off with the highest application efficiency at the present field.

## 3.9 Change-by-change: what each control moves

“Run set” means that the response is read from the simulations (Chapter 5); “engine” means that the day-by-day model of Section 5.3 responds as well.

| Control changed | Annual outputs (20-year means) | Day-by-day layers and schedules | Design layers |
|---|---|---|---|
| Soil | all (baseline run for that soil) | AWC, $(\alpha, n)$, trigger anchor ⇒ dates, depths | intake ($k$, $A$) via Saxton and Rawls and Green and Ampt; $Z_n$; the sprinkler maximum rate |
| Tillage, consolidation | drainage, evaporation, yield | pore index in the structure layers | surface storage $S_d$ of the sprinkler design |
| Crop | yield, transpiration, evaporation (Section 21.4) | crop calendar, $K_c$, $R_{\max}$, cover | peak $ET_c$, $Z_n$, row spacing |
| Mulch $\varphi$, capacity | evaporation, irrigation, drainage, yield | evaporation term | interception loss of the sprinklers |
| Rain scale | all | rain amount per event | n/a |
| Temperature | evaporation, transpiration, irrigation, yield | $E_0$ and seasonal weight | air temperature in the wind drift (Eq. 13.9) |
| Method | irrigation, drainage, yield | dose placement | the design that belongs to the method |
| Trigger | irrigation, yield, drainage | **dates of irrigation** | $Z_n$ (as $f_T \cdot \mathrm{AWC} \cdot R_{\max}$) |
| Sensor depth | irrigation, yield | **dates of irrigation** | n/a |
| Nitrogen | yield, N leaching | $R_{\max}$ | n/a |
| Weather source | through the nearest rain and temperature steps | **every daily value** | the crop water requirement and the schedules |
| Date and year | none | **which day is shown; first date in the schedule** | the schedule table |

# Chapter 4

# ◷ Dates, years and the clock

## 4.1 Why the date matters

SWAFL runs on **twenty years of daily weather** arranged on a 365-day calendar without leap days. With the built-in weather these are the twenty recorded years 1980 to 1999 of the study site (Chapter 5); with imported or online weather they are the last twenty complete years of your own record (Chapter 19). The program maps *any calendar date* you choose onto one of those twenty years. The working date therefore decides (i) which day is drawn, (ii) which of the twenty years acts as the *analogue* for that date, and (iii) the first line of every irrigation schedule.

## 4.2 The date control

The **Date** box on the toolbar and the **Year** and **Date in the season** controls in Properties ▸ Scenario all set the same working date; the year list runs from 1980 to 2100. **Today** sets the computer's date. When you change the date:

1. With the built-in weather, a year from **1980 to 1999** is its own analogue: that recorded year is drawn.
2. With imported or online weather, a year inside the twenty years of your record is its own analogue in the same way.
3. Any other year (for example 2026 or 2075) takes the **typical** year of the record as its analogue: the median of the twenty years by growing-season rain reaching the soil.
4. The engine sets the field clock to day-of-year $d_{\text{oy}}$ of the analogue year, so that 22 October 2029 shows the crop state, soil water and irrigation events of the analogue year on 22 October.

In version 1.0 a *Weather year* control let you choose the typical, the driest or the wettest year for dates outside the record. In version 1.1 the typical year is always used for those dates; to study a dry or a wet year, choose that year inside the record (the *Full water balance* layer of the Seasonal data group shows the rain of each of the twenty years), or import weather.

With $Y_0$ the calendar year of the chosen date, $a$ the analogue year index $(0, \dots, 19)$ and $k$ the model day number $(0, \dots, 7299)$ of the twenty-year run,

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

*Provenance (Eq. 4.1):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

and the calendar label of the first plotted year is $\text{CAL}_0 = Y_0 - a$. The choice of analogue year is

$$a = \begin{cases} \text{argsort}_y\,(P_{\text{thr},y})_{[10]} & \text{typical (median, 11th smallest of 20): years outside the record} \\ Y_0 - Y_{\text{rec}} & Y_{\text{rec}} \le Y_0 \le Y_{\text{rec}} + 19 \end{cases} \tag{4.2}$$

*Provenance (Eq. 4.2):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

where $P_{\mathrm{thr},y}$ is the sum of throughfall (rain reaching the soil) in year $y$ (Section 5.2) and $Y_{\mathrm{rec}}$ is the first year of the record (1980 for the built-in weather, the first of the twenty years of your weather otherwise).

**What this means for you.** A date in 2029 is *not* a forecast for 2029 unless the online weather is in use and the date falls in its 16-day forecast. Otherwise it shows how this soil, crop, trigger and method would behave on that day of the year in a typical year. Chapter 17 repeats this as a validity limit.

## 4.3 The playback bar

Under the drawing, the **playback bar** carries the play button (F5) and, for the field layers, the *Day by day*, *All 20 years* and *Average year* views and the speed buttons. In the field layers the play button animates the field clock: each real second advances 15 model days times the speed setting, and the clock *dwells* for about $3.2/\sqrt{\text{speed}}$ s on every day on which an irrigation, a wetting front of at least 20 cm, or drainage occurs. The clock wraps at the end of the twenty years. **Restart** (Shift+F5) returns to day 1 of the working year. In the three design layers the same button is **Run simulation** (Chapter 18). The year strip that sat under the drawing in version 1.0 has gone: the year is set only in Properties and on the toolbar.

## 4.4 The 14-day look-ahead

The “Next 14 days” block on the Results tab sums the same simulated daily fluxes from the working date forward. It is labelled *simulated from the weather record* because the rain on those days is the analogue year’s rain. When the online weather is in use and the days fall in its 16-day forecast, the block says *from the weather forecast* instead (Chapter 19).

## 4.5 Detail year and sensor record

The *Sensor and trigger* and *Profile evolution* layers can show either all twenty years or one year in detail. The suction traces shown there are the study’s **daily suction at 20 cm for 1980 to 1999**, stored for the Coarse Loam and the Coarse Sand; for every other soil, and for every setting other than the standard one, the trace is moved by the difference between two runs of the daily model, the run at your settings and the run at the standard settings (Eq. 6.4a). The growing season of these records is the 183 days starting 1 April.

# Chapter 5

# ⚛ The two computing engines

Every number in SWAFL comes from one of two engines, which are tied together so that they agree on the twenty-year means.

- **Engine 1, the annual meta-model.** It reads 20-year-mean results of the 10,190 Daisy simulations and scales them for the controls. It produces the Dashboard "20-year averages" and every bar, line and arrow in the annual layers. **[Data-driven]**
- **Engine 2, the daily bucket model.** A 40-layer soil-water model run for twenty years of daily weather. It produces dates, depths, schedules, root growth, water flow arrows and the animated field. It is **calibrated** so that its mean irrigation equals Engine 1's, and its fluxes are **scaled** to Engine 1's annual means (Eq. 5.24). **[Engineered, with literature components]**

The design layers (Chapters 11 to 13) are a third, separate set of *closed-form hydraulic models*; they take their soil, crop, trigger and depth from Engines 1 and 2 (Section 5.3).

## 5.1 Engine 1: the Daisy simulations and the meta-model

### 5.1.1 The simulation campaign

The repository's study ran Daisy 7.1 (Abrahamsen and Hansen, 2000; University of Copenhagen), a one-dimensional soil, plant and atmosphere model that solves Richards' equation with van Genuchten and Mualem hydraulics, coupled to the WEPP dynamic structure model for the effect of tillage and consolidation on pore structure with time. Ten soils (USDA textures from sand to silty clay loam, Ap and Bt horizons) were simulated in **10,190 runs across 17 experiments (E0 to E16)**. The families that the workbench reads are:

| Experiment | Runs | Varied | Used for |
|---|---|---|---|
| E2 | 200 | Trigger suction | Trigger correction, Eq. (5.11) |
| E3 | 200 | Sensor depth | Sensor-depth correction, Eq. (5.11) |
| E4 | 100 | Method (overhead, surface, drip at three depths) | Method correction, Eq. (5.11) |
| E5 | 300 | Tillage × structural consolidation rate | Baseline **b**, Eqs. (5.4) to (5.10) |
| E7 | 200 | Mulch vapour-flux factor × interception capacity | Soil evaporation $\bar{E}$, Eq. (5.4) |
| E11 | 1,000 | Temperature shift × evaporation factor × rain scale | Warming ratios, Eqs. (5.4) to (5.10) |
| E13 | 1,000 | Nitrogen rate × water strategy × soil | Nitrogen correction, Eq. (5.11) |

| Experiment | Runs | Varied | Used for |
|---|---|---|---|
| E14 | 1,000 | Mulch factor × capacity × rain scale | Mulch ratios, Eqs. (5.4) to (5.10) |

(The other experiments are used by the repository figures and described in the repository README. The *Regime map* and *Fate of saved water* layers are computed from the E14 and E11 grids listed above.)

$$S_e(h) = \left[1 + (\alpha|h|)^n\right]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

*Provenance (Eq. 5.1):* **[Literature]**: van Genuchten (1980).

(van Genuchten, 1980), with relative hydraulic conductivity from the Mualem (1976) model,

$$K_r(S_e) = S_e^{1/2}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2. \tag{5.2}$$

*Provenance (Eq. 5.2):* **[Literature]**: Mualem (1976) with van Genuchten (1980).

Equation (5.2) is used in the Soil, plant and air layer (Section 7.2); Eq. (5.1) is used for all water-retention drawings. The parameters $(\alpha, n)$ used by the engine are *effective* values that pass through the four anchors of the project's trigger table; for the standard textures they match the Carsel and Parrish (1988) values (Table 5.3).

### 5.1.2 Plant-available water and depletion

Depletion of plant-available water is defined between field capacity (suction 100 cm, pF 2.0) and wilting point (15,849 cm, pF 4.2):

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

*Provenance (Eq. 5.3):* **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

$f = 0$ is field capacity and $f = 1$ is wilting point; $f < 0$ (wetter than field capacity) is kept as the *signed* value only in the Trigger law layer. The inverse $h(f)$ is found by bisection on $\log_{10}|h| \in [0, 4.6]$ (60 halvings).

Table 5.3. Effective van Genuchten parameters used by the engine, plant-available water, and the soil's critical suction at the soil-independent depletion $f^* = 0.45$.

| Soil | $\alpha$ (cm$^{-1}$) | $n$ | AWC (mm cm$^{-1}$) | $h^*$ at $f^* = 0.45$ (cm) | $f$ at 300 cm |
|---|---|---|---|---|---|
| S01 Coarse Loam | 0.0961 | 1.353 | 1.5 | 387 | 0.380 |
| S02 Coarse Sand | 0.0536 | 1.600 | 0.5 | 262 | 0.496 |
| S03 Sand | 0.145 | 2.68 | 0.6 | 143 | 0.842 |
| S04 Loamy Sand | 0.124 | 2.28 | 0.9 | 160 | 0.756 |
| S05 Sandy Loam | 0.075 | 1.89 | 1.2 | 195 | 0.627 |
| S06 Loam | 0.036 | 1.56 | 1.7 | 284 | 0.468 |
| S07 Silt Loam | 0.020 | 1.41 | 2.0 | 401 | 0.367 |
| S08 Sandy Clay Loam | 0.059 | 1.48 | 1.4 | 311 | 0.439 |
| S09 Clay Loam | 0.019 | 1.31 | 1.6 | 503 | 0.314 |
| S10 Silty Clay Loam | 0.010 | 1.23 | 1.7 | 729 | 0.237 |

(AWC is "typical available water capacity by texture class" in mm of water per cm of soil; it is an engineered per-texture constant, not a measured value of the study soil.)

### 5.1.3 The meta-model

Let $\mathbf{b} = (E_b, T_b, ET_b, I_b, Y_b, D_b, N_b)$ be the 20-year-mean Daisy results for the chosen soil, tillage and consolidation rate (run set E5), and $\mathbf{b}_{\text{ref}}$ the same for conventional tillage at $\tau_c = 0.005$ $\text{d}^{-1}$. Let $\varphi$ be the mulch strength, $c$ the mulch capacity, $\rho$ the rainfall scale and $\Delta T$ the warming. Define the mulch table lookups $M(\varphi, \rho)$ (E14 at capacity $c$) and warming table lookups $W(\Delta T, \rho)$ (E11); write

$$\text{now} = M(\varphi,\rho),\ \text{ref} = M(\varphi,1),\ \text{nm} = M(1,\rho),\ \text{bare} = M(1,1),\quad \text{tN} = W(\Delta T,\rho),\ \text{tB} = W(0,\rho),\ \text{tR} = W(0,1).$$

Then the engine computes (the ratio $[x/y]$ is 1 where $|y| < 10^{-6}$):

$$E = \bar{E}(\text{soil},\varphi,c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

$\bar{E}(\text{soil},\varphi,c)$ is the linear interpolation, in $\varphi$, of the soil's simulated soil evaporation (run set E7) at capacity $c$, closed at $\varphi = 1$ by the bare-soil value. Fluxes are combined multiplicatively and drainage/leaching additively because ratios of small numbers (drainage near 0 mm) are unstable.

Four *set-specific corrections* then act in turn on the result. For each family $F \in \{$trigger (E2), sensor depth (E3), method (E4), nitrogen (E13, yield and $N_L$ only)$\}$ with current key $\kappa$ and baseline key $\kappa_0$ (trigger $-300$ cm; depth 20 cm; method Surface; nitrogen 100 %), with $r_\kappa$ the soil's row at $\kappa$ (nearest row if that exact run is absent):

$$Q \leftarrow Q\ \text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ \ (Q \in \{E,T,ET,I,Y\}),\qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D),\quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N). \tag{5.11}$$

*Provenance (Eq. 5.11):* **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

**Families whose reference run never irrigates (release 1.1).** In the sand and the loamy sand the sensor-triggered runs at the reference trigger ($-300$ cm at 20 cm, surface method) apply no irrigation at all: the coarse soil drains past the sensor before it dries to that suction. The ratio of Eq. (5.11) is then undefined, and version 1.0 returned 1, so that the trigger, the sensor depth and the method had no effect on the irrigation of these soils. Version 1.1 adds the simulated difference instead,

$$I \leftarrow I + \sum_{F \in \{\text{E2,E3,E4}\}} \max(0,\ r_\kappa.I - r_{\kappa_0}.I)\quad \text{when } r_{\kappa_0}.I < 1 \text{ mm},\qquad \Delta S = P + I - ET - D, \tag{5.12a}$$

*Provenance (Eq. 5.12a):* **[Data-driven]**: differences of Daisy run sets E2 to E4; the storage term closes the balance.

so that in the sand a trigger of $-50$ cm (122 mm more in the runs) gives 400 mm a year instead of 278 mm, $-75$ cm gives 317 mm and $-100$ cm 285 mm; drier triggers keep the reference amount, as in the runs.

Finally the model enforces non-negativity, a soil-dependent yield ceiling $Y \le Y_{\max}(\text{soil})$ (the largest simulated yield), rain $P = P_{\text{base}}\rho$, and **water-balance closure**:

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, $ET$ is clipped and $E, T$ are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

*Provenance (Eq. 5.12):* **[Engineered]**: closure constraint and bounds chosen for the workbench. $E_{\text{other}}$ is interception plus evaporation from the mulch layer, and $\Delta S$ collects the change in storage and small terms. IE converts Mg ha$^{-1}$ per mm of irrigation to kg of grain per m$^3$ of irrigation water (1 mm $= 10$ m$^3$ ha$^{-1}$).

**Crops.** The simulated crop is the cereal/oilseed rotation; its yield is shown in Mg ha$^{-1}$. The 48 crops of the library are transferred from it by the ratio of two FAO-56 daily soil-water balances, one for the library crop and one for the cereal reference, run on the same soil, weather and trigger (Section 21.4, Eqs. 21.5 to 21.9). Their yield is shown as the relative yield, in per cent of the potential.

### 5.1.4 What "20-year average at these settings" means

Every Dashboard tile in the last block of the Results tab is the output of Eqs. (5.4) to (5.12) for the current controls. Every slider of Properties is an index into a list of simulated values, so no value ever lies between runs; but two controls changed together (for example, mulch and warming) use the product of ratios above, which assumes the effects are **independent**. The yield ceiling and the closure window (Eq. 5.12) stop independent effects from compounding beyond what the water and the crop allow.

## 5.2 Engine 2: the daily bucket model

### 5.2.1 Layout and state

The soil is $N = 40$ layers of thickness $\Delta z = 2.5$ cm (100 cm total). Each layer holds water $w_i$ (mm) between 0 and the plant-available capacity $W_{\text{max}} = \text{AWC} \cdot \Delta z$. All layers start at $0.72\,W_{\text{max}}$. Relative content $s_i = w_i / W_{\text{max}}$ and depletion $f_i = 1 - s_i$. The run is 20 years of 365 days (7,300 days). The sensor sits in layer $j_s = \text{round}(z_s / \Delta z)$ for sensor depth $z_s$.

The trigger suction $|h_T|$ is converted to a depletion fraction $f_T = f_{\text{used}}(|h_T|)$ by Eq. (5.3). The sensor condition "soil at suction $h_T$" therefore becomes the equivalent statement $f_{j_s} \geq f_T$.

### 5.2.2 Crop calendar and canopy cover

The cereal crop follows the four-year rotation spring barley, winter barley, winter rape, winter wheat with sowing and harvest days (relative to 1 Jan of the year) $(100, 225)$, $(-90, 205)$, $(-125, 200)$, $(-80, 230)$; winter crops are sown in the previous autumn. Legume: sown day 110, harvested 215. Vegetable: sown day 100, harvested 185. Outside the window the field is *fallow*. Canopy cover $c(d) \in [0, 1]$ is

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

*Provenance (Eq. 5.13):* **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

*Provenance (Eq. 5.14):* **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

and after the peak day $d_{\text{pk}}$ cover declines linearly to zero at harvest: $c \leftarrow c\,\max\{0,\,1-(d-d_{\text{pk}})/(d_{\text{harv}}-d_{\text{pk}})\}$. The mid day is $d_{\text{sow}}+50$ (cereal), $d_{\text{sow}}+40$ (legume), $d_{\text{sow}}+28$ (vegetable); the peak is harvest $-50$ (cereal), sow $+65$ (legume), sow $+65$ (vegetable). These logistic curves are **[Engineered]**: they reproduce the shape of a canopy, not a measured leaf-area record.

### 5.2.3 Rainfall generator

Rain falls on a day with probability 0.28. With $\mu_0 = P/(0.28 \cdot 365)$ the mean wet-day depth of the scaled annual rain $P$, the depth on a wet day of simulated year $y$ is

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0,1), \tag{5.15}$$

*Provenance (Eq. 5.15):* **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

an exponential distribution capped at 70 mm, with the random sequence seeded by a hash of the soil name so every run is reproducible. The year factor $\gamma_y = \text{clip}\big((m_{\text{med}}/m_y)^{0.3};\,0.65,\,1.45\big)$ makes years wetter or drier according to the dryness $m_y$ of the study's 20-year rainfed suction record at 20 cm, where $m_y = 10^{\overline{\log_{10} h}}$ is the geometric-mean suction over the 183-day growing season of year $y$, and $m_{\text{med}}$ is the median over the 20 years. This is how the generated weather "follows the dryness of the real record".

The canopy and litter catch the first $I_c$ mm of every shower:

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

*Provenance (Eq. 5.16):* **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

$P_{\text{other}}$ is the interception-and-mulch loss from Eq. (5.12). Only throughfall $T_f > 0.3$ mm enters the soil. The choice of $I_c$ makes the expected interception over the exponential rain distribution equal to the annual interception of Engine 1 (the expected value of $\min(R, I_c)$ for an exponential with mean $\mu_0$ is $\mu_0(1-e^{-I_c/\mu_0})$, and $e^{-I_c/\mu_0} = r$ gives $\mu_0(1-r)$ per wet day).

### 5.2.4 Infiltration, redistribution and drainage

Water enters the top layer and cascades down: any layer exceeding $W_{\text{max}}$ passes the excess to the layer below; the excess leaving layer 40 is **drainage** $D_d$. The depth of the deepest wetted layer is the **wetting front** (cm). Between layers a slow redistribution moves water toward the drier neighbour,

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

*Provenance (Eq. 5.17):* **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

shown as *capillary rise* arrows when $q_i > 0.03\,W_{\text{max}}$ upward. Drainage and the wetting front are therefore **tipping-bucket** results, not Richards solutions; that is why the engine scales them to Engine 1 (Eq. 5.24).

### 5.2.5 Roots

The maximum rooting depth is

$$R_{\text{max}} = \text{clip}\Big(R_c\,(0.85 + 0.3\,\text{sand})\big[1 + 0.12\log_{10}(|h_T|/300)\big]\big[1 - 0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

*Provenance (Eq. 5.18):* **[Engineered]**: rooting rules with constants chosen for the workbench.

with $R_c = 95$ (cereal), 115 (legume), 45 (vegetable) cm, sand the sand fraction, and $N$ the nitrogen rate in percent. (Drier triggers push roots deeper; nitrogen-rich soils root slightly shallower.) While the canopy is growing, the rooting depth $R_d$ chases water:

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\,\mathrm{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35+\bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

*Provenance (Eq. 5.19):* **[Engineered]**: rooting rules with constants chosen for the workbench.

where $\bar{s}_{\text{wet}}$ is the mean relative water content of the four layers beneath the root front, and $\kappa = 1.3$ if the top six layers average below 0.35 (dry topsoil) and 1 otherwise. After harvest $R_d \leftarrow \max\{3, 0.965R_d\}$; drought prunes the roots by 0.9 cm per day when the plant-available-water-weighted root fraction is below 0.12 and $R_d > 8$ cm. Root density follows

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

*Provenance (Eq. 5.20):* **[Engineered]**: rooting rules with constants chosen for the workbench.

with $\lambda = 0.32$ (cereal), 0.24 (legume), 0.40 (vegetable) and $z_d$ the drip emitter depth. The bracket is **[Engineered]**: it raises root density up to fourfold around a drip emitter. These root functions are the *modelled* roots drawn in the R1 layer, "not a scan of real ones".

### 5.2.6 Transpiration, evaporation

Daily soil evaporation is the annual evaporation $E_{\text{yr}}$ of Eq. (5.4) distributed over the year with a sinusoidal weight, reduced by canopy shade,

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65\cdot\tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

*Provenance (Eq. 5.21):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is taken from the top four layers with weights $(0.55, 0.25, 0.12, 0.08)$, never drawing a layer below $0.05\,W_{\max}$. Crop transpiration demand is the annual transpiration spread by cover, times a **calibration factor** $k_t$,

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\text{ yr}} c, \tag{5.22}$$

*Provenance (Eq. 5.22):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is divided among the layers by *availability-weighted* root density,

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \mathrm{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

*Provenance (Eq. 5.23):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

limited so that no layer falls below $0.10\,W_{\max}$. The availability function $a_i$ is a linear stress function between 15 % and 75 % of plant-available water; it is **[Engineered]** in the spirit of the FAO-56 water-stress coefficient (Allen et al., 1998, Eq. 84), which is also linear in depletion.

### 5.2.7 Irrigation rule and calibration

An irrigation of dose $d$ (mm) is applied on day $d$ if *all* of these hold: canopy cover $c > 0.2$; $95 \le$ doy $\le 250$; at least $m$ days since the last irrigation ($m = 1$, or $m = \mathrm{clip}(\mathrm{round}(110\,d/\bar{I}); 2, 12)$

if $f_T < 0.1$, to space irrigations when the trigger is almost at field capacity); the yearly total would stay within the budget $6\bar{I}$ (practically no cap); the sensor layer is dry enough, $1-s_{j_s} \geq f_T$; $f_T < 0.995$; and, for drip, the layer at the emitter depth has depletion $\geq 0.12$. The dose is $d = \text{clip}(\bar{I}/10; 15, 35)$ mm, and for drip $d_{\text{eff}} = \max\{8, d/2\}$ delivered as small pulses *spread over* $\pm 3$ *layers around the emitter depth*; for other methods the dose enters at the surface. $\bar{I}$ is the annual irrigation of Eq. (5.7).

**Calibration of $k_t$.** The trigger decides *when* but the root-zone water use decides *how often*. To make the day-by-day model reproduce the annual irrigation of Engine 1, $k_t \in [0.6, 8]$ is found by eight bisection steps on the first 8 years so that the mean irrigation per year $I_{\text{sim}}$ matches $\bar{I}$ within 6 % (if the first guess $k_t = 1$ is already within 6 %, it is kept); the result is then used for all 20 years.

### 5.2.8 Scaling to the annual model

The daily results are multiplied by constants so that their 20-year means equal Engine 1:

$$S_P = \frac{P}{\overline{P}_{\text{sim}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T_{f,\text{sim}}}}, \quad S_E = \frac{E}{\overline{E}_{\text{sim}}}, \quad S_T = \frac{T}{\overline{T}_{\text{sim}}}, \quad S_D = \frac{D}{\overline{D}_{\text{sim}}}, \tag{5.24}$$

*Provenance (Eq. 5.24):* **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

(each set to 1 if the simulated mean is below $10^{-6}$), and the daily fluxes displayed on the Dashboard are $R = P_d S_P$, $T_f = \min\{R, T_{f,d} S_{\text{thr}}\}$, $E = E_d S_E$, $T = T_d S_T$, $D = D_d S_D$ and irrigation $I$ unscaled. The **storage change of the day** closes the balance:

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

*Provenance (Eq. 5.25):* **[Literature]**: water-balance identity.

### 5.2.9 Evaporation under warming (used by the E1 layer)

For the Evaporation layer a separate atmosphere factor is used (Section 9.3):

$$E_{0,\text{bare}} = 4.2\,(1 + 0.07\,\Delta T)\,\frac{0.35 + 0.65 \cdot \frac{1}{2}[1 + \sin(2\pi(\text{doy} - 100)/365)]}{0.675}\,\text{clip}(1 - 0.12(\rho - 1);\, 0.85,\, 1.12)\sqrt{\gamma_y}\ \text{mm d}^{-1}, \tag{5.26}$$

*Provenance (Eq. 5.26):* **[Engineered]**: atmosphere factor with chosen constants.

where 4.2 mm d$^{-1}$ is the engine's season-mean bare-soil demand (the seasonal weight is divided by its mean, 0.675), the factor $1 + 0.07\Delta T$ is a 7 % per °C increase (close to the Clausius and Clapeyron rate of the saturation vapour pressure; **[Engineered]** parameter), the cloud factor reduces demand in wetter years, and $\sqrt{\gamma_y}$ links demand to the dryness of the year.

## 5.3 The design layers and their link to the engines

The three design layers take from the engines:

| Quantity | Source | Used by |
|---|---|---|
| Soil texture (sand, silt, clay %) | `DATA.texture` (measured for the 10 soils) | Saxton and Rawls intake (Section 11.2) |
| Trigger depletion $f_T$ | Eq. (5.3) at $h_T$ | Required depth $Z_n$ |
| Maximum rooting depth $R_{\text{max}}$ | Eq. (5.18) | $Z_n$ |
| Plant-available water per cm AWC | Table 5.3 | $Z_n$ |
| Peak crop use $ET_c$ | engine's peak daily transpiration plus evaporation | Interval $T$ (drip, sprinkler) |

| Quantity | Source | Used by |
|---|---|---|
| Date and irrigation dates | Engine 2 | Irrigation schedule tables |
| Weather source, mulch, interception | Engine 1 and mulch capacity | sprinkler interception loss |

The default required depth is therefore

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\text{max}};\ 25,\ 150\big)\ \text{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

which is the net depth that refills the root zone from the trigger depletion back to field capacity ("*from the DSS*" on the screen). Typing a value overrides it (the label then reads *tap to reset*; click the label to return to the computed value).

## 5.4 Reproducibility

All random numbers in the engine are generated by a deterministic generator seeded by the soil name (a 32-bit hash), so the same controls always give the same dates, depths and animations. There is no external data call; the engine file contains the simulation tables as JSON.

# Chapter 6

# ☂ Irrigation control layers

The two layers of this group answer one question: **when should irrigation start?** *Sensor and trigger* shows what a sensor sees through twenty years; *Trigger law* shows why the same suction means different things in different soils and how to express the trigger so that it works for every soil.

## 6.1 Sensor and trigger

### 6.1.1 What it shows

A simulated tensiometer at the **sensor depth** (default 20 cm) records soil-water suction $h$ (cm of water, plotted on a log axis from 1 to 100,000 cm) every day. Two traces are drawn: *irrigated* (blue) and *rainfed* (orange). The red dashed line is the **trigger** you set. Pale bands are growing seasons. Under the trace, red ticks mark irrigated sensor days beyond the trigger, and the bar chart counts, for each year, the growing-season days (of 183) on which the sensor was beyond the trigger, irrigated against rainfed.

### 6.1.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Sensor and trigger*.
2. Choose **All 20 years** to see the whole record, or **1994 detail** for the one year with full-profile output (the buttons are on the playback bar under the drawing).
3. Move **Trigger (sensor suction)** in Properties; the red line and every count update.
4. Move **Sensor depth**, the soil, the mulch, the method or any other control: both traces follow it (Eq. 6.4a), and the sentence under the drawing names the record and the settings it is drawn for.
5. Press the play button under the drawing to sweep a cursor through the record; the working year is set in Properties ▸ Scenario.
6. Read the text under the plot: *"Irrigated: 85 of 183 days beyond 300 cm, mean suction 255 cm. Rainfed: 168 of 183 days beyond, mean suction 1,945 cm. Irrigation removes 39 % of the days beyond this trigger."*

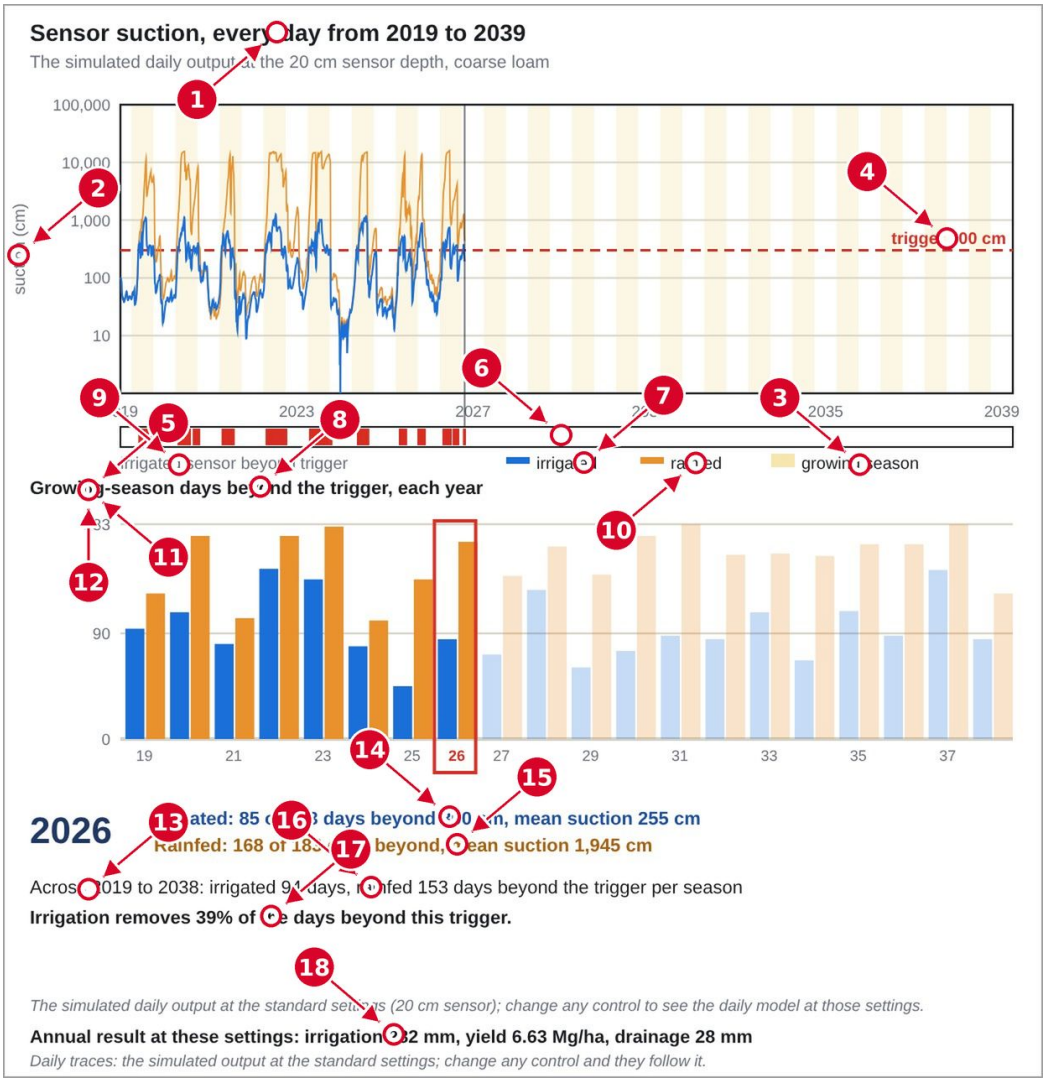



Figure 6.1: Sensor and trigger. Top: 20 years of suction at the sensor, irrigated (blue) and rainfed (orange) with the trigger (red dashed). Bottom: growing-season days beyond the trigger per year. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

### 6.1.3 How the numbers are generated

- The two suction traces start from the Daisy-simulated daily suction at 20 cm (irrigated with the study's trigger, and rainfed) for the 1980 to 1999 record, stored in the engine (`D20`) for the Coarse Loam and the Coarse Sand. At the standard settings of those two soils they are drawn as stored. For any other soil, sensor depth or setting the record is moved by the **delta method**: the daily model (Engine 2) is run twice, at your settings and at the standard settings of the same soil and weather, and the difference of the depletion of the sensor layer (smoothed over seven days) is added to the depletion of the stored record, which is then converted to suction in your soil:

$$h_d^{\mathrm{x}} = h_{\mathrm{soil}}\Big( f_{\mathrm{ref}}\big(h_d^{\mathrm{rec}}\big) + \tfrac{1}{7}\sum_{q=-3}^{3}\big[f_{d+q}^{\mathrm{set}} - f_{d+q}^{\mathrm{std}}\big]\Big), \tag{6.4a}$$

*Provenance (Eq. 6.4a):* **[Data-driven + Engineered]**: the stored simulated record carries the real weather sequence; the change made by your settings is the difference of two runs of the daily

model, converted with the van Genuchten curve of your soil (the inverse of Eq. 5.3, tabulated).

with $f_{\text{ref}}$ the depletion of the reference soil whose record is used (the nearer of the two in texture), $f^{\text{set}}$ and $f^{\text{std}}$ the daily depletion of the sensor layer in the two runs (irrigated or rainfed as the trace), and $h_{\text{soil}}(f)$ the suction of your soil at depletion $f$. * The count of days beyond the trigger is, for year $y$,

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

*Provenance (Eq. 6.1):* **[Data-driven]**: counts over the simulated daily series.

and the "Irrigation removes $X$ %" statement is

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_{y} N_y. \tag{6.2}$$

*Provenance (Eq. 6.2):* **[Data-driven]**: counts over the simulated daily series.

- The annual line at the foot ("irrigation 282 mm, yield 6.63 Mg ha$^{-1}$, drainage 28 mm") is Engine 1 (Eqs. 5.4 to 5.12) at the current settings.

### 6.1.4 Reading it

If rainfed days beyond the trigger greatly exceed irrigated days, irrigation is doing its job; if the two bars are similar the trigger is so dry that irrigation rarely starts. A trigger of $-50$ cm (very wet) produces ticks on almost every day.

## 6.2 Trigger law

### 6.2.1 The idea

The same suction does not mean the same dryness in two soils. At $-300$ cm a sand has lost 84 % of its plant-available water while a silty clay loam has lost only 24 % (Table 5.3, last column). N1 re-expresses the trigger as the **depletion fraction** $f$ of Eq. (5.3), the share of plant-available water already used, and shows that irrigation and yield responses from all ten soils collapse onto a common curve when plotted against $f$.

**[Literature + study result].** The depletion form follows the management-allowed-depletion idea of FAO-56 (Allen et al., 1998, Eq. 83 and Table 22): irrigate when a fixed share $p$ of the total available water is used. The *soil-independent* critical value $f^* = 0.45$ is the study's result (5 % yield loss), and the pooled $R^2$ rising from 0.48 to 0.75 (irrigation) and 0.23 to 0.70 (yield) is reported in the repository.

### 6.2.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Trigger law*.
2. The soil column on the left is the soil by depth on the clock's day; the blue gauge beside it shows depletion from FC (0) to WP (1). **Drag the purple trigger marker** on the gauge: the soil's own trigger suction changes, and the right-hand list of ten soils moves in step.
3. Each row on the right is one soil with its trigger suction and the net irrigation depth at that depletion ("143 cm trigger, 278 mm"); the green box gives the relative yield at the soil's trigger.
4. Click a soil name to select it (it becomes the working soil).
5. **Common and own mode.** By default one suction (the Trigger slider) applies to every soil (*common* mode). **Opening the Trigger law layer switches the program to *own* mode**: each soil then uses its own critical suction for the chosen depletion, and the Trigger slider in Properties shows the nearest value on its 20-value grid (simulated runs exist only on that

grid). Own mode stays on until you choose *Reset all settings* or a ready-made case, which restore the common mode.

6. Read the verdict box at the bottom: it states the soil's critical suction, the yield at the current trigger relative to the best trigger, and how many root-zone layers are beyond the trigger on this day.

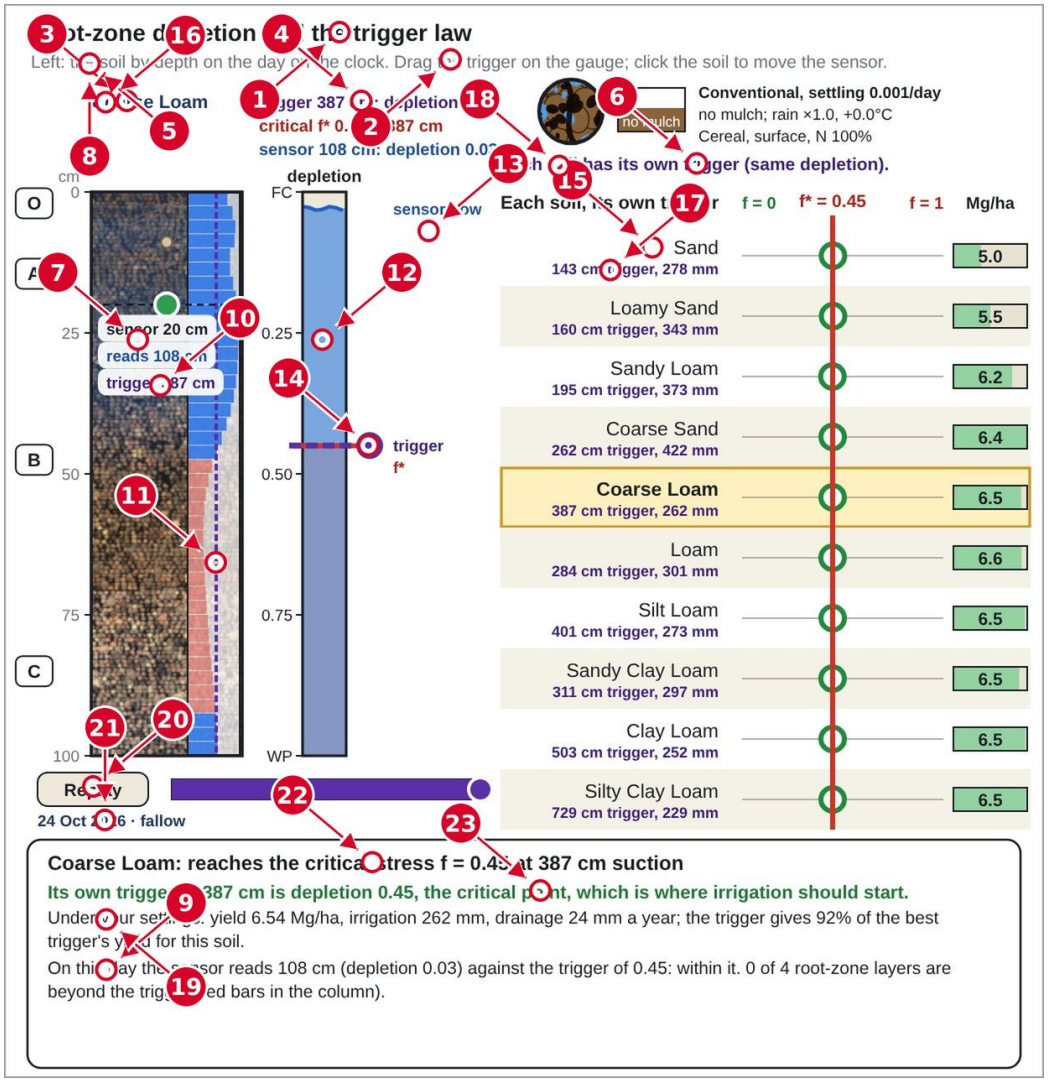



Figure 6.2: Trigger law layer: ten soils on one depletion axis, with the critical $f^* = 0.45$ (red line) and the soil's own trigger. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

### 6.2.3 Governing relations

The depletion of Eq. (5.3) is evaluated with each soil's effective van Genuchten pair $(\alpha, n)$ (Table 5.3). The suction at which a soil reaches a chosen depletion $f$ is the inverse,

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

*Provenance (Eq. 6.3):* **[Literature]**: inverse of the van Genuchten curve, solved by bisection. solved by bisection, giving for $f^* = 0.45$ the soil-specific sensor settings of Table 5.3 (143 cm for sand to 729 cm for silty clay loam). The net irrigation per event shown beside each soil is

$$Z_n = f \cdot \mathrm{AWC} \cdot R_{\max} \quad [\mathrm{mm}] \tag{6.4}$$

*Provenance (Eq. 6.4):* **[Literature]**: same relation as Eq. (5.27) without the clip.
(the same relation as Eq. 5.27 without the clip). The *relative yield at the trigger* is

$$Y_{\mathrm{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

*Provenance (Eq. 6.5):* **[Data-driven]**: ratio of Daisy run-set E2 yields.
with $Y(h)$ the soil's yield at each simulated trigger (run set E2). **[Data-driven]**.

### 6.2.4 How fast each soil dries

Each row of the soil list ends with the days from field capacity to the trigger, and the box under the drawing tells how fast the soil in focus dries. Every soil is compared on the same crop demand: the mean evapotranspiration of the peak of the season (days 150 to 220), from the daily model of a well-watered coarse loam at all your other settings. Only the water each soil holds differs:

$$r_s = \frac{ET_c}{\mathrm{TAW}_s}, \qquad \mathrm{TAW}_s = \mathrm{AWC}_s \, z_r, \qquad t_{\mathrm{trig},s} = \frac{f_{\mathrm{trig},s}}{r_s}, \tag{6.6}$$

*Provenance (Eq. 6.6):* **[Literature]**: the FAO-56 root-zone depletion rate; $\mathrm{AWC}_s$ is the available water of the soil (mm per cm), $z_r$ the root depth of the reference run and $f_{\mathrm{trig},s}$ the depletion at the soil's own trigger.

A sand holds about 0.6 mm per cm, so the same demand empties it in about a third of the time of a loam: at the default settings, sand reaches its trigger in about 7 days and coarse loam in about 18. When you press **Play**, the gauge follows the daily model day by day, and its response is tied to the soil: it answers at once in a sand and slowly in a clay (rate $= 0.09 \times 1.5/\mathrm{AWC}_s$ per frame, between 0.04 and 0.35).

Choosing another soil, in the list or in Properties, starts the replay again from the start of the season (day 95), with the gauge set to the new soil.

### 6.2.5 The core trade-off: f = 0.50 or f = 0.45

Under the explanation, a panel answers the question each grower asks: trigger at a depletion of 0.50 or of 0.45? It has two columns, and each condition is ticked when it applies to your settings.

**Choose f = 0.50 if:**

- you irrigate a standard field crop (maize, grains, alfalfa): a field crop of the library whose FAO-56 depletion fraction is $p \geq 0.5$;
- your soil is loam or clay (high water-holding capacity): less than 60 % sand and more than 0.9 mm of available water per cm;
- your system can irrigate the entire field quickly: a full round of the design of your irrigation method takes 2 days or less.

**Choose f = 0.45 if:**

- your irrigation system takes several days to finish a full round: more than 2 days. A round is the time per round of the design divided by the hours a day you irrigate (24 h for surface irrigation).
- your soil is sandy or coarse and dries out very fast: 60 % sand or more, or 0.9 mm of available water per cm or less;
- the crop is in a critical, water-sensitive growth stage such as flowering or pollination: on the date on the clock the crop is at 85 % or more of its full cover.

If any condition of the f = 0.45 column applies, the advice is f = 0.45; otherwise it is f = 0.50. A sentence gives the reasons. Each column has a button that sets the trigger of the soil in focus to

that depletion; the button of the trigger in use is green. The time per round is known once the design of your method has been opened; until then the panel says so.

### 6.2.6 Reading the sensor value

The dashboard tile *Sensor / trigger* reads two numbers: the **current depletion** $f$ at the sensor layer and the **trigger depletion** $f_T$. When the first reaches the second, irrigation is due. On 18 Oct 2026 the tile reads $0.02 / 0.38$ for the Coarse Loam: the soil is almost at field capacity (0.02), and the trigger is at depletion 0.38 ($= f$ at 300 cm).

# Chapter 7

# ⊞ Field water balance layers

## 7.1 Field balance (the home layer)

### 7.1.1 What it shows

One picture of the water budget of a field for one day, one year, or the twenty-year average: rain and irrigation arriving, transpiration, soil evaporation and interception leaving upward, drainage leaving downward, and the nitrogen leached with it. The soil profile is drawn to 100 cm with three depth marks A, B and C, the sensor at the chosen depth, the roots as they grow and die back, and the wetting front.

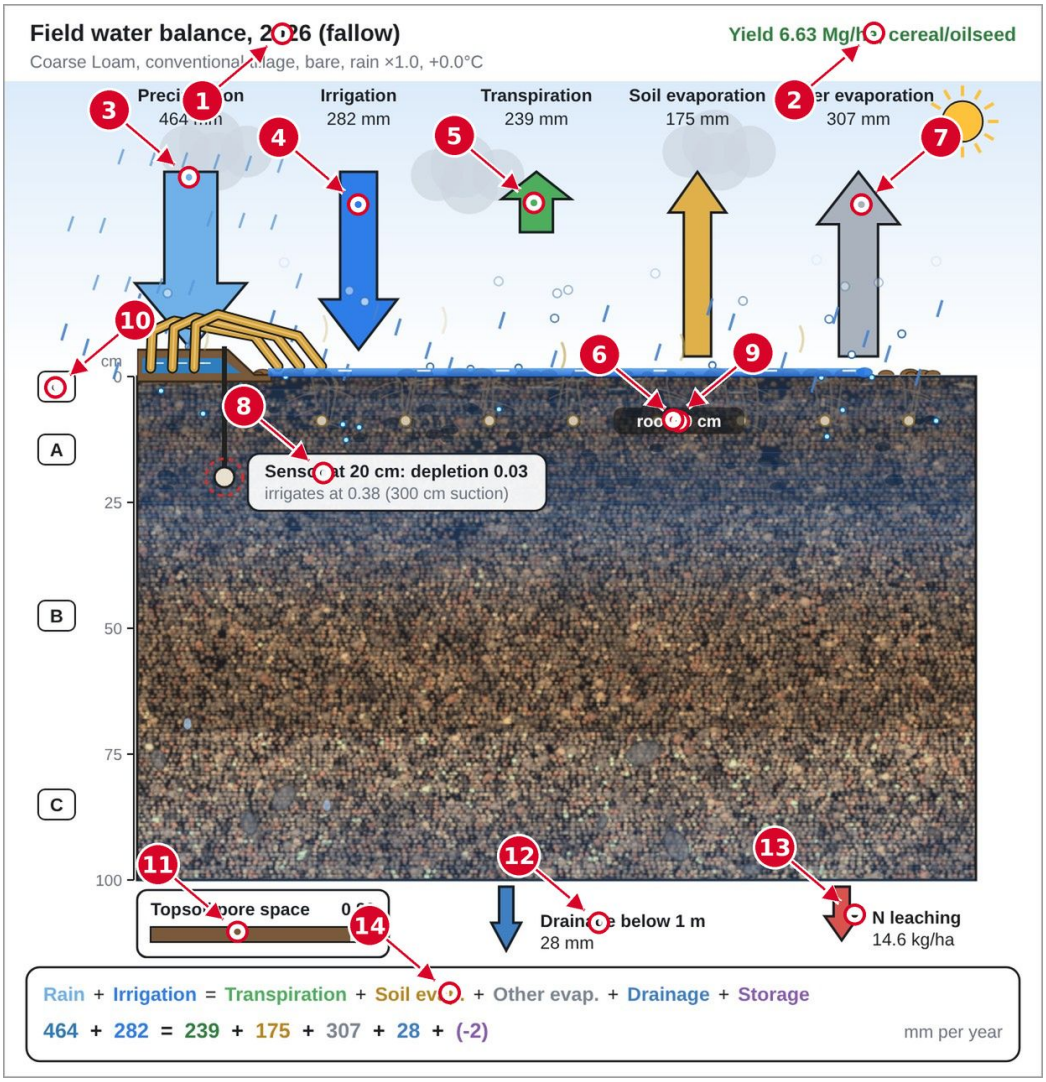



Figure 7.1: Field balance: arrows are the fluxes (mm per year in “Average year” view), the profile shows the root zone, the sensor, and drainage below 1 m. The equation strip under the picture is the closed budget. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

### 7.1.2 The control bar under the drawing

| Button | Meaning |
|---|---|
| **Day by day** | The simulated day on the field clock; arrows show that day’s fluxes ($T_f$, $I$, $T$, $E$, interception, $D$). |
| **All 20 years** | Table and bars of every simulated year (rain, irrigation, interception, evaporation, transpiration, drainage, storage change, crop, deepest roots). Click a year. |
| **Average year** | The 20-year means of Engine 1 (Eqs. 5.4 to 5.12). |
| **◀ year / year ▶** | Previous or next simulated year. |
| **1× 2× 4× 8×** and speed slider | Play speed. |
| **Play / Pause** | Animates the clock. |

### 7.1.3 How each output is generated

| On the picture | Generated by |
|---|---|
| Precipitation, Irrigation, Transpiration, Soil evaporation, Interception, Drainage (mm) in *Average year* | Engine 1: $P$, $I$, $T$, $E$, $E_{\text{other}}$, $D$ of Eqs. (5.4) to (5.12) |
| The same in *Day by day* | Engine 2 daily fluxes after scaling, Eqs. (5.15) to (5.25) |
| Equation strip "Rain + Irrigation = Transpiration + Soil evap. + Interception + Drainage + Storage" | The closure of Eq. (5.12): $P + I = T + E + E_{\text{other}} + D + \Delta S$ |
| Sensor label "depletion 0.02 / irrigates at 0.38" | Engine 2 layer $j_s$: $f_{j_s} = 1 - s_{j_s}$ against $f_T$ |
| Roots (depth, density) | Eqs. (5.19) to (5.20) |
| Wetting front | Depth of the deepest layer reached by the last infiltration event |
| Topsoil pore space bar | Pore index $p$ of Eq. (8.1) for the chosen tillage and consolidation rate |
| Drainage below 1 m | $D$ (mm), the water leaving layer 40 |
| N leaching (kg ha$^{-1}$) | $N_L$ of Eq. (5.10) |
| Yield tag (top right) | $Y \cdot f_{\text{crop}}$ from Eq. (5.8) |

### 7.1.4 Step-by-step: checking a budget

1. Open *Field balance* and choose **Average year**.
2. Read the five numbers over the arrows and the strip below. Check that rain plus irrigation minus the four sinks equals the storage change in brackets.
3. Change **Rainfall scale** to $×$1.3. The rain, drainage and N leaching arrows thicken.
4. Press **Pin results as baseline** (Ctrl+B) *before* each change to see the difference on the Dashboard tiles.
5. Click **Day by day**, Play, and watch irrigation events pause the clock (Section 4.3).

## 7.2 Soil, plant and air

### 7.2.1 What it shows

The soil, plant and atmosphere continuum as one water column "from the particles to the sky": a section of the soil with roots, the plant, and the air above; five irrigation layouts (**Furrow, Basin, Border strip, Sprinkler, Drip**) that you can tap, which set the irrigation method; and, on the right, the **water potential along the path** from the air to the deep soil. Two data sources are labelled with badges: *DAISY* (taken from your simulations) and *DERIVED* (computed by the workbench along the path).

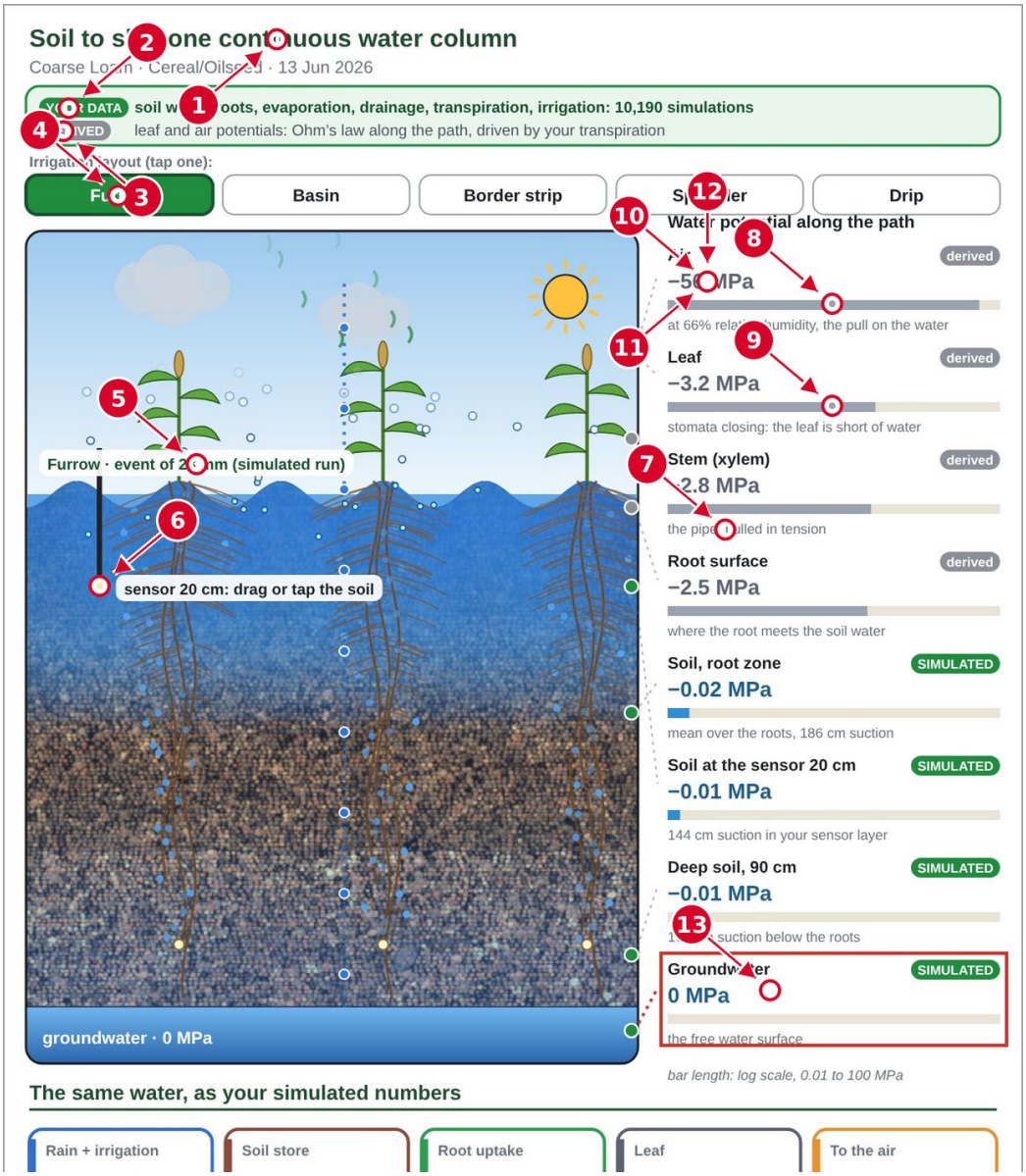



Figure 7.2: Soil, plant and air layer: potentials from the air to the deep soil. *Derived* badges mark the quantities computed along the path; *Simulated* badges mark simulation data. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

### 7.2.2 Governing equations

**Soil.** The suction in each layer is the suction that gives that layer's depletion, from the inverse of Eq. (5.3), floored at 30 cm:

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

*Provenance (Eq. 7.1):* **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

The *root-zone* suction is the root-density-weighted mean, $h_{RZ} = \sum_i \rho_i h_i \big/ \sum_i \rho_i$, and the *sensor* and *deep-soil* values are $h_{j_s}$ and $h_{N-3}$.

**Soil conductivity.** The relative conductivity at the root zone uses the Mualem and van Genuchten form, Eq. (5.2), with $S_e = S_e(h_{RZ})$ from Eq. (5.1).

**Along the path.** The plant is a chain of series resistances, in the spirit of the catenary (Ohm's-law) description of water flow through the soil, plant and atmosphere continuum (van den Honert, 1948). With $E$ the day's transpiration (mm d$^{-1}$, from Eq. 5.25) and $R_{\text{soil}}$, $R_{\text{root}}$, $R_{\text{xyl}}$, $R_{\text{leaf}}$ resistances in MPa per (mm d$^{-1}$):

$$R_{\text{soil}} = 0.03 \ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

*Provenance (Eq. 7.2 to 7.4):* **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

where $a = 1$ when canopy cover exceeds 0.05 (a crop is present) and $a = 0$ otherwise (fallow). Stomata are drawn as *closing* when $\psi_{\text{leaf}} < -1.6$ MPa.

**The air.** Water potential of the air above the canopy follows the thermodynamic relation between potential and relative humidity,

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH} \ \approx\ 135.5\,\ln \text{RH} \ \ \text{MPa} \tag{7.5}$$

*Provenance (Eq. 7.5):* **[Literature]**: Nobel (2009).

(Nobel, 2009), where $R$ is the gas constant, $T \approx 293$ K and $V_w = 18 \times 10^{-6}$ m$^3$ mol$^{-1}$ the molar volume of water. Relative humidity is RH $= \text{clip}(0.66 - 0.03\,\Delta T;\ 0.35,\ 0.90)$ with $\Delta T$ the temperature shift. At RH = 66 % the air is at $-56$ MPa, as shown.

**What is literature and what is engineered.** Eqs. (7.1), (7.5) and (5.2) are physical relations. The resistances in (7.2) to (7.4) are **[Engineered]** values chosen so that a typical transpiring cereal reaches $\psi_{\text{leaf}}$ of about $-1$ to $-1.5$ MPa and so that stress develops as the soil dries; they are *not* measured resistances, and the layer labels them "derived". **Caution.** Evaluated over mid-season crop days in all ten soils (Paper 3, Section 5.1), the relative conductivity at the root-zone suction is usually below the floor of 0.002 in Eq. (7.2), so $R_{\text{soil}}$ sits at its cap (0.92 MPa per mm d$^{-1}$) and the mean leaf potential is $-1.8$ to $-2.3$ MPa in every soil, below the intended range. The leaf, stem and root potentials of this layer therefore illustrate the series-resistance concept and should not be read as plant water status.

### 7.2.3 Step-by-step

1. Open *Soil, plant and air*.
2. Click a layout. Furrow, basin and border strip set the **Surface** method; Sprinkler sets **Overhead**; Drip sets **Drip 20 to 30 cm**.
3. Drag or click in the soil to move the **sensor depth** (the control snaps to the 5-cm list).
4. Change the **date** to a growing-season day and compare the potentials: with a crop present the leaf falls below $-1$ MPa as the soil dries; on a fallow date the path is flat (resistance terms are zero).
5. Move the **Trigger** slider and note that the *sensor* suction tile follows the same depletion as the dashboard.

## 7.3 Profile evolution

### 7.3.1 What it shows

How the pressure head profile develops through the season in a year, side by side for **irrigated** and **rainfed** soil columns, with the sensor depth shown by a dot. In *All 20 years* mode it draws the pressure head at the sensor through every year and a heat strip of the mean suction of each

growing season (darker is drier). Text states the driest and wettest rainfed season and the share of the 0 to 60 cm root zone beyond the trigger.

### 7.3.2 How it is generated

- The **1994** profile is the Daisy-simulated pressure head at every depth, irrigated and rainfed (the only year with full-depth output, as the drawing says).
- In the other years only the 20 cm sensor is available; the columns are shaded only around the sensor depth.
- *Share of the root zone beyond the trigger* is

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

*Provenance (Eq. 7.6):* **[Engineered]**: indicator defined for the workbench.

the layer-thickness-weighted fraction of the 0 to 60 cm profile whose suction exceeds the trigger, evaluated for the irrigated and the rainfed profile of 1994 (the year with full-depth output). * The mean suction of a season, used for the heat strip and for the "driest season" statements, is the geometric mean $m_y = 10^{\frac{1}{183}\sum_d \log_{10} h_d}$ over the growing season; the median across years $m_{\text{med}}$ also drives the rainfall year factor $\gamma_y$ (Eq. 5.15).

### 7.3.3 Step-by-step

1. Open *Profile evolution* and choose **All 20 years**.
2. Click a year in the heat strip to select it; click a column to put the sensor at that depth.
3. Move the trigger: the share beyond the trigger is recomputed.
4. Choose **1994 detail** to see the full profile through the season; press Play to step through it.

# Chapter 8

# ☷ Soil structure and tillage layers

## 8.1 Tillage and structure

### 8.1.1 What it shows

A modelled thin section (“microscope”) of the topsoil for the chosen soil and tillage, at three magnifications (10 mm, 2.5 mm, 0.6 mm), with pores filled by water or air according to a suction slider; the same place under the three tillage systems; the relative pore space against time; and the change in drainage and yield caused by tillage at your settings.

### 8.1.2 Pore index and consolidation

After tillage the loosened topsoil re-settles. The workbench uses a first-order relaxation of the **relative pore space** $p$ (1 = freshly loosened) to a consolidated value $p_c = 0.30$,

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

*Provenance (Eq. 8.1):* **[Engineered]**: exponential relaxation with chosen constants.

with $t$ in days since tillage and $\tau_c$ the **Structural consolidation rate** control (Table 3, 0.001 to 0.02 $\text{d}^{-1}$). **No-till** keeps a constant $p = 0.42$ (undisturbed pores). The half-life of loose pores is $\ln 2/\tau_c$: 693 d at 0.001 $\text{d}^{-1}$, 35 d at 0.02 $\text{d}^{-1}$. **[Engineered]**: the exponential form is the standard first-order consolidation relation also used in structure sub-models such as WEPP’s, on which Daisy’s tillage effect is based, but $p_0$, $p_c$ and the no-till value are *presentation* constants for the drawing; the *hydrological effect* of tillage shown in the bars is read from the Daisy runs (E5), not from Eq. (8.1).

### 8.1.3 How the section is generated

The section is a random packing of mineral grains and matrix drawn on a raster, with these parameters, all **[Engineered]**:

- Grain-size classes follow the soil’s measured sand, silt and clay percentages (`DATA.texture`).
- **Saturated water content** $\theta_s = \phi$ and **residual content** $\theta_r$ are

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

*Provenance (Eq. 8.2):* **[Engineered]**: porosity relation with chosen coefficients.

with clay and silt in percent, and the retention curve is the van Genuchten function of Eq. (5.1),

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

*Provenance (Eq. 8.3):* **[Literature]**: van Genuchten (1980).

- **Cracks and aggregates.** The number of cracks is $n_b = \text{round}(2 + 6(1 - f_{\text{till}}))$ with $f_{\text{till}} = 1,\ 0.7,\ 0.4$ for conventional, reduced, no-till; the crack width (mm) is $0.3\ \max\{0,\ p - 0.3\}^{1.3}$.
- **Which pores hold water.** A pore of radius $r$ fills with water at suction $h$ if $r \le r_c(h)$, where the capillary radius follows the Young and Laplace (Jurin) law,

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

*Provenance (Eq. 8.4):* **[Literature]**: Young-Laplace (Jurin) capillary law.

with surface tension $\sigma = 0.0728$ N m$^{-1}$, contact angle $\gamma \approx 0$ and $g = 9.81$ m s$^{-2}$ (Jurin, 1718; Hillel, 1998). In the drawing a pore is water-filled when its local half-width is at most $r_c$ **or** when it is within a film thickness $\min\{1.6,\ 0.9\,r_c/s + 0.3\}$ pixels of a grain, which draws the water films that survive drying.

- The hover probe states, for a pore, its radius, the suction at which it empties ($h = 1.49/r$) and the present suction.

### 8.1.4 Controls and steps

1. Open *Soil structure and tillage* ▸ *Tillage and structure*. Press **Play** to run a year of consolidation (the day appears under the picture).
2. Drag the **suction slider** (pF 0 to 4.2) on the microscope: at low suction the pores are water-filled; as suction rises the large pores empty first.
3. Switch the view: *Natural* (minerals), *Particle sizes* (sand/silt/clay classes), *Water and air*.
4. Change **Tillage** or the **Structural consolidation rate** and watch the curves of $p(t)$ (Eq. 8.1) and the section change.
5. Read the bars *Change in drainage at the current settings* (conventional, reduced and no-till against no-till) and the yields. These come from the Daisy runs for the selected soil: $\Delta D_{\text{conv}} = D_{\text{conv}} - D_{\text{notill}}$, $\Delta D_{\text{red}} = D_{\text{red}} - D_{\text{notill}}$, at the current $\tau_c$.
6. The sentence at the end states whether the consolidation rate changes anything: *"In this soil the consolidation rate changes nothing; tillage is the driver"* or *"Here the consolidation rate matters as well as tillage"*.

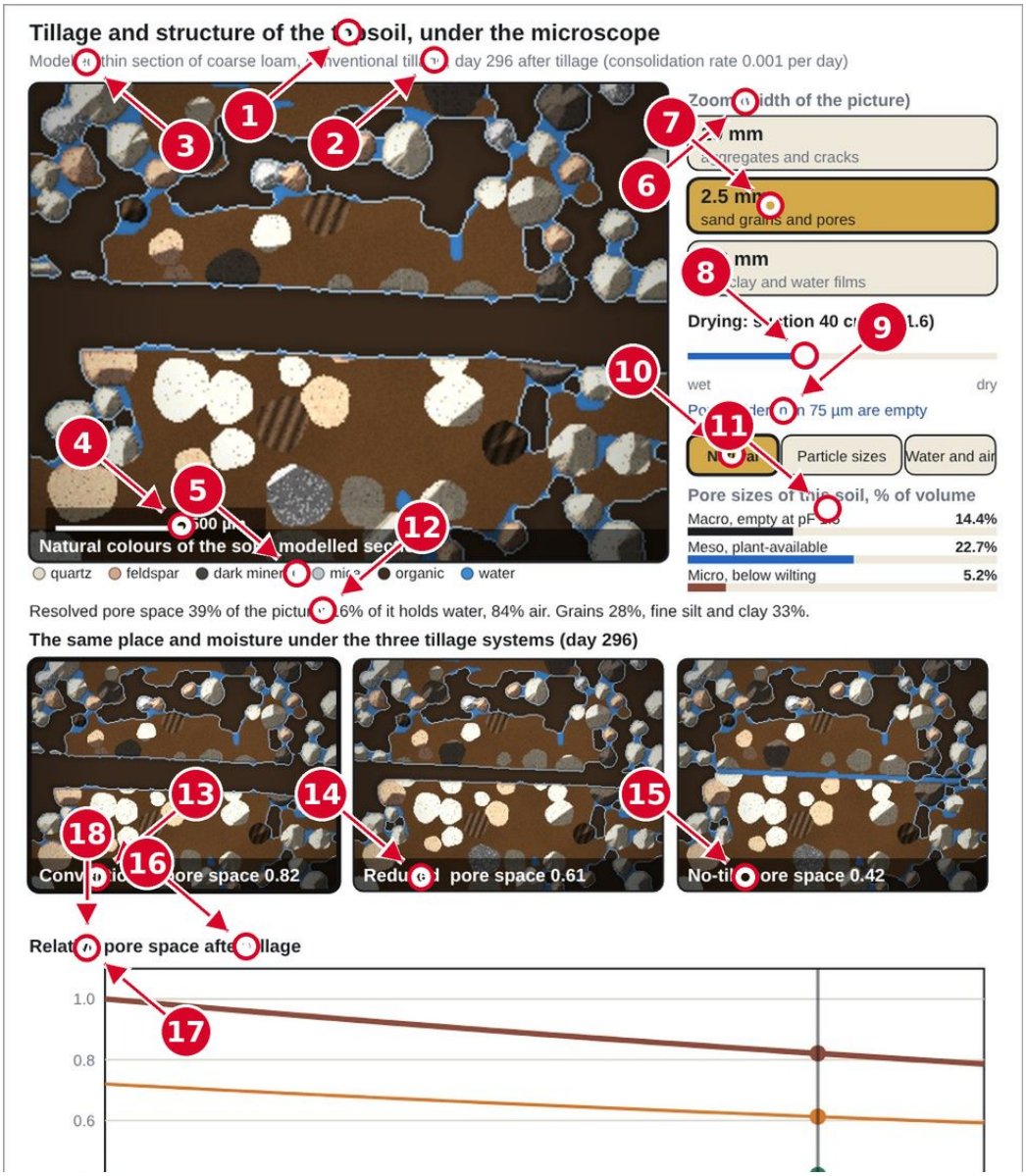



Figure 8.1: Tillage and structure layer: thin section, tillage comparison and the drainage consequences. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 8.2 Soil hydraulic drivers

### 8.2.1 What it shows

The question *"What drives change in soil hydraulic properties?"* answered for all ten soils together: for each soil sorted by clay content, a blue bar for the **tillage effect** (conventional minus no-till) and an orange bar for the **consolidation effect** (fastest minus slowest rate) on annual drainage, then scatter plots against clay content, and the 1-m soil profiles of the three tillage systems.

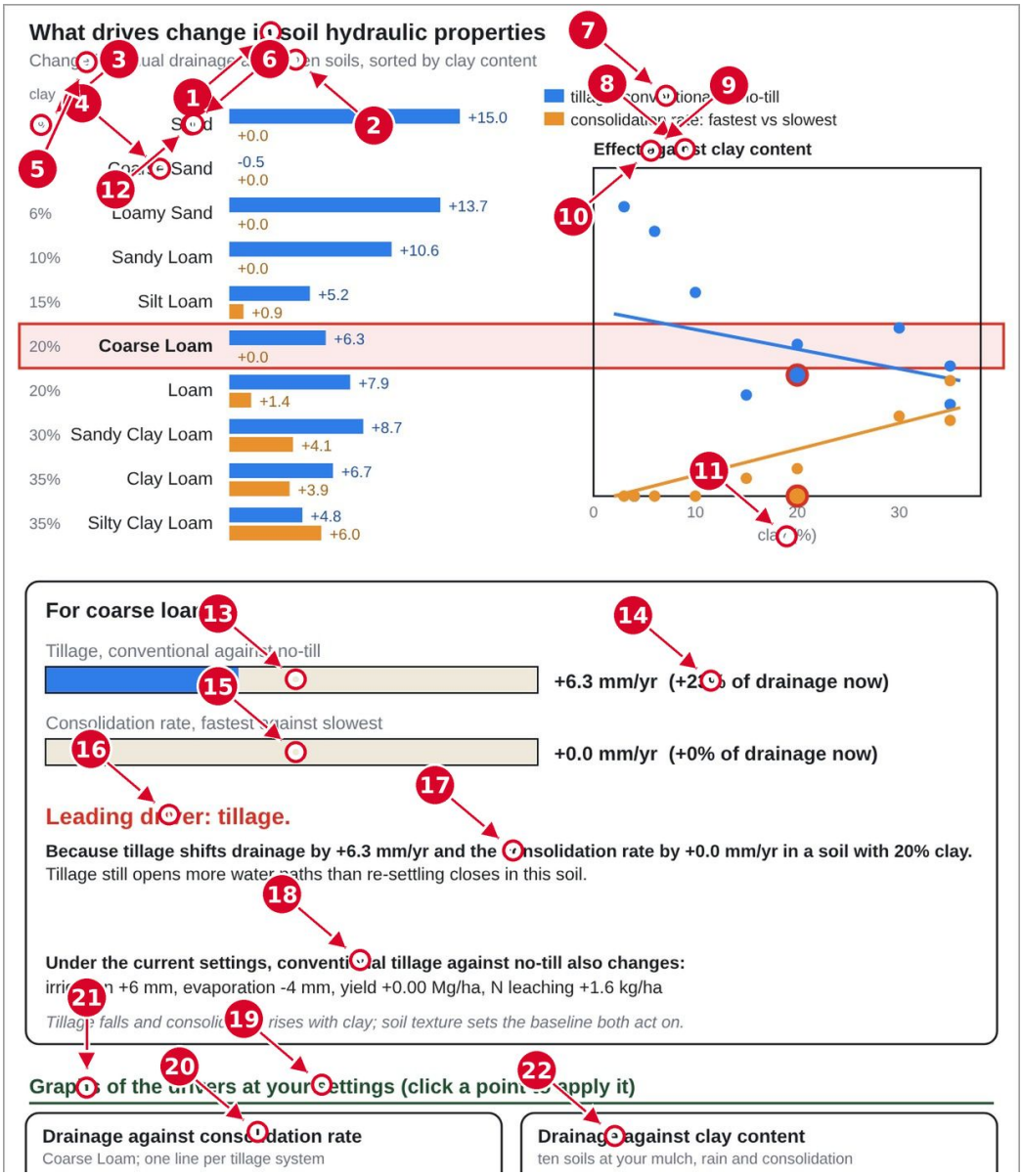



Figure 8.2: Soil hydraulic drivers. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

### 8.2.2 Generation

For each soil $s$ the Daisy run set E5 gives annual drainage $D_s(\text{tillage}, \tau_c)$. The two bars are

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

*Provenance (Eq. 8.5):* **[Data-driven]**: differences of Daisy run-set E5 results.

with the current settings as starred values; bar length is proportional up to 16 mm yr$^{-1}$. The lines in the scatter plots are ordinary least-squares fits of each effect on clay content $c$,

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

*Provenance (Eq. 8.6):* **[Data-driven]**: ordinary least squares applied to Daisy results.

The headline "Leading driver: tillage" or "consolidation rate" compares $\Delta_{\text{till}}$ with $\Delta_{\text{cons}}$ for the selected soil, and the percentage after each bar is $100\,\Delta/D_{\text{now}}$, the share of the present drainage. The last line adds, for conventional against no-till at the current mulch and rain, the changes in irrigation, evaporation, yield and N leaching, obtained by running the meta-model (Eqs. 5.4 to 5.12) twice (conventional at $\tau$ index 2 and no-till) and subtracting.

The lower part of the layer draws, for the three tillage systems, a 48 cm wide, 1 m deep face of the selected soil. The upper 24 cm (the tilled layer) is generated with the pore index $p$ of Eq. (8.1) at day 120 (adjustable), clod and crack density from $f_{\text{till}}$, and the wetness factor $\text{clip}\big((p-0.30)/0.14;\,0.3,\,1.4\big)\cdot\text{clip}(0.75+0.012\,\text{clay};\,0.8,\,1.4)$; below it the subsoil is untouched by tillage. The drainage figures printed under each column are Daisy results (E5).

### 8.2.3 Steps

1. Open the layer. Click any soil bar or any scatter point to select that soil.
2. Click a point on the *Drainage against consolidation rate* graph to apply a rate and a tillage system at once.
3. Read the verdict, and compare with the profile drawings.

## 8.3 Root structure

### 8.3.1 What it shows

**Rooting depth in all ten soils** at the same day and weather: for each soil a column with the deepest roots (20-year mean, brown dashed), and ticks at the depth that holds 50 % and 90 % of the water taken up (green ticks). Below it, a time view of rooting depth through the 20 years and the daily uptake, and a set of heat maps from the repository's **E15** runs: 1,000 Daisy runs on the Coarse Loam, rooting depth 30 to 180 cm $\times$ ten application amounts $\times$ ten irrigation methods, mapped to yield, irrigation, drainage, transpiration, N leached and water-stress days.

### 8.3.2 Generation

Roots and uptake come from Engine 2 (Eqs. 5.18 to 5.23). For each soil the engine is run in turn with that soil's parameters (and, in own mode, its own trigger). With per-layer uptake $u_i(d)$ on day $d$ the **depth holding a fraction $p$ of the uptake** is

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k:\ \frac{\sum_{i\le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d\in\text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

*Provenance (Eq. 8.7):* **[Engineered]**: diagnostic defined for the workbench.

with the layer index counted from 1 ($z_p$ is the *bottom* of layer $k_p$). It is computed for each of the 20 years from that year's uptake, then averaged over the years; the share of uptake in the top 30 cm is $\sum_{z_i\le 30} U_i \big/ \sum_i U_i$. The model stops roots at 1 m, whereas the Daisy runs reach 120 cm, as the layer notes. The statement "*Across the soils the deepest roots differ by …*" is the range of the 20-year-mean $\max_d R_d$ across soils.

**What these roots are.** The drawing states it: *"These are modelled roots, not a scan of real ones"*; root density is *relative* (Eq. 5.20), not measured root length. The E15 heat maps are **[Data-driven]** and exist for the Coarse Loam only ("they do not change with the soil you picked").

### 8.3.3 Steps

1. Open the layer. Click a column to make that soil the working soil.

2. Click or drag on the upper graph to move the day; the lower panel shows the uptake on that day.
3. In the E15 panel choose an output (yield, irrigation, …), click a rooting depth and an application amount in the heat map; the table shows the change against the present setting and tells you the best depth, for example *“Roots at 30 cm leave the crop stressed …”*.

## 8.4 Deterministic phenotyping layer

### 8.4.1 What the layer is

**P1 Deterministic phenotyping** (a layer group of its own in the Layers panel, after *Soil structure and tillage*, and the group *Phenotyping* of the layer buttons) is a virtual phenotyping experiment. One plant of the chosen crop grows day by day for 140 days after emergence in the soil, tillage, weather and rain of the project, twice, side by side: **well-watered** and **drought**. Everything the books on phenotyping let a person measure with a camera, a thermal camera, an X-ray scanner or a rhizotron is read from the simulated plant, and it can be moved: the day is chosen with the play bar or by clicking or dragging on any chart, and the plant and its roots are turned with the mouse.

The layer is **deterministic**. It has no random numbers: the same controls always give the same plant, so a change in a chart is always caused by the control that was moved. The weather of the season is built from the climate of the project (cool and humid), a warm season, or a hot and dry season, with the project rain or a rain-out shelter.

The layer is **connected to the Properties panel**: the crop and its library values (height, basal coefficient, rooting depth, sowing day), the soil, the weather and the date, the mulch (vapour-flux factor and interception), the irrigation (method, trigger suction, sensor depth, dose and efficiency), the nitrogen rate and the salt of the water all reach the plant. The **date of the toolbar** (the date control that also drives the soil trigger, set to today when the program opens) sets the season: the plant, the weather of that season, the day shown and the **drought outlook** all follow it, and changing the date computes the layer again without pressing the button. The header shows the date, and the time of day when the date is today (the model itself steps by whole days). Nothing is shown before it is computed: the layer opens with **Run simulation**, and it shows no default figure.

The layer is made for four tasks of a phenotyping programme:

1. **Quantify temporal changes in plant structure and function**: height, projected shoot area, convex hull, solidity, senescence, leaf water content, leaf potential, canopy temperature and root traits, each as a curve through the season and as a table of readouts.
2. **Investigate the relations between photosynthetic carbon assimilation, carbon allocation and the growth of roots and shoots**: the carbon section.
3. **Relate root development and stress responses to what is seen above ground**: the section *Roots from the plant above ground*, with twelve genotypes run under the same drought.
4. **Contribute experimental data to the parameterisation and validation of functional-structural plant models**: your own measurements can be loaded, compared, used to calibrate the model and exported.

*Provenance of the layer:* the forms of the equations are from the literature listed in the section *Equations* of this chapter. The two books that were the starting point define the **traits** (what is measured and how it is calculated); they do not give growth equations. The constants of the four species and the seven genotypes are typical, editable defaults of this layer. They are not data of an experiment, and no figure of this chapter is a measurement: each figure is the screen of the layer for the run named in its caption.

### 8.4.2 The controls

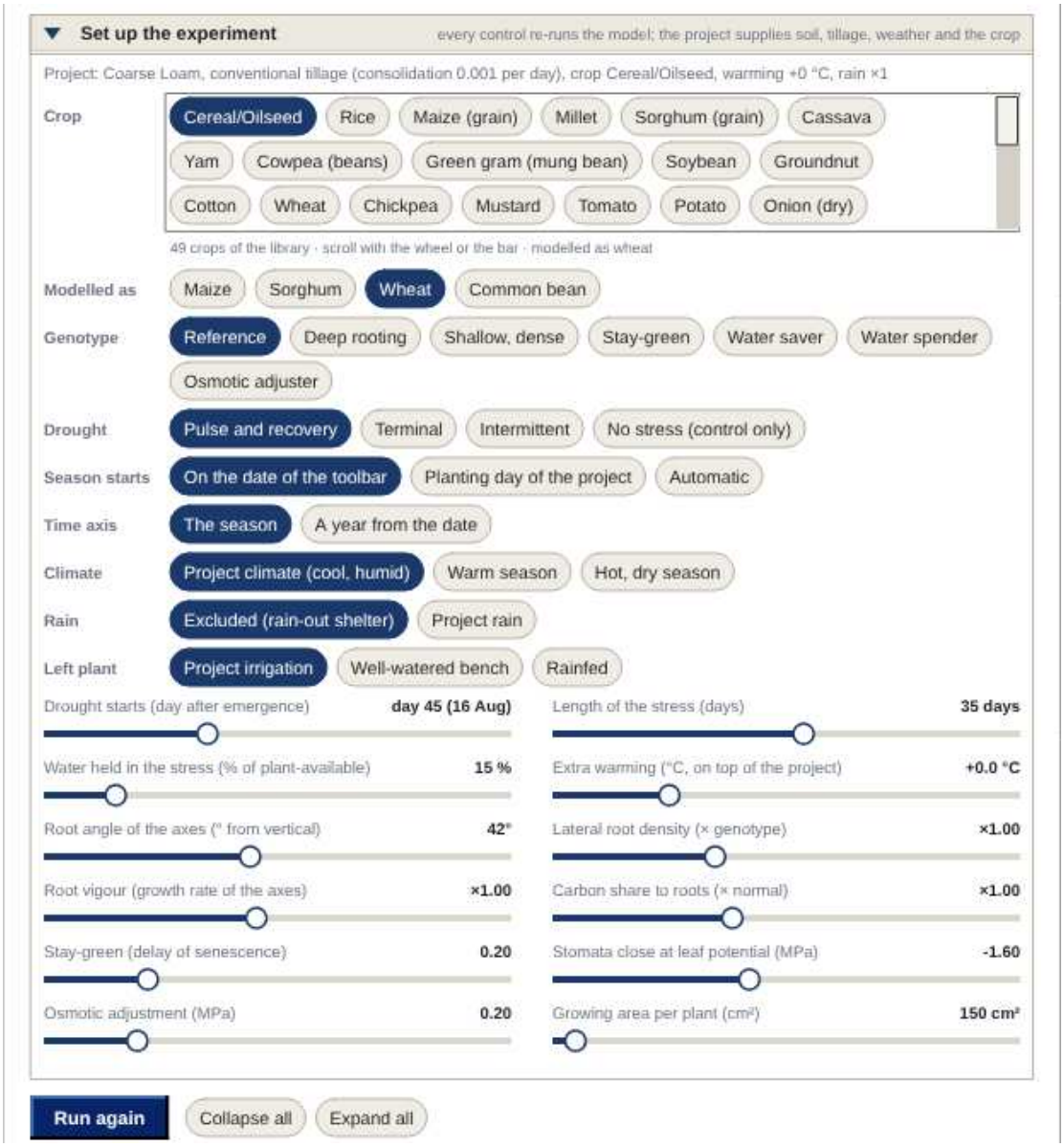



Figure 8.3: Set up the experiment. Maize, warm season, rain-out shelter, pulse of drought from day 40 for 28 days at 30 % of the plant-available water, reference genotype, Coarse Loam, conventional tillage.

**Run simulation.** The button is **under the inputs**, after the last control of the section *Set up the experiment* (the other layers of the window, whose inputs are in the Properties panel, have their own **Run simulation** bar at the top, Chapter 18). A layer shows its figures only after the button is pressed for the present controls; when a control changes, the bar asks to run again. The Play button of the toolbar does the same. **Every section of the layer has a triangle on its title: click it to collapse or expand the section**, so that a long page can be read part by part. After the run, the season **plays by itself**, from the first day to the last and again from the first, until **Pause** is pressed (the button of the toolbar, or the button *Pause* of the section *Season film*); clicking or dragging on any chart pauses it at that day.

Once run, every control re-runs the model, and every chart answers it.

| Control | What it changes in the model |
|---|---|
| Crop (a box with all the crops of the library, scrolled with the wheel or the bar; it sets the crop of the project) | height, basal crop coefficient and rooting depth of the crop of the library (Eq. P.15); the crop is run as the model species nearest in kind |
| Modelled as (maize, sorghum, wheat, common bean) | thermal time, leaf number, radiation use, light extinction, height, root axes, stomatal and turgor constants |
| Genotype (reference, deep rooting, shallow and dense, stay-green, water saver, water spender, osmotic adjuster) | angle of the axes, lateral density, stay-green, stomatal threshold, osmotic adjustment, hydraulic resistance, carbon to roots |
| Drought (pulse and recovery, terminal, intermittent, none) | the water that the drought treatment is held at, in time |
| Season starts | the first day of the season: the date of the toolbar (the start), the planting day of the project, or automatic (the section *Drought stress calendar* explains each) |
| Time axis | the season with dates, or a year from the date of the toolbar with that date marked |
| Climate and rain | temperature, radiation, vapour pressure deficit, reference evapotranspiration, rain |
| Left plant (project irrigation, well-watered bench, rainfed) | the treatment of the left plant: the irrigation of the project made by the sensor rule, a bench held wet, or the rain only; the right plant is always the drought |
| Drought starts, length, water held | day of onset, duration and the level of the stress, in per cent of the plant-available water (defaults 45 days, 35 days, 15 %) |
| Extra warming | temperature added to the whole season |
| Root angle, lateral root density, root vigour, carbon share to roots | root system architecture and the carbon it receives |
| Stay-green, stomata close at, osmotic adjustment | senescence delay, the leaf potential at which half of the stomatal opening is lost, the turgor kept in dry soil |
| Growing area per plant | the ground area of one plant, which sets the leaf area index, the light caught and the water of the column |

The **soil, the tillage and the consolidation rate** are those of the project (the V3 layer): the bulk density, the strength of the soil and therefore the growth of the roots follow the soil texture, the tillage system and the structural consolidation rate (Eq. 8.1).

### 8.4.3 Drought and the roots at a glance

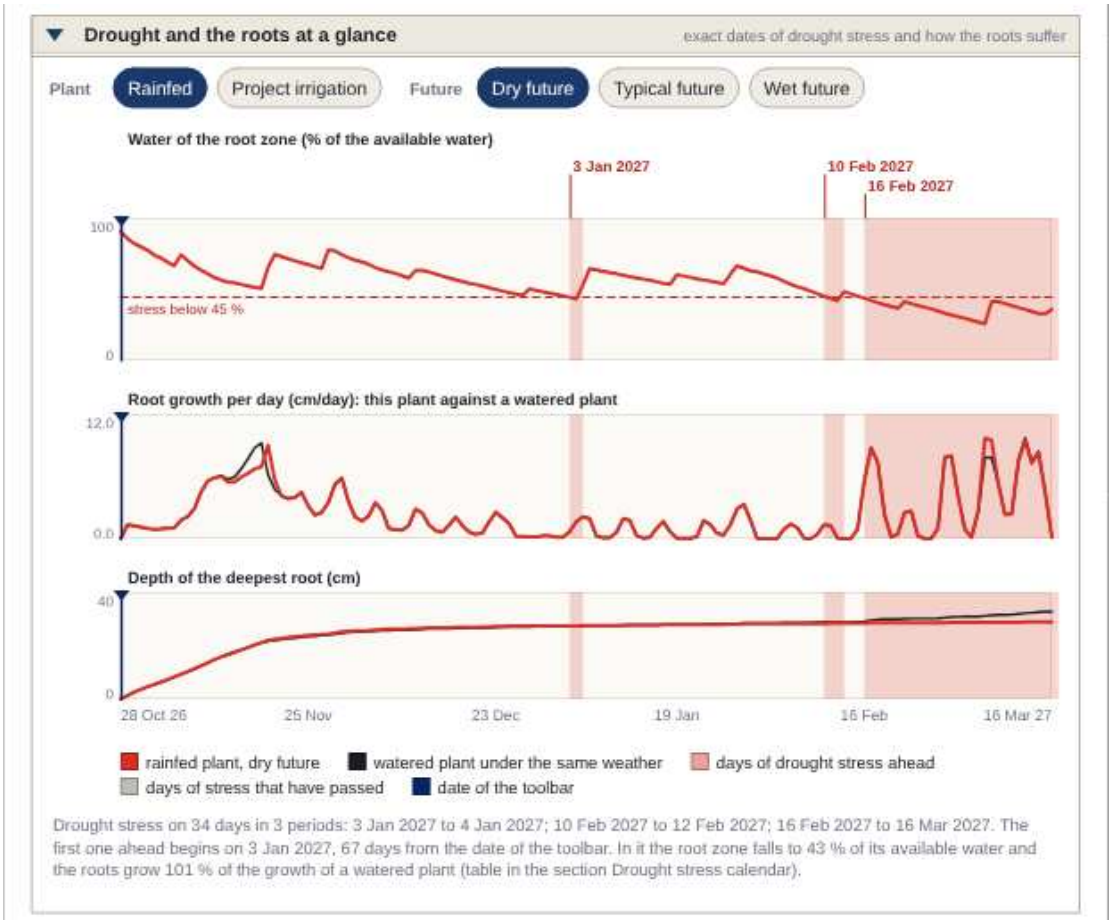



Figure 8.4: Drought and the roots at a glance: the rainfed wheat plant on the dry future, from 28 October 2026. Red bands are the days of drought stress, with the date on which each begins.

This section, which follows *Set up the experiment*, is the high-level view of the drought. Three charts share one time axis with dates. The first is the **water of the root zone** of the chosen plant as a percentage of the available water, with the stress level of Eq. P.26 as a dashed line. The second is the **root growth per day** of the plant (red, or blue for the plant irrigated by the project) against the roots of a well watered plant under the same weather (black). The third is the **depth of the deepest root** of the two. Every period of drought stress is a shaded band across all three charts, red when it is ahead of the date of the toolbar and grey when it has passed, and the **exact date on which it begins is written on the first chart**. A dark line marks the date of the toolbar. The chips choose the plant (rainfed or irrigated by the project) and the future (dry, typical or wet), and are shared with the section *Drought stress calendar*. A paragraph under the charts lists every period with its dates, the days to the next one, and what it does to the roots. When no band is drawn, the plant has no day of drought stress on that future.

### 8.4.4 Plant and roots, side by side

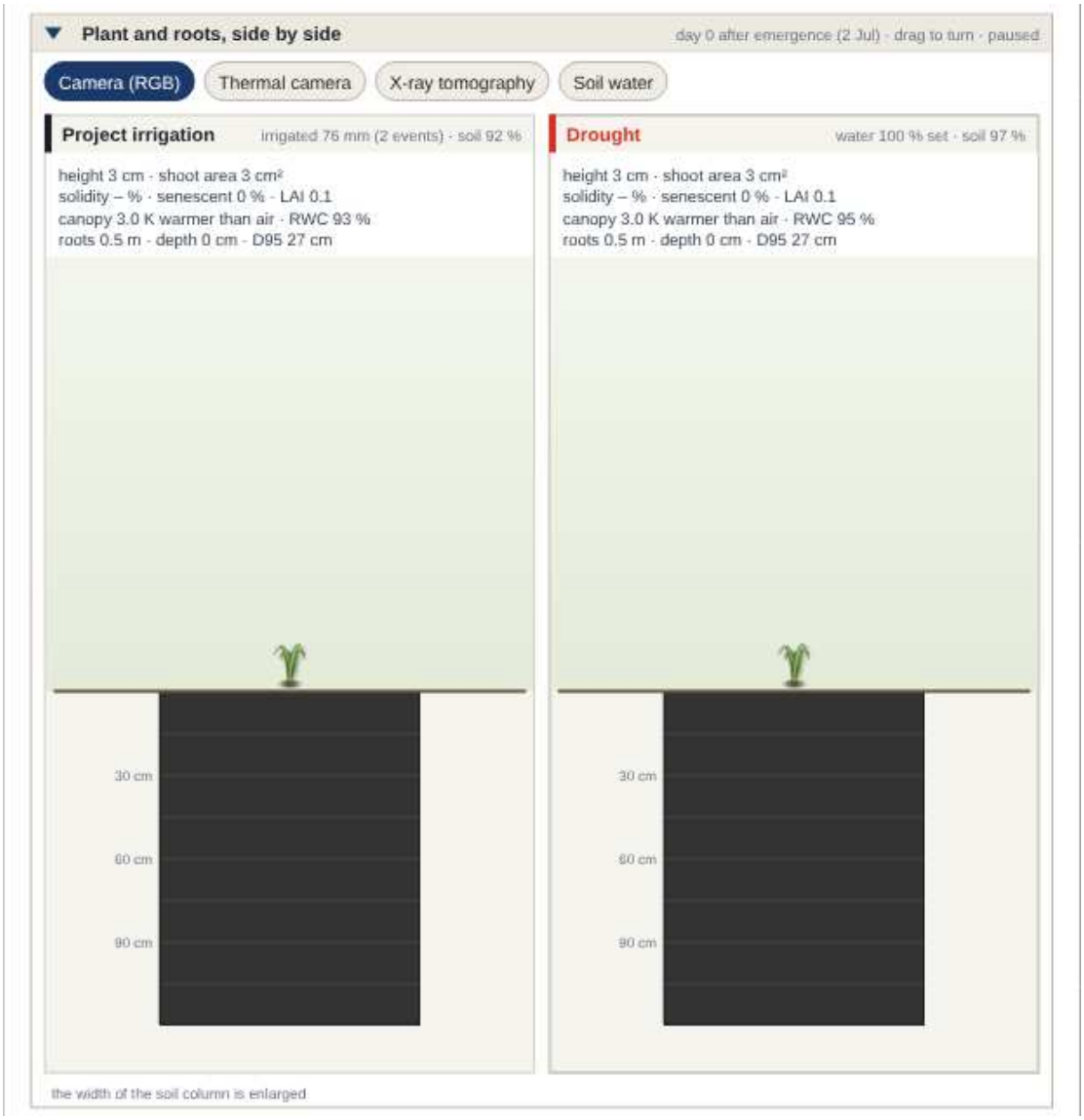



Figure 8.5: Plant and roots on day 62 (2 July) in the camera view: the shoot with the leaves of the model, the soil column with the root system of the same day. Left well-watered, right drought.

The shoot is a life-like rendering of the simulation, not a photograph: blades with a shaded surface, midrib and veins, leaf sheaths and nodes on the stem, wheat with its tillers, and the ear of maize, the spike of wheat, the panicle of sorghum or the pods of the bean, which fill as the grain mass of the model grows. Leaves that wilt (low water content) droop and pale, and the senescent part of a blade turns yellow and then brown from the tip. The arch of the leaves of maize and sorghum is drawn a little wider than the model's own geometry so that the blades can be seen; the numbers do not change. The picture is drawn from the leaf geometry of the model: each leaf has an insertion height, an azimuth, an elevation, a length, a width and a stay-green fraction, so the **projected shoot area**, the **convex hull** and the **solidity** that are printed above the plants are those of what is drawn. The roots are the segments of the root architecture model grown up to the day, turned by the rotation of the view. Four views show the same plant:

- **Camera (RGB)**: green leaves that turn brown with senescence, roots as a rhizotron sees them.

- **Thermal camera**: each leaf has the temperature of its energy balance, from 4 K below to 10 K above the air.
- **X-ray tomography**: roots in white as a scanner shows them.
- **Soil water**: the relative plant-available water of each layer with the roots on it.

Figure 8.6: Thermal camera view of the same day. The drought plant is warmer, its leaves are redder and the soil on its side is drier.

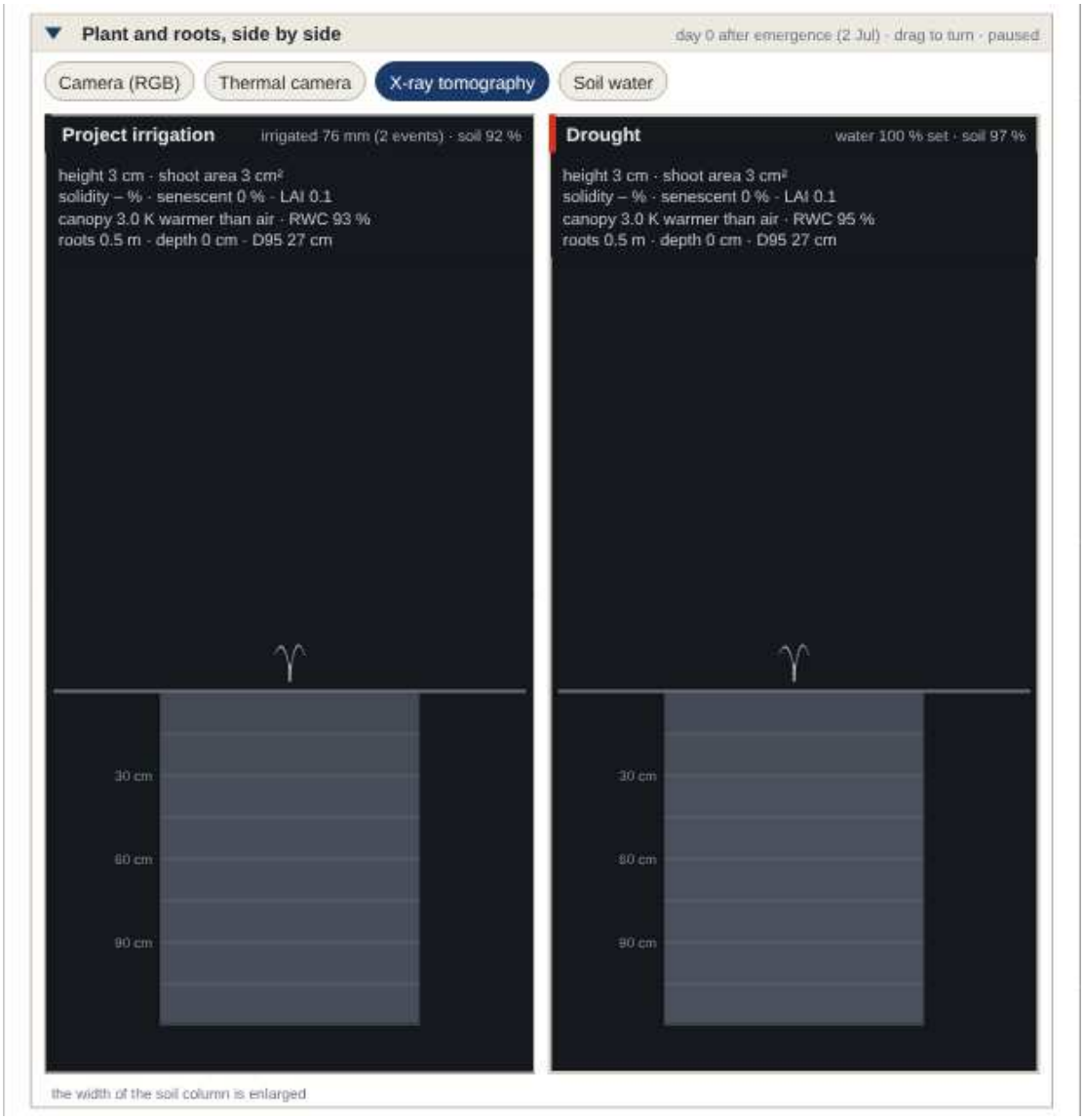



Figure 8.7: X-ray view of the same day.

Figure 8.8: Soil water view of the same day: the plant-available water of every 15 cm layer (blue is at field capacity, brown is dry) with the roots.

The **root scans through the season** show the drought plant (or the well-watered plant, by the two chips) on four days: at the top the scanner image of the root system, at the bottom the reconstruction with the first order axes, the second order laterals and the soil volume.

Figure 8.9: Root scans of the drought plant on days 35, 70, 104 and 139.

### 8.4.5 Season film, from sowing to maturity

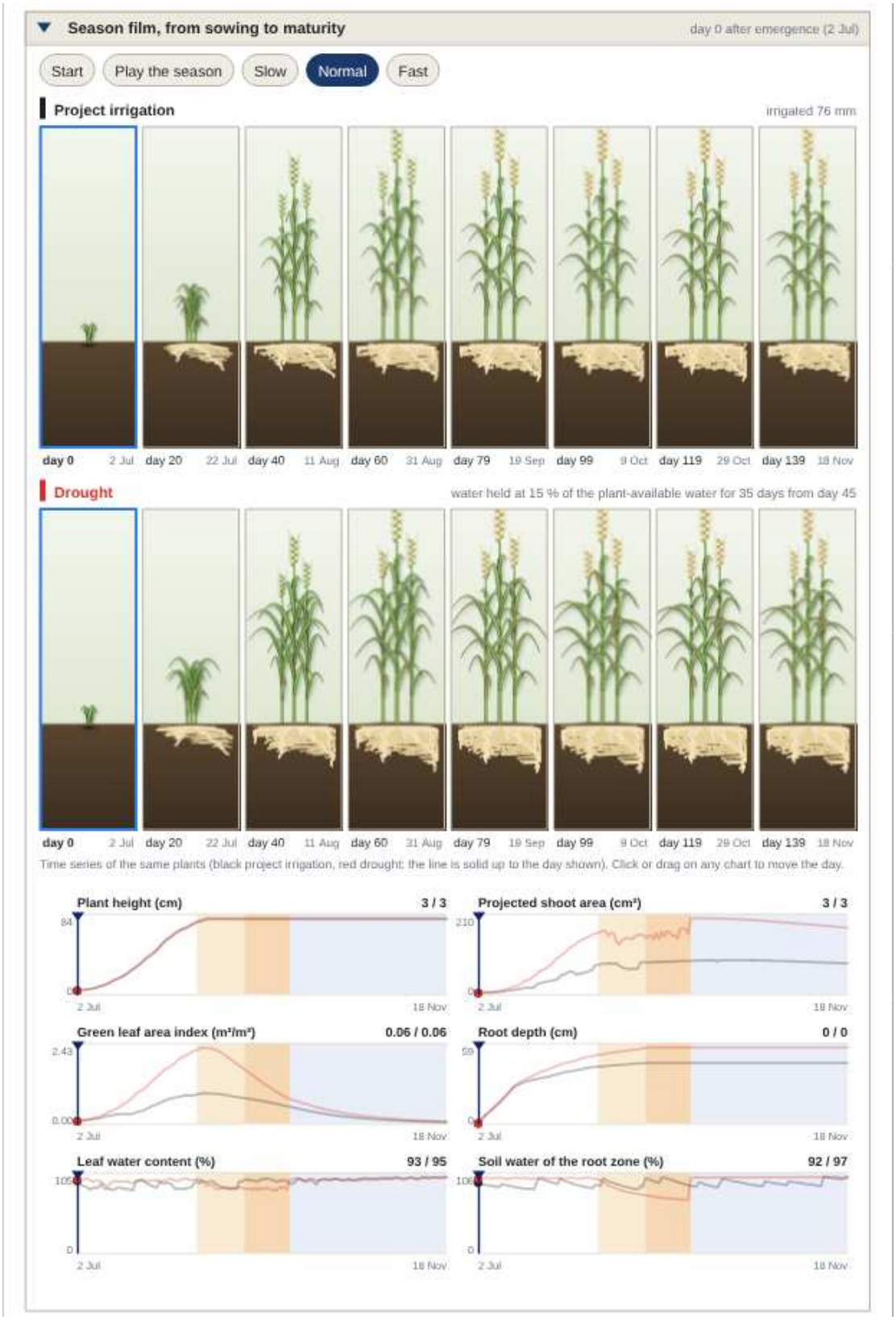



Figure 8.10: The season film: both plants and their roots on eight dates from sowing to maturity, with the time series below. Wheat, project climate, day 62 marked.

The section shows the whole life of the two plants at once. Two rows of eight pictures (the project plant above, the drought plant below) show the shoot and the roots on days 0, 20, 40, 60, 79, 99, 119 and 139 after emergence, each with its date, at one common scale, so that the growth can be compared from picture to picture. The picture nearest to the day shown has a blue frame; click a picture to go to that day. **Play the season** grows both plants from the first day to the last (the speeds are *Slow*, *Normal* and *Fast*: 28, 14 and 6 seconds for the season); **Pause** stops, and **Start** returns to the first day. Below the pictures, six time series (plant height, projected shoot area, green leaf area index, root depth, leaf water content and the water of the root zone) run through the whole season for both plants. Each line is solid up to the day shown and faint after it, with the stress and recovery bands behind them; click or drag on any chart to move the day.

### 8.4.6 Traits through time

Figure 8.11: Traits through time, four sets (shoot, water status, carbon, roots). The coloured bands are the stress (orange) and the recovery (blue); the dashed line is the day shown; the values at the right are well-watered / drought of that day.

Click or drag on any chart to move the day. The four sets are:

- **Shoot**: plant height, projected shoot area, solidity, senescence index.
- **Water status**: leaf water potential, relative water content, stomatal conductance, canopy temperature depression, crop water stress index.
- **Carbon**: assimilation, the share to roots, dry mass.
- **Roots**: root length, depth, the depths of 50 % and 95 % of the length.

The observations that you load are drawn on the same charts as squares.

### 8.4.7 Water: demand, supply and where the roots take it

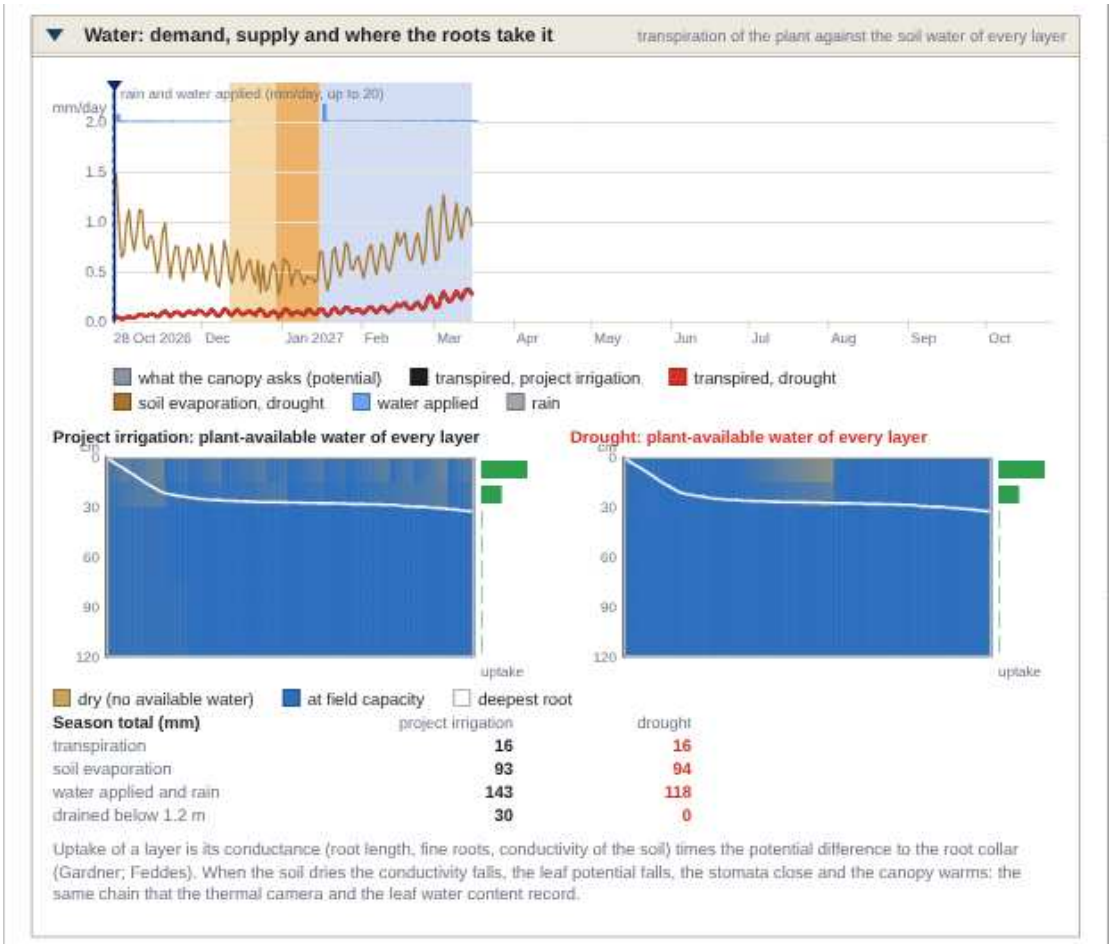



Figure 8.12: The water chart with the time axis set to a year from the date of the toolbar (28 October 2026).

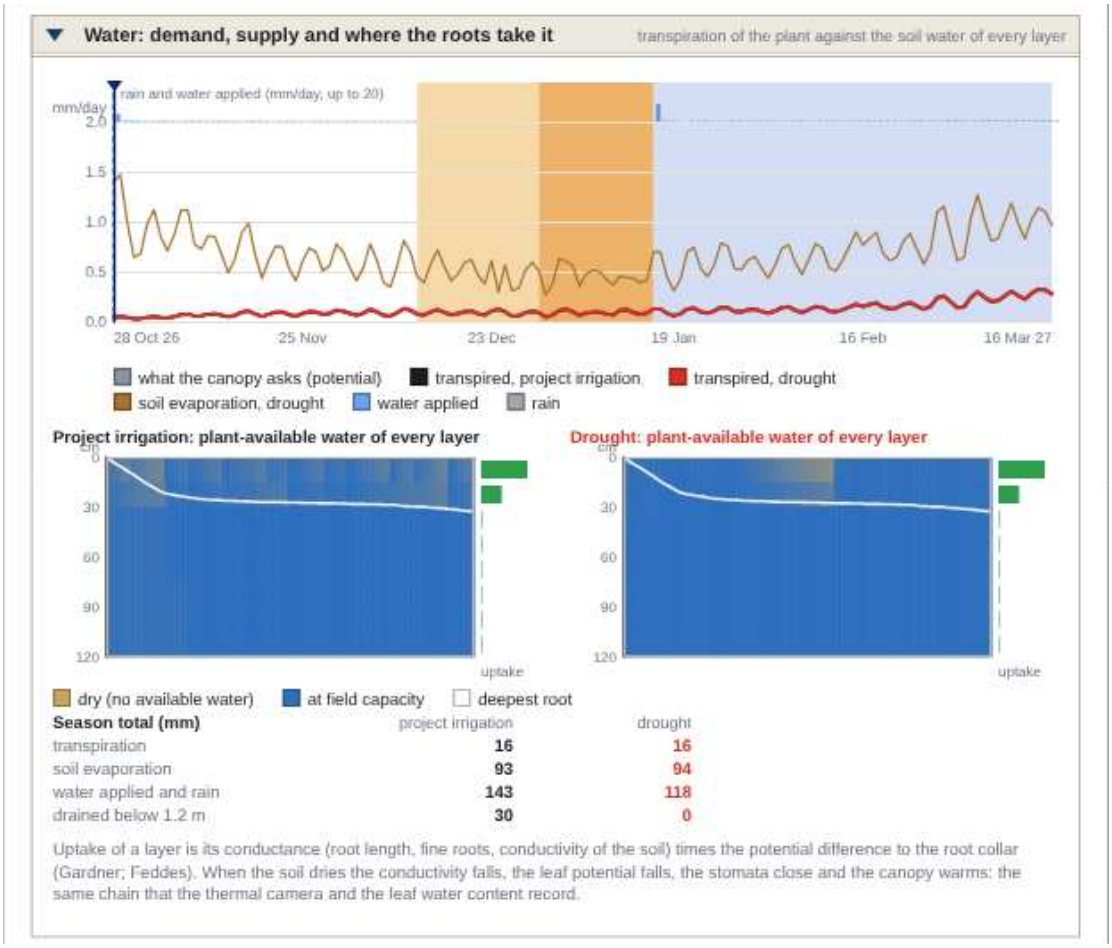



Figure 8.13: Water: transpiration, soil evaporation, rain and water applied, the plant-available water of every layer with the front of the roots, and the water taken by each layer.

The top chart shows what the canopy asks for (the potential transpiration), what it transpires in each treatment and what the soil surface loses. The two heat maps show the plant-available water of each layer through the season with the root front as a white line; the bars at the right show how much each layer gave on the day. The table gives the season totals of transpiration, soil evaporation, water in and drainage below 1.2 m, so that the water balance of every treatment is closed. In this run the well-watered plant transpired 334 mm and the drought plant 237 mm.

The path of the water is the one of the soil-plant-atmosphere chain. The uptake of a layer is its conductance (root length, fine roots, hydraulic conductivity of the soil at its water content) times the potential difference between the soil and the root collar. The collar potential is solved every day so that the layers together give the demand; when the soil dries, the conductivities fall, the leaf potential falls, the stomata close and the canopy warms. This chain is the reason why the thermal camera and the leaf water content respond to the drought before the height does.

### 8.4.8 Carbon: assimilation, allocation, root and shoot growth

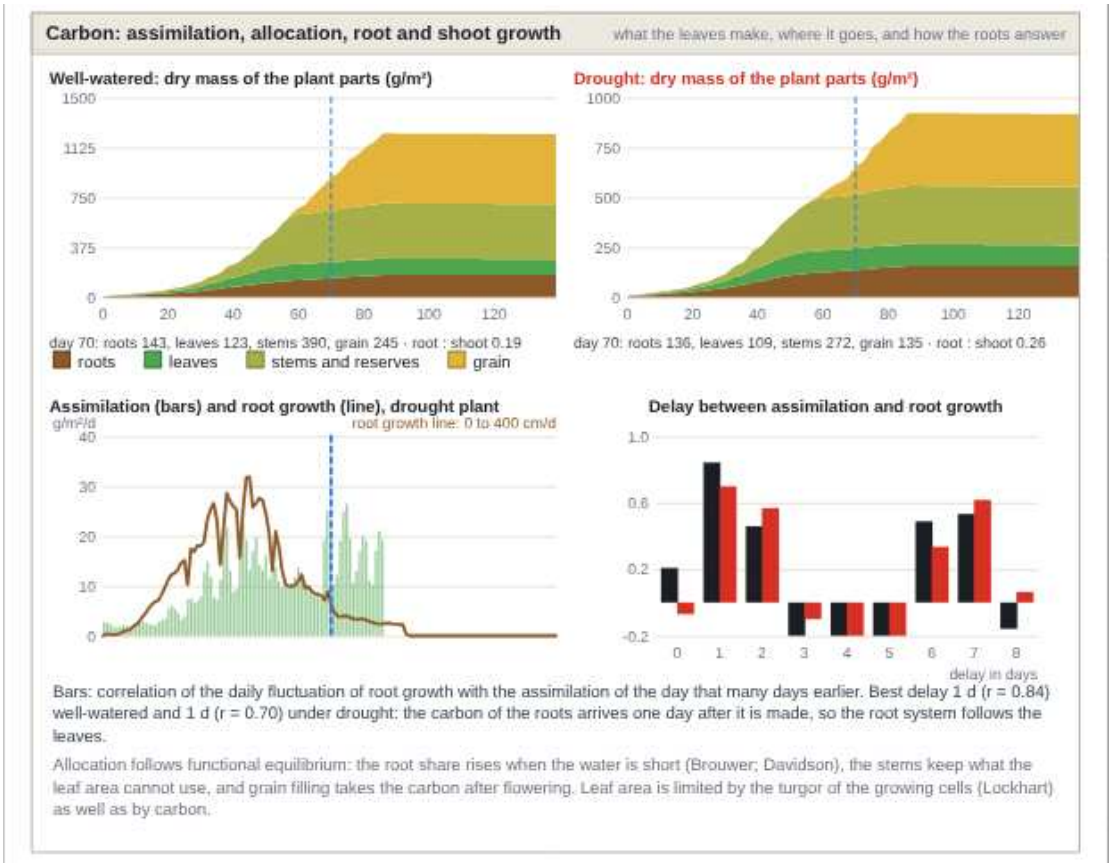



Figure 8.14: Carbon: dry mass of roots, leaves, stems and reserves, and grain for both treatments; the assimilation against the growth of the roots; and the delay between them.

The upper charts stack the dry mass of the plant parts. The lower left chart draws the daily **assimilation** as bars and the **growth of the root length** as a line. The lower right chart is the correlation between the fluctuation of the daily root growth and the assimilation of the days before, with the delay on the x axis: a bar at one day means that the carbon of one day gives roots one day later. In this run the best delay is one day in both treatments (r = 0.84 for the well-watered plant and 0.70 for the drought plant).

The carbon of the roots is a share of the daily growth. The share rises when the water is short (the functional equilibrium of Brouwer, 1963, and Davidson, 1969), the share of the leaves follows the stage, and the grain is filled after flowering. The root system of the model uses the carbon that arrives, so a day with little light gives a smaller root growth the next day. The ratio of the root mass to the shoot mass is printed under the charts (0.19 and 0.26 on day 70).

### 8.4.9 Soil structure and tillage: what the root tips meet

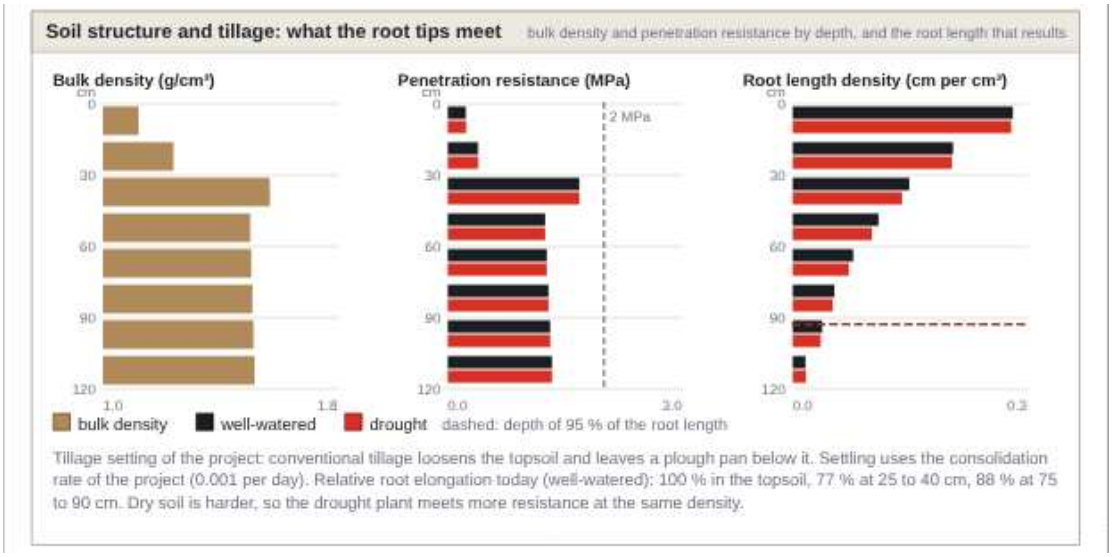



Figure 8.15: Bulk density, penetration resistance and root length density by depth.

The soil of the layer is the soil of the project. The **bulk density** of each 15 cm layer follows the texture and the tillage: the topsoil is loosened by tillage and settles with the consolidation rate of the V3 layer, and a plough pan lies below the working depth. The **penetration resistance** $Q$ rises steeply with the density and rises further when the soil is dry; the elongation of a root tip is reduced by a factor that falls from 1 to 0 around $Q = 2.5$ MPa. The right chart shows where the root length ends up. A change of the tillage, of the consolidation rate or of the soil changes the depth reached by 95 % of the roots (D95), and with it the water that the plant can use.

### 8.4.10 Roots from the plant above ground

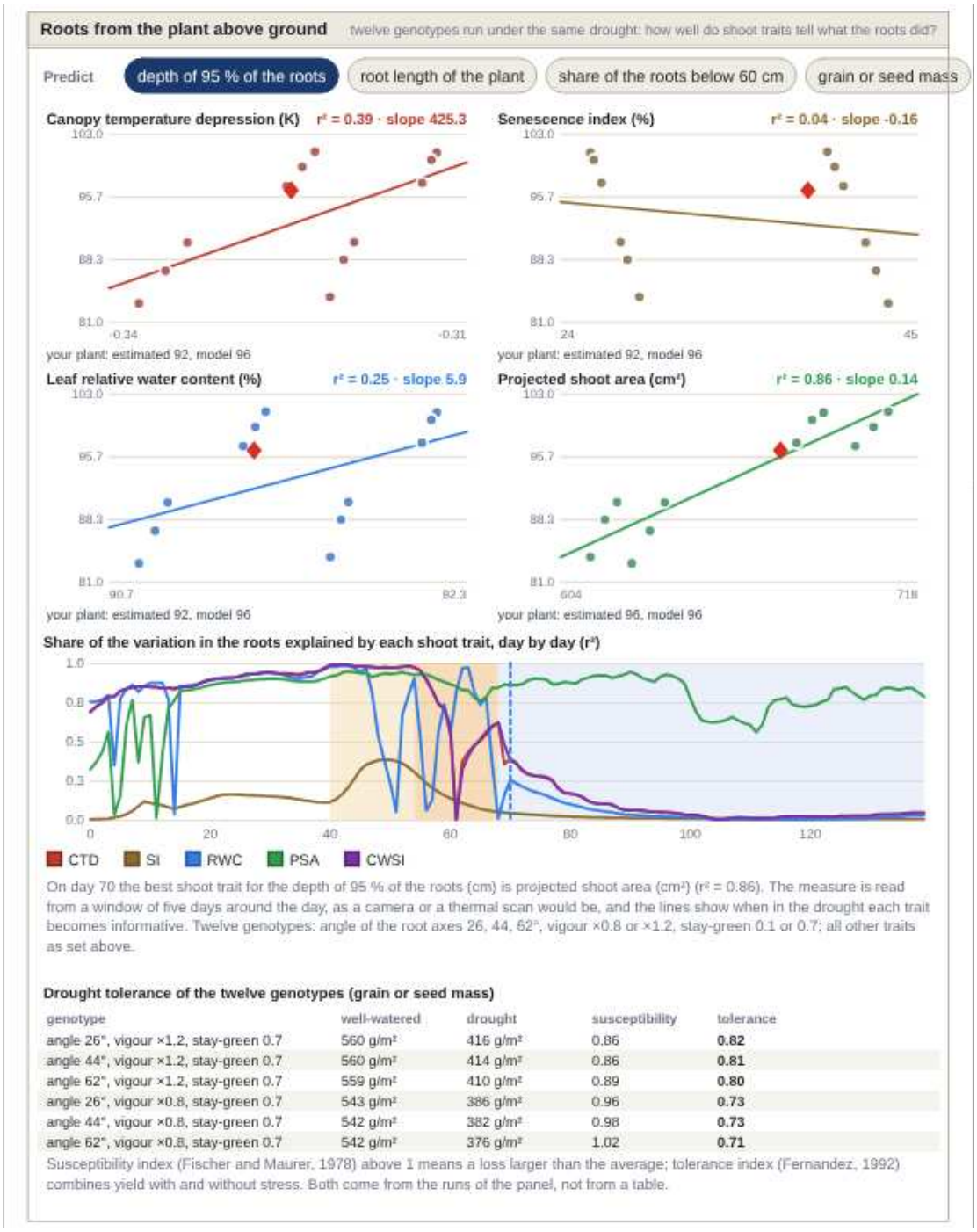



Figure 8.16: Twelve genotypes run under the same drought. Each plot relates one shoot trait to the root trait chosen above it, the lower chart shows how well each shoot trait predicts the roots day by day, and the table gives the drought tolerance of the twelve genotypes.

The panel runs twelve genotypes (angle of the axes 26, 44 and 62 degrees, vigour 0.8 and 1.2, stay-green 0.1 and 0.7; all other traits as in the reference) under the drought that is set, and relates **a root trait** (depth of 95 % of the roots, root length, share of the roots below 60 cm, or grain mass) to **four shoot traits** that a camera sees: canopy temperature depression, senescence

index, relative water content and projected shoot area. The red diamond is the plant of the layer, estimated from its own shoot trait with the regression of the panel and compared with the model. The lower chart is the share of the variation explained by each shoot trait on each day ($r^2$): it shows **when** a measurement taken above ground carries information about the roots. In this run, on day 70, the projected shoot area is the best trait for the depth of the roots ($r^2$ = 0.86), measured as the mean of a window of five days around the day.

The table gives the **drought susceptibility index** of Fischer and Maurer (1978) and the **stress tolerance index** of Fernandez (1992) of the twelve genotypes, computed from their grain mass with and without stress.

### 8.4.11 Roots day by day

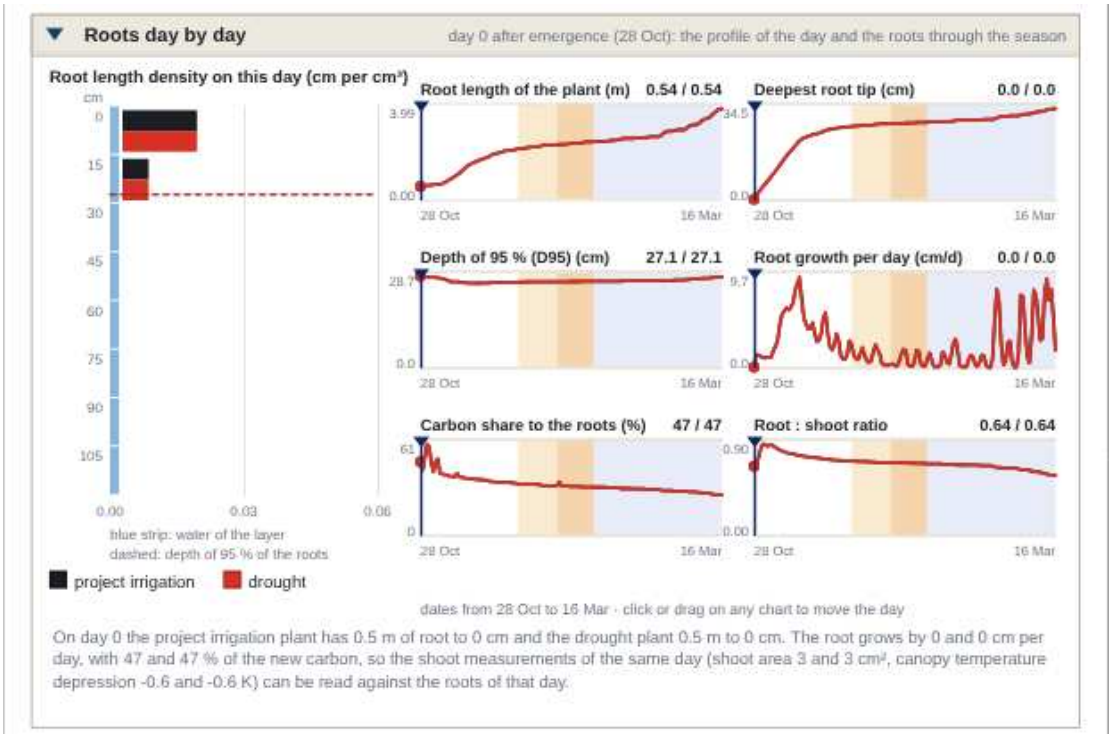



Figure 8.17: The profile of the chosen day and the roots through the season. Wheat, project climate, date of the toolbar 28 October 2026, day 0.

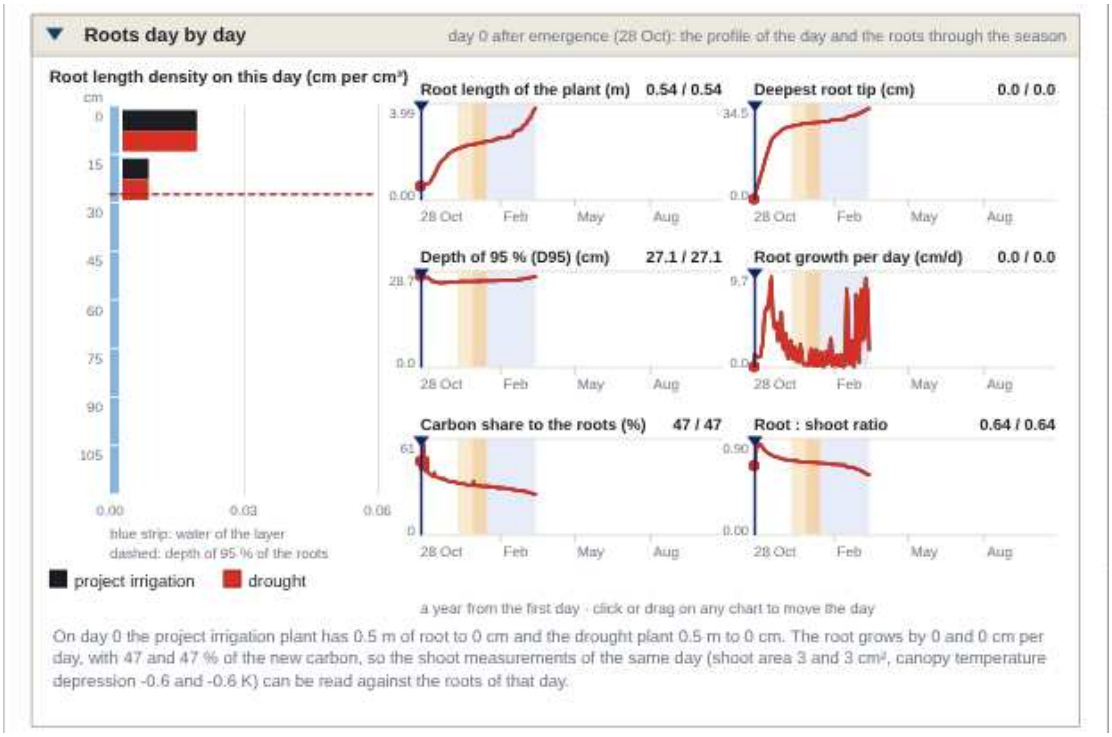
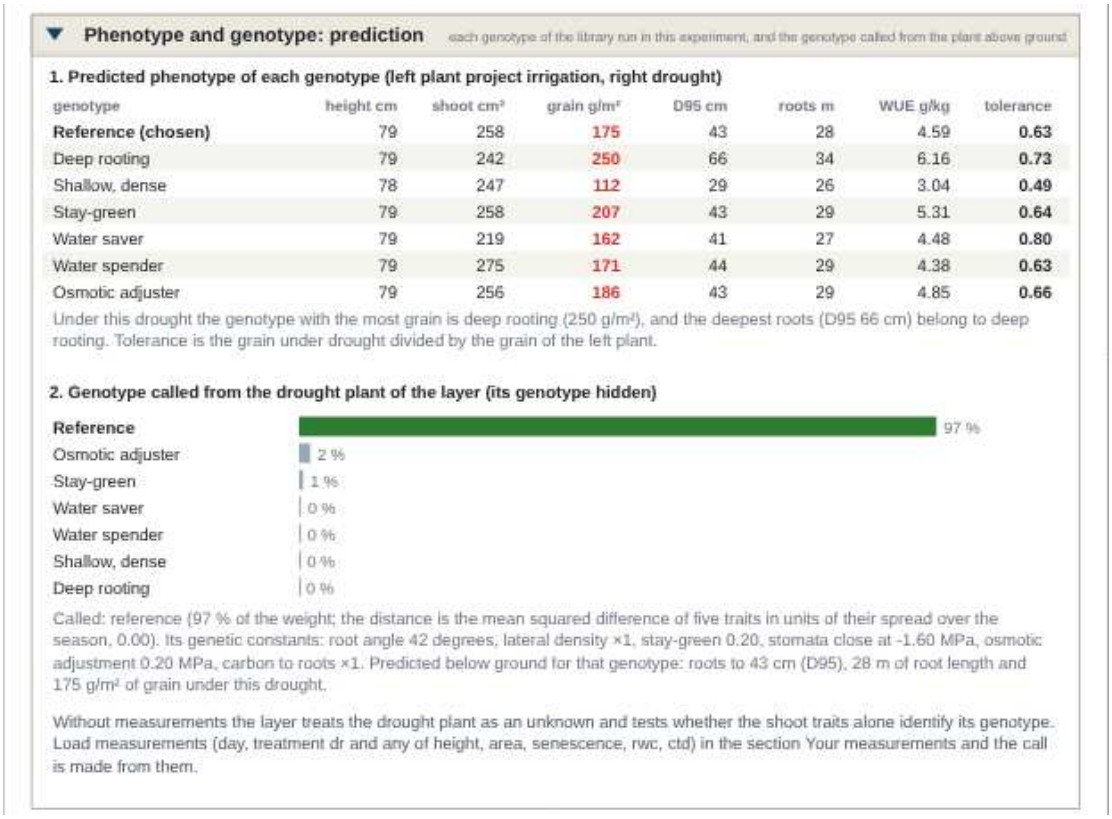



Figure 8.18: The same section with the time axis set to a year from the date.

For the chosen day the section draws the **root length density by depth** of the two plants (bars), the water of each layer (blue strip) and the depth of 95 % of the roots (dashed line). Six series follow the roots through the season, each with a cursor on the day: the root length of the plant, the deepest root tip, the depth of 95 % of the root length (D95), the growth of the roots per day, the share of the carbon that goes to the roots, and the root to shoot ratio. Click or drag on any chart to move the day; the numbers on the right of each title are the two plants on that day. The sentence under the charts reads the day: the root length, depth and growth of both plants, the new carbon they received, and the shoot measurements of the same day.

### 8.4.12 Phenotype and genotype: prediction



**Phenotype and genotype: prediction** each genotype of the library run in this experiment, and the genotype called from the plant above ground

**1. Predicted phenotype of each genotype (left plant project irrigation, right drought)**

| genotype | height cm | shoot cm² | grain g/m² | D95 cm | roots m | WUE g/kg | tolerance |
|---|---|---|---|---|---|---|---|
| **Reference (chosen)** | 79 | 258 | 175 | 43 | 28 | 4.59 | **0.63** |
| Deep rooting | 79 | 242 | 250 | 66 | 34 | 6.16 | **0.73** |
| Shallow, dense | 78 | 247 | 112 | 29 | 26 | 3.04 | **0.49** |
| Stay-green | 79 | 258 | 207 | 43 | 29 | 5.31 | **0.64** |
| Water saver | 79 | 219 | 162 | 41 | 27 | 4.48 | **0.80** |
| Water spender | 79 | 275 | 171 | 44 | 29 | 4.38 | **0.63** |
| Osmotic adjuster | 79 | 256 | 186 | 43 | 29 | 4.85 | **0.66** |

Under this drought the genotype with the most grain is deep rooting (250 g/m²), and the deepest roots (D95 66 cm) belong to deep rooting. Tolerance is the grain under drought divided by the grain of the left plant.

**2. Genotype called from the drought plant of the layer (its genotype hidden)**



Called: reference (97 % of the weight; the distance is the mean squared difference of five traits in units of their spread over the season, 0.00). Its genetic constants: root angle 42 degrees, lateral density ×1, stay-green 0.20, stomata close at -1.60 MPa, osmotic adjustment 0.20 MPa, carbon to roots ×1. Predicted below ground for that genotype: roots to 43 cm (D95), 28 m of root length and 175 g/m² of grain under this drought.

Without measurements the layer treats the drought plant as an unknown and tests whether the shoot traits alone identify its genotype. Load measurements (day, treatment dr and any of height, area, senescence, rwc, ctd) in the section Your measurements and the call is made from them.

Figure 8.19: The predicted phenotype of each genotype under the drought, and the genotype called from the plant above ground. Wheat, project climate, reference genotype.

The section has two parts. **1. Predicted phenotype**: every genotype of the library is run under the same weather, soil and drought; the table gives height, shoot area, grain, D95, root length, water use efficiency and the tolerance (grain under drought divided by grain of the left plant). **2. Genotype called from the plant above ground**: the genotype of the plant is hidden, five traits of the shoot (height, area, senescence, leaf water content and canopy temperature depression through the days) are compared with the same traits of each genotype, and the bars give the weight of each genotype (Eq. P.21). The genetic constants of the called genotype (root angle, lateral density, stay-green, stomatal threshold, osmotic adjustment, carbon to roots) and the roots predicted below ground follow. When measurements are loaded (section *Your measurements*), they replace the model plant as the thing that is called. The call is a ranking of the library, not a genome sequence: the layer predicts the **genotype of the library** (the set of trait constants) and its phenotype.

### 8.4.13 SWAFL data: the Daisy runs of the project against this plant

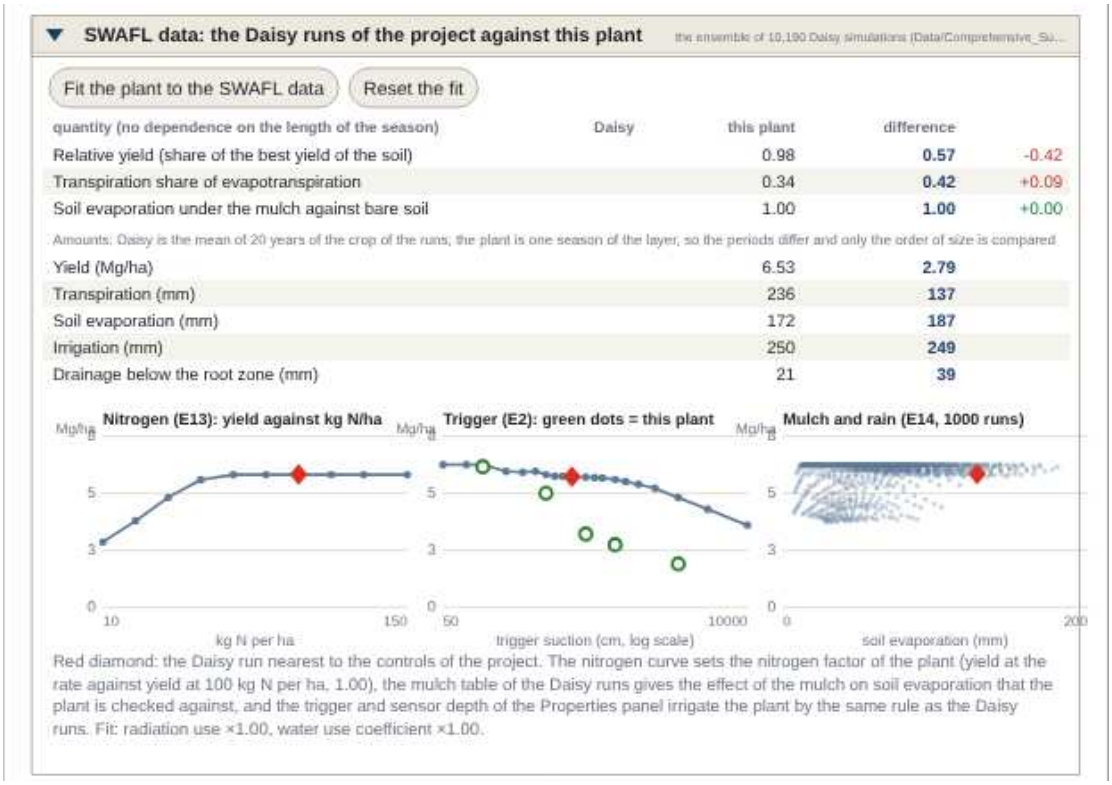



Figure 8.20: The plant against the Daisy runs of the soil and controls of the project, after the fit. Wheat, project climate.

The repository holds 10,190 Daisy simulations (*Data/Comprehensive_Summary*). The section compares the plant with the runs of the soil and the controls of the project, in terms that do not depend on the length of the season: the **relative yield** (share of the best yield of the soil, from the trigger experiment E2), the **transpiration share** of the evapotranspiration, and the **effect of the mulch on soil evaporation** (E7 and E14, 1000 runs of mulch, rain and capacity). The three charts are the nitrogen response (E13, which gives the nitrogen factor of the plant), the trigger response (E2, with the plant as green dots) and the mulch and rain runs (E14). **Fit the plant to the SWAFL data** searches the radiation use efficiency, the water use coefficient and the root conductance (a coordinate search of at most 45 runs, within 0.6 to 1.6, 0.5 to 2.2 and 0.25 to 4 times the species values) that minimise the squared difference to the three numbers and to the E2 curve (Eq. P.20). **Reset the fit** returns to the species constants. The fit is a calibration to the Daisy tables: the plant is a different model, and for a climate far from that of the Daisy runs (the warm season, for example) the agreement is weaker; the comparison is most meaningful in the project climate with the archetype nearest to the crop. When a limit of the search is reached the message says so.

### 8.4.14 Quantified traits of the run

**Quantified traits of the run** — the readouts that a phenotyping experiment reports, for the two treatments

| trait | well-watered | drought | change |
|---|---|---|---|
| Plant height at the end (cm) | 216 | 215 | 0 % |
| Largest projected shoot area (cm²) | 818 | 718 | -12 % |
| Day the shoot area reaches half of its largest | 44 | 40 | -9 % |
| Fastest relative growth of the shoot area (per day) | 0.164 | 0.164 | 0 % |
| Fastest biomass gain (g/m²/d) | 33.0 | 26.4 | -20 % |
| Largest green leaf area index | 2.50 | 1.97 | -21 % |
| Day half of the leaf area is senescent (stay-green) | 89 | 80 | -10 % |
| Green leaf area over the season (m² d per m²) | 142 | 105 | -26 % |
| Lowest leaf water content in the stress period (%) | 87 | 81 | -7 % |
| Lowest leaf water potential in the stress period (MPa) | -0.94 | -1.43 | -51 % |
| Mean canopy temperature depression in the stress (K) | -0.9 | -2.1 | -130 % |
| Highest crop water stress index in the stress period | 0.02 | 0.28 | +0.26 |
| Transpiration (mm) | 334 | 237 | -29 % |
| Soil evaporation (mm) | 265 | 280 | +6 % |
| Water applied and rain (mm) | 605 | 523 | -14 % |
| Total dry mass (g/m²) | 1225 | 917 | -25 % |
| Grain or seed mass (g/m²) | 525 | 365 | -31 % |
| Harvest index | 0.50 | 0.48 | -3 % |
| Transpiration efficiency (g dry mass per kg water) | 3.7 | 3.9 | +6 % |
| Water use efficiency of the grain (g per kg) | 0.88 | 0.71 | -19 % |
| Root depth (cm) | 120 | 120 | 0 % |
| Root length (m per plant) | 104 | 101 | -3 % |
| Depth of 50 % of the root length (cm) | 31 | 30 | -3 % |
| Depth of 95 % of the root length (cm) | 97 | 96 | 0 % |
| Roots below 60 cm (% of the length) | 22 | 22 | -2 % |
| Root : shoot ratio at the end | 0.16 | 0.20 | +30 % |

Figure 8.21: The readouts of a phenotyping experiment for the two treatments, with the change under drought.

The table lists 26 traits: the final height, the largest projected shoot area, the day the shoot area reaches half of its largest, the fastest relative growth of the shoot area, the fastest biomass gain, the largest green leaf area index, the stay-green day, the green leaf area over the season, the lowest leaf water content and potential and the highest crop water stress index in the stress period, the canopy temperature depression of the stress period, the water used (transpiration, soil evaporation, water in), dry mass, grain, harvest index, transpiration efficiency, water use efficiency, root depth and length, the depths of 50 % and 95 % of the root length, the share of roots below 60 cm and the root to shoot ratio. The column *change* is the relative change under drought, or the difference when the value of the well-watered plant is small.

### 8.4.15 Weather in use, a year from the date of the toolbar

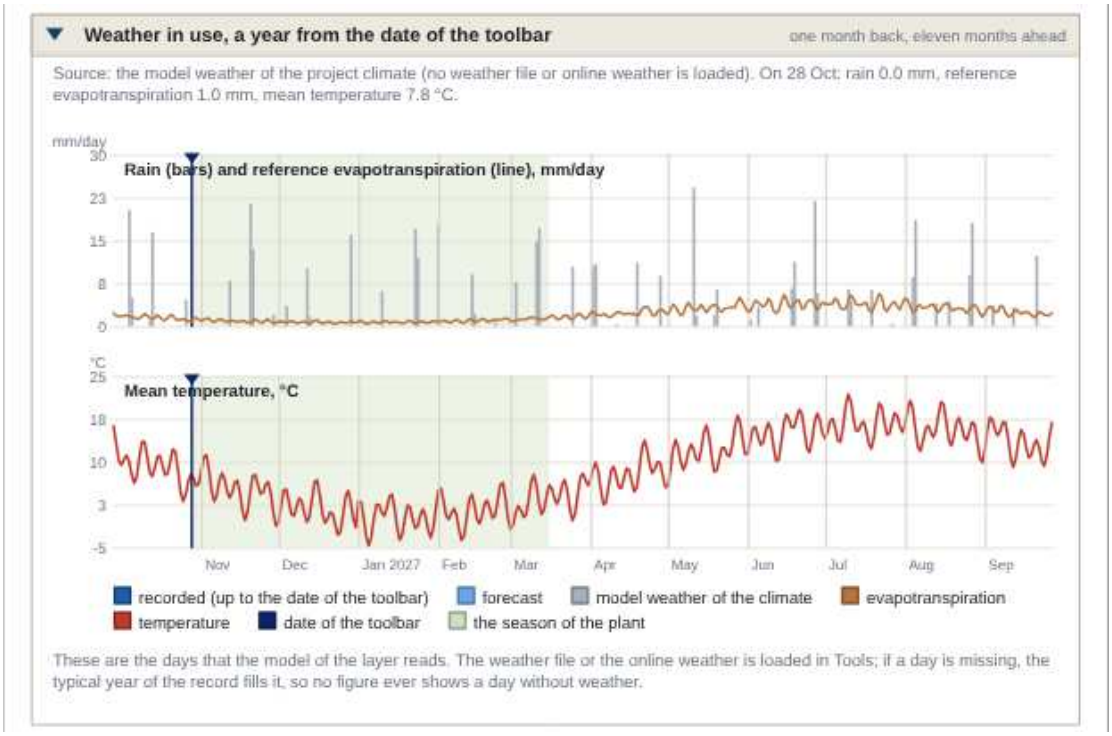



Figure 8.22: The weather in use for the year that starts a month before the date of the toolbar (28 October 2026). No weather file is loaded, so the model weather of the project climate is shown.

This section shows the weather that every figure of the layer reads, for a window of one year, from 30 days before the date of the toolbar to eleven months after it, with a dark line on that date. The window starts from the date, so no old date is shown. The header names the **source**: the weather file or the online weather of the farm when one is loaded (*Tools*), or the model weather of the project climate when none is. It states how many days of the window the record holds and which dates they cover, how many of them are the forecast after the date of the toolbar, and that the other days are the typical year of the record (the year with the median rain), and it gives the rain, the reference evapotranspiration and the mean temperature of the date of the toolbar. Two charts follow: the daily rain (bars) with the reference evapotranspiration (line), and the mean temperature. The bars are dark blue for recorded days, light blue for the forecast and grey for the days taken from the typical year or from the model. A green band marks the days of the season of the plant. When the date of the toolbar is changed the chart is drawn again for the year of the new date, so the weather in use always matches the date.

### 8.4.16 Drought outlook from the date of the toolbar

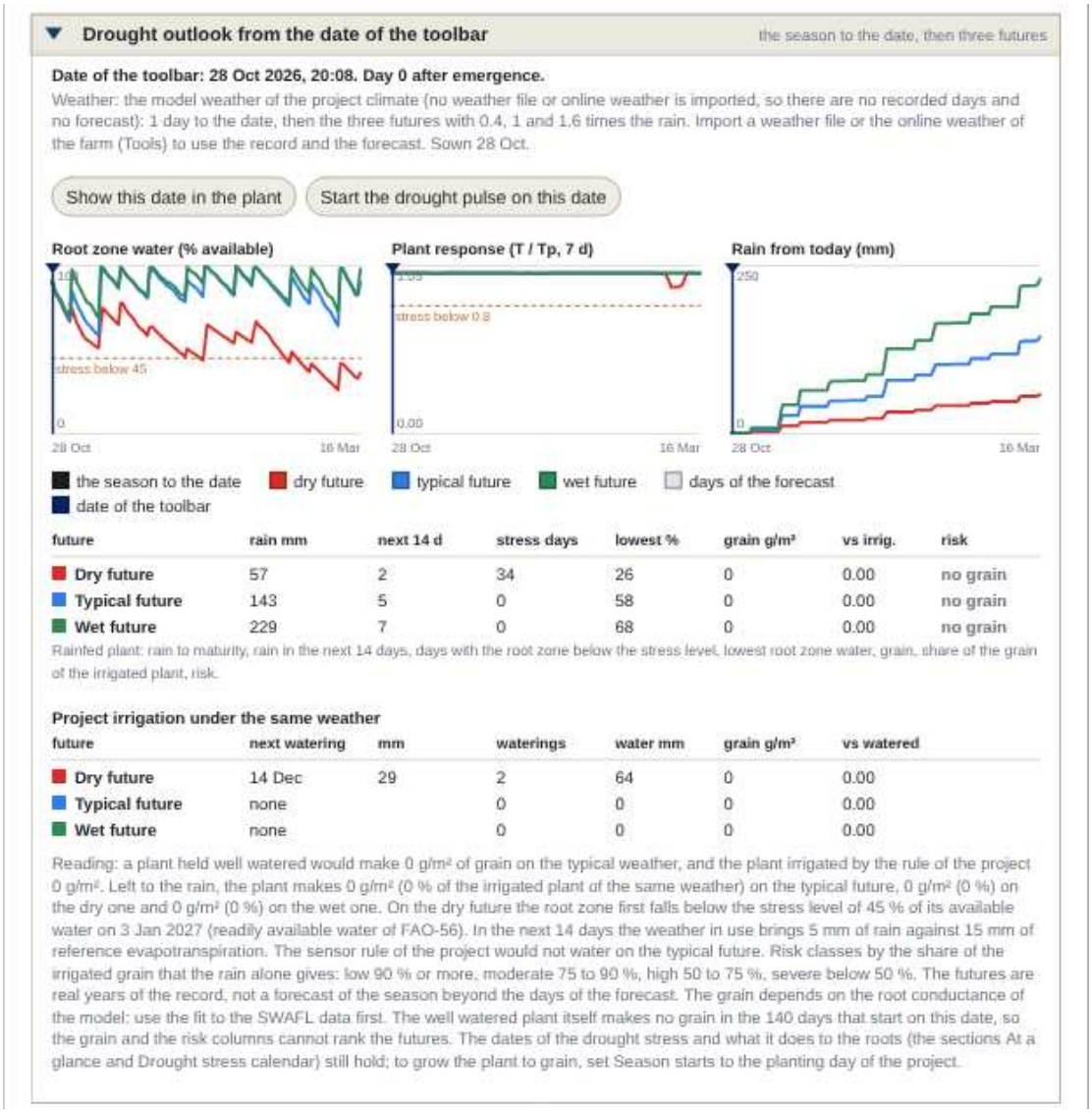



Figure 8.23: The drought outlook on 28 October 2026, the date of the toolbar and day 0 of the season. Wheat, project climate, model weather (no weather file imported), after the fit to the SWAFL data.

This section is a second drought experiment, tied to the calendar. The first (*Traits through time*) holds the right plant at a set level of water for a set time. The outlook instead lets the plant take the rain that falls: it grows the plant from sowing to **the date in the toolbar** (the *Date* control, with *Today*), and then to maturity on three futures. Change the date and the section is computed again.

**What the screen shows.**

1. The header gives the date, the day after emergence, and the weather used: how many days of the season are recorded, how many are the days of the forecast, and how many are a future. For the future, it names the three years of the record that were used.
2. Three charts run over the whole season, each with a dashed line on the date. The black line is the season up to the date; the red, blue and green lines are the dry, typical and wet futures. The light blue band marks the days of the forecast. The first chart is the **water of the root**

**zone** (per cent of the plant-available water of the layers that the roots have reached) with the stress level $1 - p$ of FAO-56; the second is the **plant response** $T_{act}/T_p$ averaged over seven days; the third is the **rain from the date**, added up.

3. A table for the rainfed plant gives, for each future, the rain to maturity, the rain of the next 14 days, the days with the root zone below the stress level, the lowest root zone water, the grain, the share of the grain of the irrigated plant, and the **drought risk** (low, moderate, high or severe; Eq. P.26).
4. A second table gives the plant irrigated by the rule of the project (method, trigger, sensor depth, dose and efficiency of the Properties panel) under the same weather: the date and depth of the next watering, the number of waterings and the water to maturity, the grain, and its share of the grain of a plant held well watered.
5. When the well watered plant itself makes no grain in the 140 days that start on the date (a winter sowing in the model weather, for example), the risk column reads *no grain* and the paragraph says so, because the grain cannot rank the futures; the dates of the stress and the roots still hold.
6. A paragraph in words reads the result for the date, including the first day on which the root zone of the dry future falls below the stress level, and the next watering.

**The two buttons.** *Show this date in the plant* moves the day of every other section (the plant and roots, the roots day by day, the traits) to the date of the toolbar. *Start the drought pulse on this date* makes the held-water drought of the first experiment begin on the date of the toolbar. Opening the layer also shows the day of the date when the date falls inside the season, so that the plant and its roots are those of the day.

**Where the weather comes from.** With an imported weather (*Tools*, a file or the online weather of the farm) the days up to the date are the recorded rain, evapotranspiration and temperature; the days that follow are the forecast rows when they exist (the online weather brings 16 days), and then the future years (Eq. P.24 and P.25). Without an imported weather the layer uses the model weather of the project climate, with no record and no forecast, and the three futures differ by the rain only (0.4, 1 and 1.6 times). The text of the header says which case applies.

**What it is and is not.** The three futures are real years of the weather record placed after the date, so each keeps the dry spells and the evaporative demand of that year. They show what a dry, an ordinary and a wet rest of the season would do to this plant; they are not a forecast of the weather. The risk class depends on the root conductance of the model: run **Fit the plant to the SWAFL data** first (section *SWAFL data*), because with the default constants the model plant is limited by the flow of water into the root more than by the water of the soil, and the grain of the rainfed plant then changes little with the rain. In the figure the fit was run before the capture.

### 8.4.17 Drought stress calendar: dates and what it does to the roots

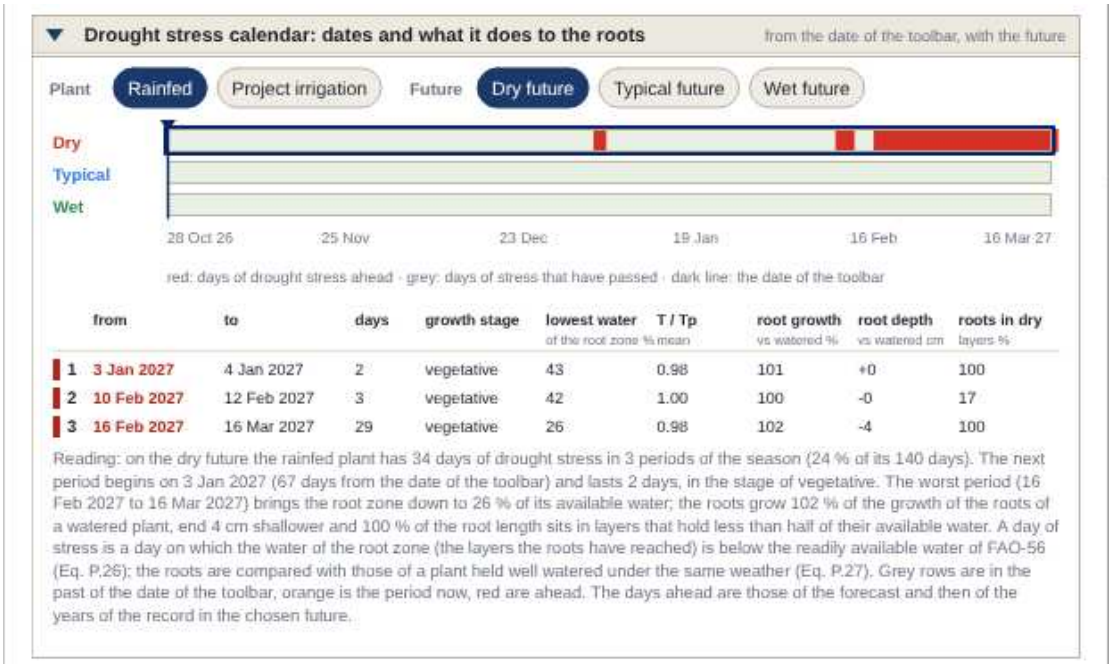



Figure 8.24: The drought stress calendar of the rainfed wheat plant on the dry future, from 28 October 2026. The dark line is the date of the toolbar.

This section turns the outlook into a calendar. It answers two questions for the plant of the project: **on which dates is it in drought stress**, and **what does that do to its roots**. It uses the same season, weather, soil and Properties inputs as the rest of the layer, so it changes when the date of the toolbar, the weather, the soil, the crop or the irrigation of the project changes.

**What the screen shows.**

1. Two rows of chips choose the plant (the rainfed plant, or the plant irrigated by the rule of the project) and the future after the date of the toolbar (dry, typical or wet).
2. Three bars, one for each future, run over the whole season and carry the dates on their axis. A red part is a period of drought stress that is still ahead, a grey part is a period that has passed, and the dark line is the date of the toolbar. The bar of the chosen future has a dark frame.
3. A table lists every period of drought stress of the chosen future with its **dates** (from and to, and "now" when the period is under way on the date of the toolbar), its length in days, the **growth stage** (vegetative, flowering or grain filling), the lowest water of the root zone, the mean ratio of transpiration to demand ($T/T_p$), and four readings of what the roots suffer: the root length they grow in the period against the roots of a well watered plant under the same weather, the depth of the roots at the end of the period against that plant (a minus sign is shallower, a plus sign deeper), and the share of the root length that sits in layers holding less than half of their plant-available water on the driest day of the period. The colour of the first column tells past (grey), now (orange) and ahead (red).
4. A paragraph in words reads the result: the number of days and of periods, the next period and how many days away it is, and the worst period with its dates and its effect on the roots.

**How a period is found.** A day is a day of stress when the water of the root zone is below the level at which transpiration starts to fall (Eq. P.26). Days of stress that are no more than two days apart form one period, and a period shorter than two days is dropped (Eq. P.27). Each period is compared with a **well watered bench plant** that is grown on the same weather, so the difference in the roots is the effect of the water and not of the weather (Eq. P.28).

**Where the dates come from.** The date in the toolbar is the key date of the whole layer, one to one: any date from 1980 to 2100 can be set (the date of today, tomorrow, or a day in 2030), the box starts on the day of the computer, and nothing in the layer is fixed to a month; the first day of every figure is that date, the day shown in the plant, the film and the charts is that date, and when the date is changed every figure starts again from the new date. The control *Season starts* in the section *Set up the experiment* has three choices. *On the date of the toolbar* (the choice at the start) sows on the date of the toolbar. *Planting day of the project* uses the planting day of the Properties panel, so the season can lie wholly in the past or wholly ahead of the date, and the date is then a day inside it. *Automatic* uses the planting day of the project when the date falls inside that season and otherwise sows on the date. The control *Time axis* chooses what the horizontal axes show: *The season* (the 140 days of the plant, with dates) or *A year from the date*: twelve months that begin on the first day of the season, so the plant occupies the first months and a dark line marks the date of the toolbar. No date older than the first day is shown. The date itself is read as the *Today* button of the toolbar reads it: the day of the computer, written into the date box of the toolbar, which you can change at any time. Every chart of the layer has its dates on the horizontal axis and the line of the date of the toolbar.

**Reading the roots.** In the figure (wheat sown on 28 October 2026, dry future, model weather) the rainfed plant has three periods of stress, all in the last weeks of the season: 3 and 4 January, 10 to 12 February, and 16 February to 16 March. In the first two the roots grow as much as those of the watered plant (101 and 100 % of its growth) because the periods are short, and in the last one, which is 29 days long, the roots reach 4 cm less depth than the roots of the watered plant, and all of the root length lies in layers that hold less than half of their available water, because the top layers dry first and most of the roots are there. These are the results of the model with the constants of the species; they are not measurements. The text under the table writes the same result in words, and the section *Drought and the roots at a glance* draws it as time series.

### 8.4.18 Your measurements: compare, calibrate and export

The last section brings experimental data to the model:

- **Load observations (CSV)**: a file with the column `day` (days after emergence), a column `treatment` (`ww` or `dr`) and any of `height`, `area`, `solidity`, `senescence`, `rwc`, `ctd`. **Download a template** writes an empty file with the columns that are read. The points appear as squares in the time series.
- The table of **errors** gives for each trait and treatment the number of points, the root mean square error, the bias and the Nash and Sutcliffe efficiency.
- **Calibrate the model** searches the radiation use efficiency, the stomatal threshold and the stay-green delay that reduce the squared error of all loaded traits (a coordinate search, one model run per frame, at most 40 runs). **Reset calibration** returns to the values of the genotype.
- **Export the results (CSV)** writes the daily traits of both treatments (height, area, solidity, senescence, water content and potential, canopy temperature depression, crop water stress index, leaf area, assimilation, root share, root depth and length, D95, transpiration and soil evaporation), as input for a functional-structural plant model or for a statistical package.

### 8.4.19 Moving about the layer

A bar of buttons stays at the top of the window while the page is scrolled (*Set up*, *Plants*, *Season film*, *Roots by day*, *Root scans*, *Traits*, *Water*, *Carbon*, *Soil*, *Roots and shoot*, *Genotype*, *Trait table*, *SWAFL data*, *Weather*, *Outlook*, *Stress calendar*, and *My data*). A button jumps to its section, and the section at the top of the window is marked. The title of every section has a triangle: click it to collapse or expand the section, or use *Collapse all* and *Expand all* at the top. The text of a hint never runs off the edge of the window: a long hint is cut into lines.

## 8.4.20 Equations

The section The layer shows no equations on the screen: they are all in this manual, numbered P.1 to P.28 in the order in which the model uses them on each day. This section explains each one: what it computes, the symbols and their units, where its result appears in the layer, and where it comes from. The model is deterministic: it contains no random numbers. A table at the end of the section gives the index. Symbols: $d$ day after emergence, $T$ air temperature (°C), LAI leaf area index (m² m⁻²), $\psi$ water potential (MPa or cm of suction as stated), $T_p$ potential transpiration (mm d⁻¹), $z$ depth (cm), $\Delta z = 15$ cm the thickness of a soil layer, $N = 140$ days.

### 8.4.20.1 Development, weather and light (P.1 to P.5)

**P.1 Thermal time.** The development of the plant is measured in degree days, not in calendar days:

$$d_g = \max\bigl(0,\ \min(T,\ T_{opt} + 6) - T_{base}\bigr), \qquad \mathrm{GDD}(d) = \sum_{i \le d} d_g(i). \tag{P.1}$$

$T_{base}$ and $T_{opt}$ are the base and optimum temperatures of the species (maize 8 and 30, wheat 0 and 20, bean 10 and 25 °C). The stage is $\mathrm{GDD}/\mathrm{GDD}_{flowering}$ before flowering and $\mathrm{GDD}/\mathrm{GDD}_{maturity}$ after it. It sets the leaf number, the start of grain filling and the end of the season. *Appears in:* the stage and the leaf count of every chart. *Source:* the degree-day method in common use in crop models (Ritchie and NeSmith, 1991).

**P.2 Weather of the day.** The reference evapotranspiration of the day is that of FAO-56 (Allen et al., 1998):

$$\mathrm{ET}_0 = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\dfrac{900}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)}. \tag{P.2}$$

$\Delta$ is the slope of the saturation vapour pressure curve (kPa °C⁻¹), $R_n$ the net radiation (MJ m⁻² d⁻¹), $G$ the soil heat flux (taken as zero on a daily step), $\gamma$ the psychrometric constant (kPa °C⁻¹), $u_2$ the wind at 2 m (m s⁻¹), and $e_s - e_a$ the vapour pressure deficit (kPa). The temperature, radiation and rain of the day come from the climate chosen in the layer (a sum of sines on the climate of the project) or from the imported weather (days of a file or of the online weather). With imported days, $\mathrm{ET}_0$ and the mean temperature are those of the file, plus the extra warming set in the layer. *Appears in:* every water chart, the demand curve, the canopy temperature and the outlook.

**P.3 Light interception.** The share of the light that the canopy catches follows Monsi and Saeki (1953):

$$f_i = 1 - e^{-k\,\mathrm{LAI}}. \tag{P.3}$$

$k$ is the light extinction coefficient of the species (0.5 to 0.65). *Appears in:* the carbon section and the leaf area charts.

**P.4 Biomass from light.** The dry matter made on a day is limited by the light caught (Monteith, 1977):

$$dW_{rad} = \mathrm{RUE}\ f_T\ f_N\ f_{sal}\ \mathrm{PAR}\ f_i. \tag{P.4}$$

RUE is the radiation use efficiency (g MJ⁻¹ of PAR), $f_T$ the beta temperature function of Yin et al. (1995) of the species, $f_N$ the nitrogen factor (P.18), $f_{sal}$ the salt factor (P.19), and PAR the photosynthetically active radiation of the day (MJ m⁻²). The fit to the SWAFL data (P.20) multiplies RUE by a factor between 0.6 and 1.6.

**P.5 Biomass from water.** The dry matter is also limited by the water that the plant transpires (Tanner and Sinclair, 1983):

$$dW_{wat} = \frac{k_{te}}{\mathrm{VPD}}\, T_{act}, \qquad dW = \min\bigl(dW_{rad},\, dW_{wat}\bigr). \tag{P.5}$$

$k_{te}$ is the transpiration efficiency coefficient of the species (Pa, 5.5 to 10), VPD the mean vapour pressure deficit of the day and $T_{act}$ the transpiration of the day (mm). Growth is the smaller of the two. *Appears in:* the carbon section and the traits (dry mass, transpiration efficiency).

#### 8.4.20.2 Soil, water and the root (P.6 to P.10)

**P.6 Soil water.** The soil of the project is divided into eight layers of 15 cm (120 cm in all). The retention curve and the conductivity are the van Genuchten (1980) and Mualem functions:

$$\theta(h) = \theta_r + \frac{\theta_s - \theta_r}{\bigl[1 + (\alpha h)^n\bigr]^m}, \qquad K(\theta) = K_s\, S_e^{l}\Bigl[1 - \bigl(1 - S_e^{1/m}\bigr)^m\Bigr]^2, \qquad m = 1 - \frac{1}{n}. \tag{P.6}$$

$\theta_r, \theta_s$ are the residual and saturated water contents, $\alpha$ and $n$ the shape parameters of the soil of the project, $K_s$ the saturated conductivity, $S_e$ the relative saturation and $l = 0.5$. Soil evaporation has two stages after Ritchie (1972): at the potential rate while the surface is wet, then in proportion to the square root of time. Water that infiltrates fills a layer to field capacity and passes the excess to the next one; the excess of the last layer is the drainage. *Appears in:* the water heat maps and the water balance.

**P.7 Root water uptake.** Each layer gives water in proportion to the root length it holds and to the difference between its water potential and that of the root (Gardner, 1960; Feddes et al., 1978):

$$S(z) = G(z)\,\bigl(\psi_{soil}(z) - \psi_{root}\bigr), \qquad G(z) = L_{root}(z)\, f_{fine}\, K(h)\, q_G, \qquad \sum_z S(z) = T_p\, g_s. \tag{P.7}$$

$L_{root}$ is the root length of the layer (cm), $f_{fine}$ the share of fine roots, $K(h)$ the conductivity of the soil at its suction, and $q_G$ the conductance of the root surface (default 0.1; the fit to the SWAFL data may change it by 0.25 to 4 times). The root potential $\psi_{root}$ is found every day by bisection so that the uptake equals the demand $T_p\, g_s$. The reduction of uptake by wet and dry soil follows the stress function of Feddes. *Appears in:* the bars at the right of the water section and the root water of the layers.

**P.8 Leaf water.** The leaf potential follows from the flow through the plant (van den Honert, 1948), and the leaf water content from the pressure and volume curve:

$$\psi_{leaf} = \psi_{root} - r\,\frac{T_{act}}{\mathrm{LAI}}, \qquad \psi = \frac{\pi_0}{\mathrm{RWC}} + P(\mathrm{RWC}), \qquad P(\mathrm{RWC}) = -\pi_0\,\frac{\mathrm{RWC} - \mathrm{RWC}_{tlp}}{1 - \mathrm{RWC}_{tlp}}. \tag{P.8}$$

$r$ is the hydraulic resistance of the plant (set by the genotype), $\pi_0$ the osmotic potential at full turgor (lowered by the osmotic adjustment), RWC the relative water content and $\mathrm{RWC}_{tlp}$ the content at which turgor is lost. The second equation is solved for RWC from $\psi_{leaf}$. *Appears in:* the water status traits (leaf potential, water content, turgor).

**P.9 Stomata.** The stomatal opening is a logistic function of the leaf potential (Jarvis, 1976):

$$g_s = \frac{1}{1 + e^{-b\,(\psi_{leaf} - \psi_{50})}}. \tag{P.9}$$

$\psi_{50}$ is the leaf potential at which half of the opening is lost (the control *Stomata close at*) and $b$ the steepness. The transpiration demand is multiplied by $g_s$. A genotype that closes its stomata early (the *water saver*) has a $\psi_{50}$ closer to zero.

**P.10 Expansion.** The growth of the leaf area and of the height is proportional to the turgor above a yield threshold (Lockhart, 1965):

$$\frac{1}{V}\frac{dV}{dt} = \phi\ \max(0,\ P - Y). \tag{P.10}$$

$V$ is the area (or the height), $P$ the turgor pressure (MPa), $Y$ the yield threshold and $\phi$ the extensibility. Because $P$ falls before photosynthesis does, expansion slows first, which is why the shoot area and the height slow before the carbon gain does.

#### 8.4.20.3 Temperature, carbon and the root system (P.11 to P.14)

**P.11 Canopy temperature.** From the energy balance of the leaf (Jackson et al., 1981):

$$T_c - T_a = \frac{r_a R_n}{\rho c_p}\,\frac{\gamma^*}{\Delta + \gamma^*} - \frac{\mathrm{VPD}}{\Delta + \gamma^*}, \qquad \gamma^* = \gamma\Big(1 + \frac{r_c}{r_a}\Big), \qquad \mathrm{CWSI} = \frac{dT - dT_{low}}{dT_{up} - dT_{low}}. \tag{P.11}$$

$T_c$ is the canopy and $T_a$ the air temperature, $r_a$ and $r_c$ the aerodynamic and canopy resistances (s m⁻¹; $r_c$ rises when the stomata close), $\rho c_p$ the volumetric heat capacity of air, and $dT = T_c - T_a$. The lower limit $dT_{low}$ is that of a fully transpiring canopy and the upper limit $dT_{up}$ that of a non-transpiring one. The canopy temperature depression is $T_a - T_c$. *Appears in:* the thermal view, the canopy temperature depression and the crop water stress index.

**P.12 Carbon allocation.** The carbon made is shared between leaf, stem, root and grain. The share to the roots rises when water is short and falls with the stage (functional equilibrium: Brouwer, 1963; Davidson, 1969):

$$f_r = \mathrm{clip}\Big[\big(f_{r0} + (f_{rA} - f_{r0})\,\mathrm{stage}\big)\,c_{root}\,(1 + 0.9\,w_s),\ 0.03,\ 0.7\Big], \qquad w_s = 1 - \frac{T_{act}}{T_p}. \tag{P.12}$$

$f_{r0}$ and $f_{rA}$ are the share of the species at emergence and at flowering (wheat 0.42 and 0.07), $w_s$ the water stress of the day (0 when the plant transpires what it demands), and $c_{root}$ the control *Carbon share to roots* times the genotype. The shoot takes the rest; after flowering the grain is filled from the new carbon and from the reserves of the stem. *Appears in:* the carbon section and the root to shoot ratio.

**P.13 Root architecture.** The roots are built with the rules of ArchiSimple (Pagès et al., 2014): axes leave at the set angle with gravitropism; laterals of the first and of the second order leave at a density per cm; each tip grows at a rate set by soil temperature, soil strength (P.14) and soil water, and the carbon share (P.12) sets how much root there is to build:

$$\frac{dl}{dt} = e_0\ 2^{(T_s - 20)/10}\ \frac{e}{e_0}(Q)\ f_w\ f_{age}, \qquad f_w = \mathrm{clip}\Big(\frac{\theta_{rel} - 0.06}{0.5},\ 0.05,\ 1\Big). \tag{P.13}$$

where $e_0$ is the elongation rate of the axis or lateral (cm d⁻¹, from the species and the genotype), $T_s$ the soil temperature (°C, factor limited to 0.05 to 1.8), $e/e_0$ the relative elongation of P.14 at the layer of the tip, $\theta_{rel}$ the relative plant-available water of the layer, and $f_{age}$ a slowing with the age of the tip. The result is the root length, the depth, the depths of 50 % and 95 % of the length, and the three-dimensional picture of the scan. *Appears in:* the root scans and the roots day by day.

**P.14 Soil strength.** The resistance that a root tip meets, and the relative elongation that remains, are

$$Q = 0.5\,e^{4.5\,(\rho_b - 1.3)}\,\big(1 + (1 - S)\big), \qquad \frac{e}{e_0} = \frac{1}{1 + (Q/2.5)^3}. \tag{P.14}$$

$\rho_b$ is the bulk density (g cm⁻³) of the layer, set by the texture, the tillage and the consolidation rate of the V3 layer; $S$ is the relative water content of the layer and $Q$ the resistance in MPa. At $Q = 2.5$ MPa the elongation is halved. *Appears in:* the section on soil structure and tillage.

#### 8.4.20.4 The Properties panel reaches the plant (P.15 to P.19)

**P.15 Project crop.** The height, the basal crop coefficient and the rooting depth of the plant are those of the crop library of the Properties panel:

$$K_{cb} = K_{c,mid} - 0.05, \qquad z_{root} = z_{root}^{library}, \qquad d_{sow} = d_{plant}. \tag{P.15}$$

The species of the model (maize, sorghum, wheat or common bean) is the one nearest in kind to the crop of the project, and the day of sowing is the planting day of the project, so that the date on the screen is the calendar date of the season. *Appears in:* the note under the controls and all dates.

**P.16 Project irrigation.** Every day the suction $h_s$ of the layer at the sensor depth is read. If $h_s \geq h_{trig}$ and the days since the last event are at least the minimum interval of the method, the root zone is refilled to field capacity, at most the dose of the method, and the water applied is the gross depth:

$$d_{net} = \min\Big(\sum_z (\theta_{fc} - \theta_z)\,\Delta z,\ D_{max}\Big), \qquad I = \frac{d_{net}}{E_a}. \tag{P.16}$$

The minimum intervals are 2, 3 and 1 days, the largest doses $D_{max}$ 35, 60 and 10 mm, and the efficiencies $E_a$ 0.75, 0.60 and 0.90 for the sprinkler, the surface and the drip methods. A drip line waters only when the layer at the drip depth is at least 12 % depleted. The trigger and the sensor depth are those of the Properties panel. *Appears in:* the left plant (*Project irrigation*) and every irrigation event of the outlook.

**P.17 Mulch.** The mulch lowers soil evaporation and keeps the first millimetres of a rain from the soil:

$$E_p' = f_m\,E_p, \qquad P' = \max(0,\ P - C_m). \tag{P.17}$$

$f_m$ is the vapour-flux factor of the mulch of the Properties panel and $C_m$ its interception capacity (mm per event). The effect on soil evaporation is compared with the Daisy runs in the SWAFL data section.

**P.18 Nitrogen.** The radiation use is multiplied by the nitrogen factor taken from the Daisy runs E13 of the soil of the project:

$$f_N = \frac{Y(N)}{Y(100\ \text{kg N ha}^{-1})}. \tag{P.18}$$

$Y(N)$ is the mean yield of the Daisy runs at the nitrogen rate $N$ of the Properties panel (interpolated between the rates simulated).

**P.19 Salt.** The solution of the root zone, of volume $V$ (twice the plant-available water of the column), takes the salt of the irrigation water $EC_w$ and of the rain, and loses it in the drainage $D$; growth is reduced above the threshold of Maas and Hoffman (1977), and the soil potential gets an osmotic term:

$$V\frac{dC}{dt} = I\,EC_w + 0.05\,P - D\,C, \quad EC_e = \frac{C}{2}, \quad f_{sal} = 1 - \frac{s}{100}\max(0, EC_e - EC_{th}), \quad \psi'_{soil} = \psi_{soil} - 0.036\,EC_e. \tag{P.19}$$

$C$ is the salt concentration (dS m⁻¹), $EC_e$ the electrical conductivity of the saturation extract, $EC_{th}$ and $s$ the threshold (dS m⁻¹) and the slope (% per dS m⁻¹) of the crop (from the salt tolerance table of the project), and $\psi$ in MPa.

#### 8.4.20.5 The SWAFL data (P.20 and P.21)

**P.20 Against the SWAFL data.** The plant is compared with the Daisy runs of the soil and the controls (E2, E7, E13 and E14, Chapter 27) in numbers that do not depend on the length of the season:

$$Y_{rel} = \frac{Y_{plant}}{Y_{well\text{-}watered}}, \qquad \tau = \frac{T}{T + E_s}, \qquad J(\theta) = \sum_k \Big( \frac{m_k(\theta) - r_k}{\tau_k} \Big)^2 . \tag{P.20}$$

$Y_{rel}$ is the relative yield, $\tau$ the share of the evapotranspiration that is transpiration, $m_k$ the model and $r_k$ the Daisy value of the relative yield, the transpiration share, the mulch effect and the five points of the E2 curve, and $\tau_k$ their tolerances. The multipliers $\theta = (f_{RUE}, f_{kte}, f_{qG})$ of the radiation use, the water use coefficient and the root conductance are searched one at a time, with a step that is halved when no move helps, within 0.6 to 1.6, 0.5 to 2.2 and 0.25 to 4 times the species values, in at most 45 runs.

**P.21 Genotype call.** For each genotype $g$ of the library, the distance to the observed traits $o$ is the mean squared difference in units of the spread $\sigma_i$ of each trait over the days $t$, and the weight is a softmax whose temperature grows with the best distance:

$$d_g = \frac{1}{nT} \sum_{i,t} \Big( \frac{x_{g,i,t} - o_{i,t}}{\sigma_i} \Big)^2 , \qquad p_g \propto \exp\Big( - \frac{d_g - d_{min}}{0.03 + 0.25\, d_{min}} \Big). \tag{P.21}$$

Five traits (shoot area, senescence, leaf water content, canopy temperature depression, height) are used. The observed traits are those of your measurements when they are loaded, else those of the drought plant of the layer with its genotype hidden. The weights add to 1.

#### 8.4.20.6 Traits and roots read out (P.22 and P.23)

**P.22 Traits read out.** The projected shoot area is the area of the leaves seen from the side, the convex hull is the smallest convex polygon around them, and the solidity is their ratio (Lorence and Medina-Jimenez, 2022). The senescence index is the senescent share of the leaf area, and the canopy temperature depression is $T_a - T_c$. The drought susceptibility index of Fischer and Maurer (1978) and the stress tolerance index of Fernandez (1992) compare the grain with stress $Y_s$ and without $Y_p$ of each genotype with the means of the set:

$$\text{solidity} = \frac{A_{proj}}{A_{hull}}, \qquad \text{DSI} = \frac{1 - Y_s/Y_p}{1 - \bar{Y}_s/\bar{Y}_p}, \qquad \text{STI} = \frac{Y_s\, Y_p}{\bar{Y}_p^{\,2}}. \tag{P.22}$$

**P.23 Root day.** The depth of 95 % of the root length is the depth where the cumulative length reaches 95 % of the total, the root growth of the day is the new root length, and the root to shoot ratio uses the dry mass of the day:

$$\int_0^{D_{95}} L(z)\, dz = 0.95 \int_0^{\infty} L(z)\, dz, \qquad \frac{dL_{root}}{dt}, \qquad \frac{W_{root}}{W_{shoot}}, \qquad \text{RLD}(z) = \frac{L(z)}{A\, \Delta z}. \tag{P.23}$$

RLD is the root length density (cm cm$^{-3}$) of the bars of the section *Roots day by day*; $A$ is the ground area of one plant.

#### 8.4.20.7 The drought outlook and the stress calendar (P.24 to P.28)

**P.24 Weather of the outlook.** Day $k$ of the season has the date $t_k = t_{start} + k$ days, where $t_{start}$ is the date of sowing in the year in which the date of the toolbar $t_0$ falls in the season. Each day takes its rain $P$, reference evapotranspiration $\text{ET}_0$ and temperature $T$ from the first of these that exists:

$$(P, \text{ET}_0, T)_k = \begin{cases} \text{record of the day} & t_k \le t_0 \text{ (recorded)} \\ \text{row of the forecast} & t_k > t_0 \text{ and a row exists} \\ \text{analogue year } i_s, \text{ same day of the year} & \text{otherwise (future } s) \end{cases} \tag{P.24}$$

The analogue year gives $P$ and $\text{ET}_0$; the temperature is the mean of the record for that day of the year. Days up to $t_0$ with no record are filled from the typical year. Without an imported weather

the model weather is used, and the rain of the days after $t_0$ is multiplied by 0.4, 1 and 1.6 (and $ET_0$ by 1.08, 1 and 0.95) for the dry, typical and wet future.

**P.25 Analogue years.** For each year $i$ of the 20 years of the weather in use, the rain of the days that have no forecast is added:

$$S_i = \sum_{k \in \text{future}} P_i(k), \qquad i_{dry} = \arg\min_i S_i, \quad i_{wet} = \arg\max_i S_i, \quad i_{typ} = \text{median}. \tag{P.25}$$

The three futures are real years of the record, so each has the rain and the evaporative demand of one year, with their dry spells in the order in which they happened. They are not a forecast of the weather beyond the days of the forecast.

**P.26 Root zone water and drought risk.** The water of the root zone is the mean of the relative plant-available water $\theta_i$ of the layers that the roots have reached, each weighted by the part $f_i$ of the layer above the deepest root. A day is a day of stress when it falls below the readily available water of FAO-56 (Allen et al., 1998):

$$a = \frac{\sum_i f_i\,\theta_i}{\sum_i f_i}, \qquad K_s^{FAO} = \min\Big(1, \frac{a}{1-p}\Big), \qquad \text{stress day if } a < 1-p, \qquad R = \frac{Y_{rainfed}}{Y_{irrigated}}. \tag{P.26}$$

$p$ is the depletion fraction below which transpiration is reduced (0.55 for maize, sorghum and wheat, 0.45 for common bean). The risk of drought is classed by $R$, the grain of the rainfed plant divided by the grain of the plant irrigated by the rule of the project, under the same future: low when $R \geq 0.90$, moderate when $0.75 \leq R < 0.90$, high when $0.50 \leq R < 0.75$, severe below 0.50. The plant response of the outlook (the ratio $T_{act}/T_p$ averaged over seven days) is the second chart and shows what the model plant does with the water that the soil holds.

**P.27 Periods of drought stress.** Let $s_k = 1$ when the root zone water $a_k < 1-p$ (Eq. P.26) and $s_k = 0$ otherwise. A period is a run of days $[k_a, k_b]$ with $s_k = 1$ on its first and last day, in which runs of $s_k = 0$ shorter than or equal to $g = 2$ days are filled, and it is kept when $k_b - k_a + 1 \geq 2$:

$$\text{period } j = [k_a, k_b], \qquad n_j = k_b - k_a + 1, \qquad a_j^{min} = \min_{k_a \leq k \leq k_b} a_k, \qquad \overline{T/T_p}_j = \frac{1}{n_j} \sum_{k=k_a}^{k_b} K_{s,k}. \tag{P.27}$$

$K_{s,k}$ is the ratio of the transpiration to the demand of the day. The date of day $k$ is $t_{start} + k$ days. A period is *past* when $k_b$ is before the day of the date of the toolbar, *now* when it holds that day and *ahead* otherwise.

**P.28 What the roots suffer.** Let $\Delta L_k$ be the root length that the plant grows on day $k$, $\Delta L_k^w$ that of the well watered bench plant under the same weather, $D_k$ the depth of the deepest root and $L_i$, $\theta_i$ the root length and relative plant-available water of layer $i$ on the driest day $k^*$ of the period:

$$G_j = 100\,\frac{\sum_{k_a}^{k_b} \Delta L_k}{\sum_{k_a}^{k_b} \Delta L_k^w}, \qquad \Delta D_j = D_{k_b}^w - D_{k_b}, \qquad F_j = 100\,\frac{\sum_{i:\ \theta_i < 0.5} L_i}{\sum_i L_i}. \tag{P.28}$$

$G_j$ is the root growth of the period against the watered plant (below 100 means the roots grew less), $\Delta D_j$ is how much shallower the roots are at the end of the period (negative if they are deeper), and $F_j$ is the share of the root length that lies in layers holding less than half of their plant-available water. $G_j$ is not given when the bench plant itself grows less than 0.4 % of its final root length in the period, because the ratio would mean nothing.

#### 8.4.20.8 Index of the equations

| No. | Computes | In the layer | Source |
|---|---|---|---|
| P.1 | thermal time, stage | all charts | degree days (Ritchie and NeSmith, 1991) |
| P.2 | reference evapotranspiration | water, outlook | Allen et al. (1998) |
| P.3 | light interception | carbon | Monsi and Saeki (1953) |
| P.4, P.5 | biomass from light, from water | carbon, traits | Monteith (1977); Tanner and Sinclair (1983) |
| P.6 | soil water, evaporation | water | van Genuchten (1980); Mualem (1976); Ritchie (1972) |
| P.7 | root water uptake | water, roots | Gardner (1960); Feddes et al. (1978) |
| P.8 to P.10 | leaf water, stomata, expansion | water status | van den Honert (1948); Jarvis (1976); Lockhart (1965) |
| P.11 | canopy temperature, CWSI | thermal view | Jackson et al. (1981) |
| P.12 | carbon allocation | carbon, roots | Brouwer (1963); Davidson (1969) |
| P.13, P.14 | root architecture, soil strength | root scans, soil | Pagès et al. (2014); Eq. 8.1 |
| P.15 to P.19 | crop, irrigation, mulch, nitrogen, salt | set up, traits | FAO-56; SWAFL E13; Maas and Hoffman (1977) |
| P.20 | comparison and fit to SWAFL data | SWAFL data | Daisy runs of the repository |
| P.21 | genotype call | genotype prediction | this layer |
| P.22, P.23 | traits, tolerance indices, root day | traits, roots day by day | Lorence and Medina-Jimenez (2022); Fischer and Maurer (1978); Fernandez (1992) |
| P.24 to P.26 | weather of the outlook, analogue years, root zone water and risk | drought outlook | Allen et al. (1998); this layer |
| P.27 and P.28 | periods of drought stress, root growth, depth and dry share in each period | stress calendar | this layer |

*Provenance (Eq. P.1 to P.28):* **[Literature]**: the forms are from the works cited; **[Engineered]**: the constants of the species and genotypes, the efficiencies, dose limits and minimum intervals of the irrigation methods, the salt volume, the tolerances of the fit, the 12 % depletion of the drip layer, the 0.4, 1 and 1.6 rain factors and the risk classes are typical defaults of this layer; **[Data]**: the nitrogen factor, the trigger curve and the mulch effect are read from the SWAFL Daisy runs.

### 8.4.21 Limits

- The model is a **daily, one plant** model. It does not simulate individual photons, the diurnal course of the leaf or the structure of the canopy beyond the leaves of one plant.
- The constants of the species and genotypes are typical values and not the data of a trial. For a real experiment, load the observations and calibrate; the calibration changes three parameters only (radiation use efficiency, stomatal threshold, stay-green). The fit to the SWAFL data changes the radiation use, the water use coefficient and the root conductance, and it can reach the limits of the search.
- The genotype call ranks the genotypes of the library; it is not a sequence of a genome. Only this layer is connected deeply to the crop, soil, weather and irrigation of the Properties panel; the other layers are computed by the engine on the same controls and are shown after **Run simulation**.
- The weather is deterministic (a sum of sines on the climate of the project), so a run is a typical season, not a year of records, except that the project rain can be used.
- The roots are drawn as the projection of the three dimensional system on the plane of the view; the root length, depth and density are those of the three dimensional system.

*References of the layer:* Allen et al. (1998), FAO Irrigation and Drainage Paper 56; Maas and Hoffman (1977); Mualem (1976); Ritchie (1972); Ritchie and NeSmith (1991); Brouwer (1963); Davidson (1969); Feddes et al. (1978); Fernandez (1992); Fischer and Maurer (1978); Gardner (1960); Jackson et al. (1981); Jarvis (1976); Lockhart (1965); Lorence and Medina-Jimenez (eds, 2022), *High-Throughput Plant Phenotyping*, Methods in Molecular Biology 2539, Springer; Monsi and Saeki (1953); Monteith (1977); Pagès et al. (2014); Panguluri and Kumar (eds, 2013), *Phenotyping for Plant Breeding*, Springer; Tanner and Sinclair (1983); van den Honert (1948); van Genuchten (1980); Yin et al. (1995).

# Chapter 9

# Mulch and evaporation layers

This group follows the water that mulch keeps in the soil: how mulch lowers evaporation (V4, V6), the physics of drying (E1), where the saved water goes (N2) and when evaporation or drainage dominates (N3).

## 9.1 Mulch against bare soil

**What it shows.** Three soil blocks side by side: bare soil, the soil under *your* mulch, and under the strongest mulch (plastic film, $\varphi = 0.1$), with bars for evaporation, irrigation and drainage, the six materials as buttons (click one to apply it), and a verdict.

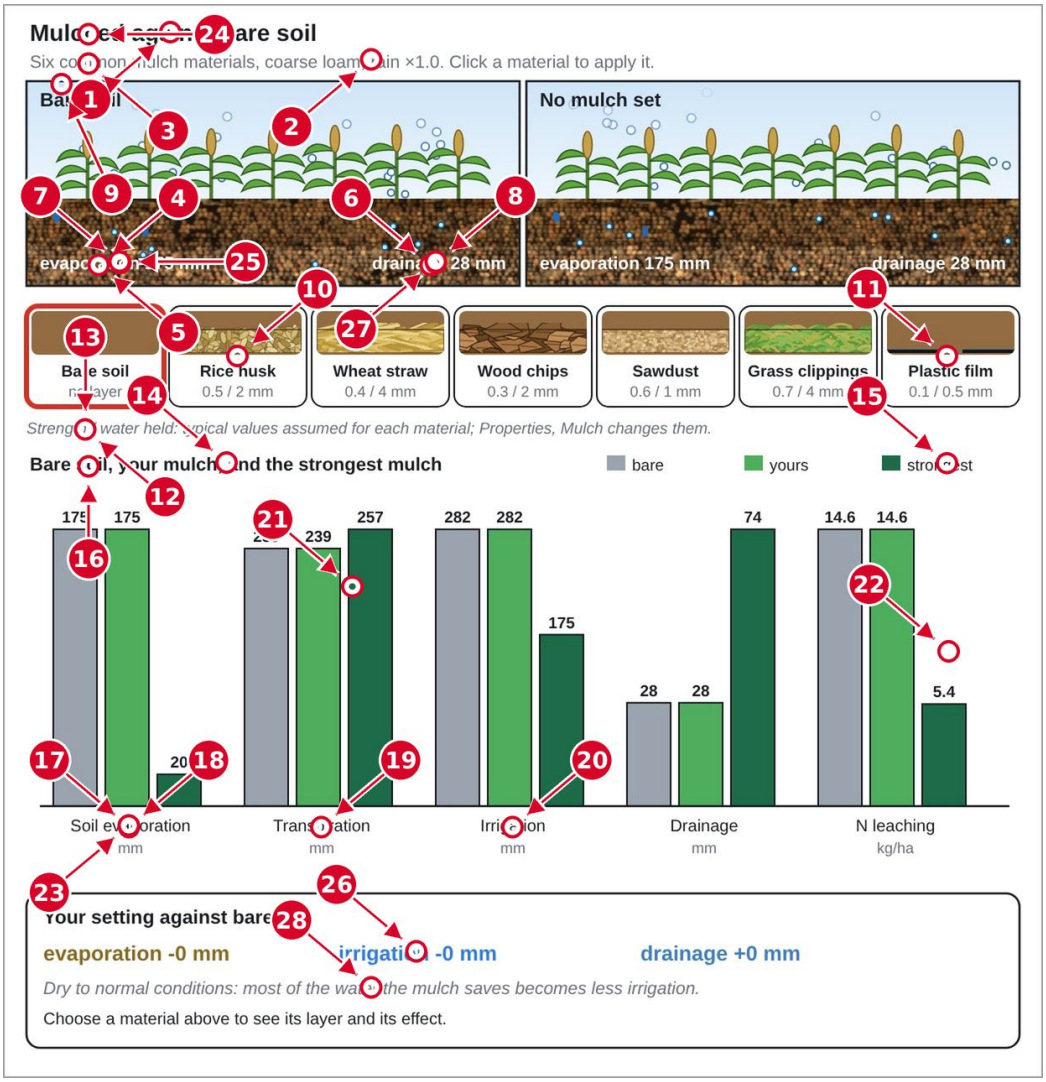



Figure 9.1: Mulch against bare soil. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

**How the numbers are generated.** All three blocks are Engine 1 evaluated at $\varphi = 1.0$ (bare), at your $\varphi$ and capacity, and at $\varphi = 0.1$, at the current rainfall. With $E_\varphi$, $I_\varphi$, $D_\varphi$ from Eqs. (5.4), (5.7), (5.9),

$$\Delta E = E_{1.0} - E_\varphi, \qquad \Delta I = I_{1.0} - I_\varphi, \qquad \Delta D = D_\varphi - D_{1.0}, \tag{9.1}$$

*Provenance (Eq. 9.1):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

are the evaporation saved, the irrigation saved and the extra drainage, in mm yr$^{-1}$, printed under the blocks. The verdict compares $\Delta I$ with $\Delta D$ (“*Wet conditions: much of the water the mulch saves now leaves as drainage*” or “*Dry to normal conditions: most of the water the mulch saves becomes less irrigation*”). The block of text for plastic film states that “*plastic film stops vapour almost completely but holds no water, so rain runs to the planting holes*”, which is the capacity of 0.5 mm in Table 3.1.

**Steps.** (1) Open *Mulch and evaporation* ▸ *Mulch against bare soil*. (2) Click a material. (3) Raise the **Rainfall scale** to $×$1.3 and watch $\Delta D$ overtake $\Delta I$. (4) Move the **Strength** slider to see the evaporation saving grow convexly.

## 9.2 Mulch control

**What it shows.** Two matrices. The first is **soil evaporation against mulch factor (ten rows, 0.1 to 1.0) by interception capacity (four columns, 0.5 to 4 mm)**. The second is **drainage against tillage system (three) by consolidation rate (five)**. Colour darkens with the value; a red outline marks your current setting. The text states what fraction of bare-soil evaporation your mulch removes and whether the soil responds to tillage.

**Generation.** Each cell of the first matrix is $E(\varphi, c)$ and each cell of the second $D(\text{till}, \tau_c)$, both from Engine 1 at the current soil, rainfall and temperature (Eqs. 5.4 and 5.9). The sentence "*No response here: this soil does not react to tillage or consolidation*" appears when the spread $\max D - \min D$ of the second matrix is below 0.5 mm yr$^{-1}$. The percentage in the text is $100\,(1 - E_{\varphi,c}/E_{1.0})$.

**Steps.** Click a cell to apply that mulch factor and capacity or that tillage and rate; the red outline moves; the Dashboard changes at once.

## 9.3 Evaporation

**What it shows.** A modelled thin section of the topsoil drying after a wetting, with liquid water in the pores (blue), vapour moving up (white), the **drying front** (orange) and, for comparison, the bare-soil curve (grey). To the right: the evaporation rate $e(t)$ and cumulative loss $Q(t)$ for 24 days, split into **stage 1** (the surface stays wet, loss at the atmospheric rate) and **stage 2** (a dry layer grows, loss is limited by the soil).

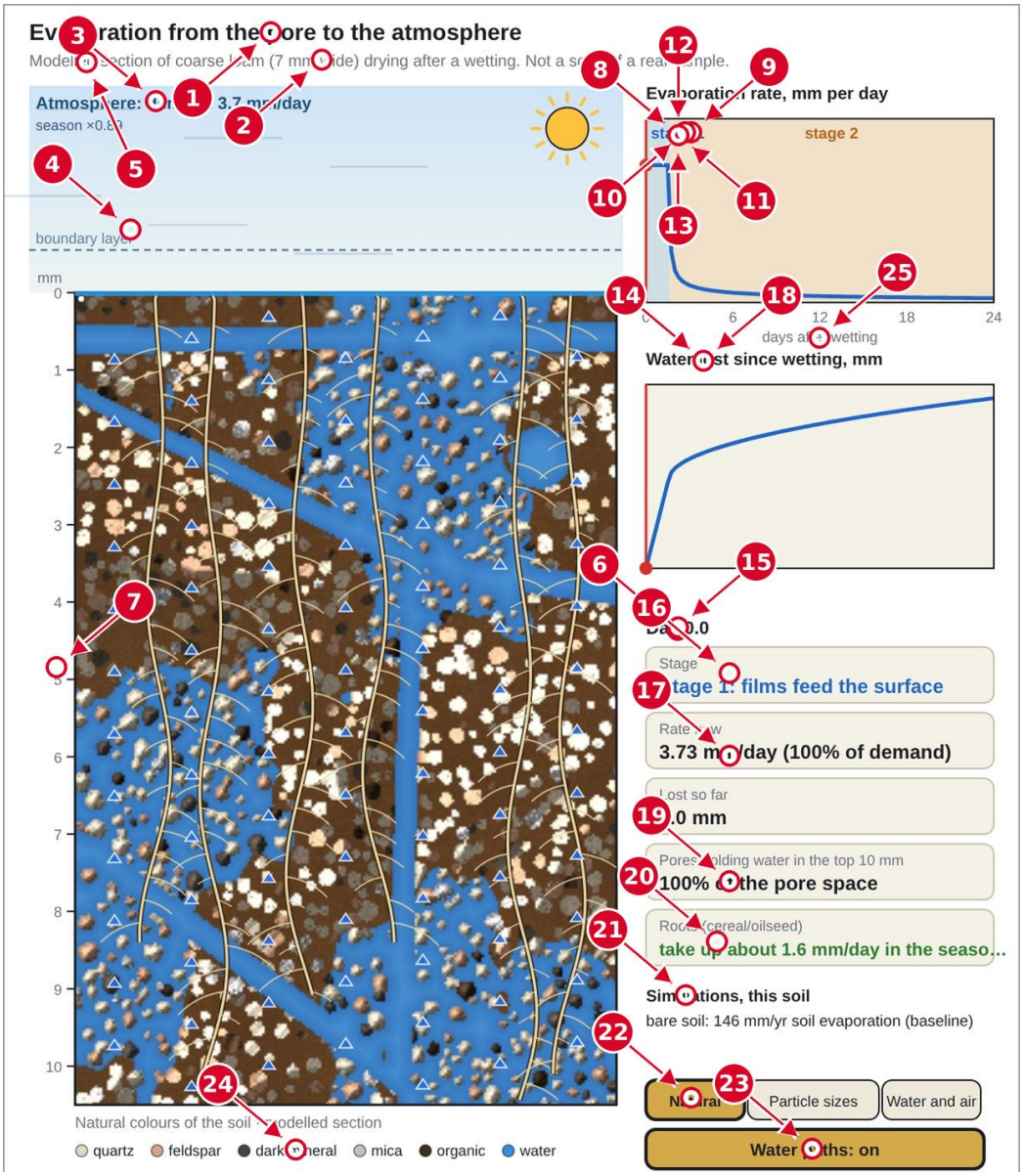



Figure 9.2: Evaporation layer. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

**Governing relations.** The loss is described by the classical two-stage model of soil drying (Philip, 1957; Ritchie, 1972): stage 1 at the atmospheric rate $e_0$, then stage 2 with a rate falling roughly as $t^{-1/2}$ (cumulative loss $\propto \sqrt{t}$). The implementation in the workbench is

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

*Provenance (Eq. 9.2):* **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

$$Q(t) = q_0 + \begin{cases} e_0\, t, & t \le t_1 \\ (Q_1 - q_0) + 2\, e_0\, \tau \left( \sqrt{1 + (t - t_1)/\tau} - 1 \right), & t > t_1 \end{cases} \tag{9.3}$$

*Provenance (Eq. 9.3):* **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

(the rate is the time derivative of the cumulative loss, so the two are consistent), with the following parameters:

| Symbol | Definition | Value or formula |
|---|---|---|
| $e_{0,b}$ | bare-soil atmospheric demand | Eq. (5.26) (mm d$^{-1}$) |
| $e_0$ | demand under mulch and canopy | $e_0 = e_{0,b}\,\varphi\,(1 - 0.4\,c)$ with canopy cover $c$ on the date |
| $Q_1$ | stage-1 storage (mm), water held in the top 40 mm between the wet and the surface-dry state | $Q_1 = \max\{2,\ [\theta(h_i) - \theta(h_s)]\,L_1\,(0.6 + 0.4\rho)\}$, $L_1 = 40$ mm |
| $h_i$, $h_s$ | initial and surface-dry suctions (cm) | $h_i = \dfrac{0.25}{\alpha}(1.5 - 0.5\rho)$; $h_s = 4/\alpha$ |
| $\tau$ | stage-2 time scale (d) | $\tau = 0.006 + 0.03\,(\text{clay} + \text{silt})/100$ |
| $q_0$ | loss already experienced when the water was placed below the surface (drip) | $q_0 = (1 - w_f)\,0.95\,Q_1$, $w_f = \text{clip}(1 - z_d/50;\ 0.05,\ 1)$, $z_d$ the drip depth in cm; 0 for surface methods |
| $\Delta\theta$ | water lost across the dry layer | $\Delta\theta = \text{clip}(\theta(h_i) - \theta(10^4);\ 0.05,\ 0.4)$ |

$\theta(h)$ is Eq. (8.3) and $\alpha$ the van Genuchten parameter of Table 5.3; $\rho$ is the rainfall scale. **[Engineered]**, every constant in the table (0.25, 1.5, 0.5, 4, 40 mm, 0.6, 0.4, 0.006, 0.03, 0.95, 50 cm, 0.05 to 1) is a presentation parameter chosen so that stage 1 lasts a few days in a loam and stage 2 gives the characteristic $\sqrt{t}$ fall; they are **not** calibrated to a laboratory evaporation experiment and the drawing says so ("*the daily rates are a two-stage drying model scaled by your mulch factor*"). The simulated annual evaporation of the workbench (Eq. 5.4) is *Daisy*, not this curve.

**The drying front.** The depth of the dry layer after stage 1 is

$$\ell(t) = \min\!\left\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\right\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

*Provenance (Eq. 9.4):* **[Engineered]**: drying-front geometry chosen for drawing.

and the suction profile with depth $z$ is an exponential approach of $\log_{10} h$ with length $\lambda = L_1/3$:

$$\log_{10} h(z) = \log_{10} h_s + (\log_{10} h_i - \log_{10} h_s)(1 - e^{-(z-\ell)/\lambda}) \quad (z \ge \ell), \qquad h = 10^5 \text{ cm for } z < \ell. \tag{9.5}$$

*Provenance (Eq. 9.5):* **[Engineered]**: drying-front geometry chosen for drawing.

**Pore picture.** Which pores are water-filled at each depth is given by the capillary rule of Eq. (8.4) with the local suction of Eq. (9.5); vapour particles are released from the drying front and rise through air-filled pores. The picture is an **illustration of the mechanism**, not a pore-network simulation.

**Steps.** (1) Open *Evaporation* and press **Play** (24 days). (2) Raise the mulch (lower $\varphi$): the grey bare-soil curve stays, your curve lengthens stage 1. (3) Change the soil: stage 1 lasts longest in the sands (largest $Q_1$, about 3 days for the Sand at the season-mean demand) and is shortest in the silty clay loam (about 1 day), because the window between $h_i$ and $h_s$ scales with $1/\alpha$; in stage 2 the fine soils lose water more slowly (larger $\tau$). (4) Set a drip method: $q_0 > 0$ shortens stage 1 because the surface has less water to lose. (5) Hover over a pore to read its radius and the suction at which it empties.

## 9.4 Fate of saved water

**What it shows.** For the soil and mulch you set (a preview of $\varphi = 0.5$ is used if you have not set mulch), four curves against rainfall scale $\times 0.5$ to $\times 1.4$: evaporation saved, irrigation saved,

extra drainage, extra transpiration. A shaded band marks the rainfall above which drainage exceeds the irrigation saved (“*drainage takes over near* $\times 1.2$”). Below, a bar divides each mm of evaporation saved among less irrigation, more drainage, more transpiration and “storage and other”.

**Generation.** For each rainfall $\rho$ in the ten-value list, with $\varphi^*$ the mulch factor,

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

*Provenance (Eq. 9.6):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

all from Engine 1 at the current soil, tillage and temperature, and the partition at the current rainfall is

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

*Provenance (Eq. 9.7):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

The crossover rainfall is the first $\rho$ with $\Delta D \geq \Delta I$ and $\Delta D > 1$ mm. The study reports about 60 % of the saved water going to reduced irrigation in dry years and about 63 % to extra drainage in wet years, with the crossover near $\rho \approx 1.2$ (repository README).

**Steps.** Choose a material; slide the **Rainfall scale**; the red dashed cursor and the bar move; read the three sentences at the bottom.

## 9.5 Regime map

**What it shows.** A map of **mulch factor (rows) against rainfall scale (columns)** coloured by which loss dominates, together with soil columns in which vapour rises (grey) or droplets fall (blue), and a verdict for your settings.

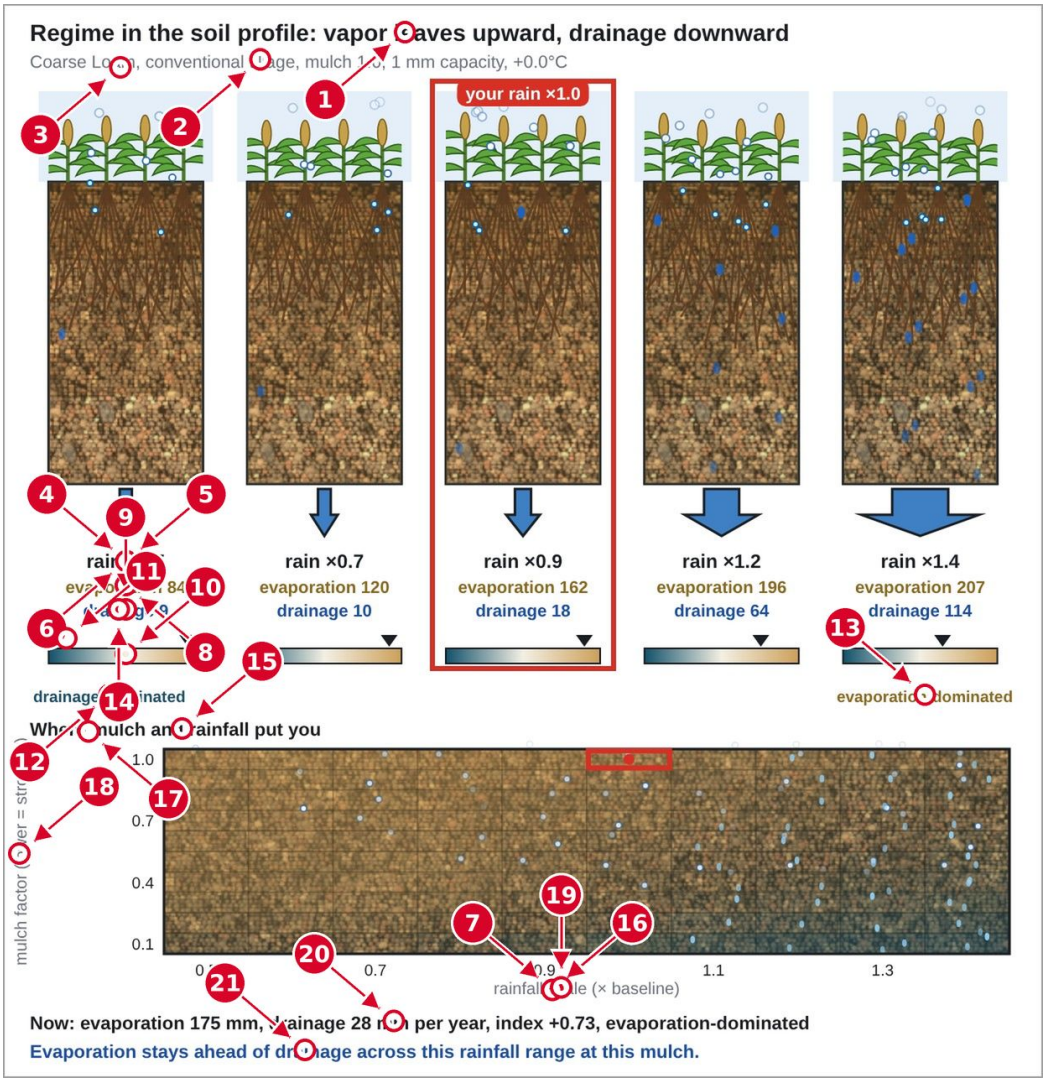



Figure 9.3: Regime map. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

**Generation.** For each cell the engine takes annual soil evaporation $E$ and drainage $D$ from Engine 1, and forms the **regime index**

$$\mathcal{I} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{I} > +0.05 \\ \text{drainage-dominated} & \mathcal{I} < -0.05 \\ \text{on the boundary} & |\mathcal{I}| \le 0.05 \end{cases} \tag{9.8}$$

*Provenance (Eq. 9.8):* **[Engineered]**: index defined for the workbench.

so $\mathcal{I} = +1$ is all evaporation and $-1$ all drainage. The statement "*Climate and mulch act non-additively … reaching 15 % of irrigation at combined extremes*" is a result of the study (the interaction of mulch and climate on irrigation); it is not recomputed by the layer.

**Steps.** Click a cell to apply that mulch factor and rainfall; the verdict says *"the main loss is evaporation (…), so mulch will pay off here"* or *"the main loss is drainage …, so mulch saves less"*.

# Chapter 10

# ♻ Feedback loops

## 10.1 What it shows

One large picture of a field, from sky to groundwater, on which red arrows join the parts that act on each other: temperature and the crop; rain and soil water; mulch and evaporation; the sensor trigger and irrigation; drainage and nitrogen leaching; nitrogen and the roots; tillage structure and drainage. Each arrow is numbered 1 to 7, and its **width grows with the size of the effect of one step** of the control behind it. Under the picture a panel lists “*What one step of each control does*”; click a row or an arrow to highlight it.

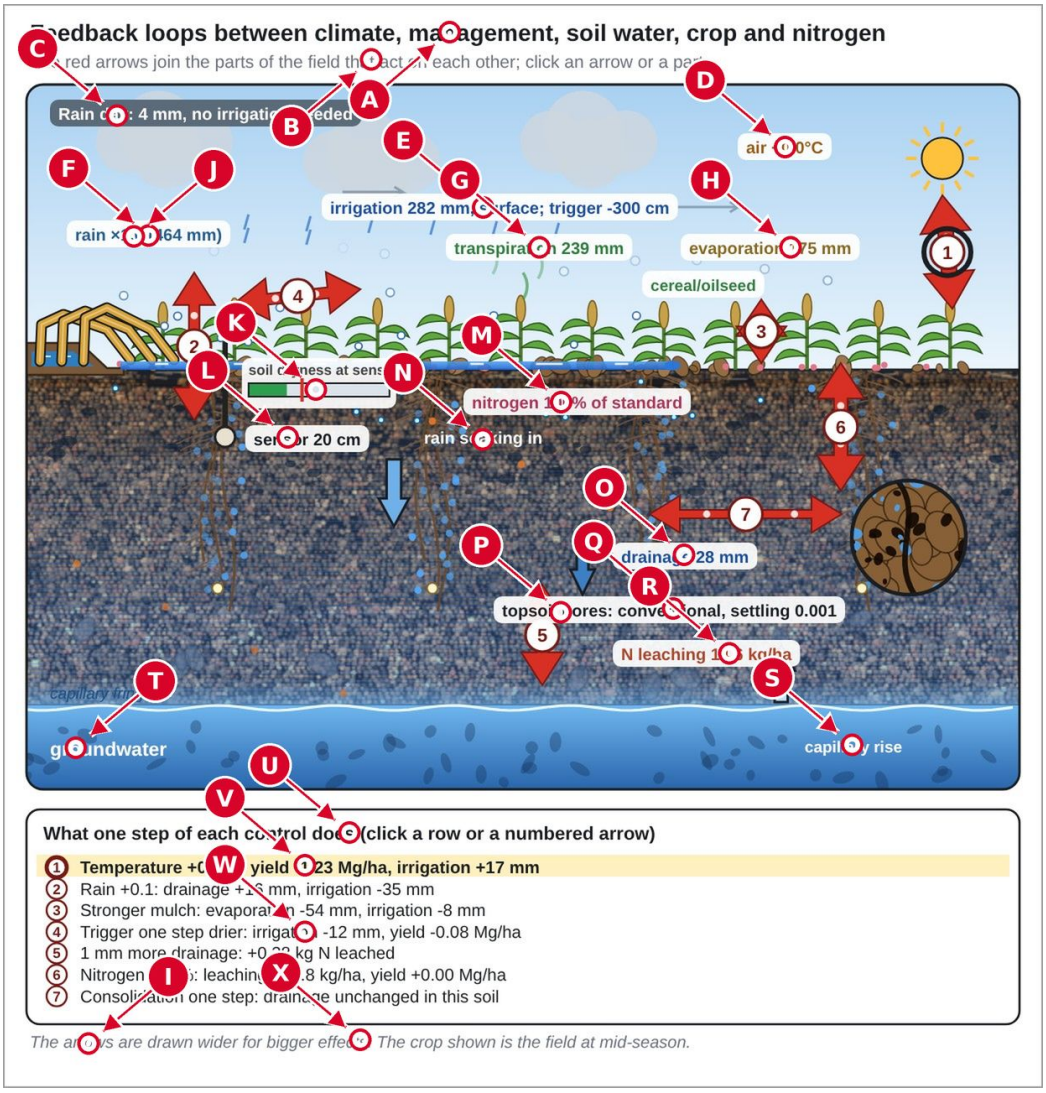



Figure 10.1: Feedback loops. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

The picture is also a set of controls: clicking the clouds cycles the rainfall, the sun the temperature, the mulch layer the material, the sensor the trigger (three grid steps drier each click, wrapping), the soil the tillage, the crop the crop category, the emitter or sprinkler the irrigation method, and the nitrogen band the nitrogen rate.

## 10.2 How the seven effects are computed

For each control the engine moves **one step** on its list (in the direction that stays inside the list: if the control is at its maximum it steps down) and re-evaluates Engine 1 (Eqs. 5.4 to 5.12). The change relative to the present result is printed. With $m$ the present result and $m^{(\pm)}$ the result after one step, and $f_{\text{crop}}$ the crop factor:

| # | Control step | Printed effect | Arrow-width input $\mu$ |
|---|---|---|---|
| 1 | Temperature $\pm 0.5$ °C | $\Delta Y\, f_{\text{crop}}$, $\Delta I$ | $8\,\lvert\Delta Y\rvert$ |
| 2 | Rain $\pm 0.1$ (scale) | $\Delta D$, $\Delta I$ | $\lvert\Delta D\rvert / 12$ |

| # | Control step | Printed effect | Arrow-width input $\mu$ |
|---|---|---|---|
| 3 | Stronger mulch (one step lower $\varphi$) | $\Delta E$, $\Delta I$ | $\lvert\Delta E\rvert/14$ |
| 4 | Trigger one step drier or wetter | $\Delta I$, $\Delta Y f_{\text{crop}}$ | $\lvert\Delta I\rvert/8$ |
| 5 | One more mm of drainage | $\Delta N_L/\Delta D$ kg N per mm (from the rain step), or "no change" if $\lvert\Delta D\rvert \le 0.5$ | $\lvert\Delta D\rvert/10$ |
| 6 | Nitrogen $\pm 15$ % | $\Delta N_L$, $\Delta Y f_{\text{crop}}$ | $\lvert\Delta N_L\rvert/8$ |
| 7 | Consolidation one step | $\Delta D$ (or "unchanged in this soil" if $\lvert\Delta D\rvert < 0.05$) | $\lvert\Delta D\rvert/2$ |

and the arrow line width in pixels is

$$w_k = 5 + \text{clip}(\mu_k;\, 0,\, 4). \tag{10.1}$$

*Provenance (Eq. 10.1):* **[Engineered]**: drawing rule.

The sensitivity printed in row 5, the **nitrate cost of drainage**, is the ratio

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D} \quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

*Provenance (Eq. 10.2):* **[Data-driven]**: one-step differences of Daisy results.

which the study reports as roughly 1.06 kg N ha$^{-1}$ per mm of drainage (README). The numbers are **[Data-driven]** one-step differences, not partial derivatives, so a step at the end of a list can be smaller than the step in the middle.

## 10.3 Hidden questions of the vadose zone

Below the seven effects the layer answers eight further questions. Every panel has the same three parts: pore-scale sections of the soil, a curve, and a sentence with the numbers.

**The sections.** Each section is the modelled thin section of the soil used in the Soil, plant and air layer and in the Evaporation layer: grains in their natural colours, the fine matrix dark, air-filled pores empty and water in blue. It is built from the texture of the selected soil, the tillage system and the settling of the project, then filled with water at the suction named under the section (a pore of radius r in micrometres empties at a suction of about 1490 divided by r in centimetres of water, and a film of water remains on the grains). It is not a scan of a real sample. Every section responds to the pointer: hover over it to read what is under the cursor (the mineral and grain size, or the radius of a pore, whether it holds water or air, and the suction at which it empties). Click a section to enlarge it. The enlarged section shows a scale bar, a bar chart of pore sizes split into water-filled and air-filled pores, the pore share of the area, the radius below which pores are full of water, the water content and the hydraulic conductivity of the soil, and a suction slider with presets (saturated, field capacity, your trigger, wilting point, air dry). Dragging the slider drains or fills the soil and every number follows. The sections are alive: white drops with a dark rim travel down through the water-filled pores, following the pore network, and pale vapour points rise through the air-filled pores. Their number follows the water and air in each section. In the enlarged section the button Play drying cycle sweeps the suction slowly from wet to dry and back, so the large pores empty first, then the medium ones, and the curves and numbers follow; touching the slider or a preset stops the cycle. The small graph of each panel is clickable too: click it to read the graph large, move the pointer over it to read the values (for the suction axes both pF and the suction in centimetres), and see the sections and the answer beside it. The percentage under a section is the share of the pores in the picture that are full of water.

**The curves.** The curves are the van Genuchten and Mualem functions of the selected soil. The compaction panel is made of runs of the model at each of the five settling steps.

**Connected to the controls.** Every panel follows the soil, tillage, settling, mulch, trigger suction, sensor depth, rain, temperature, crop and nitrogen set in Properties. For example, a drier trigger changes the suction of panels 1, 2 and 7, stronger mulch changes the demand and the stage lengths of panel 8, a different tillage or settling changes the pores of every section, and rain and irrigation change the salt balance of panel 6.

| Panel | Question | Sections | Curve | What it connects to |
|---|---|---|---|---|
| 1 | Does a sensor read the same after wetting as after drying? | At the trigger suction after drying, and after wetting (the wetting curve has twice the alpha of the drying curve) | Drying and wetting curves | Trigger suction, sensor depth, soil |
| 2 | Where does water still move when the soil is air dry? | At the trigger, at pF 4.2 and at pF 5.5 | log K against suction: capillary flow, an illustrative film term proportional to the suction to the power −1.5 anchored at $10^{-7}$ cm per day at pF 4.2, and their sum | Trigger suction, soil |
| 3 | Does compaction starve the crop or protect the groundwater? | Settling steps 0, 2 and 4 at field capacity | Drainage at each of the five steps, from runs of the model, with yield in the text | Tillage, settling, mulch, rain, crop, nitrogen |
| 4 | How far does the water table reach into the root zone? | 5, 25, 60 and 150 cm above the water table | Water content against height, and the height where saturation falls below 90 % | Soil, tillage, settling |
| 5 | How much of the water carries nearly all of the nitrate? | All pores wet, and the wide pores (those that empty at less than 31.6 cm) drained | The share of pore volume and the share of conductivity carried by wide, medium and fine pores | Soil, tillage, settling, nitrogen leaching of the model |
| 6 | How salty can the irrigation water be before the leaching it needs pollutes? | Water coloured amber with its salt: irrigation water at 2 dS/m and the drainage water | Relative yield of wheat against the salinity of the irrigation water for your drainage and for twice the drainage | Irrigation and drainage of the model (rain, mulch, trigger, soil) |
| 7 | Where on the suction scale does the trigger cost yield? | At 500 cm, at the trigger and at 8,000 cm | Reduction of root uptake with suction | Trigger suction, soil |
| 8 | How long does the soil keep up with the sun, and what does mulch change? | Just wetted, at the end of stage 1 and in stage 2 | Water the topsoil can deliver against suction, the demand reduced by mulch, and where stage 2 begins | Mulch, soil |

The film term, the salt balance (steady state, root-zone salinity taken as a quarter of the sum of the salinity of the irrigation and drainage water, wheat threshold 6 dS/m, slope 7.1 % per dS/m) and the uptake curve (breakpoints at 10, 25, 500 and 8,000 cm) are simple, stated forms for teaching and comparison. They are not calibrated to a field.

### 10.3.1 References for the hidden questions

The figures do not carry references; they are listed here.

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16. Bezerra-Coelho, C. R., Zhuang, L., Barbosa, M. C., Alfaro Soto, M. A., and van Genuchten, M. Th. (2018). Further tests of the HYPROP evaporation method for estimating the unsaturated soil hydraulic properties. Journal of Hydrology and Hydromechanics 66(2): 161 to 169.

## 10.4 The sentence under the picture

A status pill reports the day: *“Irrigating now: 28 mm (soil reached the trigger)”*, *“Rain day: 12 mm, no irrigation needed”* or *“Irrigation stopped: soil wet enough”*. The text for the trigger says whether *“the sensor has reached the trigger (depletion $f$)”*, to what depth *“the water wets the soil”* (the wetting front of the day’s irrigation, Section 5.2) and, when that is shallower than the sensor, *“the sensor may stay beyond the trigger”*. If no irrigation is due, it states how many days are free in the next 14 and that rain is needed to keep it that way.

## 10.5 Steps

1. Open *Feedback loops*. The crop shown is the field at mid-season.

2. Click an arrow (number 4, for example). The matching row is highlighted: a line of the form *“Trigger one step drier: irrigation −X mm, yield −Y Mg/ha”* (the numbers depend on your settings).
3. Click the picture parts to change controls and watch the arrow widths change.
4. Use **Pin baseline** first to compare the 20-year averages for the whole combination.

# Chapter 11

# Surface irrigation design

The surface-irrigation layer designs a **border strip, furrow or basin** for *your* soil: it derives the soil's intake from texture, solves the advance of the water front, chooses a cut-off time, computes where the water goes, and then searches for a better layout. Everything on this page is a closed-form hydraulic model; no Daisy table is read, except the crop need and the trigger, which come from the DSS (Chapter 5).

The layer is a long scrolling drawing. The sub-section bar above it (*Whole design, Layout design, Simulation, Hydrograph, Advance and opportunity, Results, Design table*) jumps to a section, as do the sub-items of the layer in the Layers panel. The twelve sections are, in order:

| # | Section (sub-item) | Purpose |
|---|---|---|
| 1 | Inputs and *Whole design* header | Choose the starting point, the system, intake, cut-off, and the value boxes; then Run simulation |
| 2 | Layout in plan: the advance front, shaded by the depth already infiltrated | Plan view of the field with the front |
| 3 | Longitudinal section: water running over your soil and soaking in | Side view with $Z_0$, $Z_L$, $Z_n$ |
| 4 | Hydrograph: inflow, infiltration and runoff through time | Flow partition with time |
| 5 | Cross-section at the inlet: the wetted bulb in your soil, and the intake curve | Intake curve $Z(\tau)$ and the soil behind it |
| 6 | Advance and opportunity time | $t_x(x)$ and $\tau(x)$ |
| 7 | Infiltrated depth along the field against the required depth $Z_n$ | $Z(x)$ against $Z_n$ |
| 8 | Performance indices and the water balance of one irrigation | $E_a$, $E_s$, DP, RO, CU, volumes |
| 9 | Design table | Iterations towards $E_a \geq 85\,\%$ |
| 10 | What the efficiency is most sensitive to | Tornado chart |
| 11 | What this means for the season and the farm | Season, sets, hours |
| 12 | Irrigation schedule and *Intervals, efficiency and the advice in short* | Dates, gaps, "When to irrigate" |

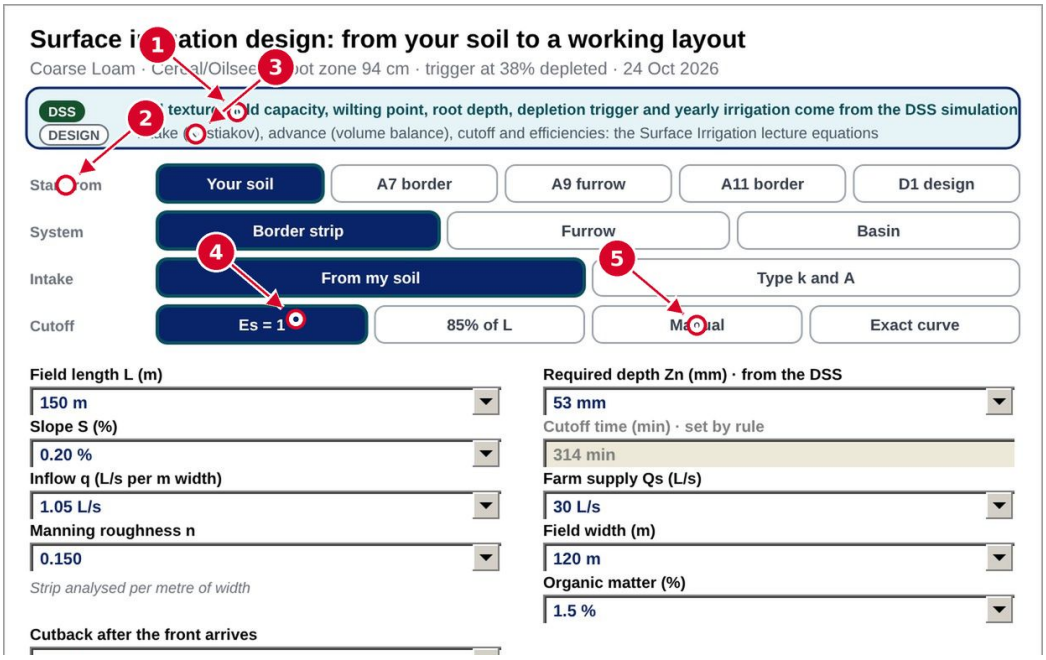



Figure 11.1: Surface irrigation design: the inputs (top) and the layout in plan. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 11.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Surface irrigation design*. The header states the soil, crop, root-zone depth, trigger depletion and date (e.g. *"Coarse Loam · Cereal/Oilseed · root zone 94 cm · trigger at 38 % depleted · 18 Oct 2026"*). The green *DSS* badge lists what is taken from the simulation; the *DESIGN* badge what the design chooses.
2. **Start from** one of five rows:

| Chip | Meaning |
|---|---|
| **Your soil** | Intake, required depth, cut-off and best inflow are all derived from your soil (default). |
| **A7 border** | Lecture worked example: border, $L = 200$ m, $S = 0.20$ %, $q = 2.5$ L s$^{-1}$ m$^{-1}$, $n = 0.15$, $k = 2$ mm min$^{-A}$, $A = 0.8$, $Z_n = 90$ mm, given advance $t_L = 120$ min, cut-off 170 min. |
| **A9 furrow** | Furrow, $L = 226$ m, $S = 0.30$ %, $q = 2.0$ L s$^{-1}$, $n = 0.04$, $w = 0.7$ m, $k = 4$, $A = 0.7$, $Z_n = 80$ mm, $\sigma_z = 0.83$. |
| **A11 border** | Border, $L = 240$ m, $S = 0.20$ %, $q = 1.667$, $n = 0.15$, $k = 7$, $A = 0.5$, $Z_n = 80$ mm, advance from field data $t_x = x^{1.35} q^{-1.5}/300$ (a power law, Eq. 11.12), cut-off at 85 % of $L$. |
| **D1 design** | Furrow, $L = 240$ m, $S = 0.30$ %, $q = 1.667$, $n = 0.04$, $w = 1$ m, $k = 2$, $A = 0.7$, $Z_n = 80$ mm, approximate advance (Eq. 11.13), $\sigma_z = 0.8154$. |

The four worked examples reproduce the course exercises so that the program can be checked against hand calculation; choosing one switches the intake to *Type k and A*.

3. **System**: *Border strip*, *Furrow* or *Basin*. This resets $L$, $S$, $q$, $n$ to the system defaults of Table 11.1.

4. **Intake**: *From my soil* (Green and Ampt from the soil's texture, fitted to Kostiakov, Section 11.2) or *Type k and A* (your own Kostiakov constants).
5. **Cutoff**: *Es = 1* (cut off when the tail has just received $Z_n$), *85 % of L* (cut off when the front reaches 85 % of the length plus the storage time), *Manual* (set the time), and the toggle *Exact curve* (draw the exact Kostiakov $Z(x)$ instead of the straight-line profile of the lecture notes).
6. Set the value boxes (Table 11.1): click a box, type the exact value or pick it from the list, choose the unit. Then press **Run simulation**; the results below are computed for these inputs (Chapter 18).

Table 11.1. Surface-design inputs (value boxes): ranges (min, max, step) and system defaults.

| Control | Border | Furrow | Basin | Notes |
|---|---|---|---|---|
| Field length $L$ (m) | 50 to 400, 5 (default 150) | 60 to 400, 5 (180) | 20 to 200, 5 (80) | |
| Slope $S$ (%) | 0.02 to 1, 0.01 (0.20) | 0.05 to 1.5, 0.01 (0.30) | 0.02 to 0.2, 0.01 (0.05) | |
| Inflow $q$ (L s$^{-1}$) | 0.2 to 20 per m width, 0.05 (6) | 0.2 to 4 per furrow, 0.05 (1.2) | 5 to 60 per m width, 0.5 (25) | In *Your soil* mode $q$ is set automatically (Section 11.9) until you move it |
| Manning $n$ | 0.02 to 0.25, 0.005 (0.15) | (0.04) | (0.15) | |
| Furrow spacing $w$ (m) | n/a | 0.5 to 1.5, 0.05 (0.75 in *Your soil*) | n/a | $w = 1$ m is used for border and basin (a strip 1 m wide is analysed) |
| Cutback after the front arrives | 0.2 to 1.0, 0.05 (1 = none) | same | same | Fraction of $q$ after $t_L$ |
| Required depth $Z_n$ (mm) | 20 to 150, 1 | same | same | Default *from the DSS* (Eq. 5.27); tap the label to reset |
| Farm supply $Q_s$ (L s$^{-1}$) | 5 to 120, 1 (30) | same | same | Farm-level flow |
| Field width $W_f$ (m) | 30 to 400, 5 (120) | same | same | |
| Organic matter (%) | 0.5 to 5, 0.1 (1.5) | same | same | Enters Saxton and Rawls |
| Kostiakov $k$ (mm min$^{-A}$) | 0.2 to 10, 0.05 | same | same | Visible in *Type k and A* mode |
| Kostiakov $A$ | 0.2 to 0.95, 0.01 | same | same | Visible in *Type k and A* mode |

## 11.2 Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)

**Step 1: soil water properties (Saxton and Rawls, 2006).** With sand fraction $S$, clay fraction $C$ (both 0 to 1) and organic matter OM (%), the 1500-kPa and 33-kPa water contents are

$$\begin{aligned} \theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031, \\ \theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02), \\ \theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299, \\ \theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015), \end{aligned} \tag{11.1}$$

*Provenance (Eq. 11.1):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

and saturation and saturated conductivity (mm h$^{-1}$) follow from

$$\begin{aligned}\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\ \theta_{(S-33)} &= \theta_{(S-33),t} + (0.636\,\theta_{(S-33),t} - 0.107), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\ B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.\end{aligned} \tag{11.2}$$

*Provenance (Eq. 11.2):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

These are the published pedotransfer equations of Saxton and Rawls (2006), implemented as written (field capacity $\theta_{fc} = \theta_{33}$, wilting point $\theta_{wp} = \theta_{1500}$). They are regressions on a large US soil database and carry its scatter; the layer uses them for design, not for prediction at a particular site.

**Step 2: wetting-front suction.** The effective suction at the wetting front (mm) is taken from the van Genuchten pair $(\alpha, n)$ of Table 5.3, $m = 1 - 1/n$,

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

*Provenance (Eq. 11.3):* **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

This is a rational-polynomial fit of the Rawls-type wetting-front suction as a function of $m$; it is an **[Engineered]** approximation (the code comment calls it the "Lisle/Rawls form") and should be read as giving the right order of magnitude (tens of mm for loams, e.g. 16 mm for the Coarse Loam), not a measured $\psi_f$.

**Step 3: available storage in the wetting zone.** The soil starts at the trigger depletion $f_T$ and is wetted to 95 % of saturation,

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

*Provenance (Eq. 11.4):* **[Engineered]**: initial and wetted water contents chosen for the workbench.

**Step 4: Green and Ampt infiltration** (Green and Ampt, 1911). Cumulative infiltration $F(t)$ (mm) solves

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

*Provenance (Eq. 11.5):* **[Literature]**: Green and Ampt (1911).

by Newton iteration from the start value $\max\{K_s t/60,\ \sqrt{2\psi_f \Delta\theta K_s t/60}\}$.

**Step 5: Kostiakov fit** (Kostiakov, 1932). The design uses the two-parameter empirical intake law

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

and $k$, $A$ are found by least squares on the log and log plot of $F(\tau)/1000$ at 25 times spaced geometrically between 10 and 600 min:

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

*Provenance (Eq. 11.7):* **[Literature]**: ordinary least squares on the log-log plot.

with coefficient of determination $R^2 = \big(\sum (x-\bar{x})(y-\bar{y})\big)^2 \big/ \big(\sum (x-\bar{x})^2 \sum (y-\bar{y})^2\big)$, shown on the layer (e.g. *"Ks 13.1 mm/h · ψf 16 mm · Δθ 0.20 · fit R² 0.997"*). This conversion of a physically

based law to a power law is what the lecture method requires; it is an approximation whose quality is reported by $R^2$.

**Step 6: required depth.** The net depth to refill the root zone is Eq. (5.27): $Z_n = \mathrm{clip}(f_T\,\mathrm{AWC}\,R_{\max};\,25,\,150)$ mm unless you type another value.

## 11.3 Inlet flow area (Manning)

The flow cross-section at the inlet, $A_0$ (m$^2$), is obtained from Manning's equation (Manning, 1891), in SI units,

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

*Provenance (Eq. 11.8):* **[Literature]**: Manning (1891).

- **Furrow.** A V-shaped section with side slope $z = 1$ (H:V), depth limit $D = 0.25$ m: $A = zy^2$, $P = 2y\sqrt{1+z^2}$. Eq. (11.8) is solved for the flow depth $y$ by 60 bisection steps for $Q = q$ (m$^3$ s$^{-1}$) and the slope $S$ (with $S \geq 10^{-4}$); $A_0 = zy^2$. If $y > D$ the layer warns that the furrow overtops.
- **Border and basin.** A wide sheet: $A \approx y$ per unit width, $R_h \approx y$, so $y_0 = \left(q\,n/\sqrt{S}\right)^{0.6}$ with $q$ in m$^2$ s$^{-1}$, and $A_0 = y_0$ (m$^2$ per m width).
- **D1 preset.** $A_0 = \left(q^2 n^2/792\right)^{0.37313}$ with $q$ in m$^3$ min$^{-1}$, the empirical furrow relation of the lecture's worked example. **[Engineered, course-specific]**

## 11.4 Advance (volume balance)

**Definition.** For an inflow $Q$ (m$^3$ min$^{-1}$ per analysed width) the position $x(t)$ of the front is obtained from a volume balance of the water on the surface and in the soil (Walker and Skogerboe, 1987; Elliott and Walker, 1982):

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^A\,w\,x \;\Longrightarrow\; x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^A\,w}. \tag{11.9}$$

*Provenance (Eq. 11.9):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

The code also accepts an optional steady-intake term, $Z = k\tau^A + f_0\tau$ (Kostiakov and Lewis), in which Eq. (11.9) gains the term $f_0\,t\,w/(1+r)$ in the denominator, with $r$ the iterated advance exponent (release 1.1; release 1.0 used a fixed $r = 0.55$); the interface does not expose $f_0$ (it is zero), and it exists so that the FAO Irrigation Paper 45 examples, which use it, can be reproduced. Here $\sigma_y = 0.8$ is the surface-storage shape factor, $w$ the width analysed (furrow spacing, or 1 m for borders and basins), and $\sigma_z$ the subsurface-storage shape factor

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

*Provenance (Eq. 11.10):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

where $r$ is the exponent of the power-law advance $x = p\,t^r$, found by iterating (10 times, starting from $\sigma_z = 0.8$) from the times $t_L$ and $t_{L/2}$ at which the front reaches $L$ and $L/2$,

$$r = \frac{\ln 2}{\ln(t_L/t_{L/2})}, \tag{11.11}$$

*Provenance (Eq. 11.11):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

with each time found by bisection on Eq. (11.9). Eq. (11.10) is the standard shape factor of the volume-balance method for the Kostiakov intake law (Elliott and Walker, 1982). The advance time to position $x$ is then $t_x(x) = x^{-1}(t)$, solved on a 300-point grid.

**Alternatives** (set by the worked examples; the first two are Eq. 11.12):

- *Given advance:* $t_x(x) = t_L\,(x/L)^{1/r}$ with $r = 0.55$ unless set (A7).
- *Field advance* (A11): $t_x(x) = c\,q^{-q_e}\,x^{x_e}$ with $c = 1/300$, $q_e = 1.5$, $x_e = 1.35$.

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

*Provenance (Eq. 11.12):* **[Engineered]**: alternative advance models set by worked examples and course notes. * *Approximate* (D1, lecture formula):

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

*Provenance (Eq. 11.13):* **[Engineered]**: alternative advance models set by worked examples and course notes.

with $t_V$ the volume-balance solution of Eq. (11.9). **[Engineered]**: this is the closed-form approximation of the course notes, not a published general law.

## 11.5 Cut-off, opportunity time and the infiltrated depth

The **time to infiltrate the required depth** follows from inverting Eq. (11.6),

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

*Provenance (Eq. 11.15):* **[Literature]**: inversion of the Kostiakov law.

The **cut-off time** $t_{co}$ is

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

*Provenance (Eq. 11.16):* **[Engineered]**: cut-off rules chosen as design options.

**A set lasts at most 48 hours.** When the inflow is too small for the soil (a sand with a small inflow), the advance can need days or even years. The inflow then stops at $t_{co} = 2880$ min, and the water front has reached $x_{\max}$ (from Eq. 11.9 at that time). The part of the run beyond $x_{\max}$ gets no water, and the deficit shows it. The design says so and suggest raising the inflow or shortening the run. *Provenance:* **[Engineered]**: a practical limit on the length of a set.

The **opportunity time** at distance $x$ is the time the water stays on the soil, $\tau(x) = t_{co} - t_x(x)$ for $t_{co} > t_x$, and the **infiltrated depth** is

$$Z(x) = k\,\big[t_{co} - t_x(x)\big]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

*Provenance (Eq. 11.17):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The *straight-line profile* of the lecture notes replaces $Z(x)$ by the line between $Z_0$ and $Z_L$ (and, for the 85 %-of-$L$ rule, a broken line through $(0.85L,\, Z_n)$); *Exact curve* draws Eq. (11.17). If $t_{co} < t_L$ the front never reaches the end of the field and the layer shows a warning (*the stream does not reach the end*).

## 11.6 Volumes, efficiencies and uniformity

With $w$ the analysed width, $N = 300$ grid intervals (90 while dragging) and the trapezoidal rule, for $Z_i = Z(x_i)$:

$$
\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0).
\end{aligned}
\tag{11.18}
$$

*Provenance (Eq. 11.18):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The performance indices (Walker and Skogerboe, 1987; Burt et al., 1997) are

$$
E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N} |Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}
$$

*Provenance (Eq. 11.19):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

$E_a$ is the **application efficiency**, $E_s$ the **storage (requirement) efficiency**, and CU the uniformity coefficient of Christiansen (1942) evaluated on the infiltrated depths. Because $V_{\text{app}} = V_{\text{inf}} + V_{\text{ro}}$ and $V_{\text{inf}} = V_{\text{st}} + V_{\text{dp}}$, the three fractions satisfy $E_a + \text{DP} + \text{RO} = 1$, which the layer prints as the closed water balance of one irrigation. The deficit $V_{\text{df}}$ is the missing part of $Z_n$. The distance at which $Z$ crosses $Z_n$, if it does, is $X = L\,(Z_0 - Z_n)/(Z_0 - Z_L)$.

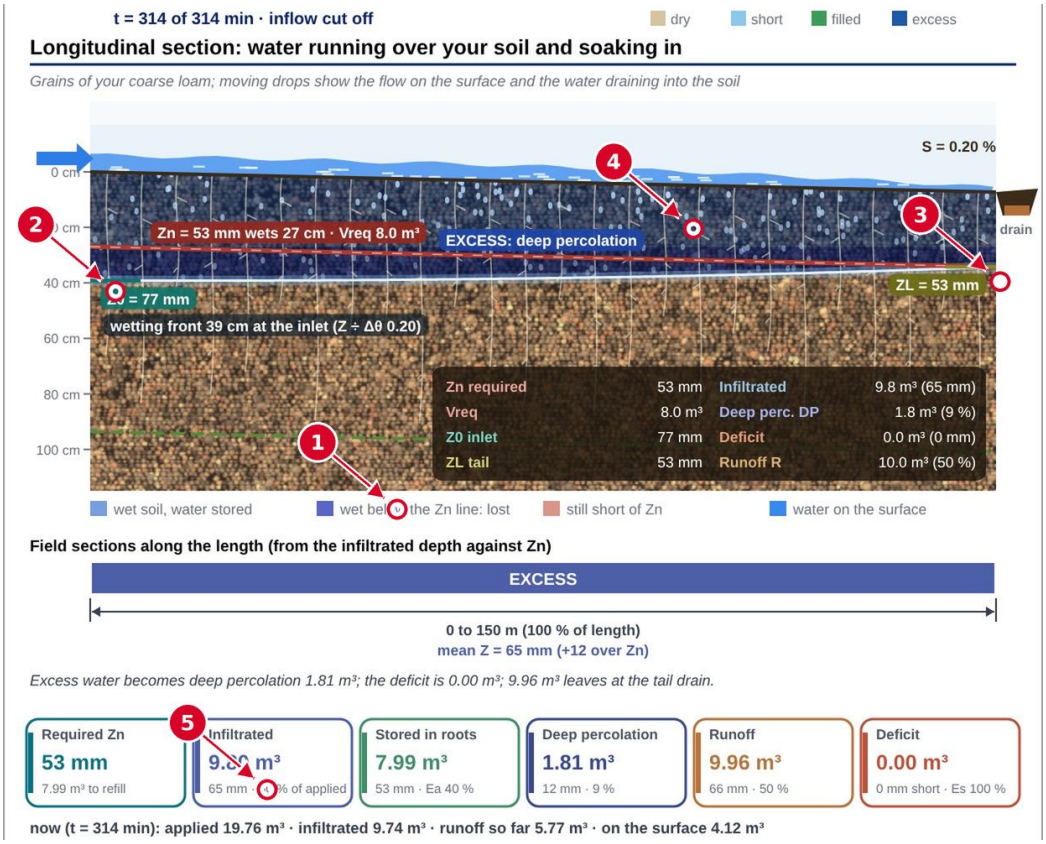



Figure 11.2: Surface design: longitudinal section with $Z_0$, $Z_L$, $Z_n$ and the water balance of one irrigation. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 11.7 Hydrograph

The hydrograph (Section 4 of the layer) divides the inflow $q$ into infiltration $I(t)$, runoff $R(t)$ and surface storage. It is built from the same advance and intake:

$$V_c(t) = w \sum_i k\left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

*Provenance (Eq. 11.20):* **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

where $\bar{t}_{x,i}$ is the advance time at the mid-point of interval $i$ and the factor $s_c$ makes the cumulative infiltration at cut-off equal to $V_{\text{inf}}$ of Eq. (11.18). While $t_L \le t \le t_{co}$ the runoff is $R(t) = \max\{0,\ q - I(t)\}$. After cut-off the surface water *recedes*: with $V_s = \max\{0,\ V_{\text{app}} - V_{\text{inf}} - \int R\,dt\}$ left on the surface, a triangular recession over $t_r = \max\{4,\ \min(0.3\,t_{co},\ 0.25\max(t_L, 10))\}$ min drains it at the tail,

$$R(t) = \frac{2V_s}{t_r}\left(1 - \frac{t - t_{co}}{t_r}\right), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

*Provenance (Eq. 11.21):* **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

The recession shape is **[Engineered]** (the course does not specify it); it conserves the volume $V_s$. Surface storage at time $t$ is $V_{\text{sto}}(t) = q\min(t, t_{co}) - V_c(t) - \int_0^t R\,dt'$.

## 11.8 Cross-section and intake curve

The *Cross-section at the inlet* section draws, at the inlet, the wetted shape in a soil section generated from the soil's texture, and the intake curve $Z(\tau)$ of Eq. (11.6) beside the Green and Ampt curve $F(\tau)$ of Eq. (11.5), so that the fit can be inspected by eye.

## 11.9 Best inflow ("Your soil" mode)

In *Your soil* mode, $q$ is chosen automatically from 25 equally spaced values between $q_{\min}$ and $q_{\max}$ of the system as the one that maximises $E_a\,E_s$. For furrows the upper limit is the non-erosive stream

$$q_{\max} = \min\left\{ q_{\max}^{\text{sys}},\ \frac{0.04}{S}\cdot\frac{1000}{60} \right\}\ \text{L s}^{-1} \quad (S \text{ in \%}), \tag{11.22}$$

*Provenance (Eq. 11.22):* **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

about $0.67/S$ L s$^{-1}$, in line with the classical Criddle et al. (1956) limit $Q_{\max} \approx 0.63/S$. The choice is stored in the inflow box, which becomes manual as soon as you change it.

## 11.10 The design table

The *Design table* automates one step at a time towards a good design. The **target** is $E_a \geq 85\,\%$ with $E_s \geq 80\,\%$ (the root zone stays at least 80 % refilled). The score of a design is

$$\text{score} = \min(E_a, 1) - 3\max(0,\ 0.8 - E_s) - 3\,[c_b < 1]\ \max(0,\ 0.05 - \text{RO}) - 5\max(0,\ E_a - 1). \tag{11.23}$$

*Provenance (Eq. 11.23):* **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

The search tries, in this order, ten single-variable moves, **cut-off, inflow, cutback, length, cut-off, cutback, inflow, length, cut-off, cutback**, and, for each move, evaluates a candidate list and keeps the best candidate **only if the score improves by more than 0.003**:

| Move | Candidates |
|---|---|
| Cut-off | $t_{co} \times$ $\{0.5, 0.58, 0.66, 0.74, 0.8, 0.85, 0.9, 0.95, 1, 1.05, 1.1, 1.2\}$ |
| Inflow | 12 values spaced geometrically from $q_{\min}$ to $q_{\max}$ |
| Cutback | $c_b \in \{1, 0.85, 0.7, 0.6, 0.5, 0.4, 0.3, 0.25, 0.2\}$ |
| Length | 11 values spaced geometrically from $\max(L_{\min}, 30)$ to $\min(L_{\max}, 1.5L)$, rounded to 5 m |

For every candidate other than a pure cut-off move, six cut-off times are also tried ($0.62$ to $1.02$ times the "$E_s = 1$" cut-off). The search stops when the target is met or after the ten moves. Each accepted row lists $q$, $L$, cutback, cut-off, $E_a$, $E_s$, DP and RO; **tap a row (green when it meets the target) to apply it** to the sliders. If the target cannot be reached the layer says so and suggests a shorter field, tailwater reuse, or drip or sprinklers. The same search is run by the toolbox item *Optimise surface layout (best design)*.

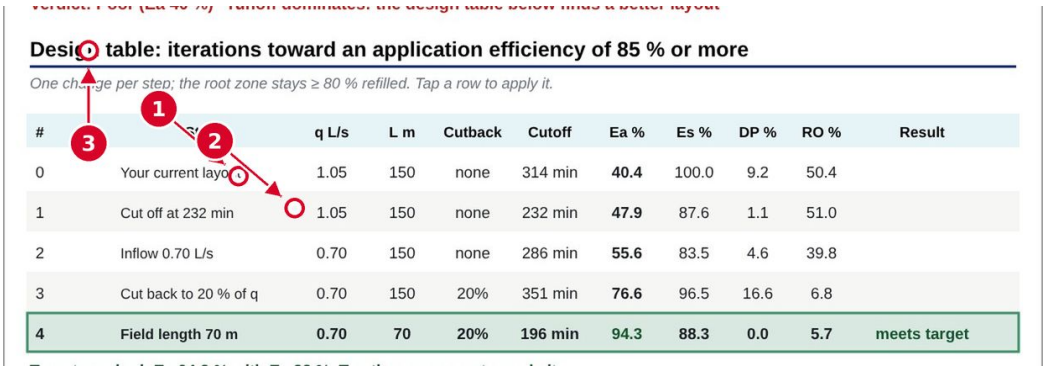



Figure 11.3: Surface design table: iterations toward $E_a \geq 85\,\%$, then the tornado chart. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 11.11 Sensitivity (tornado chart)

Each of seven inputs is moved up and down while the others are held, and the change in $E_a$ is drawn as a bar (green raises $E_a$, red lowers it):

| Input | Intake $k$ | Intake $A$ | Inflow $q$ | Length $L$ | Required depth $Z_n$ | Roughness $n$ | Slope $S$ |
|---|---|---|---|---|---|---|---|
| Perturbation | $\pm 20$ % | $\pm 10$ % | $\pm 20$ % | $\pm 20$ % | $\pm 20$ % | $\pm 30$ % | $\pm 30$ % |

The bars are sorted by the larger absolute change, and the printed value is $\max(|\Delta E_a^-|, |\Delta E_a^+|)$ in percentage points (*pts*). The test is local: it is the change around the *present* layout, not a global importance.

## 11.12 Season and farm

Using the irrigation record of Engine 2 for the year of the working date, with $\sum I$ the net irrigation of that year (mm):

$$n_{\text{ev}} = \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g,$$
$$N_{0M} = \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\left(1, \frac{N_{\text{units}}}{N_{0M}}\right), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \tag{11.24}$$

*Provenance (Eq. 11.24):* **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

and the seasonal pumping time is $n_{\text{ev}} T_{\text{tot}}$ hours. A depth of $Z_{\text{season}}$ mm equals $10\, Z_{\text{season}}$ m$^3$ per hectare. $N_{0M}$ is the number of furrows or strips that can run at once from the farm supply $Q_s$.

## 11.13 Irrigation schedule and “When to irrigate”

The last two sections are common to all three design layers, so they are described once, here.

**Schedule table.** The next eight irrigation dates after the working date are listed from the DSS (Engine 2, Section 5.2): the sensor in layer $j_s$ reaching the trigger depletion $f_T$. For each date the table gives

| Column | Definition |
|---|---|
| # | order, soonest first |
| Date | calendar date of the analogue year (Chapter 4) |
| In | *today*, *tomorrow*, or *n days* |
| Net mm | the DSS irrigation depth that day, $I_k$ |
| Pumped mm | gross depth $I_k/E_a$ with the design's application efficiency (min. $E_a = 0.05$) |
| Run time | time of one set or one round of the design |
| Gap to next | days to the following irrigation |

**Intervals.** Over the year of the working date, with irrigation days $d_1 < d_2 < \ldots$ (irrigations above 0.5 mm), the gaps $g_j = d_{j+1} - d_j$ give the mean, the shortest and the longest interval.

**Advice message.** The message *When to irrigate* is generated from these numbers: the *next irrigation date* and its net and pumped depth; *"Then irrigate about every $\bar{g}$ days (between $g_{\min}$ and $g_{\max}$ days, shorter in hot weather and longer after rain)"*; the efficiencies of the design ($E_a$, $E_s$) and the time of one round; and the rule *"If a rain above 10 mm is forecast, delay the irrigation by a day or two"*. If no irrigation is triggered, it says to *keep the sensor at $z_s$ cm and irrigate when it reaches depletion $f_T$*.

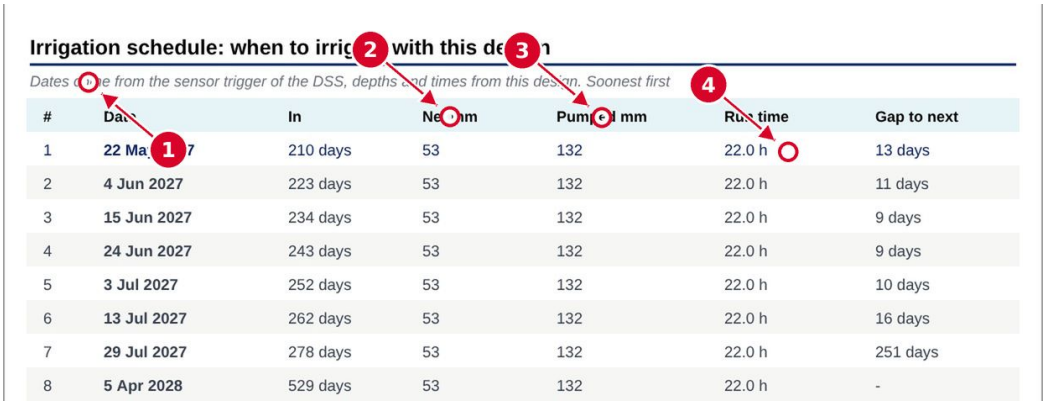



Figure 11.4: The irrigation schedule and the "When to irrigate" message at the end of the surface design. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

**Reading it.** The first row is today's decision. If the *Gap to next* is short (2 to 4 days) the trigger is wet or the soil store is small; if it is long (10+ days) the soil store is large or the weather cool. Move the **Date** to see how the schedule changes with the season, and move the **Trigger** or **Sensor depth** to see the dates shift.

# Chapter 12

# + Drip irrigation design

The drip layer designs the **emitter, its placement, the lateral and the pipe network** for the soil, crop and trigger set in the rest of the program. It answers four questions in order: how big a wet bulb does one emitter make, how much of the water it delivers stays in the root zone, how uneven is the delivery along the lateral, and what pump and pipes does the farm need. Everything is a closed-form hydraulic or geometric model, recomputed at every slider move.

The layer's sections, in order: control bar and inputs; soil moisture around the driest, average and wettest emitter (Run simulation animates the wetting); where the water goes; uniformity along the lateral; growth of the wetted bulb with time; result tiles; layout and infrastructure; irrigation schedule; the design table; the sensitivity chart; season and farm; and the "When to irrigate" advice. The last two are shared with the other design layers and are described at the end of Chapter 11.

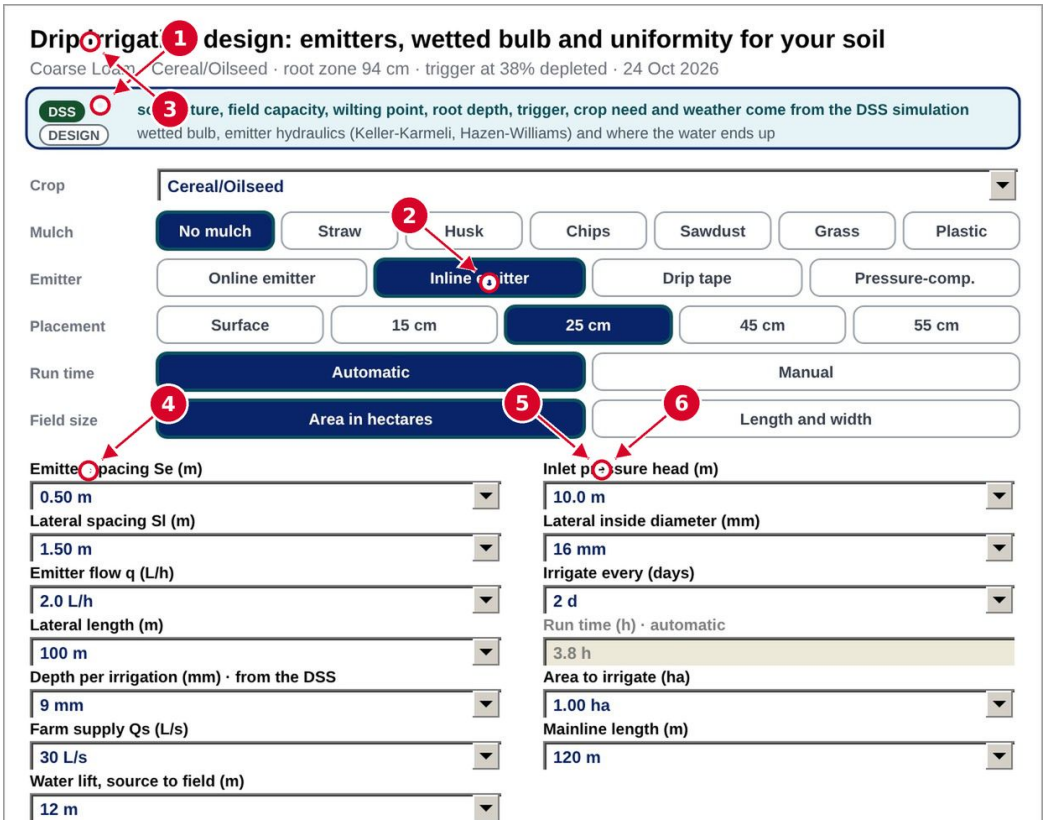



Figure 12.1: Drip design: inputs and the soil moisture around the driest, average and wettest emitter. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 12.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Drip irrigation design*. The crop, soil, root zone, trigger and date come from the DSS (the green badge); the choices below are the design (the amber badge).
2. **Emitter** (a row of four chips). Choosing one sets the nominal flow, spacing, lateral diameter and inlet head of Table 12.1.
3. **Placement**: *Surface* (design only), or a buried line at 15, 25, 45 or 55 cm. A buried choice also sets the explorer's irrigation method to the matching Daisy run (M04, M06, M09 or M10), so that the rest of the program shows the same depth.
4. **Run time**: *Automatic* (time to deliver the depth at the driest emitters, Eq. 12.17) or *Manual*.
5. Set the value boxes of Table 12.2, then press **Run simulation**. The crop, mulch and irrigation method rows at the top are the same controls as the Properties panel.

Table 12.1. Emitter types (defaults are set when the chip is chosen; $c_v$ manufacturer coefficient of variation, $x$ flow exponent, $n_e$ emitters per plant or point, $H_{\min}$ compensation threshold).

| Emitter | $q$ (L h$^{-1}$) | $S_e$ (m) | $D$ (mm) | $H_0$ (m) | $c_v$ | $x$ | $n_e$ | $H_{\min}$ (m) |
|---|---|---|---|---|---|---|---|---|
| Online emitter | 4 | 0.75 | 16 | 10 | 0.07 | 0.5 | 1 | n/a |
| Inline emitter | 2 | 0.50 | 16 | 10 | 0.04 | 0.5 | 1 | n/a |
| Drip tape | 1 | 0.20 | 16 | 10 | 0.08 | 0.6 | 2 | n/a |
| Pressure-compensating | 2 | 0.60 | 16 | 15 | 0.03 | 0.05 | 1 | 8 |

Table 12.2. Drip inputs (value boxes) (range, step, default).

| Control | Range, step | Default | Meaning |
|---|---|---|---|
| Emitter spacing $S_e$ (m) | 0.1 to 1.5, 0.05 | by emitter | Along the lateral |
| Lateral spacing $S_l$ (m) | 0.5 to 3, 0.05 | 1.5 (1.0 for cereal, legume; 1.5 for vegetable, reset when the crop changes) | Between laterals |
| Emitter flow $q$ (L h$^{-1}$) | 0.5 to 8, 0.1 | by emitter | Nominal flow at $H_0$ |
| Lateral length $L_L$ (m) | 20 to 300, 5 | 100 | |
| Inlet pressure head $H_0$ (m) | 4 to 30, 0.5 | 10 (15 for PC) | |
| Lateral inside diameter $D$ (mm) | 10 to 32, 1 | 16 | |
| Irrigate every (days) | 1 to 10, 1 | 2 | Sets the default depth |
| Run time (h) | 0.25 to 48, 0.25 | automatic | Greyed while *Automatic* |
| Depth per irrigation $Z_n$ (mm) | 3 to 80, 1 | from the DSS | Tap the label to reset |
| Area to irrigate $A_f$ (ha) | 0.2 to 10, 0.1 | 1 | |
| Farm supply $Q_s$ (L s$^{-1}$) | 2 to 120, 1 | 30 | |
| Mainline length $L_m$ (m) | 20 to 600, 10 | 120 | |
| Water lift, source to field (m) | 0 to 60, 1 | 12 | |

**Depth per irrigation.** By default the depth is the crop need over the interval, capped by what the root zone can hold at the trigger,

$$Z_n = \min\{ Z_n^{\mathrm{DSS}},\ \max(4,\ \mathrm{ET}_{\mathrm{pk}}\,\Delta t)\}\ \ \mathrm{mm}, \tag{12.1}$$

*Provenance (Eq. 12.1):* **[Engineered]**: depth and demand rules chosen for the workbench.

where $\Delta t$ is the chosen interval, $Z_n^{\mathrm{DSS}} = \mathrm{clip}(f_T\,\mathrm{AWC}\,R_{\max}; 25, 150)$ is Eq. (5.27), and $\mathrm{ET}_{\mathrm{pk}}$ is the peak five-day moving mean of simulated evaporation plus transpiration over the year, bounded to 2.5 to 9 mm d$^{-1}$,

$$\mathrm{ET}_{\mathrm{pk}} = \mathrm{clip}\Big\{ \max_d \tfrac{1}{5}\sum_{j=0}^{4} (E_{d+j} + T_{d+j});\ 2.5,\ 9 \Big\}. \tag{12.2}$$

*Provenance (Eq. 12.2):* **[Engineered]**: depth and demand rules chosen for the workbench.

The mean evaporative demand used for surface drying, $E_0$, is the mean of the simulated daily evaporation on days 141 to 250 of the year (when positive), clipped to 1 to 5 mm d$^{-1}$.

## 12.2 Lateral hydraulics: head loss and flow along the lateral

An emitter lateral of length $L_L$ carries $N = \mathrm{round}(L_L/S_e)$ intervals and $N+1$ emitters, indexed $j = 0 \ldots N$ from the inlet. The friction head loss in the segment feeding emitter $j$ is the Hazen and Williams equation (Hazen and Williams, 1905), in SI form with a roughness coefficient $C = 140$ for plastic pipe,

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i\ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

*Provenance (Eq. 12.3):* **[Literature]**: Hazen and Williams (1905).

The head at emitter $j$ is $h_j = h_{j-1} - h_{f,j-1}$ with $h_0 = H_0$ (floored at 0.5 m). Emitter discharge depends on the local head through the emitter equation (Keller and Karmeli, 1974),

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\ h_j/H_0 \geq 0.01), \tag{12.4}$$

*Provenance (Eq. 12.4):* **[Literature]**: Keller and Karmeli (1974) emitter law.

with $x = 0.5$ for a fully turbulent orifice or long-path emitter and $x = 0.6$ for the drip tape of Table 12.1. The pressure-compensating (PC) emitter is modelled as

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

*Provenance (Eq. 12.5):* **[Engineered]**: pressure-compensating emitter model with chosen constants.

i.e. a nearly flat discharge above $H_{\min} = 8$ m (a 4 % change in flow per 100 % change in head) and the orifice law below it. The flows depend on the heads and the heads on the flows, so the pair (12.3) to (12.4) is **iterated to convergence** from the uniform start $q_j = q$, $h_j = H_0$ (the first three passes are plain substitution, then the flows are averaged with the previous pass; the iteration stops when the largest relative change in any emitter flow is below $10^{-7}$, at most 500 passes). Version 1 used four passes, which is sufficient only when the head loss is small; the converged solver agrees with an independent solution to $3 \times 10^{-7}$ for head losses up to 60 % of the inlet head. When the friction loss exceeds 20 % of the inlet head, or the iteration fails to converge, the layer prints a red warning, because uniformity then falls and the design is outside the usual rule of thumb. The mean and minimum discharge are then

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600}\ \ (\mathrm{L\,s^{-1}}), \tag{12.6}$$

*Provenance (Eq. 12.6):* **[Literature]**: definitions of mean and minimum discharge.

and the loss along the whole lateral is $h_f = H_0 - h_N$, with $H_{\text{end}} = h_N$ the head at the far end (Section “Uniformity along the lateral”).

**Emission uniformity.** The design emission uniformity follows Keller and Karmeli (1974),

$$E_u = \text{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

*Provenance (Eq. 12.7):* **[Literature]**: Keller and Karmeli (1974).

The first factor is the manufacturing variation; the second is the hydraulic variation along the lateral (here $q_{\min}$ replaces the classical quarter-lowest mean, a conservative choice). The *combined* relative standard deviation of the delivery, which drives the spread of bulbs below, is

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

*Provenance (Eq. 12.8):* **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Reading the lateral plot: the solid line is $q_j/q$ in per cent against distance, the dashed line is the head as a percentage of $H_0$. A lateral that is too long or too narrow shows a steep fall of both. For a non-compensating emitter a 10 % loss of head costs about 5 % of flow ($x = 0.5$); the PC emitter stays flat until the head drops below $H_{\min}$.

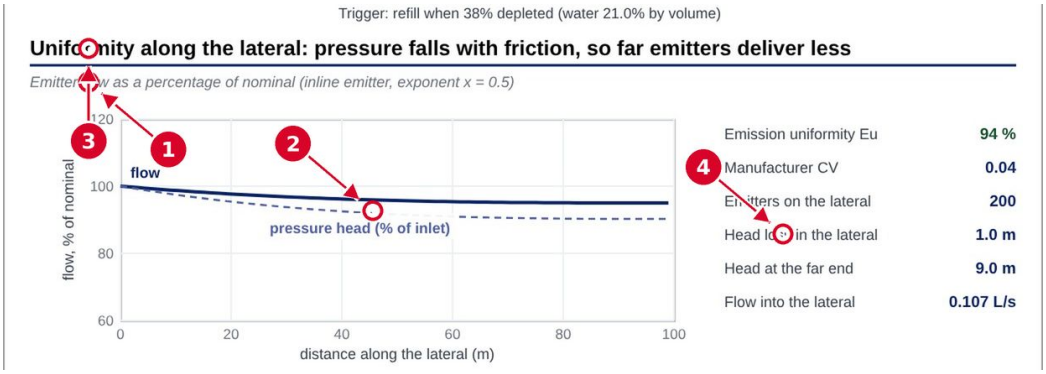



Figure 12.2: Drip design: uniformity along the lateral and the growth of the wetted bulb with run time. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 12.3 The wetted bulb

Water from an emitter spreads into soil that is initially at the trigger water content and is refilled up to field capacity. The layer models the wetted volume around a *buried* point source as a **half-ellipsoid** below the emitter and a short one above it (an **[Engineered]** simplification of the wetted-volume shapes used in drip design; the aspect ratio is derived from the soil, not read from a table).

**Water to be stored per unit soil volume.**

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

*Provenance (Eq. 12.9):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Shape.** With emitter depth $z_d$ (mm), horizontal radius $r$ (mm) at the emitter level, lower semi-axis $a\,r$ and upper semi-axis $u\,r$ with $u = 0.6$, the wetted radius at depth $z$ is

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \geq z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \geq 1. \tag{12.10}$$

*Provenance (Eq. 12.10):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Aspect ratio.** The shape factor $a$ (depth over radius) follows the soil: gravity dominates in a coarse soil of high saturated conductivity, so the bulb is narrow and deep; capillarity dominates in a fine soil, so the bulb is as wide as it is deep or wider. The form is a scaling argument and the constants are **[Engineered]**,

$$a = \mathrm{clip}\Big\{\Big[0.7 + 1.6\Big(\frac{K_s}{K_s + 15}\Big)^{1.1}\Big]\sqrt{\mathrm{clip}(2/q;\ 0.7,\ 1.4)};\ 0.75,\ 2.4\Big\}, \tag{12.11}$$

*Provenance (Eq. 12.11):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

with $K_s$ in mm h$^{-1}$ (from the pedotransfer functions of the soil, Section “Intake from your soil” of Chapter 11) and $q$ the emitter flow in L h$^{-1}$. The result changes with the soil you choose: about 2.1 for sand and coarse sand, 1.9 for sandy loam, 1.3 to 1.4 for loam and silt loam, and 0.9 to 1.0 for clay loam and silty clay loam (a bulb wider than deep). A larger emitter flow spreads the bulb sideways. Change the soil in Properties and run the design again to see the bulb of each soil.

**Radius.** For an emitter that delivers $V$ litres, the wetted volume is $V_w = V \cdot 10^6/\Delta\theta$ mm$^3$

(1 L $= 10^6$ mm$^3$). The radius $r$ is found by bisection (38 steps between 5 and 3500 mm) so that the volume of revolution of Eq. (12.10), integrated in 56 layers,

$$V_w = \int \pi\, \rho(z)^2\, dz, \tag{12.12}$$

*Provenance (Eq. 12.12):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

equals $V_w$. When the volume is *line-like*, the cross-section instead of the volume is conserved: for drip tape ($S_e < 0.34$ m) or when neighbouring bulbs overlap ($2r > S_e$ with $S_e < 1.2$ m), the water is spread along the line, $A_w = V \cdot 10^6/(S_e \cdot 1000)/\Delta\theta$ mm$^2$ per unit length of lateral and the profile area is $2\int \rho\, dz$. The panel is then labelled "line (bulbs merge)" rather than "point". The wetted share of the soil surface is

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\, S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

*Provenance (Eq. 12.13):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Where the water goes.** Of the $V$ litres, the part that falls inside the bulb below the root-zone depth $R_{\max}$ is lost to deep percolation, a fraction $B_{\text{below}}/V_w$ of $V$:

$$V_{\text{dp}}^{(1)} = V\, \frac{\int_{z>R_{\max}} \pi\rho^2\, dz}{V_w}, \tag{12.14}$$

*Provenance (Eq. 12.14):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

and the part evaporated from the wet surface patch is limited to half the delivery,

$$V_{\text{ev}} = \min\big\{0.5\, V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\big\} \quad (\text{L, area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

*Provenance (Eq. 12.15):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

(for a line source the area is $2\rho(0)S_e \cdot 10^{-6}$ m$^2$ per emitter). The factor 2 corresponds to about two days of evaporation from a wet surface patch of mean demand $E_0$; this is an **[Engineered]** simplification of the two-stage evaporation of Chapter 9. For a surface line ($z_d = 0$) the bulb is half-spherical and the evaporating patch is largest, which is why surface drip stores less than buried drip in the table. The remainder is available to the roots, $V_{\text{root}} = V - V_{\text{dp}}^{(1)} - V_{\text{ev}}$.

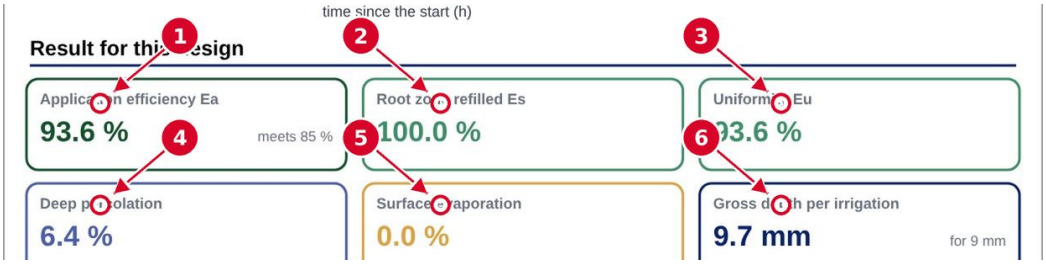



Figure 12.3: Drip design: result tiles and the water partition of one irrigation. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 12.4 Spread between emitters and the efficiencies

Emitters do not all deliver the same volume. The layer evaluates five representative emitters, at standard-normal quantiles $z \in \{-1.28, -0.52, 0, 0.52, 1.28\}$ (the 10th, 30th, 50th, 70th and 90th

percentiles, equal weights), each delivering

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1+\sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

*Provenance (Eq. 12.16):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

where $t$ is the run time (h). For each, the bulb, deep percolation and evaporation are computed with Eqs. (12.10) to (12.15). The water that the root zone can accept is limited by the need,

$$V_{\text{req}} = Z_n\,S_e\,S_l\ \ (\text{L}), \qquad V_{\text{st},i} = \min\{V_{\text{root},i},\ V_{\text{req}}\}, \qquad V_{\text{ex},i} = V_{\text{root},i} - V_{\text{st},i}. \tag{12.17}$$

*Provenance (Eq. 12.17):* **[Literature]**: definition of the requirement over the area served by one emitter.

Because 1 mm over 1 m$^2$ is 1 L, $V_{\text{req}}$ is the depth needed over the area $S_e S_l$ served by one emitter. **Automatic run time** is the time at which the *driest* emitters deliver the need,

$$t = \text{clip}\Big\{\frac{V_{\text{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \text{h}. \tag{12.18}$$

*Provenance (Eq. 12.18):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

The efficiencies are means over the five quantiles ($\langle\cdot\rangle$):

$$E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

*Provenance (Eq. 12.19):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

so that $E_a + \text{DP} + \text{EV} = 1$. Reading them: $E_a$ is how much of the pumped water ends up stored in the root zone; $E_s$ is how well the root zone is refilled (a value below 85 % turns the “Root zone refilled” line red and says *short*); $E_u$ is the lateral uniformity; $Z_{\text{gross}}$ is the depth to pump per irrigation. The layer colours $E_a \geq 85$ % green, 65 to 85 % amber and below 65 % red; $E_u \geq 90$ % is green and 80 to 90 % amber.

## 12.5 Layout and infrastructure

The farm side of the design uses the area $A_f$ and the supply $Q_s$.

**Laterals and zones.**

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \Big\lfloor \frac{W_f}{S_l} \Big\rfloor, \quad n_{\text{zone}} = \Big\lceil \frac{N_L\,Q_{\text{lat}}}{Q_s} \Big\rceil, \quad n_l = \Big\lceil \frac{N_L}{n_{\text{zone}}} \Big\rceil, \quad Q_z = n_l\,Q_{\text{lat}}, \tag{12.20}$$

*Provenance (Eq. 12.20):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

where $W_f$ is the width of the field perpendicular to the laterals, $n_{\text{zone}}$ the number of zones that must run in turn so that $Q_s$ is not exceeded, and $Q_z$ (L s$^{-1}$) the flow of one zone. The total number of emitters is $N_L(\text{round}(L_L/S_e)+1)$.

**Pipe sizing.** Submain length $L_{sm} = n_l S_l$, mainline length $L_m$ (value box). The pipe diameter is the smallest of the commercial series

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

*Provenance (Eq. 12.21):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

for which the mean velocity $v = 4Q_z/(\pi D^2) \le 1.5$ m s$^{-1}$ (a common design limit that keeps friction and water-hammer moderate; see Keller and Bliesner, 1990, for pipe design practice). The submain loss is $0.4$ of the full-length Hazen and Williams loss, Eq. (12.3) with $Q_j = Q_z$ (the Christiansen (1942) reduction factor $F$ for a pipe with many equally spaced outlets is $F \to 1/2.852 = 0.35$ for $m = 1.852$, so $0.4$ is slightly conservative), and the mainline loss is the full value:

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

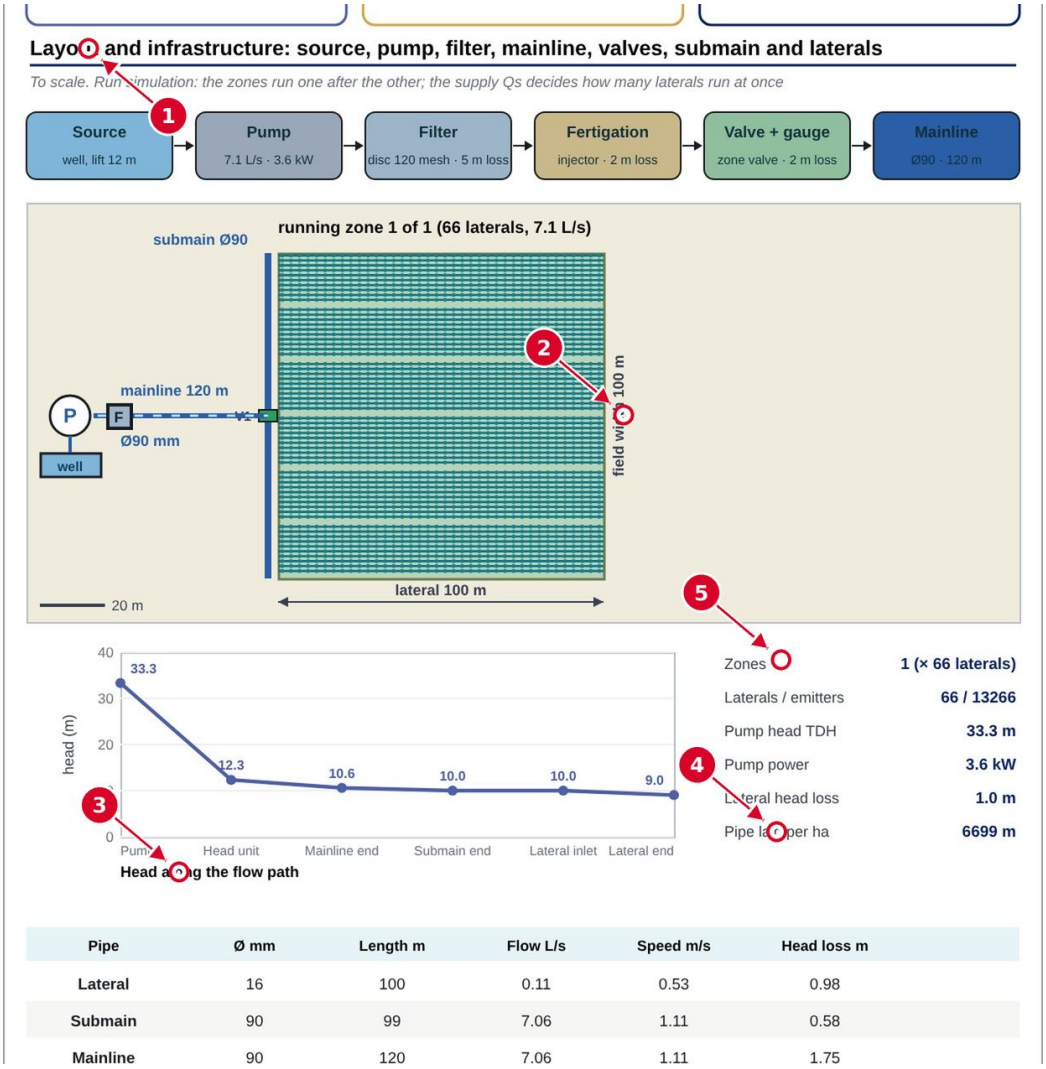
*Provenance (Eq. 12.22):* **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

**Pump.** The total dynamic head and the shaft power are

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\text{ m}, \qquad P = \frac{\rho g\,Q_z\,\text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

*Provenance (Eq. 12.23):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The constant 9 m covers the filter, valves and fittings, and $\eta = 0.65$ is the overall pump-and-motor efficiency; both are **[Engineered]** design allowances, not measured values. The laterals' own loss is already inside $H_0$.




| Pipe | Ø mm | Length m | Flow L/s | Speed m/s | Head loss m |
|---|---|---|---|---|---|
| Lateral | 16 | 100 | 0.11 | 0.53 | 0.98 |
| Submain | 90 | 99 | 7.06 | 1.11 | 0.58 |
| Mainline | 90 | 120 | 7.06 | 1.11 | 1.75 |

Figure 12.4: Drip design: layout and infrastructure, to scale, with the zones running in turn. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

The drawing shows the source, pump, filter, mainline, valves, submain and laterals to scale; during the run the zones run one after the other, and the head along the flow path is drawn underneath. The table beside it gives zones (× laterals per zone), laterals and emitters, TDH, pump power, lateral head loss and pipe laid per hectare, $(N_L L_L + L_{sm})/A_f$.

## 12.6 Season and farm

With $y_I$ the net seasonal irrigation of the DSS for the year of the working date, $E_u$ and $E_a$ as above,

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

*Provenance (Eq. 12.24):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

*Provenance (Eq. 12.25):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The layer prints these as a table (net irrigation need and number of events, gross water per irrigation and per season in mm and $m^3$ $ha^{-1}$, peak crop need, flow per hectare, sets for $A_f$ with $Q_s$, hours of pumping and energy). Then the *Irrigation schedule* and the *When to irrigate* advice, described at the end of Chapter 11, close the layer.

## 12.7 The design table

The optimiser takes the current design and tries one change at a time, in this order: **placement depth, run time, emitter flow, emitter spacing, placement depth, run time, emitter kind, lateral spacing, emitter flow, run time**. For each move the candidates are:

| Move | Candidates |
|---|---|
| Placement depth | Surface, 15, 25, 45, 55 cm |
| Run time | $t_0 \times \{0.55, 0.65, 0.75, 0.85, 0.95, 1, 1.1, 1.25\}$, with $t_0$ the automatic run time of the current design |
| Emitter flow $q$ | 0.6, 1, 1.5, 2, 3, 4, 6, 8 L $h^{-1}$ (run time reset to automatic) |
| Emitter spacing $S_e$ | 0.2, 0.3, 0.4, 0.5, 0.6, 0.75, 1, 1.25 m |
| Emitter kind | the four emitters of Table 12.1, each with its own default $q$, $S_e$ |
| Lateral spacing $S_l$ | 0.75, 1, 1.25, 1.5, 1.8, 2.2, 2.6 m |

The score is

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

*Provenance (Eq. 12.26):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

so a design is penalised when the root zone is not refilled to 90 %. A move is accepted when its best candidate improves the score by more than 0.002; the search stops when $E_a \geq 85$ % **and** $E_s \geq 85$ %, or after ten moves. One move is performed per screen update, and the search pauses while a slider is being dragged. Each accepted row shows the step, $q$, $S_e$, depth, run time, $E_a$, $E_s$ and DP; a green row meets the target, and **tapping any row applies that design** to the sliders (including the placement depth and the explorer's irrigation method). If the target is not met, the layer reports the best found and suggests a shorter interval or a different emitter.

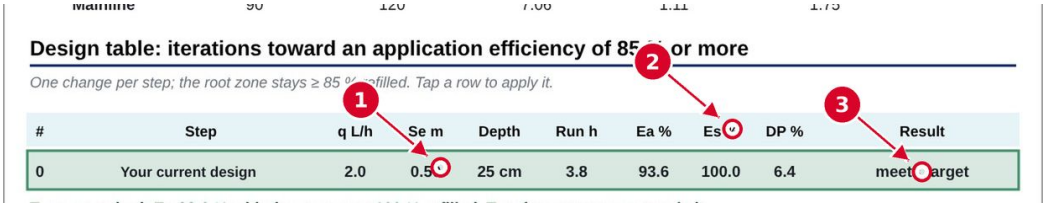



Figure 12.5: Drip design table and the sensitivity chart. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 12.8 Sensitivity

As for the surface layer, each input is moved up and down and the change of $E_a$ is shown as a bar: emitter flow $q$ ($\pm 25$ %), emitter spacing ($\pm 25$ %), lateral spacing ($\pm 25$ %), placement depth ($\pm 30$ %, at least 10 cm), depth per irrigation ($\pm 25$ %) and run time ($\pm 20$ %). The result is a local sensitivity at the present design.

## 12.9 How to use the layer

- **To choose an emitter for a soil.** Select the soil in the Properties panel, open the layer and compare the four chips with the same run time: the tile of $E_a$ and the line chart of the bulb show which emitter keeps the bulb inside the root zone.
- **To check a lateral length.** Increase $L_L$ until $E_u$ turns amber; the lateral plot shows where the head falls below the pressure at which the emitter holds its flow.
- **To size a pump.** Fix $A_f$, $Q_s$, $L_m$ and the lift; read TDH and kW from the layout table.
- **To see the effect of the trigger.** Move *Trigger* in the Properties panel: $f_T$ changes $\Delta\theta$ (Eq. 12.9) and $Z_n$ (Eq. 12.1), and the bulb and the run time adjust.

# Chapter 13

# ❄ Sprinkler irrigation design

The sprinkler layer designs the **nozzle, pressure, spacing and layout** of a sprinkler system for the soil, crop, slope and wind of the case, and then checks the result against the standard course rules (spacing, application rate, uniformity, pipes and pump). It has two parts that are built on the same run: a *simulation* part (the overlapped spray pattern, the wind drift, the soil intake that causes runoff, and the efficiencies) and a *textbook* part (the ten design checks, the pipe and pump sizing, the kinds of sprinkler system).

The layer reads top to bottom: **A** site, **B** sprinkler and layout, **C** operation and **D** farm and pipes (the inputs); the layout in plan; the profile along the lateral; the intake against the application rate; result tiles and the water partition; the textbook checks and system plans; layout and infrastructure; the irrigation schedule; the design table with the sensitivity chart; the season and the advice.

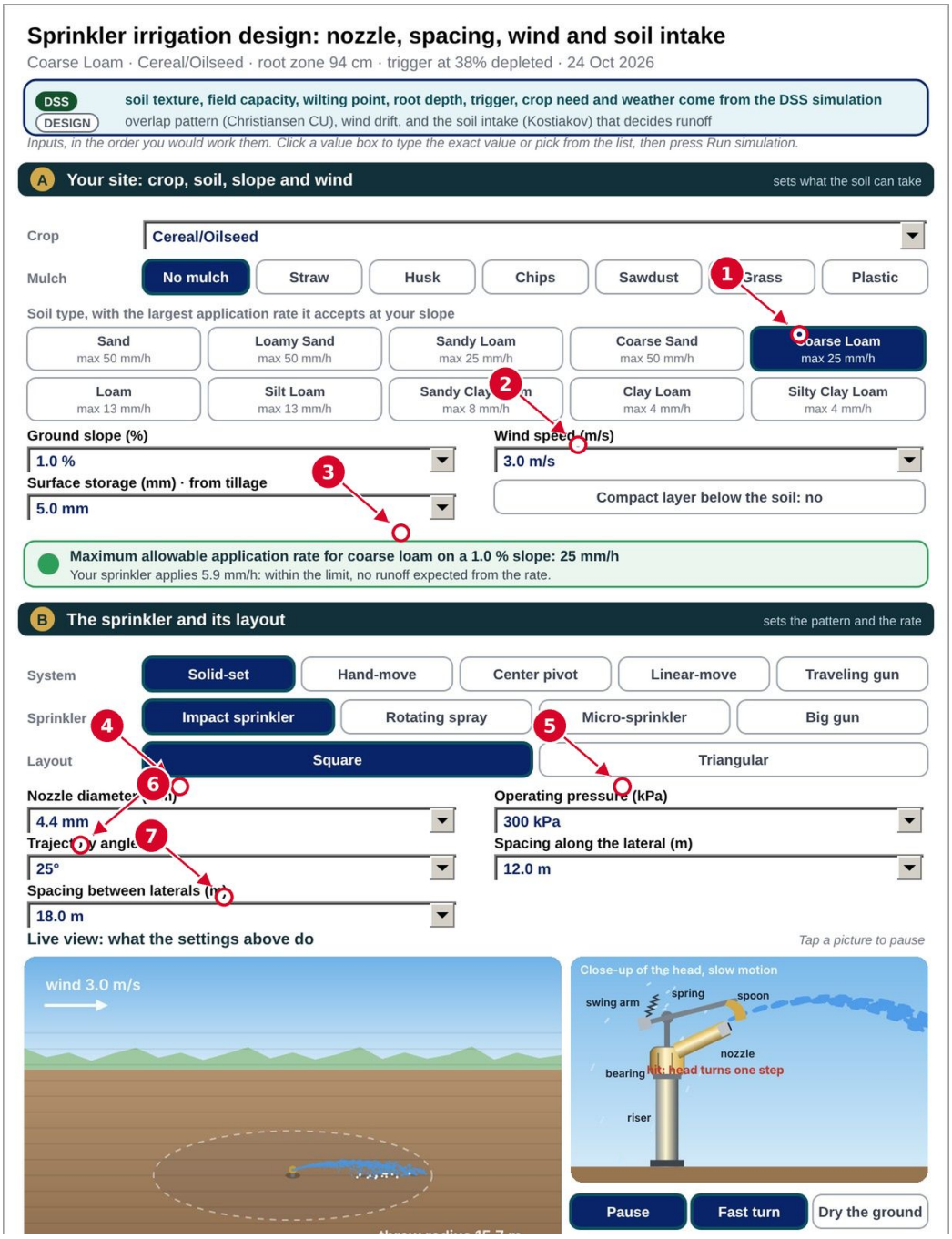



Figure 13.1: Sprinkler design: inputs A to D and the layout in plan with the overlapping spray patterns. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 13.1 Inputs, step by step

**A. Your site.** Crop, mulch and irrigation method are the same controls as in the Properties panel. The soil row shows each soil with the largest application rate it accepts at the present slope (Table 13.3). Sliders: ground slope $S$ (0 to 16 %, step 0.5, default 1), wind speed $u$ (0 to 12 m s$^{-1}$, step 0.5, default 3), surface storage $S_d$ (0 to 20 mm, step 0.5; default from tillage, below) and a toggle *Compact layer below the soil* (which lowers the allowed rate, Table 13.3). A banner states the

maximum allowable application rate for the soil and slope and whether the present sprinkler is within it.

**B. The sprinkler and its layout.** *System* (five chips, see *Kinds of sprinkler system* below), *Sprinkler* (four types, Table 13.1), *Layout* (square or triangular), then value boxes for the nozzle diameter $d$, operating pressure $P$, trajectory angle $\alpha$ (8 to 45°, default 25°), spacing along the lateral $S_s$ and between laterals $S_l$. Choosing a type resets $d, P, S_s, S_l$ to its defaults and run time to automatic.

Table 13.1. Sprinkler types: ranges (min and max, step) and defaults; $c_R$ range coefficient, $m$ profile exponent, $F$ drift factor. The impact value $c_R = 1.35$ is the Keller and Bliesner (1990) coefficient for impact sprinklers (changed from 1.15 in version 1; the earlier value under-predicted the wetted diameter of their worked example by 14 %). The values for the other three types are **[Engineered]** and have not been verified against published data.

| Type | $d$ (mm) | $P$ (kPa) | $S_s$ (m) | $S_l$ (m) | $c_R$ | Profile | $F$ |
|---|---|---|---|---|---|---|---|
| Impact | 2.4 to 8, 0.1 (4.4) | 200 to 500, 10 (300) | 6 to 30, 1 (12) | 6 to 36, 1 (18) | 1.35 | $1 - x^{1.3}$ | 1.0 |
| Rotating spray | 1.5 to 5, 0.1 (3.2) | 70 to 250, 5 (170) | 3 to 15, 0.5 (6) | 3 to 15, 0.5 (6) | 0.80 | ring | 1.6 |
| Micro-sprinkler | 0.8 to 2.5, 0.05 (1.3) | 100 to 250, 5 (180) | 1.5 to 8, 0.25 (3) | 1.5 to 8, 0.25 (3) | 0.70 | ring-2 | 0.7 |
| Big gun | 10 to 25, 0.5 (16) | 350 to 700, 10 (500) | 20 to 80, 1 (30) | 20 to 90, 1 (36) | 0.95 | $1 - x$ | 1.3 |

**C. How you run it.** *Run time* automatic or manual; depth per irrigation $Z_n$ (5 to 100 mm; default Eq. 5.27, tap to reset); run time $T$ (10 to 1200 min, step 5); operating hours per day (4 to 24, default 16).

**D. Farm, water supply and pipes.** Area $A_f$ (0.5 to 40 ha, step 0.5, default 5), farm supply $Q_s$ (5 to 200 L s$^{-1}$, default 40), lateral length $L_L$ (40 to 500 m, step 10, default 150), mainline length $L_m$ (20 to 600 m, default 120) and lift (0 to 60 m, default 12).

**Tillage and surface storage.** Unless set by hand, the depression storage $S_d$ follows the tillage of the Properties panel: 5 mm conventional, 3.5 mm reduced, 2 mm none. On a slope the effective storage is reduced,

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

*Provenance (Eq. 13.1):* **[Engineered]**: depression storage reduction with slope.

This is an **[Engineered]** relation: storage in depressions decreases as the slope rises (compare the slope dependence of depression storage in Hillel, 1998).

## 13.2 Nozzle discharge and wetted radius

For a nozzle of diameter $d$ (mm) at operating pressure $P$ (kPa) the pressure head is $h = P/9.81$ m. The discharge follows the orifice equation (Torricelli) with a discharge coefficient of 0.95,

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4 \;\; (\text{L min}^{-1}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \;\text{m}^2, \tag{13.2}$$

*Provenance (Eq. 13.2):* **[Literature]**: Torricelli orifice equation.

and the wetted radius is (an empirical range law of the form used in sprinkler manuals, Keller and Bliesner, 1990, with the coefficient $c_R$ of Table 13.1 chosen per type)

$$R = c_R\sqrt{d\,h}\;f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

*Provenance (Eq. 13.3):* **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

with $d$ in mm, $h$ in m and $\alpha$ in degrees ($3.1\alpha$ as an angle in degrees). The angle function peaks near $\alpha = 29^\circ$ and equals 1 at the default $25^\circ$ ($\sin 77.5^\circ = 0.9763$). **[Engineered]**: the exponent 0.45 and the $3.1\alpha$ term are chosen to give a mild optimum, not fitted to a data set. The reach $R$ is printed in the plan as a percentage of the spacing along the lateral.

## 13.3 The spray pattern and the overlap

**Radial profile.** A single sprinkler deposits water at radius $r$ with a shape $\hat{p}(x)$, $x = r/R_e$, taken from Table 13.1:

$$\hat{p}(x) = \begin{cases} 1 - x^m, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^6), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^5), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

*Provenance (Eq. 13.4):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

These are **[Engineered]** shapes that reproduce the qualitative patterns published for each class: a single sprinkler with a rapidly falling profile (impact, gun), and a pattern that peaks away from the centre (spray and micro types).

**Wind.** Wind elongates the pattern downwind and shifts it:

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

*Provenance (Eq. 13.5):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

where $\theta$ is the angle from the wind direction and $u$ is the wind speed in m s$^{-1}$. The pattern is computed on a $45 \times 45$ grid over $\pm 1.4R$ and normalised so that its volume equals the discharge, $\iint p\,dA = q$, with $p$ in mm min$^{-1}$ (1 mm over 1 m$^2$ is 1 L).

**Overlap.** The application rate at a point of the repeating cell is the sum over neighbouring sprinklers (square grid $S_s \times S_l$; triangular layout: every second lateral shifted by $S_s/2$, repeating cell $S_s \times 2S_l$),

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_k a_k, \tag{13.6}$$

*Provenance (Eq. 13.6):* **[Literature]**: overlap by superposition of single-sprinkler patterns.

evaluated on a $14 \times 14$ grid of the cell ($14 \times 24$ for the triangular layout). The mean *gross* application rate is AR $= 60\bar{a}$ mm h$^{-1}$ and the **Christiansen coefficient of uniformity** (Christiansen, 1942) is

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

*Provenance (Eq. 13.7):* **[Literature]**: Christiansen (1942).

A CU $\geq 85$ % is the usual requirement for a sprinkler system (Burt et al., 1997; Keller and Bliesner, 1990); the tile is green from 84 %. The *Layout in plan* shows the circles of reach, the wind arrow and the depth infiltrated so far (Play animates the time): green is the depth needed, red-brown is short, blue is more than needed, and brown hatching is water running off.

## 13.4 Wind drift and evaporation loss

The fraction of the water lost between nozzle and soil is

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2 + 0.5u + 0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}\big[1 + 0.025\,(T_a - 20)\big]\Big[0.6 + 0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \quad (13.8)$$

*Provenance (Eq. 13.8):* **[Engineered]**: loss and ponding rules chosen for the workbench.

where $F$ is the type factor of Table 13.1, $P$ is in kPa, $u$ in m s$^{-1}$ and the air temperature $T_a = 24\,^{\circ}\mathrm{C} + \Delta T$, with $\Delta T$ the warming of the Properties panel. The form follows the known trends, loss rises with wind, with pressure (finer droplets), with temperature, and falls with a flatter trajectory, in line with the ranges of 5 to 15 % usually quoted in textbooks (Keller and Bliesner, 1990); the numerical coefficients are **[Engineered]**. $w_d$ is clipped to 0.5 to 45 %.

## 13.5 The soil intake, ponding and runoff

At each cell the water balance is computed with the Kostiakov intake of the soil (Eq. 11.6, from the texture; $K = k \times 1000$ mm min$^{-A}$). The applied rate after drift is $a_k(1 - w_d)$ and for a run of $T$ min the supply is $\mathrm{sup} = a\,T$. Ponding begins when the application rate exceeds the intake rate, which the layer takes as $a = K t^{A-1}$, i.e. at

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \mathrm{sup}, & T \le t_p \\ K\,T^A, & T > t_p \end{cases} \qquad R = \mathrm{sup} - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \mathrm{sup} - \mathrm{RO}. \quad (13.9)$$

*Provenance (Eq. 13.9):* **[Engineered]**: loss and ponding rules chosen for the workbench.

This is a **simplified point balance** (it uses the cumulative law $F = K\,T^A$ after ponding, and it ignores re-infiltration of water that has run to a lower cell, which makes the layer slightly pessimistic about runoff in a field with surface storage and slightly optimistic on a steep slope). A portion of the infiltrated depth is retained on the crop canopy and mulch and evaporates,

$$I_c = \min\{I,\ I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \quad (13.10)$$

*Provenance (Eq. 13.10):* **[Engineered]**: loss and ponding rules chosen for the workbench.

where CAP is the mulch storage capacity (mm, Table 3.1), the indicator is 1 when a mulch is present, and $\mathrm{LAI}_{\mathrm{live}}$ is the live leaf-area index of the simulated crop on the working date. The depth that reaches the soil at the cell is $Z_k = I - I_c$.

The intake figure shows the rate the soil accepts, $\dot{I}(t) = 60\,KA\,t^{A-1}$ mm h$^{-1}$, the mean applied rate $\overline{AR}(1 - w_d)$, the shaded area where the applied rate exceeds the intake (runoff), and the time at which ponding starts. (The plotted curve is the derivative of $Kt^A$; the point balance uses the ponding condition $a = Kt^{A-1}$, so the two differ by the constant factor $A$ in the intake rate, a documented simplification.)

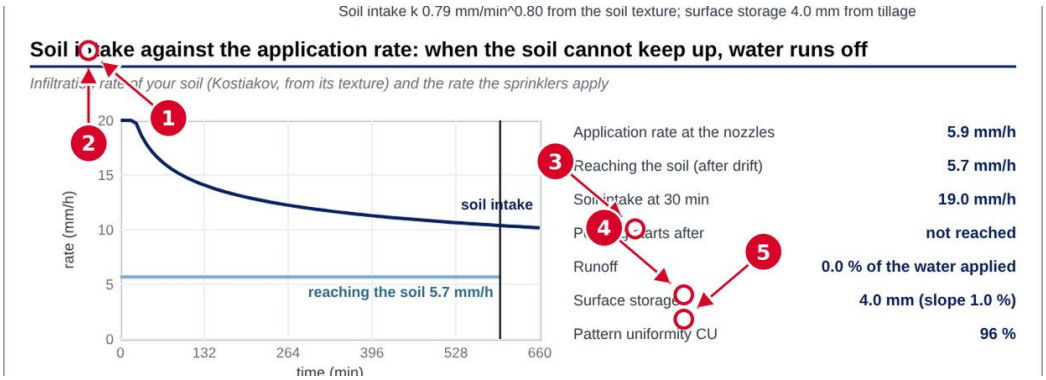



Figure 13.2: Sprinkler design: depth applied and infiltrated along the lateral, and the soil intake against the application rate. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 13.6 Efficiencies and run time

With $Z_k$ the infiltrated depth at cell $k$ and gross $= \bar{a}\,T$ the mean depth discharged:

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

*Provenance (Eq. 13.11):* **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

where $\overline{Z}_{25}$ is the mean of the lowest quarter of the $Z_k$ (the distribution uniformity of Merriam and Keller, 1978). The partition is closed, $E_a + \text{DP} + \text{RO} + \text{WD} = 1$. The tiles show $E_a$ (green from 85 %), $E_s$, CU, runoff, deep percolation and the combined drift, evaporation and interception loss, with a stacked bar of the partition.

**Automatic run time.** The time $T$ is the smallest value for which the driest quarter of the field receives the required depth,

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\ 2400] \text{ min}, \tag{13.12}$$

*Provenance (Eq. 13.12):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

found by 18 bisection steps. If the soil cannot take the water as fast as it is applied, longer runs only raise runoff and the layer shows the run time and low $E_a$ together.

## 13.7 Maximum allowable application rate

The allowable rate by soil and slope follows the standard table of sprinkler manuals (Keller and Bliesner, 1990, Table of maximum application rates; reproduced here for the textbook check).

Table 13.3. Maximum application rate (mm h$^{-1}$) by soil type and slope.

| Soil group | 0 to 5 % | 5 to 8 % | 8 to 12 % | 12 to 16 % |
|---|---|---|---|---|
| Deep coarse sandy soils | 50 | 38 | 25 | 13 |
| Coarse sandy, compact layer below | 38 | 25 | 19 | 10 |
| Deep sandy loam | 25 | 20 | 15 | 10 |

| Soil group | 0 to 5 % | 5 to 8 % | 8 to 12 % | 12 to 16 % |
|---|---|---|---|---|
| Sandy loam, compact layer below | 19 | 13 | 10 | 8 |
| Deep silt loam | 13 | 10 | 8 | 5 |
| Silt loam, compact layer below | 8 | 6 | 4 | 2.5 |
| Clay or clay loam, strong structure | 4 | 2.5 | 2 | 1.5 |

The ten soils of the program are assigned to rows: Coarse Sand, Sand, Loamy Sand $\rightarrow$ 1; Sandy Loam, Coarse Loam $\rightarrow$ 3; Loam, Silt Loam $\rightarrow$ 5; Sandy Clay Loam $\rightarrow$ 6; Clay Loam and Silty Clay Loam $\rightarrow$ 7. The toggle *Compact layer below the soil* moves an odd row to the next (compact) row. The slope selects the column.

## 13.8 The ten textbook checks

The "Does your design pass?" panel runs the design through the course rules. Let $h_s = P/9.81$ (m), $q_h = 0.06\,q$ (m$^3$ h$^{-1}$), $R_w = R$, the wind in km h$^{-1}$ $W = 3.6\,u$, and the wind factor

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

*Provenance (Eq. 13.13):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

The spacing limits, the field layout and the time budget are

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x/S_s \rfloor, \quad N_m = \lfloor L_y/S_l \rfloor, \tag{13.14}$$

*Provenance (Eq. 13.14):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{days}, \tag{13.15}$$

*Provenance (Eq. 13.15):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

where $L_x = L_L$ is the lateral length, $h_{\text{op}}$ the operating hours per day and $Z_{\text{gross}}$ the gross depth per irrigation. For $n = 1 \ldots 4$ laterals running at once, the days needed and the inflow are

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\,N\,q_h, \tag{13.16}$$

*Provenance (Eq. 13.16):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

and the program picks the smallest $n$ with $N_d \le T_{\text{int}}$ and $Q_{\text{in}} \le 3.6\,Q_s$ (m$^3$ h$^{-1}$); if none satisfies both it picks the smallest $n$ meeting the time, else $n = 4$.

**Pipes.** Friction losses use the Blasius form (Blasius, 1913) for smooth plastic pipe,

$$J = \frac{0.472\,Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

*Provenance (Eq. 13.17):* **[Literature]**: Blasius friction law for smooth pipe.

with the Christiansen reduction factor $F = 1/2.75 + 1/(2N)$ for $N$ equally spaced outlets and a 10 % allowance for local losses. The lateral pipe is the smallest of the standard internal diameters $\{46.4,\ 58,\ 63,\ 65.8,\ 75,\ 84.9,\ 100\}$ mm for which

$$h_{f,\mathrm{lat}} = 1.1\,F\,J(Q_L, D)\,(L_x - S_s/2) \le 0.2\,h_s, \qquad Q_L = 1000\,N\,q_h, \tag{13.18}$$

*Provenance (Eq. 13.18):* **[Literature]**: 20 % rule of Keller and Bliesner (1990).

the classical 20 %-of-operating-head rule that keeps discharge variation along the lateral under about 10 % (Keller and Bliesner, 1990). The mainline (flow $Q_m = n\,Q_L$, length $L_y - S_l/2$) is the smallest diameter not below the lateral's with $J < 0.05$ m m$^{-1}$.

**Manometric head and pump.**

$$H_{\mathrm{man}} = z_{\mathrm{lift}} + J\,L_{\mathrm{conv}} + J\,L_{\mathrm{main}} + h_s + 0.75\,h_{f,\mathrm{lat}}, \tag{13.19}$$

*Provenance (Eq. 13.19):* **[Engineered]**: head and sets bookkeeping.

$$P_{\mathrm{CV}} = \frac{1000\,(Q_m/3.6\times10^{6})\,H_{\mathrm{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\mathrm{kW}} = 0.7355\,P_{\mathrm{CV}}, \tag{13.20}$$

*Provenance (Eq. 13.20):* **[Literature]**: hydraulic power definition.

with $L_{\mathrm{conv}}$ the conveyance length (the mainline length box) and $0.75\,h_{f,\mathrm{lat}}$ the head at the lateral inlet above $h_s$ (the textbook inlet-head rule). The ten checks (spacing along the lateral, spacing between laterals, wind below 16 km h$^{-1}$, rate below the allowed value, rate below $K_s$, CU $\ge 85$ %, lateral loss $\le 20$ % of $h_s$, mainline gradient, the whole field within the interval, flow within the farm supply) are shown with a tick or an exclamation point and the numbers; "N of 10 checks pass" closes the panel. The textbook sequence in the layer also plots the class of sprinkler (low, medium or high pressure) from $h_s$: $< 20$ m, 20 to 45 m, $> 45$ m.

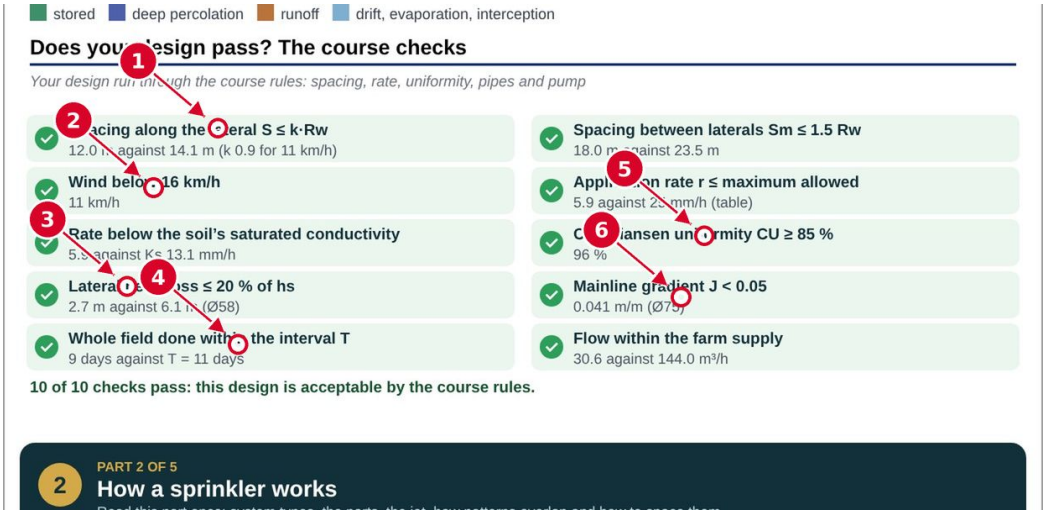



Figure 13.3: Sprinkler design: the ten textbook checks and the kinds of sprinkler system. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

## 13.9 Kinds of sprinkler system

Five systems are offered, each with its efficiency range (Keller and Bliesner, 1990; Burt et al., 1997 report the range of field values):

| System | Principle | Typical $E_a$ |
|---|---|---|
| Solid-set | Fixed buried laterals; valves open some at a time | 70 to 75 % |
| Hand-move | The lateral is moved by hand after every set (1 to 4 sets a day) | 70 to 75 % |
| Center pivot | A rotating lateral on towers about 30 m apart; the corners stay dry | 65 to 70 % |
| Linear-move | A lateral travelling in a straight line | 65 to 70 % |
| Traveling gun | One gun on a hose reel covers a 220° sector, strip by strip | 65 to 70 % |

The plan of the selected system is drawn to scale for the present field; the run moves it (the active lateral, the pivot arm, the travelling lateral or the gun strip). A bar shows where the design's $E_a$ lies against the range for the system.

## 13.10 Layout and infrastructure

The *Layout and infrastructure* block of this layer uses the same hydraulic tools as the drip layer, with these differences. With $N_s = \lfloor L_L / S_s \rfloor$ sprinklers per lateral and $Q_{\text{lat}} = N_s q / 60$ (L s$^{-1}$), the number of laterals that can run together is $\lfloor Q_s / Q_{\text{lat}} \rfloor$ and the sets

$$N_L = \left\lfloor \frac{10^4 A_f / L_L}{S_l} \right\rfloor, \quad n_l = \min\!\left(N_L,\ \left\lfloor \frac{Q_s}{Q_{\text{lat}}} \right\rfloor\right), \quad n_{\text{set}} = \left\lceil \frac{N_L}{n_l} \right\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

*Provenance (Eq. 13.21):* **[Engineered]**: head and sets bookkeeping.

The lateral loss uses the Hazen and Williams equation (Eq. 12.3, $C = 140$) with the Christiansen factor for $N_s$ outlets and the exponent $m = 1.852$,

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m \, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

*Provenance (Eq. 13.22):* **[Literature]**: Christiansen reduction factor with Hazen-Williams exponent.

and the lateral pipe is the smallest of the series (12.21) with $h_{f,\text{lat}} \le 0.2\,h$ and $v \le 2.0$ m s$^{-1}$. The lateral inlet head is $H_0 = h + 1.5 + h_{f,\text{lat}}$ m (1.5 m of riser). The submain, mainline and pump then follow Eqs. (12.22) to (12.23) exactly as for drip, with $H_0$ from this paragraph. The block's table gives the sets ($\times$ laterals), laterals and sprinklers, TDH, power, lateral loss ($\le 20$ %) and pipe laid per hectare.

(Two parallel calculations therefore appear in this layer: the **simulation block** sizes the pipe by Hazen and Williams with $C = 140$, while the **textbook block** uses the Blasius equation and the commercial plastic-pipe sizes of the course. They are separate presentations of the same design question; the numbers will differ by a few per cent, and the textbook block is the one to compare with hand calculations from the course.)

## 13.11 The design table and sensitivity

The optimiser works like the drip one (Eq. 12.26): score $\min(E_a, 1) - 2.5\max(0, 0.9 - E_s)$, target $E_a \ge 85$ % and $E_s \ge 85$ %, accept a move only if the score rises by more than 0.002. The ten moves are **pressure, spacing along the lateral, run time, nozzle, spacing between laterals, run time, pressure, triangular layout, run time, spacing along the lateral**. For pressure, nozzle and the two spacings the candidates are nine equally spaced values over the range of Table 13.1 (rounded to the step); for the layout the two options; for the run time $T_0 \times$

$\{0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.25\}$ with $T_0$ the automatic time. Tapping a row applies it. If the target is not met, the layer suggests a lower application rate (a smaller nozzle or wider spacing) or another sprinkler type.

The sensitivity chart moves eight inputs and shows the change of $E_a$: soil intake $k$ ($\pm 20$ %), wind ($\pm 50$ %), pressure ($\pm 20$ %), nozzle ($\pm 15$ %), the two spacings ($\pm 20$ %), the depth per irrigation ($\pm 25$ %) and the ground slope ($\pm 50$ %).

## 13.12 Season and farm

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60} \text{ h.} \tag{13.23}$$

*Provenance (Eq. 13.23):* **[Engineered]**: head and sets bookkeeping.

The gross water per irrigation is gross $= \bar{a}\, T$ mm, per season $n_{\text{ev}}$ gross mm ($= 10\, n_{\text{ev}}$ gross m$^3$ ha$^{-1}$). The table also gives one sprinkler's discharge and reach, the sprinklers for $A_f$, the sets with $Q_s$, and the pumping hours and energy of the season. The layer closes with the *Irrigation schedule* and the *When to irrigate* message (end of Chapter 11).

## 13.13 How to use the layer

- **To pick a sprinkler for a soil.** Choose the soil, set the slope and the wind, then compare the types: the banner tells you the largest allowed rate, the intake plot shows where runoff starts, and the checks name the failing rule.
- **To reduce runoff.** Lower the application rate: a smaller nozzle or lower pressure reduces $q$; a wider spacing spreads it over a larger area; a triangular layout improves CU at the same rate.
- **To reduce drift.** Reduce the pressure, use a flatter angle, or use a micro-sprinkler ($F = 0.7$); the wind arrow and the $w_d$ share of the partition show the effect.
- **To size the system for the farm.** Set the area, farm supply and operating hours; read the sets, the pipe diameters and the pump power from the layout block, and the textbook block for the hand-calculation version.

# Chapter 14

# ▤ The Dashboard: how every number is produced

The Dashboard is the panel to the right of the drawing. It has four tabs (*Results*, *Design*, *Charts*, *Messages*) and the buttons **Wider** (enlarges or shrinks the panel), **Pin** (stores the present 20-year averages as a baseline; Ctrl+B) and **Clear** (removes the baseline). It is laid out like the Properties panel: a white sheet with grey group bars. **Every block can be folded and unfolded on its own** by clicking its bar; the folding is remembered while the program runs, also when the numbers refresh. The dashboard re-reads the engine each time the state, the date or the weather changes, so it always agrees with the drawing.

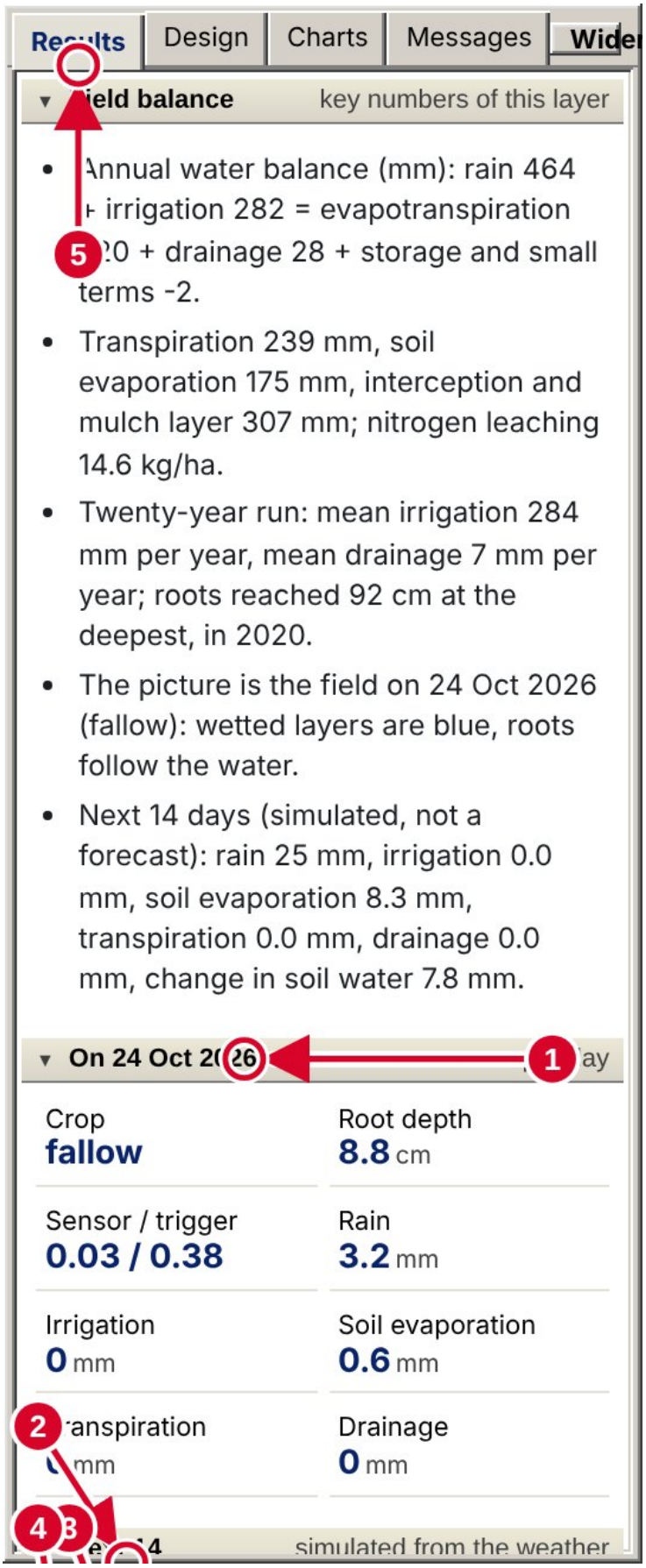



Figure 14.1: The Dashboard, Results tab, with the Field balance layer open.

## 14.1 Results

The first block of the Results tab carries the name of the open layer and its **key numbers** (the same lines as in the report). For example, on *Soil hydraulic drivers* it shows the change in drainage due to tillage and to the consolidation rate on your soil, and the leading driver. A design layer shows its key numbers after its run.

On *Trigger law* this block shows only the soil you are working on. It lists the trigger (cm) and its depletion, the critical $f^*$, the sensor reading and its depletion now, the days from field capacity to the trigger, the drying speed, the water held in the root zone, the yield, the irrigation and drainage, the share of the best trigger's yield, and the advice $f = 0.50$ or $f = 0.45$ with its reasons. Click another soil in the drawing to see its numbers. Under the table, the chart of the season shows the depletion at the sensor day by day, the trigger (dashed) and the irrigation days (blue), with both axes named; then the advice, and the two lists *Choose f = 0.50 if* and *Choose f = 0.45 if*, each condition ticked or not with the value that decides it.

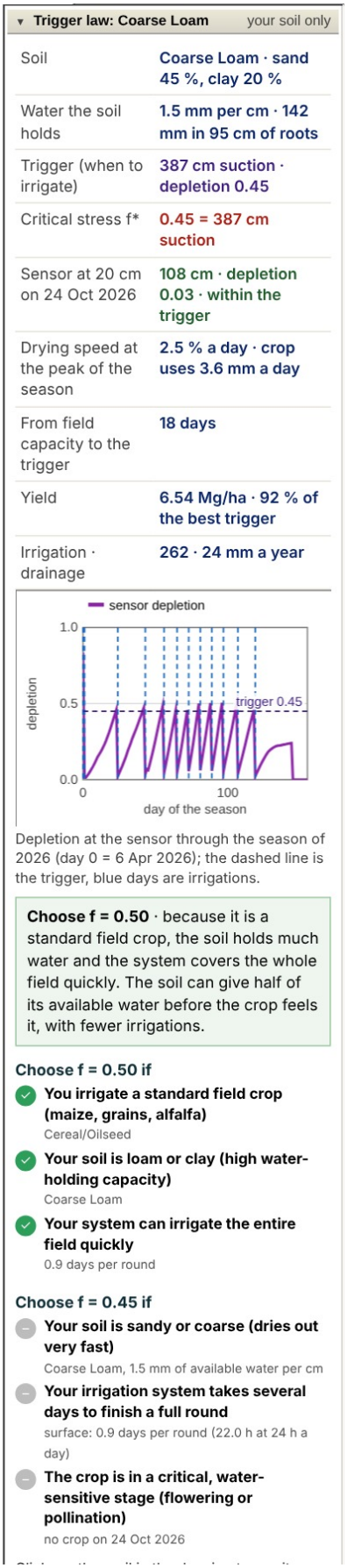



Figure 14.2: The Trigger law block of the Dashboard: the soil, its numbers, the season chart and the advice.

On *Soil hydraulic drivers* this block shows your soil: its texture; your tillage, consolidation rate and topsoil pore space; its drainage below 1 m, nitrate leaching, yield and irrigation at these settings; the effect of tillage (conventional against no-till) and of the consolidation rate (fastest against slowest); the leading driver and the rank of the soil among the ten. A chart gives the drainage against the consolidation rate for the three tillage systems with your rate marked, a table the drainage for every setting, the advice says which tillage gives the least drainage and leaching, and bars compare the two effects in the ten soils.

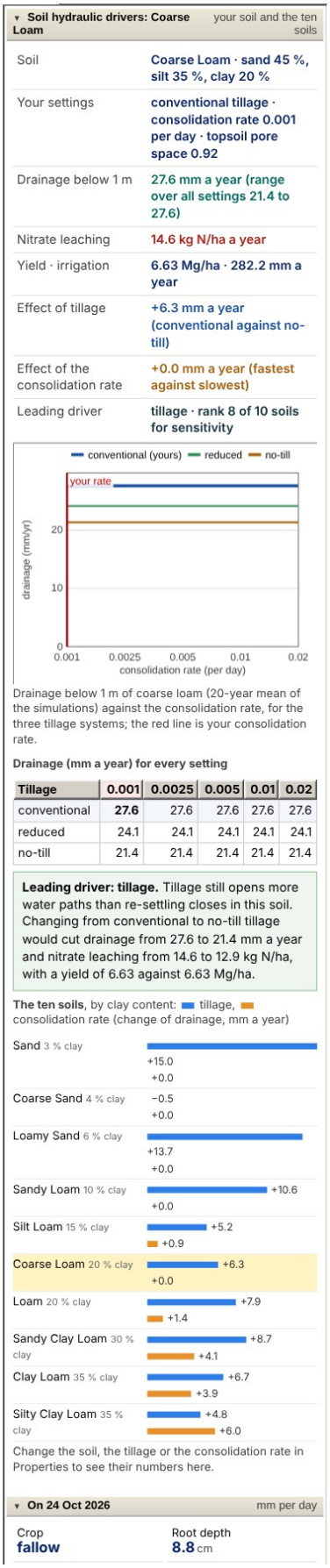



Figure 14.3: The Soil hydraulic drivers block of the Dashboard.

The tab has four blocks.

**Block 1: On the working date** (mm per day unless stated). Crop, root depth (cm), *Sensor / trigger* (the simulated relative depletion $f$ of the sensor layer and the trigger $f_T$; irrigation fires when $f \geq f_T$, Eq. 6.1), rain, irrigation, soil evaporation, transpiration and drainage on that day. They are the Engine 2 daily series of Chapter 5, read at the day index $k$ of the working date:

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm}\,\text{d}^{-1}). \tag{14.1}$$

*Provenance (Eq. 14.1):* **[Engineered]**: bookkeeping of Engine 2 series.

**Block 2: Next 14 days.** Sums of the same series over the following $N = 14$ days,

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

*Provenance (Eq. 14.2):* **[Engineered]**: bookkeeping of Engine 2 series.

The bar of the block says *simulated from the weather record* when the days come from the analogue year, and *from the weather forecast* when the online weather is in use and the days fall in its 16-day forecast (Chapter 19).

**Block 3: Whole year.** For the calendar year of the working date: rain, irrigation, other evaporation (interception, litter, ponded water and snow), soil evaporation, transpiration, drainage, the change in storage and the deepest roots (cm). The water balance of the year closes as

$$\Delta S = P + I - E_{\text{other}} - E - T - D, \tag{14.3}$$

*Provenance (Eq. 14.3):* **[Literature]**: water-balance identity.

and the tile *Storage change* is $\Delta S$.

**Block 4: 20-year averages at these settings.** The annual-model outputs of Engine 1 (Chapter 5): soil evaporation, transpiration, evapotranspiration ET $= E + T$, drainage, irrigation, yield (Mg/ha for the simulated cereal, per cent of the potential for a library crop), irrigation efficiency, nitrogen leaching, rainfall, and *Irrigation without mulch* (the same case with no mulch, for the saving).

### 14.1.1 The pinned baseline

Press **Pin** (or Ctrl+B): the tiles of Block 4 are stored. After any change the tiles show the difference from the pinned values,

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

*Provenance (Eq. 14.4):* **[Engineered]**: bookkeeping.

with ▲ (increase) or ▼ (decrease), green when the change is in the better direction and red when it is worse, and "= no change" when $|\Delta_i| < 10^{-9}$. Lower is better for soil evaporation, drainage, irrigation and nitrogen leaching; higher is better for yield and irrigation efficiency; transpiration, evapotranspiration, rainfall and *Irrigation without mulch* are neutral.

## 14.2 Design

With a design layer open, the tab shows the results of its **last run**. Before the first run, or after an input changed, it says that the design waits for a run and carries the two buttons **Run simulation** and **Find the most efficient inputs** (Chapter 18). After a run it shows, in folding blocks:

| Block | Contents |
|---|---|
| Key results | Surface: $E_a$, $E_s$, deep percolation, runoff, CU, cut-off, fill and advance times, the volumes. Drip: $E_a$, $E_s$, $E_u$, deep percolation, wetted fraction $P_w$, gross application, irrigation time, volumes per emitter. Sprinkler: $E_a$, $E_s$, DU, CU, deep percolation, runoff, gross and net application, set time, discharge, application rate. |
| Warning | For a surface design whose stream does not reach the end of the field in a practical time. |
| Most efficient inputs | The best combination found by the search of Section 18.4, with its $E_a$, $E_s$, deep percolation and runoff, and **Apply these inputs and run**; or *Already the most efficient* when no combination tried beats your inputs. |

The definitions are Eqs. (11.18) to (11.19), (12.19) and (13.11). The Design tab and the *Design results (CSV)* export read the same object, so the numbers agree digit for digit.

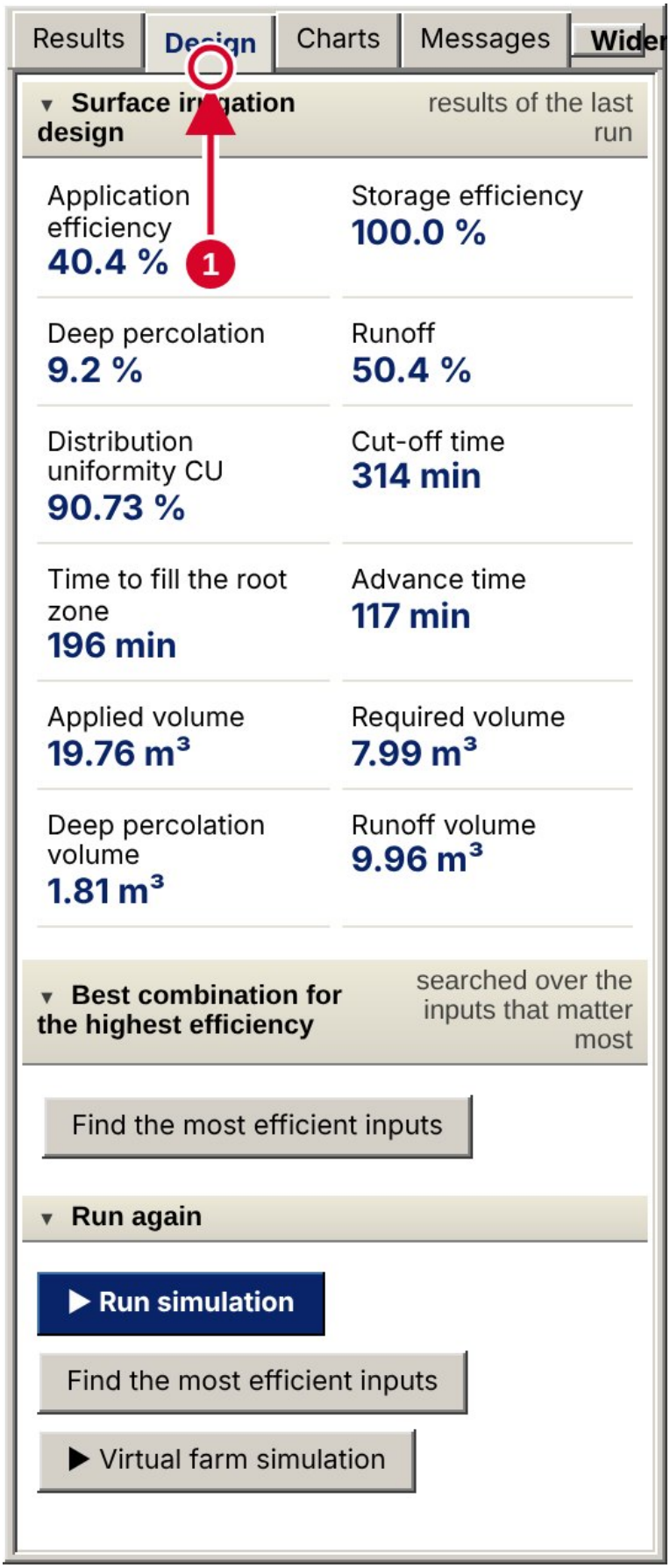



Figure 14.4: The Dashboard, Design tab, after a run of the surface design.

## 14.3 Charts

On the seasonal layers the Charts tab has one chart for each part of the layer, over the twenty seasons, with the year on the clock marked and the rainfed seasons dashed. Under each chart a line tells how the quantity varies: its mean, the 10 to 90 % range, the lowest and the highest season, and the variation from year to year.

- *Reliability over 20 seasons:* irrigation, drainage below 1 m, grain yield, nitrate leaching, evapotranspiration, days of crop water stress.
- *Full water balance:* rain and irrigation, transpiration and soil evaporation, other evaporation, drainage and drainage through biopores, runoff.
- *Nitrogen budget:* nitrogen supplied, taken up and lost.

With a design layer open and run, the tab draws the charts of the run:

| Design | Charts |
|---|---|
| Surface | Hydrograph: inflow, infiltration and runoff against time, with the advance time $t_L$ and the cut-off $t_{co}$ marked; advance, cut-off and recession along the field; infiltrated depth along the field against the required depth; the times of the irrigation ($t_L$, $t_s$ to fill the root zone, $t_{co}$, the recession time $t_r$ and the total); where the water goes. |
| Drip | Hydrograph at the pump (the zones run one after the other at the zone flow); the times of the irrigation (run time of a zone, the whole field, the interval); pressure head along the lateral; emitter flow along the lateral; where the water goes. |
| Sprinkler | Hydrograph at the pump (the sets run one after the other); the times of the irrigation; soil intake rate against the application rate; depth caught at each point, sorted (uniformity); where the water goes. |

With any other layer open, three charts of the twenty years around the working date (calendar years on the horizontal axis, mm per year): *Rainfall and irrigation*, *Drainage below the root zone* and *Evaporation and transpiration*. They are the annual sums of the Engine 2 daily series,

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

*Provenance (Eq. 14.5):* **[Engineered]**: bookkeeping of Engine 2 series.

Every chart sits in its own folding block; a chart that is unfolded is drawn again at once. The pump hydrograph of drip and sprinklers is the step series

$$Q_{\text{pump}}(t) = Q_z \quad \text{for } t \in [\,0,\ n_z\, t_{\text{run}}\,), \qquad 0 \text{ afterwards}, \tag{14.6}$$

*Provenance (Eq. 14.6):* **[Engineered]**: zones of the layout (Eqs. 12.20 to 12.23 and the sprinkler analogue) run in turn.

with $Q_z$ the flow of one zone or set, $n_z$ the number of zones or sets and $t_{\text{run}}$ the run time of one zone (drip) or the set time (sprinklers).

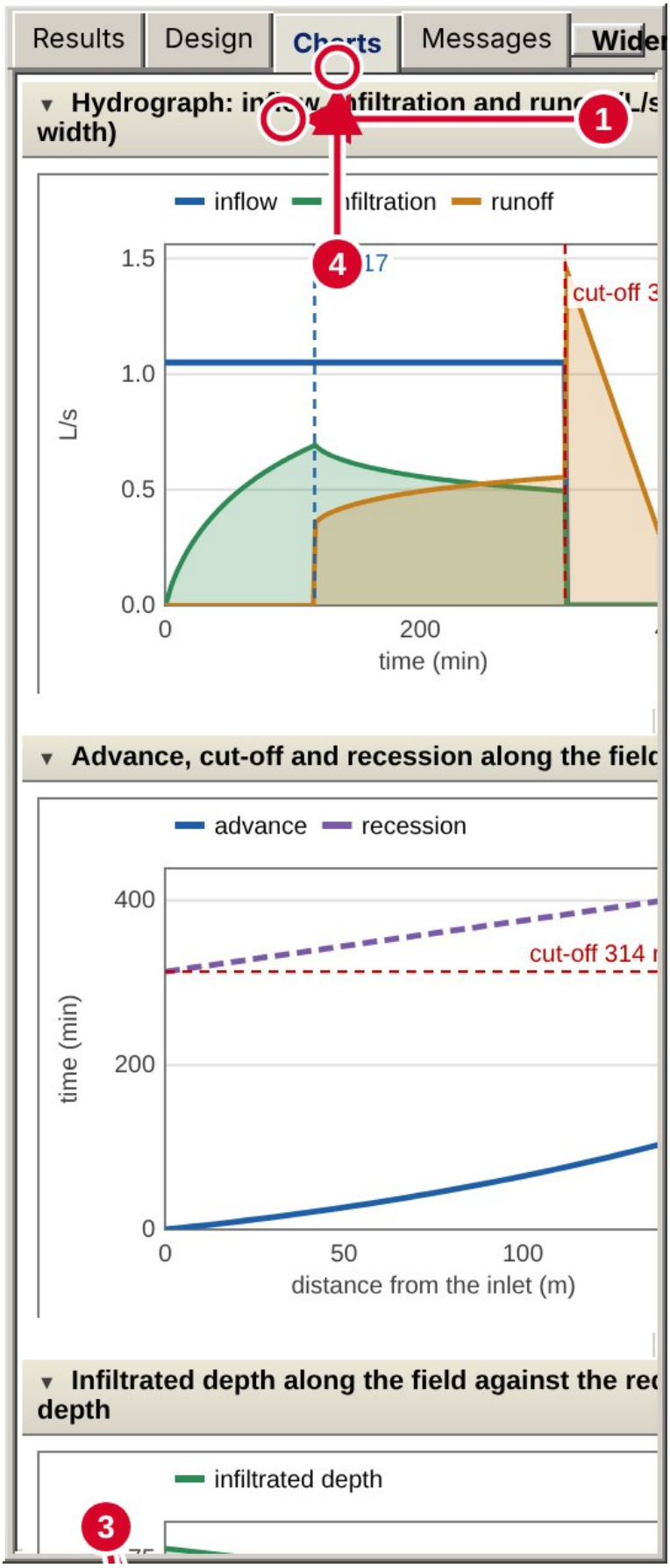



Figure 14.5: The Dashboard, Charts tab, after a run of the drip design.

## 14.4 Messages

A time-stamped log of what the program did (weather imported, design run, report saved, baseline pinned), newest first, in two folding blocks: *Warnings* (only when there are any) and *All messages*. Use it to find the path of a saved file.

# Chapter 15

# ✔ Worked workflows

Each workflow is a complete sequence of clicks, what to expect, and the equations behind it.

## 15.1 Workflow 1: How much does straw mulch save?

1. Properties ▸ Scenario ▸ *Ready-made case* ▸ *Default* (Coarse Loam, bare, normal climate).
2. Press **Pin** above the dashboard (Ctrl+B). The tiles of the 20-year block are stored.
3. In Properties ▸ Mulch choose **Wheat straw**; the strength becomes 0.4 and the capacity 4 mm. Read the arrows: soil evaporation falls (green), irrigation falls (green).
4. Open *Mulch and evaporation* ▸ *Fate of saved water* to see where the saved water goes (less irrigation, more drainage, more transpiration).

*Why:* the strength $\varphi$ scales the vapour flux through the mulch, and the capacity the water the mulch holds (Chapter 9); *Irrigation without mulch* minus *Irrigation* is the saving on the whole-year balance.

## 15.2 Workflow 2: When should I irrigate?

1. Set the working date to today and choose the soil and crop.
2. Open *Irrigation design* ▸ *Surface irrigation design* (or drip, or sprinklers), set the inputs and press **Run simulation**. Scroll to the *Irrigation schedule*: the first row is the next irrigation date; the *When to irrigate* message at the end repeats it with the mean interval.
3. Change the **Trigger** in Properties to a drier value (for example $-300 \rightarrow -1000$ cm): the run is cancelled; run again: the interval lengthens and the number of irrigations falls (Eq. 6.3 shifts $f_T$).
4. Change the **Sensor depth**: a shallow sensor dries first and triggers earlier.

## 15.3 Workflow 3: Design a border strip for a loam

1. Set the soil to Loam; open *Surface irrigation design*; *Start from: Your soil*; *System: Border*.
2. Type the run length of your field in the *Field length* box (for example 200 m), the width and the slope.
3. Press **Run simulation**. Read $E_a$ and $E_s$ in *Result*, and the hydrograph and the times on the Charts tab.
4. Press **Find the most efficient inputs**. The search tries inflows, cut-off rules, cutbacks and shorter runs (Section 18.4); if it finds a better combination, the Design tab offers **Apply these inputs and run**.
5. Read the sensitivity chart: if the intake constants dominate, the soil intake is the main uncertainty, so measure it in the field.

## 15.4 Workflow 4: Choose and size a drip system

1. Choose the crop, the soil and the emitter (inline).
2. *Field size*: **Length and width**; type the length and width of the field. The area and the lateral length follow (Eq. 18.1).
3. *Placement*: compare *Surface*, *25 cm* and *45 cm*: each change cancels the run; run each and compare $E_a$ and the losses.
4. Set the farm supply $Q_s$; read zones, pump head and power in the layout block (Eqs. 12.20 to 12.23), and the pump hydrograph on the Charts tab.
5. Open *Tools* ▸ *Field on the map...* to read the real field: its indices, its land cover, the water sources around it and the system that suits it (Chapter 20).

## 15.5 Workflow 5: Check a sprinkler against the course rules

1. Choose the soil and the slope (for example Sandy Loam, 3 %); the banner states the allowed rate (Table 13.3).
2. Choose *Impact sprinkler*; set the nozzle, the pressure and the spacing in their value boxes; press **Run simulation**; read CU and the application rate.
3. Open *Does your design pass?*; fix the red checks in the order the text names; run again.
4. Compare the system chips (solid-set, hand-move, center pivot, linear-move, travelling gun) and their efficiency ranges.

## 15.6 Workflow 6: My own weather, a warmer future

1. *File* ▸ *Import weather data* and choose your daily weather file, or *File* ▸ *Online weather for a location* (Chapter 19). Every layer now runs on it; Properties shows *Imported weather file* (or *Online weather of a location*) as the weather source.
2. Pin the baseline.
3. Set the temperature shift to +2 °C and the rainfall scale to ×0.8. The arrows show the effect of the change on your own climate.
4. Switch back to *Built-in weather record* in Properties ▸ Scenario at any time: your earlier rainfall and temperature settings return.

## 15.7 Workflow 7: A report for a client

1. *File* ▸ *Save report as PDF (choose the layers)...*.
2. Untick the groups you do not need (for example *Seasonal data* and *Feedback loops*), keep the irrigation design.
3. **Make the PDF**, choose where to save it. Each layer gets your inputs, its key numbers and its full drawing; long drawings continue on the next pages, cut at their section headings (Chapter 23).

## 15.8 Workflow 8: Save, share and reproduce

Save the project (*File* ▸ *Save project*) after each case, export the yearly CSV and the design CSV, and note the version shown in the status bar (*SWAFL 1.1.0*). Reopen the `.swafl` file to restore the state exactly.

# Chapter 16

# ✎ Files, exports and troubleshooting

## 16.1 What can be saved

| Item | Menu / toolbox | File | Contents |
|---|---|---|---|
| Report | *File* ▸ *Save report as PDF (choose the layers)...*, Ctrl+Shift+P, toolbar *Report* | `SWAFL_full_report_<soil>_<date>.pdf` | The layers you ticked, each with your inputs, its key numbers and its full drawing (Chapter 23) |
| Picture of the window | *File* ▸ *Export* ▸ *Picture of the window (PNG)*, Ctrl+E | `SWAFL_<layer>.png` | The whole drawing of the open layer, top to bottom |
| Yearly results | *File* ▸ *Export* ▸ *Yearly results (CSV)* | `SWAFL_yearly_results.csv` | Two comment lines with soil, tillage, crop, mulch and method; then `Year, Rain_mm, Irrigation_mm, Drainage_mm, Evaporation_mm, Transpiration_mm` |
| Design results | *File* ▸ *Export* ▸ *Design results (CSV)* | `SWAFL_design_<id>.csv` | Two columns `Quantity,Value`: every number the open design layer computed |
| Project | *File* ▸ *Save project* | `SWAFL_project.swafl` | JSON: app name and version, the open layer, the state, the year and the date |
| Field report | *Export the report* in *Tools* ▸ *Field on the map...* | `SWAFL_field_<place>_<date>.pdf` or `.png` | The field on the map, its indices, zones, land cover, water sources and the suitable system, at 96 to 600 dpi, with the coordinates and the author (Chapter 20); also GeoJSON and CSV |
| Print | *File* ▸ *Print / save this view as PDF*, Ctrl+P | PDF | The open layer through the system print dialog |

In the yearly CSV, *Year* is the index 1 to 20 of the 20-year run starting at the calendar year shown on the Charts tab; the values are the annual sums of Eq. (14.5), rounded to 0.01 mm. In the design CSV the quantity names are the internal names of the model (for example `Ea`, `Es`, `dp`, `ro` for the surface layer; `Eu`, `gross`, `t` for drip; `CU`, `DU`, `AR`, `T` for sprinkler); the efficiencies are fractions (0 to 1).

The imported weather is not stored in the project file; import it again after opening a project, or keep the file beside the project.

## 16.2 Troubleshooting

| Problem | Fix |
|---|---|
| “Running scripts is disabled” in PowerShell | Use a Command Prompt terminal, or run `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned`, or type `cmd /c npm install`. |
| `npm` is not recognised | Install Node.js, then close and reopen the terminal. |
| `ENOENT … package.json not found` | You are in the wrong folder; the zip makes a folder inside a folder. `cd` into the one that contains `package.json`. |
| Install or build is slow or fails inside OneDrive | Copy the project to `C:\SWAFL_Desktop` and run it there. |
| “Windows protected your PC” | Click *More info* ▸ *Run anyway* (the app is not code-signed). |
| The window still shows version 1.0 | An old `SWAFL.exe` is still on the Desktop or pinned. Delete it and run `BUILD_AND_INSTALL_SWAFL.bat` again: the build removes the old `dist` folder first and the program clears its cache on start. |
| White or empty window | Wait 10 seconds (the engine loads), then *Refresh* ▸ *Reload simulation engine*. Close any second SWAFL window. |
| A design shows *Run the design* instead of results | The inputs or a setting changed since the last run. Press **Run simulation**. |
| The field map says the satellite data or the photo cannot be reached | These need the internet (Chapter 20). Without it the program uses the simulated indices and says so. |
| A design layer shows “the stream does not reach the end” | The cut-off time is shorter than the advance time: increase the cut-off or the inflow (Eq. 11.16), or shorten the run. |
| $E_a$ is low and runoff is high (sprinklers) | The application rate is above the soil intake: reduce the nozzle or the pressure, widen the spacing (Chapter 13). |
| A weather file is refused | Read the message: the file needs a date column (or year, month and day) and a rain column with their names in a header row. A Daisy setup file (`.dai`) holds no weather: the message names the weather file it uses; import that file (Chapter 19). |
| The online weather is not reachable | The computer is offline or the service does not answer. The current weather stays in use; try again later or import a file. |

If a problem remains, open an issue on the project page, and attach the `.swafl` file and the text of the *Messages* tab.

# Chapter 17

# ⚠ Validity, limits and honest use

The program is a **teaching and design-exploration tool** built on a research data set. This chapter says what each part can and cannot support.

**Engine 1 (the annual meta-model)** reproduces, by interpolation, the Daisy 7.1 runs it was fitted to (run sets E2, E3, E4, E5, E7, E11, E13 and E14; the study as a whole has 10,190 runs). Within the ranges of the controls it is an *emulator* of those runs: it is not a new physical model, and it does not extrapolate outside the soils, mulches, climates and irrigation methods that were simulated. It was calibrated to one location's weather record (1980 to 1999) and its projections to 2100 are **scenarios built from analogue years**, not forecasts.

**Engine 2 (the daily bucket model)** is a 40-layer, 1-D, daily water balance. It uses closed-form root, evaporation and transpiration functions with a single calibrated coefficient $k_t$ so that its seasonal transpiration matches Engine 1. It does not solve the Richards equation, has no lateral flow, no runoff on slopes and no water table (upward flow is a redistribution between layers only). Its day-to-day irrigation dates are therefore indicative: use them to compare scenarios, not to schedule a real irrigation to the day.

**Design layers (surface, drip, sprinkler)** are textbook models (Chapters 11 to 13). They are suitable for teaching, for comparing layouts, and for a first sizing. They are **not a substitute for field evaluation** of intake, advance, pressure and flow, or for a hydraulic design by a qualified engineer. In particular:

- the intake of the soil is derived from texture by pedotransfer functions with their own scatter; measure it in the field for real design;
- the sprinkler pattern is a parametric shape, not a measured catch-can test; the wind-drift loss is a parametric estimate;
- the drip bulb is a geometric simplification of two-dimensional soil water redistribution;
- pump power and pipe sizes use stated allowances (9 m of fittings, 0.65 pump efficiency, $C = 140$) that you should replace with your own;
- the "Engineered" items of the label scheme (Section "About this manual") are modelling choices, not published laws.

**Data and reproducibility.** The same inputs and version give the same outputs (no random numbers in the design layers; the rainfall generator and all animations of the engine use a deterministic generator seeded by the soil name). Save the `.swafl` file and the version number with any result you report.

**Your weather (version 1.1).** Imported or online weather drives the daily model and the crop water requirement day by day. The annual model, however, can only move along the simulated steps of rainfall ($×0.5 to 1.4) and warming (-1$ to +3.5 °C) of the study site (Eqs. 19.4 to 19.5): a climate far outside those steps (for example a humid tropical climate with 2,000 mm of rain) is represented by the nearest step, and the annual results should then be read as the direction of a change, not as its size. The online forecast covers 16 days; dates after it are the typical year of your record.

**The crop library (version 1.1).** The 48 library crops are not simulated: they are transferred from the simulated cereal rotation by the ratio of two FAO-56 balances (Section 21.4). The transfer is reliable for the water need and the irrigation (which follow from $K_c$ and $ET_0$), less so for drainage and nitrogen, and the yield is a relative yield (FAO-33), not a harvest in tonnes.

**The real field (version 1.1).** Sentinel-2 indices are 10 m pixels of one date and need a scene without cloud over the field. The satellite photo is a mosaic of dates that varies by place and has no near-infrared band, so it gives the visible-band indices and the land cover only; the land cover of Eq. (20.4) is an image analysis by colour and texture, not a trained model, and mistakes dark soil for water or bright soil for roofs now and then. The water sources are those mapped in OpenStreetMap and those seen in the photo; groundwater depth is not known. The DEM is 90 m, so the slope of a small field is approximate. The ranking of the systems is guidance, to be checked on the ground.

**Not included.** The package does not contain a soil-erosion model, a crop-pest model, an economic optimiser, or a groundwater model.

When you cite results from the program, state the version, the case (soil, crop, mulch, method, trigger, date and weather source) and whether the numbers come from Engine 1, Engine 2 or a design layer.

# Chapter 18

# ⚑ Running a design and finding the most efficient inputs

Version 1.1 changes how the three irrigation designs are used. Their inputs are set first, and nothing is computed while you type; the results, charts, design table, schedule and soil-water film appear only when you press **Run simulation**. This chapter describes the run, the value boxes, the field size, the search for the most efficient inputs, the field sketch and the comparison of the three systems.

## 18.1 The run

When a design layer opens, its inputs are drawn as usual and, below them, the results are replaced by the **Run the design** box:

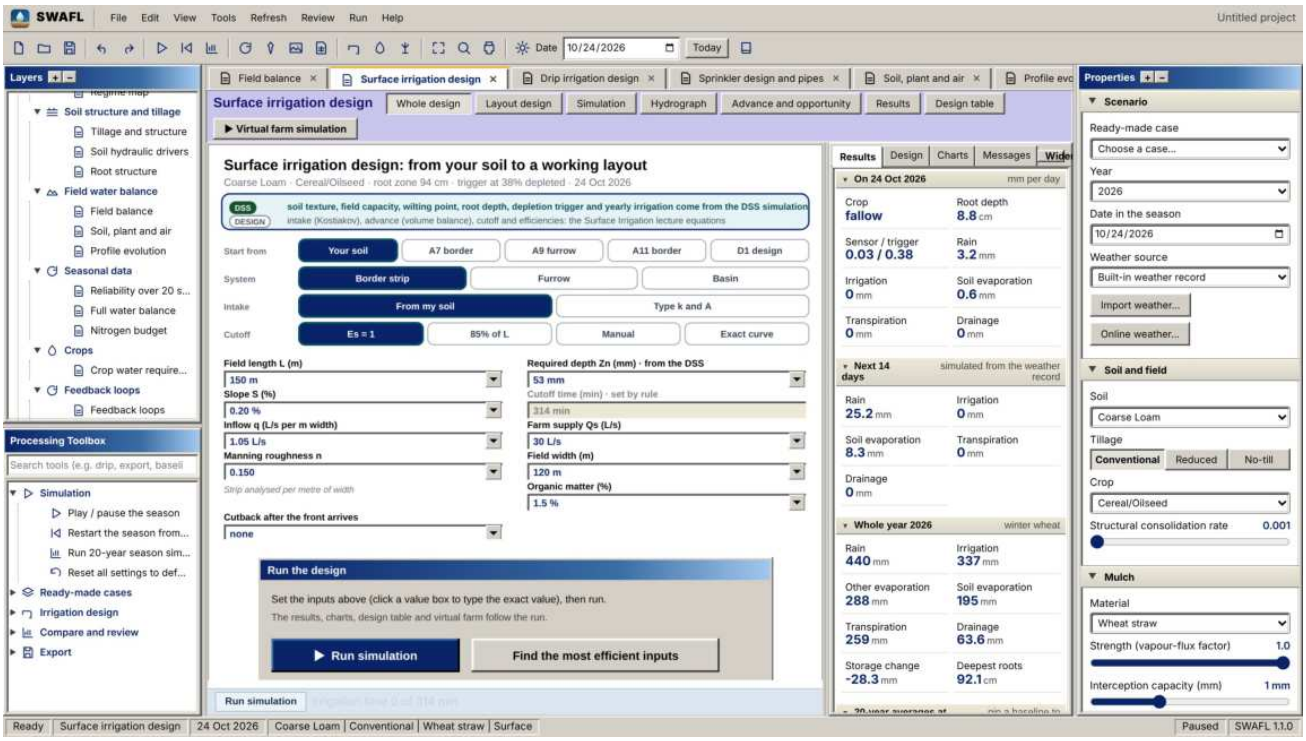



Figure 18.1: The surface design before a run: the inputs are set, the results wait for the run.

- **Run simulation** computes the design with the inputs above it. The run animates the irrigation from its start to its end (the time is shown on the playback bar, for example *irrigation time 0 of 314 min*), and then starts again from the beginning, so it keeps running until you press Pause.
- **Find the most efficient inputs** searches the inputs for the highest application efficiency (Section 18.4), puts the best combination into the value boxes and runs it.
- The play button under the drawing is labelled **Run simulation** in the design layers and does the same.

The last run stays valid until something that changes the result changes: a value box, a chip of the design (system, emitter, placement, sprinkler type, layout, run-time mode, mulch, crop, field size), or a setting of Properties other than the irrigation method (soil, tillage, consolidation, crop, mulch, rain, temperature, trigger, sensor depth, nitrogen, year or date). The box then returns with the reason, for example *The settings in Properties changed: the results wait for a new run*. The Design and Charts tabs of the dashboard follow the same rule.

Formally, a run stores the key $\kappa_{\text{run}}$ of the state with which it was made; the results are shown only while

$$\kappa(\text{state now}) = \kappa_{\text{run}}, \qquad \kappa = (\text{every Properties control except the irrigation method, weather mode}), \tag{18.0}$$

*Provenance (Eq. 18.0):* **[Engineered]**: bookkeeping of the run.

and no input of the design has been changed since.

## 18.2 Value boxes and units

Every input of a design is a value box. Clicking it opens a small window:

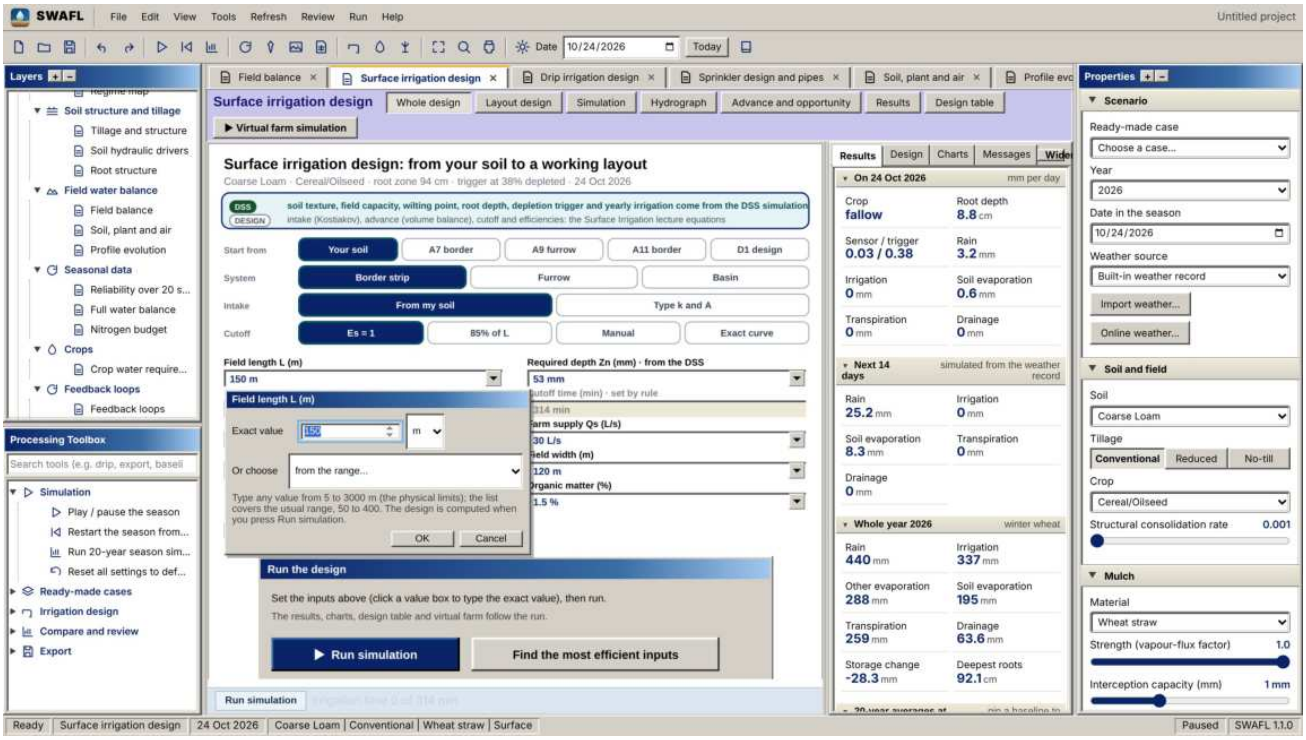



Figure 18.2: The value window of a surface-design input: exact value, unit, list of the allowed range.

- type the exact value, in the unit chosen beside it;

- or pick a value from the list, which runs across the allowed range;
- lengths can be given in m, km or ft; flows in L/s, m³/h or L/min (emitter flow in L/h or L/s); areas in ha, m² or acre; pressures in kPa, bar, m of water or psi; pressure heads in m, kPa, bar or ft; times in min or h; spacings in m or cm.

The value is kept exactly as typed (2.13 stays 2.13); it is not rounded to a step of the list, and the design runs on it. You may type any value up to the physical limits of the system: for example a run of 5 to 3000 m, an inflow up to 300 L/s per m, an emitter flow up to 100 L/h, a pressure up to 1200 kPa. The list covers only the usual range. A value beyond the physical limits is set to the nearest limit, and the window gives both ranges. The value is converted to the unit of the model, $x_{\text{model}} = x_{\text{typed}}\, c_u$, with $c_u$ the factor of the chosen unit (for example 1000 for km, 1/3.6 for m³/h to L/s, 9.80665 for m of water to kPa).

## 18.3 Field size: an area, or a length and a width

The drip and sprinkler designs need the area to irrigate and the length of the laterals. The row **Field size** offers two ways of giving them:

- **Area in hectares**, type the area; the lateral length is its own value box.
- **Length and width**, type the length and the width of the field in metres. The area and the lateral length then follow, and the lateral-length box is greyed out with the note *from the field length*:

$$A_f = \frac{L\,W}{10^4}\ \ [\text{ha}], \qquad L_{\text{lat}} = \text{clip}\Big(\frac{L}{\lceil L/L_{\max}\rceil};\ L_{\min},\ L_{\max}\Big), \tag{18.1}$$

*Provenance (Eq. 18.1):* **[Engineered]**: geometry; laterals longer than $L_{\max}$ are split into equal runs from a central submain.

with $L_{\min} = 20$ m and $L_{\max} = 300$ m for drip laterals and $L_{\min} = 40$ m and $L_{\max} = 500$ m for sprinkler laterals. The number of laterals follows from the width, $n_l = \lfloor W_f / S_l \rfloor$ with $W_f = A_f \cdot 10^4 / L_{\text{lat}}$ (Eq. 12.20 and its sprinkler analogue in Chapter 13), and with it the zones, the pipes and the pump.

The surface design is given by its run length and width directly (*Field length* L, *Field width*), and its area is their product.

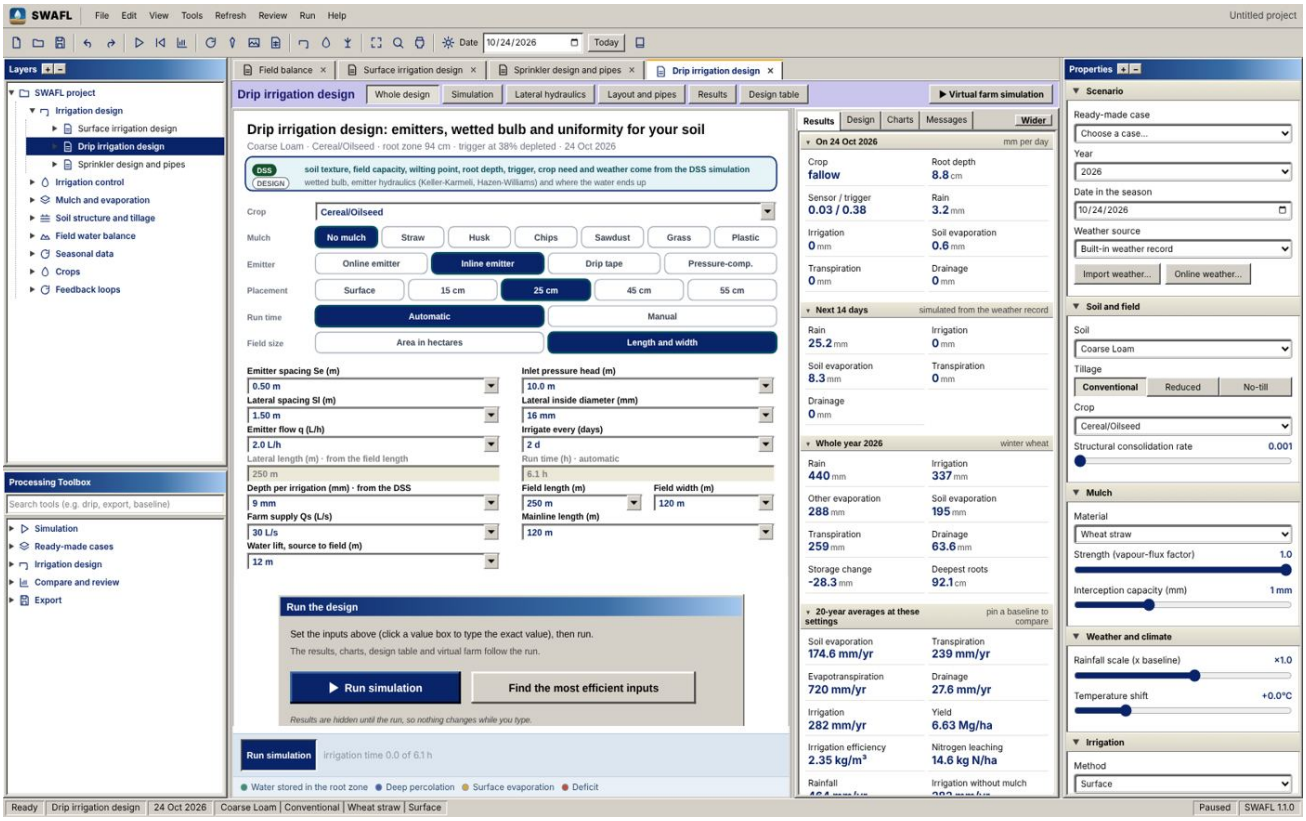



Figure 18.3: The drip design with the field given by its length and width.

## 18.4 The search for the most efficient inputs

**Find the most efficient inputs** (in the Run box, on the Design tab, or Tools ▸ *Find the most efficient design inputs*) tries a grid of the inputs that matter most and keeps the one with the highest score

$$s = E_a - 3\,\max\!\left(0,\ 0.95 - E_s\right) + \begin{cases} 0 & \text{surface} \\ 0.05\,E_u & \text{drip} \\ 0.05\,\mathrm{CU} - 2\,R_o & \text{sprinkler} \end{cases} \tag{18.2}$$

*Provenance (Eq. 18.2):* **[Engineered]**: the application efficiency, with a penalty when the root zone is refilled to less than 95 % (an efficient design that leaves the crop short is not efficient) and a small reward for uniformity.

| Design | Grid tried |
|---|---|
| Surface | 11 inflows spaced geometrically across the allowed range, $q_i = q_{\min}(q_{\max}/q_{\min})^{i/10}$; the cut-off rules *when the tail is filled* and *at 85 % of the length*; cutback to 100, 75 or 50 % of the inflow; and the run length $L/n$ for $n = 1$ to 4 (the field split into $n$ runs, each at least the shortest length of the system). |
| Drip | emitter flow 1, 1.6, 2, 3, 4 or 8 L/h; emitter spacing 0.3, 0.5, 0.75 or 1 m; interval 1 to 4 days. |
| Sprinkler | five nozzle diameters and four pressures across the range of the sprinkler type; for each, a square spacing of $0.4$ to $0.65$ times the wetted diameter. |

**Your present inputs take part in the search.** A combination is recommended only when its score is higher than the score of your present inputs; otherwise the Design tab says *Already the most efficient: your present inputs* and offers nothing to apply. The search takes about 0.1 to 3.5 s and stops after that time.

Applying a combination writes its values into the value boxes, so you can see and change each of them, and runs the design.

## 18.5 The field sketch

*View* or *Tools* ▸ *Field sketch to design…* opens a window with a grid on the left and the questions on the right.

**On the grid.** A red crosshair follows the mouse and gives its position in metres.

- **Field corners:** click to place a corner. Click near a side to add a corner on it, drag a corner to move it, and double-click a corner to remove it.
- **Side lengths:** each side shows its length just outside the field. Click a length to type it exactly; the next corner moves along the side, and the grid changes its scale if the field no longer fits.
- **Water source:** in this mode, click where the water is and drag the blue marker to move it.
- **Inlet:** in this mode, drag the yellow square along the edge of the field to where the pipe or the canal enters.
- **Grid square:** sets the scale, from 1 to 100 m per square.
- **Hand, zoom in and out, Fit:** the hand button moves the drawing; the right or middle mouse button, or Shift and drag, also move it in any mode. **+** and **−**, or the mouse wheel over the grid, zoom in and out around the cursor (10 % to 1200 %). **Fit** shows the whole field. The scale bar and the north arrow stay on the screen and follow the zoom.
- **Locate on the map:** opens the field on the satellite map (Chapter 20), to find the farm, demarcate it on the image and compute its vegetation and water indices.

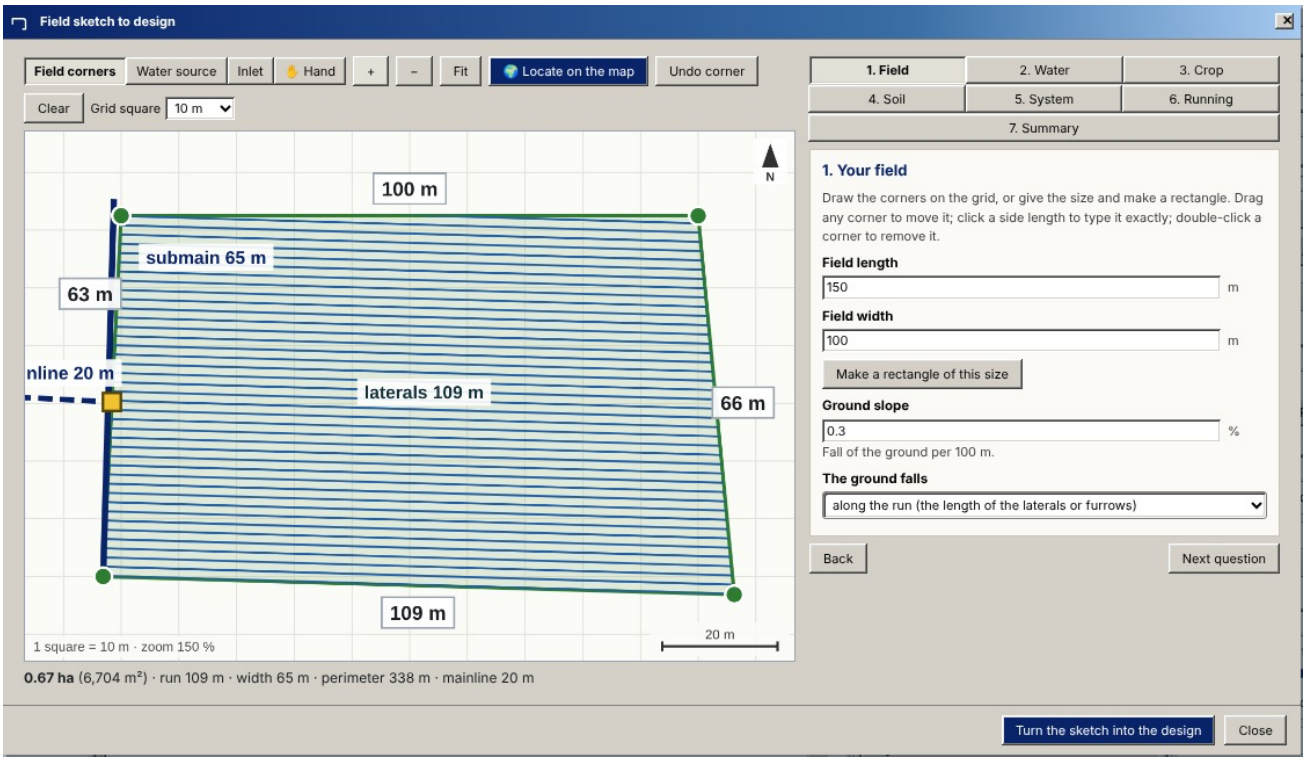



Figure 18.4: The sketch zoomed in, with the hand, zoom and map buttons.

**CAD tools under the grid.** Below the grid is a drawing toolbar in the manner of AutoCAD. Choose a tool, or type its short name in the command box (a letter such as L, PL, REC, C, A, D, T, M, CO, RO, MI, SC, O, TR, F, X, AR, E) and press Enter. The line under the toolbar always says what the tool asks for next.

- **Select:** click an object, or drag a window. A window dragged from left to right takes the objects wholly inside it; dragged from right to left it takes every object it touches. Shift adds or removes an object. The blue squares of a selected object are grips: drag one to change that point. Delete erases the selection, Ctrl+A selects all.
- **Draw:** Line (chained), Polyline (C closes the shape, Enter or double-click finishes), Rectangle (two corners, or type width,height in metres), Circle (centre and radius, or type D and the diameter), Arc (three points), Text (click, then type the note) and Dimension (two points, then where the dimension line goes; it reads the length in metres).
- **Modify:** Move, Copy, Rotate, Mirror, Scale, Offset (a parallel copy at a typed distance), Trim (cuts a line at the lines that cross it), Corner (joins two lines in a sharp corner), Explode (a rectangle or polyline into lines), Array (rows and columns of copies, for example 3,5,2,0.75 for three rows 2 m apart and five columns 0.75 m apart) and Erase. Select first, then choose the tool, as in AutoCAD; with nothing selected the tool asks you to select.
- **Inquiry:** Distance gives the distance, the angle and the two offsets between two points; Area gives the area in square metres and hectares and the perimeter of a shape.
- **Exact input:** with a point already placed, type a length to go that far toward the cursor, *length*<*angle* (for example 12<30), or @*dx,dy* in metres. Without a point, type *x*,*y* in metres, with y upward. The length and angle also show beside the cursor.
- **Snap, Ortho, Polar:** Snap (F3) pulls the cursor to endpoints, middles, centres, crossings, the nearest point of a line and the grid; the small button Snaps chooses which. Ortho (F8, or hold Shift) keeps lines horizontal or vertical; Polar clicks to every 15 degrees.
- **Layers:** the Layers button lists the layers (0, Pipes, Dimensions, Text, Construction). Tick to show or hide, change the colour, choose the layer for new objects, or add one.
- **Use the selected shape as:** a closed shape drawn with these tools becomes the field, an area to irrigate, a no-irrigation zone, a greenhouse or a plant bed; an open line or polyline becomes a mainline, submain or lateral pipe. The design then reads it as if it had been drawn with the green and blue tools.
- **Save DXF and Open DXF:** Save DXF writes the drawing, the field and the objects of the design in metres for AutoCAD or any CAD program. Open DXF reads lines, polylines, circles, arcs and text; a file that looks like millimetres is scaled to metres. Zoom all shows the whole drawing. Ctrl+Z and Ctrl+Y undo and redo every CAD step.

**Greenhouses, beds, areas to irrigate, exclusions, plants and traps.** Beside the modes above, the sketch has six more tools for drawing what is on the field. Choose the tool, then click on the grid (the buttons are in the row above the grid):

| Tool | How to draw it | What it is used for |
|---|---|---|
| Greenhouse | Click two opposite corners | A greenhouse or polytunnel, drawn as a blue rectangle with its ribs; its area is written inside. |
| Plant bed | Click two opposite corners | A raised or ploughed bed, drawn as a brown rectangle with its rows; its area is written inside. |
| Area to irrigate | Click the corners; double-click, or click the first corner again, to close | The part of the field that the design must irrigate (green). |
| No-irrigation zone | Click the corners; double-click, or click the first corner again, to close | A part that is left dry, such as a track, a building, a pond or a rock outcrop (red, dashed). |
| Plant | Click the place | One plant or tree (a green circle). |
| Monitoring trap | Click the place | One trap or monitoring post (a purple triangle). |

Double-click on a drawn object, with its tool chosen, to remove it. *Undo* (Ctrl+Z) and *Redo* (Ctrl+Y) go back and forward through every change of the drawing, one step at a time: a corner placed, a corner moved, an object added or removed, and *Clear*. Press *Escape* or choose another tool to stop an area you have not closed.

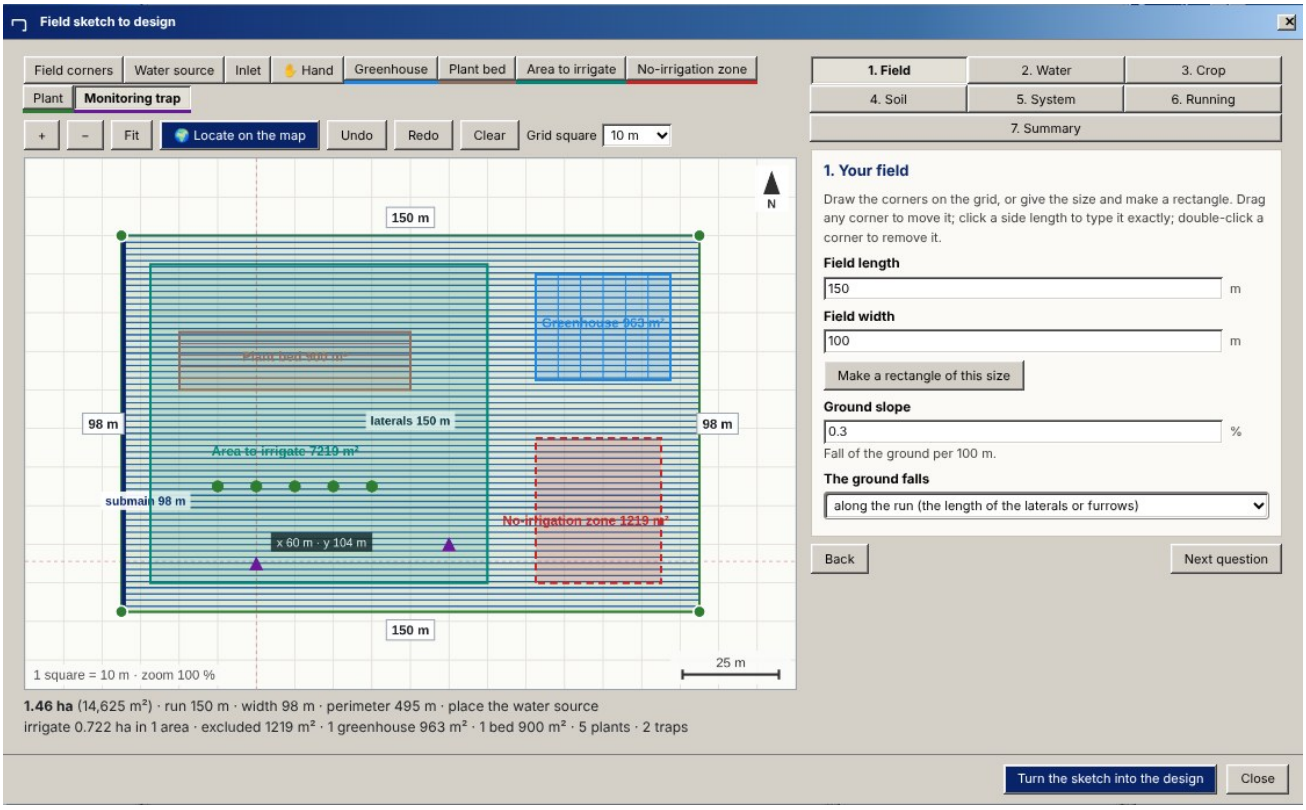



Figure 18.5: The sketch with an area to irrigate, a no-irrigation zone, a greenhouse, a plant bed, plants and traps.

The line under the grid sums what is drawn: the hectares to irrigate and the number of areas, the square metres excluded, the number and area of greenhouses and beds, and the numbers of plants and traps. The same line appears in the summary of the seventh question as *Drawn on the field*.

**The area that is irrigated.** When the design is filled from the sketch, the area it uses is the sum of the areas to irrigate. If none is drawn, it is the area of the field less the no-irrigation zones. This area sets the area of the surface, drip and sprinkler designs, and therefore the volume, the pumping time and the cost of the run.

While you draw, the window draws the layout of the system you choose:

- furrows, borders or basins;
- drip laterals at the row spacing;
- sprinkler laterals with the sprinklers and their wetted circles;
- the submain or head ditch along the inlet end;
- the mainline from the source to the inlet, with its length.

**The questions**, in seven steps (*Next question* and *Back* move through them; the step buttons jump):

1. *Field:* length and width (*Make a rectangle of this size*), ground slope, and the direction in which the ground falls.
2. *Water:* where the water comes from (well, borehole, canal, river, pond or tank), the lift to the field, the flow the source can give, and the hours a day you can irrigate.
3. *Crop:* the crop, the row spacing and the plant spacing.
4. *Soil:* the soil, and whether there is a compact layer below the topsoil.
5. *System:* border strips, furrows, basins, drip or sprinklers; whether the laterals run along the field or across it; for drip, the depth of the laterals and the emitter; for sprinklers, the system, the sprinkler and the two spacings.

6. *Running:* how often to irrigate (drip), and whether the farmer or an automatic controller opens the valves.
7. *Summary:* everything that will be set.

The sketch gives

$$A = \tfrac{1}{2}\Big|\sum_i (x_i y_{i+1} - x_{i+1} y_i)\Big|, \qquad L_{\text{run}} = \max_{\text{along the longest side}} - \min, \qquad W = \text{extent across it}, \quad (18.3)$$

*Provenance (Eq. 18.3):* **[Literature]**: shoelace formula and projection on the direction of the longest side.

and the mainline length from the source to the inlet. Here $L_{\text{run}}$ is measured along the direction of the laterals: along the longest side, or across it.

**Turn the sketch into the design** sets the crop and the soil in Properties and fills the chosen design:

- *Surface:* the system, the run length, the field width, the slope and the farm supply, and the furrow spacing (from the rows).
- *Drip:* the lateral length, the lateral spacing (the rows), the emitter spacing (the plants), the field length and width, the supply, the interval, the placement and the emitter.
- *Sprinklers:* the area, the slope, the supply, the two spacings, the sprinkler, the system and the lateral length.

It also sets the mainline length, the lift, the source, the hours a day, the compact layer and the operator of the valves. Then it opens the design, which waits for **Run simulation**.

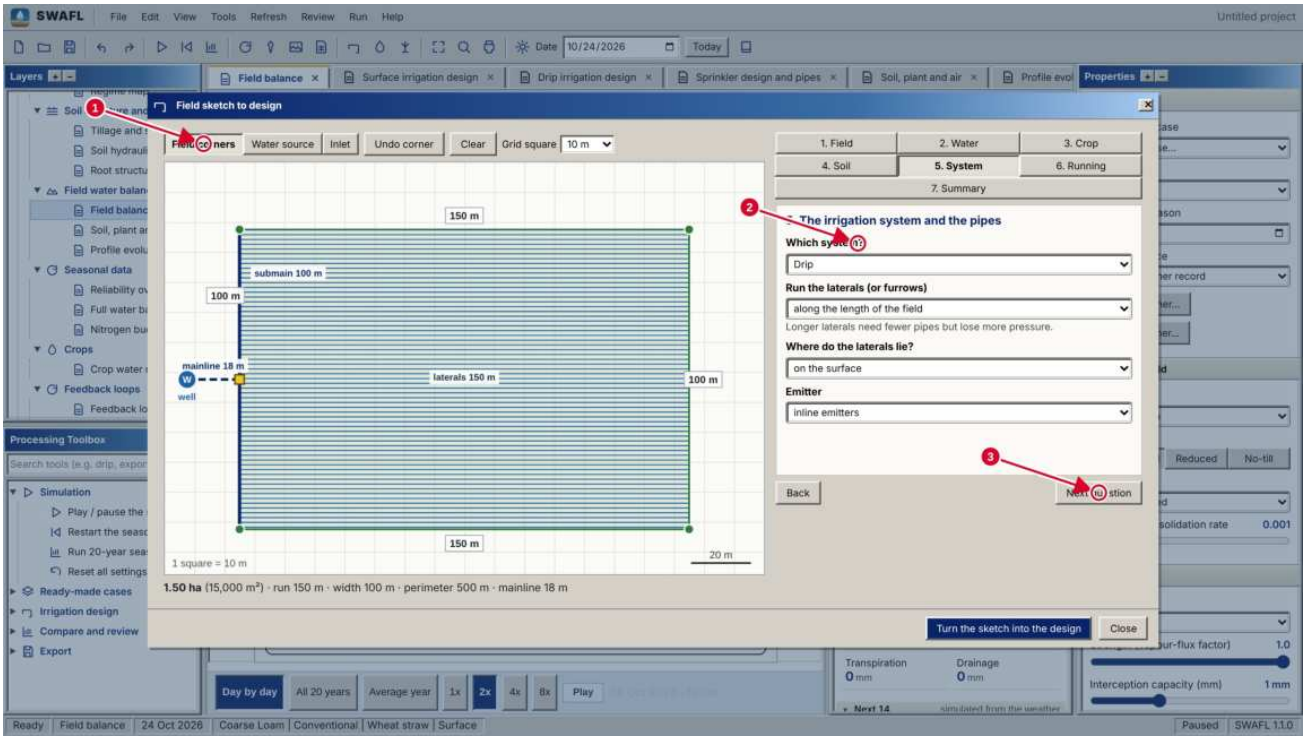



Figure 18.6: The field sketch.

## 18.6 The soil water in the days ahead

Each design layer (surface, drip and sprinklers) ends with a film of the soil water: **Soil water in the days ahead (DSS)**. It plays day by day; click it to pause or play.

**How the dates and depths are computed.** The film and the schedule under it come from one FAO-56 daily root-zone balance of the farm (Allen et al., 1998), the *irrigation plan*:

- the **crop water use** of every day, $ET_c = K_c\,ET_0$: $K_c$ from the four-stage curve of the crop in Properties (its stages, $K_{c,ini}$, $K_{c,mid}$ and $K_{c,end}$, its sowing date and, for trees, its density; Chapter 21), or $0.3 + 0.85\,c$ from the crop cover $c$ for the simulated cereal rotation; the part of $K_c$ that is evaporation from the bare ground between the plants, $0.8\ \min(K_c, 1)(1 - c)$, is cut by the mulch to its vapour-flux fraction;
- the **reference evapotranspiration** $ET_0$ **and the rain** of every day of the weather in use: the online weather of the farm with its 16-day forecast, a weather file, or the recorded weather, with the rainfall scale and the temperature shift of Properties;
- the **root zone**: the roots of every day of the daily model (Chapter 5) and the available water of the soil, $\text{TAW} = \text{AWC}\,z_r$;
- the **trigger** of the soil (Chapter 6) and the **design**: its net depth per irrigation $Z_n$, its pumped depth and its run time.

An irrigation is due when the root zone has lost the trigger depth $f_T$ TAW or the design depth $Z_n$, whichever is smaller; it puts back $\min(Z_n, D_r)$ (a shorter run when the trigger comes first). The balance starts at the sowing (or 150 days before the working date) from the water of the daily model, so the water of the working date carries the rain and the irrigations before it.

$$D_r(d) = D_r(d-1) + K_s\,K_c\,ET_0(d) - (P(d) - RO(d)) - I(d), \qquad K_s = \min\Big(1, \frac{\text{TAW} - D_r}{(1-p)\,\text{TAW}}\Big),$$
$$I(d) = \min(Z_n, D_r) \quad \text{when} \quad D_r \geq \min(f_T\,\text{TAW},\, Z_n), \tag{18.4}$$

*Provenance (Eq. 18.4):* **[Literature]** FAO-56 root-zone balance and water stress coefficient (Allen et al., 1998), $p = p_0 + 0.04(5 - ET_c)$ with $p_0$ of the crop; runoff $RO = 0.3(P - 35)$ above 35 mm (none for bunded rice); water above field capacity drains. The irrigation rule is **[Engineered]**.

**The season of your crop.** The crop chosen in Properties is the crop of the decision. When the working date falls in its season, the film runs from the working date to 30 days after the harvest. When the working date has no crop (before sowing or after harvest), the film starts at the **next sowing of that crop** and runs through its whole season, and a yellow note above the film says so; the schedule under the film follows the same season. With the online weather, whose record ends in the present year, the next season is predicted with the weather of the last season of the record, its dates moved on by one year. The crops of the library are sown on their sowing date (Properties), which may be any day of the year; their season may run past 31 December, and the daily model irrigates them whenever they are in the field.

**The weather of the decision.** The banner at the top of every design has a **WEATHER** line: the weather in use (the recorded weather of Taastrup, Denmark, 1980 to 1999; a weather file; or the online weather of a place with the 16-day forecast) and the button **Online weather of the farm...** (or *Change the weather...*), which opens the online-weather window (Chapter 19). The weather of your farm calibrates everything that follows: the soil water of every day, the irrigation dates and depths, the film, the schedule and the planting advice.

- **Left: the profile of the day.** The water content of each layer of the real soil, 0 to 100 cm, with lines at the wilting point, the trigger and field capacity, the depth of the roots and the sensor. The frame and the label below take the colour of the phase of the day.
- **Right: the days, coloured by phase.** The band at the top colours every day: irrigation day (blue), rain over 5 mm (light blue), comfortable (green), nearing the trigger (yellow), crop stress, $K_s < 1$ (orange, also shaded across the chart), near the wilting point (red), and no crop (grey). The chart also shows:
  - the water held in the root zone in mm (left axis), against the full level (field capacity over the roots of that day) and the level at which this design irrigates (the trigger, or the design depth when it is smaller); both levels rise and fall with the roots, so the axis changes with the soil and the crop;
  - the water left at the sensor, in % of the available water (right axis), from the profile of

the daily model scaled to the root zone of the plan;
    - rain as ticks at the top and drainage as ticks at the bottom;
    - a dashed red line for what would happen **without irrigation from the day shown**: the root zone dries on with the weather of the record until it reaches the wilting point. It starts again from each day of the film.
- **Below: the analysis.**
    - The day in words: the date, the crop, the phase, the water left, and its crop water use, $K_c \times ET_0 = ET_c$.
    - The next irrigation after the day shown, with its date, the days to it, the water the root zone has lost by then, the net and pumped depths of this design (and the part that drains, if any), and the date the wilting point would be reached without irrigation from that day. Both are counted again for every day of the film.
    - For the whole span: the number of irrigations and their water, the rain, the crop water use, the drainage, and the share of days in each phase.

Every part of the film is labelled.

- **The profile** has its depth axis (cm) and its water-content axis (% by volume, with the values at the wilting point and at field capacity). Labels mark the wilting point (WP), the trigger and field capacity (FC).
- **The chart** has its axis of the water in the root zone (mm, left), of the water at the sensor (% of available, right) and its axis of dates (the first of each month, with the year at January). The day shown is marked by a line with its date and its water above the chart; arrows mark each irrigation; *next irrigation* names the date and the days to it, and the cross of the wilting point names its date and the days to it without irrigation. These labels move with the film.
- **The colours** are distinct: irrigation dark blue, rain light blue, comfortable green, nearing the trigger yellow, stress orange, near wilting red, no crop grey. The root zone line is dark blue, the sensor purple, the trigger black dashed, the dry-down without irrigation red dotted, the rain ticks blue and the drainage ticks teal.

**In the Dashboard**, the block *Soil water now: the film of this design* follows the film day by day: the date and day of the film, the crop and phase, the water in the root zone (mm and %), the crop water use of the day ($K_c \times ET_0$), the depletion of the root zone against the level at which the design irrigates, the next irrigation (date, days to it, net and pumped depths) and the wilting point without irrigation (date, days to it).

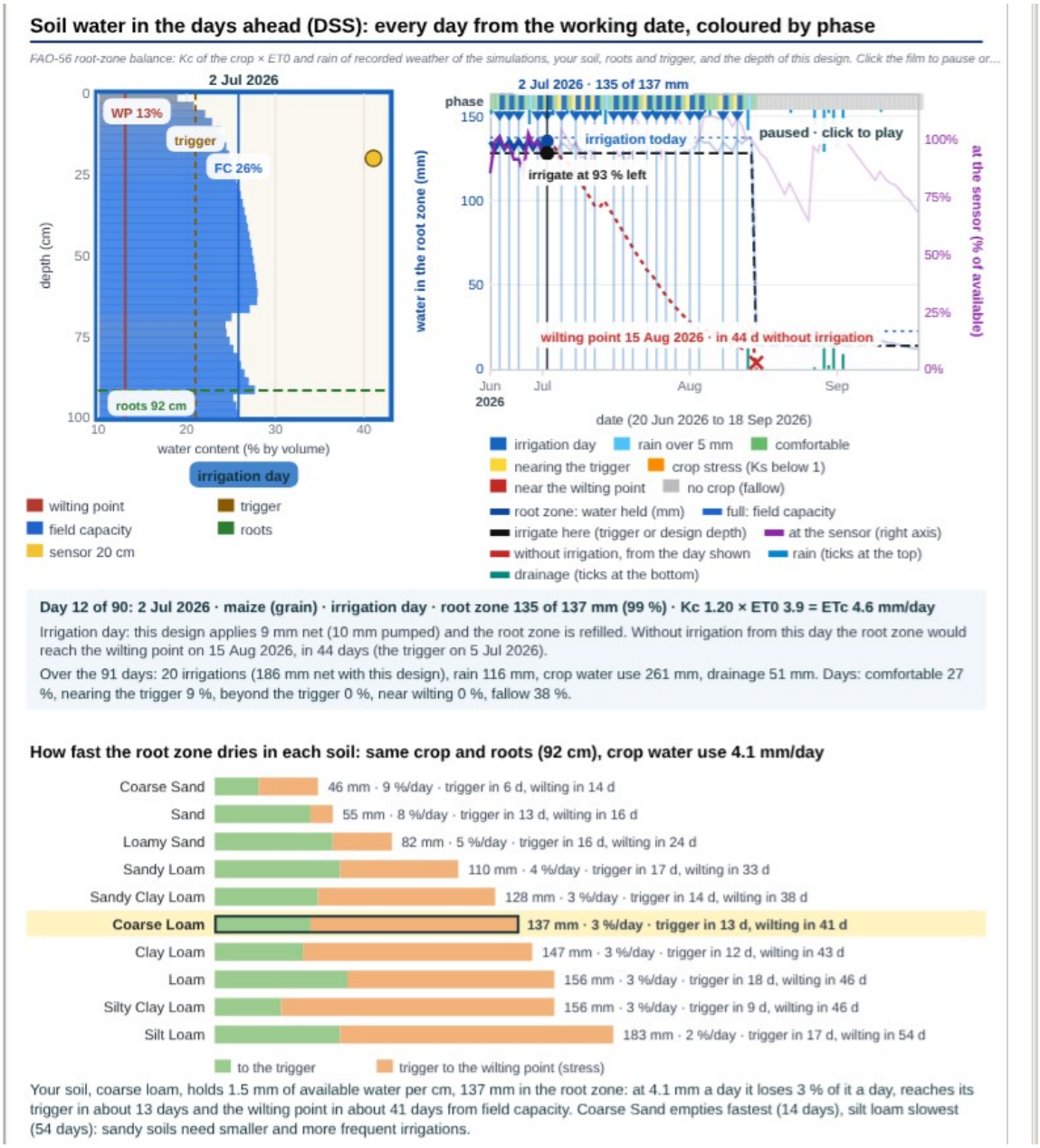



Figure 18.7: The soil water in the days ahead: the FAO-56 plan of the drip design for maize, an irrigation day, with the crop water use of the day (Kc × ET0) and the next irrigation.

**When to plant.** Under the film, the chart *When to plant …* gives, for every sowing week of the year, the irrigation the crop needs in a season (bars, mm) and its rain-fed yield (red line, % of potential), from the FAO-56 root-zone balance of the crop library over the 20 years of the weather in use, with your soil, trigger and method. Weeks too cold or too hot for the crop are grey: warm-season crops need a minimum air temperature at sowing and over the season (maize 10 and 14 °C, soybean 10 and 15 °C, tomato 12 and 15 °C, pepper 14 and 17 °C, sorghum 14 and 18 °C, rice, millet, cotton and cassava 15 and 20 °C) and no frost; cool-season crops (wheat, chickpea, mustard, potato, onion, cabbage) a season mean between 5 and 24 °C. The temperatures are those of the weather in use when it has them (online weather, or a file with temperatures), otherwise the typical climate of Taastrup, Denmark, the site of the recorded weather. The best week is the feasible week with the least irrigation (within 10 %) and the highest rain-fed yield; the best window is the run of feasible weeks within 15 % of its need. The text gives the window, the need,

the rain-fed yield and the need at your own sowing date; **Sow on … (best)** sets the sowing date in Properties. When no week is warm enough, the warmest window is shown and the text says so. For the simulated cereal rotation, whose dates are fixed, the advice is for wheat from the library (soybean for the legume, tomato for the vegetable) with a button to use that crop. Trees, vines and palms stand all year and have no sowing date.

The Dashboard block *Soil water now* also gives the planting window and the weather in use.

Everything follows the settings. Change the trigger, the sensor depth, the soil, the crop, the mulch, the rain or the weather and the film is computed again.

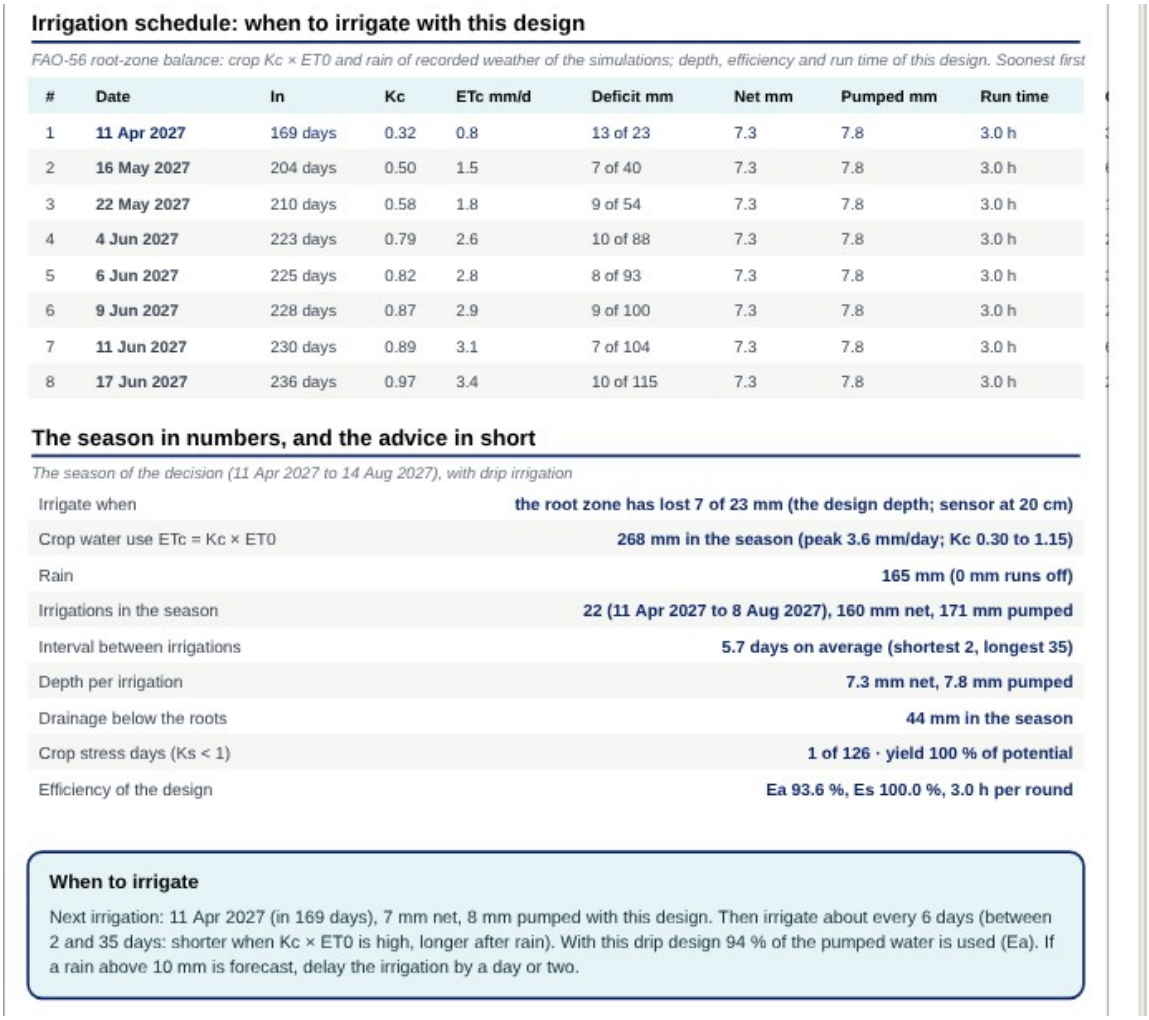
The phase of a day is set in this order: an irrigation day; a day with no crop; near wilting when the root zone has lost 85 % of its available water; crop stress when $K_s < 1$; a rain day (over 5 mm); nearing the trigger when the depletion reaches 80 % of the level at which the design irrigates; otherwise comfortable.

The water left in the root zone on day $d$ is TAW $- D_r$ of Eq. 18.4. Without irrigation from the day shown, the depletion goes on with the same $K_c\,ET_0$ and rain:

$$D_r(d+1) = \text{clip}\big(D_r(d) + K_c\,ET_0(d)\,[0.35 + 0.65\,K_s] - (P - RO)(d);\ 0,\ \text{TAW}\big), \qquad (18.5)$$

*Provenance (Eq. 18.5):* **[Literature]** with an **[Engineered]** floor: the crop and the soil keep losing about a third of $ET_c$ as the soil dries. The wilting point is reached when $D_r \geq 0.97$ TAW.

**The schedule** under the film lists the next eight irrigations of this design: the date and the days to it, $K_c$ and $ET_c$ of that day, the water the root zone has lost (of its TAW), the net and pumped depths and the run time (shorter when the trigger comes before the design depth), and the gap to the next one. *The season in numbers* gives, for the season of the decision, when to irrigate, the crop water use ($K_c \times ET_0$, with its peak), the rain and its runoff, the irrigations (their number, dates, net and pumped water), the mean, shortest and longest interval, the drainage below the roots, the days of crop stress with the yield as a share of the potential ($1 - K_y(1 - ET_a/ET_c)$), and the efficiency of the design. The season rows of each design (*Net irrigation in the season*, the gross water and the pumping hours) use the same plan, and so does the zone-by-zone irrigation of the field map (Chapter 20).



**Irrigation schedule: when to irrigate with this design**

*FAO-56 root-zone balance: crop Kc × ET0 and rain of recorded weather of the simulations; depth, efficiency and run time of this design. Soonest first*

| # | Date | In | Kc | ETc mm/d | Deficit mm | Net mm | Pumped mm | Run time |
|---|---|---|---|---|---|---|---|---|
| 1 | **11 Apr 2027** | 169 days | 0.32 | 0.8 | 13 of 23 | 7.3 | 7.8 | 3.0 h |
| 2 | **16 May 2027** | 204 days | 0.50 | 1.5 | 7 of 40 | 7.3 | 7.8 | 3.0 h |
| 3 | **22 May 2027** | 210 days | 0.58 | 1.8 | 9 of 54 | 7.3 | 7.8 | 3.0 h |
| 4 | **4 Jun 2027** | 223 days | 0.79 | 2.6 | 10 of 88 | 7.3 | 7.8 | 3.0 h |
| 5 | **6 Jun 2027** | 225 days | 0.82 | 2.8 | 8 of 93 | 7.3 | 7.8 | 3.0 h |
| 6 | **9 Jun 2027** | 228 days | 0.87 | 2.9 | 9 of 100 | 7.3 | 7.8 | 3.0 h |
| 7 | **11 Jun 2027** | 230 days | 0.89 | 3.1 | 7 of 104 | 7.3 | 7.8 | 3.0 h |
| 8 | **17 Jun 2027** | 236 days | 0.97 | 3.4 | 10 of 115 | 7.3 | 7.8 | 3.0 h |

**The season in numbers, and the advice in short**

*The season of the decision (11 Apr 2027 to 14 Aug 2027), with drip irrigation*

| | |
|---|---|
| Irrigate when | **the root zone has lost 7 of 23 mm (the design depth; sensor at 20 cm)** |
| Crop water use ETc = Kc × ET0 | **268 mm in the season (peak 3.6 mm/day; Kc 0.30 to 1.15)** |
| Rain | **165 mm (0 mm runs off)** |
| Irrigations in the season | **22 (11 Apr 2027 to 8 Aug 2027), 160 mm net, 171 mm pumped** |
| Interval between irrigations | **5.7 days on average (shortest 2, longest 35)** |
| Depth per irrigation | **7.3 mm net, 7.8 mm pumped** |
| Drainage below the roots | **44 mm in the season** |
| Crop stress days (Ks < 1) | **1 of 126 · yield 100 % of potential** |
| Efficiency of the design | **Ea 93.6 %, Es 100.0 %, 3.0 h per round** |

**When to irrigate**

Next irrigation: 11 Apr 2027 (in 169 days), 7 mm net, 8 mm pumped with this design. Then irrigate about every 6 days (between 2 and 35 days: shorter when Kc × ET0 is high, longer after rain). With this drip design 94 % of the pumped water is used (Ea). If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 18.8: The schedule of the drip design and the season in numbers.

## 18.7 Comparing the three systems

*Tools* ▸ *Compare the irrigation systems on this field* computes the surface, drip and sprinkler designs with the inputs they have now, on the same soil, crop and trigger, and lists their application efficiency, root zone filled, deep percolation, runoff or evaporation and the time per irrigation side by side.

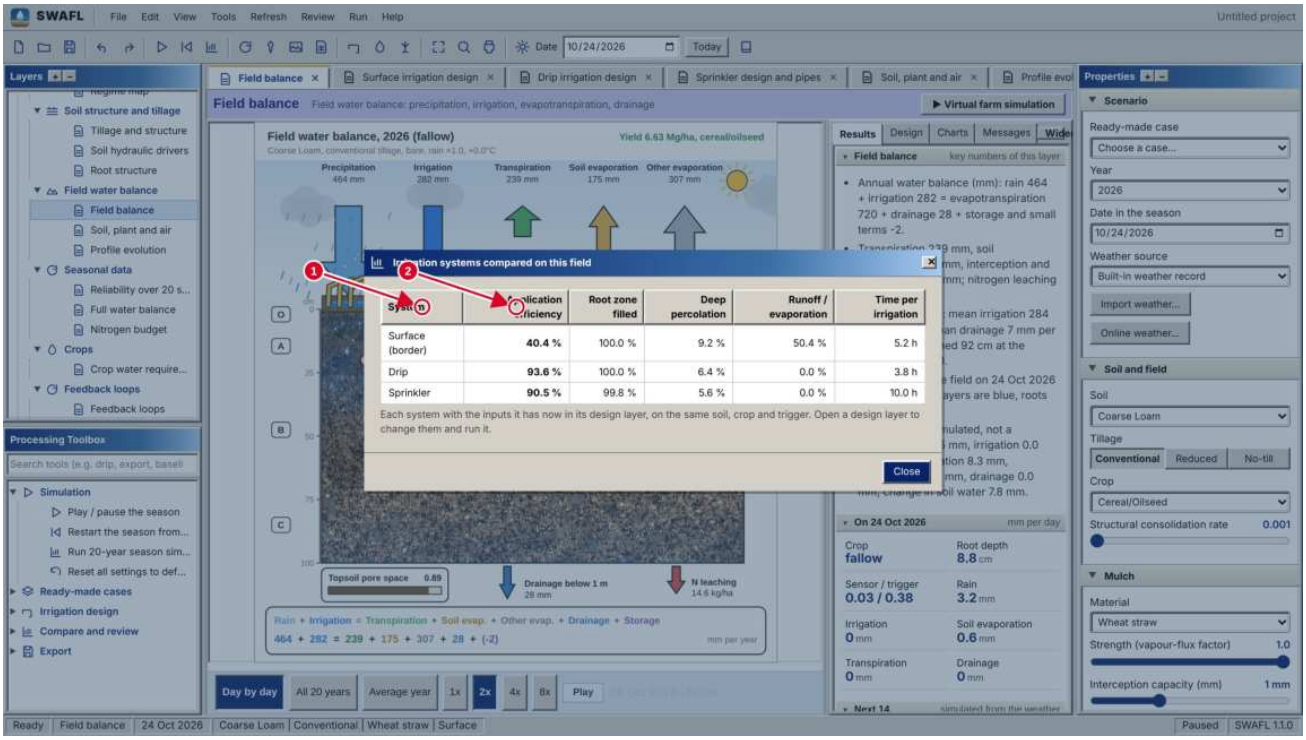



Figure 18.9: The comparison of the three systems.

# Chapter 19

# ☀ Weather data: your own file, or the weather of a place

Without weather data of your own, SWAFL runs on the twenty recorded years of its study site. Version 1.1 lets every layer run on **your weather**: a daily weather file, or the measured daily weather of any place on Earth with its 16-day forecast. The change is seen at once in every layer, the dashboard, the schedules and the crop water requirement.

## 19.1 Where to start

- **File ▸ Import weather data (CSV, Excel, DWF, DLF, DAI)…**, Tools ▸ *Import weather data…*, or the **Import weather…** button in Properties ▸ Scenario.
- **File ▸ Online weather for a location…**, Tools ▸ *Online weather for a location…*, or the **Online weather…** button.
- Properties ▸ Scenario ▸ **Weather source** switches between *Built-in weather record*, *Imported weather file* and *Online weather of a location*. The last two can be chosen only after a file has been imported or the weather of a place fetched.

## 19.2 Importing a file

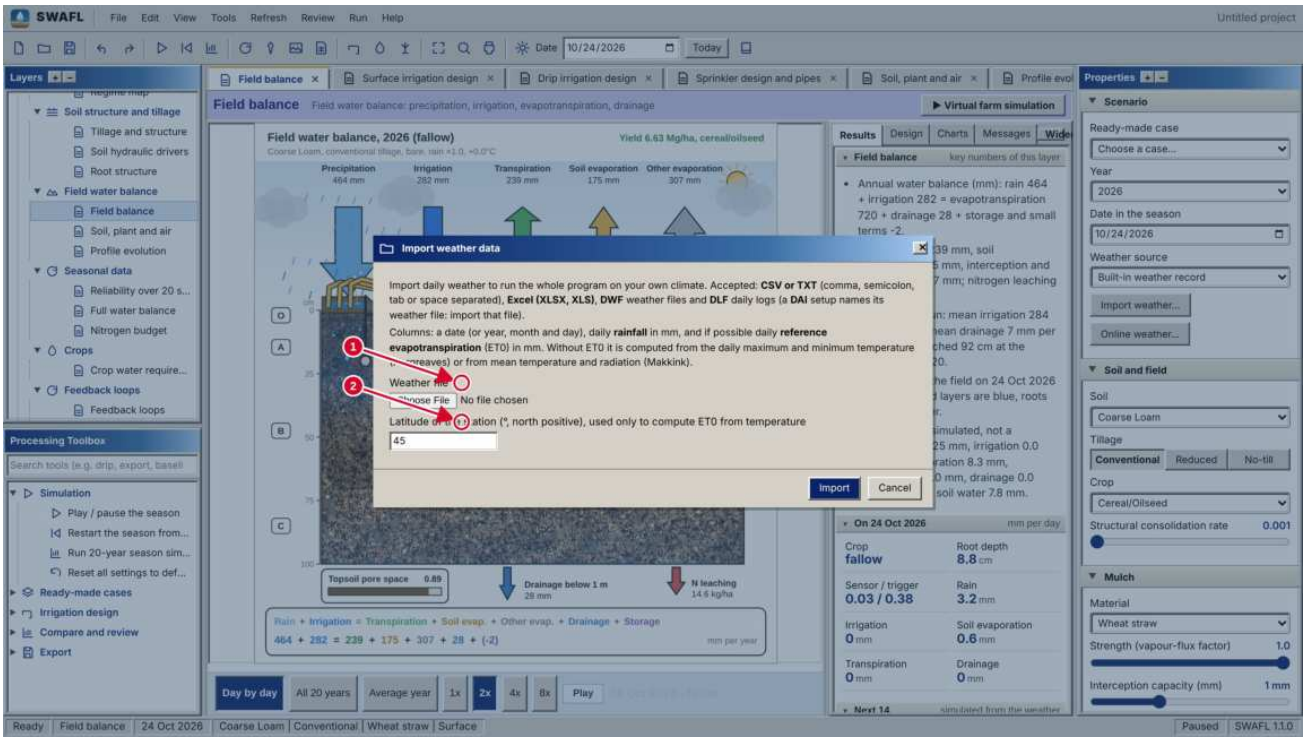



Figure 19.1: The weather import dialog.

**Accepted files.**

| Format | Reading |
|---|---|
| CSV or TXT | comma, semicolon, tab or space separated; the separator is found from the first lines |
| Excel (XLSX, XLS, ODS) | the first sheet |
| DWF (Daisy weather file) | header lines, a line of dashes, the column names, a units line, then the data; hourly values are summed to days |
| DLF (Daisy daily log) | as DWF |
| DAI (Daisy setup file) | holds no weather; the program reads the name of the weather file it uses and asks you to import that file |

**Columns.** The columns are found by their names, in any order and with or without units:

| Quantity | Names recognised |
|---|---|
| Date | `date`, `datetime`, `time`; or `year`, `month`, `day`; or `year` and `doy` (day of year) |
| Rainfall (mm) | `precip…`, `rain…`, `prcp`, `ppt`, `P`; values in inches when the header says `in` or `inch` |
| Reference evapotranspiration (mm) | `ET0`, `ETo`, `ref…evap…`, `PET`, `potential evap…` |
| Temperature (°C) | `Tmax`, `Tmin`, `Tmean`, `air temp…`, `Tavg` |
| Global radiation | `rad`, `srad`, `globrad`, `solar` (MJ $m^{-2}$ $d^{-1}$, or W $m^{-2}$ when the values exceed 60) |

**Reference evapotranspiration.** When the file has no ET0 column, it is computed. With daily maximum and minimum temperature, by Hargreaves:

$$ET_0 = 0.0023\,(T_{\text{mean}} + 17.8)\,\sqrt{T_{\text{max}} - T_{\text{min}}}\;R_a\;0.408 \quad [\text{mm}\,\text{d}^{-1}], \tag{19.1}$$

*Provenance (Eq. 19.1):* **[Literature]**: Hargreaves and Samani (1985), FAO-56 Eq. 52; $R_a$ the extraterrestrial radiation (MJ $\text{m}^{-2}$ $\text{d}^{-1}$) at the latitude typed in the dialog (FAO-56 Eq. 21).
With daily mean temperature and global radiation $R_s$, by Makkink:

$$ET_0 = 0.61\,\frac{\Delta}{\Delta+\gamma}\,\frac{R_s}{2.45} - 0.12, \qquad \Delta = \frac{4098 \cdot 0.6108\,e^{17.27T/(T+237.3)}}{(T+237.3)^2}, \quad \gamma = 0.066\ \text{kPa}\,\text{K}^{-1}, \tag{19.2}$$

*Provenance (Eq. 19.2):* **[Literature]**: Makkink (1957), as used in the Netherlands and Denmark.
Without either, the seasonal pattern and the annual demand of the built-in climate are used, and the dialog says so.

## 19.3 Putting the weather on the twenty-year grid

The program works on twenty years of 365 days. The weather is put on that grid as follows:

1. The **last twenty complete years** of the file are used (a year is complete with at least 300 days). With fewer years, the available years are repeated in turn, and the dialog says how many.
2. Leap days are dropped. A missing day of rain is filled from the same day of another year of the file; a missing day of ET0 from the mean of that day of the year.
3. The mean annual rainfall $\bar{P}$ and reference evapotranspiration $\overline{ET_0}$ are computed.

The daily model (Engine 2) then uses the daily rain $P_k$ of your weather, scaled to the rainfall scale of Properties, and the daily evaporative demand in proportion to your daily ET0:

$$P_k^{\text{model}} = P_k\,\frac{\rho}{\rho_{\text{ref}}}, \qquad e_k = \frac{ET_{0,k}}{\overline{ET_0}/365} \quad (\text{the shape of the demand through the year}), \tag{19.3}$$

*Provenance (Eq. 19.3):* **[Engineered]**: the daily sequence of your weather replaces the generated rainfall (Eqs. 5.15 to 5.16) and the seasonal demand weight.
with $\rho$ the rainfall scale of Properties and $\rho_{\text{ref}}$ the step chosen for your weather by Eq. (19.4). The annual model (Engine 1) works on the simulated steps of rainfall and warming; it is set to the steps nearest your weather,

$$\rho_{\text{ref}} = \underset{\rho\in\{0.5,\ldots,1.4\}}{\text{nearest}}\ \frac{\bar{P}}{P_{\text{rec}}}, \tag{19.4}$$

*Provenance (Eq. 19.4):* **[Engineered]**: nearest simulated step; $P_{\text{rec}}$ the mean annual rain of the built-in record (464 mm).

$$\Delta T_{\text{ref}} = \underset{\Delta T\in\{-1,\ldots,3.5\}}{\text{nearest}}\ \frac{\overline{ET_0}/E_{\text{rec}} - 1}{0.035}, \tag{19.5}$$

*Provenance (Eq. 19.5):* **[Engineered]**: the evaporative demand expressed as the warming that gives it (3.5 % per °C, Eq. 5.26); $E_{\text{rec}}$ the mean annual reference ET of the built-in record (about 870 mm).

When your weather lies outside the simulated range (for example 1,500 mm of rain, more than 1.4 times the record), the annual results use the nearest step and the dialog says so; the daily layers use your weather as it is.

The crop water requirement (Chapter 21) uses your daily rain and ET0 directly.

**The years.** With your weather, the twenty years of the grid are the calendar years of your record, so a working date inside them shows that very day of your weather (Eq. 4.2).

**After the import** a report lists the years used, the days in the file, the mean annual rain and ET0, how ET0 was obtained, the rainfall and temperature steps of the annual model, the columns read, and any notes (repeated years, filled gaps, values outside the simulated range).

## 19.4 The weather of a place

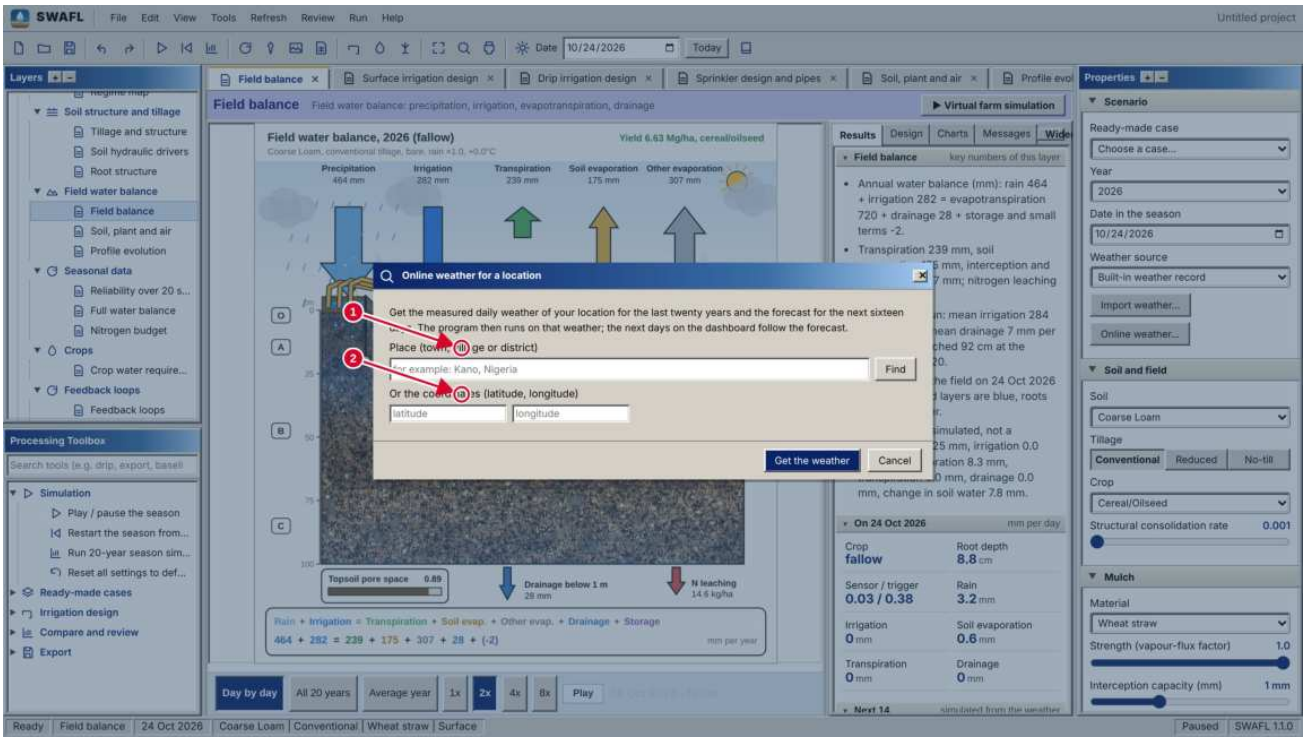



Figure 19.2: The online weather dialog.

Type the name of a town, village or district and press **Find**, then choose the place from the list; or type its latitude and longitude. **Get the weather** fetches:

- the measured daily rain, reference evapotranspiration (FAO Penman and Monteith) and maximum and minimum temperature of the place from 1 January twenty years ago to two days ago;
- the forecast for the next 16 days;
- for the rest of the current year after the forecast, the same days of the most typical year of the record (median rain).

The weather is then put on the grid as above. On the dashboard, the *Next 14 days* block says *from the weather forecast* while the days fall in the forecast.

The online weather needs the internet; everything else does not. If the place search or the weather service cannot be reached, the dialog says so and **the current weather stays in use**; nothing else changes. The service is a free public weather archive that needs no key.

## 19.5 Switching back

Choose *Built-in weather record* in Properties ▸ Scenario ▸ *Weather source*. The rainfall scale and temperature shift you had before the import come back. The imported file and the online weather stay available in the list until the program is closed.

# Chapter 20

# ⊞ The real field: satellite indices, image analysis, water sources and the irrigation system

*Tools* ▸ *Field on the map…*, or **Locate on the map** in the field sketch, opens the field on the real map. There you find the farm, demarcate the field, and the program reads the real field from open data:

- the **vegetation and water indices** NDVI, EVI, SAVI, NDRE and NDWI from the **Sentinel-2** satellite (Copernicus);
- an **image analysis (computer vision) of the satellite photo** of the field: its land cover (dense and sparse vegetation, bare soil, water, buildings and roads, shadow) and the visible-band indices VARI, GLI, ExG and NGRDI;
- **the site**: the water features around the field (OpenStreetMap), water seen in the photo, the ground elevation and slope (Copernicus DEM), and the soil texture (SoilGrids);
- **suggestions**: where to take the water from, and which irrigation system suits the field, with the reasons.

Everything is computed for the demarcated outline only, after it is drawn, and follows the outline when you change it. All of it needs the internet; without it the program says what it could not reach and uses the simulated indices of SWAFL.

> The pictures of this chapter were made without internet, on a generated test image and test answers of the open-data services, so that every path of the program could be shown. On your computer the real satellite photo and the real data of your field appear in the same places.

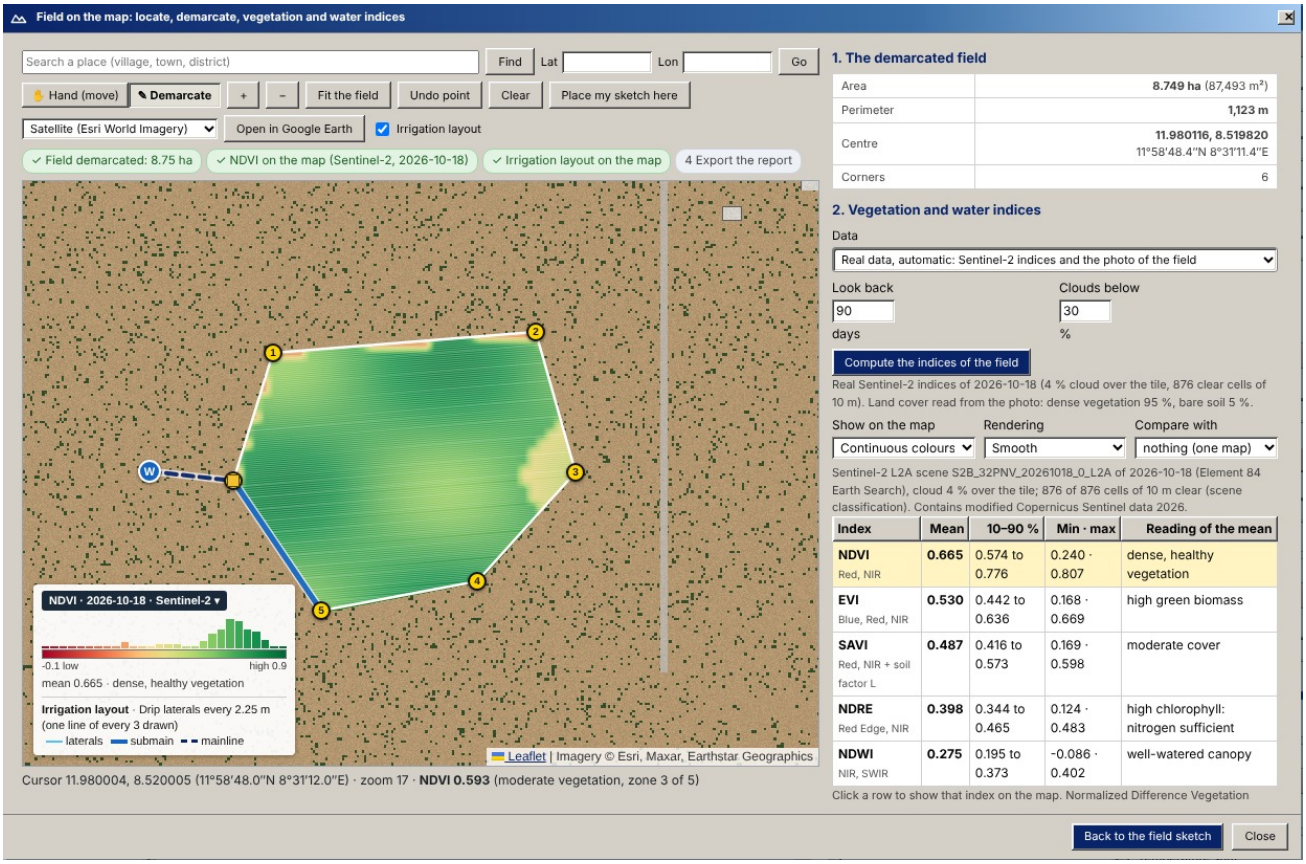



Figure 20.1: The field on the map with its real Sentinel-2 NDVI cut to the outline, the legend on the map (index, date, scale, spread of the field), the irrigation layout and the value under the cursor (test image).

## 20.1 Finding and demarcating the field

**Find the farm.** Type a village, town or district and press *Find* (place search of Open-Meteo, the same as for the online weather), or type the latitude and longitude and press *Go*. If the online weather of a place is in use, the map opens there. The background is the satellite image (Esri World Imagery, with place names), the satellite image with roads and names, or the map (OpenStreetMap). **Open in Google Earth** opens Google Earth on the centre of the field, to look at it in 3D.

**2. Demarcate the field.** In *Demarcate* mode, click the corners of the field on the image. Drag a corner to move it; right-click a corner to remove it; *Undo point* and *Clear* work as in the sketch. The *Hand* moves the map; **+** and **−** or the mouse wheel zoom; *Fit the field* shows the whole field. *Place my sketch here* puts the field of the sketch, at its true size, on the centre of the map. The panel gives the area, the perimeter and the centre, in decimal degrees and in degrees, minutes and seconds (WGS 84). On the sphere of radius $R = 6\,378\,137$ m,

$$A = \frac{R^2}{2}\left|\sum_i (\lambda_{i+1} - \lambda_i)(2 + \sin\varphi_i + \sin\varphi_{i+1})\right|, \qquad P = \sum_i 2R \arcsin\sqrt{\sin^2\tfrac{\Delta\varphi}{2} + \cos\varphi_i \cos\varphi_{i+1} \sin^2\tfrac{\Delta\lambda}{2}}, \quad (20.1)$$

*Provenance (Eq. 20.1):* **[Literature]**: spherical polygon area (Chamberlain and Duquette, 2007) and the haversine distance; $\varphi$ latitude, $\lambda$ longitude in radians.

The bar of steps above the map shows what is done (with a tick) and what comes next: the field demarcated, the index on the map, the irrigation layout on the map, the report.

## 20.2 The indices of the field

Choose the data and press **Compute the indices of the field**:

| Data | What it gives |
|---|---|
| **Real data, automatic** (the default) | The Sentinel-2 indices of the most recent clear scene, and the image analysis of the satellite photo (land cover). If Sentinel-2 cannot be read, the indices of the photo; if neither, the simulated indices. The message under the button says what was used and what could not be reached. |
| Sentinel-2 only | NDVI, EVI, SAVI, NDRE and NDWI. |
| Satellite photo only | The land cover and VARI, GLI, ExG and NGRDI. |
| Simulated by SWAFL | No internet needed (below). |

**Sentinel-2 L2A** (Copernicus open data). The catalogue of Element 84 Earth Search is searched for the scenes over the field of the last *Look back* days with less than *Clouds below* % cloud over the tile; if it cannot be reached, the catalogue of Microsoft Planetary Computer is used. The most recent scenes are tried in turn until one is clear over the field. The bands B02 (blue), B04 (red), B05 (red edge), B08 (NIR) and B11 (SWIR) are read for the field only (cloud-optimised GeoTIFF, in the UTM zone of the scene), the offset of processing baseline 4 is removed, and cloud, cloud shadow, snow and no-data pixels are left out with the scene classification (SCL). Each cell of 10 m inside the outline gets

$$\begin{aligned} \mathrm{NDVI} &= \frac{\rho_N - \rho_R}{\rho_N + \rho_R}, \qquad \mathrm{EVI} = 2.5\,\frac{\rho_N - \rho_R}{\rho_N + 6\rho_R - 7.5\rho_B + 1}, \quad \mathrm{SAVI} = \frac{(1+L)(\rho_N - \rho_R)}{\rho_N + \rho_R + L},\ L = 0.5, \\ \mathrm{NDRE} &= \frac{\rho_N - \rho_{RE}}{\rho_N + \rho_{RE}}, \quad \mathrm{NDWI} = \frac{\rho_N - \rho_{SW}}{\rho_N + \rho_{SW}}. \end{aligned} \tag{20.2}$$

*Provenance (Eq. 20.2):* **[Literature]**: Rouse et al. (1974), Huete et al. (2002), Huete (1988), Barnes et al. (2000), Gao (1996). NDWI after Gao is also called NDMI.

| Index | Bands | What it follows |
|---|---|---|
| NDVI | Red, NIR | Drought tracking, baseline crop health |
| EVI | Blue, Red, NIR | Resists saturation in thick canopies (dense crops) |
| SAVI | Red, NIR, soil factor $L$ | Removes the soil background (dry land, early crop) |
| NDRE | Red edge, NIR | Deep canopy; nitrogen management, precision farming |
| NDWI (NDMI) | NIR, SWIR | Water inside the canopy; crop water stress, fire risk |

**The satellite photo** (Esri World Imagery, about 0.3 to 1 m per pixel; the capture date varies by place). The photo of the field is put together from its tiles at the finest zoom that fits about 1700 pixels across; tiles without imagery are left out. Its pixels have only red, green and blue, so it gives the visible-band indices (shadows left out), averaged over the cells of the field:

$$\mathrm{VARI} = \frac{G - R}{G + R - B}, \qquad \mathrm{GLI} = \frac{2G - R - B}{2G + R + B}, \qquad \mathrm{ExG} = 2g - r - b\ \left(r = \tfrac{R}{R+G+B}, \ldots\right), \qquad \mathrm{NGRDI} = \frac{G - R}{G + R}. \tag{20.3}$$

*Provenance (Eq. 20.3):* **[Literature]**: Gitelson et al. (2002), Louhaichi et al. (2001), Woebbecke et al. (1995), Tucker (1979).

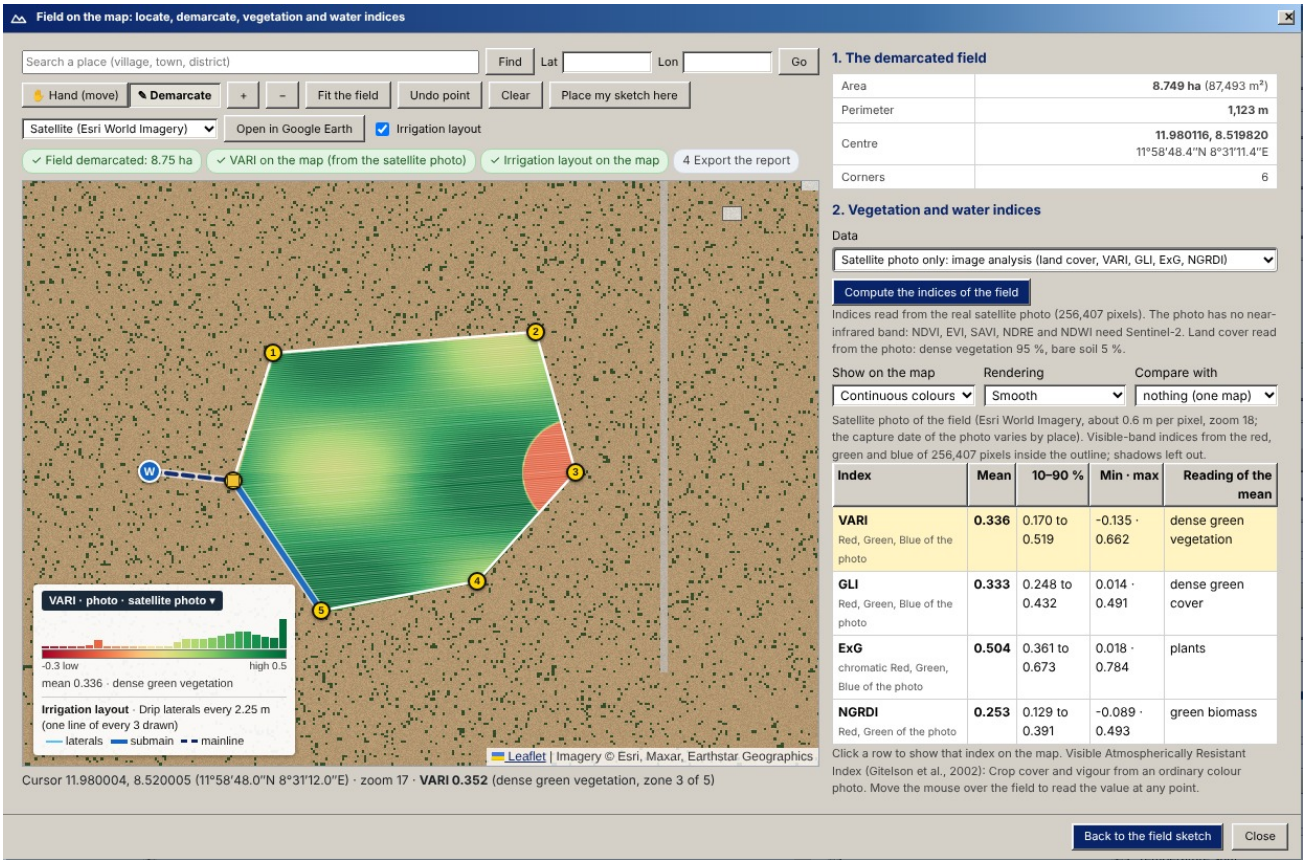



Figure 20.2: The indices read from the real satellite photo: VARI cut to the field, with the table of VARI, GLI, ExG and NGRDI (test image).

**The land cover: image analysis of the photo.** Every pixel of the photo is described by six features: its brightness $L$, saturation $S$, excess green ExG, VARI, blueness $b-(r+g)/2$ and texture (the standard deviation of the brightness over a $5\times5$ window). The pixels are grouped into seven groups by *k*-means clustering of the standardised features (k-means++ start, 14 iterations), and each group is named from its mean features:

$$\text{class} = \begin{cases} \text{shadow} & L < 0.10 \\ \text{water} & \left(b - \frac{r+g}{2} > 0.025,\ \text{ExG} < 0.03,\ L < 0.55\right) \text{ or } \left(L < 0.2,\ \text{smooth},\ \text{ExG} < 0.02\right) \\ \text{dense vegetation} & \text{ExG} > 0.085 \text{ or VARI} > 0.16 \\ \text{sparse vegetation} & \text{ExG} > 0.035 \text{ or VARI} > 0.05 \\ \text{buildings, roads} & \left(S < 0.13,\ L > 0.45\right) \text{ or } \left(\text{rough},\ S < 0.2\right) \\ \text{bare soil} & \text{otherwise} \end{cases} \tag{20.4}$$

*Provenance (Eq. 20.4):* **[Engineered]**: classical computer vision (colour indices, texture, *k*-means, rules) and a $3 \times 3$ majority filter; not a trained neural network. The shares and areas are those of the pixels inside the outline.

Choose *Land cover read from the photo* under *Show on the map* to see it; the table gives the area and share of each cover. Buildings, roads and water inside the outline are worth leaving out of the irrigated area.

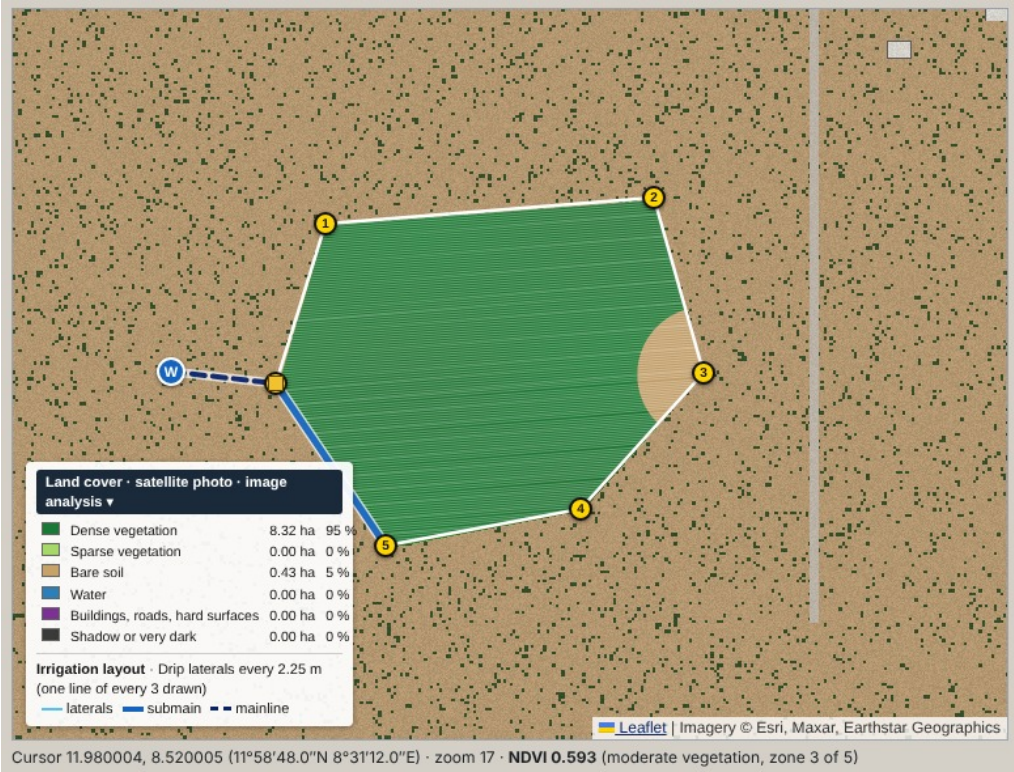



Figure 20.3: The land cover read from the photo, cut to the field, with the area of each cover in the legend (test image).

**Simulated by SWAFL** (no internet), on the working date or at the peak of the crop: each cell mixes the reflectance of the crop and of the soil by the crop cover $c$ of the daily model, $\rho = c\,\rho_{\text{veg}} + (1 - c)\,\rho_{\text{soil}}$; the soil is darker when wet, the canopy reflects less NIR and more SWIR under stress and less red edge when nitrogen is short; vigour and soil water vary in patches of 25 to 40 m tied to the ground, and the wetting falls from the inlet end to the tail with the uniformity of the design. **[Engineered]**.

The table gives, for each index, the mean, the 10 and 90 % values, the minimum and maximum over the field, and what the mean means. Click a row to show that index on the map.

**The index follows your outline.** The colours are cut exactly to the outline and laid over the real map. When you add, drag or remove a corner, the simulated indices are computed again at once; satellite and photo data are read again when you press *Compute*.

**What the map shows.**

- *Continuous colours* or *Management zones (5 classes)*: the zones split the values of the field by natural breaks (Jenks), drawn from the field smoothed over about 9 m so that each zone is a patch you can manage. The table under the indices gives, for each zone, its range, mean, area (ha) and share of the field.
- *Continuous colours, smooth interpolation* (the default) or *Pixels and classes, as measured*. With the default every map of the window (the indices, the monitoring change, trend and stability, the soil loss and gully risk, the soil water retention and the sensor maps) is drawn as one continuous colour ramp, interpolated bilinearly between the cells, with a gradient legend. The second choice draws each 10 m Sentinel-2 pixel as a square and the retention, sensor and erosion maps in their classes. The values themselves are the same in both: only the drawing changes, and the cursor readout and the pixel CSV always give the measured pixel.
- *Compare with*: a second index side by side; the two maps move and zoom together.
- The legend in the lower left corner of the map gives the index, the date and the source, the

colour scale (or the zones with their area), the spread of the field as a histogram and the mean. Click its dark title to fold it.

- Move the mouse over the field: the line under the map gives the coordinates and the value of the index at that point, its reading and its zone (and the value of the second index when two are shown).
- *Irrigation layout* draws the system of the field sketch on the map: the furrows, borders, basins, drip laterals or sprinklers (light lines; one in every few when they are very close), the submain or head ditch along the edge that holds the inlet (thick blue), the mainline from the water source **W** to the inlet (dashed), and the inlet (yellow square). It appears after *Use this field in the sketch and the designs* (or *Place my sketch here*), and it follows the outline when you move a corner. Choose the system, the spacing and the source in the sketch.

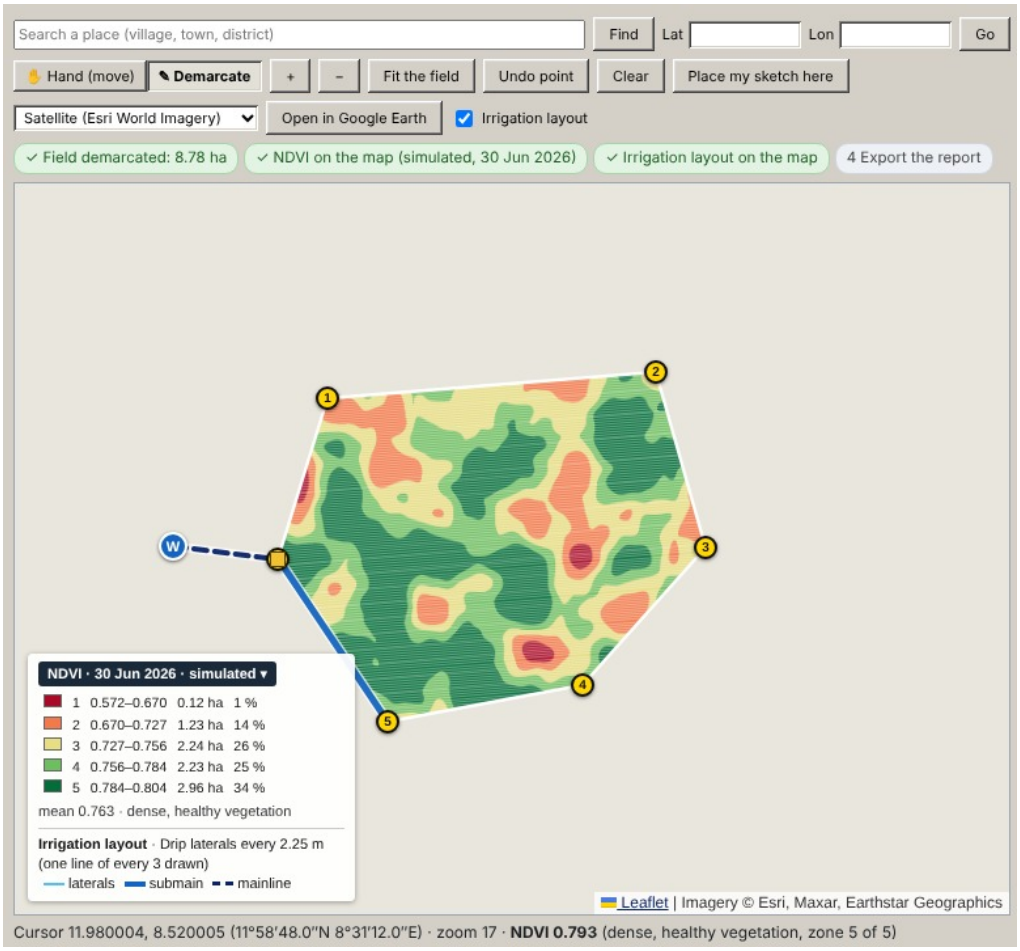



Figure 20.4: Management zones of NDVI (simulated, without internet).

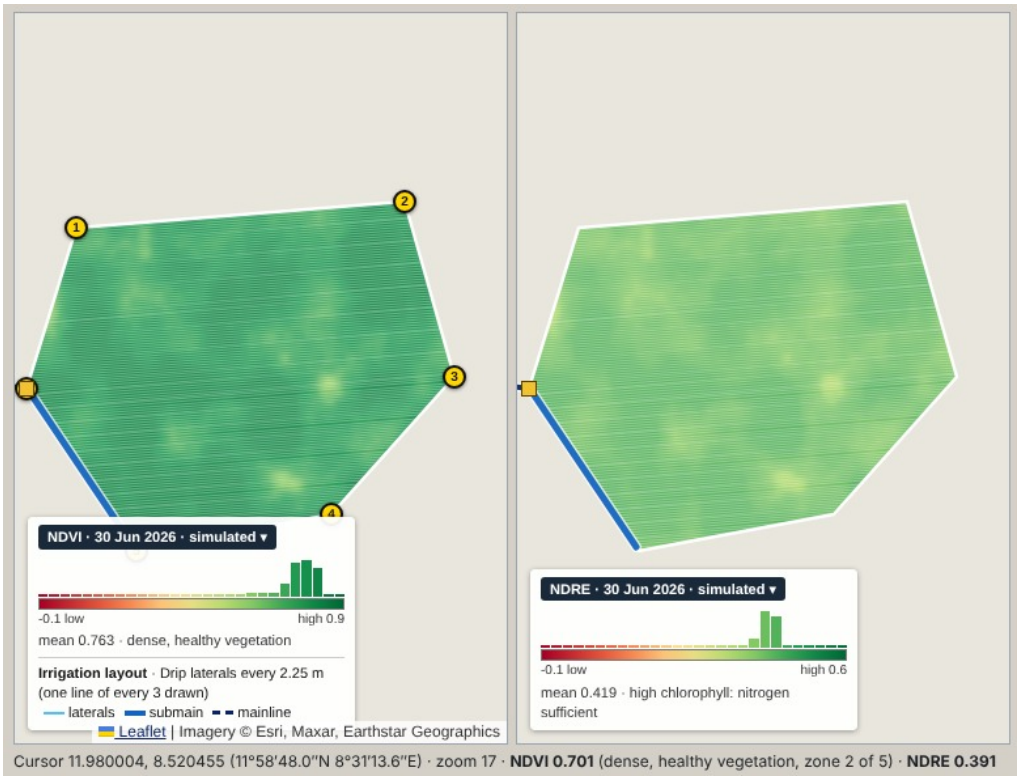



Figure 20.5: Two indices side by side (simulated, without internet).

## 20.3 The site: water sources and the irrigation system

**Analyse the site (real open data)** reads, around the demarcated field:

- the **water features** within 3 km from OpenStreetMap (Overpass API): rivers, streams, canals, ditches and drains, lakes, ponds and reservoirs, wells, boreholes, springs and water points;
- **water seen in the photo** within 1.5 km: groups of water pixels of the image analysis (Eq. 20.4) larger than about 600 m², not already mapped;
- the **ground elevation** (Copernicus DEM GLO-90, through Open-Meteo) on a grid over the field, at its corners and at the nearest point of each source;
- the **soil** at the centre of the field from ISRIC SoilGrids (sand, silt, clay and organic carbon, 0 to 60 cm), with its USDA texture class. **Use the nearest SWAFL soil** sets the soil of the project to the soil of the program closest in sand and clay.

The slope of the field is that of the plane fitted to the elevations by least squares,

$$z = a + b\,x + c\,y, \qquad s = 100\sqrt{b^2 + c^2}\ \ [\%], \tag{20.5}$$

*Provenance (Eq. 20.5):* **[Literature]**: least-squares plane; $x$, $y$ in metres east and north of the centre. The direction the ground falls to is that of $-(b, c)$.

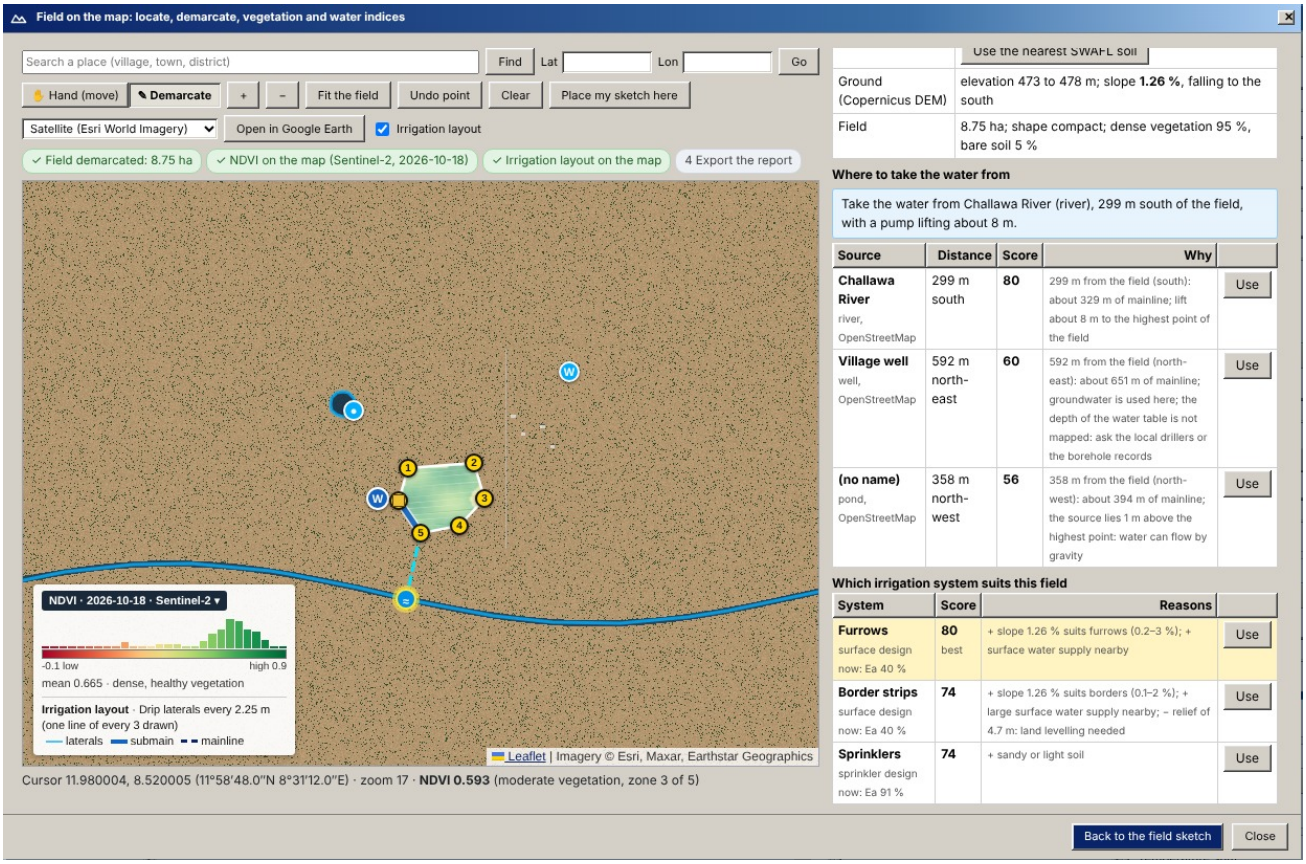



Figure 20.6: The site analysed: the water sources on the map (the best one joined to the field), the soil, the slope, the advice on the water and the ranking of the systems (test data).

**Where to take the water from.** Each source gets a score

$$\text{score} = B_{\text{type}} - 30\,[\text{seasonal}] - \min\!\left(45,\ d/60\right) - \min\!\left(20,\ 0.8\,\Delta z\right), \tag{20.6}$$

*Provenance (Eq. 20.6):* **[Engineered]**: $B$ = 92 river, 90 canal, 86 reservoir, 82 lake, 74 borehole, 70 well, 62 pond, 60 spring, 58 stream, 50 water seen in the photo, 42 ditch, 30 drain; $d$ the distance from the field edge (m); $\Delta z$ the lift from the source to the highest point of the field (m), not counted for wells, whose water table depth is not mapped.

The advice names the best surface source within 1.5 km (with its distance, direction and lift, or *by gravity*); without one, a borehole at the edge of the field when wells are mapped nearby, or a hydrogeological survey first. **Use** puts the source into the field sketch, at its place, with its kind and lift, so the mainline length and the pump head go into the designs.

**Which irrigation system suits this field.** Each system starts from a base score and gains or loses points for the conditions of the field, after the FAO and USDA guidance for choosing an irrigation system:

| System | Gains | Loses |
|---|---|---|
| Basins | level ground ($s \leq 0.3$ %), clay, large surface supply, rice, orchards | slope over 1 %, sand over 65 %, groundwater only, irregular shape |
| Border strips | slope 0.1 to 2 %, loam to clay loam, close-growing crops, surface supply | slope over 2 %, sand, groundwater only, irregular shape, relief to level |
| Furrows | slope 0.2 to 3 %, row crops, surface supply | slope over 3 %, sand, close-growing crops, irregular shape |

| System | Gains | Loses |
|---|---|---|
| Sprinklers | rolling ground (2 to 15 %), light soil, close-growing crops, irregular shape, borehole supply | slope over 20 %, heavy clay, tall trees, level fields with plenty of surface water |
| Drip | sandy soil, scarce or pumped water, slopes, row crops and orchards | close-growing crops, very large fields, hard surfaces in the outline |

The table lists each system with its score, the reasons, and the application efficiency of the SWAFL design of that system as it stands. **Use** sets the system (and the slope of the site) in the field sketch; then *Turn the sketch into the design* and run it.

## 20.4 The maps of this field, as layers

Every map computed for the demarcated field is kept as a layer in the panel **Maps of this field** at the top right of the map: the five indices and the management zones of the index shown, the land cover read from the photo, the four maps of the season (monitoring zones, change, trend, stability), the irrigation depth and date of each zone, the ephemeral gullies and the soil loss, and, on top, the water sources and the irrigation layout. Click a layer to show it again: nothing is computed again. The layers stay when the window is closed and opened again (with the results of the site), and go only when the outline of the field changes. Click the dark title to fold the panel.

In the list of water sources of the site, the map moves to a source only when its **Show** button is pressed (the mouse passing over the list leaves the map where it is); **Use** sets the source in the sketch and the design.

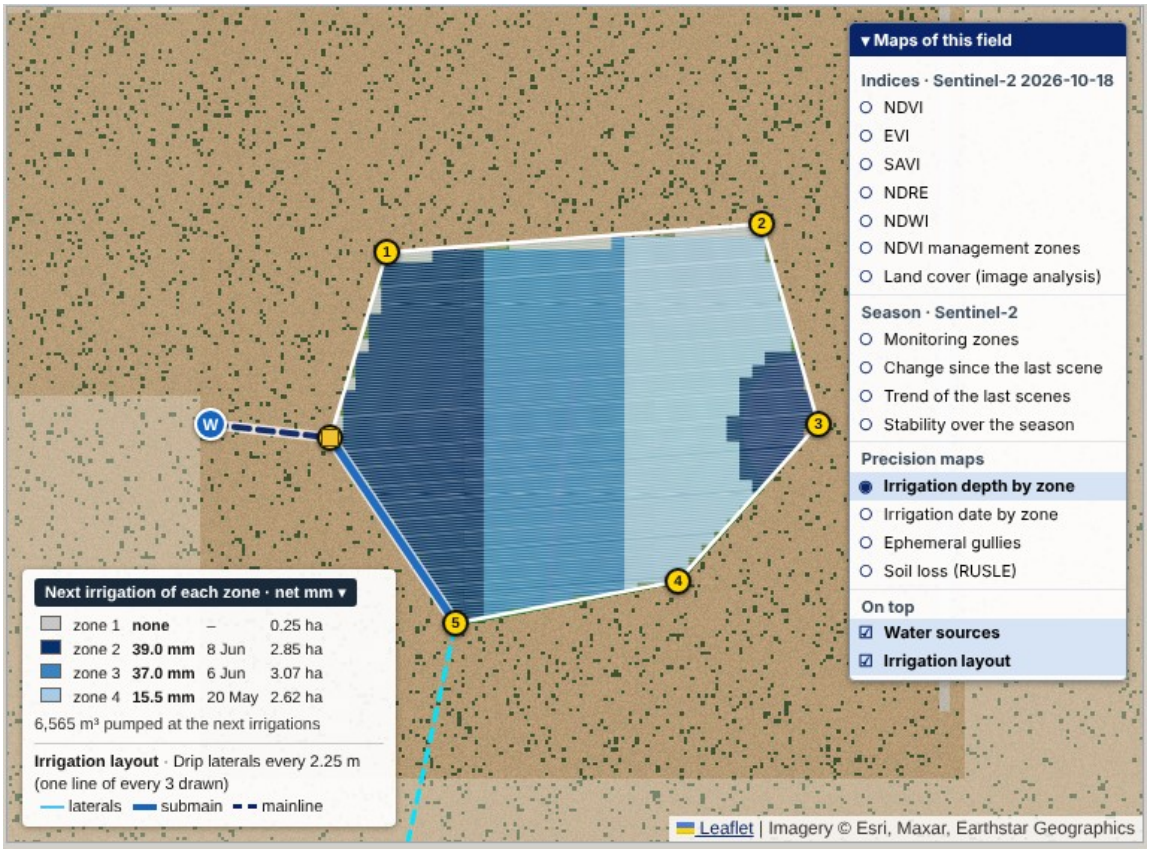



Figure 20.7: The maps of this field as layers, after the indices, the site, the season, the zone-by-zone irrigation and the erosion were computed (test data).

## 20.5 Full screen, and the colours of the maps (Symbology)

The window of the field opens in full screen, like the Daisy window: the map fills the window, and the tools (*Map tools*) scroll beside it. The button with the corners on the title bar, or a double click on the title bar, returns to a smaller window; **A−** and **A+** change the size of the text.

The panel **Symbology** (top left of the map; *Opened maps* is at the top right) sets the colours of the map in view, as the symbology tab of QGIS does. Both panels can be **dragged anywhere on the map by their title** (a double click on the title puts a panel back in its corner); their place is kept for the next time.

- **Coloured map** or **Real colour (photo)**: the second hides the coloured maps so that the satellite photo itself is seen under the outline; **Opacity** sets how much of the photo shows through a coloured map.
- **Every map starts in its original SWAFL colours** (marked with a star): the vegetation indices and their zones from red (little vegetation) through yellow to green (dense canopy), the water index from brown to teal, the elevation from blue to red, the wetness index from yellow to blue, soil water, retention, suction and days to the trigger in classes from red to cream to blue, the change of the season from red through white to green, the zones from red to dark green, the irrigation zones in the Spectral colours, and erosion from green (low risk) to red. **Original colours of this map** returns to them.
- **Colour ramp**: a scrollable list of ramps, each drawn as a bar. At the top, **suggested for this map**: the styles that maps of the same kind use in the literature and in GIS (for example red-yellow-green, the NDVI of Sentinel Hub and greens for vegetation; blues, brown-blue-green and the NDMI of Sentinel Hub for water; hypsometric tints and greys for elevation; yellow-green-blue for wetness and field capacity; red-yellow-blue for available water and suction; red-yellow-green for erosion). Then the original colours of every map, and the QGIS library: remote-sensing ramps, diverging (RdYlGn, Spectral, RdYlBu, RdBu, BrBG, PuOr), sequential (YlGn, YlGnBu, Greens, Blues, YlOrBr, YlOrRd, Reds, Greys) and perceptual (Viridis, Magma, Inferno, Plasma, Cividis, Turbo). **Invert the ramp** turns it round.
- For the **land cover** read from the photo, the class colours of the land cover products: SWAFL natural colours, ESA WorldCover 10 m, Google Dynamic World, CORINE Land Cover, USGS NLCD, or **Real colour** (each class in the mean colour of its pixels in the photo). A click on a colour changes that class alone.
- **Ctrl+Z** undoes the last change of the colours or of the outline of the field (a corner added, moved or removed, Clear), and **Ctrl+Y** brings it back.

The legend on the map, the tables of the zones, the 3D view and the exported report take the same colours. The choices are kept on this computer for the next time.

**Opened maps** groups the maps of the field (indices, season, precision maps, terrain and flow, sensors and satellite, soil retention, on top). A click on the name of a group folds or unfolds it; **Expand all** and **Collapse all** do it for every group.

**Corners in any order.** A new corner goes at the end of the outline, unless that would make the outline cross itself: it is then put between the two corners where it fits without a crossing, and the program says which corner it became.

## 20.6 The analysis of the indices, step by step

**Compute the indices of the field** shows its work as numbered steps that fold and unfold (each step with a one-line summary and a tick, and a button to the next step; *Walk through from step 1*, *Open all* and *Fold all* above them):

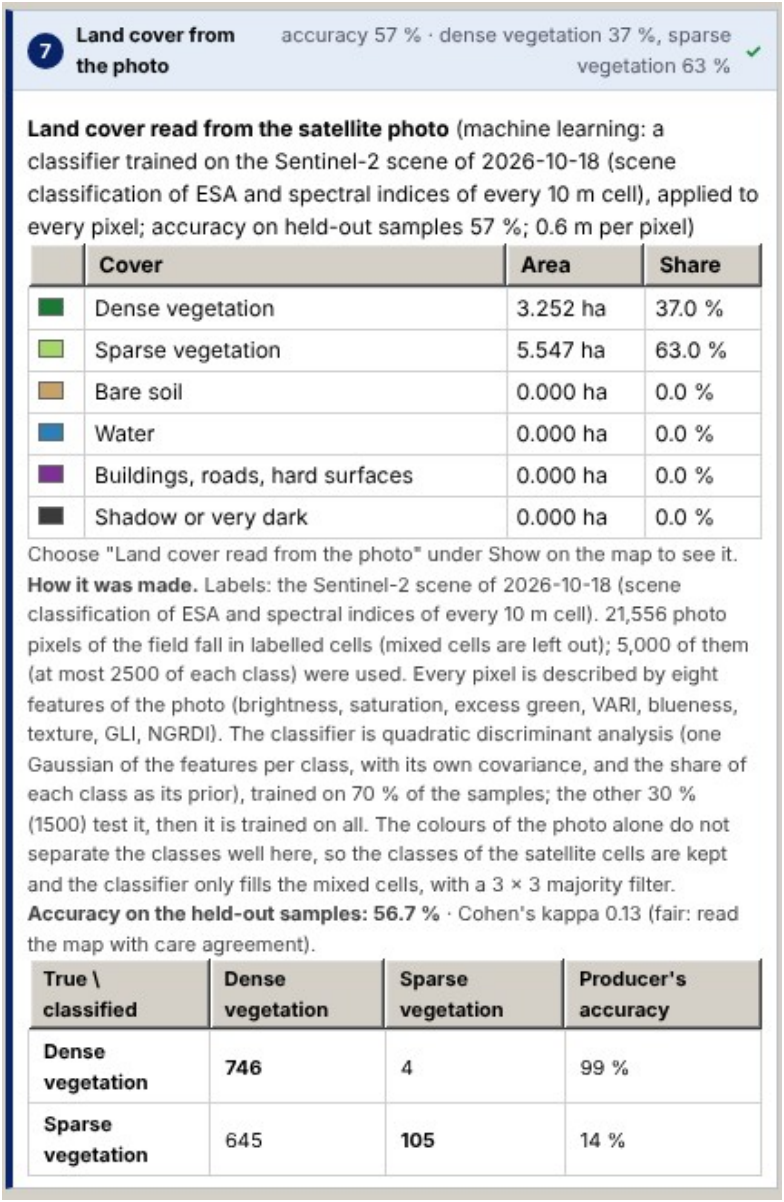
1. **Scenes found**: the Sentinel-2 Level-2A catalogue searched over the outline (the dates and clouds of the scenes, the one used, and those tried and found cloudy over the field).
2. **Bands read** (every cell of the map is one real 10 m pixel of the scene: the cells are laid on the pixel grid of the scene, none read twice and none skipped; *Download the value of every*

*pixel (CSV)* gives each pixel's centre, class, the surface reflectance of every band and its indices, and *Check this scene in the Copernicus Browser of ESA* opens the same date and place with the NDVI of ESA, so every number can be checked against the source; the map shows the pixels as measured, and *Smoothed* is offered only as an option): B02 blue, B03 green, B04 red, B05 red edge, B08 near infrared, B11 and B12 shortwave infrared and the scene classification, with their wavelength, pixel size and what each tells.

3. **Clouds and shadows removed**: the classes of the scene classification over the field (vegetation, bare soil, water, cloud, shadow, cirrus, snow) in the colours of ESA, and how many cells are used.
4. **Spectral signature of the field**: the reflectance band by band (mean of the field, 10 to 90 % of the cells, and the greenest and least green third), with a reading: the green peak of chlorophyll, the red edge (near infrared against red), and the absorption of water in the shortwave infrared.
5. **Indices of the field, one at a time**: NDVI opens first. **Next index** opens the next one (EVI, SAVI, NDRE, NDWI, then, from the green and the second shortwave band, GNDVI, MSAVI2, NBR, the bare soil index BSI and the chlorophyll index CIre), with a line on what it shows; *Open all* opens them all. Only the indices you have opened are listed in *Opened maps*. The 20 m bands (red edge, shortwave infrared) are resampled to the 10 m cells by bilinear interpolation, as SNAP and GDAL do.
6. **Management zones** of the index shown.
7. **Land cover from the photo, by machine learning**: a classifier is trained on the field itself. The labels are real: the scene classification of ESA and the spectral indices of the Sentinel-2 cells of the field (NDVI for dense and sparse vegetation, MNDWI for water, NDBI for built surfaces; mixed cells are left out), or, when no scene is read, the ESA WorldCover 2021 map. Every photo pixel is described by eight features (brightness, saturation, excess green, VARI, blueness, texture, GLI, NGRDI); the classifier is quadratic discriminant analysis, trained on 70 % of the samples and tested on the other 30 %: the step shows the accuracy, Cohen's kappa and the confusion matrix. When the classifier separates the classes well (kappa 0.4 or more) it draws every pixel at the detail of the photo; otherwise the classes of the satellite cells are kept and the classifier fills only the mixed cells. Without any real labels the photo is grouped by k-means and named by colour rules, and the step says the map is rough.

**One step at a time.** *Compute the indices of the field* reads only the satellite and the photo: the terrain, the watershed, the soil retention and the sensor sites are made only when their own buttons are pressed (or when *Do everything at once* is ticked). In *Opened maps* each group says where its maps come from: *Terrain and flow · Copernicus elevation* (measured), *Soil retention · estimated from SoilGrids* (field capacity, wilting point, available water and the trigger water content, from the measured texture by a pedotransfer function), and *SWAFL model · not measured* (the matric potential now and the days to the trigger, from the water balance of the model; folded at first, and marked on the map). When SoilGrids cannot be reached, no retention map is drawn unless you type your own soil samples.

**Only real data.** In the modes that read the satellite, nothing simulated is drawn when the satellite or the photo cannot be read: the program says what could not be reached. The estimate of the model is shown only when *Simulated by SWAFL* is chosen under Data.



**7 Land cover from the photo** accuracy 57 % · dense vegetation 37 %, sparse vegetation 63 %

**Land cover read from the satellite photo** (machine learning: a classifier trained on the Sentinel-2 scene of 2026-10-18 (scene classification of ESA and spectral indices of every 10 m cell), applied to every pixel; accuracy on held-out samples 57 %; 0.6 m per pixel)

| | Cover | Area | Share |
|---|---|---|---|
| | Dense vegetation | 3.252 ha | 37.0 % |
| | Sparse vegetation | 5.547 ha | 63.0 % |
| | Bare soil | 0.000 ha | 0.0 % |
| | Water | 0.000 ha | 0.0 % |
| | Buildings, roads, hard surfaces | 0.000 ha | 0.0 % |
| | Shadow or very dark | 0.000 ha | 0.0 % |

Choose "Land cover read from the photo" under Show on the map to see it.

**How it was made.** Labels: the Sentinel-2 scene of 2026-10-18 (scene classification of ESA and spectral indices of every 10 m cell). 21,556 photo pixels of the field fall in labelled cells (mixed cells are left out); 5,000 of them (at most 2500 of each class) were used. Every pixel is described by eight features of the photo (brightness, saturation, excess green, VARI, blueness, texture, GLI, NGRDI). The classifier is quadratic discriminant analysis (one Gaussian of the features per class, with its own covariance, and the share of each class as its prior), trained on 70 % of the samples; the other 30 % (1500) test it, then it is trained on all. The colours of the photo alone do not separate the classes well here, so the classes of the satellite cells are kept and the classifier only fills the mixed cells, with a 3 × 3 majority filter.

**Accuracy on the held-out samples: 56.7 %** · Cohen's kappa 0.13 (fair: read the map with care agreement).

| True \ classified | Dense vegetation | Sparse vegetation | Producer's accuracy |
|---|---|---|---|
| **Dense vegetation** | **746** | 4 | 99 % |
| **Sparse vegetation** | 645 | **105** | 14 % |

Figure 20.8: Step 7: the land cover classified by machine learning, with how it was trained and its accuracy (test data).

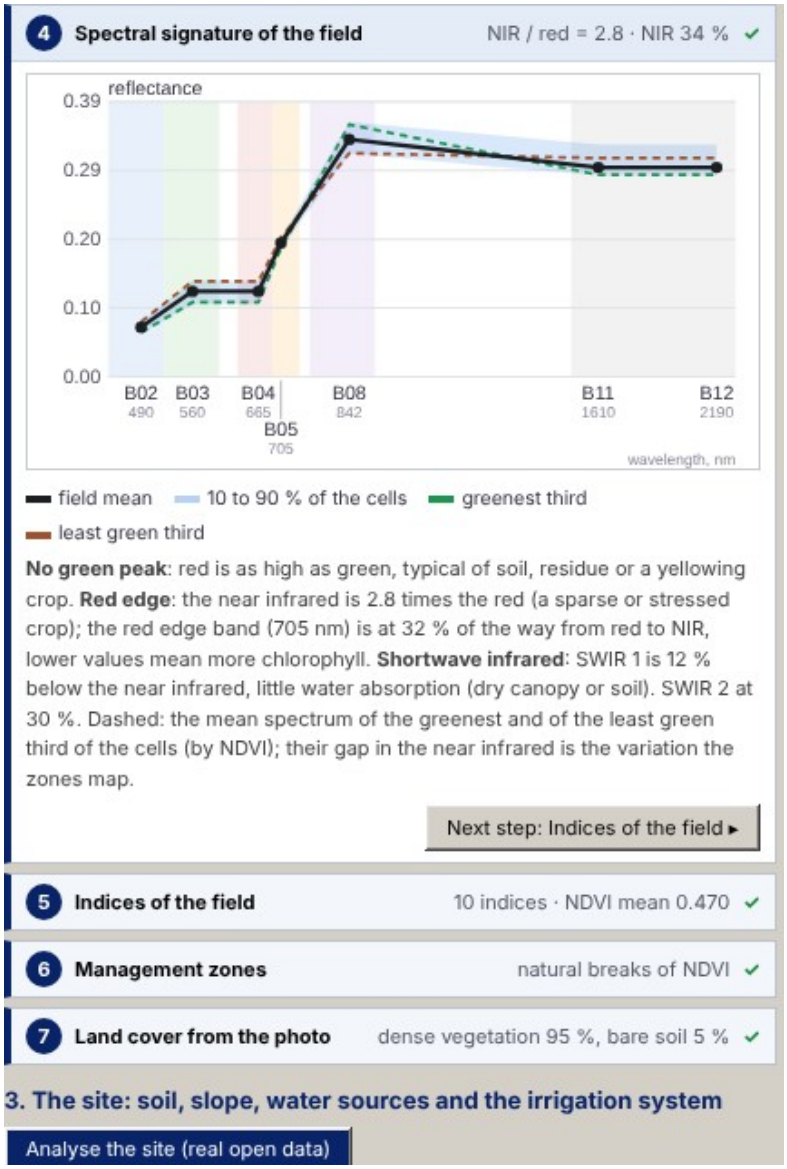



Figure 20.9: The analysis step by step: step 4, the spectral signature of the field with its reading (test data).

## 20.7 The soil of the field, from SoilGrids

The soil of the site and of the retention maps comes from SoilGrids 2.0 (ISRIC, 250 m). The point service is tried three times (it is often busy); when it does not answer, the map service of SoilGrids is read instead. Without soil samples the 250 m maps of sand and clay (mean of 0 to 60 cm) are read around the field and interpolated over its cells, so the retention maps show the real variation of the soil; your own soil samples, typed or pasted as *latitude, longitude, sand %, clay %, organic matter %*, replace them by kriging. The indicative trend on the wetness index of the terrain is off unless you tick it. In the desktop app these requests go through the app itself.

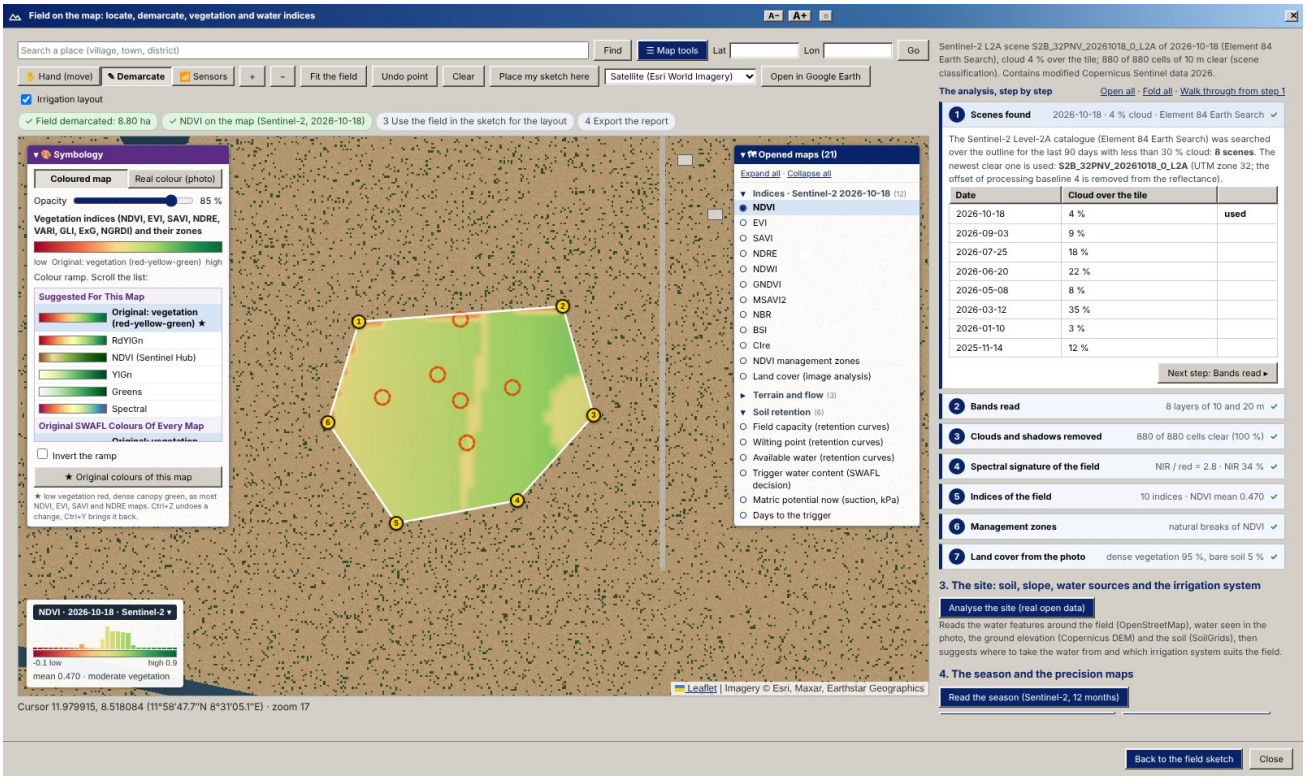



Figure 20.10: The field in full screen with the Symbology panel open: the ramps of the QGIS library and the remote-sensing standard of the map in view (test data).

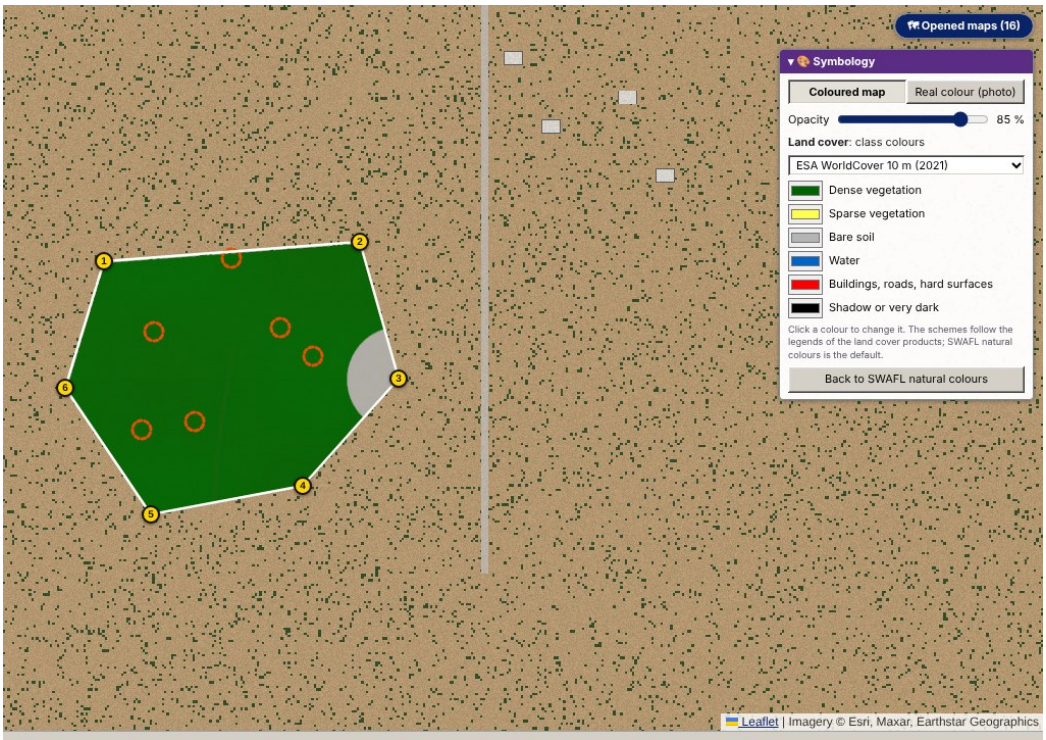



Figure 20.11: The land cover of the field in the class colours of ESA WorldCover (test data).

## 20.8 The season of the field, and the precision maps

Section 4 of the panel reads the season of the real field and makes the precision maps from it and from the SWAFL simulation of the farm. Every map is a choice of *Show on the map* and a layer of *Maps of this field*, has its legend on the map, gives its values under the cursor, and goes into the report (a page for the season and the irrigation, a page for the erosion).

### 20.8.1 The season seen by Sentinel-2

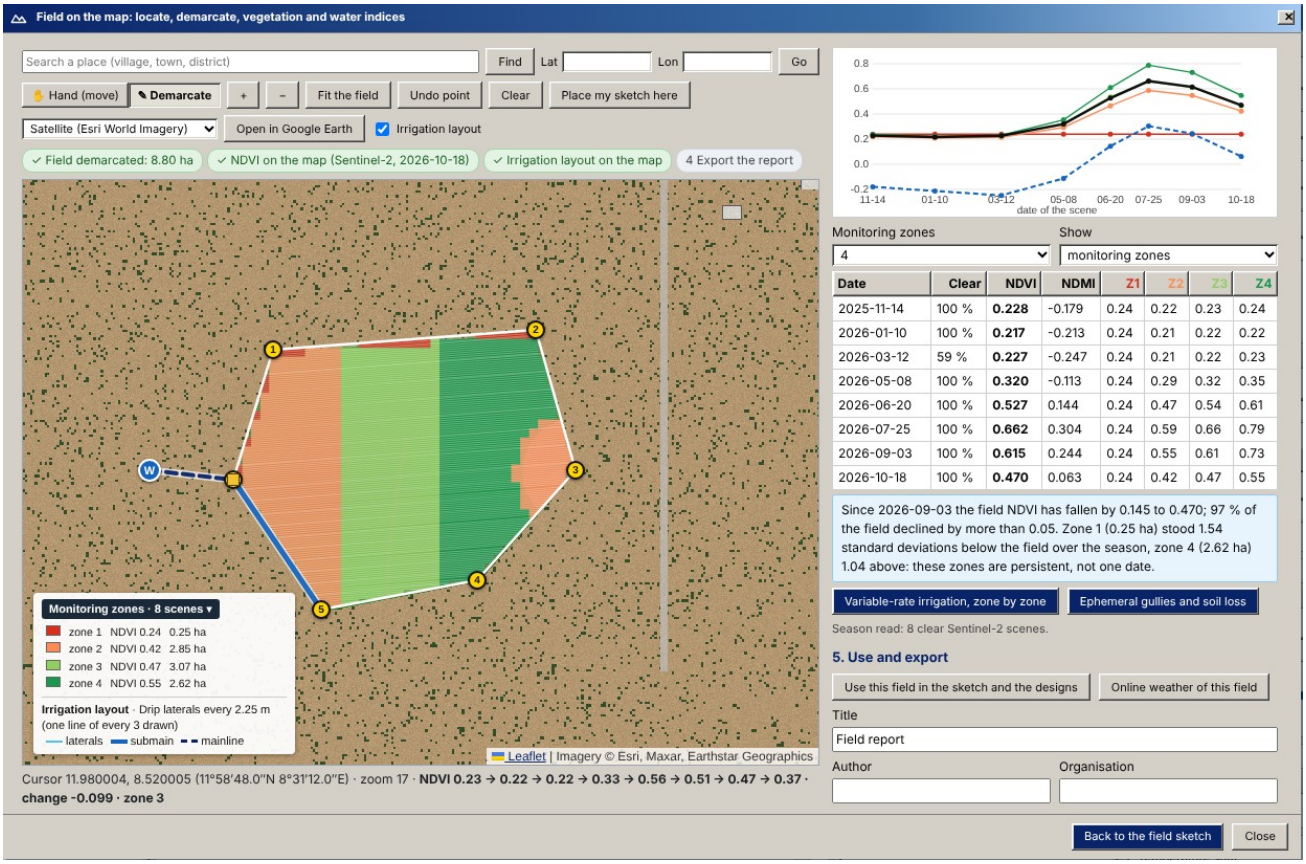
**Read the season (Sentinel-2, 12 months)** searches the Sentinel-2 L2A scenes of the last twelve months over the field (clouds below 60 % over the tile), keeps the least cloudy scene of each month, and reads its red, NIR and SWIR bands and its scene classification cell by cell (10 m), leaving out cloud, shadow and no-data cells. A scene with less than 30 % of the field clear is dropped. For every date the panel gives the share of the field that is clear, the NDVI and NDMI of the field and the NDVI of each monitoring zone, and a chart shows the curves.

From the series every cell gets:

$$z_{i,t} = \frac{\mathrm{NDVI}_{i,t} - \overline{\mathrm{NDVI}}_{t}}{\sigma_t}, \qquad \text{stability}_i = \frac{1}{n_i}\sum_{t} z_{i,t}, \qquad \Delta_i = \mathrm{NDVI}_{i,t_n} - \mathrm{NDVI}_{i,t_{n-1}}, \tag{20.7}$$

*Provenance (Eq. 20.7):* **[Literature]**: the standardised index of each date against the field, as in yield-stability and management-zone delineation from image series.

and the trend of the last three scenes (least squares, per 30 days). The **monitoring zones** (3 to 6) group the cells whose standardised series behaved alike over the season (*k*-means on the series, ordered from the weakest to the strongest), so a zone is a part of the field that was weaker or stronger on every date, not on one. The maps: *monitoring zones*, *change since the previous scene*, *trend of the last scenes* and *stability over the season*. The text says by how much the field changed since the previous scene, the share of the field that declined, and how far the weakest and strongest zones stood from the field over the season.




| Date | Clear | NDVI | NDMI | Z1 | Z2 | Z3 | Z4 |
|---|---|---|---|---|---|---|---|
| 2025-11-14 | 100 % | **0.228** | -0.179 | 0.24 | 0.22 | 0.23 | 0.24 |
| 2026-01-10 | 100 % | **0.217** | -0.213 | 0.24 | 0.21 | 0.22 | 0.22 |
| 2026-03-12 | 59 % | **0.227** | -0.247 | 0.24 | 0.21 | 0.22 | 0.23 |
| 2026-05-08 | 100 % | **0.320** | -0.113 | 0.24 | 0.29 | 0.32 | 0.35 |
| 2026-06-20 | 100 % | **0.527** | 0.144 | 0.24 | 0.47 | 0.54 | 0.61 |
| 2026-07-25 | 100 % | **0.662** | 0.304 | 0.24 | 0.59 | 0.66 | 0.79 |
| 2026-09-03 | 100 % | **0.615** | 0.244 | 0.24 | 0.55 | 0.61 | 0.73 |
| 2026-10-18 | 100 % | **0.470** | 0.063 | 0.24 | 0.42 | 0.47 | 0.55 |

Figure 20.12: The season read from Sentinel-2: the monitoring zones on the map, the curves of the field and of each zone, the table of the dates (test data).

### 20.8.2 Variable-rate irrigation, simulated zone by zone

**Variable-rate irrigation, zone by zone** runs each zone through the FAO-56 daily root-zone balance with:

- **its soil**: SoilGrids at the centre of the zone (sand and clay), the nearest soil of SWAFL and its available water $\mathrm{AWC}_z$;
- **its observed cover**: the crop development of the SWAFL daily model, scaled by the zone's cover from its last NDVI against the field, $f_{c,z}(t) = \min\!\left(1,\ c(t)\, f_c(\mathrm{NDVI}_z)/\overline{f_c}\right)$ with $f_c = (\mathrm{NDVI} - 0.15)/0.7$, and the crop coefficient $K_c = K_{c,ini} + (K_{c,mid} - K_{c,ini})\, K_d$, $K_d = \min(1,\ 1.5 f_c,\ f_c^{0.5})$ (Allen and Pereira, 2009), with the coefficients of the crop in Properties;
- **the farm of the simulation**: the roots of every day, the trigger, the depletion fraction and the past irrigations of the irrigation plan of Chapter 18 (Eq. 18.4), the season of the decision of Chapter 18, and the application efficiency of the design in use;
- **the weather in use**: the daily reference ET and rain (the online weather of the farm with its forecast, a weather file, or the recorded weather).

The balance starts 45 days before the decision with the water of the SWAFL model on that day:

$$D_{r,z}(t+1) = D_{r,z}(t) + K_{c,z}(t)\, K_s\, ET_0(t) - (P(t) - RO(t)) - I(t), \qquad I_z(t) = D_{r,z}(t) \text{ when } f_{c,z} > 0.2 \text{ and } D_{r,z} \geq f_T\, \mathrm{AWC}_z\, z_r(t), \tag{20.8}$$

*Provenance (Eq. 20.8):* **[Literature]** FAO-56 root-zone balance (Allen et al., 1998) with the trigger and irrigation rule of the SWAFL daily model; $RO = 0.3(P - 35)$ above 35 mm as in the crop balance of Chapter 21.

**Cell by cell.** The same balance (Eq. 20.8) is also run for every 10 m cell of the field, with the NDVI of that cell (the last clear Sentinel-2 scene, or the computed indices) and its available

water interpolated between the zones (inverse squared distance to the zone centres). The two maps, *depth of the next irrigation* and *date of the next irrigation*, are drawn from these cells as continuous colours with smooth interpolation: the depth on the chosen palette, stretched over the 2nd to 98th percentile of the field so that differences stand out, and the date from red (soon) to blue (late). Cells that need no water within the 60 days (bare soil, or enough water) are drawn in pale sand. The legend gives the range in mm or in dates, the share of the field that needs no water, and the zone table below it. The cursor reads the date, net depth, water left, NDVI and available water of the cell under it, and *Cells CSV (10 m)* saves all of them.

**When the crop of the project is not in season** on the working date (for example after harvest, in the dry season), the model crop would not use water and no zone would ever be irrigated. The balance then uses the vegetation Sentinel-2 sees in the field now: the cover $f_c(\mathrm{NDVI})$ of each cell is kept over the 60 days and the roots are at the full depth of the crop. The legend and the advice say which of the two is in use.

So every zone gets its own **dates and depths**: the table gives its soil and AWC, its cover now (NDVI, NDMI, $K_c$), the water left in its root zone, its next irrigation (date, net and gross mm) and the irrigations of the next 60 days. The maps show the depth of the next irrigation of each zone or its date. The text names the zone that needs water first and the one that needs it last, and compares the volume with the single depth of the SWAFL model of the whole field. The depths can be changed with **−** and **+**; **Prescription GeoJSON** (every cell with its zone, date, net and gross rate, AWC and NDVI) and **Zones and dates CSV** (the zones and every irrigation date of each zone) go to the controller or a GIS. Without the season, the zones are classes of the index of one date.

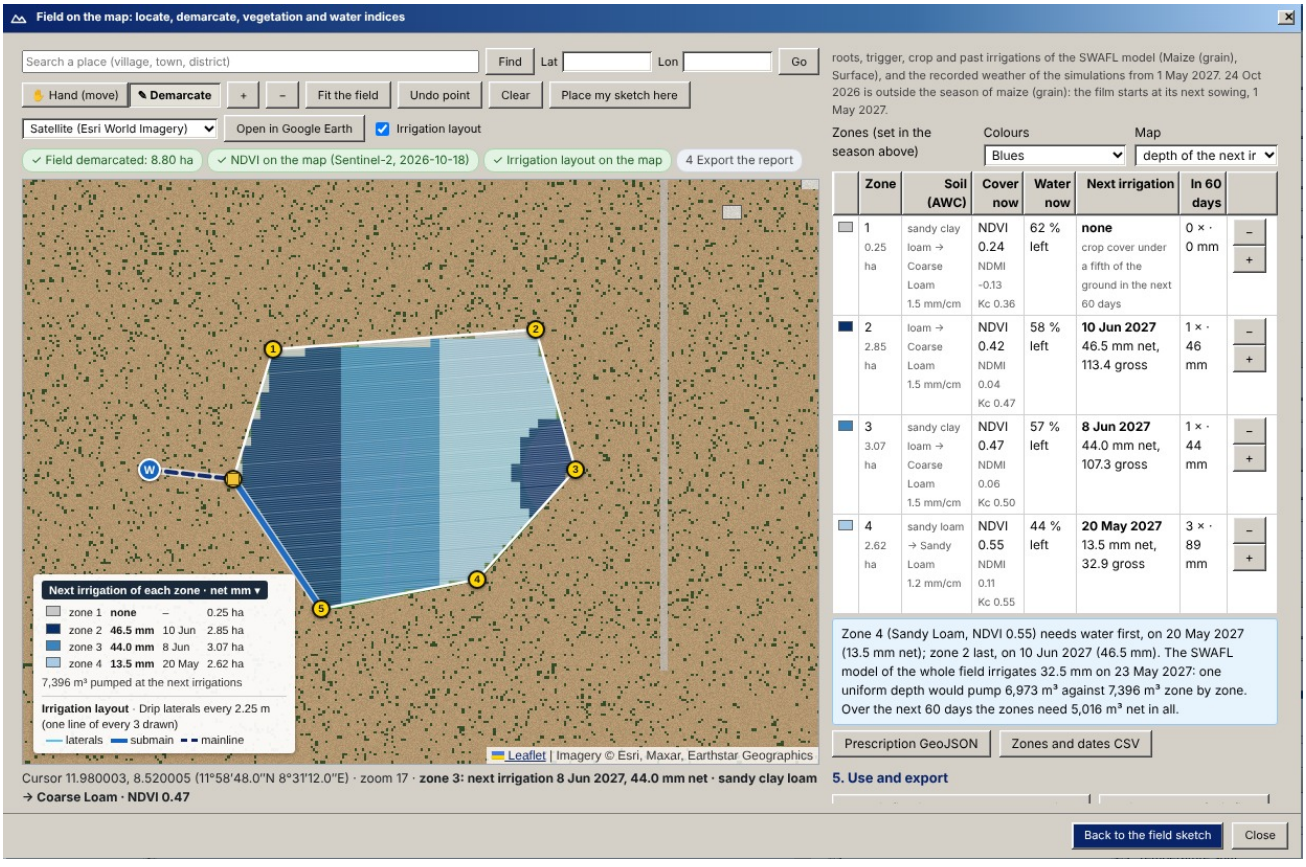



Figure 20.13: Variable-rate irrigation simulated zone by zone: the depth of the next irrigation, and the table of soils, cover, water and dates (test data).

### 20.8.3 Ephemeral gullies and soil loss

**Ephemeral gullies and soil loss** reads the terrain of the field and 250 m around it from open elevation tiles (AWS Terrain Tiles; about 5 m cells at the equator, from SRTM, Copernicus and national lidar where available), smooths it, fills its sinks (priority flood, Barnes et al., 2014) and gives every cell its D8 flow direction, upslope area $A$ and slope $S$. An ephemeral gully is likely where

$$G = \frac{S\,A^{0.4}}{0.025} \;\geq 1 \qquad (A \text{ in ha},\ S \text{ in m/m}), \tag{20.9}$$

*Provenance (Eq. 20.9):* **[Literature]**: the topographic threshold of ephemeral gullies in cultivated fields (Vandaele et al., 1996).

From every cell above the threshold with 0.1 ha or more upslope the gully follows the flow lines down the field. Each predicted gully is checked in two ways:

- **in the photo**: a Hessian ridge and valley filter at two scales (Frangi et al., 1998); a gully with a linear feature on 40 % of its cells is *seen in the photo*;
- **in the crop**: the Sentinel-2 season; a gully along which the crop stood more than half a standard deviation below the field over the season (Eq. 20.7) is *seen in the crop*.

Both are drawn in purple. **The rain and the runoff** come from the daily rain of the weather in use (twenty years): the runoff of every day on this soil and this cover by the SCS curve number (TR-55: soil group from the SoilGrids texture, row crop in good condition, bare soil when the NDVI of the month is below 0.3), the runoff a year, the number of runoff events of 10 mm or more a year, and the largest rain of the record with its runoff and the volume it sends down each gully.

**Soil loss (RUSLE)**, $A_s = R\,K\,LS\,C\,P$, with every factor from the farm:

$$R = 0.0483\,P_a^{1.61}, \qquad R_m = R\,\frac{\sum_{d\in m,\ P_d\geq 12.7} P_d^{1.81}}{\sum_{d,\ P_d\geq 12.7} P_d^{1.81}}, \qquad C_i = \sum_m \frac{R_m}{R}\,e^{-2\,\mathrm{NDVI}_{i,m}/(1-\mathrm{NDVI}_{i,m})}, \qquad LS = \left(\frac{A/w}{22.13}\right)^{0.4}\left(\frac{\sin\beta}{0.0896}\right)^{1.3}, \tag{20.10}$$

*Provenance (Eq. 20.10):* **[Literature]**: $R$ from the mean annual rain of the series (Renard and Freimund, 1994; $587.8 - 1.219P_a + 0.004105P_a^2$ above 850 mm), shared over the months by the erosive daily rains (exponent of Richardson et al., 1983); $C$ of every cell in every month from the NDVI of the Sentinel-2 scene of that month (van der Knijff et al., 1999), the crop cover of the SWAFL model where a month has no scene; $K$ from the SoilGrids sand, silt, clay and organic carbon of the field (EPIC, Sharpley and Williams, 1990); $LS$ after Moore and Wilson (1992); $P$ chosen in the panel.

A month table gives the share of the erosive rain, the NDVI and the runoff of each month. The advice names the **risk window** (the months when bare soil, NDVI below 0.35, meets a large share of the erosive rain), the gullies and the runoff they carry, the soil loss against the tolerable 2 t/ha/yr, and what to do.

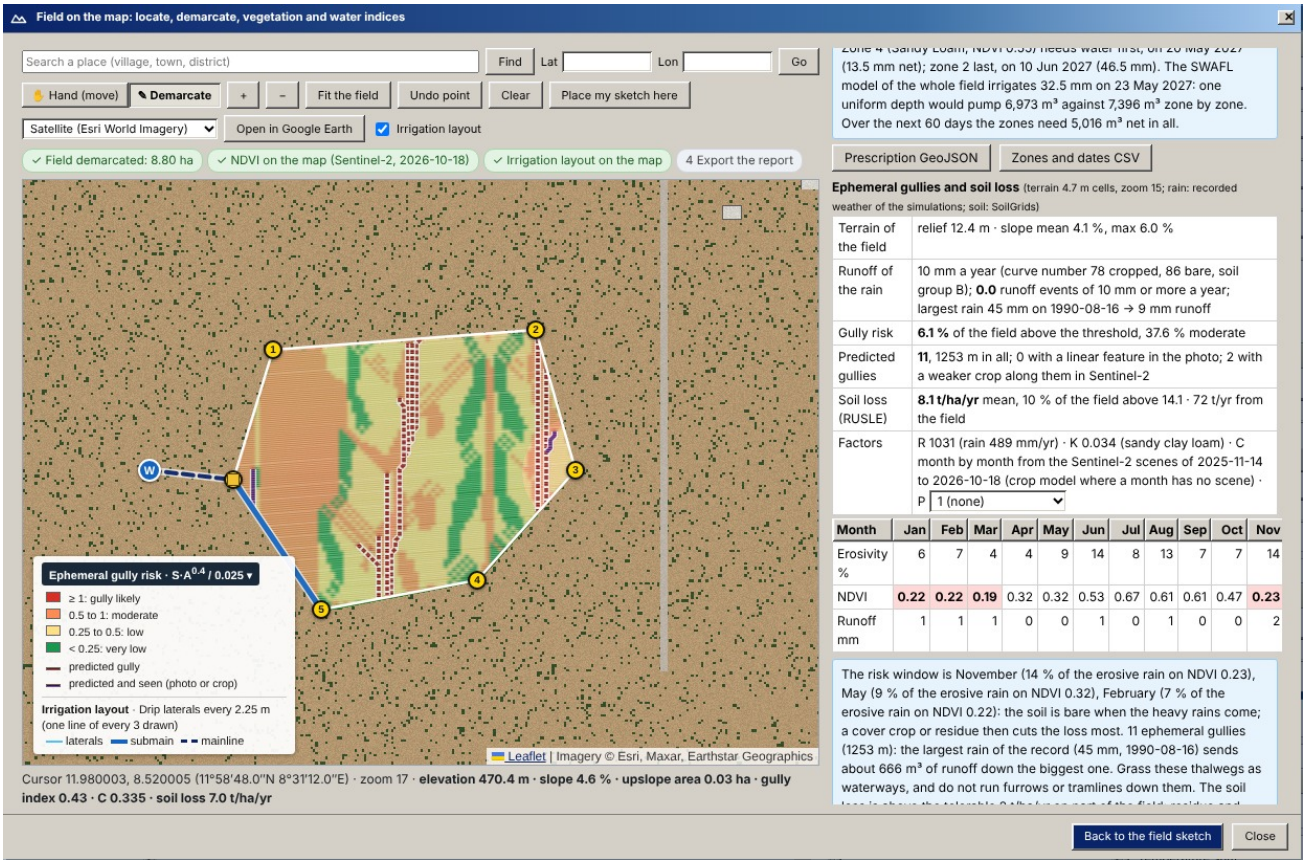



Figure 20.14: Ephemeral gully risk with the predicted gullies, the runoff of the rain and the month table (test data).

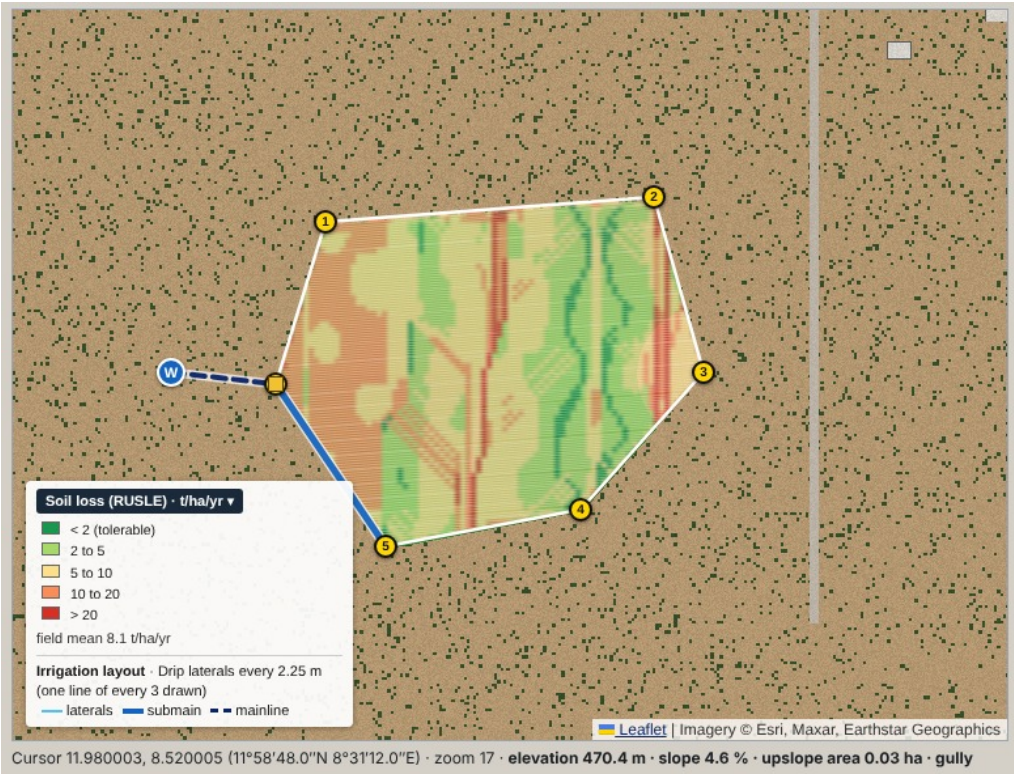



Figure 20.15: Soil loss (RUSLE) with the cover of every month (test data).

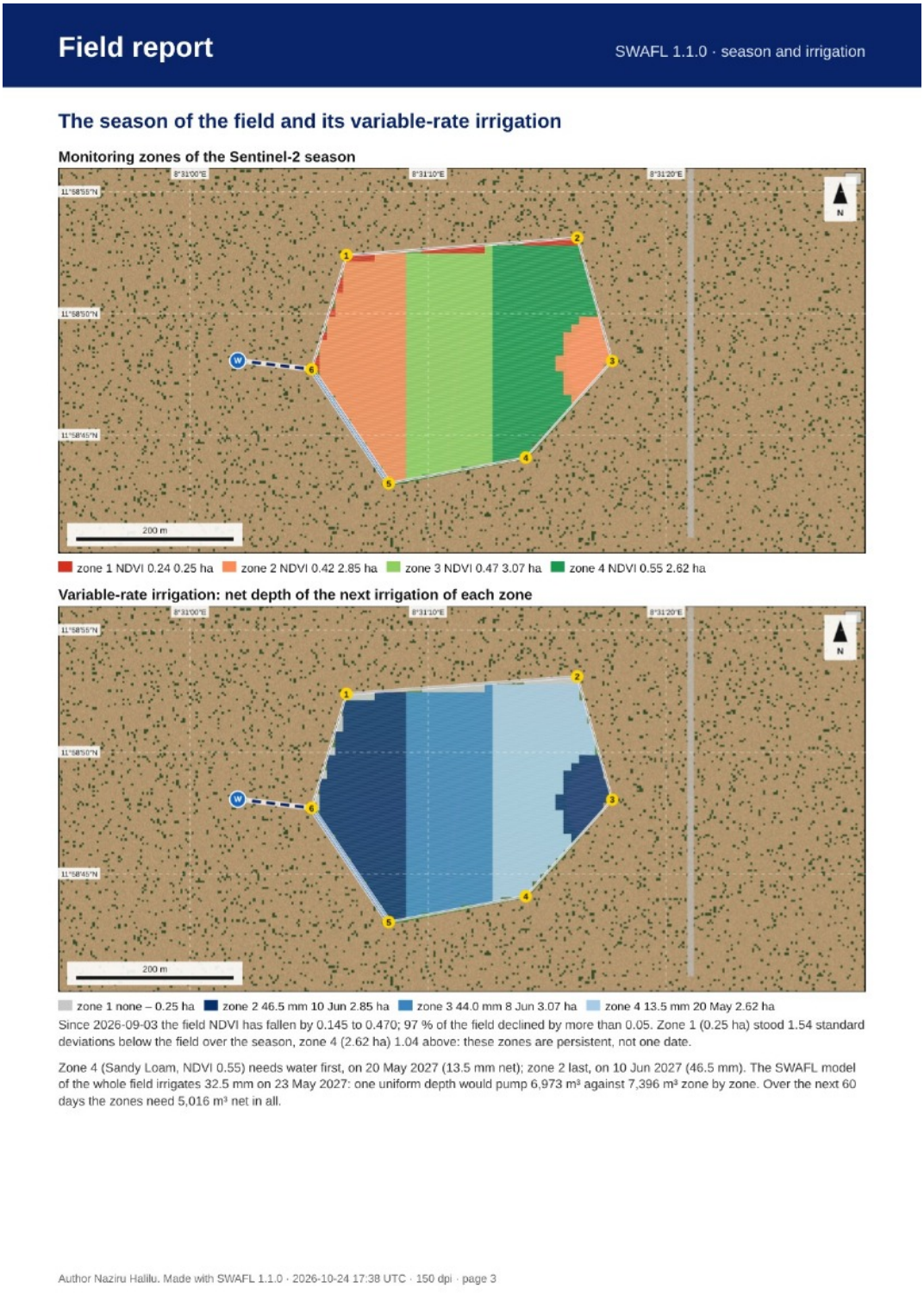



Figure 20.16: The report: the season and the variable-rate irrigation (test data).

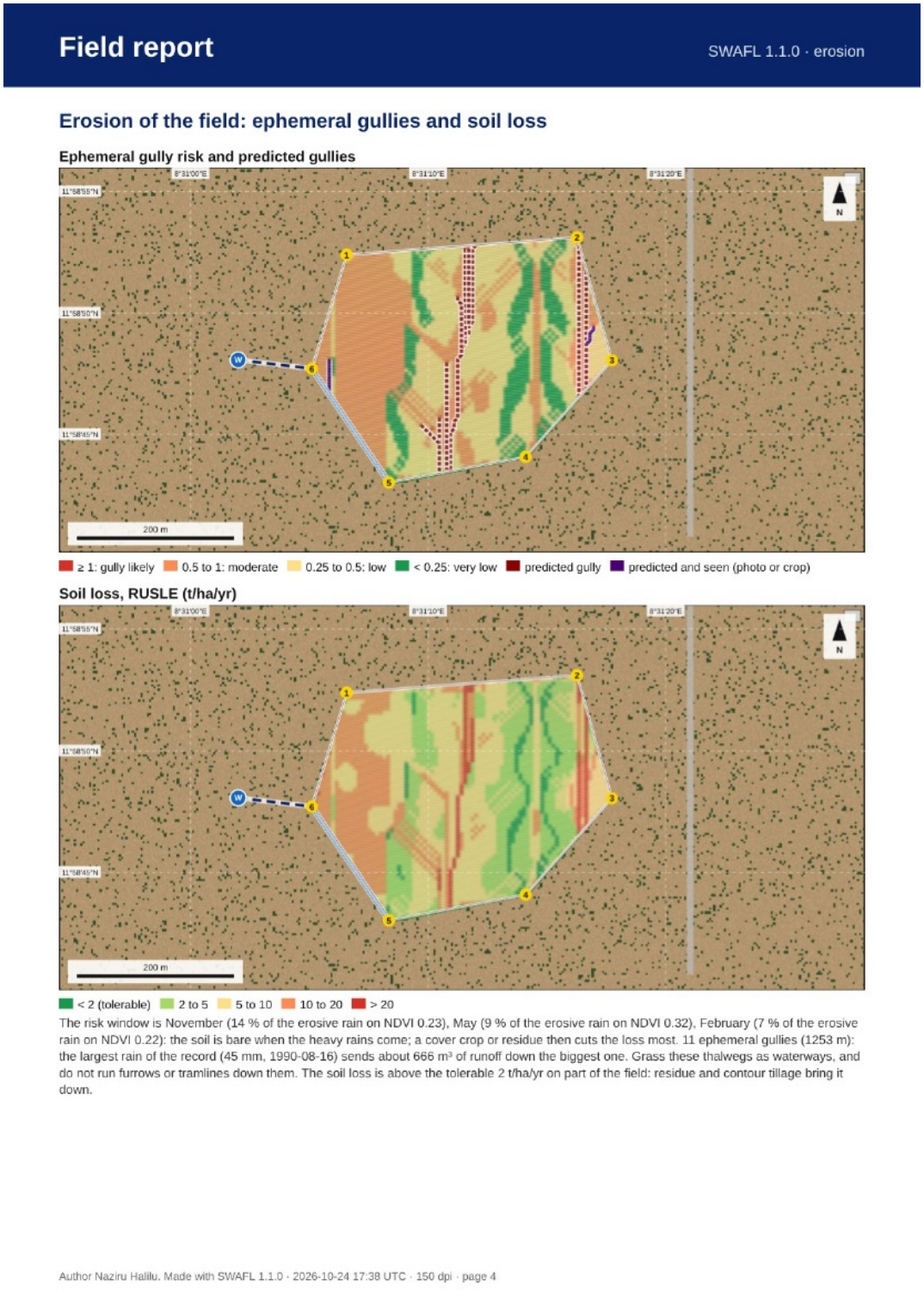



Figure 20.17: The report: the erosion (test data).

## 20.9 Terrain, flow, watershed, soil retention and sensors (section 4b)

Section 4b of the panel reads the terrain of the demarcated field and builds four things from it: the contours and flow lines, the watershed, the maps of the soil retention curves, and the soil water sensors. Everything is computed on the cells of the terrain inside the field, and every map is a layer of *Maps of this field* with its legend on the map.

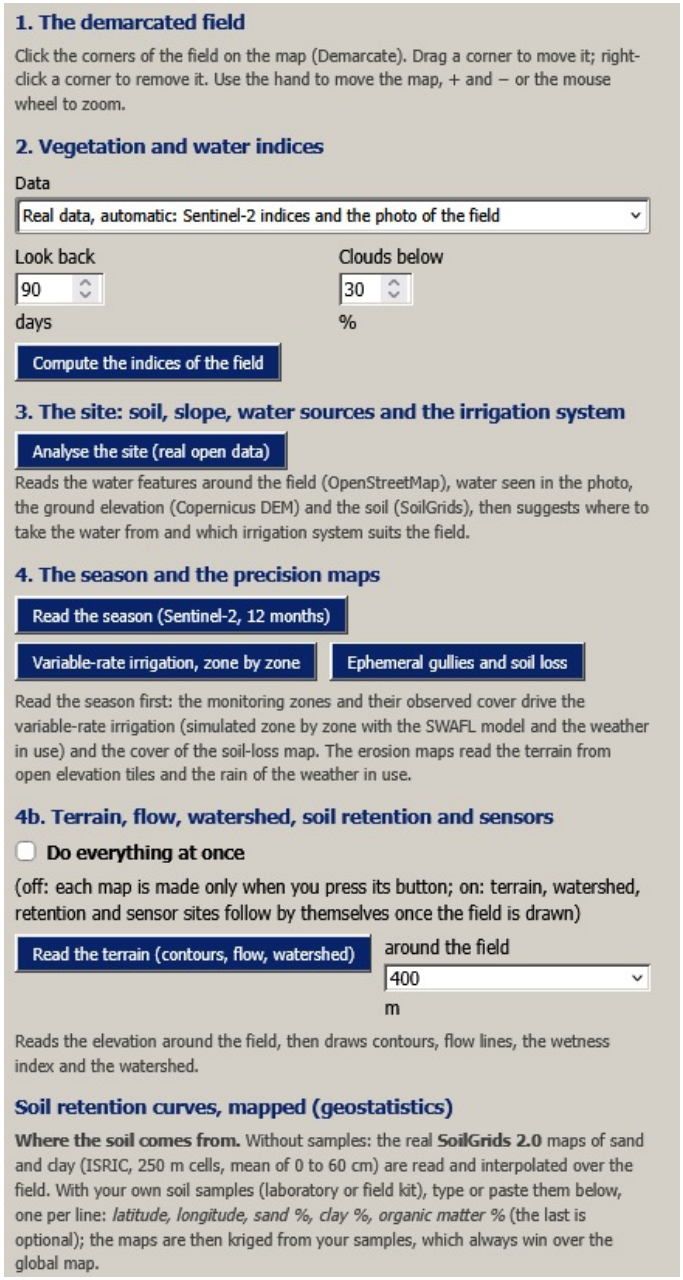



Figure 20.18: The panel of the real field as it appears in the program: 1. the demarcated field, 2. vegetation and water indices (real Sentinel-2 data, look back and clouds), 3. the site (soil, slope, water sources, irrigation system), 4. the season and the precision maps (variable-rate irrigation, ephemeral gullies and soil loss), 4b. terrain, flow, watershed, soil retention and sensors, with the start of the soil retention curves mapped by geostatistics.

**Automatic.** When the box *Automatic* is ticked, the satellite indices, the terrain, the watershed, the retention maps and the sensor sites are all made once the field is drawn, one after the other, so that a field only has to be demarcated.

**Read the terrain (contours, flow, watershed).** The button reads the elevation of the field and of its surroundings from open elevation tiles; *around the field* (150, 250, 400, 700 or 1000 m) sets how much of the surroundings is read, and a larger window is needed when the catchment of the field reaches the edge. The terrain is smoothed, its sinks are filled by priority flood, and the flow directions and the upslope area are found by the D8 method, the same as in the erosion maps. The line under the button tells how many cells of what size were read and from which source. Then five buttons choose what the map shows:

- *Contours and flow*: contour lines, the direction in which the water moves (white arrows), the channels (blue) and a hillshade. The line *Terrain of the field* gives the lowest and the highest elevation of the window and the interval of the contours.
- *Wetness index*: the topographic wetness index, which is high where water gathers (hollows, the foot of slopes) and low on crests.
- *Watershed*: the catchment and its channel network.
- *3D view*: the terrain as a surface that turns with the mouse (drag to turn, wheel to zoom), coloured by elevation, flow accumulation, wetness index, available water, soil water now or matric potential now, with a slider for the vertical exaggeration (1 to 12), the water running down the flow lines, and the sensors. *Close the 3D view* returns to the map.
- *channels from* (0.05 to 2 ha): the smallest upslope area at which a cell is drawn as a channel.

**The watershed.** *Outlet* is the lowest outlet of the field, or a point that you click on the map (*Click the outlet*). *Delineate the watershed* then finds the catchment draining to it, as a terrain portal does from an elevation model. The table gives

- the area, with the area of the field and the part that drains onto it;
- the perimeter and the longest flow path;
- the relief and the mean slope;
- the length of the channels, the highest Strahler order and the drainage density, in km per km$^2$;
- the time of concentration by the formula of Kirpich (1940).

When the catchment reaches the edge of the terrain that was read, a red note asks for a larger window in *around the field*.

**Soil retention curves, mapped (geostatistics).** The button *Map the retention curves* maps the sand and the clay of the field, and from them the water retention of every cell. The soil comes from SoilGrids; it is varied across the field with the wetness index of the terrain, and the satellite index is used as well when it has been computed. You may type your own soil samples in the box, one per line: latitude, longitude, sand per cent, clay per cent and, if known, organic matter per cent (for example *9.0831, 7.4012, 62, 18*).

- Without samples, the soil is varied with the wetness index as an indicative trend; the two numbers *clay* and *sand*, in points per standard deviation of the wetness index, set how strongly (the box *indicative terrain trend* switches it on or off).
- With samples, sand and clay are mapped by regression kriging: the trend on the wetness index and the satellite index is fitted to the samples when there are enough of them, and the residuals are kriged with a spherical variogram fitted to them.
- The retention curve of every cell follows from the equations of Saxton and Rawls (2006).

The list under the button records every step as it is read and mapped. When it is done, the table *Retention of the field* gives the mean, the 10 per cent and the 90 per cent values of the field capacity, the wilting point, the available water and the days to the trigger (with the depletion fraction, the root depth and the crop evapotranspiration used). A line says when the driest cells reach the trigger and compares it with the average cell and with the DSS, which uses one soil for the whole field. Six buttons put the maps on the field: *Field capacity*, *Wilting point*, *Available water*, *Trigger* (the water content at the critical depletion of the SWAFL decision), *Matric potential now*

and *Days to the trigger*. Two charts follow: the retention curves of the cell at the 10 per cent (driest), the median and the 90 per cent (most retentive) of the available water, and of the first three sensors (water content against suction on a log scale, with the point of now marked), and the variogram of the residuals, with its nugget, sill and range, marked as fitted or, when there are too few points to fit, as default.

**Use this field in the DSS.** The button sets the soil of the program to the class whose sand and clay are closest to the field means, and the area of the program to the area of the field. Every irrigation design, the trigger and the schedule then follow this field.

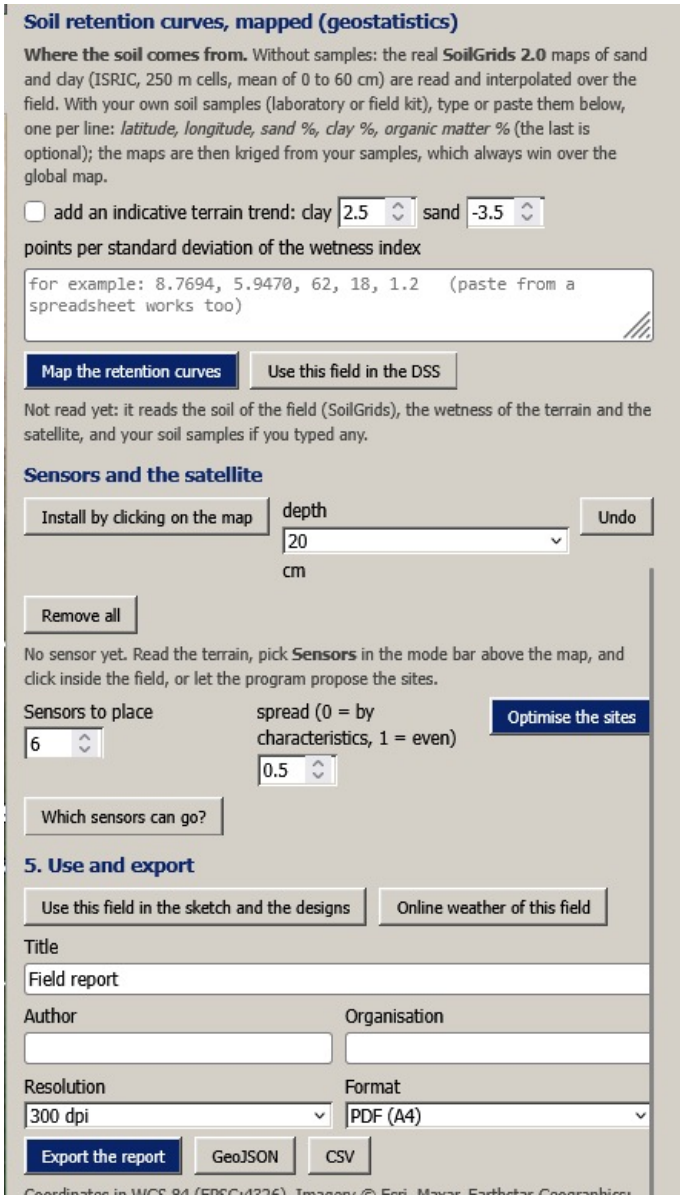



Figure 20.19: The rest of the panel as it appears in the program: the soil retention curves (SoilGrids 2.0 or your own samples, the optional terrain trend, Map the retention curves, Use this field in the DSS), the sensors and the satellite (install by clicking on the map, depth, the number to place, the spread, Optimise the sites, Which sensors can go?), and 5. use and export (the sketch and the designs, the online weather of the field, the title, author, organisation, resolution and format of the field report, GeoJSON and CSV).

**Sensors and the satellite.** Soil water sensors are installed with the mode *Sensors* of the bar

above the map, or with the button *Install by clicking on the map*: click inside the field, at the depth chosen in *depth* (10, 20, 30, 45, 60 or 90 cm); click a sensor to remove it. *Undo* takes back the last one and *Remove all* clears them. The terrain must have been read first. The table lists each sensor with its position, depth and the reading in per cent by volume; leave the reading blank and the program models it from the SWAFL water balance (the water left in the root zone on the working date, shifted by the wetness index and by the satellite index of the cell), or type the real reading.

- *Fusion*: the sensors and the satellite are fused by regression kriging. The readings are regressed on the wetness index and the satellite index, the residuals are kriged, and the line under the title gives the regression (in per cent by volume per standard deviation), its R squared and the variogram. The buttons *Soil water map* and *Uncertainty map* show the map of the soil water and its kriging standard deviation.
- *Sensors to place* (2 to 20) and *spread* (0 puts the sensors by the characteristics of the cells, 1 spreads them evenly) set the number and the character of the proposed sites. *Optimise the sites* (after the retention curves have been mapped) groups the cells by k-means in the space of elevation, wetness index, available water, the satellite index and position, and proposes one site for each group, drawn as an orange ring. The table gives the share of the field that each site represents and why it was chosen, and compares the mean kriging deviation with these sites, with your sensors and with the same number placed at random. *Install these sensors* replaces the present ones by the proposed ones.
- *Which sensors can go?* ranks the sensors by the rise of the mean kriging deviation if each is removed, the least useful first, with a *Remove* button for each.
- *Sensors CSV* saves the sensors with their position, depth, reading and modelled value.

## 20.10 A real field: Zaria, Nigeria

The pictures of this chapter marked *test data* were made without a connection. The same panel on a real field, a 5.85 ha field (perimeter 992 m, 17 corners) near Zaria, Kaduna State, Nigeria, with a connection: the Sentinel-2 L2A catalogue held ten scenes below 30 % cloud in the last 90 days, the newest (30 October 2026) had no cloud over the tile and all 589 cells of the field were clear; NDVI averaged 0.296 and GNDVI 0.415. SoilGrids gave a sandy clay loam (sand 52 %, silt 25 %, clay 23 %), the Copernicus DEM 612 to 620 m with a slope of 1.93 % to the south-west, and the best water source was the Galma River, 288 m south with a lift of about 9 m. The soil loss averaged 8.0 t/ha a year (tolerable 2), with nine ephemeral gullies; the retention curves gave a field capacity of 25.8 %, a wilting point of 14.6 % and an available water of 11.2 % by volume. None of these maps was checked on the ground.

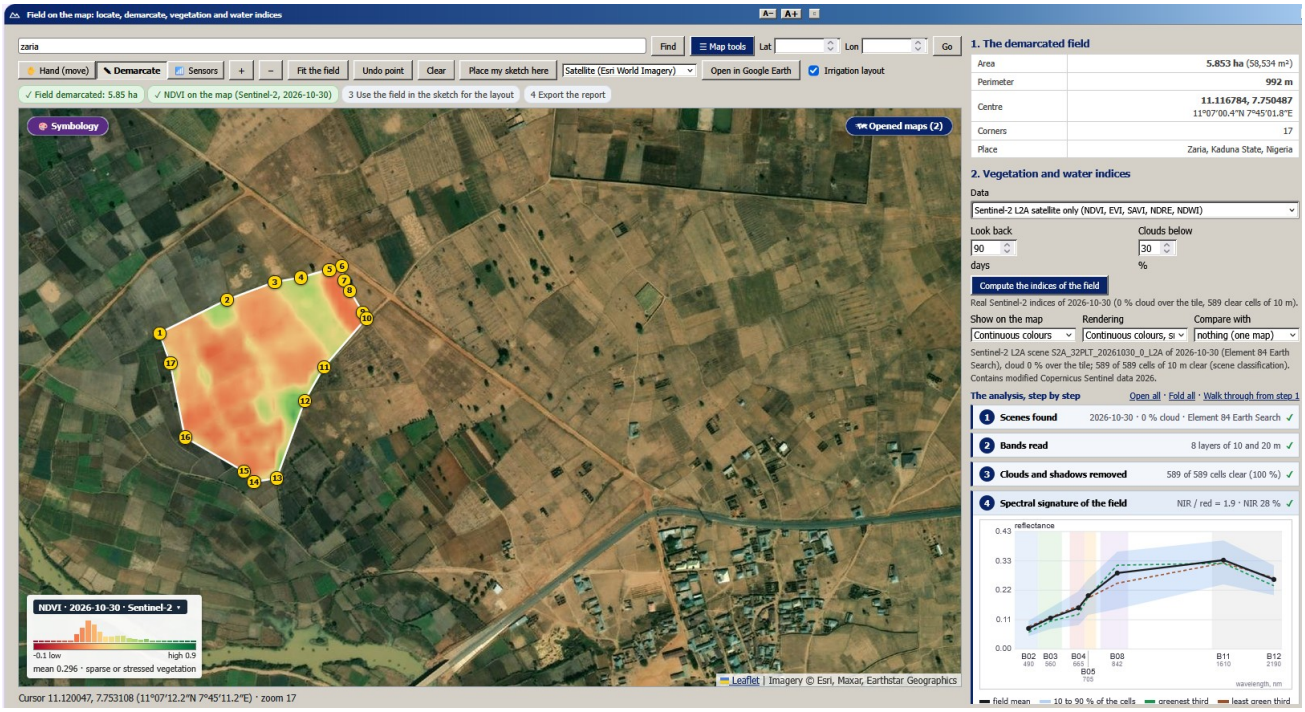



Figure 20.20: The real field: NDVI of the Sentinel-2 scene of 30 October 2026 cut to the outline, and the analysis step by step with the spectral signature of the field.

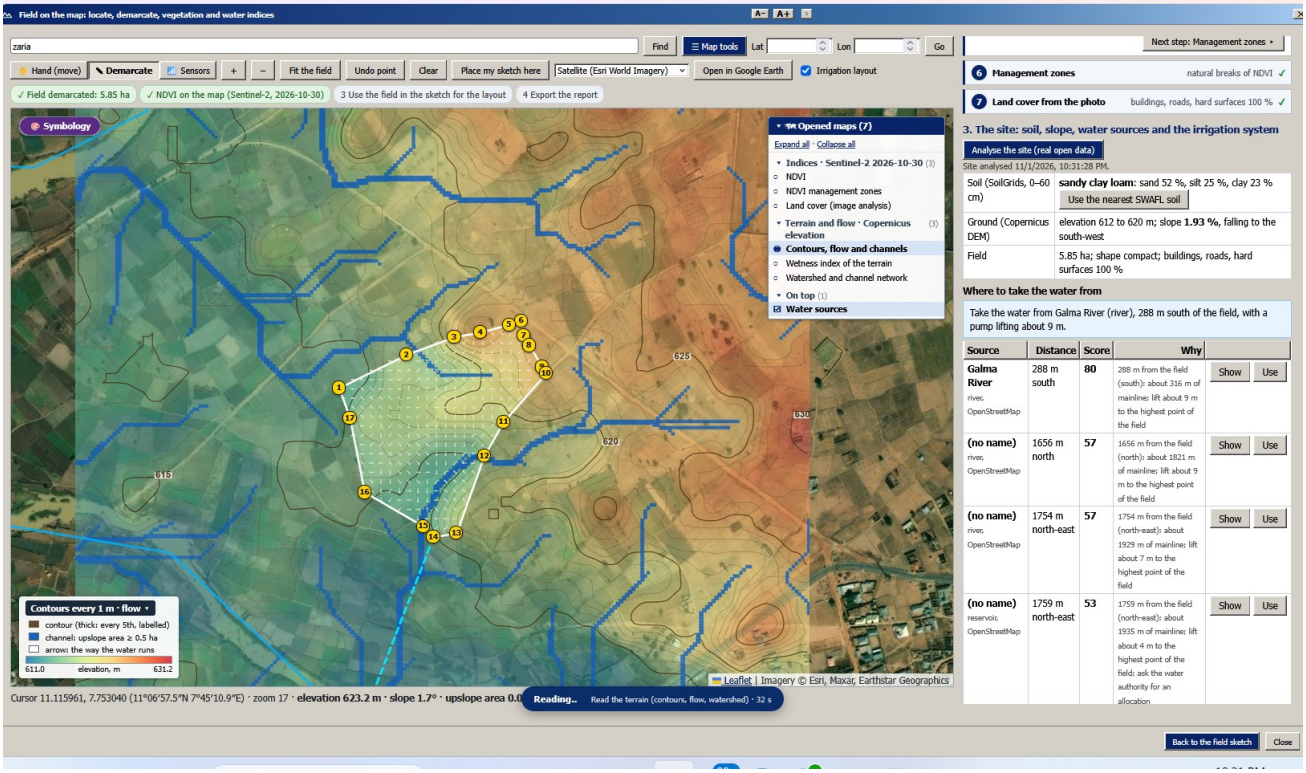



Figure 20.21: The terrain (contours every metre, flow and channels) and the water sources ranked by distance, lift and score.

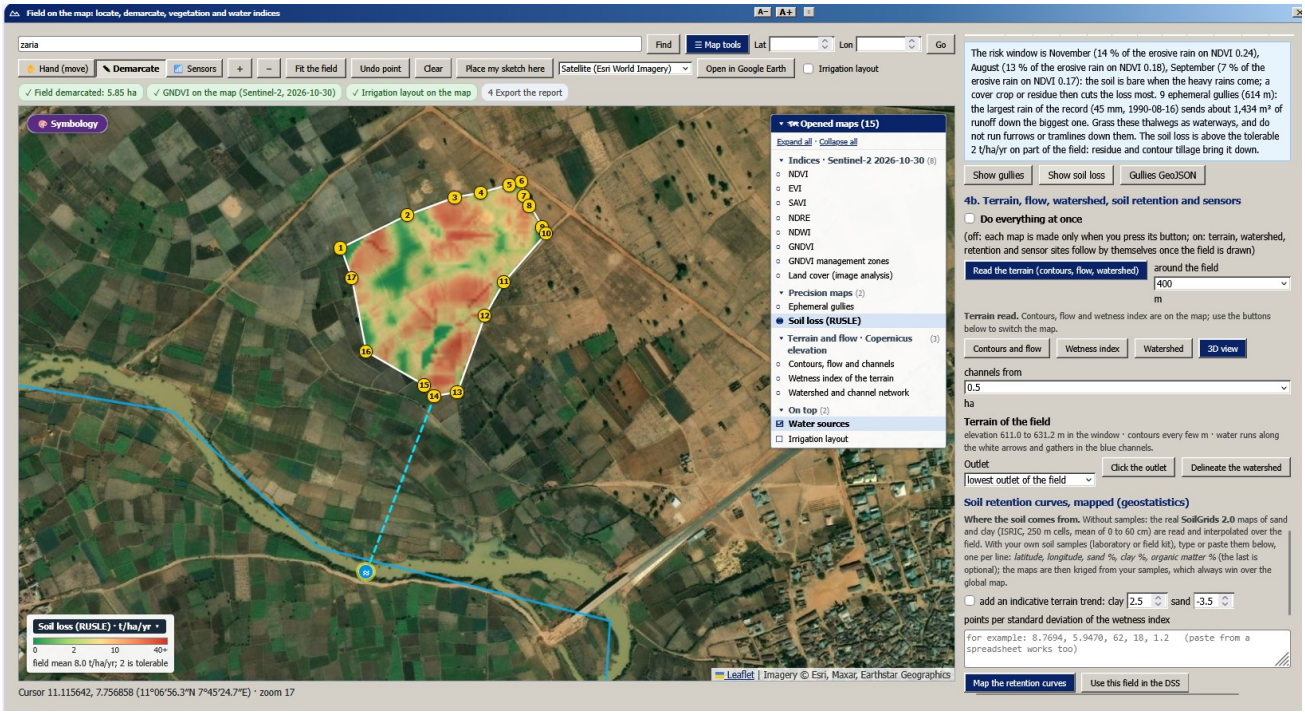



Figure 20.22: Soil loss (RUSLE) of the field and the months of risk.

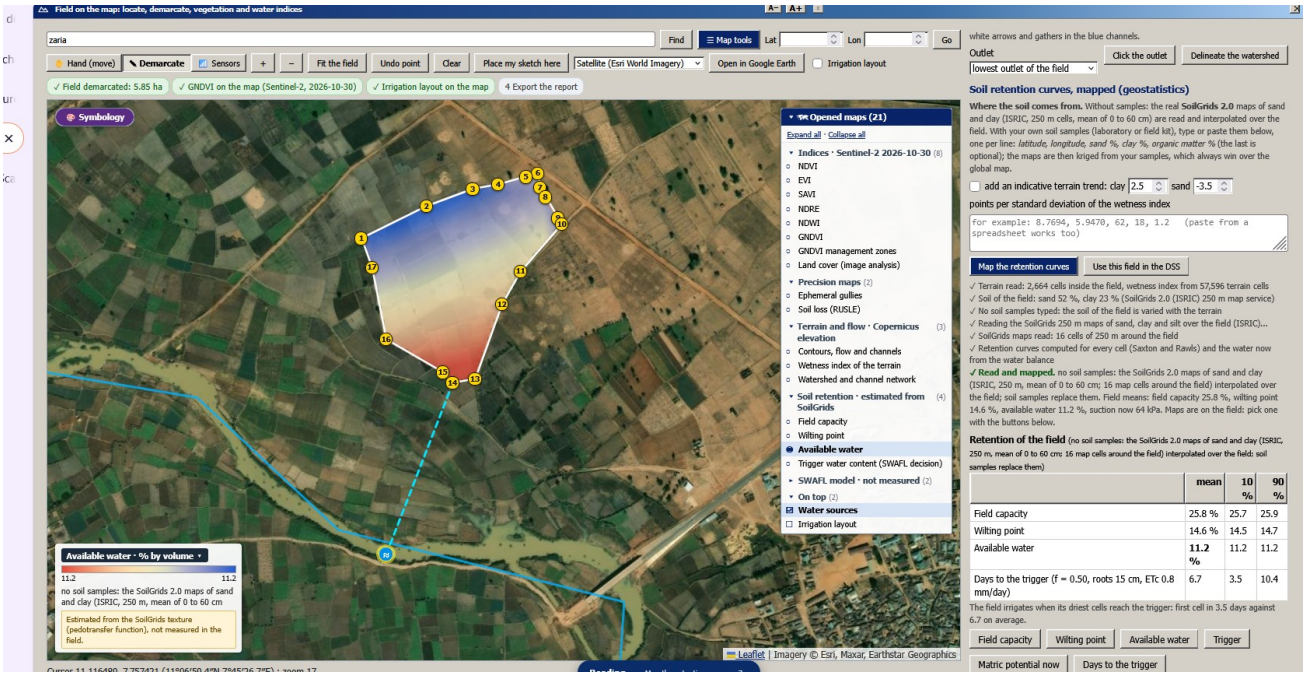



Figure 20.23: Available water from the soil retention curves mapped from SoilGrids, and the days to the trigger.

## 20.11 Use and export

Section 5 of the panel is *Use and export*. *Use this field in the sketch and the designs* makes the demarcated field the sketch (true lengths, centre coordinates kept), and *Irrigation layout* draws

its system on the map. *Online weather of this field* downloads the daily weather of the centre of the field and runs the program on it. The report has a **title, an author and an organisation**, a **resolution of 96, 150, 300 or 600 dpi** and a format:

- **PDF (A4)**: page 1, the map of the chosen index (continuous, zones or land cover) on the satellite photo, cut to the field, with the irrigation layout, a grid of latitude and longitude in degrees, minutes and seconds, north arrow and scale bar, the area, perimeter, centre and corner coordinates, the table of the indices and the zones; page 2, the indices side by side with their spread and what they say together; page 3 (after an image analysis or a site analysis), the land cover, the soil, the slope, the water sources on the map with the advice, and the ranking of the systems. The title, author, coordinates and keywords are also in the properties of the PDF.
- **PNG (A4)**, one image per page, with the resolution written in the file so that it prints at the chosen dpi.
- **GeoJSON** (the field polygon in WGS 84 with the area, the author and the means of the indices) and **CSV** (the corners and every statistic), for a GIS or a spreadsheet.

The report is made with the boxes *Title*, *Author* and *Organisation* (written on the pages), *Resolution* (the dpi of the pages) and *Format* (PDF A4 or PNG A4); **Export the report** writes it, **GeoJSON** and **CSV** write the field for a GIS or a spreadsheet. *Back to the field sketch* returns to the sketch with the field drawn, and *Close* closes the window.

Every page carries its sources: photo © Esri, Maxar, Earthstar Geographics; map and water features © OpenStreetMap contributors; contains modified Copernicus Sentinel data; Copernicus DEM; ISRIC SoilGrids.

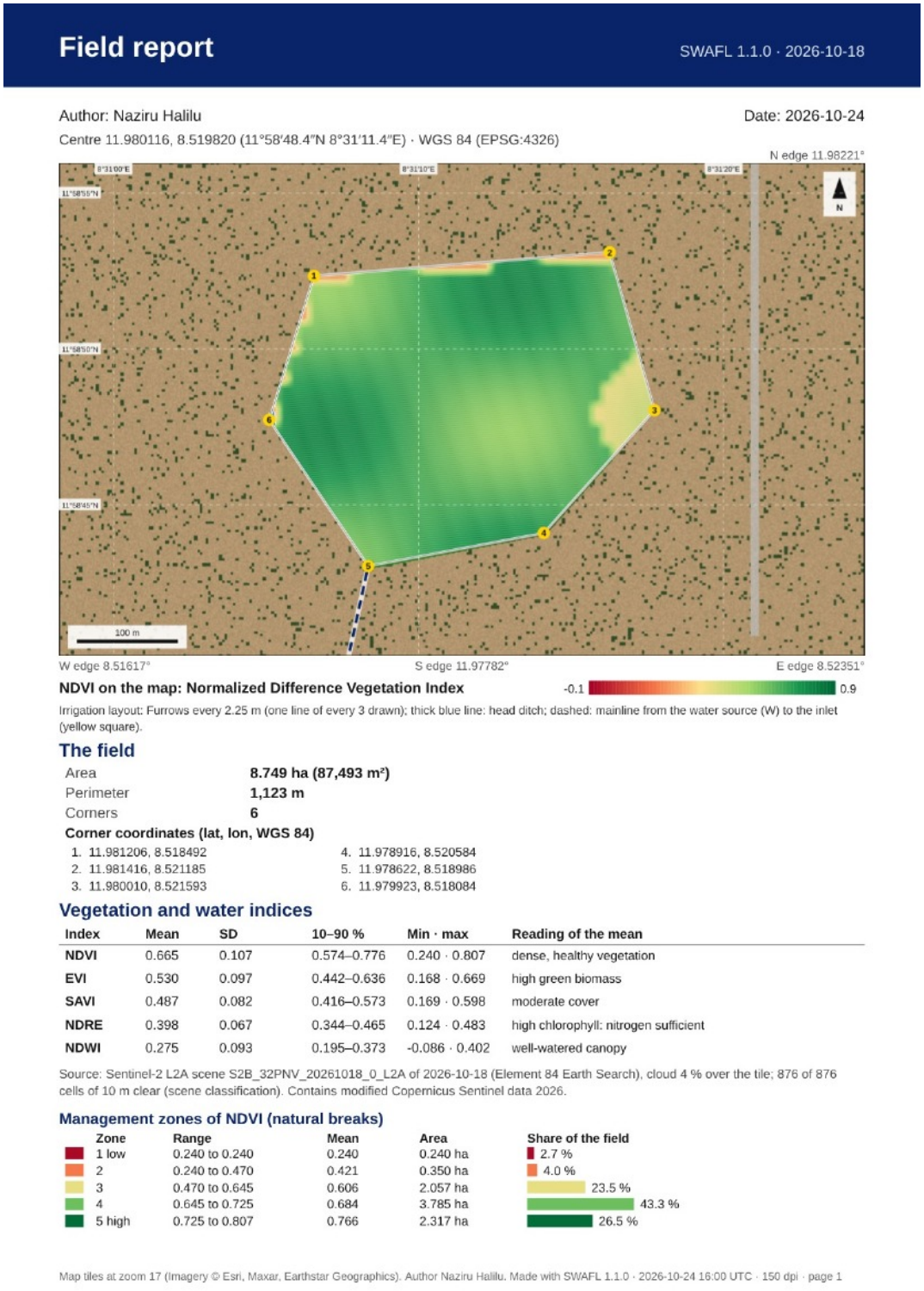



Figure 20.24: The field report, page 1: NDVI of Sentinel-2 on the photo, the coordinates, the indices and the zones (test data).

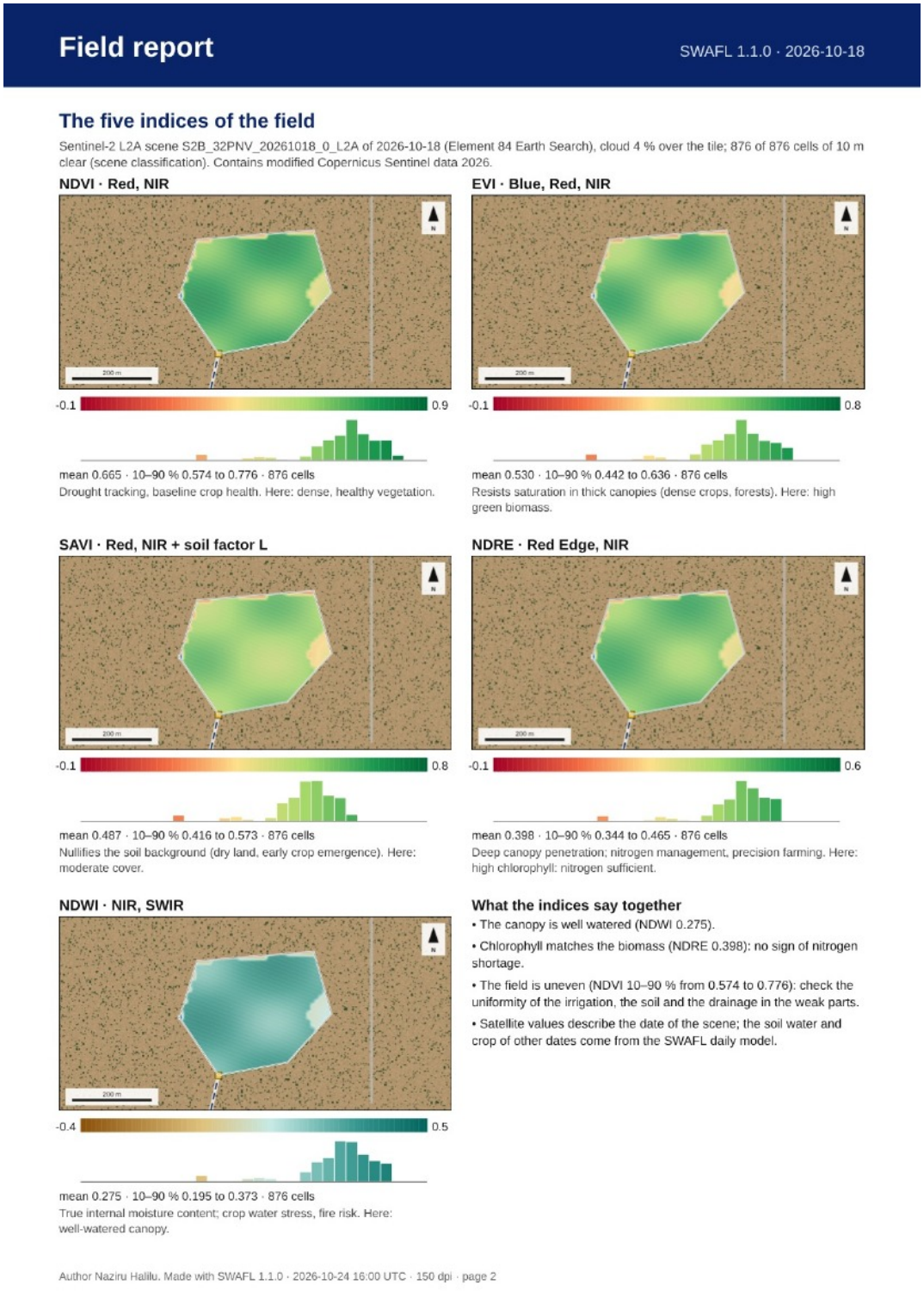
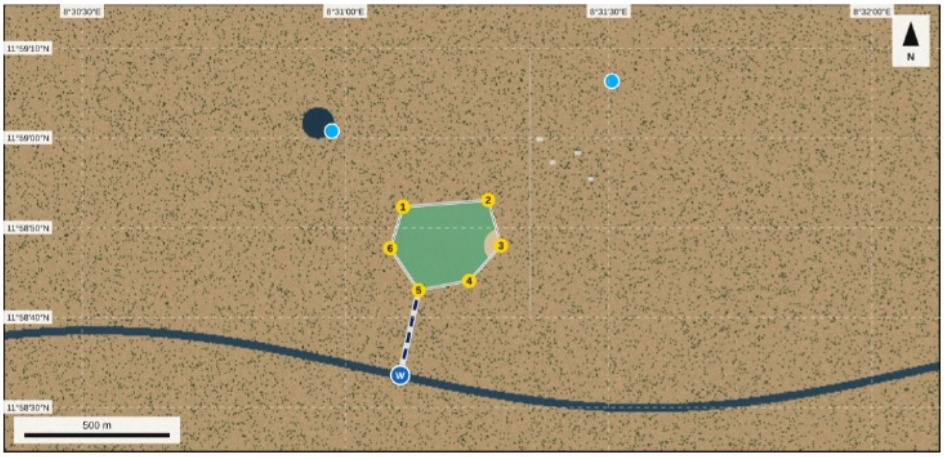



Figure 20.25: The field report, page 2: the five indices.

## Field report

SWAFL 1.1.0 · the site

### The site: land cover, soil, slope, water sources and the irrigation system




Land cover read from the satellite photo by image analysis (colour and texture of every pixel, k-means clustering, named by rules, majority filter; 0.6 m per pixel):

- Dense vegetation: 95.1 % (8.32 ha)
- Sparse vegetation: 0.0 % (0.00 ha)
- Bare soil: 4.9 % (0.43 ha)
- Water: 0.0 % (0.00 ha)
- Buildings, roads, hard surfaces: 0.0 % (0.00 ha)
- Shadow or very dark: 0.0 % (0.00 ha)

| | |
|---|---|
| **Soil (SoilGrids)** | sandy loam: sand 64 %, silt 19 %, clay 17 % |
| **Ground (Copernicus DEM)** | elevation 473 to 478 m, slope 1.26 % falling to the south |

### Where to take the water from

**Take the water from Challawa River (river), 299 m south of the field, with a pump lifting about 8 m.**

| Source | Distance | Score | Why |
|---|---|---|---|
| **Challawa River** river | 299 m south | **80** | 299 m from the field (south): about 329 m of mainline; lift about 8 m to the highest point of the field |
| **Village well** well | 592 m north-east | **60** | 592 m from the field (north-east): about 651 m of mainline; groundwater is used here; the depth of the water table is not mapped: ask the local drillers or the borehole records |
| **(no name)** pond | 358 m north-west | **56** | 358 m from the field (north-west): about 394 m of mainline; the source lies 1 m above the highest point: water can flow by gravity |

### Which irrigation system suits this field

| | | |
|---|---|---|
| **Furrows (best)** | **80** | + slope 1.26 % suits furrows (0.2–3 %); + surface water supply nearby |
| **Border strips** | **74** | + slope 1.26 % suits borders (0.1–2 %); + large surface water supply nearby; − relief of 4.7 m: land levelling needed |
| **Sprinklers** | **74** | + sandy or light soil |
| **Drip** | **70** | + sandy or light soil: drip keeps the water in the root zone |
| **Basin (level basins)** | **32** | − slope 1.3 %: basins need level ground; + large surface water supply nearby |

Open data: water features © OpenStreetMap contributors (Overpass API); elevation Copernicus DEM GLO-90 (Open-Meteo); soil ISRIC SoilGrids 250 m; photo © Esri, Maxar, Earthstar Geographics. Scores follow the FAO and USDA guidance for choosing an irrigation system; check on the ground.

Map tiles at zoom 15 (Imagery © Esri, Maxar, Earthstar Geographics). Author Naziru Halilu. Made with SWAFL 1.1.0 · 2026-10-24 16:00 UTC · 150 dpi · page 3

Figure 20.26: The field report, page 3: the land cover, the soil, the slope, the water sources and the systems (test data).

# Chapter 21

# ✿ The crop library and the crop water requirement

## 21.1 The crops

Version 1.1 carries **48 crops** besides the simulated cereal/oilseed rotation. They are chosen in Properties ▸ Soil and field ▸ **Crop**, in four groups: *Staple food crops*, *Cash and oil crops*, *Vegetables*, and *Trees, vines and palms*. For each crop the library holds the crop coefficients $K_{c,\mathrm{ini}}$, $K_{c,\mathrm{mid}}$ and $K_{c,\mathrm{end}}$, the lengths of the four growth stages (initial, development, mid-season, late season), the plant height, the maximum rooting depth, the depletion fraction $p$, the yield response factor $K_y$, the ground cover and, for trees and vines, the planting densities or training systems. Field crops take their coefficients and stages from a field study that computed the daily crop water requirement with the FAO Penman and Monteith reference evapotranspiration; trees, vines and berries from the updated FAO-56 coefficients of fruit trees and vines. Rooting depth, $p$ and $K_y$ are the FAO-56 and FAO-33 tabulated values; where a crop is not tabulated the group value is used and the table says *group default*.

The staple and cash crops asked for in this release carry these coefficients:

| Crop | $K_{c,\mathrm{ini}}$ | $K_{c,\mathrm{mid}}$ | $K_{c,\mathrm{end}}$ | Note |
|---|---|---|---|---|
| Cassava | 0.30 | 0.80 | 0.30 | |
| Yam | 0.50 | 1.10 | 0.85 | stages and rooting depth: group default |
| Rice | 1.05 | 1.20 | 0.90 | bunded: no runoff in the season |
| Maize | 0.30 | 1.20 | 0.60 | |
| Sorghum | 0.30 | 1.00 | 0.55 | |
| Millet | 0.30 | 1.00 | 0.30 | |
| Cowpeas | 0.40 | 1.05 | 0.50 | |
| Groundnut | 0.40 | 1.15 | 0.60 | |
| Soybean | 0.40 | 1.15 | 0.50 | |
| Oil palm | 0.90 | 0.95 | 0.95 | perennial |
| Cocoa | 1.00 | 1.05 | 1.05 | perennial, high canopy shade |

The complete library is listed in Table 21.2 at the end of this chapter.

## 21.2 The Crop water requirement layer

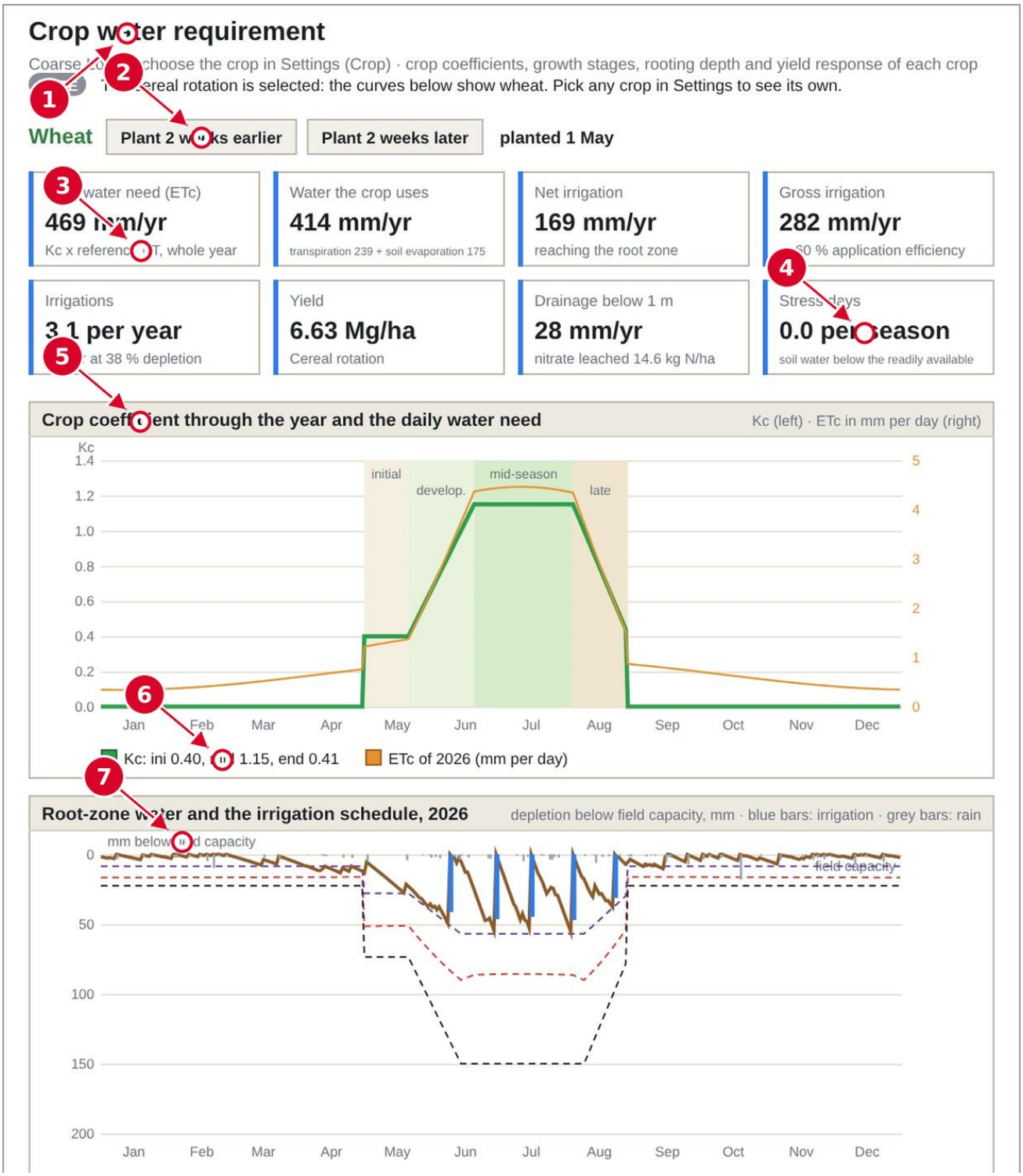



Figure 21.1: The Crop water requirement layer.

The layer *Crops* ▸ *Crop water requirement* shows, for the chosen crop on the soil, weather, trigger and method of the project:

- the **crop coefficient through the year** and the daily water need $ET_c$ of the working year (the label reads *ETc of 2026*, following the year in Properties);
- the **root-zone water and the irrigation schedule** of the working year: the depletion below field capacity against the readily available water and the trigger, with every irrigation of the year;
- buttons to plant two weeks earlier or later (field crops) or to choose the planting density or training system (trees and vines);
- **every crop of the library on this soil**: the yearly crop water need, net irrigation, irrigations per year and relative yield, so that crops can be compared on the same field.

## 21.3 The daily balance of a crop

The layer computes, for each of the twenty years of the weather in use, the FAO-56 daily balance of the root zone. The crop coefficient follows the four stages,

$$K_c(t) = \begin{cases} K_{c,\text{ini}} & t < L_1 \\ K_{c,\text{ini}} + (K_{c,\text{mid}} - K_{c,\text{ini}})\frac{t-L_1}{L_2} & L_1 \le t < L_1 + L_2 \\ K_{c,\text{mid}} & L_1 + L_2 \le t < L_1 + L_2 + L_3 \\ K_{c,\text{mid}} + (K_{c,\text{end}} - K_{c,\text{mid}})\min(1, \frac{t-L_1-L_2-L_3}{L_4}) & \text{after} \end{cases} \tag{21.1}$$

*Provenance (Eq. 21.1):* **[Literature]**: FAO-56 crop coefficient curve (Allen et al., 1998, Fig. 25); $t$ days after planting; evergreen crops repeat it every year.

and outside the season the bare soil has $K_c = 0.25$. The daily crop water need is

$$ET_{c,d} = K_c(d)\, ET_{0,d}, \tag{21.2}$$

*Provenance (Eq. 21.2):* **[Literature]**: FAO-56 single crop coefficient approach.

with $ET_{0,d}$ the daily reference evapotranspiration of the weather in use (your own weather when imported; otherwise the annual reference ET of each recorded year shaped by the seasonal weight, adjusted by 3.5 % per °C of the temperature shift). The roots of an annual crop grow with the canopy, perennial roots are at full depth:

$$z_r = \text{clip}\Big(15 + (R_{\max} - 15)\min\Big(1, \frac{1.15\,K_c}{\max(0.3, K_{c,\text{mid}})}\Big);\ 15,\ R_{\max}\Big)\ \ [\text{cm}], \qquad R_{\max} \le 100\ \text{cm}, \tag{21.3}$$

*Provenance (Eq. 21.3):* **[Engineered]**: root growth tied to the crop coefficient, limited to the 1 m of the model.

The total and readily available water and the depletion $D_r$ of the root zone follow FAO-56,

$$\begin{aligned} &\text{TAW} = \text{AWC}\, z_r, \qquad p_d = \text{clip}(p + 0.04\,(5 - K_c\, ET_0);\ 0.1,\ 0.8), \qquad \text{RAW} = p_d\, \text{TAW}, \\ &K_s = \begin{cases} 1 & D_r \le \text{RAW} \\ \dfrac{\text{TAW} - D_r}{(1-p_d)\,\text{TAW}} & D_r > \text{RAW} \end{cases}, \qquad ET_a = K_s\, ET_c, \\ &D_{r,d} = D_{r,d-1} - (P_d - RO_d) + ET_{a,d}, \qquad RO_d = 0.3\ \max(0, P_d - 35), \end{aligned} \tag{21.4}$$

*Provenance (Eq. 21.4):* **[Literature]**: FAO-56 Eqs. 82 to 85 (water stress coefficient, adjustment of $p$ with the demand) and the root-zone balance Eq. 85; the runoff of heavy rain is **[Engineered]**.

with any negative $D_r$ draining below the roots. Irrigation refills the root zone to field capacity on the day the depletion reaches the trigger of the project, $D_r \ge f_T\,\text{TAW}$, with $f_T$ the depletion at the trigger suction (Eq. 5.3): the net depth is $I_n = D_r$ and the gross depth $I = I_n / E_{a,m}$, with the method efficiency $E_{a,m}$ = 0.60 (surface), 0.75 (overhead) or 0.90 (drip). The relative yield of a year is

$$Y_{\text{rel}} = \text{clip}\Big(1 - K_y\Big(1 - \frac{\sum ET_a}{\sum ET_c}\Big);\ 0,\ 1\Big), \tag{21.5}$$

*Provenance (Eq. 21.5):* **[Literature]**: FAO-33 yield response (Doorenbos and Kassam, 1979); $K_y = 1$ when not tabulated, as the table says.

## 21.4 How a library crop changes every other layer

The annual model (Engine 1) is the result of the simulations of the cereal rotation. A library crop is transferred from it by the ratio of two runs of the balance above on the same soil, weather, trigger and method: one for the crop ($c$) and one for the cereal reference ($r$, FAO-56 small grains, $K_c$ 0.30/1.15/0.25). With $m$ the annual-model result of the cereal,

$$I = \begin{cases} I_m \ \text{clip}(I_c / I_r;\ 0,\ 4) & I_r > 5 \text{ mm} \\ I_c & \text{otherwise} \end{cases} \tag{21.6}$$

$$\begin{aligned} T &= \min\Big(0.9\,ET,\ \ T_m \ \text{clip}\Big(\frac{ET_{a,c}}{ET_{c,c}} \cdot \frac{ET_{c,c}}{ET_{c,r}} \cdot \Big(\frac{f_{c,c}}{f_{c,r}}\Big)^{0.3};\ 0.2,\ 3\Big)\Big), \\ E^* &= E_m \ \text{clip}\Big(\frac{1 - 0.7\,f_{c,c}}{1 - 0.7\,f_{c,r}};\ 0.6,\ 1.6\Big), \end{aligned} \tag{21.7}$$

$$D = \max\big(0.2\,D_m,\ D_m + 0.5\,(DP_c - DP_r)\big), \qquad ET = \text{clip}\big(P + I - D - \Delta S_m;\ 50,\ 5000\big), \tag{21.8}$$

$$(E,\ E_{\text{other}}) = (E^*,\ E^*_{\text{other}})\,\frac{ET - T}{E^* + E^*_{\text{other}}}, \qquad N_L = N_{L,m}\,\frac{D + 5}{D_m + 5}, \qquad Y = 100\,Y_{\text{rel},c}\ [\%], \tag{21.9}$$

*Provenance (Eqs. 21.6 to 21.9):* **[Engineered]**: transfer by the ratio of two FAO-56 balances; the clips keep the transfer within the range in which the cereal result is informative; the remainder of the balance is shared between soil evaporation and the other evaporation terms in proportion to their targets, so that the balance closes without inflating either.

with $f_c$ the ground cover, $E^*_{\text{other}} = E_{\text{other},m}\,ET/ET_m$ and $\Delta S_m$ the storage change of the cereal result. The yield of a library crop is therefore shown as the relative yield in per cent, and the differences of yield in the sensitivity and feedback layers are in *% points*.

## 21.5 Table 21.2. The crop library

Table 21.2. The 48 crops of the library: crop coefficients, stage lengths (initial/development/mid/late, days), plant height (m), maximum rooting depth (cm), depletion fraction $p$ and yield response factor $K_y$ (n/a when not tabulated).

| Crop | Group | $K_{c,ini}$ | $K_{c,mid}$ | $K_{c,end}$ | Stages (d) | Height (m) | Root (cm) | $p$ | $K_y$ |
|---|---|---|---|---|---|---|---|---|---|
| Rice | field crop | 1.05 | 1.20 | 0.90 | 15/30/45/30 | 0.8 | 75 | 0.20 | n/a |
| Maize (grain) | field crop | 0.30 | 1.20 | 0.60 | 15/30/40/20 | 1.5 | 135 | 0.55 | 1.25 |
| Millet | field crop | 0.30 | 1.00 | 0.30 | 15/25/40/25 | 1.5 | 150 | 0.55 | n/a |
| Sorghum (grain) | field crop | 0.30 | 1.00 | 0.55 | 20/35/40/30 | 1.5 | 150 | 0.55 | 0.90 |

| Crop | Group | $K_{c,ini}$ | $K_{c,mid}$ | $K_{c,end}$ | Stages (d) | Height (m) | Root (cm) | $p$ | $K_y$ |
|---|---|---|---|---|---|---|---|---|---|
| Cassava | field crop | 0.30 | 0.80 | 0.30 | 20/40/90/60 | 1.0 | 65 | 0.35 | n/a |
| Yam | field crop | 0.50 | 1.10 | 0.85 | 30/60/90/60 | 1.5 | 60 | 0.40 | n/a |
| Cowpea (beans) | field crop | 0.40 | 1.05 | 0.50 | 20/30/30/20 | 0.4 | 80 | 0.45 | n/a |
| Green gram (mung bean) | field crop | 0.40 | 0.96 | 0.51 | 17/23/34/16 | 0.4 | 75 | 0.45 | n/a |
| Soybean | field crop | 0.40 | 1.15 | 0.50 | 20/30/45/25 | 0.6 | 95 | 0.50 | 0.85 |
| Groundnut | field crop | 0.40 | 1.15 | 0.60 | 20/35/40/25 | 0.4 | 75 | 0.50 | 0.70 |
| Cotton | field crop | 0.35 | 1.15 | 0.50 | 30/45/65/50 | 1.1 | 135 | 0.65 | 0.85 |
| Wheat | field crop | 0.40 | 1.15 | 0.41 | 20/30/45/25 | 0.8 | 165 | 0.55 | 1.05 |
| Chickpea | field crop | 0.40 | 1.00 | 0.40 | 15/35/45/25 | 0.4 | 80 | 0.50 | n/a |
| Mustard | field crop | 0.35 | 1.15 | 0.35 | 15/35/45/25 | 1.3 | 125 | 0.60 | n/a |
| Tomato | field crop | 0.60 | 1.15 | 0.80 | 30/40/40/25 | 0.6 | 100 | 0.40 | 1.05 |
| Potato | field crop | 0.50 | 1.15 | 0.75 | 25/30/45/30 | 0.6 | 50 | 0.35 | 1.10 |
| Onion (dry) | field crop | 0.70 | 1.05 | 0.75 | 15/25/70/40 | 0.4 | 45 | 0.30 | 1.10 |
| Cabbage | field crop | 0.70 | 1.05 | 0.95 | 40/60/50/15 | 0.4 | 65 | 0.45 | 0.95 |
| Sweet pepper | field crop | 0.60 | 1.05 | 0.90 | 25/35/40/20 | 0.7 | 75 | 0.30 | 1.10 |
| Table grapes | tree, vine or palm | n/a | n/a | n/a | 20/40/120/60 | n/a | 150 | 0.35 | n/a |
| Wine grapes | tree, vine or palm | n/a | n/a | n/a | 20/40/120/60 | n/a | 150 | 0.45 | n/a |
| Kiwi | tree, vine or palm | n/a | n/a | n/a | 20/40/120/60 | n/a | 100 | 0.35 | n/a |
| Highbush blue-berry | tree, vine or palm | n/a | n/a | n/a | 20/50/90/20 | n/a | 90 | 0.50 | n/a |
| Hop | tree, vine or palm | n/a | n/a | n/a | 25/40/80/10 | n/a | 110 | 0.50 | n/a |
| Citrus | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 135 | 0.50 | n/a |
| Olive | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 145 | 0.65 | n/a |
| Apple | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Pear | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Cherry | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |

| Crop | Group | $K_{c,ini}$ | $K_{c,mid}$ | $K_{c,end}$ | Stages (d) | Height (m) | Root (cm) | $p$ | $K_y$ |
|---|---|---|---|---|---|---|---|---|---|
| Peach | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Apricot | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Almond | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.40 | n/a |
| Hazelnut | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Walnut | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 205 | 0.50 | n/a |
| Pecan | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Pistachio | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 125 | 0.40 | n/a |
| Fig tree | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Persimmon | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Pomegranate | tree, vine or palm | n/a | n/a | n/a | 20/70/90/30 | n/a | 150 | 0.50 | n/a |
| Avocado | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 75 | 0.70 | n/a |
| Mango | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 120 | 0.50 | n/a |
| Date palm | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 200 | 0.50 | n/a |
| Coconut | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 120 | 0.50 | n/a |
| Papaya | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 90 | 0.50 | n/a |
| Guava | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 120 | 0.50 | n/a |
| Oil palm | tree, vine or palm | 0.90 | n/a | n/a | 60/90/120/95 | n/a | 90 | 0.65 | n/a |
| Cocoa | tree, vine or palm | 1.00 | n/a | n/a | 60/90/120/95 | n/a | 85 | 0.30 | n/a |
| Cherimoya | tree, vine or palm | n/a | n/a | n/a | 60/90/120/95 | n/a | 120 | 0.50 | n/a |

# Chapter 22

# ◷ Seasonal data: twenty seasons, year by year

The group *Seasonal data* of the Layers panel holds three layers built on the **year-by-year results of the simulations**: the twenty growing seasons of each run, not only their mean. They show how reliable a setting is from one year to the next. In version 1.0 the group was called *Year by year (Daisy data)*; the name of the simulation model has gone from the program and is given here.

## 22.1 Where the yearly values come from

For each soil the simulations hold the twenty yearly values of rain, irrigation, evaporation, transpiration, drainage, storage change, yield and nitrogen of the reference runs (irrigated and rainfed). A layer reads the run nearest your settings and moves each year by the response of the annual model to your settings,

$$P'_y = \rho\, P_y, \qquad X'_y = X_y \,\mathrm{clip}\Big(\frac{X^{\mathrm{model}}(\text{your settings})}{X^{\mathrm{model}}(\text{reference})};\ 0,\ 6\Big)\ \ (X \in \{I, T, E, E_{\mathrm{other}}, N_L, Y\}),$$

$$D'_y = \begin{cases} D_y \dfrac{D^{\mathrm{model}}}{D^{\mathrm{model}}_{\mathrm{ref}}} & \bar{D} > 5,\ D^{\mathrm{model}}_{\mathrm{ref}} > 5 \\ D_y + \left(D^{\mathrm{model}} - D^{\mathrm{model}}_{\mathrm{ref}}\right) & \text{otherwise} \end{cases}, \qquad ET'_0 = ET_0\,(1 + 0.035\,\Delta T), \tag{22.1}$$

*Provenance (Eq. 22.1):* **[Data-driven]**: the yearly values of the simulations, moved by the ratio (or, for small drainage, the difference) of Engine 1 between your settings and the reference of the run (your mulch, rain ×1.0, no warming, nitrogen 100 %); for a library crop the yield is the relative yield of Eq. (21.5) year by year and the irrigation follows the ratio of the two crop balances of Section 21.4.

so that every year follows the crop, rain, temperature, nitrogen, mulch, trigger and method you set, while keeping the weather sequence of the record. A sentence under each layer names the run used and the adjustment made. The calendar years on the axes end in the working year.

## 22.2 Reliability over 20 seasons

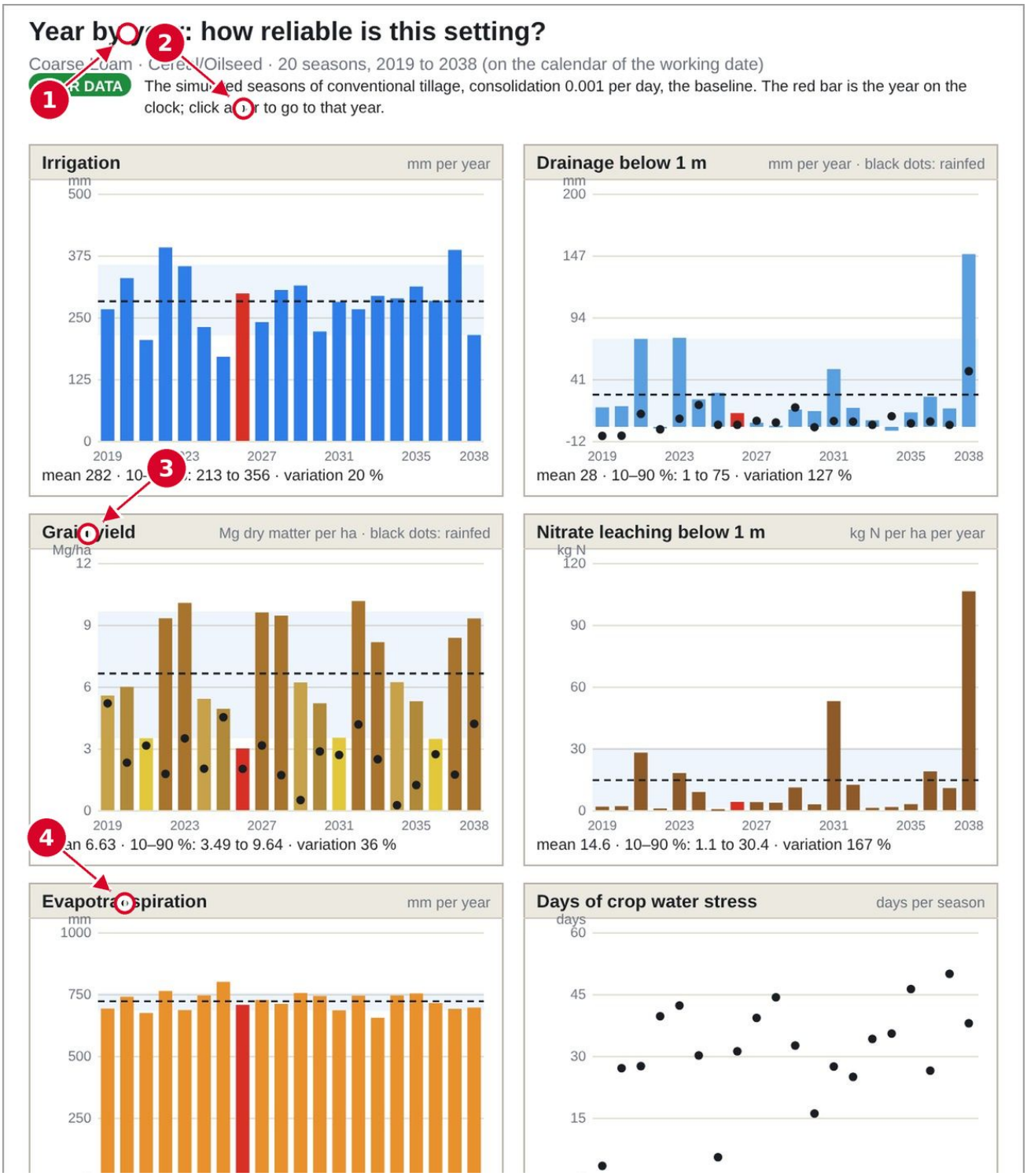



Figure 22.1: Reliability over 20 seasons.

*Reliability of this setting over the 20 seasons*: the irrigation, drainage, yield (or relative yield) and water stress of each season, with their mean and spread. *How often a given irrigation is exceeded*: the twenty seasonal irrigation amounts sorted into an exceedance curve, so that the

amount needed in four years out of five can be read off. Click a year to set the working year to it.

## 22.3 Full water balance

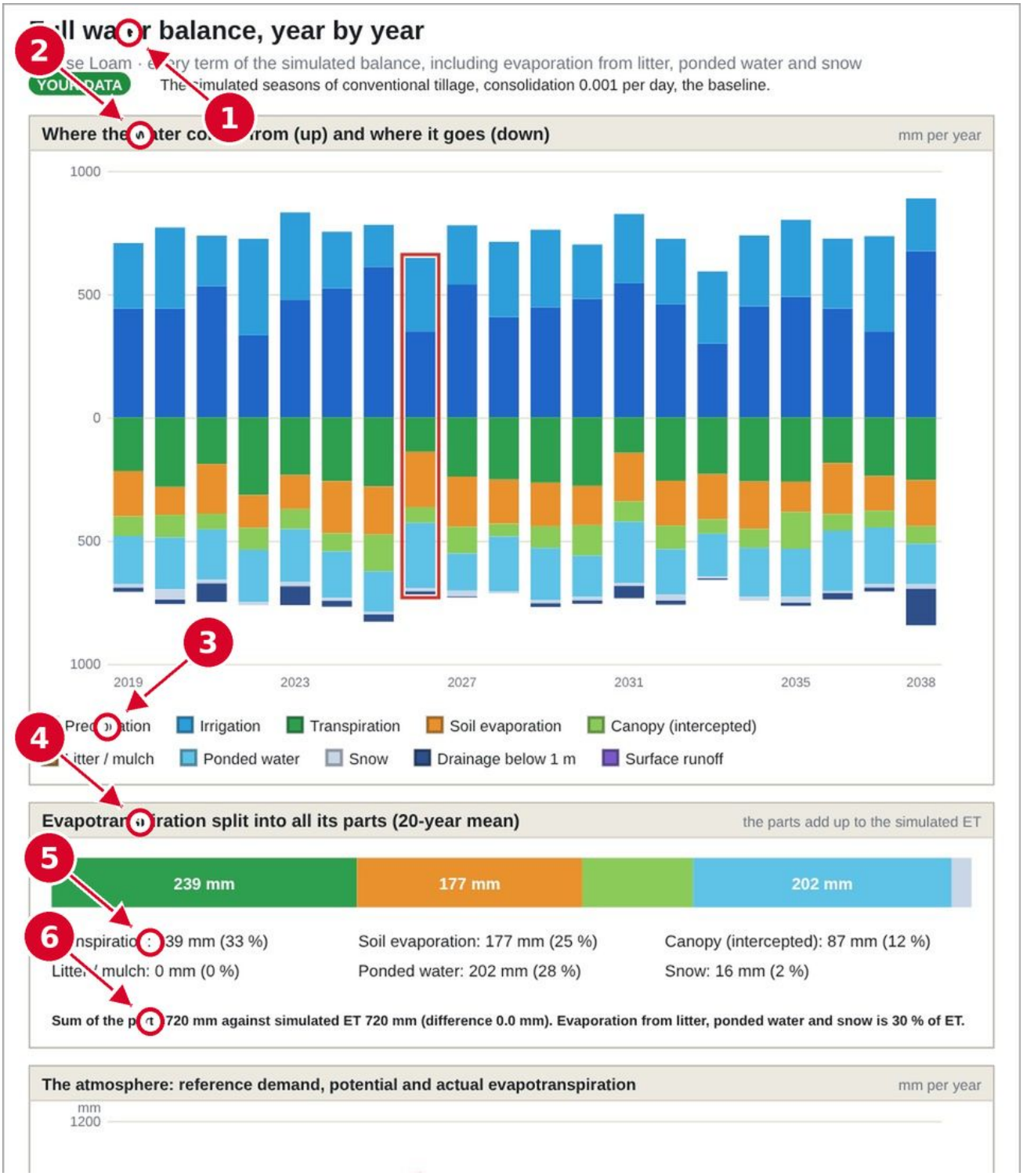



Figure 22.2: Full water balance.

*Where the water comes from (up) and where it goes (down)*: the complete balance of each season, $P + I = E + T + E_{\text{other}} + D + \Delta S$, stacked year by year, so that a dry season (high irrigation, little drainage) and a wet season can be told apart at a glance. *Evapotranspiration split into all its parts*: transpiration, soil evaporation, canopy, litter, ponded and snow evaporation (20-year mean). *The atmosphere*: reference demand, potential and actual evapotranspiration, year by year. *How the water enters and leaves the soil*: infiltration through the soil matrix and the macropores, runoff and drainage.

## 22.4 Nitrogen budget

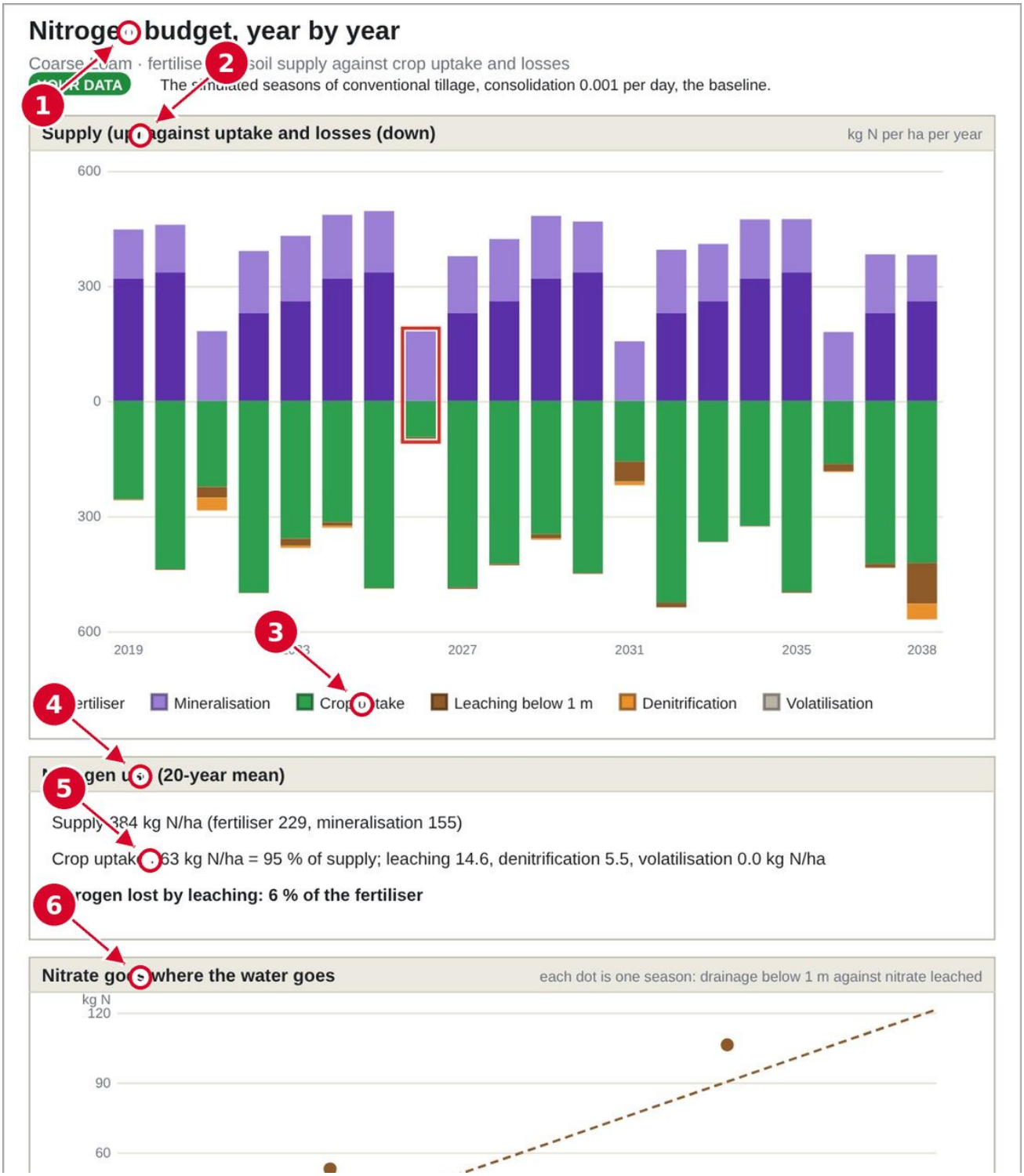



Figure 22.3: Nitrogen budget.

*Supply (up) against uptake and losses (down)*: fertiliser and mineralisation against crop uptake, leaching, denitrification and volatilisation, season by season; *Nitrogen use (20-year mean)*; and *Nitrate goes where the water goes*: the leaching of every season against its drainage, with the

regression line. The slope of that line is printed under the chart: *each extra millimetre of drainage carries … kg N/ha of nitrate*.

## 22.5 The dashboard, the charts and the design layers on screen

This chapter shows three screens of version 1.1.0 with numbered arrows, and gives the law or the principle behind every figure that the arrows point to. The screens were captured from the program with the benchmark weather, the soil Coarse Loam and the crop Cereal/Oilseed.

### 22.5.1 The dashboard results

The dashboard is the panel on the right of the layer. It has the colours of the Daisy workbench: a grey ground, cards with a thin border and beige title bars. Every layer fills it only after **Run simulation**; before that it says so and tells where to press the button.

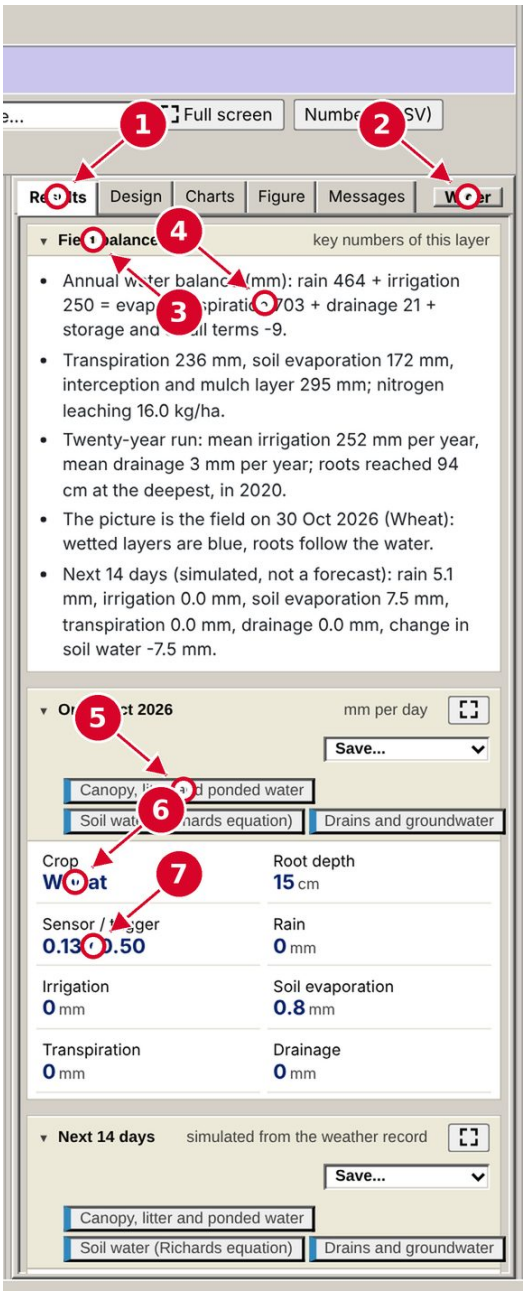



Figure 22.4: The Results tab of the layer Field balance after a run.

**What the arrows point to.**

1. **Results.** The tab of the key numbers. Design, Charts, Figure and Messages are the other tabs of the same run.
2. **Wider.** Widens the dashboard over the picture; the charts open wide by themselves.
3. **A title bar.** Each card has a title, a small line saying what it is, a button for full screen and a Save menu for the picture and the numbers.
4. **The sentences.** Each sentence gives the numbers of the season in words, in the order of the balance below.
5. **The process chips.** The chips name the processes of the model that produced the numbers; clicking one opens the process in the Daisy workbench.
6. **A value.** The large number is the value of the day shown in the picture, with its unit.
7. **Sensor and trigger.** The first number is the depletion at the sensor, the second the trigger; irrigation starts when the first reaches the second.

**Governing principle.** The annual water balance of the first sentence is the balance of the soil column,

$$P + I = ET + D + \Delta S, \qquad ET = E_s + E_i + T,$$

with rain $P$, irrigation $I$, evapotranspiration $ET$ (soil evaporation, interception and mulch evaporation, transpiration), drainage $D$ and storage change $\Delta S$, all in mm. The small term printed after drainage is $\Delta S$ together with the closing error of the model.

## 22.5.2 The charts of a layer

The Charts tab draws the working season day by day, the water in the root zone against the twenty seasons, and the twenty seasons year by year. The figures follow the style of a journal: a black frame with outward ticks, a light dotted grid, markers on the series, lines and bars with a shared legend, and no panel letters.

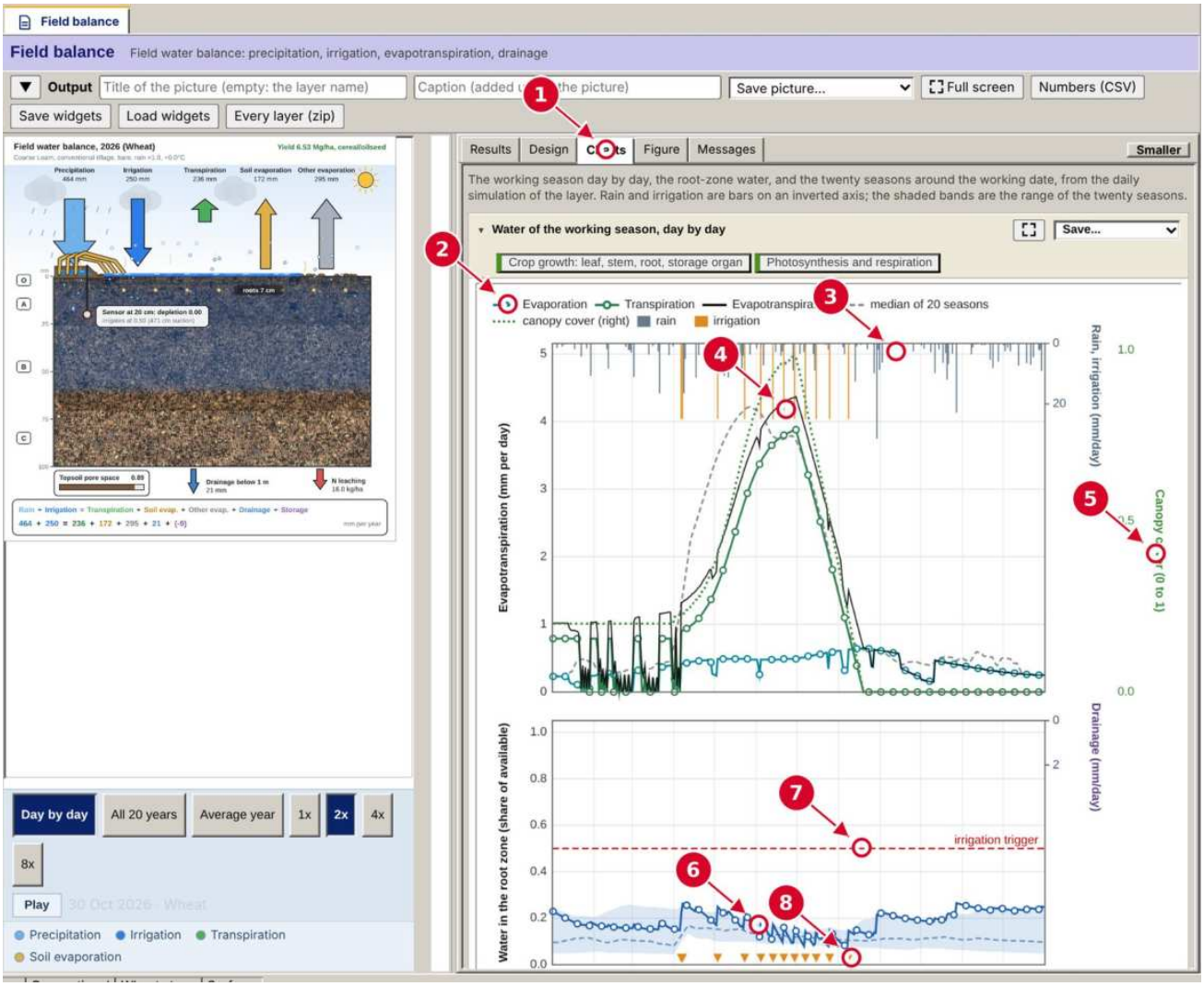



Figure 22.5: The Charts tab of the layer Field balance.

**What the arrows point to.**

1. **Charts.** The tab. The dashboard widens by itself and the figures redraw whenever its width changes.
2. **The legend.** Evaporation $E$, transpiration $T$ and their sum, the evapotranspiration, with the median of the twenty seasons as a dashed line, the canopy cover on the right axis, and the bars for rain and irrigation.
3. **Rain and irrigation.** Bars that hang from the top, on an inverted axis in mm per day, so that the water that arrives is drawn above the water that leaves.
4. **Evapotranspiration.** The black line is $ET = E + T$ per day; its peak is the demand of the crop at full cover.
5. **Canopy cover.** The right axis, from 0 to 1, is the share of the ground covered by the leaves; it sets how the demand is divided between $E$ and $T$.

6. **Root-zone water.** The blue line is the water in the root zone as a share of the available water, with the shaded 10 to 90 % range of the twenty seasons and the dashed median.
7. **The irrigation trigger.** The red dashed line is the share at which irrigation starts.
8. **Irrigation events.** Each triangle at the bottom is a day on which the layer irrigated.

**Governing principles.** The quantities of the figure are

$$w = \frac{S - S_{WP}}{S_{FC} - S_{WP}}, \qquad \text{irrigate when } w \le 1 - f,$$

where $S$ is the water stored in the root zone, $S_{FC}$ and $S_{WP}$ its value at field capacity and at the wilting point, and $f$ the depletion fraction of the crop (the critical depletion of Properties). The daily demand is the reference evapotranspiration times the crop coefficient, reduced by the water stress coefficient when the root zone is drier than the trigger,

$$ET_c = K_c\, ET_0, \qquad ET = K_s\, ET_c \;\; (K_s \in [0,1],\; K_s = 1 \text{ while } w > 1 - f),$$

and the canopy cover $c$ divides it: the transpiration $T$ grows with $c$ and the evaporation $E$ of the soil surface falls as the canopy closes, $E + T = ET$. The year by year chart sums these daily values over each season; the means in the legend are the means of the twenty seasons, and the shaded band is the range between the 10th and the 90th percentile. Chapter 5 gives the full equations of the two engines.

### 22.5.3 An irrigation design layer before the run

The three irrigation design layers (surface, drip, sprinkler) open with their inputs at the top of the picture. Nothing is computed until the design is run, so that the results never change while the inputs are typed.

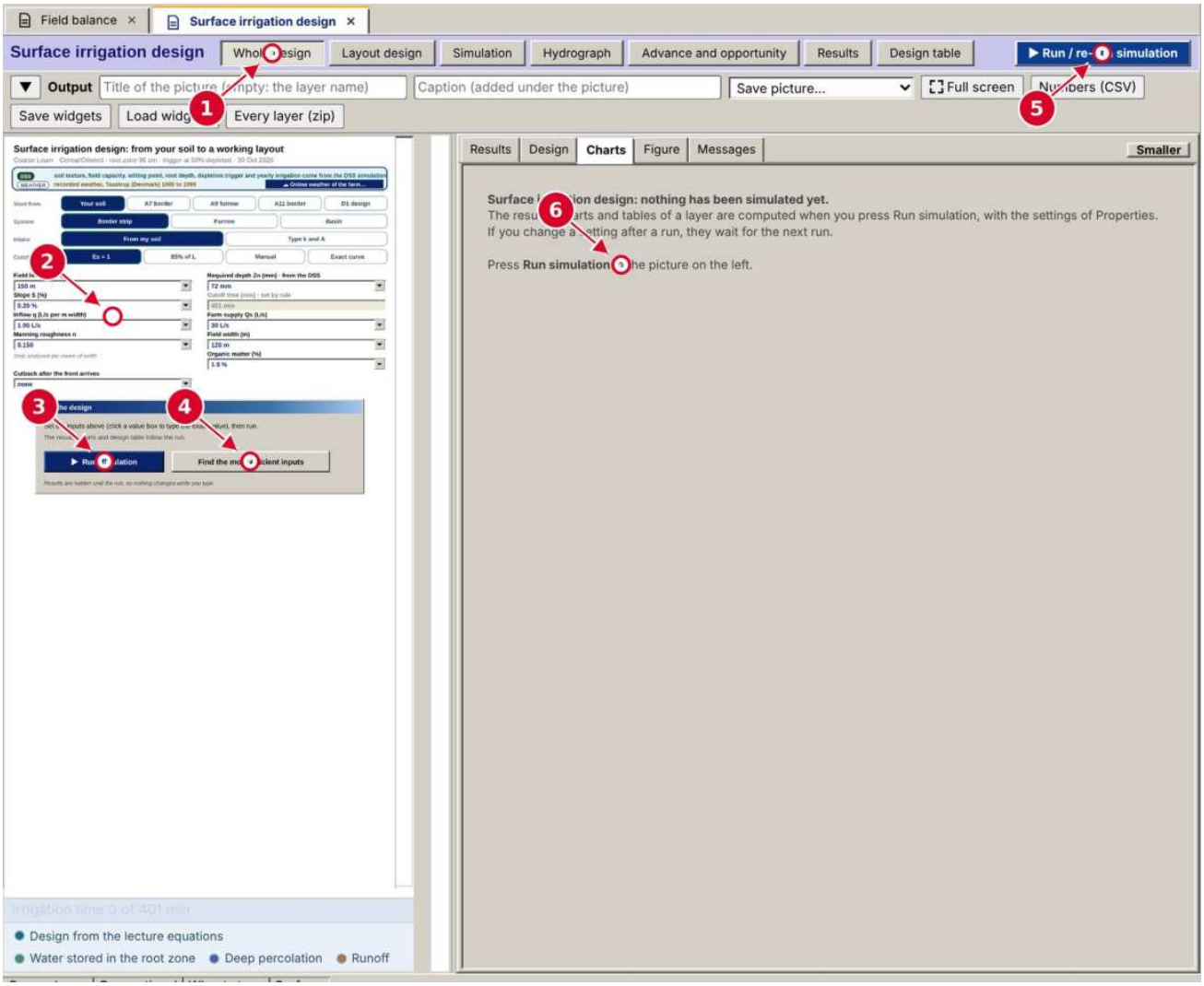



Figure 22.6: The layer Surface irrigation design before the run.

**What the arrows point to.**

1. **Whole design.** The first of the views of the layer; the other buttons jump to one part of the picture.
2. **The inputs.** Click a value box to type the exact value or to pick one from the list; buttons choose the system, the intake and the cutoff.
3. **Run simulation.** Computes the design with the inputs above it. In these layers the button is in the picture, in the dashboard and in the header; the bar under the picture shows Pause and Play after the run.
4. **Find the most efficient inputs.** Searches the inputs for the highest application efficiency, applies them and runs the design.
5. **Run / re-run simulation.** The same command in the header of the layer, so that the picture does not need to be scrolled.
6. **The waiting message.** The dashboard says that nothing has been simulated and points to

the button.

**Governing principles.** The design of a surface layout balances the volume of water with the time it needs to soak in. With the infiltrated depth $Z(\tau) = k\,\tau^{A}$ (the Kostiakov equation, with $k$ and $A$ fitted to the soil or typed by you) and the required depth $Z_n$, the layer finds the opportunity time $\tau_n$ at which $Z(\tau_n) = Z_n$, and judges the design by the application efficiency and the distribution uniformity,

$$E_a = \frac{V_{stored}}{V_{applied}}, \qquad DU = \frac{Z_{lq}}{Z_{avg}},$$

where $V_{stored}$ is the water kept in the root zone, $Z_{lq}$ the mean depth of the lowest quarter of the field and $Z_{avg}$ the mean depth. Chapters 11 to 13 give the equations of each system.

## 22.6 Projecting the seasons after the last simulated one

The year charts of the three layers, the charts of the dashboard for them and the yearly figure of the Charts tab can continue after the twenty simulated seasons. In Properties, the group *Projection* sets **Seasons projected** (none, 10 or 20 years), the **warming per decade** (default +0.3 °C) and the **rainfall change per decade** (default 0 %).

Each projected season is one of the twenty simulated seasons, taken in a fixed random order (the same order every time for the same soil and crop), and moved by the trend of the settings: the reference evapotranspiration rises by 3.5 % per degree of added warming; irrigation follows it (1.2 times that change) and falls with added rain (0.8 times the rainfall change); drainage and the nitrate leached follow the rain (1.5 times the change) and fall with the warming (0.6 times 3.5 % per degree); evapotranspiration and transpiration change by 0.4 of the warming effect and 0.3 of the rainfall change; the yield falls by 2 % per degree and rises by 0.1 of the rainfall change; the days of water stress rise with the warming and fall with the rain. The trend is measured from the middle of the twenty simulated seasons.

Projected seasons are drawn lighter, right of a dashed line marked *projection*, on the same calendar as the simulated seasons, so that the red bar of the working date stays on its year. The mean, the 10 to 90 % range and the reliability figures are those of the twenty simulated seasons only; the layer adds a line that gives the change of the projected seasons against them. A projection shows where a setting is heading under a stated trend; it is not a weather forecast. Setting **Seasons projected** to *None* returns the charts to the twenty simulated seasons.

# Chapter 23

# ☷ Reference of every layer: controls, properties and data sources

This chapter gathers, for each layer of the *Layers* panel, three things in the same order: the **controls** on the layer itself, the **Properties** that move it, and the **data sources** behind its numbers. Whatever you set in Properties reaches every layer, the dashboard, the report and the irrigation designs, because they all read the same set of inputs. Press **Run simulation** in the bar under the picture to compute a layer after a change.

## 23.1 The Properties that every layer shares

| Group of Properties | What you set | Layers it moves |
|---|---|---|
| Scenario | Ready made case, year, date in the season, weather source (built in record, imported file, online weather of a place) | all |
| Soil and field | Soil (nine classes), tillage (conventional, reduced, no till), structural consolidation rate | soil, mulch, control, seasonal, crop, designs |
| Crop | One list of 48 crops: field crops, vegetables, trees, vines and palms | field, seasonal, crop, designs |
| Mulch | Material, strength (vapour flux factor), interception capacity | mulch, loops, field, seasonal |
| Weather and climate | Rainfall scale, temperature shift | all that use the weather |
| Projection | Seasons projected after the last simulated one (none, 10 or 20 years), warming per decade, rainfall change per decade | the seasonal layers, the dashboard charts of the seasons, the yearly figure of the Charts tab |
| Irrigation | Method, decision horizon, crop in the decision, soil water on the date, field area, accumulated salt, salinity of the water, critical depletion, trigger and sensor depth | control, designs, field, seasonal |
| Nutrients | Nitrogen rate as a percentage of the standard | seasonal (nitrogen budget), feedback loops |
| Design shortcuts | Opens the design of the chosen method, the field sketch with its pipes, and the three designs | designs |

The toolbar **Date** is the key date: the working year, the red bar of every year chart, the season

shown and the irrigation advice all follow it.

## 23.2 Irrigation design: surface, drip and sprinkler with pipes

**Controls.** Every number of the drawing is a value box: click it to open a picker with the exact value, its unit (metres or kilometres, litres per second or cubic metres per hour, hectares or square metres) and a list of values across the allowed range. **Run simulation** computes the design; **Find the most efficient inputs** searches the inputs for the highest application efficiency; *Optimise surface* applies the best surface design; the tabs **Results, Design, Charts, Figure, Messages** show the numbers, the inputs, the hydrograph and depth charts, the picture and the log. The sprinkler layer draws the pipe network (source, pump, filter, fertigation, valve, mainline, submain, laterals) with the diameter, length, flow, speed and head loss of each pipe.

**How it is generated.** Surface: the intake of your soil is built from its texture (Saxton and Rawls to Green and Ampt to Kostiakov), the inflow is turned into a flow depth by Manning, the advance by a volume balance, and the cut off time, depths, efficiencies and uniformity follow from the advance and the intake (Chapter 11). Drip: the emitter flow along the lateral comes from the head loss of the lateral (Hazen and Williams, Keller and Karmeli); the water of one emitter fills a wetted bulb whose volume is the water divided by the water the soil can still store, and whose shape depends on the soil (narrow and deep in sand, wide and shallow in clay loam, Eq. 12.11); the part of the bulb below the roots, the evaporation from its wet surface and the water stored in the root zone give the efficiencies (Chapter 12). Sprinkler: the application rate and the overlap of the heads, the wind and evaporation losses, and the pipe sizes from the flow and the allowed speed (Chapter 13). In all three the crop water need comes from the crop library and the weather.

**Properties.** Soil, crop, method (which decides the design layer), field area (the same area in the three designs), salt, trigger and weather. The drip depth of the method sets the depth of the laterals. The mainline length, the mainline and submain diameters and the lateral diameter can come from the field sketch (see below).

**Data sources.** Intake from the soil: Saxton and Rawls pedotransfer functions to Green and Ampt to Kostiakov. Flow in the inlet: Manning. Advance and recession: volume balance (Walker and Skogerboe). Pipe friction: Hazen and Williams, with C = 140 for the main and submain and 150 for the laterals. Standard pipe diameters from 32 to 315 mm. Crop water: FAO paper 56 (Allen and others, 1998). Weather: the weather in use.

### 23.2.1 Designing the pipes in the field sketch

The field sketch (**Tools, Field sketch to design**) has pipe tools beside the tools for greenhouses, beds, irrigated areas and plants. Choose **Mainline pipe**, **Submain pipe** or **Lateral pipe**, click the points of the route and double click (or press Enter) to finish; a list of diameters (in mm) appears while a pipe tool is chosen and changes every pipe of that kind. Place a **Valve**, **Pump**, **Filter** or **Tank** with one click. Every pipe shows its diameter and length on the drawing.

The Summary step lists the pipe network: for each pipe the diameter, length, flow, speed and friction loss (Hazen and Williams), whether the speed is good (too fast above 1.5 m/s in the main and submain and 2 m/s in a lateral, very slow below 0.3 m/s), and the pump head the network asks for (lift, friction, and the working head and fittings) with the power at 65 % efficiency. **Turn the sketch into the design** hands the drawn mainline length, the mainline and submain diameters and, for drip, the lateral diameter to the design, which then uses them in place of its own sizing; its layout then says “sketch” beside the mainline.

## 23.3 Irrigation control: sensor and trigger, trigger law

**Controls.** The sensor and the trigger are sliders in Properties (trigger suction and sensor depth); the layer plays the season by itself and **Play** and **Stop** control it. The soil row and the shaded band mark the soil only, not the chart beside it.

**How it is generated.** The soil column is the daily soil water model of the program; the sensor reads the suction of the layer at its depth from the retention curve of the soil; the trigger is a suction (or a depletion fraction of the available water, with the critical depletion f of the crop), and an irrigation is called on the day the sensor passes it. The yield, irrigation and drainage of each soil at each trigger are the results of the Daisy experiments, read by the model (Chapter 6).

**Properties.** Soil, trigger, sensor depth, critical depletion, crop, mulch, weather.

**Data sources.** The Daisy experiments on the trigger (E2) and on the sensor depth (E3) for the nine soils; the retention curve of each soil (Saxton and Rawls); the weather in use.

## 23.4 Mulch and evaporation: mulch against bare soil, mulch control, evaporation, fate of saved water, regime map

**Controls.** The mulch sliders (strength, capacity) and the material list in Properties; the sliders of the layer for the physics of evaporation; Play, Stop and the year clicks.

**How it is generated.** The mulch lowers the vapour flux from the soil by the strength you set and holds a depth of rain (the capacity) that evaporates from the mulch. The effect on evaporation, drainage, irrigation and yield is the ratio of the Daisy runs with and without the mulch, taken from the grid of mulch strength, rain and capacity, and moved by the temperature grid (Chapter 9).

**Properties.** Mulch group, soil, tillage, rainfall scale, temperature shift.

**Data sources.** The Daisy experiments on the mulch (E14 and E7) over strength, rainfall and capacity; the temperature and rainfall grid (E11); the equations of vapour diffusion and surface resistance given in the chapter of the layer.

## 23.5 Soil structure and tillage: tillage and structure, soil hydraulic drivers, root structure

**Controls.** The tillage buttons and the consolidation slider in Properties; click a soil row of the table to select it (the highlight covers the soil table only); Play and Stop for the consolidation in time.

**How it is generated.** Tillage sets the pore index of the topsoil and the consolidation rate says how fast it re settles; the drainage, evaporation and yield of each tillage and rate are the Daisy results for the nine soils, and the soil table ranks the soils by the effect of each driver (Chapter 8).

**Properties.** Soil, tillage, structural consolidation rate, crop.

**Data sources.** The Daisy experiment on tillage and consolidation (E5); the hydraulic parameters of the nine soils (Saxton and Rawls, with the Daisy parameters of the soil classes); root distribution of the crop kind of the library.

## 23.6 Deterministic phenotyping

**Controls.** The sowing mode, species, genotype and climate lists of the layer, its sliders and its Run button.

**How it is generated.** Each genotype is a set of traits; the crop model grows canopy, roots and yield day by day on the weather and the soil water, and the table reports the traits that result. The same inputs always give the same output (Chapter 8b).

**Properties.** Soil, tillage, rainfall, temperature and the weather in use.

**Data sources.** The phenotyping models of the program (crop growth, canopy, roots and traits) driven by the weather in use; the traits typed into the layer.

## 23.7 Field water balance: field balance, soil plant and air, profile evolution

**Controls.** The control bar under the drawing (play, stop, speed, the year and the day), the click on a year bar to move the clock, and the decision panel of Properties (horizon, crop in the decision, soil water on the date, field area).

**How it is generated.** Rain and irrigation enter, the crop and the soil evaporate and transpire, the rest drains or runs off; the soil water is stepped one day at a time, the crop follows the crop library, and the picture is drawn from the day on the clock (Chapter 7).

**Properties.** Every group of Properties.

**Data sources.** The daily simulation of the program (soil water, crop, mulch and irrigation) on the weather in use; the soil and crop libraries; the irrigation advice uses the area to give volumes.

## 23.8 Seasonal data: reliability over the seasons, full water balance, nitrogen budget

**Controls.** Click a bar to move the clock to that year; the red bar is the year on the clock. Hover a bar for its value. Dashed lines are the rainfed runs. **Seasons projected** in Properties (none, 10 or 20) adds seasons after the last simulated one.

**Projection.** Seasons after the twenty simulated ones are drawn lighter, right of a dashed line marked *projection*, on the calendar that follows the working date. Each projected season is one of the twenty simulated seasons, taken in a fixed random order, and moved by the climate trend of Properties: the warming per decade (default +0.3 °C) raises the reference evapotranspiration by 3.5 % per degree, which raises irrigation and crop water use and lowers yield by 2 % per degree; the rainfall change per decade (default 0 %) scales the rain and, through it, the drainage, the nitrate leached and the irrigation. The means, the 10 to 90 % range and the reliability figures always use the twenty simulated seasons only. A projection shows where the setting is heading; it is not a weather forecast.

**How it is generated.** For each soil the Daisy runs hold twenty years of rain, irrigation, evaporation, transpiration, drainage, yield and nitrogen. Your settings move these years by the ratio of the annual model at your settings against its reference; for a library crop the daily FAO balance of that crop replaces them season by season (Chapter 22).

**Properties.** Every group; crop and weather move the seasons most.

**Data sources.** The yearly values of the Daisy reference runs (irrigated and rainfed) for each soil, held in the program for the twenty seasons; the ratios of the annual model between your settings and the reference; the crop library for a library crop (daily FAO paper 56 balance, season by season).

## 23.9 Crops: crop water requirement

**Controls.** The crop list, the year and the dates; the crop curve and the daily balance of the season in the picture.

**How it is generated.** The crop coefficient curve of the crop (initial, mid and end values over four growth stages) multiplies the reference evapotranspiration of each day; rain, soil storage and the depletion fraction give the irrigation need and the stress (Chapter 21).

**Properties.** Crop, soil, trigger, weather, salt, method.

**Data sources.** The crop library of 48 crops: crop coefficients, stage lengths, height, rooting depth, depletion fraction and yield response factor (FAO papers 56 and 33, and a field study of daily crop water requirement with the FAO Penman and Monteith evapotranspiration); the weather in use. Crops not tabulated take the default of their group and say so.

## 23.10 Feedback loops

**Controls.** The sliders of mulch, rain and temperature in Properties and the click on an arrow of the loop to read its effect.

**How it is generated.** Seven effects of the mulch (on evaporation, temperature, drainage, nitrogen, structure, roots and yield) are computed as ratios of the Daisy runs and shown as loops that feed each other (Chapter 10).

**Properties.** Mulch, weather and climate, soil, nitrogen, crop.

**Data sources.** The Daisy experiments E14 and E11 (mulch, rain, capacity, temperature) and the annual model built on them.

## 23.11 Daisy model workbench

**Controls.** The tabs of the workbench (data, library, experiments, scenarios, process map, log plots, mass balance, comparison, set up editor, model builder, soil column, denitrification, learning, documentation, reference manual); every tab has a **Manual p. N** button. The Set up editor opens the folders and files of the library, formats, saves and runs; a very large file or an output log (.dlf, .dwf, .csv) is sent to Log plots instead of the editor, so the editor never has to draw it.

**How it is generated.** The Daisy tabs write the files of a Daisy run (set up, weather, soil, management), run Daisy on the desktop (or save the folder to run elsewhere), and read its logs for the figures; the decision support layers are built from the logs of your own runs (Chapters 26 to 38).

**Properties.** The Daisy tabs read the weather and the soil of the project when you press *Use the project*; the Decision support layers (chemical fate, decision table, water balance, nitrogen balance, yield risk, chemicals and drinking water, trade offs, return of irrigation, nitrous oxide) read the runs you made. Their year charts also carry the projection of Properties.

**Data sources.** The Daisy model (Hansen and others, 2012) and its reference manual; the parameter library (63 soil horizons, 84 crops, 79 fertilisers, 35 chemicals, 82 managements, 55 programs, 176 logs); the weather files and the runs you produce.

## 23.12 Weather, field and map data

| Source | Used for | How it is reached |
|---|---|---|
| Built in benchmark record, 1980 to 1999, projected to 2100 | the default weather of every layer | inside the program |
| Imported weather file (CSV, Excel, DWF or DLF) | your own field weather | Import weather in Properties |
| Open-Meteo forecast, archive and geocoding | online weather of a place, twenty years of measured weather and the forecast | Online weather in Properties |
| Esri World Imagery, OpenStreetMap | the map and the satellite picture of the field | Field on the map |

| Source | Used for | How it is reached |
|---|---|---|
| Sentinel 2 L2A (Microsoft Planetary Computer, Element 84) | vegetation and water indices of the field | Field on the map |
| SoilGrids 2.0 (ISRIC) | the soil of the field and its retention curves | Field on the map |
| ESA WorldCover 2021, Overpass (OpenStreetMap) | land cover, roads, water sources near the field | Field on the map |
| Open-Meteo elevation, terrain tiles | the slope and the terrain of the field | Field on the map |

Online sources need a connection; every other number of the program is computed on your computer.

## 23.13 Drip wetting by soil

The wetted bulb of the drip design depends on the soil: the shape factor of Eq. 12.11 rises with the saturated conductivity, so a sand wets a narrow, deep bulb and a clay loam a wide, shallow one, and the volume of the bulb is the water of the emitter divided by the water the soil can store. Choose another soil in Properties and run the design to compare.

## 23.14 Ending a pipe in the field sketch

While a pipe tool is chosen, a pipe ends when you click its last point again (the dead end), when you click on any other pipe (it joins that pipe at the point clicked), or when you double click or press Enter.

# Chapter 24

# ✎ Reports

## 24.1 Choosing the layers

*File* ▸ *Save report as PDF (choose the layers)…* (Ctrl+Shift+P, or the *Report* button of the toolbar) opens a window with every layer of the Layers panel, in its groups. **Every layer is ticked to start with.** Untick a layer, or a whole group with its box, or everything with *All layers*; the count at the bottom tells how many layers are ticked. **Make the PDF** builds the report and asks where to save it.

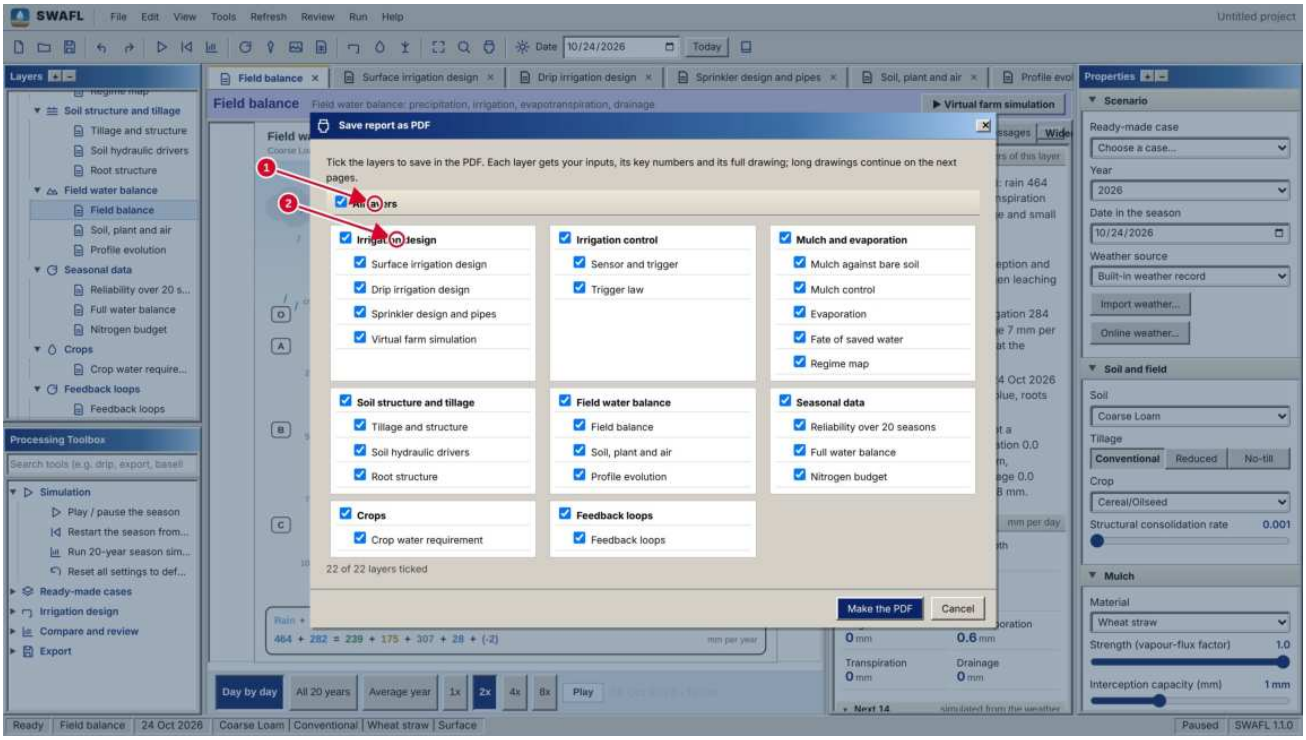



Figure 24.1: The report window: every layer ticked.

## 24.2 What the report holds

- **The first page**: your inputs (soil, tillage, crop, consolidation rate, mulch, rain, temperature, trigger, sensor depth, method, nitrogen), the results at these inputs (the 20-year averages), the water balance of the year, and the list of the layers included.

- **For each layer**: its name and purpose, the sentence that sums it up, the key numbers at your inputs, and **the whole drawing from top to bottom**. When the key numbers run past the end of the page they continue on the next page. When the drawing is longer than the page it continues on the next pages, cut at its section headings (or, where a layer has none, at a blank band between two parts), so no part of a picture or of its text is hidden or cut through.
- **The design layers** are drawn at the end of the irrigation, so every result is complete.
- **The field balance** appears three times: day by day, all twenty years, and the average year.

A report of every layer has about 60 to 70 pages and takes a few seconds to build.

## 24.3 The page and its numbers

The pages are A4. The drawing of a layer is placed at the full text width (515 points) and cut into parts of at most

$$h_{\max} = 748 \text{ pt} \times \frac{W}{515} \ \ [\text{px of the drawing}], \tag{23.1}$$

*Provenance (Eq. 23.1):* **[Engineered]**: page geometry; $W$ is the width of the drawing in pixels, so a wide drawing is cut into taller parts than a narrow one.

and the first part fills the room left under the key numbers when it is at least 220 points.

## 24.4 Saving the outputs of a layer

The bar *Output* above the simulation window works for every layer in the same way as the figures of the Daisy workbench.

1. **Title and caption.** Write the title that the picture should carry (empty: the name of the layer) and a caption. Under the picture the program adds the settings it was made with (the soil, the tillage, the crop, the date) so that the picture can be understood alone.
2. **Save picture.** Screen resolution, 300 dpi, 600 dpi or the highest resolution the computer allows. The layer is drawn again at that size, so lines and letters stay sharp; tall layers are limited by the memory of the computer, and the size in pixels is reported when the file is saved. *Copy the picture* puts it on the clipboard.
3. **Numbers (CSV)** saves the settings and the live results of the layer. **Save widgets** keeps the position of every control in a small file, and **Load widgets** sets them again.
4. **Every layer (zip)** draws all the layers at 300 dpi and saves them with their numbers in one zip.
5. The triangle at the left of the bar collapses the picture and expands it again.

Every section of the dashboard (Results and Charts) has a *Save* menu at the right of its header: its picture at 300 or 600 dpi, a copy of the picture, and the numbers of its tables and charts as a CSV file. The tab **Figure** of the dashboard shows the map of the processes of Daisy with the processes the controls of the layer act on in orange and those its results come from in blue. Click a process and the outputs that depend on it are highlighted in the list under the map; click an output and its processes are shown and the layer scrolls to it. The map can be enlarged and saved as a 300 dpi picture or as an SVG.

# Chapter 25

# ★ What changed from version 1.0 to version 1.1

## 25.1 The window

- The program, its title bar, the status bar and the About window show **1.1.0**. The build removes any older program before building, and the program clears its cache on start, so an old version cannot be shown by mistake.
- The **Layers panel** has a new order: Irrigation design, Irrigation control, Mulch and evaporation, Soil structure and tillage, Deterministic phenotyping, Field water balance, Seasonal data, Crops, Feedback loops. The short codes of the layers (S2, V1, N4, …) are gone; layers are named by their names only.
- The **year strip** and the date bar under the drawings are gone; the year (1980 to 2100) and the date are set in Properties and on the toolbar.
- The **dashboard** is laid out like the Properties panel and every block folds on its own; the Charts tab draws the hydrograph, the times of the irrigation and where the water goes for each design (Chapter 14).
- **Properties**: the *Weather of years after 1999* selector and the explanatory notes under it, under the consolidation rate and under the design shortcuts have gone; their content is in this manual. *Weather source* chooses the built-in record, an imported file or the online weather.
- The name of the simulation model is no longer shown anywhere in the program; badges that said so now say *Simulated*.
- Labels are drawn after the pictures and moved apart when they would overlap; text near the edge of a drawing is moved inwards instead of being cut; long drawings take their full height at once.
- On screens narrower than 1500 pixels the docks give way so that the drawing keeps its size; the drawing always fits the window, nothing is cut off on the right.
- The selected layer is marked in light blue; the Play/Pause button is small.
- The first block of the Results tab gives the key numbers of the open layer (Chapter 14).
- Layers that play by changing a setting (each soil in turn, rainfall from dry to wet) wait for **Play** in the program, so Properties and the dashboard keep your values.
- Variable-rate irrigation is also computed cell by cell (10 m): the depth and the date of the next irrigation vary continuously over the field, in smooth colours stretched to the range of the field; out of the crop season the vegetation seen by Sentinel-2 drives the water use, so the maps are no longer all grey.

- All the maps of the field are drawn in continuous colours with smooth interpolation by default (indices, monitoring, erosion, retention, sensors); pixels and classes remain one choice away.
- Daisy: a new tab *Learn the Daisy files* (Documentation) teaches the files of a Daisy experiment one by one (the folder, the grammar, the soil, the management, the logs, the weather, the run file, running and reading the results, Daisy's messages, templates) with a complete sample experiment. Every file carries numbered labels explained in a table; the sample was run with Daisy 7.1.14 (water balance error 0.0 mm), its logs are included, and every error message shown was produced by Daisy itself (chapter Learn the Daisy files).
- Daisy: the process map keeps the original Fig. 1 as its default and adds a standalone figure with every component of Daisy: Fig. 1 unchanged, and under it every other component as a box in the same style, grouped by module, with ten new ones from the reference manual. Either figure saves as SVG or PNG (300 or 600 dpi).
- Daisy: Log plots makes one independent figure for every quantity of a log, with its own controls (kind of line, zoom by dragging, values under the cursor, titles, saving), a caption and its key numbers.
- Daisy: Run comparison gives a verdict in plain words, a table with the means, the differences, the RMS difference, the largest difference and its date, r and the Nash-Sutcliffe efficiency, three figures for every quantity, and a double-blind comparison with a sealed key.
- New SWAFL layers from Daisy runs: D2 Daisy water balance, D3 Daisy nitrogen balance and D4 Yield risk and limiting factor.

## 25.2 Figures of experiments and the Daisy layers

- **To the SWAFL layers is clearer.** A status panel at the top says whether you have runs, opens the best layer for your question with one button, and every layer card is marked *ready* (green) or says in one line what it still *needs* (amber), with a button that makes those runs. The topic list and the list of all layers are folded at the bottom.
- **The data of the four papers.** The per-experiment outputs of all 10 190 Daisy runs (folder `Data/Comprehensive_Summary` of the repository) are the data of the four papers of the project; the fourth gives a closed form for the reach of a suction trigger, which the Trigger law layer shows for every soil.
- The **file viewer and editor** (Notepad++ colours) colours and numbers only the lines in view, so logs of several megabytes open, scroll and are edited at once without the program hanging; wrap is kept for files below 300 kB.
- The Layers panel and the panel of the real field are shown in this manual as they appear in version 1.1.0.
- **Why a run gave no results**: the root cause of every Daisy error in plain words (unknown name, numerical problem, value out of range, missing file, weather, syntax), with the line of the set-up and what to do; runs with the same cause together.
- Horizons taken from the Daisy samples no longer bring the logs of their own run (the error Unknown 'log' model 'crop' is gone).
- **See all the Daisy output files**, at the bottom of Experiments: every experiment, run and file in order, in the Notepad++ viewer.
- Log plots follow the runs: new runs replace those of the previous experiment.
- **Summary of an experiment**: what was varied, the range of every result, the effect of every input on every result as a coloured table and in words; saved as summary.csv.
- **Log plots with several runs**: every run with the same log together, or two runs against each other, in any of the figures; a new *Full range over time* view.
- **Log plots: twenty kinds of figure** in six groups (over time, totals and means, the seasons, distribution, relationship, over depth), each explained in one line, with only the controls it

needs.

- **View / edit the Daisy outputs**: every file Daisy wrote (.dlf logs, daisy.log) opens in a full-screen editor in the style of Notepad++ (tabs, line numbers, find), one file at a time or all runs together; the edited text is used in every save and the text Daisy wrote is kept beside it.
- The guide of Experiments starts with choosing a ready experiment (Best combinations, By topic, Prepared experiments, Import, Load this experiment).
- Daisy runs are kept in **SWAFL runs next to Daisy** (the Daisy folder when it can be written, otherwise next to the Daisy of SWAFL), no longer in Documents.
- Figures of every log: **monthly, yearly and cumulative** besides through the year and every day; the Log plots tab has monthly and cumulative values too.
- **Compare similar runs** (Log plots): the same log of several runs in one figure and each run minus a reference.
- **Which combination is best**: every combination ranked with weights the person sets, and the best value of each input.
- **All the outputs of every run**: every run in a folding group, files ticked one by one or by kind, saved together or all at once.
- Results in **folding groups**.
- Five new **soil hydraulic sensitivity factors**: K_sat, alpha, n, Theta_sat and l of the van Genuchten-Mualem curves of every horizon.
- The **request box, *Make a plan from my request* and the online AI are removed** from Experiments: plans start from the Daisy reference (best combinations, topics), the prepared experiments, the experiments of the study or imported set-ups.
- **Every log of every run**: every quantity of every log Daisy wrote becomes a journal-style figure (panels by one factor, lines by another, a band for the spread): daily logs through the year or over the whole period, annual logs year by year, depth logs as depth-time maps and profiles.
- Experiments write daily crop, water and soil-water logs by default, so the time series are always there.
- **Save both**: the outputs of the experiment and the raw files of Daisy (with the set-up) in one .zip.
- The table of raw files names each run once.
- The figures of an experiment begin with **every run as one row** (each combination with its own results) and **the main effect of each factor** with its standard error and number of runs. When only some combinations were run, the figures by factor give way to the run-by-run figures.
- Water and nitrogen balances are drawn **run by run** as stacked bars when the design is not complete.
- The processes of an experiment or a scenario are a **three-column diagram**: factors, the processes of Daisy involved in the colours of Fig. 1, results.
- No figure says *not drawn*: figures and process chips without data are left out.
- In the layers D1 to D8, the runs of an experiment are named by **short labels** of their values; every label is cut to its room with an ellipsis, chips wrap by their measured width, and the tags of the scatter plots are placed where they do not touch a point or another tag.

## 25.3 The designs

- Inputs are **value boxes** with exact values, units and a list of the allowed range (Section 18.2).
- **Nothing runs while you type**: results appear after **Run simulation** and are withdrawn when an input or a setting changes (Section 18.1).
- **Find the most efficient inputs**, with your present inputs competing (Section 18.4); the surface search also tries shorter runs.
- **Field size** as an area or as a length and a width, for drip and sprinklers (Section 18.3).
- The **field sketch** is a guided drawing: corners you drag, side lengths you type, the source and the inlet, the pipe layout drawn as you choose, and seven steps of questions (field, water, crop, soil, system, running, summary) (Chapter 20). **Compare the irrigation systems** (Section 18.7).
- Each design ends with a **film of the soil water** covering the days ahead and the rest of the season. Every day is coloured by its phase (irrigation, rain, comfortable, nearing the trigger, stress, near wilting, no crop). The film shows the root zone drying towards the trigger and, without irrigation, towards the wilting point, and gives the totals and the share of days in each phase (Chapter 20). The schedule takes its depths and times from the design.
- The seasonal layers have **one dashboard chart per part of the layer**, with how each quantity varies over the twenty seasons (Chapter 14).
- *Trigger law* and *Mulch against bare soil* stand still until **Play**; the season plays at two days a second.
- Values may be typed up to the **physical limits** of the system, beyond the usual range of the list (Section 18.2).
- A surface set lasts at most 48 hours; when the inflow is too small for the soil, the design says so (Chapter 11).
- The surface layer iterates the advance exponent in the steady-intake term (Chapter 11); the slow-advance cases of the textbook example are now within 2 %.

## 25.4 The data and the models

- **Weather**: import of daily weather files and the online weather of a place, with every layer following (Chapter 19).
- **Crops**: 48 crops with their FAO-56 coefficients; the *Crop water requirement* layer (Chapter 21).
- **Seasonal data**: the twenty seasons of the simulations, following every setting (Chapter 22).
- **Sensor records** follow every setting and every soil by the delta method (Eq. 6.4a).
- **Sand and loamy sand** now respond to the trigger, the sensor depth and the method (Eq. 5.12a).
- The water balance of every crop closes (the limit on evapotranspiration was raised from 2,000 to 5,000 mm).
- Yield is in Mg/ha for the simulated cereal and in per cent of the potential for a library crop, with differences in *% points*.

## 25.5 Reports

- The report asks which layers to include, all ticked to start with; notes and drawings continue on the next pages; designs are drawn complete (Chapter 23).

## 25.6 The field on the map

- The field sketch has a hand to move the drawing, zoom in and out (buttons and mouse wheel), *Fit*, and a north arrow and scale bar that follow the zoom.

- The field sketch draws greenhouses and polytunnels, plant beds, areas to irrigate, no-irrigation zones, single plants and monitoring traps; the area to irrigate feeds the design; *Undo* and *Redo* (Ctrl+Z, Ctrl+Y) cover every change (Chapter 18).
- The real field has a section 4b: terrain (contours, flow, wetness index, 3D view), watershed with Strahler orders and time of concentration, soil retention curves mapped by geostatistics with the days to the trigger cell by cell, soil water sensors fused with the satellite by regression kriging, optimised sites, and *Use this field in the DSS* (Chapter 20).
- *Tools* ▸ *Field on the map…* finds the farm (place search, coordinates, satellite image, Google Earth), demarcates it on the image, computes NDVI, EVI, SAVI, NDRE and NDWI of the demarcated field from Sentinel-2 or simulated by SWAFL, and exports the report (PDF or PNG at 96 to 600 dpi, with coordinates and authorship), GeoJSON and CSV (Chapter 20).
- The virtual farm simulation has been removed; the real field on the map (Chapter 20) takes its place.
- The Trigger law block of the Dashboard shows the current soil as a table, its season chart and the advice with each condition ticked.
- The index map is cut exactly to the demarcated outline on the real map and follows every change of the outline; it shows continuous colours or five management zones with their areas, two indices side by side, the value under the cursor, and the irrigation layout of the sketch.
- The soil-water film of every design follows the season of the crop in Properties (starting at its next sowing when the working date has no crop), gives the advice of when to plant over the whole year, and each design has the weather line with the button for the online weather of the farm.
- The field map makes a variable-rate irrigation map (zones of net depth from the satellite indices and the simulation, editable, exported as a prescription), the ephemeral gully risk with the predicted gullies (terrain threshold and a ridge and valley filter of the photo) and the soil loss (RUSLE); the Soil hydraulic drivers dashboard is complete.
- The field map reads the season of the real field from twelve months of Sentinel-2 (curves, change, trend, stability and monitoring zones); the variable-rate irrigation simulates every zone with its own soil (SoilGrids), its observed cover and the SWAFL model and weather of the farm, giving each zone its dates and depths; the erosion uses the daily rain of the weather in use, the runoff on that soil and the cover of every month seen by Sentinel-2.
- The irrigation schedule of every design is the FAO-56 plan of the farm: the crop coefficient Kc of every day times the reference evapotranspiration of the weather in use, the rain, the roots and the soil, the trigger, the mulch, and the net depth, efficiency and run time of the design (Eq. 18.4). The film, the schedule, the season rows of the designs, the Dashboard and the zone-by-zone irrigation of the field map use it, and every control changes it.
- The field map keeps every computed map as a layer (*Maps of this field*) to show again without computing; the site keeps its results when the window is opened again; the map moves to a water source only on **Show**.
- The Soil hydraulic drivers layer shows its whole height, down to the notes under the soil profiles.
- The web explorer of the repository is built from the engine of version 1.1.0 (tools/build_explorer.py).
- The new layer **Deterministic phenotyping** (a layer group of its own in the Layers panel, after Soil structure and tillage, Chapter 8) is a virtual phenotyping experiment: one plant of maize, sorghum, wheat or common bean grown twice (well-watered and drought) in the soil, tillage, weather and rain of the project, with a camera, thermal, X-ray and soil water view that can be turned, traits through time, the water and carbon paths, the soil that the root tips meet, twelve genotypes that relate root traits to shoot traits, a table of 26 traits, and the loading, comparison, calibration and export of your own measurements.
- The layer has a **drought stress calendar** (periods of stress with their dates, growth stage, lowest water and what each does to the roots, for the past and for a dry, typical or wet future); every time chart of the layer shows calendar dates with a line on the date of the toolbar; the date of the toolbar is the first day of every figure and the day shown, the control *Season starts* and the control *Time axis* (the season, or a year from the date) set the rest; the new section *Weather in use* shows the weather file, the online weather or the model weather that

the layer reads; the water chart keeps rain and irrigation in a band of their own (Chapter 8)
- **Daisy 7.1.14 for Windows is built into SWAFL** (the folder `daisy` of the zip, copied into the program by the build): the Daisy runs start by themselves, nothing has to be installed or chosen; the Daisy installer file is recognised and ignored if it was chosen as the program
- After **Run simulation** every layer keeps playing, and starts again at the end of its season, until **Pause** is pressed (the designs, the other layers and the layer Deterministic phenotyping)
- The layer Deterministic phenotyping has a section **Drought and the roots at a glance**: time series of the water of the root zone, the root growth and the root depth with the exact date on which each period of drought stress begins
- In the Layers panel the **Daisy model** and all its groups start closed and open only when clicked; *Collapse all* closes them too
- Every layer has a **Run simulation** bar and shows its figures only after it is pressed for the present controls; the toolbar Play button does the same. The Deterministic phenotyping layer has its Run simulation button under its inputs, follows the date of the toolbar (and shows the time of day when it is today), shows no equations on the screen (they are all in Chapter 8), and is connected to the crop library, soil, weather, date, mulch, irrigation trigger, sensor and method, nitrogen and salt of the Properties panel, has collapsible sections, a section *Roots day by day*, a prediction of the phenotype and of the genotype of the library, a comparison and a fit to the SWAFL Daisy runs (E2, E7, E13, E14), the equations P.1 to P.26 explained one by one (Chapter 8), life-like plants with a **season film** (both plants and roots at eight dates, a play button that grows them from sowing to maturity, and the time series below), a bar of jump buttons that stays at the top of the window, and a **Drought outlook from the date of the toolbar**: the plant grown to the date on the recorded weather and then on the forecast and on a dry, a typical and a wet future taken from the years of the weather record, with the root zone water, the days of stress, the grain, the next watering and the drought risk.

## 25.7 The Daisy workbench

- The tabs of the Daisy workbench are arranged in five groups by what a person does: Your data, Ask a question, Look at results, Build and test, and Learn.
- The weather of every Daisy simulation is chosen once in one strip: the SWAFL project, the built-in weather, a bundled station, online weather of a place, or a file. The latitude needed for radiation is found from the place name.
- The Experiments tab can load several designs at once. Each is designed on its own and its runs are listed in order of preference; the folders are written as Experiments_N with one folder E1, E2 and so on for each design.
- Start from the Daisy reference offers 27 best combinations and 107 reference topics, in words that need no knowledge of modelling.
- The Processes tab is a map of the model with the figures of each process, drawn from a Daisy run with all process logs.
- The Process curves tab draws any function of the Daisy library.
- The Daisy workbench is arranged as a guided path with Back and Next, from the data of the site to the layers of the SWAFL project, which read the Daisy runs: the layers *Chemical fate (Daisy runs)* and *Daisy decision table* (Chapter 31). Online weather of any place can be used for the analysis, the Daisy runs and the figures, and every weather figure carries its citation.
- The Daisy workbench has a new look: a slate and ink window with a spectral accent line, rounded cards, pill tabs and a calmer Experiments tab with three steps (Ask, Check the plan, Run and read), short text and example questions as chips. Figure captions carry a short source line only; the long reference lines are gone.
- The figures of the Daisy workbench were redrawn in the style of a journal figure and of the figures of the repository (Chapter 30). No citation or caption is added any more to a figure, a saved picture or a saved report. Several panels of one figure are saved as one picture. The tabs Plot and Soil column draw their figure at once.

- The raw output of Daisy is given for every run: the files exactly as Daisy wrote them, one by one or as a zip, with the set-up, and the folder on the computer where the desktop app keeps every run (Chapter 30).
- Experiments: the rainfed reference is made for every soil and weather of the design, so the gain from irrigation is always set against the same soil; a rainfed run is no longer repeated for each irrigation setting and no longer counts as a level of an irrigation factor; the reference runs are left out of the main effects; a one at a time design includes its baseline run. The nitrogen scenario has five fertiliser levels, from none to 50 % more.
- Denitrification was read from a column name that Daisy 7 does not use; the gaseous loss of nitrogen (denitrification and N2O) is now read from the right columns.
- Chapters 26 to 39 show all of these screens with numbered arrows and the data behind them.
- Daisy experiments: every experiment of the study has its own figures (lines with a colour bar, bars, smooth surfaces with contour lines, where the water and the nitrogen go, a ranking of the factors) drawn again from its runs when results arrive; the user chooses which figures to draw and can add others; the map of the processes is redrawn for each experiment with the changed and the measured processes marked, and the caption under it was removed (Chapter 30). Scenarios are a part of Experiments.
- Each experiment keeps its own folder of runs and its own Outputs folder (README, plan, design table, results table, every figure and its numbers); experiments E5 (tillage and consolidation of the tilled layer) and E6 (nitrogen rate in a Mediterranean climate) were added.
- The crops of the rotation are chosen by the user from the whole Daisy library, with their own dates and nitrogen (Chapter 30).
- The titles, the axis texts and the panel letters of every Daisy figure can be edited; time axes show every year; files saved from the program carry the date and the time of the computer in their names.
- SWAFL layers: a bar above the simulation window saves the picture of a layer at 300 dpi, 600 dpi or the highest resolution the computer allows (the layer is drawn again at that size) with a title and a caption, copies it, saves the numbers, saves and loads the position of the widgets, and saves every layer at once in one zip (Chapter 23). Every section of the dashboard has a Save menu for its picture and its numbers, and a new tab *Figure* shows the process map of Daisy marked for the layer and linked to its outputs.
- The program carries its own Node.js on Windows, so that npm is not needed on the computer; the moving widget of the deterministic phenotyping layer no longer extends over the dashboard.
- The process figures are complete: the map is no longer cut at the left, and the processes Daisy has but Fig. 1 has no box for (soil heat and ice, denitrification) are boxes of their own under the map. Click a process in an experiment, a scenario or a SWAFL layer and the program shows what Daisy does there and the figures of that process from your runs; in a SWAFL layer the map is marked for that layer, the processes it does not use are pale and the layer scrolls to the matching section. Start files for the Mac were added (START_SWAFL.command) and the Windows start no longer stops on a PATH with parentheses.
- Every figure can be shown on the whole screen: a full screen button on each Daisy figure, on the layer picture, on each section of the dashboard and on the process figure of the layer. Esc leaves it.
- Every figure lists its own processes as buttons, and a click shows what Daisy does in that process and its figures. This holds for the figures of the experiments, the scenarios, the processes and the sections of the layers.
- The note under each figure is written from that figure: the factor and the result it shows, the number of runs and the range of the result. The process figure of each layer starts with a chain of the steps of that layer in its own words, then the map of Daisy with the controls and the results of the layer.
- Every popup window, the Daisy window included, has a full screen button in its title bar (a double click on the title bar does the same).

- The Figure tab of each layer now starts with the own figure of that layer, drawn like the figures of the manuscripts: the driving variables, the outputs above the soil surface line, the chain of processes of the layer, the state of the soil, and the controls below, with a legend. The map of the Daisy model follows it, folded.
- A click on a process, or on an output, in that figure shows only the results of the layer that belong to that process, taken from the Results, Design and Charts tabs, and hides the rest. The button Show all processes brings them back.
- The results, charts and tables of a layer are shown only after Run simulation, as the picture is. Before the run the Results and Charts tabs say so and offer a Run simulation button; after a change of a setting they wait for the next run.
- The full screen button now uses the usual full screen sign (four corners) everywhere: popups, Daisy figures, the layer picture, the sections of the dashboard and the Figure tab.
- Pictures are drawn at twice the screen resolution at least (the charts of the dashboard, the figures of Daisy and the picture of the layer), and the pictures of the Figure tab at three times.
- The figure of a layer now carries every result of the dashboard (Results, Design and Charts) as a box above the process it comes from, with bold colours. A click on a result shows only that result, a click on a process shows the results of that process. The long lists and notes of the earlier Figure tab were removed.
- Every layer, the phenotyping layer and the three irrigation designs included, shows nothing in the Results, Design, Charts and Figure tabs until Run simulation is pressed. Each tab says so and offers a Run simulation button.
- The Daisy window opens over the whole window. Full screen of a figure or a window now comes back with Esc.
- In full screen the Daisy window enlarges its text to fit the screen and fills the height of the screen. The buttons A- and A+ in its title bar make the text smaller or larger.
- Deterministic phenotyping: in the panel Roots from the plant above ground, when the twelve genotypes give the same value of a shoot trait the fitted line used to turn vertical and run across the dashboard. The line is now cut at the frame of its chart, becomes flat when the values are equal, and the red diamond of your plant stays inside the chart.
- The picture of a layer, deterministic phenotyping included, now grows to fill the whole width of its window, left and right, instead of staying in a narrow column. The process buttons of a dashboard section sit on their own line under its title.
- The figure of a layer now fits the width of the window, without sideways scrolling. A click on a process or on a result opens its results in a window of their own, with the description of the process and its Daisy figures; the window has the full screen and text size buttons.
- The figure of a layer fits the width of the window without sideways scrolling, and the dashboard widens by itself while the Figure tab is open. A click on a process or on a result opens its results in a window of their own, with the description of the process and its Daisy figures; the window has the full screen and text size buttons. The results are spread over the processes of the figure.
- The map of the Processes tab of the Daisy window is one figure: the fluxes on the arrows of Fig. 1 (evaporation from the canopy, the litter, the pond and the soil, transpiration, runoff, infiltration, leaching, mineralisation and the others) are boxes of their own, in place, and a click on any box shows what Daisy does there, the Daisy component it comes from and the figures drawn from the logs of your run. The components of the Daisy reference manual that have no arrow (45) are buttons in a closed list under the figure.
- The text of the whole program grows with the window, so that it stays easy to read on a large or high resolution screen. View, Text size has larger, smaller and automatic (Ctrl and plus, Ctrl and minus); the choice is remembered.
- The text of the program keeps its original size. View, Text size makes it larger or smaller at any time (Ctrl and plus, Ctrl and minus) and the choice is remembered; Automatic returns to the original size.
- In the map of the Processes tab each flux box now stands next to its own arrow (evaporation from the snowpack, the canopy, the litter and the pond, percolation, throughfall, drip off,

runoff, infiltration, soil evaporation, biopore flow) and no box covers an arrow or another box.

- In the map of the Processes tab, Percolation is drawn only where it happens, deep in the soil at the bottom of the profile. The water that leaves the snowpack is called melt water and the water that passes from the litter to the pond is called drains to pond, so no percolation box stands among the processes in the air.
- The map "Processes of Daisy in this experiment" is tidy: the processes the factors act on carry the tags F1, F2 and the processes the results come from carry R1, R2, and a key centred under the map gives their names in full. Nothing is written over the map any more and the map stands in the middle of its window.
- After an experiment has been run, the button "Clear the runs" (beside Run all and in the Results header) removes the results from the screen and keeps the design, so that the experiment can be run again.
- Daisy runs are faster: a run that is asked for again with exactly the same set-up files returns its logs at once (the last 30 runs are kept in memory), and the number of runs at the same time follows the number of cores of the computer.
- In the figure of a layer, a click on a process shows exactly the results that are drawn above that process, so that the figure and the results agree. In the map of Daisy, a box of a flux or a pool shows the figures of that flux or pool when the process has such a figure.
- The text of the window of Daisy no longer grows more than a quarter on a wide screen.
- The tabs of the window of Daisy are arranged as the path of the work: 1 Your data, 2 Experiments, 3 Look at results, 4 Decide in the SWAFL project, in one row joined by arrows, with the tools (build and test, learn) in a separate band below. The window opens on My data, the first step. All the pages were checked for sideways overflow and fit the window.
- In the window of Daisy, the pages My data and Weather stations are one page, My data and weather: the weather and the soil of the site come first, then the Daisy stations, the online weather of any place and the Danish weather data, one below the other. The path has ten steps.
- The Charts tab of the layers shows scientific figures instead of three plain line charts. Figure (a) is the working season day by day: evaporation, transpiration and evapotranspiration with the median of the twenty seasons, the canopy cover on a second axis, and rain and irrigation as bars on an inverted axis. Figure (b) is the water of the root zone with its 10 to 90 % range over the twenty seasons, the irrigation trigger, the irrigation events and the drainage as inverted bars. Figure (c) is the twenty seasons year by year, with the rain, irrigation, drainage and evapotranspiration of each season and their means. The dashboard widens by itself for this tab and the figures are redrawn when its width changes. The year-by-year charts of the seasonal layers have markers on every season, the mean of each series as a dotted line and the 10 to 90 % range as a shaded band.
- In the window of Daisy, the page Files is part of the page Plot. The page starts with the data of the plot: the output of your own experiments and scenarios can be added with one press (Use my experiments, Use my scenarios), files and folders of your own can be loaded or dropped, and the bundled examples can be loaded; the loaded logs are listed, and the plot follows below on the same page. The path has nine steps.
- The Daisy model in the Layers panel and the tabs of the Daisy window now have one careful naming: 1 Input data, 2 Experiments and scenarios, 3 Process structure (Fig. 1), 4 Results and analysis, 5 Decision support (SWAFL layers), 6 Model development, 7 Documentation. The two entries that had been merged into other pages (Weather stations, Collect Daisy files) are gone from the tree, scenario entries start with "Scenario:", and the tab names of the window follow the same words (Input data, Parameter library, Log plots, Run comparison, Calibration).
- The charts of the design layers (surface, drip and sprinkler: hydrograph, advance and recession, depth along the field, pressure and emitter flow, intake and application rate, catch, times of the irrigation) are drawn in the style of a journal figure like the season charts: lettered panels (a), (b), …, black frame with outward ticks, light dotted grid, markers on short series and bar charts with an axis and value labels.

- The season figures of the layers no longer carry the panel labels (a), (b), (c) or the year in the frame, and the same holds for the charts of the design layers; the titles of the sections say what each chart is. The right-hand axis titles are shorter so that they are never cut.
- The layers C2 Chemical fate and D1 Daisy decision table now say “no Daisy run yet” in the dashboard as well as in the picture, with a button that opens the Daisy scenarios; they no longer show the results of the SWAFL model in place of Daisy. As soon as a question has been run (also the own set-up), the dashboard and the picture fill from the run.
- A layer always opens at the top of its picture (the position of the layer before it is no longer kept, which had left the design layers blank until the person scrolled up).
- There is one Run simulation button for every layer: the one in the bar under the picture, now navy and bold while it runs the layer. The second button inside the picture, the button in the dashboard and the “Run / re-run” button of the layer header are gone; the picture and the dashboard say where to press. Find the most efficient inputs stays for the design layers.
- The groups of the Daisy model in the Layers panel and in the Daisy window carry no numbers any more, and the group is called Process structure (without the figure number).
- The three irrigation design layers keep their Run simulation buttons next to Find the most efficient inputs (in the picture, in the dashboard and in the header of the layer); the single Run button of the bar under the picture applies to the other layers.
- In the three irrigation design layers the Run simulation button of the bar under the picture is hidden until the design has run (it then becomes Pause or Play); the design layers run from the button in the picture, the dashboard or the header.
- The prepared experiments of the study (the list with its Load this experiment button) are a tab inside Start from the Daisy reference on the Experiments page, next to Best combinations, By topic and Import data and set-ups; the separate row “Experiments of the study” and its text are gone. In the Layers panel the two entries are simply Experiments and Scenarios.
- The map of the processes of Daisy (Process map tab) and the list of components below it are centred on the screen.
- The dashboard of every SWAFL layer has the colours of the Daisy workbench: grey ground, cards with a thin border and beige title bars.
- A layer holds the top of its picture while it draws (until the person scrolls or presses a key), so the inputs of the irrigation design layers are in view at once; the waiting message of the design layers points to the Run simulation button in the picture.
- The manual has new sections at the end of Chapter 26, Daisy: governing equations and guiding principles (Richards equation with van Genuchten and Mualem hydraulics and Feddes uptake, the water balance, evaporative demand, soil heat, solute transport, organic matter and nitrification, denitrification and nitrous oxide, and the crop chain from light to yield), and new sections at the end of Chapter 22 with three labelled screens of the dashboard, the charts and the design layers, each with its governing principle.
- The screens of the Daisy workbench (Input data, Experiments, Log plots, Mass balance, Run comparison, Set-up editor, Soil column, Calibration, Documentation, Process map) were taken again from the current program; the names of the tabs and groups in the text follow the program (Input data, Parameter library, Experiments, Scenarios, Process map, Log plots, Mass balance, Run comparison, To the SWAFL layers, Model development, Documentation). The screens of the processes with figures of real Daisy runs need the Daisy program and are kept from the earlier capture.
- The denitrification function of the Daisy calculator (response curves) now takes its inputs as already corrected by the pressure response, as Daisy does, and reproduces all 792 stored Daisy outputs to rounding error (the earlier form differed by a median of 40 %); the raw form of the extension remains available by an option.
- Daisy workbench: **Model builder**, a form for every model of the Daisy reference manual (683 models of 95 components) that writes, checks and saves the definition; **Fill in a form for this model** in the reference manual; the reference manual tab checked against the manual, with every attribute and its kind (3249 attributes) and the attributes grouped as in the book.
- Daisy workbench: step numbers on the tabs of the guided path (1 to 9, left to right), a

sentence on every tab when the mouse is over it, and a line *What to do here* on every tab.
- Daisy experiments: *Save the experiment folder* and *Save the ticked as one folder* first show the folder (folders, files, sizes and what is in each file), then save the zip.
- Weather: the online weather or the weather file loaded in SWAFL is named in the weather source of Properties and is the first choice of the weather stations of the Daisy window.
- The irrigation designs no longer leave an empty white area after a control is changed: while a design waits for a run, its picture is only the inputs and the Run card.
- Faster drawing: the pictures are drawn at the resolution of the screen with a 25 per cent margin (pictures saved at 300 or 600 dpi are unchanged), and the measure of repeated text is kept.
- Field on the map: opens in full screen like the Daisy window; a **Symbology** panel on the map gives every map its remote-sensing standard colours and the colour ramps of the QGIS library (with invert, opacity and the real colour of the photo), and the land cover the class colours of ESA WorldCover, Dynamic World, CORINE, NLCD or the photo.
- Set-up editor: every placeholder has a card that says what it means and how to fill it, with a scrollable list of its possible values and an explanation of each.
- Deterministic phenotyping: no text is drawn over a chart or over other text.
- Daisy experiments: the runs, the run files of the saved folder and its README use the years typed in the form, kept inside the weather in use (with online weather of 2014 to 2025, the runs simulate 1 January 2015 to 1 April 2025; Daisy starts one year after the weather begins). A line under the form says which years are simulated and why, and every run file starts with the same line.
- Trigger law and root-zone depletion: the season plays by itself when the layer opens and again and again until **Stop** is pressed; **Play** starts it again.
- Crop: one Crop list for every crop, field crops, vegetables, trees, vines and palms; the separate long crop and the crop of the 20 seasons are gone.
- Field on the map: the original colours of every map are the default again; the Symbology panel suggests the styles of the literature for each map and keeps the QGIS library; the panels can be dragged anywhere on the map; Ctrl+Z and Ctrl+Y undo and redo the colours and the outline; the groups of Opened maps fold and unfold; corners clicked in any order give an outline that does not cross itself.
- The indices of the field, step by step: scenes found, bands read, clouds and shadows removed, spectral signature, indices (with GNDVI, MSAVI2, NBR, BSI and CIre), zones and land cover.
- The soil of the field: SoilGrids point service tried three times, then the SoilGrids 250 m maps, which are interpolated over the field for the retention maps.
- Experiments: the whole Daisy library (63 soil horizons, 84 crops, 79 fertilisers, 35 chemicals, 82 managements, 55 programs and 176 logs) can be tested; Test it in an experiment in the Parameter library.
- Indices one at a time: after Compute, NDVI opens first and Next index opens the others; Opened maps lists only the indices opened. The 20 m bands are resampled bilinearly.
- Land cover by machine learning: a classifier (quadratic discriminant analysis) trained on the real Sentinel-2 classes of the field (or ESA WorldCover), with its accuracy, kappa and confusion matrix; no simulated values are drawn when the real data cannot be read.
- Field on the map: Compute the indices makes only the indices; terrain, retention and sensors come only from their own buttons; the groups of Opened maps say whether a map is measured, estimated from measured data or a model estimate; no retention map from the soil of the project when SoilGrids is not reached.
- Sentinel-2 indices checkable: each cell is one pixel of the scene (cells laid on the pixel grid), the map shows the pixels as measured (no filling of cloudy cells), the cursor gives the reflectance of the pixel, every pixel can be saved as CSV, and the scene opens in the Copernicus Browser of ESA for comparison.

### 25.7.1 Scenarios, experiments, comparison and layers of the Daisy workbench (November 2026)

- **Scenarios**: 24 questions in five themes instead of five, each built on the parameter library (mineral fertilisers, slurries and manures, the Danish soil classes JB1 to JB7, eight USDA textures, tillage, rooting depth, the aquitard, six crops, six climates of Europe, radiation, six pesticides, five PFAS, sorption); a search box; *Add a run* with any other value of the library (Section 29.1.3).
- **The library in a run**: a slurry, a chemical or a crop of the library is now read inside the rotation file, after the files the rotation reads and before its definitions, and its copy in Common/ leaves out what the run reads already. Before, Daisy stopped on these runs (*Unknown 'am' model*, *Unknown 'chemical' model*, *already defined*); all 24 questions were run with Daisy 7.1.14.
- **The number of runs** of an experiment is now kept exactly, the rainfed references included; the runs are chosen in a balanced way (a Latin hypercube over the levels), at random, evenly spaced or as the first ones, with a seed; *See the runs* lists them (Section 29.2.8).
- **Every row of an experiment** says what the factor is, where Daisy reads it, the typical values, what to type and what changes; Chapter 37 gives it for all 58 factors.
- **The folder of an experiment** opens every file in the colours of VS Code, with find, wrap and a bracket check; a file can be changed and the zip holds the change. The README says how the combinations were made, how each .dai file is made, which lines change with each factor and how the data reach every run; *runs.csv* lists the runs (Section 29.2.9).
- **Parameter library**: the text of every item in colour, editable, opened in the Set-up editor or downloaded; the meaning of every value of a fertiliser or chemical (Section 38.1).
- **Run comparison**: two logs, two whole runs (every log) or many runs on one quantity, from Files, Experiments and Scenarios; years and sums or means per year, per month or cumulative; mean absolute difference and Kling-Gupta efficiency; CSV of every table; the double-blind comparison takes the runs of Experiments and Scenarios too.
- **Four new layers** read the Daisy runs: D5 Chemicals and drinking water, D6 Trade-offs and the best edge, D7 The return of irrigation, D8 Nitrous oxide and gaseous nitrogen; the runs of Experiments are passed on as well, and the last step lists every layer.
- **The manual**: every chapter has an icon in the colour of its part; every display equation is set within the width of the page (a long one is shrunk to fit instead of running off the page).
- **Daisy is found by itself**: the folder that comes with SWAFL, its copy in the folder of the user, an installed Daisy, or the downloaded installer run once; no file has to be chosen (Section 1.2).
- **The manual inside the program**: a *Manual p. N* button on every tab of the Daisy workbench and above every layer opens this manual at its section; the tab *The SWAFL manual* lists and searches every section.
- **Run comparison**: the run library (runs of Experiments and Scenarios, every run saved on the computer, files and folders, the samples), every log of every run in the lists of Log A and Log B.
- **Experiment folders**: a README in every run folder, with how its .dai file was created and how Daisy calls everything, line by line.
- **Parameter library**: a steady view of an item, drawn once, without the page jumping (Section 38.1).

### 25.7.2 Projections, pipe design, linked irrigation settings and a steadier Daisy editor (November 2026)

- **Projection of the seasons**: the year charts of Reliability over 20 seasons, Full water balance and Nitrogen budget, the Daisy layers that show years, the charts of the dashboard and the yearly figure of the Charts tab continue 10 or 20 years after the last simulated season (Properties, Projection), lighter and marked, with the red bar of the working date kept.
- **Shading**: in Soil hydraulic drivers the selected row shades the soil table only; in the Trigger

law the highlight covers the row of the soil only and no longer runs into the chart.

- **Pipe design in the field sketch**: mainline, submain and lateral pipes with a diameter, and valve, pump, filter and tank symbols; a table of length, flow, speed and head loss for each pipe and the pump head asked; the drawn lengths and diameters pass to the design (Chapter 22b).
- **Linked irrigation settings**: the method and the field area of Properties pass to the three designs, the Design shortcuts group shows the link and the last result of the design, and the sketch passes the pipes to it.
- **Daisy editor**: the buttons are ready before the file list loads, a stale tab can no longer break a later one, a failed tab shows a message with a retry button, and very large files and output logs are sent to Log plots; the Daisy scripts load after the first screen, so the program opens faster.
- **Reference of every layer**: Chapter 22b lists the controls, the Properties and the data sources of each layer.

### 25.7.3 Final touches: literature figures, more questions, saving and speed (November 2026)

- **Efficiencies, budgets and trade-offs, from the literature**: 21 new figures under every experiment, and under every scenario of three or more runs, measured from the runs: nitrogen use efficiency and surplus, the nitrogen budget, the nitrogen response curve, agronomic efficiency, recovery and physiological efficiency, nitrate in the drainage against the EU limit, the water balance, the water production function, Ky, the Pareto front, correlations, factor sensitivity, distributions, a combined score and the GUS index of a chemical (Section 31.12).
- **New results to choose**: nitrogen use efficiency, nitrogen surplus, reactive nitrogen lost, nitrate per Mg of yield, AE, RE and PE of nitrogen, water-use efficiency, yield per m3 of rain and irrigation, harvest index, relative evapotranspiration, nitrate in the drainage and the share of a chemical leached or degraded.
- **Scenarios**: 37 questions instead of 24, with the 13 new ones on irrigation intensity and window, nitrogen with water, nitrogen on sand and loam, K_sat, pore size, topsoil settling, rain, warming, rain timing, the evapotranspiration model, half-life and the GUS classes (Section 29.1.3).
- **Scenario results from your own runs**: the map of the processes shows only what the variants changed and what the logs of the runs measured, with the range of every result.
- **Interactions and time series**: the interaction map, the interaction plot, the effect of a factor through the year, every run through the year and runs against days (Section 31.13).
- **You choose where results are saved**: nothing is written to a hidden folder; the save buttons ask for a folder (Section 31.14).
- **Faster and steadier**: the Daisy outputs open without a pause (a file is read only when it is opened), figures are drawn when they come into view, the root-structure layer spaces out its background work, and the sensor label of the root-structure layer is clear.
- **Easier to read**: formulas and units are set as in a journal (subscripts, superscripts, Greek letters and minus signs) in every figure and every label, the layout follows small windows, and every button has a hover text.
- **Feedback loops, eight hidden questions**: under the arrows of the feedback loops layer, eight panels, each with pore-scale sections of the soil (grains, matrix, air and water, as in the Evaporation layer) beside a curve, all following the controls of Properties, with the sources listed in the manual and not on the figures: whether a sensor reads the same after wetting as after drying (hysteresis), where water moves when the soil is air dry (film flow), whether compaction starves the crop or protects the groundwater, how far the water table reaches into the root zone, how much of the water carries the nitrate (wide pores), how salty the irrigation water can be before the leaching it needs pollutes, where on the suction scale the trigger costs yield, and how long the soil keeps up with the sun and what mulch changes. The topics follow the work of M. Th. van Genuchten and co-authors.
- **Irrigation designs and all layers open at once**: the waiting of 80 to 260 milliseconds and

the slow animated scroll after a click are gone.

- The pore-scale sections of the feedback loops layer are interactive: hover for the pore size, grain size, water or air and emptying suction, click to enlarge with a pore-size distribution and a suction slider.
- The small graphs of the hidden questions are clickable: click to enlarge, hover to read values.
- The pore-scale sections move: water drops travel through the wet pores and vapour rises through the air-filled pores; the enlarged section has a drying cycle.
- The equations of the surface irrigation design layer are typeset as in a paper: italic symbols, true subscripts and superscripts.
- The field sketch has AutoCAD style tools: draw (line, polyline, rectangle, circle, arc, text, dimension), modify (move, copy, rotate, mirror, scale, offset, trim, corner, explode, array, erase), object snaps, ortho and polar, layers, a command line with exact input, grips, DXF save and open, and a button that turns a drawn shape into the field, an area, a greenhouse or a pipe of the design.
- The PDF report keeps every figure whole: the pages are cut at the real resolution of the drawings (they were cut from a wrong corner), the phenotyping layer is built completely before it is captured, and layers of the Daisy workbench that has not been run give one line instead of an empty page.
- Sharper layers on every computer: the drawing is never kept at fewer pixels than the screen shows, however slow the computer is (a slow computer now draws less often instead of blurring), and tall layers get the same sharpness as short ones.
- Daisy experiments now show every factor and every level in the graphs: each factor gets its own graphs on the results it actually moves (the sorption and half-life of a chemical on the chemical leached and degraded, soil and irrigation method on yield, irrigation and percolation), a new graph shows where the chemical goes, and the tornado, main effect and interaction maps use the right result for each factor.
- Sharper simulation pictures in the dashboard: the picture of a layer is drawn with exactly two pixels for each screen pixel (it was drawn at a fixed density and then shrunk by the browser, which made it soft when shown small), the process figure of a layer is wide enough to read its words (it scrolls sideways in a narrow dashboard, and Full screen shows it whole), and the dashboard tab buttons no longer overlap on a medium sized screen.
- The Experiments page no longer stops with the message that the reference library is undefined: the library is set on the window in every case, a file that did not run at the first load is loaded a second time, and if it still fails the message now says what to do.
- Experiment time series: the results of every experiment now open with the time series of the runs. By default each factor you ticked gets its own panel and each value you ticked is one line (the other factors are averaged), for every quantity Daisy wrote, in the lettered panel style of a journal figure; you can still choose the panels and lines yourself and the time scale (through the year, every day, monthly, yearly, cumulative).
- Figures of the decision support layers in the style of a journal: black axes with outward tick marks, dark tick labels, the quantity and unit written on the axis, a title for the horizontal axis, and bars that can go below zero (the season by season gain in yield of The return of irrigation no longer hides a loss). The result text of the dashboard and of the Daisy window is darker and a little bolder, so it is easier to read.
- A layer now appears at once when you click it: the picture was drawn only every 0.3 to 0.6 seconds while the program around it had the focus (to spare the processor), so it showed late. A click or a key press now wakes the picture, and it rests only when the whole window is in the background.
- Online weather no longer stops with “too many requests”: a request that is refused for a moment is repeated after a wait, a request refused for an hour or a day passes at once to a second service (NASA POWER, from 1981, with the evapotranspiration computed from temperature), the place search has a second service too, and the weather that was fetched once is kept on the computer, so the same place and years need no new request and work without the internet. The window says which service answered.
- Numbers on the axes of the decision support figures no longer run into each other: the year

labels under the bars are placed by their measured width (a change of the axes had printed every year under the bars of the layers Year by year and Seasonal data), and the irrigation numbers above the bars of The return of irrigation are spaced the same way.

- Soil textural triangle: the small triangle button beside Soil in Properties, beside each horizon in My soil and on the soil factor of the Daisy experiments opens a clickable USDA triangle. Click a point to see the class and percentages, then pick the nearest program soil or use the percentages.
- Soil textural triangle, own mixture: sand, silt and clay can also be typed. The third value completes to 100 automatically, the point is placed on the triangle, the class is named and the nearest soil of the program is chosen for the SWAFL layers.
- Next irrigations chart of the irrigation design: the labels inside the picture (baseline touch, AW bracket, day of the film, refill, irrigation dates) are now placed one after the other so that none covers another, the axis or the values at the right edge.
- Heavier text in every layer picture: the plain text of all charts, tables and notes drawn in the layers is slightly bolder for reading, while bold headings stay as they were.
- Tillage and structure layer: the numbers under the three soil profiles now wrap inside their boxes, and the explanation lines below them wrap to the width of the screen. A full pass over all 142 layers and the 18 Daisy tabs found no script errors.
- Root zone depletion and the trigger law: every soil in the list now has an orange bar and a number for its speed of depletion, in percent of the available water used per day (coarse sand fastest, silt loam slowest), so the soils can be told apart at a glance.
- Full screen of the results and figures: the simulation keeps running. The results and figures now cover the whole window instead of the browser’s own full screen mode, which stopped the picture of the simulation from being drawn. The picture itself still uses the real full screen.
- Daisy reference, more experiments: six new factors from the actions of the reference (chapter 15): stubble height and straw at harvest (15.14), temperature of the irrigation water (15.18), depth the fertiliser is worked into (15.12 and 15.13), fertiliser by the soil nitrogen (15.11) and surface detention capacity (15.40). Tillage now has five systems (plowing, rotavation, disk harrowing, stubble cultivation, none). The Best combinations list grows from 27 to 43, among them method with trigger and dose, speed of application for each method, the irrigation season against the trigger, fertiliser form with irrigation, fertiliser date with the rain of the year, rooting depth with the rain, warming with nitrogen, straw with tillage, soil with irrigation method and depth of the drip line.

# Chapter 26

# ▥ Screenshot atlas: every screen with its governing equations

This part of the manual shows every screen of the workbench, one after another, each accompanied by the equations that produce what it displays. The screenshots were taken from the SWAFL workbench itself (the window in `app/index.html`) with the default settings (Coarse Loam, conventional tillage, cereal/oilseed rotation, wheat-straw mulch of strength 1.0, Surface irrigation, typical year). For each screen the first picture is the whole workbench and the second is the drawing area at higher magnification. For the fourteen single layers, whose drawing is longer than the drawing area and is scrolled inside it, the pictures of the whole drawing from top to bottom follow, together with its playback states and with one picture for each control that changes the drawing (every mulch material, every zoom and view of the soil section, every irrigation layout, every day button, and so on). Every picture carries numbered arrows, each ending on the item that its legend entry names. Below the pictures every equation that the screen draws on is printed in full, with its number, the chapter that explains it and its provenance label, so that no number on a screen is without its equation. A reverse index at the end lists, for each of the 134 numbered equations, the screens that use it.

Equation numbers are those of the chapters (for example Eq. 5.4 is the fourth equation of the chapter on the engines). The symbols are defined where the equation first appears; the *Where explained* line gives that place.

## 26.1 The window and its controls

### 26.1.1 The welcome dialog

Shown on first start; it carries no equations.

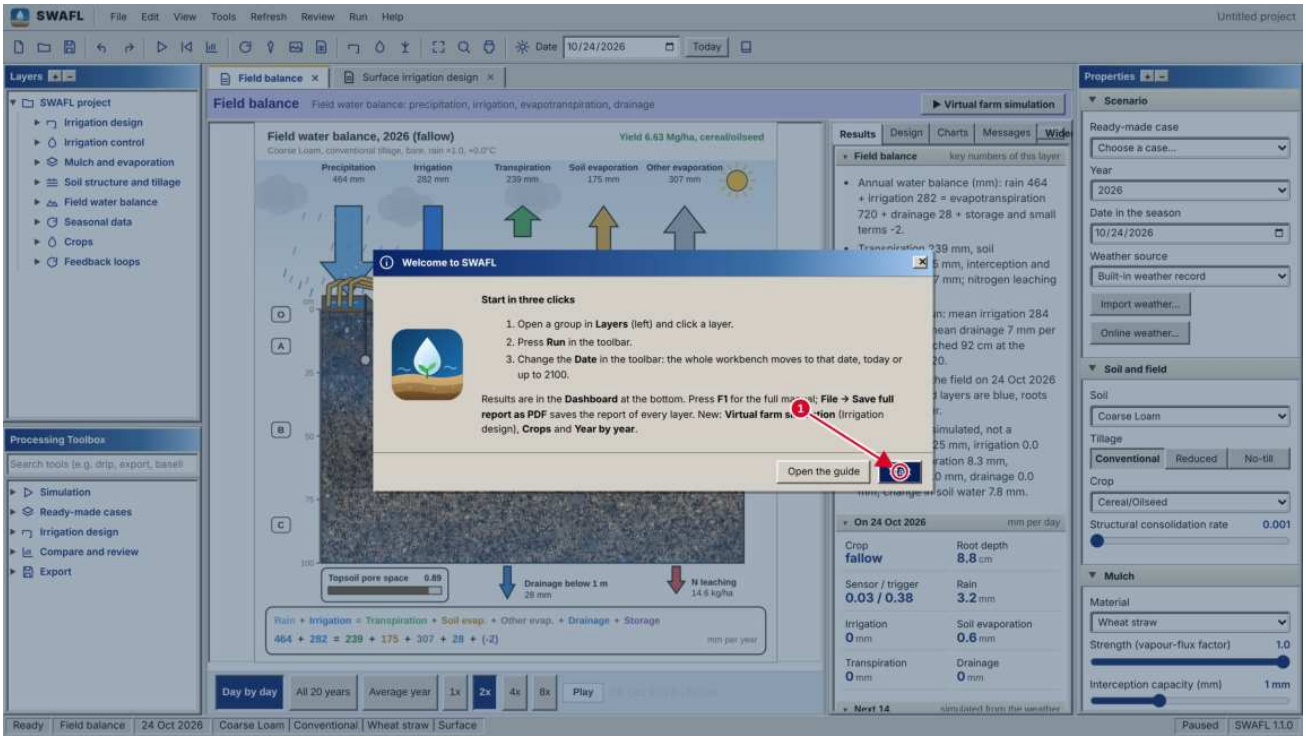



Figure 26.1: The welcome dialog.

**What the arrows point to.**

- **1** The Start button closes the dialog and opens the workbench; the dialog carries no equation.

## 26.1.2 The workbench

Layers (left), drawing area (centre), Dashboard (below and right of the drawing) and Properties (right). The date shown in the toolbar sets the whole workbench to that day.

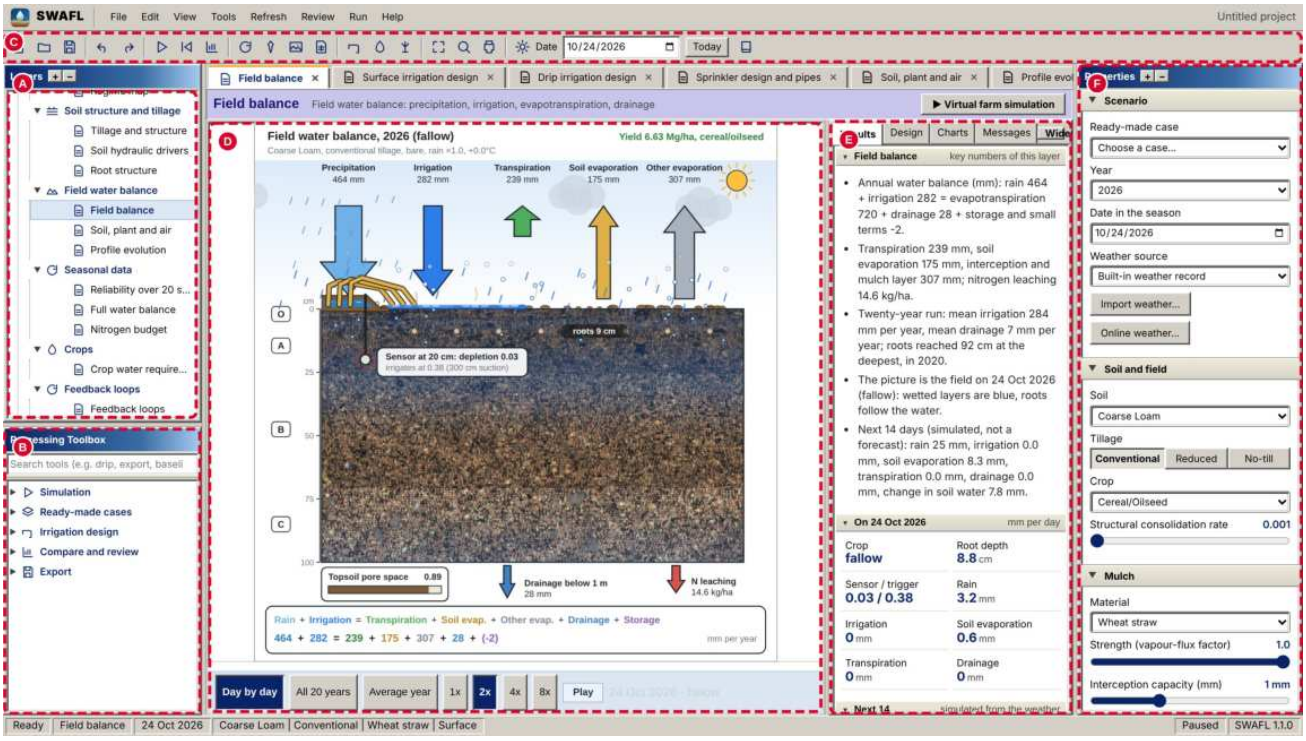



Figure 26.2: The workbench.

**What the arrows point to.**

- **L** Layers panel: choose a layer or one of its sections. Every choice is one screen of this atlas.
- **X** Processing Toolbox: run, cases, design, compare and export commands.
- **T** Toolbar with the Date control: the date sets the whole workbench to that day (Eqs. 4.1 and 4.2).
- **D** Drawing area: the picture produced by the equations printed under each screen.
- **R** Dashboard: the numerical results of the same equations (Engine 1, Eqs. 5.4 to 5.12; Engine 2, Eqs. 5.13 to 5.25).
- **C** Properties panel: the controls that move the equations (soil, tillage, mulch, weather, irrigation, nutrients).

**Equations behind this screen.** Eq. 4.1, Eq. 4.2.

*Eq. 4.1. Where explained: chapter "Dates", section "The date control".*

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \left(Y_0 + \lfloor k/365 \rfloor - a\right) + \left(k - 365\lfloor k/365 \rfloor\right)\ \text{days}, \tag{4.1}$$

*Provenance (Eq. 4.1):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

*Eq. 4.2. Where explained: chapter "Dates", section "The date control".*

$$a = \begin{cases} \text{argsort}_y \left(P_{\text{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

*Provenance (Eq. 4.2):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

### 26.1.3 The File menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

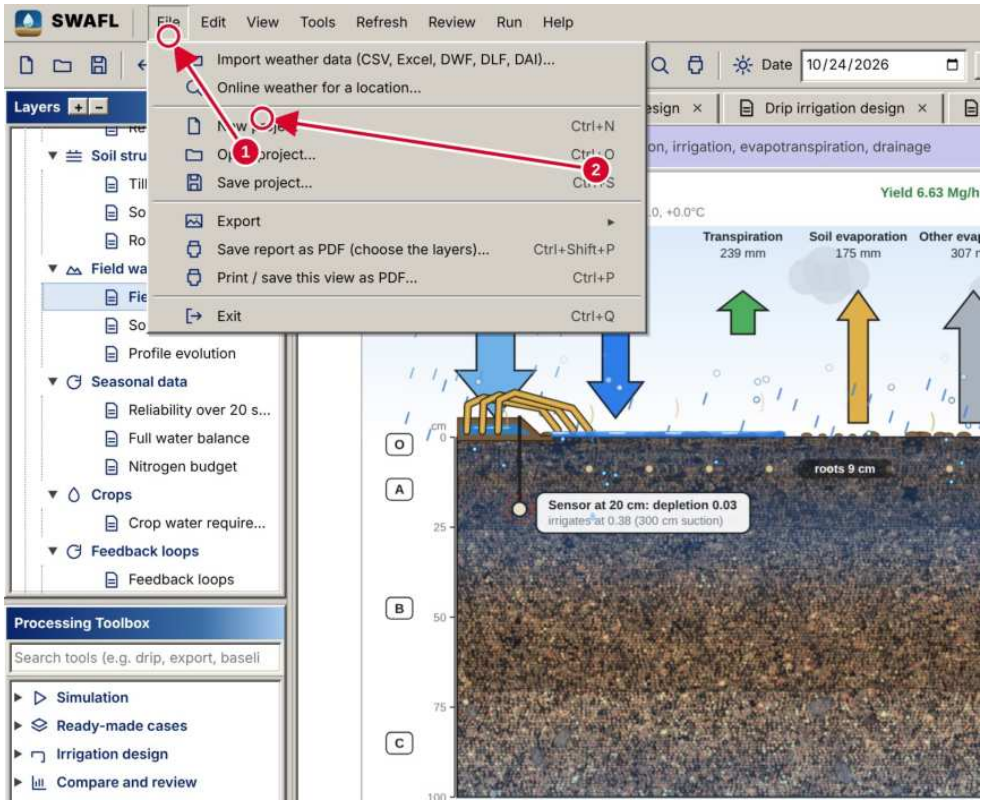



Figure 26.3: The File menu.

**What the arrows point to.**

- **1** The File menu in the menu bar.
- **2** Its list of commands.

### 26.1.4 The Edit menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

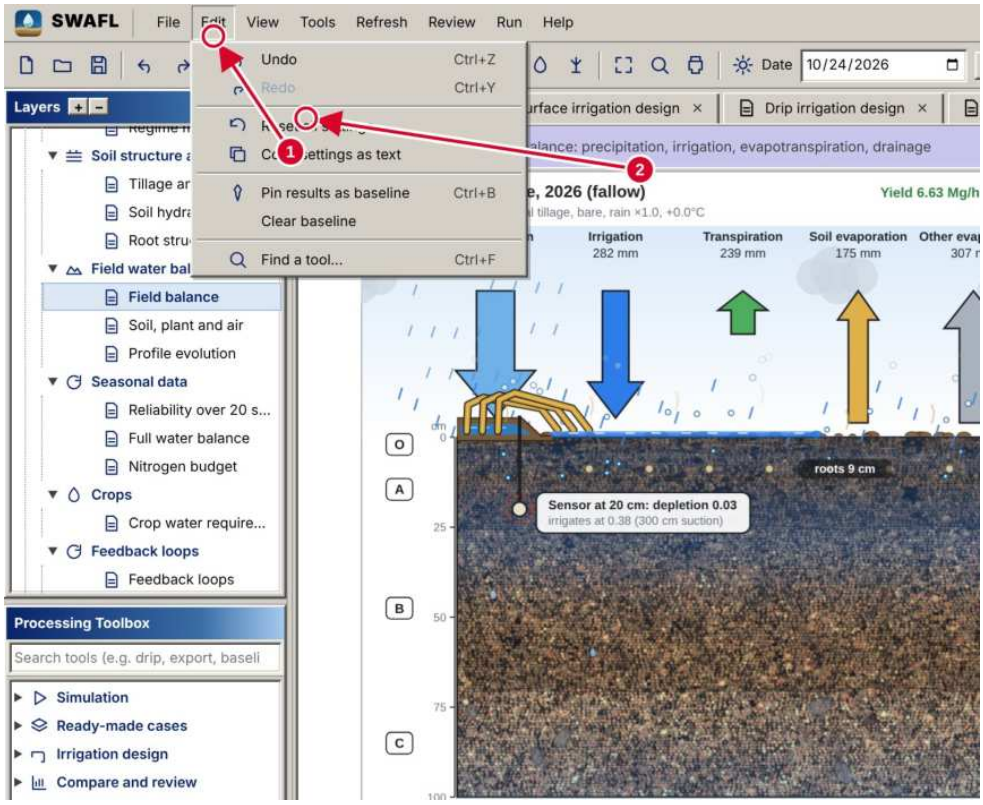



Figure 26.4: The Edit menu.

**What the arrows point to.**

- **1** The Edit menu in the menu bar.
- **2** Its list of commands.

### 26.1.5 The View menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

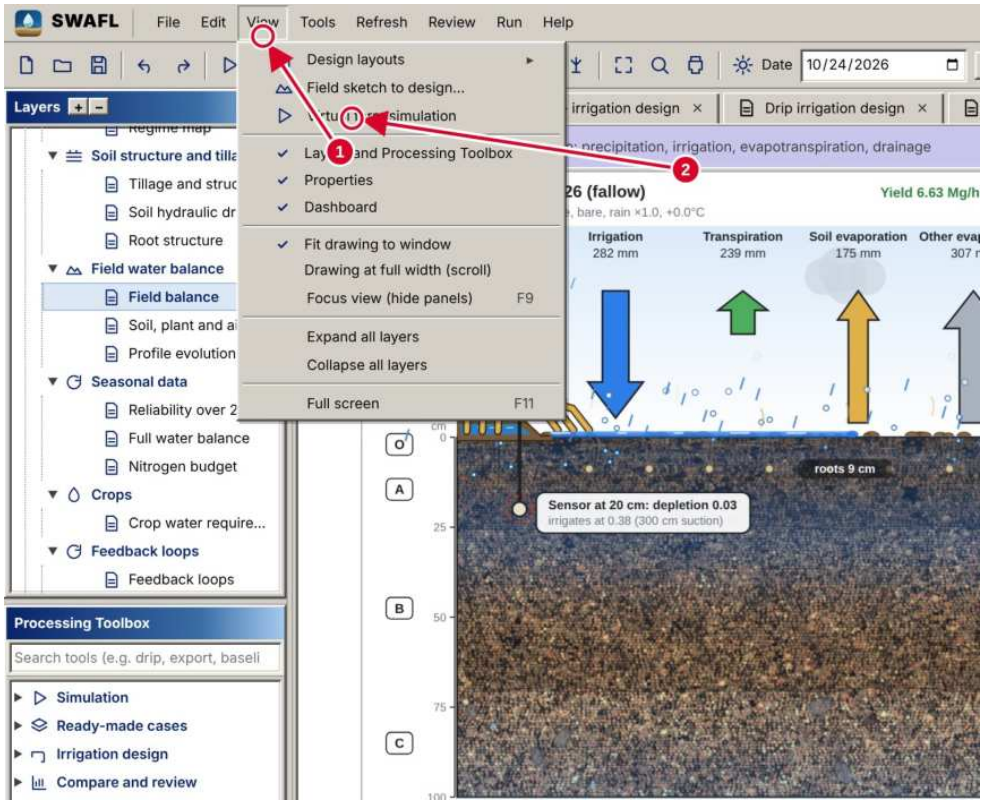



Figure 26.5: The View menu.

**What the arrows point to.**

- **1** The View menu in the menu bar.
- **2** Its list of commands.

### 26.1.6 The Tools menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

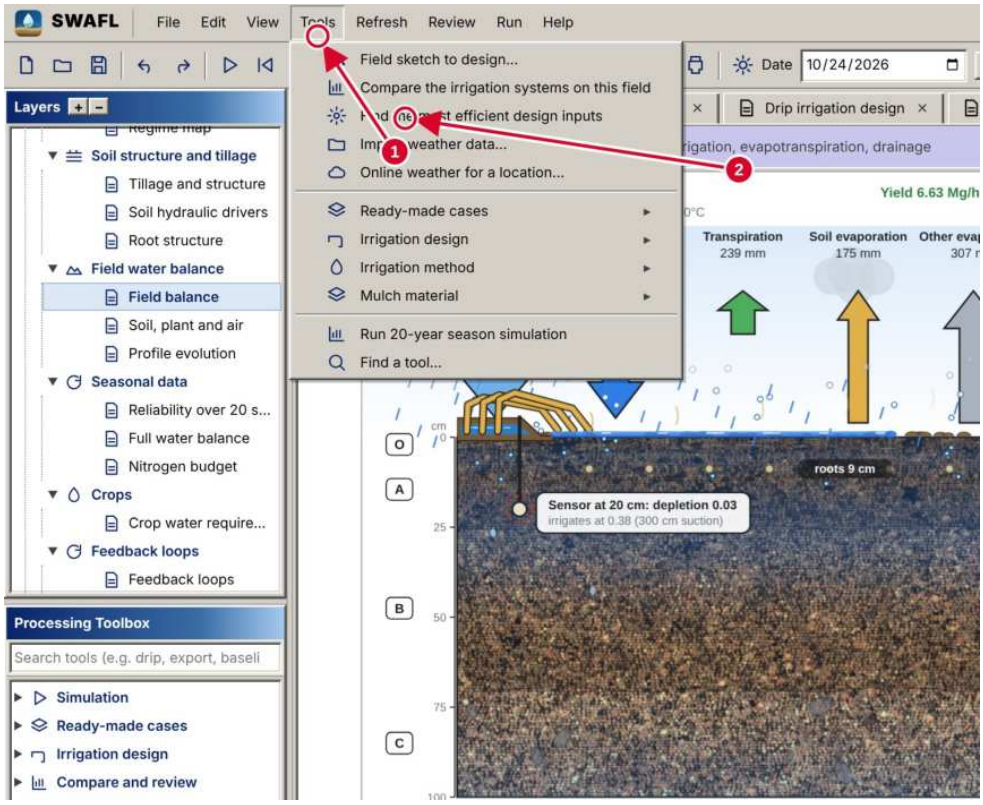



Figure 26.6: The Tools menu.

**What the arrows point to.**

- **1** The Tools menu in the menu bar.
- **2** Its list of commands.

### 26.1.7 The Refresh menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

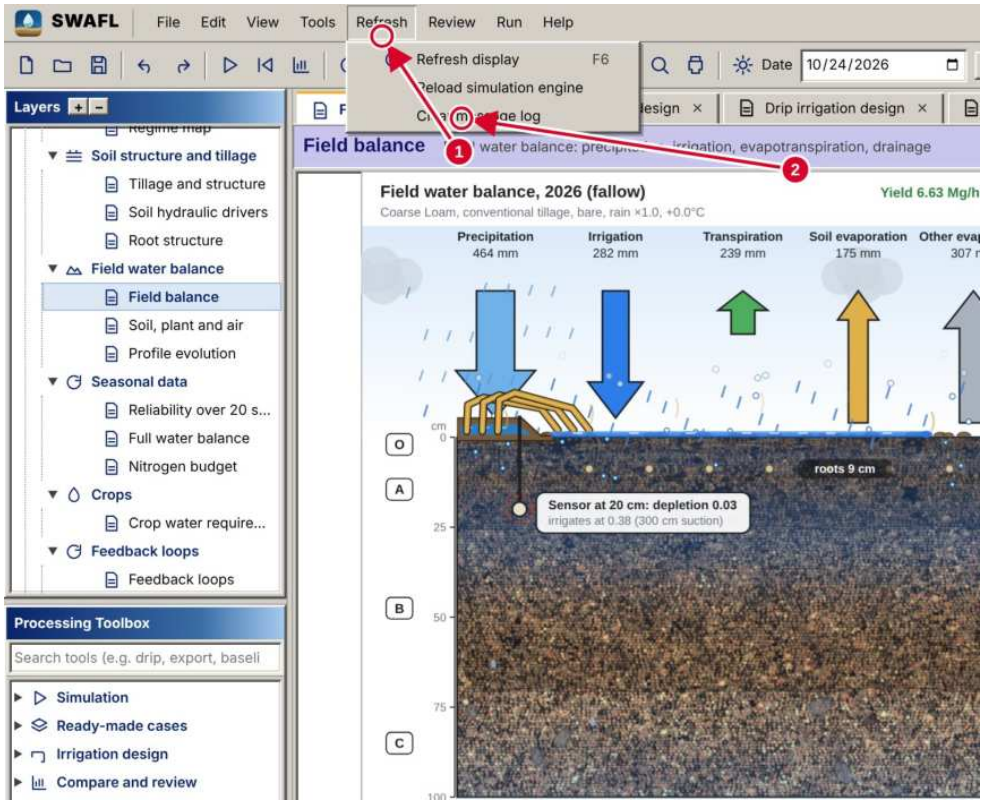



Figure 26.7: The Refresh menu.

**What the arrows point to.**

- **1** The Refresh menu in the menu bar.
- **2** Its list of commands.

### 26.1.8 The Review menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

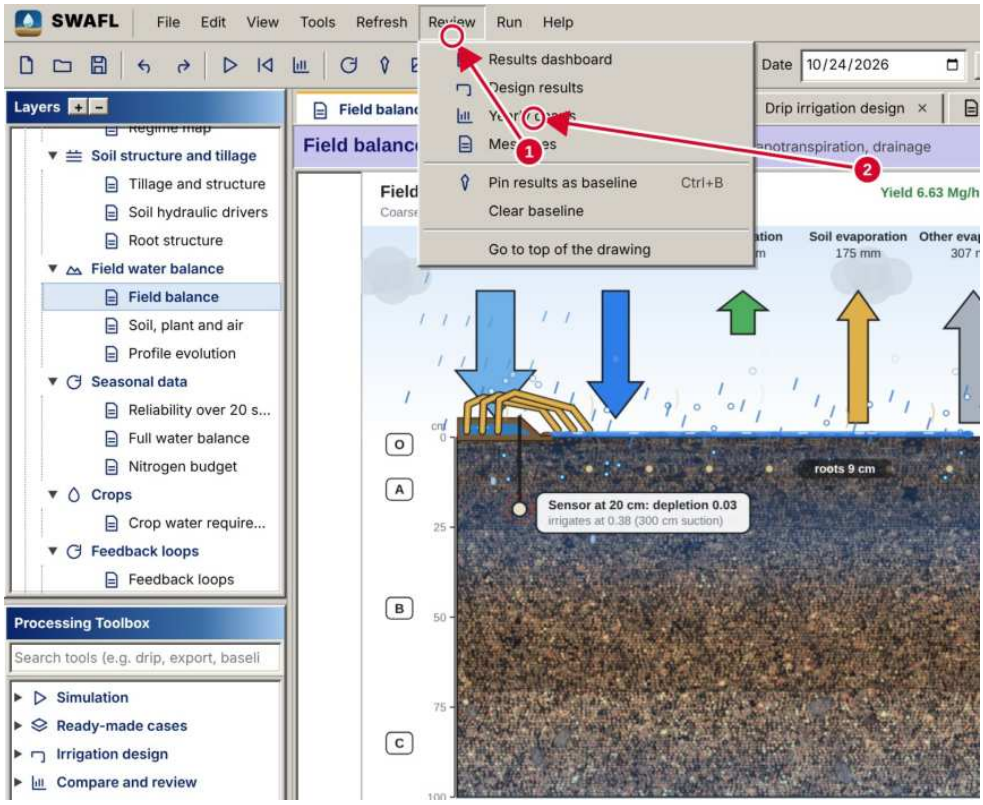



Figure 26.8: The Review menu.

**What the arrows point to.**

- **1** The Review menu in the menu bar.
- **2** Its list of commands.

### 26.1.9 The Run menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

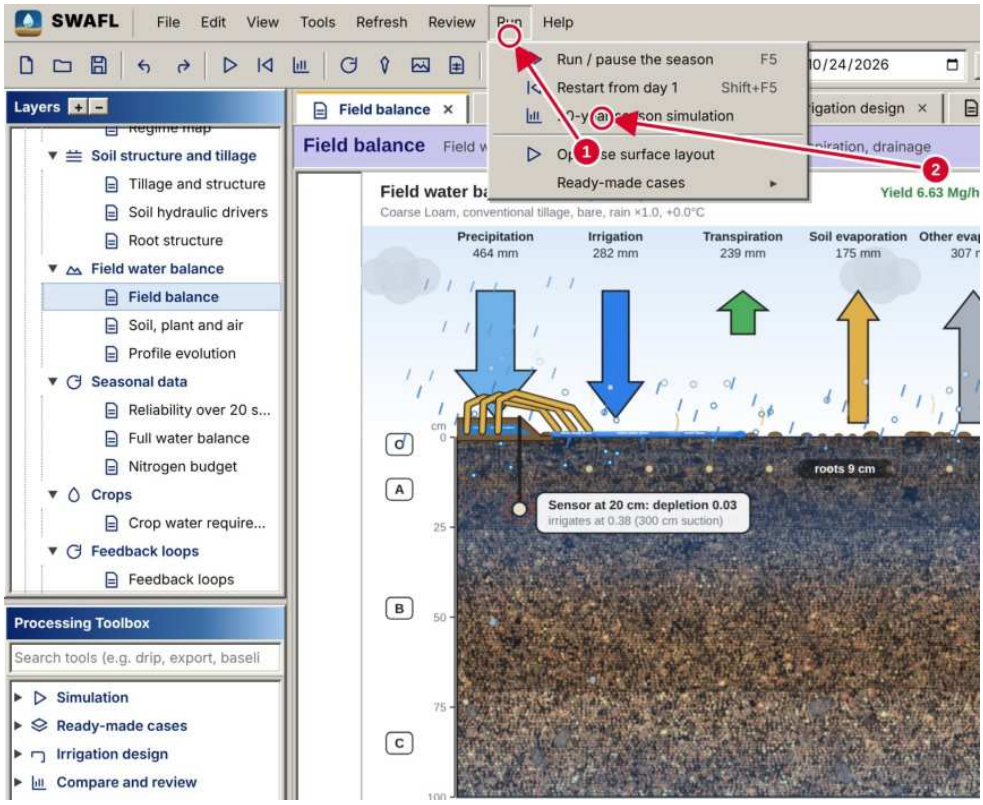



Figure 26.9: The Run menu.

**What the arrows point to.**

- **1** The Run menu in the menu bar.
- **2** Its list of commands.

### 26.1.10 The Help menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

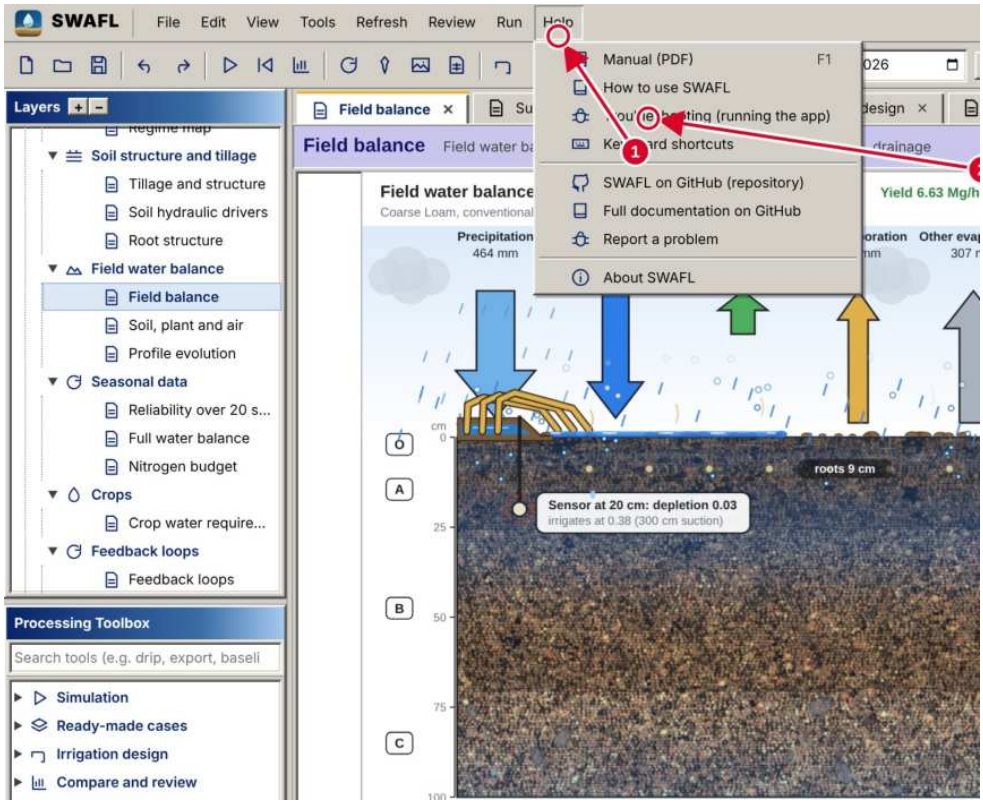



Figure 26.10: The Help menu.

**What the arrows point to.**

- **1** The Help menu in the menu bar.
- **2** Its list of commands.

### 26.1.11 The command palette (Ctrl+K)

Searches every layer, control and tool by name.

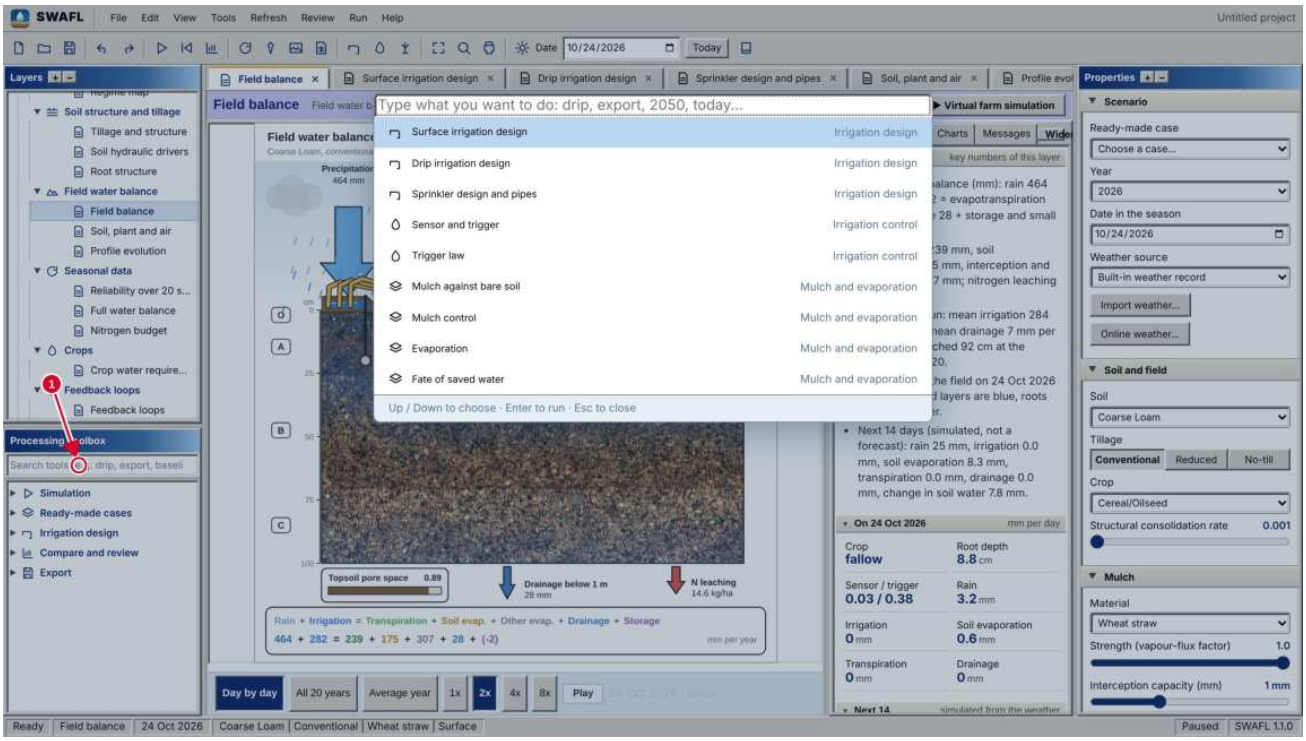



Figure 26.11: The command palette (Ctrl+K).

**What the arrows point to.**

- **1** Search box of the command palette (Ctrl+K): type a layer, control or tool and press Enter.

### 26.1.12 The Processing Toolbox

Simulation, ready-made cases, irrigation design, compare and review, export.

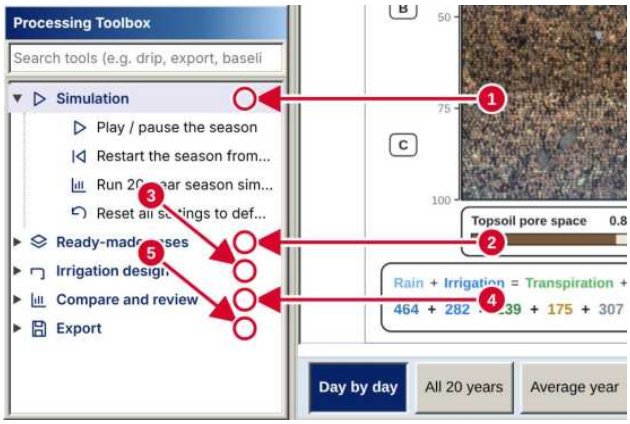



Figure 26.12: The Processing Toolbox.

**What the arrows point to.**

- **1** Simulation: play or pause the season, restart, 20-year run (Engine 2, Eqs. 5.13 to 5.25).
- **2** Ready-made cases: load a stored combination of soil, tillage, mulch, weather and irrigation.
- **3** Irrigation design: open the surface, drip or sprinkler design layer (Eqs. 11.1 to 13.23).
- **4** Compare and review: pin the current results as the baseline (Eqs. 14.4 and 14.5), clear it, show the dashboard.
- **5** Export: picture, yearly results (CSV), design results (CSV), project, print.

### 26.1.13 The toolbar

Run, restart, 20-year run, refresh, baseline, picture, focus, search and the Date control (Eqs. 4.1, 4.2).

Figure 26.13: The toolbar.

**What the arrows point to.**

- **1** Run or pause the season (Engine 2).
- **2** Run the 20-year simulation.
- **3** Pin results as baseline (Eq. 14.4).
- **4** Open a design layer: Surface, Drip or Sprinklers.
- **5** Date control: the working date, Eqs. 4.1 and 4.2.

**Equations behind this screen.** Eq. 4.1, Eq. 4.2.

*Eq. 4.1. Where explained: chapter "Dates", section "The date control".*

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

*Provenance (Eq. 4.1):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

*Eq. 4.2. Where explained: chapter "Dates", section "The date control".*

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\text{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

*Provenance (Eq. 4.2):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

### 26.1.14 The status bar

The current layer, date, soil, tillage, mulch and irrigation method.

Figure 26.14: The status bar.

**What the arrows point to.**

- **1** Layer and section on screen.
- **2** Working date.
- **3** Soil, tillage, mulch and irrigation method in force.

### 26.1.15 The Dashboard, Results tab

Working day, next 14 days, whole year and 20-year averages.

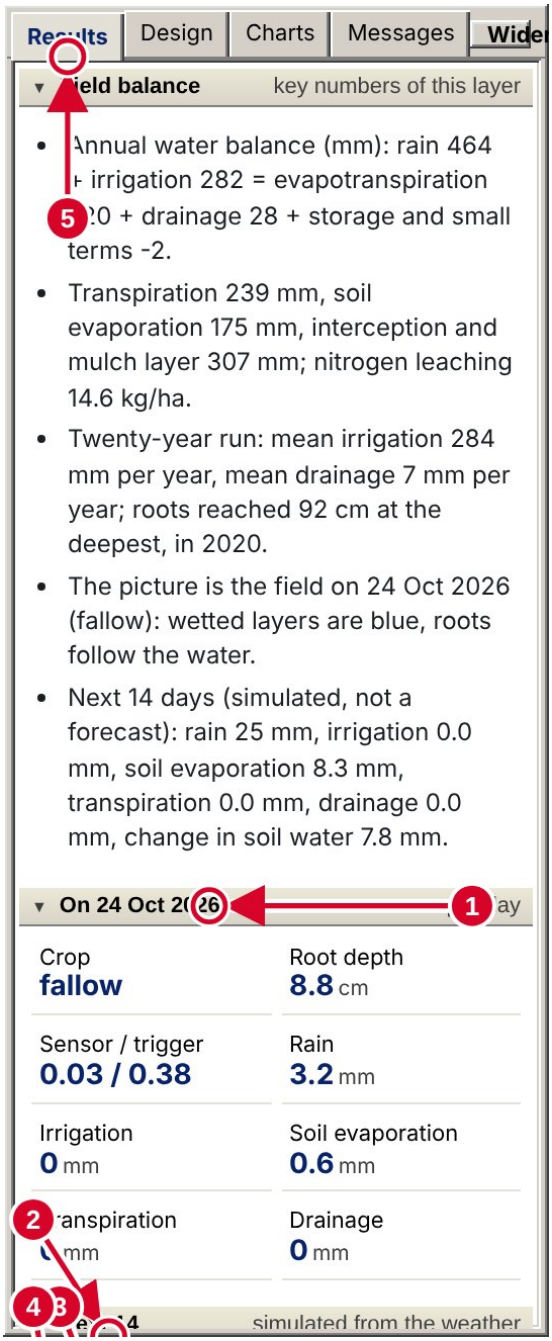



Figure 26.15: The Dashboard, Results tab.

**What the arrows point to.**

- **1** Working day: crop, root depth, sensor against trigger (Eq. 5.3), rain, irrigation, evaporation, transpiration and drainage of that day, Eqs. 14.1 to 14.3.
- **2** Next 14 days: simulated, not a forecast, from Engine 2.
- **3** Whole year: the annual fluxes of Engine 1, Eqs. 5.4 to 5.12, and the closure of Eq. 5.12.
- **4** Tabs: Results, Design, Charts and Messages.

**Equations behind this screen.** Eq. 6.1, Eq. 14.1, Eq. 14.2, Eq. 14.3, Eq. 14.4.

*Eq. 6.1. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".*

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

*Provenance (Eq. 6.1):* **[Data-driven]**: counts over the simulated daily series.

*Eq. 14.1. Where explained: chapter "The Dashboard", section "Results".*

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm}\,\text{d}^{-1}). \tag{14.1}$$

*Provenance (Eq. 14.1):* **[Engineered]**: bookkeeping of Engine 2 series.

*Eq. 14.2. Where explained: chapter "The Dashboard", section "Results".*

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

*Provenance (Eq. 14.2):* **[Engineered]**: bookkeeping of Engine 2 series.

*Eq. 14.3. Where explained: chapter "The Dashboard", section "Results".*

$$\Delta S = P + I - E_{\text{int}} - E - T - D, \tag{14.3}$$

*Provenance (Eq. 14.3):* **[Literature]**: water-balance identity.

*Eq. 14.4. Where explained: chapter "The Dashboard", section "The pinned baseline".*

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

*Provenance (Eq. 14.4):* **[Engineered]**: bookkeeping of Engine 2 series.

### 26.1.16 The Dashboard, Design tab

Results of the open design layer.

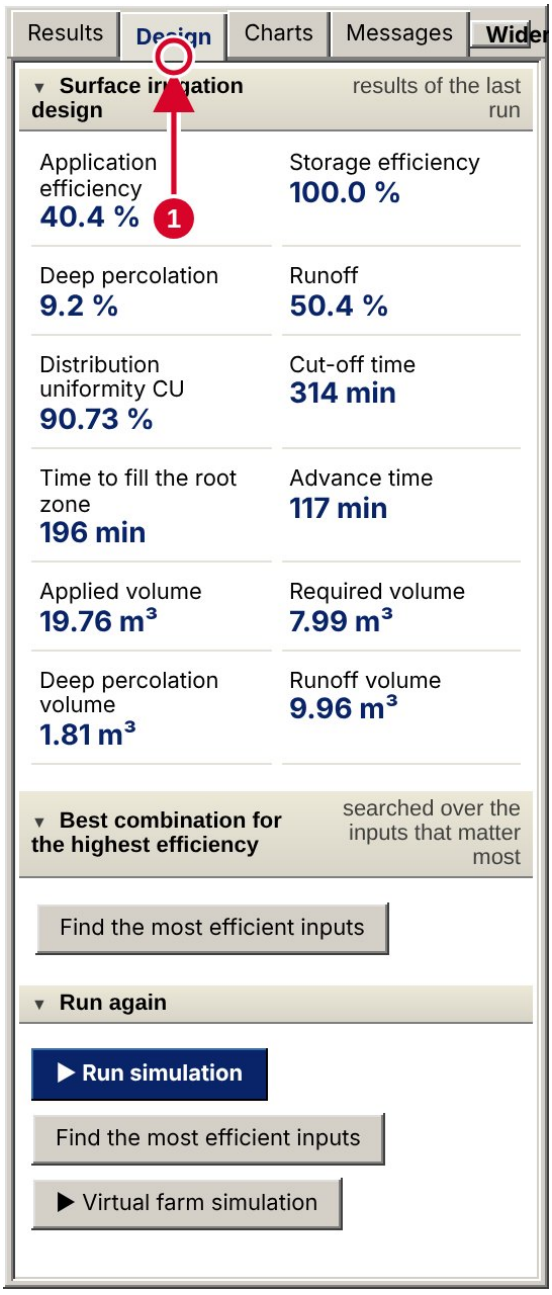



Figure 26.16: The Dashboard, Design tab.

**What the arrows point to.**

- **1** Design tab with a design layer open: efficiencies, uniformity, times and volumes of the system being designed.
- **2** Tabs.

**Equations behind this screen.** Eq. 11.18, Eq. 11.19, Eq. 12.19, Eq. 13.11.

*Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".*

$$
\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b<1:\ Q\,t_L + c_b\,Q\,(t_{co}-t_L)), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z-Z_n,0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n-Z,0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0).
\end{aligned}
\tag{11.18}
$$

*Provenance (Eq. 11.18):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

*Eq. 11.19. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".*

$$
E_a = \frac{V_{\text{st}}}{V_{\text{app}}},\quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}},\quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}},\quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}},\quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i-\bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}
$$

*Provenance (Eq. 11.19):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

*Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".*

$$
E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle},\quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}},\quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle},\quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle},\quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}
$$

*Provenance (Eq. 12.19):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

*Eq. 13.11. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".*

$$
\begin{aligned}
&E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, && E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n},\quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},\\
&\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}},\quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, && \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}},
\end{aligned}
\tag{13.11}
$$

*Provenance (Eq. 13.11):* **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

### 26.1.17 The Dashboard, Charts tab

Yearly charts and the pinned baseline.

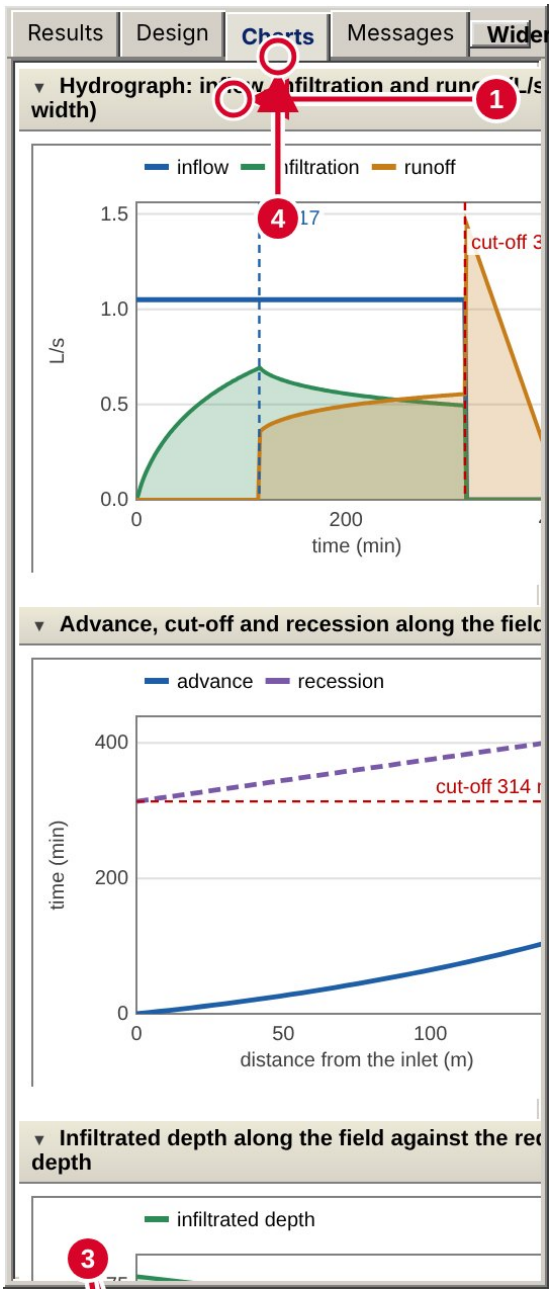



Figure 26.17: The Dashboard, Charts tab.

**What the arrows point to.**

- **1** Rainfall and irrigation for twenty years (Engine 1 annual values, Eqs. 5.4 to 5.12), with the pinned baseline of Eq. 14.5.
- **2** Drainage below the root zone, D of Eq. 5.9.
- **3** Evaporation and transpiration, E and T of Eqs. 5.4 to 5.12.

**Equations behind this screen.** Eq. 14.5.

*Eq. 14.5. Where explained: chapter “The Dashboard”, section “The pinned baseline”.*

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

*Provenance (Eq. 14.5):* **[Engineered]**: bookkeeping of Engine 2 series.

### 26.1.18 The Dashboard, Messages tab

Plain-language messages.

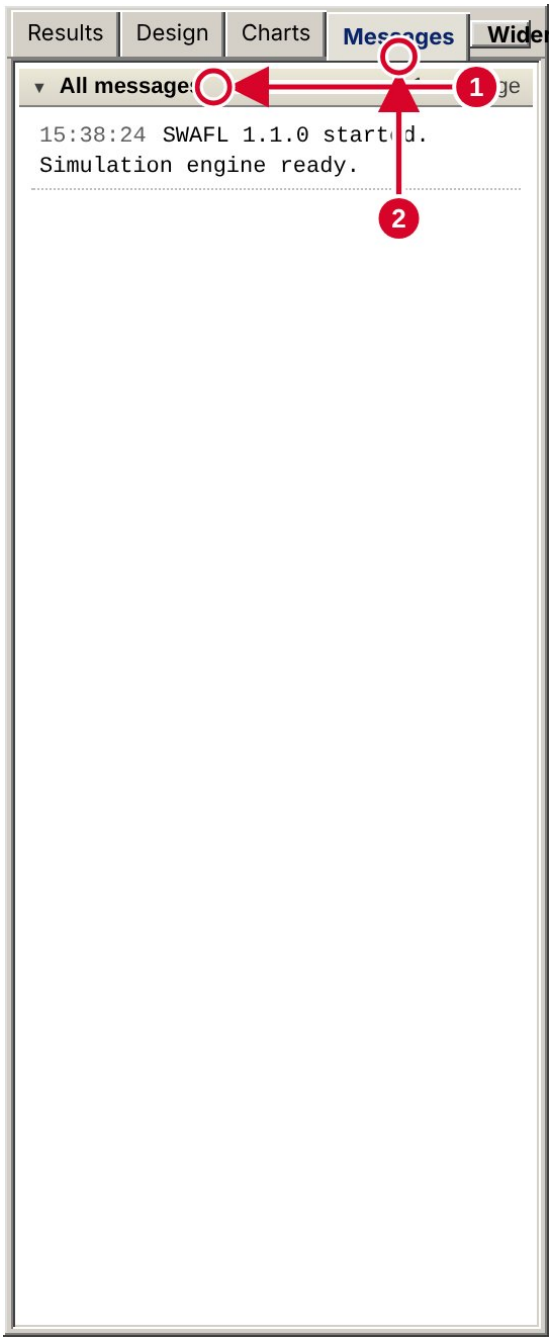



Figure 26.18: The Dashboard, Messages tab.

**What the arrows point to.**

- **1** Message log: engine start and plain-language notices.

### 26.1.19 The Properties panel

Every control; the equations are those that the controls move.

**What the arrows point to.**

- **1** Scenario: ready-made case, weather year and date (Eqs. 4.1 and 4.2).
- **2** Soil and field: soil, tillage, crop and consolidation rate tau_c (Eqs. 5.1, 5.2, 8.1).
- **3** Mulch: material, strength phi and interception capacity c (Eqs. 5.4 to 5.10, 9.1).
- **4** Weather and climate: rainfall scale rho and temperature shift (Eqs. 5.4 to 5.10).
- **5** Irrigation: method, trigger suction h_T and sensor depth (Eqs. 5.3, 5.11, 5.27).
- **6** Nutrients: nitrogen rate (Eqs. 5.10, 5.11).

**Equations behind this screen.** Eq. 5.18, Eq. 5.26, Eq. 8.1, Eq. 13.9.

*Eq. 5.18. Where explained: chapter “The two computing engines”, section “Roots”.*

$$R_{\max} = \operatorname{clip}\Big( R_c\,(0.85 + 0.3\,\text{sand})\big[1 + 0.12\log_{10}(|h_T|/300)\big]\big[1 - 0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \quad (5.18)$$

*Provenance (Eq. 5.18):* **[Engineered]**: rooting rules with constants chosen for the workbench.

*Eq. 5.26. Where explained: chapter “The two computing engines”, section “Evaporation under warming (used by the E1 layer)”.*

$$E_{0,\text{bare}} = 4.2\,(1 + 0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\tfrac{1}{2}[1 + \sin(2\pi(\text{doy} - 100)/365)]}{0.675}\,\operatorname{clip}(1 - 0.12(\rho - 1);\,0.85,\,1.12)\sqrt{\gamma_y}\ \text{mm d}^{-1}, \quad (5.26)$$

*Provenance (Eq. 5.26):* **[Engineered]**: atmosphere factor with chosen constants.

*Eq. 8.1. Where explained: chapter “Soil structure and tillage layers”, section “Pore index and consolidation”.*

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \quad (8.1)$$

*Provenance (Eq. 8.1):* **[Engineered]**: exponential relaxation with chosen constants.

*Eq. 13.9. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.*

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \text{sup}, & T \le t_p \\ K\,T^A, & T > t_p \end{cases} \qquad R = \text{sup} - F, \quad \text{RO} = \max(0, R - S_d^{\text{eff}}), \quad I = \text{sup} - \text{RO}. \quad (13.9)$$

*Provenance (Eq. 13.9):* **[Engineered]**: loss and ponding rules chosen for the workbench.

## 26.2 Surface irrigation design

### 26.2.1 The whole layer: inputs and overview

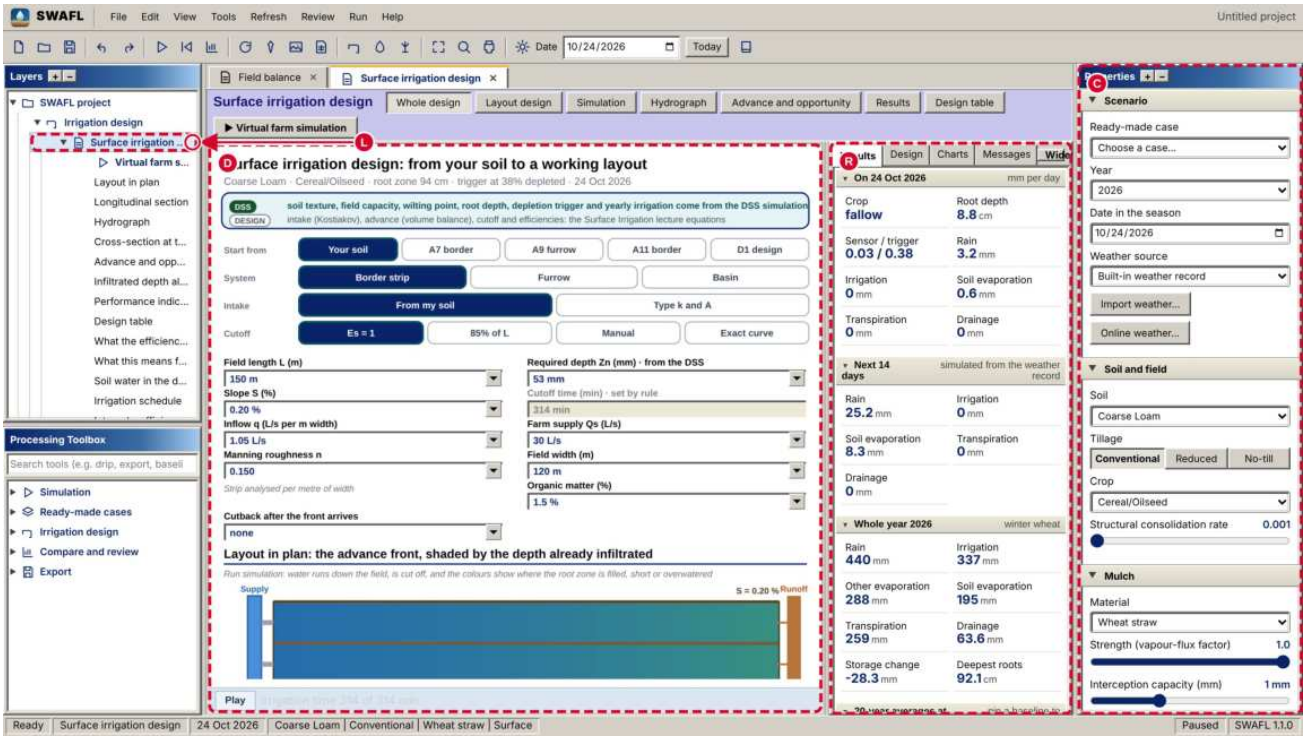



Figure 26.19: Surface irrigation design: the whole workbench.

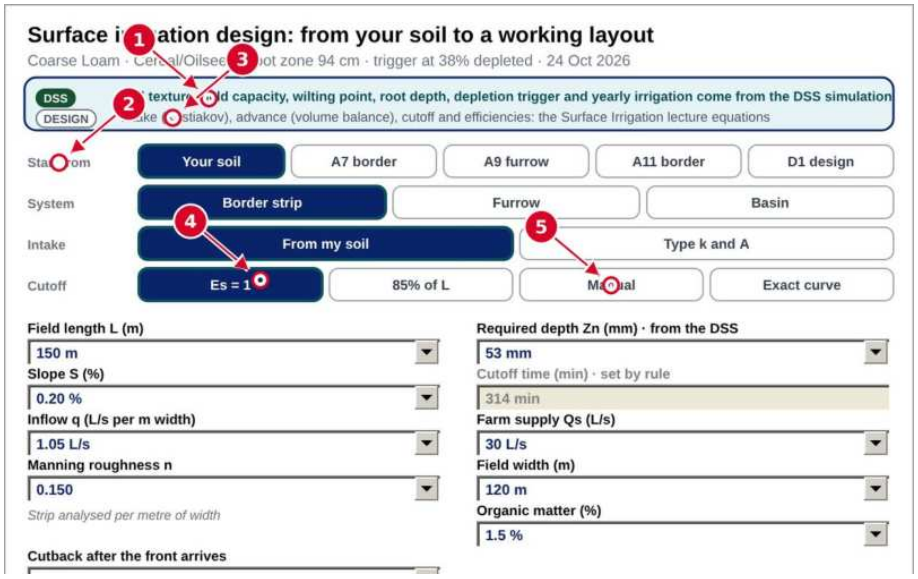



Figure 26.20: Surface irrigation design: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** DSS badge: soil, root zone, trigger and yearly irrigation come from the Engine 1 and 2 results (Eqs. 5.1 to 5.12, 5.27).
- **2** "Start from": in Your soil mode intake, Z_n and the best inflow are derived from the soil, Eqs. 11.1 to 11.7, 5.27 and 11.22.
- **3** Intake "From my soil": Saxton-Rawls, Green-Ampt and Kostiakov, Eqs. 11.1 to 11.7.
- **4** Cut-off "Es = 1": cut-off time t_co, Eq. 11.16.
- **5** Field length L, slope S, inflow q and Manning n: the inputs of Eqs. 11.8 and 11.9.

**Equations behind this screen.** Eq. 5.27, Eq. 11.1, Eq. 11.2, Eq. 11.3, Eq. 11.4, Eq. 11.5, Eq. 11.6, Eq. 11.7, Eq. 11.8, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13.

*Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".*

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

*Eq. 11.1. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\begin{aligned} \theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031, \\ \theta_{1500} &= \theta_{1500,t} + \big(0.14\,\theta_{1500,t} - 0.02\big), \\ \theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299, \\ \theta_{33} &= \theta_{33,t} + \big(1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015\big), \end{aligned} \tag{11.1}$$

*Provenance (Eq. 11.1):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

*Eq. 11.2. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\begin{aligned} \theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078, \\ \theta_{(S-33)} &= \theta_{(S-33),t} + \big(0.636\,\theta_{(S-33),t} - 0.107\big), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043, \\ B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}. \end{aligned} \tag{11.2}$$

*Provenance (Eq. 11.2):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

*Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

*Provenance (Eq. 11.3):* **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

*Eq. 11.4. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

*Provenance (Eq. 11.4):* **[Engineered]**: initial and wetted water contents chosen for the workbench.

*Eq. 11.5. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

*Provenance (Eq. 11.5):* **[Literature]**: Green and Ampt (1911).

*Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 11.7. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

*Provenance (Eq. 11.7):* **[Literature]**: ordinary least squares on the log-log plot.

*Eq. 11.8. Where explained: chapter "Surface irrigation design", section "Inlet flow area (Manning)".*

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

*Provenance (Eq. 11.8):* **[Literature]**: Manning (1891).

*Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

*Provenance (Eq. 11.9):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

*Provenance (Eq. 11.10):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

*Provenance (Eq. 11.11):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

*Provenance (Eq. 11.12):* **[Engineered]**: alternative advance models set by worked examples and course notes.

*Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

*Provenance (Eq. 11.13):* **[Engineered]**: alternative advance models set by worked examples and course notes.

### 26.2.2 Layout in plan: the advance front, shaded by the depth already infiltrated

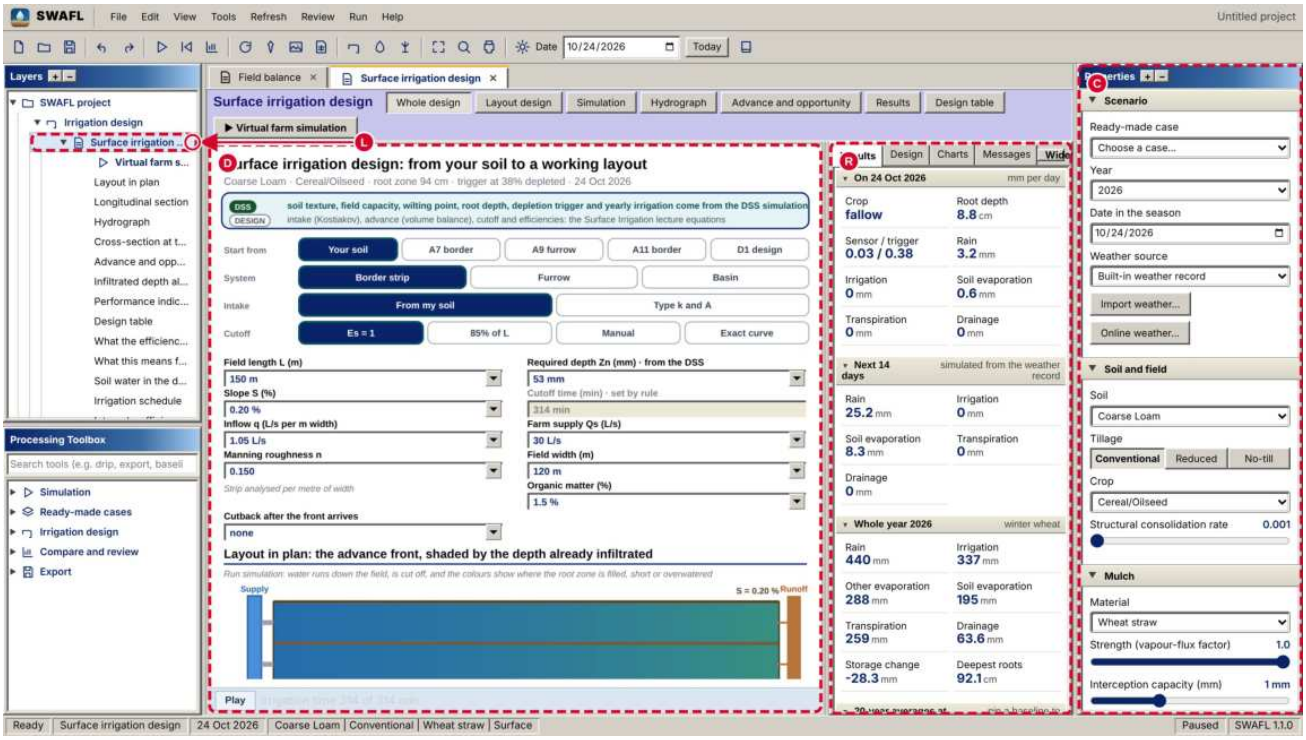



Figure 26.21: Surface irrigation design / Layout in plan: the whole workbench.

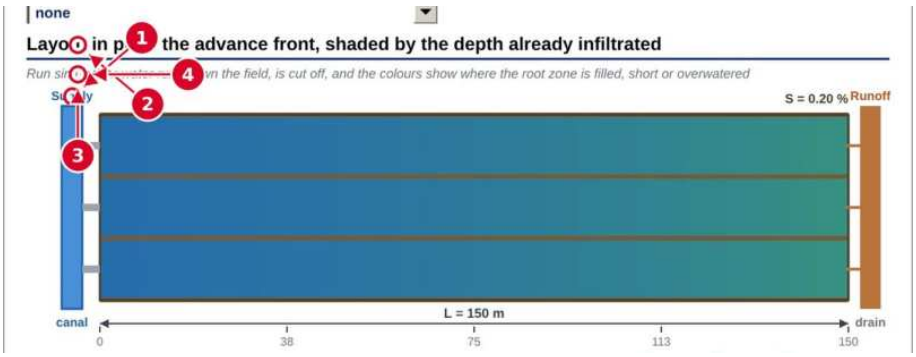



Figure 26.22: Surface irrigation design / Layout in plan: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Supply canal: inflow q enters the strip (flow area A_0 from Manning, Eq. 11.8).
- **2** Strip in plan: front position x(t) from Eq. 11.9; colours show infiltrated depth Z(x) against Z_n, Eq. 11.17.
- **3** Field length L (input).
- **4** Legend dry / short / filled / excess: classes of Z(x) against Z_n, Eq. 11.17.

**Equations behind this screen.** Eq. 5.27, Eq. 11.8, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13.

*Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".*

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

*Eq. 11.8. Where explained: chapter "Surface irrigation design", section "Inlet flow area (Manning)".*

$$Q = \frac{1}{n}\, A\, R_h^{2/3}\, S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

*Provenance (Eq. 11.8):* **[Literature]**: Manning (1891).

*Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$Q\,t = \sigma_y\, A_0\, x + \sigma_z\, k\, t^{A}\, w\, x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\, k\, t^{A}\, w}. \tag{11.9}$$

*Provenance (Eq. 11.9):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

*Provenance (Eq. 11.10):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

*Provenance (Eq. 11.11):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\, q^{-q_e}\, x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

*Provenance (Eq. 11.12):* **[Engineered]**: alternative advance models set by worked examples and course notes.

*Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\, k\, w\, L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

*Provenance (Eq. 11.13):* **[Engineered]**: alternative advance models set by worked examples and course notes.

### 26.2.3 Longitudinal section: water running over your soil and soaking in

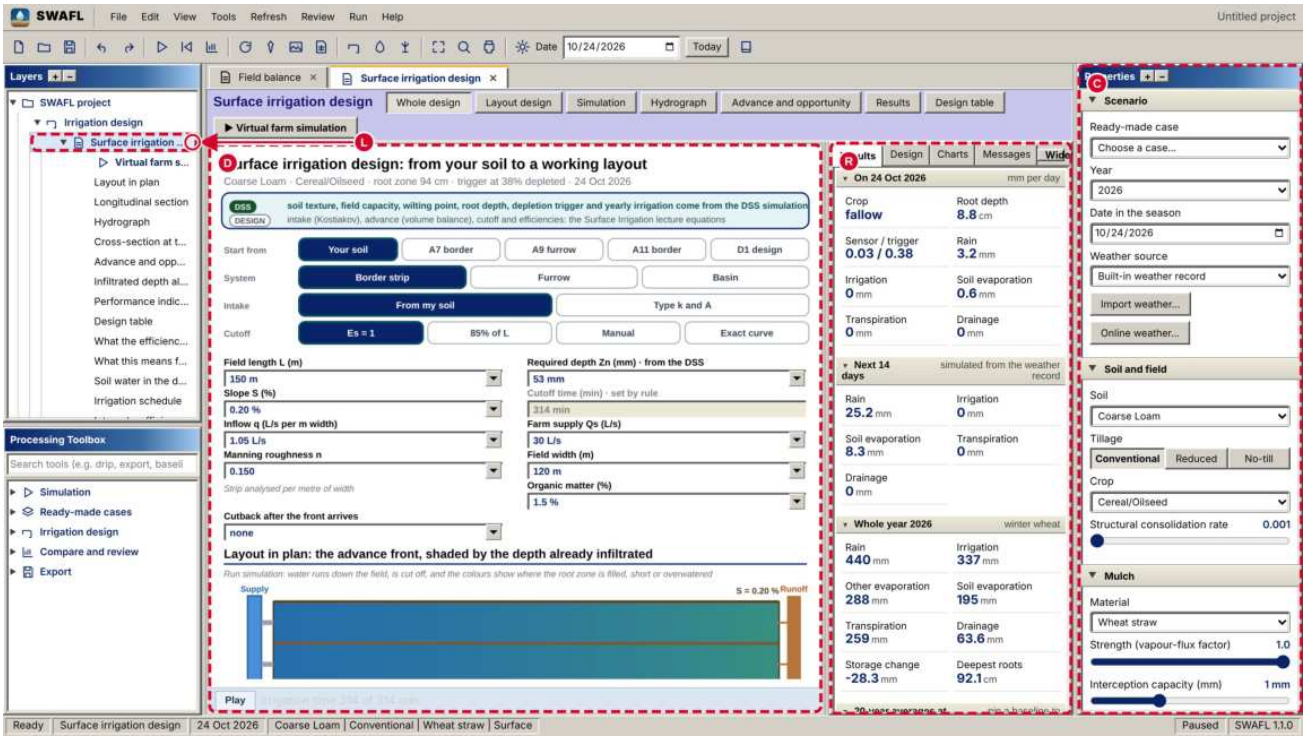



Figure 26.23: Surface irrigation design / Longitudinal section: the whole workbench.

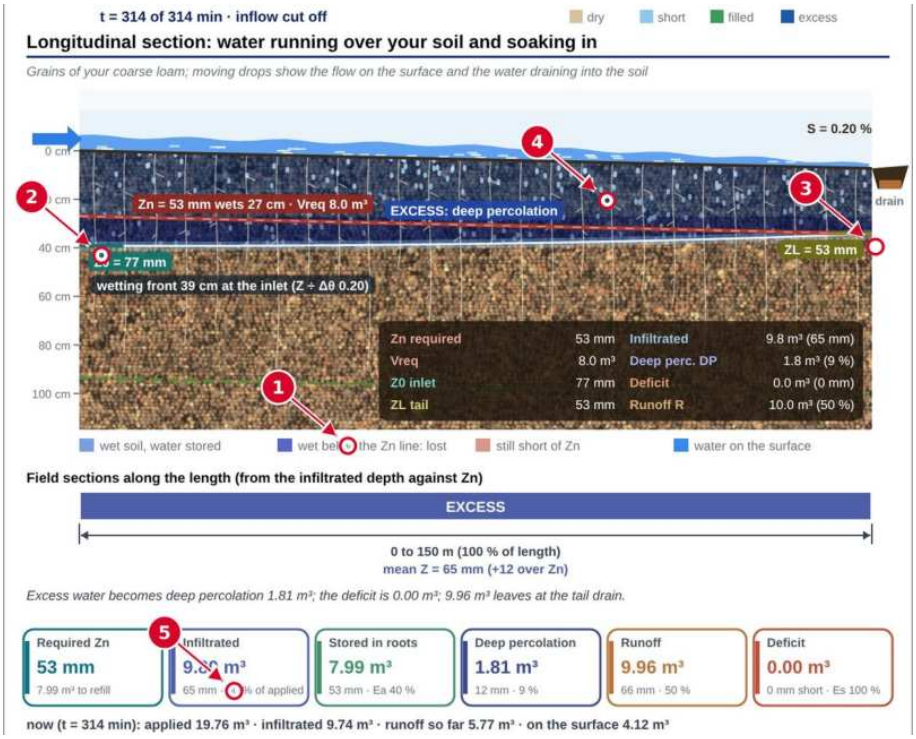



Figure 26.24: Surface irrigation design / Longitudinal section: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Z_n line: required depth from the DSS (Eq. 5.27) drawn as depth of wetted soil.
- **2** Z_0: depth infiltrated at the inlet, Eq. 11.17.
- **3** Z_L: depth infiltrated at the tail, Eq. 11.17.
- **4** EXCESS: water below the Z_n line is lost to deep percolation, V_dp of Eq. 11.18.
- **5** Infiltrated volume V_inf (tile at the foot), Eq. 11.18.

**Equations behind this screen.** Eq. 11.6, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13, Eq. 11.15, Eq. 11.16, Eq. 11.17.

*Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

*Provenance (Eq. 11.9):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

*Provenance (Eq. 11.10):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

*Provenance (Eq. 11.11):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

*Provenance (Eq. 11.12):* **[Engineered]**: alternative advance models set by worked examples and course notes.

*Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

*Provenance (Eq. 11.13):* **[Engineered]**: alternative advance models set by worked examples and course notes.

*Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$t_{st} = \left(\frac{Z_n}{k}\right)^{1/A}. \tag{11.15}$$

*Provenance (Eq. 11.15):* **[Literature]**: inversion of the Kostiakov law.

*Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

*Provenance (Eq. 11.16):* **[Engineered]**: cut-off rules chosen as design options.

*Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

*Provenance (Eq. 11.17):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

### 26.2.4 Hydrograph: inflow, infiltration and runoff through time

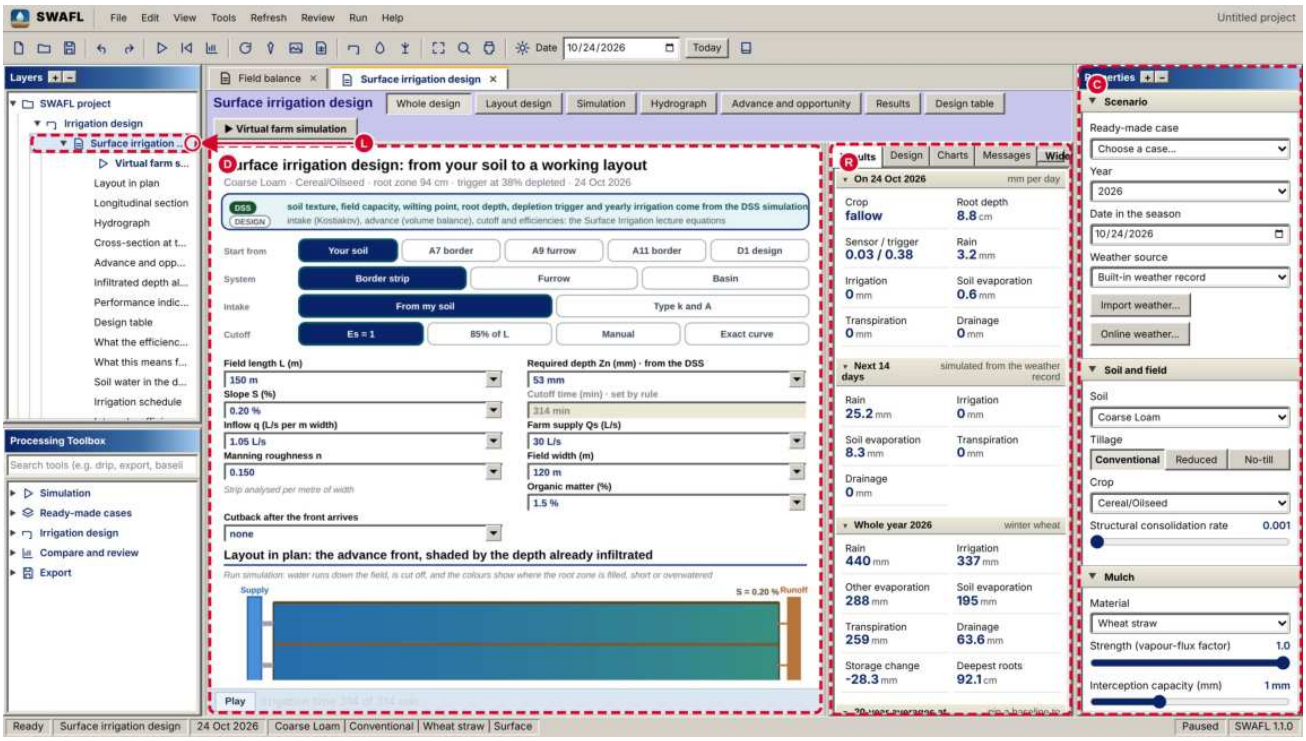



Figure 26.25: Surface irrigation design / Hydrograph: the whole workbench.

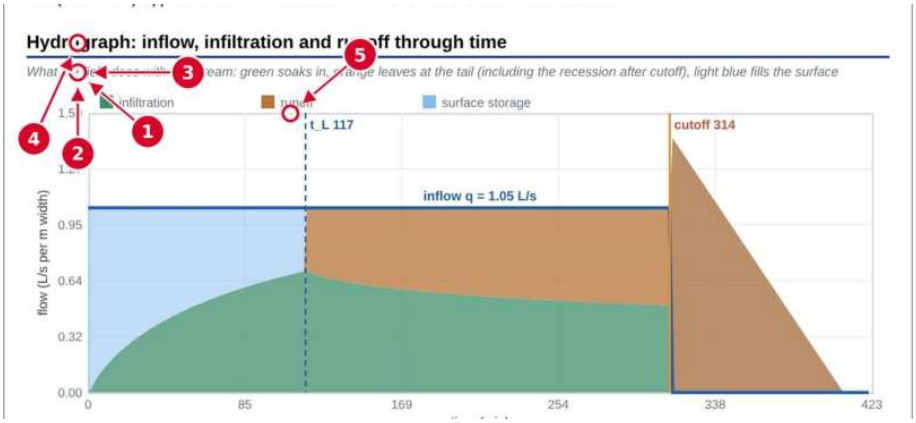



Figure 26.26: Surface irrigation design / Hydrograph: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Green: infiltration I(t): Eq. 11.20.
- **2** Orange: runoff R(t) with its recession after cut-off: Eqs. 11.20 and 11.21.
- **3** Blue: water stored on the surface: Eq. 11.21.
- **4** Cut-off time t_co: Eq. 11.16.
- **5** t_L: advance time to the tail, Eq. 11.9.

**Equations behind this screen.** Eq. 11.18, Eq. 11.20, Eq. 11.21.

*Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".*

$$
\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b<1:\ Q\,t_L + c_b\,Q\,(t_{co}-t_L)), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z,Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z-Z_n,0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n-Z,0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}}-V_{\text{inf}},\,0).
\end{aligned}
\tag{11.18}
$$

*Provenance (Eq. 11.18):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

*Eq. 11.20. Where explained: chapter "Surface irrigation design", section "Hydrograph".*

$$
V_c(t) = w\sum_i k\left[\min(t,t_{co})-\bar{t}_{x,i}\right]_+^A \Delta x\ \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}
$$

*Provenance (Eq. 11.20):* **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

*Eq. 11.21. Where explained: chapter "Surface irrigation design", section "Hydrograph".*

$$
R(t) = \frac{2V_s}{t_r}\Big(1-\frac{t-t_{co}}{t_r}\Big), \quad t_{co}<t<t_{co}+t_r. \tag{11.21}
$$

*Provenance (Eq. 11.21):* **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

### 26.2.5 Cross-section at the inlet: the wetted bulb in your soil, and the intake curve

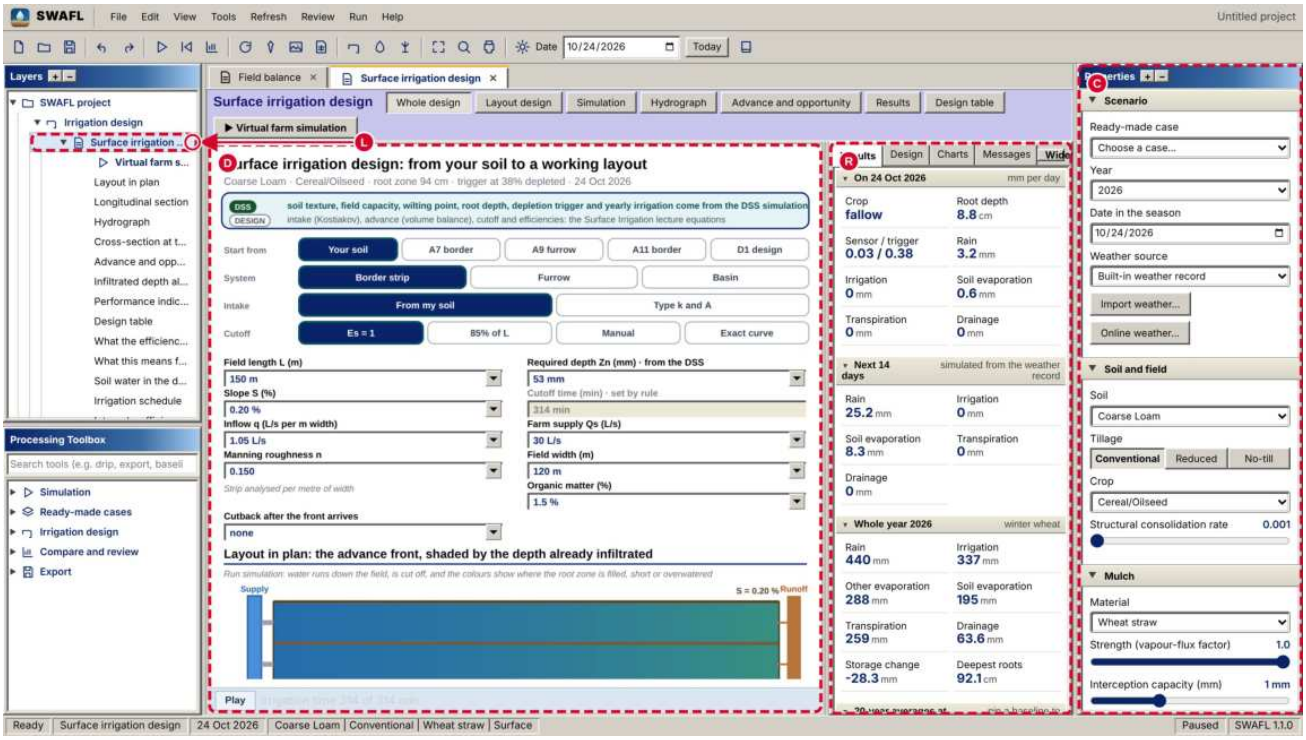



Figure 26.27: Surface irrigation design / Cross-section at the inlet: the whole workbench.

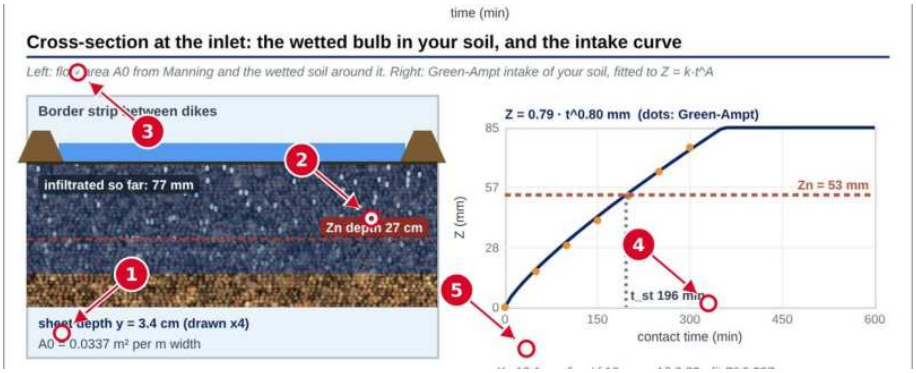



Figure 26.28: Surface irrigation design / Cross-section at the inlet: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sheet depth and flow area A_0 from Manning: Eq. 11.8.
- **2** Z_n depth: the required depth Z_n as a depth of wetted soil (storage of Eq. 11.4).
- **3** Intake curve Z = k tau^A fitted to Green-Ampt: Eqs. 11.5 to 11.7.
- **4** t_st: time to infiltrate the required depth, Eq. 11.15.
- **5** Ks, psi_f and Delta theta of the soil: Eqs. 11.2 to 11.4.

**Equations behind this screen.** Eq. 5.27, Eq. 11.1, Eq. 11.2, Eq. 11.3, Eq. 11.4, Eq. 11.5, Eq. 11.6, Eq. 11.7.

*Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".*

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

*Eq. 11.1. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + \big(0.14\,\theta_{1500,t} - 0.02\big),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + \big(1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015\big),
\end{aligned} \tag{11.1}$$

*Provenance (Eq. 11.1):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

*Eq. 11.2. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + \big(0.636\,\theta_{(S-33),t} - 0.107\big), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

*Provenance (Eq. 11.2):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

*Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

*Provenance (Eq. 11.3):* **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

*Eq. 11.4. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

*Provenance (Eq. 11.4):* **[Engineered]**: initial and wetted water contents chosen for the workbench.

*Eq. 11.5. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

*Provenance (Eq. 11.5):* **[Literature]**: Green and Ampt (1911).

*Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 11.7. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\big(F(\tau_i)/1000\big), \tag{11.7}$$

*Provenance (Eq. 11.7):* **[Literature]**: ordinary least squares on the log-log plot.

## 26.2.6 Advance and opportunity time

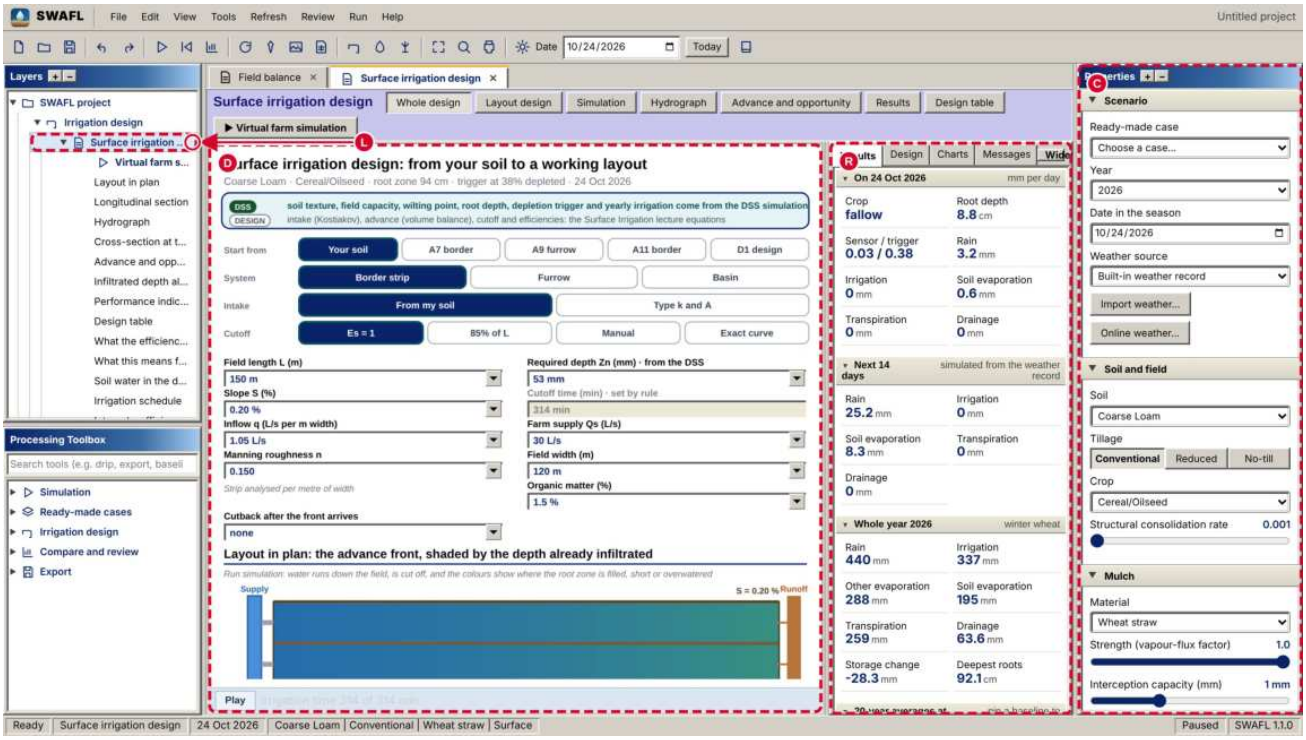



Figure 26.29: Surface irrigation design / Advance and opportunity time: the whole workbench.

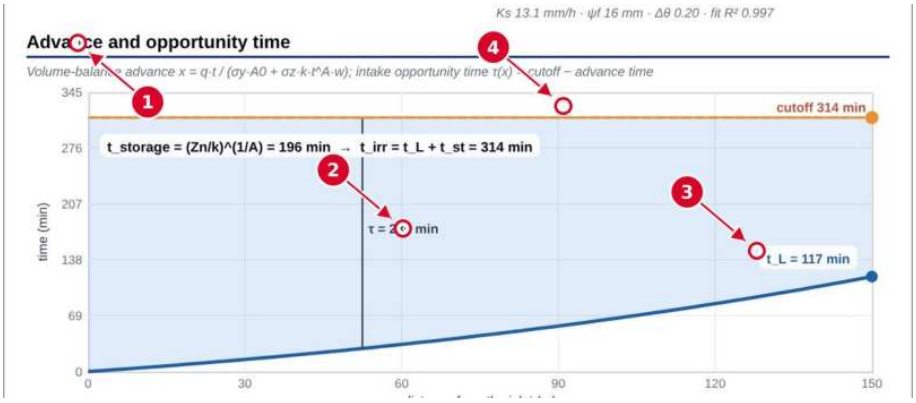



Figure 26.30: Surface irrigation design / Advance and opportunity time: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Opportunity time tau = t_co minus t_x: Eq. 11.17.
- **2** Advance curve x(t) from the volume balance: Eq. 11.9.
- **3** t_L: time for the front to reach the tail.
- **4** Cut-off 314 min = t_st + t_L: Eqs. 11.15 and 11.16.

**Equations behind this screen.** Eq. 11.6, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13, Eq. 11.15, Eq. 11.16, Eq. 11.17.

*Eq. 11.6. Where explained: chapter “Surface irrigation design”, section “Intake from your soil*

*(Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

*Provenance (Eq. 11.9):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

*Provenance (Eq. 11.10):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

*Provenance (Eq. 11.11):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

*Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

*Provenance (Eq. 11.12):* **[Engineered]**: alternative advance models set by worked examples and course notes.

*Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".*

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

*Provenance (Eq. 11.13):* **[Engineered]**: alternative advance models set by worked examples and course notes.

*Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

*Provenance (Eq. 11.15):* **[Literature]**: inversion of the Kostiakov law.

*Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

*Provenance (Eq. 11.16):* **[Engineered]**: cut-off rules chosen as design options.

*Eq. 11.17. Where explained: chapter “Surface irrigation design”, section “Cut-off, opportunity time and the infiltrated depth”.*

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

*Provenance (Eq. 11.17):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

## 26.2.7 Infiltrated depth along the field against the required depth Zn

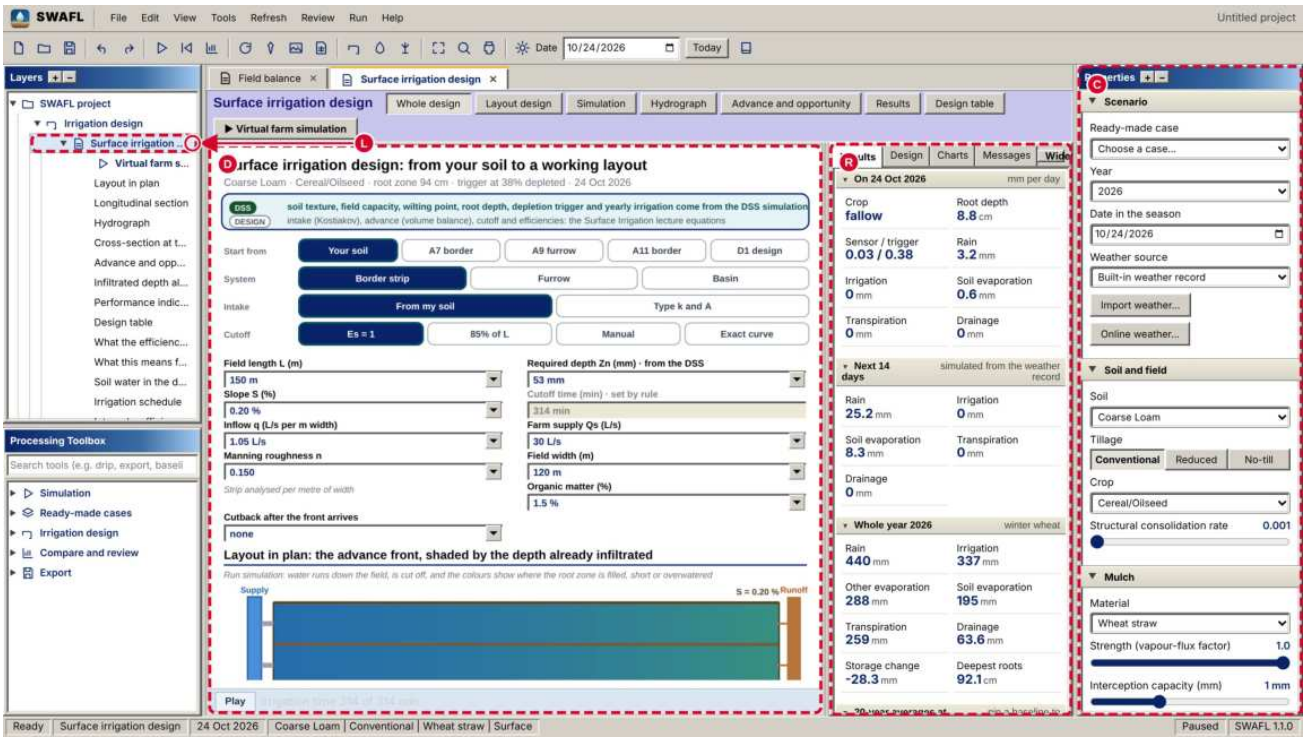



Figure 26.31: Surface irrigation design / Infiltrated depth along the field against the required depth Zn: the whole workbench.

Figure 26.32: Surface irrigation design / Infiltrated depth along the field against the required depth Zn: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Z_0: depth at the inlet, Eq. 11.17.
- **2** Z_L: depth at the tail, Eq. 11.17.
- **3** Z_n: required depth, Eq. 5.27.
- **4** Indigo: depth above Z_n, lost to deep percolation, V_dp of Eq. 11.18.
- **5** Green: water stored in the root zone, V_st of Eq. 11.18.

**Equations behind this screen.** Eq. 11.6, Eq. 11.15, Eq. 11.16, Eq. 11.17.

*Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$t_{st} = \left(\frac{Z_n}{k}\right)^{1/A}. \tag{11.15}$$

*Provenance (Eq. 11.15):* **[Literature]**: inversion of the Kostiakov law.

*Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$t_{co} = \begin{cases} t_L + t_{st} & \text{"}E_s = 1\text{": the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{"85 \% of L"} \\ t_{co}^{\text{set}} & \text{"Manual"} \end{cases} \tag{11.16}$$

*Provenance (Eq. 11.16):* **[Engineered]**: cut-off rules chosen as design options.

*Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".*

$$Z(x) = k\left[t_{co} - t_x(x)\right]^{A}, \quad Z_0 = k\,t_{co}^{A}\ (\text{inlet}), \qquad Z_L = k\,(t_{co} - t_L)^{A}\ (\text{tail}). \tag{11.17}$$

*Provenance (Eq. 11.17):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

## 26.2.8 Performance indices and the water balance of one irrigation

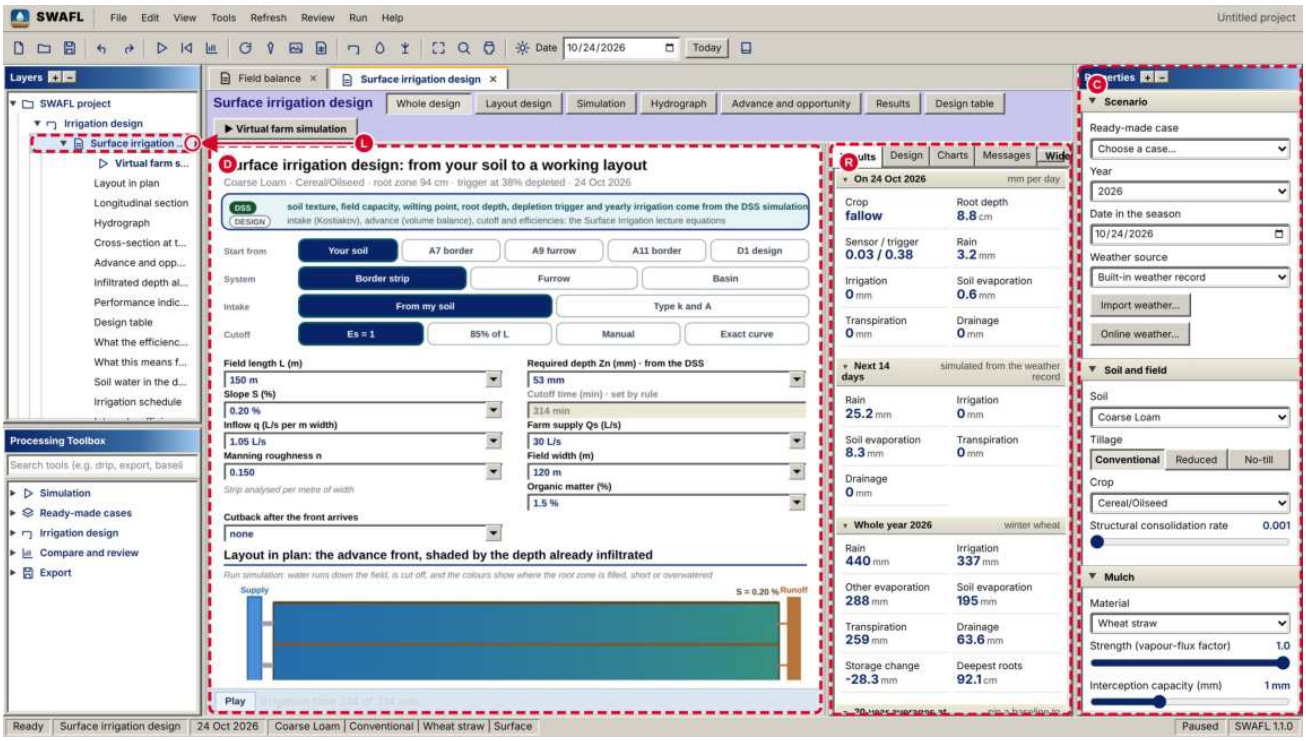



Figure 26.33: Surface irrigation design / Performance indices and the water balance of one irrigation: the whole workbench.

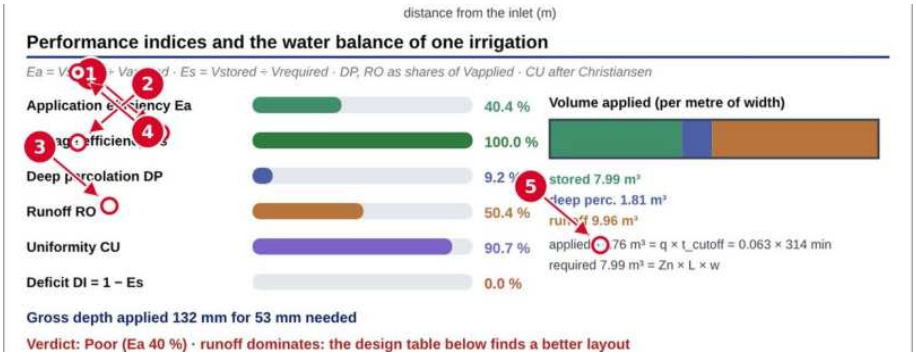



Figure 26.34: Surface irrigation design / Performance indices and the water balance of one irrigation: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a = V_st / V_app: Eq. 11.19.
- **2** Storage efficiency E_s = V_st / V_req: Eq. 11.19.
- **3** Deep percolation DP and runoff RO as shares of the applied volume: Eq. 11.19.
- **4** Uniformity CU after Christiansen: Eq. 11.19.
- **5** Volumes applied, stored, percolated and run off: Eq. 11.18.

**Equations behind this screen.** Eq. 11.18, Eq. 11.19.

*Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".*

$$
\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b<1:\ Q\,t_L + c_b\,Q\,(t_{co}-t_L)), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z,Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z-Z_n,0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n-Z,0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}}-V_{\text{inf}},\,0).
\end{aligned}
\tag{11.18}
$$

*Provenance (Eq. 11.18):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

*Eq. 11.19. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".*

$$
E_a = \frac{V_{\text{st}}}{V_{\text{app}}},\quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}},\quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}},\quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}},\quad \text{CU} = 100\Big(1-\frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i-\bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}
$$

*Provenance (Eq. 11.19):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

### 26.2.9 Design table: iterations toward an application efficiency of 85 % or more

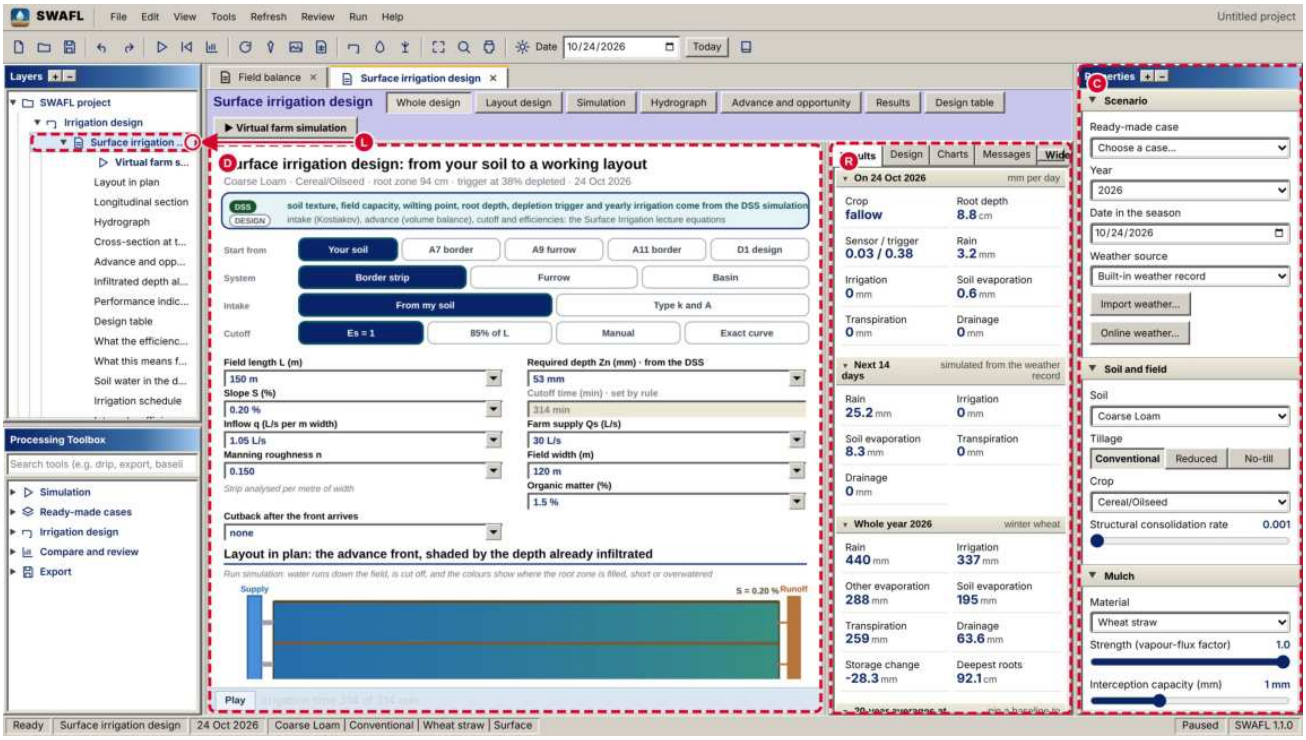



Figure 26.35: Surface irrigation design / Design table: the whole workbench.

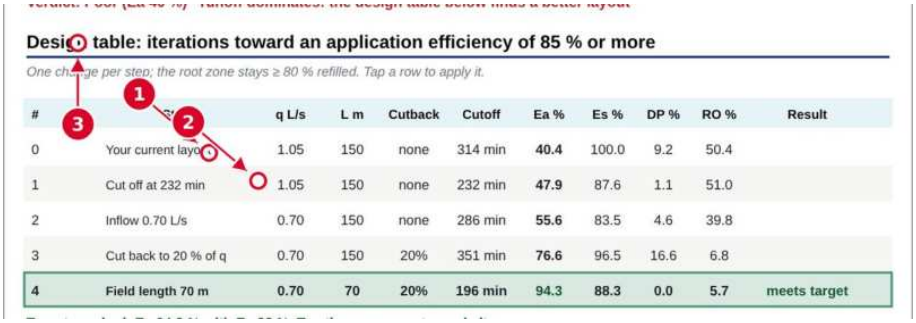



Figure 26.36: Surface irrigation design / Design table: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current layout; its score is Eq. 11.23.
- **2** Candidate moves in cut-off time, Eq. 11.16, then inflow (Eq. 11.22), cutback and length.
- **3** Winning design: E_a of at least 85 % with E_s of at least 80 %, chosen by the score of Eq. 11.23.

**Equations behind this screen.** Eq. 11.22, Eq. 11.23.

*Eq. 11.22. Where explained: chapter “Surface irrigation design”, section “Best inflow (”Your soil” mode)“.*

$$q_{\max} = \min\!\left\{ q^{\text{sys}}_{\max},\ \frac{0.04}{S} \cdot \frac{1000}{60} \right\} \ \mathrm{L\,s^{-1}} \quad (S \text{ in \%}), \tag{11.22}$$

*Provenance (Eq. 11.22):* **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

*Eq. 11.23. Where explained: chapter “Surface irrigation design”, section “The design table”.*

$$\text{score} = \min(E_a, 1) - 3\max(0,\, 0.8 - E_s) - 3\,[c_b < 1]\ \max(0,\, 0.05 - \text{RO}) - 5\max(0,\, E_a - 1). \tag{11.23}$$

*Provenance (Eq. 11.23):* **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

## 26.2.10 What the efficiency is most sensitive to

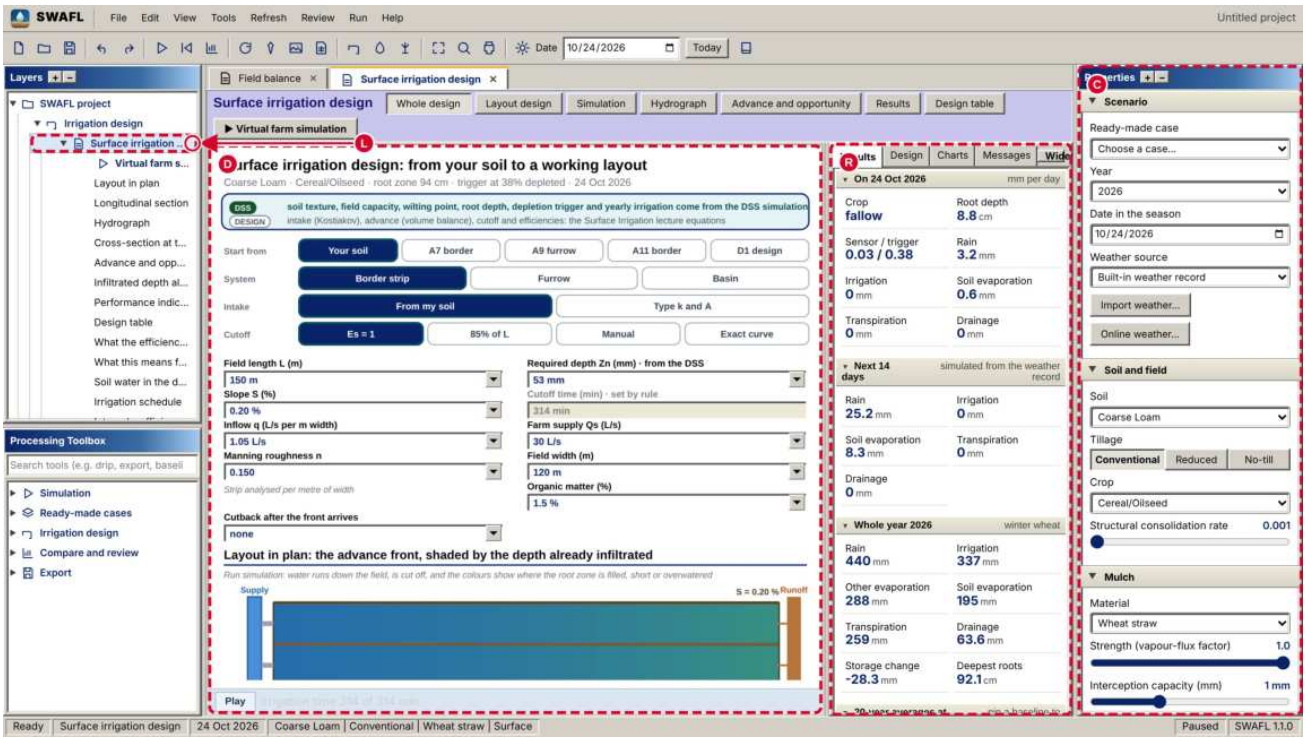



Figure 26.37: Surface irrigation design / What the efficiency is most sensitive to: the whole workbench.

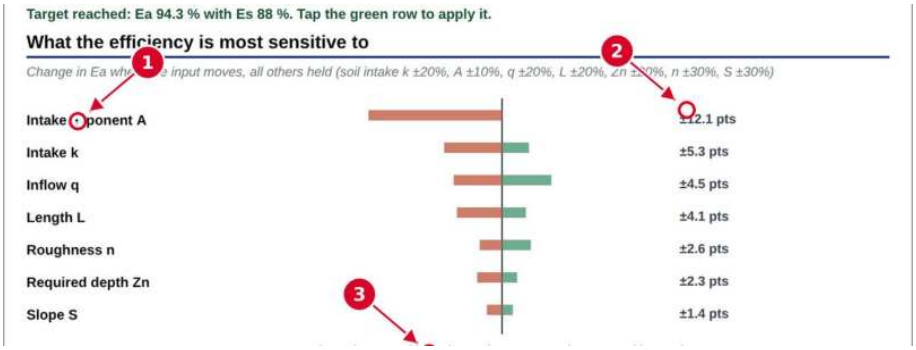



Figure 26.38: Surface irrigation design / What the efficiency is most sensitive to: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 11.19) when this input moves by 10 to 30 %.
- **2** Size of the change in percentage points.
- **3** Green raises E_a, red lowers it.

**Equations behind this screen.** none (the screen shows stored results).

## 26.2.11 What this means for the season and the farm

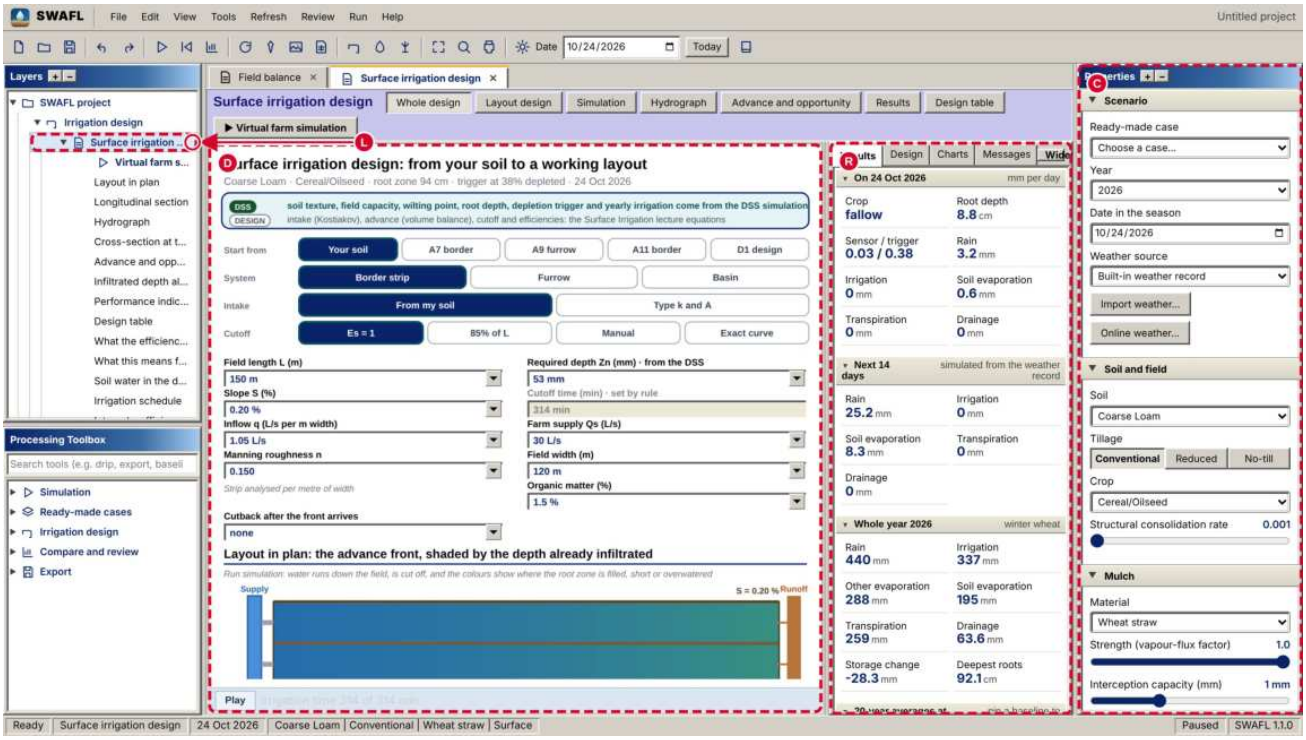



Figure 26.39: Surface irrigation design / What this means for the season and the farm: the whole workbench.

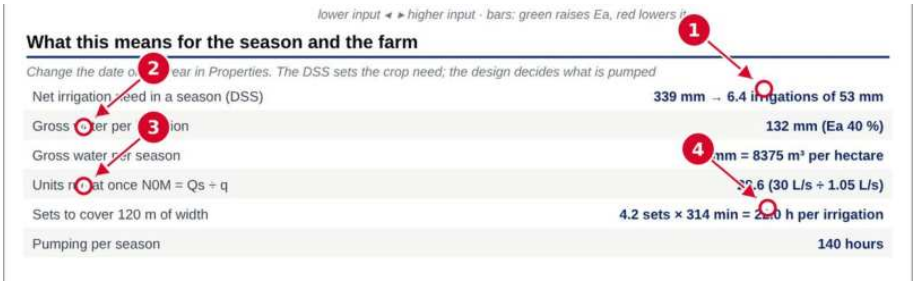



Figure 26.40: Surface irrigation design / What this means for the season and the farm: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season, n_ev = sum I / Z_n: Eq. 11.24.
- **2** Gross depth per irrigation, Z_g = Z_n / E_a: Eq. 11.24.
- **3** Units run at once, Q_s / q: Eq. 11.24.
- **4** Sets and hours per irrigation: Eq. 11.24.

**Equations behind this screen.** Eq. 11.24.

*Eq. 11.24. Where explained: chapter “Surface irrigation design”, section “Season and farm”.*

$$n_{\text{ev}} = \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g,$$
$$N_{0M} = \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \tag{11.24}$$

*Provenance (Eq. 11.24):* **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

## 26.2.12 Irrigation schedule: when to irrigate with this design

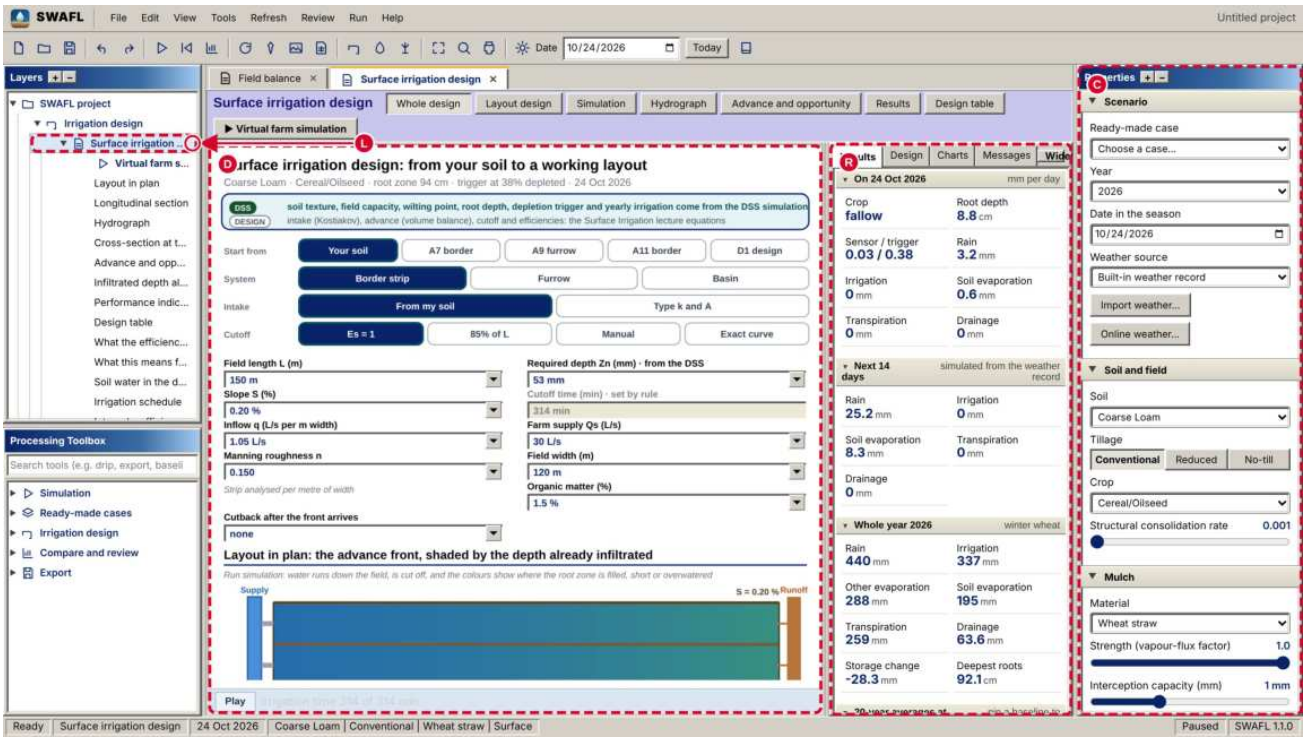


Figure 26.41: Surface irrigation design / Irrigation schedule: the whole workbench.

Figure 26.42: Surface irrigation design / Irrigation schedule: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates: from the Engine 2 irrigation record of the analogue year (Eqs. 4.1 and 4.2).
- **2** Net mm, I_k: the DSS irrigation depth that day.
- **3** Pumped mm = I_k / E_a: Eq. 11.24.
- **4** Run time of one round: Eq. 11.24.

**Equations behind this screen.** none (the screen shows stored results).

## 26.2.13 Intervals, efficiency and the advice in short

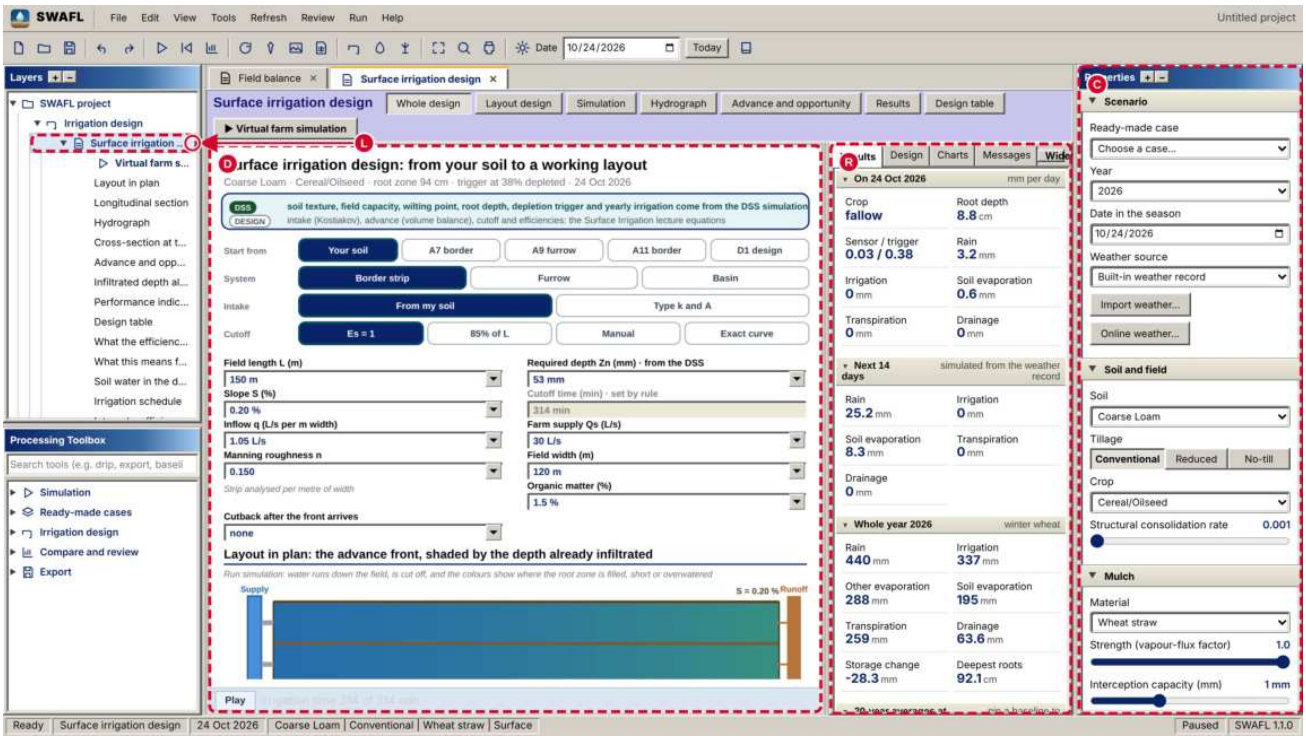


Figure 26.43: Surface irrigation design / Intervals, efficiency and the advice in short: the whole workbench.

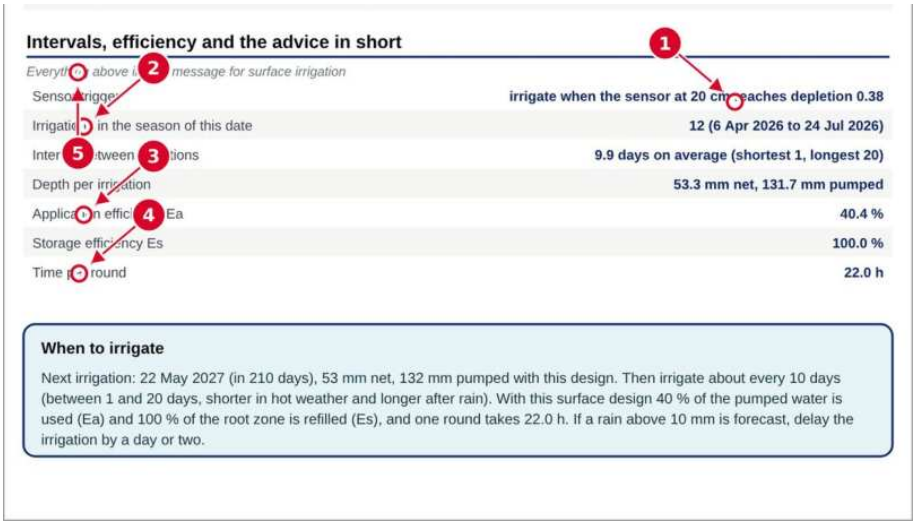



Figure 26.44: Surface irrigation design / Intervals, efficiency and the advice in short: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: depletion f_T, Eqs. 5.3 and 5.27.
- **2** Irrigations in the season: n_ev, Eq. 11.24.

- **3** Application efficiency E_a: Eq. 11.19.
- **4** Time per round: Eq. 11.24.
- **5** “When to irrigate”: a message assembled from the schedule and from E_a and E_s (Eqs. 11.19 and 11.24).

**Equations behind this screen.** Eq. 11.24.

*Eq. 11.24. Where explained: chapter “Surface irrigation design”, section “Season and farm”.*

$$n_{\text{ev}} = \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g,$$
$$N_{0M} = \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \tag{11.24}$$

*Provenance (Eq. 11.24):* **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

## 26.3 Drip irrigation design

### 26.3.1 The whole layer: inputs and overview

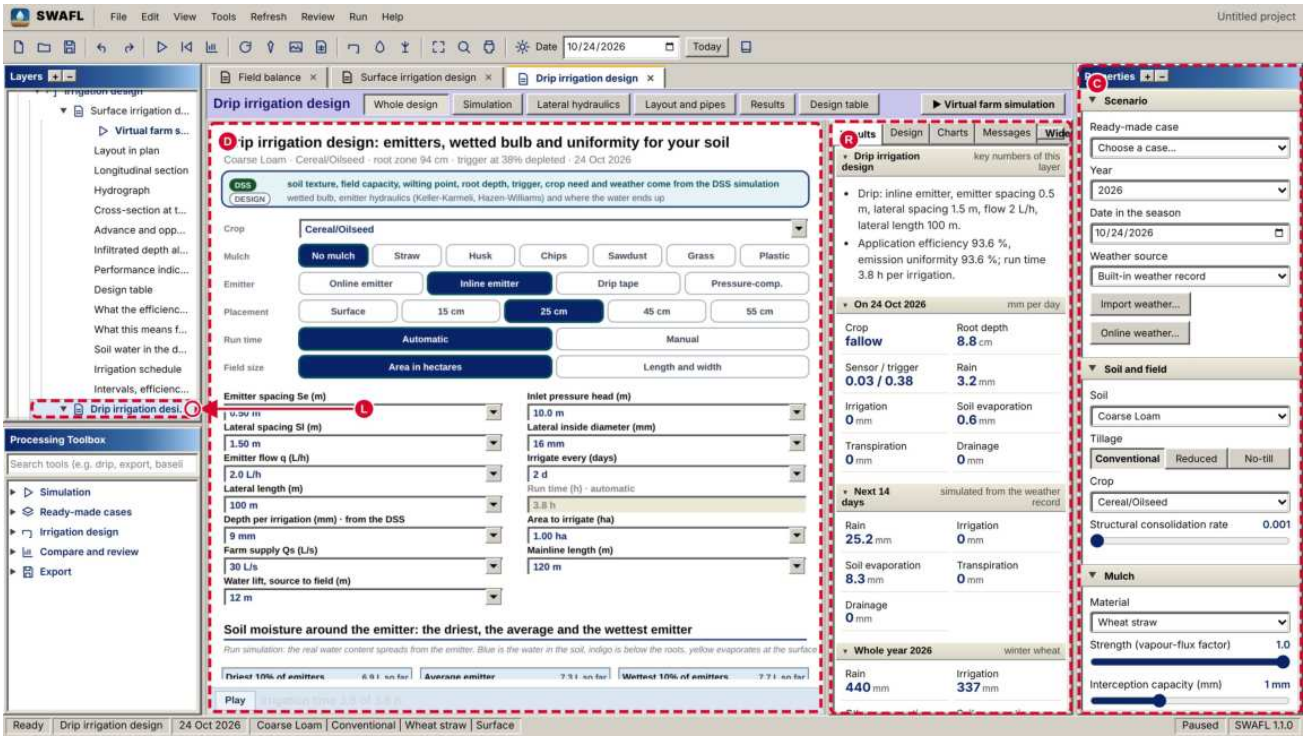



Figure 26.45: Drip irrigation design: the whole workbench.

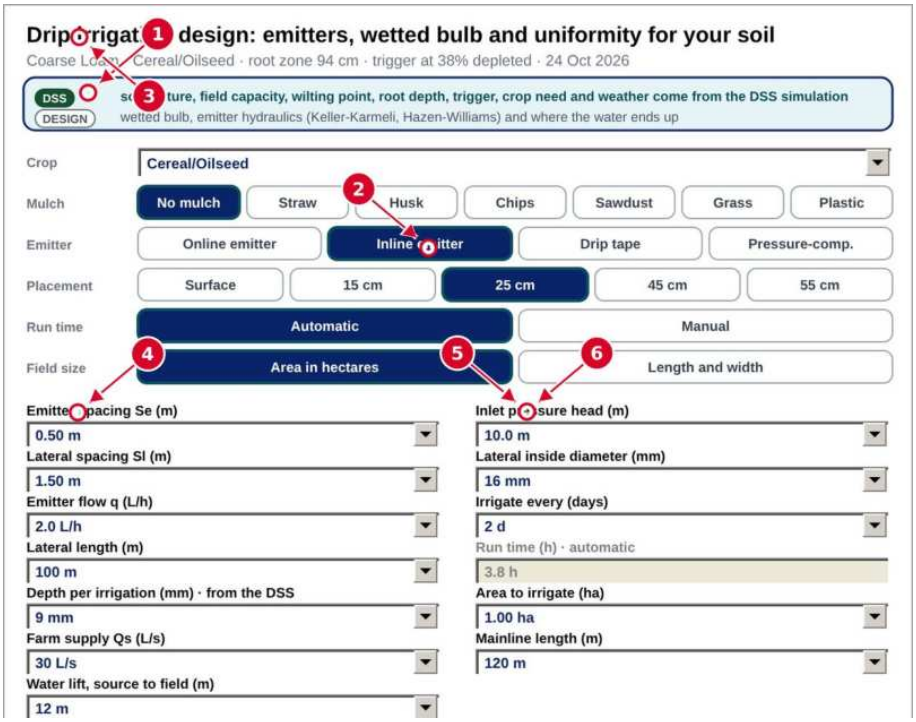



Figure 26.46: Drip irrigation design: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** DSS badge: soil, root depth, trigger and crop need come from Engines 1 and 2.
- **2** Emitter type sets the discharge exponent x: Eqs. 12.4 and 12.5.
- **3** Emitter depth z_d sets the bulb geometry: Eq. 12.10.
- **4** Emitter and lateral spacing, S_e and S_l: area served by one emitter, Eq. 12.17.
- **5** Emitter flow q at the inlet head H_0: Eq. 12.4.
- **6** Inlet pressure head H_0: Eqs. 12.3 and 12.4.

**Equations behind this screen.** Eq. 5.27, Eq. 11.3, Eq. 12.1, Eq. 12.2, Eq. 12.3, Eq. 12.4, Eq. 12.5, Eq. 12.6, Eq. 12.7, Eq. 12.8, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

*Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".*

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

*Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

*Provenance (Eq. 11.3):* **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

*Eq. 12.1. Where explained: chapter "Drip irrigation design", section "Inputs, step by step".*

$$Z_n = \min\big\{Z_n^{\mathrm{DSS}},\ \max(4,\ \mathrm{ET}_{\mathrm{pk}}\,\Delta t)\big\}\ \ \mathrm{mm}, \tag{12.1}$$

*Provenance (Eq. 12.1):* **[Engineered]**: depth and demand rules chosen for the workbench.

*Eq. 12.2. Where explained: chapter "Drip irrigation design", section "Inputs, step by step".*

$$\mathrm{ET}_{\mathrm{pk}} = \mathrm{clip}\Big\{\max_d \tfrac{1}{5}\sum_{j=0}^{4}(E_{d+j} + T_{d+j});\ 2.5,\ 9\Big\}. \tag{12.2}$$

*Provenance (Eq. 12.2):* **[Engineered]**: depth and demand rules chosen for the workbench.

*Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$h_{f,j} = \frac{10.67\,S_e\,Q_j^{1.852}}{C^{1.852}\,D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i\ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

*Provenance (Eq. 12.3):* **[Literature]**: Hazen and Williams (1905).

*Eq. 12.4. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^x \quad (\text{non-compensating; } h_j/H_0 \geq 0.01), \tag{12.4}$$

*Provenance (Eq. 12.4):* **[Literature]**: Keller and Karmeli (1974) emitter law.

*Eq. 12.5. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \ge H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

*Provenance (Eq. 12.5):* **[Engineered]**: pressure-compensating emitter model with chosen constants.

*Eq. 12.6. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600}\ \ (\mathrm{L\,s^{-1}}), \tag{12.6}$$

*Provenance (Eq. 12.6):* **[Literature]**: definitions of mean and minimum discharge.

*Eq. 12.7. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

*Provenance (Eq. 12.7):* **[Literature]**: Keller and Karmeli (1974).

*Eq. 12.8. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

*Provenance (Eq. 12.8):* **[Engineered]**: combined spread chosen to drive the bulb quantiles.

*Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

*Provenance (Eq. 12.9):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\rho(z) = r\,\sqrt{1 - \zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

*Provenance (Eq. 12.10):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

*Provenance (Eq. 12.11):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

*Provenance (Eq. 12.12):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.13. Where explained: chapter “Drip irrigation design”, section “The wetted bulb”.*

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

*Provenance (Eq. 12.13):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.14. Where explained: chapter “Drip irrigation design”, section “The wetted bulb”.*

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

*Provenance (Eq. 12.14):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.15. Where explained: chapter “Drip irrigation design”, section “The wetted bulb”.*

$$V_{\mathrm{ev}} = \min\big\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\big\}\quad (\mathrm{L},\ \text{area in m}^2,\ E_0\ \text{in mm d}^{-1}), \tag{12.15}$$

*Provenance (Eq. 12.15):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.16. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$V_i = \max\big\{0.02\,\bar V,\ \bar V\,(1+\sigma_q z_i)\big\},\qquad \bar V = \bar q\,t, \tag{12.16}$$

*Provenance (Eq. 12.16):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

*Eq. 12.17. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}),\qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\},\qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

*Provenance (Eq. 12.17):* **[Literature]**: definition of the requirement over the area served by one emitter.

*Eq. 12.18. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar q\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

*Provenance (Eq. 12.18):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

*Eq. 12.19. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle},\quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}},\quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle},\quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle},\quad Z_{\mathrm{gross}} = \frac{\bar V}{S_e S_l}, \tag{12.19}$$

*Provenance (Eq. 12.19):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

### 26.3.2 Soil moisture around the emitter: the driest, the average and the wettest emitter

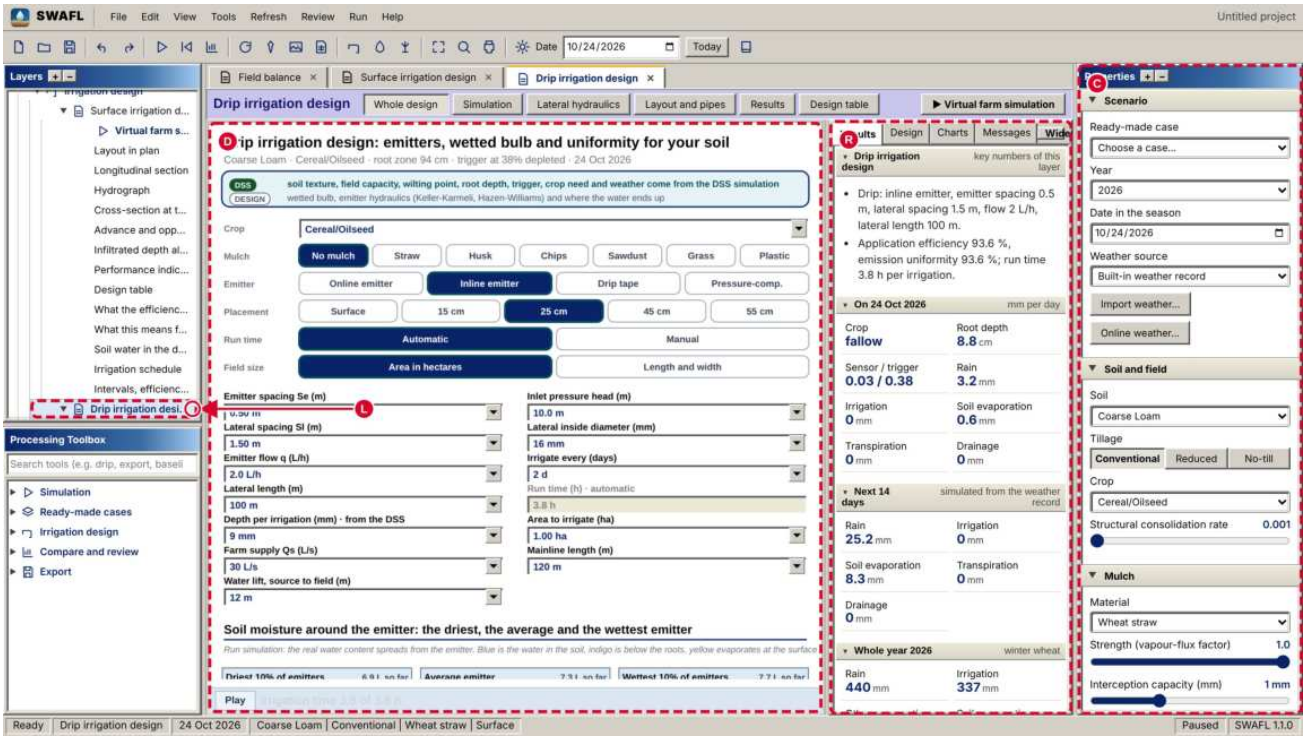



Figure 26.47: Drip irrigation design / Soil moisture around the emitter: the whole workbench.

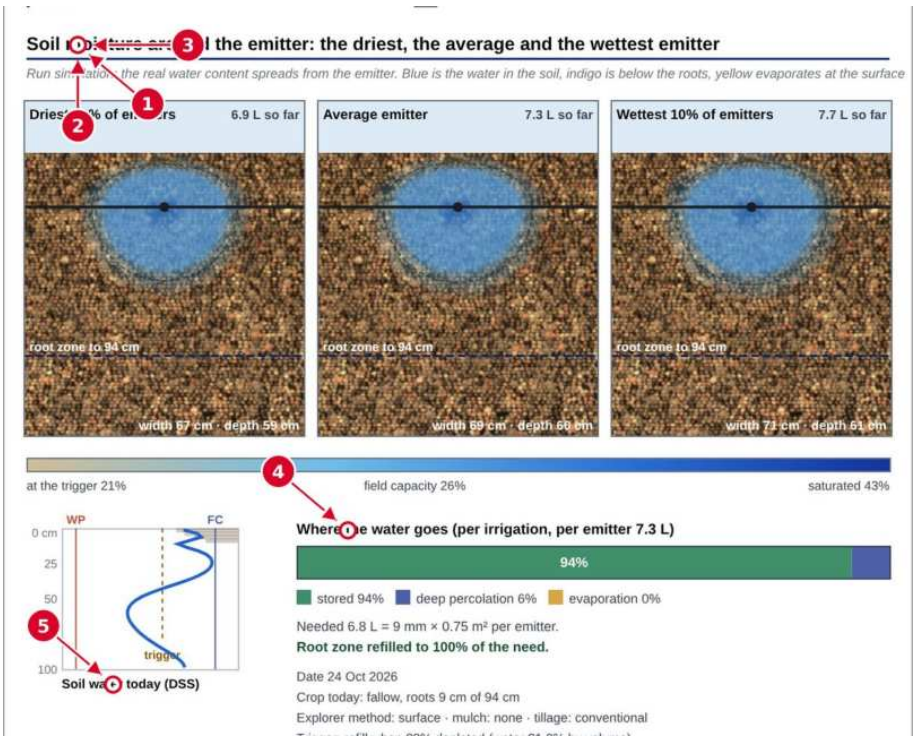



Figure 26.48: Drip irrigation design / Soil moisture around the emitter: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bulb under the driest 10 % of emitters (low quantile of Eq. 12.16).
- **2** Bulb under the average emitter: Eqs. 12.10 to 12.12.
- **3** Bulb under the wettest 10 % of emitters (high quantile of Eq. 12.16).
- **4** Where the water goes: stored, deep percolation (Eq. 12.14) and evaporation (Eq. 12.15).
- **5** Soil water today from the DSS profile.

**Equations behind this screen.** Eq. 11.3, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15.

*Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

*Provenance (Eq. 11.3):* **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

*Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

*Provenance (Eq. 12.9):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

*Provenance (Eq. 12.10):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

*Provenance (Eq. 12.11):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

*Provenance (Eq. 12.12):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

*Provenance (Eq. 12.13):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

*Provenance (Eq. 12.14):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.15. Where explained: chapter “Drip irrigation design”, section “The wetted bulb”.*

$$V_{\text{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\text{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

*Provenance (Eq. 12.15):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

### 26.3.3 Uniformity along the lateral: pressure falls with friction, so far emitters deliver less

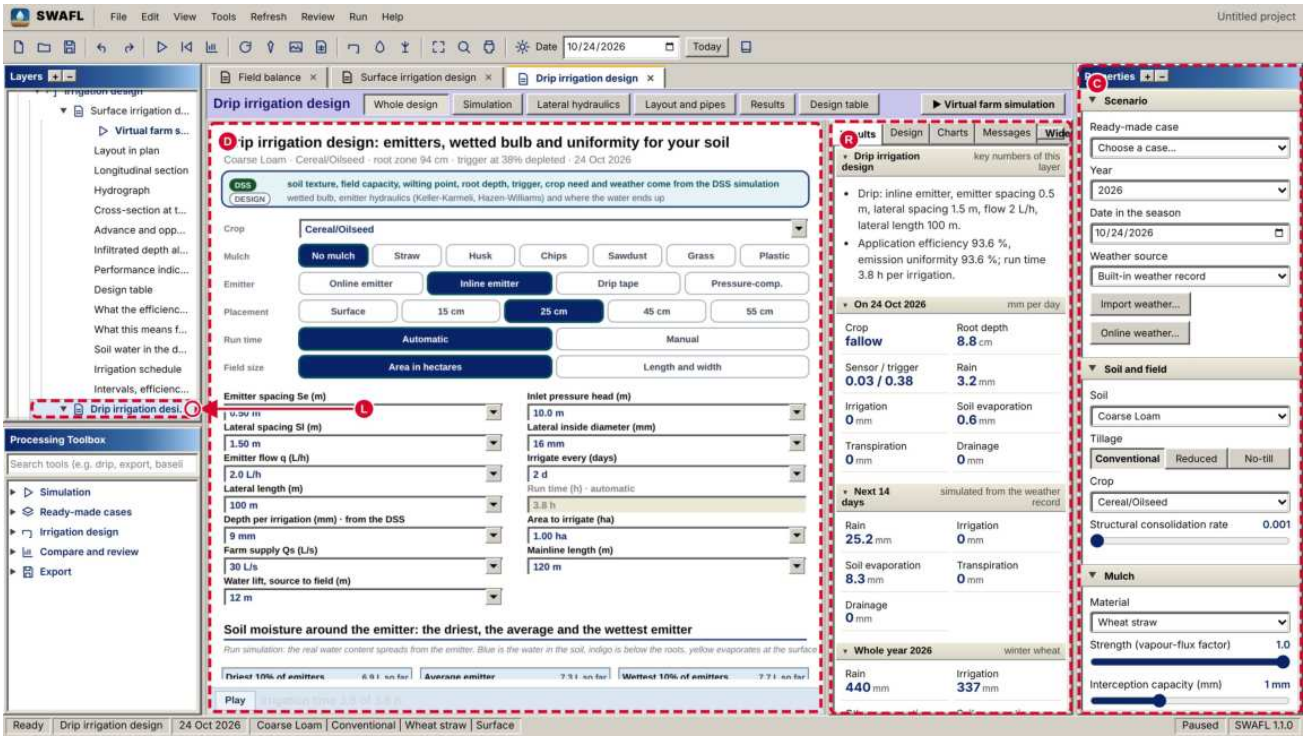



Figure 26.49: Drip irrigation design / Uniformity along the lateral: the whole workbench.

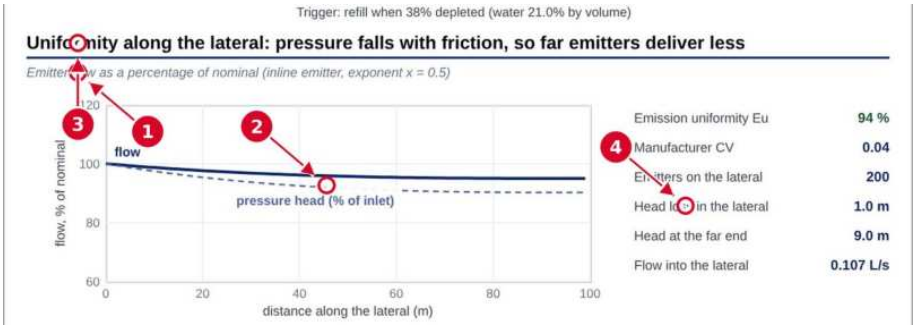



Figure 26.50: Drip irrigation design / Uniformity along the lateral: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Emitter flow q_j along the lateral: Eq. 12.4.
- **2** Pressure head h_j along the lateral, falling with friction: Eq. 12.3.
- **3** Emission uniformity E_u: Eq. 12.7.
- **4** Head loss in the lateral, sum of h_f: Eq. 12.3.

**Equations behind this screen.** Eq. 12.3, Eq. 12.4, Eq. 12.5, Eq. 12.6, Eq. 12.7, Eq. 12.8, Eq. 12.10, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

*Eq. 12.3. Where explained: chapter “Drip irrigation design”, section “Lateral hydraulics: head loss and flow along the lateral”.*

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

*Provenance (Eq. 12.3):* **[Literature]**: Hazen and Williams (1905).

*Eq. 12.4. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\ h_j/H_0 \geq 0.01), \tag{12.4}$$

*Provenance (Eq. 12.4):* **[Literature]**: Keller and Karmeli (1974) emitter law.

*Eq. 12.5. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

*Provenance (Eq. 12.5):* **[Engineered]**: pressure-compensating emitter model with chosen constants.

*Eq. 12.6. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\mathrm{lat}} = \frac{N\,\bar{q}}{3600} \;\; (\mathrm{L\,s^{-1}}), \tag{12.6}$$

*Provenance (Eq. 12.6):* **[Literature]**: definitions of mean and minimum discharge.

*Eq. 12.7. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

*Provenance (Eq. 12.7):* **[Literature]**: Keller and Karmeli (1974).

*Eq. 12.8. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

*Provenance (Eq. 12.8):* **[Engineered]**: combined spread chosen to drive the bulb quantiles.

*Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\rho(z) = r\,\sqrt{1 - \zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \geq z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \geq 1. \tag{12.10}$$

*Provenance (Eq. 12.10):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_{\mathrm{ev}} = \min\big\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\big\} \quad (\mathrm{L},\ \text{area in } \mathrm{m^2},\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

*Provenance (Eq. 12.15):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.16. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1+\sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

*Provenance (Eq. 12.16):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

*Eq. 12.17. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}), \qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\}, \qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

*Provenance (Eq. 12.17):* **[Literature]**: definition of the requirement over the area served by one emitter.

*Eq. 12.18. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

*Provenance (Eq. 12.18):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

*Eq. 12.19. Where explained: chapter “Drip irrigation design”, section “Spread between emitters and the efficiencies”.*

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}}, \quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle}, \quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle}, \quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

*Provenance (Eq. 12.19):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

### 26.3.4 How the wetted bulb grows with the run time

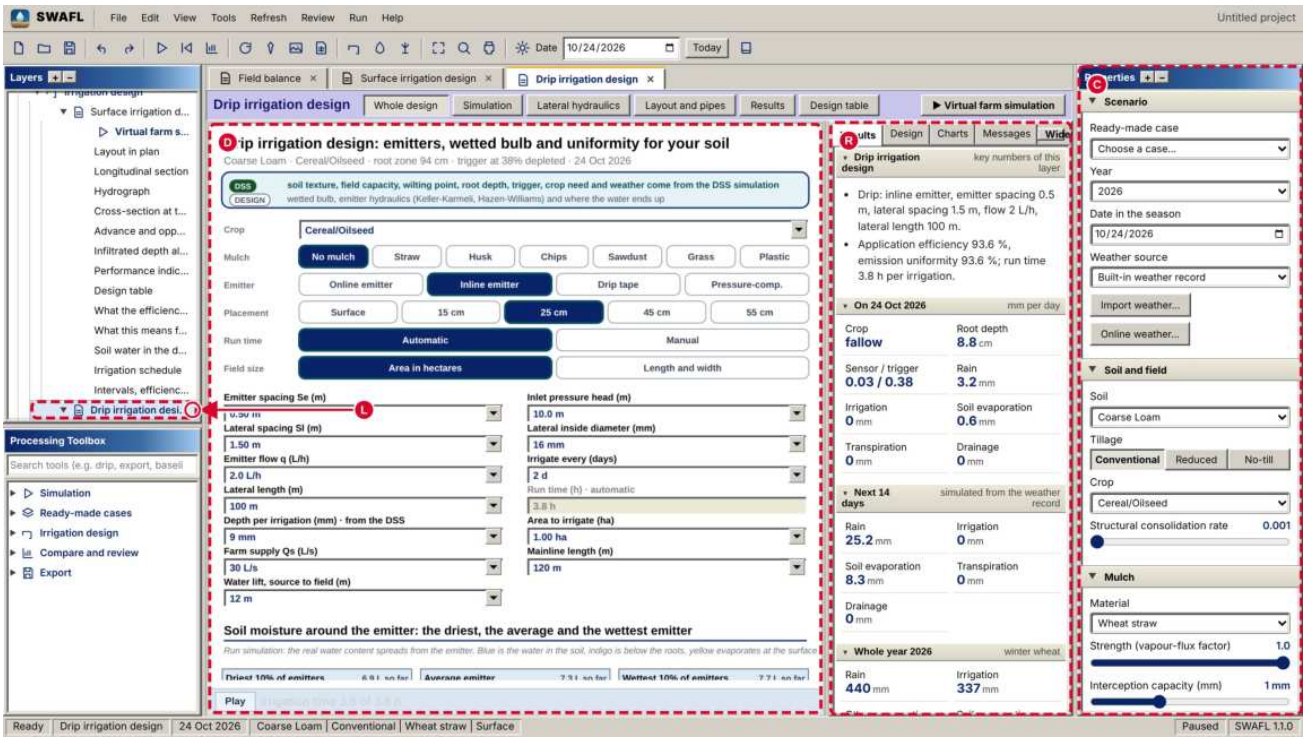



Figure 26.51: Drip irrigation design / How the wetted bulb grows with the run time: the whole workbench.

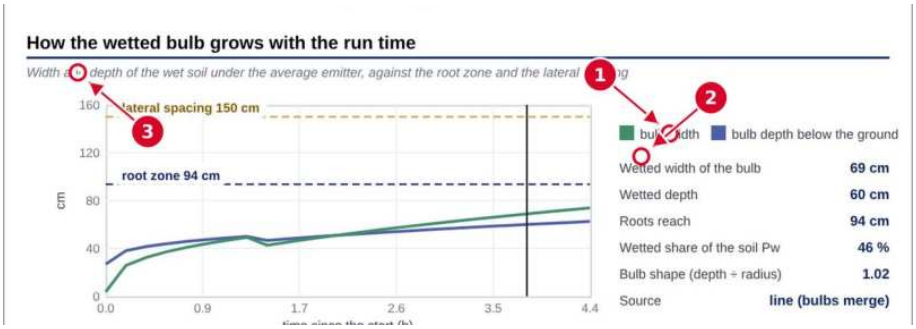



Figure 26.52: Drip irrigation design / How the wetted bulb grows with the run time: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bulb width (green) and depth (indigo) against time: Eqs. 12.10 to 12.12.
- **2** Root-zone depth R_max, against which the bulb depth gives deep percolation: Eq. 12.14.
- **3** Lateral spacing S_l: the bulbs merge when the width reaches it, P_w of Eq. 12.13.

**Equations behind this screen.** Eq. 11.3, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15.

*Eq. 11.3. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)”.*

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

*Provenance (Eq. 11.3):* **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

*Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

*Provenance (Eq. 12.9):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

*Provenance (Eq. 12.10):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

*Provenance (Eq. 12.11):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

*Provenance (Eq. 12.12):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

*Provenance (Eq. 12.13):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

*Provenance (Eq. 12.14):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\} \quad (\mathrm{L},\ \text{area in } \mathrm{m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

*Provenance (Eq. 12.15):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

### 26.3.5 Result for this design

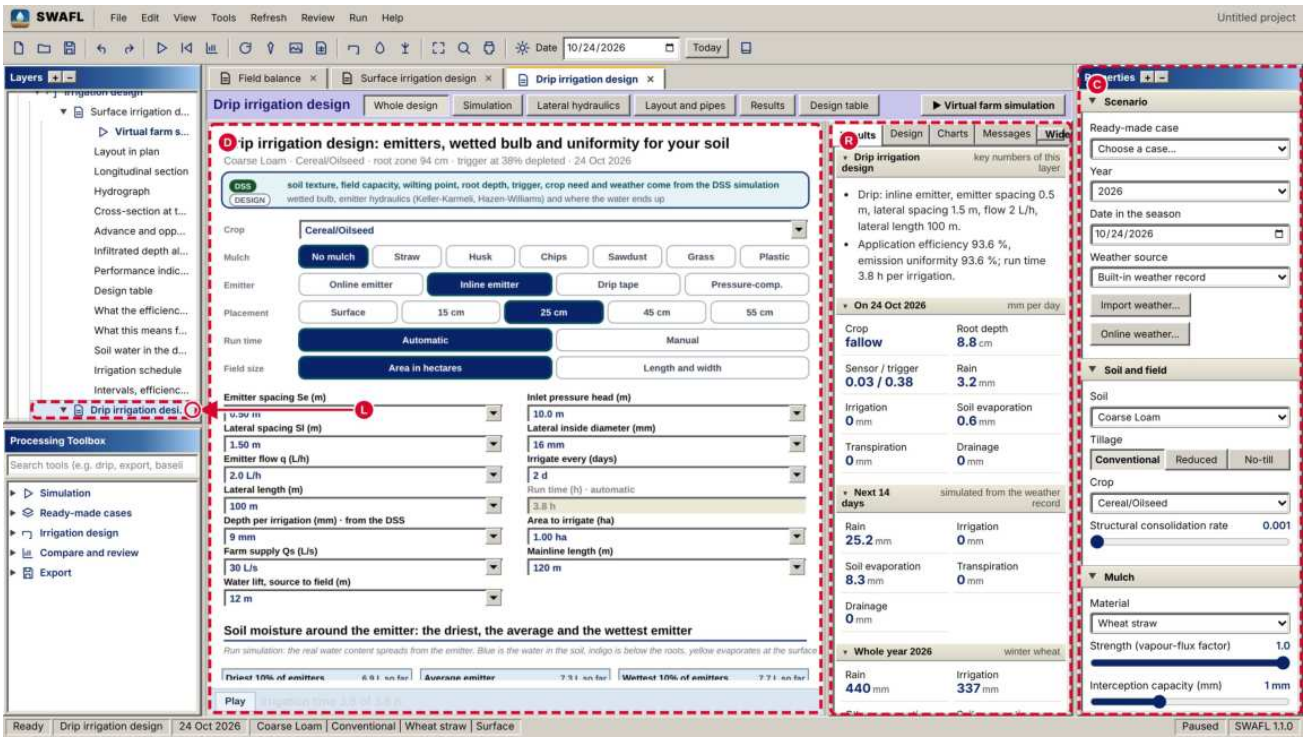



Figure 26.53: Drip irrigation design / Result for this design: the whole workbench.

Figure 26.54: Drip irrigation design / Result for this design: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a: Eq. 12.19.
- **2** Root-zone refilled E_s: Eq. 12.19.
- **3** Emission uniformity E_u: Eq. 12.7.
- **4** Deep percolation: Eq. 12.14.
- **5** Surface evaporation: Eq. 12.15.
- **6** Gross depth per irrigation: Eqs. 12.17 and 12.18.

**Equations behind this screen.** Eq. 12.10, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

*Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$\rho(z) = r\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \geq z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \geq 1. \tag{12.10}$$

*Provenance (Eq. 12.10):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".*

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

*Provenance (Eq. 12.15):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

*Eq. 12.16. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".*

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1 + \sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

*Provenance (Eq. 12.16):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

*Eq. 12.17. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".*

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}), \qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\}, \qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

*Provenance (Eq. 12.17):* **[Literature]**: definition of the requirement over the area served by one emitter.

*Eq. 12.18. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".*

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

*Provenance (Eq. 12.18):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

*Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".*

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}}, \quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle}, \quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle}, \quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

*Provenance (Eq. 12.19):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

### 26.3.6 Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals

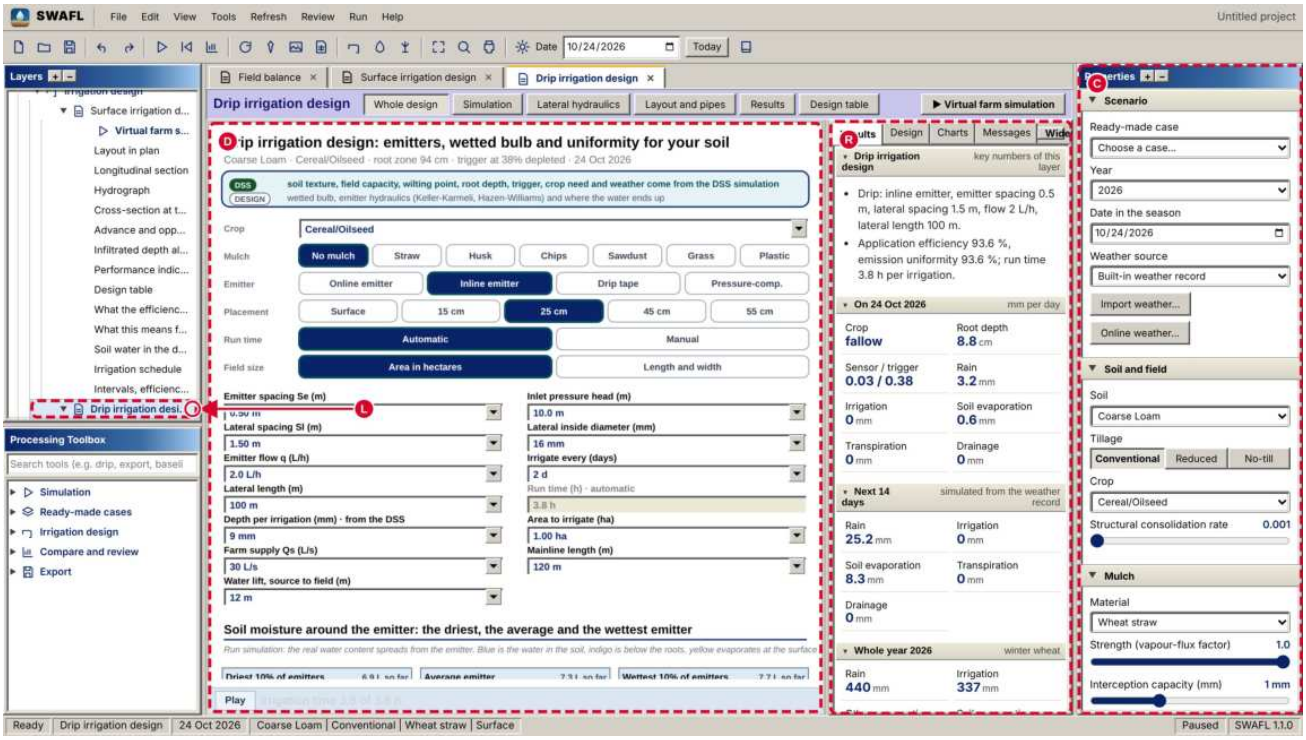



Figure 26.55: Drip irrigation design / Layout and infrastructure: the whole workbench.

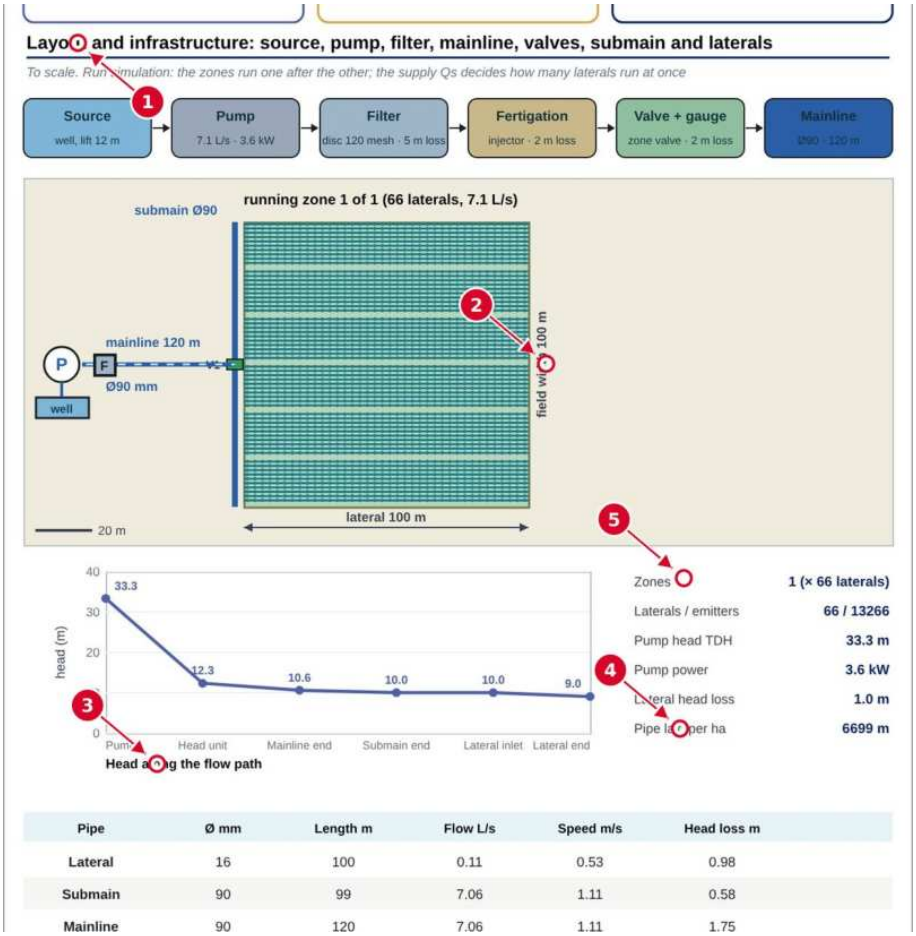



Figure 26.56: Drip irrigation design / Layout and infrastructure: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Source, pump, filter and valves: pump power from the head of Eq. 12.23.
- **2** Laterals across the field, N_L = field width / S_l: Eq. 12.20.
- **3** Head along the flow path: TDH of Eq. 12.23.
- **4** Pipe table: smallest commercial diameter, losses by Eq. 12.22 (Hazen-Williams, Eq. 12.3).
- **5** Zones, laterals and emitters: Eq. 12.20.

**Equations behind this screen.** Eq. 12.3, Eq. 12.20, Eq. 12.21, Eq. 12.22, Eq. 12.23.

*Eq. 12.3. Where explained: chapter “Drip irrigation design”, section “Lateral hydraulics: head loss and flow along the lateral”.*

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

*Provenance (Eq. 12.3):* **[Literature]**: Hazen and Williams (1905).

*Eq. 12.20. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.*

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \left\lfloor \frac{W_f}{S_l} \right\rfloor, \quad n_{\text{zone}} = \left\lceil \frac{N_L\, Q_{\text{lat}}}{Q_s} \right\rceil, \quad n_l = \left\lceil \frac{N_L}{n_{\text{zone}}} \right\rceil, \quad Q_z = n_l\, Q_{\text{lat}}, \tag{12.20}$$

*Provenance (Eq. 12.20):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 12.21. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.*

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\} \text{ mm} \tag{12.21}$$

*Provenance (Eq. 12.21):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 12.22. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.*

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

*Provenance (Eq. 12.22):* **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

*Eq. 12.23. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.*

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9 \text{ m}, \qquad P = \frac{\rho g\, Q_z\, \text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65} \text{ kW}. \tag{12.23}$$

*Provenance (Eq. 12.23):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

### 26.3.7 Design table: iterations toward an application efficiency of 85 % or more

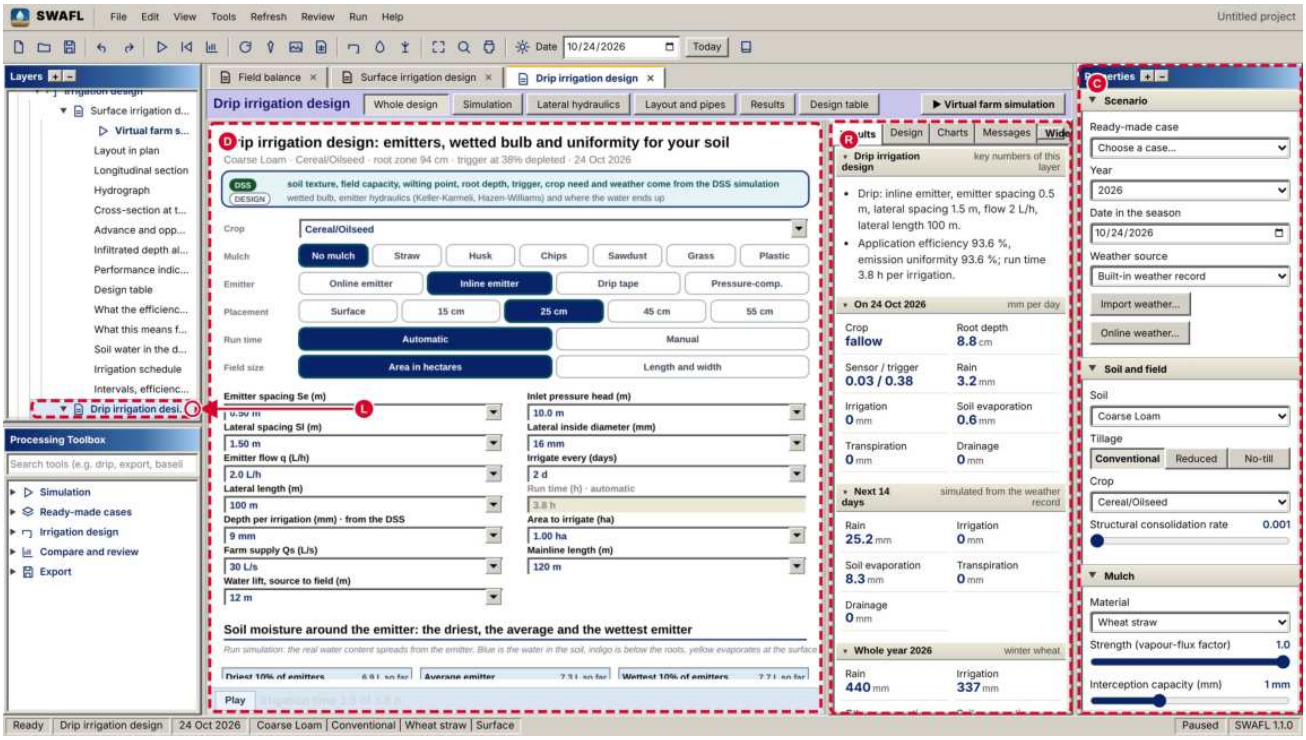



Figure 26.57: Drip irrigation design / Design table: the whole workbench.

Figure 26.58: Drip irrigation design / Design table: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current design; its score is Eq. 12.26.
- **2** Column E_a: Eq. 12.19.
- **3** Target reached: score = min(E_a, 1) minus 2.5 max(0, 0.9 minus E_s), Eq. 12.26.

**Equations behind this screen.** Eq. 12.26.

*Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.*

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

*Provenance (Eq. 12.26):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

## 26.3.8 What the efficiency is most sensitive to

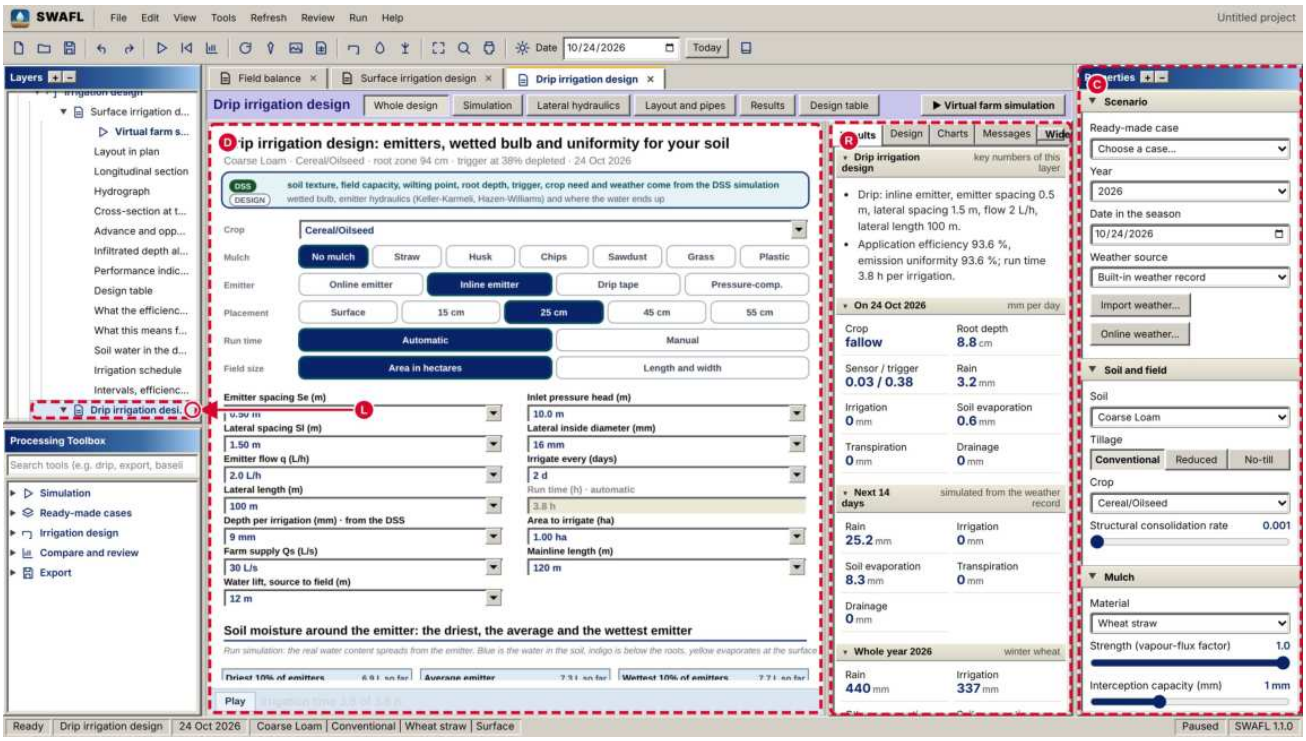



Figure 26.59: Drip irrigation design / What the efficiency is most sensitive to: the whole workbench.

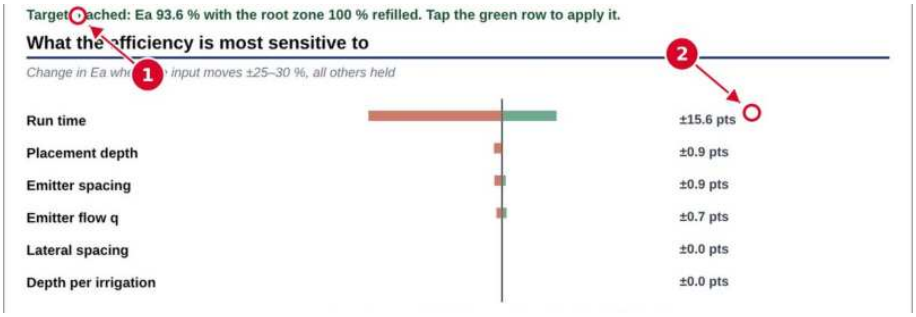



Figure 26.60: Drip irrigation design / What the efficiency is most sensitive to: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 12.19) when the run time moves by 25 to 30 %.
- **2** Size of the change in percentage points.

**Equations behind this screen.** none (the screen shows stored results).

## 26.3.9 What this means for the season and the farm

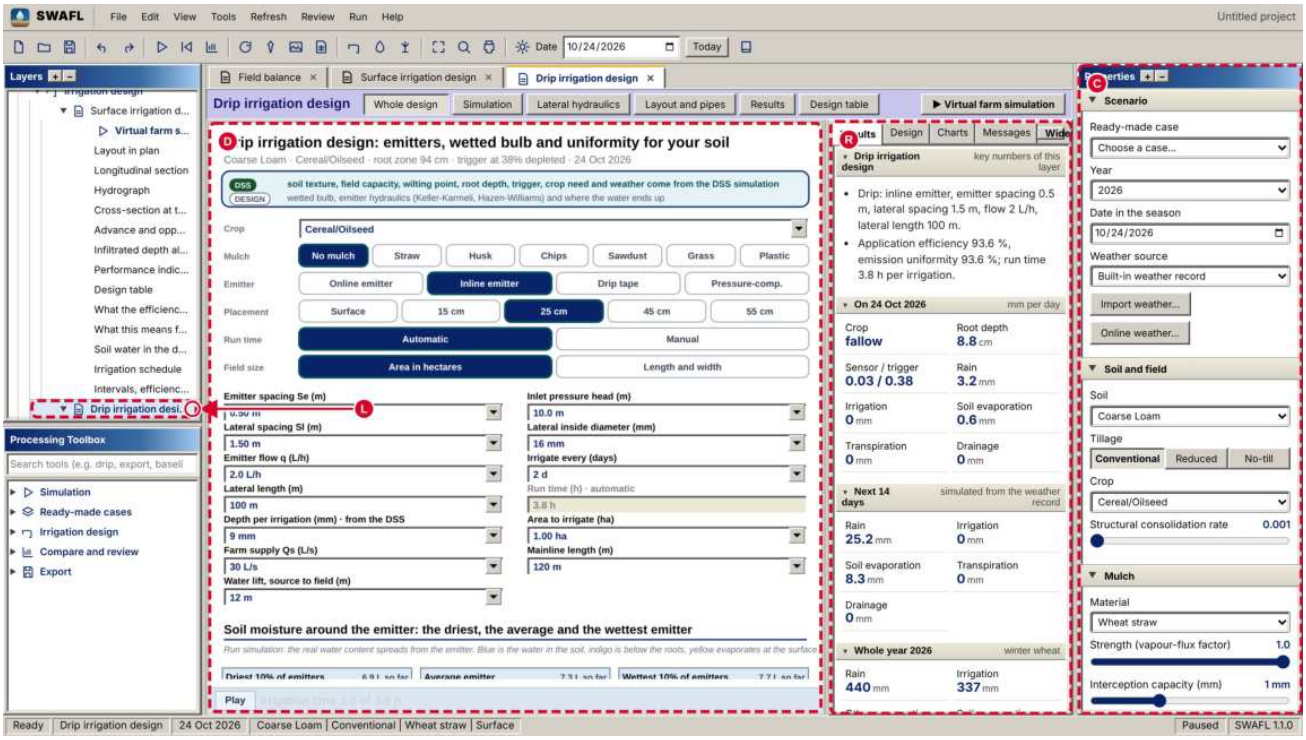



Figure 26.61: Drip irrigation design / What this means for the season and the farm: the whole workbench.

Figure 26.62: Drip irrigation design / What this means for the season and the farm: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season, n_ev = y_I / Z_n: Eq. 12.24.
- **2** Gross depth per irrigation: Eq. 12.24.
- **3** System flow per hectare: Eq. 12.24.
- **4** Sets and hours: Eq. 12.25.

**Equations behind this screen.** Eq. 12.24, Eq. 12.25.

*Eq. 12.24. Where explained: chapter “Drip irrigation design”, section “Season and farm”.*

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\mathrm{L\,s^{-1}\,ha^{-1}}), \tag{12.24}$$

*Provenance (Eq. 12.24):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.*

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

*Provenance (Eq. 12.25):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

### 26.3.10 Irrigation schedule: when to irrigate with this design

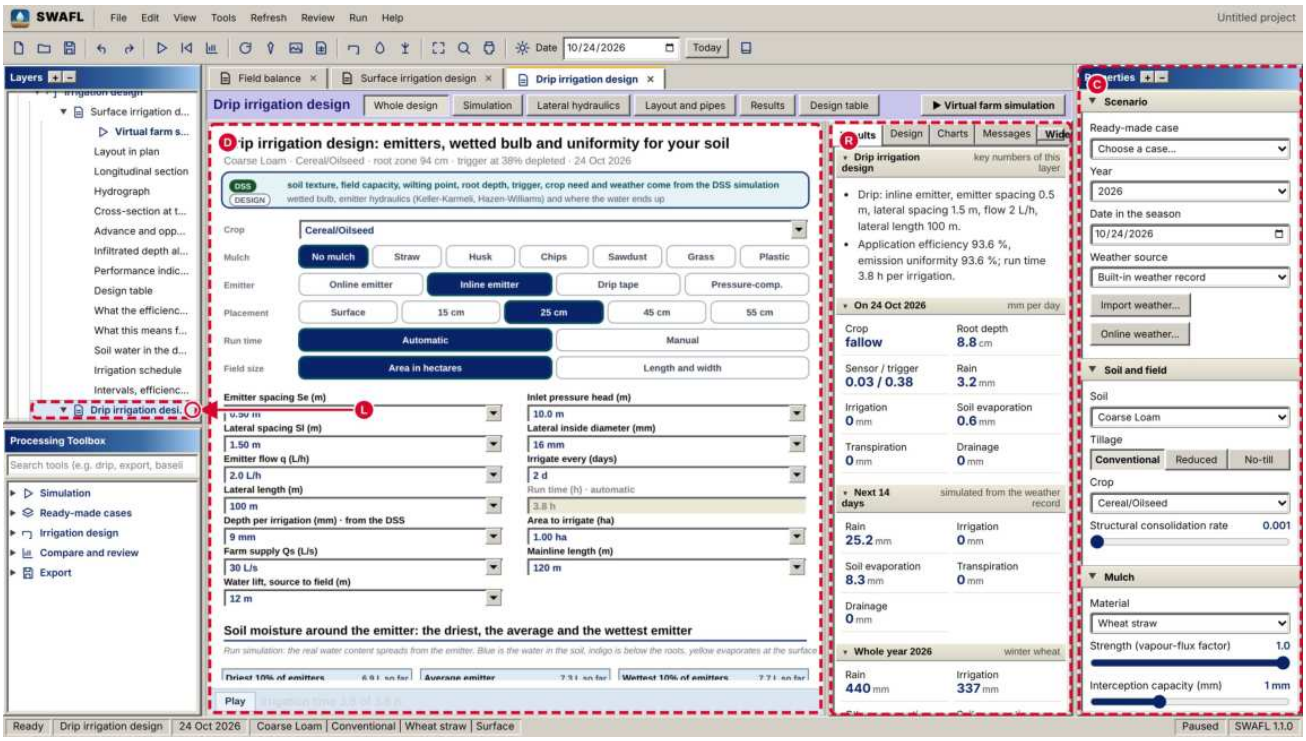



Figure 26.63: Drip irrigation design / Irrigation schedule: the whole workbench.

| # | Date | In | Net mm | Pumped mm | Run time | Gap to next |
|---|---|---|---|---|---|---|
| 1 | 22 May 2027 | 210 days | 9 | 10 | 3.8 h | 13 days |
| 2 | 4 Jun 2027 | 223 days | 9 | 10 | 3.8 h | 11 days |
| 3 | 15 Jun 2027 | 234 days | 9 | 10 | 3.8 h | 9 days |
| 4 | 24 Jun 2027 | 243 days | 9 | 10 | 3.8 h | 9 days |
| 5 | 3 Jul 2027 | 252 days | 9 | 10 | 3.8 h | 10 days |
| 6 | 13 Jul 2027 | 262 days | 9 | 10 | 3.8 h | 16 days |
| 7 | 29 Jul 2027 | 278 days | 9 | 10 | 3.8 h | 251 days |
| 8 | 5 Apr 2028 | 529 days | 9 | 10 | 3.8 h | - |

Figure 26.64: Drip irrigation design / Irrigation schedule: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates from the Engine 2 irrigation record.
- **2** Pumped mm = I_k / E_a: Eq. 12.24.
- **3** Run time: Eq. 12.18 (automatic run time).

**Equations behind this screen.** Eq. 12.24, Eq. 12.25.

*Eq. 12.24. Where explained: chapter “Drip irrigation design”, section “Season and farm”.*

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

*Provenance (Eq. 12.24):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.*

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

*Provenance (Eq. 12.25):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

## 26.3.11 Intervals, efficiency and the advice in short

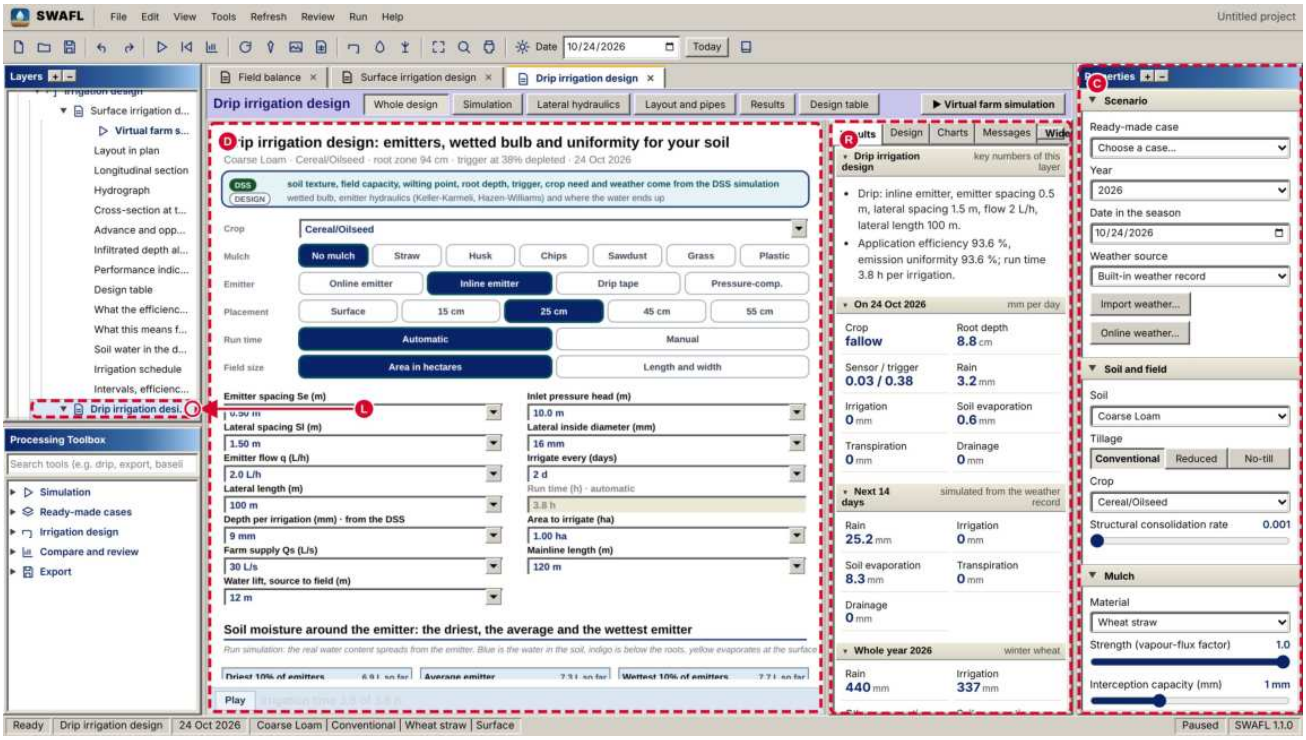


Figure 26.65: Drip irrigation design / Intervals, efficiency and the advice in short: the whole workbench.

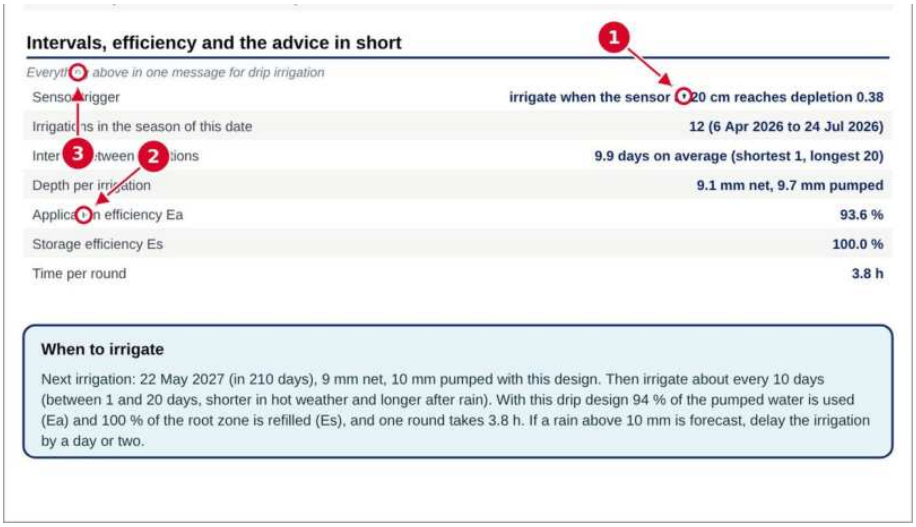



Figure 26.66: Drip irrigation design / Intervals, efficiency and the advice in short: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: Eqs. 5.3 and 5.27.
- **2** Application efficiency E_a: Eq. 12.19.

- **3** “When to irrigate”: message built from the schedule and from E_a and E_s.

**Equations behind this screen.** Eq. 12.24, Eq. 12.25.

*Eq. 12.24. Where explained: chapter “Drip irrigation design”, section “Season and farm”.*

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

*Provenance (Eq. 12.24):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.*

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

*Provenance (Eq. 12.25):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

## 26.4 Sprinkler design and pipes

### 26.4.1 The whole layer: inputs and overview

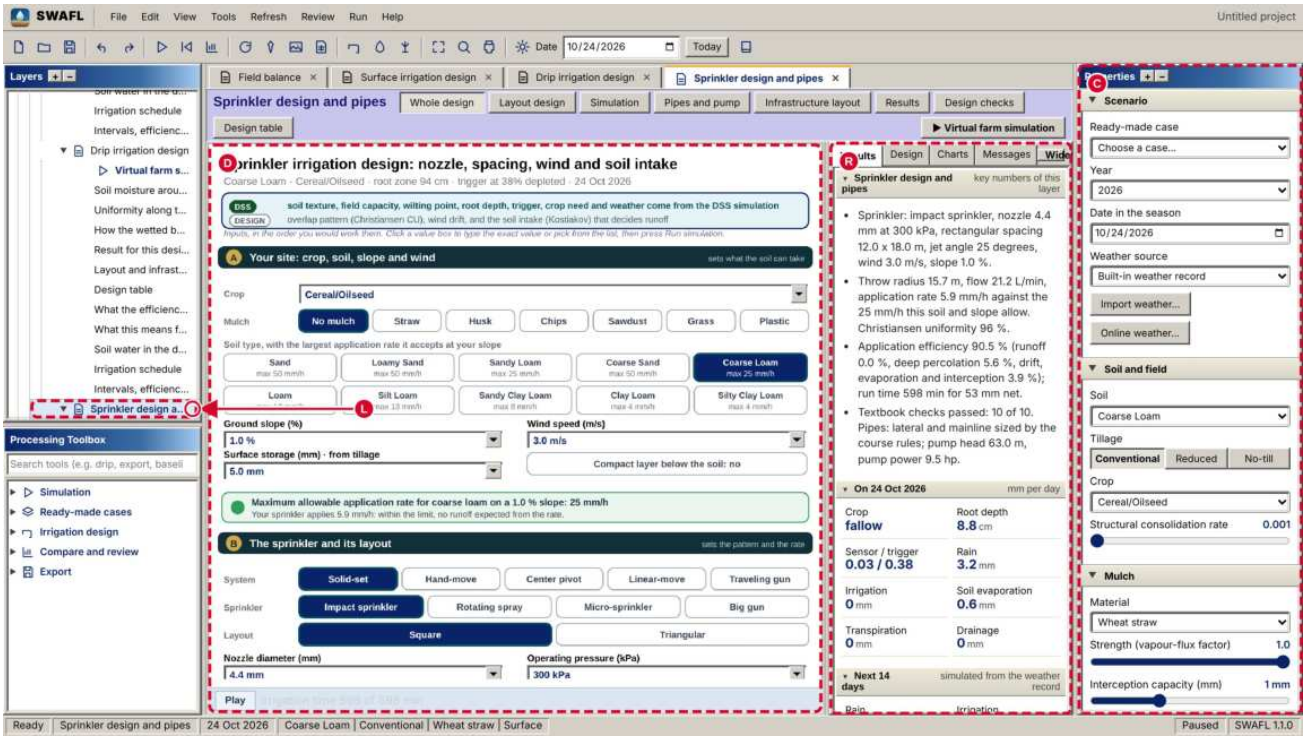



Figure 26.67: Sprinkler design and pipes: the whole workbench.

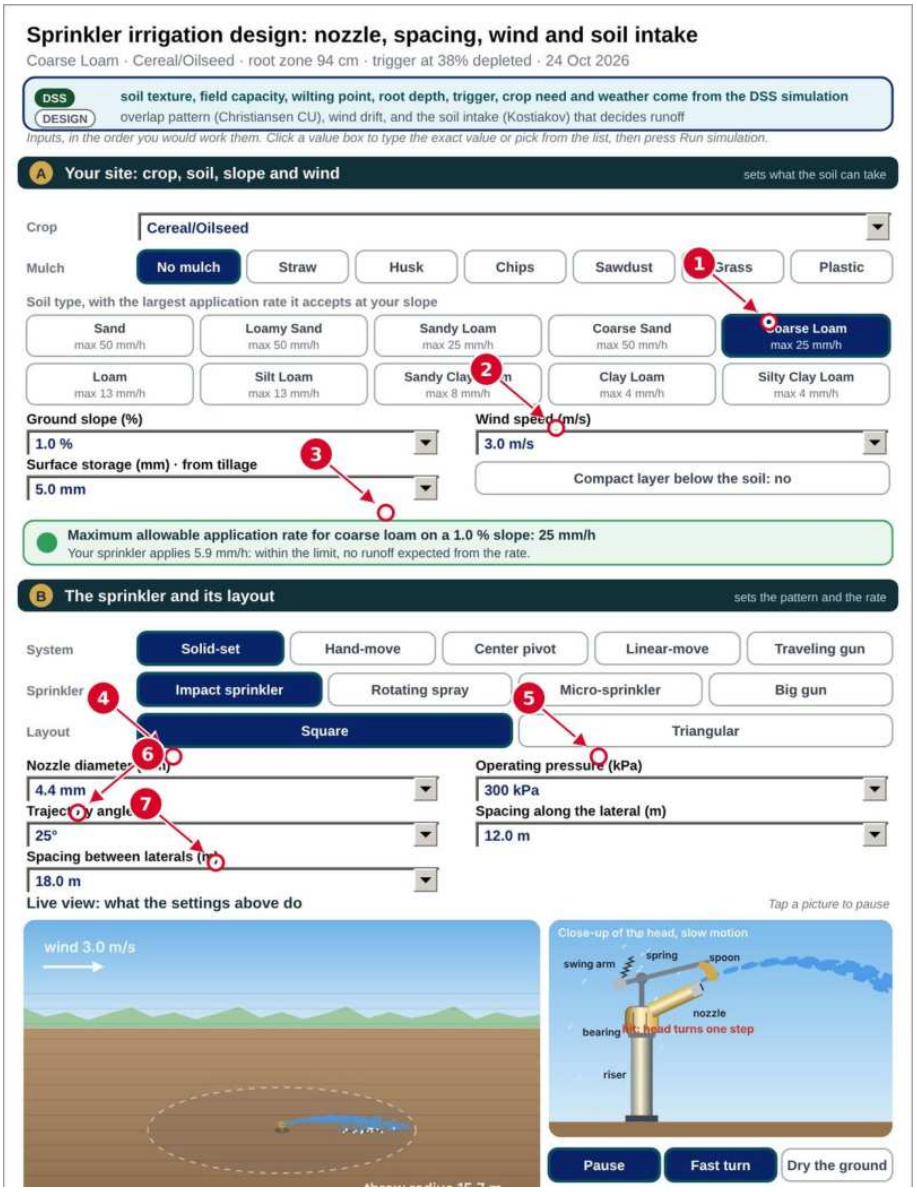



Figure 26.68: Sprinkler design and pipes: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Selected soil: intake K and A and the maximum allowable application rate, Eq. 13.9 and the course table.
- **2** Wind speed u: Eqs. 13.5 and 13.8.
- **3** Banner: application rate against the maximum allowed, Eqs. 13.9 and 13.10.
- **4** Nozzle diameter d: Eqs. 13.2 and 13.3.
- **5** Operating pressure P gives the head h = P / 9.81: Eq. 13.2.
- **6** Trajectory angle alpha: f_alpha in Eq. 13.3.
- **7** Spacing between laterals S_l: Eq. 13.6.

**Equations behind this screen.** Eq. 5.27, Eq. 11.6, Eq. 13.1, Eq. 13.2, Eq. 13.3, Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7, Eq. 13.9, Eq. 13.10.

*Eq. 5.27. Where explained: chapter “The two computing engines”, section “Evaporation under*

*warming (used by the E1 layer)".*

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

*Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 13.1. Where explained: chapter "Sprinkler design", section "Inputs, step by step".*

$$S_d^{\mathrm{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

*Provenance (Eq. 13.1):* **[Engineered]**: depression storage reduction with slope.

*Eq. 13.2. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".*

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^{4}\ \ (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^{2}\ \mathrm{m^2}, \tag{13.2}$$

*Provenance (Eq. 13.2):* **[Literature]**: Torricelli orifice equation.

*Eq. 13.3. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".*

$$R = c_R\sqrt{d\,h}\ f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

*Provenance (Eq. 13.3):* **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

*Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

*Provenance (Eq. 13.4):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

*Provenance (Eq. 13.5):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\ y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

*Provenance (Eq. 13.6):* **[Literature]**: overlap by superposition of single-sprinkler patterns.

*Eq. 13.7. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.*

$$\mathrm{CU} = 1 - \frac{\sum_{k} |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

*Provenance (Eq. 13.7):* **[Literature]**: Christiansen (1942).

*Eq. 13.9. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.*

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \mathrm{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \mathrm{sup} - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \mathrm{sup} - \mathrm{RO}. \tag{13.9}$$

*Provenance (Eq. 13.9):* **[Engineered]**: loss and ponding rules chosen for the workbench.

*Eq. 13.10. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.*

$$I_c = \min\{I,\ I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

*Provenance (Eq. 13.10):* **[Engineered]**: loss and ponding rules chosen for the workbench.

### 26.4.2 Layout in plan: depth infiltrated so far, from the overlapping spray patterns

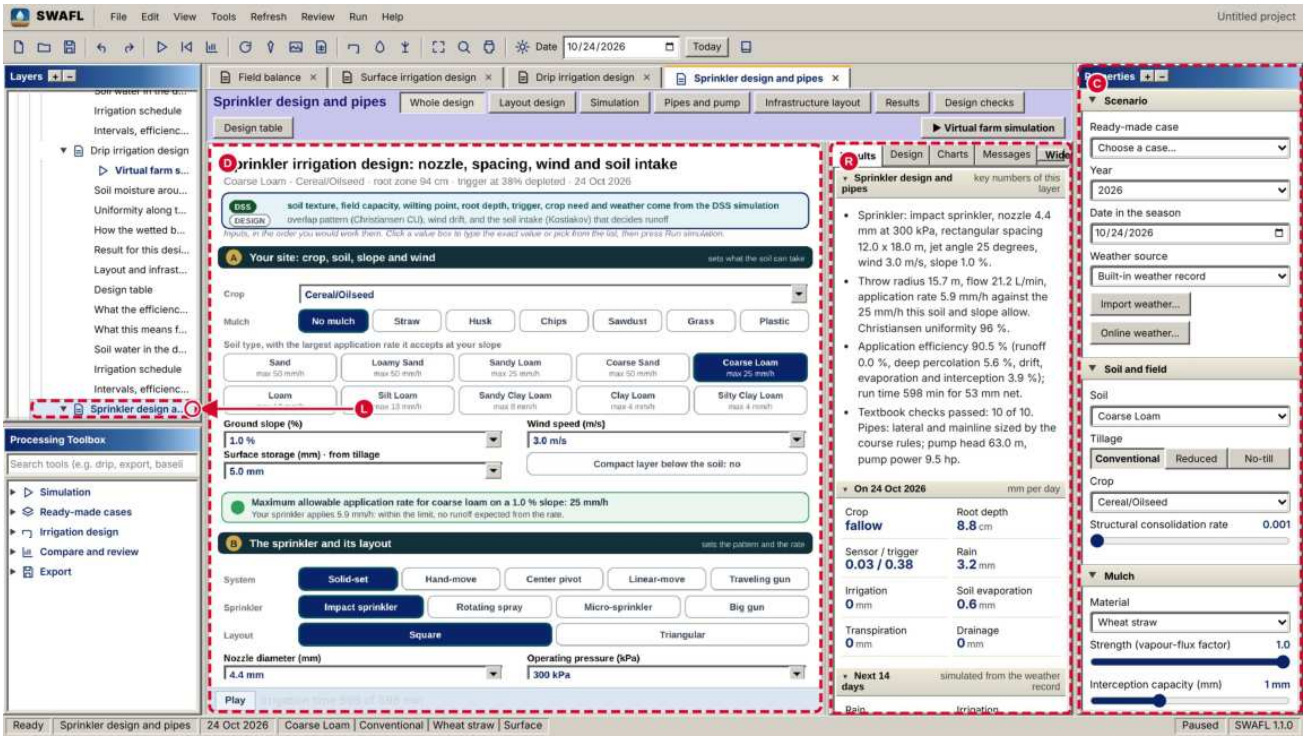



Figure 26.69: Sprinkler design and pipes / Layout in plan: the whole workbench.

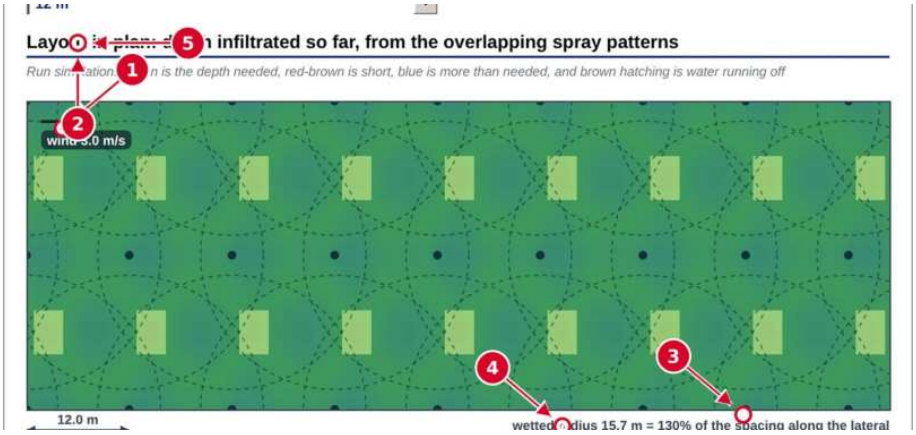



Figure 26.70: Sprinkler design and pipes / Layout in plan: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Wind arrow: wind lengthens and shifts the pattern, Eq. 13.5.
- **2** Overlapping spray patterns: sum over neighbours, Eq. 13.6, with the radial profile of Eq. 13.4.
- **3** Wetted radius R: Eq. 13.3.
- **4** Spacing along the lateral S_s.

- **5** Colour shows infiltrated depth against Z_n: Eq. 13.11.

**Equations behind this screen.** Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

*Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

*Provenance (Eq. 13.4):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

*Provenance (Eq. 13.5):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

*Provenance (Eq. 13.6):* **[Literature]**: overlap by superposition of single-sprinkler patterns.

*Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

*Provenance (Eq. 13.7):* **[Literature]**: Christiansen (1942).

### 26.4.3 Along the lateral: depth applied, depth infiltrated and where it ends up in the soil

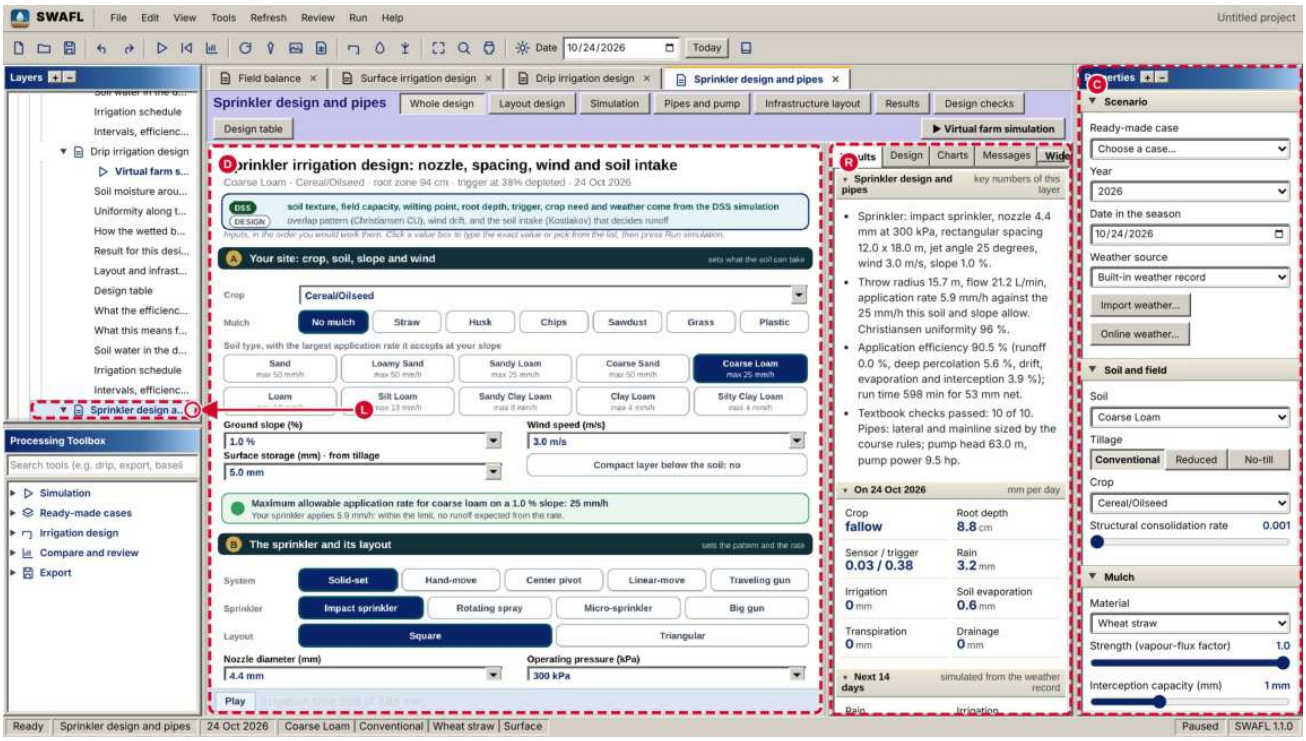



Figure 26.71: Sprinkler design and pipes / Along the lateral: the whole workbench.

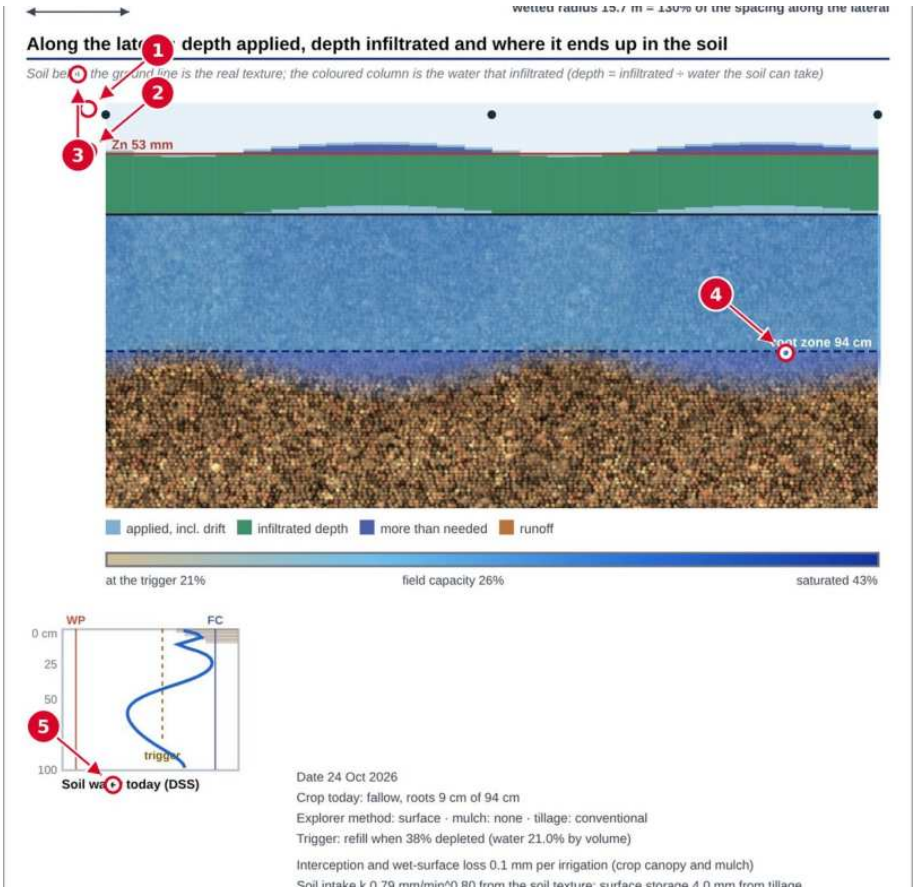



Figure 26.72: Sprinkler design and pipes / Along the lateral: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sprinkler positions.
- **2** Z_n: required depth, Eq. 5.27.
- **3** Infiltrated depth in each cell, Z_k: Eq. 13.11.
- **4** Root-zone depth (94 cm).
- **5** Soil water today from the DSS profile.

**Equations behind this screen.** Eq. 11.6, Eq. 13.9, Eq. 13.10, Eq. 13.11, Eq. 13.12.

*Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)".*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 13.9. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".*

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \sup, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \sup - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \sup - \mathrm{RO}. \tag{13.9}$$

*Provenance (Eq. 13.9):* **[Engineered]**: loss and ponding rules chosen for the workbench.

*Eq. 13.10. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".*

$$I_c = \min\{I,\ I_{\text{int}}\}, \qquad I_{\text{int}} = 0.6\,\text{CAP}\,[\text{VFF} < 0.95] + 1.2\,\text{LAI}_{\text{live}}, \tag{13.10}$$

*Provenance (Eq. 13.10):* **[Engineered]**: loss and ponding rules chosen for the workbench.

*Eq. 13.11. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".*

$$\begin{aligned} E_a &= \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, & E_s &= \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, & \text{DU} &= \frac{\overline{Z}_{25}}{\overline{Z}}, \\ \text{DP} &= \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, & \text{RO} &= \frac{\overline{\text{RO}_k}}{\text{gross}}, & \text{WD} &= w_d + \frac{\overline{I_c}}{\text{gross}}, \end{aligned} \tag{13.11}$$

*Provenance (Eq. 13.11):* **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

*Eq. 13.12. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".*

$$T = \min\{T :\ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\ 2400]\ \text{min}, \tag{13.12}$$

*Provenance (Eq. 13.12):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

### 26.4.4 Soil intake against the application rate: when the soil cannot keep up, water runs off

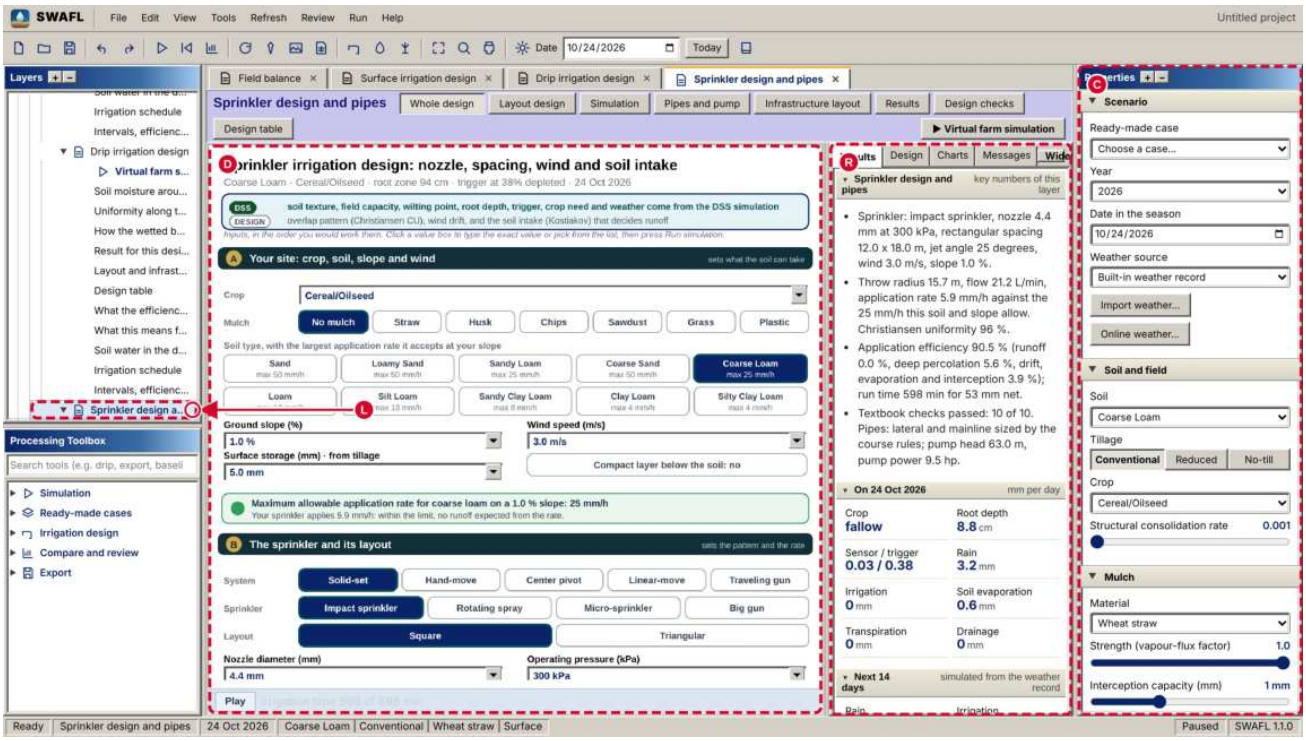


Figure 26.73: Sprinkler design and pipes / Soil intake against the application rate: the whole workbench.

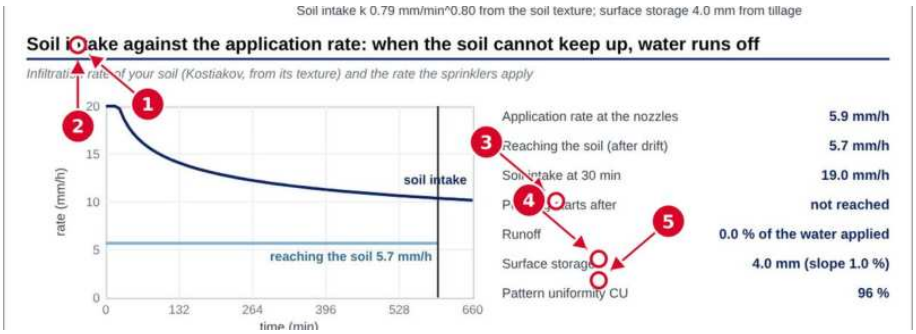



Figure 26.74: Sprinkler design and pipes / Soil intake against the application rate: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Soil intake rate K t^(A-1) (Kostiakov): Eq. 13.9.
- **2** Applied rate a = 60 times the mean depth: Eq. 13.7.
- **3** Ponding starts after t_p: Eq. 13.9.
- **4** Runoff share: Eqs. 13.9 and 13.10.
- **5** Surface storage S_d: Eq. 13.1.

**Equations behind this screen.** Eq. 11.6, Eq. 13.9, Eq. 13.10.

*Eq. 11.6. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton and Rawls → Green and Ampt → Kostiakov)”.*

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

*Eq. 13.9. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.*

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \text{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \text{sup} - F, \quad \text{RO} = \max(0, R - S_d^{\text{eff}}), \quad I = \text{sup} - \text{RO}. \tag{13.9}$$

*Provenance (Eq. 13.9):* **[Engineered]**: loss and ponding rules chosen for the workbench.

*Eq. 13.10. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.*

$$I_c = \min\{I,\ I_{\text{int}}\}, \qquad I_{\text{int}} = 0.6\,\text{CAP}\,[\text{VFF} < 0.95] + 1.2\,\text{LAI}_{\text{live}}, \tag{13.10}$$

*Provenance (Eq. 13.10):* **[Engineered]**: loss and ponding rules chosen for the workbench.

### 26.4.5 Result for this design

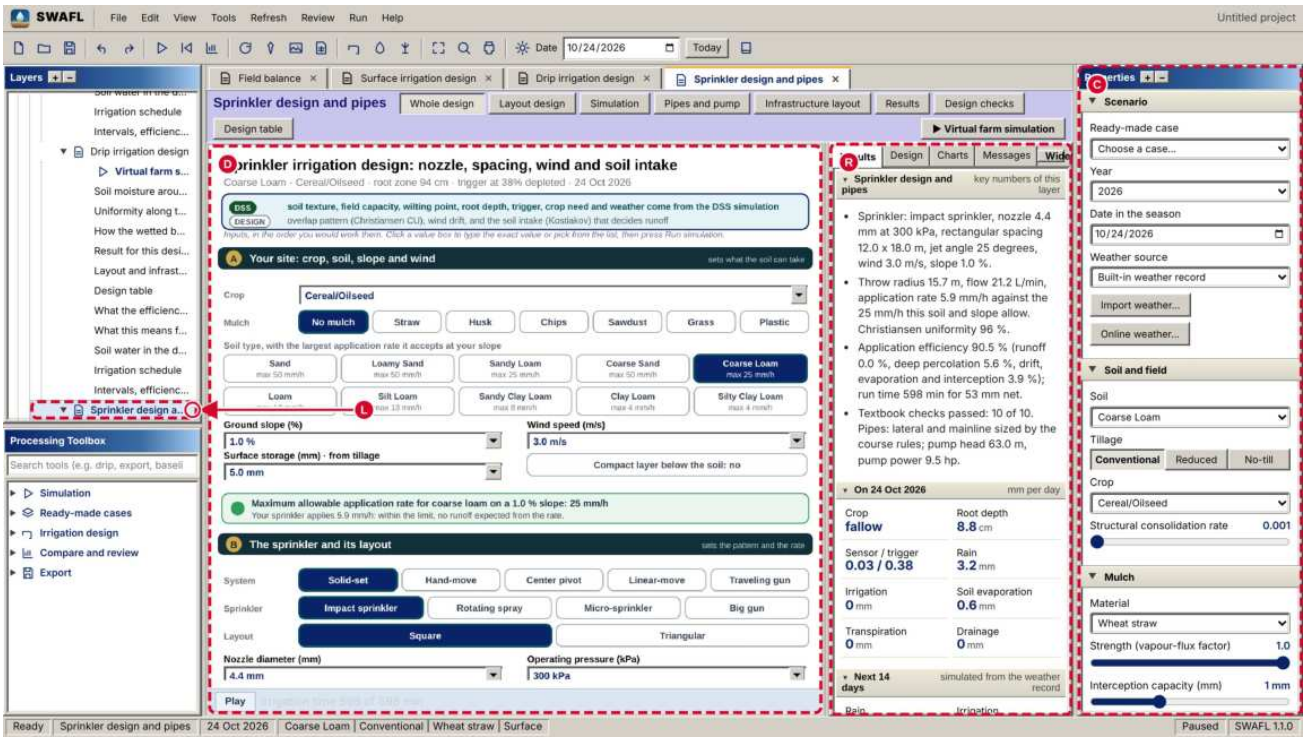


Figure 26.75: Sprinkler design and pipes / Result for this design: the whole workbench.

Figure 26.76: Sprinkler design and pipes / Result for this design: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a: Eq. 13.11.
- **2** Root-zone refilled E_s: Eq. 13.11.
- **3** Christiansen CU: Eq. 13.7.
- **4** Runoff: Eqs. 13.9 and 13.10.
- **5** Deep percolation: Eq. 13.11.
- **6** Drift, evaporation and interception: Eqs. 13.8 and 13.10.

**Equations behind this screen.** Eq. 13.11, Eq. 13.12.

*Eq. 13.11. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.*

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

*Provenance (Eq. 13.11):* **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

*Eq. 13.12. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".*

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \ \text{min}, \tag{13.12}$$

*Provenance (Eq. 13.12):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

### 26.4.6 Does your design pass? The course checks

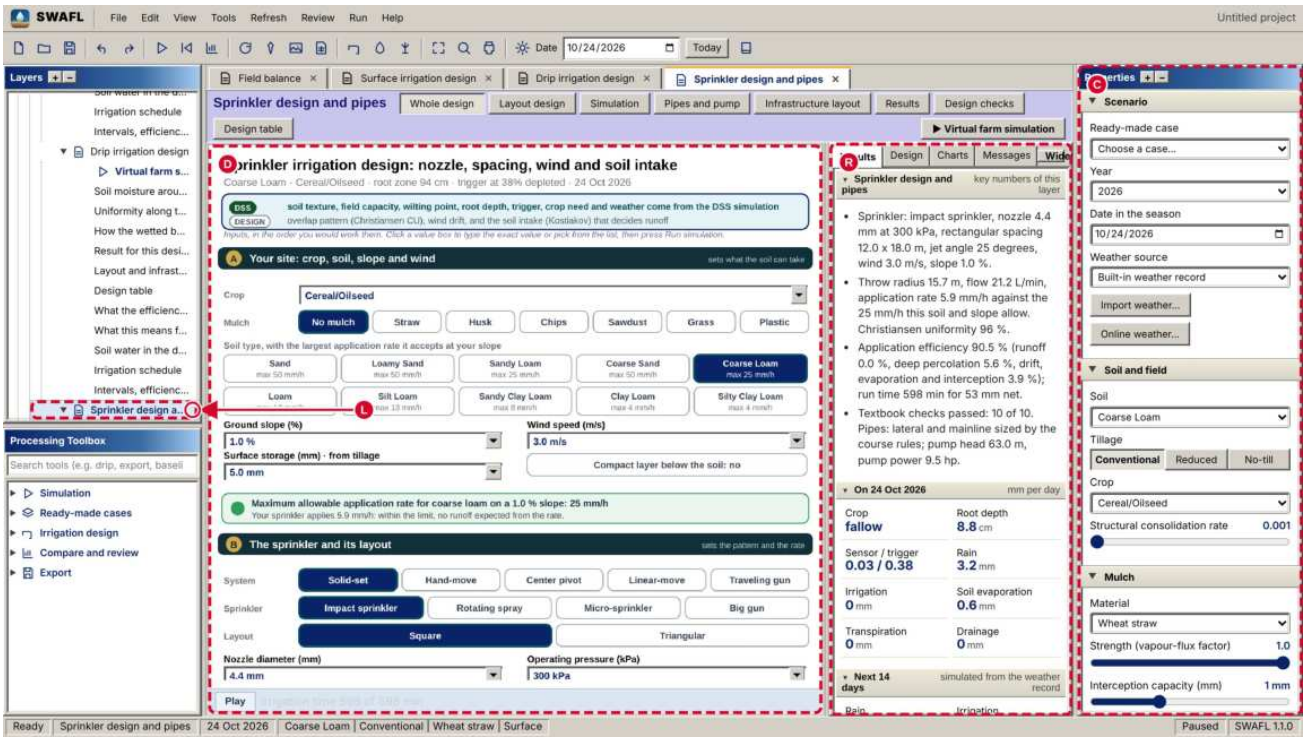


Figure 26.77: Sprinkler design and pipes / Does your design pass? The course checks: the whole workbench.

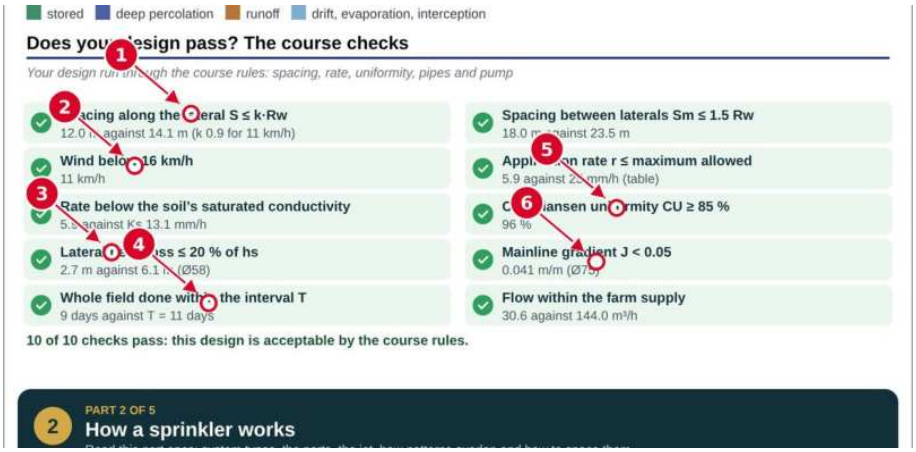



Figure 26.78: Sprinkler design and pipes / Does your design pass? The course checks: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Spacing along the lateral S_s <= k R_w: Eqs. 13.13 and 13.14.
- **2** Wind below 16 km/h: Eq. 13.13.
- **3** Lateral head loss <= 20 % of the head: Eq. 13.18.
- **4** Whole field done within the interval T: Eqs. 13.14 to 13.16.

- **5** Christiansen CU >= 85 %: Eq. 13.7.
- **6** Mainline gradient J < 0.05: Eq. 13.17.

**Equations behind this screen.** Eq. 13.13, Eq. 13.14, Eq. 13.15, Eq. 13.16, Eq. 13.17, Eq. 13.18, Eq. 13.19, Eq. 13.20.

*Eq. 13.13. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

*Provenance (Eq. 13.13):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

*Eq. 13.14. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x/S_s \rfloor, \quad N_m = \lfloor L_y/S_l \rfloor, \tag{13.14}$$

*Provenance (Eq. 13.14):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

*Eq. 13.15. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{ days}, \tag{13.15}$$

*Provenance (Eq. 13.15):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

*Eq. 13.16. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\,N\,q_h, \tag{13.16}$$

*Provenance (Eq. 13.16):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

*Eq. 13.17. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$J = \frac{0.472\,Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

*Provenance (Eq. 13.17):* **[Literature]**: Blasius friction law for smooth pipe.

*Eq. 13.18. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$h_{f,\text{lat}} = 1.1\,F\,J(Q_L, D)\,(L_x - S_s/2) \le 0.2\,h_s, \qquad Q_L = 1000\,N\,q_h, \tag{13.18}$$

*Provenance (Eq. 13.18):* **[Literature]**: 20 % rule of Keller and Bliesner (1990).

*Eq. 13.19. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$H_{\text{man}} = z_{\text{lift}} + J\,L_{\text{conv}} + J\,L_{\text{main}} + h_s + 0.75\,h_{f,\text{lat}}, \tag{13.19}$$

*Provenance (Eq. 13.19):* **[Engineered]**: head and sets bookkeeping.

*Eq. 13.20. Where explained: chapter "Sprinkler design", section "The ten textbook checks".*

$$P_{\text{CV}} = \frac{1000\,(Q_m/3.6\times10^6)\,H_{\text{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\text{kW}} = 0.7355\,P_{\text{CV}}, \tag{13.20}$$

*Provenance (Eq. 13.20):* **[Literature]**: hydraulic power definition.

## 26.4.7 Kinds of sprinkler system

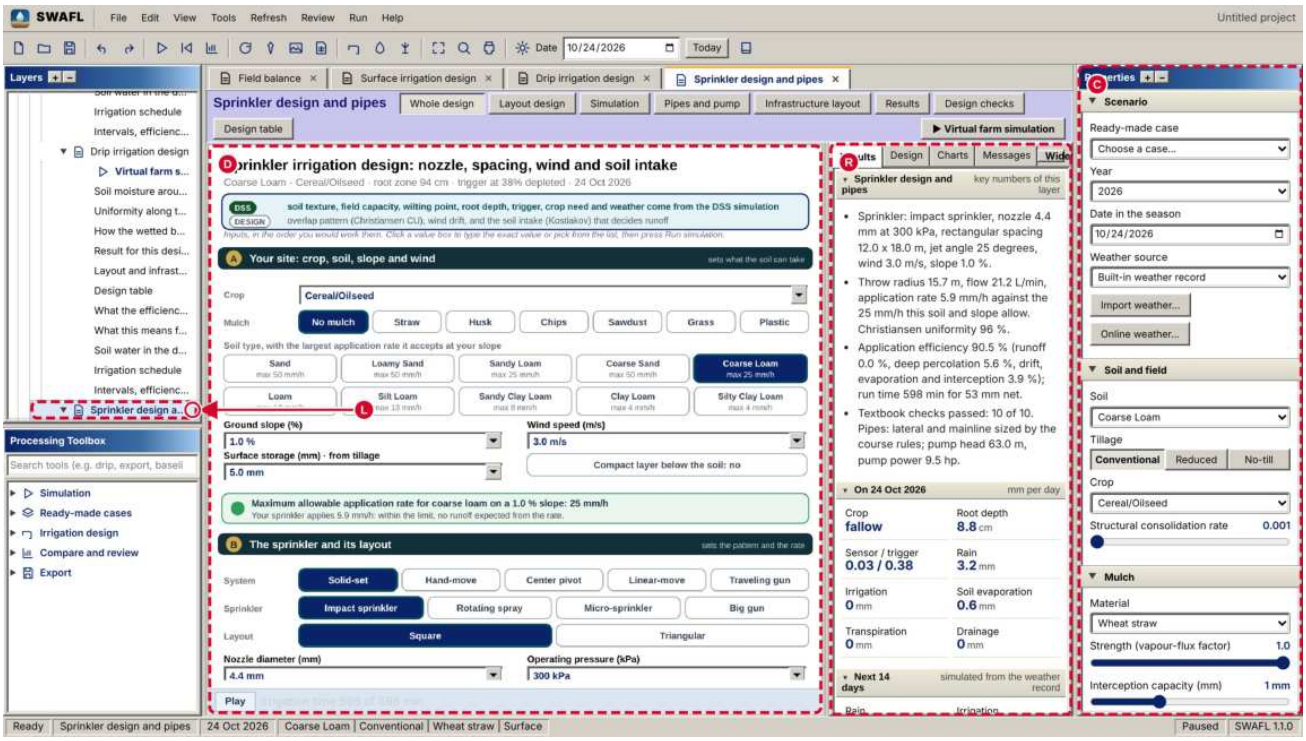



Figure 26.79: Sprinkler design and pipes / Kinds of sprinkler system: the whole workbench.

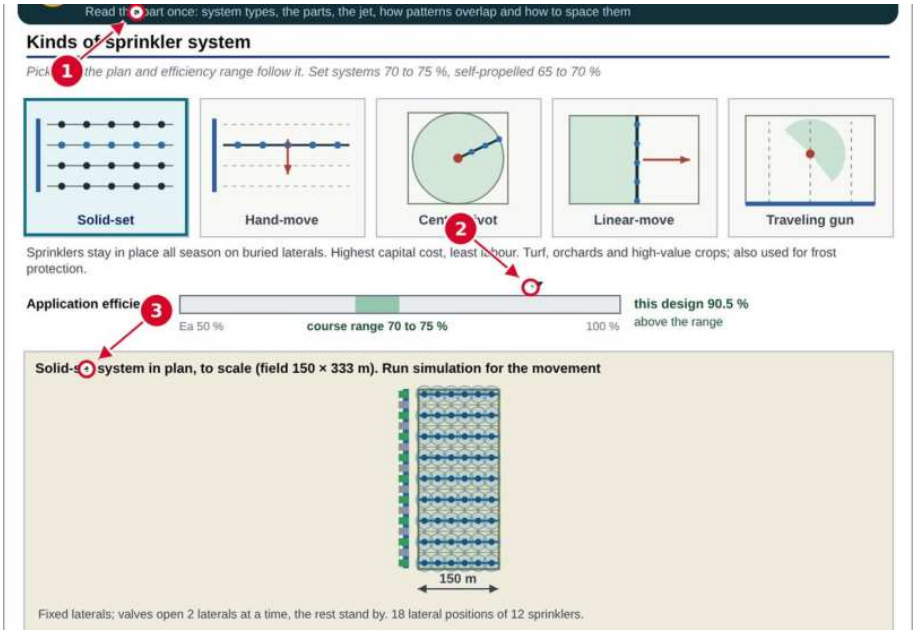



Figure 26.80: Sprinkler design and pipes / Kinds of sprinkler system: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** System type (solid-set selected): sets the efficiency range of the course table.
- **2** This design’s E_a against the course range: Eq. 13.11.
- **3** Field plan to scale: laterals and sets, Eq. 13.21.

**Equations behind this screen.** none (the screen shows stored results).

## 26.4.8 The sprinkler: parts and pressure class

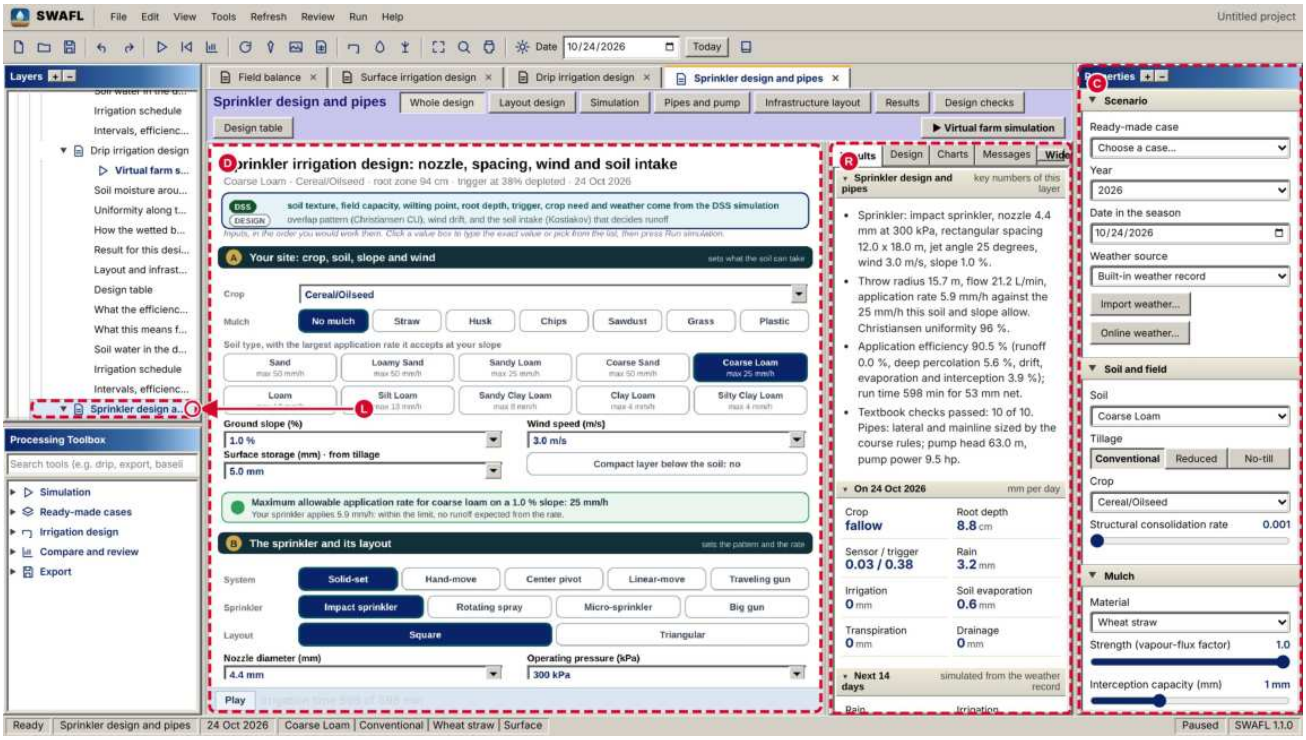



Figure 26.81: Sprinkler design and pipes / The sprinkler: the whole workbench.

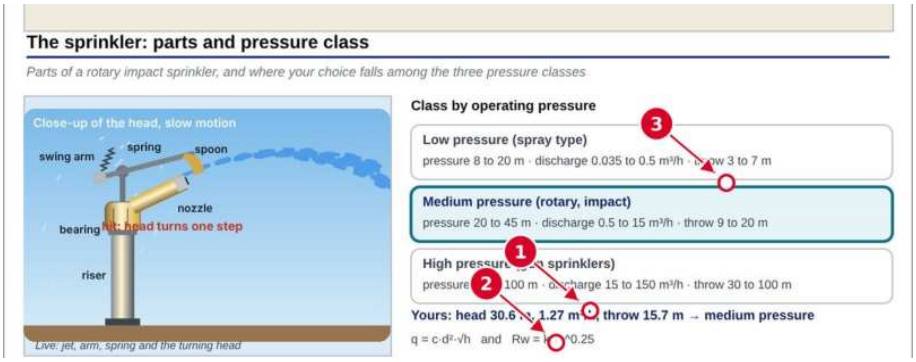



Figure 26.82: Sprinkler design and pipes / The sprinkler: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** “Yours”: head h = P / 9.81 and discharge q (Eq. 13.2), throw R (Eq. 13.3).
- **2** Formulas q = c d^2 sqrt(h) and R = k_r h^0.25: Eqs. 13.2 and 13.3.
- **3** Pressure classes (low, medium, high): table, no equation.

**Equations behind this screen.** Eq. 13.2, Eq. 13.3.

*Eq. 13.2. Where explained: chapter “Sprinkler design”, section “Nozzle discharge and wetted radius”.*

$$q = 0.95\, A_n \sqrt{2g\,h} \times 6 \times 10^4 \;\; (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \;\mathrm{m^2}, \tag{13.2}$$

*Provenance (Eq. 13.2):* **[Literature]**: Torricelli orifice equation.

*Eq. 13.3. Where explained: chapter “Sprinkler design”, section “Nozzle discharge and wetted radius”.*

$$R = c_R \sqrt{d\,h}\; f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \left[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\right]^{0.45}, \tag{13.3}$$

*Provenance (Eq. 13.3):* **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

### 26.4.9 The jet: trajectory angle, throw radius and droplet size

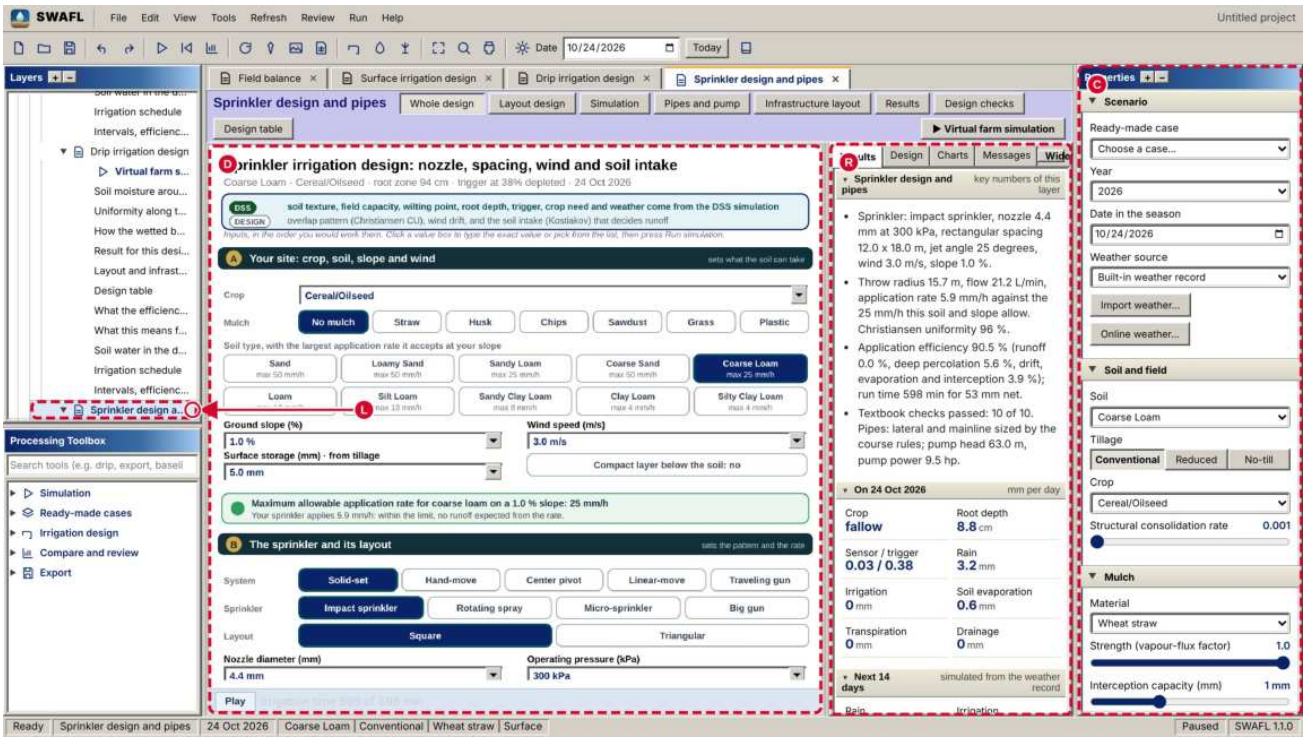



Figure 26.83: Sprinkler design and pipes / The jet: the whole workbench.

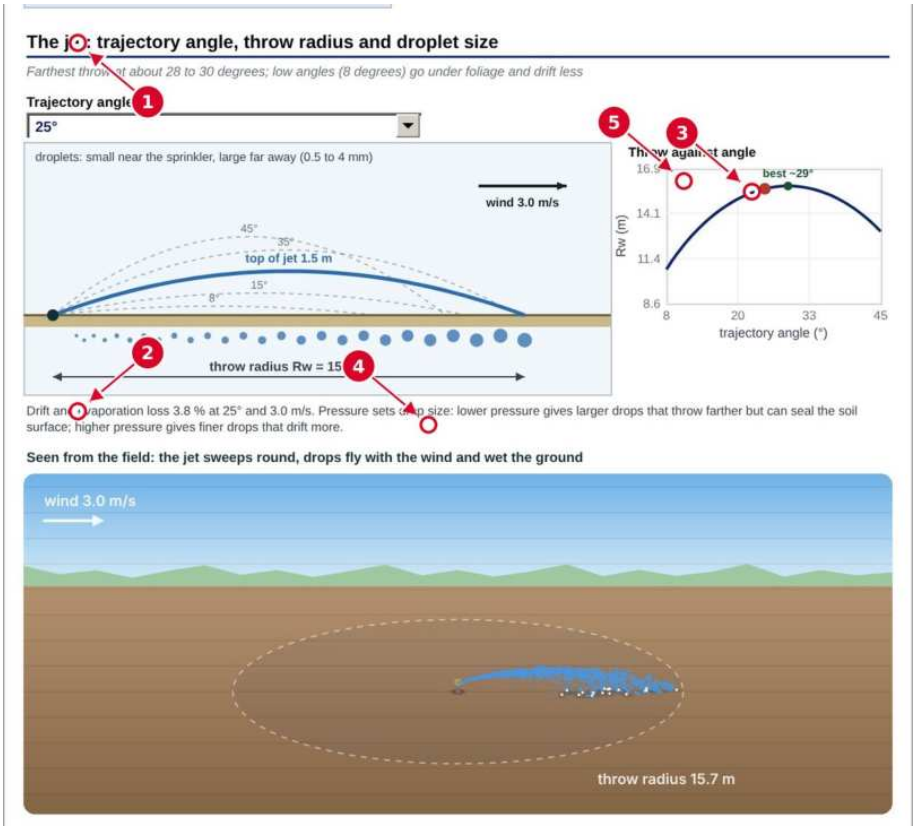



Figure 26.84: Sprinkler design and pipes / The jet: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Trajectory angle alpha: f_alpha(alpha) in Eq. 13.3.
- **2** Throw radius R_w: Eq. 13.3.
- **3** Throw against angle, best near 29 degrees: Eq. 13.3.
- **4** Drift and evaporation loss w_d: Eq. 13.8.
- **5** Wind speed u: Eqs. 13.5 and 13.8.

**Equations behind this screen.** Eq. 13.2, Eq. 13.3, Eq. 13.8.

*Eq. 13.2. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".*

$$q = 0.95\, A_n \sqrt{2g\,h} \times 6 \times 10^4 \;\; (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \;\mathrm{m^2}, \tag{13.2}$$

*Provenance (Eq. 13.2):* **[Literature]**: Torricelli orifice equation.

*Eq. 13.3. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".*

$$R = c_R \sqrt{d\,h}\; f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

*Provenance (Eq. 13.3):* **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

*Eq. 13.8. Where explained: chapter "Sprinkler design", section "Wind drift and evaporation loss".*

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{\left(1.2 + 0.5u + 0.04u^2\right) F\Big(\frac{P}{250}\Big)^{0.7}\left[1 + 0.025\,(T_a - 20)\right]\Big[0.6 + 0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

*Provenance (Eq. 13.8):* **[Engineered]**: loss and ponding rules chosen for the workbench.

### 26.4.10 The water pattern of one sprinkler, and how overlap evens it out

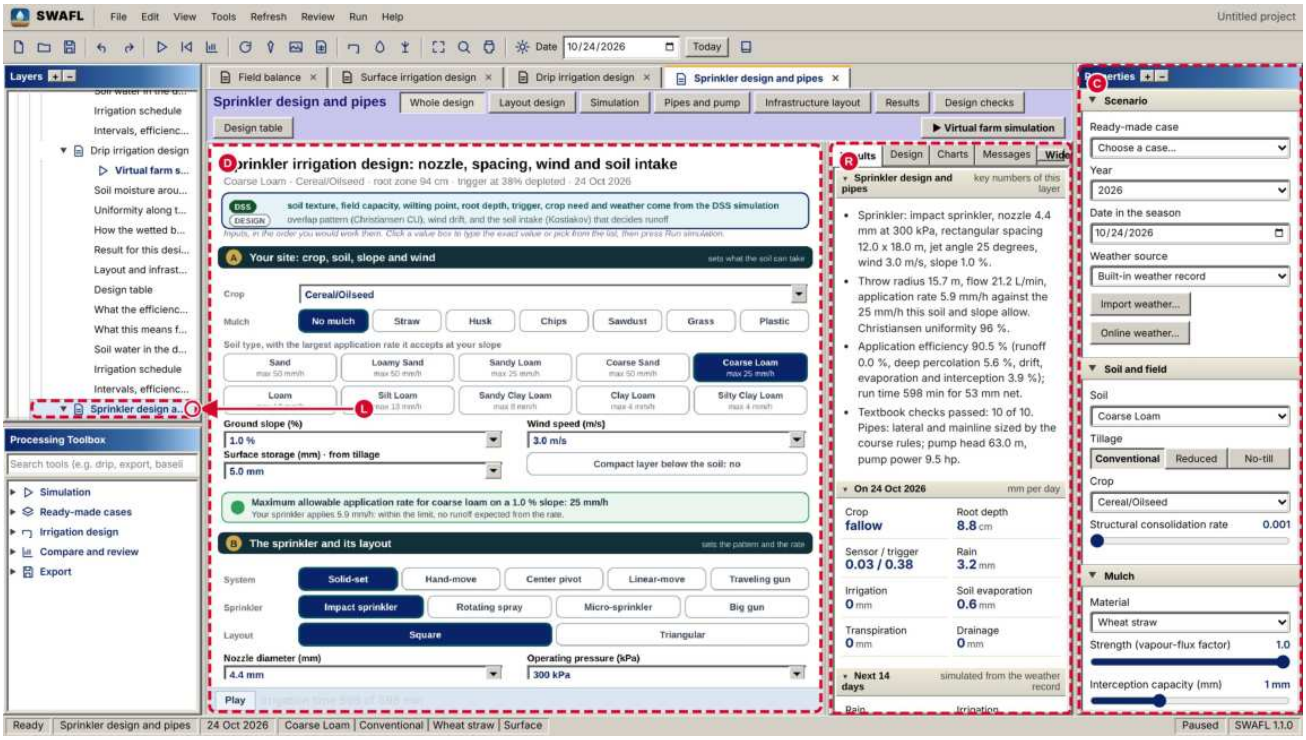



Figure 26.85: Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out: the whole workbench.

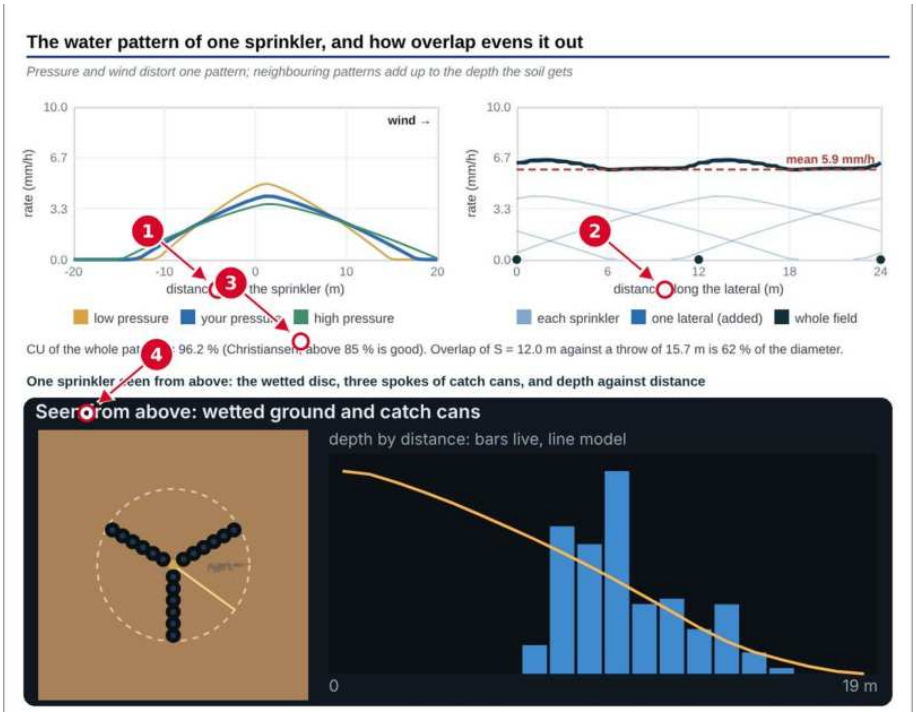



Figure 26.86: Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Radial profile of one sprinkler, p(x): Eq. 13.4, skewed by the wind of Eq. 13.5.
- **2** Sum over neighbouring sprinklers, a(x, y): Eq. 13.6; the dashed line is the mean rate.
- **3** Christiansen coefficient of the whole pattern: Eq. 13.7.
- **4** One sprinkler seen from above with three spokes of catch cans.

**Equations behind this screen.** Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

*Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

*Provenance (Eq. 13.4):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$R_{e}(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

*Provenance (Eq. 13.5):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$a(x, y) = \sum_{i,j} p\big(x - x_{ij},\, y - y_{ij}\big) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_{k}, \tag{13.6}$$

*Provenance (Eq. 13.6):* **[Literature]**: overlap by superposition of single-sprinkler patterns.

*Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$\mathrm{CU} = 1 - \frac{\sum_{k} |a_{k} - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

*Provenance (Eq. 13.7):* **[Literature]**: Christiansen (1942).

### 26.4.11 Layouts: rectangular, square and triangular spacing for this sprinkler

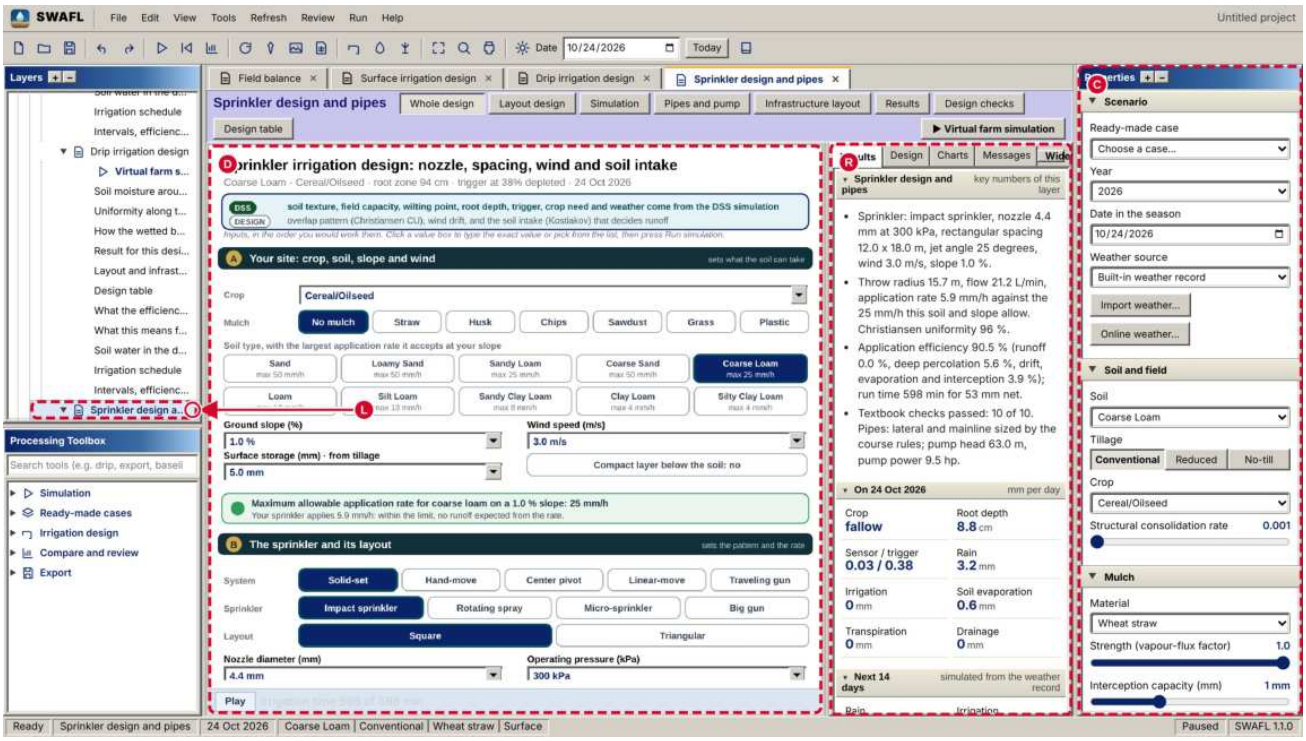



Figure 26.87: Sprinkler design and pipes / Layouts: the whole workbench.

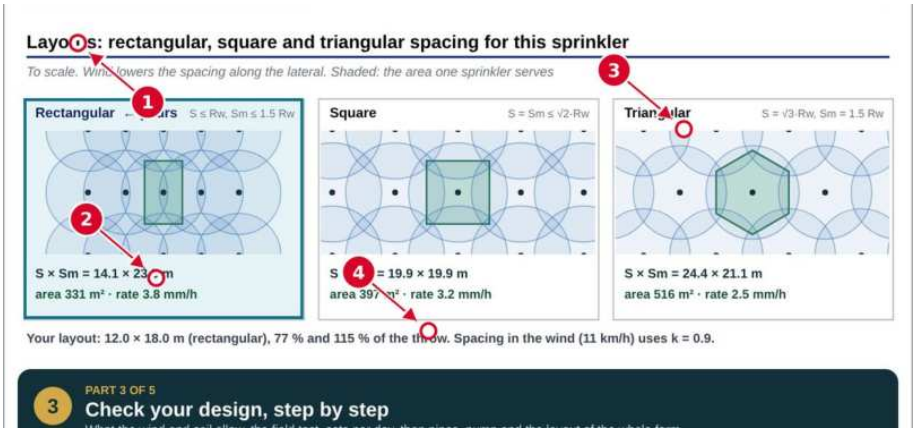



Figure 26.88: Sprinkler design and pipes / Layouts: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Your layout: S_s <= k R_w and S_l <= 1.5 R_w, Eq. 13.14.
- **2** Area per sprinkler and application rate: Eq. 13.7.
- **3** Triangular layout: every second lateral shifted by S_s / 2, Eq. 13.6.
- **4** Wind factor k: Eq. 13.13.

**Equations behind this screen.** Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

*Eq. 13.4. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.*

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

*Provenance (Eq. 13.4):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.5. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.*

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

*Provenance (Eq. 13.5):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.6. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.*

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

*Provenance (Eq. 13.6):* **[Literature]**: overlap by superposition of single-sprinkler patterns.

*Eq. 13.7. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.*

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

*Provenance (Eq. 13.7):* **[Literature]**: Christiansen (1942).

## 26.4.12 What wind and soil allow

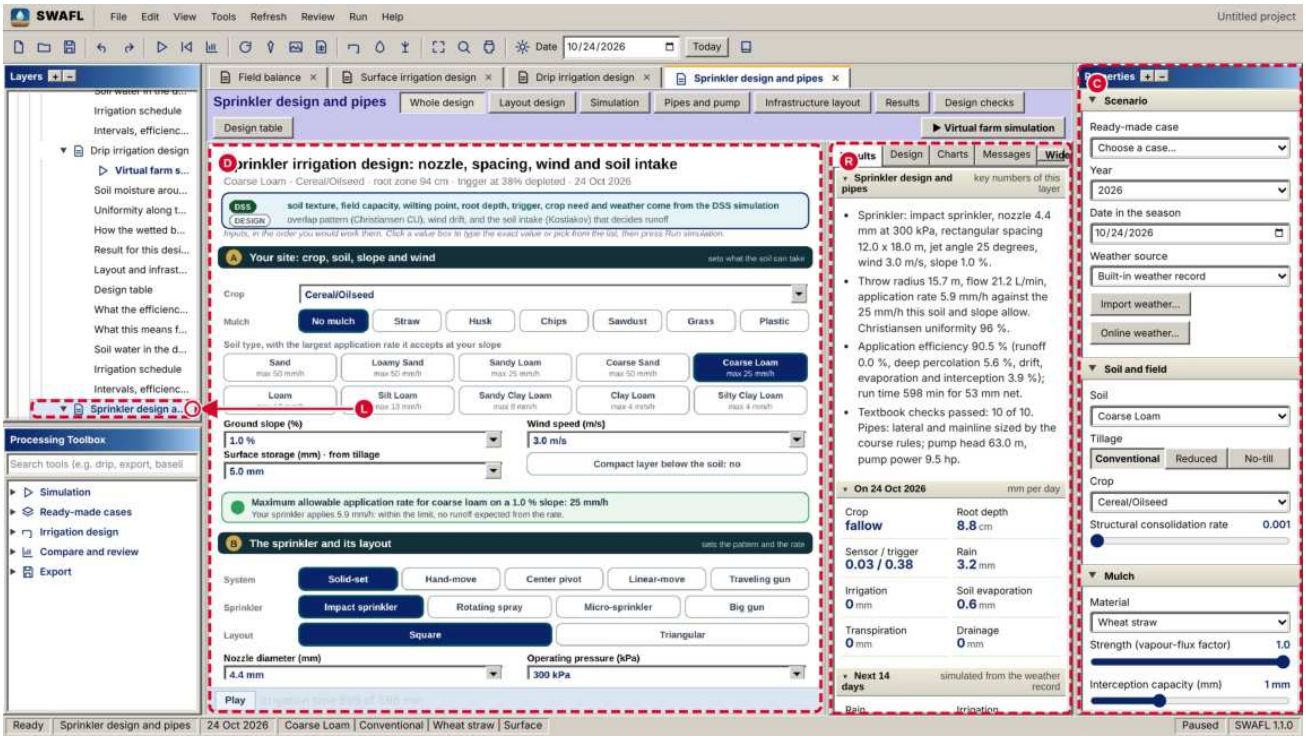


Figure 26.89: Sprinkler design and pipes / What wind and soil allow: the whole workbench.

Figure 26.90: Sprinkler design and pipes / What wind and soil allow: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Step line of the wind factor k: Eq. 13.13.
- **2** “Yours”: your wind speed and spacing.
- **3** Maximum allowable application rate by soil and slope (course table), used in Eqs. 13.9 and 13.13.

**Equations behind this screen.** Eq. 13.8.

*Eq. 13.8. Where explained: chapter “Sprinkler design”, section “Wind drift and evaporation loss”.*

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{\left(1.2 + 0.5u + 0.04u^2\right) F\Big(\frac{P}{250}\Big)^{0.7}\left[1 + 0.025\,(T_a - 20)\right]\Big[0.6 + 0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

*Provenance (Eq. 13.8):* **[Engineered]**: loss and ponding rules chosen for the workbench.

### 26.4.13 Catch-can test: measuring the uniformity the way it is done in the field

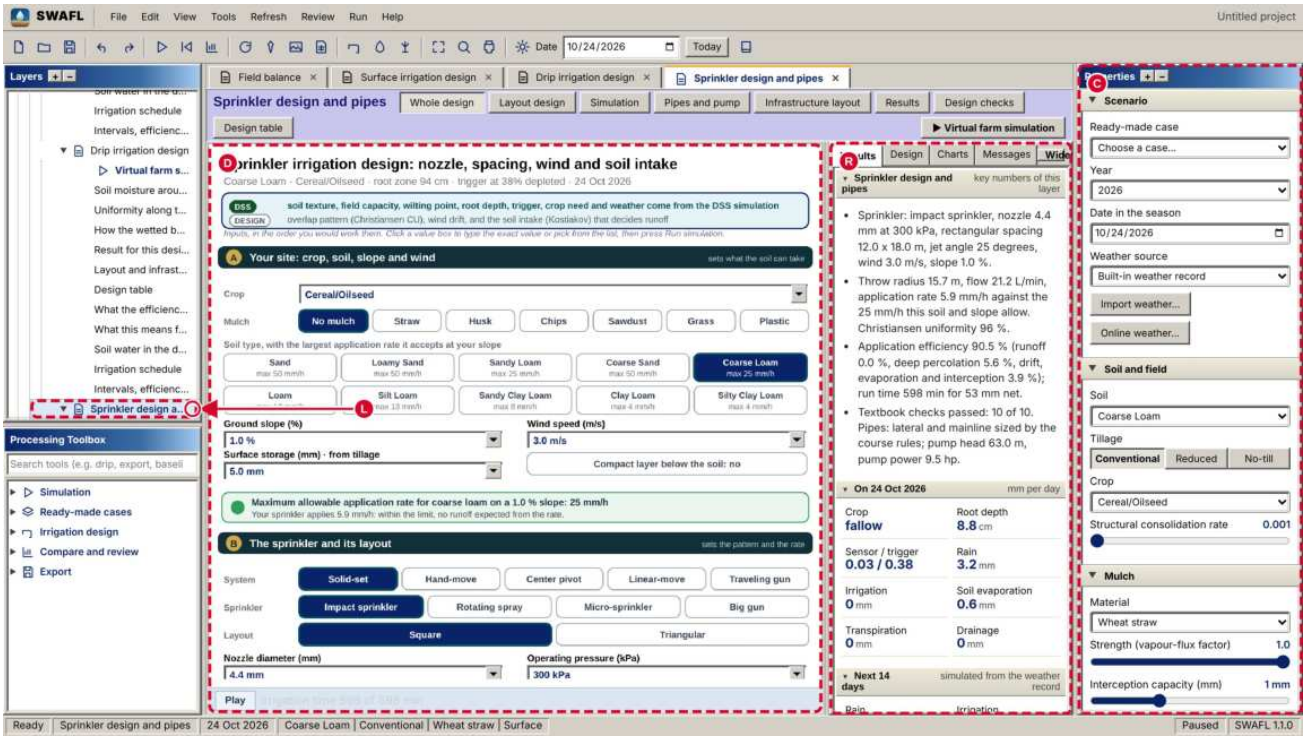


Figure 26.91: Sprinkler design and pipes / Catch-can test: the whole workbench.

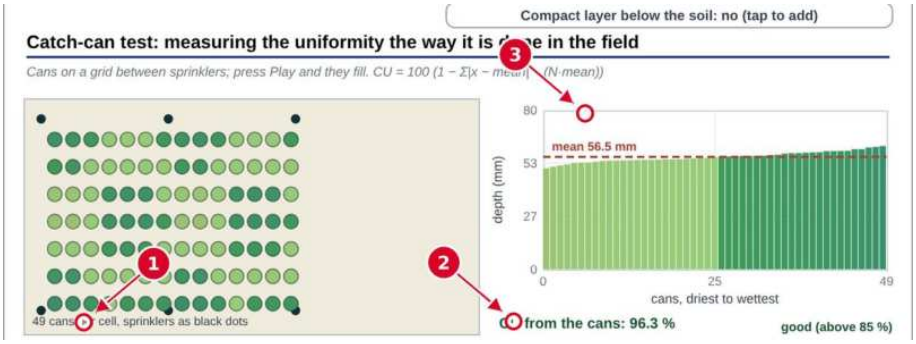



Figure 26.92: Sprinkler design and pipes / Catch-can test: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Catch cans: depth a_k in each can from the pattern of Eq. 13.6.
- **2** CU from the cans = 1 minus sum |a_k minus mean| / (n mean): Eq. 13.7.
- **3** Cans sorted from driest to wettest, with the mean depth.

**Equations behind this screen.** Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

*Eq. 13.4. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.*

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

*Provenance (Eq. 13.4):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$R_{e}(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

*Provenance (Eq. 13.5):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

*Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\ y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_{k}, \tag{13.6}$$

*Provenance (Eq. 13.6):* **[Literature]**: overlap by superposition of single-sprinkler patterns.

*Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".*

$$\mathrm{CU} = 1 - \frac{\sum_{k} |a_{k} - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

*Provenance (Eq. 13.7):* **[Literature]**: Christiansen (1942).

## 26.4.14 Soil water as a reservoir: interval, sets per day and the flow the field needs

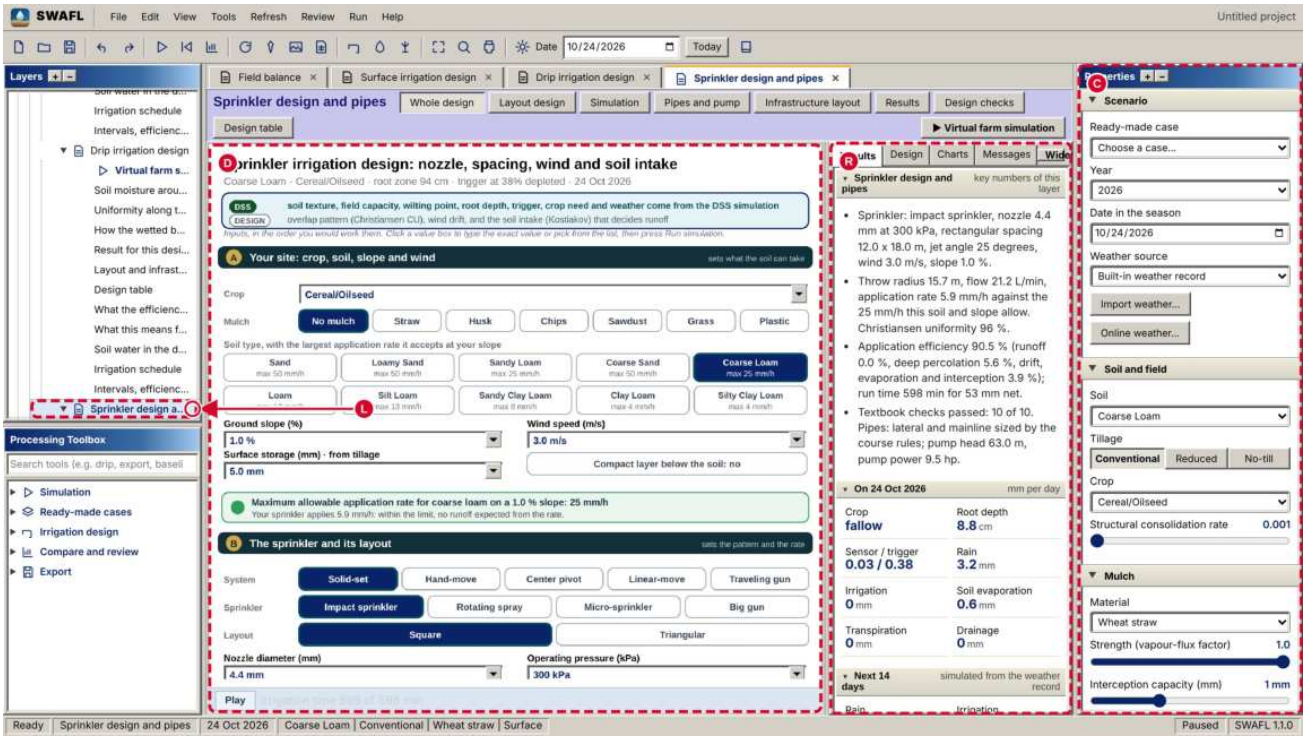



Figure 26.93: Sprinkler design and pipes / Soil water as a reservoir: the whole workbench.

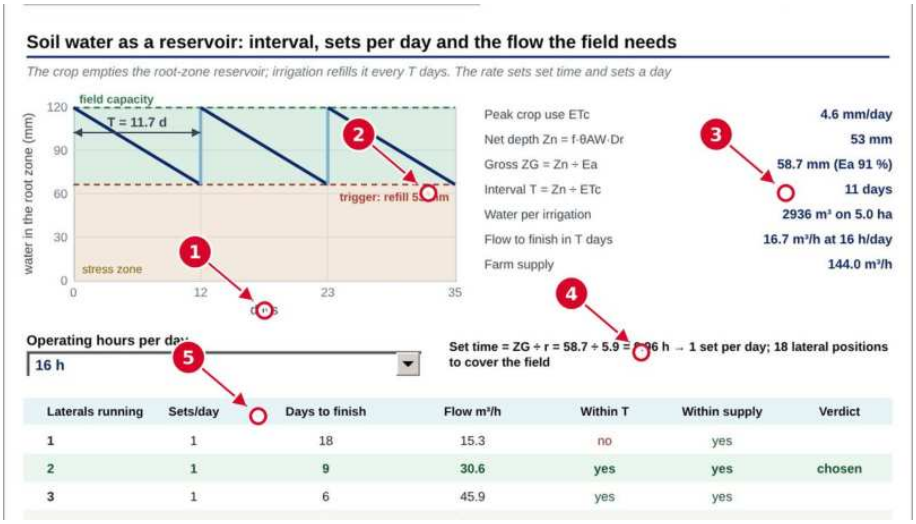



Figure 26.94: Sprinkler design and pipes / Soil water as a reservoir: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Root-zone store drawn down by ET_c and refilled every T days.
- **2** Refill at Z_n from the DSS: Eq. 5.27.
- **3** Gross depth Z_G = Z_n / E_a: Eq. 13.15.

- **4** Set time t = Z_G / r: Eq. 13.15.
- **5** Laterals running together, days to finish and inflow: Eq. 13.16.

**Equations behind this screen.** Eq. 13.11, Eq. 13.12, Eq. 13.23.

*Eq. 13.11. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.*

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

*Provenance (Eq. 13.11):* **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

*Eq. 13.12. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.*

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \text{ min}, \tag{13.12}$$

*Provenance (Eq. 13.12):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

*Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.*

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60} \text{ h}. \tag{13.23}$$

*Provenance (Eq. 13.23):* **[Engineered]**: head and sets bookkeeping.

## 26.4.15 Pipes, pump and the hydraulic grade line (the course sizing)

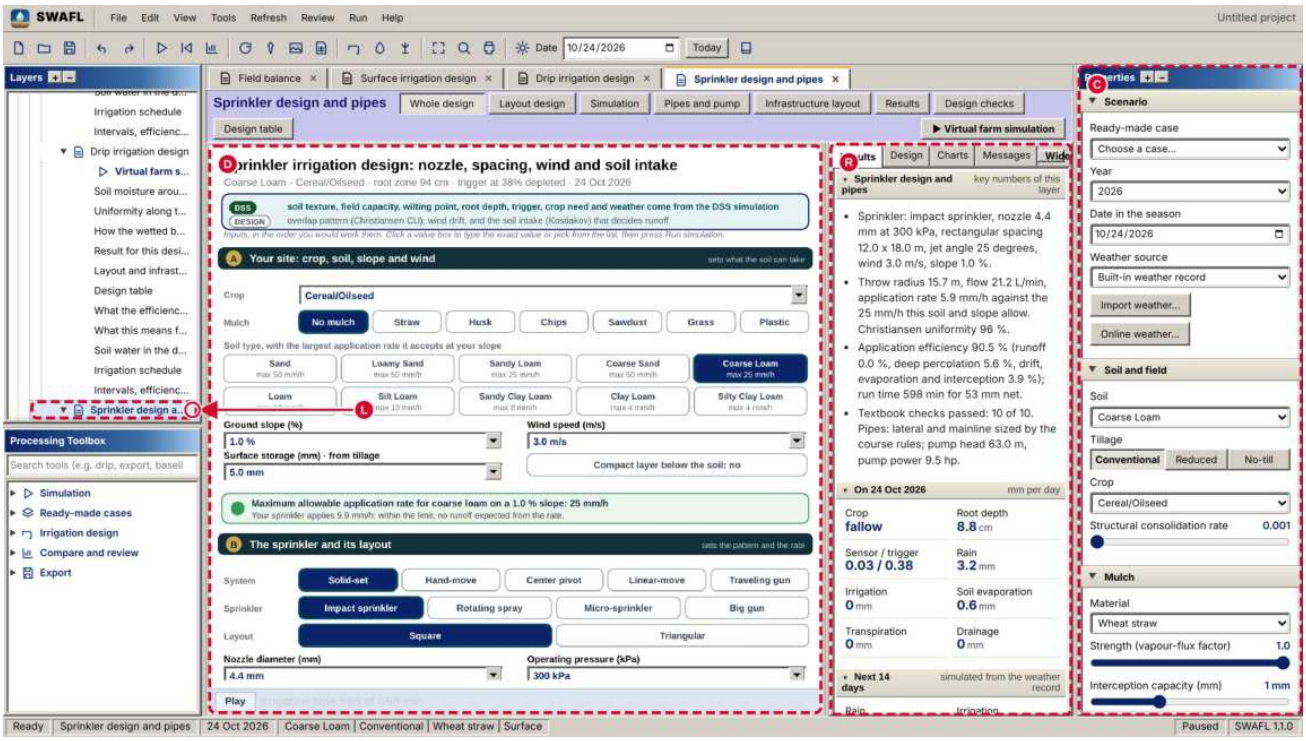



Figure 26.95: Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing): the whole workbench.

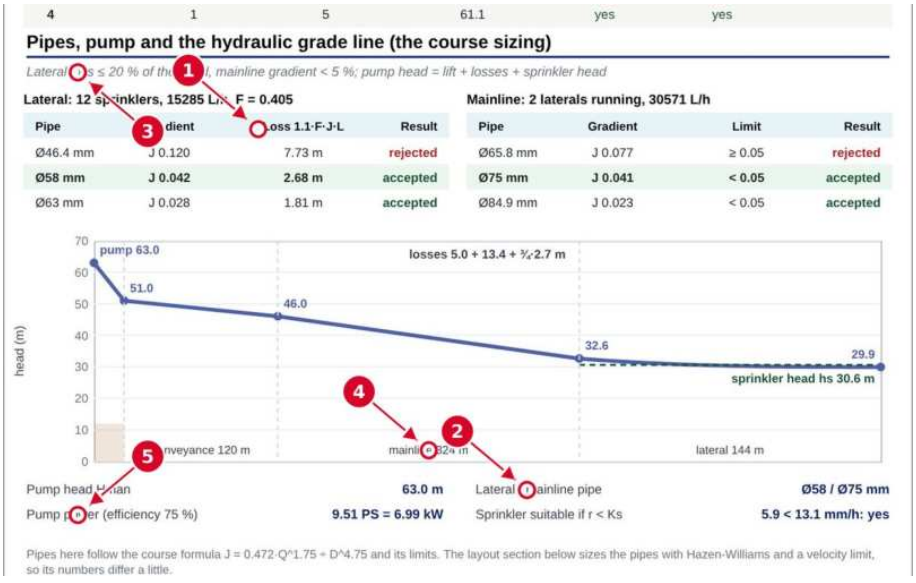



Figure 26.96: Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing): the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Gradient J of the pipe (Blasius form): Eq. 13.17.
- **2** Lateral pipe: smallest standard diameter meeting the loss limit, Eq. 13.18.
- **3** Mainline: gradient below 0.05, Eq. 13.17.
- **4** Head along the flow path to the pump, H_man: Eq. 13.19.
- **5** Pump power: Eq. 13.20.

**Equations behind this screen.** Eq. 12.3, Eq. 12.21, Eq. 12.22, Eq. 12.23, Eq. 13.21, Eq. 13.22.

*Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

*Provenance (Eq. 12.3):* **[Literature]**: Hazen and Williams (1905).

*Eq. 12.21. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".*

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\} \text{ mm} \tag{12.21}$$

*Provenance (Eq. 12.21):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 12.22. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".*

$$h_{f,sm} = 0.4\, \frac{10.67\, L_{sm}\, Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\, L_m\, Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

*Provenance (Eq. 12.22):* **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

*Eq. 12.23. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".*

$$\mathrm{TDH} = z_{\mathrm{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9 \text{ m}, \qquad P = \frac{\rho g\, Q_z\, \mathrm{TDH}}{\eta} = \frac{9.81\, (Q_z/1000)\, \mathrm{TDH}}{0.65} \text{ kW}. \tag{12.23}$$

*Provenance (Eq. 12.23):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 13.21. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".*

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\mathrm{lat}}} \Big\rfloor\Big), \quad n_{\mathrm{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\mathrm{lat}}. \tag{13.21}$$

*Provenance (Eq. 13.21):* **[Engineered]**: head and sets bookkeeping.

*Eq. 13.22. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".*

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\mathrm{lat}} = F_m\, h_{HW}(Q_{\mathrm{lat}}, D, L_L), \tag{13.22}$$

*Provenance (Eq. 13.22):* **[Literature]**: Christiansen reduction factor with Hazen-Williams exponent.

### 26.4.16 Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

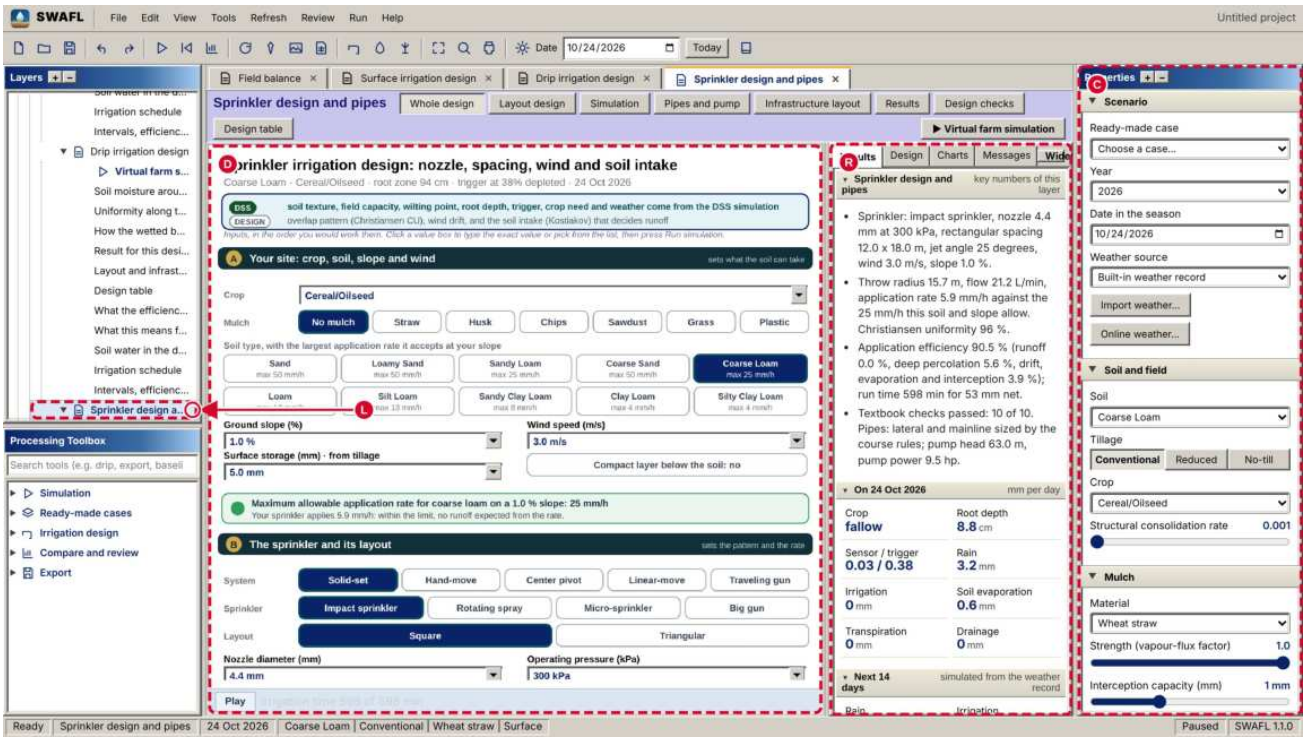



Figure 26.97: Sprinkler design and pipes / Layout and infrastructure: the whole workbench.

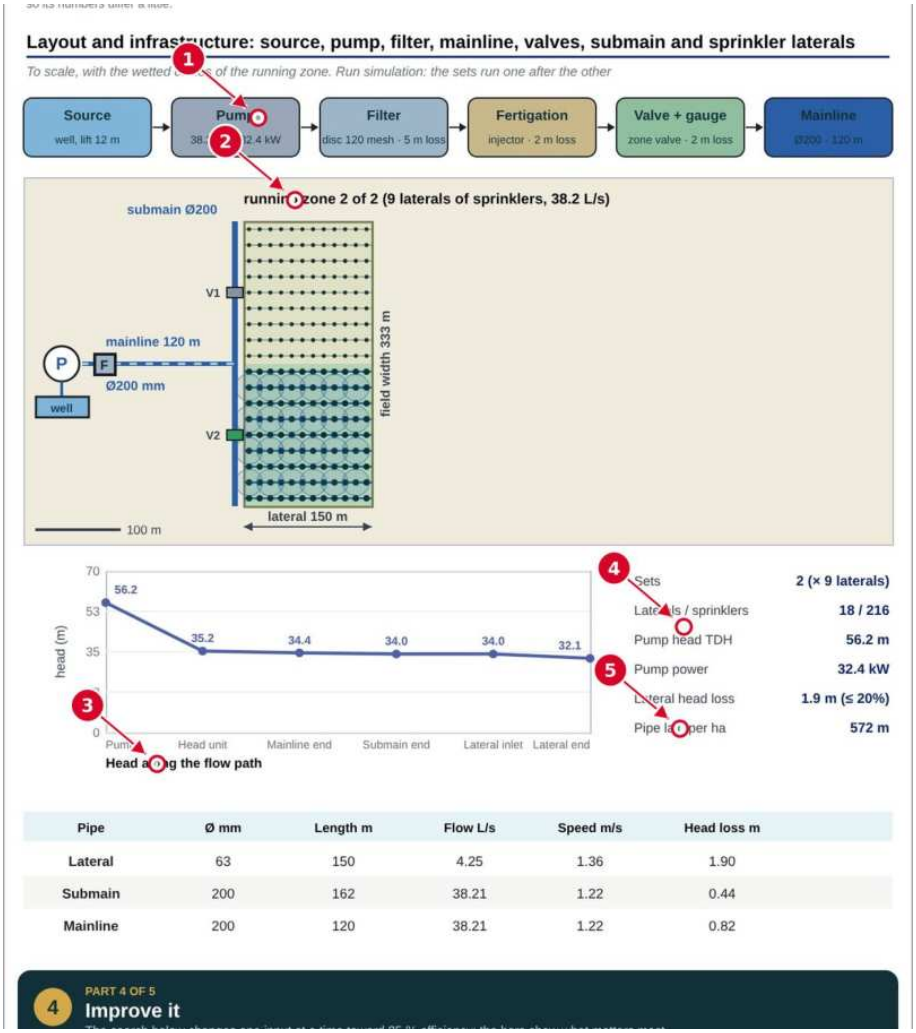



Figure 26.98: Sprinkler design and pipes / Layout and infrastructure: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Source, pump, filter, fertigation and valves in order.
- **2** Laterals, N_L = field length / S_l, and laterals running together: Eq. 13.21.
- **3** Head along the flow path: Eqs. 13.19 and 13.22.
- **4** Lateral head loss with the Christiansen factor F_m: Eq. 13.22.
- **5** Pipe table: Eq. 13.18.

**Equations behind this screen.** Eq. 12.3, Eq. 12.21, Eq. 12.22, Eq. 12.23, Eq. 13.21, Eq. 13.22.

*Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".*

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

*Provenance (Eq. 12.3):* **[Literature]**: Hazen and Williams (1905).

*Eq. 12.21. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".*

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

*Provenance (Eq. 12.21):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 12.22. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".*

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

*Provenance (Eq. 12.22):* **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

*Eq. 12.23. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".*

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\,Q_z\,\text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

*Provenance (Eq. 12.23):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

*Eq. 13.21. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".*

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\text{lat}}} \Big\rfloor\Big), \quad n_{\text{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

*Provenance (Eq. 13.21):* **[Engineered]**: head and sets bookkeeping.

*Eq. 13.22. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".*

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m\,h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

*Provenance (Eq. 13.22):* **[Literature]**: Christiansen reduction factor with Hazen-Williams exponent.

### 26.4.17 Design table: iterations toward an application efficiency of 85 % or more

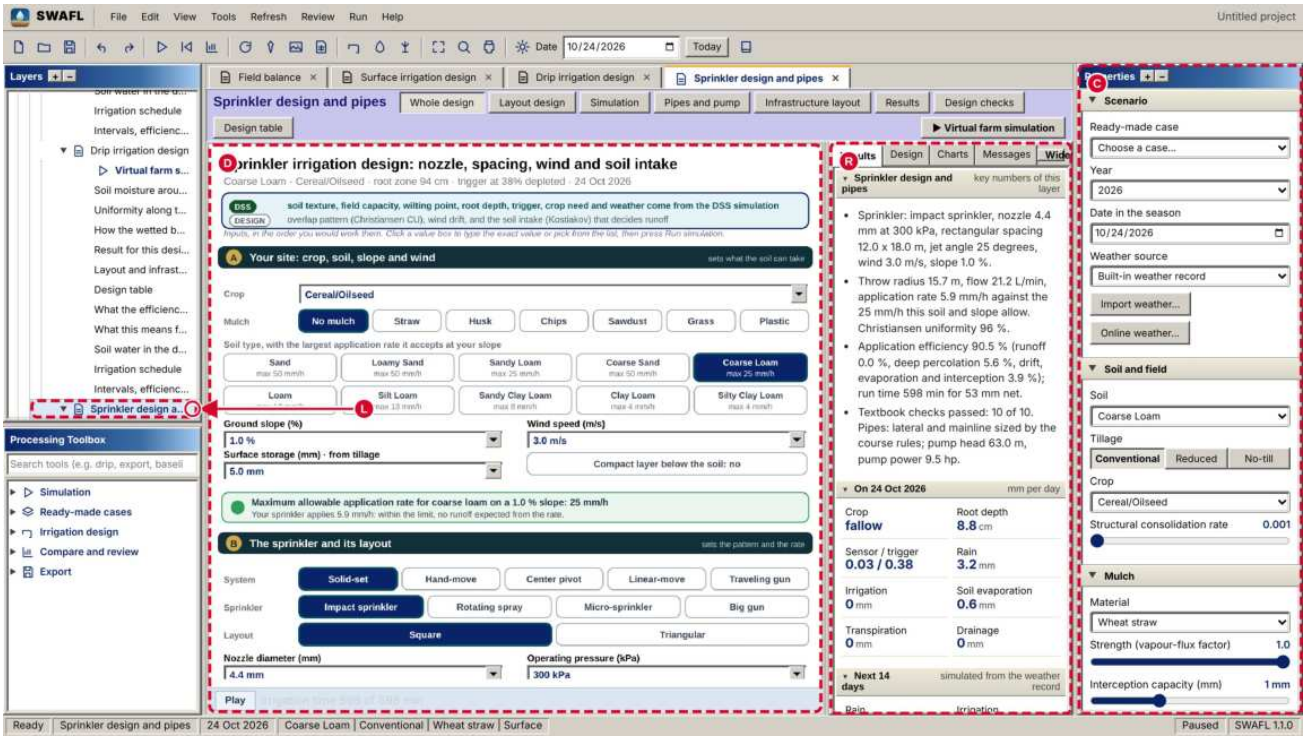


Figure 26.99: Sprinkler design and pipes / Design table: the whole workbench.

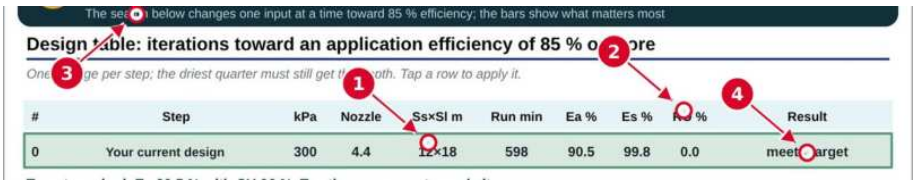


Figure 26.100: Sprinkler design and pipes / Design table: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current design (pressure, nozzle, spacing, run time, E_a, E_s).
- **2** Column E_a: Eq. 13.11.
- **3** Run time T: Eq. 13.12.
- **4** Target reached: E_a of at least 85 % with CU of at least 85 %, Eqs. 13.7 and 13.11.

**Equations behind this screen.** Eq. 12.26.

*Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.*

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

*Provenance (Eq. 12.26):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

## 26.4.18 What the efficiency is most sensitive to

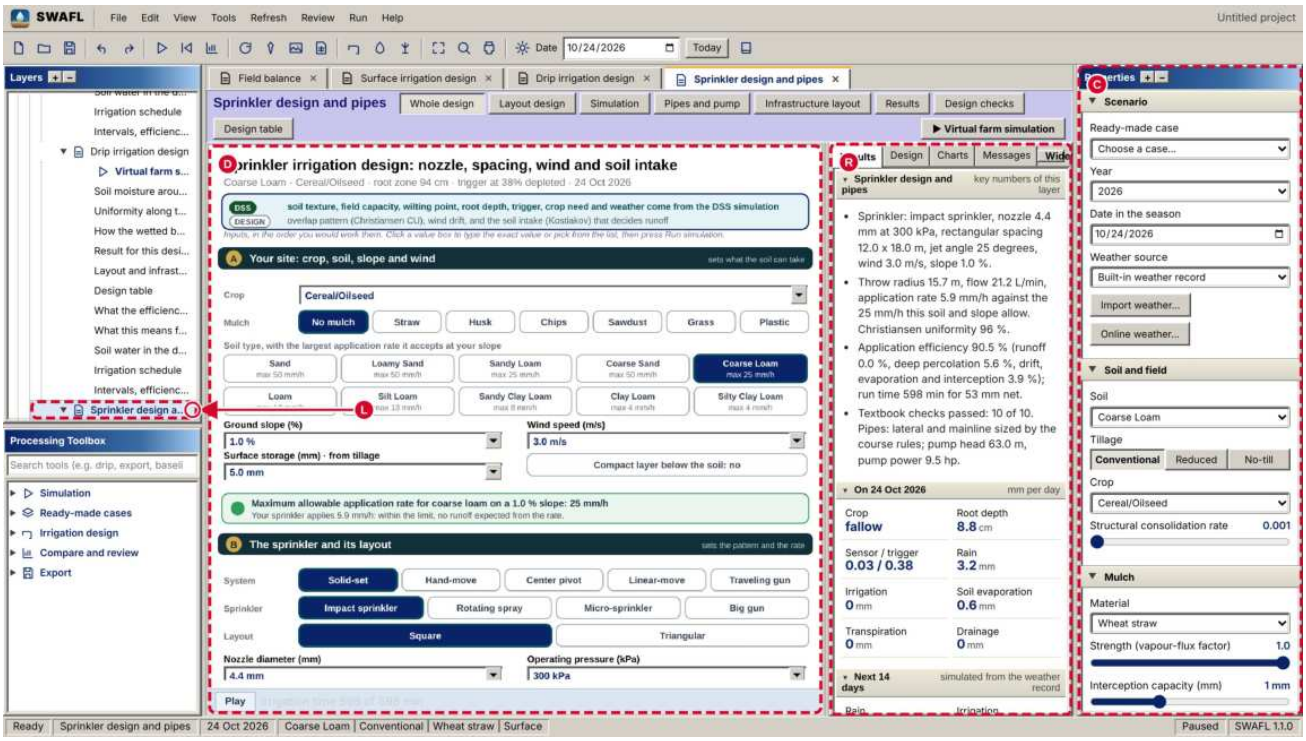


Figure 26.101: Sprinkler design and pipes / What the efficiency is most sensitive to: the whole workbench.

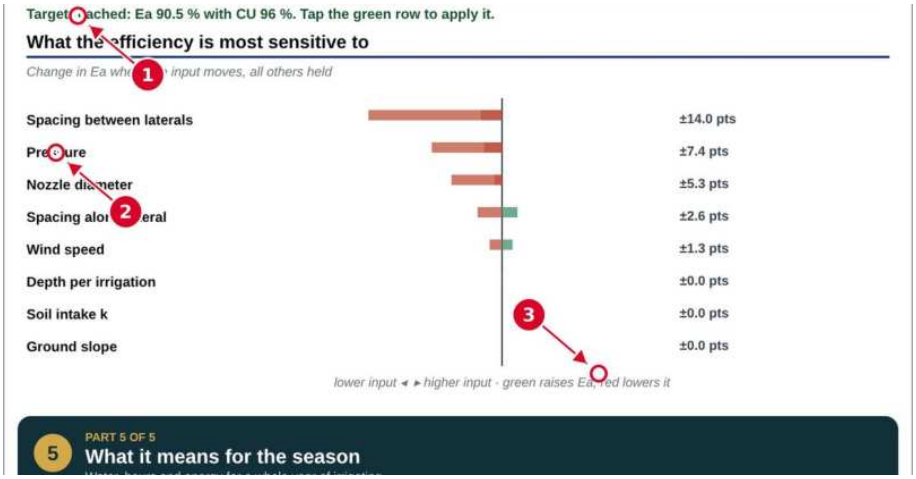



Figure 26.102: Sprinkler design and pipes / What the efficiency is most sensitive to: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 13.11) when the input moves.
- **2** Pressure: second most sensitive input.
- **3** Green raises E_a, red lowers it.

**Equations behind this screen.** Eq. 12.26.

*Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.*

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

*Provenance (Eq. 12.26):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

## 26.4.19 What this means for the season and the farm

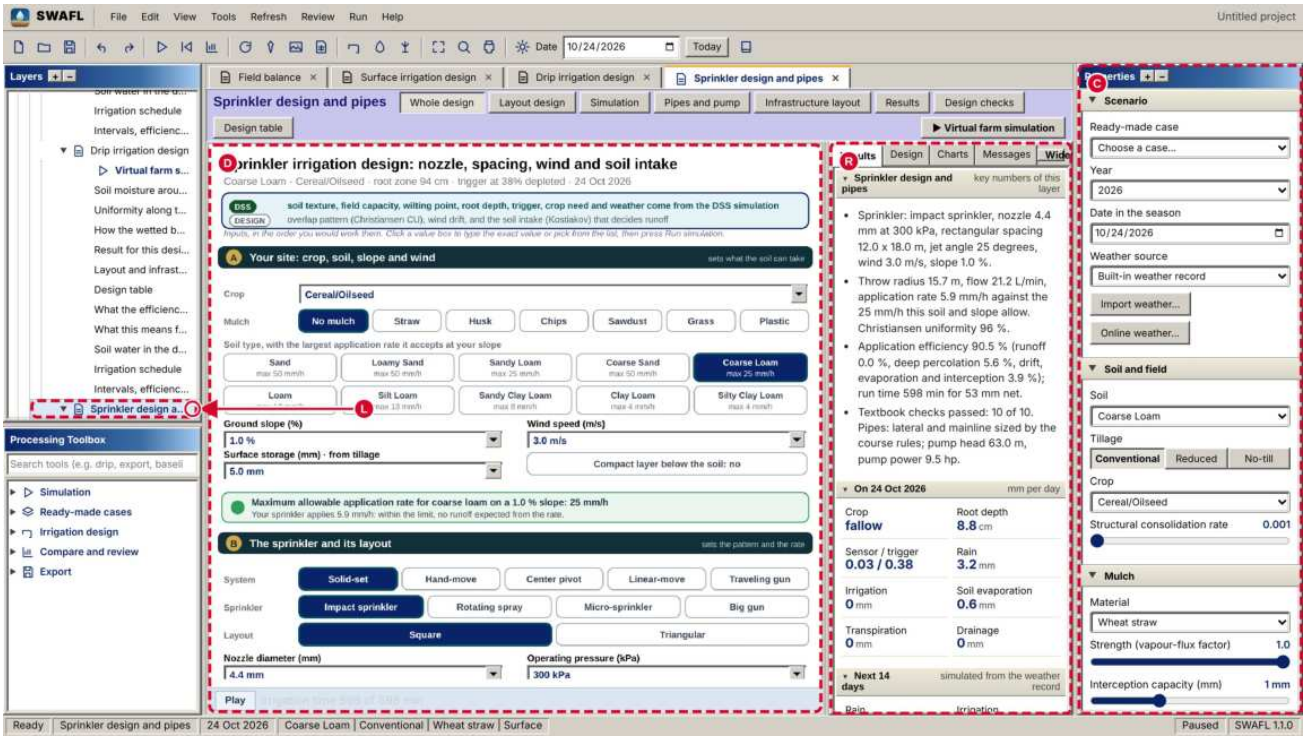



Figure 26.103: Sprinkler design and pipes / What this means for the season and the farm: the whole workbench.

Figure 26.104: Sprinkler design and pipes / What this means for the season and the farm: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season: Eq. 13.23.
- **2** Gross depth per irrigation: Eq. 13.23.
- **3** One sprinkler discharge q: Eq. 13.2.
- **4** Sprinklers for the farm area, N_s: Eq. 13.21.
- **5** Sets and run time: Eqs. 13.12 and 13.21.

**Equations behind this screen.** Eq. 13.23.

*Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.*

$$n_{\mathrm{ev}} = \frac{y_{I}}{Z_{n}}, \qquad n_{\mathrm{sp}} = \mathrm{round}\Big(\frac{10^{4} A_{f}}{S_{s} S_{l}}\Big), \qquad Q_{\mathrm{tot}} = \frac{n_{\mathrm{sp}}\, q}{60}, \qquad n_{\mathrm{set}} = \Big\lceil \frac{Q_{\mathrm{tot}}}{Q_{s}} \Big\rceil, \qquad t_{\mathrm{round}} = \frac{n_{\mathrm{set}}\, T}{60}\ \mathrm{h}. \tag{13.23}$$

*Provenance (Eq. 13.23):* **[Engineered]**: head and sets bookkeeping.

## 26.4.20 Irrigation schedule: when to irrigate with this design

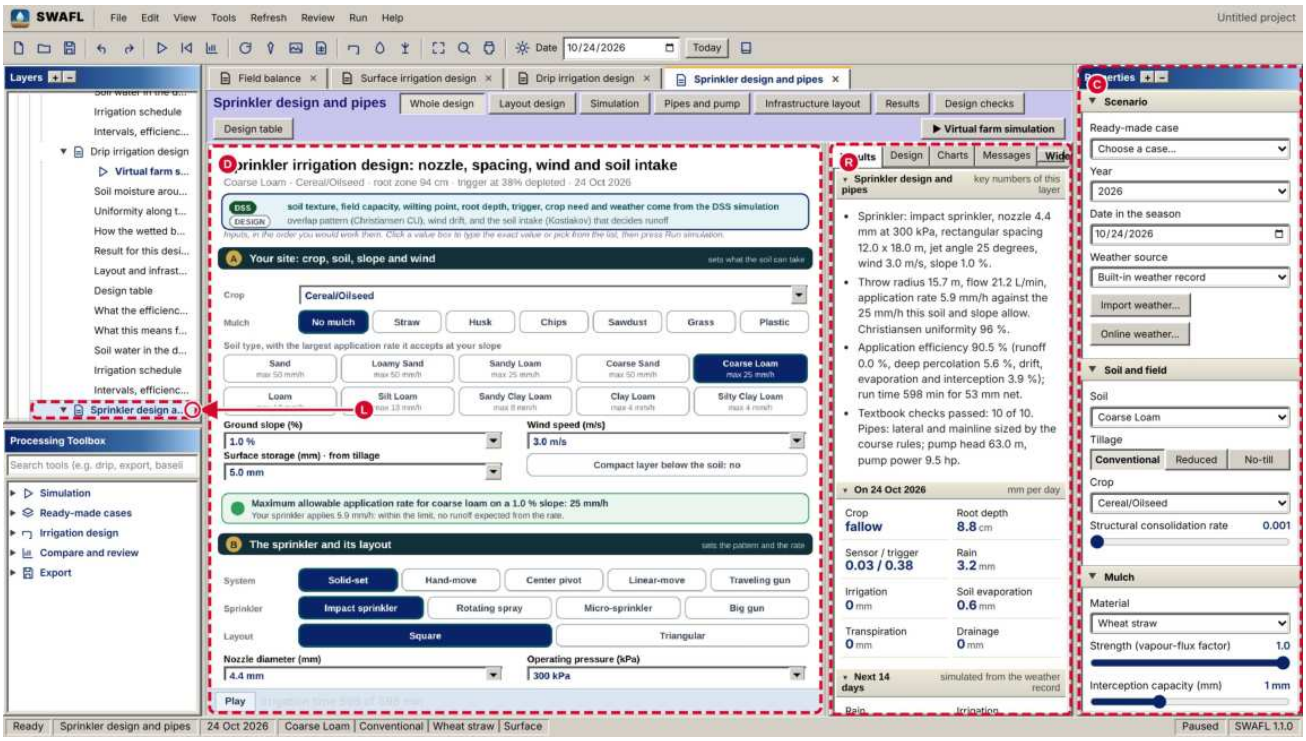



Figure 26.105: Sprinkler design and pipes / Irrigation schedule: the whole workbench.

| # | Date | In | Net mm | Pumped mm | Run time | Gap to next |
|---|---|---|---|---|---|---|
| 1 | 22 May 2027 | 210 days | 53 | 59 | 29.9 h | 13 days |
| 2 | 4 Jun 2027 | 223 days | 53 | 59 | 29.9 h | 11 days |
| 3 | 15 Jun 2027 | 234 days | 53 | 59 | 29.9 h | 9 days |
| 4 | 24 Jun 2027 | 243 days | 53 | 59 | 29.9 h | 9 days |
| 5 | 3 Jul 2027 | 252 days | 53 | 59 | 29.9 h | 10 days |
| 6 | 13 Jul 2027 | 262 days | 53 | 59 | 29.9 h | 16 days |
| 7 | 29 Jul 2027 | 278 days | 53 | 59 | 29.9 h | 251 days |
| 8 | 5 Apr 2028 | 529 days | 53 | 59 | 29.9 h | - |

Figure 26.106: Sprinkler design and pipes / Irrigation schedule: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates from the Engine 2 irrigation record.
- **2** Pumped mm = I_k / E_a: Eq. 13.23.
- **3** Run time of one round: Eq. 13.12.

**Equations behind this screen.** Eq. 13.23.

*Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.*

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60}\ \text{h}. \tag{13.23}$$

*Provenance (Eq. 13.23):* **[Engineered]**: head and sets bookkeeping.

## 26.4.21 Intervals, efficiency and the advice in short

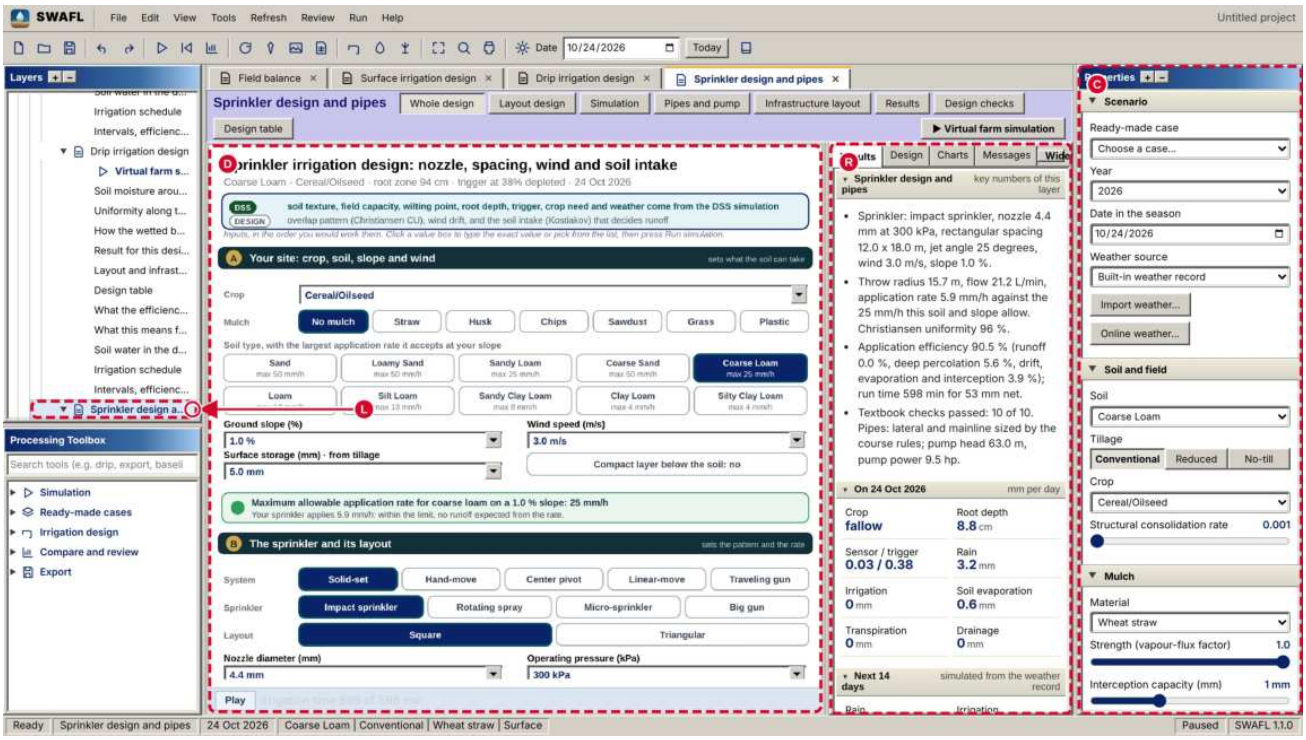


Figure 26.107: Sprinkler design and pipes / Intervals, efficiency and the advice in short: the whole workbench.

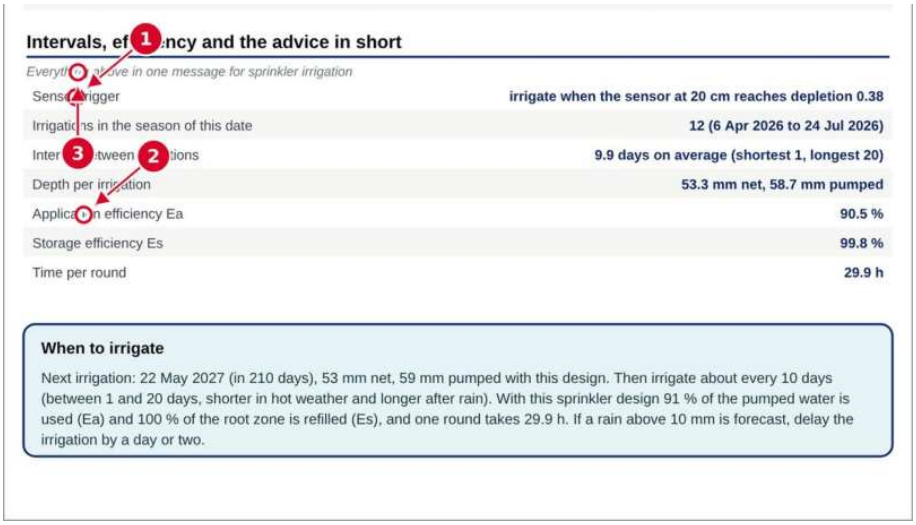



Figure 26.108: Sprinkler design and pipes / Intervals, efficiency and the advice in short: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: Eqs. 5.3 and 5.27.
- **2** Application efficiency E_a: Eq. 13.11.

- **3** “When to irrigate”: message built from the schedule and from E_a and E_s.

**Equations behind this screen.** Eq. 13.23.

*Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.*

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60} \text{ h.} \tag{13.23}$$

*Provenance (Eq. 13.23):* **[Engineered]**: head and sets bookkeeping.

# 26.5 Field balance

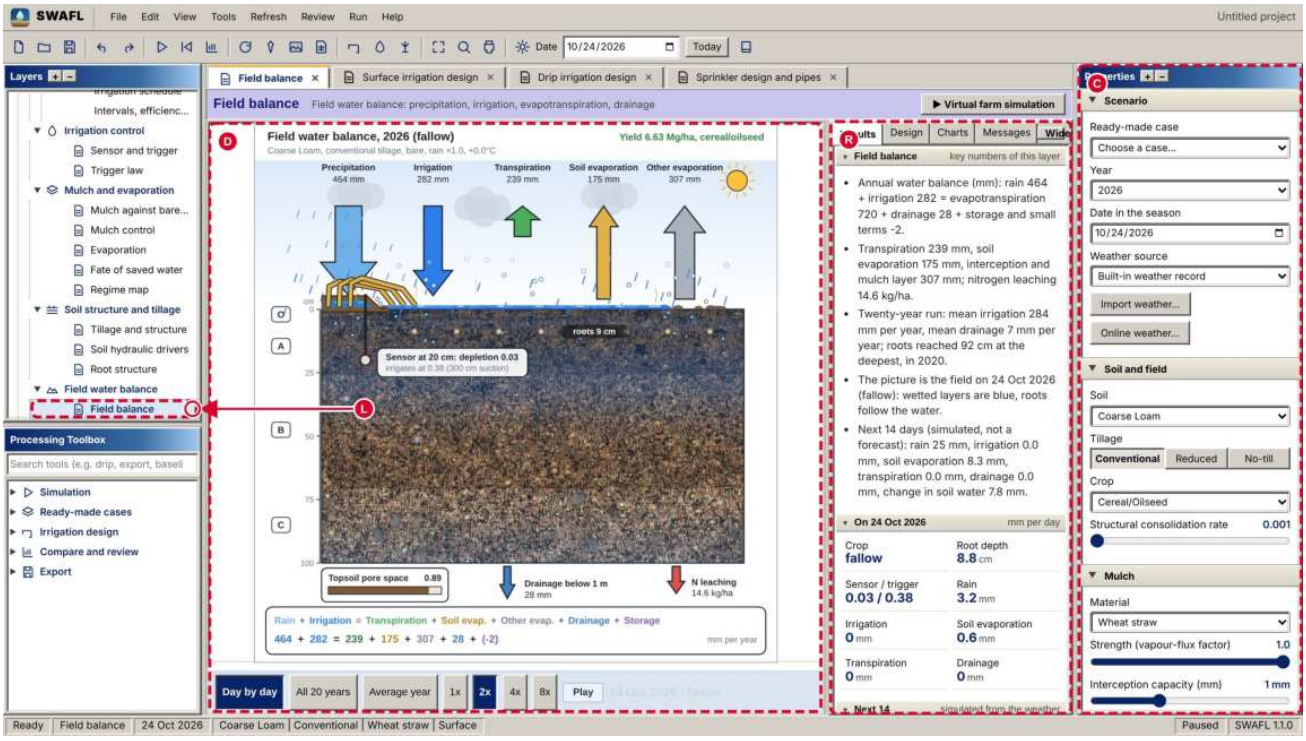



Figure 26.109: Field balance: the whole workbench.

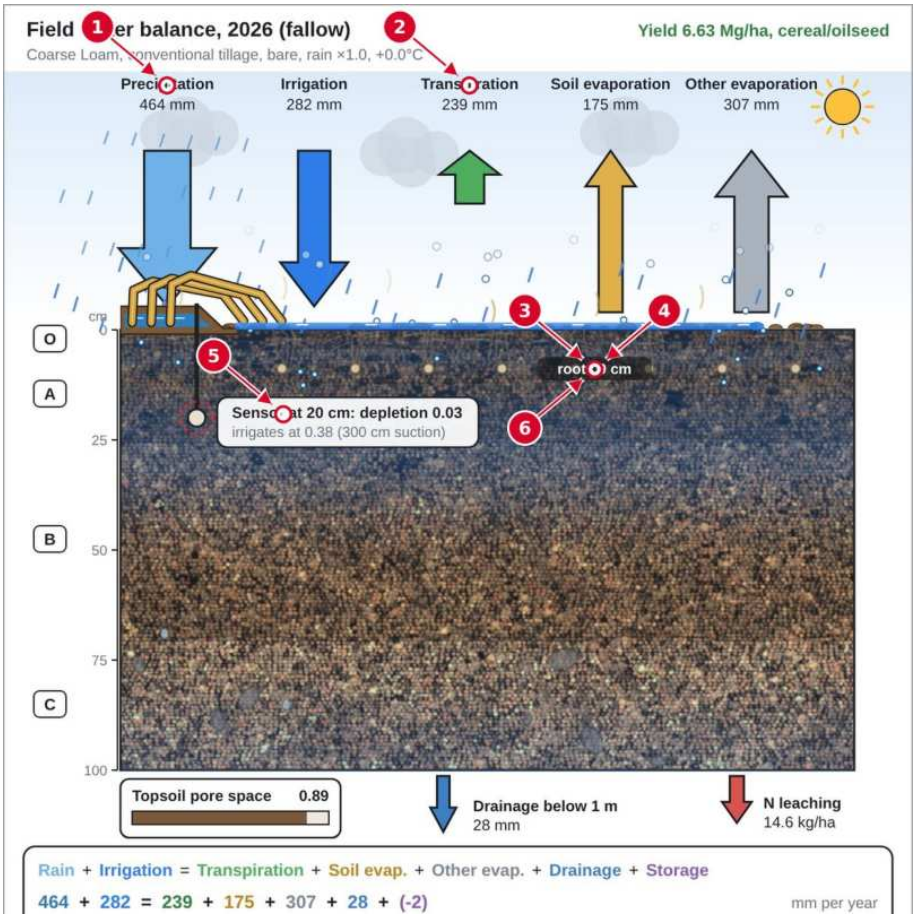



Figure 26.110: Field balance: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Precipitation arrow, P: annual value from Eqs. 5.4 to 5.12 (Average year); daily value from the rainfall generator, Eqs. 5.15 and 5.16 (Day by day).
- **2** Transpiration arrow, T: Eqs. 5.4 to 5.12 (annual) and Eqs. 5.21 to 5.25 (daily).
- **3** Soil evaporation arrow, E: Eqs. 5.4 to 5.12 (annual) and Eqs. 5.21 to 5.26 (daily).
- **4** Interception arrow, E_other: the closure term of Eq. 5.12.
- **5** Sensor tag: depletion f of the sensor layer against the trigger f_T, Eq. 5.3 (Engine 2 layer j_s).
- **6** Roots (depth and density): Eqs. 5.19 and 5.20.

**Equations behind this screen.** Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 5.13, Eq. 5.14, Eq. 5.15, Eq. 5.16, Eq. 5.17, Eq. 5.18, Eq. 5.19, Eq. 5.20, Eq. 5.21, Eq. 5.22, Eq. 5.23, Eq. 5.24, Eq. 5.25, Eq. 8.1.

*Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.5. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.6. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.7. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.8. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.9. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.10. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".*

$$Q \leftarrow Q\ \text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ \ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N). \tag{5.11}$$

*Provenance (Eq. 5.11):* **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

*Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".*

$$\begin{gathered} ET \in [\,P + I - D - 30,\ P + I - D + 15\,] \text{ (if not, } ET \text{ is clipped and } E, T \text{ are scaled by the same factor)}, \\ E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \end{gathered} \tag{5.12}$$

*Provenance (Eq. 5.12):* **[Engineered]**: closure constraint and bounds chosen for the workbench.

*Eq. 5.13. Where explained: chapter "The two computing engines", section "Crop calendar and canopy cover".*

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

*Provenance (Eq. 5.13):* **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

*Eq. 5.14. Where explained: chapter "The two computing engines", section "Crop calendar and canopy cover".*

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

*Provenance (Eq. 5.14):* **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

*Eq. 5.15. Where explained: chapter "The two computing engines", section "Rainfall generator".*

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

*Provenance (Eq. 5.15):* **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

*Eq. 5.16. Where explained: chapter "The two computing engines", section "Rainfall generator".*

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

*Provenance (Eq. 5.16):* **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

*Eq. 5.17. Where explained: chapter "The two computing engines", section "Infiltration, redistribution and drainage".*

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\;\; w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

*Provenance (Eq. 5.17):* **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

*Eq. 5.18. Where explained: chapter "The two computing engines", section "Roots".*

$$R_{\max} = \operatorname{clip}\Big(R_c\,(0.85 + 0.3\,\text{sand})\big[1 + 0.12\log_{10}(|h_T|/300)\big]\big[1 - 0.18\,\tfrac{N-100}{100}\big];\; 25,\; 100\Big)\ \text{cm}, \tag{5.18}$$

*Provenance (Eq. 5.18):* **[Engineered]**: rooting rules with constants chosen for the workbench.

*Eq. 5.19. Where explained: chapter "The two computing engines", section "Roots".*

$$R_d \leftarrow \max\Big\{R_d,\; \min\big\{R_{\max}\operatorname{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\; R_{\max},\; R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

*Provenance (Eq. 5.19):* **[Engineered]**: rooting rules with constants chosen for the workbench.

*Eq. 5.20. Where explained: chapter "The two computing engines", section "Roots".*

$$\rho_i \propto \exp\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

*Provenance (Eq. 5.20):* **[Engineered]**: rooting rules with constants chosen for the workbench.

*Eq. 5.21. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".*

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65\cdot\tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

*Provenance (Eq. 5.21):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

*Eq. 5.22. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".*

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\text{ yr}} c, \tag{5.22}$$

*Provenance (Eq. 5.22):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

*Eq. 5.23. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".*

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \operatorname{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

*Provenance (Eq. 5.23):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

*Eq. 5.24. Where explained: chapter "The two computing engines", section "Scaling to the annual model".*

$$S_P = \frac{P}{\overline{P_{\text{sim}}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T_{f,\text{sim}}}}, \quad S_E = \frac{E}{\overline{E_{\text{sim}}}}, \quad S_T = \frac{T}{\overline{T_{\text{sim}}}}, \quad S_D = \frac{D}{\overline{D_{\text{sim}}}}, \tag{5.24}$$

*Provenance (Eq. 5.24):* **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

*Eq. 5.25. Where explained: chapter "The two computing engines", section "Scaling to the annual model".*

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

*Provenance (Eq. 5.25):* **[Literature]**: water-balance identity.

*Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".*

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

*Provenance (Eq. 8.1):* **[Engineered]**: exponential relaxation with chosen constants.

### 26.5.1 The whole layer, part by part

The picture below is the complete drawing of the layer Field balance, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

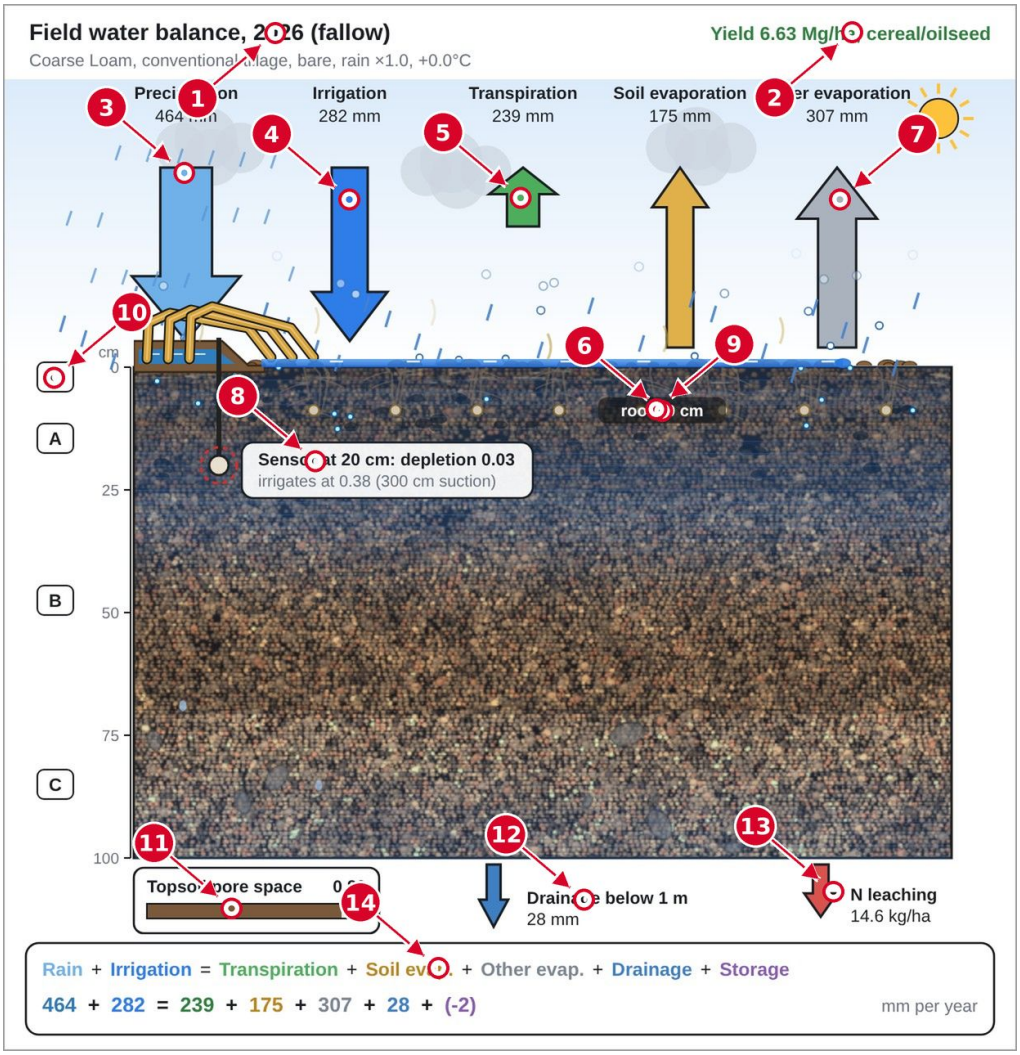



Figure 26.111: Field balance, part 1 of 1.

**What the arrows point to.**

- **1** The title gives the year, the crop state and the scenario in force (soil, tillage, mulch, rain scaling and warming).
- **2** The yield of the year, with the crop that produced it.
- **3** Precipitation, the wet input of the budget, in mm for the year.
- **4** Irrigation, the applied water decided by the sensor trigger, in mm for the year.
- **5** Transpiration, the water taken up through the roots and released by the leaves.
- **6** Soil evaporation, the water lost directly from the soil surface.
- **7** Interception, the rain held on leaves and mulch and evaporated before reaching the soil.
- **8** The soil sensor at 20 cm, with its depletion and the depletion at which irrigation starts.
- **9** The depth of the roots at this moment of the season.
- **10** The horizons O, A, B and C of the soil profile, with the depth scale in cm.
- **11** The topsoil pore space, the fraction of the topsoil volume that is air and water.
- **12** Drainage below 1 m, the water that leaves the root zone downward.
- **13** Nitrogen leaching carried out of the root zone with the drainage.
- **14** The closed budget strip, in which the inputs equal the outputs plus the change in storage.

## 26.5.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

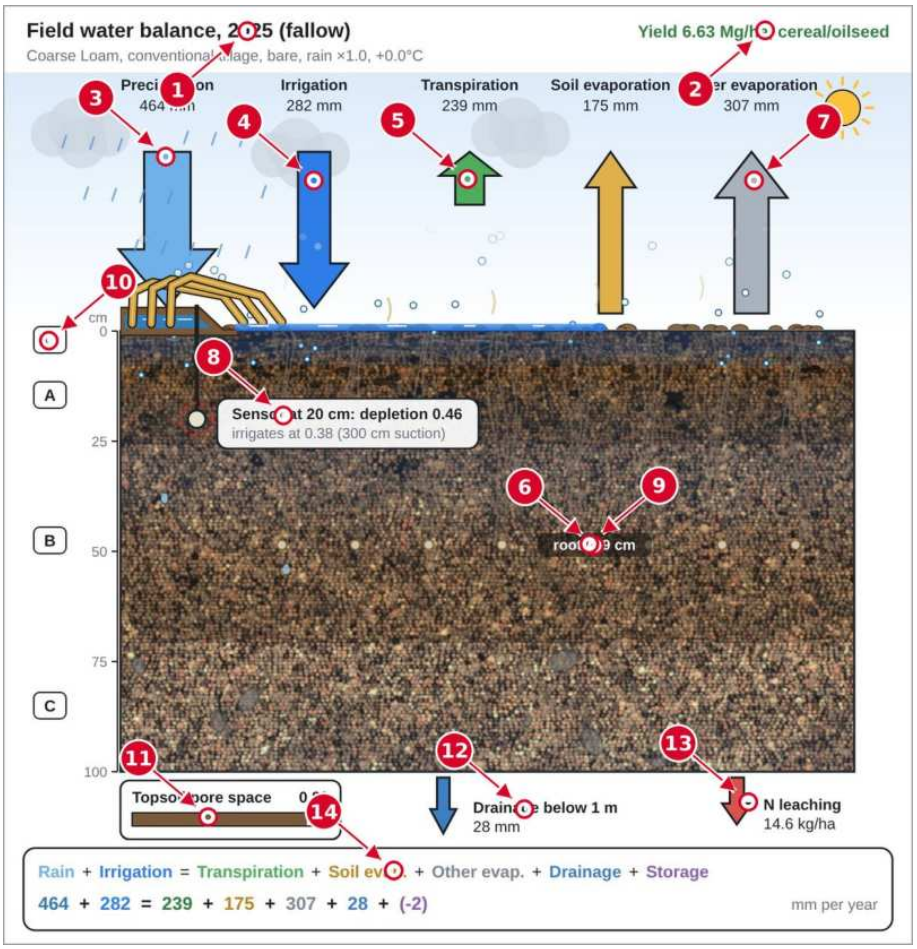



Figure 26.112: Field balance, part 1, playback a third of the way (7 Aug 2025 · fallow).

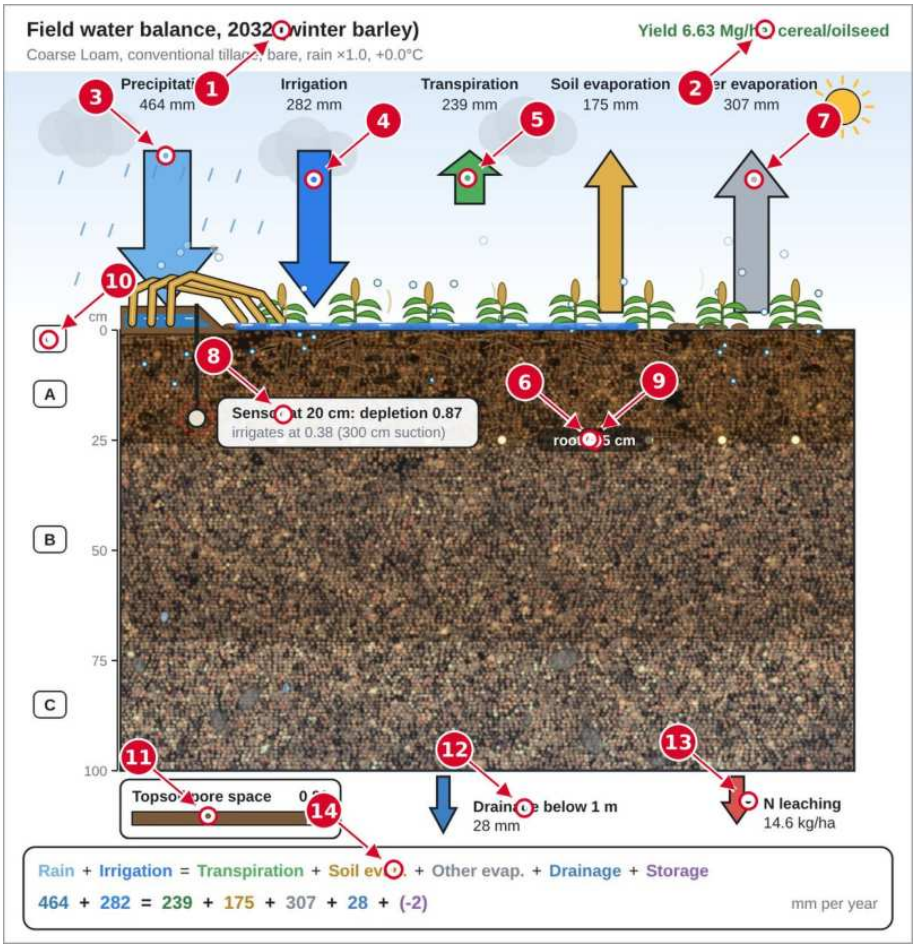



Figure 26.113: Field balance, part 1, playback two thirds of the way (13 Mar 2032 · winter barley).

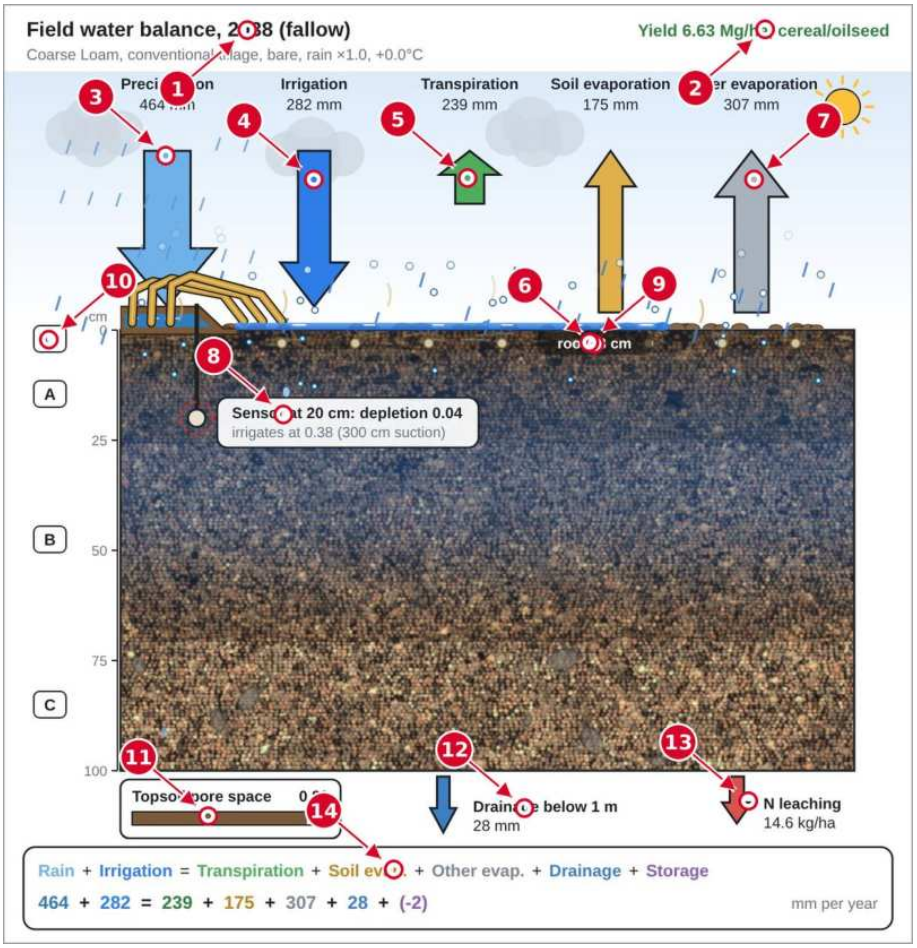



Figure 26.114: Field balance, part 1, playback the end (31 Dec 2038 · fallow).

### 26.5.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

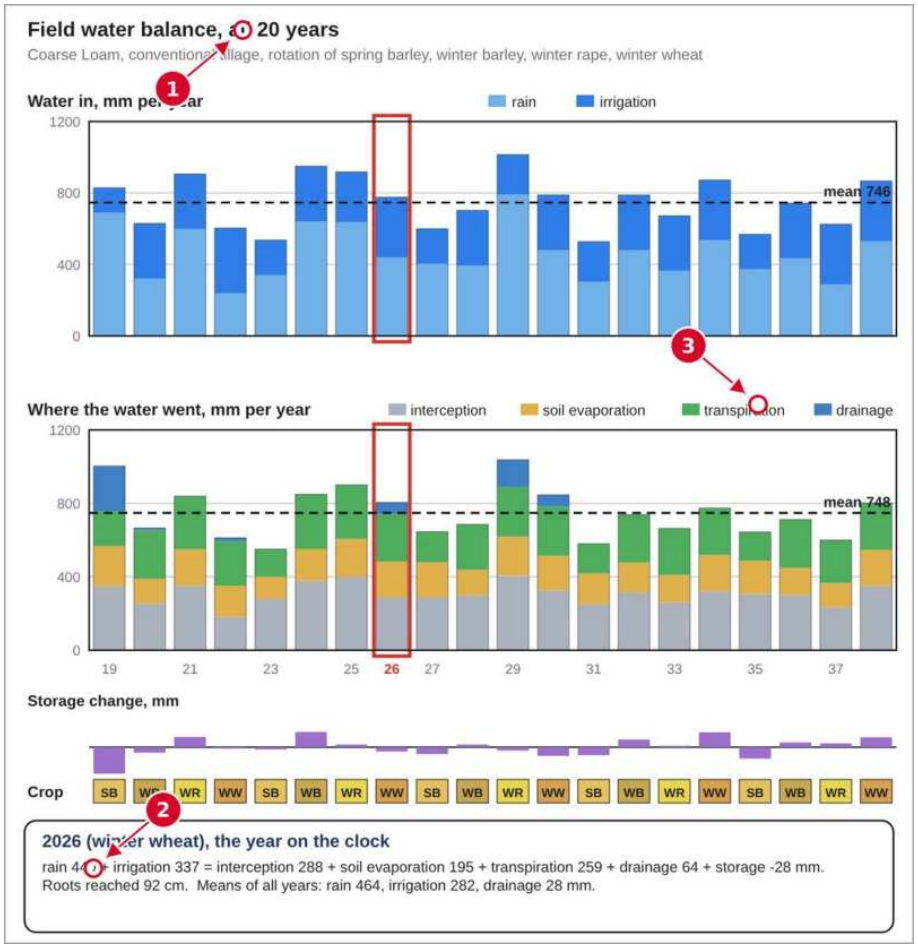



Figure 26.115: Field balance, part 1. The All 20 years button pressed, so the arrows and the budget show the whole twenty-year record in one picture.

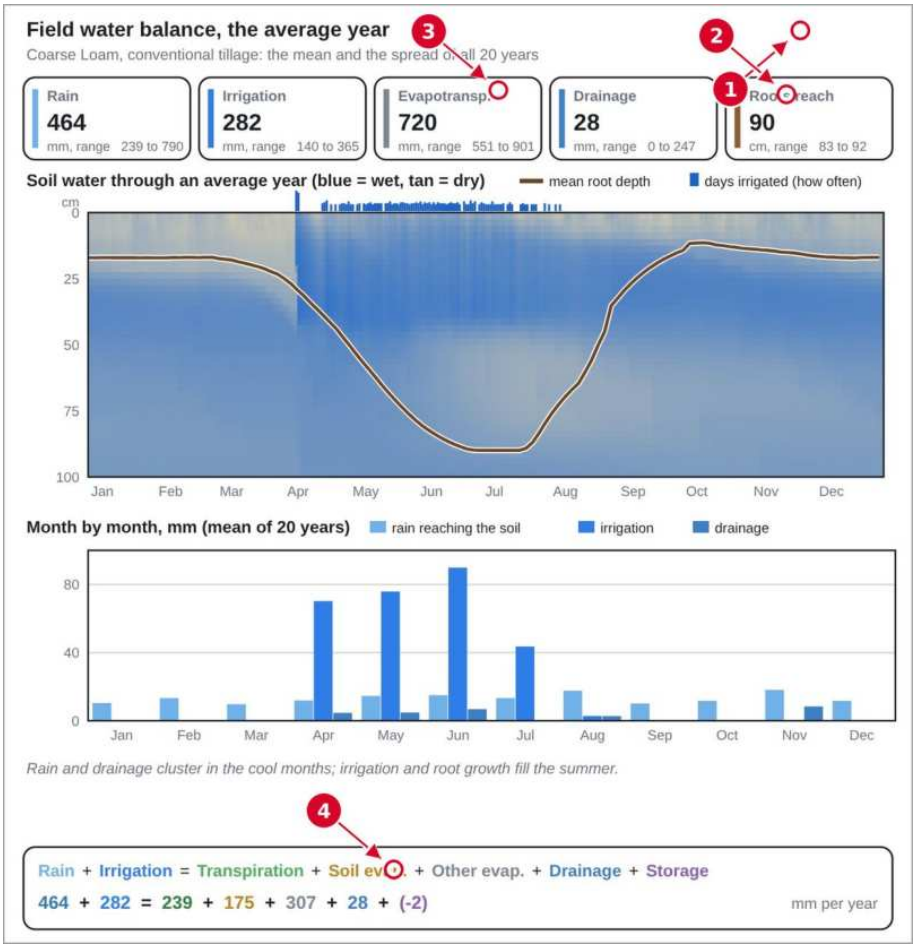



Figure 26.116: Field balance, part 1. The Average year pressed, so every flux is the mean of the twenty simulated years.

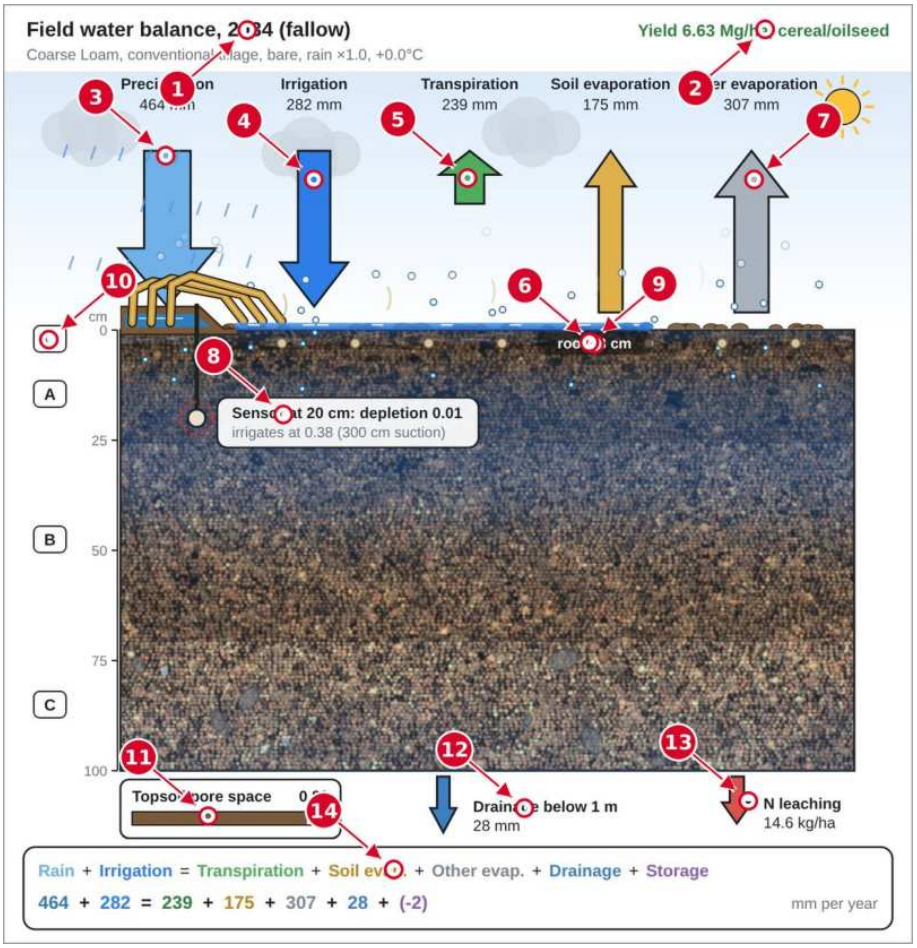



Figure 26.117: Field balance, part 1. The year 2030 pressed, so the picture shows the budget of that single year.

# 26.6 Soil, plant and air

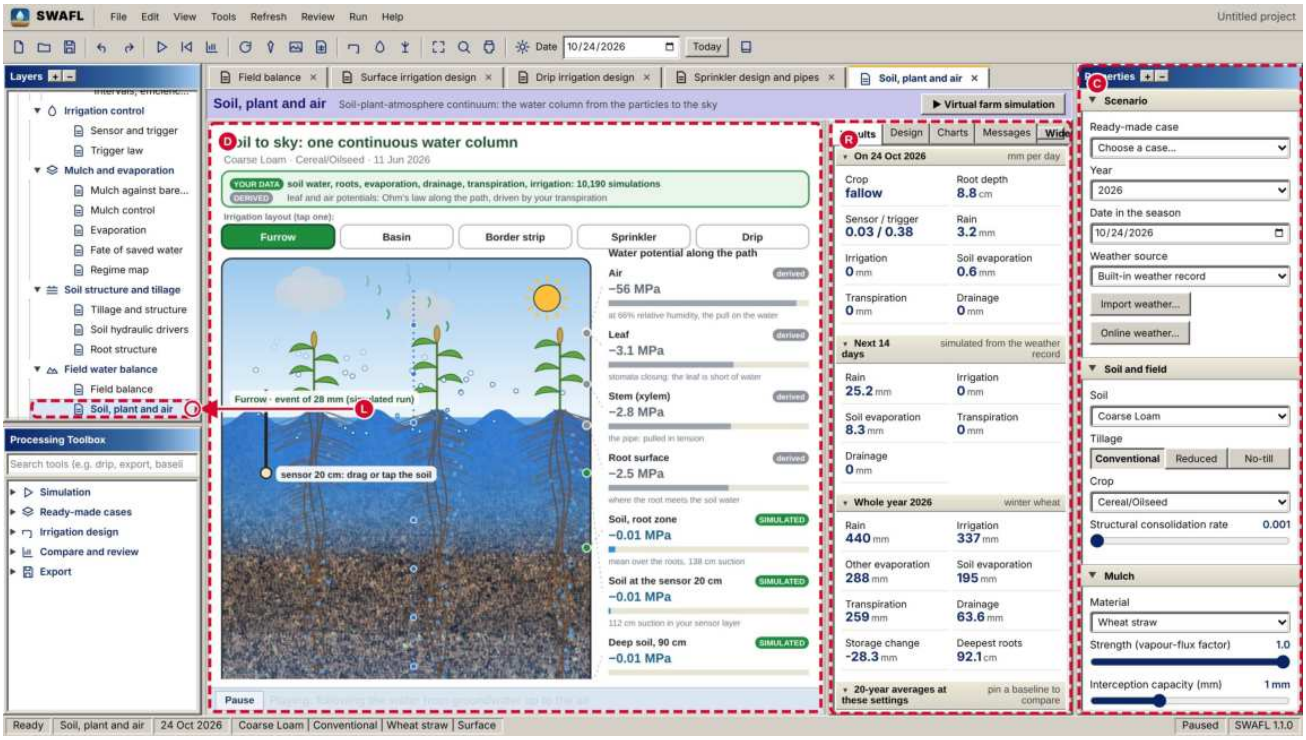



Figure 26.118: Soil, plant and air: the whole workbench.

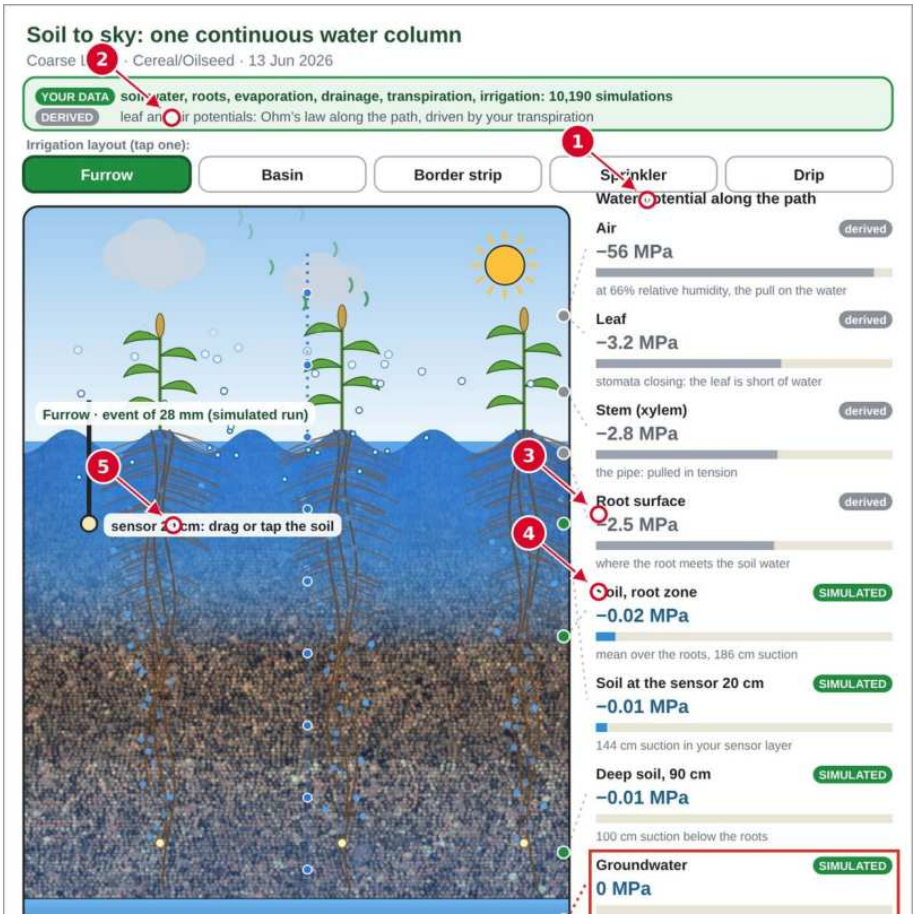



Figure 26.119: Soil, plant and air: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Water potential of the air, psi_air = 135.5 ln RH: Eq. 7.5.
- **2** Leaf potential, psi_leaf: Eq. 7.4.
- **3** Stem and root-surface potentials, psi_stem and psi_root: Eqs. 7.2 and 7.3.
- **4** Root-zone soil potential from the layer suctions: Eq. 7.1 (inverse of Eq. 5.3), with Eqs. 5.1 and 5.2 for the conductivity.
- **5** Sensor at 20 cm: the suction of layer j_s, Eq. 7.1.

**Equations behind this screen.** Eq. 5.1, Eq. 5.2, Eq. 5.3, Eq. 5.25, Eq. 7.1, Eq. 7.2, Eq. 7.3, Eq. 7.4, Eq. 7.5.

*Eq. 5.1. Where explained: chapter "The two computing engines", section "The simulation campaign".*

$$S_e(h) = \left[1 + (\alpha|h|)^n\right]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

*Provenance (Eq. 5.1):* **[Literature]**: van Genuchten (1980).

*Eq. 5.2. Where explained: chapter "The two computing engines", section "The simulation campaign".*

$$K_r(S_e) = S_e^{1/2}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2. \tag{5.2}$$

*Provenance (Eq. 5.2):* **[Literature]**: Mualem (1976) with van Genuchten (1980).

*Eq. 5.3. Where explained: chapter "The two computing engines", section "Plant-available water and depletion".*

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

*Provenance (Eq. 5.3):* **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

*Eq. 5.25. Where explained: chapter "The two computing engines", section "Scaling to the annual model".*

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

*Provenance (Eq. 5.25):* **[Literature]**: water-balance identity.

*Eq. 7.1. Where explained: chapter "Field water balance layers", section "Governing equations".*

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

*Provenance (Eq. 7.1):* **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

*Eq. 7.2. Where explained: chapter "Field water balance layers", section "Governing equations".*

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

*Provenance (Eq. 7.2 to 7.4):* **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

*Eq. 7.3. Where explained: chapter "Field water balance layers", section "Governing equations".*

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

*Provenance (Eq. 7.2 to 7.4):* **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

*Eq. 7.4. Where explained: chapter "Field water balance layers", section "Governing equations".*

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

*Provenance (Eq. 7.2 to 7.4):* **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

*Eq. 7.5. Where explained: chapter "Field water balance layers", section "Governing equations".*

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH} \;\approx\; 135.5\,\ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

*Provenance (Eq. 7.5):* **[Literature]**: Nobel (2009).

### 26.6.1 The whole layer, part by part

The picture below is the complete drawing of the layer Soil, plant and air, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

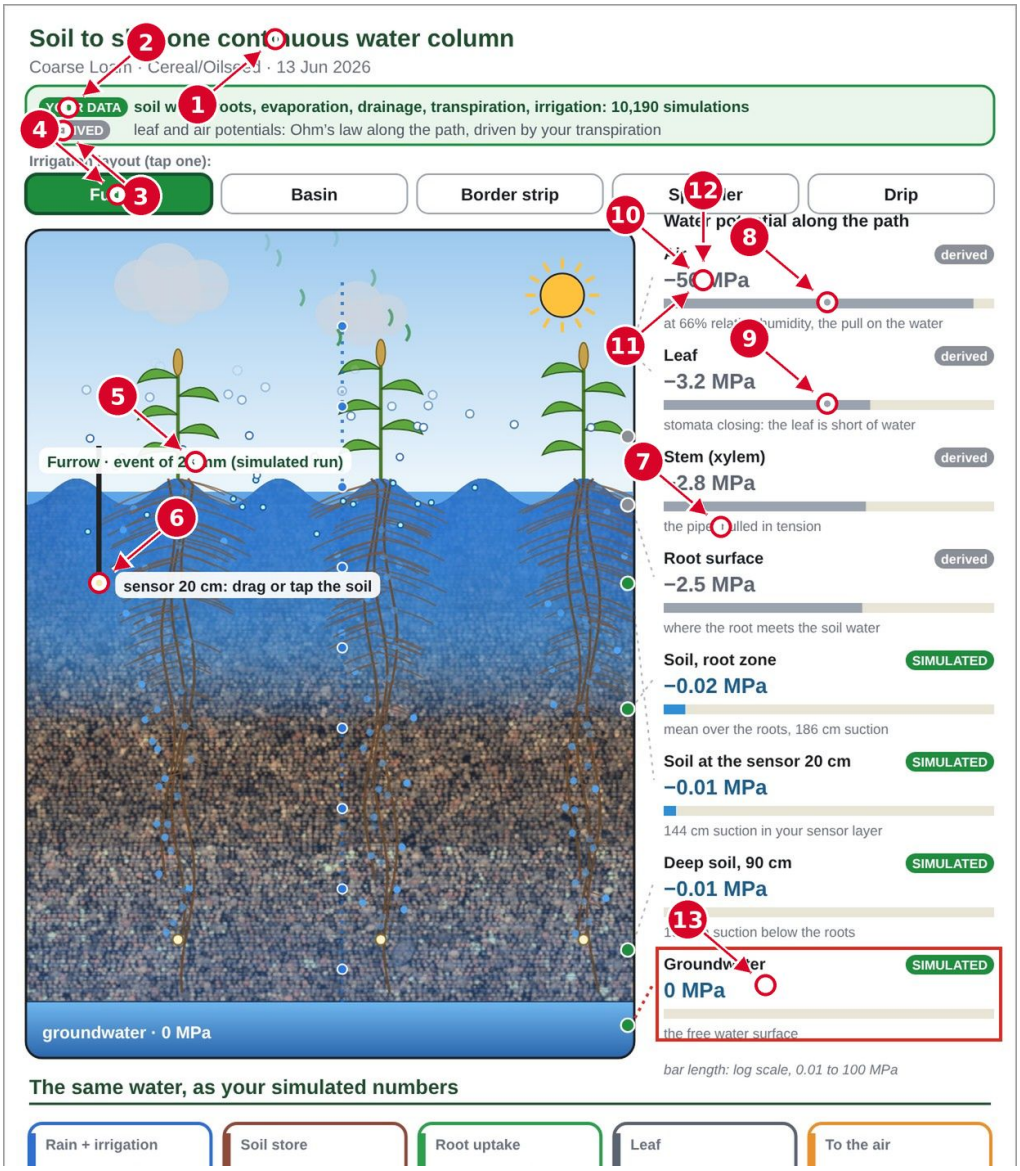



Figure 26.120: Soil, plant and air, part 1 of 3.

**What the arrows point to.**

- **1** The title and the date of the water column shown, from the groundwater up to the air.
- **2** The SIMULATED badge marks quantities read from the simulations of the ten soils.
- **3** The DERIVED badge marks quantities computed along the path from Ohm's law, the leaf and the air potentials.
- **4** The irrigation layout buttons, here Furrow; the other layouts are Basin, Border strip, Sprinkler and Drip.
- **5** The label of the irrigation event with the depth applied in the simulated run.
- **6** The soil sensor at 20 cm, which can be dragged or tapped to another depth.
- **7** The roots of the crop, with the depth they have reached.

- **8** The potential of the air at the current relative humidity, the strongest pull on the water.
- **9** The potential of the leaf, set by the stomata and the transpiration.
- **10** The potential of the soil in the root zone, mean over the roots.
- **11** The potential of the soil at the sensor depth.
- **12** The potential of the deep soil below the roots.
- **13** The groundwater, the free water surface at zero potential.

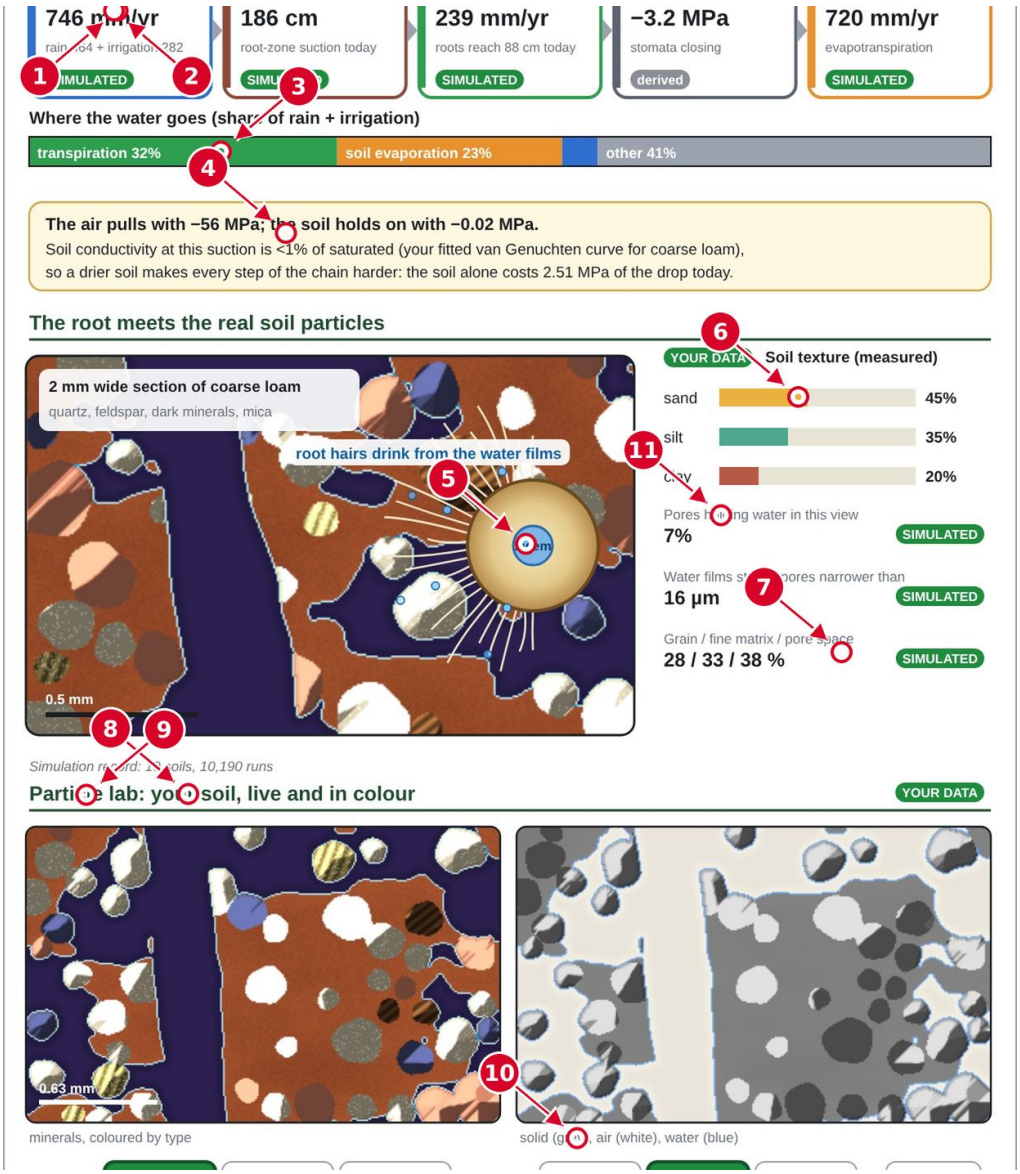



Figure 26.121: Soil, plant and air, part 2 of 3.

**What the arrows point to.**

- **1** The cards give the same water as numbers, from rain and irrigation to the soil store.
- **2** The root uptake card, the transpiration drawn through the roots in mm per year.
- **3** The partition of rain and irrigation into transpiration, soil evaporation and the other losses.
- **4** The message that compares the pull of the air with the hold of the soil.
- **5** A section of the root surrounded by the real soil particles, with root hairs in the water films.
- **6** The measured texture of the soil, the sand, silt and clay fractions.

- **7** The grain, fine matrix and pore fractions of this section.
- **8** The particle lab, the soil shown live and in colour.
- **9** The left view colours the grains by mineral type.
- **10** The right view shows the solid in grey, the air in white and the water in blue.
- **11** The suction, the water content and the pores holding water at the chosen state.

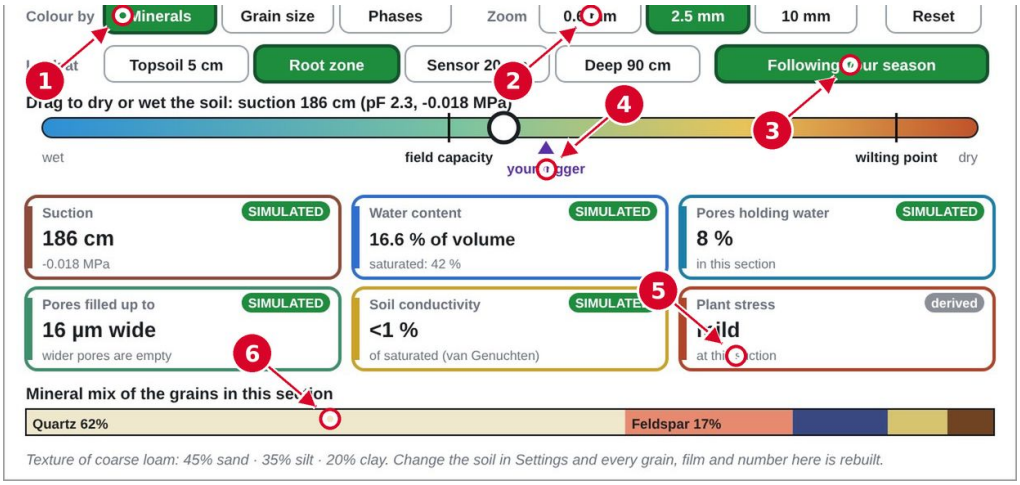



Figure 26.122: Soil, plant and air, part 3 of 3.

**What the arrows point to.**

- **1** The Colour by buttons choose minerals, grain size or phases.
- **2** The Zoom buttons set the width of the section to 0.6, 2.5 or 10 mm.
- **3** The Look at buttons choose the depth, or follow your season.
- **4** The value box dries or wets the soil, with the field capacity, your trigger and the wilting point marked.
- **5** The plant stress flag, past or inside your trigger at this suction.
- **6** The mineral mix of the grains in this section.

## 26.6.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

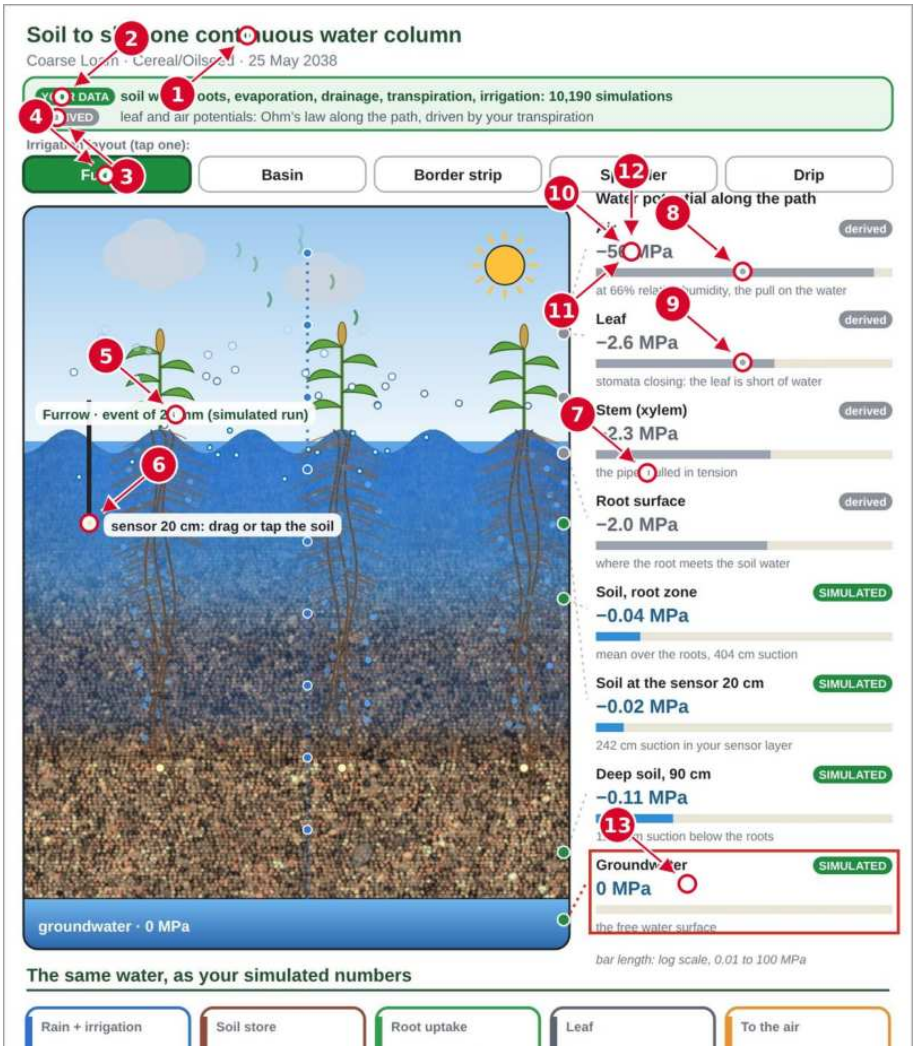



Figure 26.123: Soil, plant and air, part 1, playback a third of the way (Paused: following the water from groundwater up to the air).

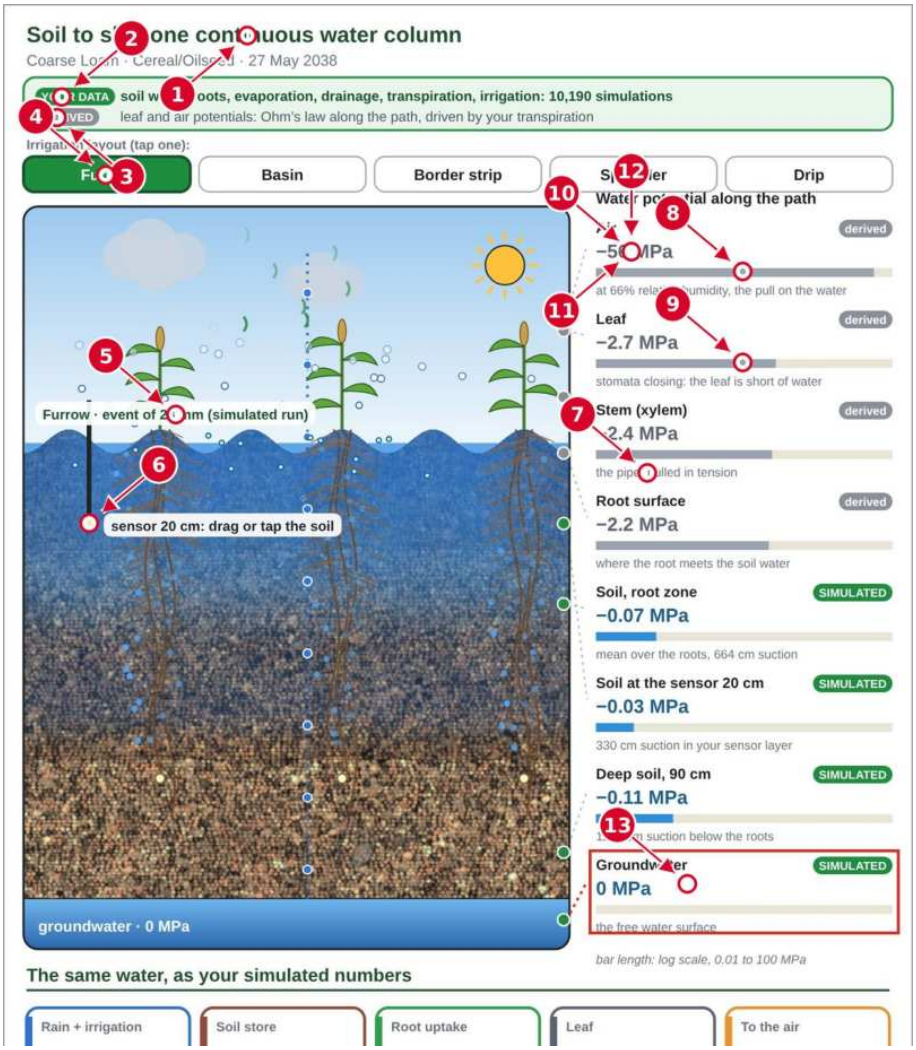



Figure 26.124: Soil, plant and air, part 1, playback two thirds of the way (Paused: following the water from groundwater up to the air).

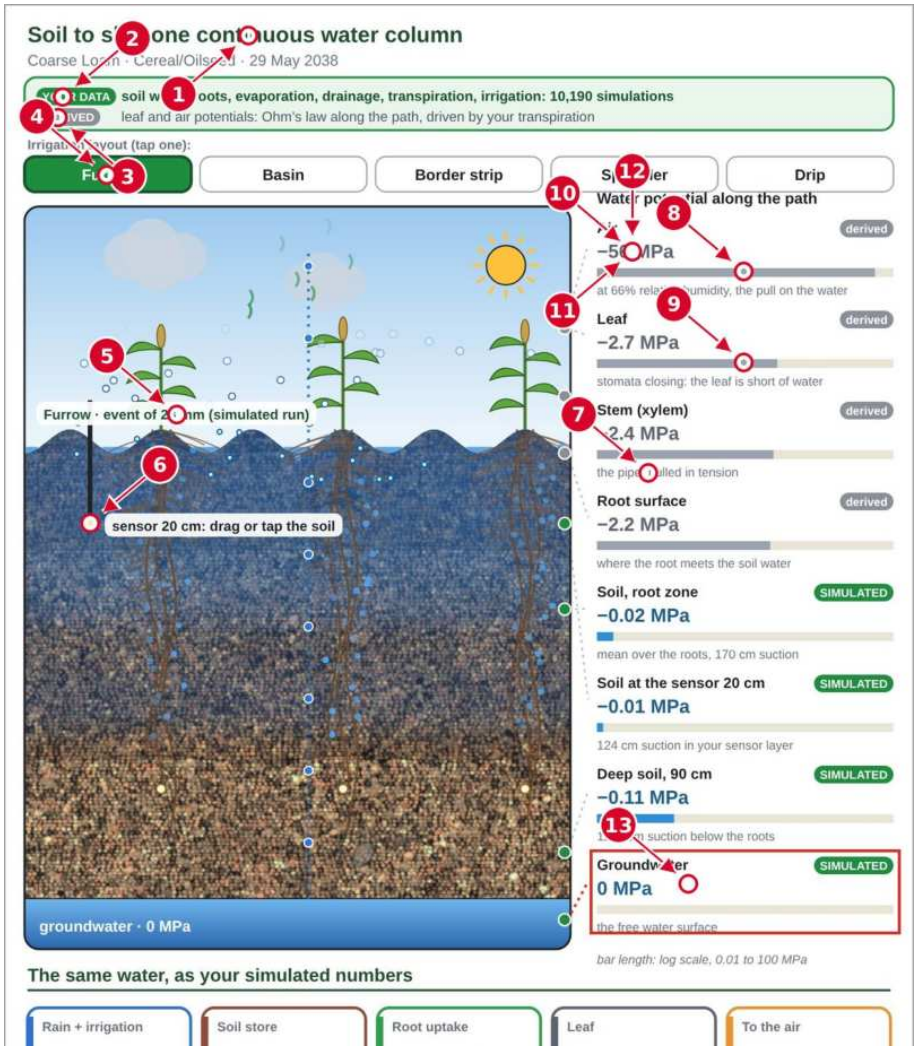



Figure 26.125: Soil, plant and air, part 1, playback the end (Paused: following the water from groundwater up to the air).

### 26.6.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

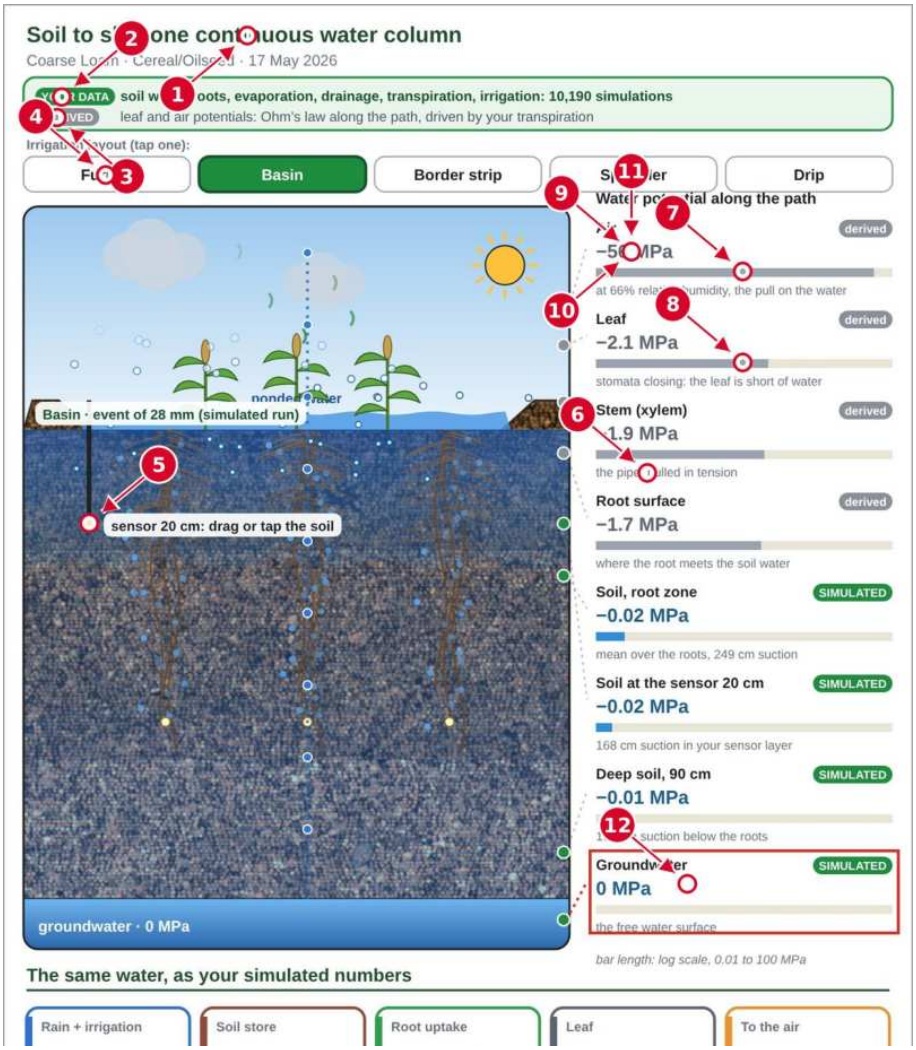



Figure 26.126: Soil, plant and air, part 1. The Basin layout chosen, so the water ponds over the whole level plot.

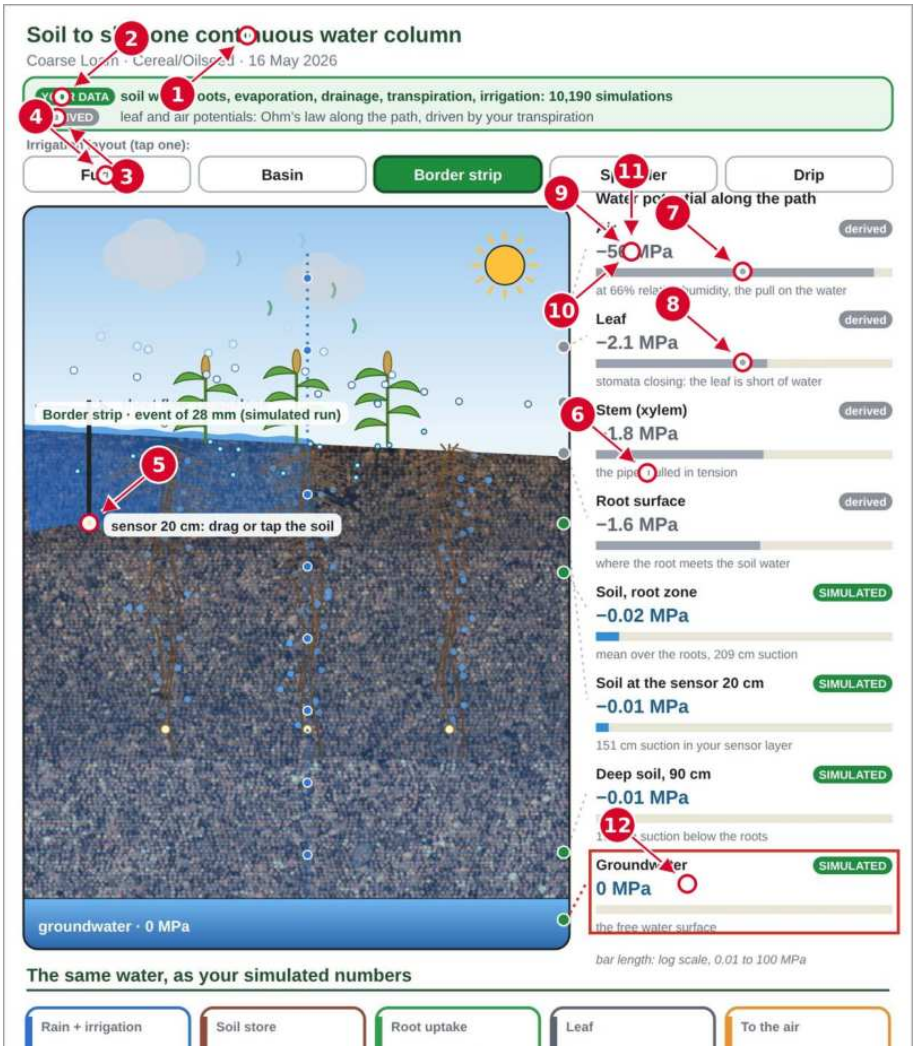



Figure 26.127: Soil, plant and air, part 1. The Border strip layout chosen.

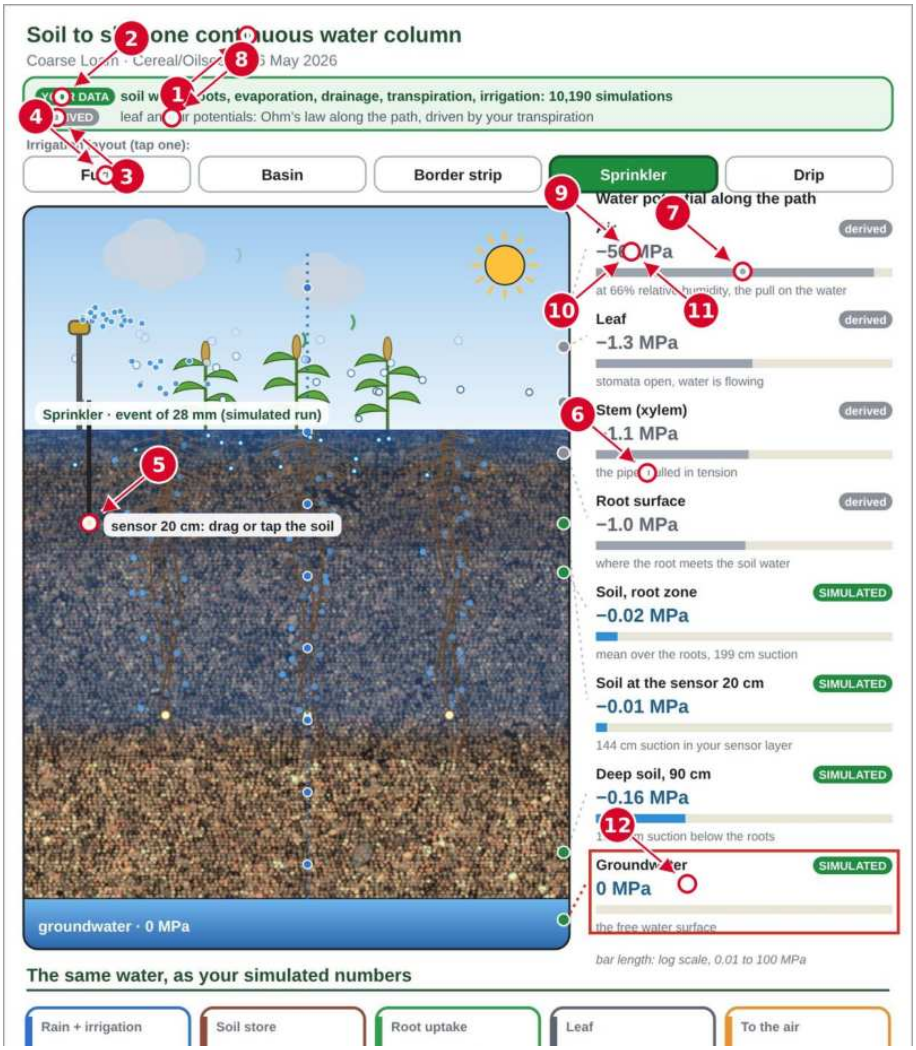



Figure 26.128: Soil, plant and air, part 1. The Sprinkler layout chosen, so the water arrives as spray over the plants.

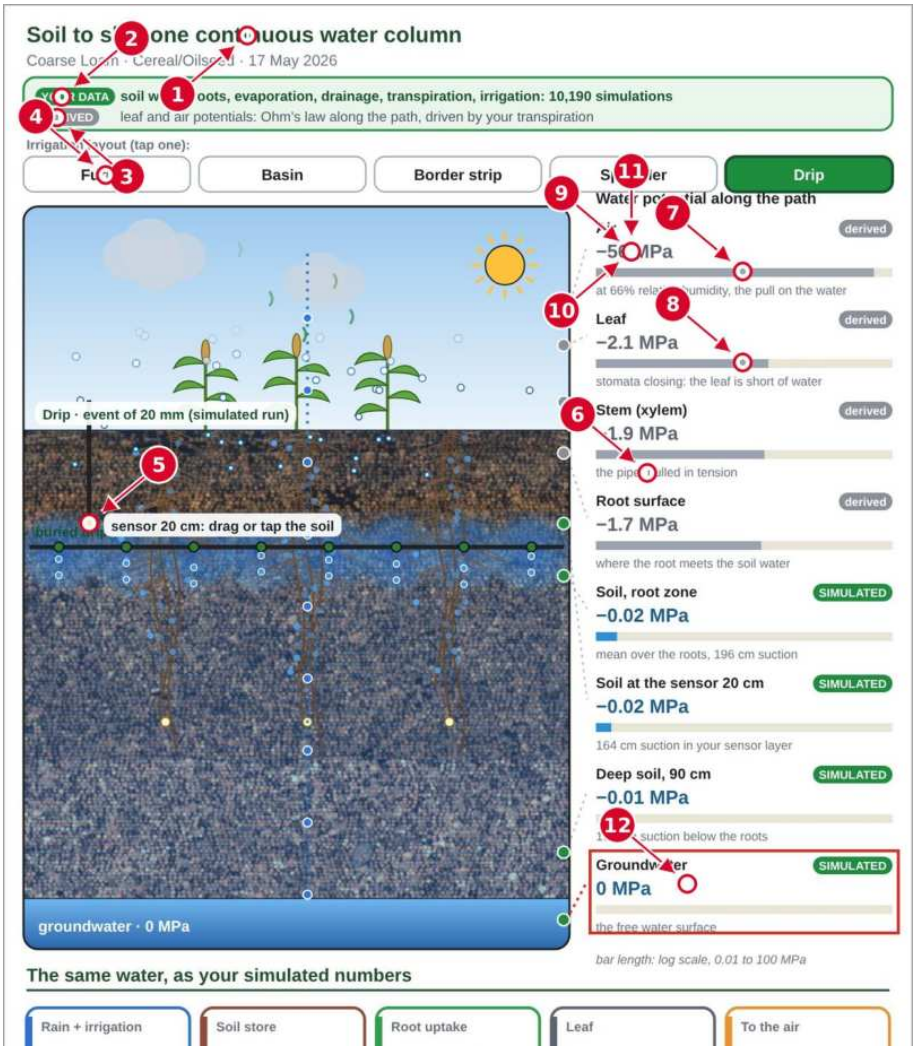



Figure 26.129: Soil, plant and air, part 1. The Drip layout chosen, so the water arrives at the plant row only.

Figure 26.130: Soil, plant and air, part 3. The Colour by Grain size button pressed, so the grains of the section are coloured by their size.

Figure 26.131: Soil, plant and air, part 3. The Colour by Phases button pressed, so the section shows the solid, the air and the water.

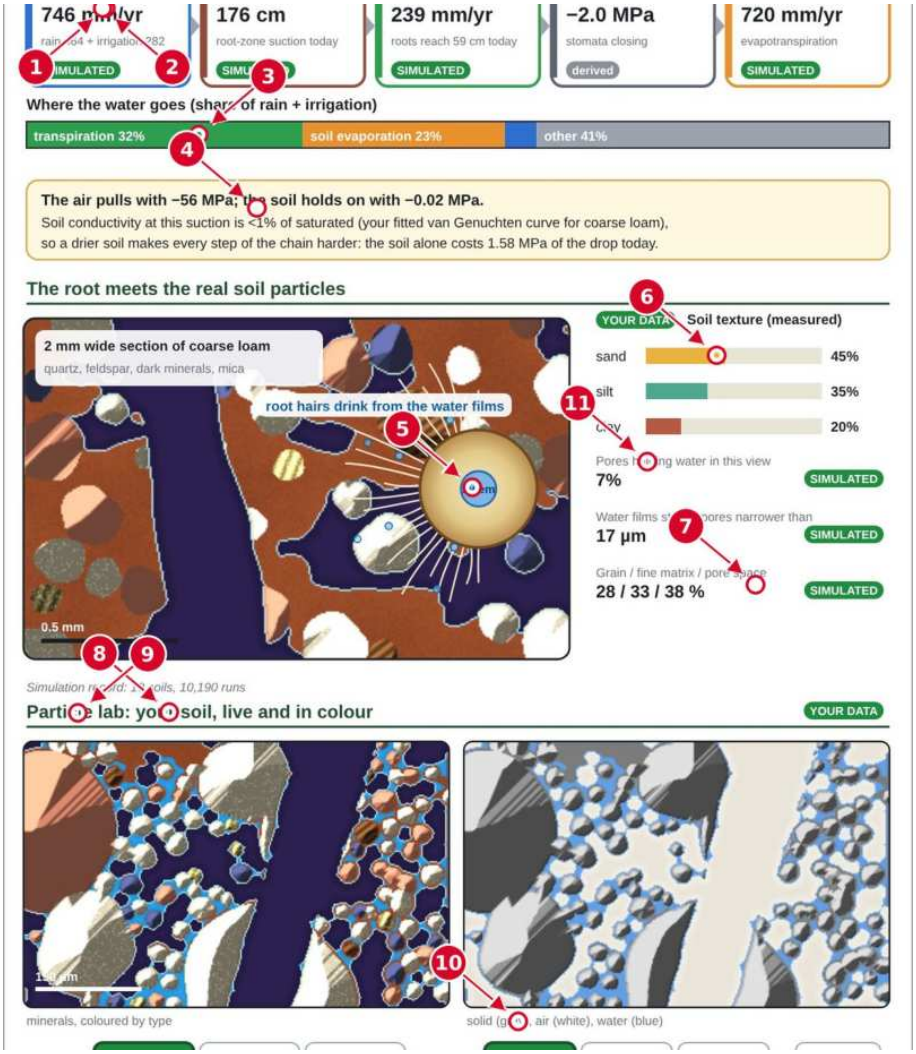



Figure 26.132: Soil, plant and air, part 2. The Zoom 0.6 mm button pressed, so the section shows silt, clay and the water films.

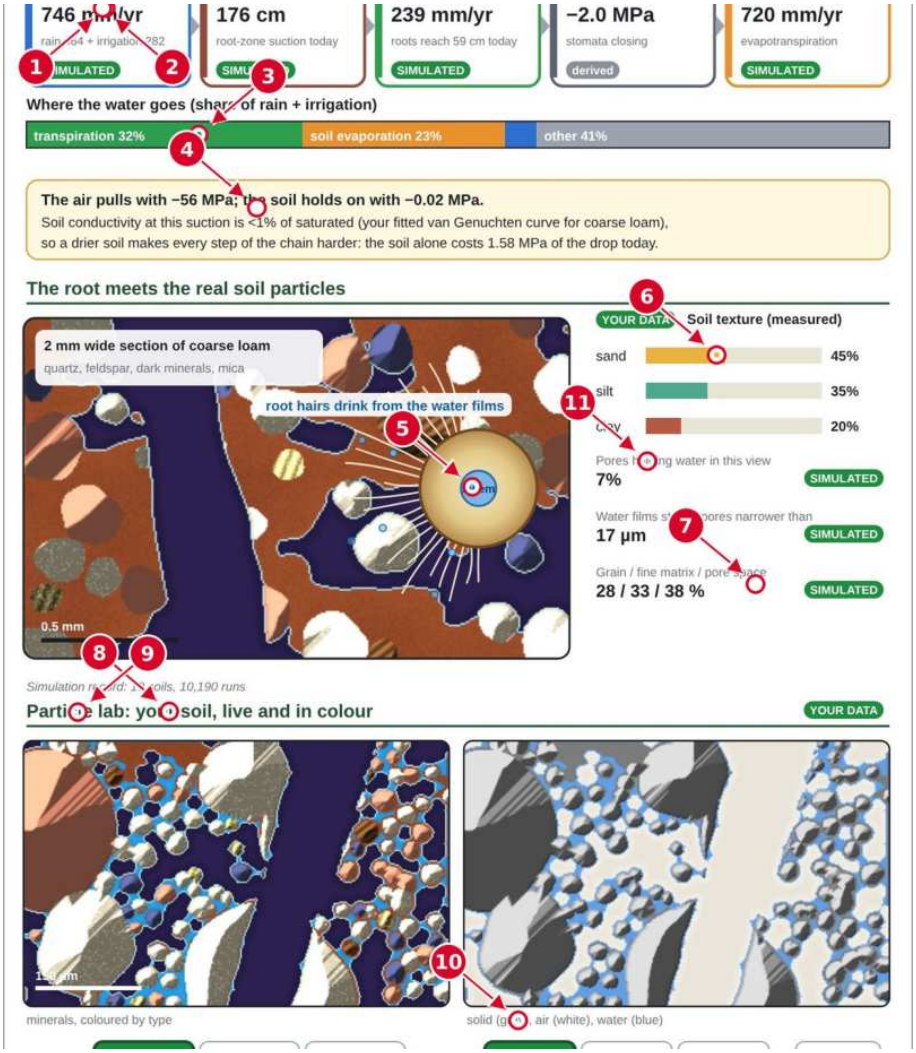



Figure 26.133: Soil, plant and air, part 2. The Zoom 10 mm button pressed, so the section shows aggregates and cracks.

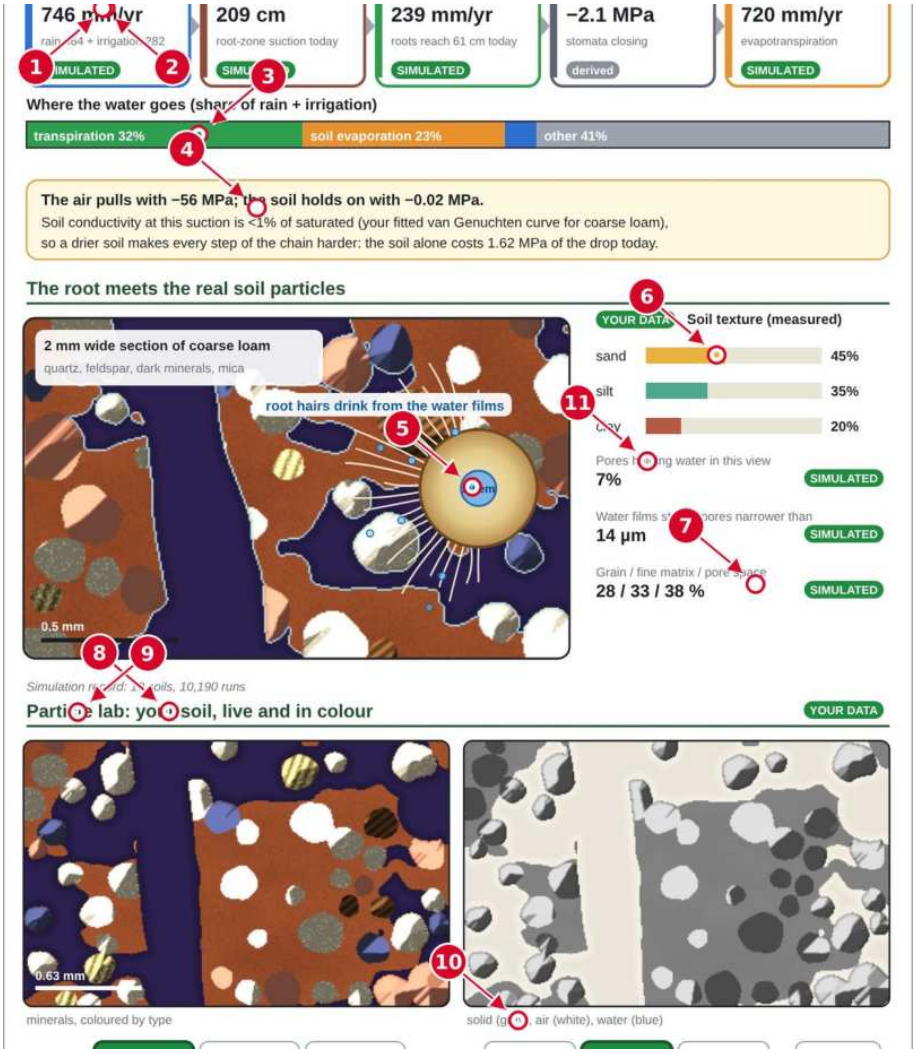



Figure 26.134: Soil, plant and air, part 2. The Look at Topsoil 5 cm button pressed, so the section and the readouts refer to the topsoil.

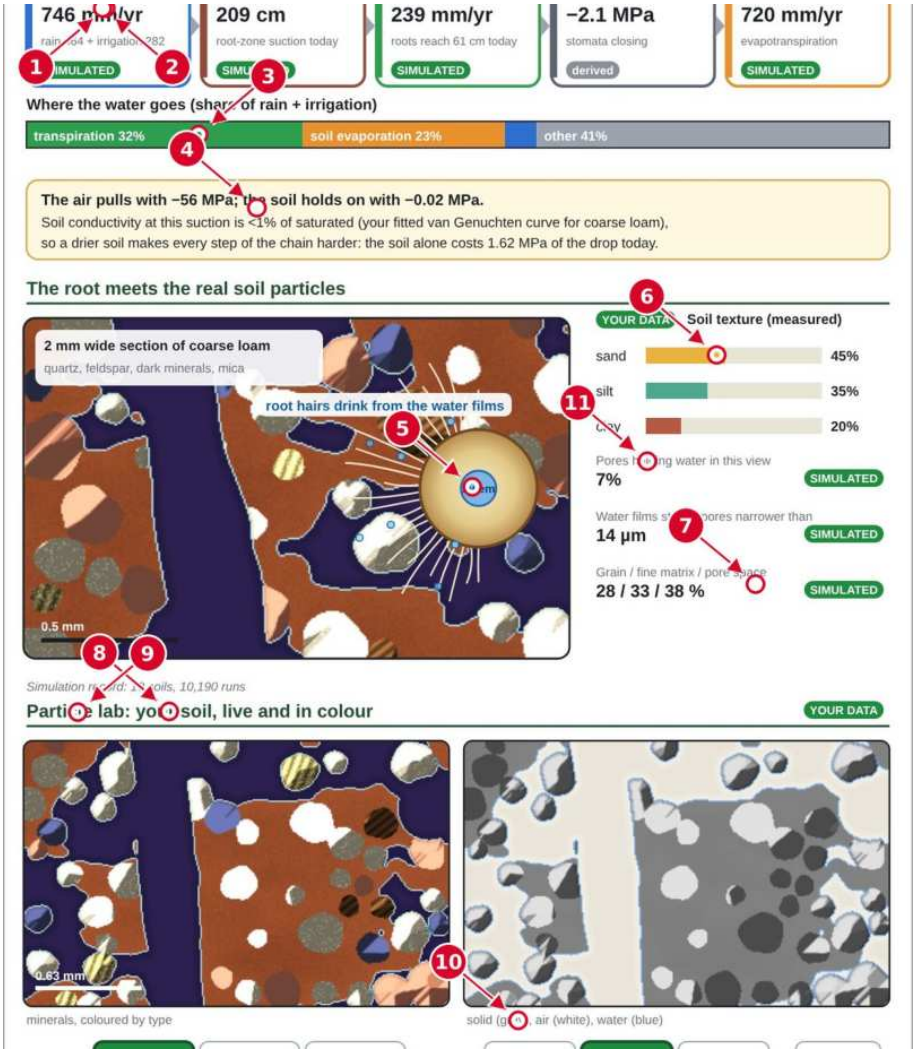



Figure 26.135: Soil, plant and air, part 2. The Look at Sensor 20 cm button pressed.

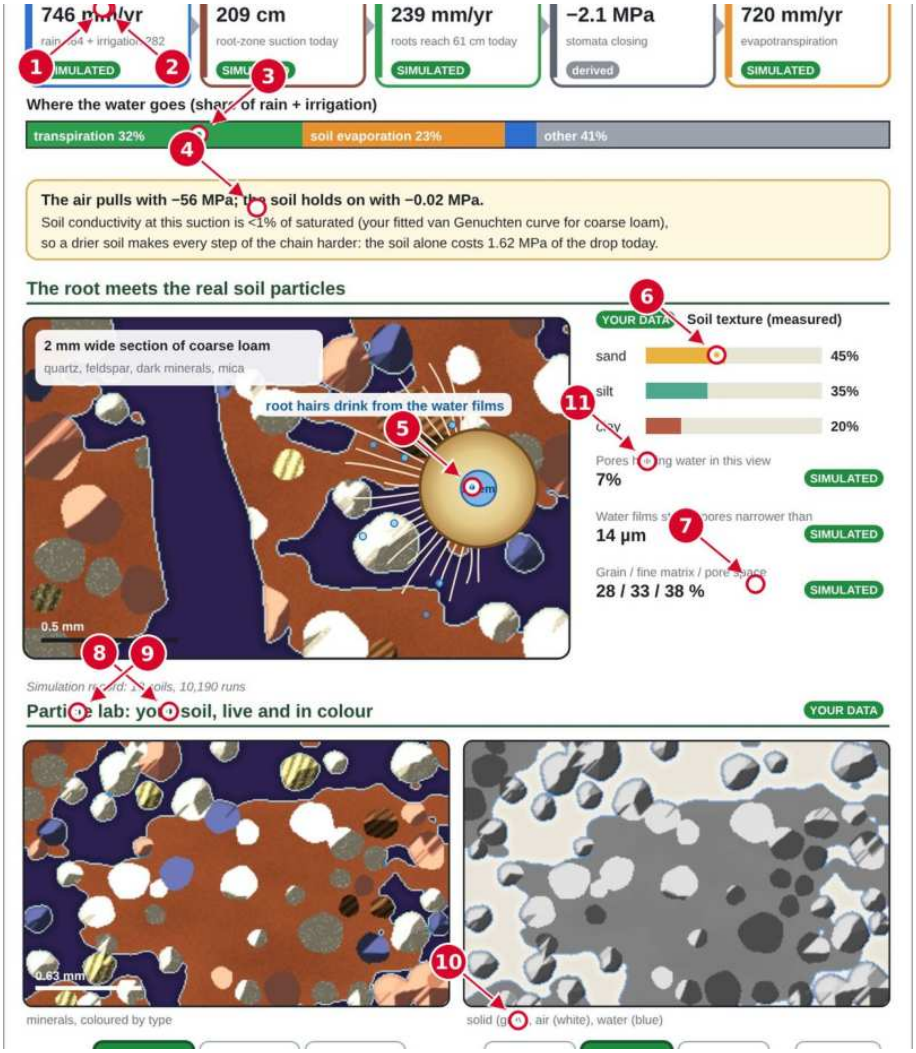



Figure 26.136: Soil, plant and air, part 2. The Look at Deep 90 cm button pressed, so the section and the readouts refer to the deep soil.

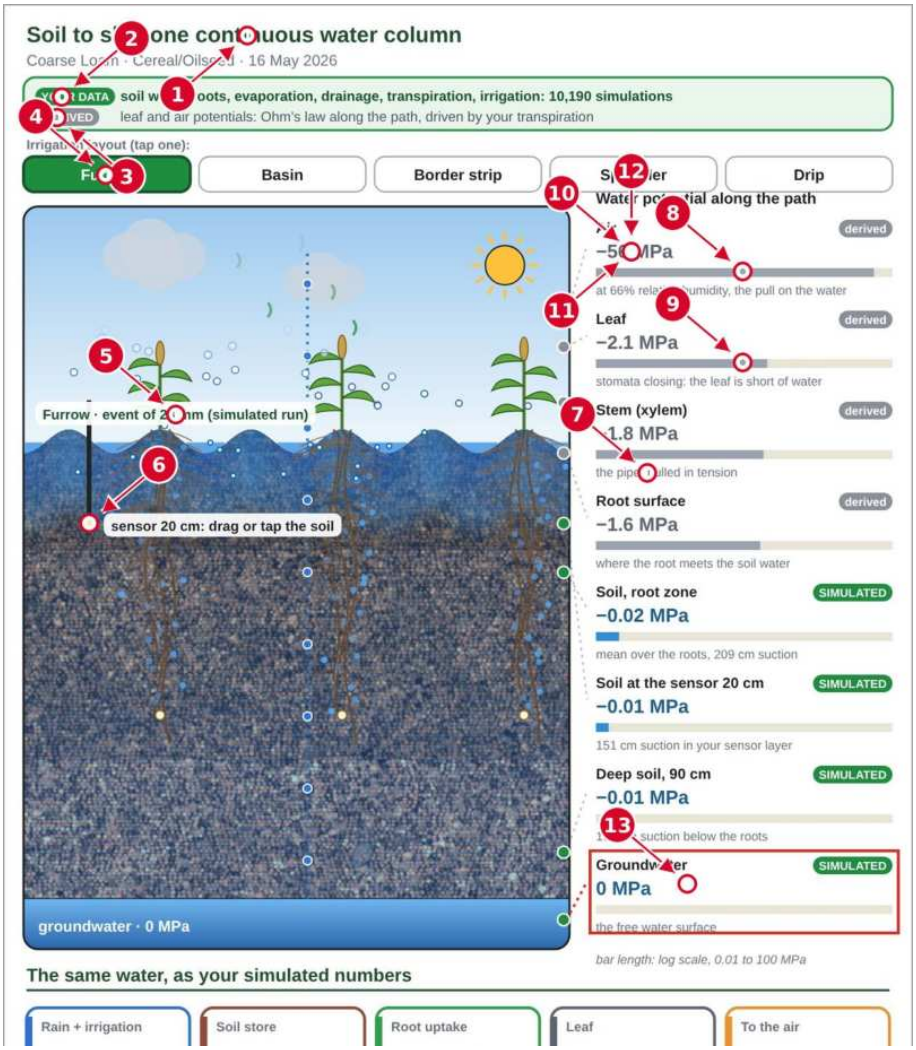



Figure 26.137: Soil, plant and air, part 1. The suction value box dragged to the wet end, so the soil is shown near saturation.

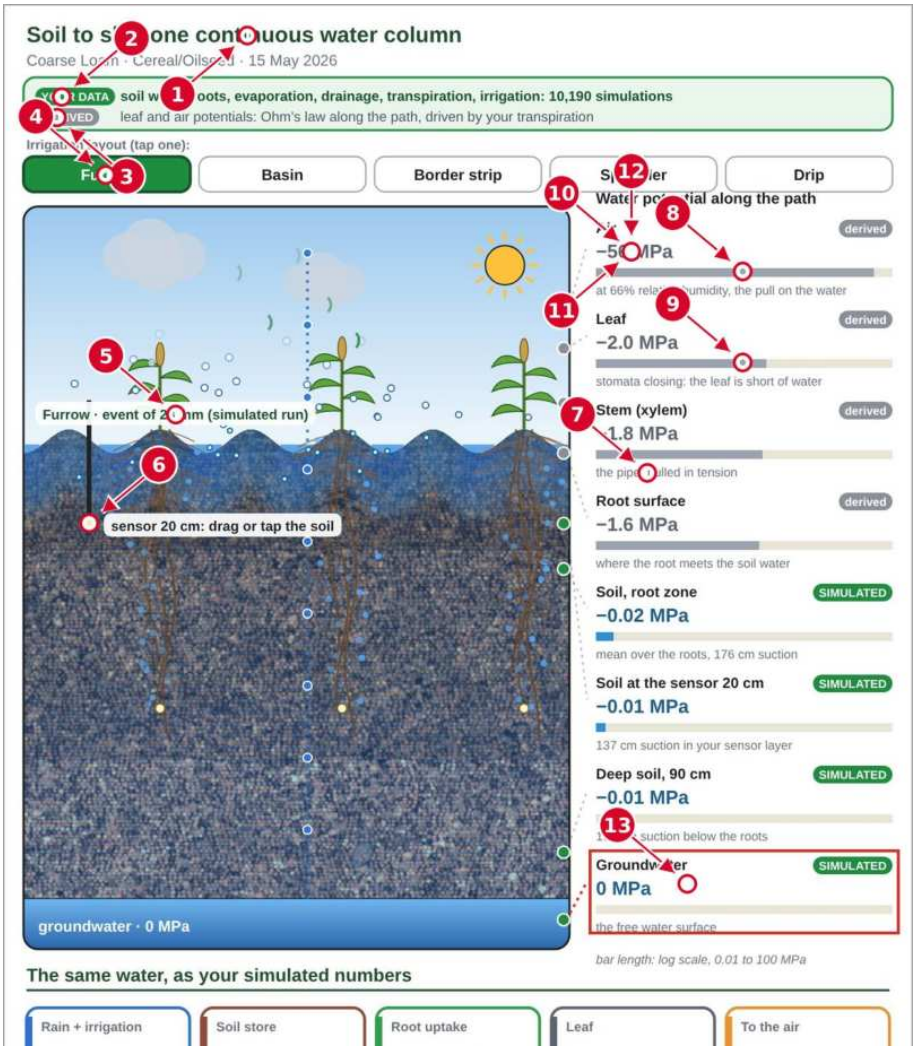



Figure 26.138: Soil, plant and air, part 1. The suction value box dragged to the dry end, so the soil is shown near the wilting point.

# 26.7 Profile evolution

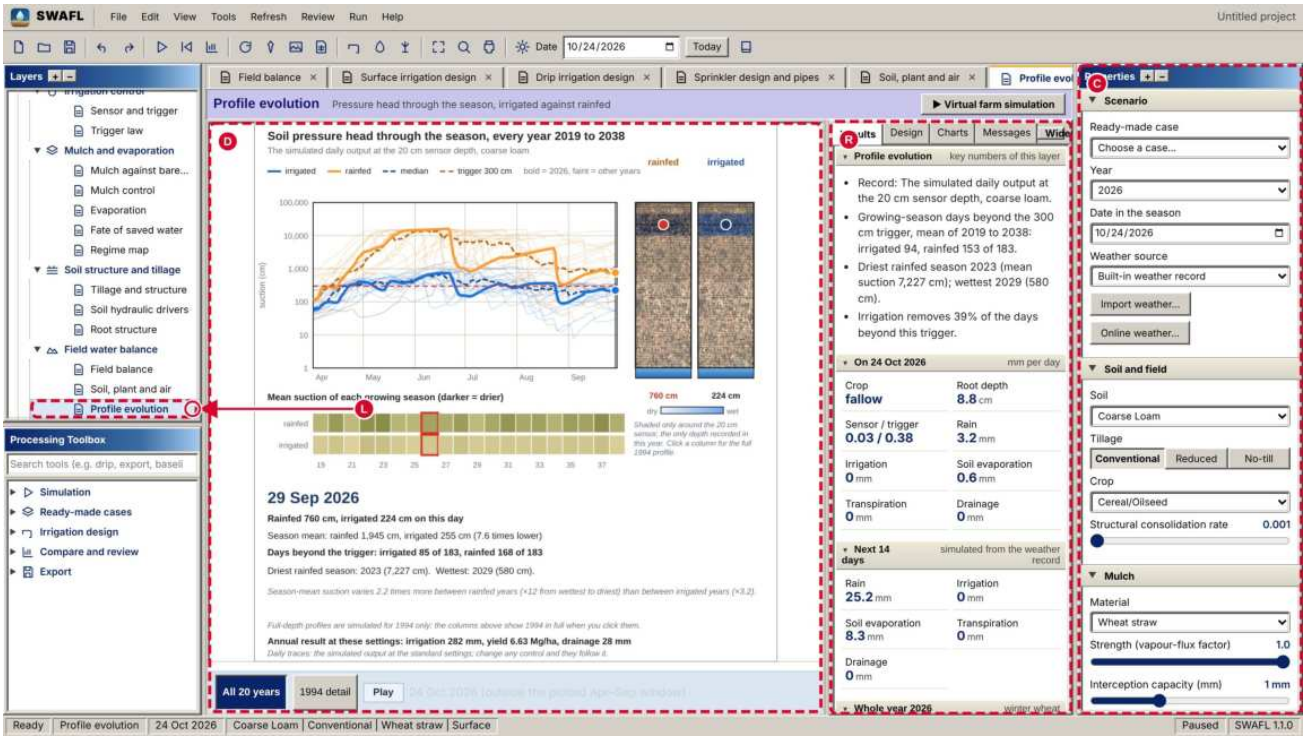



Figure 26.139: Profile evolution: the whole workbench.

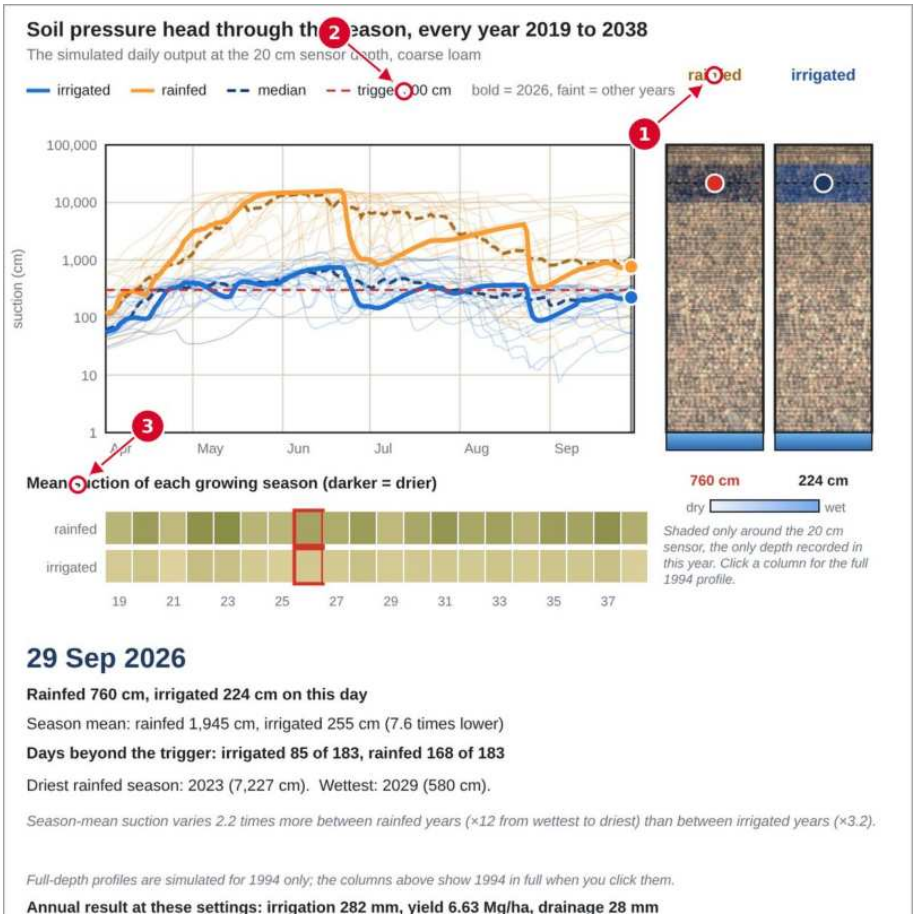



Figure 26.140: Profile evolution: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Daily suction at the 20 cm sensor: stored Daisy series (data-driven, no equation).
- **2** Trigger h_T (dashed red line), set by the value box; the share of the root zone beyond it is Eq. 7.6.
- **3** Heat strip: geometric mean suction of each growing season, text under Eq. 7.6.

**Equations behind this screen.** Eq. 5.15, Eq. 5.17, Eq. 7.6.

*Eq. 5.15. Where explained: chapter "The two computing engines", section "Rainfall generator".*

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0,1), \tag{5.15}$$

*Provenance (Eq. 5.15):* **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

*Eq. 5.17. Where explained: chapter "The two computing engines", section "Infiltration, redistribution and drainage".*

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

*Provenance (Eq. 5.17):* **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

*Eq. 7.6. Where explained: chapter "Field water balance layers", section "How it is generated".*

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

*Provenance (Eq. 7.6):* **[Engineered]**: indicator defined for the workbench.

### 26.7.1 The whole layer, part by part

The picture below is the complete drawing of the layer Profile evolution, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

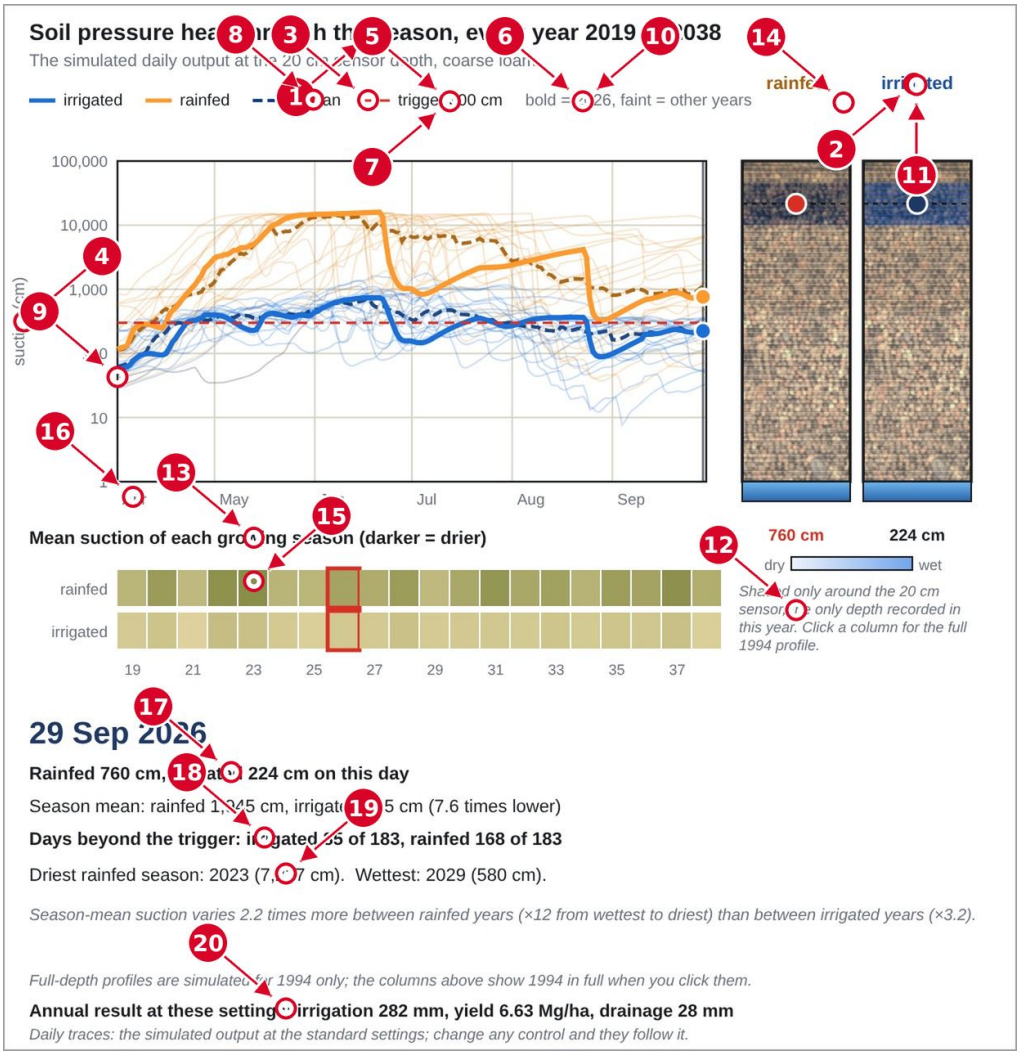



Figure 26.141: Profile evolution, part 1 of 1.

**What the arrows point to.**

- **1** The title, the daily suction at the 20 cm sensor for every year of the record.
- **2** The legend of the traces, irrigated in blue and rainfed in orange.
- **3** The trigger legend, the dashed red line at the suction that starts an irrigation.
- **4** The suction axis in cm of water on a log scale.
- **5** The bold blue trace, the irrigated season of the year shown.
- **6** The bold orange trace, the rainfed season of the same year.

- **7** The trigger line at 300 cm.
- **8** The faint traces, the other years of the record.
- **9** The marker of the day shown, moving along the year as the animation plays.
- **10** The rainfed soil column, shaded by the suction around the sensor.
- **11** The irrigated soil column.
- **12** The sensor ring at 20 cm, the only depth recorded in this year.
- **13** The strip of the mean suction of every growing season, darker for drier.
- **14** The selected year in the strip; clicking a column selects that year.
- **15** The dark cells are the dry rainfed seasons, far darker than the irrigated row.
- **16** The date of the day shown.
- **17** The suction on this day for the rainfed and the irrigated soil.
- **18** The days beyond the trigger in the growing season, irrigated against rainfed.
- **19** The driest and the wettest rainfed season of the record.
- **20** The annual result at the settings in force, the irrigation, the yield and the drainage.

### 26.7.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

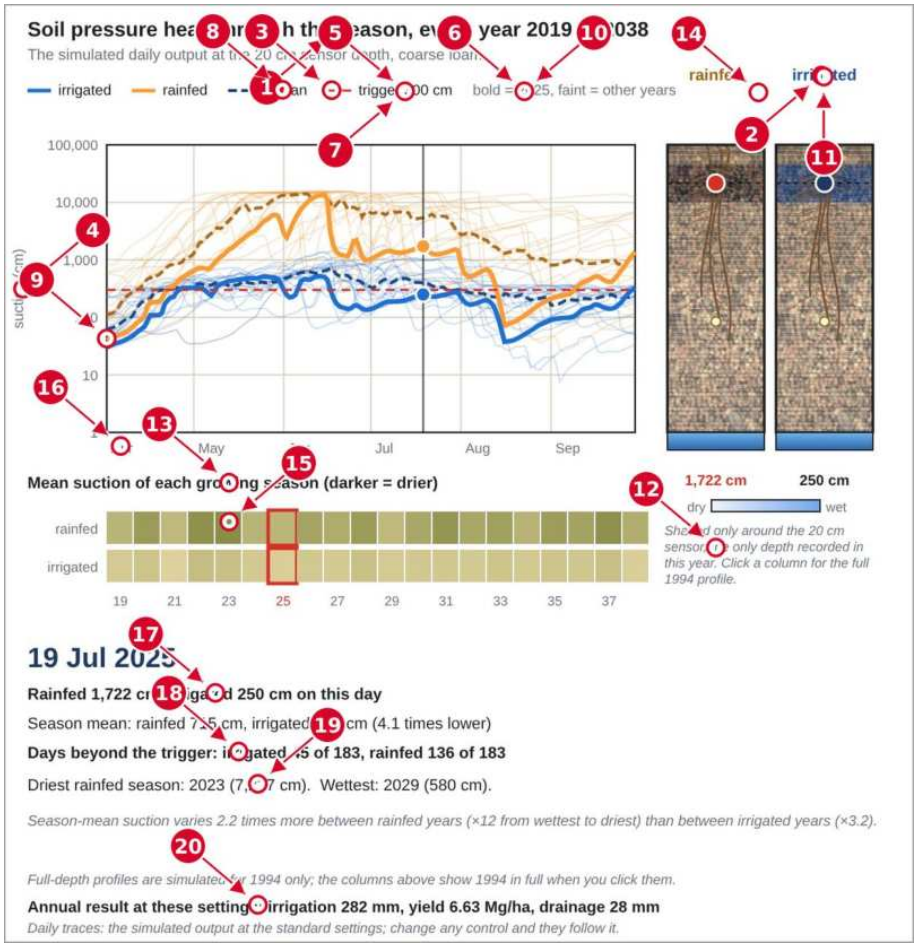



Figure 26.142: Profile evolution, part 1, playback a third of the way (31 Dec 2038 (outside the plotted Apr and Sep window)).

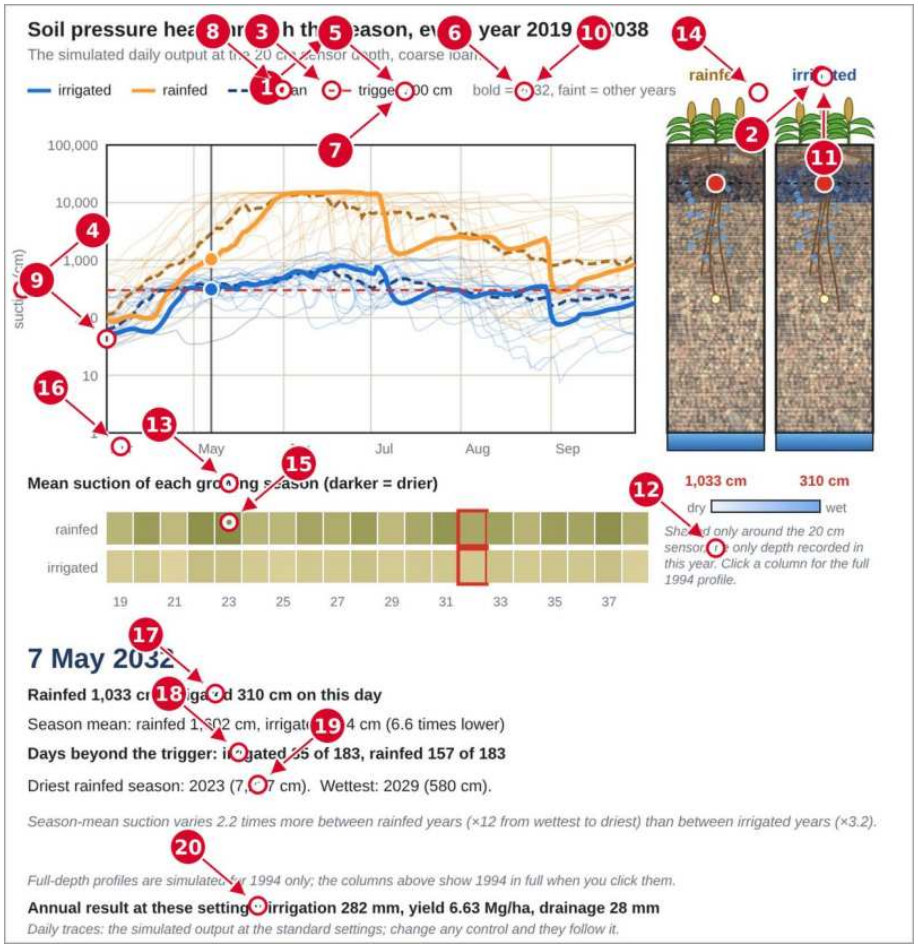



Figure 26.143: Profile evolution, part 1, playback two thirds of the way (31 Dec 2038 (outside the plotted Apr and Sep window)).

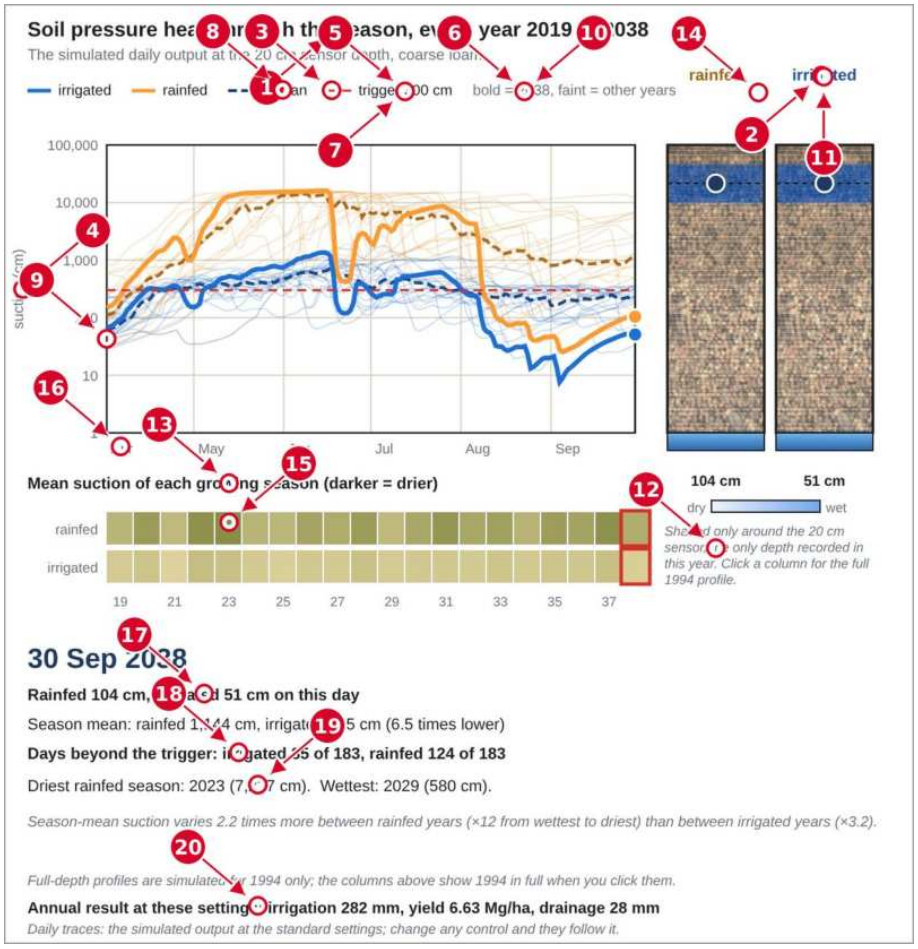



Figure 26.144: Profile evolution, part 1, playback the end (31 Dec 2038 (outside the plotted Apr and Sep window)).

### 26.7.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

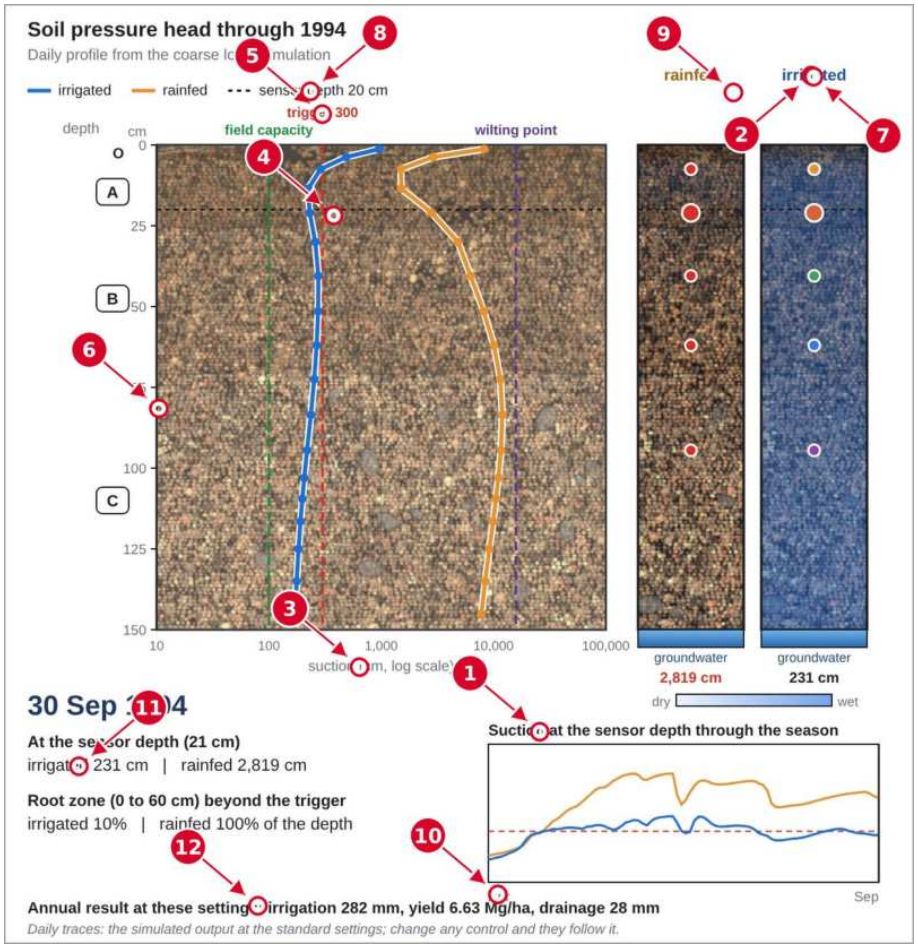



Figure 26.145: Profile evolution, part 1. The 1994 detail button pressed, so the chart shows the daily record of the year 1994 at full depth.

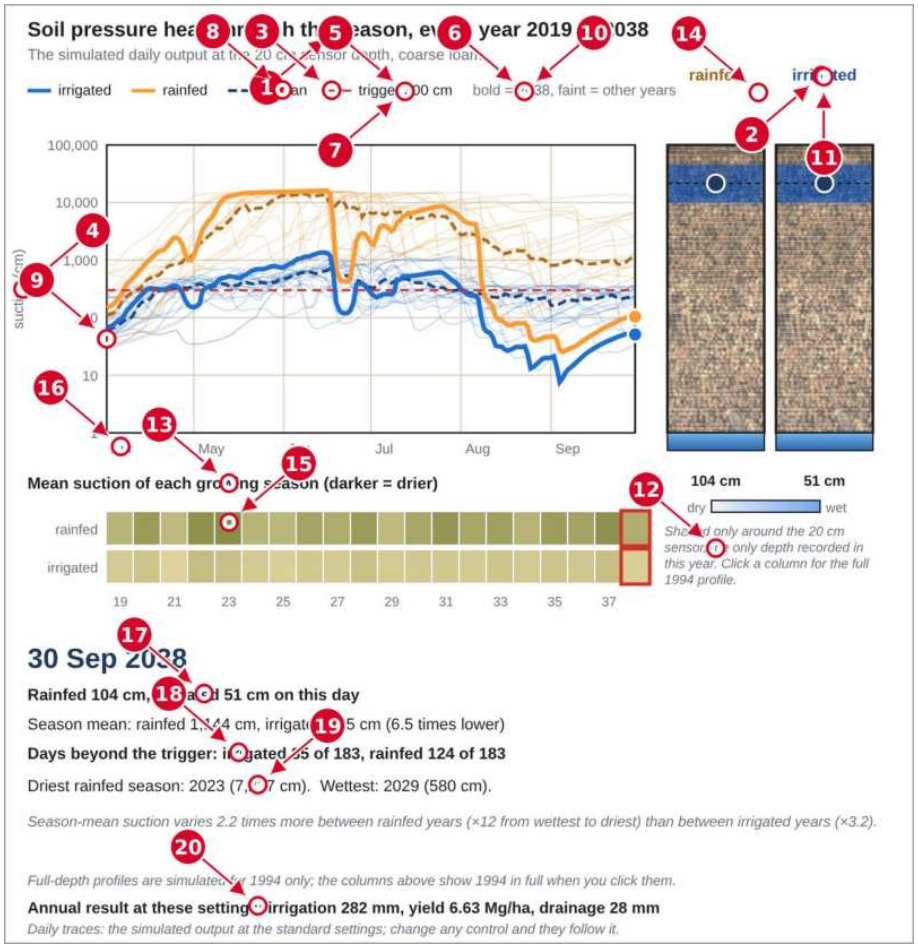



Figure 26.146: Profile evolution, part 1. A column of the season strip clicked, so the chart and the columns show the selected year.

# 26.8 Sensor and trigger

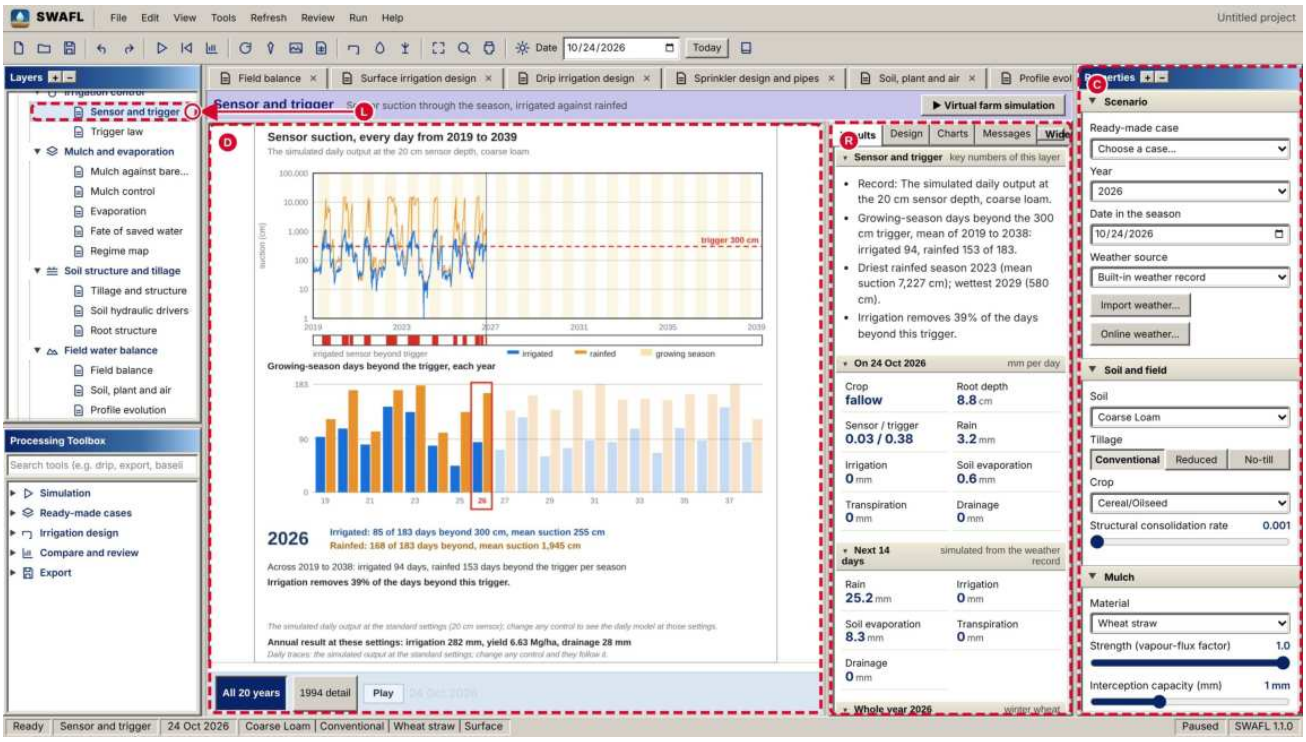



Figure 26.147: Sensor and trigger: the whole workbench.

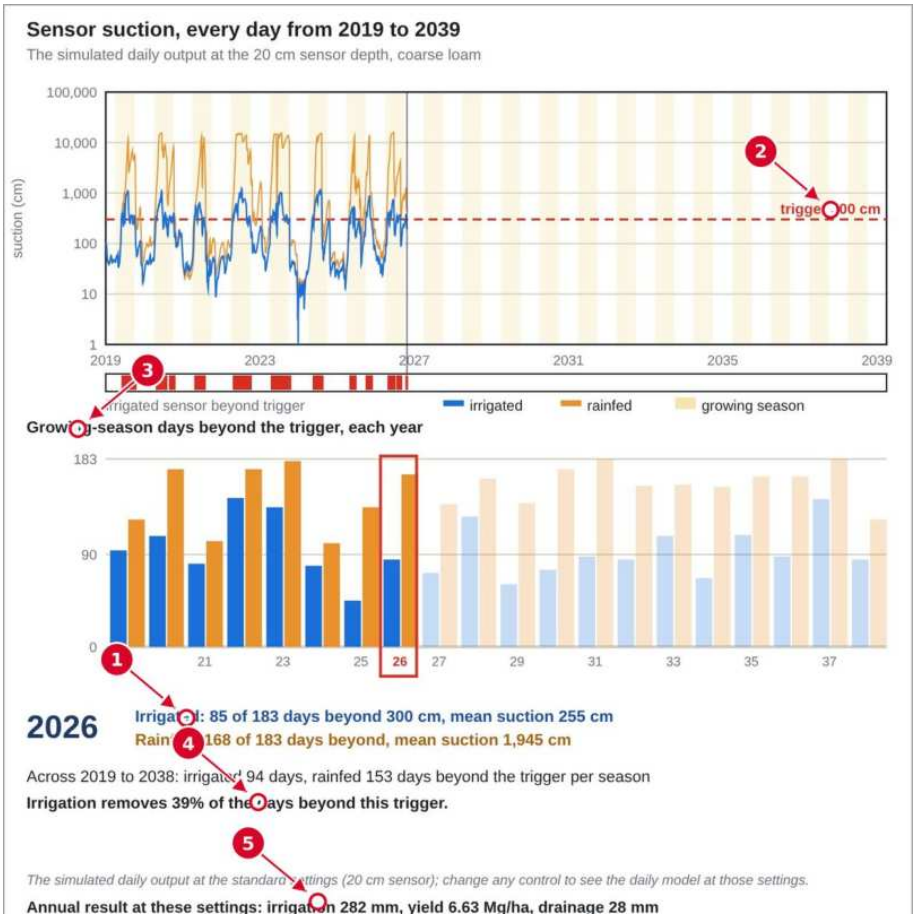



Figure 26.148: Sensor and trigger: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Daily suction at 20 cm, irrigated (blue) and rainfed (orange): stored Daisy series (data-driven).
- **2** Trigger h_T (red dashed line), set in Properties; days beyond it are counted by Eq. 6.1.
- **3** Growing-season days beyond the trigger for each year, N_y(h_T): Eq. 6.1.
- **4** “Irrigation removes 39 % of the days”: X = 100(1 minus N_irr / N_rainfed), Eq. 6.2.
- **5** Annual result at these settings (irrigation, yield, drainage): Engine 1, Eqs. 5.4 to 5.12.

**Equations behind this screen.** Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 6.1, Eq. 6.2.

*Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.5. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.6. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.7. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.8. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.9. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.10. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".*

$$Q \leftarrow Q\ \text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ \ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N). \tag{5.11}$$

*Provenance (Eq. 5.11):* **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

*Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".*

$$ET \in [\,P + I - D - 30,\ P + I - D + 15\,] \text{ (if not, } ET \text{ is clipped and } E, T \text{ are scaled by the same factor)},$$
$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

*Provenance (Eq. 5.12):* **[Engineered]**: closure constraint and bounds chosen for the workbench.

*Eq. 6.1. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".*

$$N_y(h_T) = \#\big\{\, d \in \text{growing season of year } y : h_d > h_T \,\big\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

*Provenance (Eq. 6.1):* **[Data-driven]**: counts over the simulated daily series.

*Eq. 6.2. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".*

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_{y} N_y. \tag{6.2}$$

*Provenance (Eq. 6.2):* **[Data-driven]**: counts over the simulated daily series.

### 26.8.1 The whole layer, part by part

The picture below is the complete drawing of the layer Sensor and trigger, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

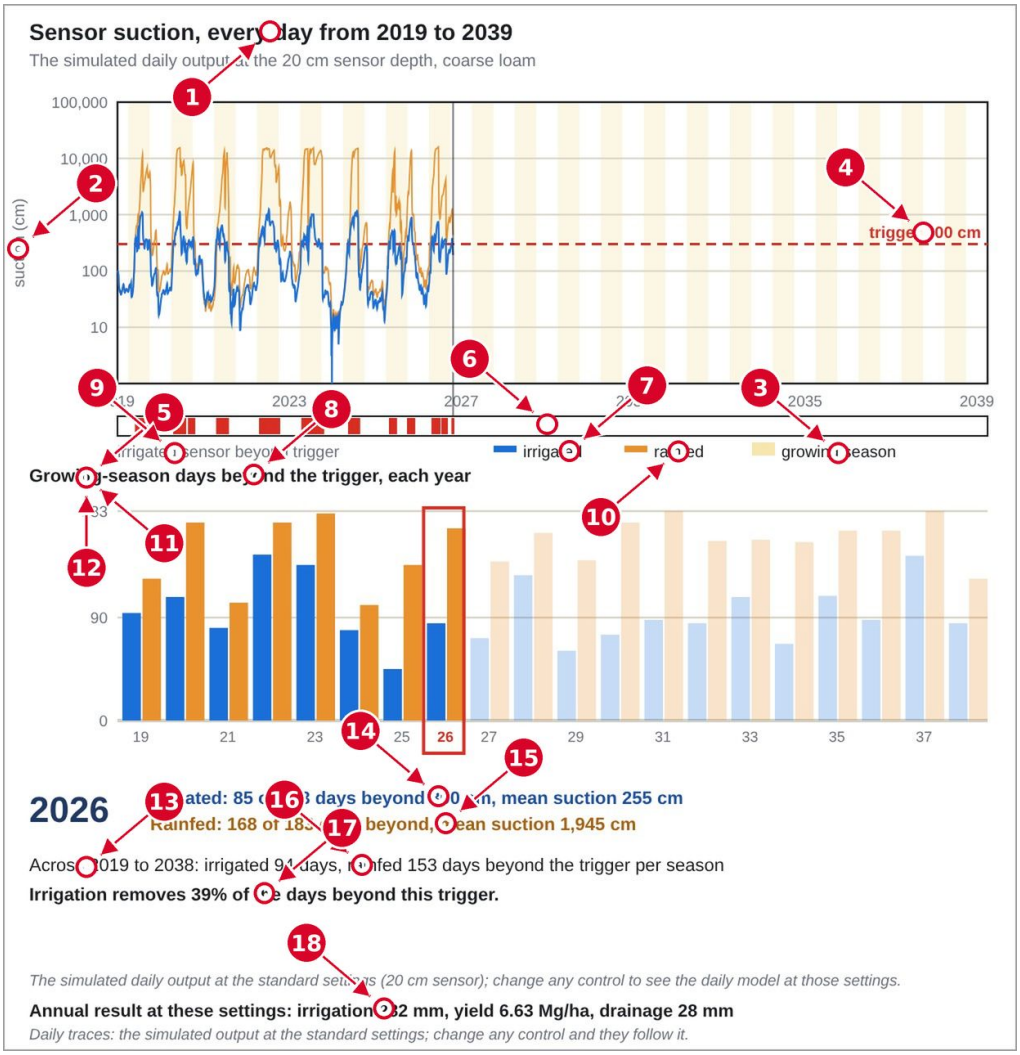



Figure 26.149: Sensor and trigger, part 1 of 1.

**What the arrows point to.**

- **1** The title, the daily sensor suction for every day of the twenty years.
- **2** The suction axis in cm on a log scale.
- **3** The shaded bands are the growing seasons.
- **4** The trigger line, the suction at which the sensor calls for irrigation.
- **5** The time axis in years.
- **6** The strip that marks the days on which the irrigated sensor is beyond the trigger.
- **7** The legend, irrigated in blue, rainfed in orange and the growing season in tan.
- **8** The chart of the growing-season days beyond the trigger, one pair of bars per year.
- **9** The irrigated bar of the selected year.
- **10** The rainfed bar of the selected year.
- **11** The faint bars are the other years.
- **12** The scale of the days, from 0 to the 183 days of the season.
- **13** The year selected.
- **14** The irrigated days beyond the trigger and the mean suction of the season.
- **15** The rainfed days beyond the trigger and the mean suction of the season.
- **16** The mean over the whole record, irrigated against rainfed.
- **17** The share of the days beyond the trigger that irrigation removes.
- **18** The annual result at the settings in force.

### 26.8.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

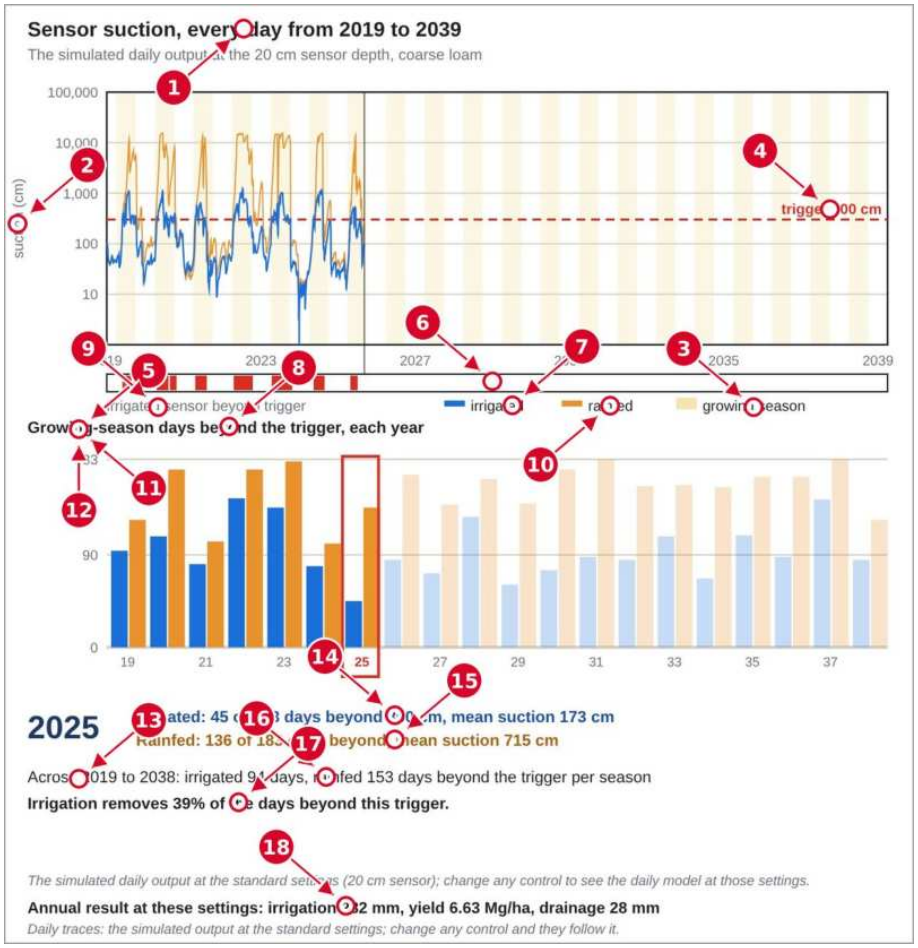



Figure 26.150: Sensor and trigger, part 1, playback a third of the way (7 Sep 2025).

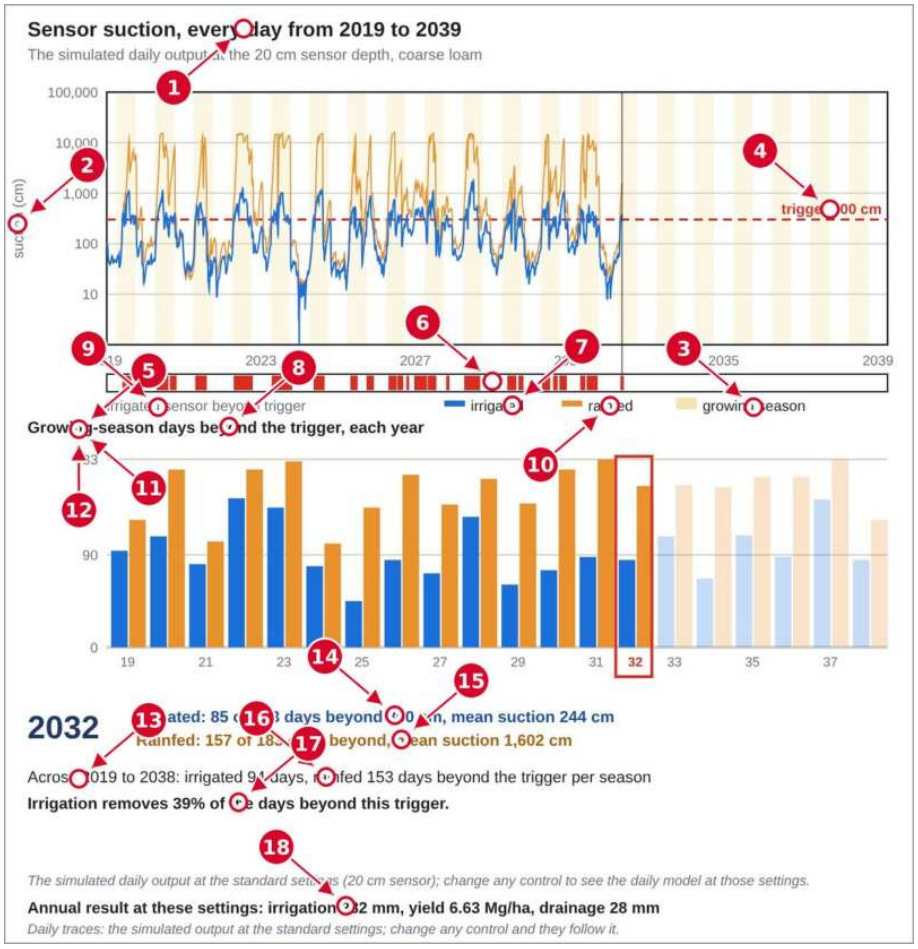



Figure 26.151: Sensor and trigger, part 1, playback two thirds of the way (13 May 2032).

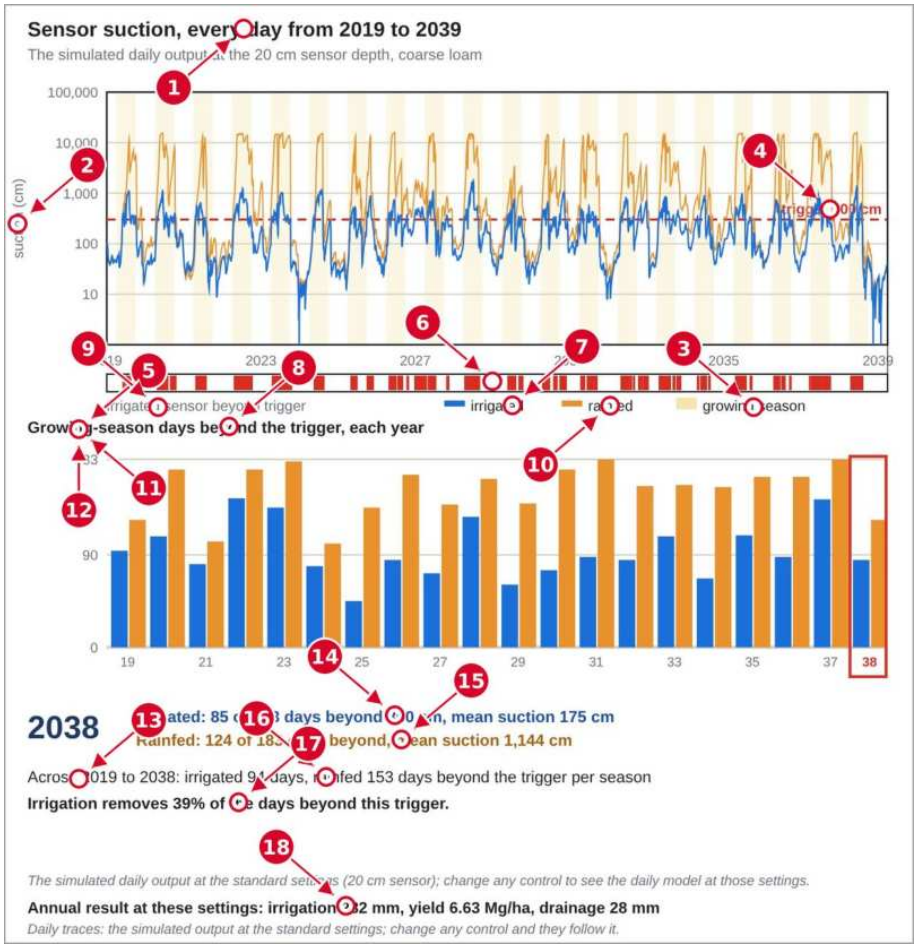



Figure 26.152: Sensor and trigger, part 1, playback the end (1 Apr 2039).

### 26.8.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

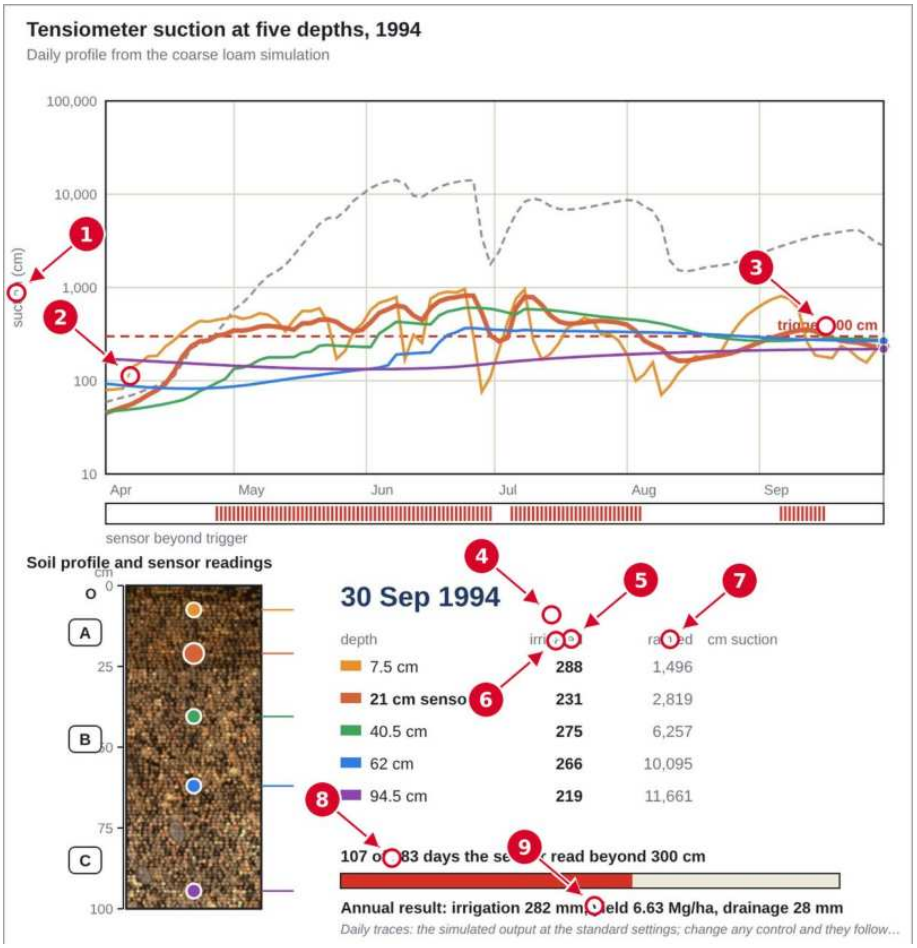



Figure 26.153: Sensor and trigger, part 1. The 1994 detail button pressed, so the chart shows the daily sensor record of the year 1994.

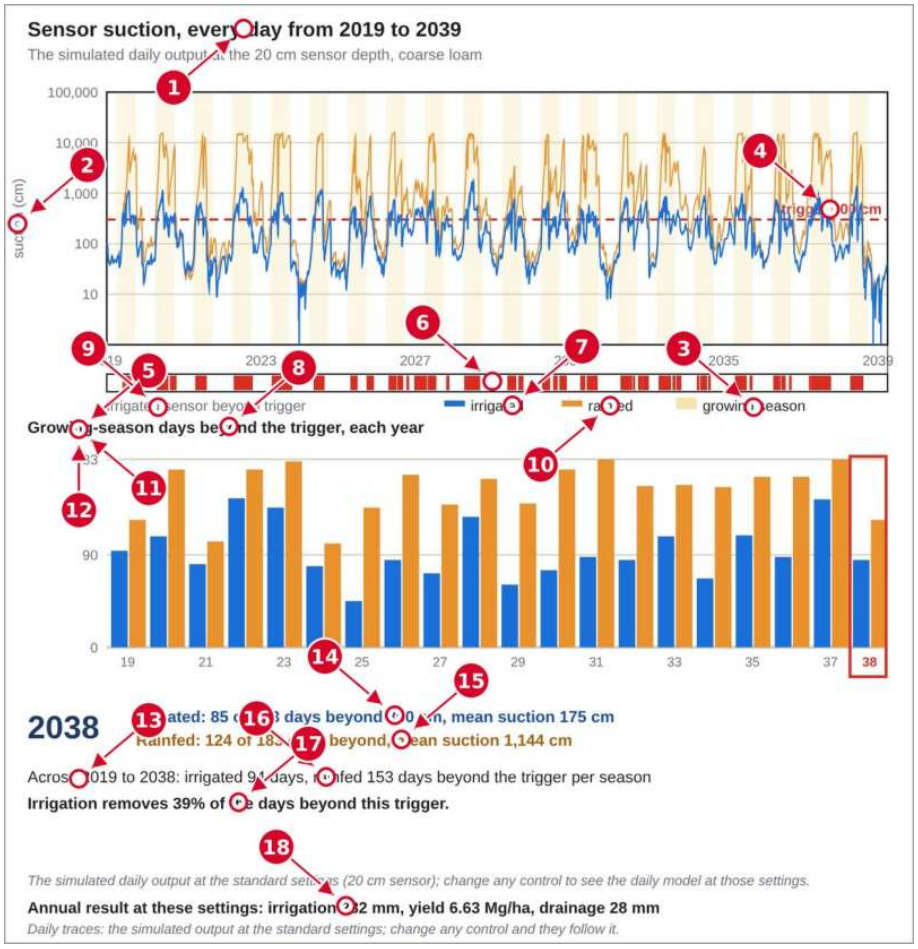



Figure 26.154: Sensor and trigger, part 1. A bar of the lower chart clicked, so the readout below shows the selected year.

# 26.9 Trigger law

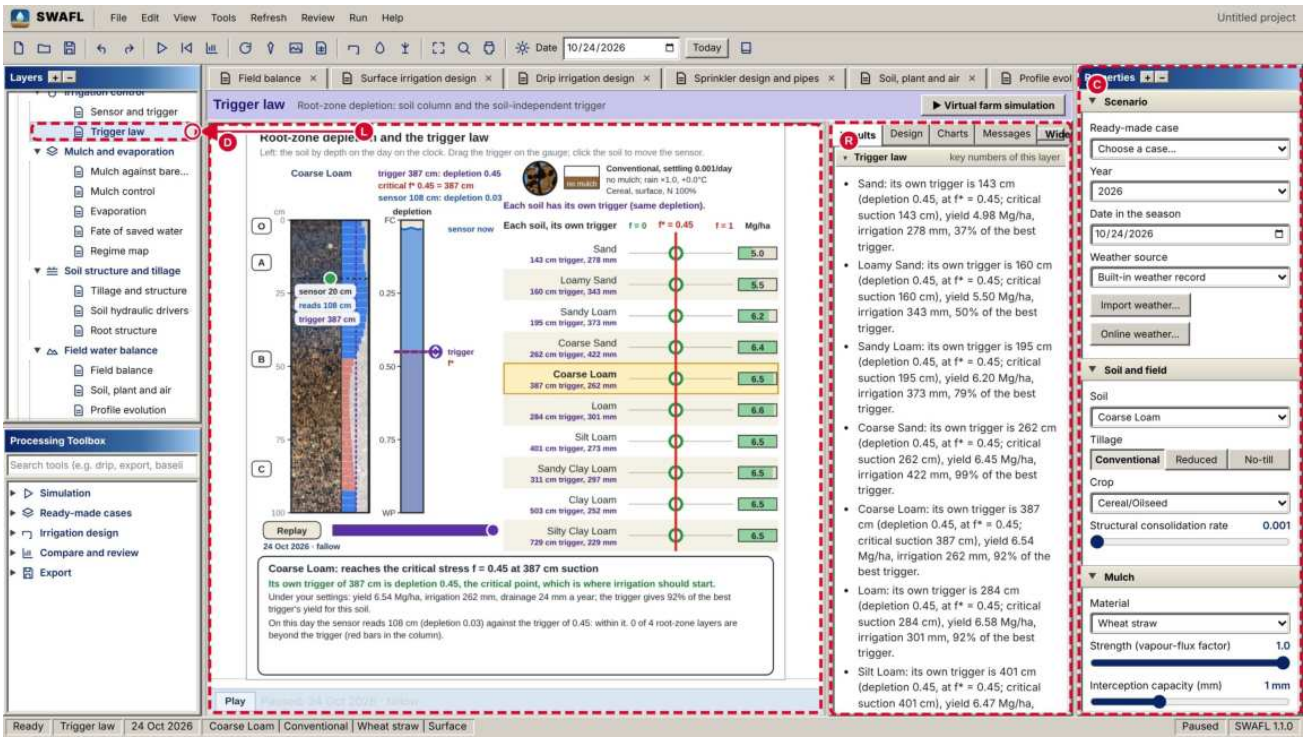


Figure 26.155: Trigger law: the whole workbench.

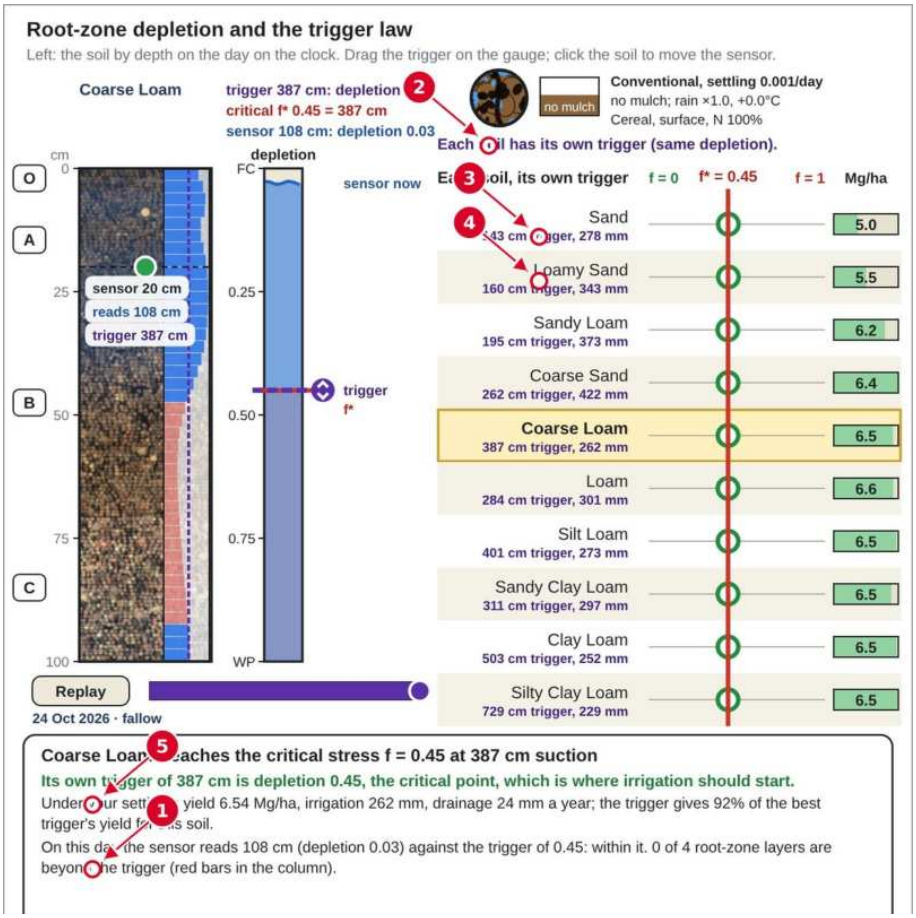



Figure 26.156: Trigger law: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Soil column: each layer coloured by its depletion f, Eq. 5.3 (red = beyond the trigger).
- **2** Critical depletion f* = 0.45, the same for every soil (Eq. 5.3).
- **3** Soil-specific trigger suction h(f*), the inverse of the van Genuchten curve: Eq. 6.3.
- **4** Net irrigation per event Z_n = f AWC R_max: Eq. 6.4.
- **5** Relative yield at the trigger, Y_rel (Eq. 6.5) and the trigger suction (Eq. 6.3) for the selected soil.

**Equations behind this screen.** Eq. 5.3, Eq. 5.27, Eq. 6.3, Eq. 6.4, Eq. 6.5.

*Eq. 5.3. Where explained: chapter "The two computing engines", section "Plant-available water and depletion".*

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

*Provenance (Eq. 5.3):* **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

*Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".*

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \text{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an

engineered clip.

*Eq. 6.3. Where explained: chapter “Irrigation control layers”, section “Governing relations”.*

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

*Provenance (Eq. 6.3):* **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

*Eq. 6.4. Where explained: chapter “Irrigation control layers”, section “Governing relations”.*

$$Z_n = f \cdot \mathrm{AWC} \cdot R_{\max} \quad [\mathrm{mm}] \tag{6.4}$$

*Provenance (Eq. 6.4):* **[Literature]**: same relation as Eq. (5.27) without the clip.

*Eq. 6.5. Where explained: chapter “Irrigation control layers”, section “Governing relations”.*

$$Y_{\mathrm{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

*Provenance (Eq. 6.5):* **[Data-driven]**: ratio of Daisy run-set E2 yields.

### 26.9.1 The whole layer, part by part

The picture below is the complete drawing of the layer Trigger law, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

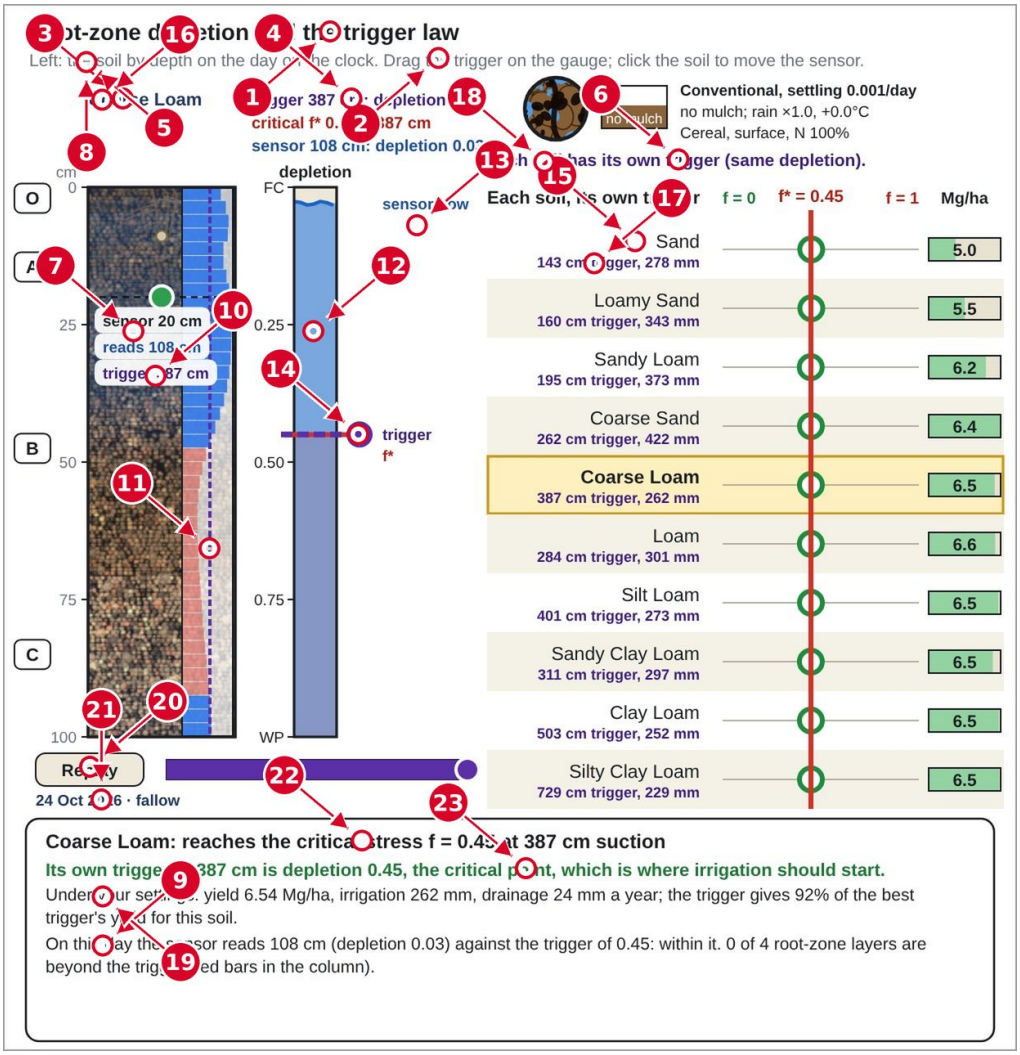



Figure 26.157: Trigger law, part 1 of 1.

**What the arrows point to.**

- **1** The title, the depletion of the root zone and the trigger law.
- **2** The instruction line, drag the trigger on the gauge and click the soil to move the sensor.
- **3** The soil whose column and gauge are shown.
- **4** The readout of the trigger, the critical depletion f* and the sensor depletion on the day.
- **5** The scenario in force, the tillage, mulch, rain scaling, warming, crop and nitrogen.
- **6** The statement that each soil has its own trigger at the same depletion.
- **7** The soil column by depth with the layers of the root zone.
- **8** The sensor tag, its depth, its reading and the trigger in cm of suction.
- **9** The blue bars are the layers inside the trigger on the day shown.
- **10** The red bars are the layers beyond the trigger.
- **11** The dashed line is the trigger carried down the column.
- **12** The depletion gauge from field capacity (FC) to wilting point (WP).
- **13** The sensor level on the gauge today.
- **14** The handle of the trigger, which is dragged to another depletion.
- **15** The soils ranked in the list, each with its own trigger in cm and its seasonal irrigation.
- **16** The soil selected, highlighted in the list.
- **17** The critical depletion f* as the red line across all ten soils.
- **18** The ring on each soil gives where its own trigger sits between f = 0 and f = 1.

- **19** The simulated yield of each soil at its own trigger, in Mg per ha.
- **20** The Play and Stop button and the time value box: the season plays by itself until Stop is pressed.
- **21** The date shown on the clock.
- **22** The conclusion, the stress at which the selected soil needs water.
- **23** The sentence that links the soil's own trigger to the critical point.

### 26.9.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

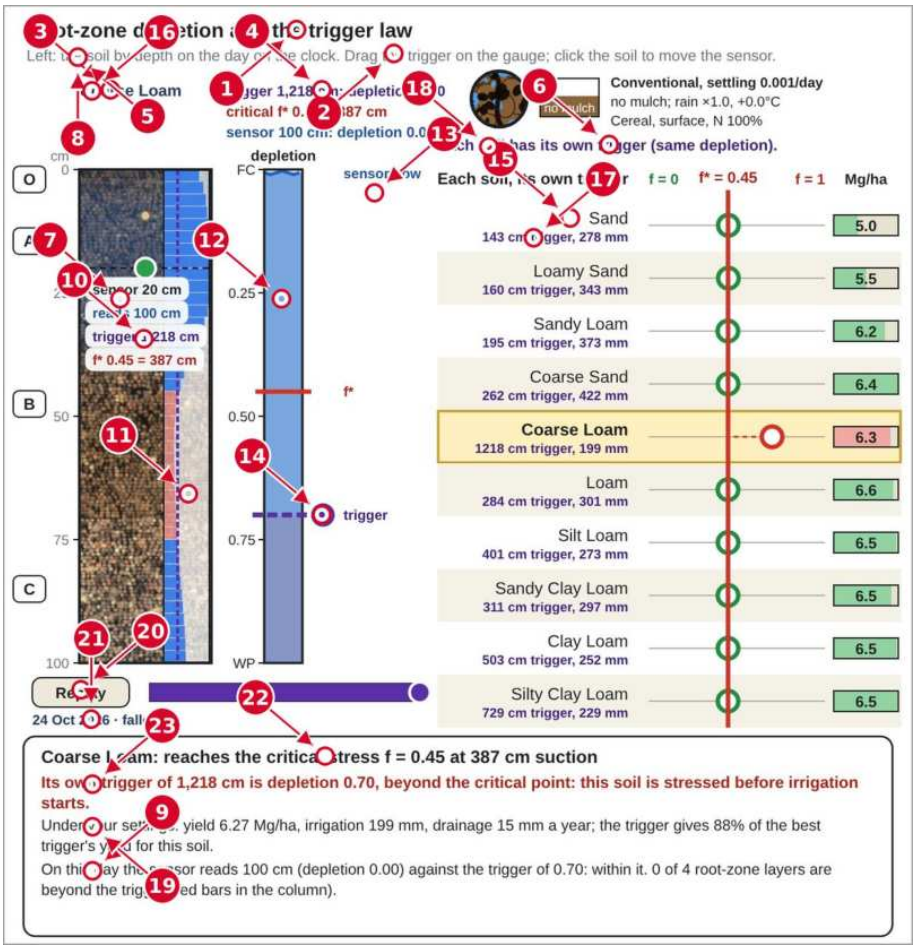



Figure 26.158: Trigger law, part 1. The trigger handle dragged to a deeper depletion on the gauge, so the trigger moves and the number of layers beyond it changes.

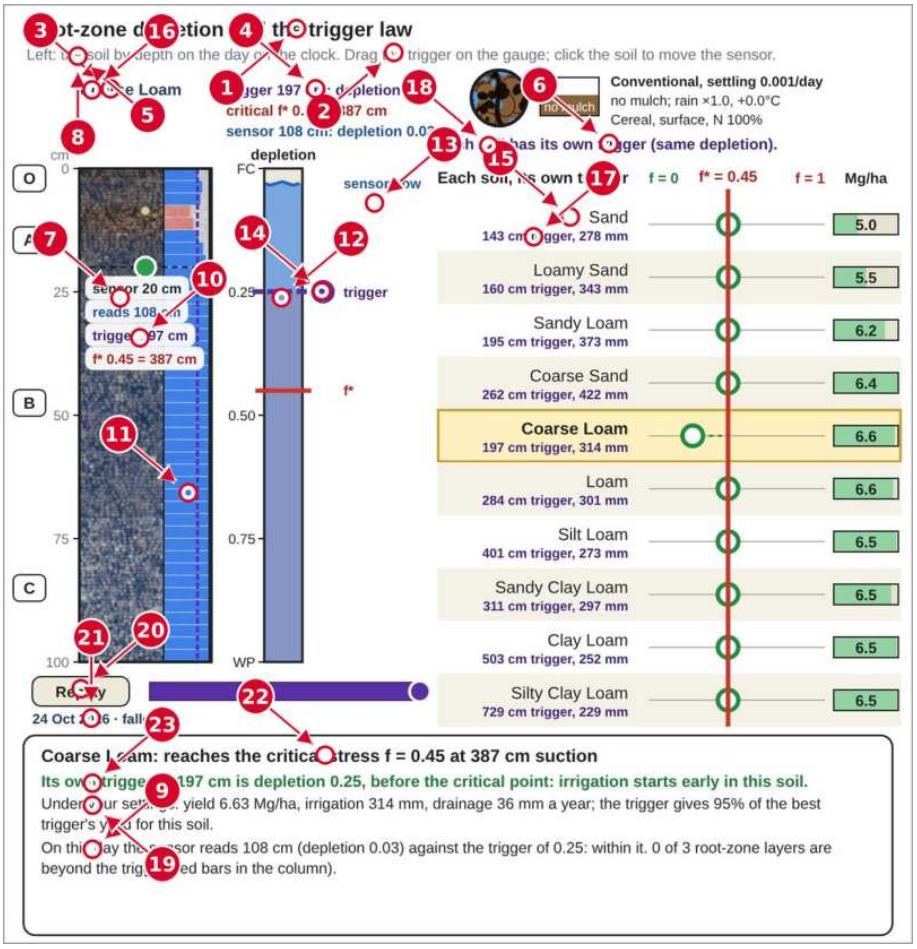



Figure 26.159: Trigger law, part 1. The trigger handle dragged to a shallow depletion on the gauge.

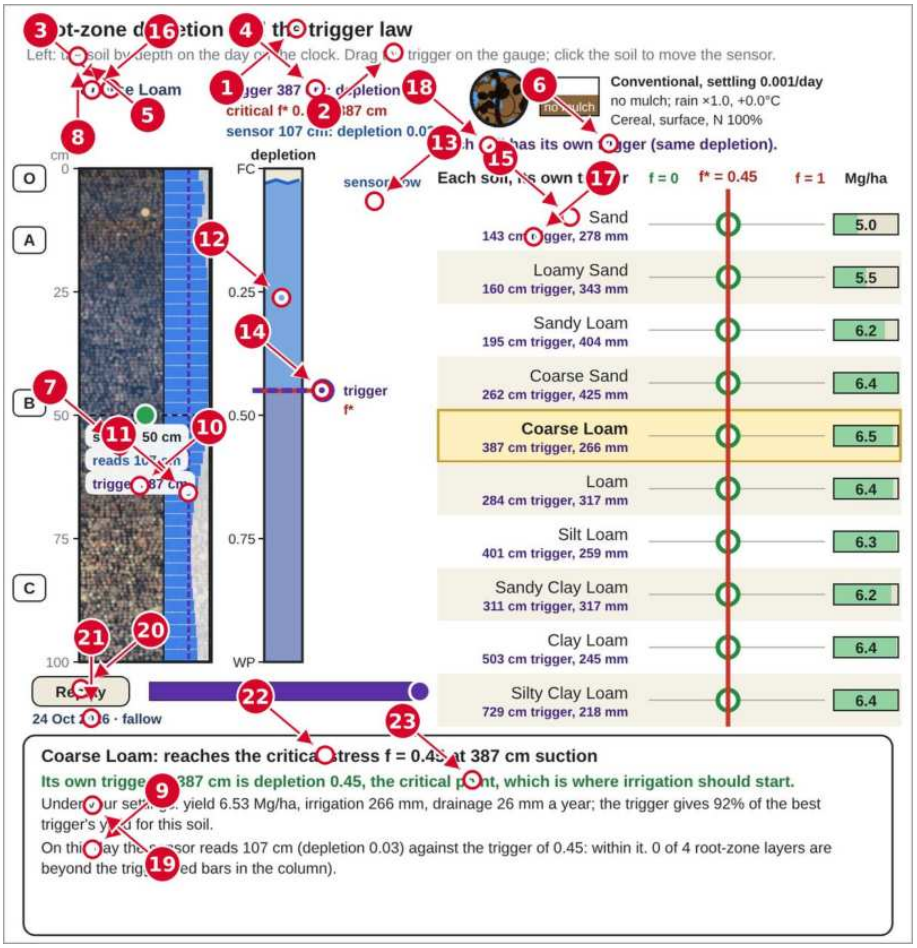



Figure 26.160: Trigger law, part 1. The soil column clicked at 60 cm, so the sensor moves to that depth.

# 26.10 Tillage and structure

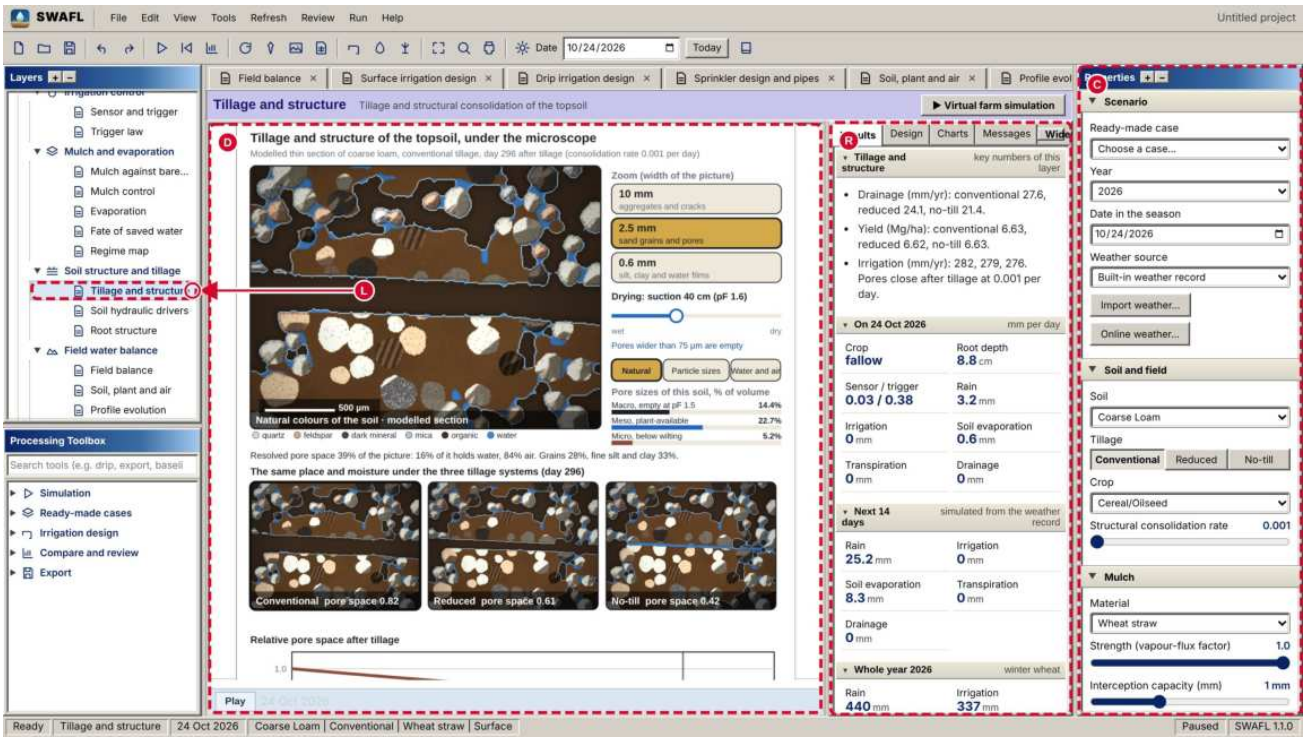



Figure 26.161: Tillage and structure: the whole workbench.

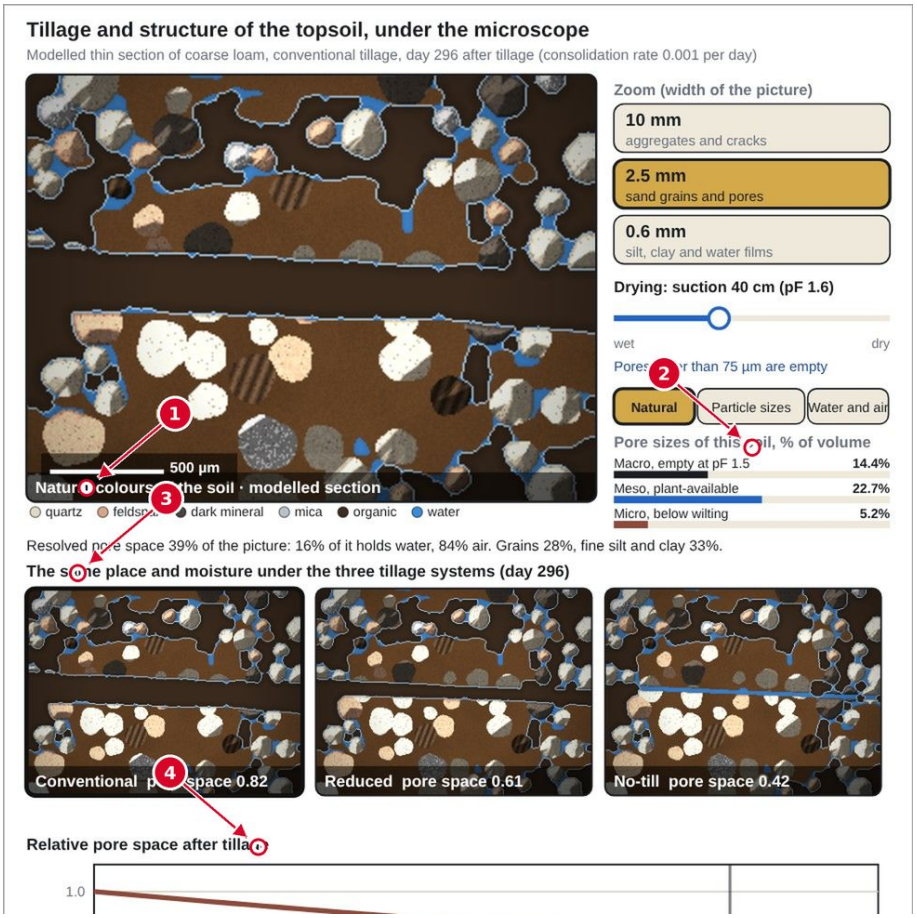



Figure 26.162: Tillage and structure: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Modelled thin section of the topsoil; the share of open pores follows the pore index p(t), Eq. 8.1.
- **2** Pore-size classes from the retention curve, theta(h): Eqs. 8.3 and 8.4 (capillary radius).
- **3** The same place under the three tillage systems; pore space 0.82, 0.61 and 0.42 from Eq. 8.1.
- **4** Relative pore space after tillage, p(t) = p_c + (p_0 minus p_c) exp(-tau_c t): Eq. 8.1.

**Equations behind this screen.** Eq. 5.1, Eq. 8.1, Eq. 8.2, Eq. 8.3, Eq. 8.4.

*Eq. 5.1. Where explained: chapter “The two computing engines”, section “The simulation campaign”.*

$$S_e(h) = \left[1 + (\alpha|h|)^n\right]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

*Provenance (Eq. 5.1):* **[Literature]**: van Genuchten (1980).

*Eq. 8.1. Where explained: chapter “Soil structure and tillage layers”, section “Pore index and consolidation”.*

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

*Provenance (Eq. 8.1):* **[Engineered]**: exponential relaxation with chosen constants.

*Eq. 8.2. Where explained: chapter “Soil structure and tillage layers”, section “How the section is generated”.*

$$\phi = \mathrm{clip}\big(0.36 + 0.0016\,\mathrm{clay} + 0.0009\,\mathrm{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\mathrm{clay}, \tag{8.2}$$

*Provenance (Eq. 8.2):* **[Engineered]**: porosity relation with chosen coefficients.

*Eq. 8.3. Where explained: chapter “Soil structure and tillage layers”, section “How the section is generated”.*

$$\theta(h) = \theta_r + (\phi - \theta_r)\,S_e(h). \tag{8.3}$$

*Provenance (Eq. 8.3):* **[Literature]**: van Genuchten (1980).

*Eq. 8.4. Where explained: chapter “Soil structure and tillage layers”, section “How the section is generated”.*

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h} \;\approx\; \frac{0.149\ \mathrm{cm}^2}{h} \quad (\text{or } r_c[\mathrm{mm}] \approx 1.49/h[\mathrm{cm}]), \tag{8.4}$$

*Provenance (Eq. 8.4):* **[Literature]**: Young-Laplace (Jurin) capillary law.

### 26.10.1 The whole layer, part by part

The picture below is the complete drawing of the layer Tillage and structure, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

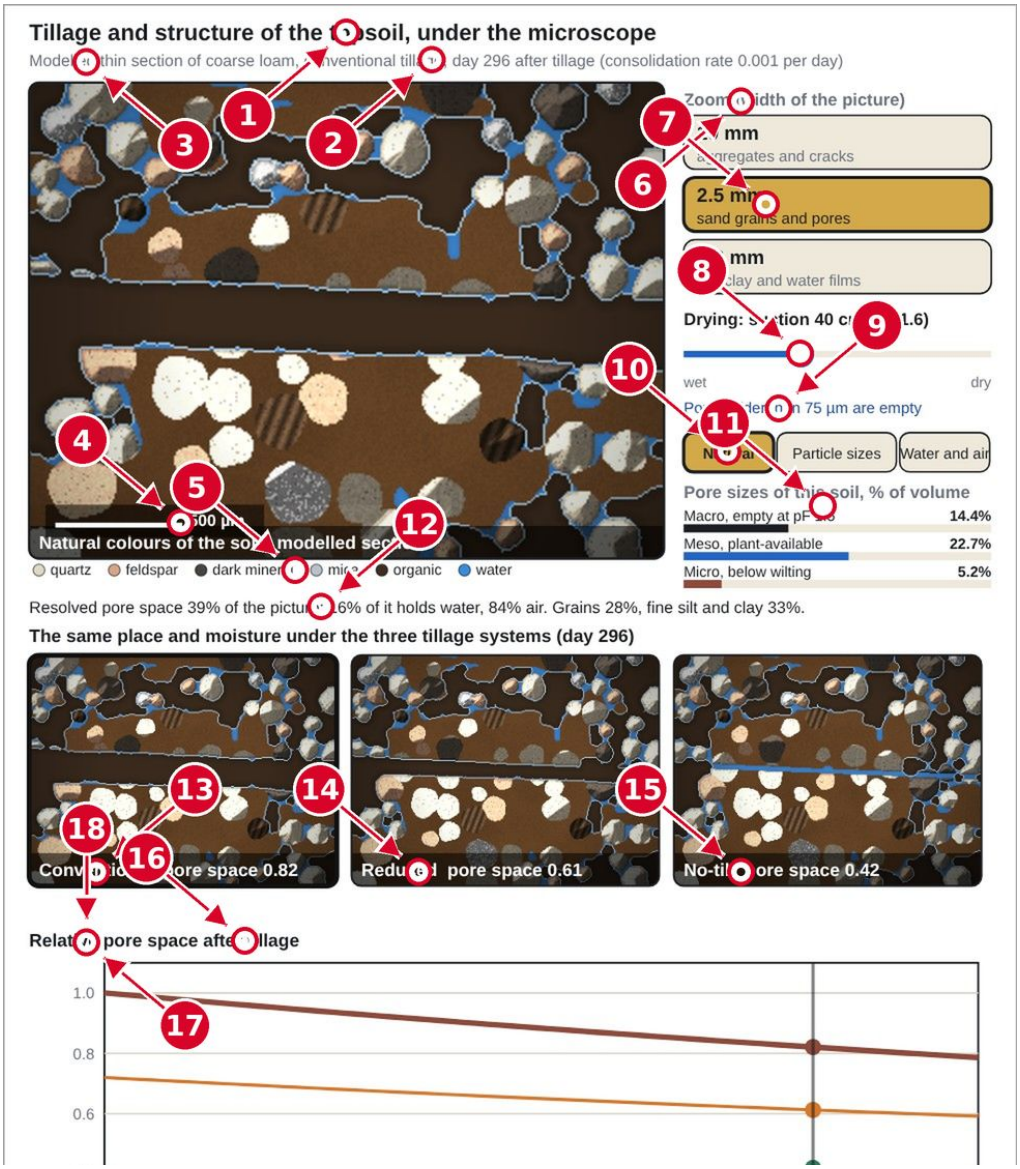



Figure 26.163: Tillage and structure, part 1 of 2.

**What the arrows point to.**

- **1** The title, tillage and the structure of the topsoil under the microscope.
- **2** The scenario of the section, the soil, the tillage, the day after tillage and the consolidation rate.
- **3** The modelled thin section, with grains, pores and water films drawn at the chosen zoom.
- **4** The scale bar of the section.
- **5** The legend of the colours, quartz, feldspar, dark mineral, mica, organic matter and water.
- **6** The Zoom buttons, 10 mm, 2.5 mm and 0.6 mm wide pictures.
- **7** The zoom selected, here the 2.5 mm picture of sand grains and pores.
- **8** The drying value box, which sets the suction and so which pores hold water.
- **9** The note on the pore width that empties at this suction.
- **10** The view buttons, Natural, Particle sizes and Water and air.
- **11** The pore-size classes of this soil, macro, meso and micro, in percent of the volume.
- **12** The resolved pore space of the picture and the share of it holding water.
- **13** The conventional tillage tile, with its relative pore space of 1.00.
- **14** The reduced tillage tile, with a smaller relative pore space.

- **15** The no-till tile, with the smallest relative pore space.
- **16** The chart of the relative pore space after tillage against the days since tillage.
- **17** The conventional line, which starts at 1.0 and consolidates with time.
- **18** The reduced tillage line.

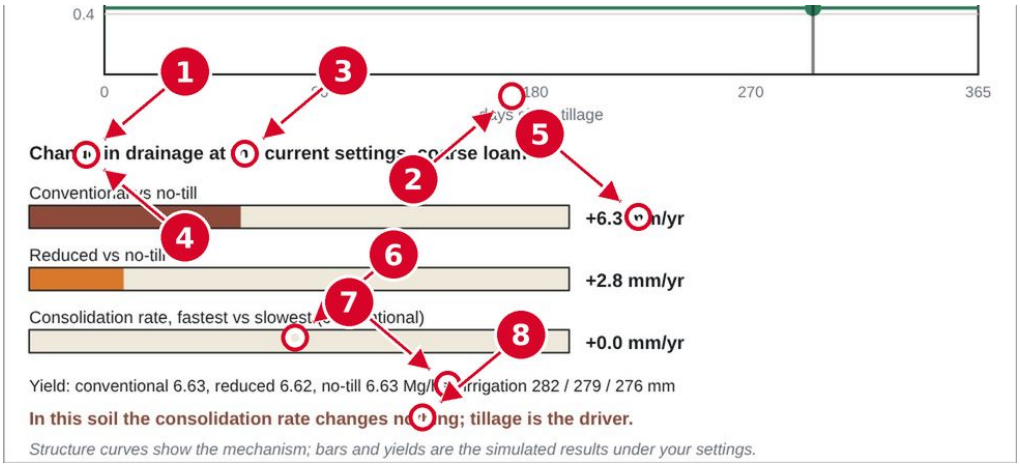



Figure 26.164: Tillage and structure, part 2 of 2.

**What the arrows point to.**

- **1** The no-till line, flat because the soil is not loosened.
- **2** The time axis in days since tillage.
- **3** The change in drainage at the current settings.
- **4** The bar of drainage for conventional against no-till tillage.
- **5** The value of that difference in mm per year.
- **6** The bar for the consolidation rate, fastest against slowest.
- **7** The yield and the irrigation of the three tillage systems.
- **8** The conclusion, which factor drives the drainage in this soil.

## 26.10.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

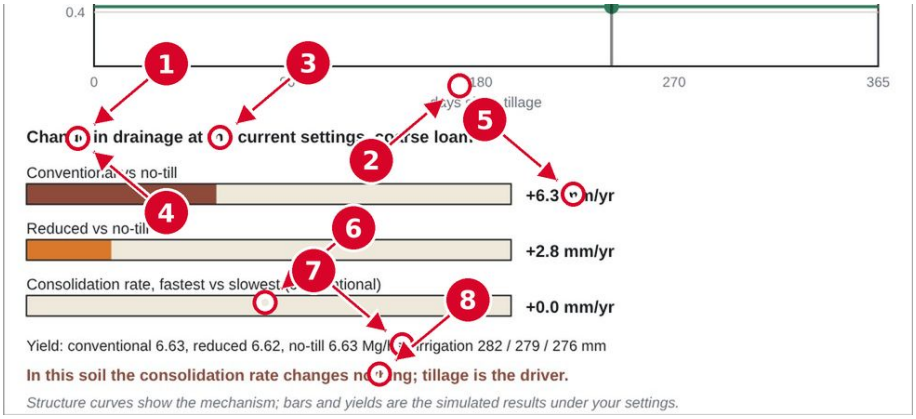



Figure 26.165: Tillage and structure, part 2, playback two thirds of the way (day 241).

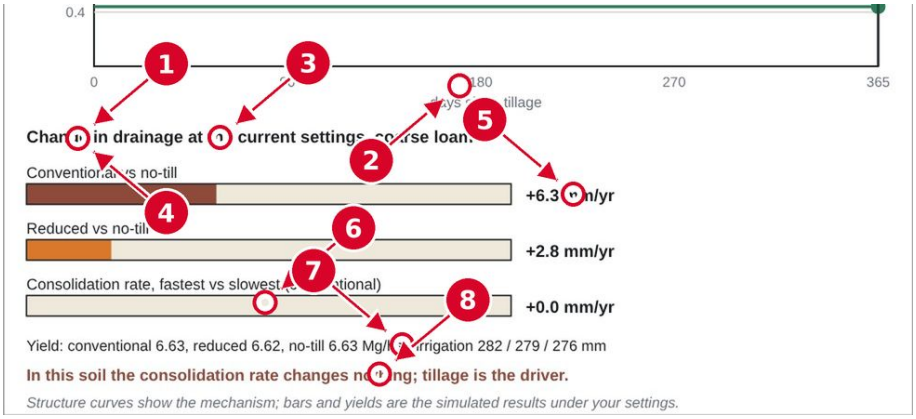



Figure 26.166: Tillage and structure, part 2, playback the end (day 365).

### 26.10.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

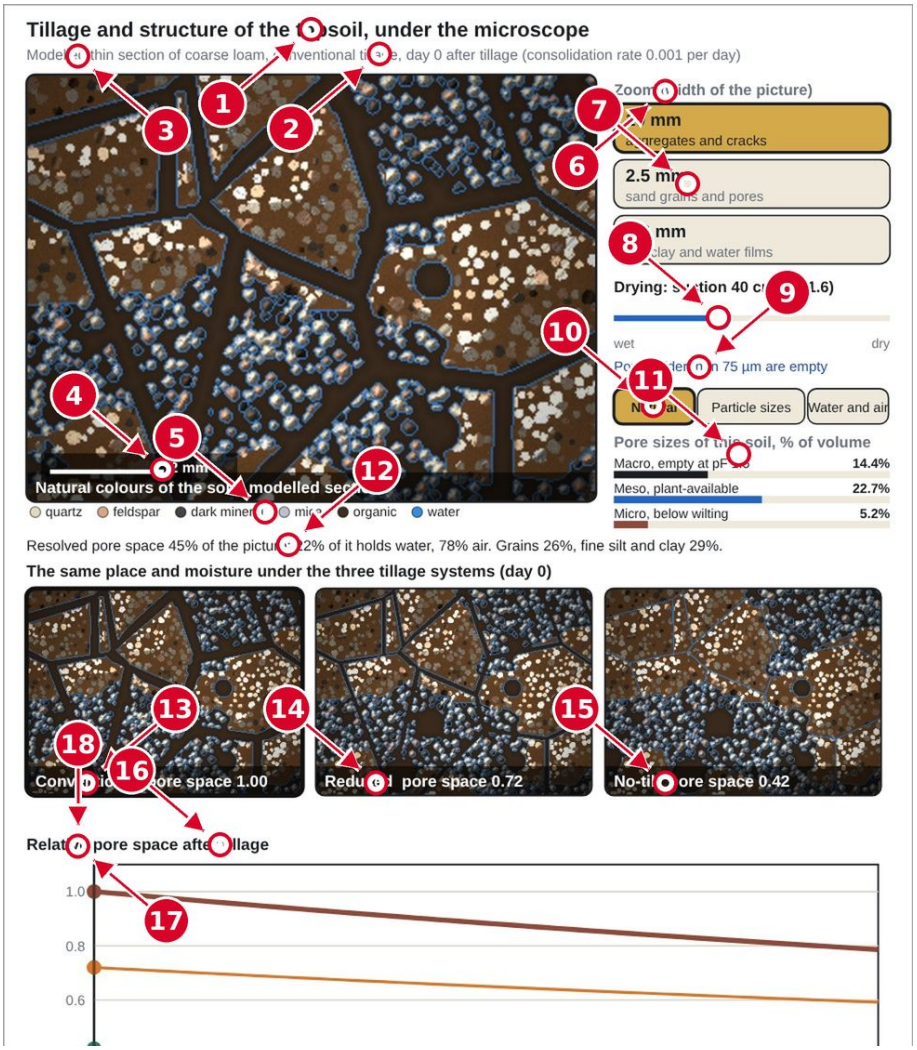



Figure 26.167: Tillage and structure, part 1. The 10 mm zoom chosen, so the picture shows aggregates and cracks.

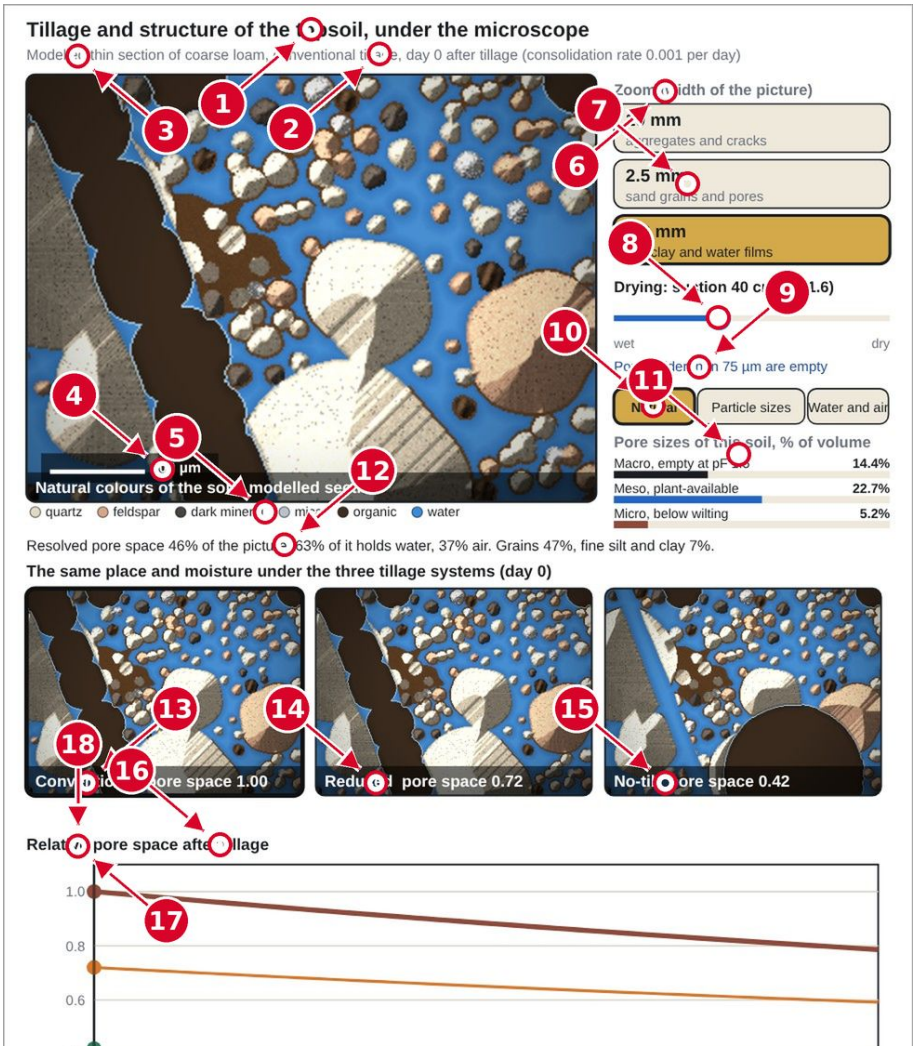



Figure 26.168: Tillage and structure, part 1. The 0.6 mm zoom chosen, so the picture shows silt, clay and the water films.

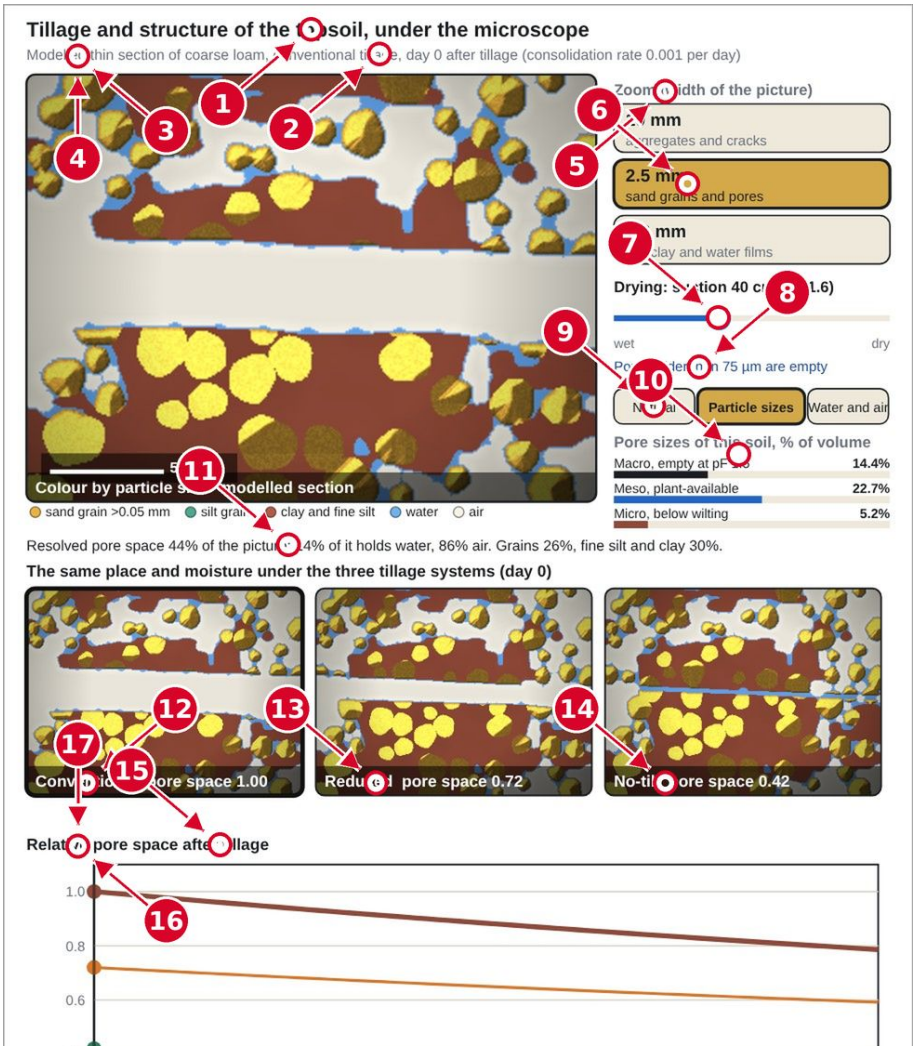



Figure 26.169: Tillage and structure, part 1. The Particle sizes view chosen, so the grains are coloured by size.

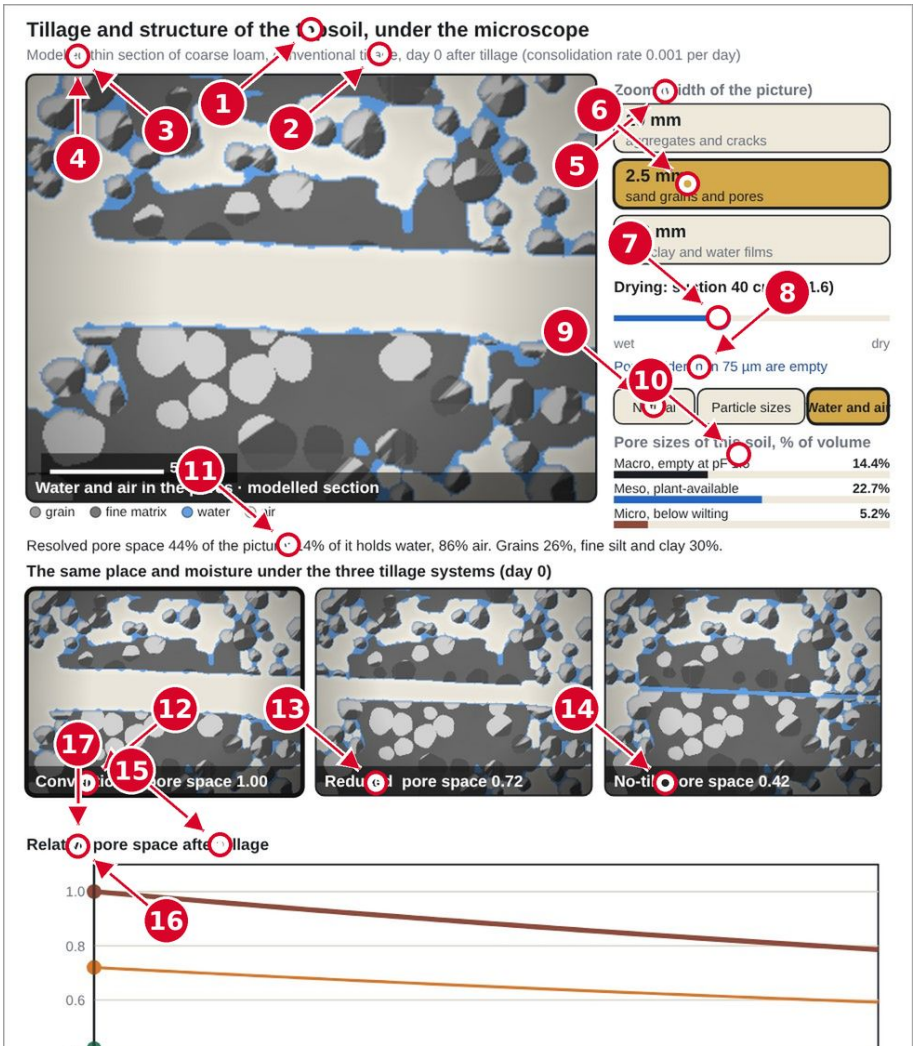



Figure 26.170: Tillage and structure, part 1. The Water and air view chosen, so the pores are coloured by whether they hold water or air.

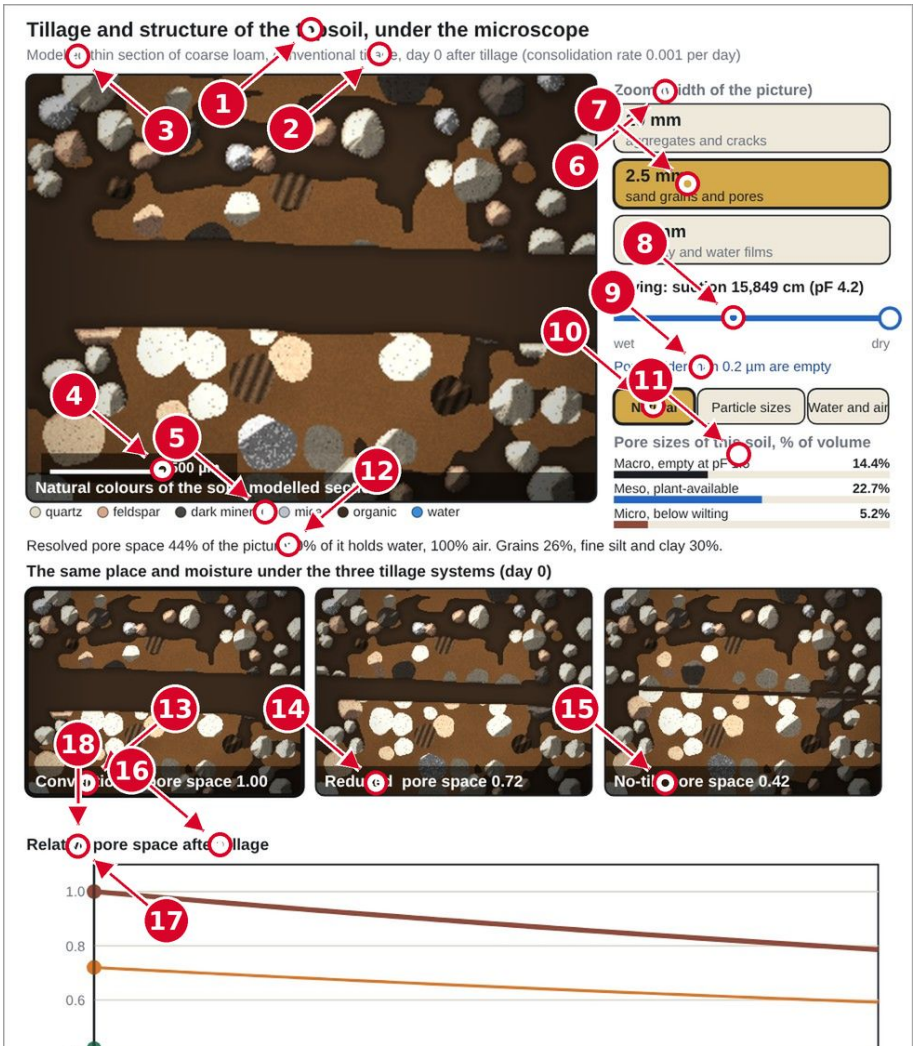



Figure 26.171: Tillage and structure, part 1. The drying value box dragged to the dry end, so most pores are empty.

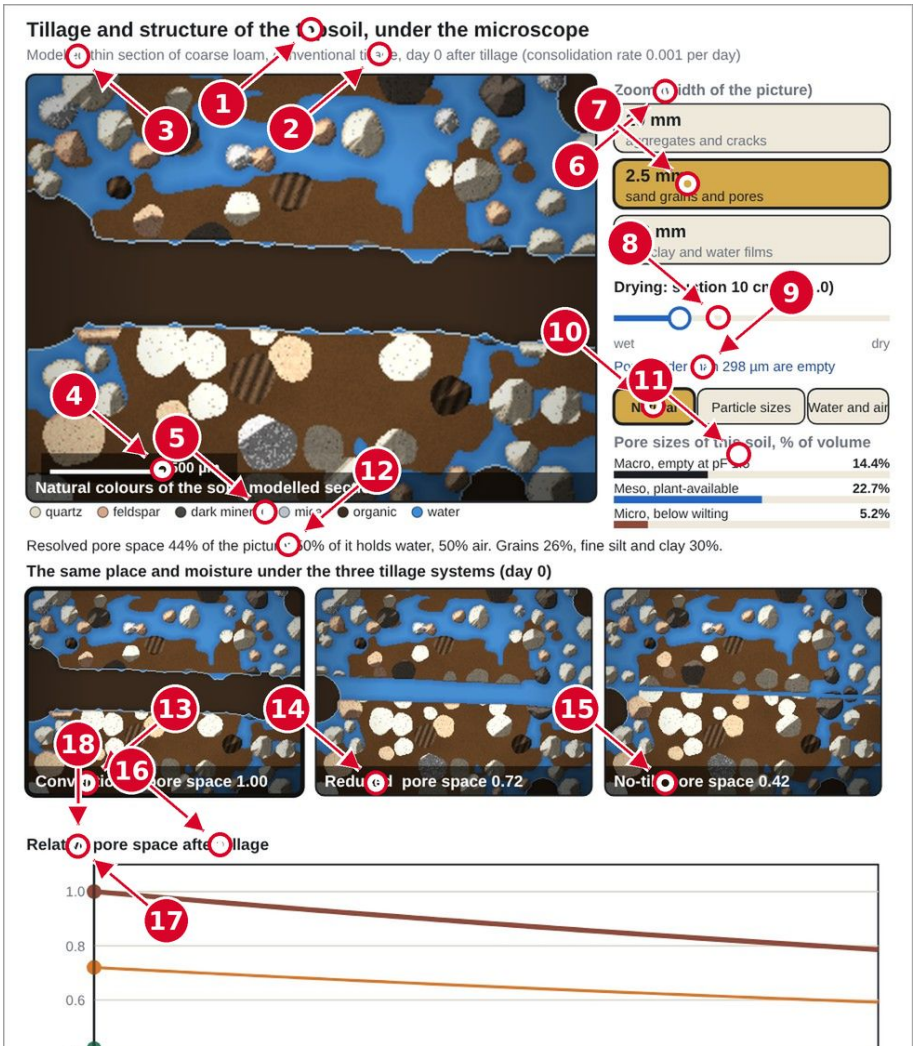



Figure 26.172: Tillage and structure, part 1. The drying value box dragged to the wet end, so most pores hold water.

# 26.11 Soil hydraulic drivers

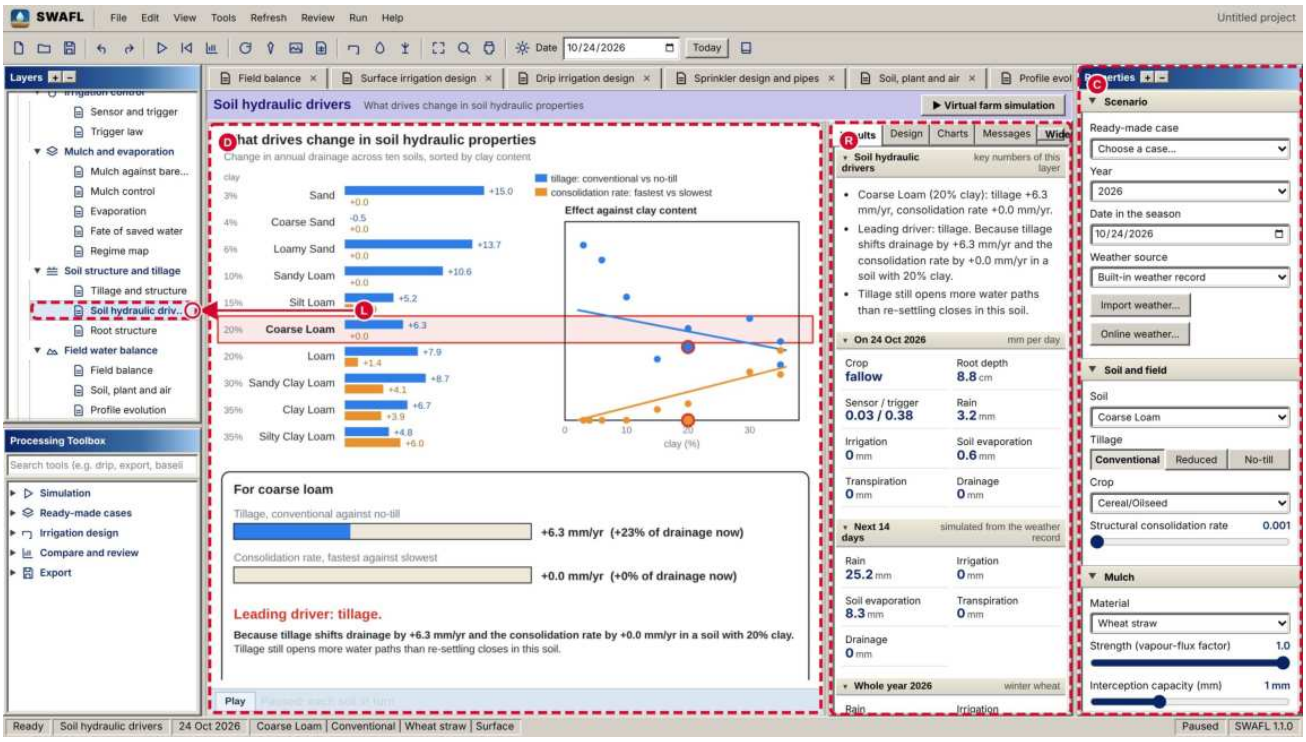



Figure 26.173: Soil hydraulic drivers: the whole workbench.

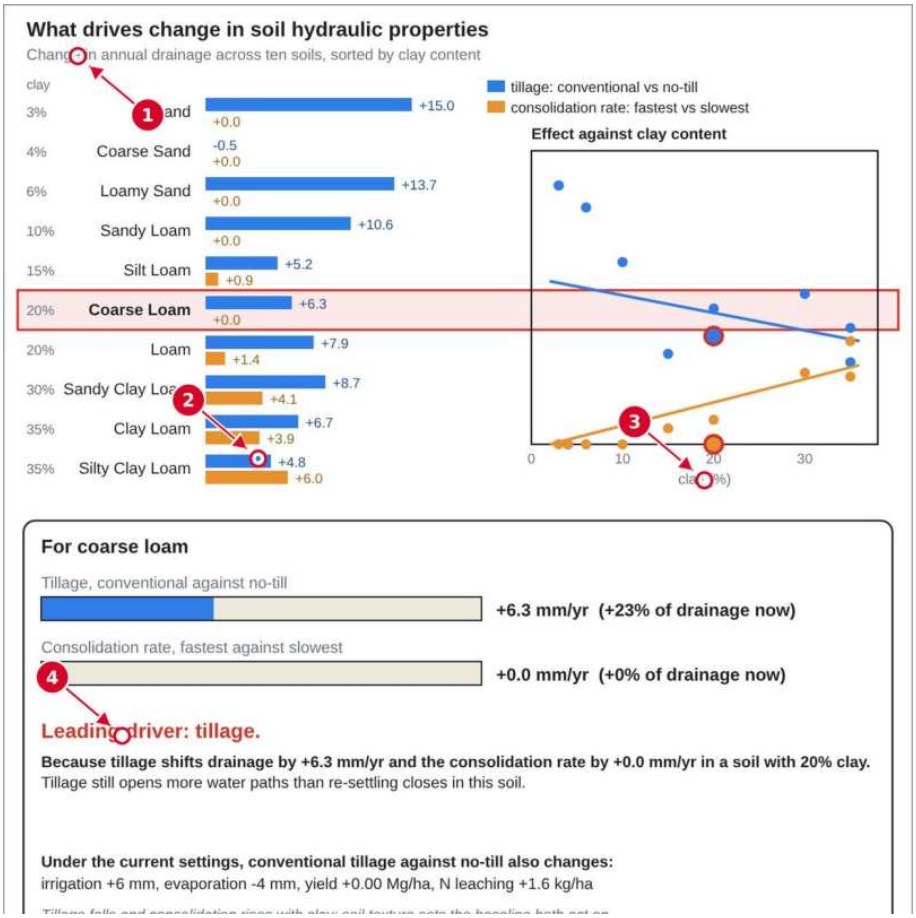



Figure 26.174: Soil hydraulic drivers: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Blue bar: effect of tillage on drainage, Delta_till = D(conventional) minus D(no-till): Eq. 8.5.
- **2** Orange bar: effect of the consolidation rate on drainage, Delta_cons: Eq. 8.5.
- **3** Scatter of each effect against clay content with the least-squares fit: Eq. 8.6.
- **4** “Leading driver”: the larger of Delta_till and Delta_cons for the selected soil, Eq. 8.5.

**Equations behind this screen.** Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 8.1, Eq. 8.5, Eq. 8.6.

*Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.5. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.6. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.7. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.8. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.9. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.10. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".*

$$Q \leftarrow Q\ \text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ \ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N). \tag{5.11}$$

*Provenance (Eq. 5.11):* **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

*Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".*

$$ET \in [\,P + I - D - 30,\ P + I - D + 15\,] \text{ (if not, } ET \text{ is clipped and } E, T \text{ are scaled by the same factor)},$$
$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

*Provenance (Eq. 5.12):* **[Engineered]**: closure constraint and bounds chosen for the workbench.

*Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".*

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

*Provenance (Eq. 8.1):* **[Engineered]**: exponential relaxation with chosen constants.

*Eq. 8.5. Where explained: chapter "Soil structure and tillage layers", section "Generation".*

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

*Provenance (Eq. 8.5):* **[Data-driven]**: differences of Daisy run-set E5 results.

*Eq. 8.6. Where explained: chapter "Soil structure and tillage layers", section "Generation".*

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

*Provenance (Eq. 8.6):* **[Data-driven]**: ordinary least squares applied to Daisy results.

### 26.11.1 The whole layer, part by part

The picture below is the complete drawing of the layer Soil hydraulic drivers, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

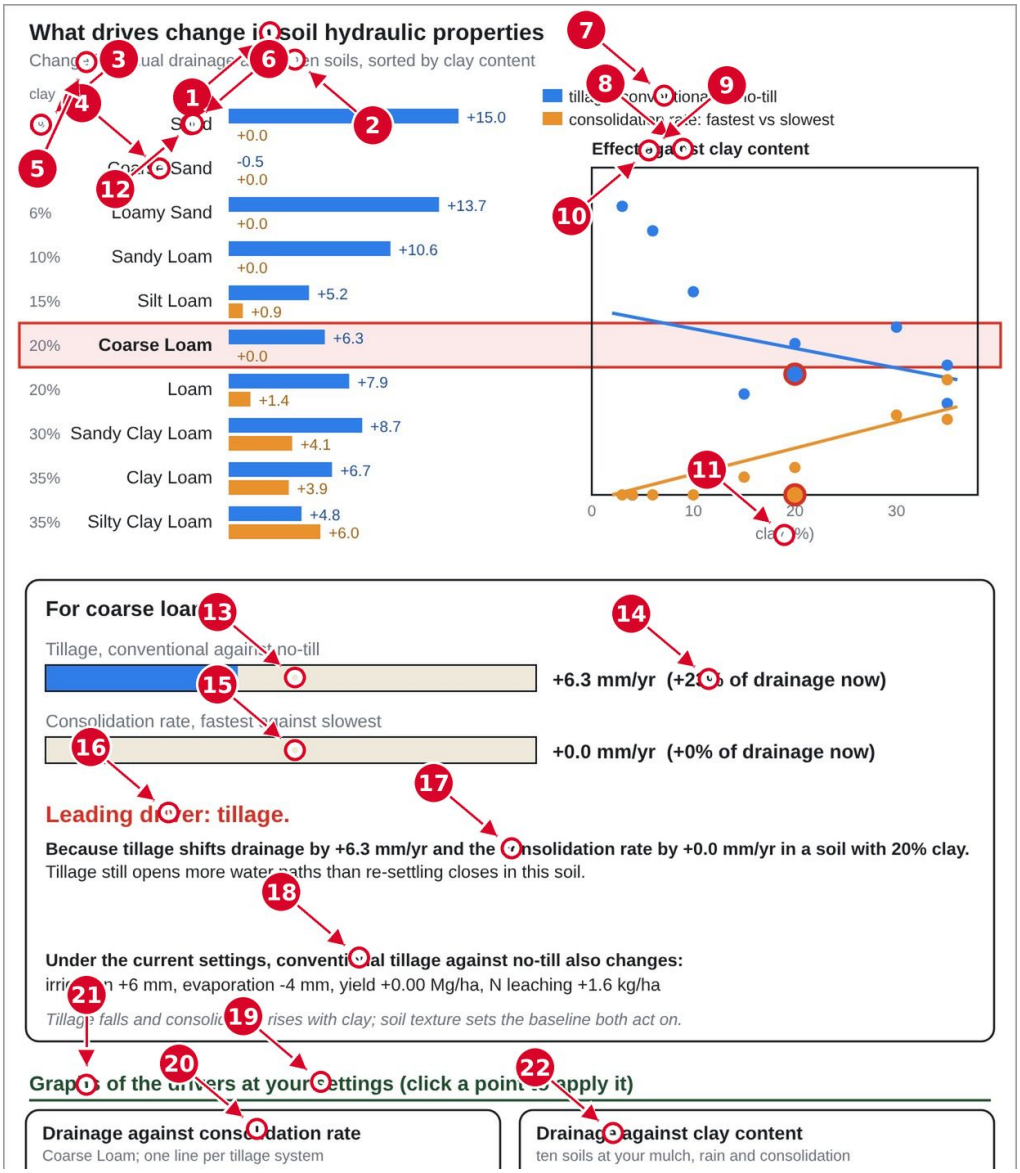



Figure 26.175: Soil hydraulic drivers, part 1 of 3.

**What the arrows point to.**

- **1** The title, what drives the change in the hydraulic properties of the soil.
- **2** The subtitle, the change in annual drainage across ten soils sorted by clay content.
- **3** The clay content of each soil, the sorting key.
- **4** The soil highlighted in the list, here sand, which every panel below refers to.
- **5** The blue bar, the change in drainage from tillage, conventional against no-till.
- **6** The orange bar, the change in drainage from the consolidation rate.
- **7** The legend of the two bars.
- **8** The scatter chart of the two effects against the clay content.
- **9** The highlighted soil on the scatter chart.
- **10** The blue trend line, the tillage effect falls with clay.
- **11** The orange trend line, the consolidation effect rises with clay.
- **12** The card for the selected soil, with the two effects in mm per year.
- **13** The bar of the tillage effect against its largest value.
- **14** The value of the tillage effect and its share of the drainage now.

- **15** The bar of the consolidation effect.
- **16** The leading driver, tillage or the consolidation rate.
- **17** The reasoning behind the leading driver for this clay content.
- **18** The other changes of the conventional tillage against no-till, irrigation, evaporation, yield and nitrogen leaching.
- **19** The graphs of the drivers; clicking a point applies it to the settings.
- **20** The graph of the drainage against the consolidation rate, one line per tillage system.
- **21** A point of the graph, which applies that value when it is clicked.
- **22** The graph of the drainage against the clay content of the ten soils.

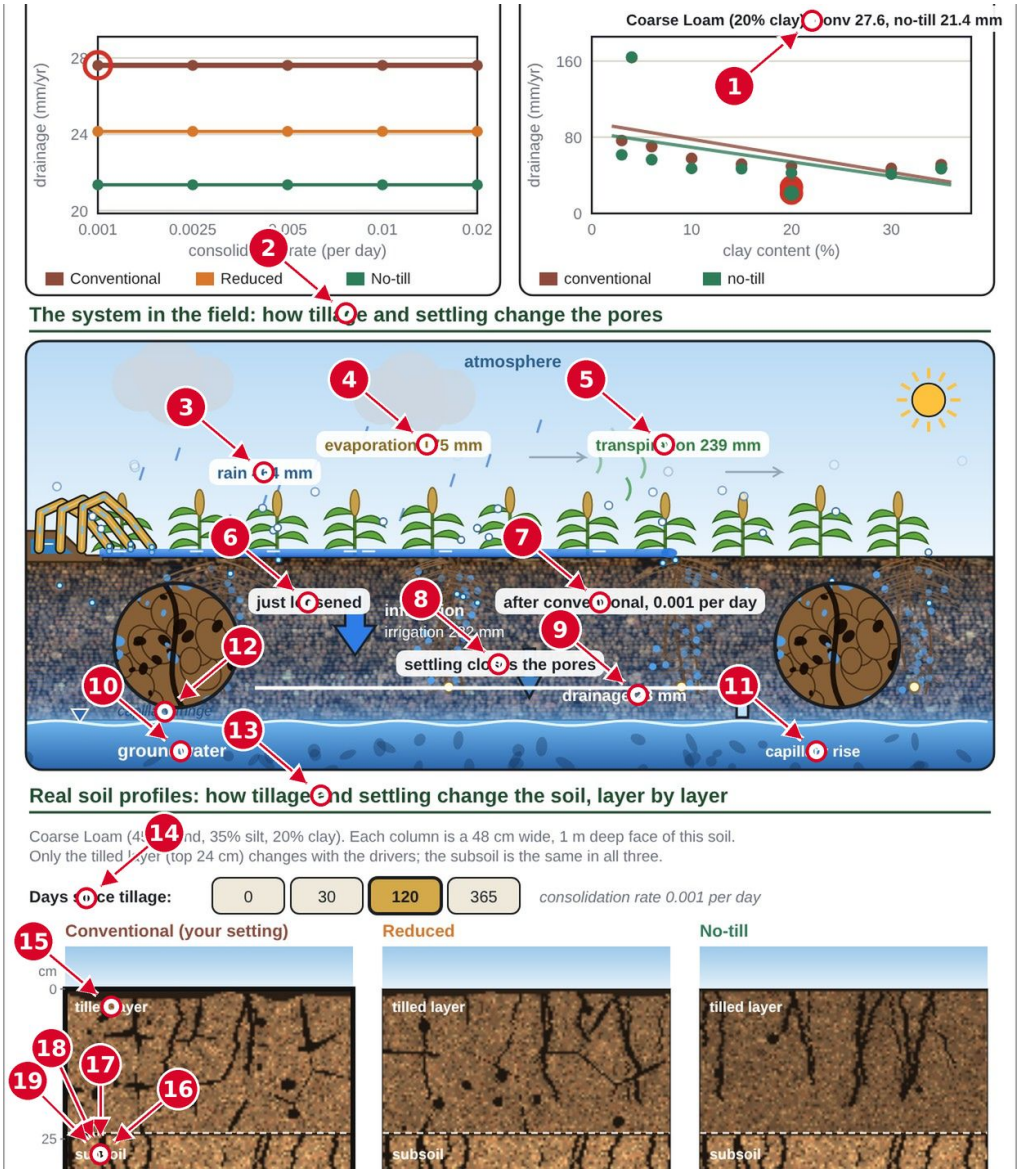



Figure 26.176: Soil hydraulic drivers, part 2 of 3.

**What the arrows point to.**

- **1** The readout of the soil selected, drainage of the conventional and no-till systems.
- **2** The system in the field, how tillage and settling change the pores.
- **3** The rain of the year.
- **4** The evaporation of the year.

- **5** The transpiration of the year.
- **6** The loosened layer just after the tillage.
- **7** The state of the layer after tillage at the consolidation rate in force.
- **8** The settling that closes the pores with time.
- **9** The drainage through the profile, from the Daisy simulation.
- **10** The groundwater below the profile.
- **11** The capillary rise from the groundwater.
- **12** The aggregate drawn inside the tilled layer.
- **13** The real soil profiles, the three tillage systems layer by layer.
- **14** The buttons of the days since tillage, 0, 30, 120 and 365.
- **15** The tilled layer, the top 24 cm, the only layer that changes.
- **16** The boundary of the tilled layer and the subsoil at 25 cm.
- **17** The conventional column, with the largest clods and cracks.
- **18** The reduced tillage column.
- **19** The no-till column, with fine cracks near the surface.

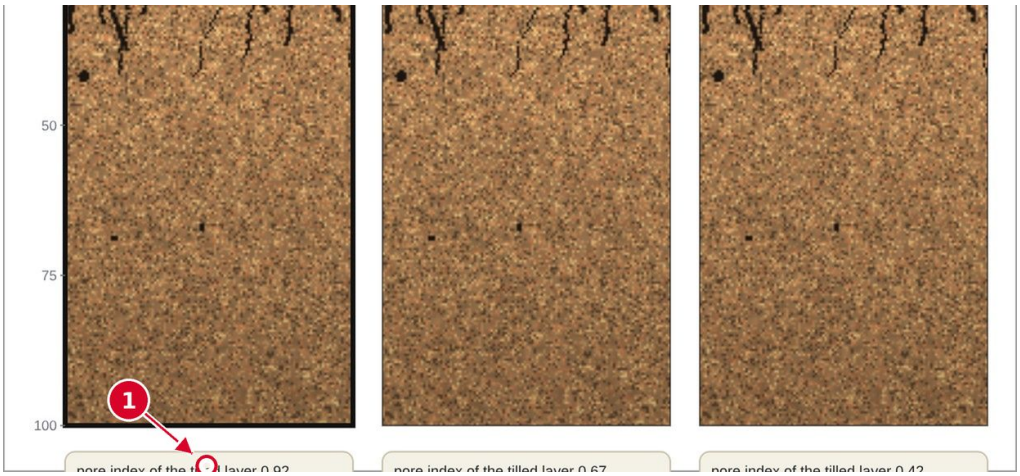



Figure 26.177: Soil hydraulic drivers, part 3 of 3.

**What the arrows point to.**

- **1** The pore index, the voids on the face and the share open to air.

### 26.11.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

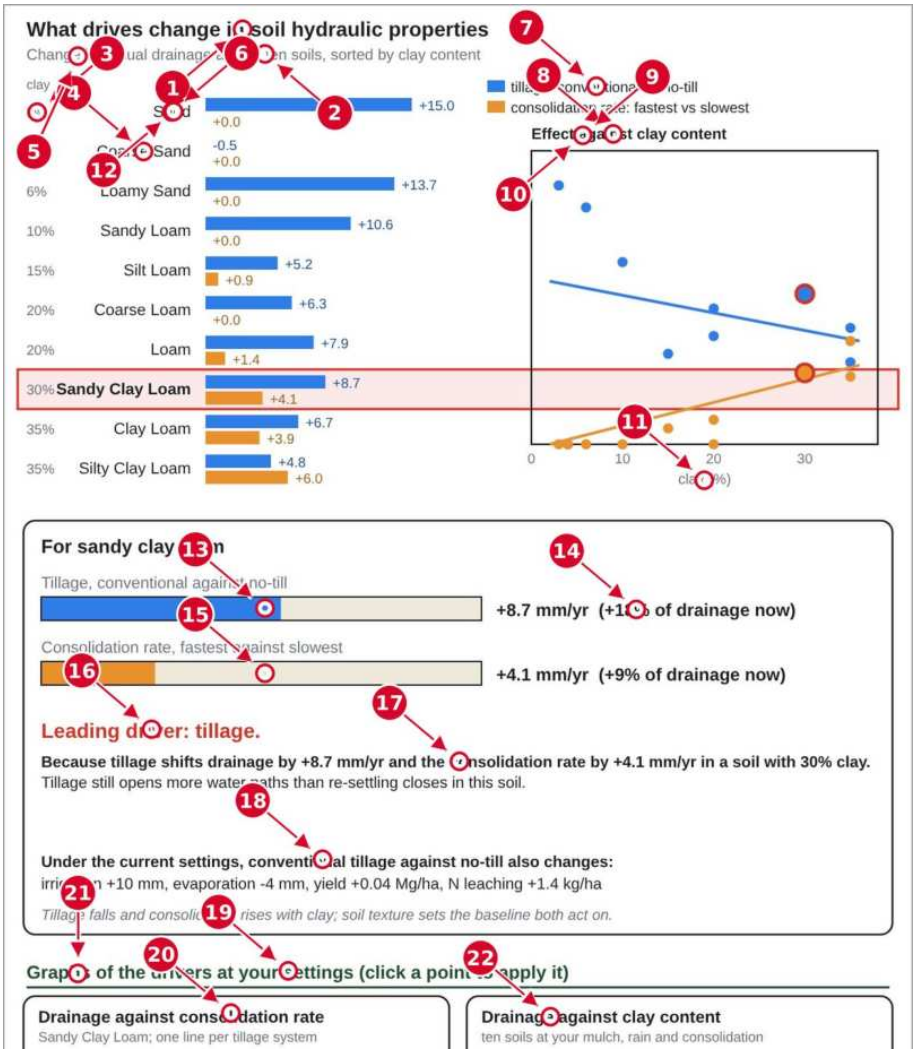



Figure 26.178: Soil hydraulic drivers, part 1. A different soil clicked in the list, so every panel refers to that soil.

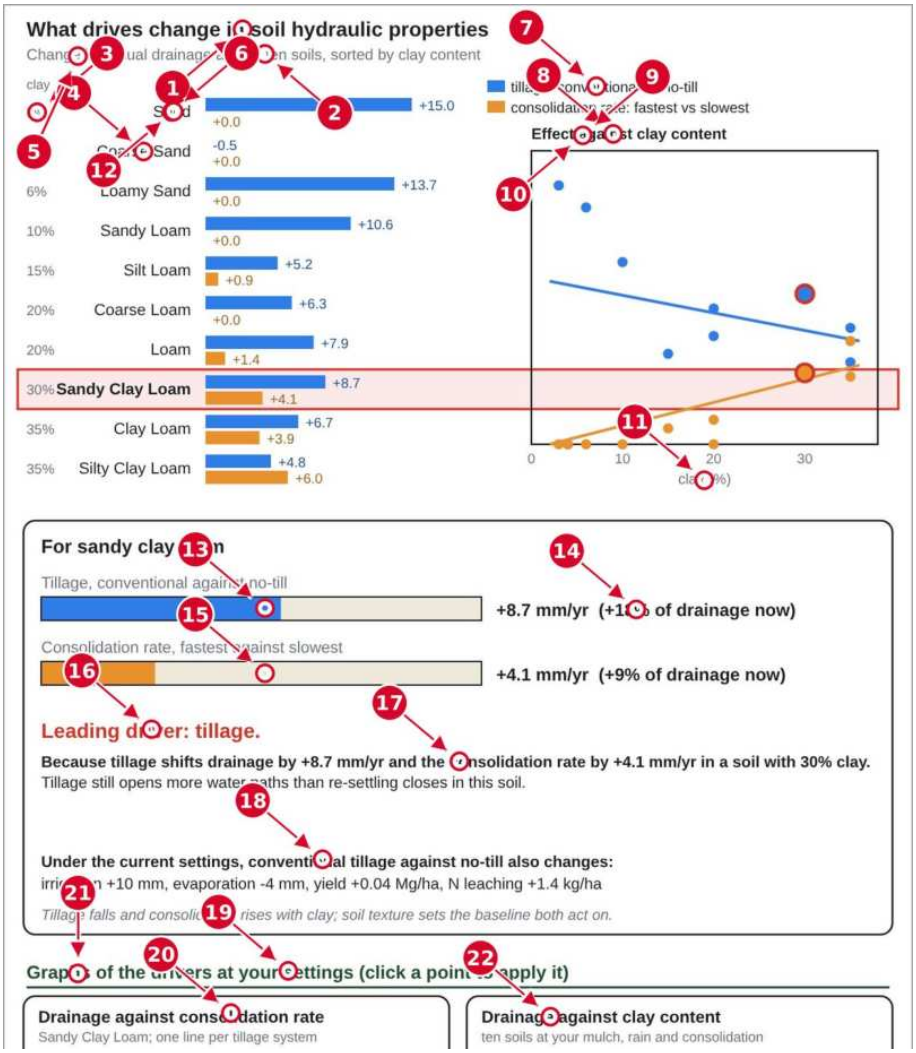



Figure 26.179: Soil hydraulic drivers, part 1. A different soil clicked in the list, so every panel refers to that soil.

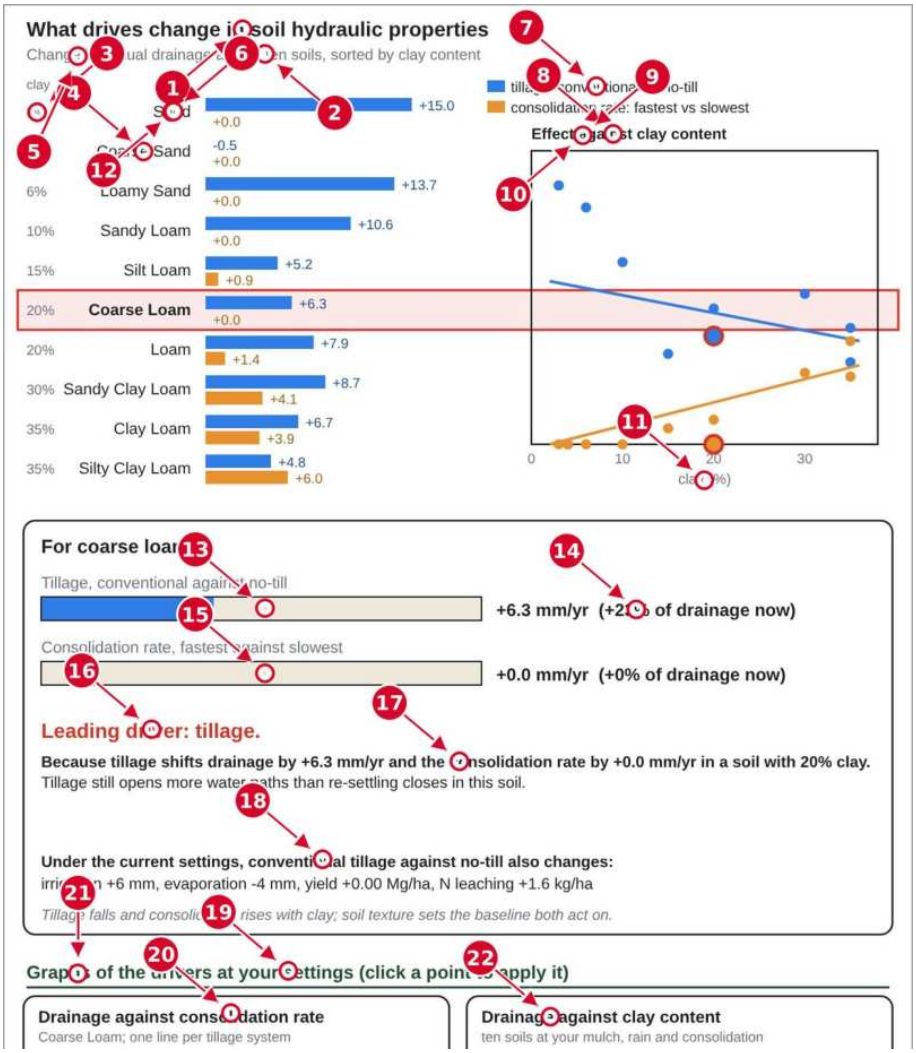



Figure 26.180: Soil hydraulic drivers, part 1. The 0 days since tillage button pressed, so the tilled layer is shown just after tillage.

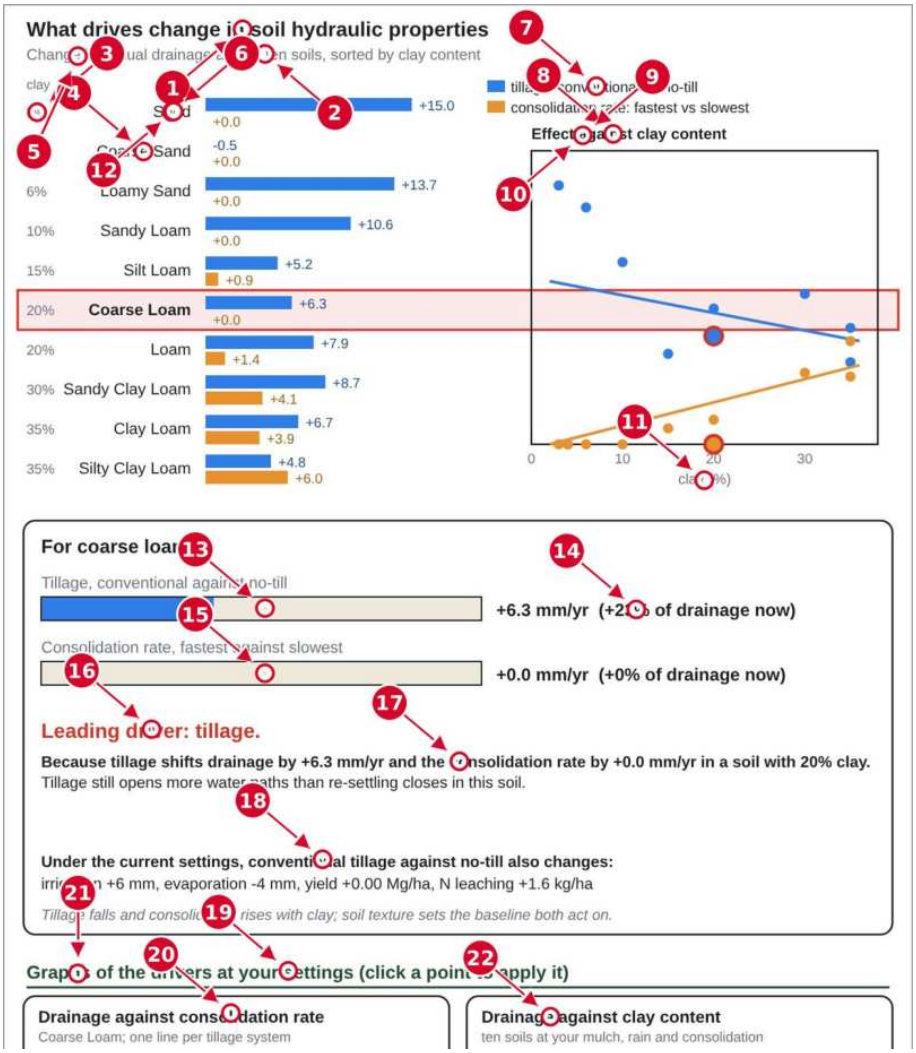



Figure 26.181: Soil hydraulic drivers, part 1. The 30 days since tillage button pressed.

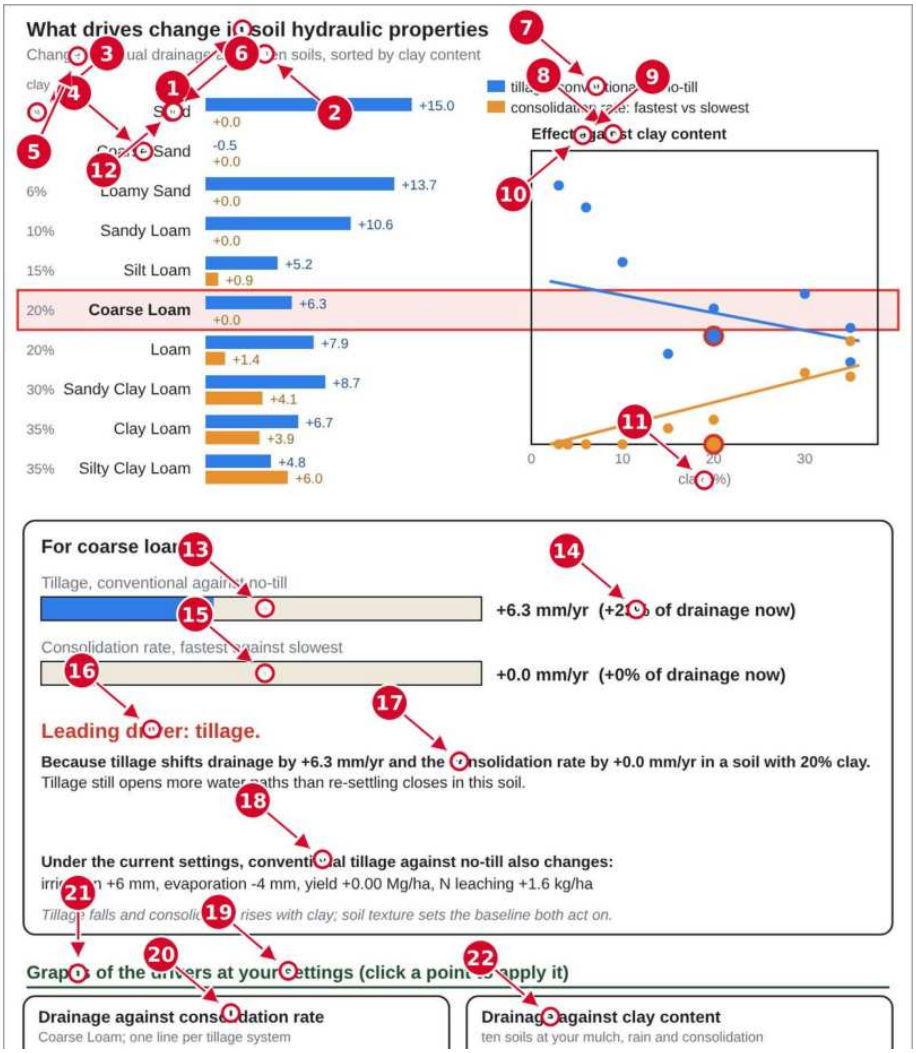



Figure 26.182: Soil hydraulic drivers, part 1. The 365 days since tillage button pressed, so the tilled layer has settled for a year.

# 26.12 Root structure

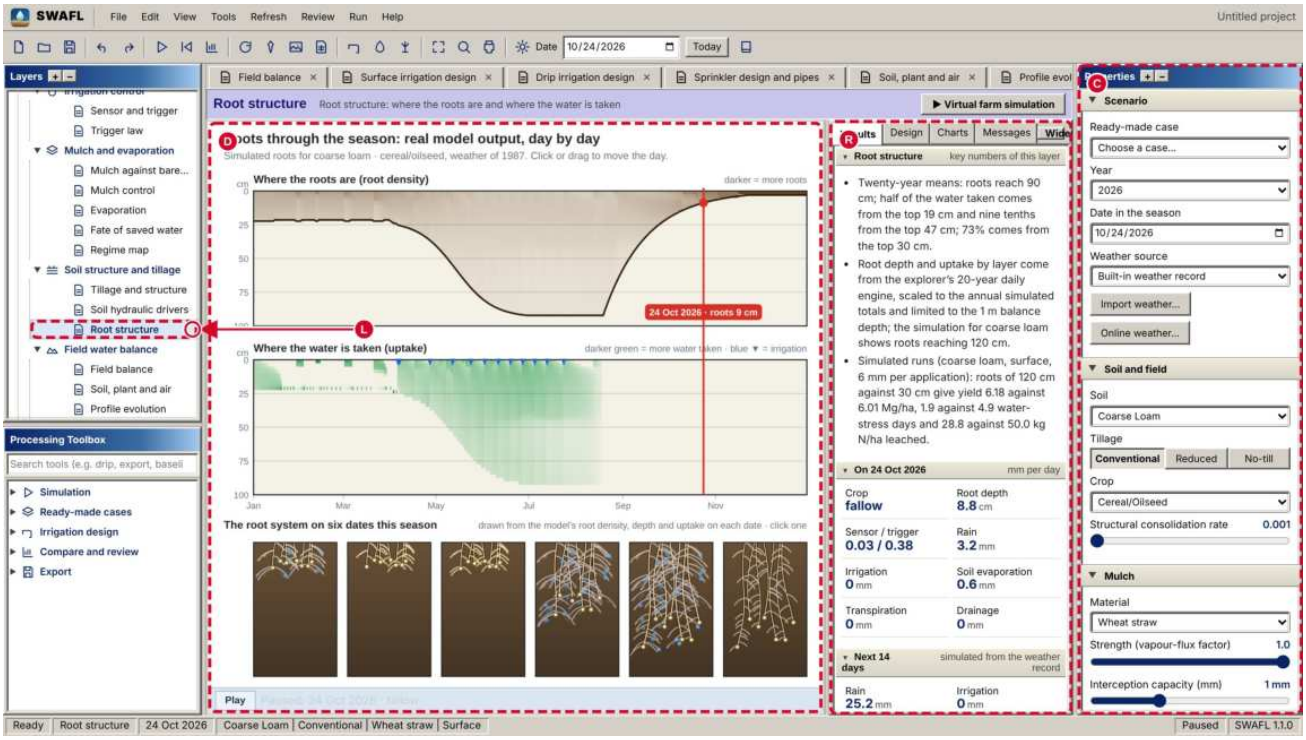



Figure 26.183: Root structure: the whole workbench.

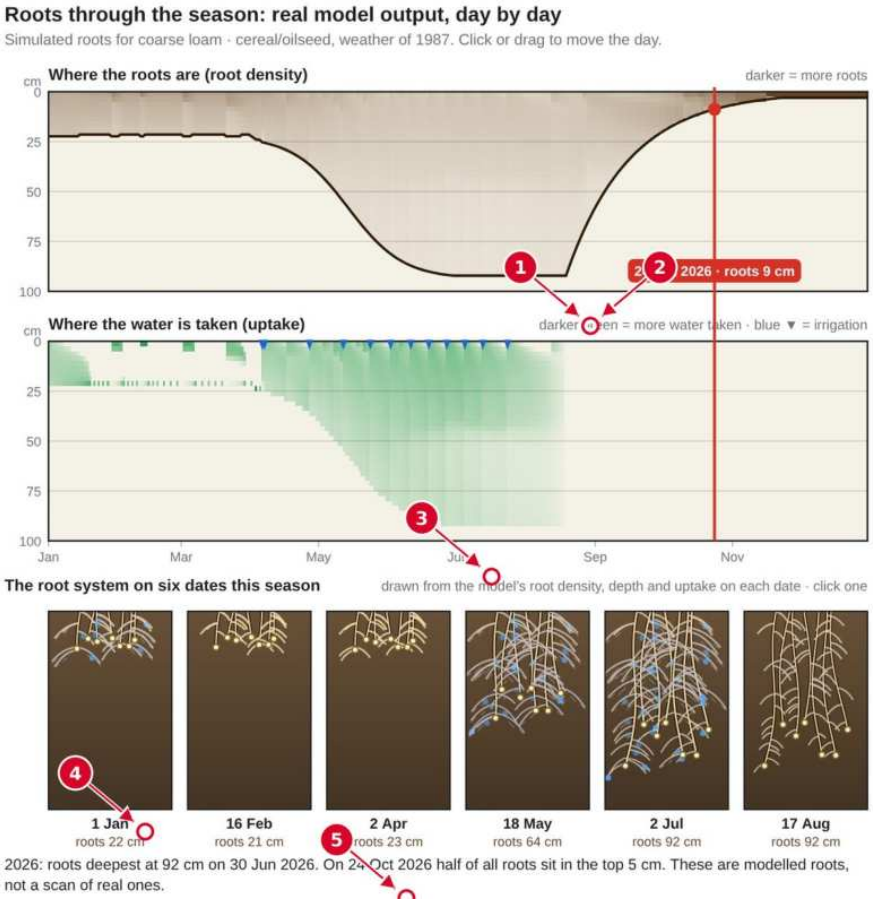



Figure 26.184: Root structure: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Heat map of relative root density by depth and day: Eq. 5.20.
- **2** Rooting depth through the season (black line): Eq. 5.18.
- **3** Heat map of water uptake per layer, u_i(d): Eqs. 5.21 to 5.23.
- **4** Root systems drawn on six dates from the same density: Eqs. 5.19 and 5.20.
- **5** Depth holding half of the uptake, z_p: Eq. 8.7.

**Equations behind this screen.** Eq. 5.18, Eq. 5.19, Eq. 5.20, Eq. 5.21, Eq. 5.22, Eq. 5.23, Eq. 8.7.

*Eq. 5.18. Where explained: chapter “The two computing engines”, section “Roots”.*

$$R_{\max} = \mathrm{clip}\Big( R_c\,(0.85 + 0.3\,\mathrm{sand})\big[1 + 0.12\log_{10}(|h_T|/300)\big]\big[1 - 0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \mathrm{cm}, \quad (5.18)$$

*Provenance (Eq. 5.18):* **[Engineered]**: rooting rules with constants chosen for the workbench.

*Eq. 5.19. Where explained: chapter “The two computing engines”, section “Roots”.*

$$R_d \leftarrow \max\Big\{ R_d,\ \min\big\{ R_{\max}\,\mathrm{clip}\big(\tfrac{c-0.04}{0.96}; 0, 1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35 + \bar{s}_{\mathrm{wet}})\,\kappa \big\} \Big\}, \quad (5.19)$$

*Provenance (Eq. 5.19):* **[Engineered]**: rooting rules with constants chosen for the workbench.

*Eq. 5.20. Where explained: chapter “The two computing engines”, section “Roots”.*

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)\,(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

*Provenance (Eq. 5.20):* **[Engineered]**: rooting rules with constants chosen for the workbench.

*Eq. 5.21. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".*

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65 \cdot \tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy} - 100)}{365}\Big], \tag{5.21}$$

*Provenance (Eq. 5.21):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

*Eq. 5.22. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".*

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

*Provenance (Eq. 5.22):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

*Eq. 5.23. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".*

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \text{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

*Provenance (Eq. 5.23):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

*Eq. 8.7. Where explained: chapter "Soil structure and tillage layers", section "Generation".*

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k :\ \frac{\sum_{i \le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d \in \text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

*Provenance (Eq. 8.7):* **[Engineered]**: diagnostic defined for the workbench.

### 26.12.1 The whole layer, part by part

The picture below is the complete drawing of the layer Root structure, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

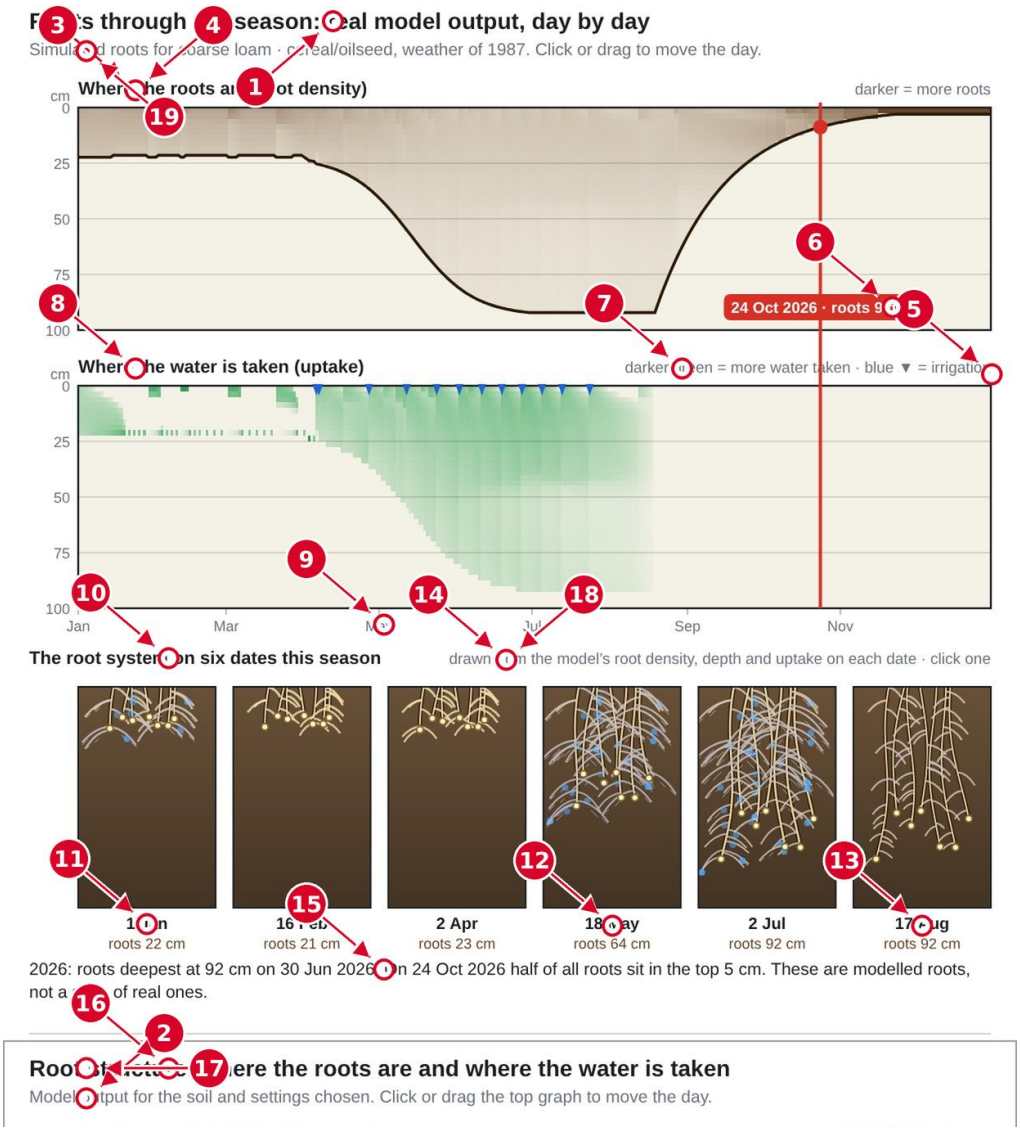



Figure 26.185: Root structure, part 1 of 4.

**What the arrows point to.**

- **1** The title, the roots through the season as real model output, day by day.
- **2** The instruction, the day is moved by clicking or dragging on the graphs.
- **3** The root density by depth and day, darker where there are more roots.
- **4** The black line is the rooting depth, which falls and grows with the season.
- **5** The red day cursor, the day of the clock.
- **6** The label of the cursor, the date and the rooting depth on that day.
- **7** The uptake of water by depth and day, darker green for more water taken.
- **8** The blue triangles mark the irrigations.
- **9** The month axis of the season.
- **10** The six dates on which the root system is drawn, each clickable.
- **11** The root system on the first date, still small.
- **12** The root system in the middle of the season, deep and branched.
- **13** The root system late in the season with the uptake in blue.
- **14** The date and the rooting depth under each drawing.
- **15** The sentence that reports the deepest roots and where half of all the roots sit.

- **16** The second section, the root structure where the roots are and where the water is taken.
- **17** The soil, roots and water of the day.
- **18** The relative root density by depth on the day.
- **19** The note that the runs exist for coarse loam only.

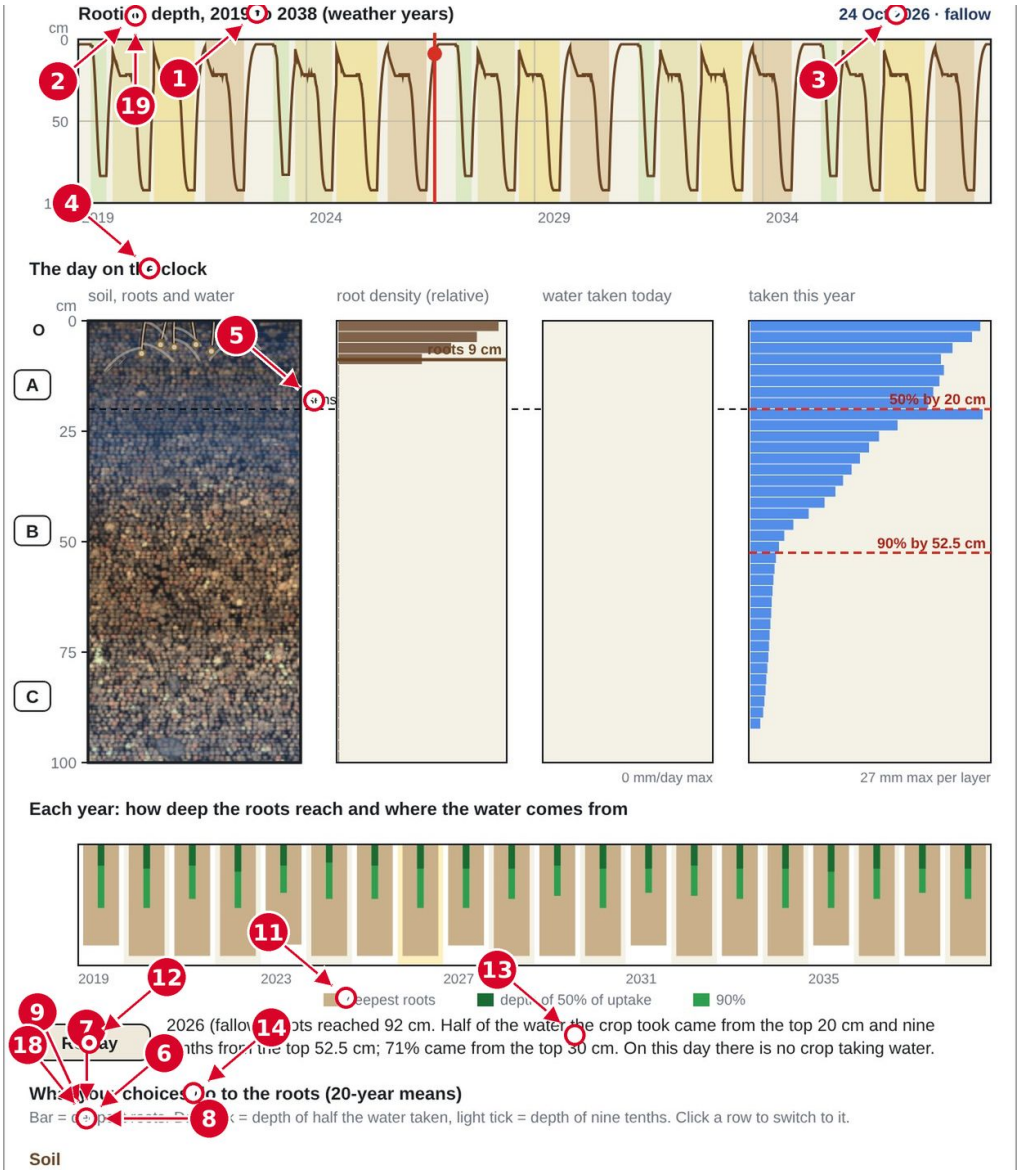



Figure 26.186: Root structure, part 2 of 4.

**What the arrows point to.**

- **1** The rooting depth for every year of the record, with the season colours.
- **2** The brown line is the rooting depth of each year.
- **3** The cursor of the selected day on the multi-year graph.
- **4** The day on the clock, four panels for the selected day.
- **5** The dashed line carries the sensor depth across the panels.
- **6** The water taken on the day by depth.
- **7** The water taken in the year by depth.
- **8** The depth by which half of the water was taken.
- **9** The depth by which nine tenths of the water was taken.

- **10** The deepest roots of each year and the depth of half and nine tenths of the uptake.
- **11** The legend of the deepest roots and of the two uptake depths.
- **12** The Play and Stop button of the season (it plays by itself until Stop is pressed).
- **13** The summary of the year, the depth of the roots and where the water came from.
- **14** What your choices do to the roots, as 20-year means; a row is clicked to switch to it.
- **15** The bar of the deepest roots of the soil.
- **16** The dark tick, the depth of half of the water taken.
- **17** The light tick, the depth of nine tenths of the water taken.
- **18** The green ticks, the depths of half and nine tenths of the uptake.
- **19** The red line, the rooting depth set for this run.

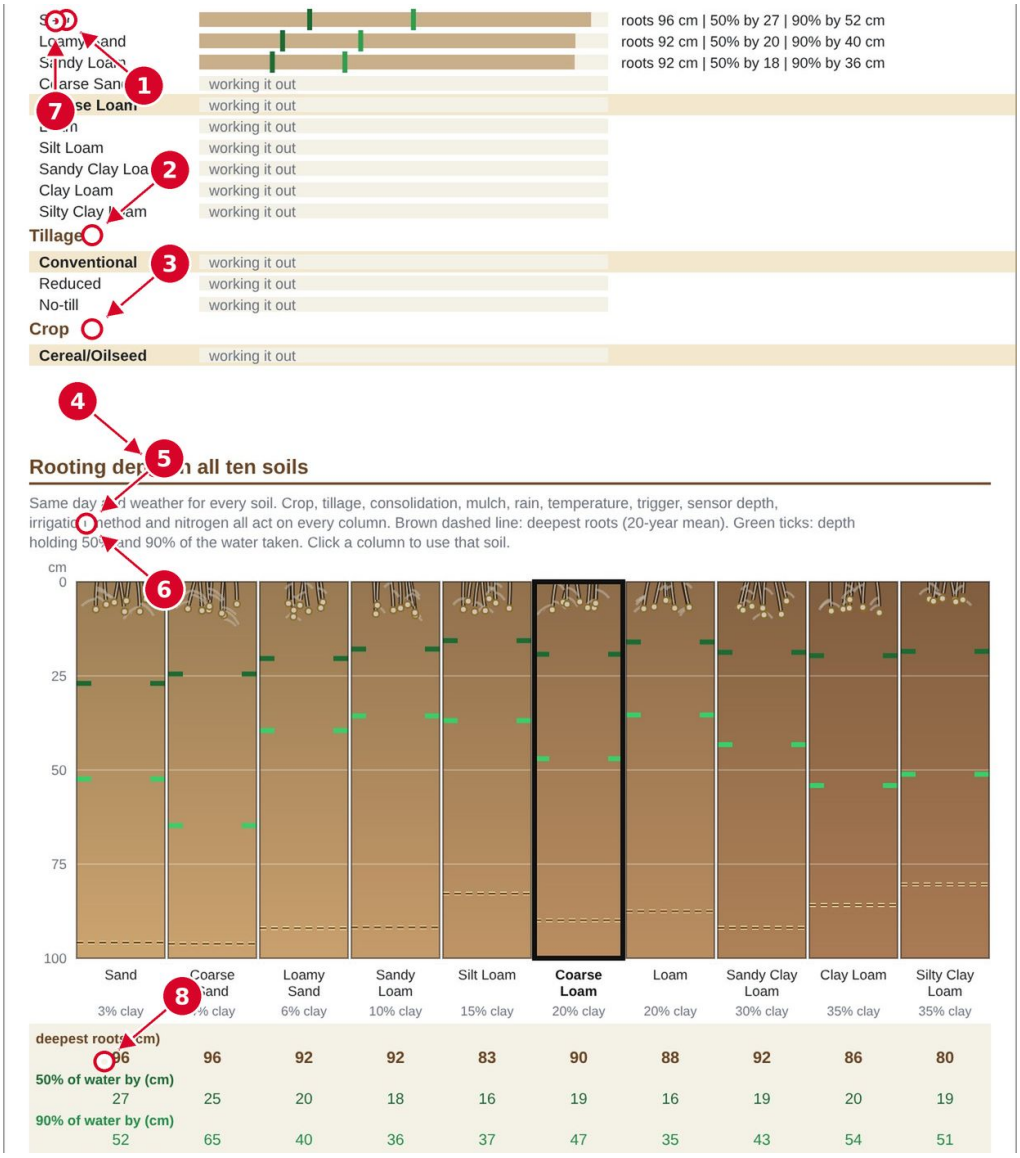



Figure 26.187: Root structure, part 3 of 4.

**What the arrows point to.**

- **1** The soil selected, highlighted in the list.
- **2** The tillage systems compared, conventional, reduced and no-till.
- **3** The crops compared, cereal, legume and vegetable.
- **4** The rooting depth in all ten soils, on the same day and weather.

- **5** The soil column of the selected soil.
- **6** The dashed line, the deepest roots of the 20-year mean.
- **7** The soil name and its clay content under the column.
- **8** The table of the deepest roots and of the uptake depths of the ten soils.

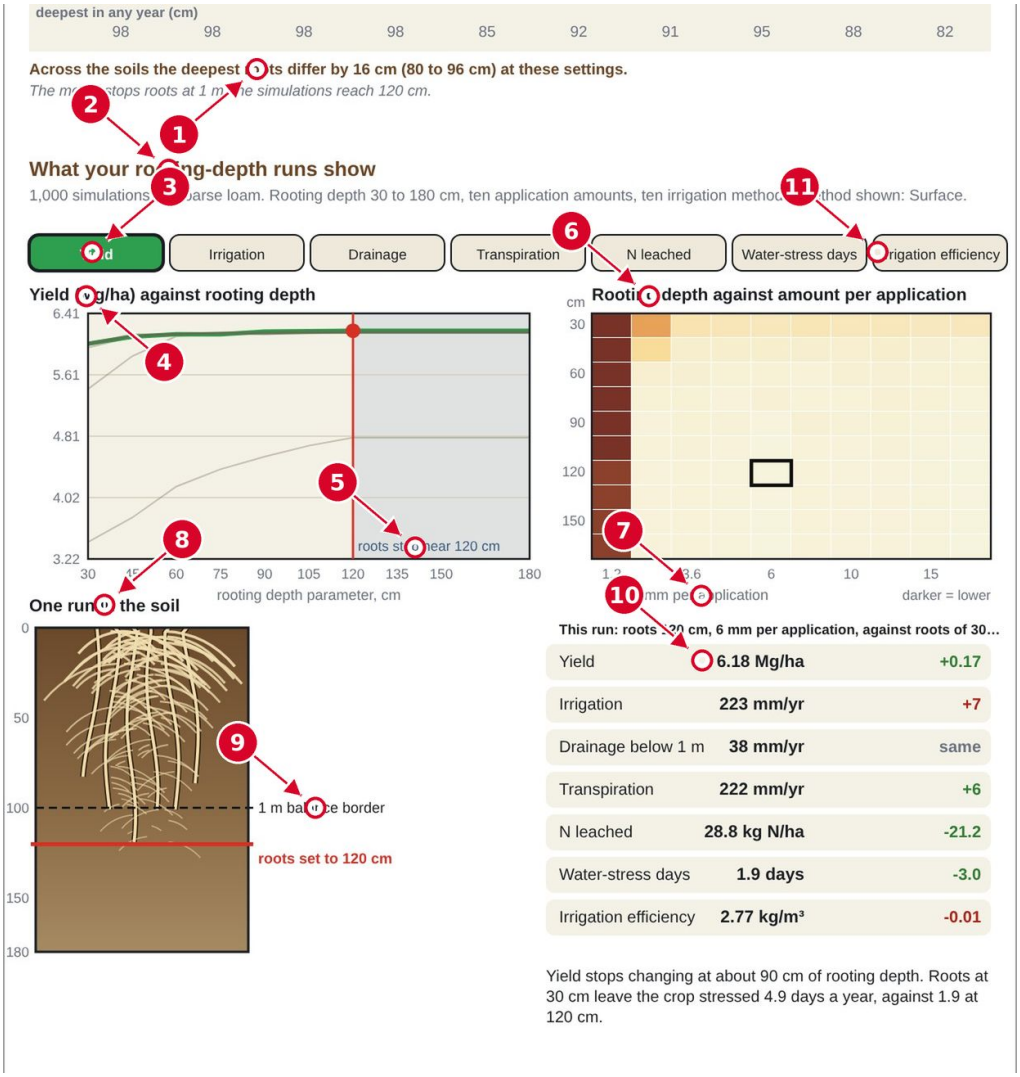



Figure 26.188: Root structure, part 4 of 4.

**What the arrows point to.**

- **1** The range of the deepest roots across the soils at these settings.
- **2** The rooting-depth runs, 1,000 Daisy runs on coarse loam.
- **3** The metric buttons, Yield, Irrigation, Drainage, Transpiration, N leached, Water-stress days and Irrigation efficiency.
- **4** The yield against the rooting depth for ten application amounts.
- **5** The red line, where the roots stop near 120 cm.
- **6** The map of the rooting depth against the amount per application.
- **7** The black square, the run selected.
- **8** One run in the soil, with the roots drawn to scale.
- **9** The dashed line at 1 m, the border of the balance.
- **10** The table of the run against the reference run with roots of 30 cm.
- **11** The water-stress days of the run and their change.

### 26.12.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

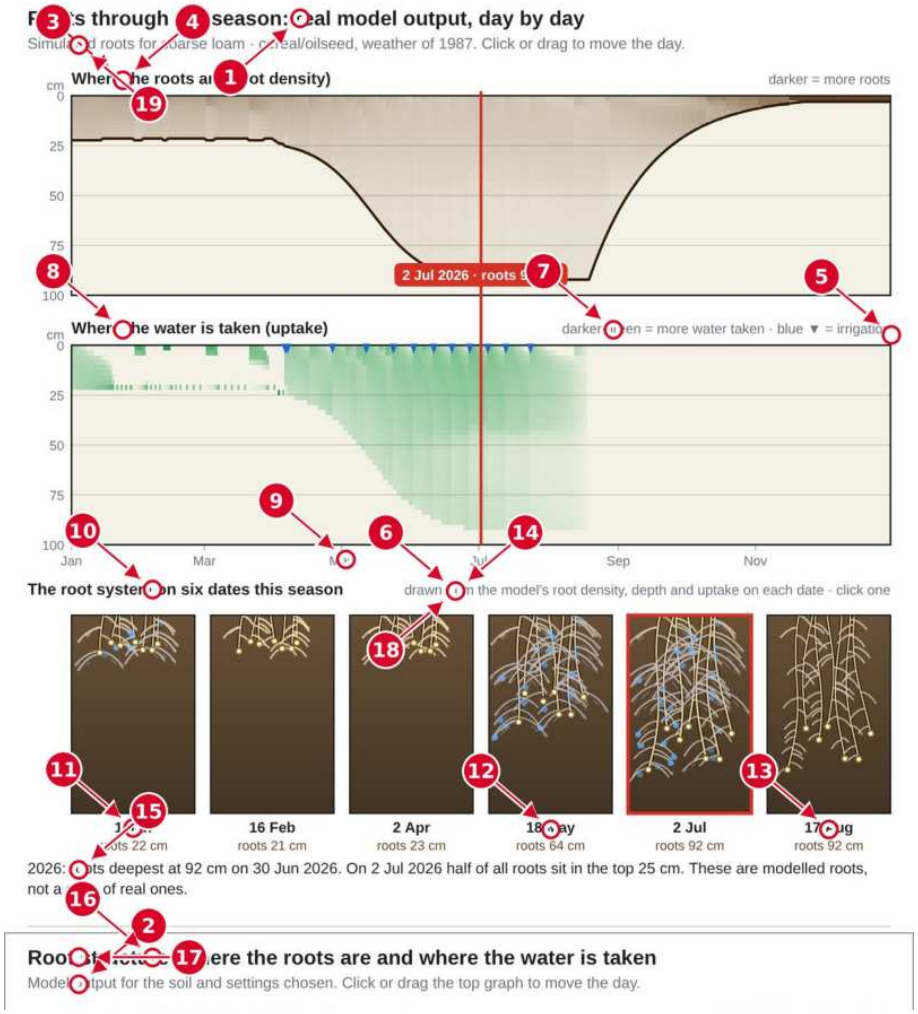



Figure 26.189: Root structure, part 1. The top graph clicked in the middle of the season, so the day on the clock moves there.

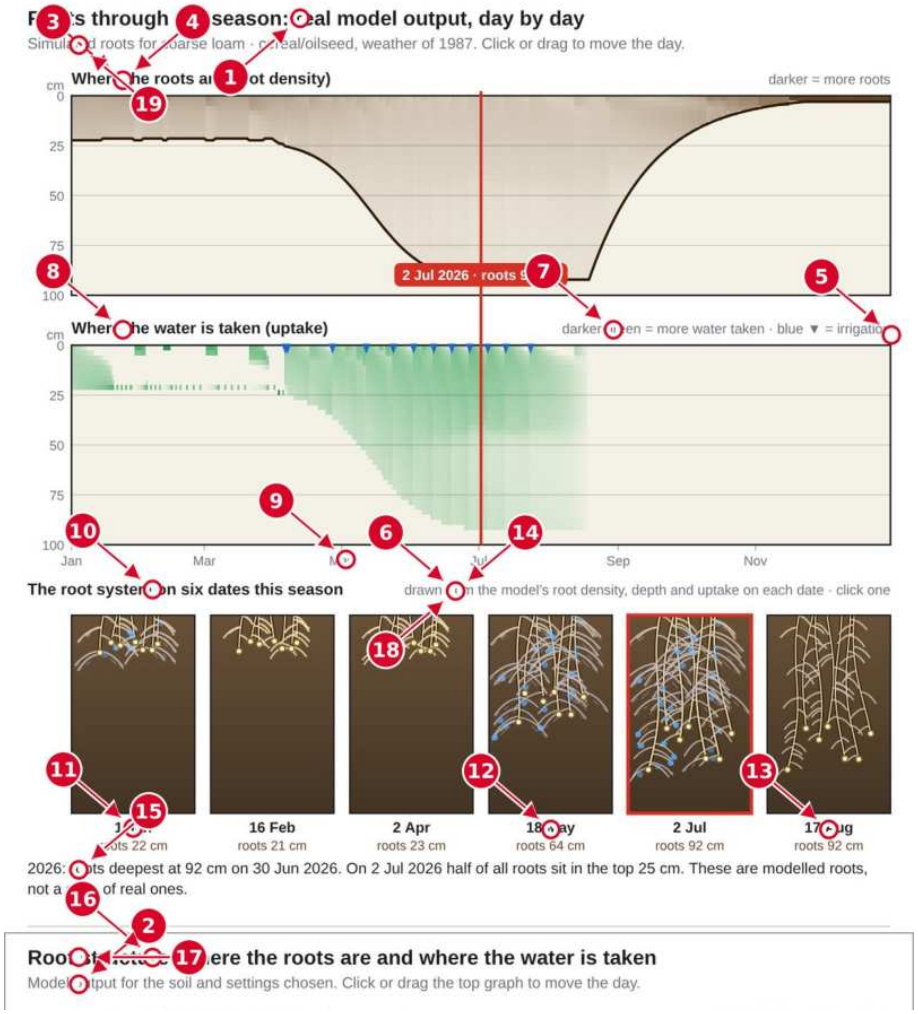



Figure 26.190: Root structure, part 1. The top graph clicked in the middle of the season, so the day on the clock moves there.

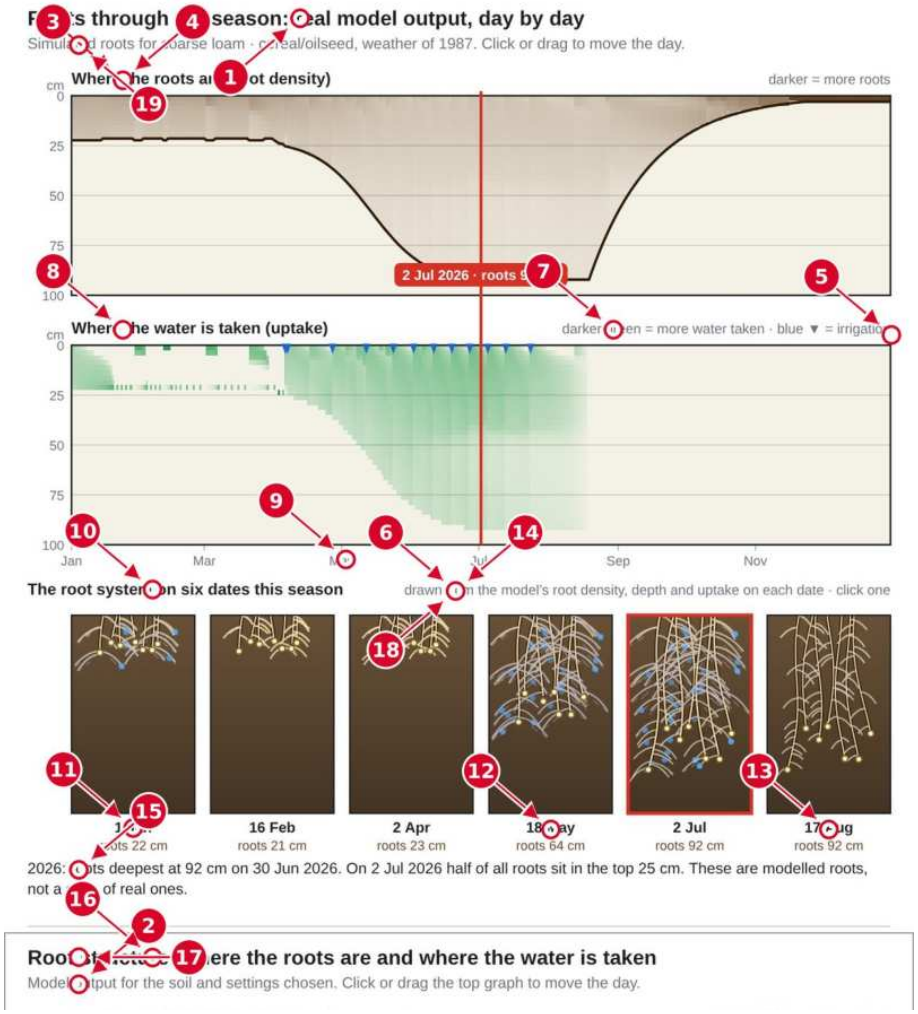



Figure 26.191: Root structure, part 1. The top graph clicked late in the season, so the day on the clock moves there.

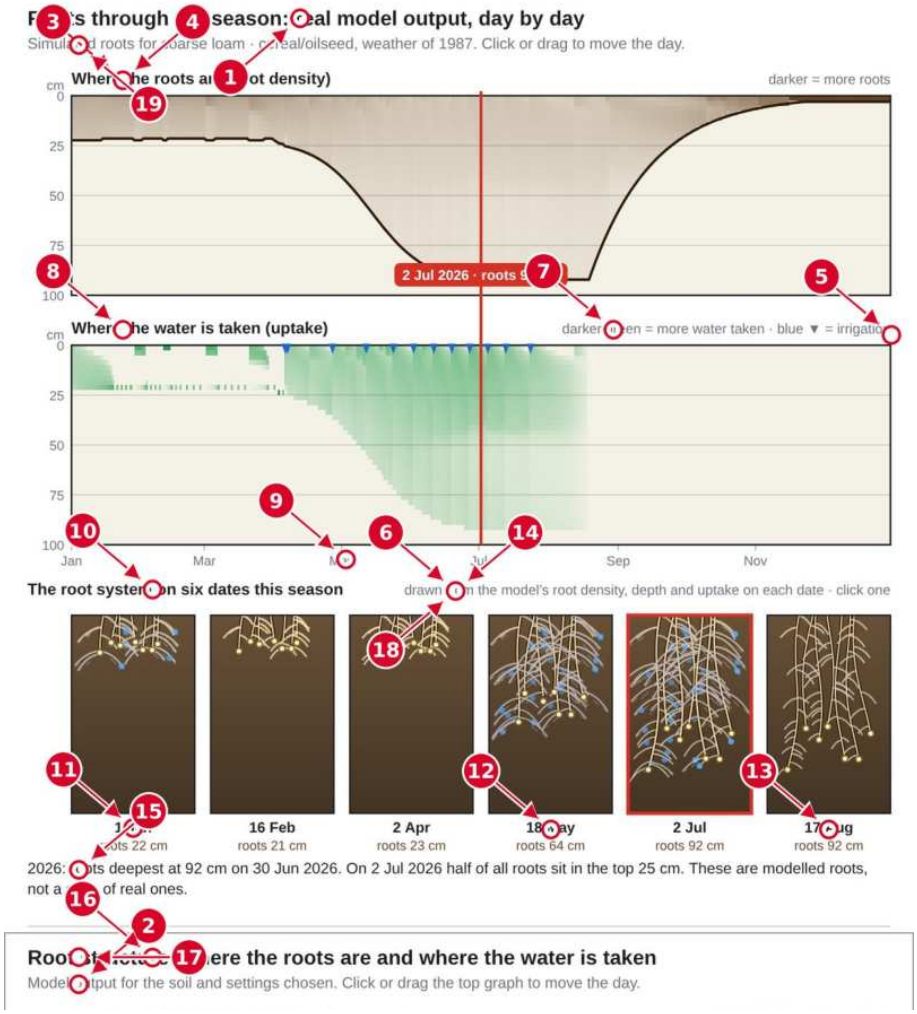



Figure 26.192: Root structure, part 1. The top graph clicked late in the season, so the day on the clock moves there.

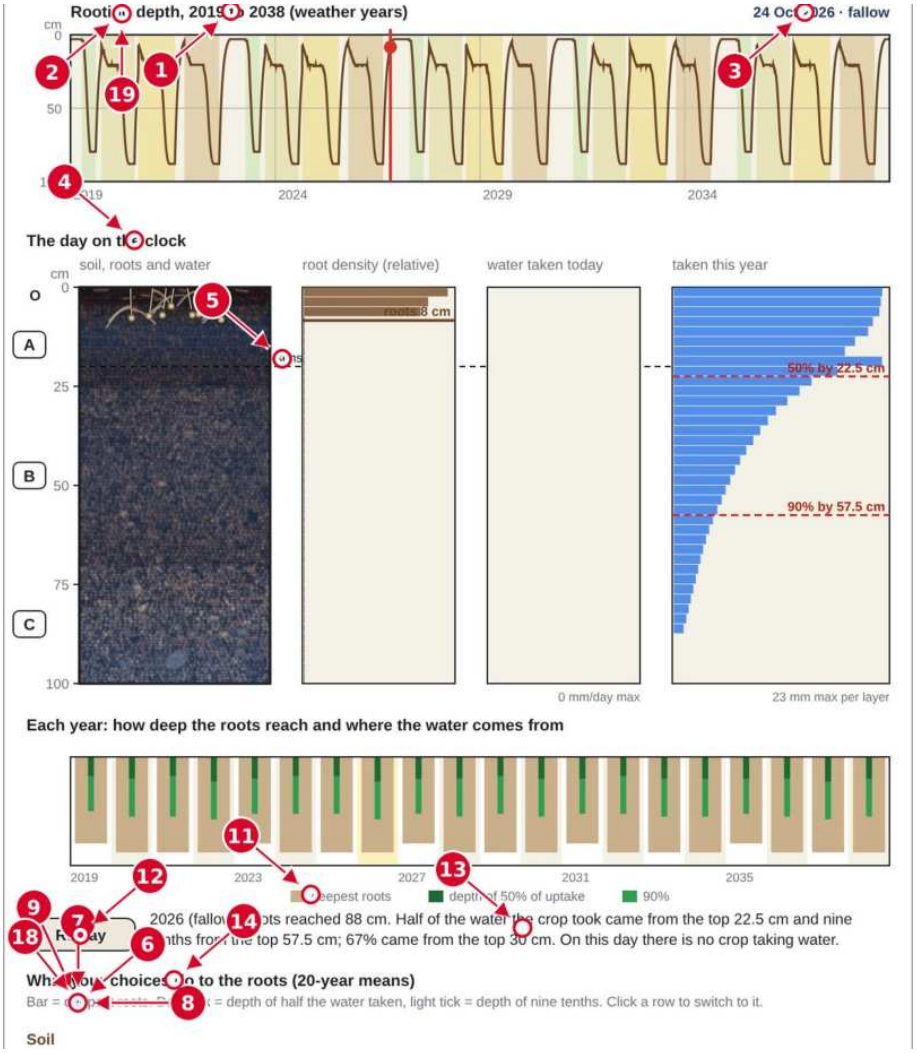



Figure 26.193: Root structure, part 2. A soil row of the list clicked, so the page switches to that soil.

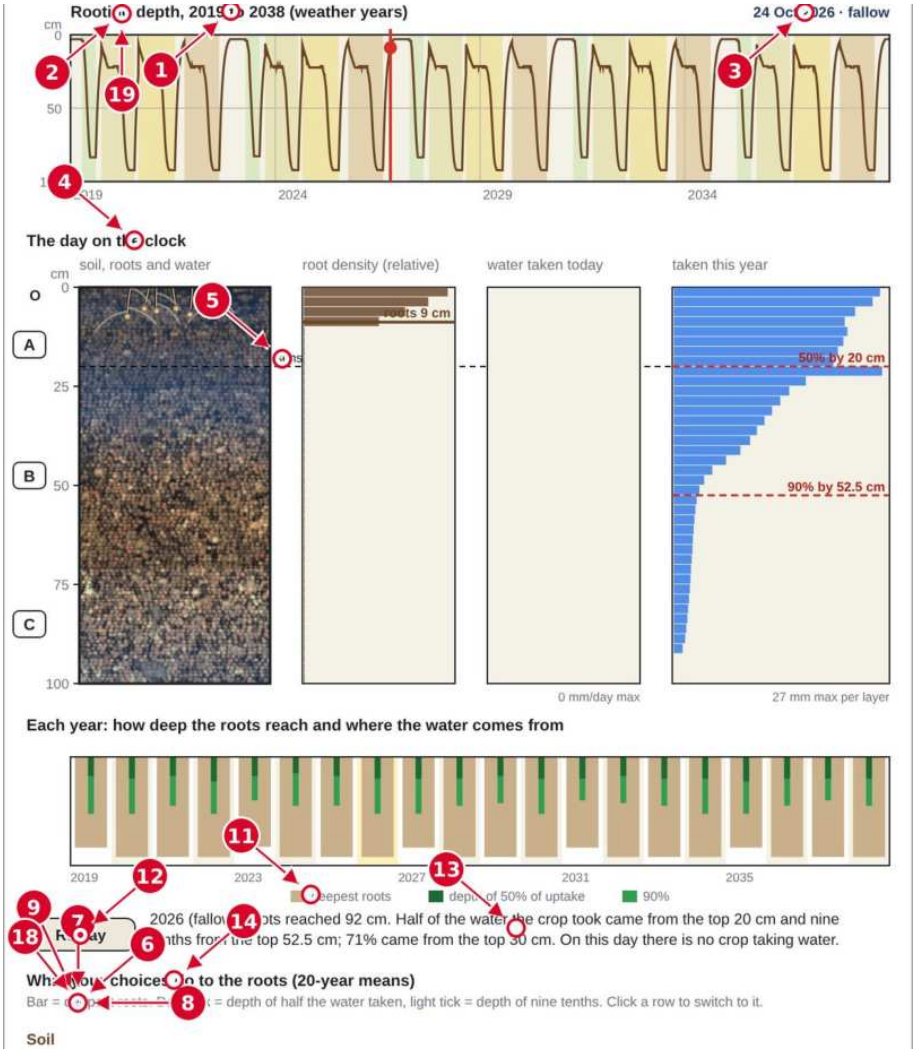



Figure 26.194: Root structure, part 2. The Irrigation button pressed, so the runs show the irrigation against the rooting depth.

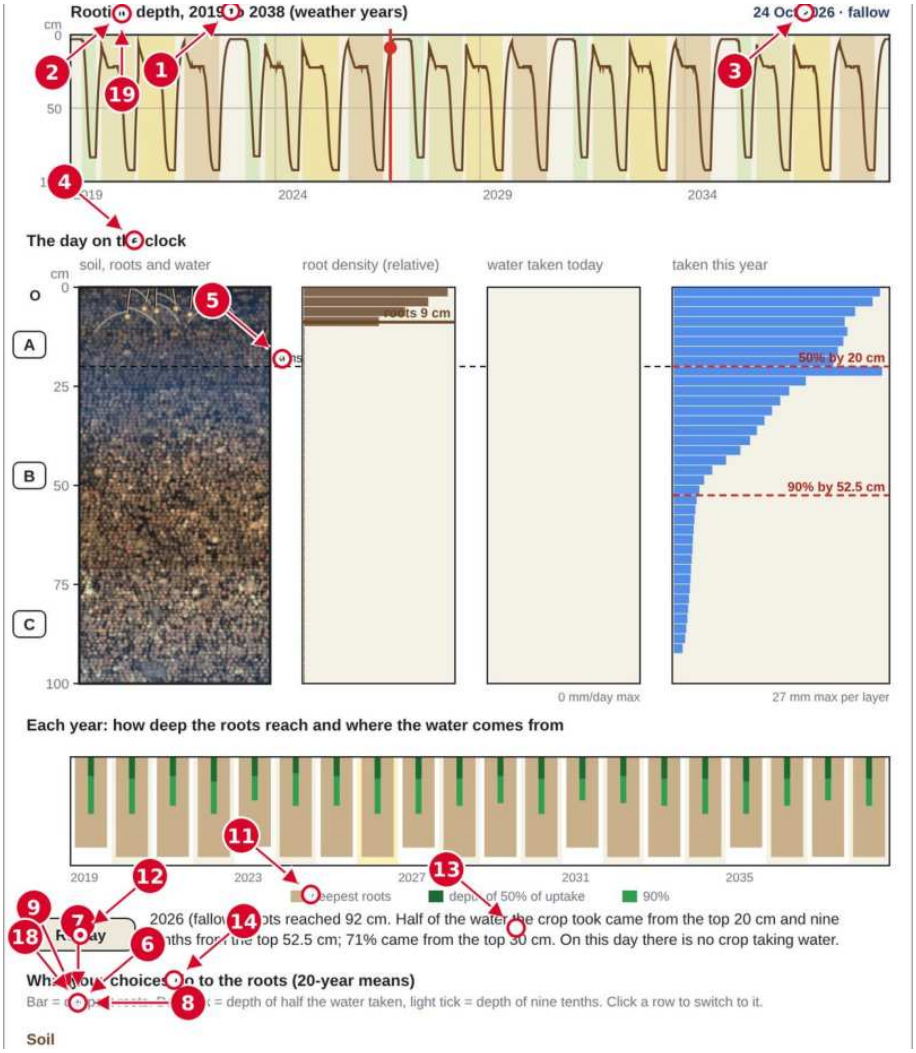



Figure 26.195: Root structure, part 2. The Drainage button pressed, so the runs show the drainage against the rooting depth.

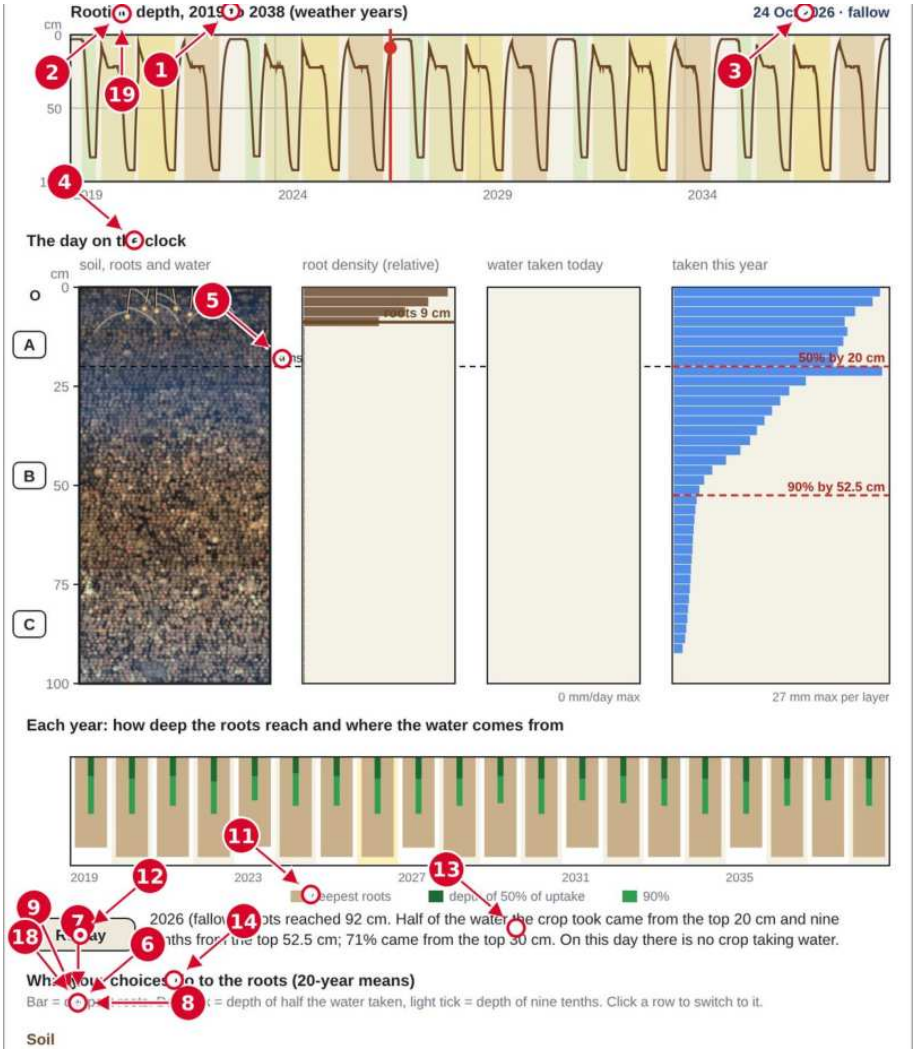



Figure 26.196: Root structure, part 2. The Transpiration button pressed.

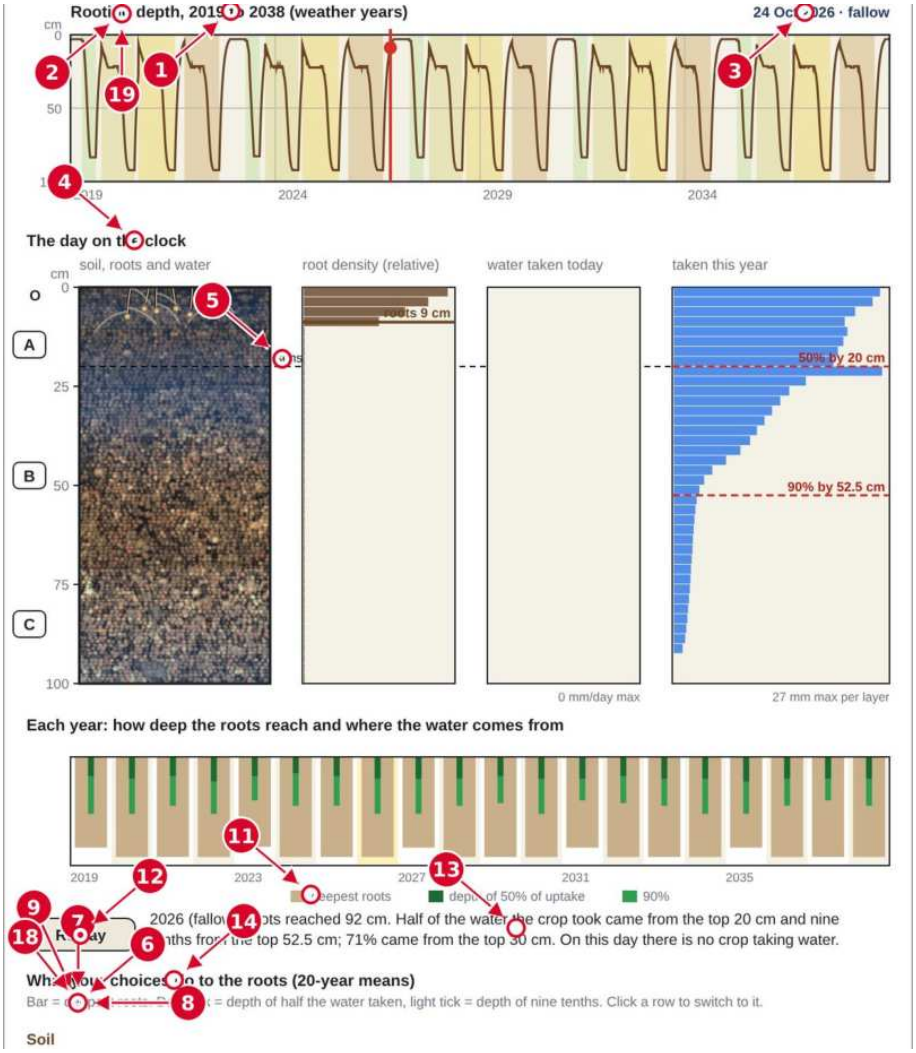



Figure 26.197: Root structure, part 2. The N leached button pressed.

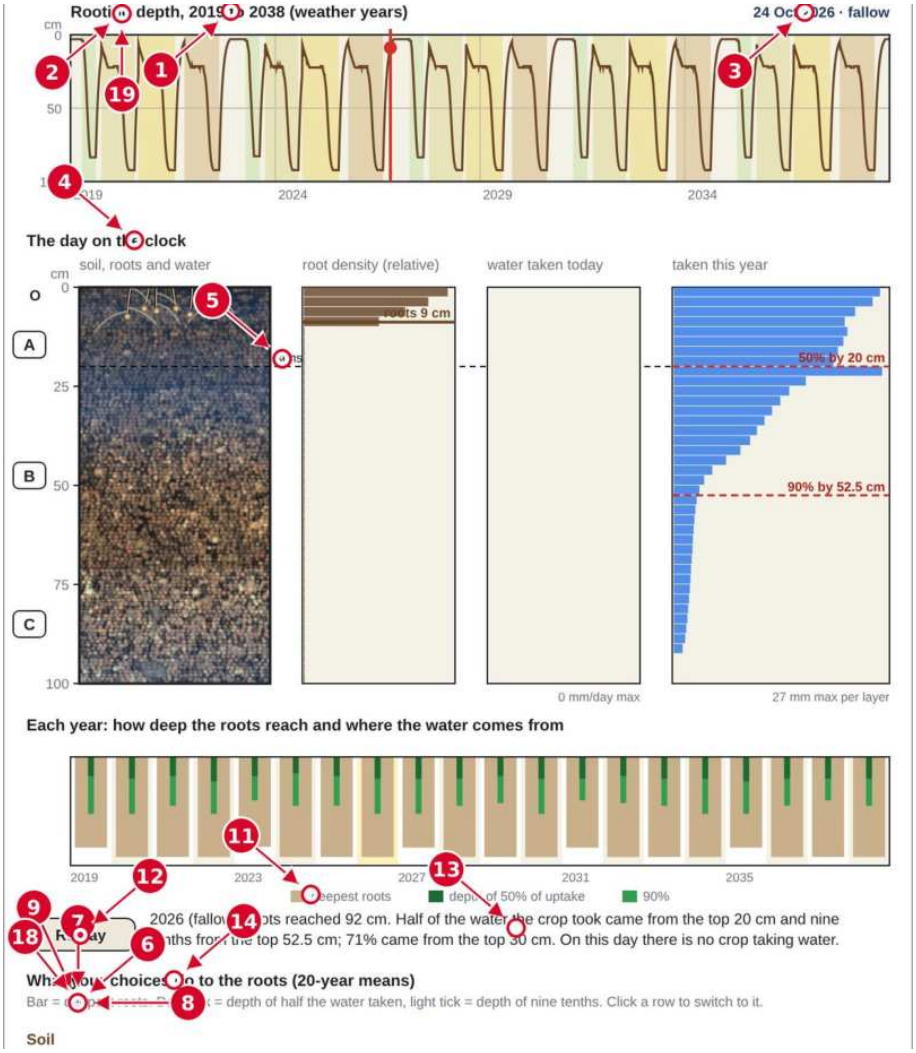



Figure 26.198: Root structure, part 2. The Water-stress days button pressed.

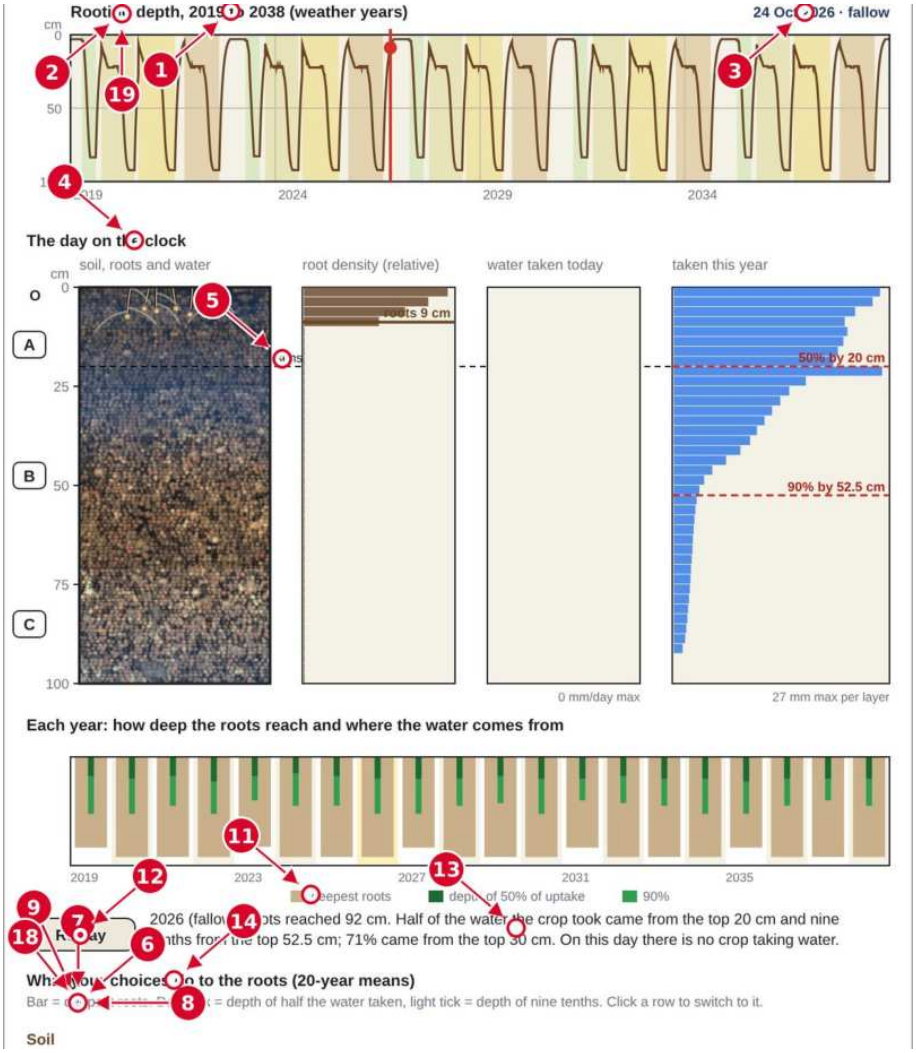



Figure 26.199: Root structure, part 2. The Irrigation efficiency button pressed.

# 26.13 Mulch against bare soil

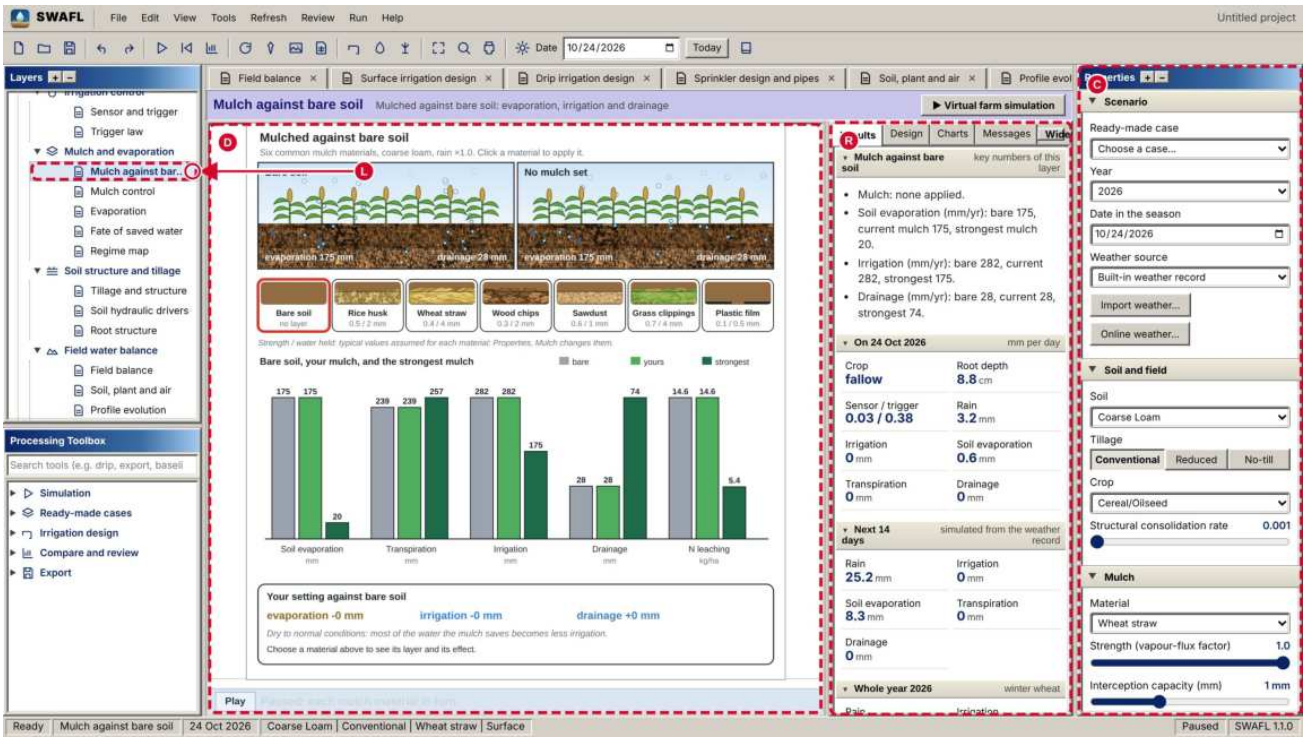



Figure 26.200: Mulch against bare soil: the whole workbench.

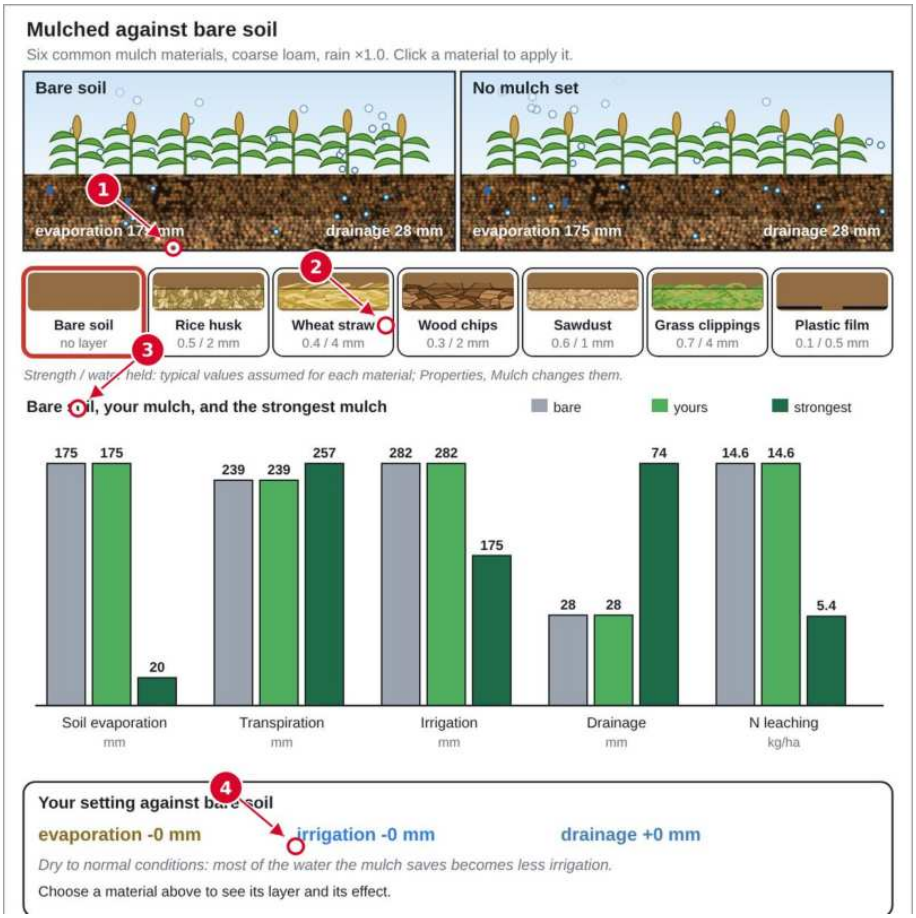



Figure 26.201: Mulch against bare soil: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bare-soil reference: Engine 1 at mulch factor 1, Eq. 5.4.
- **2** Selected mulch (strength phi, capacity c): the input of Eq. 9.1.
- **3** Evaporation, transpiration, irrigation, drainage and nitrogen leaching of bare soil, your mulch and the strongest mulch: Eqs. 5.4, 5.7, 5.9 and 5.10.
- **4** Differences between bare soil and the selected mulch, Delta E, Delta I, Delta D: Eq. 9.1.

**Equations behind this screen.** Eq. 5.4, Eq. 5.7, Eq. 5.9, Eq. 9.1.

*Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.7. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.9. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 9.1. Where explained: chapter "Mulch and evaporation layers", section "Mulch against bare soil".*

$$\Delta E = E_{1.0} - E_\varphi, \qquad \Delta I = I_{1.0} - I_\varphi, \qquad \Delta D = D_\varphi - D_{1.0}, \tag{9.1}$$

*Provenance (Eq. 9.1):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

### 26.13.1 The whole layer, part by part

The picture below is the complete drawing of the layer Mulch against bare soil, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

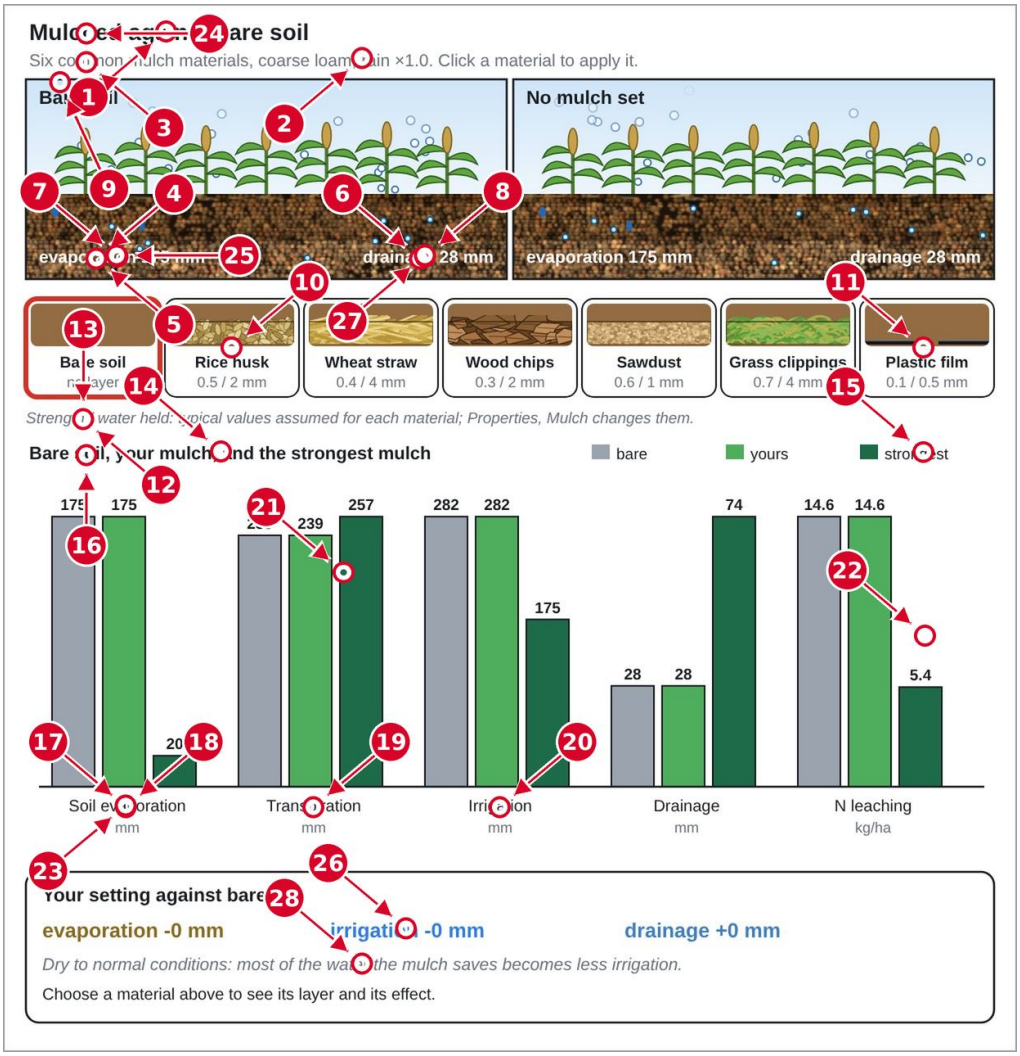



Figure 26.202: Mulch against bare soil, part 1 of 1.

**What the arrows point to.**

- **1** The title, the mulched soil against the bare soil.
- **2** The instruction, a material is clicked to apply it.
- **3** The bare soil panel, with its evaporation and drainage for the year.
- **4** The panel of the mulch selected, here rice husk.
- **5** The mulch layer drawn on top of the soil.
- **6** The close-up lens that shows the mulch material.
- **7** The evaporation of the bare soil in mm.
- **8** The drainage of the mulched soil in mm.

- **9** The bare soil card, no layer.
- **10** The material selected, highlighted in red.
- **11** The plastic film card, the last of the six materials.
- **12** The strength and the water held by the material, in mm of rain.
- **13** The note that the typical values are assumed and the value boxs change them.
- **14** The chart of the bare soil, your mulch and the strongest mulch.
- **15** The legend of the three bars.
- **16** The bar of the bare soil, grey.
- **17** The bar of the mulch selected, light green.
- **18** The bar of the strongest mulch, dark green.
- **19** The transpiration group, nearly the same for the three.
- **20** The irrigation group, lower under mulch.
- **21** The drainage group, higher under mulch.
- **22** The nitrogen leaching group, in kg per ha.
- **23** The unit of each group.
- **24** The result box, the mulch against the bare soil.
- **25** The change of the evaporation.
- **26** The change of the irrigation.
- **27** The change of the drainage.
- **28** The note that explains where the saved water goes.

### 26.13.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

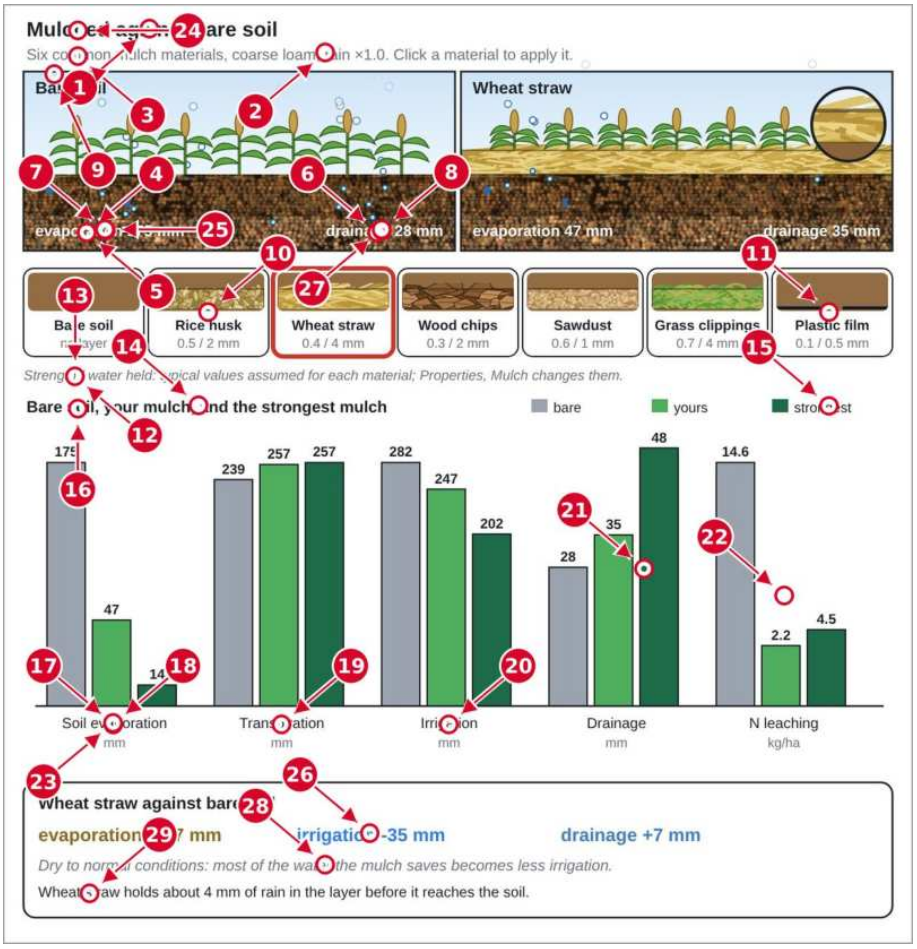



Figure 26.203: Mulch against bare soil, part 1. Wheat straw chosen as the material.

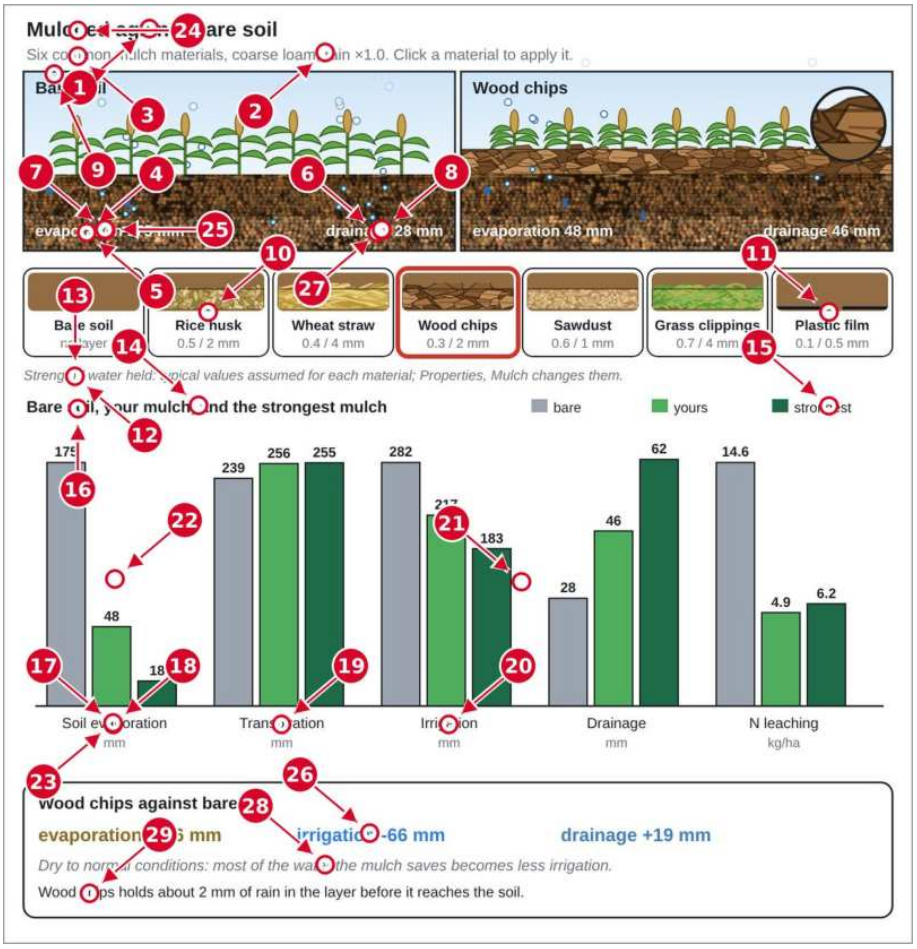



Figure 26.204: Mulch against bare soil, part 1. Wood chips chosen as the material.

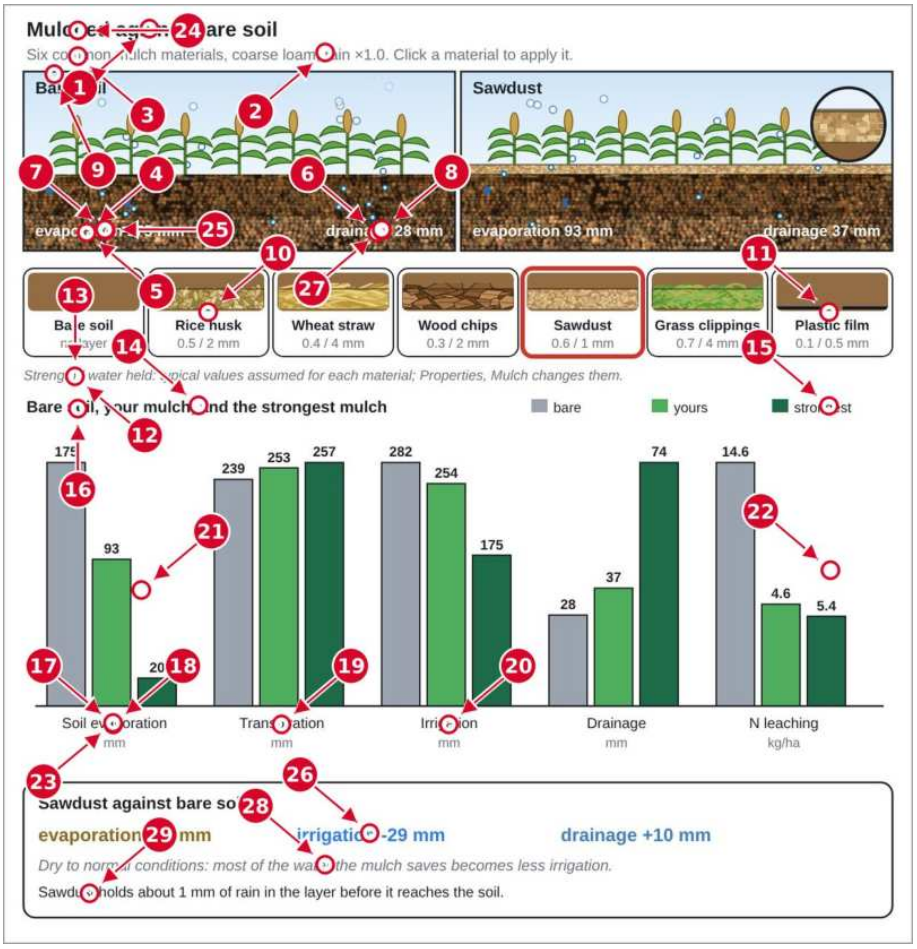



Figure 26.205: Mulch against bare soil, part 1. Sawdust chosen as the material.

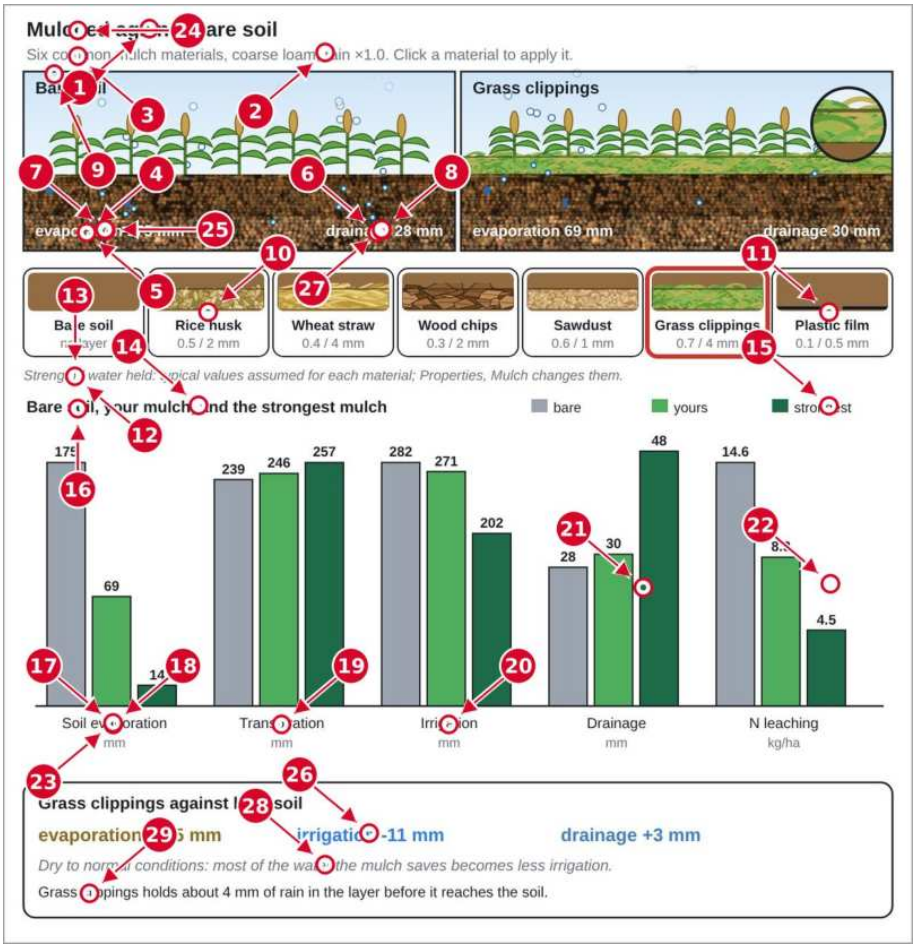



Figure 26.206: Mulch against bare soil, part 1. Grass clippings chosen as the material.

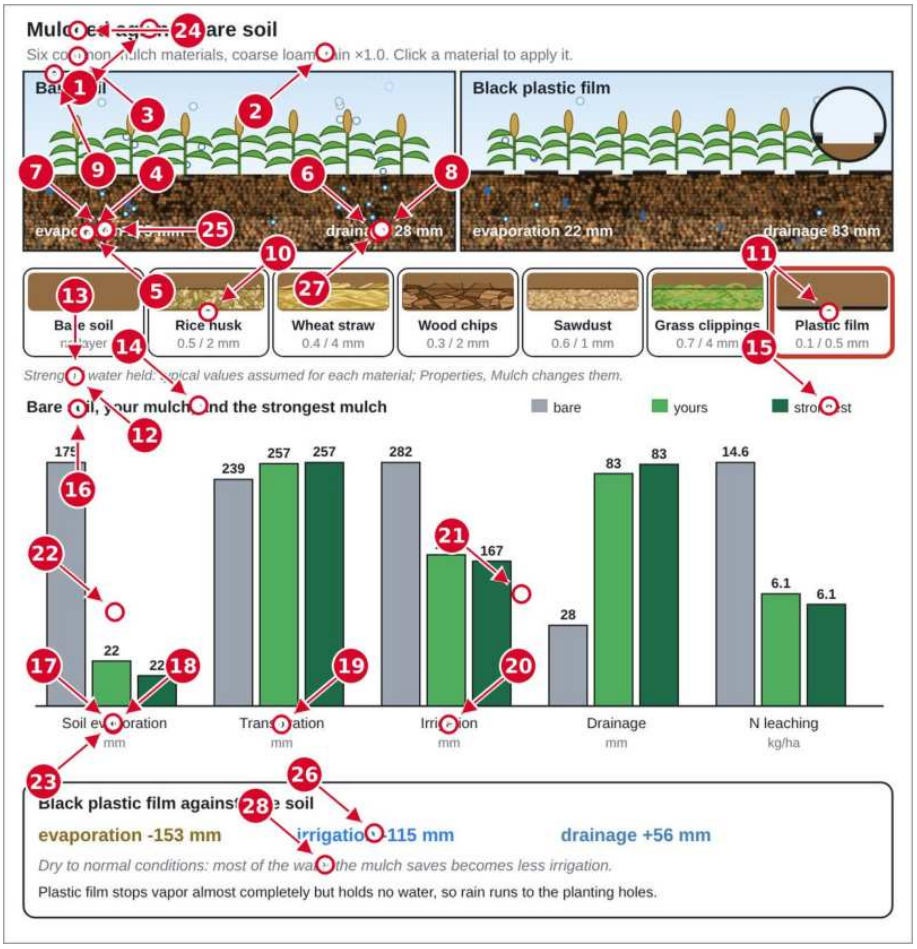



Figure 26.207: Mulch against bare soil, part 1. Plastic film chosen as the material.

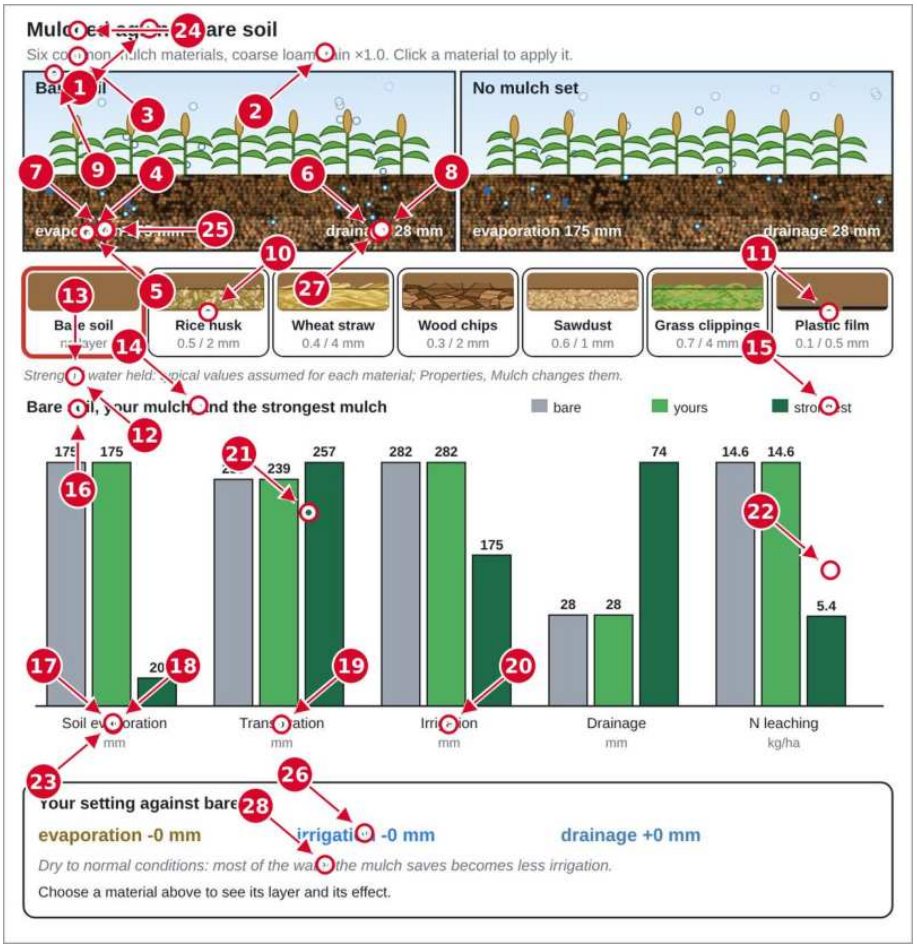



Figure 26.208: Mulch against bare soil, part 1. Bare soil chosen, so the comparison shows no change.

# 26.14 Mulch control

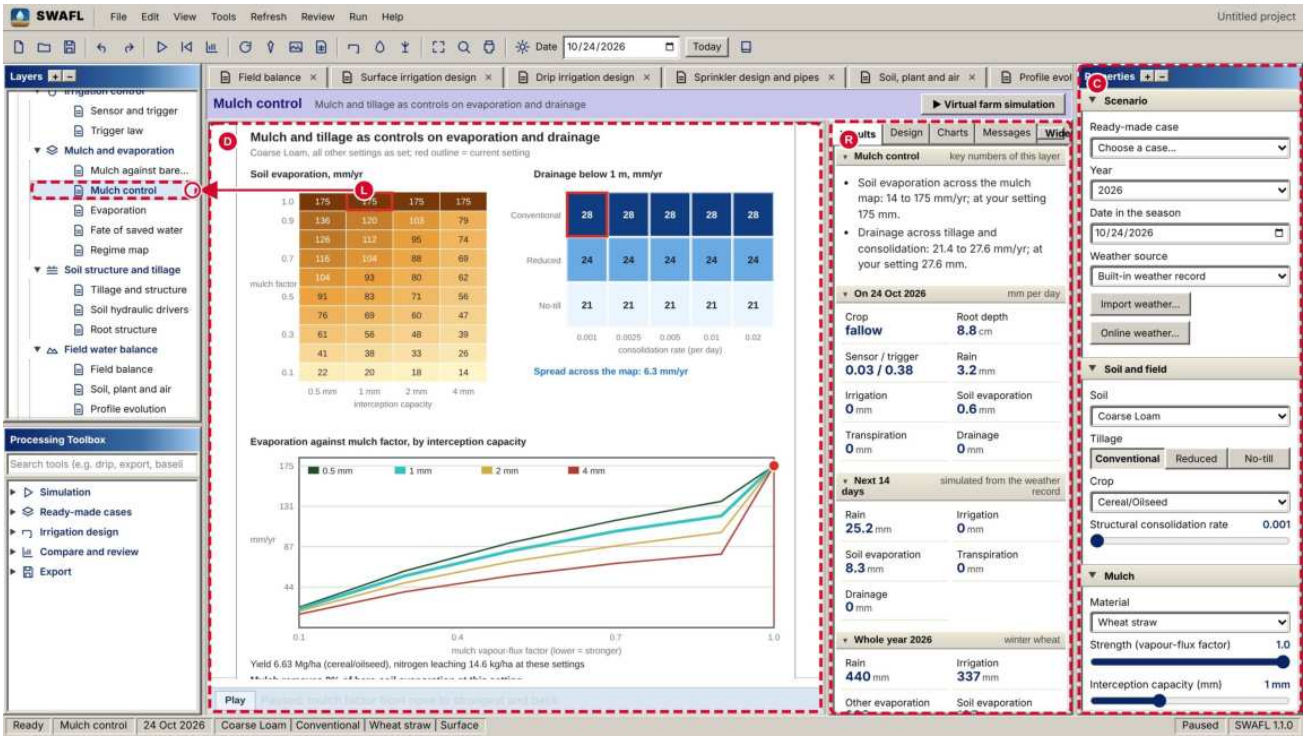



Figure 26.209: Mulch control: the whole workbench.

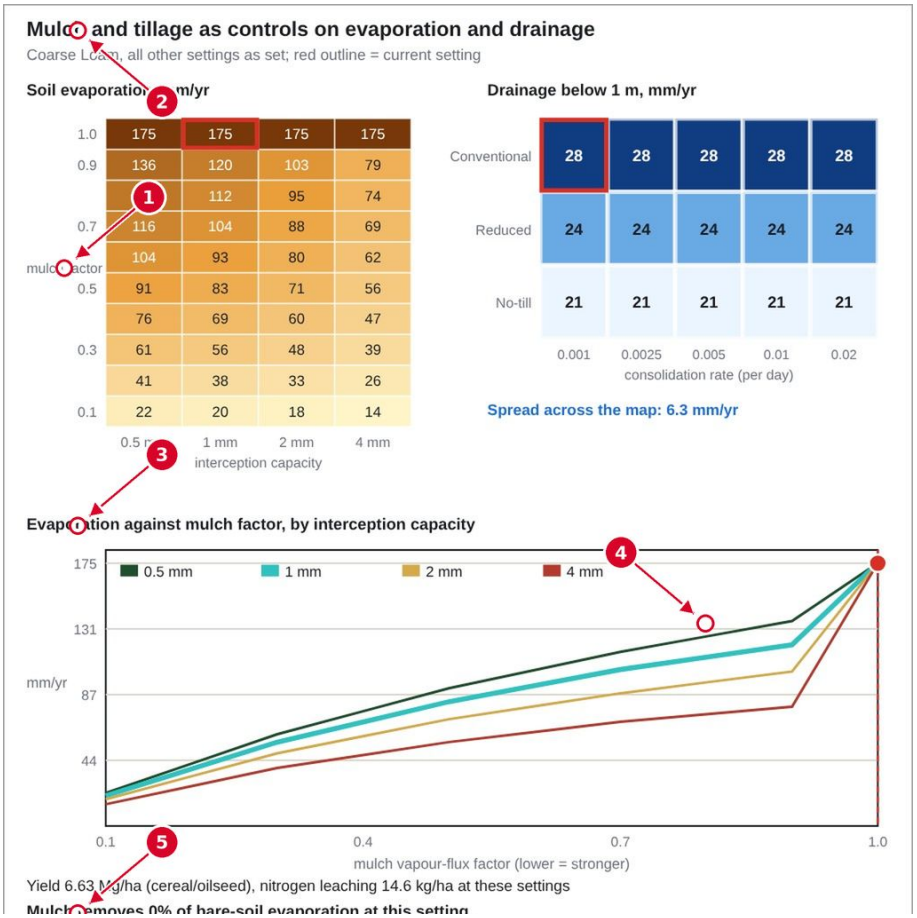



Figure 26.210: Mulch control: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Matrix of soil evaporation E(phi, c) for each mulch factor and interception capacity: Eq. 5.4.
- **2** Matrix of drainage D(tillage, consolidation rate): Eq. 5.9.
- **3** Evaporation against mulch factor for four capacities: Eq. 5.4.
- **4** Red dot: your current setting.
- **5** “Mulch removes 31 % of bare-soil evaporation”: Eq. 9.1.

**Equations behind this screen.** Eq. 5.4, Eq. 5.9.

*Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

### 26.14.1 The whole layer, part by part

The picture below is the complete drawing of the layer Mulch control, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

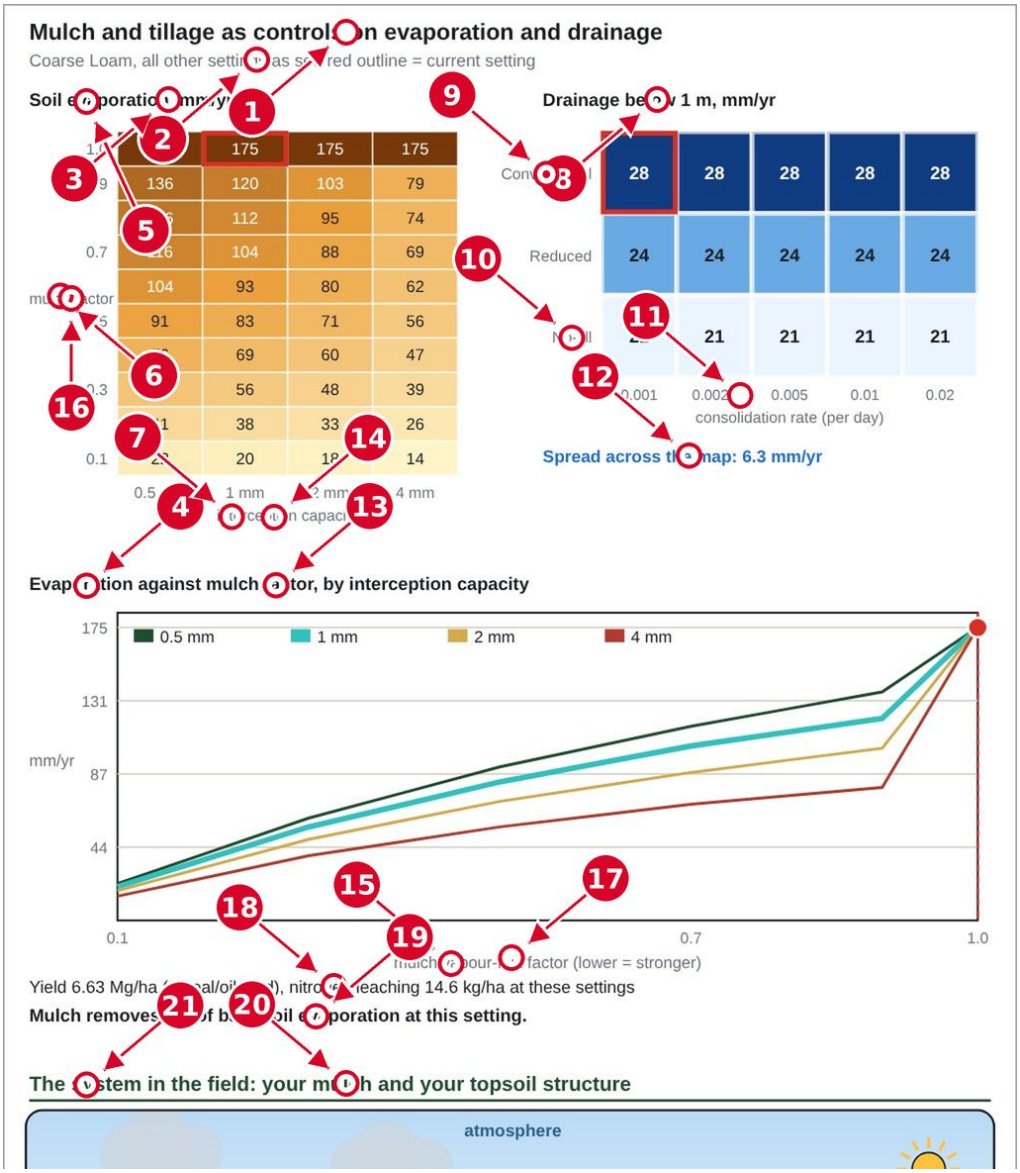



Figure 26.211: Mulch control, part 1 of 2.

**What the arrows point to.**

- **1** The title, mulch and tillage as controls on evaporation and drainage.
- **2** The note that the red outline marks the current setting.
- **3** The map of the soil evaporation in mm per year.
- **4** The red outlined cell, the evaporation at the current mulch factor and interception capacity.
- **5** The top row, no mulch effect, the bare soil evaporation.
- **6** The mulch factor axis, from 1.0 for no mulch down to 0.1 for the strongest.
- **7** The interception capacity axis, from 0.5 to 4 mm.
- **8** The map of the drainage below 1 m in mm per year.
- **9** The red outlined cell, the drainage under the current tillage and consolidation rate.
- **10** The no-till row, with the smallest drainage.
- **11** The consolidation rate axis, per day.
- **12** The spread of the drainage across the map.
- **13** The curves of the evaporation against the mulch factor.
- **14** The legend of the interception capacities.

- **15** The thick line, the curve of the capacity in force.
- **16** The red dashed line, the current mulch factor.
- **17** The mulch vapour-flux factor axis.
- **18** The yield and the nitrogen leaching at these settings.
- **19** The share of the bare-soil evaporation that the mulch removes.
- **20** The system in the field, your mulch and your topsoil structure.
- **21** The lens that shows the mulch material.

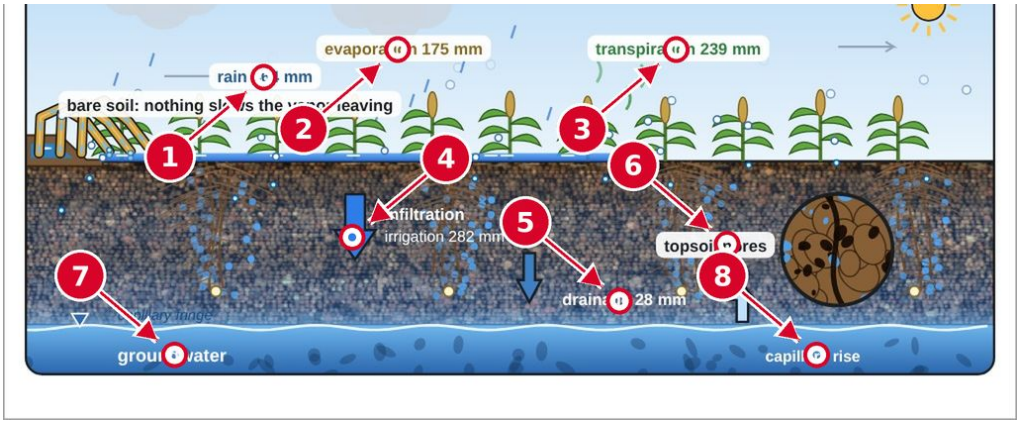



Figure 26.212: Mulch control, part 2 of 2.

**What the arrows point to.**

- **1** The rain of the year.
- **2** The evaporation of the year under the mulch.
- **3** The transpiration of the year.
- **4** The infiltration arrow with the irrigation applied.
- **5** The drainage arrow.
- **6** The topsoil pores, drawn as an aggregate.
- **7** The groundwater.
- **8** The capillary rise.

## 26.14.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

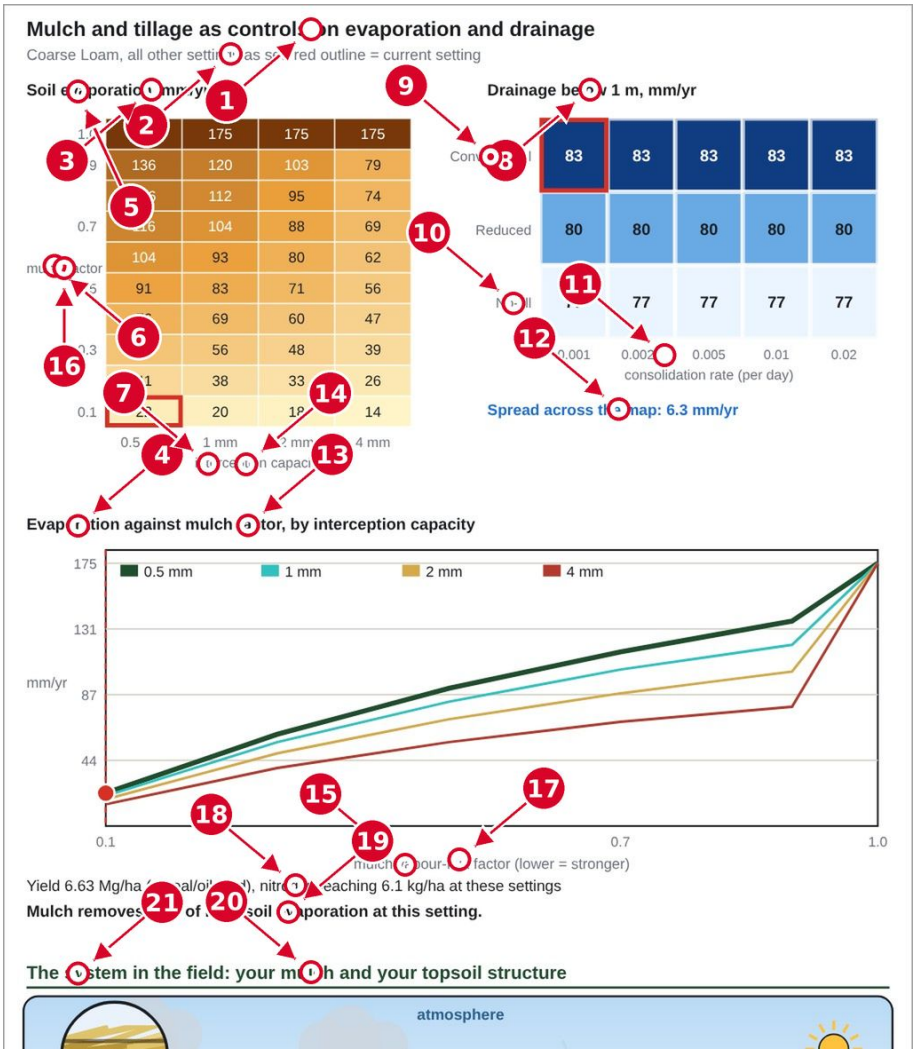



Figure 26.213: Mulch control, part 1. A cell of the evaporation map clicked, so that mulch factor and capacity become the current setting.

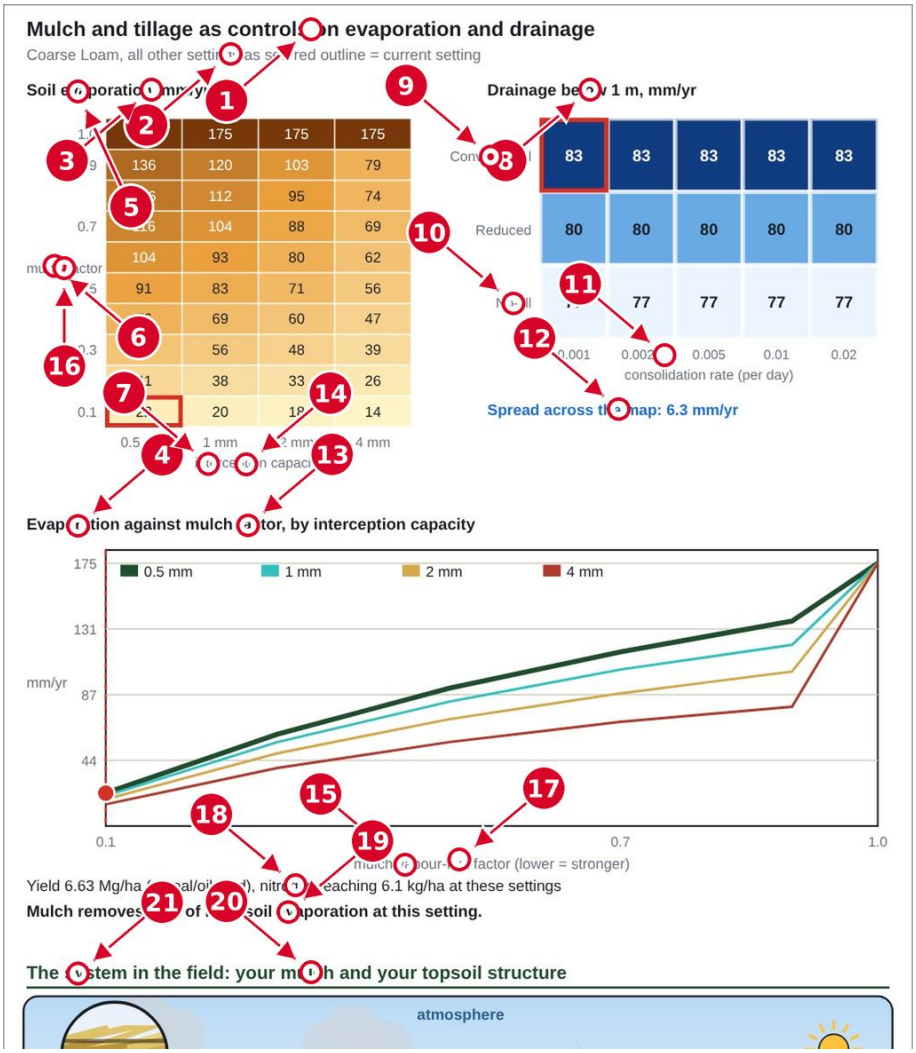



Figure 26.214: Mulch control, part 1. A cell of the drainage map clicked, so that tillage and consolidation rate become the current setting.

# 26.15 Evaporation

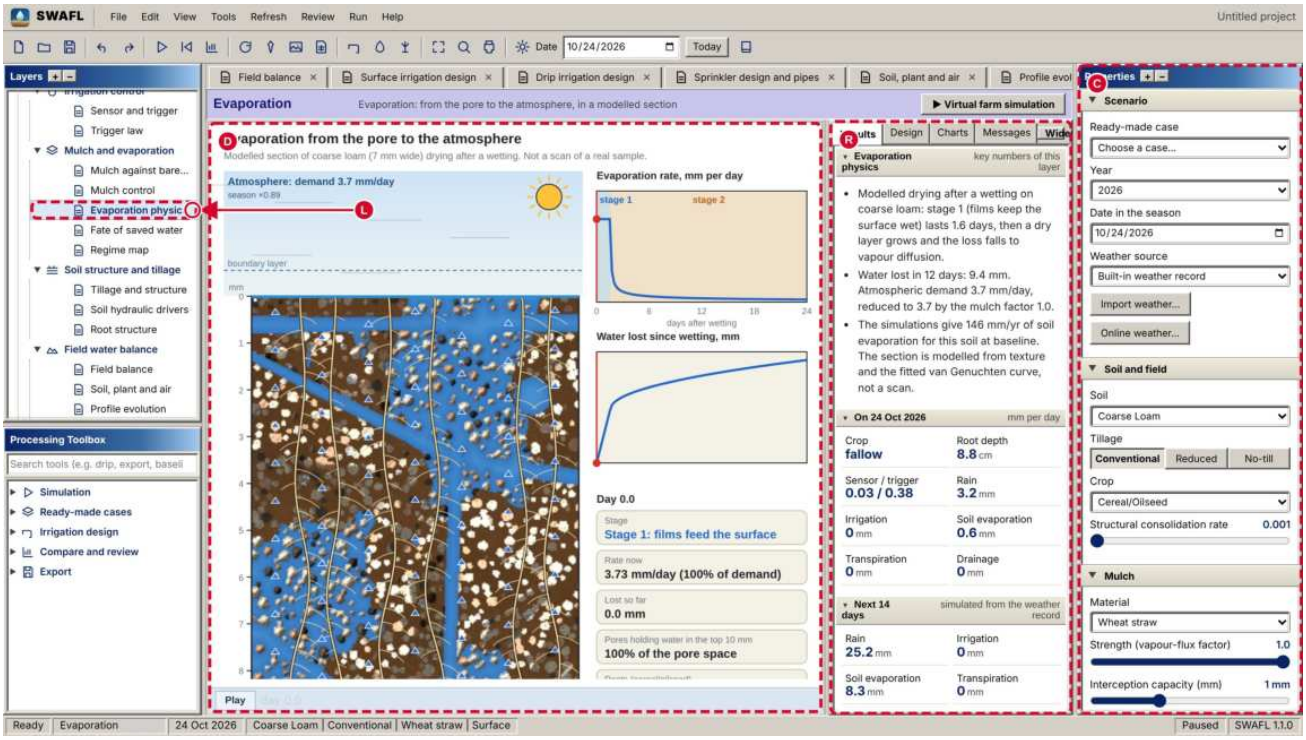



Figure 26.215: Evaporation: the whole workbench.

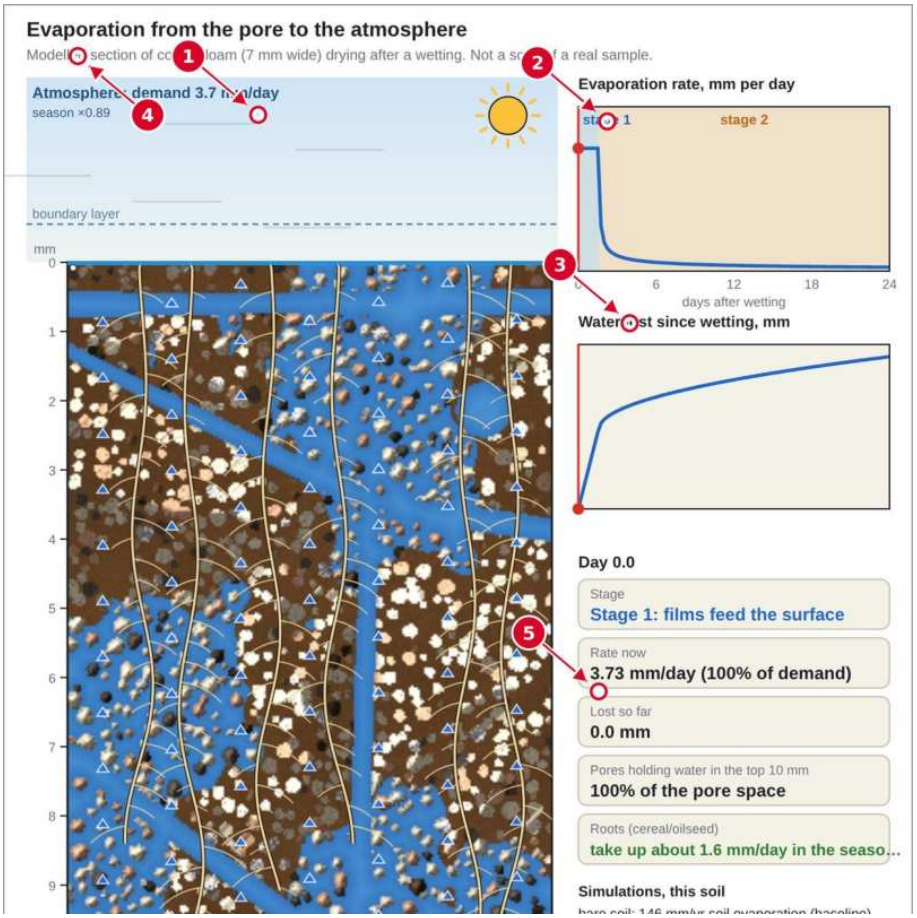



Figure 26.216: Evaporation: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Atmospheric demand under mulch and canopy, e_0 = e_0,b phi (1 minus 0.4 c): Eq. 5.26.
- **2** Evaporation rate against time since wetting, stage 1 then stage 2: Eq. 9.2.
- **3** Cumulative water lost since wetting, Q(t): Eq. 9.3.
- **4** Section of soil: dry layer depth (Eq. 9.4) and water-filled pores from the capillary rule (Eq. 8.4) with the suction profile of Eq. 9.5.
- **5** Panel with the stage, the rate now and the water lost so far: Eqs. 9.2 and 9.3.

**Equations behind this screen.** Eq. 5.4, Eq. 5.26, Eq. 8.3, Eq. 8.4, Eq. 9.2, Eq. 9.3, Eq. 9.4, Eq. 9.5.

*Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.26. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".*

$$E_{0,\text{bare}} = 4.2\,(1 + 0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\frac{1}{2}[1 + \sin(2\pi(\text{doy} - 100)/365)]}{0.675}\,\text{clip}(1 - 0.12(\rho - 1);\ 0.85,\ 1.12)\sqrt{\gamma_y}\ \text{mm}\,\text{d}^{-1}, \tag{5.26}$$

*Provenance (Eq. 5.26):* **[Engineered]**: atmosphere factor with chosen constants.

*Eq. 8.3. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".*

$$\theta(h) = \theta_r + (\phi - \theta_r)\,S_e(h). \tag{8.3}$$

*Provenance (Eq. 8.3):* **[Literature]**: van Genuchten (1980).

*Eq. 8.4. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".*

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h}\ \approx\ \frac{0.149\ \text{cm}^2}{h}\quad(\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

*Provenance (Eq. 8.4):* **[Literature]**: Young-Laplace (Jurin) capillary law.

*Eq. 9.2. Where explained: chapter "Mulch and evaporation layers", section "Evaporation".*

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

*Provenance (Eq. 9.2):* **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

*Eq. 9.3. Where explained: chapter "Mulch and evaporation layers", section "Evaporation".*

$$Q(t) = q_0 + \begin{cases} e_0\,t, & t \le t_1 \\ (Q_1 - q_0) + 2\,e_0\,\tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

*Provenance (Eq. 9.3):* **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

*Eq. 9.4. Where explained: chapter "Mulch and evaporation layers", section "Evaporation".*

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm}\quad(Q > Q_1), \tag{9.4}$$

*Provenance (Eq. 9.4):* **[Engineered]**: drying-front geometry chosen for drawing.

*Eq. 9.5. Where explained: chapter “Mulch and evaporation layers”, section “Evaporation”.*

$$\log_{10} h(z) = \log_{10} h_s + (\log_{10} h_i - \log_{10} h_s)(1 - e^{-(z-\ell)/\lambda}) \quad (z \geq \ell), \qquad h = 10^5 \text{ cm for } z < \ell. \tag{9.5}$$

*Provenance (Eq. 9.5):* **[Engineered]**: drying-front geometry chosen for drawing.

### 26.15.1 The whole layer, part by part

The picture below is the complete drawing of the layer Evaporation, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

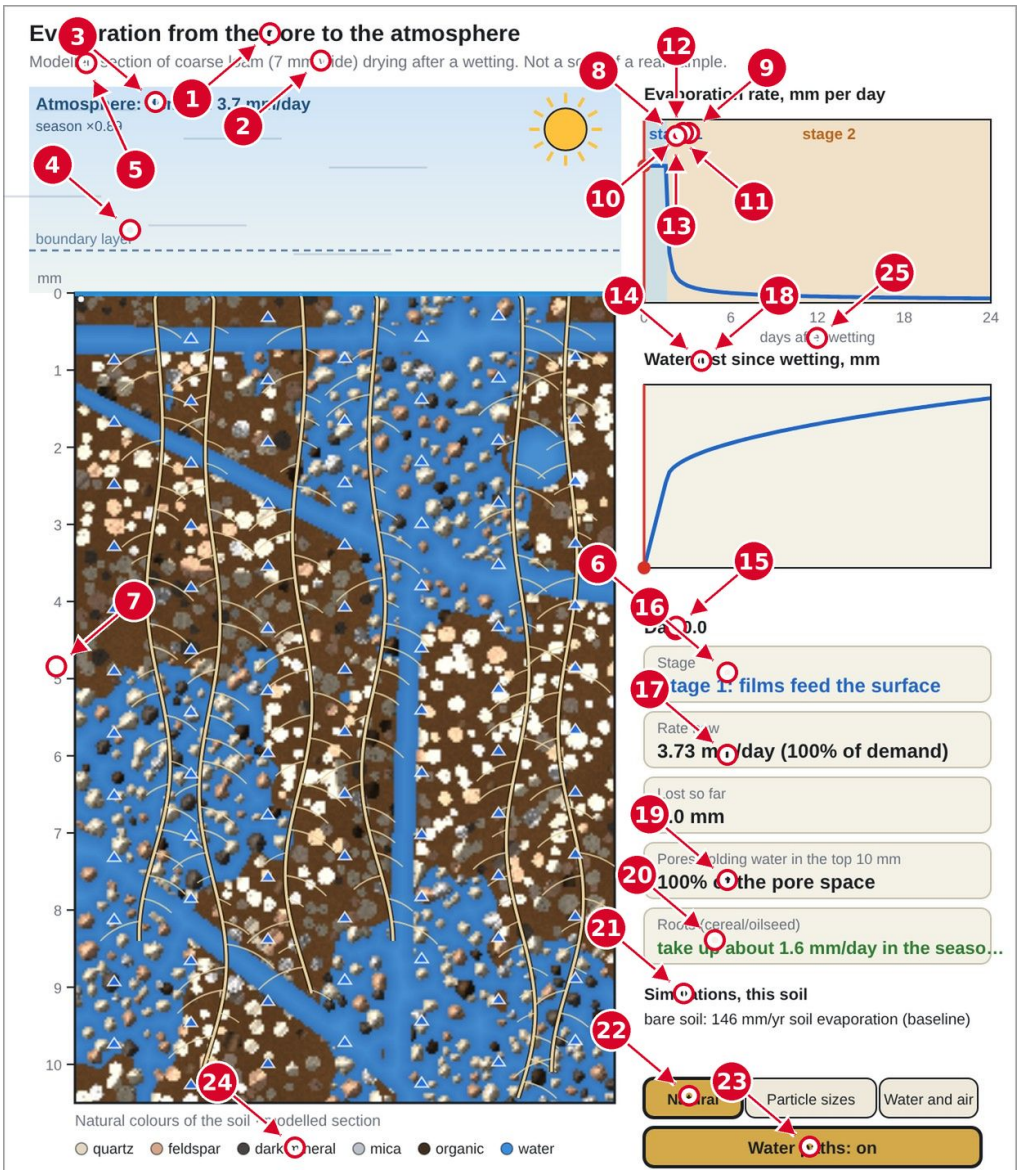



Figure 26.217: Evaporation, part 1 of 3.

**What the arrows point to.**

- **1** The title, the evaporation from the pore to the atmosphere.
- **2** The subtitle, a modelled section of the soil drying after a wetting.

- **3** The atmospheric demand in mm per day, with the season and weather scaling.
- **4** The boundary layer through which the vapour must pass.
- **5** The sun, the energy that drives the demand.
- **6** The modelled section of the soil, 7 mm wide, with grains, films and the water in blue.
- **7** The depth scale of the section in mm.
- **8** The chart of the evaporation rate in mm per day against the days after wetting.
- **9** The stage 1 band, the films feed the surface.
- **10** The stage 2 band, the dry layer limits the loss.
- **11** The blue line, the rate with the mulch.
- **12** The grey dashed line, the same soil without mulch.
- **13** The red dot, the day shown, which moves with the play bar.
- **14** The chart of the water lost since wetting in mm.
- **15** The day shown on the clock.
- **16** The stage card, whether the films feed the surface or the dry layer limits the loss.
- **17** The rate now in mm per day and its share of the demand.
- **18** The water lost so far and the water saved by the mulch.
- **19** The share of the pore space still holding water in the top 10 mm.
- **20** The roots card, the uptake of the crop.
- **21** The yearly figures from your Daisy simulations for this soil.
- **22** The colour buttons of the section, Natural, Particle sizes and Water and air.
- **23** The Water paths button, which draws the flow of the water and the vapour.
- **24** The legend of the minerals and the water.
- **25** The Day shown buttons, 0.5, 1, 1.75, 2.5, 4 and 7 days.

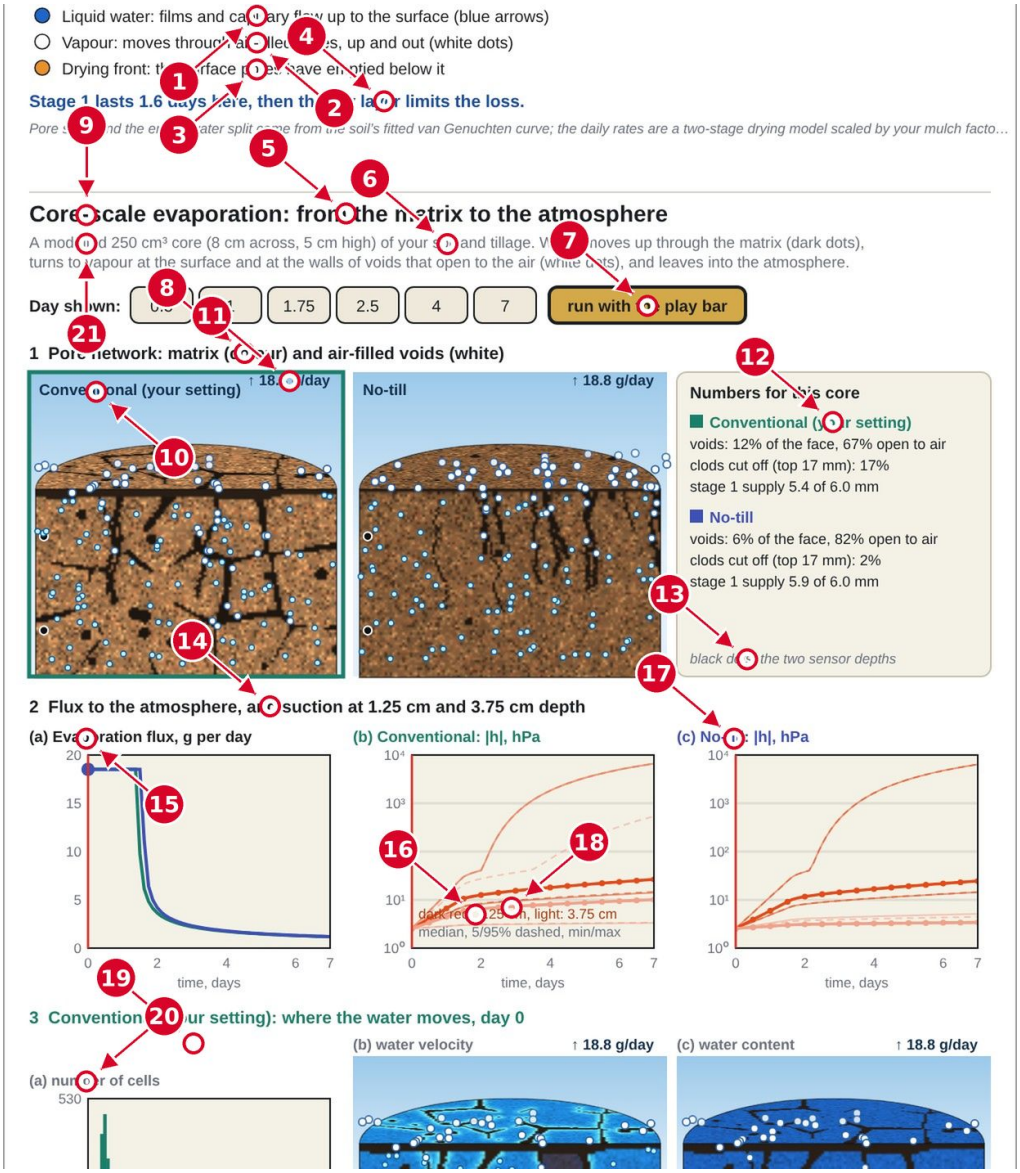



Figure 26.218: Evaporation, part 2 of 3.

**What the arrows point to.**

- **1** The blue arrows, the liquid water moving up by films and capillary flow.
- **2** The white dots, the vapour in the air-filled pores.
- **3** The drying front, below which the surface pores have emptied.
- **4** The duration of stage 1 with the mulch and without it.
- **5** The core-scale evaporation, from the matrix to the atmosphere.
- **6** The description of the modelled 250 cm³ core.
- **7** The button that runs the core with the play bar.
- **8** The pore network, the matrix in colour and the air-filled voids in white.
- **9** The flux of the core to the atmosphere in g per day.
- **10** The conventional core, with large clods and open cracks.
- **11** The no-till core, with fine open voids.
- **12** The numbers of the two cores, voids on the face, open to air and the stage 1 supply.
- **13** The black dots, the two sensor depths of the core.
- **14** The flux to the atmosphere and the suction at 1.25 and 3.75 cm depth.

- **15** The evaporation flux of the two cores against the days.
- **16** The suction of the conventional core at the two depths.
- **17** The suction of the no-till core at the two depths.
- **18** The dark red line, the median suction at 1.25 cm, with the light line at 3.75 cm.
- **19** The conventional core, where the water moves.
- **20** The histogram of the number of cells by water speed.
- **21** The histogram of the no-till core.

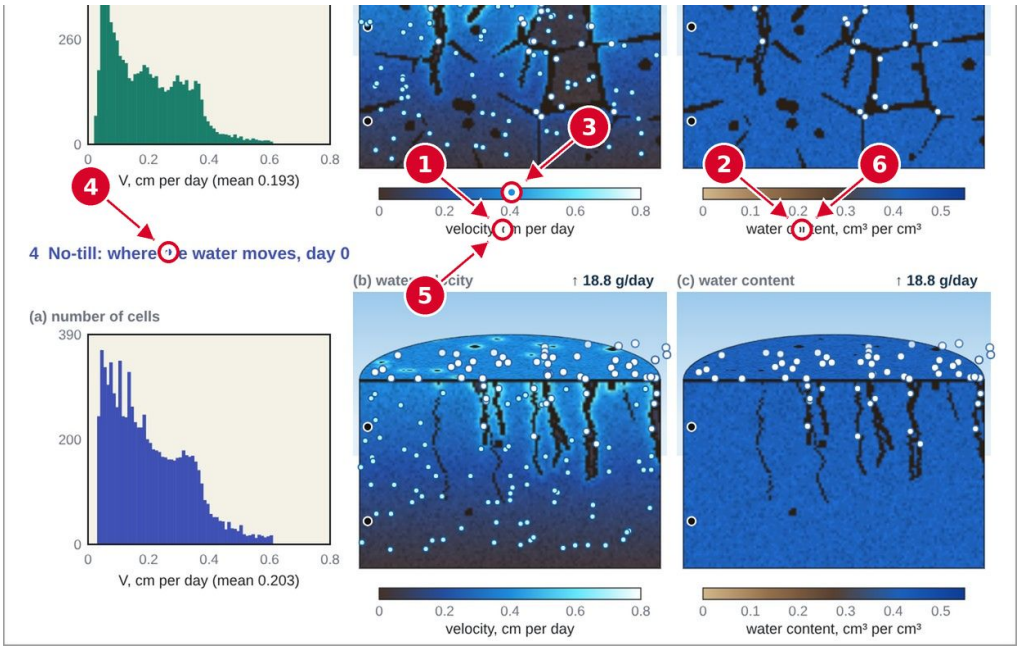



Figure 26.219: Evaporation, part 3 of 3.

**What the arrows point to.**

- **1** The map of the water velocity.
- **2** The map of the water content.
- **3** The colour scale of the velocity in cm per day.
- **4** The no-till core, where the water moves.
- **5** The velocity map of the no-till core.
- **6** The water content map of the no-till core.

## 26.15.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

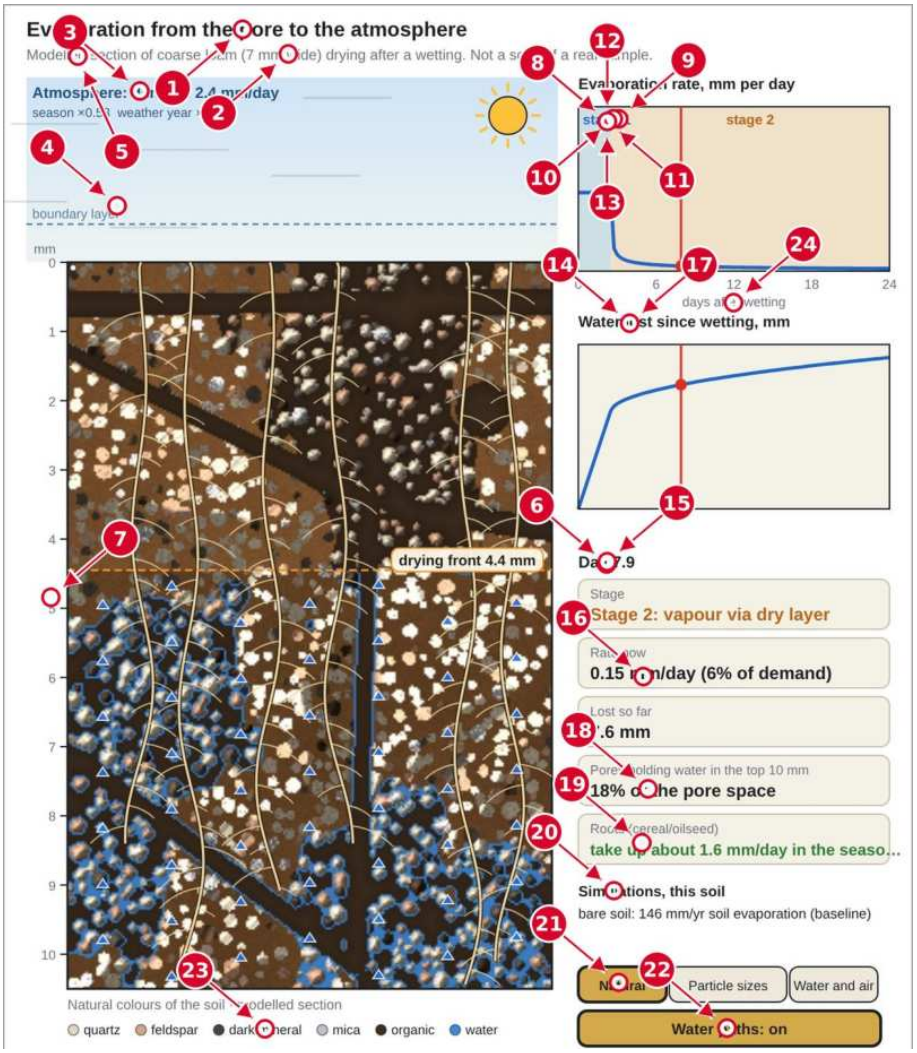



Figure 26.220: Evaporation, part 1, playback a third of the way (day 7.9).

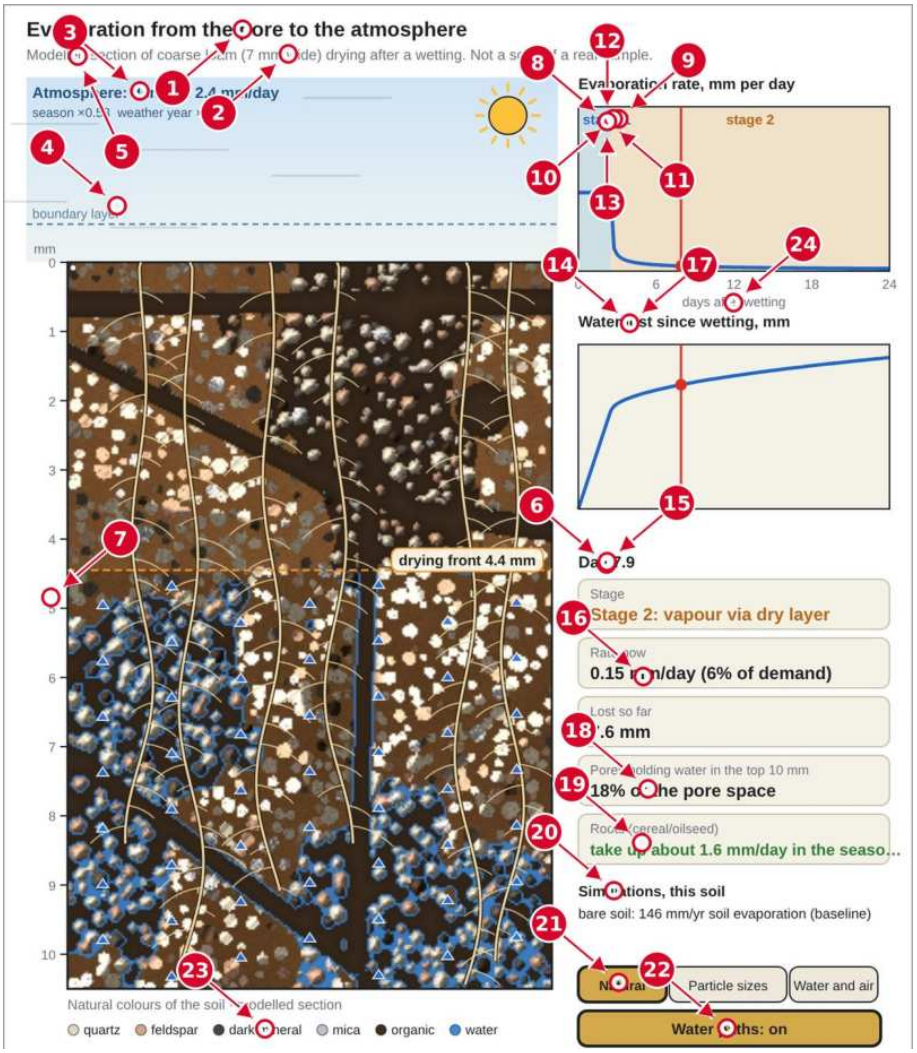



Figure 26.221: Evaporation, part 1, playback a third of the way (day 7.9).

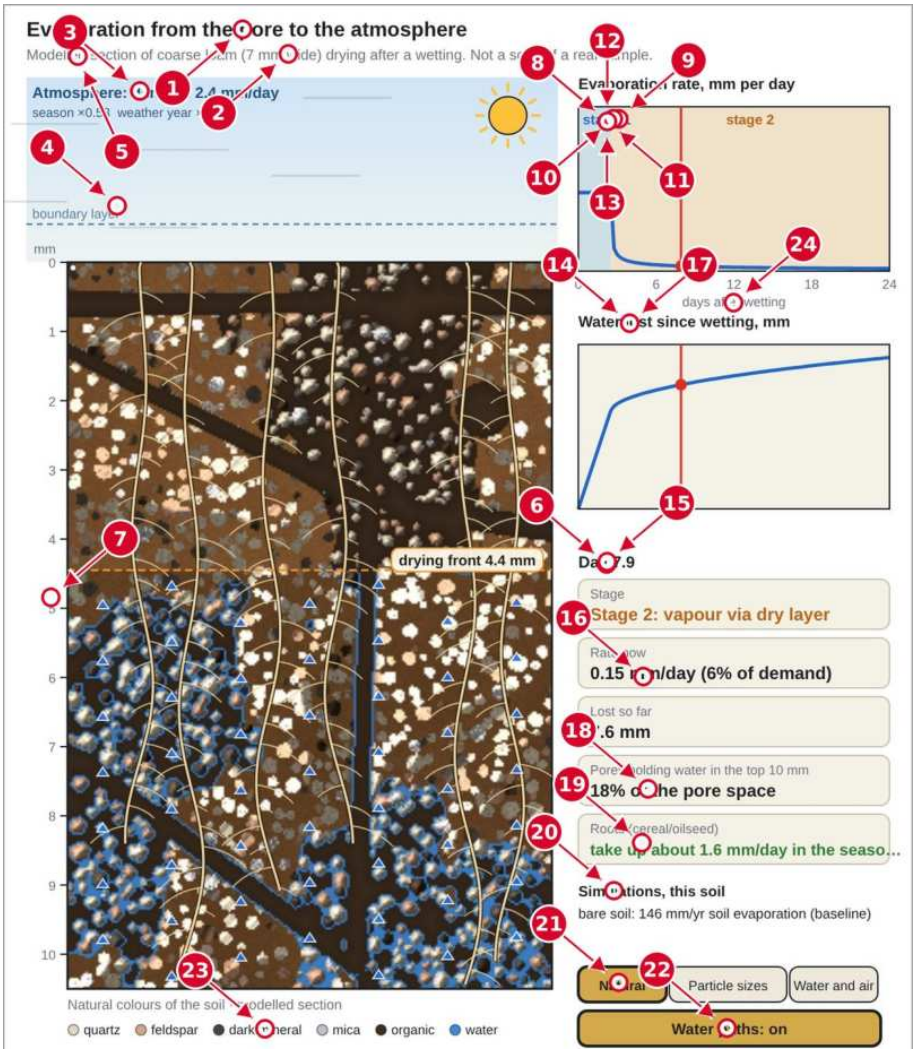



Figure 26.222: Evaporation, part 1, playback a third of the way (day 7.9).

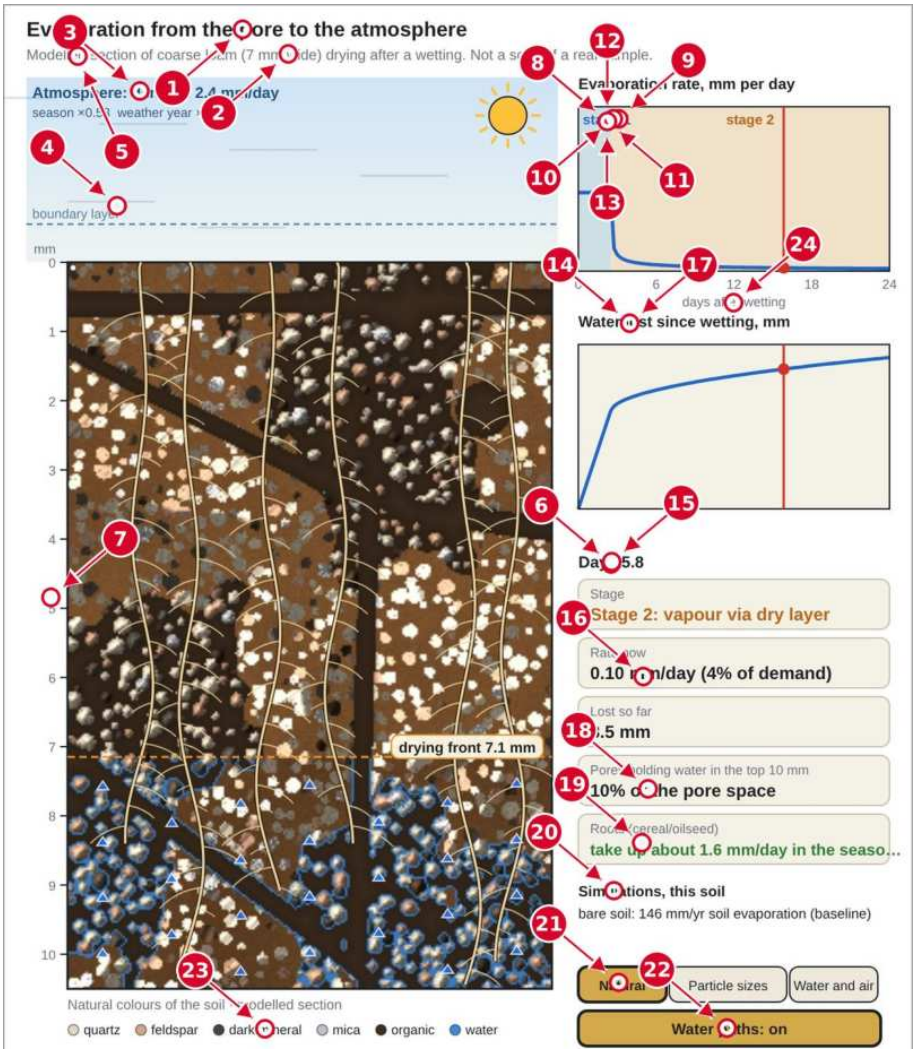



Figure 26.223: Evaporation, part 1, playback two thirds of the way (day 15.8).

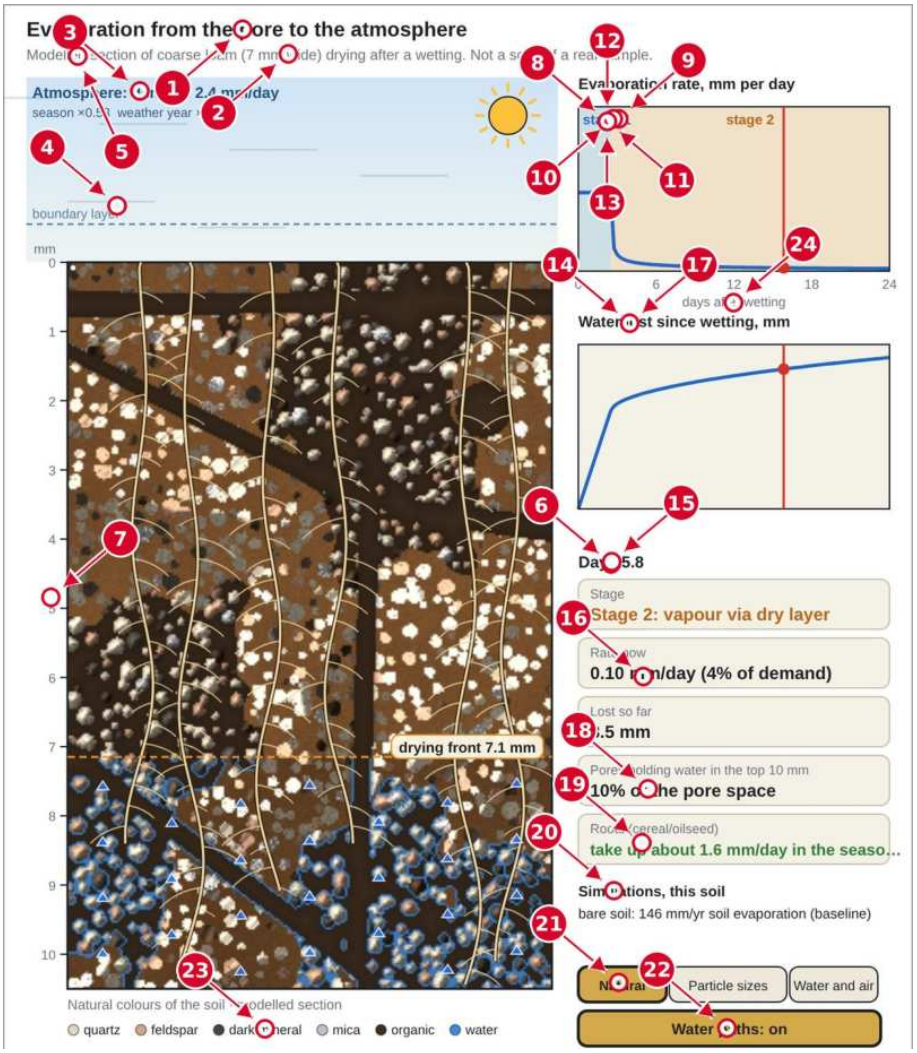



Figure 26.224: Evaporation, part 1, playback two thirds of the way (day 15.8).

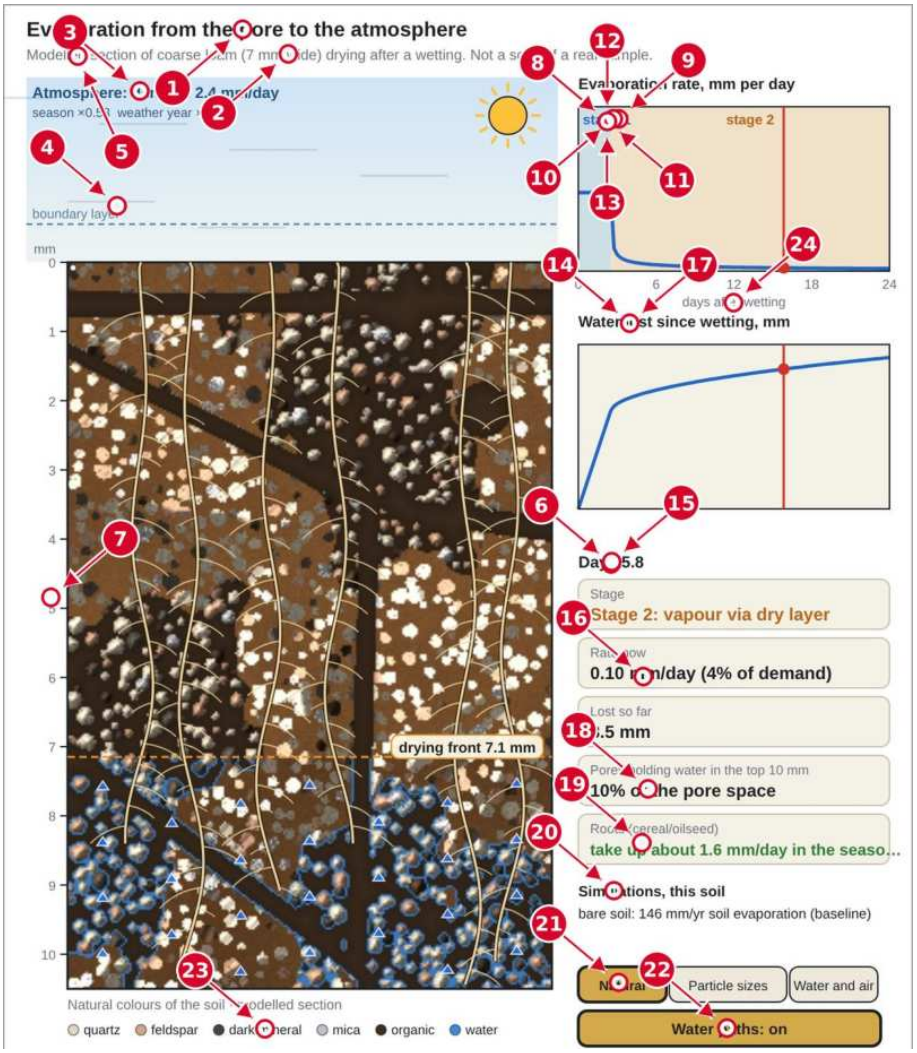



Figure 26.225: Evaporation, part 1, playback two thirds of the way (day 15.8).

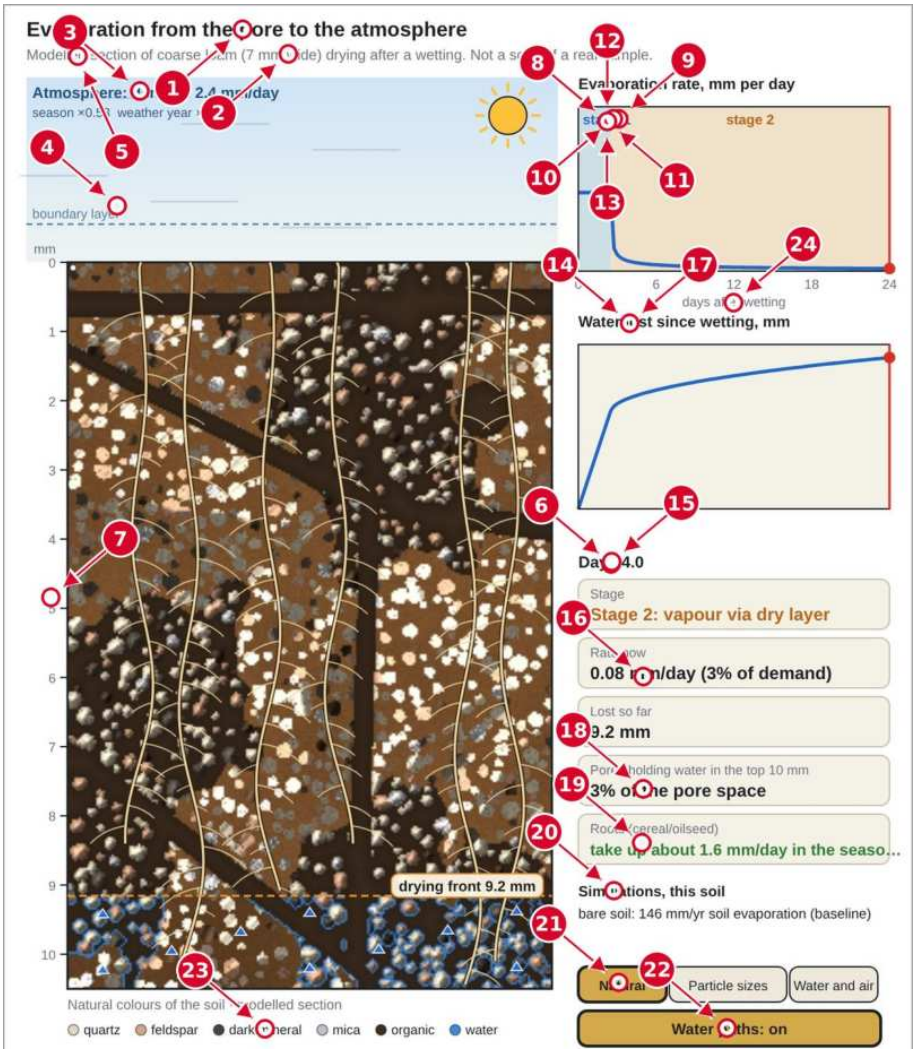



Figure 26.226: Evaporation, part 1, playback the end (day 24.0).

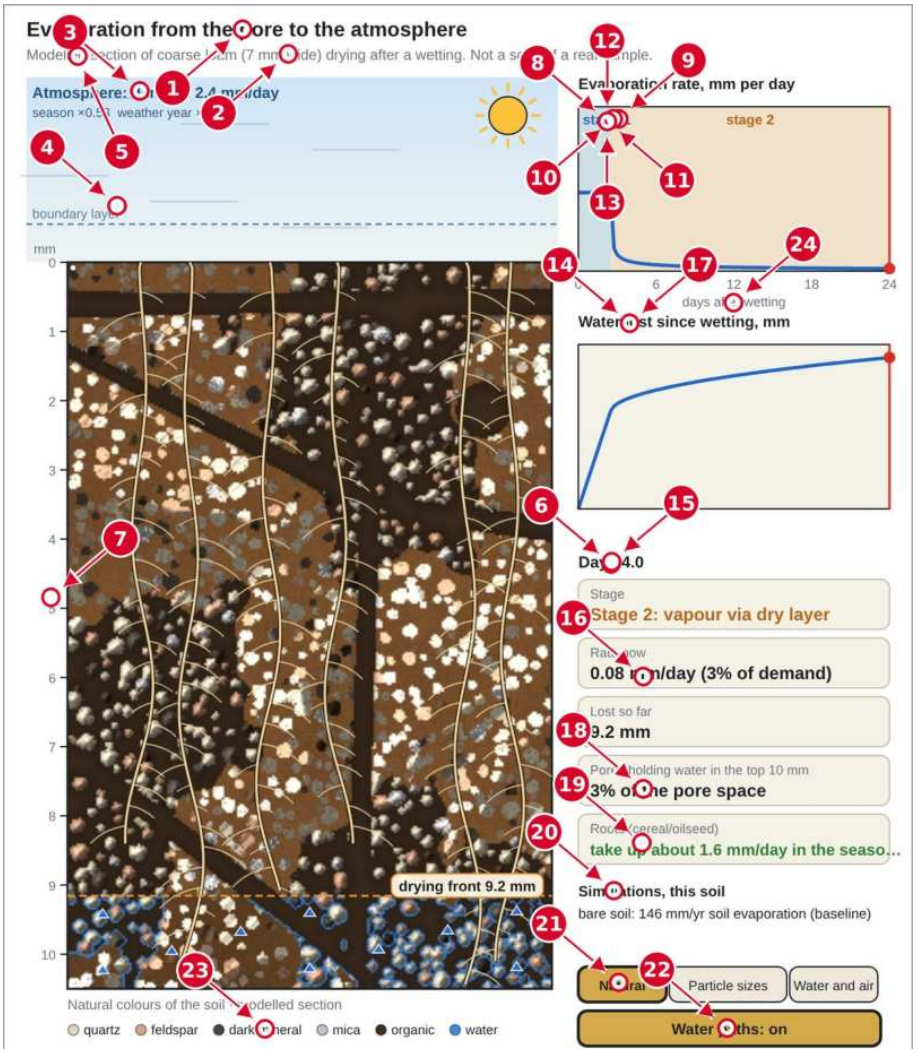



Figure 26.227: Evaporation, part 1, playback the end (day 24.0).

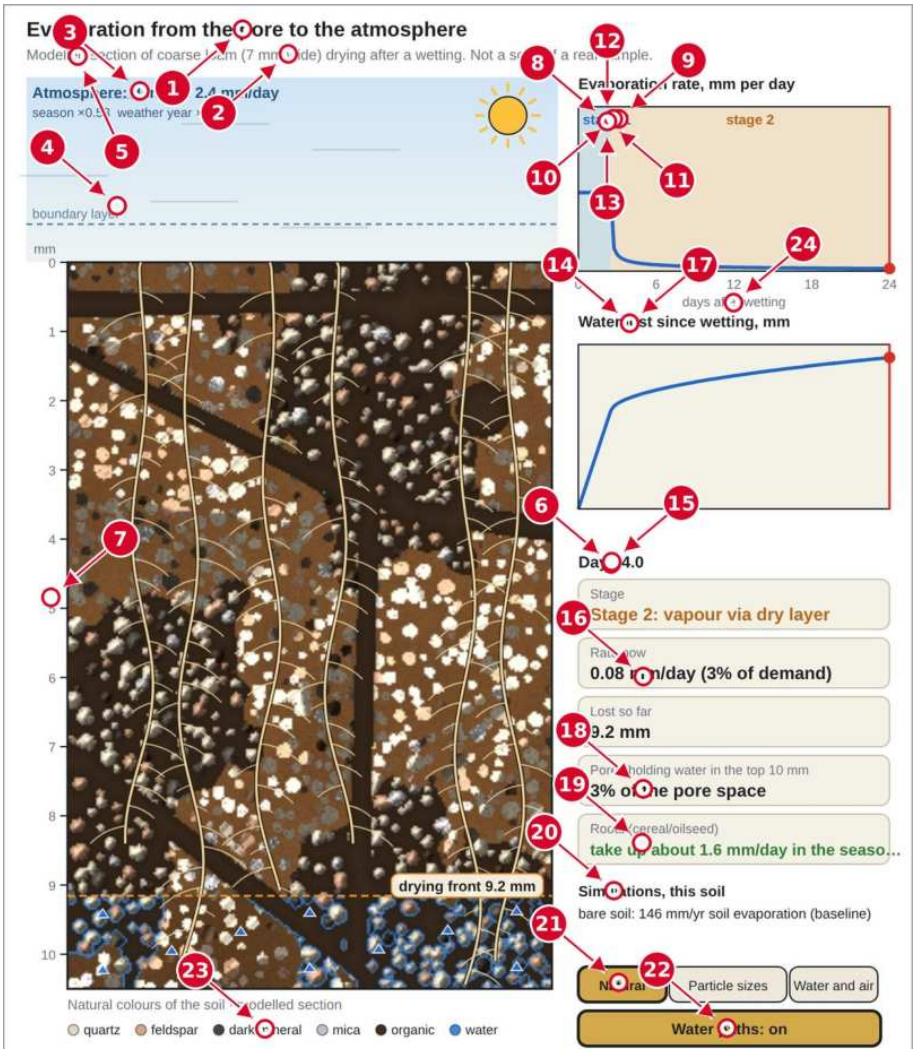



Figure 26.228: Evaporation, part 1, playback the end (day 24.0).

### 26.15.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

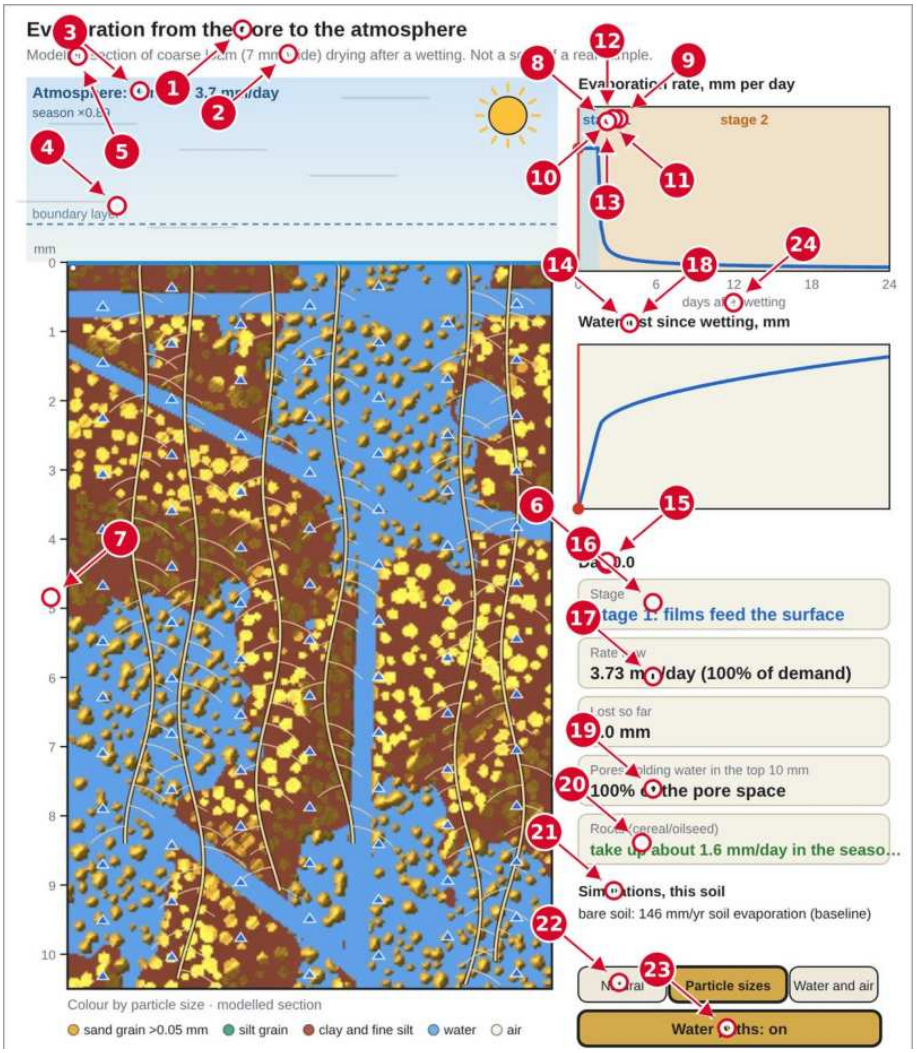



Figure 26.229: Evaporation, part 1. The Particle sizes view of the section chosen.

Figure 26.230: Evaporation, part 1. The Water and air view of the section chosen.

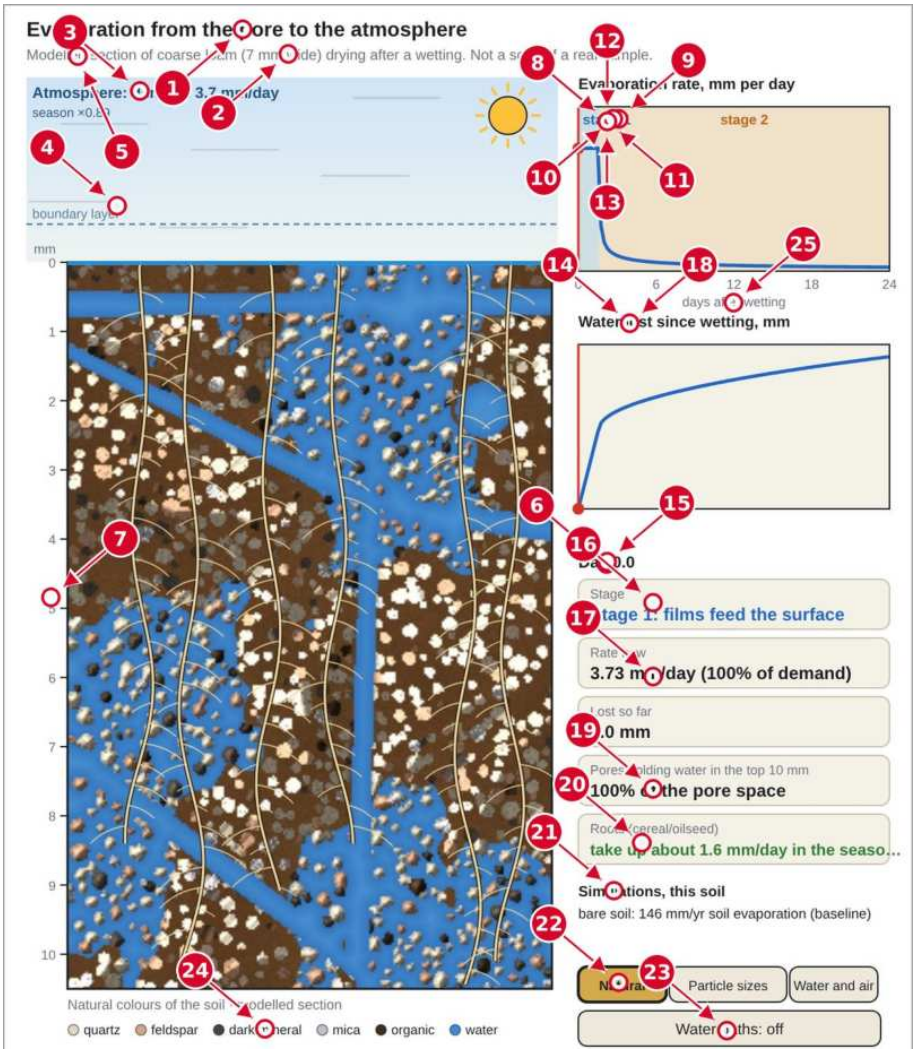



Figure 26.231: Evaporation, part 1. The Water paths button switched off, so the arrows and dots of the flow are hidden.

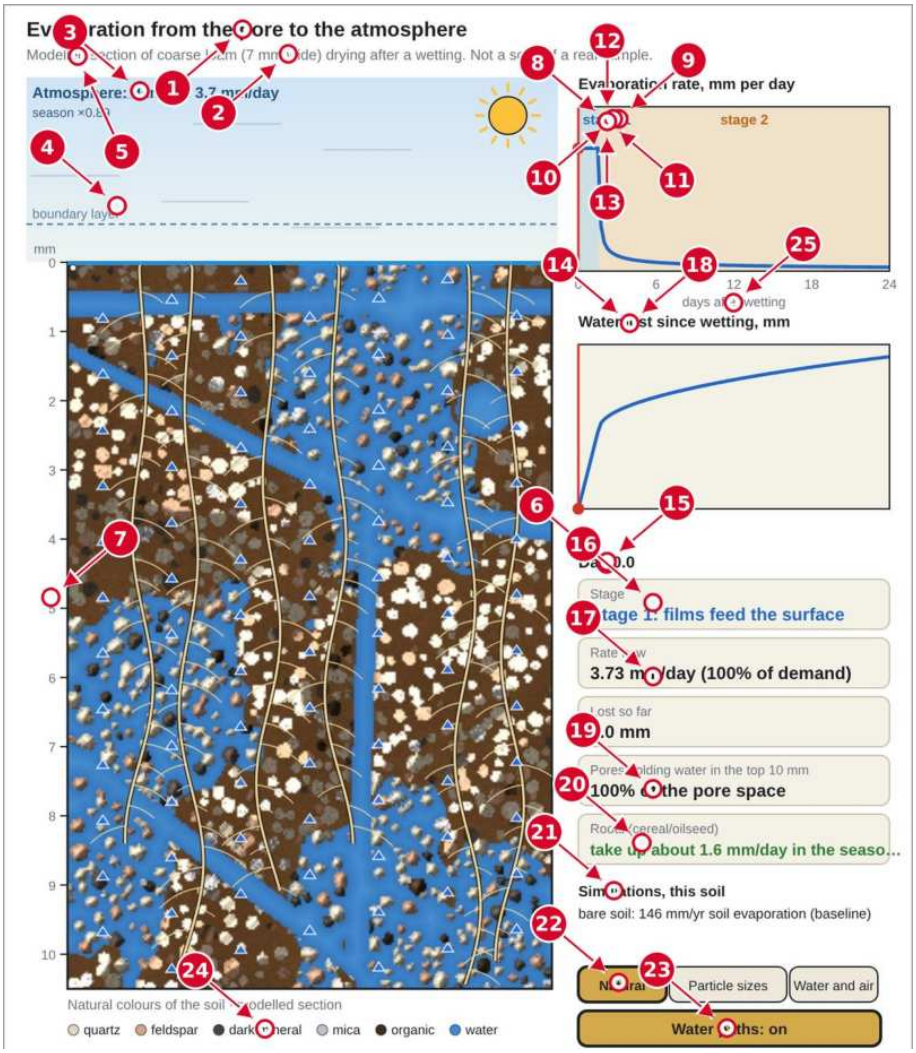



Figure 26.232: Evaporation, part 1. The day 0.5 button pressed, so the cores are shown at half a day.

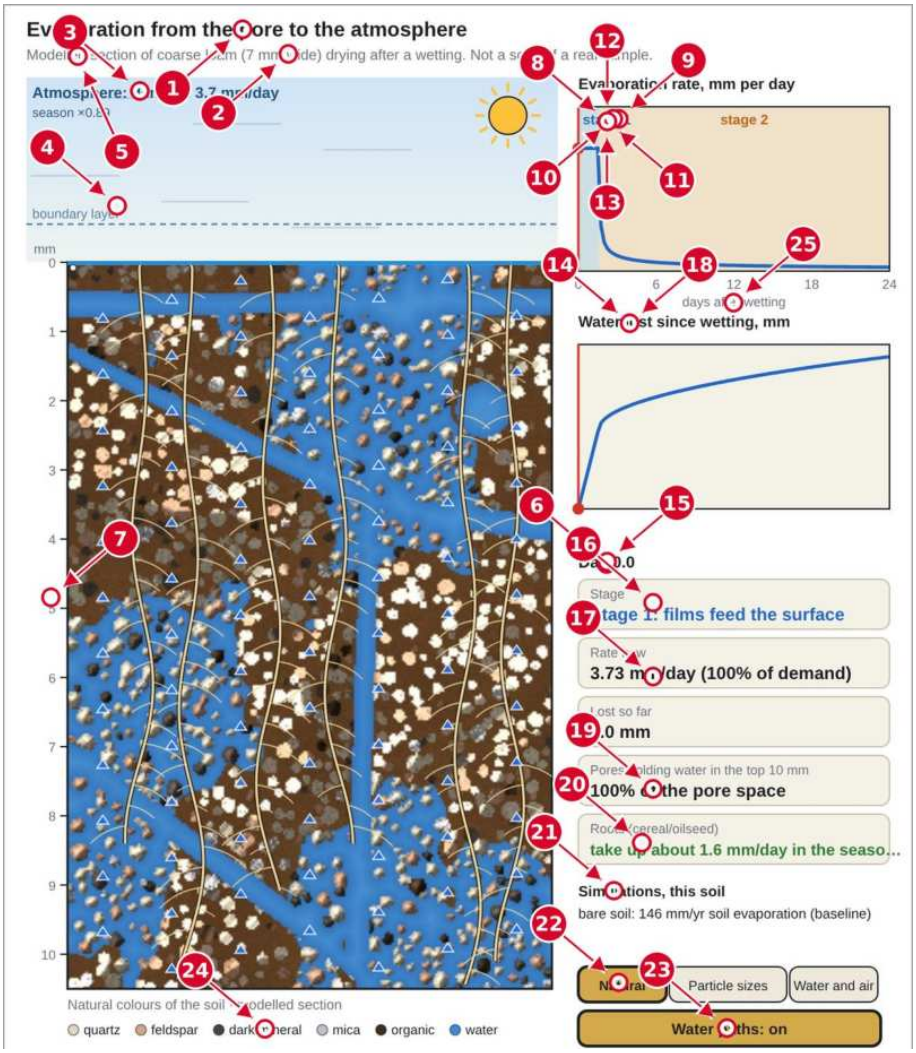



Figure 26.233: Evaporation, part 1. The day 0.5 button pressed, so the cores are shown at half a day.

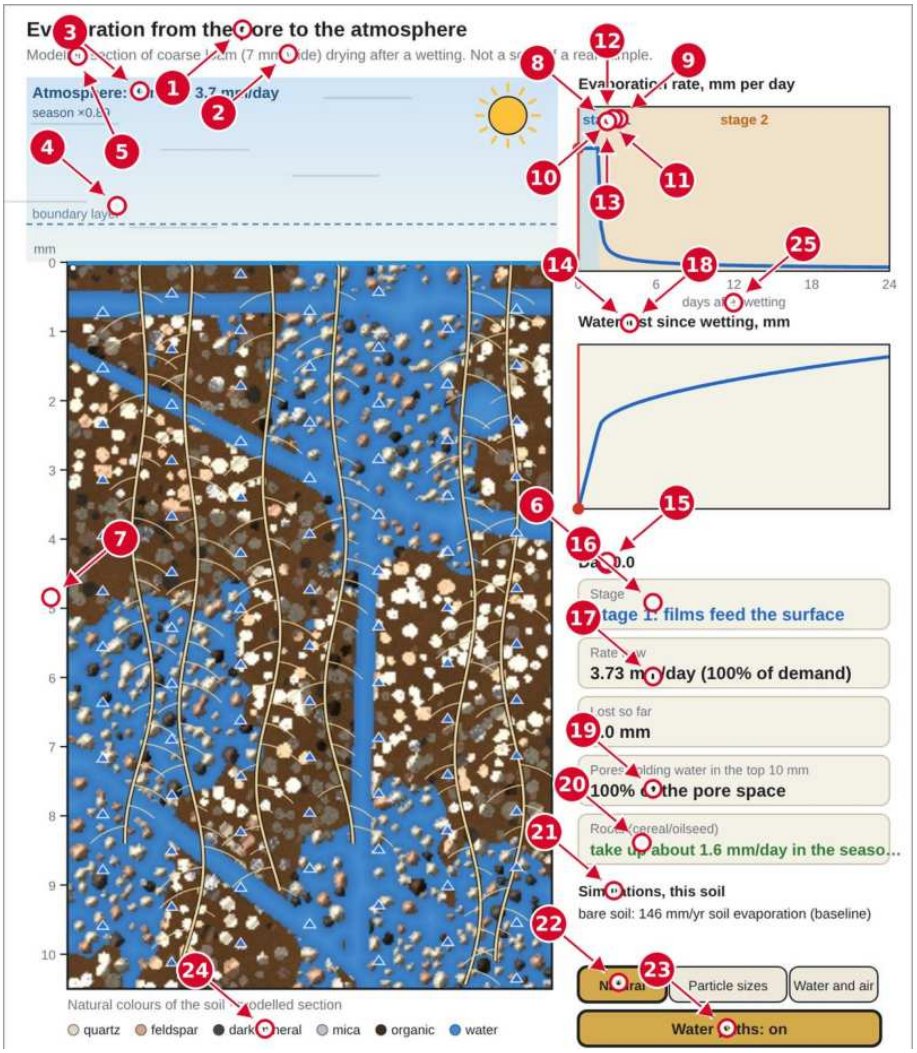



Figure 26.234: Evaporation, part 1. The day 1.75 button pressed.

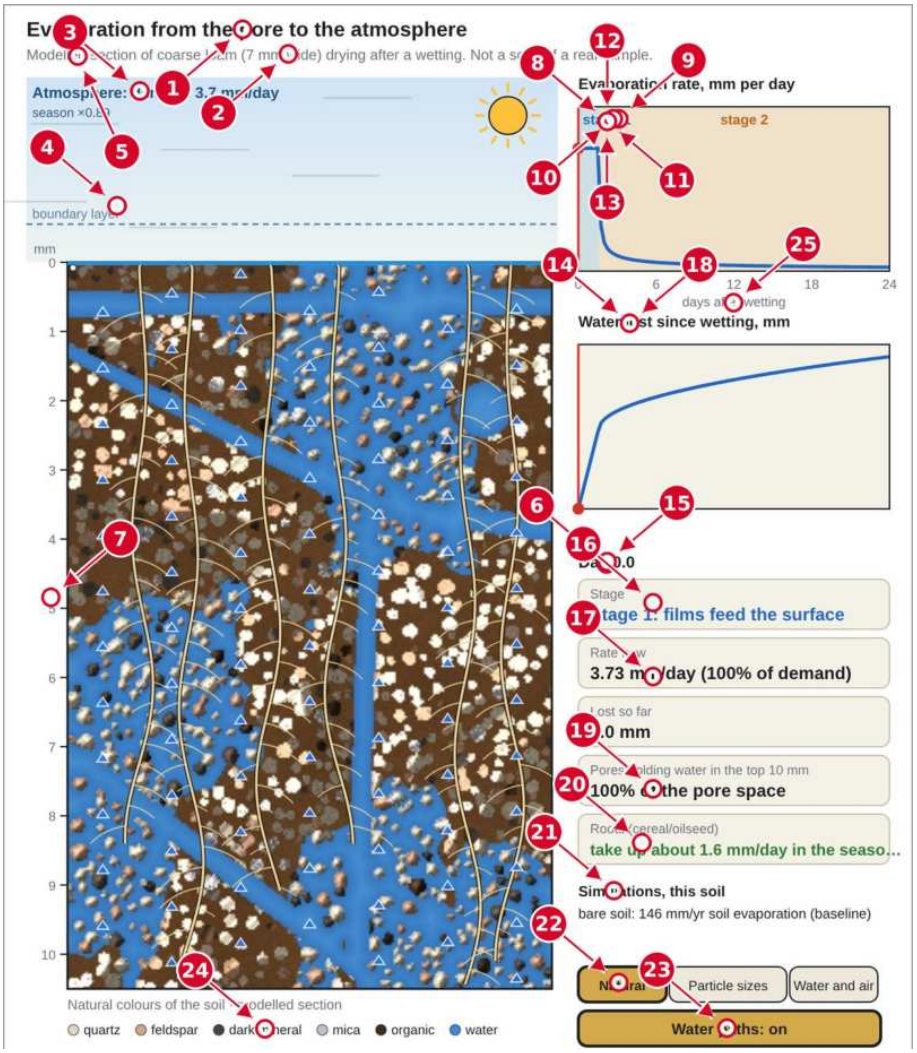



Figure 26.235: Evaporation, part 1. The day 1.75 button pressed.

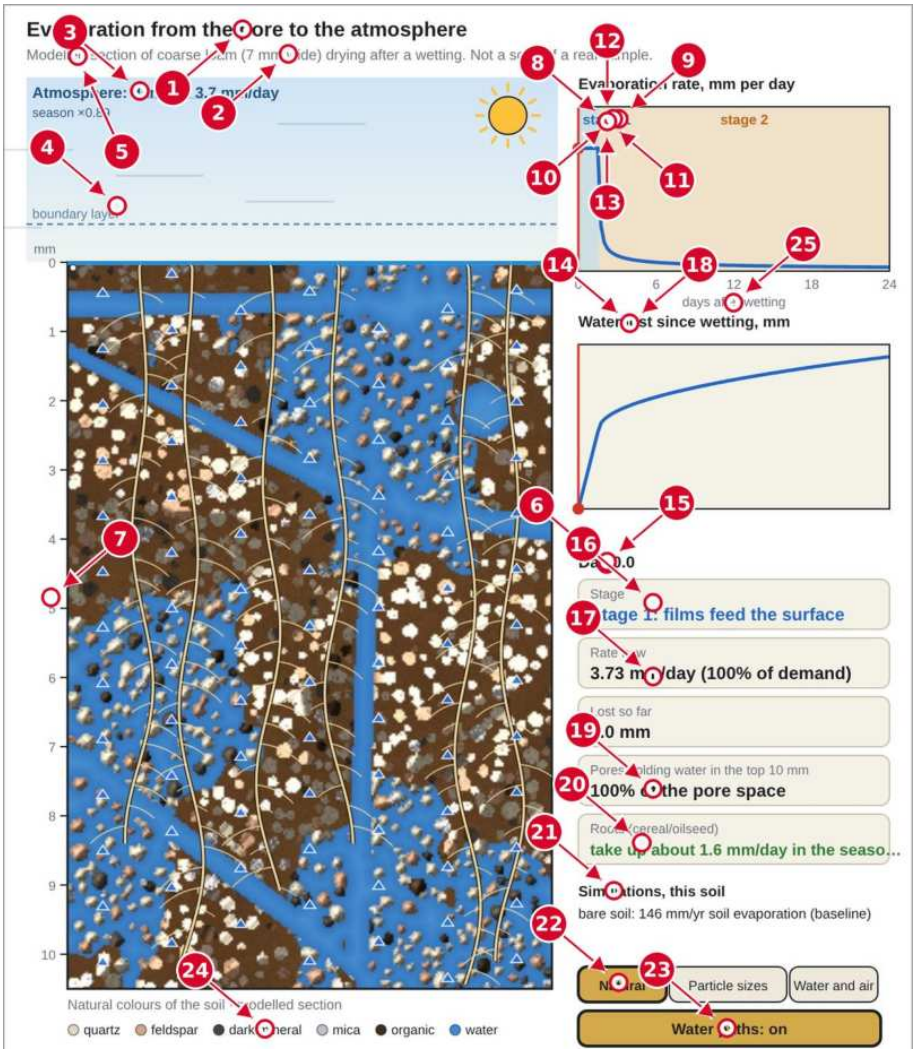



Figure 26.236: Evaporation, part 1. The day 4 button pressed.

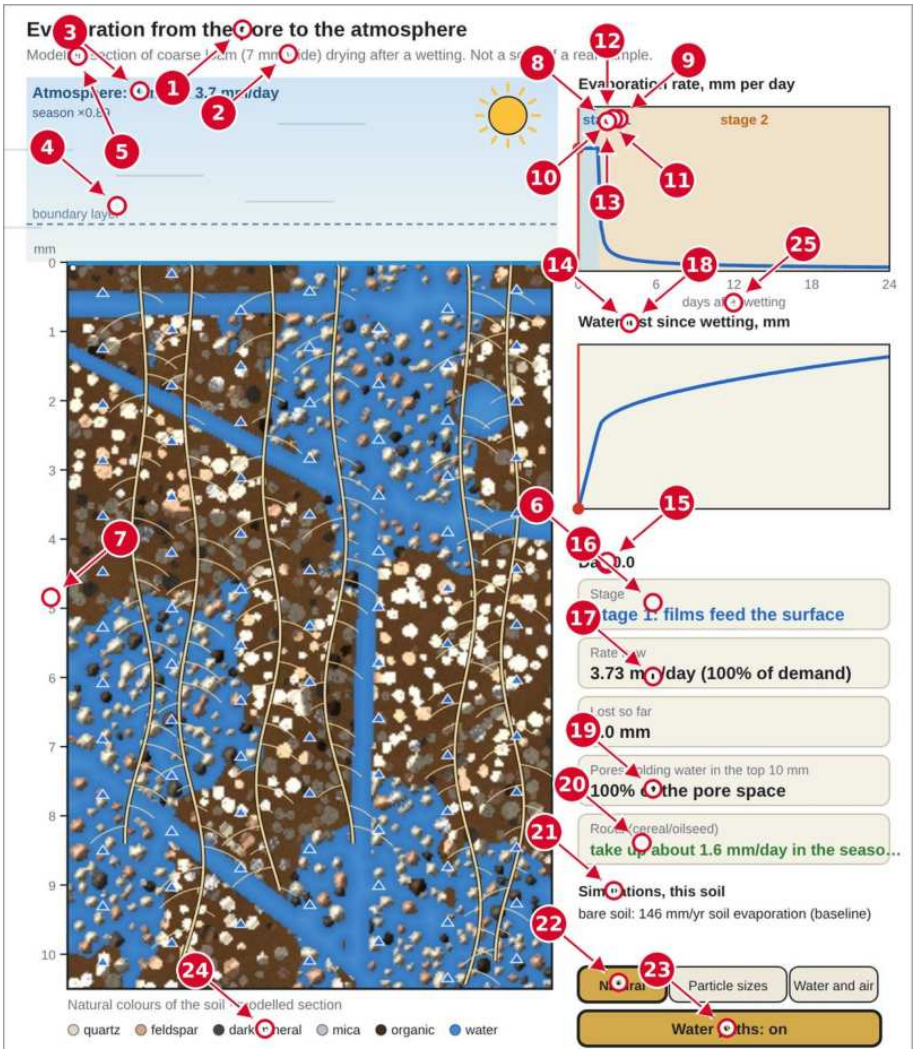



Figure 26.237: Evaporation, part 1. The day 4 button pressed.

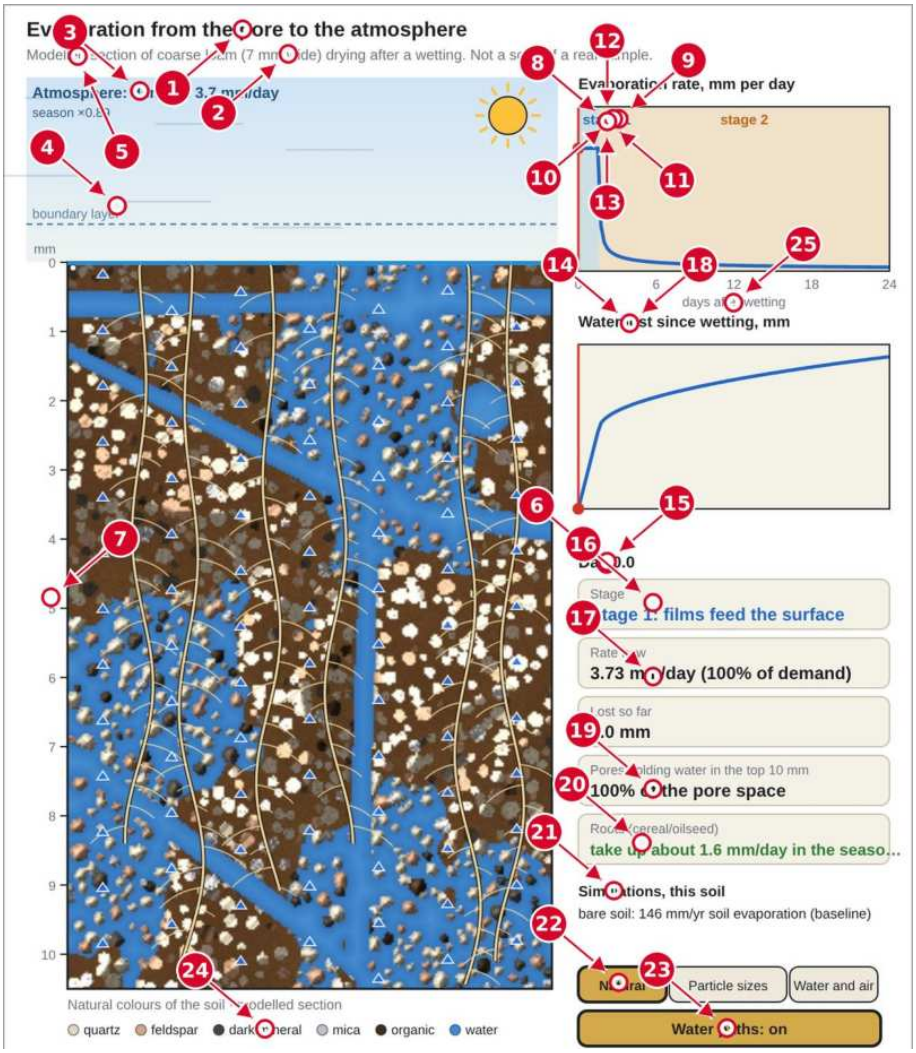



Figure 26.238: Evaporation, part 1. The day 7 button pressed, so the cores are shown at the end of stage 1.

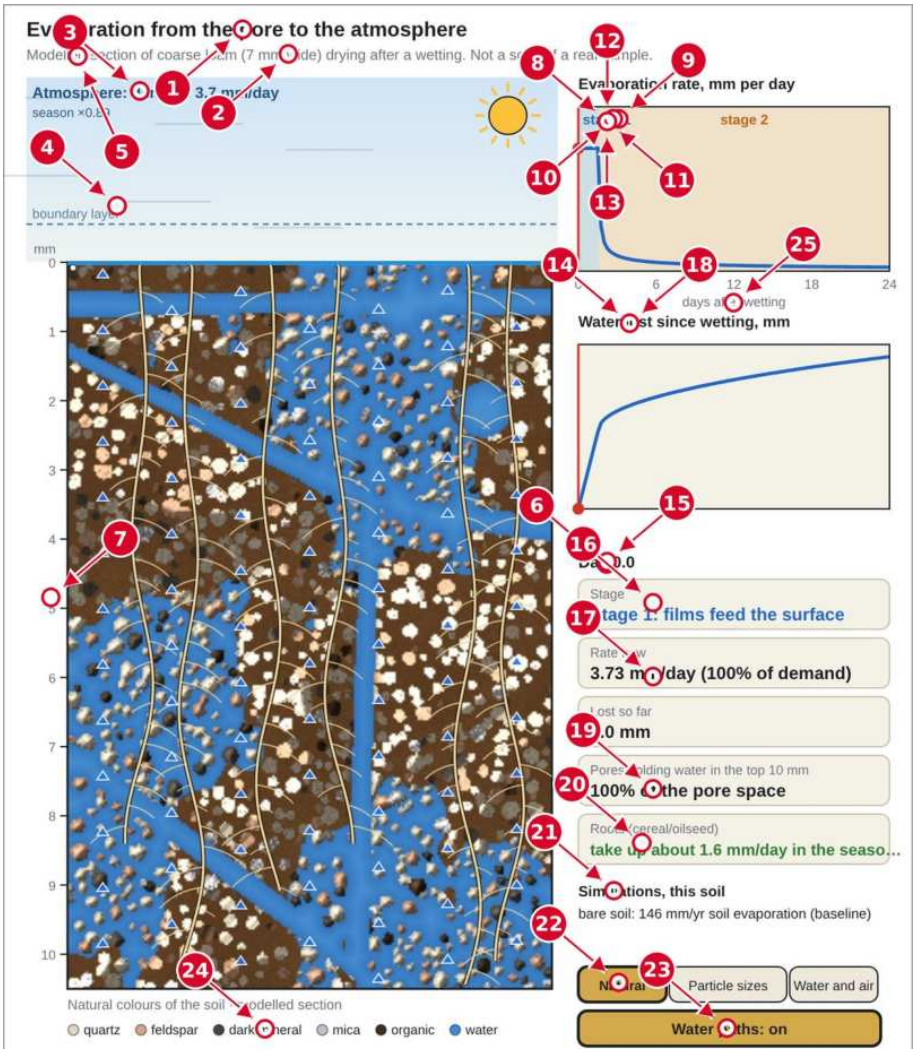



Figure 26.239: Evaporation, part 1. The day 7 button pressed, so the cores are shown at the end of stage 1.

# 26.16 Fate of saved water

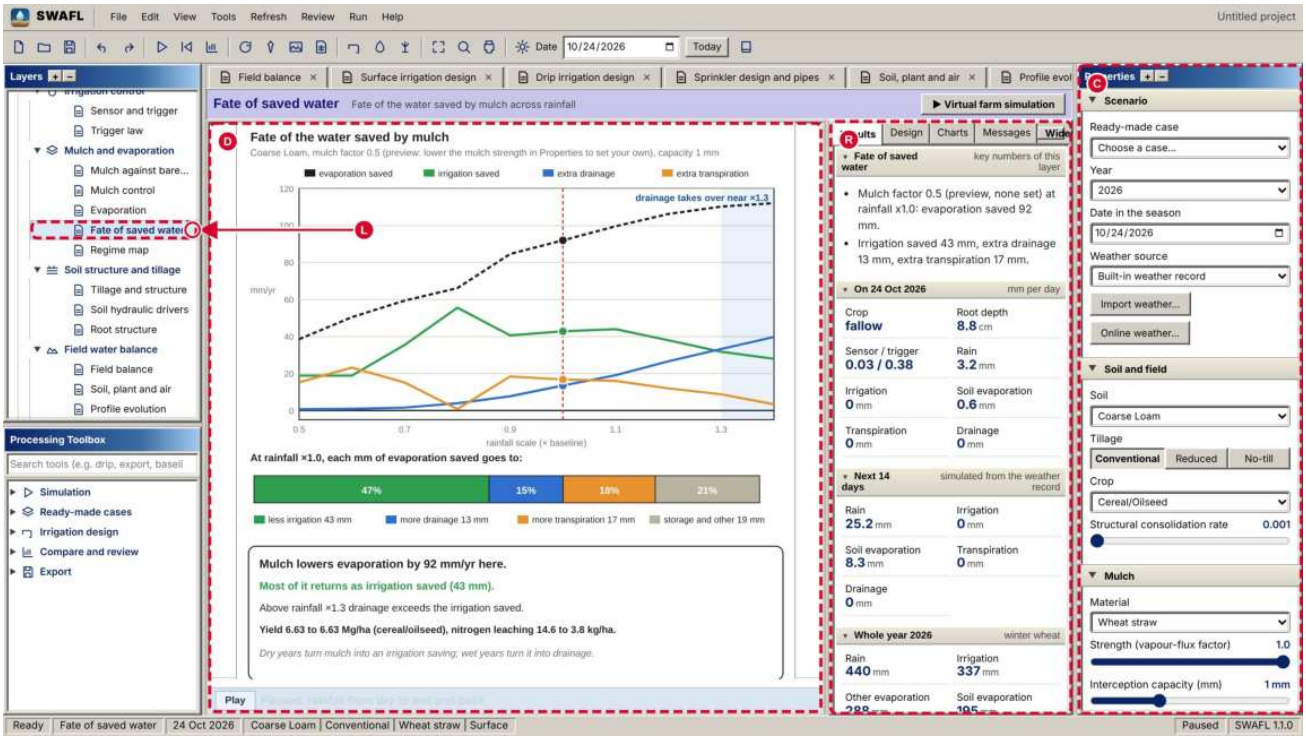



Figure 26.240: Fate of saved water: the whole workbench.

Figure 26.241: Fate of saved water: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Evaporation saved, irrigation saved, extra drainage and extra transpiration against rainfall scale: Eq. 9.6.
- **2** Rainfall at which drainage overtakes the irrigation saved: from the curves of Eq. 9.6.
- **3** Where each mm of evaporation saved goes (irrigation, drainage, transpiration, storage): Eq. 9.7.
- **4** The sentence under the picture is built from the shares of Eq. 9.7.

**Equations behind this screen.** Eq. 9.6, Eq. 9.7.

*Eq. 9.6. Where explained: chapter "Mulch and evaporation layers", section "Fate of saved water".*

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

*Provenance (Eq. 9.6):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

*Eq. 9.7. Where explained: chapter "Mulch and evaporation layers", section "Fate of saved water".*

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

*Provenance (Eq. 9.7):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

### 26.16.1 The whole layer, part by part

The picture below is the complete drawing of the layer Fate of saved water, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

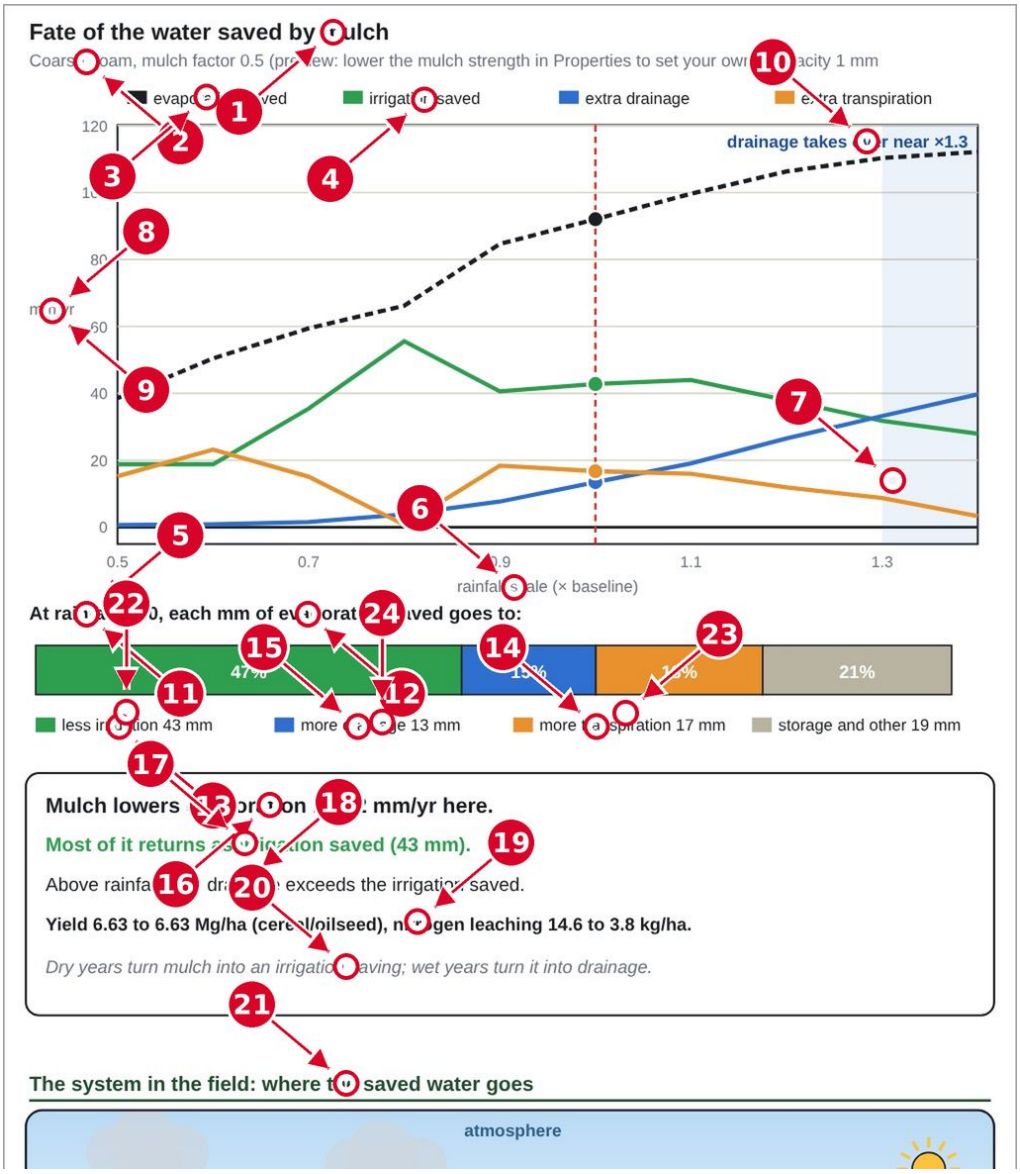



Figure 26.242: Fate of saved water, part 1 of 2.

**What the arrows point to.**

- **1** The title, the fate of the water saved by mulch.
- **2** The settings of the chart, the soil, the mulch factor and the capacity.
- **3** The legend of the four lines, evaporation saved, irrigation saved, extra drainage and extra transpiration.
- **4** The black dotted line, the evaporation saved by the mulch at each rainfall scale.
- **5** The orange line, the extra transpiration.
- **6** The green line, the irrigation saved.
- **7** The blue line, the extra drainage, which grows with the rainfall.
- **8** The red dashed line, the rainfall scale in force.

- **9** The point of the evaporation saved at that rainfall scale.
- **10** The rainfall scale above which the drainage takes over.
- **11** The rainfall scale axis, as a multiple of the baseline.
- **12** The partition of each mm of evaporation saved at the rainfall in force.
- **13** The share that returns as less irrigation, in green.
- **14** The share that stays as storage and other terms, in grey.
- **15** The legend of the partition with the values in mm.
- **16** The box with the conclusion for these settings.
- **17** The main destination of the saved water.
- **18** The rainfall above which the drainage exceeds the irrigation saved.
- **19** The yield and the nitrogen leaching before and after the mulch.
- **20** The message that dry years turn the mulch into irrigation saving and wet years into drainage.
- **21** The system in the field, where the saved water goes.
- **22** The irrigation saved.
- **23** The extra transpiration.
- **24** The extra drainage.

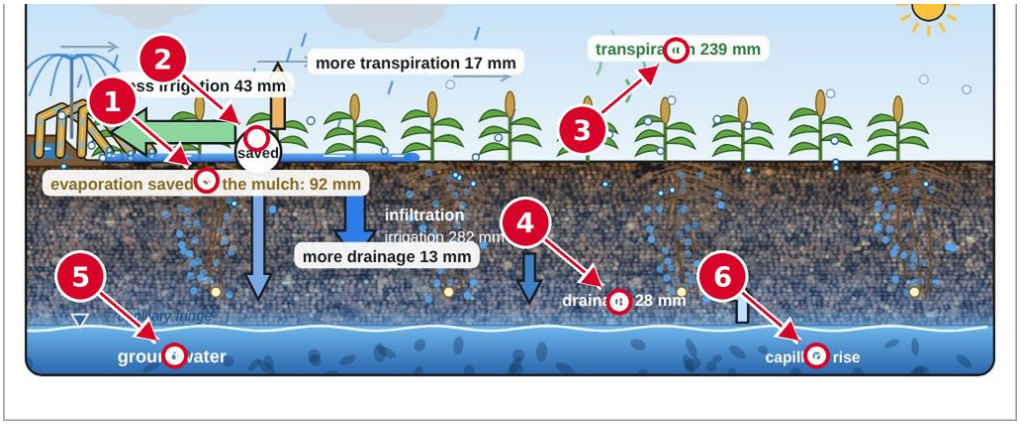



Figure 26.243: Fate of saved water, part 2 of 2.

**What the arrows point to.**

- **1** The tag of the evaporation saved by the mulch.
- **2** The saved water leaving the surface, drawn as the circle labelled saved.
- **3** The transpiration of the year.
- **4** The drainage arrow.
- **5** The groundwater.
- **6** The capillary rise.

# 26.17 Regime map

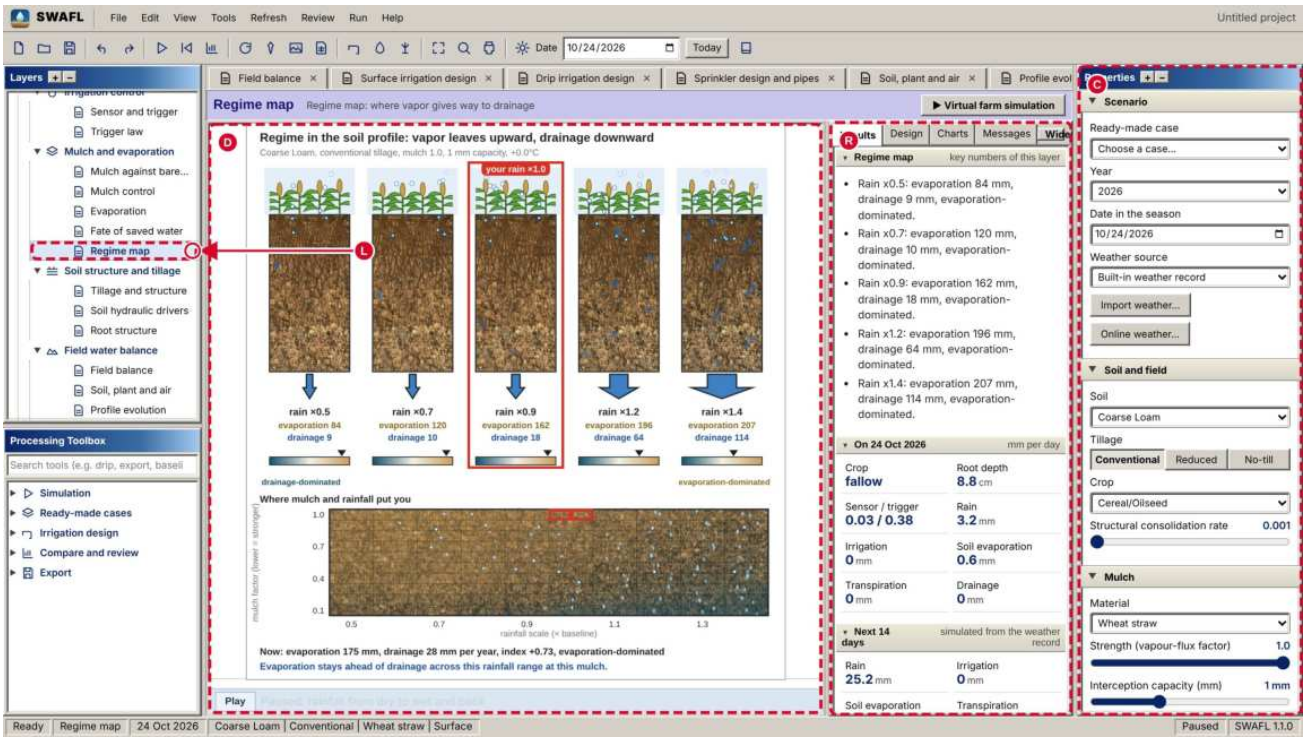



Figure 26.244: Regime map: the whole workbench.

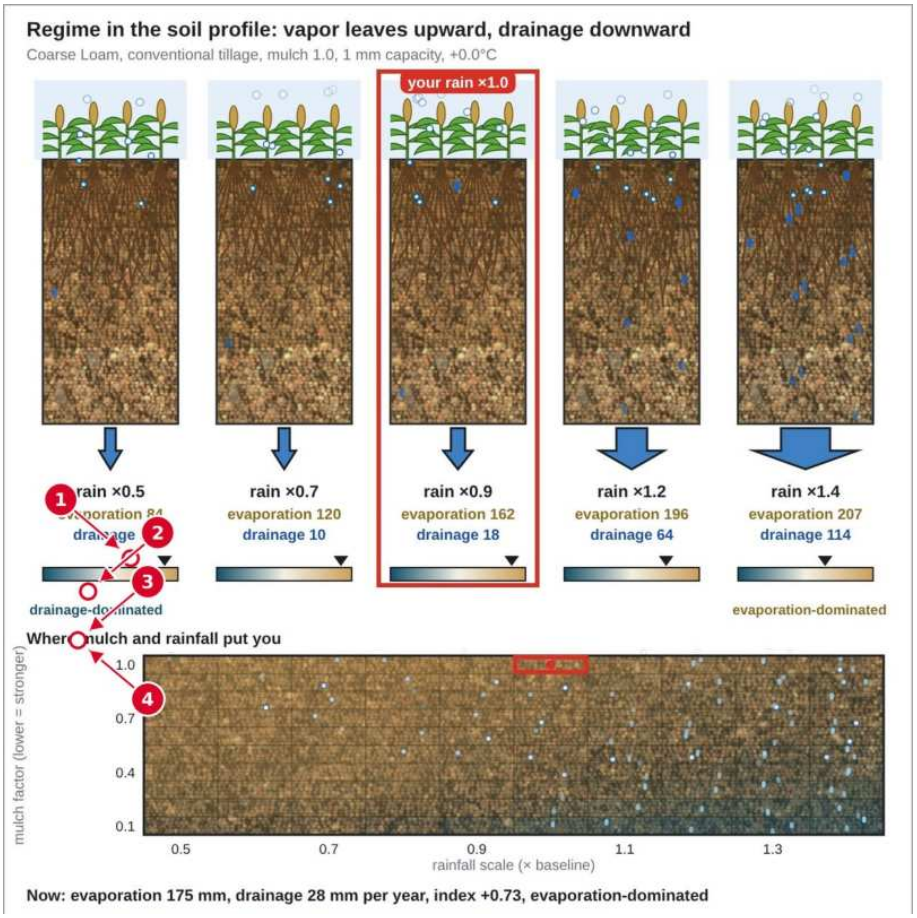



Figure 26.245: Regime map: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Rainfall columns: evaporation and drainage of Engine 1 at each rainfall scale, Eqs. 5.4 and 5.9.
- **2** Colour bar under each column: regime index I = (E minus D) / max(1, E + D): Eq. 9.8.
- **3** Regime map over rainfall scale and mulch factor: Eq. 9.8 in every cell.
- **4** Red box: your current setting on the map.

**Equations behind this screen.** Eq. 9.8.

*Eq. 9.8. Where explained: chapter “Mulch and evaporation layers”, section “Regime map”.*

$$\mathcal{I} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{I} > +0.05 \\ \text{drainage-dominated} & \mathcal{I} < -0.05 \\ \text{on the boundary} & |\mathcal{I}| \le 0.05 \end{cases} \tag{9.8}$$

*Provenance (Eq. 9.8):* **[Engineered]**: index defined for the workbench.

### 26.17.1 The whole layer, part by part

The picture below is the complete drawing of the layer Regime map, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

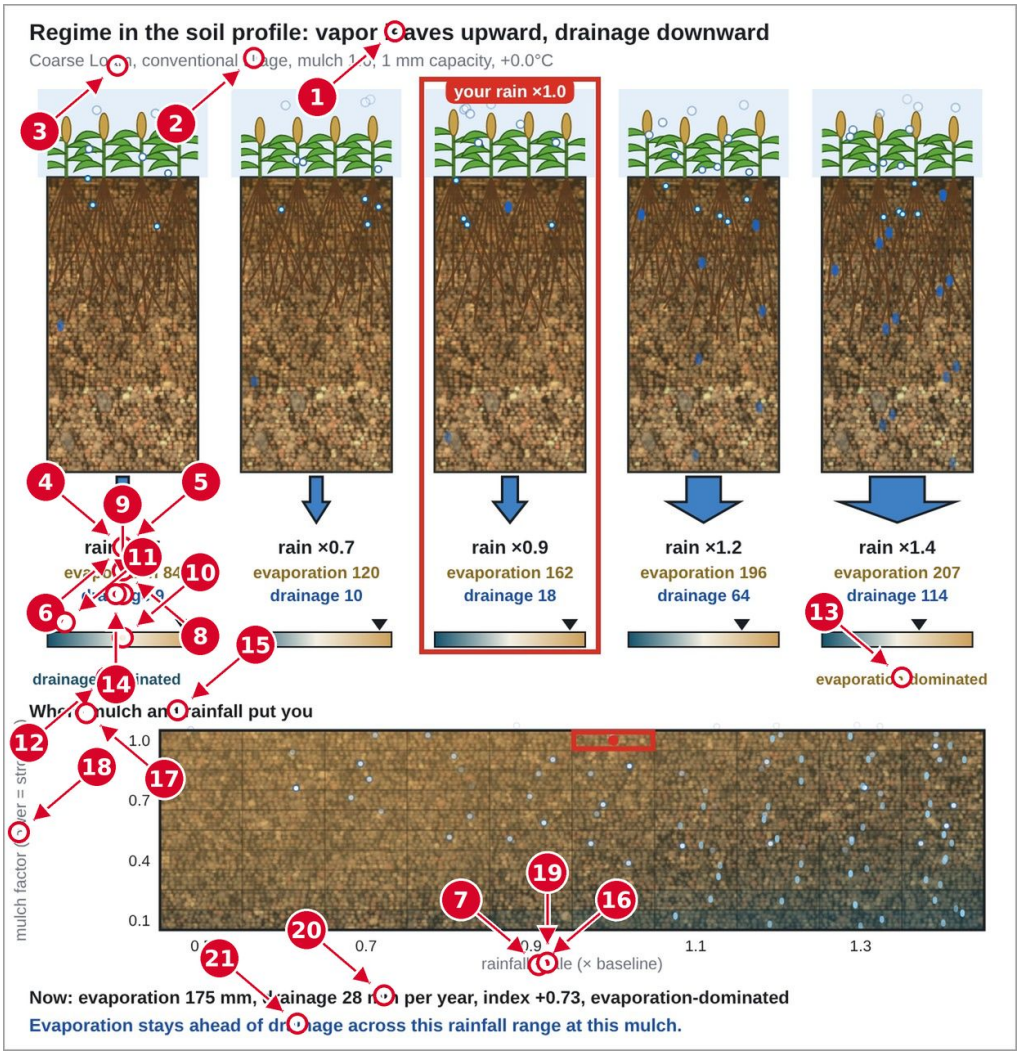



Figure 26.246: Regime map, part 1 of 1.

**What the arrows point to.**

- **1** The title, the regime of the soil profile, vapour upward and drainage downward.
- **2** The settings of the five profiles, the soil, tillage, mulch factor, capacity and warming.
- **3** The tag of the profile that matches your own rain scaling.
- **4** The profile outlined in red, the rainfall scale in force.
- **5** The next profile, a wetter rainfall scale.
- **6** The rain arrow, thicker for more rain.
- **7** The rainfall scale of the profile.
- **8** The evaporation of the year at that rainfall scale, in mm.
- **9** The drainage of the year at that rainfall scale, in mm.
- **10** The regime scale from drainage-dominated to evaporation-dominated.
- **11** The marker of the regime index on that scale.
- **12** The label of the drainage-dominated end.
- **13** The label of the evaporation-dominated end.
- **14** The drainage of the wettest profile, far above its evaporation.
- **15** The map of where the mulch and the rainfall put you.
- **16** The regime map over the rainfall scale and the mulch factor.
- **17** The red outlined cell, your own setting on the map.
- **18** The mulch factor axis, lower means stronger.

- **19** The rainfall scale axis, as a multiple of the baseline.
- **20** The evaporation, the drainage and the regime index of your setting.
- **21** The sentence on the rainfall at which the drainage overtakes the evaporation.

## 26.18 Feedback loops

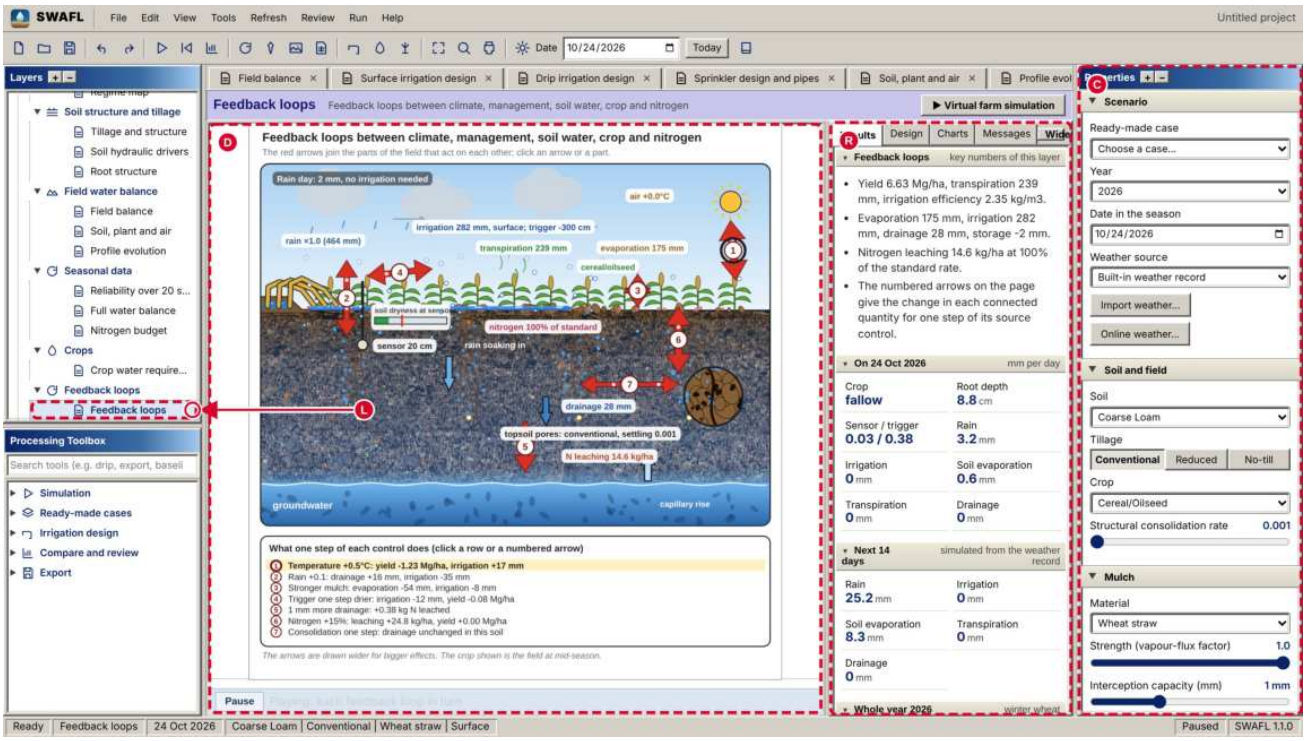



Figure 26.247: Feedback loops: the whole workbench.

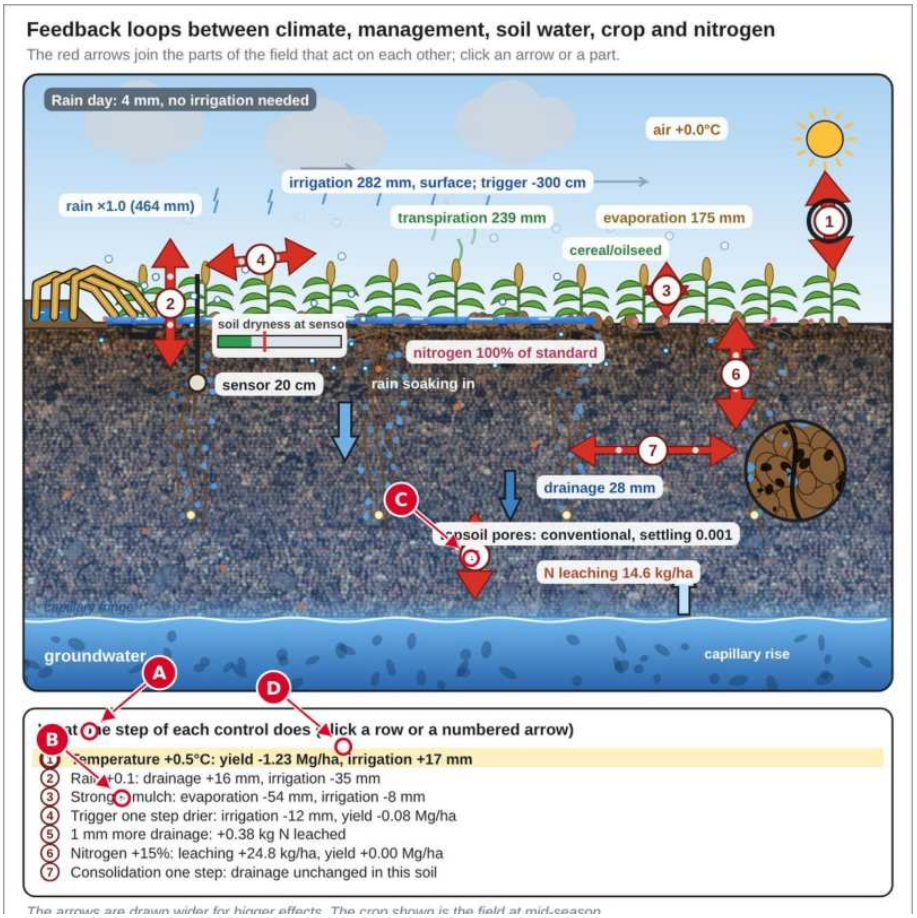



Figure 26.248: Feedback loops: the drawing area.

**What the arrows point to.** The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **A** Arrow 1, temperature step: Engine 1 (Eqs. 5.4 to 5.12) re-evaluated; arrow width from Eq. 10.1.
- **B** Arrow 3, stronger mulch: change in evaporation and irrigation, Eqs. 5.4, 5.7 and 9.1.
- **C** Arrow 5, nitrate cost of drainage, kappa_N: Eq. 10.2.
- **D** List of the change produced by one step of each control; widths of the arrows: Eq. 10.1.

**Equations behind this screen.** Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 10.1, Eq. 10.2.

*Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.5. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.6. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.7. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.8. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.9. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.10. Where explained: chapter "The two computing engines", section "The meta-model".*

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

*Eq. 5.11. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$Q \leftarrow Q\ \text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ \ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N). \tag{5.11}$$

*Provenance (Eq. 5.11):* **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

*Eq. 5.12. Where explained: chapter “The two computing engines”, section “The meta-model”.*

$$ET \in [\,P + I - D - 30,\ P + I - D + 15\,] \text{ (if not, } ET \text{ is clipped and } E, T \text{ are scaled by the same factor)},$$
$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

*Provenance (Eq. 5.12):* **[Engineered]**: closure constraint and bounds chosen for the workbench.

*Eq. 10.1. Where explained: chapter “Feedback loops”, section “How the seven effects are computed”.*

$$w_k = 5 + \text{clip}(\mu_k;\ 0,\ 4). \tag{10.1}$$

*Provenance (Eq. 10.1):* **[Engineered]**: drawing rule.

*Eq. 10.2. Where explained: chapter “Feedback loops”, section “How the seven effects are computed”.*

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D}\quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

*Provenance (Eq. 10.2):* **[Data-driven]**: one-step differences of Daisy results.

### 26.18.1 The whole layer, part by part

The picture below is the complete drawing of the layer Feedback loops, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

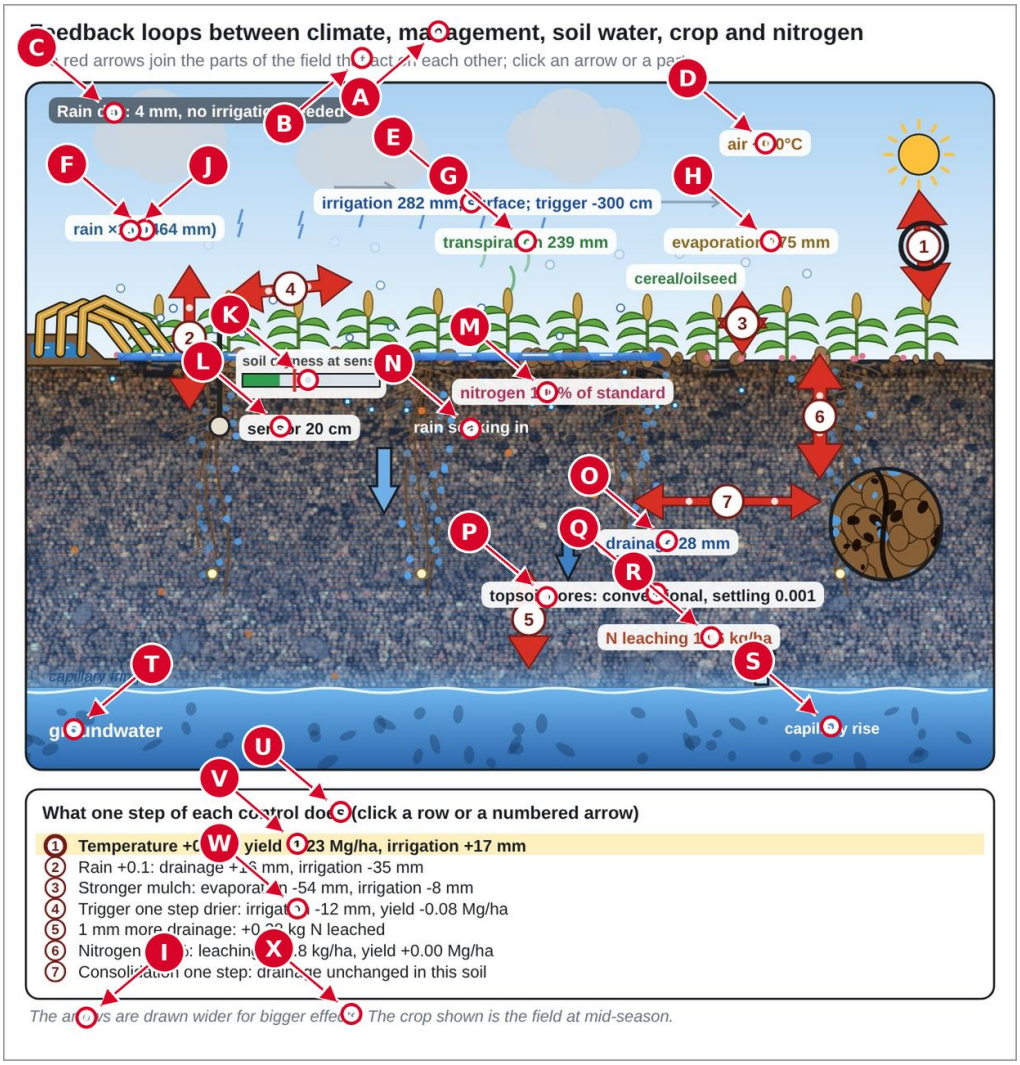



Figure 26.249: Feedback loops, part 1 of 1.

**What the arrows point to.**

- **A** The title, the feedback loops between climate, management, soil water, crop and nitrogen.
- **B** The instruction, the red arrows join the parts that act on each other, and an arrow or a part is clicked.
- **C** The banner of the rain on the day shown and whether irrigation is needed.
- **D** The air temperature change in force.
- **E** The irrigation of the year, its method and the trigger.
- **F** The rain scaling and the rain of the year.
- **G** The transpiration of the year.
- **H** The evaporation of the year and its change against bare soil.
- **I** The crop shown, the field at mid-season.
- **J** The water held by the mulch.
- **K** The gauge of the soil dryness at the sensor.
- **L** The sensor at 20 cm.
- **M** The nitrogen supply as a share of the standard.
- **N** The rain soaking into the soil.
- **O** The drainage of the year.
- **P** The aggregate that stands for the structure of the topsoil.
- **Q** The topsoil pores, the tillage and the settling rate.

- **R** The nitrogen leaching of the year.
- **S** The capillary rise from the groundwater.
- **T** The groundwater.
- **U** The table of what one step of each control does.
- **V** The row of the loop selected, highlighted and clickable.
- **W** The other rows, each loop with its effect on the yield, irrigation and drainage.
- **X** The note that the arrows are wider for bigger effects.

### 26.18.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

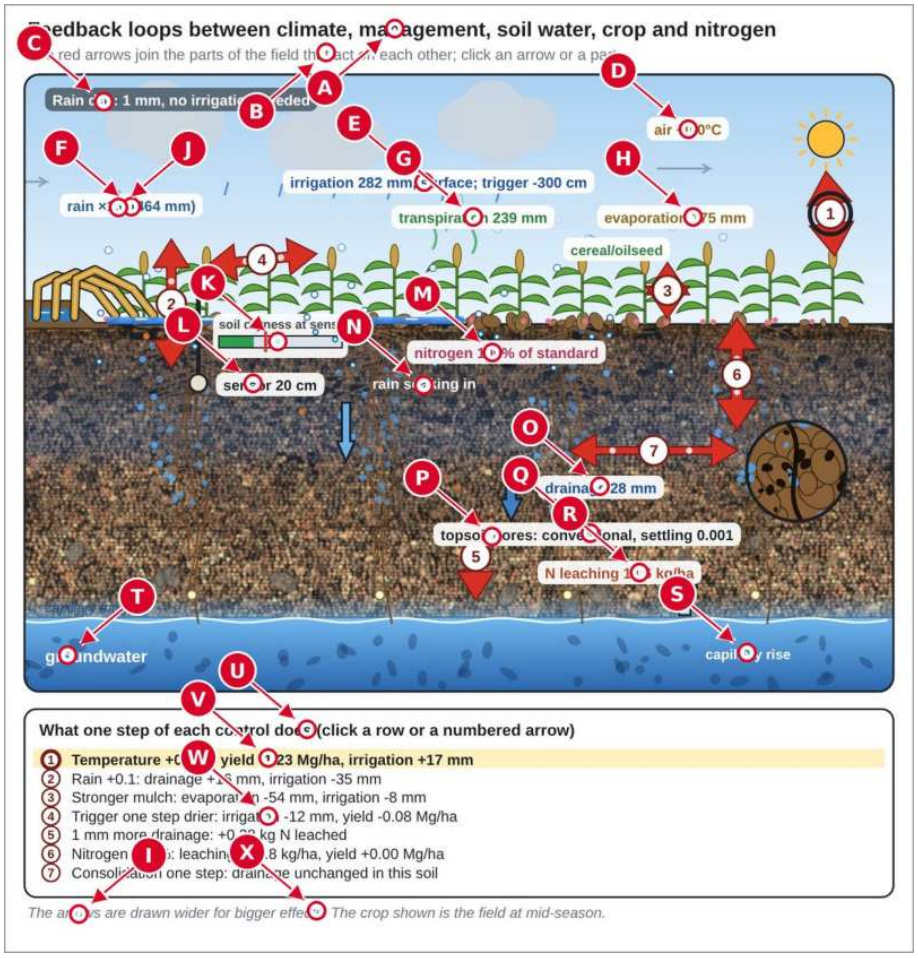



Figure 26.250: Feedback loops, part 1, playback a third of the way (Paused: each feedback loop in turn).

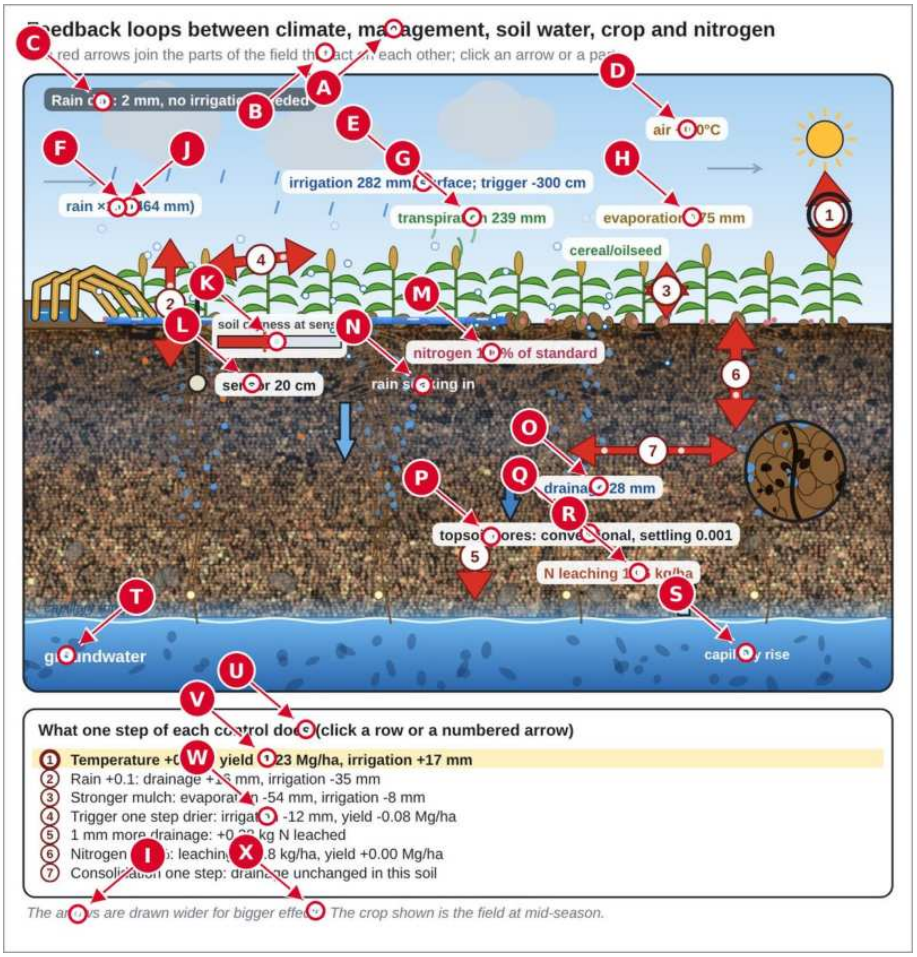



Figure 26.251: Feedback loops, part 1, playback two thirds of the way (Paused: each feedback loop in turn).

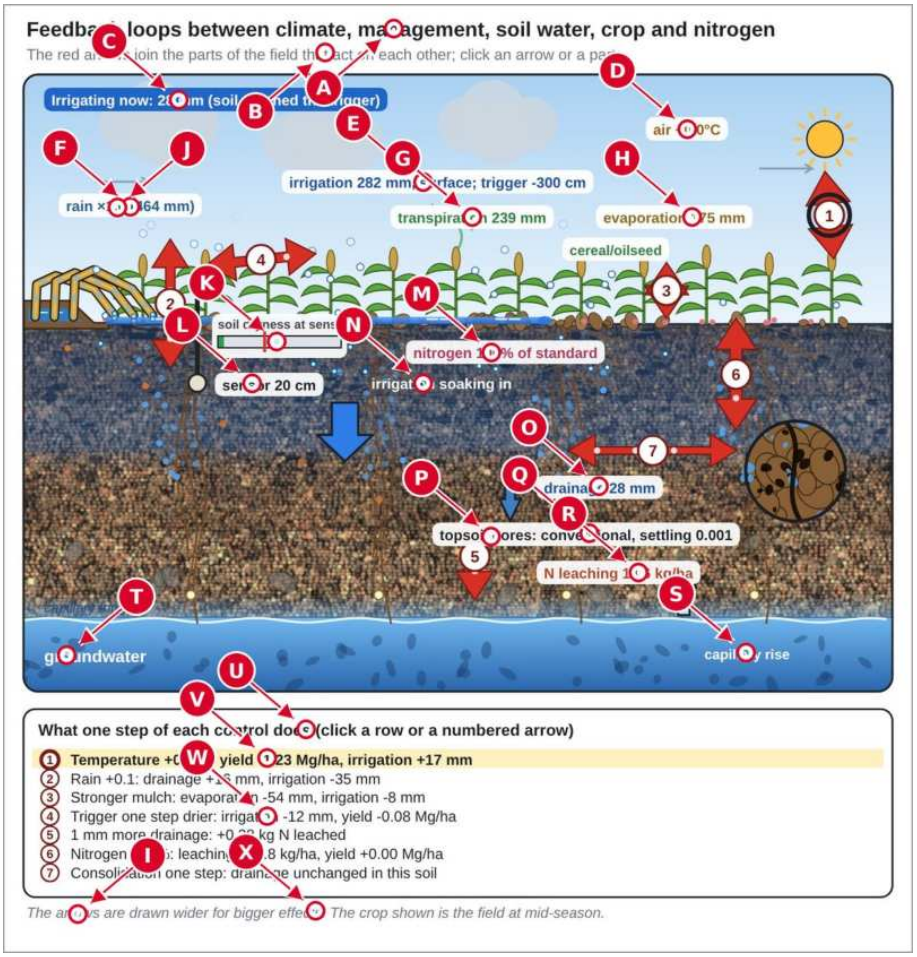



Figure 26.252: Feedback loops, part 1, playback the end (Paused: each feedback loop in turn).

## 26.19 Screens added in version 1.1

The screens below are new in version 1.1. The numbered arrows point to the headings of each part; the chapters named in each caption explain the screen and its equations.

### 26.19.1 A design before its run

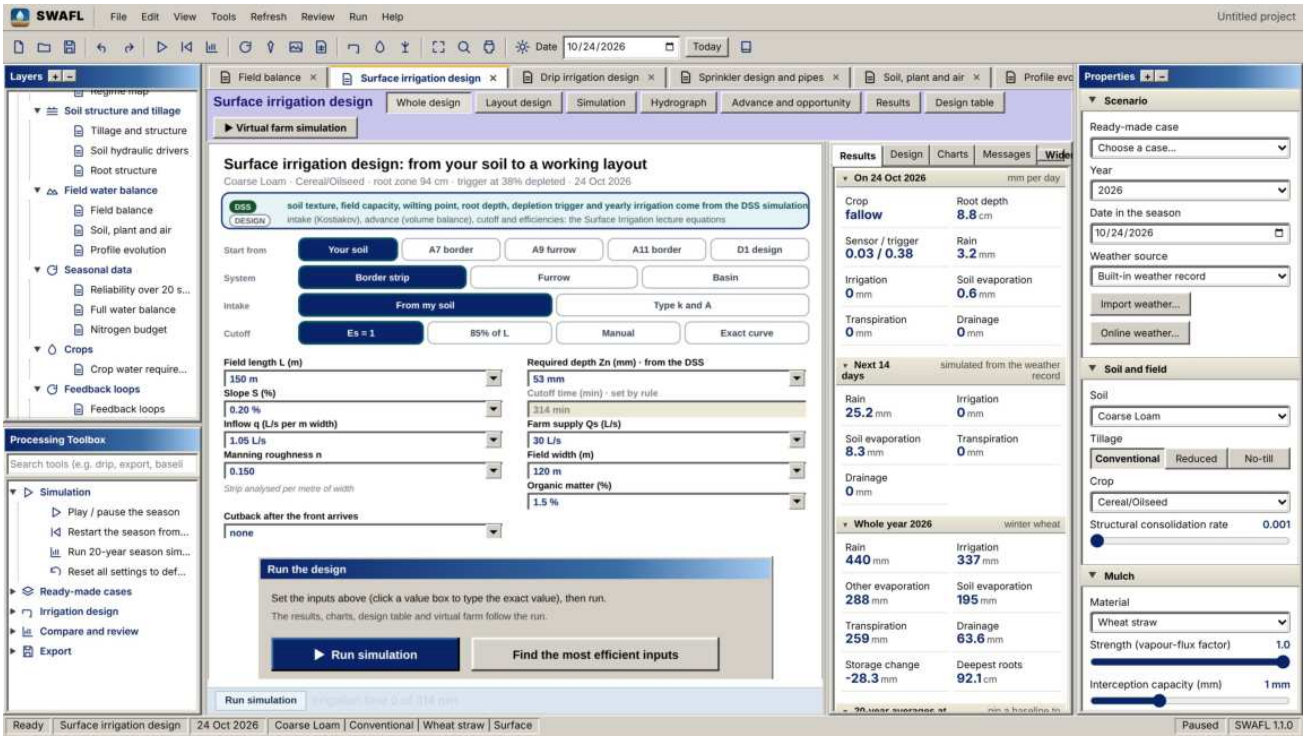



Figure 26.253: A design before its run (Chapter 18).

## 26.19.2 The value window of a design input

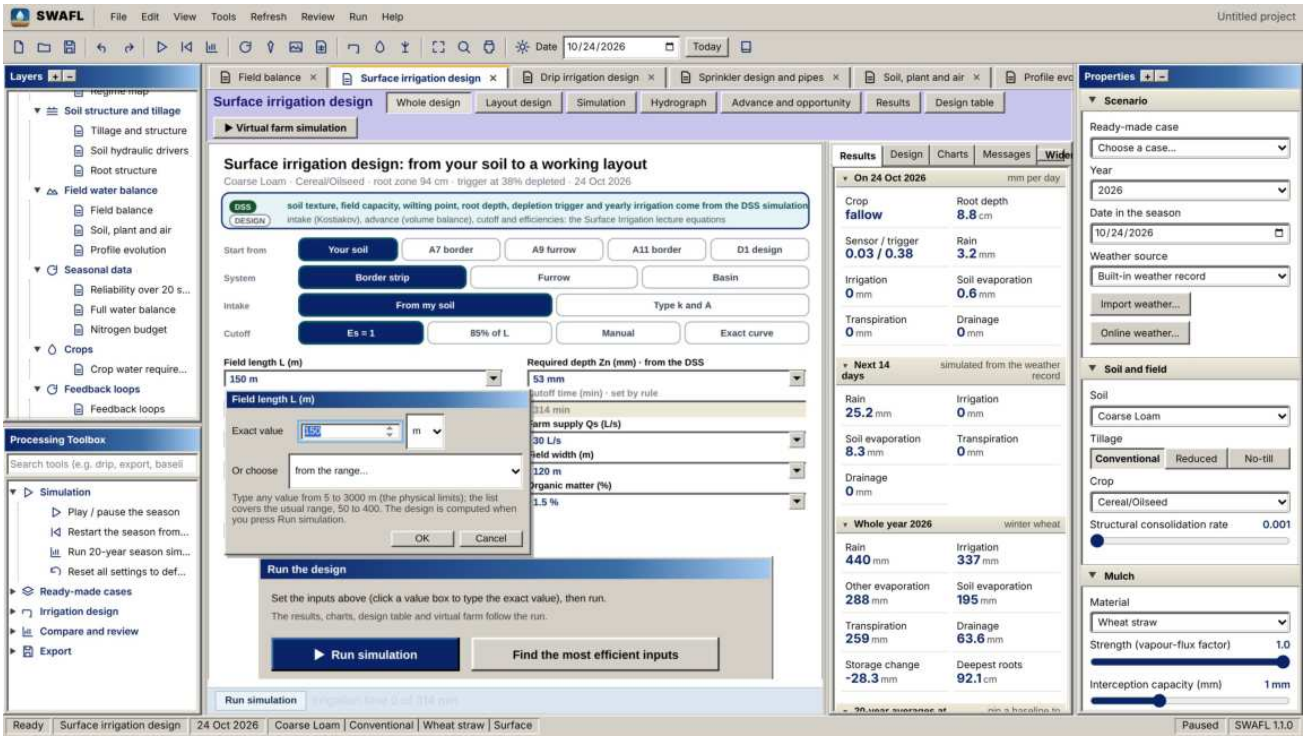



Figure 26.254: The value window of a design input (Chapter 18).

## 26.19.3 Import weather data

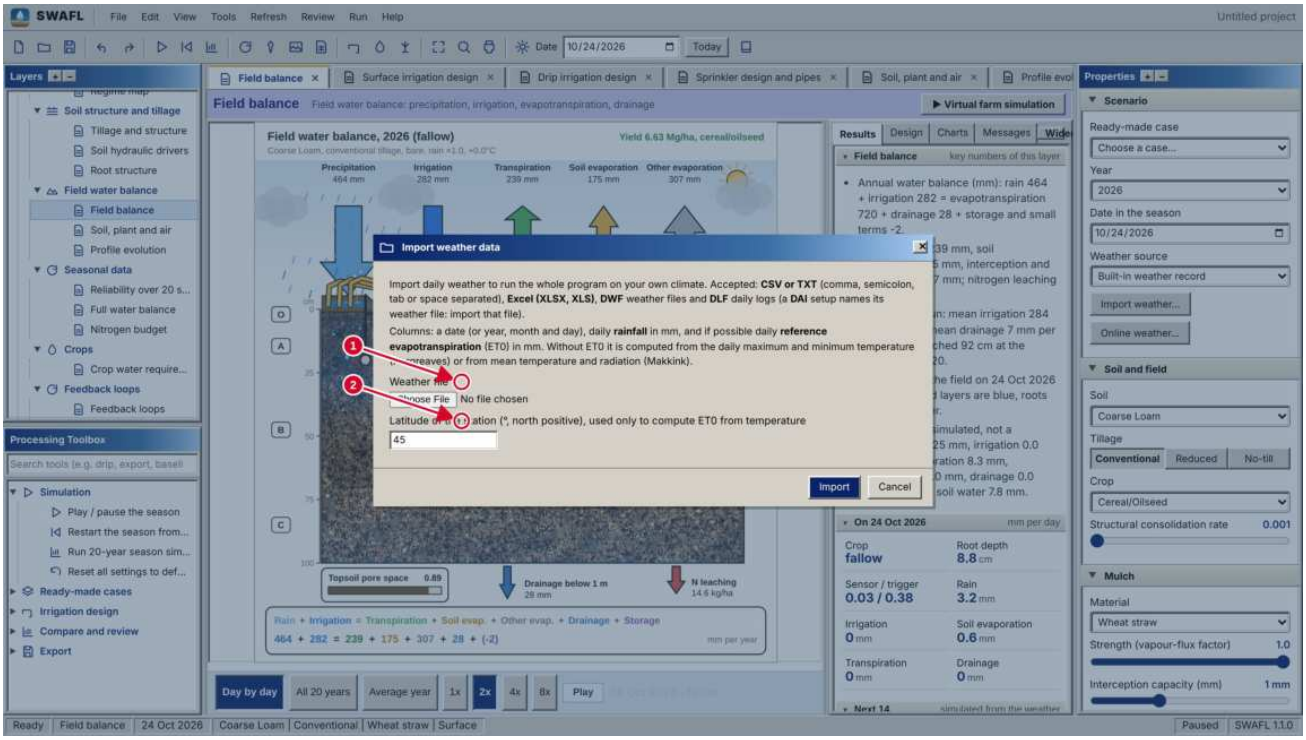



Figure 26.255: Import weather data (Chapter 19).

## 26.19.4 Online weather for a location

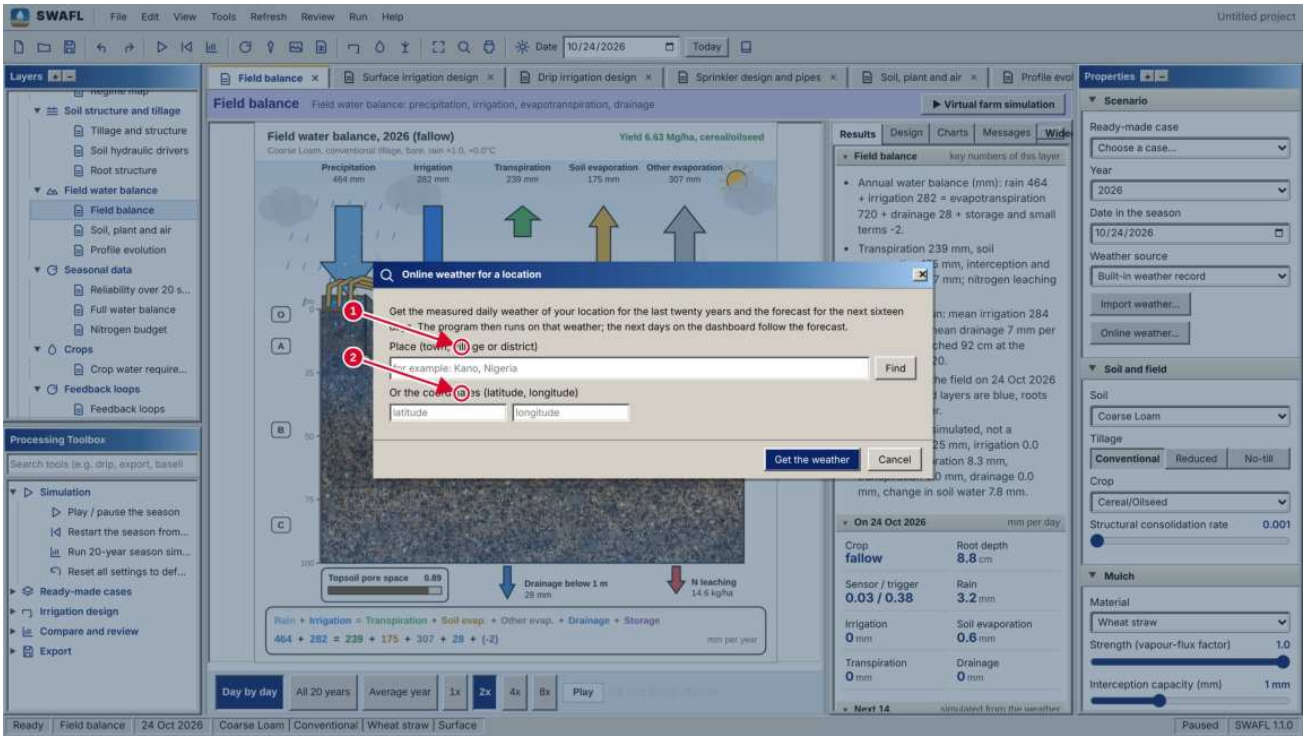



Figure 26.256: Online weather for a location (Chapter 19).

## 26.19.5 Save report as PDF: choose the layers

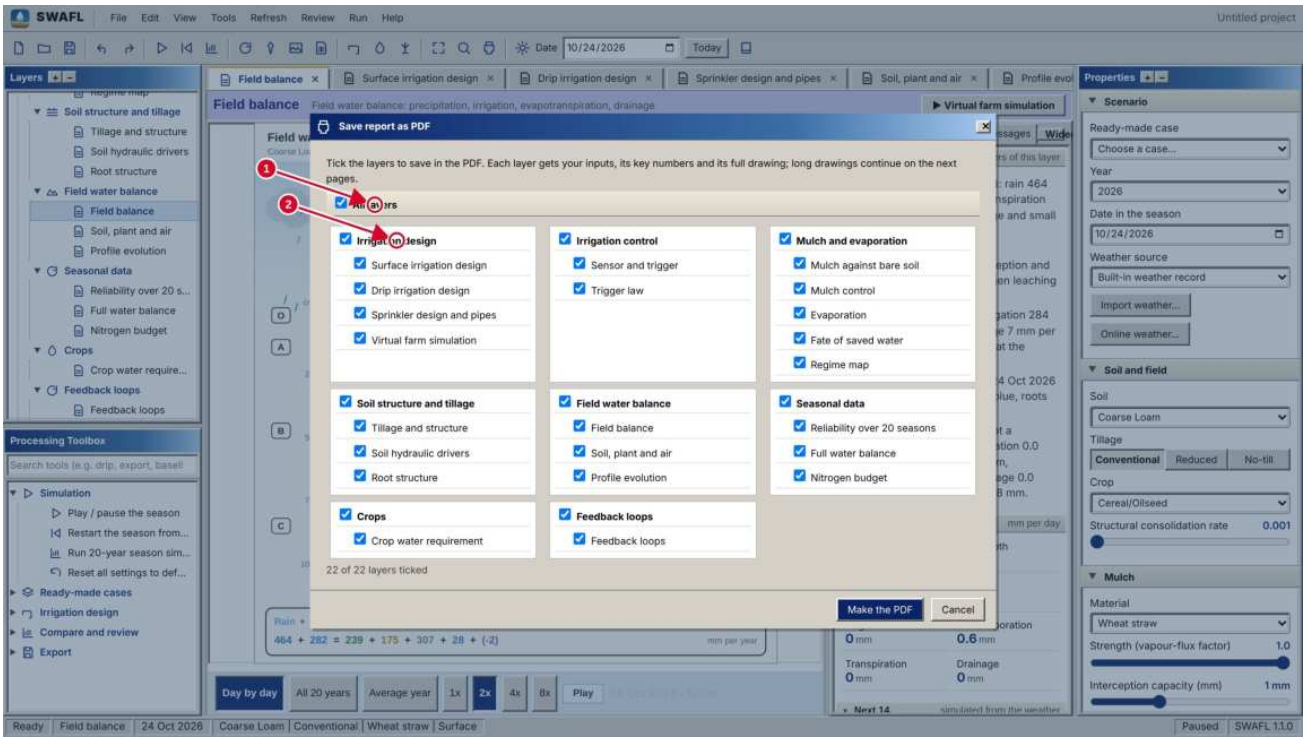



Figure 26.257: Save report as PDF: choose the layers (Chapter 23).

## 26.19.6 Field sketch to design

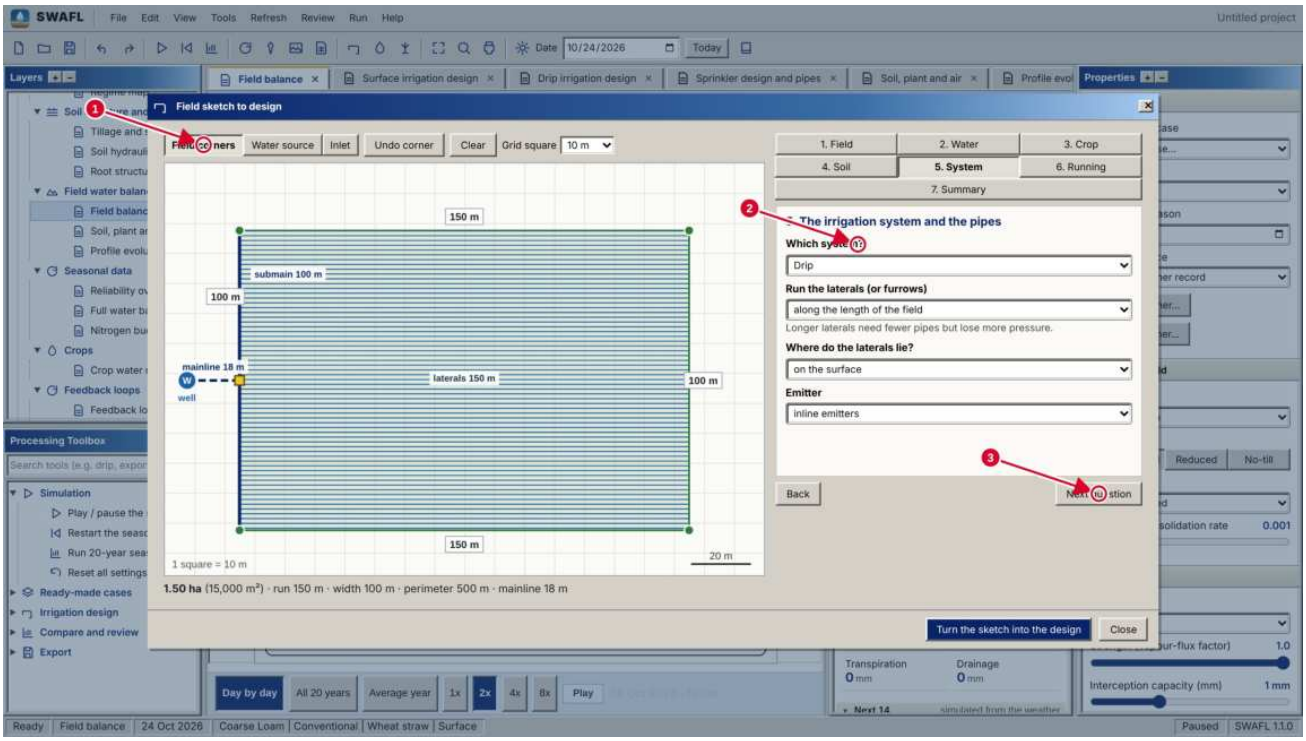



Figure 26.258: Field sketch to design (Chapter 18).

## 26.19.7 The irrigation systems compared on this field

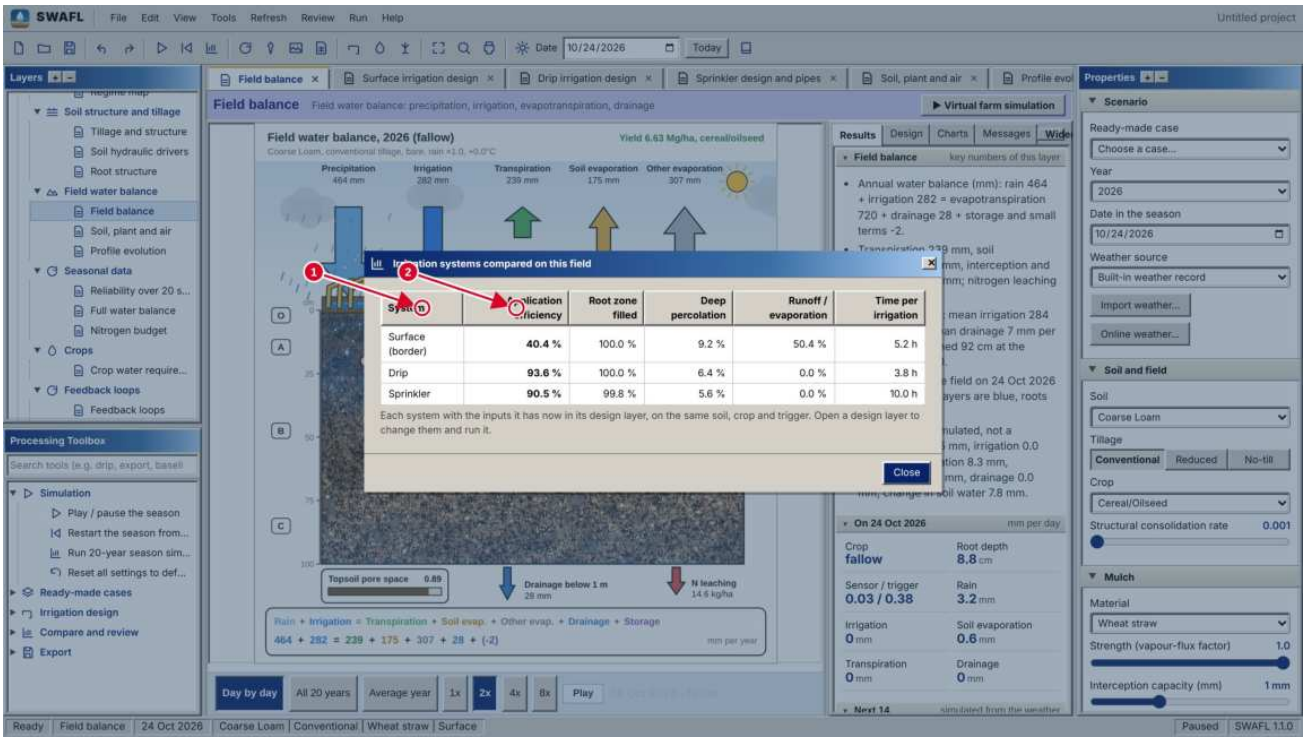



Figure 26.259: The irrigation systems compared on this field (Chapter 18).

### 26.19.8 The Charts tab of a design after its run

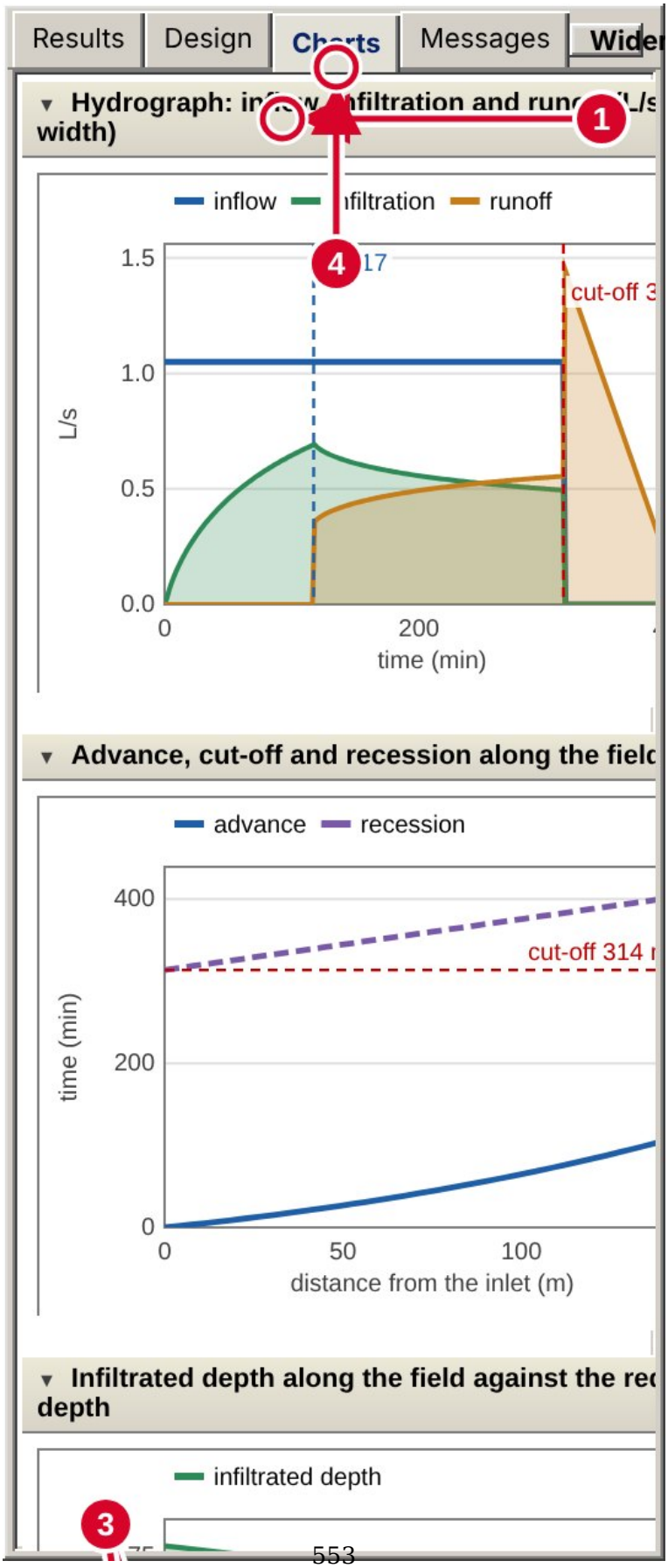





Figure 26.260: The Charts tab of a design after its run (Chapter 14).

### 26.19.9 Dashboard: the Results tab

The results at the working date and the 20-year averages at your inputs. Click a block header to fold or unfold it (Chapter 14).

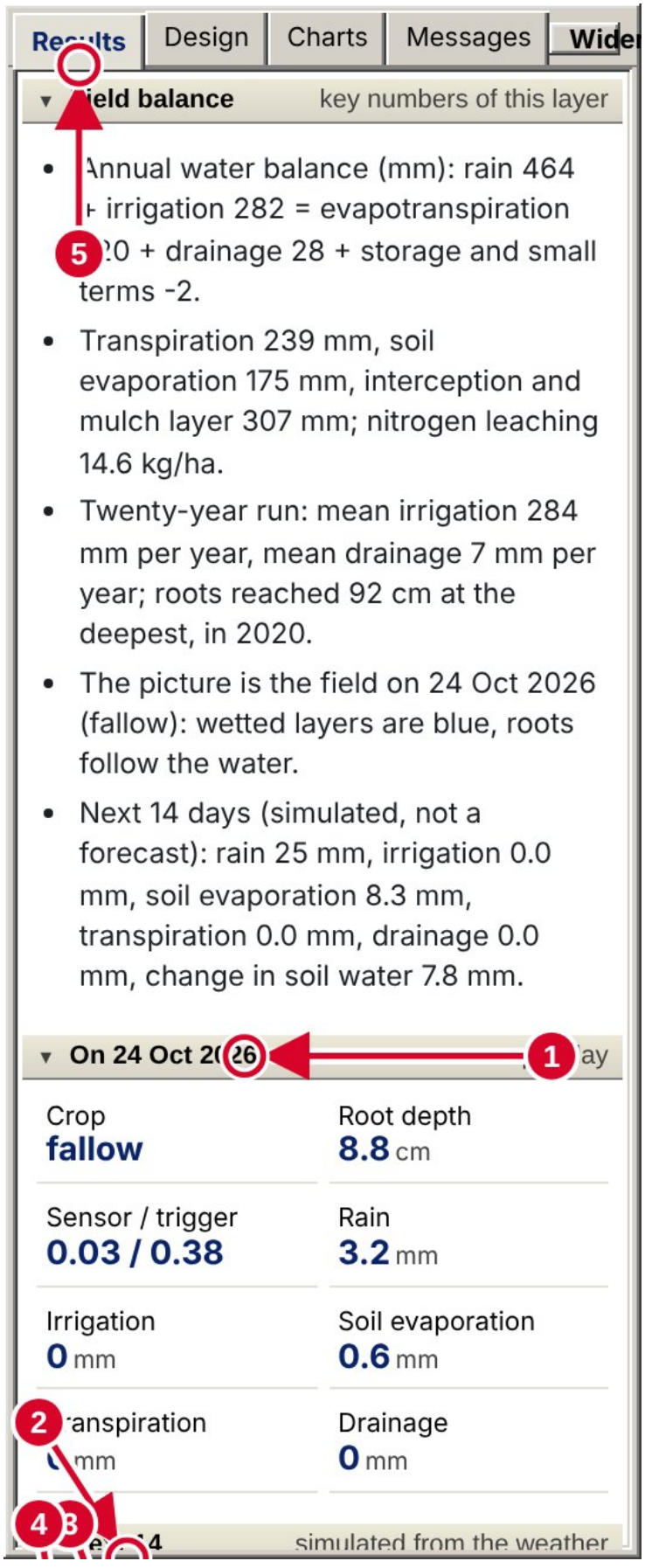



Figure 26.261: Dashboard: the Results tab.

### 26.19.10 Dashboard: the Design tab

The results of the last run of the open design: efficiencies, depths, times, the pump or the gate and the warnings of the run (Chapters 11 to 13 and 18).

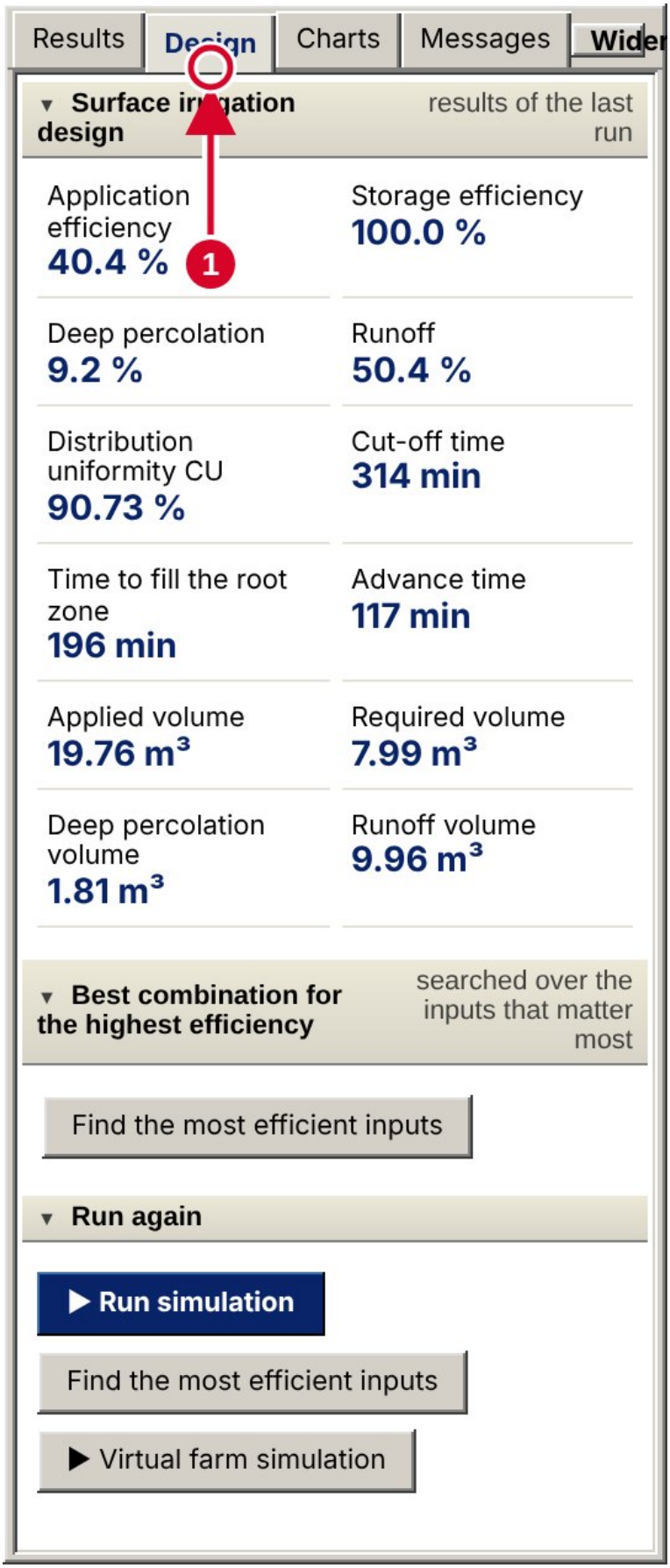



Figure 26.262: Dashboard: the Design tab.

### 26.19.11 Dashboard: the Charts tab of the Field balance

The yearly charts of the twenty seasons; every chart can be folded (Chapter 14).

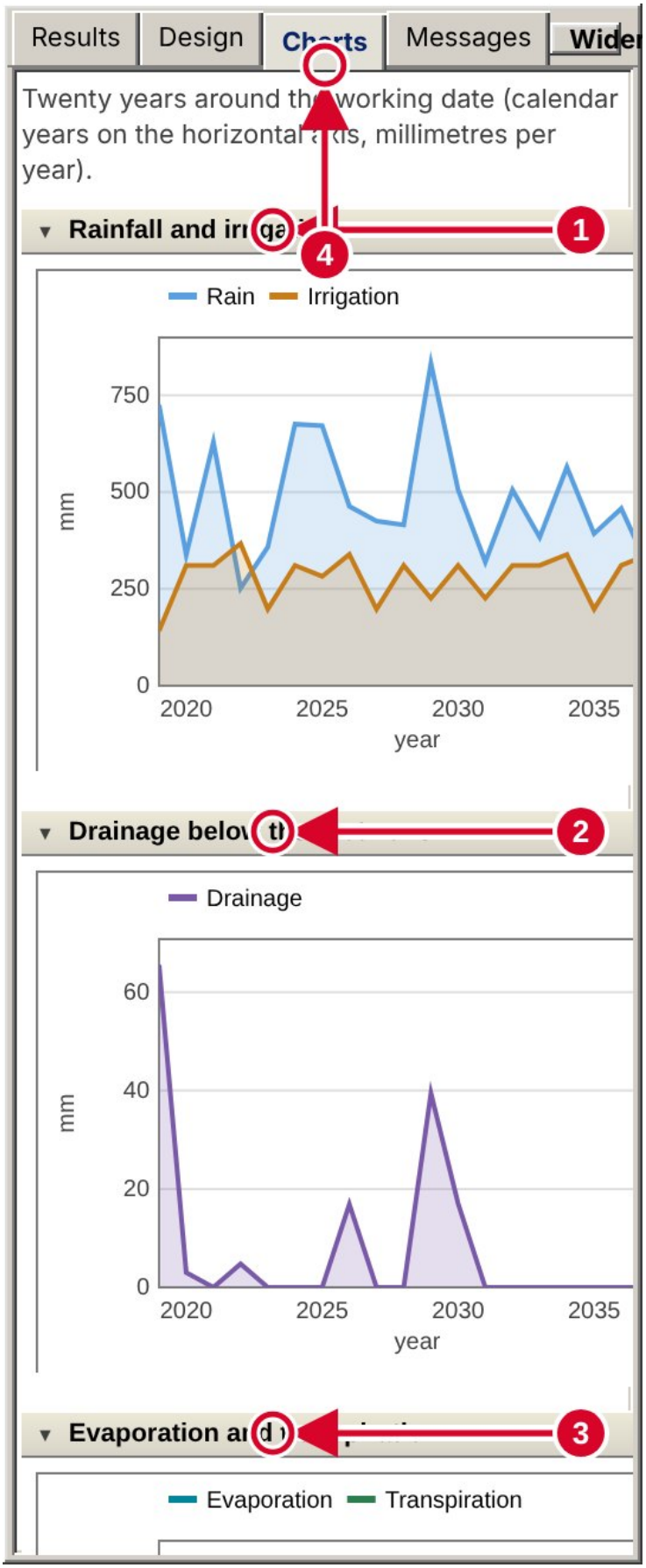



Figure 26.263: Dashboard: the Charts tab of the Field balance.

### 26.19.12 Dashboard: the Messages tab

The warnings first, then all messages of the session (Chapter 14).

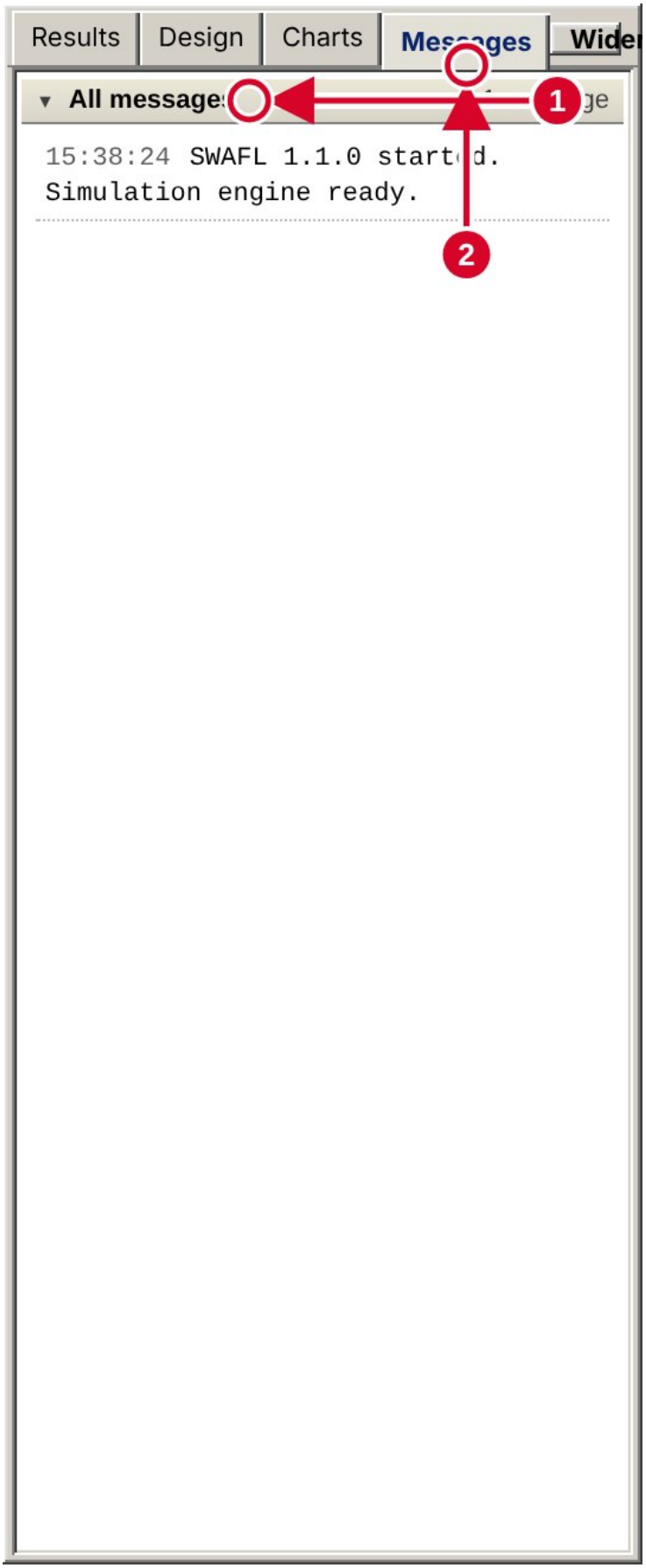



Figure 26.264: Dashboard: the Messages tab.

## 26.19.13 The surface irrigation design after its run

The whole workbench with the surface design open after Run simulation (Chapters 11 and 18).

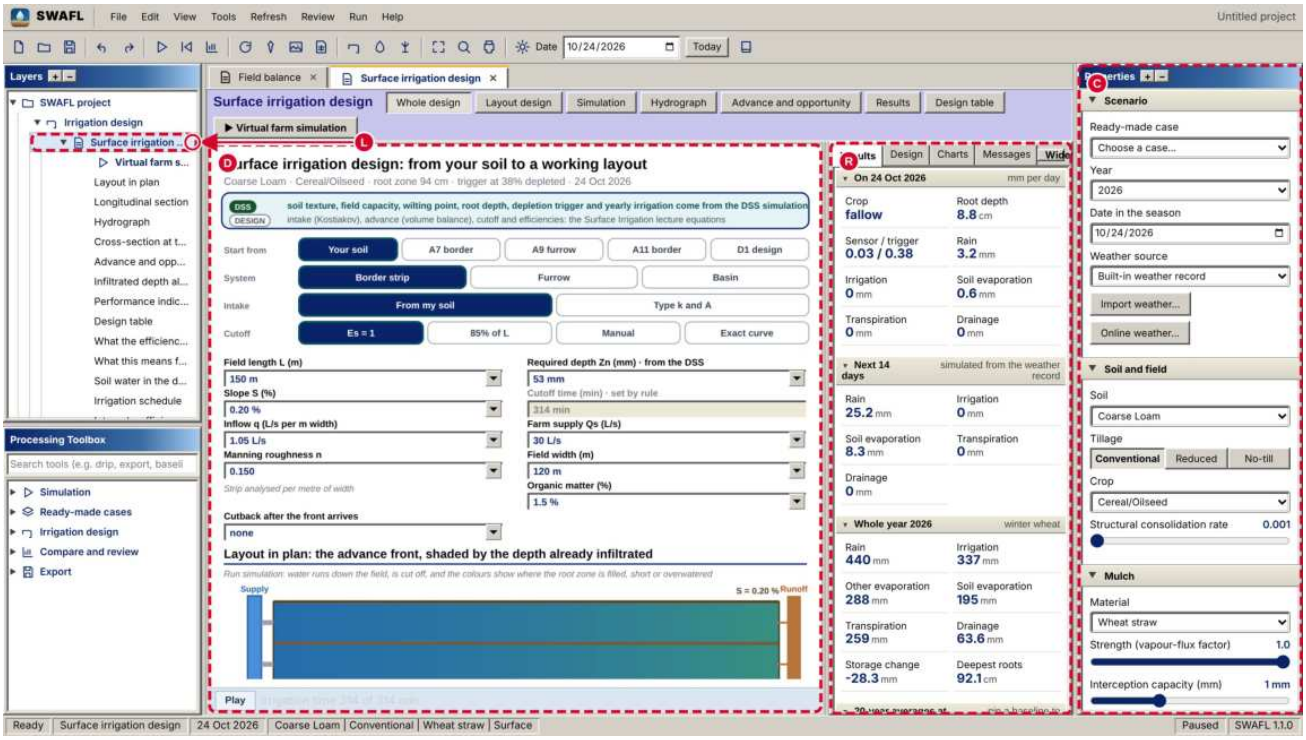



Figure 26.265: The surface irrigation design after its run.

## 26.19.14 The drip irrigation design after its run

The whole workbench with the drip design open after Run simulation (Chapters 12 and 18).

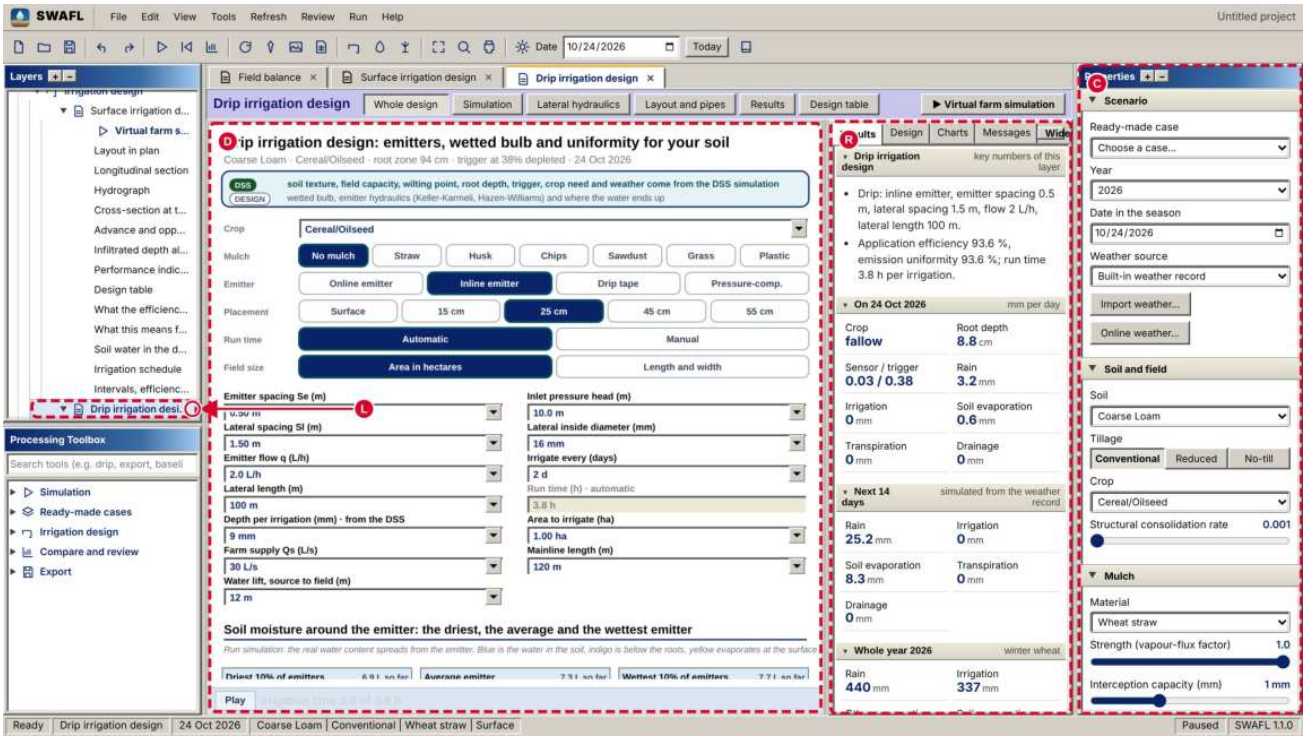



Figure 26.266: The drip irrigation design after its run.

## 26.19.15 The sprinkler design after its run

The whole workbench with the sprinkler design open after Run simulation (Chapters 13 and 18).

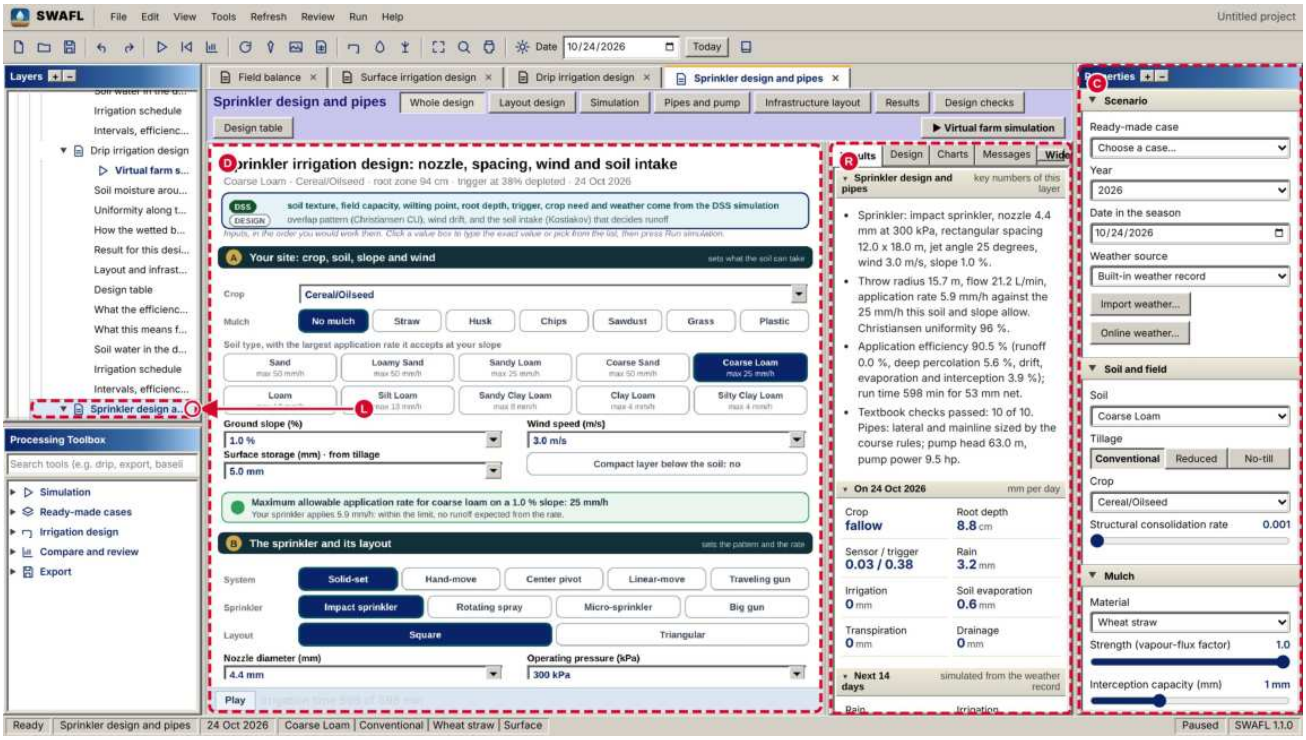



Figure 26.267: The sprinkler design after its run.

## 26.19.16 Crop water requirement

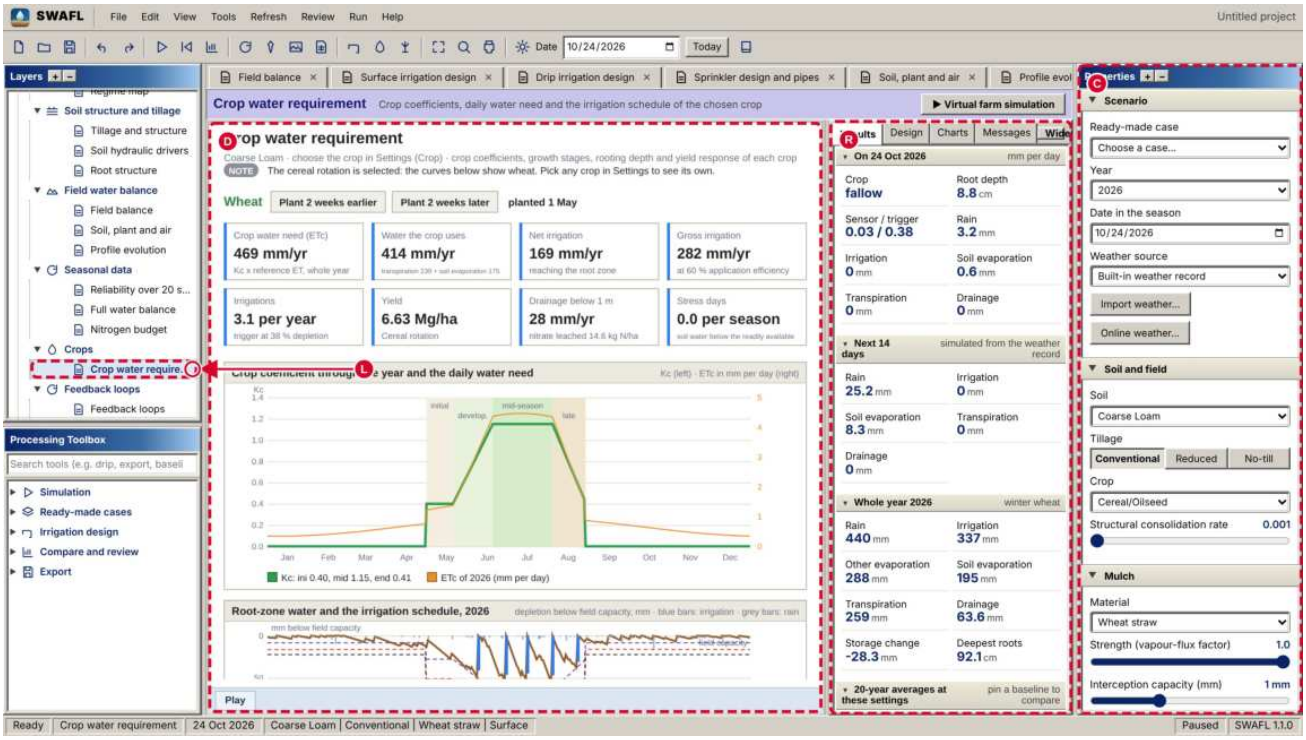



Figure 26.268: Crop water requirement: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 21; its equations are Eqs. 21.1 to 21.9.

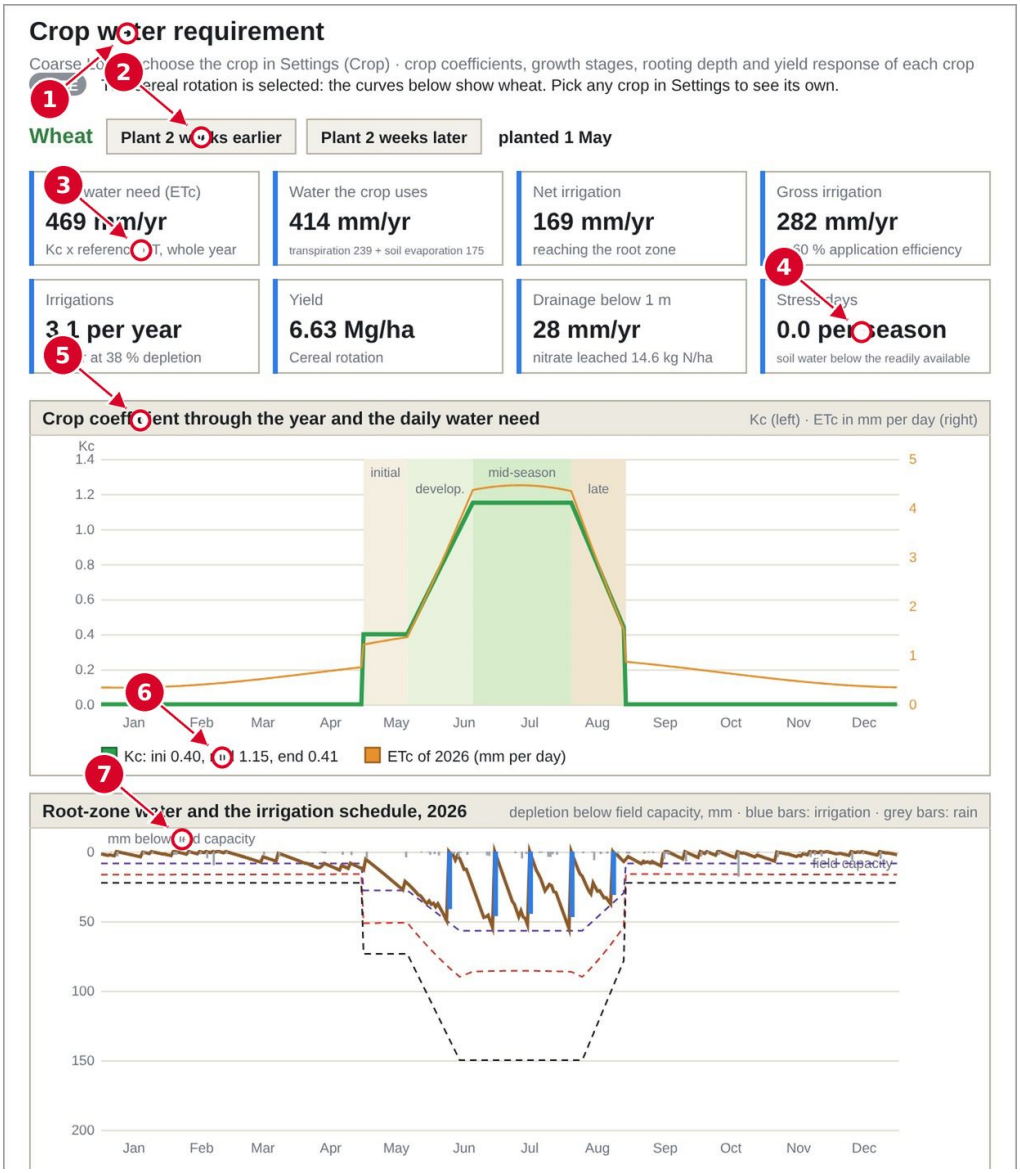
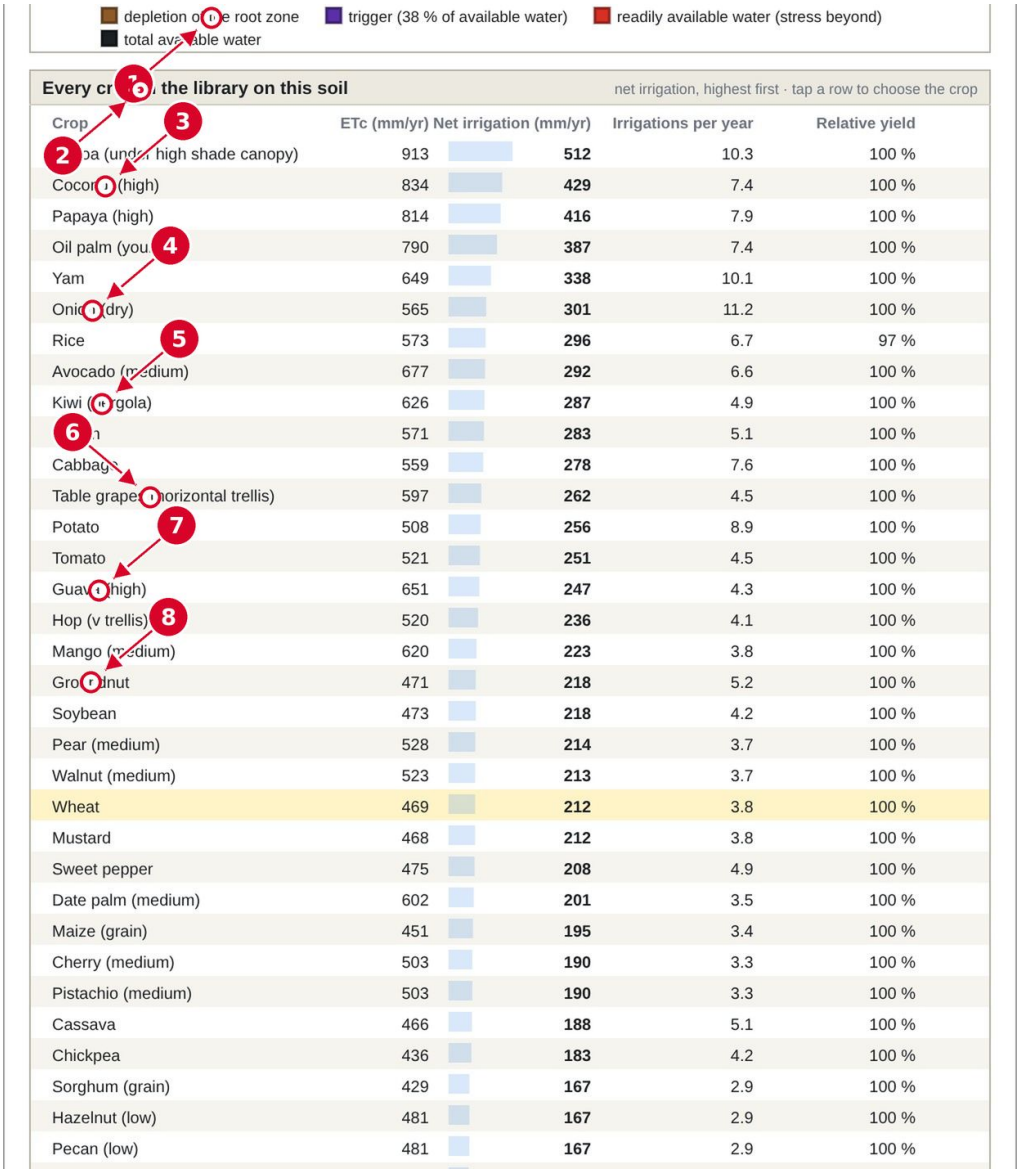



Figure 26.269: Crop water requirement, part 1 of 3.

**What the arrows point to.**

- **1** The heading or label *Crop water requirement*.
- **2** The heading or label *Plant 2 weeks earlier*.
- **3** The heading or label *Kc x reference ET, whole year*.
- **4** The heading or label *0.0 per season*.
- **5** The heading or label *Crop coefficient through the year and the daily water need*.
- **6** The heading or label *Kc: ini 0.40, mid 1.15, end 0.41*.
- **7** The heading or label *mm below field capacity*.




| Crop | ETc (mm/yr) | Net irrigation (mm/yr) | Irrigations per year | Relative yield |
|---|---|---|---|---|
| [illegible] high shade canopy) | 913 | **512** | 10.3 | 100 % |
| Coconut (high) | 834 | **429** | 7.4 | 100 % |
| Papaya (high) | 814 | **416** | 7.9 | 100 % |
| Oil palm (young) | 790 | **387** | 7.4 | 100 % |
| Yam | 649 | **338** | 10.1 | 100 % |
| Onion (dry) | 565 | **301** | 11.2 | 100 % |
| Rice | 573 | **296** | 6.7 | 97 % |
| Avocado (medium) | 677 | **292** | 6.6 | 100 % |
| Kiwi (pergola) | 626 | **287** | 4.9 | 100 % |
| [illegible] | 571 | **283** | 5.1 | 100 % |
| Cabbage | 559 | **278** | 7.6 | 100 % |
| Table grapes (horizontal trellis) | 597 | **262** | 4.5 | 100 % |
| Potato | 508 | **256** | 8.9 | 100 % |
| Tomato | 521 | **251** | 4.5 | 100 % |
| Guava (high) | 651 | **247** | 4.3 | 100 % |
| Hop (v trellis) | 520 | **236** | 4.1 | 100 % |
| Mango (medium) | 620 | **223** | 3.8 | 100 % |
| Groundnut | 471 | **218** | 5.2 | 100 % |
| Soybean | 473 | **218** | 4.2 | 100 % |
| Pear (medium) | 528 | **214** | 3.7 | 100 % |
| Walnut (medium) | 523 | **213** | 3.7 | 100 % |
| Wheat | 469 | **212** | 3.8 | 100 % |
| Mustard | 468 | **212** | 3.8 | 100 % |
| Sweet pepper | 475 | **208** | 4.9 | 100 % |
| Date palm (medium) | 602 | **201** | 3.5 | 100 % |
| Maize (grain) | 451 | **195** | 3.4 | 100 % |
| Cherry (medium) | 503 | **190** | 3.3 | 100 % |
| Pistachio (medium) | 503 | **190** | 3.3 | 100 % |
| Cassava | 466 | **188** | 5.1 | 100 % |
| Chickpea | 436 | **183** | 4.2 | 100 % |
| Sorghum (grain) | 429 | **167** | 2.9 | 100 % |
| Hazelnut (low) | 481 | **167** | 2.9 | 100 % |
| Pecan (low) | 481 | **167** | 2.9 | 100 % |

Figure 26.270: Crop water requirement, part 2 of 3.

**What the arrows point to.**

- **1** The heading or label *depletion of the root zone*.
- **2** The heading or label *Every crop in the library on this soil*.
- **3** The heading or label *Coconut (high)*.
- **4** The heading or label *Onion (dry)*.
- **5** The heading or label *Kiwi (pergola)*.
- **6** The heading or label *Table grapes (horizontal trellis)*.
- **7** The heading or label *Guava (high)*.
- **8** The heading or label *Groundnut*.

| | | | | | |
|---|---|---|---|---|---|
| Cherimoya (medium) | 545 | | **161** | 2.8 | 100 % |
| Citrus (medium [illegible]y, short) | 552 | | **152** | 2.6 | 100 % |
| [illegible]ple (medium) | 459 | | **149** | 2.6 | 100 % |
| Cowpea (beans) | 398 | | **148** | 3.4 | 100 % |
| [illegible] | 395 | | **144** | 2.6 | 100 % |
| Wine grapes (pergola) | 457 | | **125** | 2.2 | 100 % |
| Green gram (mung bean) | 372 | | **123** | 2.9 | 100 % |
| Peach (low) | 435 | | **119** | 2.1 | 100 % |
| Almond (medium) | 435 | | **119** | 2.1 | 100 % |
| Apricot (young) | 380 | | **76** | 1.3 | 100 % |
| [illegible]live (traditional, medium density) | 446 | | **61** | 1.1 | 100 % |
| Persimmon (young, low) | 358 | | **58** | 1.0 | 100 % |
| Pomegranate (low) | 358 | | **58** | 1.0 | 100 % |
| Highbush blueberry (low) | 338 | | **55** | 1.1 | 100 % |
| Fig tree (young, low) | 327 | | **24** | 0.4 | 100 % |

Figure 26.271: Crop water requirement, part 3 of 3.

**What the arrows point to.**

- **1** The heading or label *Cherimoya (medium)*.
- **2** The heading or label *Cowpea (beans)*.
- **3** The heading or label *Green gram (mung bean)*.
- **4** The heading or label *Apricot (young)*.
- **5** The heading or label *Pomegranate (low)*.

## 26.19.17 Reliability over 20 seasons

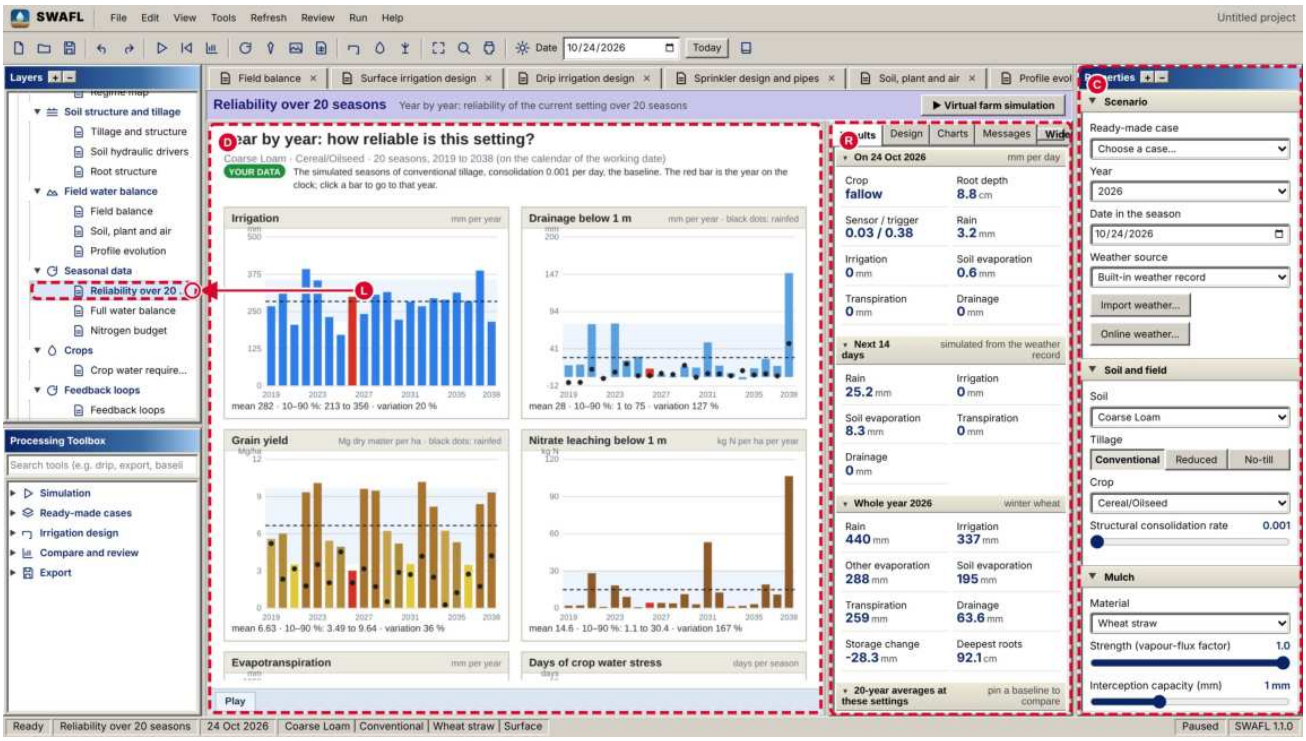



Figure 26.272: Reliability over 20 seasons: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 22; its equations are Eqs. 22.1.

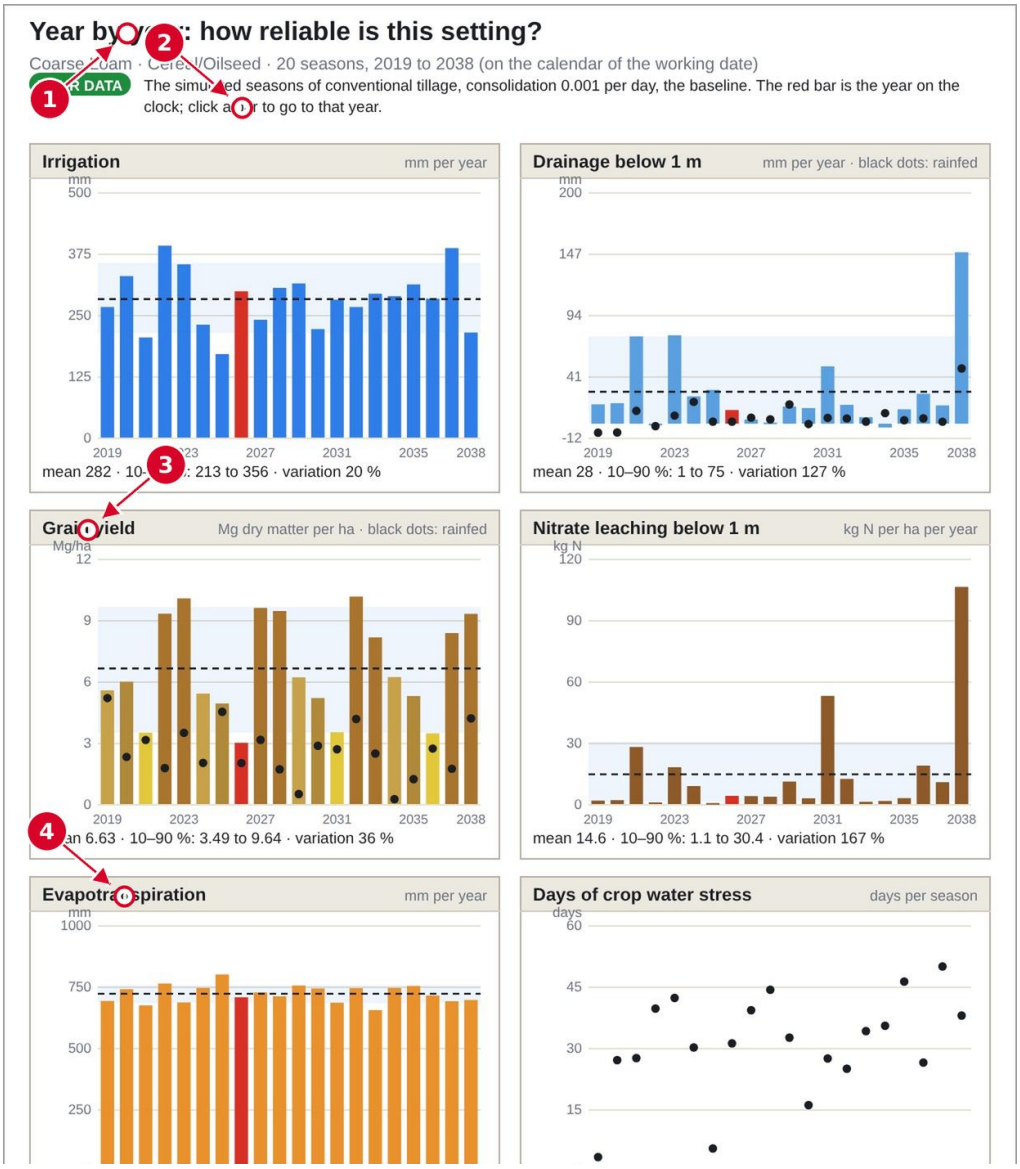



Figure 26.273: Reliability over 20 seasons, part 1 of 2.

**What the arrows point to.**

- **1** The heading or label *Year by year: how reliable is this setting?*.
- **2** The heading or label *clock; click a bar to go to that year.*.
- **3** The heading or label *Grain yield*.
- **4** The heading or label *Evapotranspiration*.

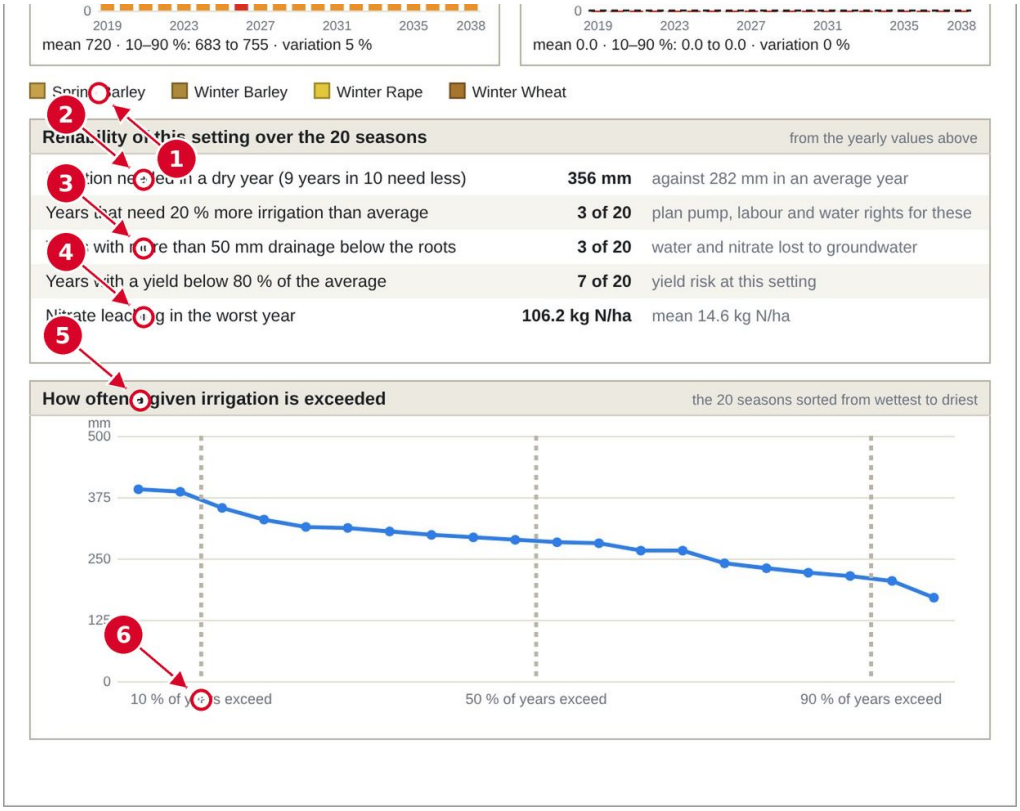



Figure 26.274: Reliability over 20 seasons, part 2 of 2.

**What the arrows point to.**

- **1** The heading or label *Spring Barley*.
- **2** The heading or label *Irrigation needed in a dry year (9 years in 10 need less)*.
- **3** The heading or label *Years with more than 50 mm drainage below the roots*.
- **4** The heading or label *Nitrate leaching in the worst year*.
- **5** The heading or label *How often a given irrigation is exceeded*.
- **6** The heading or label *10 % of years exceed*.

## 26.19.18 Full water balance

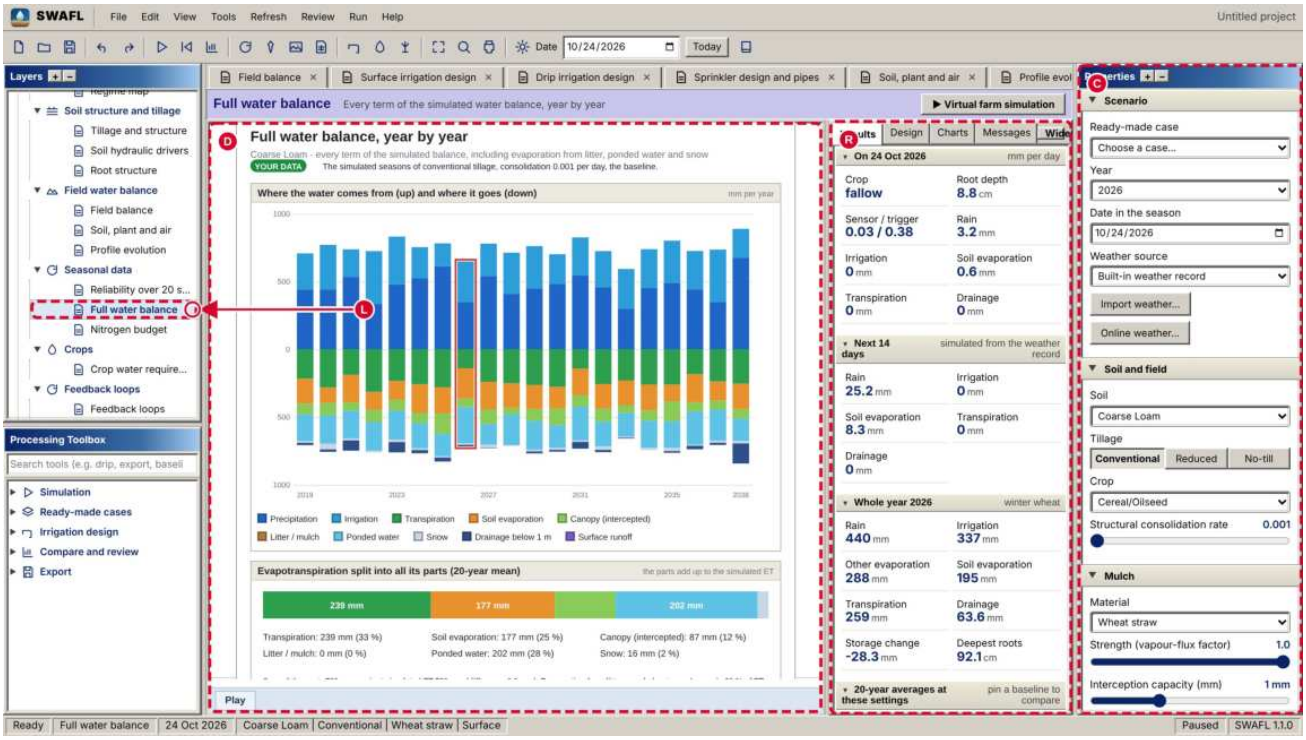



Figure 26.275: Full water balance: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 22; its equations are Eqs. 22.1.

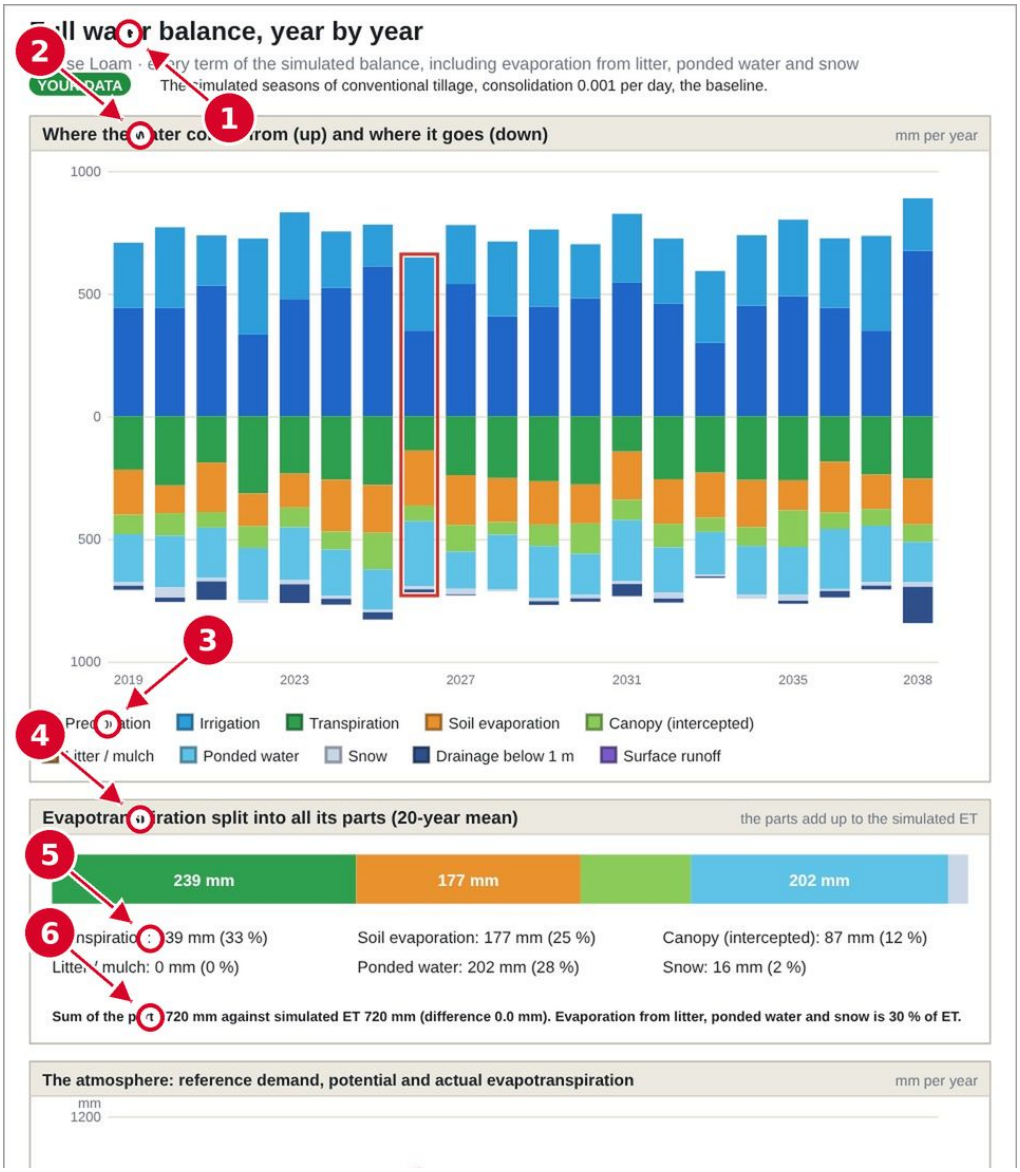



Figure 26.276: Full water balance, part 1 of 2.

**What the arrows point to.**

- **1** The heading or label *Full water balance, year by year*.
- **2** The heading or label *Where the water comes from (up) and where it goes (down)*.
- **3** The heading or label *Precipitation*.
- **4** The heading or label *Evapotranspiration split into all its parts (20-year mean)*.
- **5** The heading or label *Transpiration: 239 mm (33 %)*.
- **6** The heading or label *Sum of the parts 720 mm against simulated ET 720 mm (difference 0.0 mm). Evaporation from litter, ponded water and snow is 30 % of ET.*.

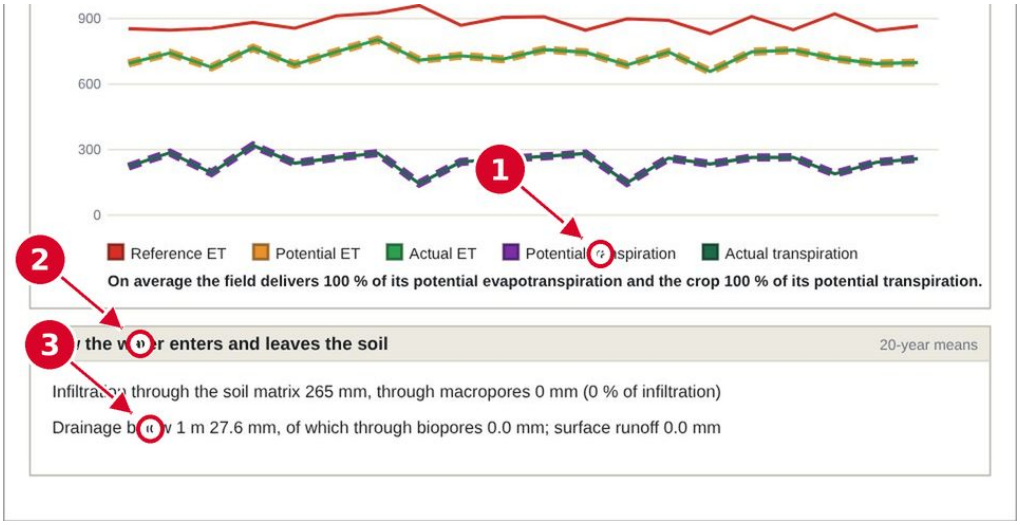



Figure 26.277: Full water balance, part 2 of 2.

**What the arrows point to.**

- **1** The heading or label *Potential transpiration*.
- **2** The heading or label *How the water enters and leaves the soil*.
- **3** The heading or label *Drainage below 1 m 27.6 mm, of which through biopores 0.0 mm; surface runoff 0.0 mm*.

## 26.19.19 Nitrogen budget

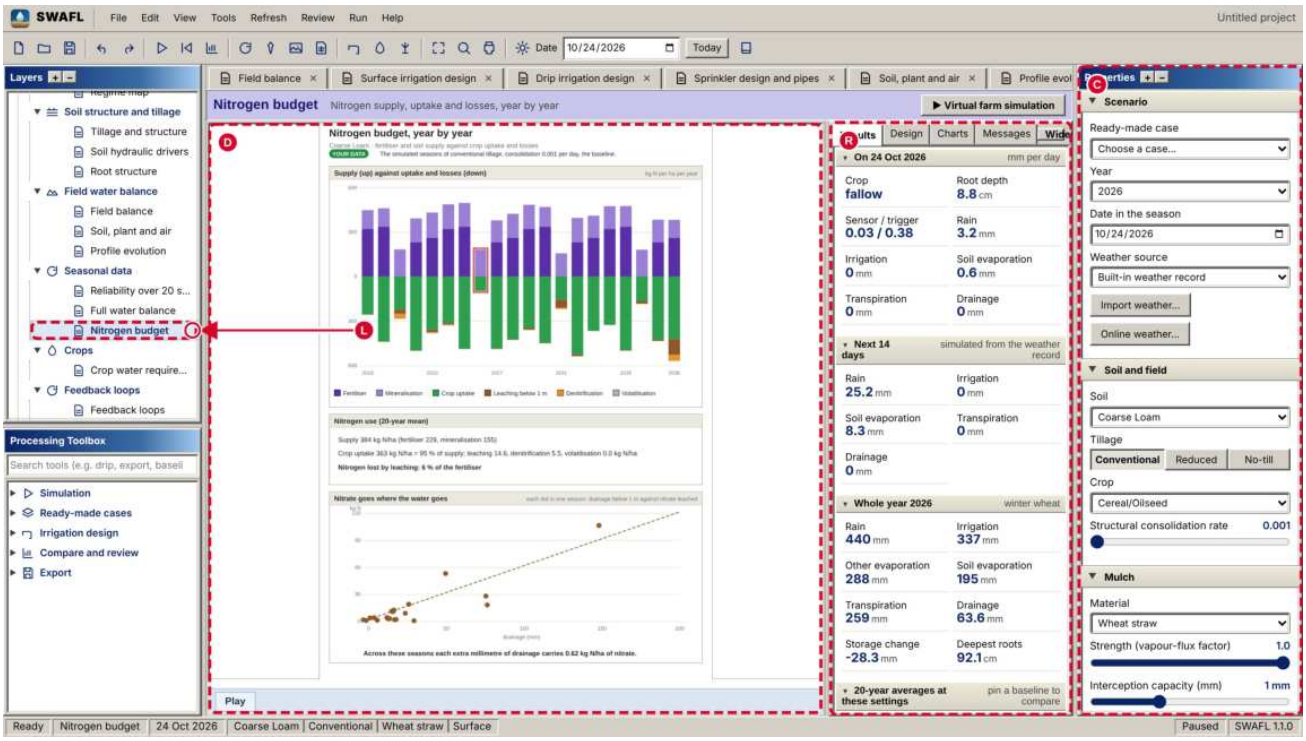



Figure 26.278: Nitrogen budget: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 22; its equations are Eqs. 22.1.

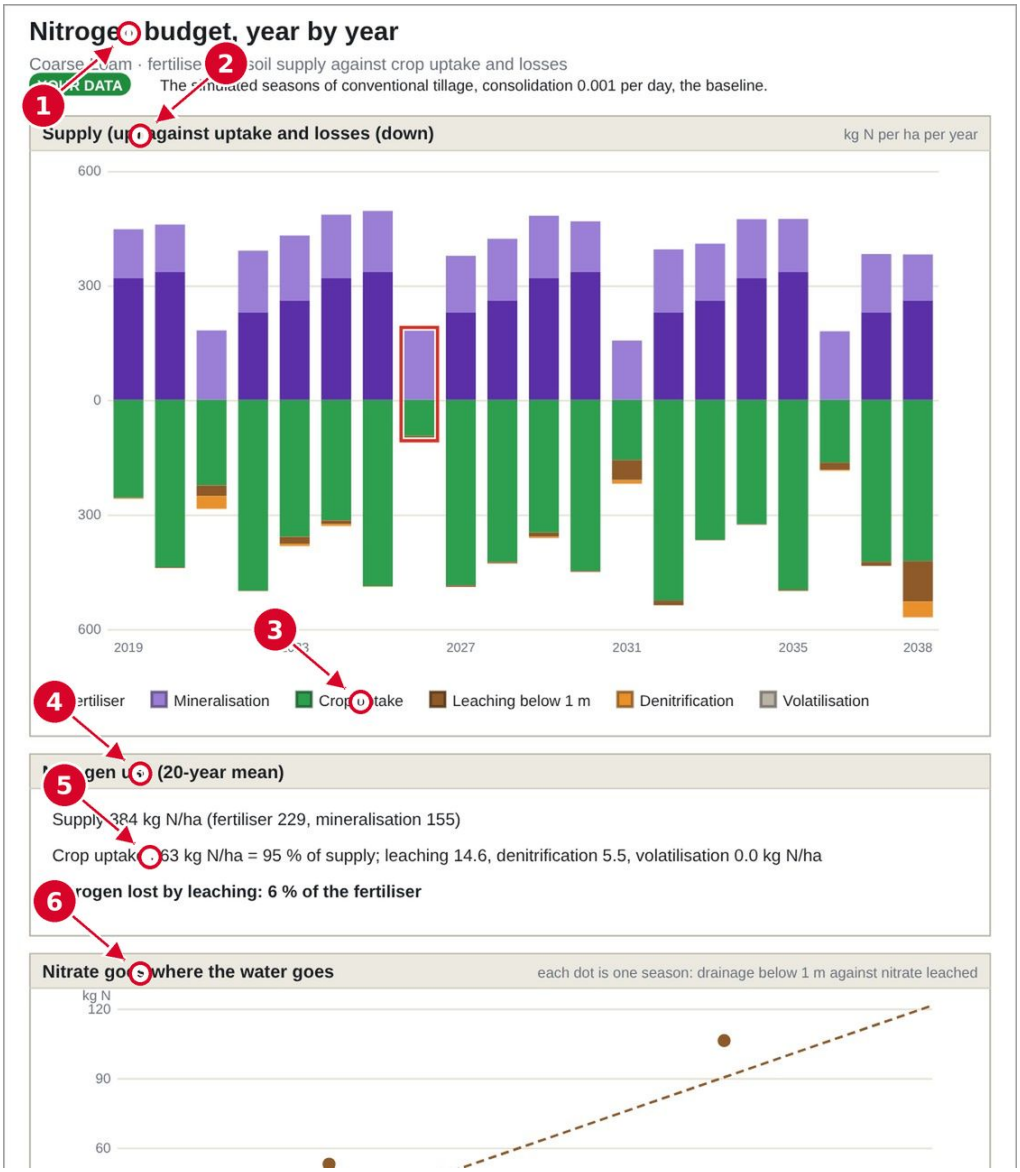



Figure 26.279: Nitrogen budget, part 1 of 2.

**What the arrows point to.**

- **1** The heading or label *Nitrogen budget, year by year*.
- **2** The heading or label *Supply (up) against uptake and losses (down)*.
- **3** The heading or label *Crop uptake*.
- **4** The heading or label *Nitrogen use (20-year mean)*.
- **5** The heading or label *Crop uptake 363 kg N/ha = 95 % of supply; leaching 14.6, denitrification 5.5, volatilisation 0.0 kg N/ha*.
- **6** The heading or label *Nitrate goes where the water goes*.

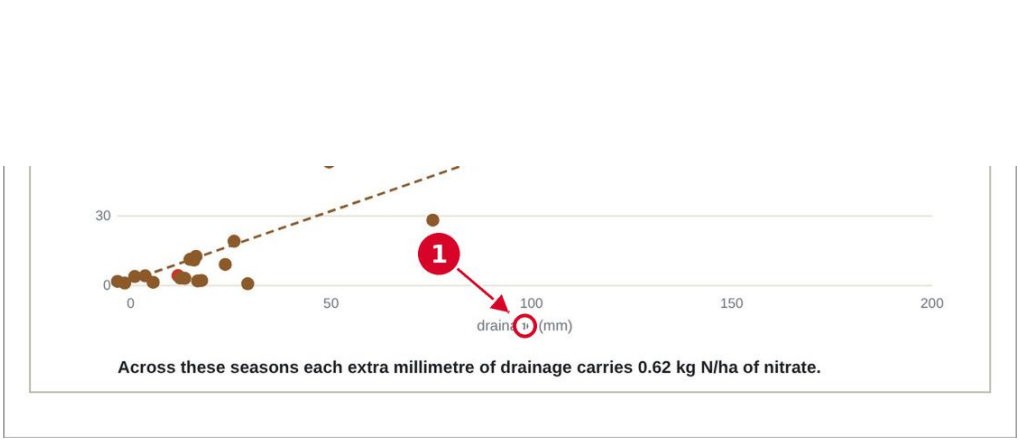



Figure 26.280: Nitrogen budget, part 2 of 2.

**What the arrows point to.**

- **1** The heading or label *drainage (mm)*.

## 26.20 Reverse index: where each equation appears on screen

| Equation | Explained in chapter | Screens that use it |
|---|---|---|

4.1 | Dates | The toolbar; The workbench |

4.2 | Dates | The toolbar; The workbench |

5.1 | The two computing engines | Soil, plant and air; Tillage and structure |

5.2 | The two computing engines | Soil, plant and air |

5.3 | The two computing engines | Soil, plant and air; Trigger law |

5.4 | The two computing engines | Evaporation; Feedback loops; Field balance; Mulch against bare soil; Mulch control; Sensor and trigger; Soil hydraulic drivers |

5.5 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.6 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.7 | The two computing engines | Feedback loops; Field balance; Mulch against bare soil; Sensor and trigger; Soil hydraulic drivers |

5.8 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.9 | The two computing engines | Feedback loops; Field balance; Mulch against bare soil; Mulch control; Sensor and trigger; Soil hydraulic drivers |

5.10 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.11 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.12 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.13 | The two computing engines | Field balance |

5.14 | The two computing engines | Field balance |

5.15 | The two computing engines | Field balance; Profile evolution |

5.16 | The two computing engines | Field balance |

5.17 | The two computing engines | Field balance; Profile evolution |

5.18 | The two computing engines | Field balance; Root structure; The Properties panel |

5.19 | The two computing engines | Field balance; Root structure |

5.20 | The two computing engines | Field balance; Root structure |

5.21 | The two computing engines | Field balance; Root structure |

5.22 | The two computing engines | Field balance; Root structure |

5.23 | The two computing engines | Field balance; Root structure |

5.24 | The two computing engines | Field balance |

5.25 | The two computing engines | Field balance; Soil, plant and air |

5.26 | The two computing engines | Evaporation; The Properties panel |

5.27 | The two computing engines | Surface irrigation design; Surface irrigation design / Cross-section at the inlet; Surface irrigation design / Layout in plan; Drip irrigation design; Sprinkler design and pipes; Trigger law |

6.1 | Irrigation control layers | Sensor and trigger; The Dashboard, Results tab |

6.2 | Irrigation control layers | Sensor and trigger |

6.3 | Irrigation control layers | Trigger law |

6.4 | Irrigation control layers | Trigger law |

6.5 | Irrigation control layers | Trigger law |

7.1 | Field water balance layers | Soil, plant and air |

7.2 | Field water balance layers | Soil, plant and air |

7.3 | Field water balance layers | Soil, plant and air |

7.4 | Field water balance layers | Soil, plant and air |

7.5 | Field water balance layers | Soil, plant and air |

7.6 | Field water balance layers | Profile evolution |

8.1 | Soil structure and tillage layers | Field balance; Soil hydraulic drivers; The Properties panel; Tillage and structure |

8.2 | Soil structure and tillage layers | Tillage and structure |

8.3 | Soil structure and tillage layers | Evaporation; Tillage and structure |

8.4 | Soil structure and tillage layers | Evaporation; Tillage and structure |

8.5 | Soil structure and tillage layers | Soil hydraulic drivers |

8.6 | Soil structure and tillage layers | Soil hydraulic drivers |

8.7 | Soil structure and tillage layers | Root structure |

9.1 | Mulch and evaporation layers | Mulch against bare soil |

9.2 | Mulch and evaporation layers | Evaporation |

9.3 | Mulch and evaporation layers | Evaporation |

9.4 | Mulch and evaporation layers | Evaporation |

9.5 | Mulch and evaporation layers | Evaporation |

9.6 | Mulch and evaporation layers | Fate of saved water |

9.7 | Mulch and evaporation layers | Fate of saved water |

9.8 | Mulch and evaporation layers | Regime map |

10.1 | Feedback loops | Feedback loops |

10.2 | Feedback loops | Feedback loops |

11.1 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.2 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.3 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet; Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

11.4 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.5 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.6 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Cross-section at the inlet; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section; Sprinkler design and pipes; Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Soil intake against the application rate |

11.7 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.8 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Layout in

plan |

11.9 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.10 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.11 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.12 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.13 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.15 | Surface irrigation design | Surface irrigation design / Advance and opportunity time; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section |

11.16 | Surface irrigation design | Surface irrigation design / Advance and opportunity time; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section |

11.17 | Surface irrigation design | Surface irrigation design / Advance and opportunity time; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section |

11.18 | Surface irrigation design | Surface irrigation design / Hydrograph; Surface irrigation design / Performance indices and the water balance of one irrigation; The Dashboard, Design tab |

11.19 | Surface irrigation design | Surface irrigation design / Performance indices and the water balance of one irrigation; The Dashboard, Design tab |

11.20 | Surface irrigation design | Surface irrigation design / Hydrograph |

11.21 | Surface irrigation design | Surface irrigation design / Hydrograph |

11.22 | Surface irrigation design | Surface irrigation design / Design table |

11.23 | Surface irrigation design | Surface irrigation design / Design table |

11.24 | Surface irrigation design | Surface irrigation design / Intervals, efficiency and the advice in short; Surface irrigation design / What this means for the season and the farm |

12.1 | Drip irrigation design | Drip irrigation design |

12.2 | Drip irrigation design | Drip irrigation design |

12.3 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Layout and infrastructure; Drip irrigation design / Uniformity along the lateral; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.4 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.5 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.6 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.7 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.8 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.9 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.10 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Result for this design; Drip irrigation design / Soil moisture around the emitter; Drip irrigation design / Uniformity along the lateral |

12.11 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.12 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.13 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.14 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.15 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Result for this design; Drip irrigation design / Soil moisture around the emitter; Drip irrigation design / Uniformity along the lateral |

12.16 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral |

12.17 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral |

12.18 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral |

12.19 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral; The Dashboard, Design tab |

12.20 | Drip irrigation design | Drip irrigation design / Layout and infrastructure |

12.21 | Drip irrigation design | Drip irrigation design / Layout and infrastructure; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.22 | Drip irrigation design | Drip irrigation design / Layout and infrastructure; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.23 | Drip irrigation design | Drip irrigation design / Layout and infrastructure; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.24 | Drip irrigation design | Drip irrigation design / Intervals, efficiency and the advice in short; Drip irrigation design / Irrigation schedule; Drip irrigation design / What this means for the season and the farm |

12.25 | Drip irrigation design | Drip irrigation design / Intervals, efficiency and the advice in short; Drip irrigation design / Irrigation schedule; Drip irrigation design / What this means for the season and the farm |

12.26 | Drip irrigation design | Drip irrigation design / Design table; Sprinkler design and pipes / Design table; Sprinkler design and pipes / What the efficiency is most sensitive to |

13.1 | Sprinkler design | Sprinkler design and pipes |

13.2 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / The jet; Sprinkler design and pipes / The sprinkler |

13.3 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / The jet; Sprinkler design and pipes / The sprinkler |

13.4 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test;

Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.5 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test; Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.6 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test; Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.7 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test; Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.8 | Sprinkler design | Sprinkler design and pipes / The jet; Sprinkler design and pipes / What wind and soil allow |

13.9 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Soil intake against the application rate; The Properties panel |

13.10 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Soil intake against the application rate |

13.11 | Sprinkler design | Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Result for this design; Sprinkler design and pipes / Soil water as a reservoir; The Dashboard, Design tab |

13.12 | Sprinkler design | Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Result for this design; Sprinkler design and pipes / Soil water as a reservoir |

13.13 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.14 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.15 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.16 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.17 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.18 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.19 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.20 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.21 | Sprinkler design | Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

13.22 | Sprinkler design | Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

13.23 | Sprinkler design | Sprinkler design and pipes / Intervals, efficiency and the advice in short; Sprinkler design and pipes / Irrigation schedule; Sprinkler design and pipes / Soil water as a reservoir; Sprinkler design and pipes / What this means for the season and the farm |

14.1 | The Dashboard | The Dashboard, Results tab |

14.2 | The Dashboard | The Dashboard, Results tab |

14.3 | The Dashboard | The Dashboard, Results tab |

14.4 | The Dashboard | The Dashboard, Results tab |

14.5 | The Dashboard | The Dashboard, Charts tab |

# Chapter 27

# ⚘ The Daisy workbench: overview

The Daisy workbench is the part of SWAFL that runs the Daisy 7.1.14 model, draws its results and tests it. It opens from the button Daisy model of the toolbar. The tabs are in groups that follow what a person does: give the data, ask a question, look at the results, decide in the SWAFL project, and, apart from that path, build and test and learn. A bar above every tab on the path, with the buttons Back and Next, leads through the steps in this order (Chapter 31).

| Group | Tab | What it is for |
|---|---|---|
| Input data | Input data | Set the weather (online stations, a file, the stations bundled with Daisy) and the soil of the site once; every simulation uses them. |
| Input data | Parameter library | Read the horizons, crops, fertilisers and other entries of the Daisy library. |
| Experiments | Experiments | Design your own set of runs from 53 factors, or start from 27 best combinations. |
| Experiments | Scenarios | 24 ready questions in five themes (water, nitrogen, soil, crop and climate, chemicals), each open to any other value of the parameter library. |
| Results and analysis | Process map | A map of the model processes with the figures of each one. |
| Results and analysis | Log plots | Load logs (or take the output of your runs) and draw any column of them. |
| Results and analysis | Mass balance | Check that inputs, outputs and content agree. |
| Results and analysis | Run comparison | Compare two logs against the smallest meaningful level. |
| Decision support | To the SWAFL layers | Take the results to the layers of the SWAFL project that turn them into decisions. |
| Model development | Set-up editor | Edit and run a Daisy set-up file. |
| Model development | Soil column | Solve the Richards equation for a soil column. |
| Model development | Response curves | Draw any function of the Daisy library. |
| Model development | Calibration | Search parameters with three optimisers. |
| Documentation | Model and documentation | The model, the questions and where each is answered, the downloads. |

The chapters that follow take the groups in this order. Every screen is shown with numbered arrows; the list under it says what each arrow points to.

**How the weather is chosen.** The weather of a run is chosen once in the strip at the top of the

Scenarios, Experiments and Process map tabs, and its source is set in Input data. The choices are the weather of the SWAFL project, the built-in Taastrup weather, one of the bundled stations, online weather of a place and a file of your own. All simulations then use that weather, so a figure and the project always agree.

**What runs where.** The scenarios, experiments, set-up editor and calibration against the Daisy program need the desktop app. Daisy 7.1.14 for Windows comes with SWAFL (the folder `daisy` next to the program) and starts by itself, so nothing has to be installed or downloaded. The soil column, the process curves and the denitrification calibration run in the web version too. The figures of results that are drawn here come from real Daisy runs of the built-in farm for the years 1990 to 1993.

## 27.1 Daisy: governing equations and guiding principles

This chapter gathers the equations and the principles behind the tabs of the Daisy workbench, so that a figure can be read with the law that produced it. The forms below are the ones the workbench itself uses where it computes (Soil column, Process curves, Calibration) and the standard forms of the Daisy reference manual where Daisy computes (the runs of Scenarios and Experiments). Section numbers of the Daisy reference are given in the tab Model and documentation. Symbols: $z$ is the depth below the surface (cm, positive downward), $t$ is time, $\theta$ the volumetric water content ($\mathrm{cm}^3\,\mathrm{cm}^{-3}$), $h$ the soil water pressure head (cm, negative when the soil is unsaturated) and $K$ the hydraulic conductivity.

**Guiding principles.** Daisy follows four principles that explain how its figures behave.

1. *Conservation.* Water, nitrogen, carbon and heat are tracked as balances: what enters, what leaves and what changes in storage must agree. The tab Mass balance tests this on any log.
2. *One dimension, many processes.* The soil is a vertical column. Water, heat and solutes move up and down; the processes of Fig. 1 (snow, canopy, litter, pond, biopores, drains, organic matter, crop) are coupled to that column.
3. *Rates with limiting factors.* Most biological rates are a potential rate multiplied by factors between 0 and 1 for temperature, water, oxygen or nutrients, so a figure changes shape when one factor becomes limiting.
4. *Demand and supply.* The atmosphere sets the demand for water and the soil sets the supply; the crop transpires the smaller of the two, and the shortage shows as stress.

### 27.1.1 Soil water: the Richards equation

Water flow in the unsaturated column follows the Richards equation, with the Darcy flux $q = K(h)\,(1 - \partial h/\partial z)$ downward:

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\left(\frac{\partial h}{\partial z} - 1\right)\right] - S(z,t).$$

Here $S$ is the uptake of water by the roots (1/day). The tab Soil column solves this equation with an implicit finite volume scheme and Picard iteration with the mass conserving capacity term $C(h) = \mathrm{d}\theta/\mathrm{d}h$. Its boundaries are rain and evaporation at the top and a free drain, a closed bottom or a groundwater table at the bottom.

**Retention and conductivity (van Genuchten and Mualem).** With the effective saturation $S_e$ and $m = 1 - 1/n$:

$$S_e = \left[1 + (\alpha\,|h|)^n\right]^{-m}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\,S_e,$$

$$K(h) = K_s\,S_e^{\,l}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2.$$

The parameters $\theta_s, \theta_r, \alpha, n, K_s, l$ come from the horizons of the library. Field capacity is read at $h = -100$ cm and the wilting point at $h = -15{,}000$ cm, and the available water of a horizon is $\theta_{FC} - \theta_{WP}$. Daisy also offers the Brooks and Corey and Campbell forms; the tab Process curves draws all of them.

**Root uptake (Feddes).** The potential transpiration $E_p$ is spread over the roots in proportion to root density, and reduced by a stress factor $\alpha_s(h)$ that is 1 in the comfortable range of the pressure head and falls to 0 near saturation and towards the wilting point:

$$S(z) = \alpha_s(h)\,S_p(z), \qquad \int S_p(z)\,\mathrm{d}z = E_p.$$

### 27.1.2 Water balance of the season

Every layer and every season obeys the balance that the tab Mass balance checks and the dashboards print:

$$P + I = E_s + E_i + T + D + R + \Delta S,$$

with rain $P$, irrigation $I$, soil evaporation $E_s$, evaporation of intercepted water $E_i$, transpiration $T$, drainage out of the column $D$, runoff $R$ and the change of storage $\Delta S$ (all in mm). The balance residual should be small compared with the largest term.

### 27.1.3 Evaporative demand

The atmosphere sets the demand. The reference evapotranspiration follows the Penman and Monteith equation with the daily weather; the potential rate of a crop is the reference rate times a crop coefficient (or, in Daisy, a canopy resistance derived from the leaf area):

$$ET_0 = \frac{0.408\,\Delta\,(R_n - G) + \gamma\,\dfrac{900}{T + 273}\,u_2\,(e_s - e_a)}{\Delta + \gamma\,(1 + 0.34\,u_2)}, \qquad ET_c = K_c\,ET_0.$$

The canopy shares the demand between soil evaporation and transpiration with the light that reaches the soil, $\exp(-k\,\mathrm{LAI})$ (the Beer and Lambert law with the extinction coefficient $k$).

### 27.1.4 Soil heat and ice

Heat moves by conduction and with the water flux:

$$C_h\,\frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\left(\lambda\,\frac{\partial T}{\partial z}\right) - C_w\,q\,\frac{\partial T}{\partial z} + L_f\,\frac{\partial \theta_i}{\partial t},$$

where $C_h$ is the volumetric heat capacity, $\lambda$ the thermal conductivity, $C_w$ the heat capacity of water and $L_f$ the latent heat of freezing. The temperature sets the factors of the biological rates below and the freezing of the water.

### 27.1.5 Solutes: nitrate, ammonium and chemicals

A solute is carried by the water and spread by dispersion, with sorption to the soil:

$$\frac{\partial(\theta c + \rho_b s)}{\partial t} = \frac{\partial}{\partial z}\left(\theta D\,\frac{\partial c}{\partial z}\right) - \frac{\partial(qc)}{\partial z} - \mu\,(\theta c + \rho_b s) - U,$$

$$s = K_d\,c \text{ (linear sorption)}, \qquad D = \lambda_d\,\frac{|q|}{\theta} + D_m\,\tau.$$

Here $c$ is the concentration in the soil water, $s$ the sorbed amount, $\rho_b$ the bulk density, $\lambda_d$ the dispersivity, $D_m$ the molecular diffusion, $\tau$ the tortuosity, $\mu$ the first order decay and $U$ the uptake by the crop. A mobile tracer has $K_d = 0$ and $\mu = 0$; the three chemicals of the scenario Pesticide or tracer fate differ in exactly these parameters. For a pesticide, $K_d = K_{OC}\,f_{OC}$ with the organic carbon fraction $f_{OC}$.

### 27.1.6 Organic matter, mineralisation and nitrification

The organic matter pools (AOM1, AOM2, SMB1, SMB2, SOM1, SOM2, SOM3 in Fig. 1) decay by first order kinetics, with a part of the decayed carbon built into other pools (the efficiency $E$) and the rest released as carbon dioxide:

$$\frac{\mathrm{d}C_i}{\mathrm{d}t} = -k_i\, f_T(T)\, f_\theta(h)\, C_i + \sum_j E_{ji}\, k_j\, f_T\, f_\theta\, C_j + \text{inputs},$$

and the nitrogen follows from the carbon to nitrogen ratio of the receiving pool, so that the soil either releases ammonium (mineralisation) or takes it up (immobilisation). Ammonium is oxidised to nitrate by nitrification, which saturates with the ammonium concentration:

$$r_{nit} = f_T\, f_\theta\, \frac{V_{max}\,[\mathrm{NH_4^+}]}{K_m + [\mathrm{NH_4^+}]}.$$

The temperature factor is a piecewise linear function of $T$ or a $Q_{10}$ form, $f_T = Q_{10}^{(T-T_{ref})/10}$.

### 27.1.7 Denitrification and nitrous oxide

The tab Process curves and the denitrification screens use the function of the Daisy default model. A potential rate follows the carbon dioxide produced by the fast and the slow organic matter, $r_{pot} = f_T\left(\alpha_F\,\mathrm{CO_2^{fast}} + \alpha\,(\mathrm{CO_2} - \mathrm{CO_2^{fast}})\right)$. It is limited by the water filled pore space, $f_w(\mathrm{WFPS})$, a rising function from 0 at 40 % to 1 at 90 %, and by the nitrate present:

$$r_{act} = \min\!\left(f_w\, r_{pot,\,slow},\; K_d\,[\mathrm{NO_3^-}]\right) + \min\!\left(f_w\, r_{pot,\,fast},\; K_d\,[\mathrm{NO_3^-}]\right).$$

The gas formed is split between nitrous oxide and dinitrogen with the ratio $f_r = \mathrm{N_2/N_2O}$ that falls with nitrate, rises with carbon dioxide and with the wetness:

$$\mathrm{N_2O} = \frac{r_{act}}{1+f_r}, \qquad \mathrm{N_2} = \frac{r_{act}}{1+1/f_r}.$$

### 27.1.8 The crop: light, assimilates and growth

The crop module follows the chain of the right half of Fig. 1.

1. *Light interception.* The absorbed light is $I_a = I_0\left(1 - e^{-k\,\mathrm{LAI}}\right)$ with the leaf area index summed down the canopy.
2. *Photosynthesis.* The gross assimilation of a leaf layer rises with light and saturates, $A = A_{max}\left(1 - e^{-\varepsilon I/A_{max}}\right)$ with the quantum efficiency $\varepsilon$, and is reduced by water stress (through the stomata), by temperature, by senescence and by the nitrogen content of the leaves (the boxes Stress factors and Senescence).
3. *Respiration.* The assimilate pool loses maintenance respiration, $R_m = \sum_o r_o\, f_T\, W_o$ over the organs $o$ (root, stem, leaf, storage organ) with a $Q_{10}$ temperature factor, and growth respiration as a fraction of what is built into the organs.
4. *Partitioning.* What remains is shared between the organs by fractions that depend on the development stage $DS$ (0 at emergence, 1 at flowering, 2 at maturity); the storage organ takes the largest share after flowering and gives the yield at harvest.
5. *Development.* The development stage advances with the temperature sum (and, for some crops, with day length): $\mathrm{d}DS/\mathrm{d}t = f(T, \text{photoperiod})$.

### 27.1.9 How the equations reach the screens

| Tab | Law behind the figure | Where it is computed |
|---|---|---|
| Scenarios and Experiments | All of the above, in Daisy 7.1.14 | Daisy in the desktop app |
| Process map | The processes of Fig. 1 and the factors that act on them | Map and tables of the workbench |

| Tab | Law behind the figure | Where it is computed |
|---|---|---|
| Log plots, Run comparison | The columns of the logs, differences and the smallest meaningful level | The workbench |
| Mass balance | inputs $-$ outputs $-$ $\Delta$content $=$ residual per quantity | The workbench |
| Soil column | Richards equation, van Genuchten and Mualem, Feddes | The workbench (own solver) |
| Process curves | Retention, conductivity, $Q_{10}$, Michaelis and Menten, denitrification | The workbench |
| Calibration | Loss MSE $= \frac{1}{n}\sum(y_i - \hat{y}_i)^2$ minimised by three optimisers | The workbench with Daisy runs |

# Chapter 28

# ☀ Input data

The first group holds everything that describes the site: weather, soil and the files of earlier runs.

### 28.0.1 Input data: the starting screen

Input data is the first tab of the group Input data. It is where the weather and the soil of the site are set once; Scenarios, Experiments and Processes all read them from here. With nothing changed, the built-in weather of Taastrup and the built-in loam and sand are used.

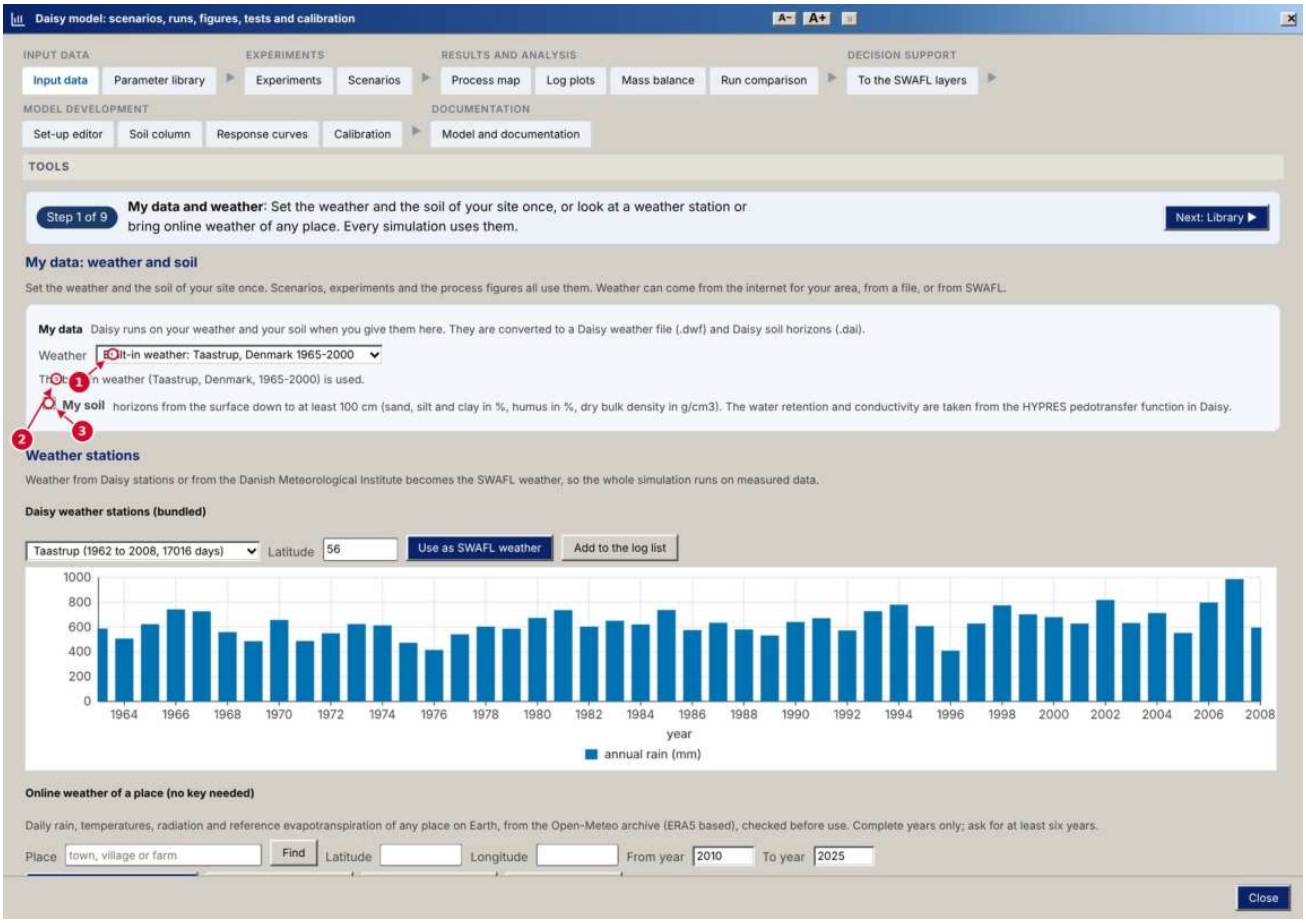



Figure 28.1: Input data: the starting screen. Screen 1 of the Daisy workbench.

**What the arrows point to.**

1. **Weather choice.** Where the weather comes from: the built-in weather, the weather of the SWAFL project, online weather of a place, or a file of your own.
2. **Weather in use.** A summary of the weather that will be used: its name, years, mean rain and mean temperature, and the checks that were made.
3. **My soil.** Tick to describe your own soil horizons instead of the two built-in soils.

### 28.0.2 Input data: online weather of a place

With Online weather of my area the place is typed once; Find looks it up and fills the site boxes, and Get the weather reads the daily data of the chosen years from the internet. Only complete years up to last year are asked for.

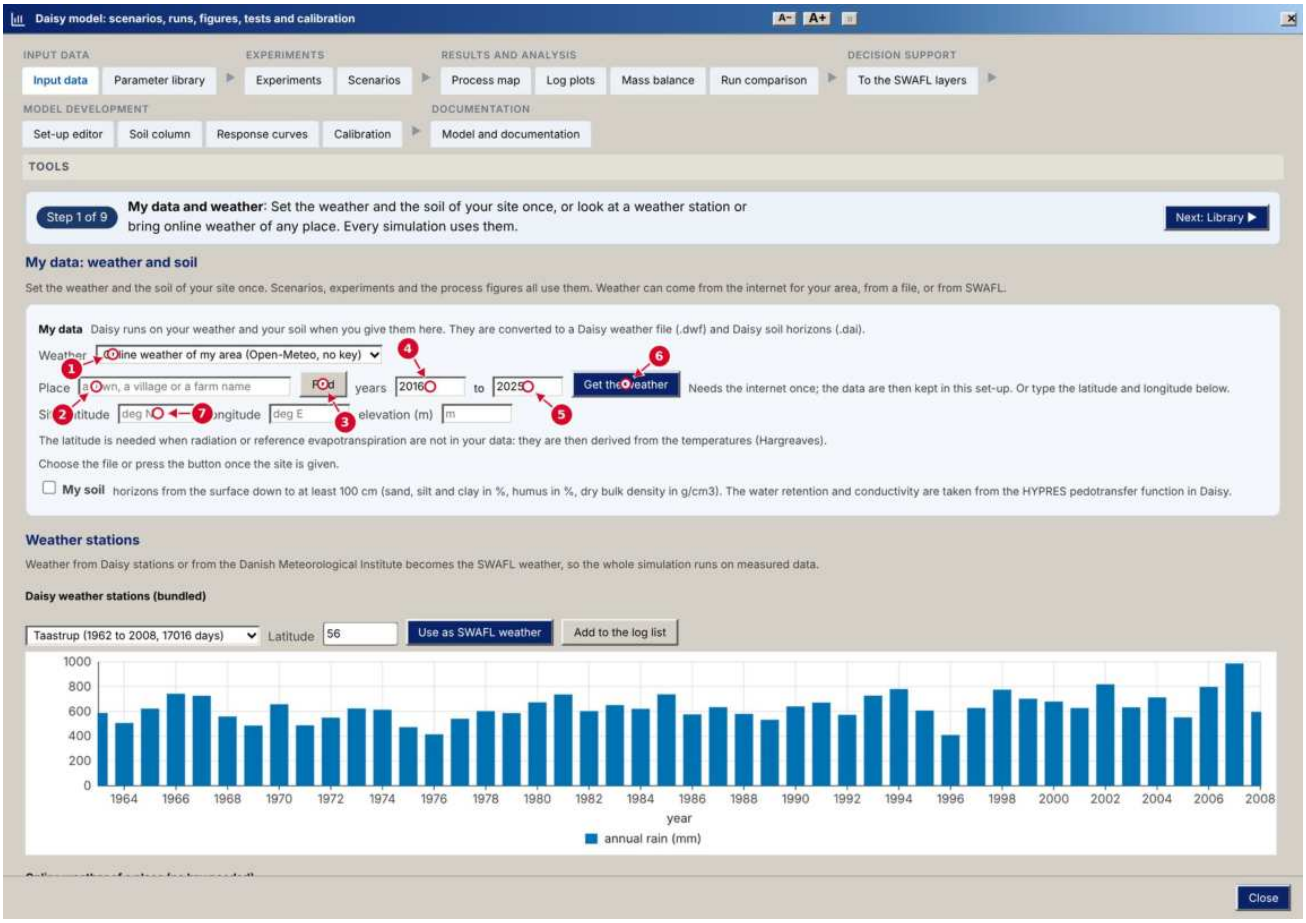



Figure 28.2: Input data: online weather of a place. Screen 2 of the Daisy workbench.

**What the arrows point to.**

1. **Online weather.** The choice that shows the place controls below.
2. **Place.** A town, a village or a farm name.
3. **Find.** Looks the place up and fills the latitude, longitude and elevation.
4. **First year.** First year of the weather record to read.
5. **Last year.** Last complete year of the record; the current year is not available yet.
6. **Get the weather.** Reads the daily rain, temperatures, radiation and reference evapotranspiration and checks them before use.
7. **Latitude.** Needed to work out radiation when it is not in the data; found automatically from the place name when possible.

### 28.0.3 Input data: a weather file of your own

A weather file may be a Daisy weather file (.dwf) or a table (CSV, TXT) with a date column and the columns rain and temperature. The radiation unit is chosen or found out from the values.

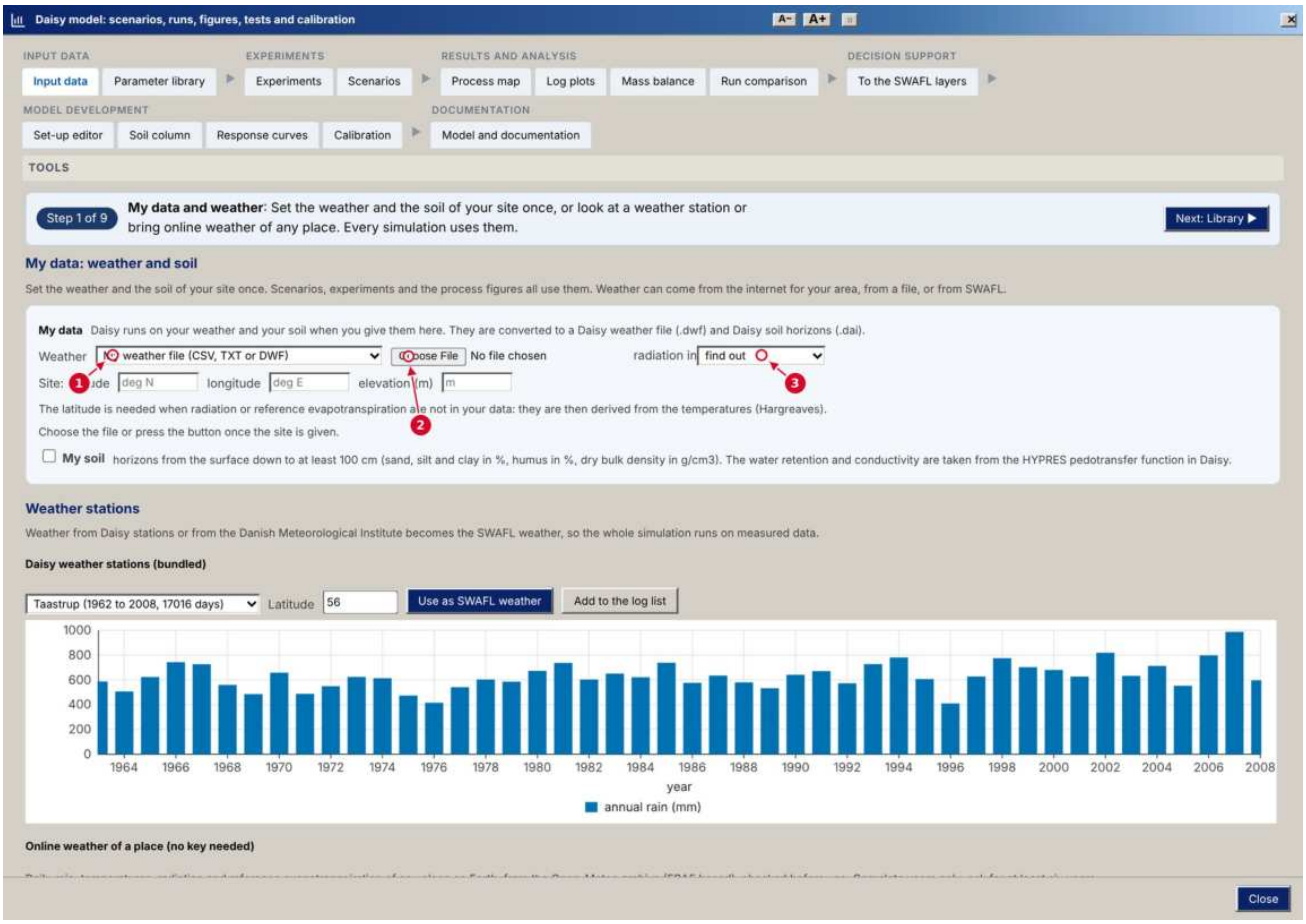



Figure 28.3: Input data: a weather file of your own. Screen 3 of the Daisy workbench.

**What the arrows point to.**

1. **My weather file.** The choice that shows the file chooser.
2. **File chooser.** Choose a .dwf, .csv, .txt, .dat or .tsv file.
3. **Radiation unit.** MJ per square metre and day, kWh per square metre and day, or W per square metre; leave on find out when unsure.

### 28.0.4 Input data: a Daisy weather station file read and checked

Here the Foulum station file of the bundled data was chosen as a file of your own. The line under the choice reports the years, the mean rain and temperature, the columns that were used and the notes of the checks.

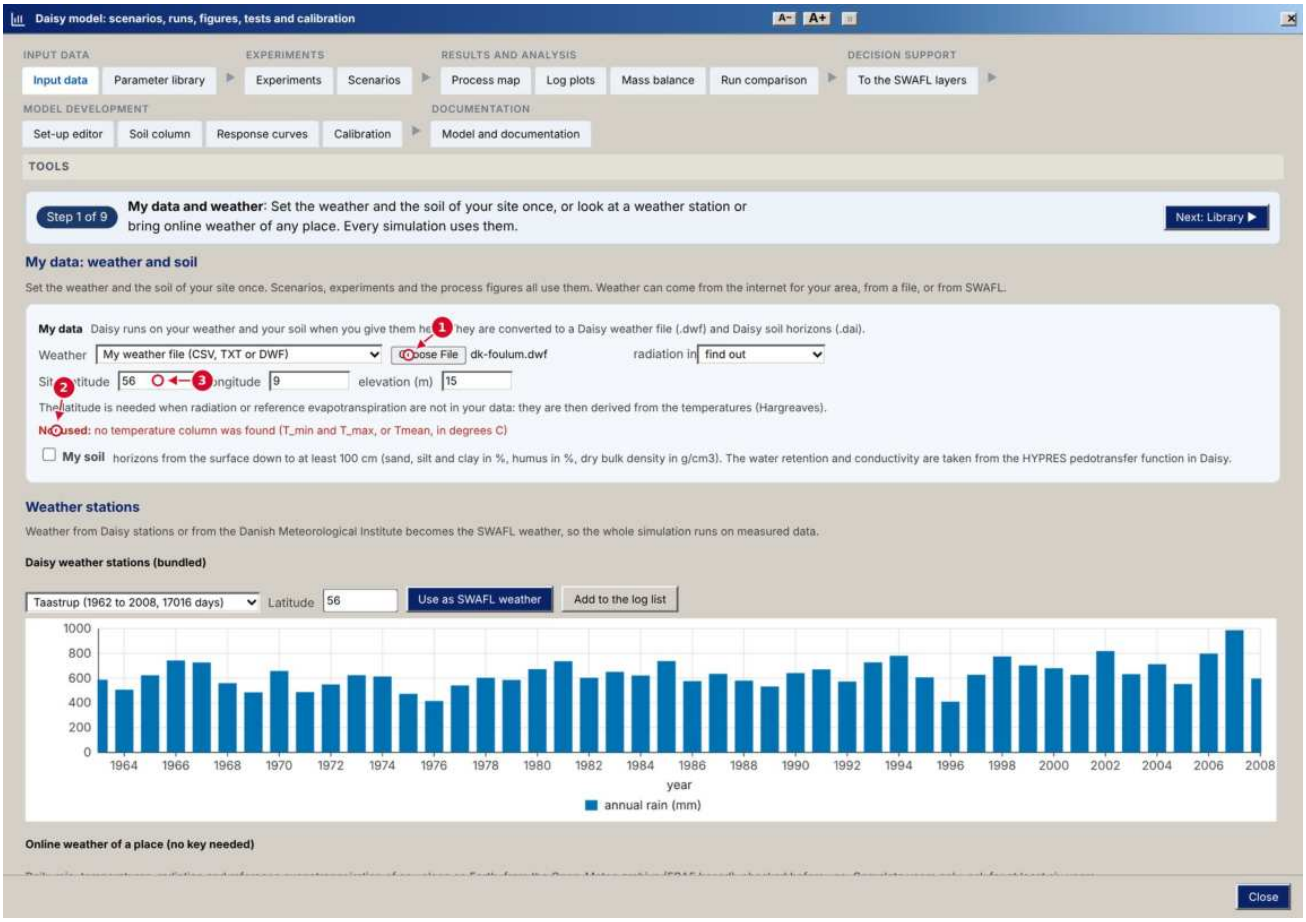



Figure 28.4: Input data: a Daisy weather station file read and checked. Screen 4 of the Daisy workbench.

**What the arrows point to.**

1. **File read.** The file that was chosen.
2. **Result of the reading.** Name, first and last year, complete years, mean annual rain, mean temperature and the columns used.
3. **Site.** The latitude, longitude and elevation read from the header of the file.

### 28.0.5 Input data: my own soil

My soil takes the horizons from the surface down to at least 100 cm: lower limit, sand, silt and clay in per cent, humus in per cent and dry bulk density. Water retention and conductivity are then taken from the HYPRES pedotransfer function in Daisy.

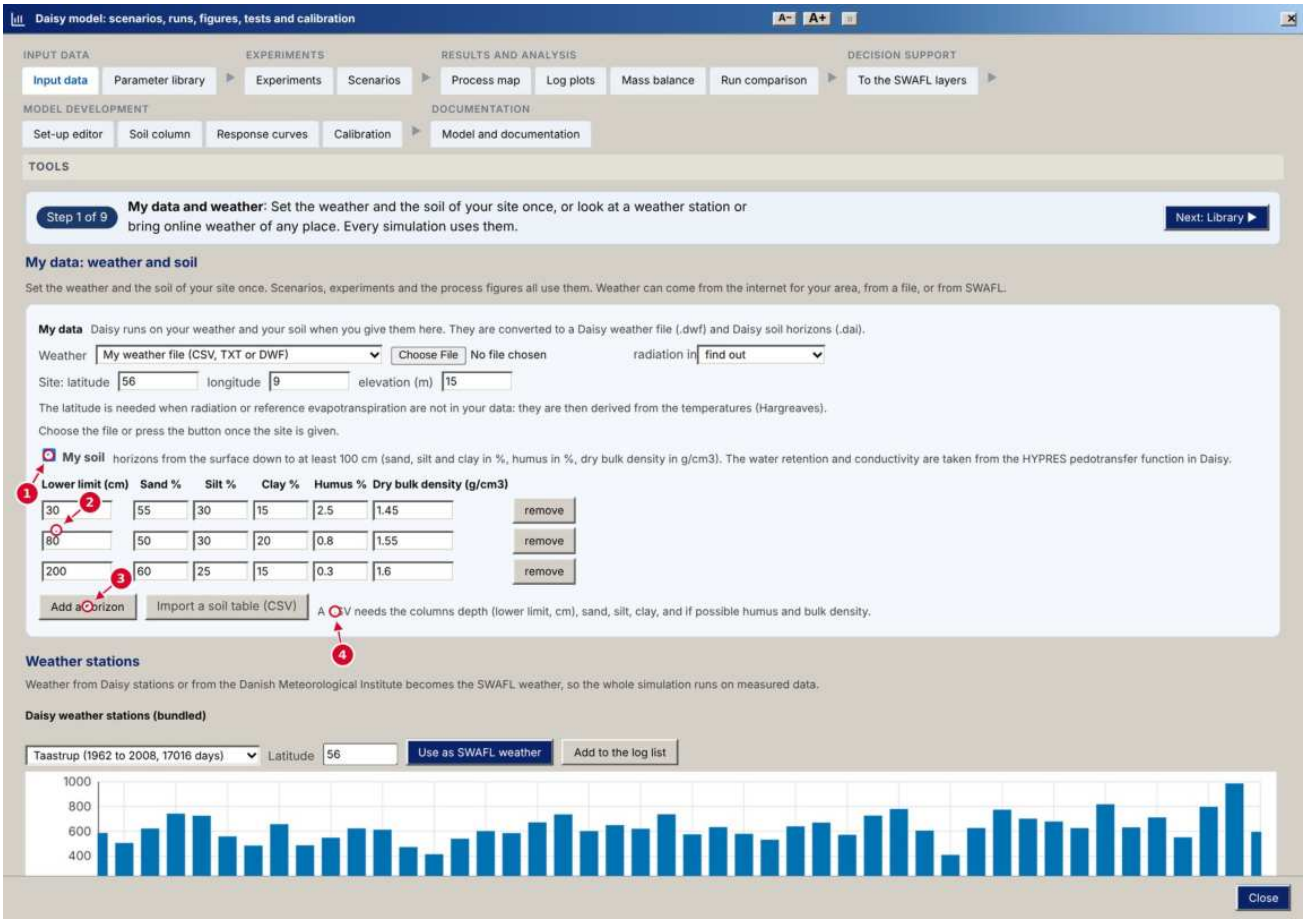



Figure 28.5: Input data: my own soil. Screen 5 of the Daisy workbench.

**What the arrows point to.**

1. **My soil.** Switches the table of horizons on; the soil choice of Scenarios and Experiments is then replaced by this soil.
2. **Horizons.** One row for each horizon: lower limit, sand, silt, clay, humus and bulk density.
3. **Add a horizon.** Adds a row at the bottom.
4. **Soil table import.** A CSV with the columns depth, sand, silt, clay and, if possible, humus and bulk density can be imported instead.

### 28.0.6 Files: collecting Daisy output and set-up files

Files collects everything Daisy writes or reads: output logs (.dlf), weather files (.dwf), tables (.csv) and set-up files (.dai). Whatever is loaded here is available to the plots, the mass balance, the comparison, the calibration and the soil column.

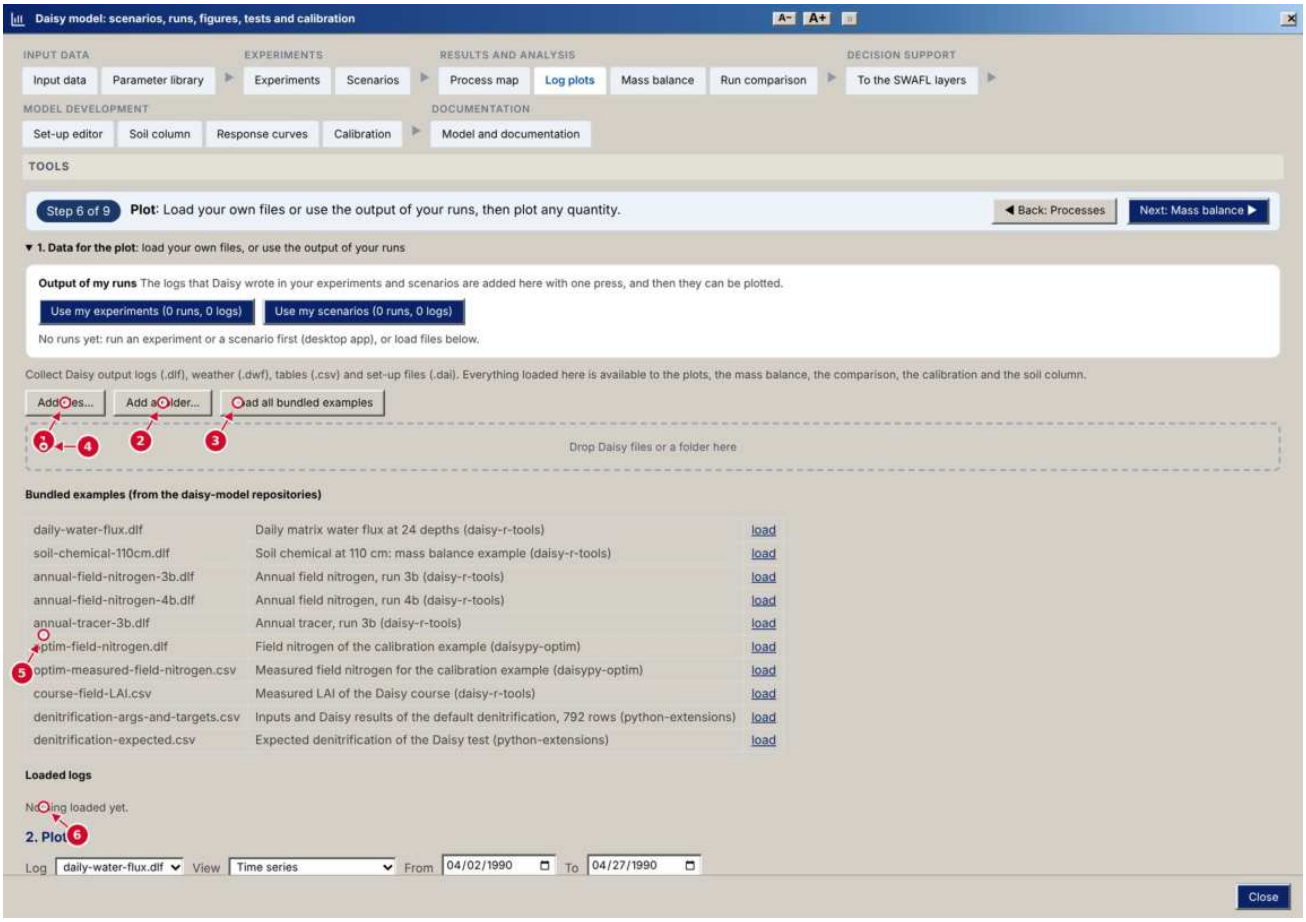



Figure 28.6: Files: collecting Daisy output and set-up files. Screen 6 of the Daisy workbench.

**What the arrows point to.**

1. **Add files.** Chooses one or several files.
2. **Add a folder.** Reads every Daisy file of a folder, for example a whole experiment folder.
3. **Load all bundled examples.** Loads every example log that comes with the program.
4. **Drop area.** Files and folders can also be dropped here.
5. **Bundled examples.** The example logs from the repositories of the Daisy model; load them one by one.
6. **Loaded logs.** The logs that are loaded now, with their columns.

### 28.0.7 Files: the bundled logs loaded

After Load all bundled examples every log appears in the list of loaded logs with its number of rows and columns.

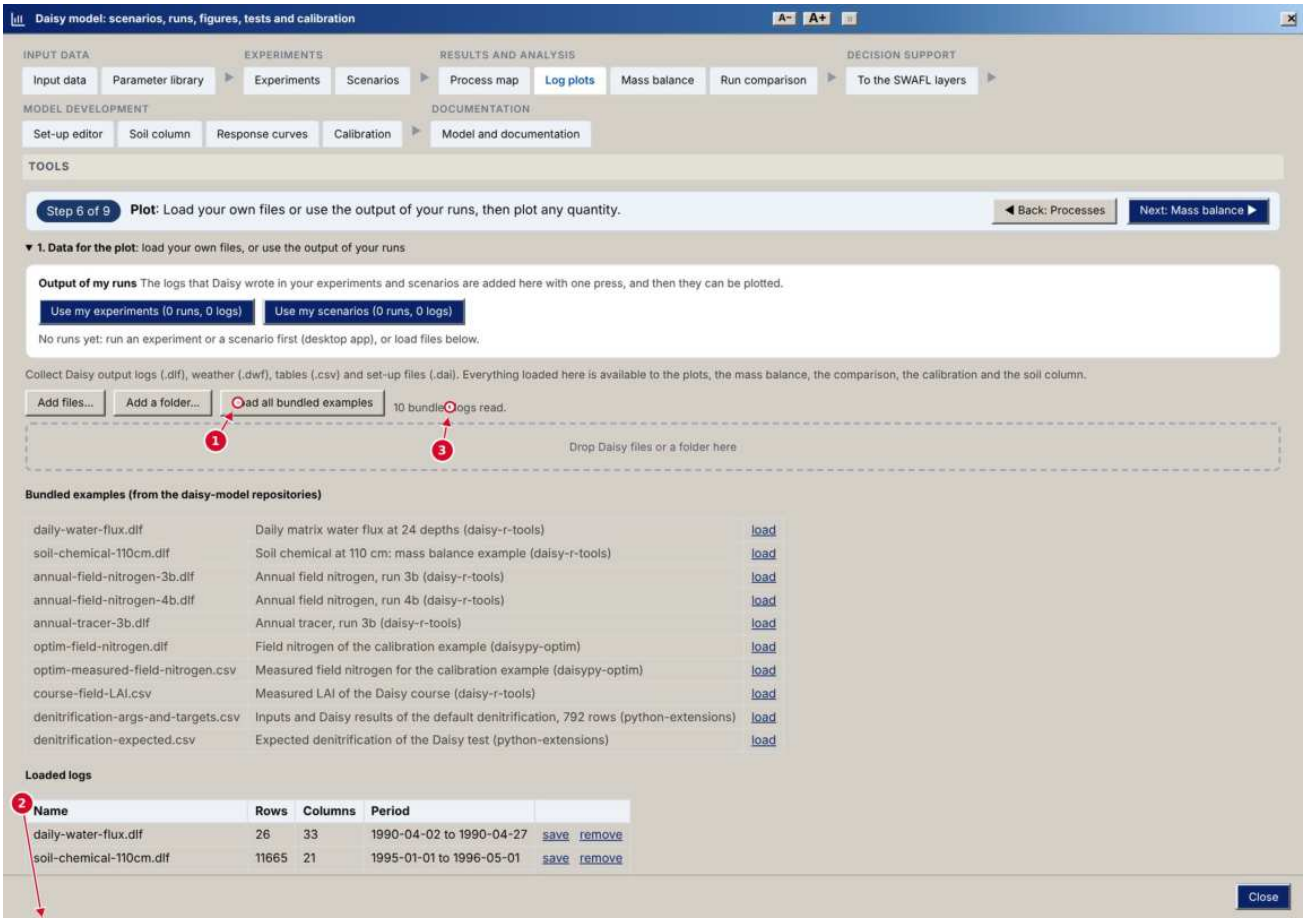



Figure 28.7: Files: the bundled logs loaded. Screen 7 of the Daisy workbench.

**What the arrows point to.**

1. **Load all.** The button that was pressed.
2. **Loaded logs.** Each loaded log with its size; click a name to see its columns.
3. **Status.** A message that says what was read.

### 28.0.8 Library: soil horizons

The Library holds the soil horizons, crops, fertilisers, chemicals and management actions of the Daisy library files, read in full. A soil horizon can be opened to see its water retention and conductivity curves.

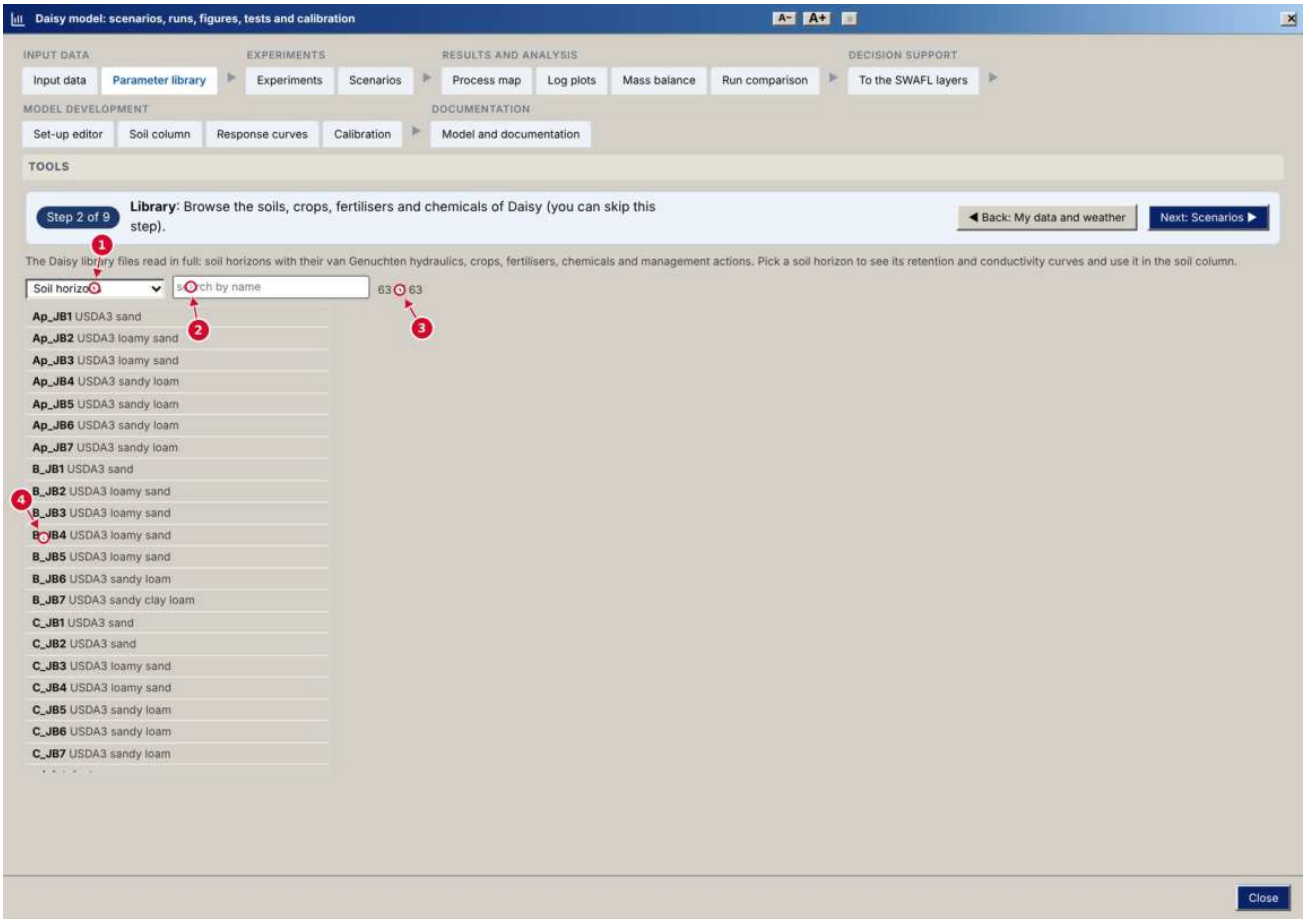



Figure 28.8: Library: soil horizons. Screen 8 of the Daisy workbench.

**What the arrows point to.**

1. **Kind of entry.** Soil horizons, crops, fertilisers, chemicals and management actions.
2. **Search.** Filters the list by name.
3. **Count.** The number of entries shown out of all.
4. **List.** The entries; click one to see its parameters.

### 28.0.9 Library: a soil horizon opened

Choosing a horizon shows its texture class and, below the list, the retention and conductivity curves drawn from its parameters.

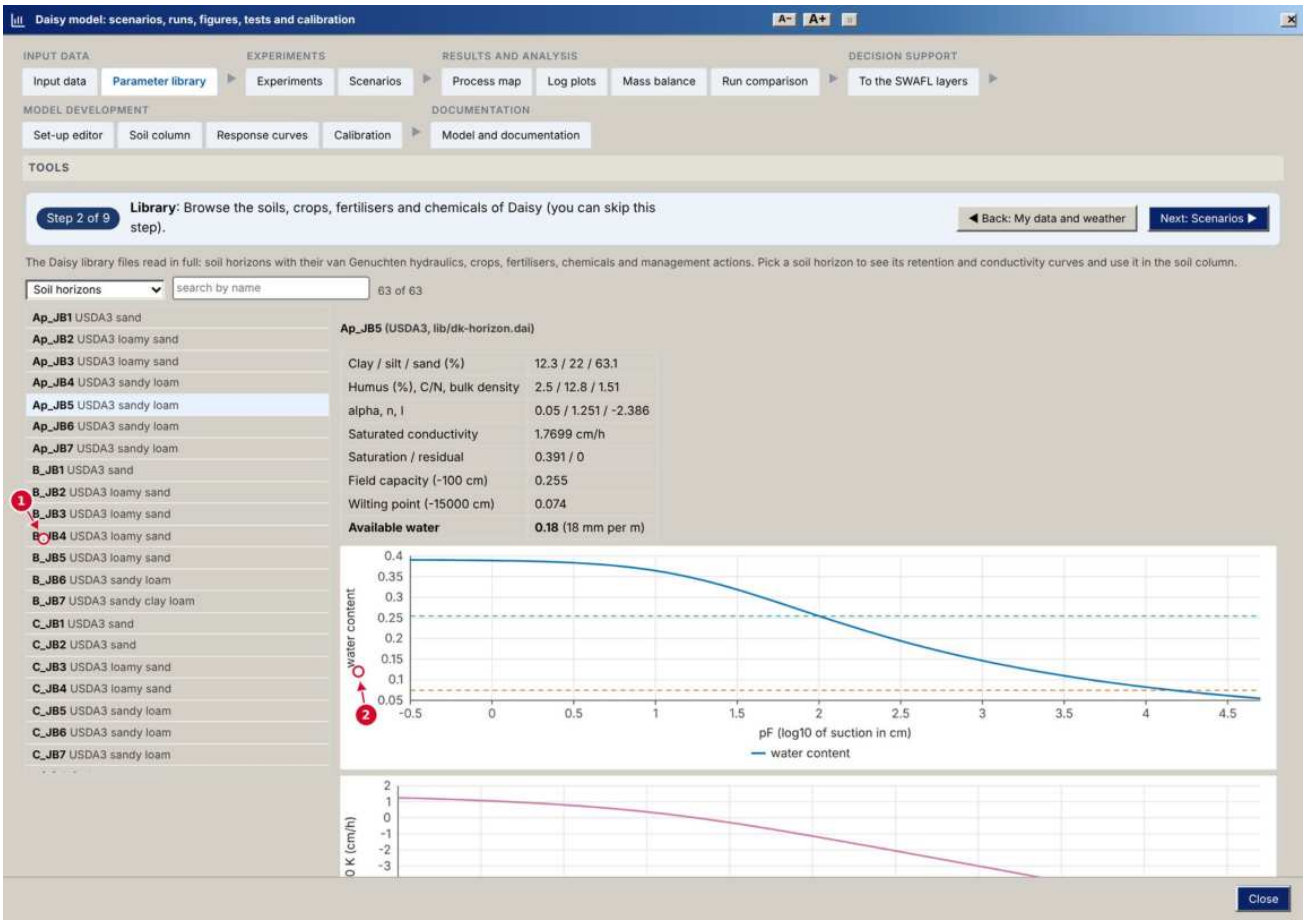



Figure 28.9: Library: a soil horizon opened. Screen 9 of the Daisy workbench.

**What the arrows point to.**

1. **Selected horizon.** The horizon whose parameters are shown.
2. **Parameters and curves.** Parameters of the horizon and the curves of water content and conductivity against suction.

### 28.0.10 Library: the crops of the Daisy library

The same screen with the crops chosen in the first list.

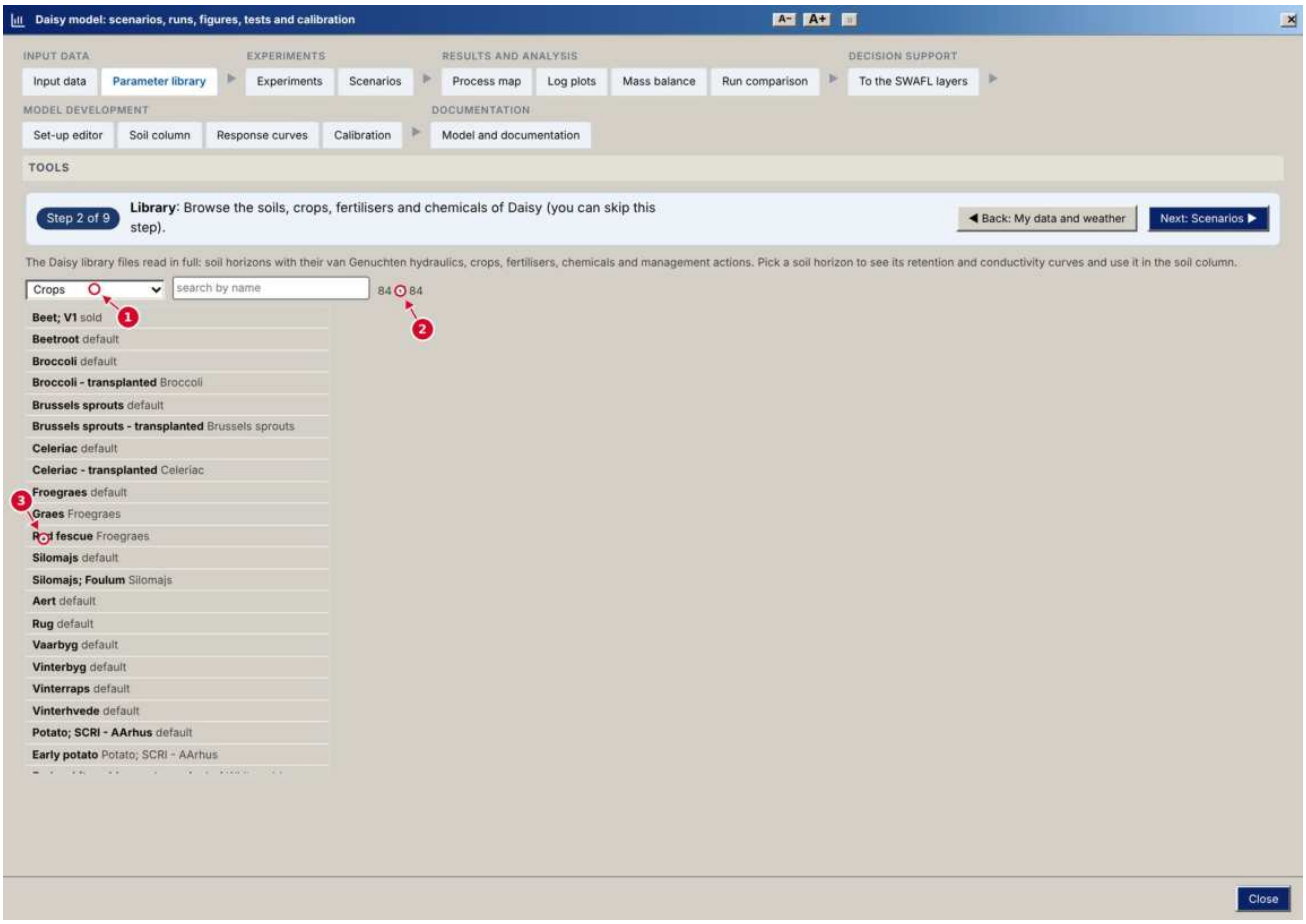



Figure 28.10: Library: the crops of the Daisy library. Screen 10 of the Daisy workbench.

**What the arrows point to.**

1. **Kind of entry.** Now set to crops.
2. **Count.** Number of crops in the library.
3. **Crops.** Each crop with the model that describes it.

### 28.0.11 Weather stations: the weather loaded in SWAFL is in the list

When SWAFL has an online weather of a place or a weather file loaded, the list of the weather stations starts with it, under *Weather loaded in SWAFL* (for example *Online weather: Zaria, Kaduna, Nigeria (2007 to 2026), in use*), followed by the bundled Daisy stations. Choosing it draws its annual rain and puts its latitude in the box; **Use as SWAFL weather** makes it the weather of SWAFL again. In Properties, the weather source also names the loaded weather: *Online weather: place (years)* and *Imported weather file: name (years)*.

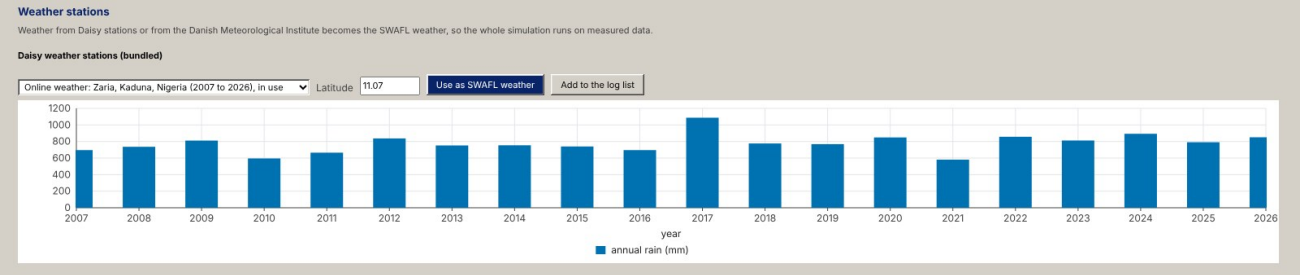



Figure 28.11: The weather stations: the online weather loaded in SWAFL at the top of the list, with its annual rain.

### 28.0.12 Weather stations: the bundled stations and the annual rain

Weather stations lists the stations that come with Daisy. Choosing one draws its annual rain; the station can be used as the SWAFL weather or added to the list of logs for the plots.

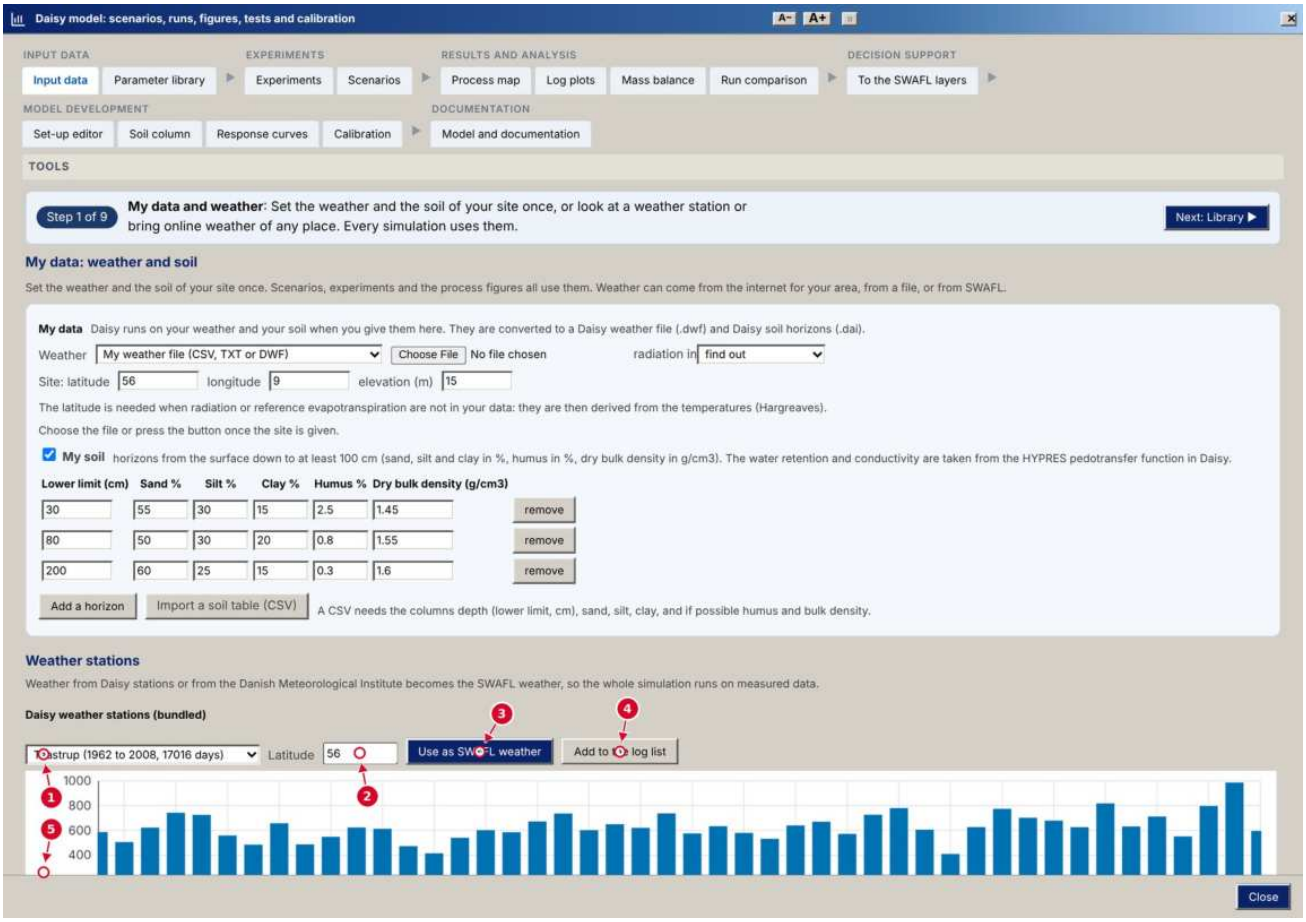



Figure 28.12: Weather stations: the bundled stations and the annual rain. Screen 11 of the Daisy workbench.

**What the arrows point to.**

1. **Station.** The bundled stations, each with its years and number of days.
2. **Latitude.** Latitude of the station, used to work out reference evapotranspiration when it is not recorded.
3. **Use as SWAFL weather.** Sends the station to the main workbench as its weather.
4. **Add to the log list.** Puts the station among the logs that the plot tabs can draw.
5. **Annual rain.** Sum of the daily rain of each year of the record.

### 28.0.13 Weather stations: another station and the open data of the Danish Meteorological Institute

The lower part of the tab reads open weather data of the Danish Meteorological Institute (a free key of your own is needed) and a groundwater series for the lower boundary of the soil column.

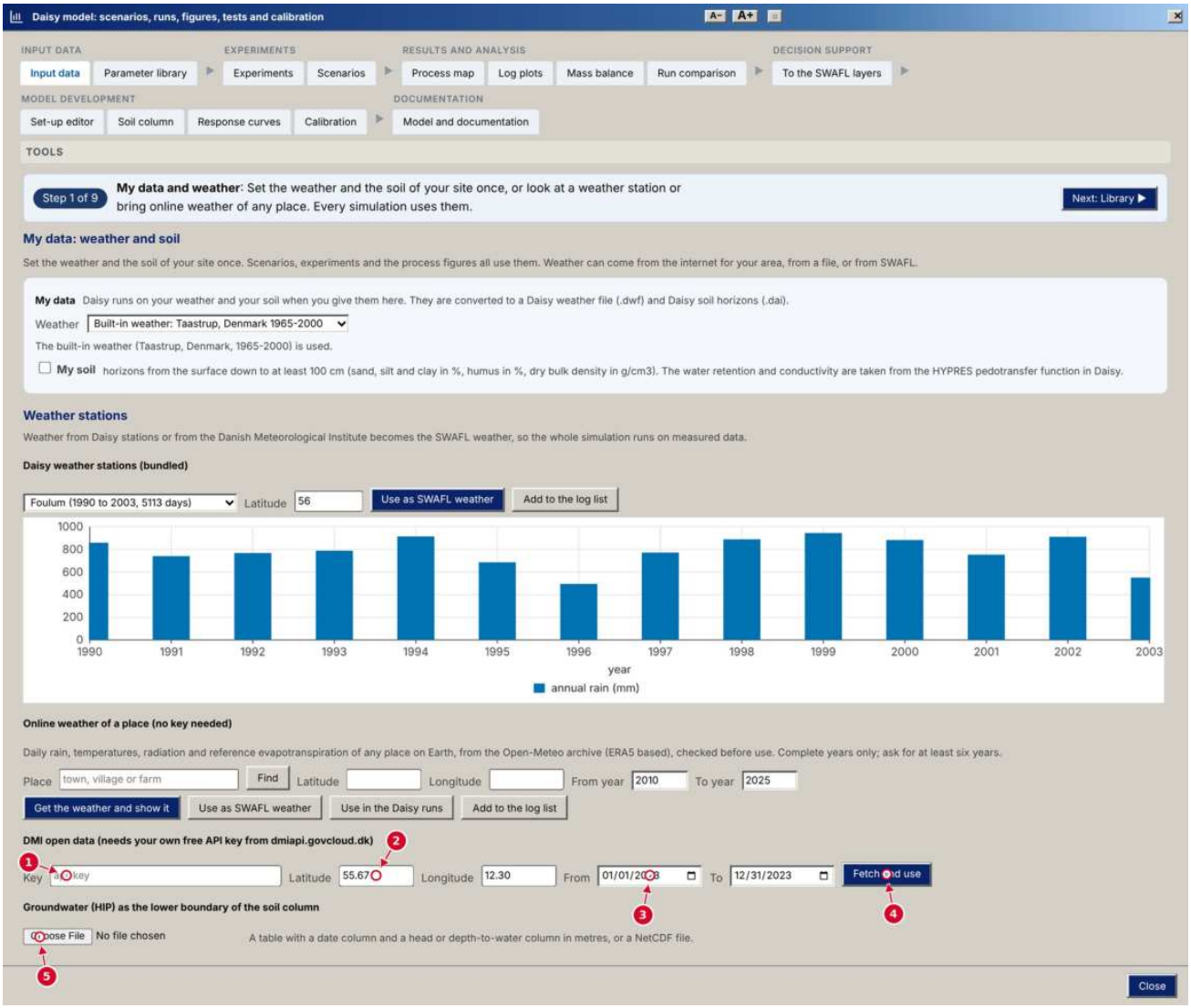



Figure 28.13: Weather stations: another station and the open data of the Danish Meteorological Institute. Screen 12 of the Daisy workbench.

**What the arrows point to.**

1. **Interface key.** The free interface key of the Danish open data service.
2. **Latitude.** Latitude of the place.
3. **From.** First day to read.
4. **Fetch and use.** Reads the daily series and makes it the weather in use.
5. **Groundwater file.** A table with a date column and a depth to water column in metres, or a NetCDF file, used as the lower boundary of the soil column.

### 28.0.14 Weather stations: online weather of any place, for the analysis and the runs

The section Online weather of a place, at the foot of Weather stations, finds a place by name, reads its daily weather of the years you choose from the internet (Open-Meteo, ERA5 reanalysis) and then offers three uses: the weather of the SWAFL project, the weather of every Daisy run, and a log in the list of the plots. The same record is therefore used in the analysis of the project, in the Daisy runs and in the figures.

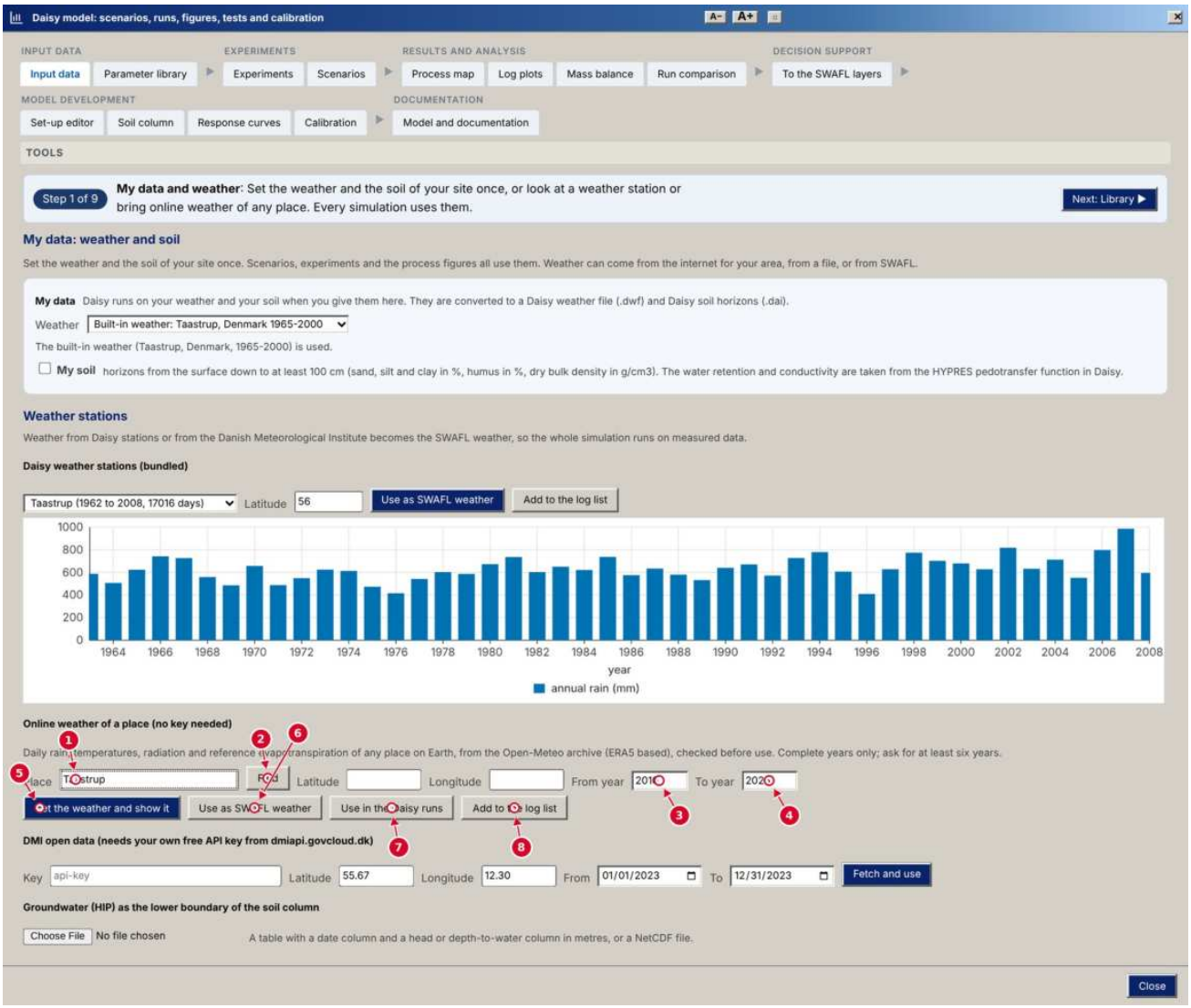



Figure 28.14: Weather stations: online weather of any place, for the analysis and the runs. Screen 13 of the Daisy workbench.

**What the arrows point to.**

1. **Place.** A town, a village or a farm name; the name is searched on the internet.
2. **Find.** Lists the places of that name; choose one and the latitude and longitude are filled.
3. **First year.** First year of the record to read.
4. **Last year.** Last complete year; the current year is not complete yet.
5. **Get the weather.** Reads the daily rain, temperature and radiation and checks them.
6. **Use as the SWAFL weather.** Sets the record as the weather of the SWAFL project, so every layer of the project uses it.

7. **Use in the Daisy runs.** Sets the record as the weather of every simulation of Scenarios, Experiments and Processes.
8. **Add to the list of logs.** Adds the record to the logs, so that it can be drawn in Plot.

## 28.1 The weather stations of Daisy

The stations that come with Daisy are listed here with the years that they cover. The first one is the built-in weather used when nothing else is chosen.

| Station | Label | First year | Last year |
|---|---|---|---|
| course | Taastrup, Denmark (built-in, 1965-2000) | 1965 | 2000 |
| dk-taastrup | Taastrup, Denmark (Daisy sample, 1963-2007) | 1963 | 2007 |
| dk-foulum | Foulum, Denmark (1990-2003) | 1990 | 2003 |
| dk-jyndevad | Jyndevad, Denmark (1997-2004) | 1997 | 2004 |
| it-bologna | Bologna, Italy (1994-2004) | 1994 | 2004 |
| cz-valecov | Valecov, Czechia (1991-2004) | 1991 | 2004 |
| pl-jadwisin | Jadwisin, Poland (1992-2005) | 1992 | 2005 |
| sk-bratislava | Bratislava, Slovakia (1990-2005) | 1990 | 2005 |

# Chapter 29

# ❖ Experiments and scenarios: ask a question

The second group answers a question by running Daisy: five ready scenarios and a free design of experiments.

## 29.1 Scenarios

Each scenario is a question about the built-in farm. Its runs differ only in the option that the question is about.

| Scenario | Question | Runs compared |
|---|---|---|
| irrigation | Irrigation: does it pay off, and what does it cost the soil? | Rainfed; Irrigated |
| nitrogen | Nitrogen: how much fertiliser, how much leaching? | No fertiliser (control); Half the fertiliser; Standard fertiliser; 30 % more fertiliser; 50 % more fertiliser |
| climate | Climate: warmer, drier or wetter | Weather file as it is; Warmer by 2 C; Drier (rain x 0.7); Warmer and drier; Wetter (rain x 1.3) |
| chemical | A pesticide or tracer: sorption and degradation | Mobile tracer (no sorption, no decay); Sorbing (K_OC 100); Sorbing and degrading (K_OC 100, half-life 30 d) |
| soil | Soil: how much does texture matter? | Sand; Loam |

### 29.1.1 The options of a scenario

| Option | Meaning | Kind |
|---|---|---|
| soil | Soil | select |
| start | First year | number |
| stop | Last year | number |
| precip | Rain, factor on the weather file | number |
| temp | Temperature offset (C) | number |
| nfac | Nitrogen fertiliser, factor on the rotation | number |
| irrig | Irrigation | select |
| trig | Irrigate when the potential at 20 cm drops below (cm) | number |
| mm | Water per irrigation (mm) | number |
| station | Weather of | select |

| Option | Meaning | Kind |
|---|---|---|
| method | Irrigation method | select |
| z | Sensor depth (cm) | number |
| hours | Duration of one irrigation (hours) | number |
| dfrom | Drip from depth (cm) | number |
| dto | Drip to depth (cm) | number |
| limit | Water limit per crop season (mm, 0 = none) | number |
| w1 | Irrigation window opens (month-day) | text |
| w2 | Irrigation window closes (month-day) | text |
| root | Maximum rooting depth (cm, 0 = soil default) | number |
| koc | Sorption of the chemical, K_OC (L/kg) | number |
| half | Half-life of the chemical in soil (days, 0 = none) | number |
| flux | Application rate of irrigation (mm/h, 0 = from dose and duration) | number |
| waitd | Days between irrigations | number |
| spin | Warm-up years before the results are counted | number |
| till | Tillage before sowing | select |
| ntype | Fertiliser | select |
| ntime | Fertiliser dates moved by (days) | number |
| bio | Evapotranspiration model (Bioclimate) | select |
| radscale | Global radiation, scale | number |
| pmonths | Rain scale per month (12 numbers, Jan to Dec; blank = none) | text |

### 29.1.2 Scenarios: choose the question

The Scenarios tab starts with the questions that the built-in farm can answer: water, nitrogen, climate, a chemical and soil. Two more cards lead to your own set-up and to two tools that need no Daisy program.

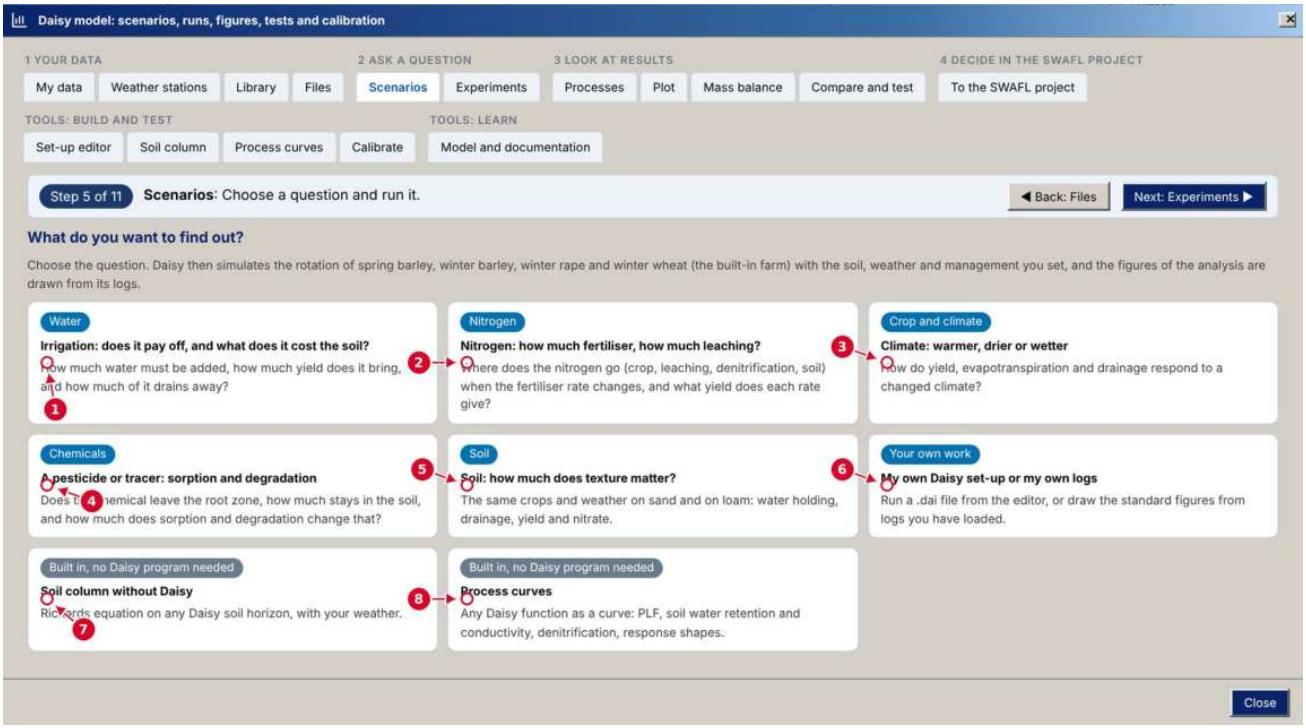



Figure 29.1: Scenarios: choose the question. Screen 14 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation question.** Does irrigation pay off, and what does it cost the soil.
2. **Nitrogen question.** How much fertiliser, how much leaching.
3. **Climate question.** A warmer, drier or wetter climate.
4. **Chemical question.** Sorption and degradation of a pesticide or tracer.
5. **Soil question.** How much does texture matter.
6. **Your own work.** Run a set-up of your own or draw the figures of logs you loaded.
7. **Soil column.** A built-in tool that needs no Daisy program.
8. **Process curves.** A built-in tool that draws any Daisy function.

### 29.1.3 Scenarios: the 37 questions, and one more run from the library

The Scenarios tab holds 37 questions in five themes: water, nitrogen, soil, crop and climate, and chemicals. *Find a question* filters the cards by a word (drip, slurry, PFAS, clay, maize). Each card says how many runs it compares and, when its runs differ in one option, how many choices of the parameter library can be added. The five questions of the earlier version are kept as they were; the others vary one option each over the values of the parameter library:

| Theme | Question | Runs compared | One more run from |
|---|---|---|---|
| Water | Irrigation: does it pay off, and what does it cost the soil? | Rainfed; Irrigated | n/a |
| Water | Irrigation method: overhead, surface or drip? | Rainfed (reference); Overhead (sprinkler); Surface; Drip at 10-20 cm | 3 choices of the library |
| Water | When to irrigate: the trigger suction | Wet: -150 cm (pF 2.2); Moist: -300 cm (pF 2.5); Drier: -600 cm (pF 2.8); Dry: -1000 cm (pF 3) | any value |
| Water | How much water per irrigation? | 10 mm per event; 20 mm per event; 30 mm per event; 45 mm per event | any value |
| Water | Deficit irrigation: a water limit per season | Rainfed; At most 60 mm a season; At most 120 mm a season; At most 240 mm a season; No limit | any value |
| Water | Where to put the soil moisture sensor? | Sensor at 10 cm; Sensor at 20 cm; Sensor at 40 cm; Sensor at 60 cm | any value |
| Nitrogen | Nitrogen: how much fertiliser, how much leaching? | No fertiliser (control); Half the fertiliser; Standard fertiliser; 30 % more fertiliser; 50 % more fertiliser | any value |
| Nitrogen | Which mineral fertiliser? | N25S (the farm); Ammonium nitrate; Calcium nitrate; Nitrate; Ammonia | 79 choices of the library |
| Nitrogen | Mineral fertiliser or slurry and manure? | Mineral (N25S); Cattle slurry; Pig slurry (fattening pigs); Cattle deep litter; Poultry manure | 79 choices of the library |
| Nitrogen | When to fertilise? | 3 weeks earlier; 1 week earlier; As the farm; 2 weeks later; 4 weeks later | any value |
| Soil | Soil: how much does texture matter? | Sand; Loam | 73 choices of the library |
| Soil | Eight soil textures (USDA) | Sand; Loamy sand; Sandy loam; Loam; Silt loam; Sandy clay loam; Clay loam; Silty clay loam | 73 choices of the library |
| Soil | Danish soil classes JB1 to JB7 of the library | JB1 (Ap_JB1); JB2 (Ap_JB2); JB3 (Ap_JB3); JB4 (Ap_JB4); JB5 (Ap_JB5); JB6 (Ap_JB6); JB7 (Ap_JB7) | 73 choices of the library |
| Soil | Tillage: plough, rotavate, harrow or none | Ploughing (the farm); Rotavation; Disk harrowing; Stubble cultivation; No tillage | 6 choices of the library |
| Soil | How deep can the roots go? | Roots to 40 cm; Roots to 70 cm; Roots to 100 cm; Roots to 150 cm | any value |
| Soil | The aquitard below the profile | Tight: K 0.0002 cm/h; The farm: K 0.001 cm/h; Leaky: K 0.005 cm/h | any value |
| Crop and climate | Climate: warmer, drier or wetter | Weather file as it is; Warmer by 2 C; Drier (rain x 0.7); Warmer and drier; Wetter (rain x 1.3) | n/a |
| Crop and climate | Which crop? One crop of the library every year | Spring barley; Winter wheat; Silage maize; Potato; Sugar beet; Grass | 84 choices of the library |
| Crop and climate | The same farm in other climates of Europe | Taastrup, Denmark; Foulum, Denmark; Jadwisin, Poland; Valecov, Czechia; Bratislava, Slovakia; Bologna, Italy | 8 choices of the library |
| Crop and climate | Sunnier or cloudier | Cloudier (radiation x 0.85); As it is; Sunnier (radiation x 1.15) | any value |
| Chemicals | A pesticide or tracer: sorption and degradation | Mobile tracer (no sorption, no decay); Sorbing (K_OC 100); Sorbing and degrading (K_OC 100, half-life 30 d) | 35 choices of the library |
| Chemicals | Pesticides of the Daisy library | Bentazon; MCPP (mecoprop); Atrazine; Isoproturon (IPU); Pendimethalin; 2,4-D | 35 choices of the library |
| Chemicals | PFAS: the forever chemicals | PFOS; PFOA; PFHxA; PFBS; PFBA | 35 choices of the library |

| Theme | Question | Runs compared | One more run from |
|---|---|---|---|
| Chemicals | How strongly the chemical is held (K_OC) | K_OC 0 L/kg; K_OC 10 L/kg; K_OC 50 L/kg; K_OC 200 L/kg; K_OC 1000 L/kg | any value |
| Water | How fast is the water applied? Irrigation intensity | 0.1 mm per hour; 0.5 mm per hour; 2 mm per hour; 5 mm per hour | any value |
| Water | When in the year may irrigation run? The irrigation window | From April 1 to October 1; From May 1 to October 1; From June 1 to October 1; From May 1 to September 1 | any value |
| Nitrogen | Nitrogen and water together: the water-nitrogen interaction | Rainfed and irrigated, each at nitrogen x 0.5, x 1 and x 1.3 (six runs) | any value |
| Nitrogen | The same nitrogen on a sand and on a loam | Sand and loam, each at nitrogen x 0.7, x 1 and x 1.3 (six runs) | any value |
| Soil | Saturated hydraulic conductivity of the soil | K_sat x 0.25; x 0.5; x 1; x 2; x 4 | any value |
| Soil | Pore-size distribution of the soil (van Genuchten n) | n minus 1 x 0.7; x 0.85; x 1; x 1.15; x 1.3 | any value |
| Soil | How fast does the tilled topsoil settle? | Consolidation 0.001; 0.005; 0.01; 0.02 per day | any value |
| Crop and climate | Dry and wet years: the rain of the weather file scaled | Rain x 0.6; x 0.8; x 1; x 1.2; x 1.4 | any value |
| Crop and climate | Warming: temperature raised by 0 to 3 degrees | As the weather file; +1 C; +2 C; +3 C | any value |
| Crop and climate | When the rain falls: dry spring, dry summer, wet winter | As the weather file; Dry summer; Dry spring; Wet winter | any value |
| Crop and climate | Which evapotranspiration model? | FAO56; Daisy energy balance | n/a |
| Chemicals | How long the chemical lasts (half-life) | Half-life 5; 15; 30; 90; 365 days at K_OC 100 | any value |
| Chemicals | Leachers and non-leachers: the GUS classes | Five chemicals with GUS 1.0, 1.5, 2.2, 3.0 and 4.8 | any value |

The thirteen questions added in the last release come from the literature on irrigation, nitrogen, soil hydraulics, climate and pesticide leaching: they let a first-time user ask a complete question (for example *does water make nitrogen pay?*) without building a design. After the runs, every question with three or more runs also shows the efficiency figures of Section 31.12 calculated from its own runs, and the map of the processes of Daisy shows only what the variants changed and what the runs measured.

**Add a run.** In the panel of a question with a varied option, the row *Add a run* offers every other value of that option: a list with a search box for a choice of the library (79 fertilisers and slurries, 84 crops, 35 chemicals, 73 soils, 8 weather stations), or a number box for a number option (with the range it accepts). *Add* makes one more run with that value and every other option as set above; it is ticked like the others and run with them. *Remove the added runs* takes them out again.

**The library in a run.** A run that uses a slurry, a chemical or a crop of the library reads the file of the library inside its rotation file, after the files the rotation reads itself (fertilizer.dai, crop.dai, tillage.dai) and before its own definitions, which use it. The copy of the file in Common/ leaves out what the run already reads, because Daisy stops on a definition made twice. All 24 questions were run with Daisy 7.1.14 and give their logs.

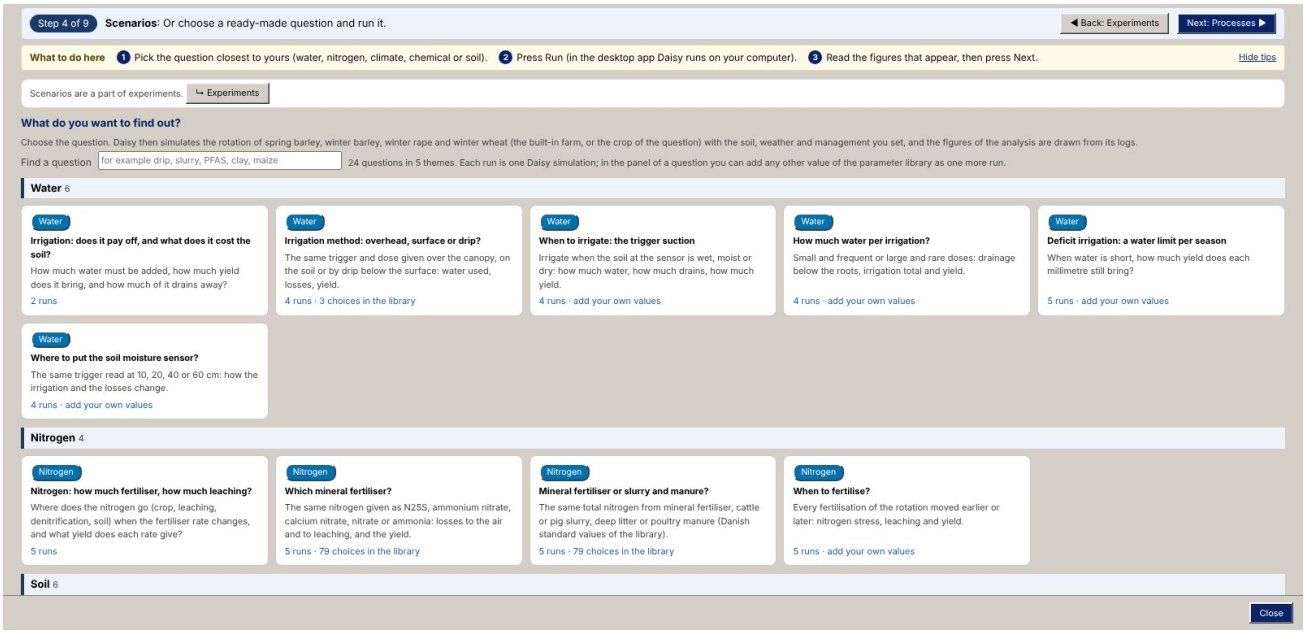



Figure 29.2: Questions of the Scenarios tab in their five themes.

## 29.1.4 Scenarios: irrigation

The irrigation question compares rainfed and irrigated crops of the built-in rotation. The trigger is the potential at 20 cm below which the overhead irrigation starts; the water per irrigation is fixed.

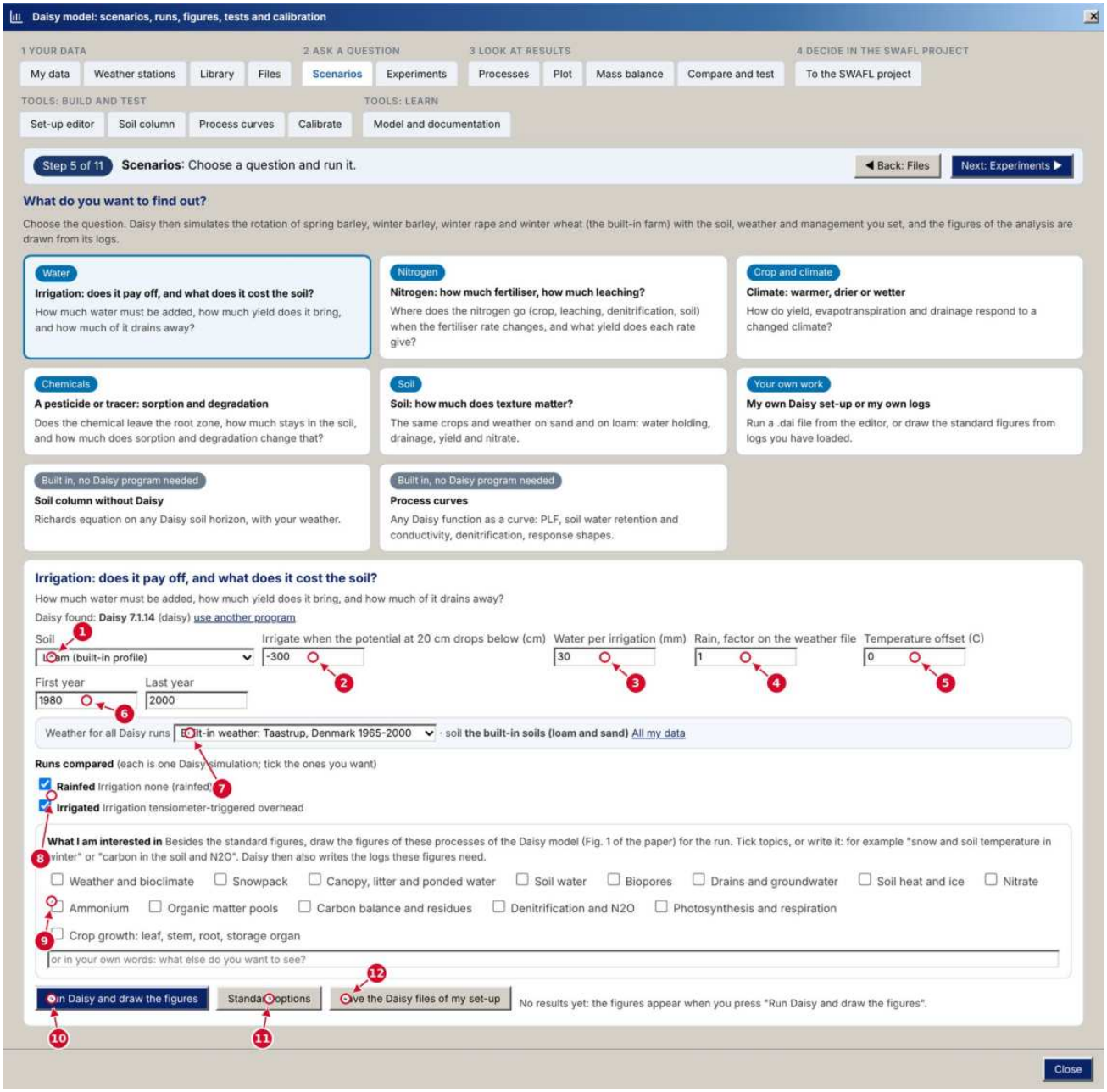



Figure 29.3: Scenarios: irrigation. Screen 15 of the Daisy workbench.

**What the arrows point to.**

1. **Soil.** The soil profile of the run: the built-in loam or sand, or your own soil when it is set in Input data.
2. **Trigger.** Irrigation starts when the soil water potential at 20 cm drops below this value (cm).
3. **Water per irrigation.** Depth of water applied each time (mm).
4. **Rain factor.** Multiplies the rain of the weather file (1 leaves it as recorded).

5. **Temperature offset.** Added to every daily temperature (degrees Celsius).
6. **First year.** First year of the simulation; it must lie at least one year after the start of the weather.
7. **Weather for all runs.** The one weather choice of every Daisy simulation: the weather of the SWAFL project when one is loaded, the built-in Taastrup weather, a built-in station, online weather or a file.
8. **Runs compared.** Each ticked run is one Daisy simulation; the figures compare them.
9. **What I am interested in.** Tick the processes of Fig. 1 whose figures should be drawn as well, or write them in words.
10. **Run Daisy.** Runs every ticked simulation and draws the figures.
11. **Standard options.** Puts every option back to its standard value.
12. **Save the Daisy files.** Saves the set-up files of the first ticked run, to run Daisy anywhere.

### 29.1.5 Scenarios: nitrogen

The nitrogen question compares three fertiliser levels, with or without irrigation, on the built-in rotation. It reports yield, nitrate leached, denitrification and the nitrogen balance.

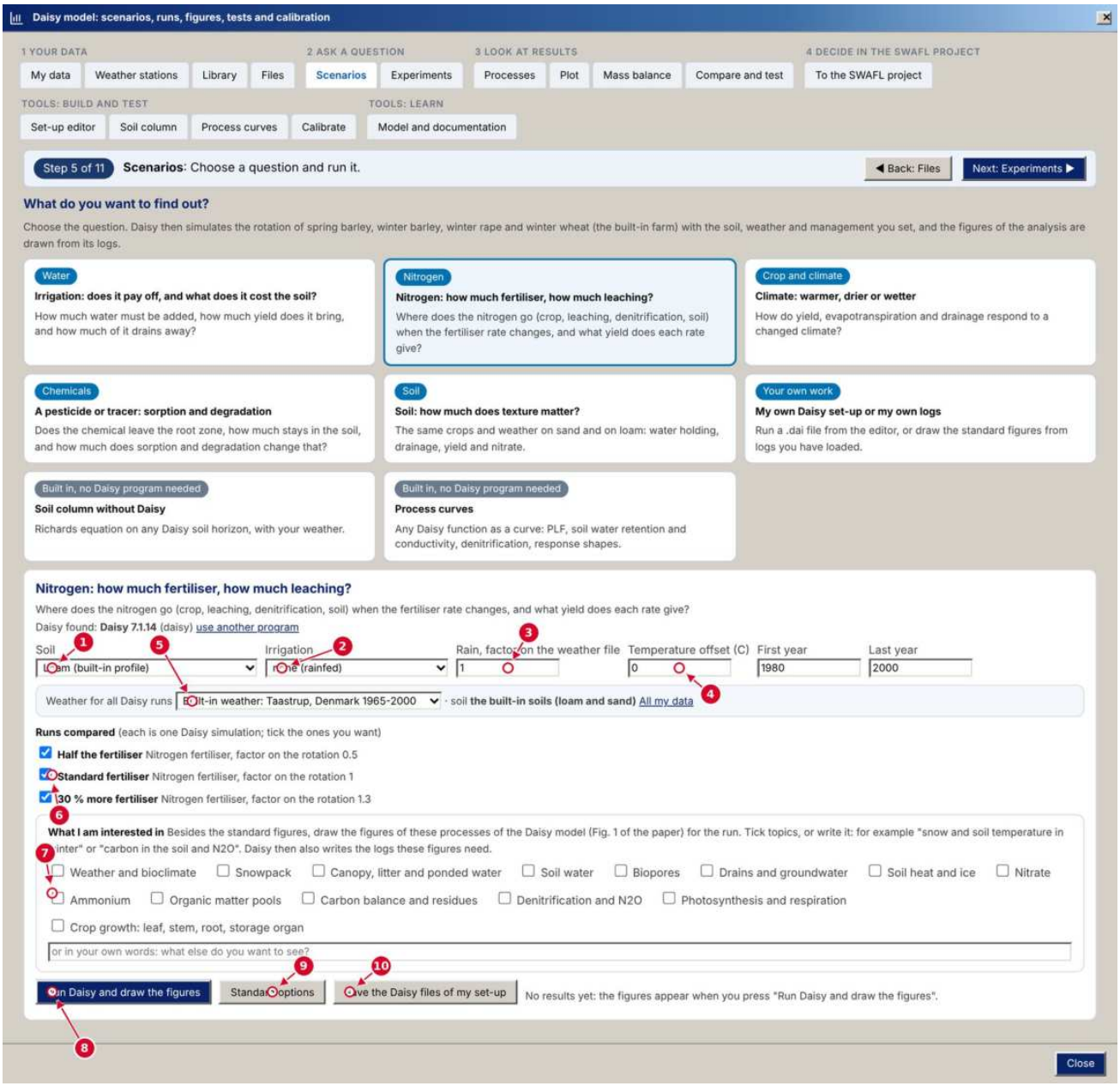



Figure 29.4: Scenarios: nitrogen. Screen 16 of the Daisy workbench.

**What the arrows point to.**

1. **Soil.** The soil profile of the run: the built-in loam or sand, or your own soil when it is set in Input data.
2. **Irrigation.** Rainfed or irrigated.
3. **Rain factor.** Multiplies the rain of the weather file.
4. **Temperature offset.** Added to the daily temperatures.
5. **Weather for all runs.** The one weather choice of every Daisy simulation: the weather of the SWAFL project when one is loaded, the built-in Taastrup weather, a built-in station, online

weather or a file.

6. **Runs compared.** Each ticked run is one Daisy simulation; the figures compare them.
7. **What I am interested in.** Tick the processes of Fig. 1 whose figures should be drawn as well, or write them in words.
8. **Run Daisy.** Runs every ticked simulation and draws the figures.
9. **Standard options.** Puts every option back to its standard value.
10. **Save the Daisy files.** Saves the set-up files of the first ticked run, to run Daisy anywhere.

### 29.1.6 Scenarios: climate

The climate question runs the same farm under the weather as it is and under a warmer, a drier, a warmer and drier and a wetter weather.

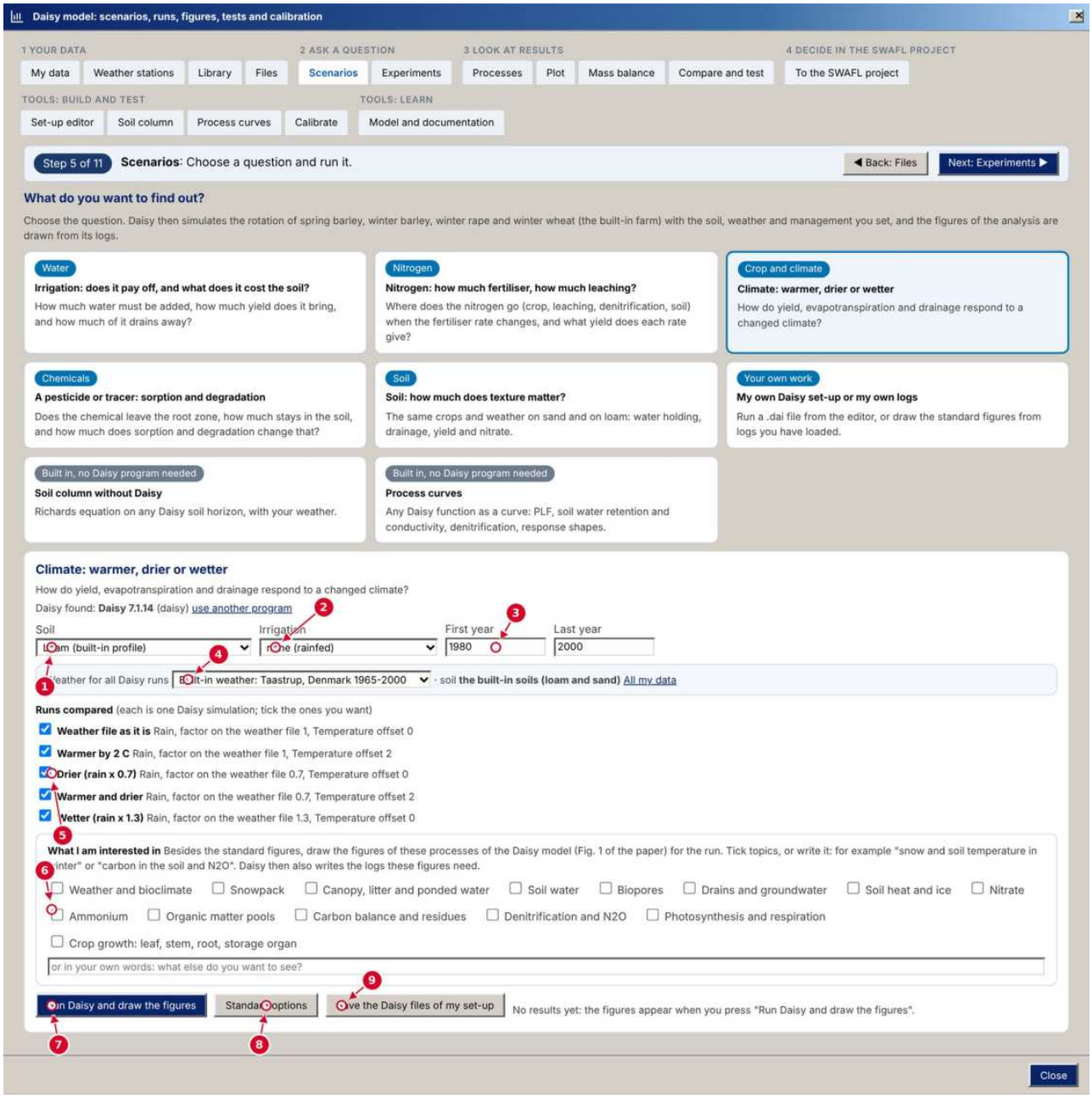



Figure 29.5: Scenarios: climate. Screen 17 of the Daisy workbench.

**What the arrows point to.**

1. **Soil.** The soil profile of the run: the built-in loam or sand, or your own soil when it is set in Input data.
2. **Irrigation.** Rainfed or irrigated.
3. **First year.** First year of the runs.
4. **Weather for all runs.** The one weather choice of every Daisy simulation: the weather of the SWAFL project when one is loaded, the built-in Taastrup weather, a built-in station, online

weather or a file.

5. **Runs compared.** Each ticked run is one Daisy simulation; the figures compare them.
6. **What I am interested in.** Tick the processes of Fig. 1 whose figures should be drawn as well, or write them in words.
7. **Run Daisy.** Runs every ticked simulation and draws the figures.
8. **Standard options.** Puts every option back to its standard value.
9. **Save the Daisy files.** Saves the set-up files of the first ticked run, to run Daisy anywhere.

### 29.1.7 Scenarios: a pesticide or tracer

The chemical question follows a chemical applied to the soil: a mobile tracer, a strongly sorbed chemical and a quickly decaying one. It shows how much leaves the root zone and how much stays in the soil.

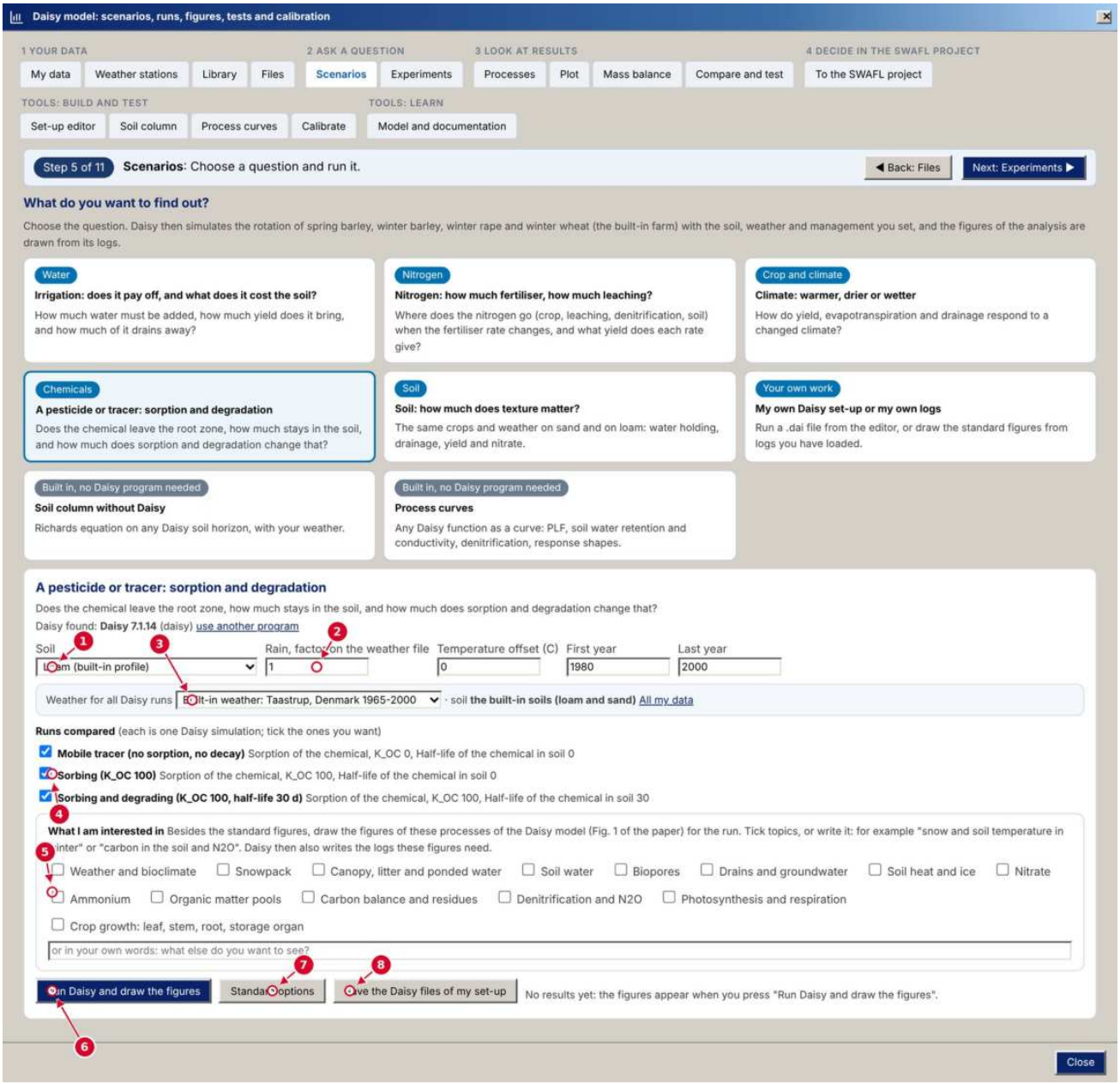



Figure 29.6: Scenarios: a pesticide or tracer. Screen 18 of the Daisy workbench.

**What the arrows point to.**

1. **Soil.** The soil profile of the run: the built-in loam or sand, or your own soil when it is set in Input data.
2. **Rain factor.** Multiplies the rain of the weather file.
3. **Weather for all runs.** The one weather choice of every Daisy simulation: the weather of the SWAFL project when one is loaded, the built-in Taastrup weather, a built-in station, online weather or a file.

4. **Runs compared.** Each ticked run is one Daisy simulation; the figures compare them.
5. **What I am interested in.** Tick the processes of Fig. 1 whose figures should be drawn as well, or write them in words.
6. **Run Daisy.** Runs every ticked simulation and draws the figures.
7. **Standard options.** Puts every option back to its standard value.
8. **Save the Daisy files.** Saves the set-up files of the first ticked run, to run Daisy anywhere.

## 29.1.8 Scenarios: soil texture

The soil question runs the same crops, weather and management on the built-in sand and loam, and shows water holding, drainage, yield and nitrate.

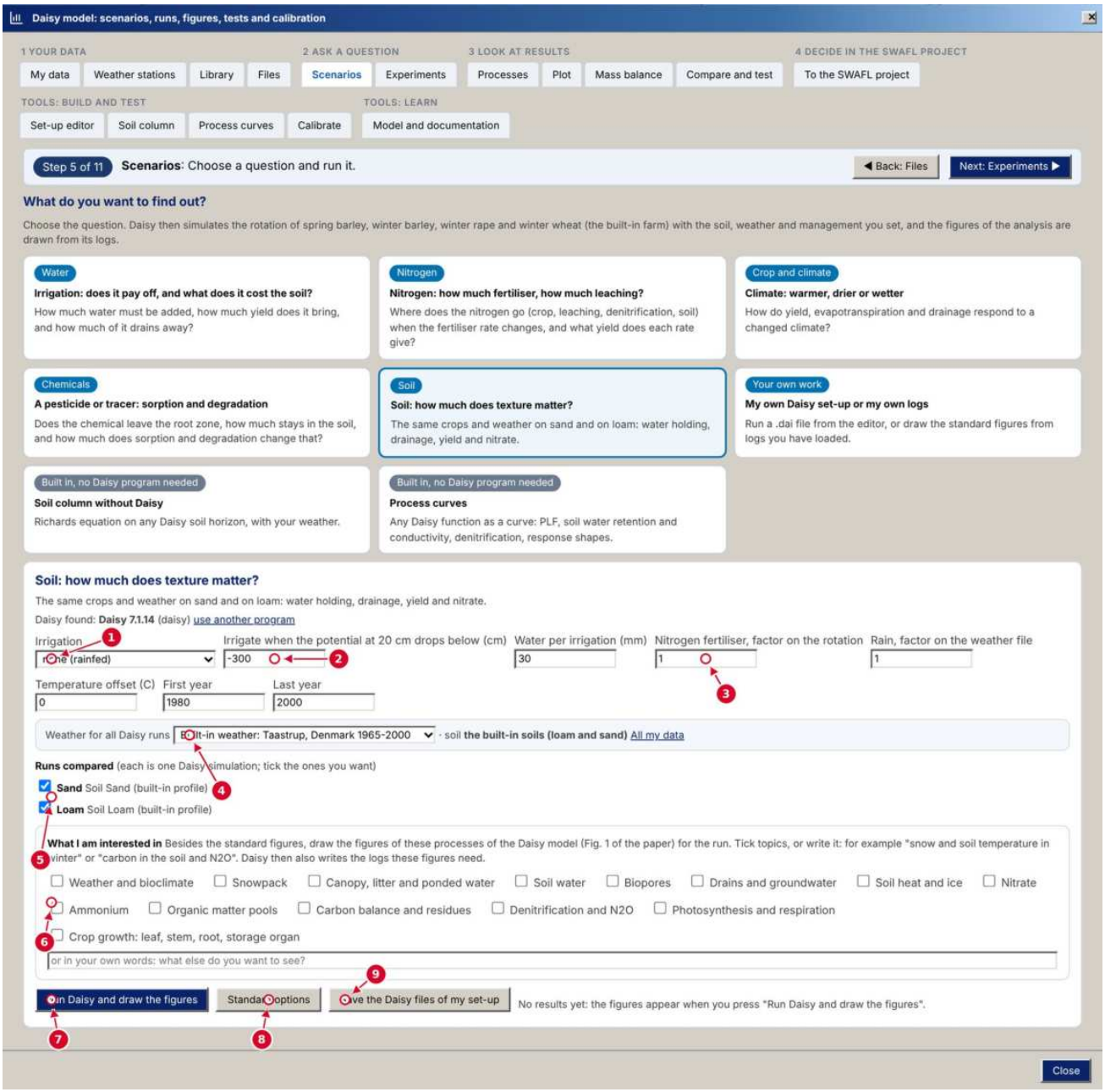



Figure 29.7: Scenarios: soil texture. Screen 19 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation.** Rainfed or irrigated.
2. **Trigger.** Potential at 20 cm that starts irrigation.
3. **Fertiliser factor.** Multiplies the fertiliser of the rotation.
4. **Weather for all runs.** The one weather choice of every Daisy simulation: the weather of the SWAFL project when one is loaded, the built-in Taastrup weather, a built-in station, online weather or a file.

5. **Runs compared.** Each ticked run is one Daisy simulation; the figures compare them.
6. **What I am interested in.** Tick the processes of Fig. 1 whose figures should be drawn as well, or write them in words.
7. **Run Daisy.** Runs every ticked simulation and draws the figures.
8. **Standard options.** Puts every option back to its standard value.
9. **Save the Daisy files.** Saves the set-up files of the first ticked run, to run Daisy anywhere.

### 29.1.9 Scenarios: your own work

This card leads to the set-up editor, where a .dai file of your own is run, and to the figures that are drawn from logs you have loaded.

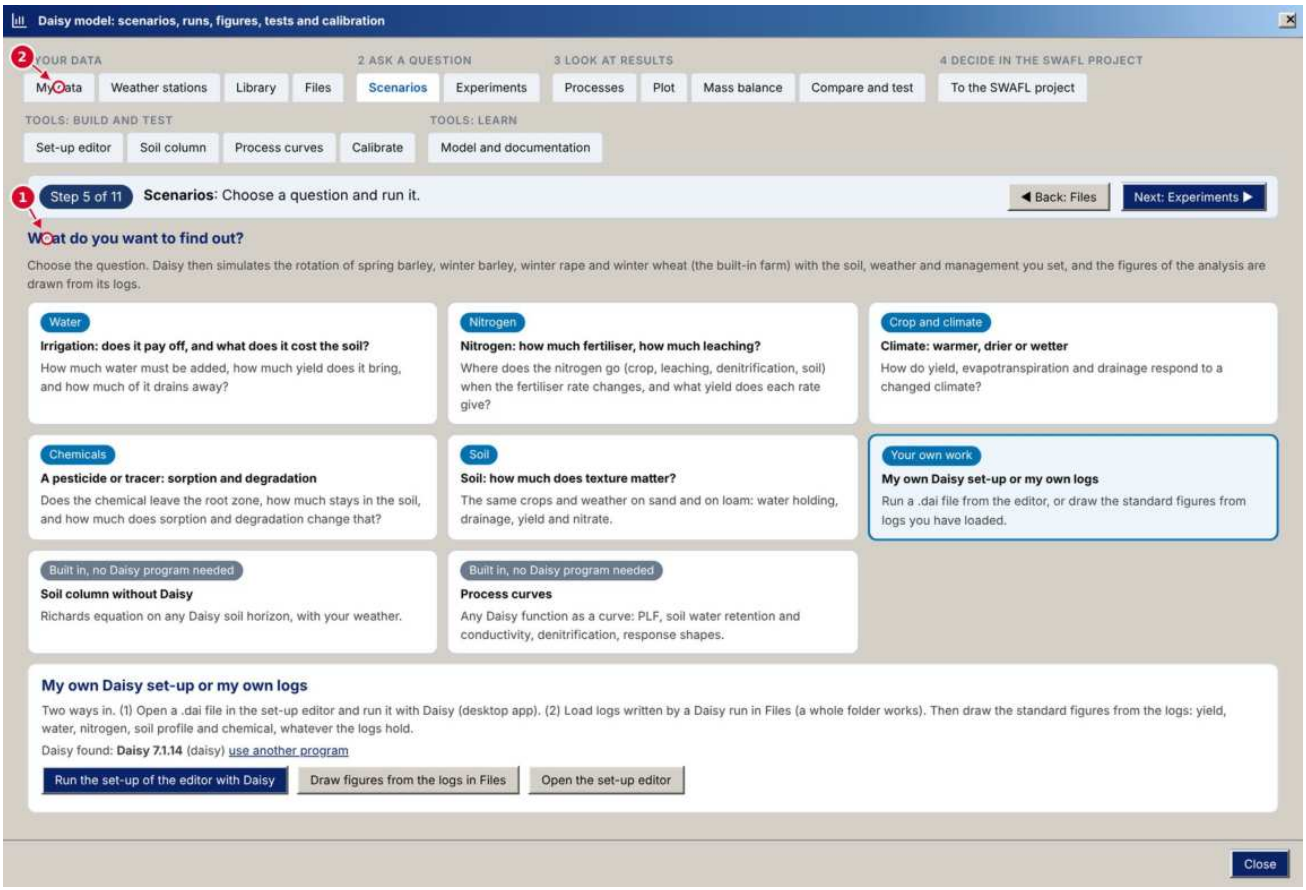



Figure 29.8: Scenarios: your own work. Screen 20 of the Daisy workbench.

**What the arrows point to.**

1. **Heading.** The work you bring: a set-up of your own or logs of earlier runs.
2. **First action.** Opens the matching tab.

## 29.1.10 Scenarios: processes to draw as well

Under the runs a list of the processes of Fig. 1 of the Daisy paper can be ticked, or described in words. Daisy then writes the logs these figures need.

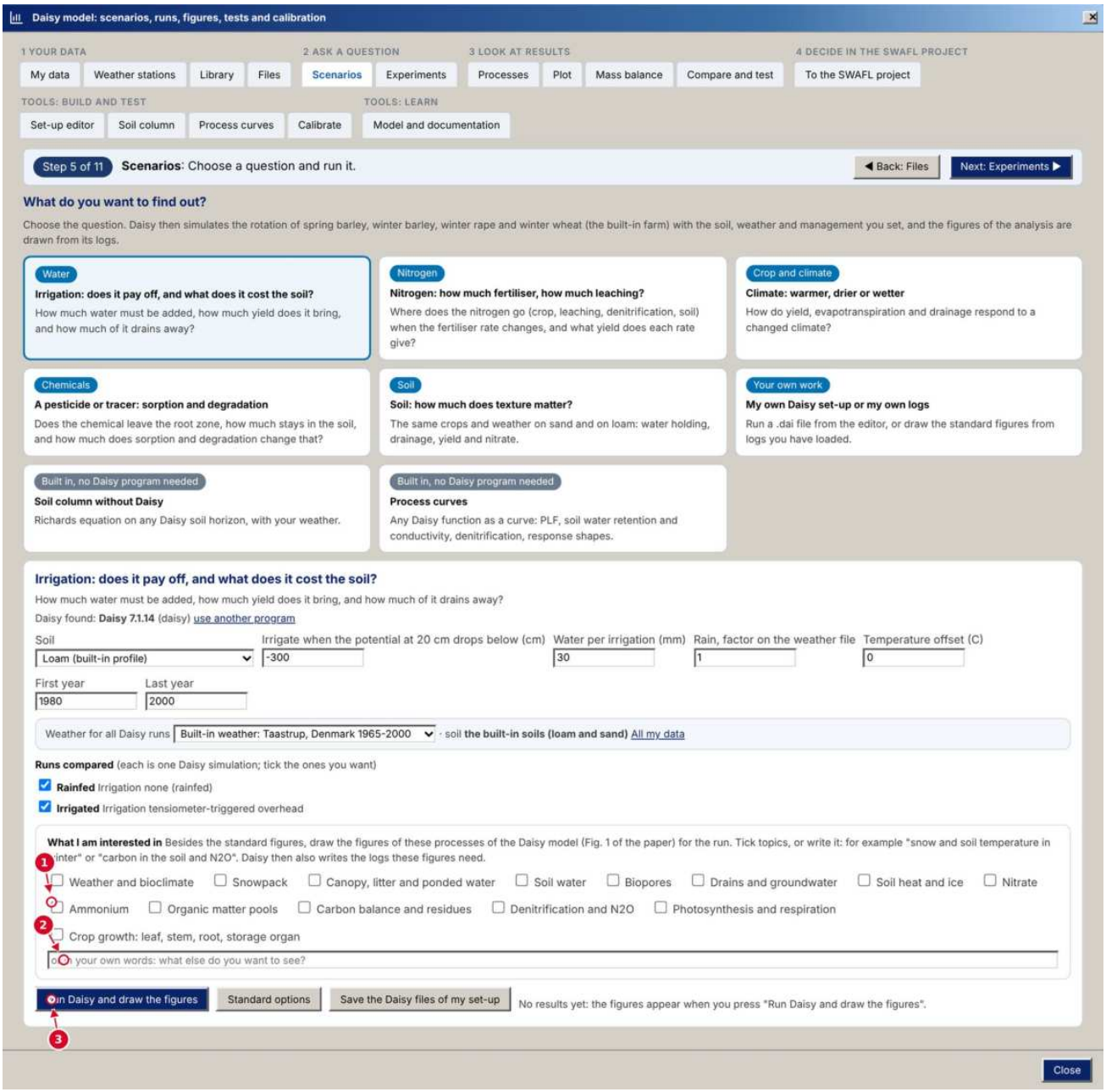



Figure 29.9: Scenarios: processes to draw as well. Screen 21 of the Daisy workbench.

**What the arrows point to.**

1. **Processes of Fig. 1.** Weather, snow, canopy, soil water, biopores, drains, heat and ice, nitrate, ammonium, organic matter, carbon, denitrification, photosynthesis and crop growth.
2. **In words.** Write for example snow and soil temperature in winter; the matching processes are ticked.
3. **Run Daisy.** Runs the simulations and draws the standard figures and these.

## 29.2 Experiments

Experiments is the free design. Runs are chosen from the factors, the plan is made as a one at a time, factorial, Latin hypercube or other design, and the runs are listed in order of preference.

### 29.2.1 Experiments: the starting screen

The Experiments tab turns a choice into a set of Daisy runs: start from a ready made combination of the Daisy reference, choose a topic of the reference, load an experiment of the study, or bring set-ups of your own. Every plan can then be changed row by row.

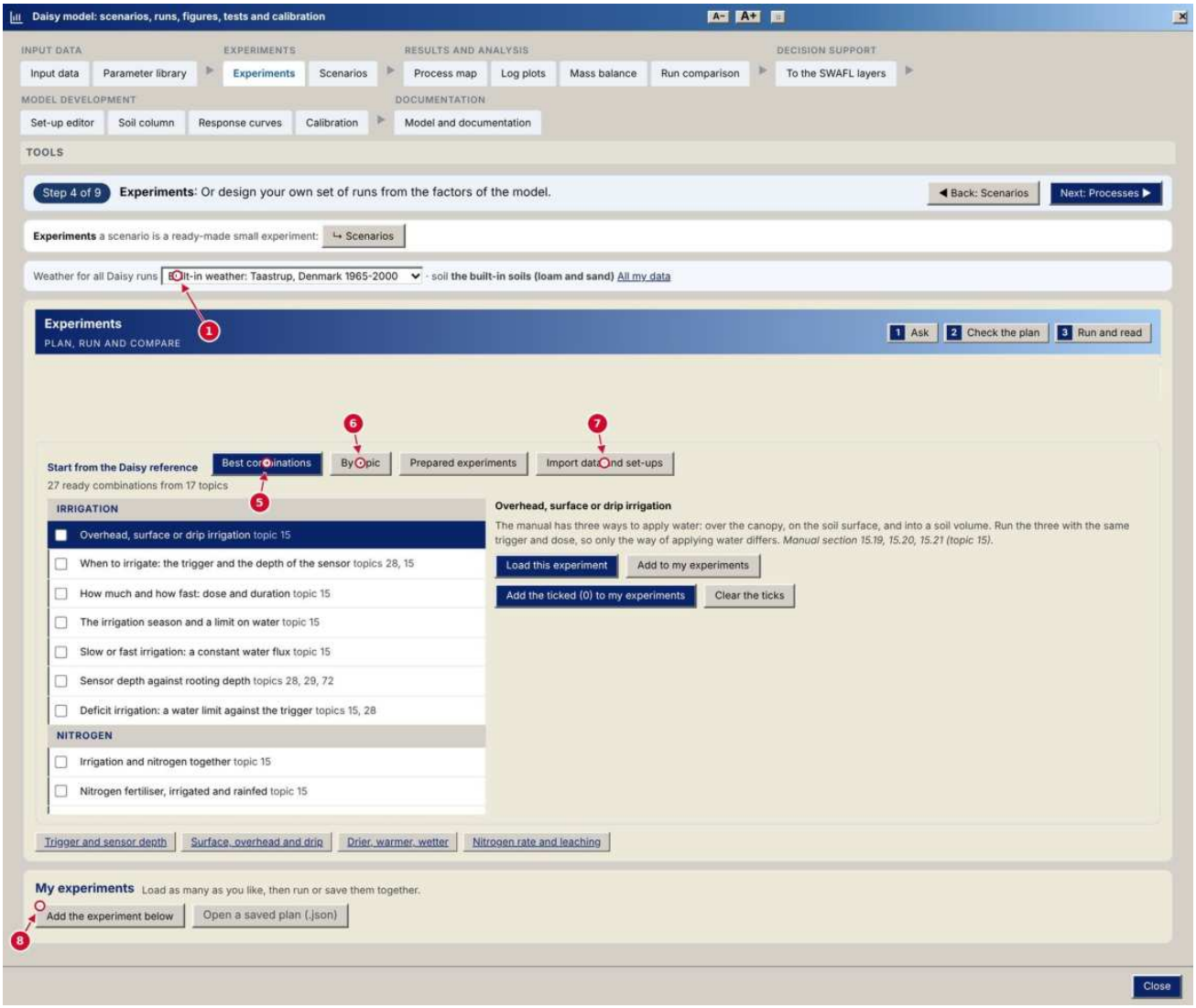



Figure 29.10: Experiments: the starting screen. Screen 22 of the Daisy workbench.

**What the arrows point to.**

1. **Weather for all runs.** The weather of every simulation, shared with Scenarios and Processes.

5. **Best combinations.** Ready made experiments worked out from the Daisy reference.
6. **By topic.** The 107 topics of the Daisy reference, each leading to experiments.
7. **Import.** Bring an experiment folder, set-ups, weather, soil tables or logs.

8. **My experiments.** The list of experiments that are run or saved together.

### 29.2.2 Soil hydraulic sensitivity

Five factors change the van Genuchten-Mualem curves of every horizon of the built-in soils (ResFarm.dai), as in the sensitivity studies of soil water retention and conductivity: **K_sat** (multiplied, for example 0.25, 1, 4), **alpha** (multiplied: 0.5, 1, 2), **n** (n - 1 multiplied, so n stays above 1: 0.7, 1, 1.3), **Theta_sat** (multiplied, kept below 0.7: 0.9, 1, 1.1) and the tortuosity **l** (added: -1, 0, 1). Only the numbers inside each (hydraulic M_vG ...) block of the soil file of the run change; the README of each run shows them. In a test with Daisy 7.1.14 on the loam, K_sat x 4, alpha x 2, n x 1.3, Theta_sat x 1.1 and l + 1 together raised matrix percolation from 177 to 222 mm and removed the runoff of the season. Combined with *Which combination is best*, they show which soil property matters most for yield, drainage and leaching.

### 29.2.3 Experiments: best combinations to start with

The list holds 27 combinations in groups: Irrigation, Nitrogen, Climate, Soil and crop, Chemicals and Parameters. Each shows the topics of the reference it draws on. Choosing one shows why it is worth running; the tick boxes select several to add at once.

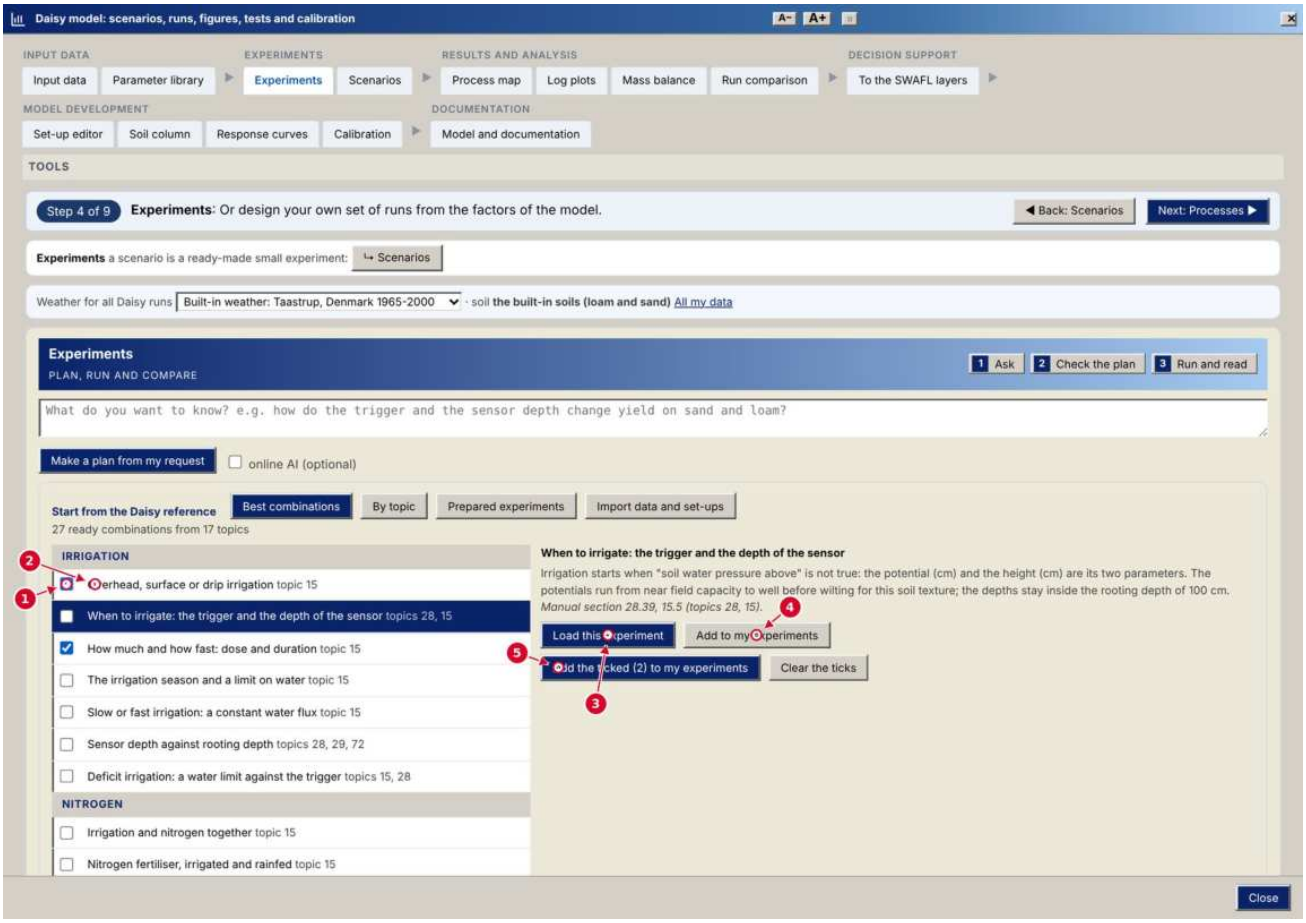



Figure 29.11: Experiments: best combinations to start with. Screen 23 of the Daisy workbench.

**What the arrows point to.**

1. **Tick box.** Ticks the combination so that several can be added together.
2. **Combination.** A combination with the topics of the Daisy reference it draws on.
3. **Load this experiment.** Loads the plan of the chosen combination for checking and running.
4. **Add to my experiments.** Adds the chosen combination to the list of experiments without loading it.
5. **Add the ticked.** Adds every ticked combination, each as an experiment of its own.

### 29.2.4 Experiments: the topics of the Daisy reference

The second tab lists the 107 topics of the Daisy reference manual in three parts. A topic shows its models, the parameters that Daisy accepts for experiments and the related experiments. The size of the experiment is set before loading.

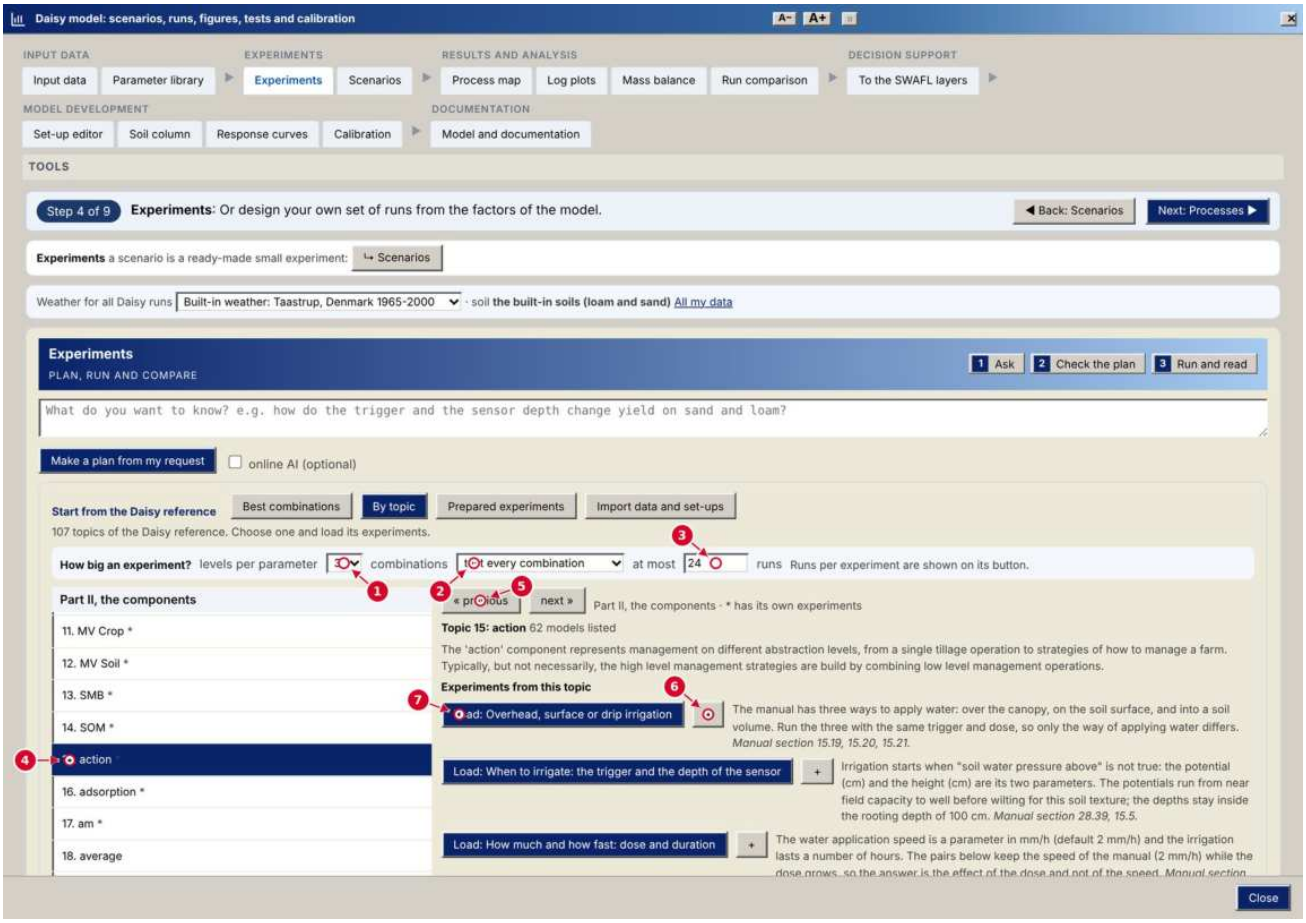



Figure 29.12: Experiments: the topics of the Daisy reference. Screen 24 of the Daisy workbench.

**What the arrows point to.**

1. **Levels per parameter.** How many values of each parameter are tested.
2. **How to combine.** Every combination, one thing at a time, or paired.
3. **At most.** The largest number of runs.
4. **List of topics.** Scroll the list; a star marks a topic that has its own experiments.
5. **Previous and next.** Moves through the topics in order.
6. **Add.** Adds the experiment to my experiments.
7. **Load.** Loads the experiment for checking and running.

### 29.2.5 Experiments: a topic with parameters you can vary

Topics whose parameters Daisy accepts show them with a tick box. Each parameter was tested in Daisy; the levels are fractions of the value the farm uses.

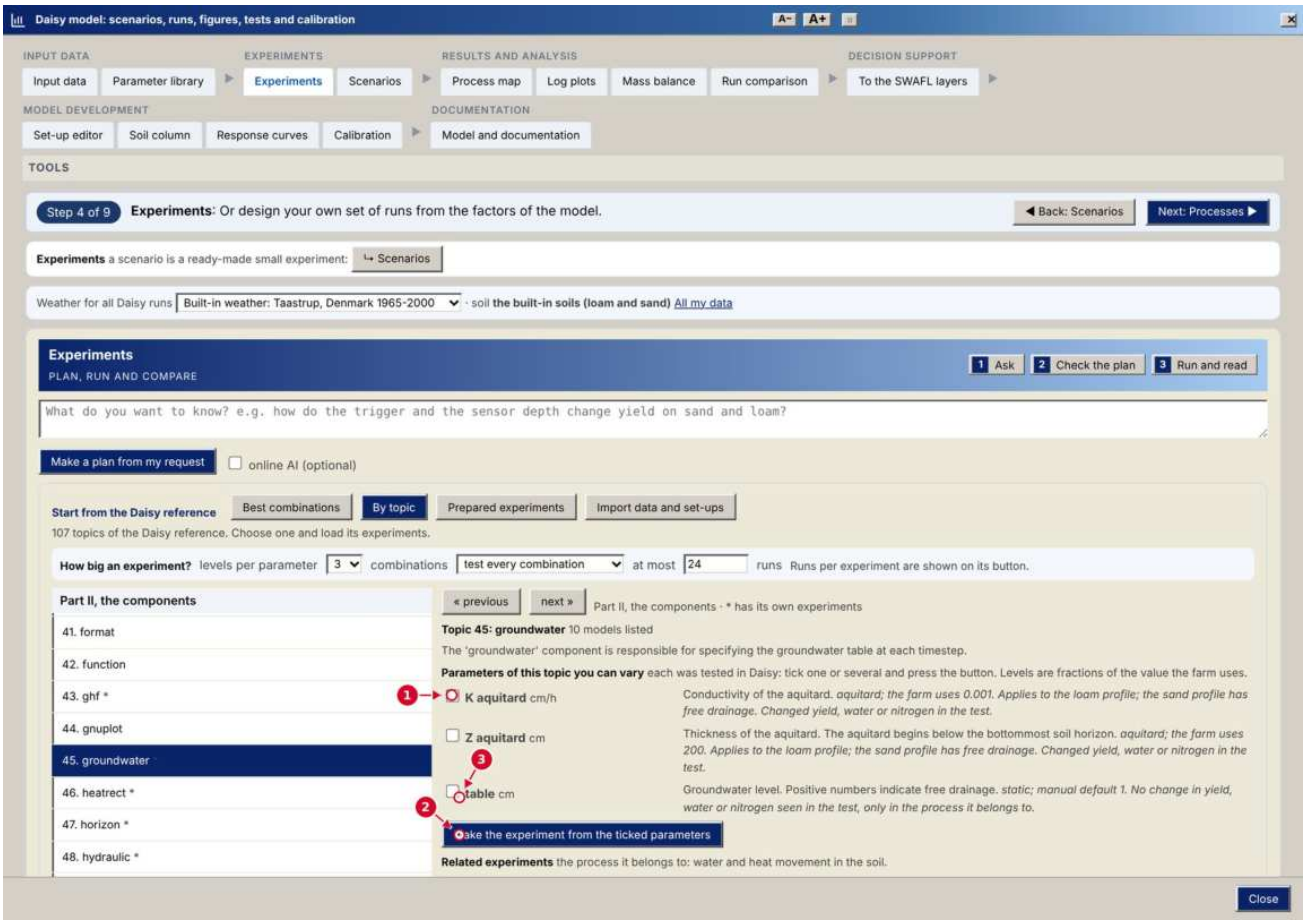



Figure 29.13: Experiments: a topic with parameters you can vary. Screen 25 of the Daisy workbench.

**What the arrows point to.**

1. **Parameter.** Tick one or several parameters to vary.
2. **Topic details.** The introduction of the topic, its parameters and the models it lists.

### 29.2.6 Experiments: import data and set-ups

The third tab brings in what you already have: an experiment folder as a zip, Daisy set-up files, weather, soil tables or logs. Each goes where it is used; set-ups are run as experiments.

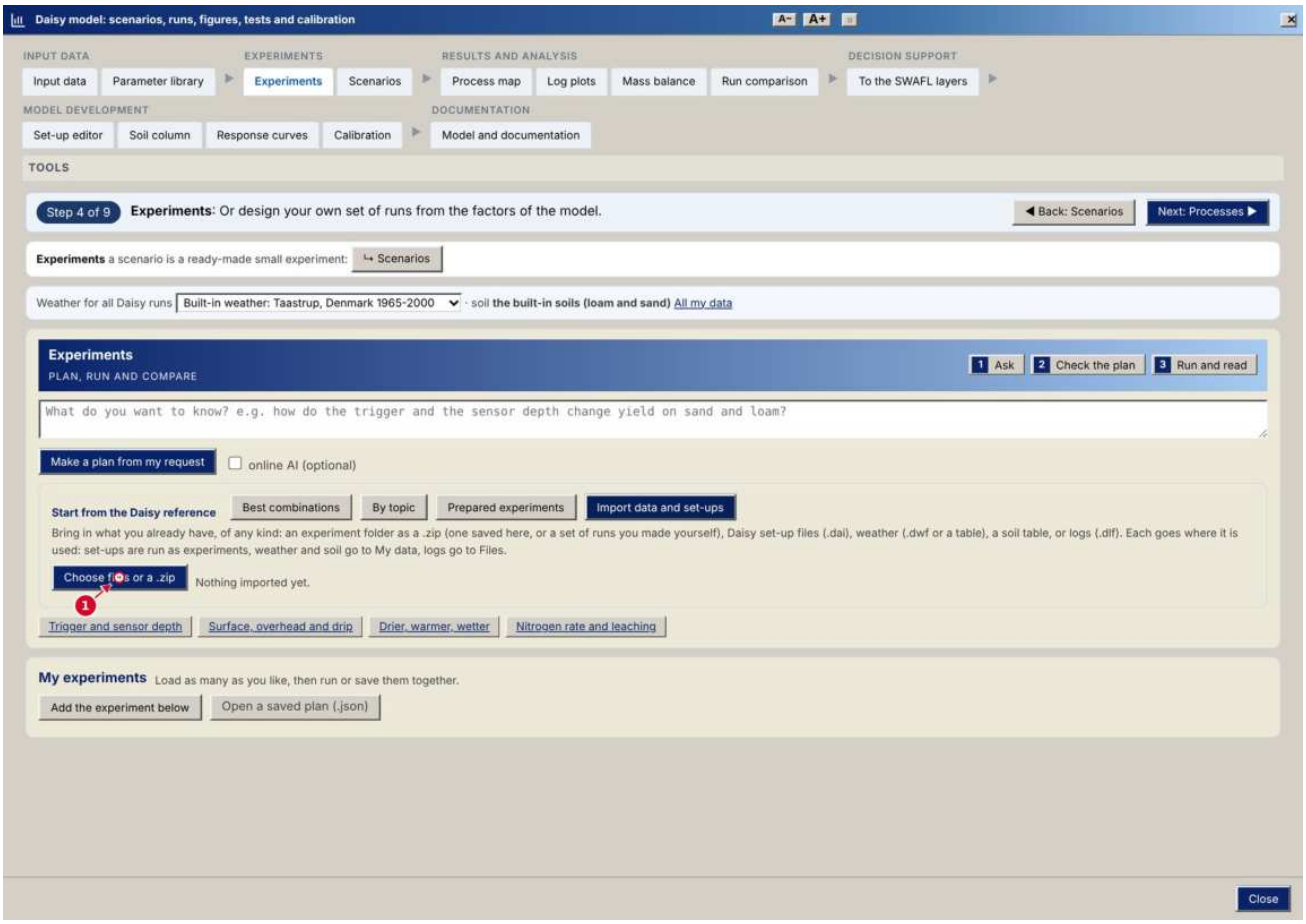



Figure 29.14: Experiments: import data and set-ups. Screen 26 of the Daisy workbench.

**What the arrows point to.**

1. **Choose a folder.** Reads every Daisy file of a folder.

### 29.2.7 Experiments: the plan before running

Loading a combination fills the plan: what changes, the values tested, how they are combined, the weather and soil, the years and the results saved. The size of the experiment is shown before anything is run.

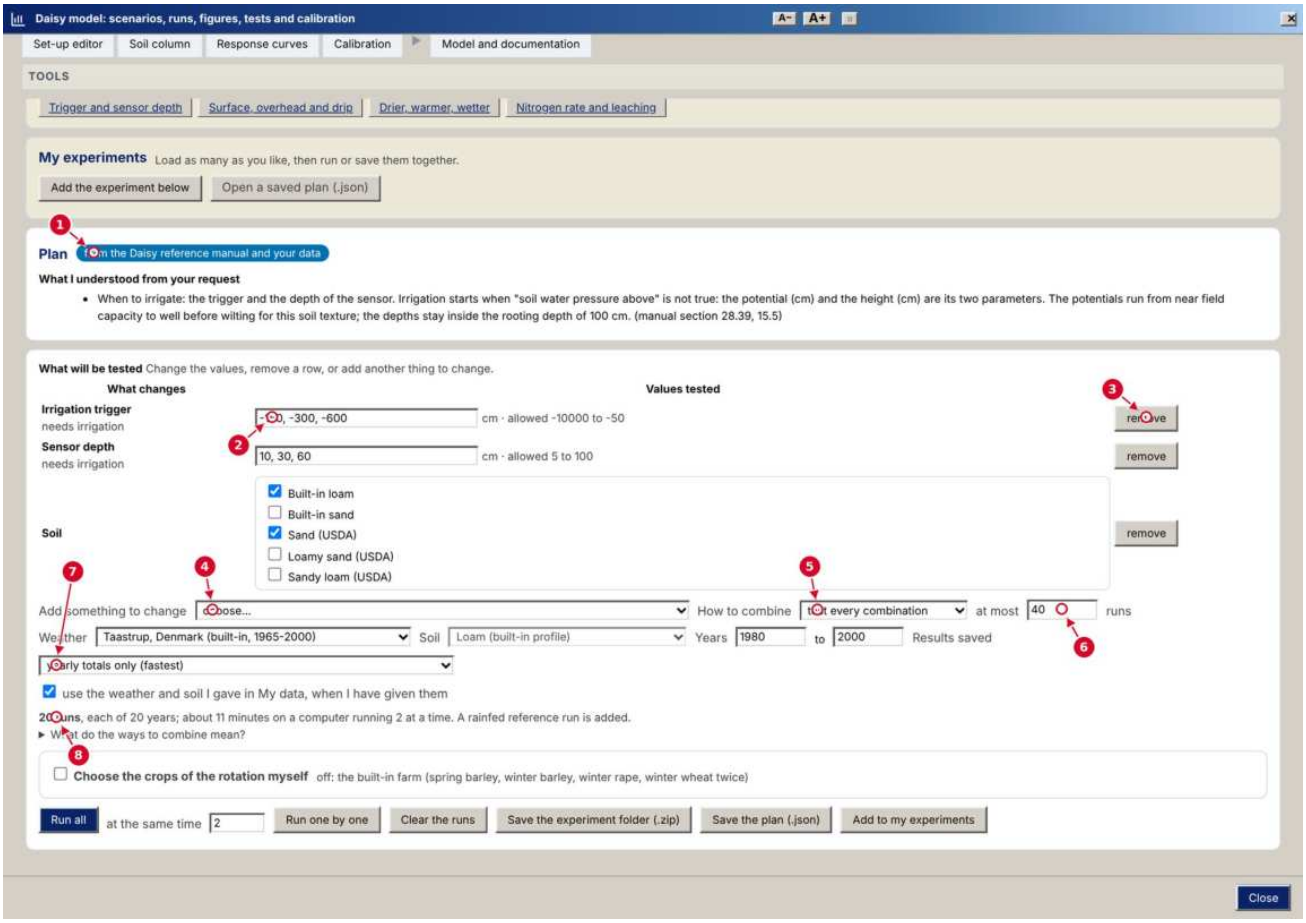



Figure 29.15: Experiments: the plan before running. Screen 27 of the Daisy workbench.

**What the arrows point to.**

1. **Source of the plan.** Where the plan comes from: the reference manual, a prepared experiment or an experiment of the study.
2. **Values tested.** The values can be typed over; the allowed range is shown beside them.
3. **Remove.** Takes the row out of the experiment.
4. **Add something to change.** Adds a further factor.
5. **How to combine.** Every combination, one thing at a time, or paired.
6. **At most.** The largest number of runs.
7. **Results saved.** How much Daisy writes: yearly totals only, with harvest, with daily values, or everything.
8. **Size.** Number of runs and the time they take.

## 29.2.8 Experiments: what each row means, and the number of runs

Under every row of *What will be tested* a sentence says what the factor is; for a parameter of the Daisy reference it also gives the Daisy default and the value of the farm. *more* opens four lines: where Daisy reads the value (the line of the run file or of the rotation), the typical values, what to type, and what changes in the results. The note after a number box, *allowed a to b*, is what the box accepts (0 to ten times the value used), not a limit of Daisy; the typical values are much narrower. Chapter 37 gives the same for every factor.

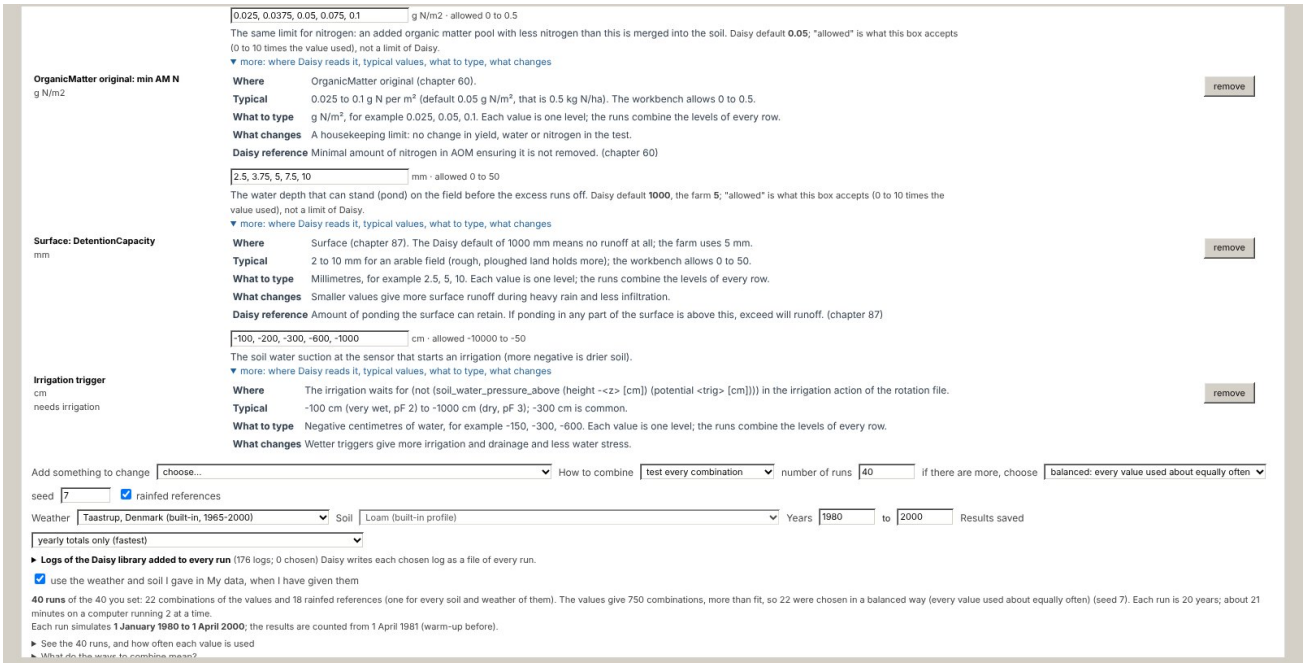



Figure 29.16: Three rows with their explanations: the minimum nitrogen of an added organic matter pool, the detention capacity of the surface and the trigger of the irrigation.

**The number of runs.** The box *number of runs* is the number of runs Daisy makes, the rainfed references included, and it is never exceeded. When the values give more combinations, the runs are chosen in the way set in *if there are more, choose*: balanced (each value used about equally often: a Latin hypercube over the levels), at random, evenly spaced through the list, or the first ones in the list. The *seed* fixes the choice: the same plan and seed always give the same runs. *rainfed references* adds a rainfed run for every soil and weather of the irrigated runs; untick it to get exactly the number set. The line under the form says how many runs there are, how many combinations they come from and how they were chosen; *See the runs* lists them and counts how often each value is used.

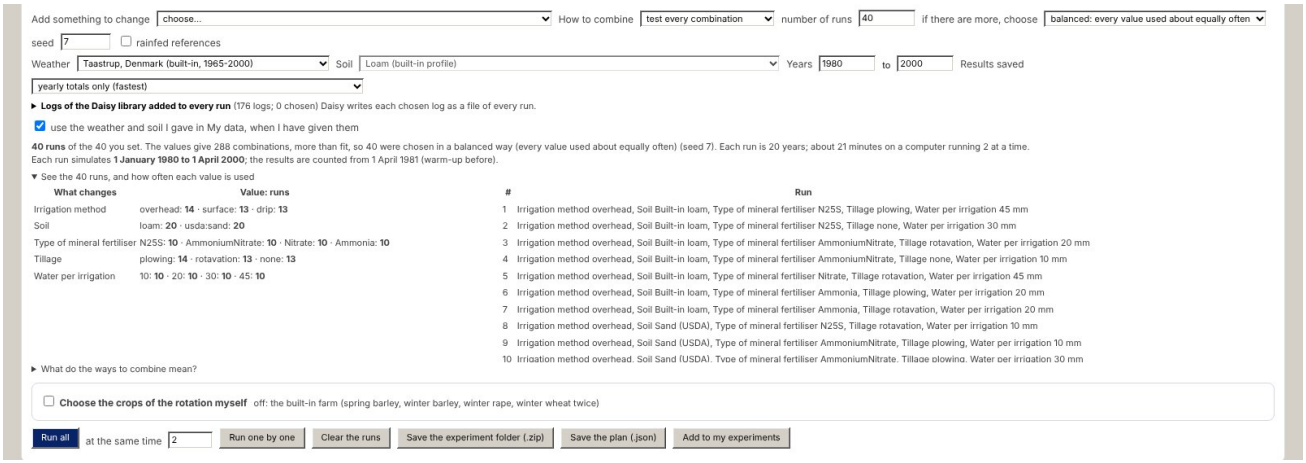



Figure 29.17: The number of runs: 40 of 288 combinations, chosen in a balanced way, with the count of every value.

### 29.2.9 Experiments: the folder before it is saved, in colour

*Save the experiment folder (.zip)* and *Save the ticked as one folder (.zip)* first show the folder: the tree on the left, the file on the right. Every file opens in an editor with the colours of VS Code (Dark+ or Light+, button *light* / *dark*): comments green, keywords blue, names of parameters light blue, numbers light green, text in quotes orange, units teal, definitions purple. Line numbers, *find* (Ctrl+F), *wrap*, the cursor position and, for a .dai file, the check of the brackets are in the bars above and below. Change the text and press *Save changes* (Ctrl+S): the zip then holds the changed file, and the file is marked *changed* in the tree. *undo all* returns to the text as it was opened, *download* saves the one file, and *open in set-up editor* opens a .dai file in the Set-up editor.

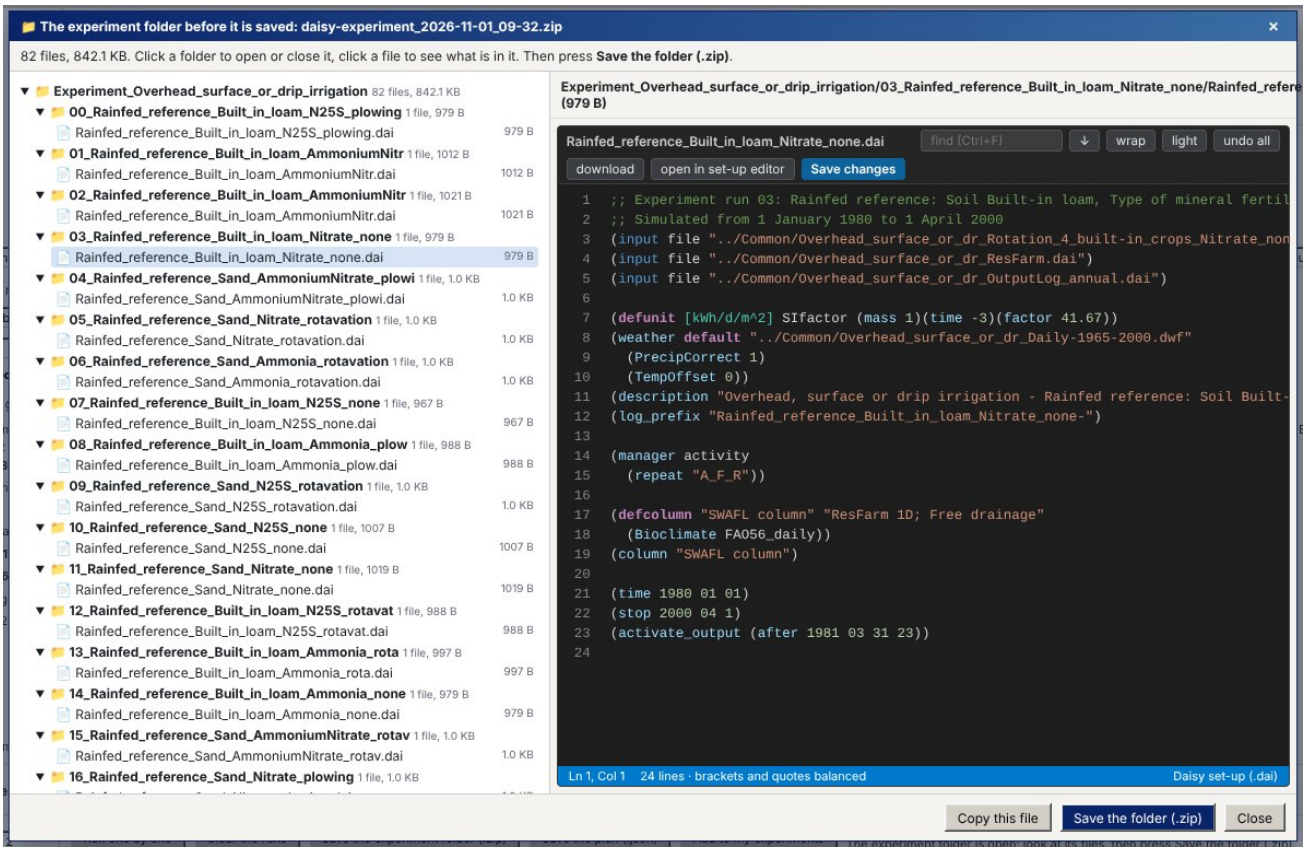



Figure 29.18: The folder of an experiment: a run file in the colours of VS Code.

**The README of the folder** says how the folder was made:

1. *HOW THE COMBINATIONS WERE MADE*: the way of combining, the number of combinations (for example 3 x 2 x 4 x 3 x 4 = 288), the number of runs set, how many were chosen and in which way, with the seed, and the rainfed references.
2. *HOW EACH .dai FILE IS MADE*: one run file per combination from the same template; the weather, soil, rotation and output files read with (input file "../Common/..."); each value written where Daisy reads it; files that several runs share written once in Common/ and named after the values they carry.
3. *WHERE EACH VALUE IS WRITTEN*: for every factor, two runs that differ in that factor only, and the lines that change between them, in the run file and inside the Common file it points to (for example (irrigate_overhead 3.333 [mm/h] (hours 3)) against (irrigate_overhead 6.667 [mm/h] (hours 3)) for 10 and 20 mm).
4. *HOW THE DATA REACH EVERY RUN*: the weather (one .dwf in Common/; rain, temperature and radiation factors as PrecipCorrect, TempOffset and GlobRadScale in the run file), the

soil, the crops, the output and the relative paths.

*runs.csv* lists every run folder with the value of every factor. A rainfed reference is named after the soil and weather it belongs to.

5. *HOW THE FOLDER AND THE CODE ARE ARRANGED*: the tree of the folder, and the run file of one run explained part by part with its line numbers: the comments, each (input file ...) and what that Common file holds, the unit of the radiation, the weather with its factors, the description and log prefix, the management, the column with the parameters the run varies, and the dates.

**How a .dai file is created and how everything is called.** Every README (of the experiment and of each run) starts with *IN SHORT*, four plain sentences, and then gives two sections in easy steps. *HOW THIS .dai FILE WAS CREATED*: (1) the values you chose, (2) the combinations and the choice with its seed, (3) the template of the farm, (4) each value of the run and where Daisy reads it, (5) the shared files written once in Common/, (6) the comments, description and log prefix. *HOW EVERYTHING IS CALLED WHEN DAISY RUNS IT*: a tree with the real names of the run, read from top to bottom:

```
daisy overhead_Built_in_loam.dai
 |-- reads ../Common/..._Rotation_..._overhead.dai
 |     |-- reads tillage.dai, crop.dai, fertilizer.dai (Daisy library)
 |   `-- defines the actions "sbarley_A_F_R" ... "irrigate_30_tensiometer_overhead", "A_F_R"
 |-- reads ../Common/..._ResFarm.dai: the horizons and the column "ResFarm 1D; Free drainage"
 |-- reads ../Common/..._OutputLog_annual.dai: the logs Annual-Harvest.dlf ...
 |-- (weather default "...dwf"): the weather of every day
 |-- (defcolumn "SWAFL column" "ResFarm 1D; Free drainage"): uses the column defined above
 |-- (manager ... (repeat "A_F_R")): calls the action "A_F_R" every year
 |-- (column "SWAFL column"): calls the column defined above
 `-- simulates hour by hour from 1980 01 01 to 2000 04 1, writes <run>-<log>.dlf
```

A name must be defined before it is called, so the files in Common/ come first.

**Every run folder has its own README_run.txt**: the values of the run (and which factors stay at the value of the farm), its place in the design (run 4 of 8, made from 6 of 6 combinations, chosen how and with which seed), its rainfed reference, the files of the folder, its run file explained line by line, the Common files it reads with what differs in them from the other runs, and how to run it alone on Linux, macOS or Windows. When several experiments are saved as one folder, *README_experiments.txt* lists every experiment with what it changes, how its values were combined and how many runs were chosen, and draws the arrangement of the whole folder. Both the folder window and the editor have a *Full screen* button (Esc leaves it).

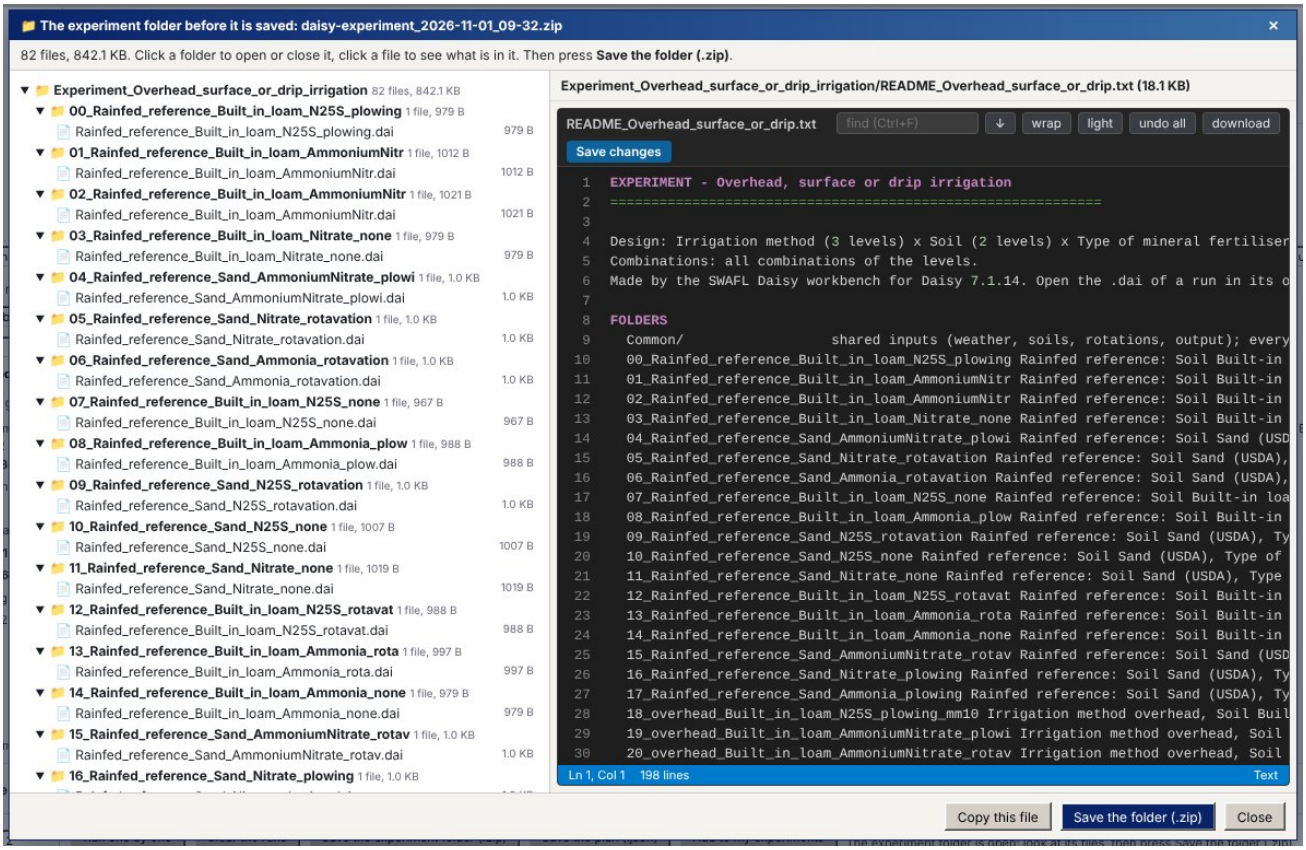



Figure 29.19: The README in the light colours.

## 29.2.10 Experiments: weather, soil and years of the experiment

The weather list of an experiment follows the choice at the top of the tab and offers the built-in stations as well. The soil, the years and what is saved are set beside it.

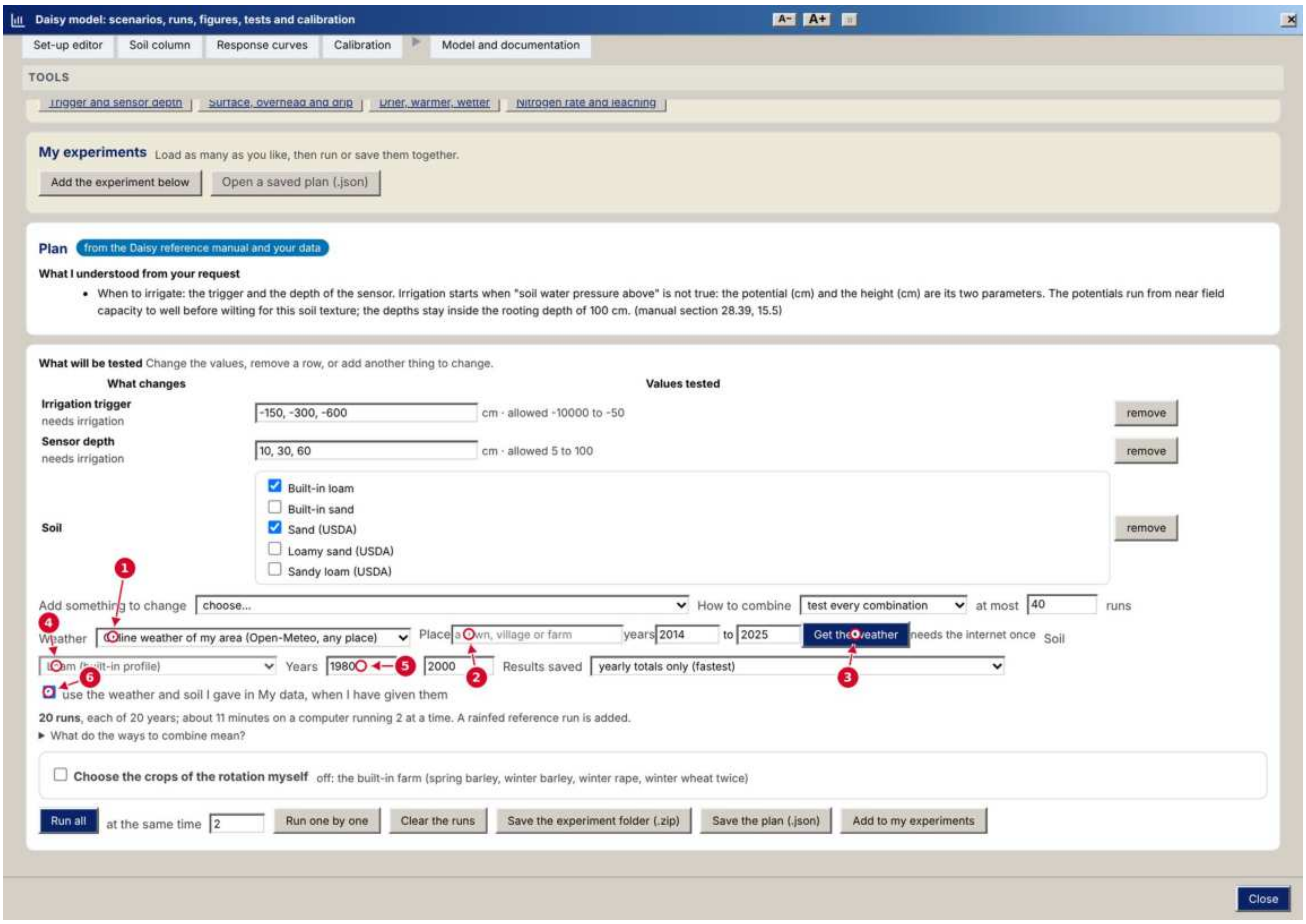



Figure 29.20: Experiments: weather, soil and years of the experiment. Screen 28 of the Daisy workbench.

**What the arrows point to.**

1. **Weather.** Online weather of my area, the weather of the SWAFL project, my weather, or a built-in station.
2. **Place.** The place whose weather is read from the internet.
3. **Get the weather.** Reads and checks the weather of the place.
4. **Soil.** The soil of the experiment when it is not a factor.
5. **Years.** First and last year of every run.
6. **Use my data.** Uses the weather and soil given in Input data when they are set.

### 29.2.11 Experiments: my experiments, ordered and ticked

Experiments are added to a list in the order of preference. Each has its own name, design and largest number of runs; only the ticked ones are run, from the top down, or saved together in one folder.

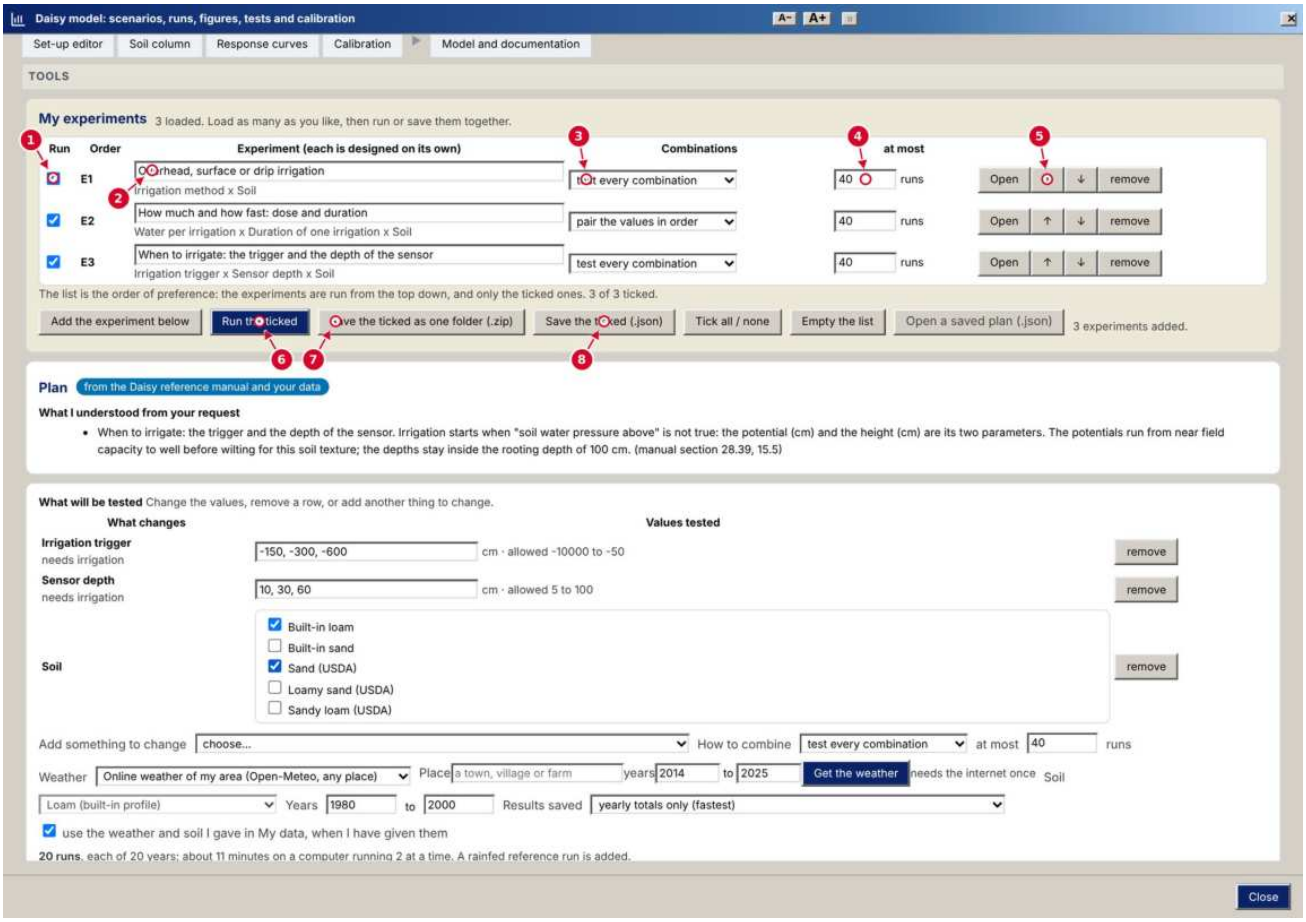



Figure 29.21: Experiments: my experiments, ordered and ticked. Screen 29 of the Daisy workbench.

**What the arrows point to.**

1. **Run.** Ticks the experiments that are run or saved.
2. **Name.** The name of the experiment; it becomes the name of its folder.
3. **Combinations.** The design of this experiment on its own.
4. **At most.** The largest number of runs of this experiment.
5. **Move up.** Moves the experiment earlier in the list.
6. **Run the ticked.** Runs the ticked experiments one after another.
7. **Save as one folder.** Saves the ticked experiments as E1, E2 and so on in one zip.
8. **Save as plan.** Saves the ticked plans as one JSON file.

# Chapter 30

# ⚖ The best combinations

Start from the Daisy reference offers 27 combinations of factors that the Daisy reference manual makes worth comparing. Each is shown as it opens in the Experiments tab, with the reason it is offered and the place of the reference manual where its parameters are described.

### 30.0.1 Overhead, surface or drip irrigation

The manual has three ways to apply water: over the canopy, on the soil surface, and into a soil volume. Run the three with the same trigger and dose, so only the way of applying water differs.

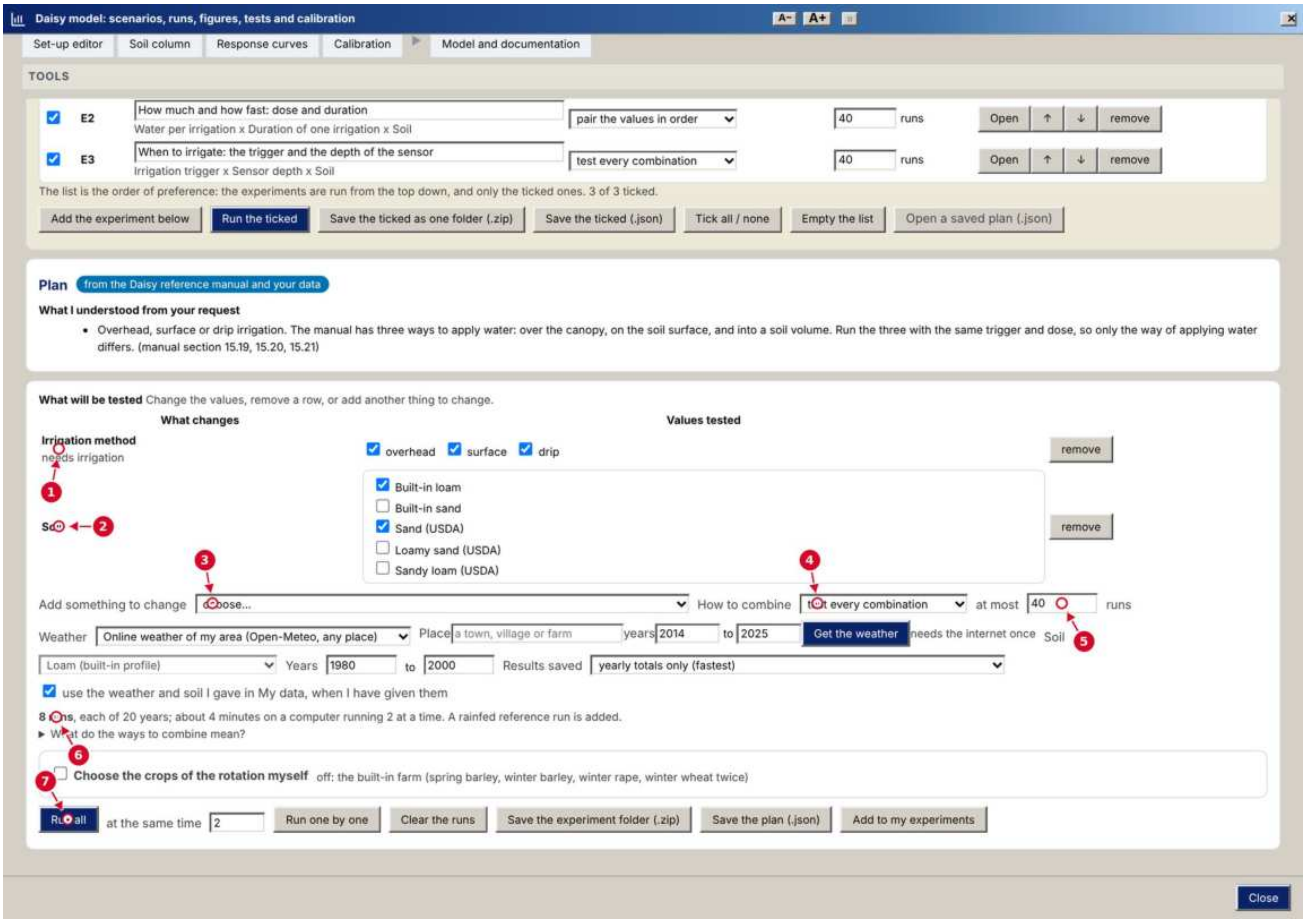



Figure 30.1: Overhead, surface or drip irrigation. Screen 30 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation method.** Varied over overhead, surface, drip.
2. **Soil.** Varied over usda:sand, loam.
3. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
4. **How to combine.** Every combination, one thing at a time, or the values paired in order.
5. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
6. **Size of the experiment.** The number of runs and the time they take on this computer.
7. **Run all.** Runs every combination, several at a time.

### 30.0.2 Irrigation and nitrogen together

Water moves nitrate and decides how much of the fertiliser the crop can use. Irrigation against rainfed, at three fertiliser doses, shows whether more water raises the yield only when the nitrogen is there, and what it costs in nitrate below the roots.

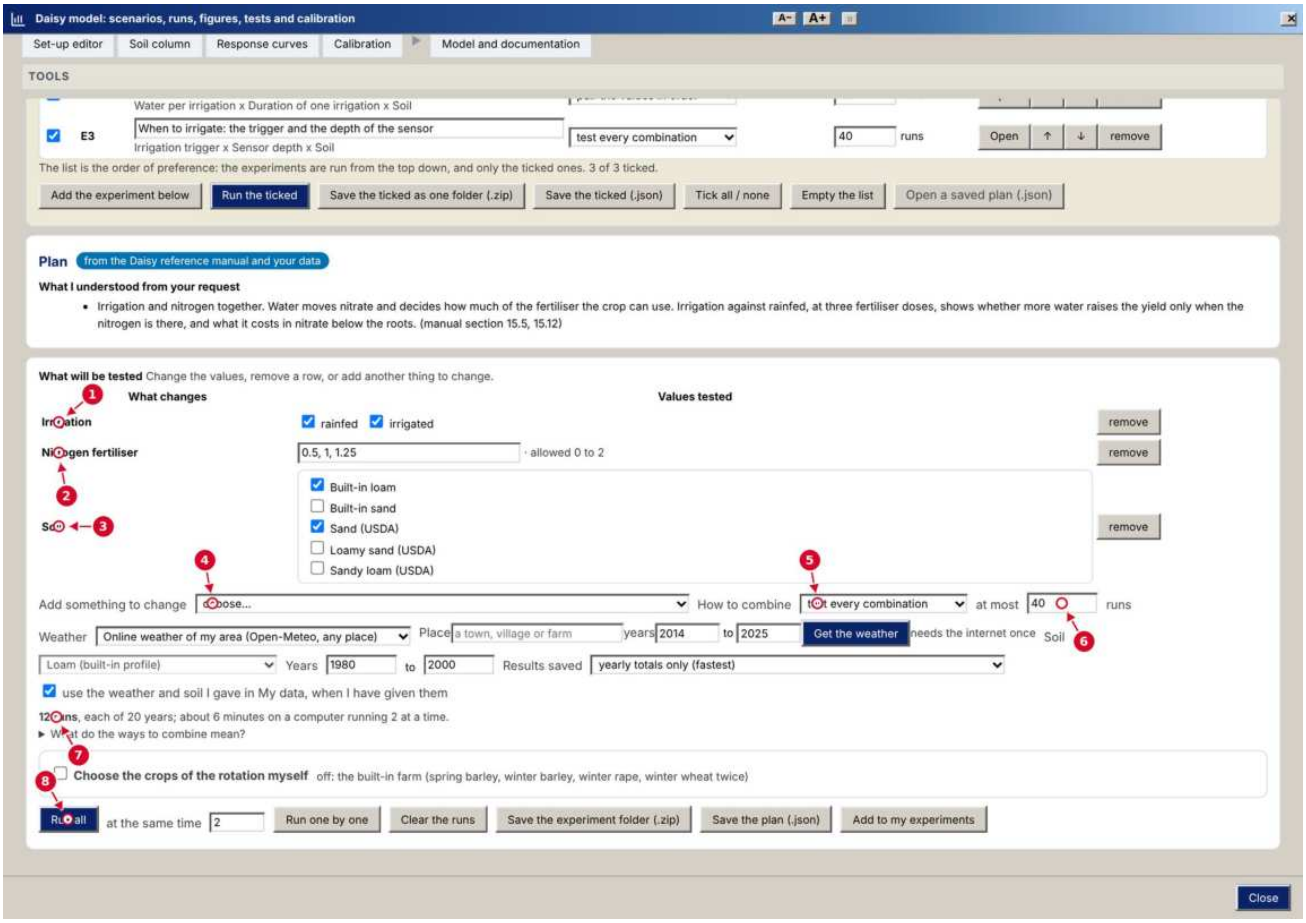



Figure 30.2: Irrigation and nitrogen together. Screen 31 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation.** Varied over 0, 1.
2. **Nitrogen fertiliser, factor on the rotation.** Varied over 0.5, 1, 1.25.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.3 When to irrigate: the trigger and the depth of the sensor

Irrigation starts when “soil water pressure above” is not true: the potential (cm) and the height (cm) are its two parameters. The potentials run from near field capacity to well before wilting for this soil texture; the depths stay inside the rooting depth of 100 cm.

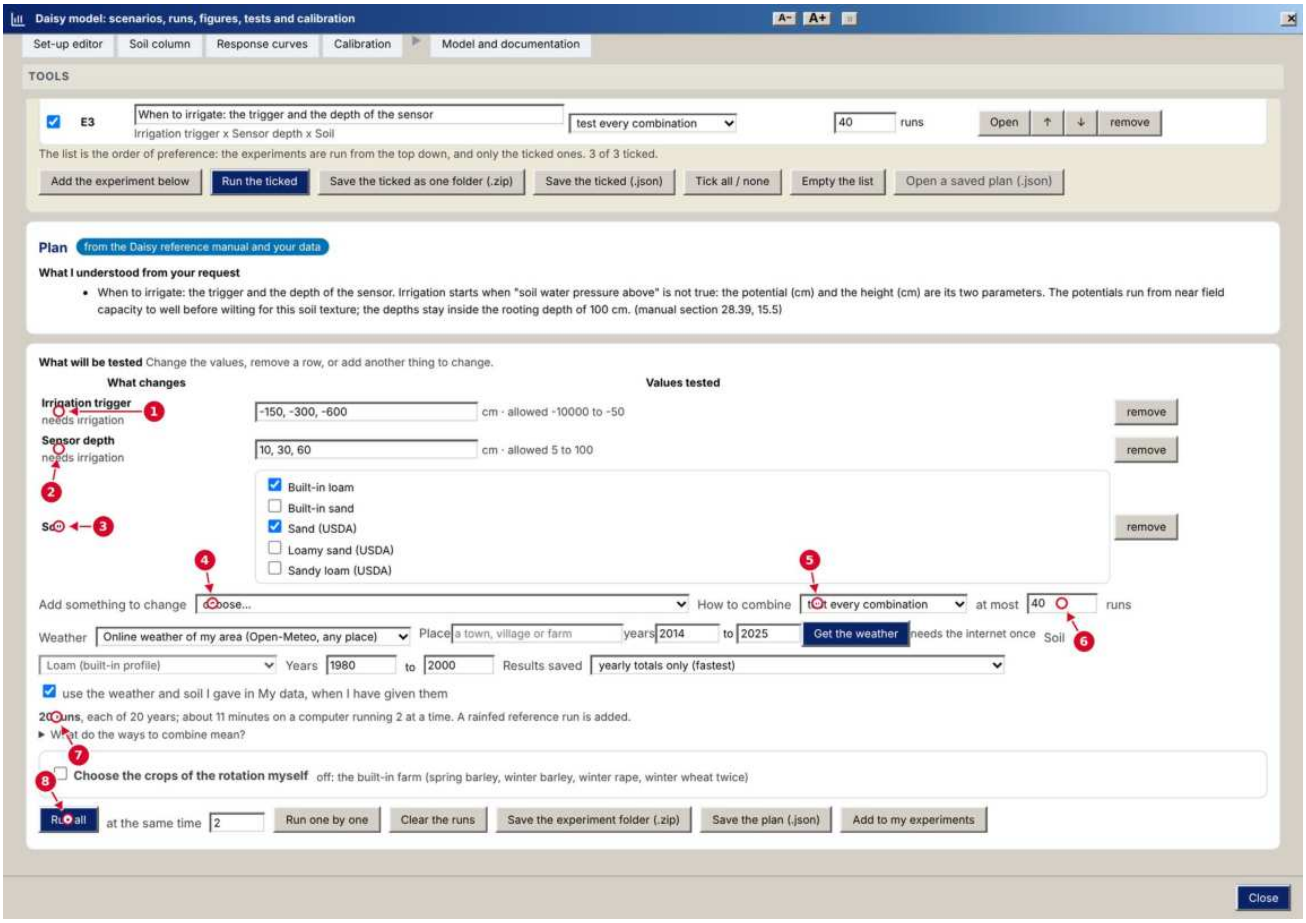



Figure 30.3: When to irrigate: the trigger and the depth of the sensor. Screen 32 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation trigger (soil suction).** Varied over -150, -300, -600.
2. **Sensor depth.** Varied over 10, 30, 60.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.4 How much and how fast: dose and duration

The water application speed is a parameter in mm/h (default 2 mm/h) and the irrigation lasts a number of hours. The pairs below keep the speed of the manual (2 mm/h) while the dose grows, so the answer is the effect of the dose and not of the speed.

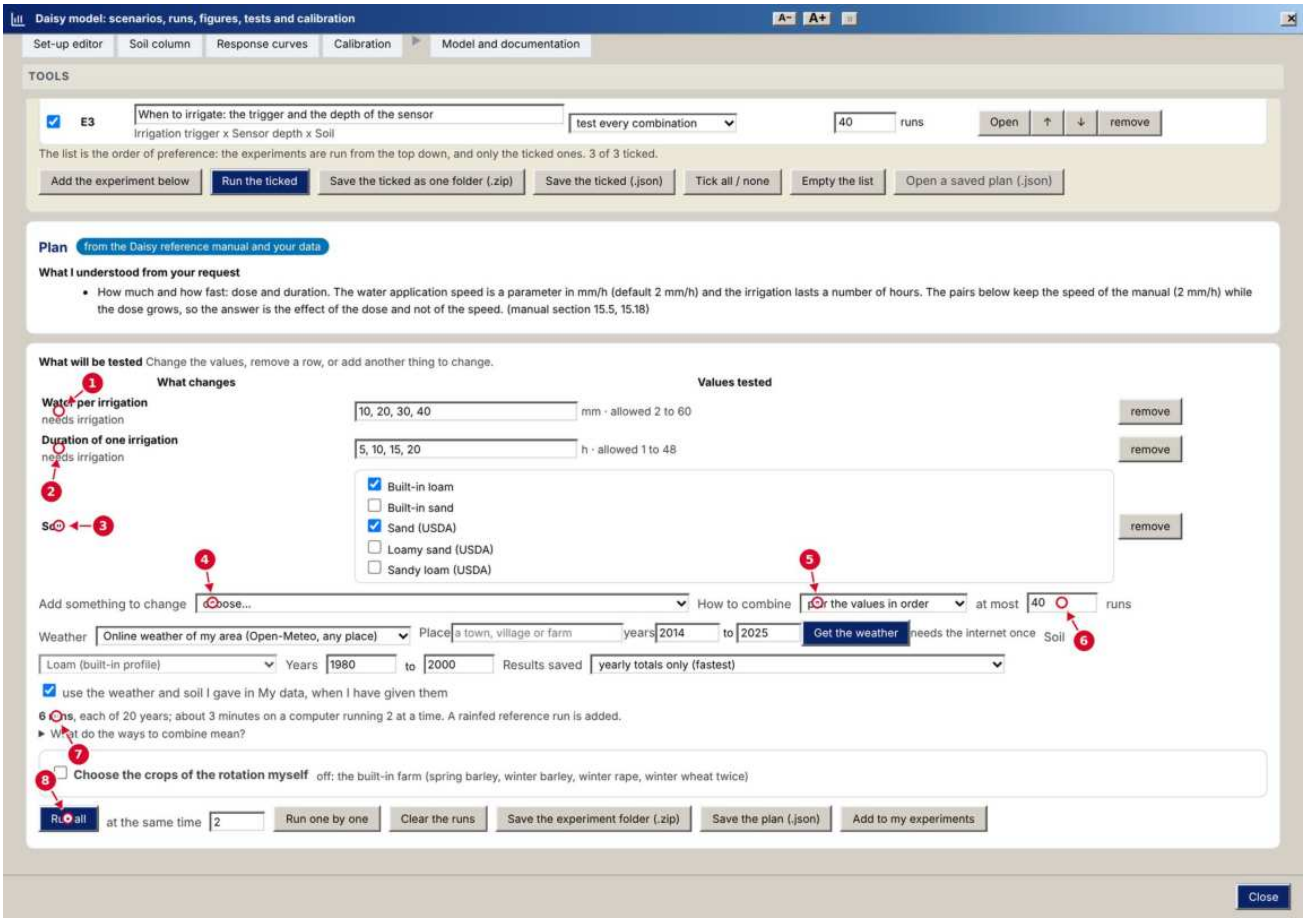



Figure 30.4: How much and how fast: dose and duration. Screen 33 of the Daisy workbench.

**What the arrows point to.**

1. **Water per irrigation.** Varied over 10, 20, 30, 40.
2. **Duration of one irrigation.** Varied over 5, 10, 15, 20.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.5 Nitrogen fertiliser, irrigated and rainfed

Fertiliser is applied as an amount in kg N/ha. Scaling the rotation between half and one and a quarter of its dose, with and without irrigation, shows the dose where yield stops rising and the nitrate leached below the roots starts to climb.

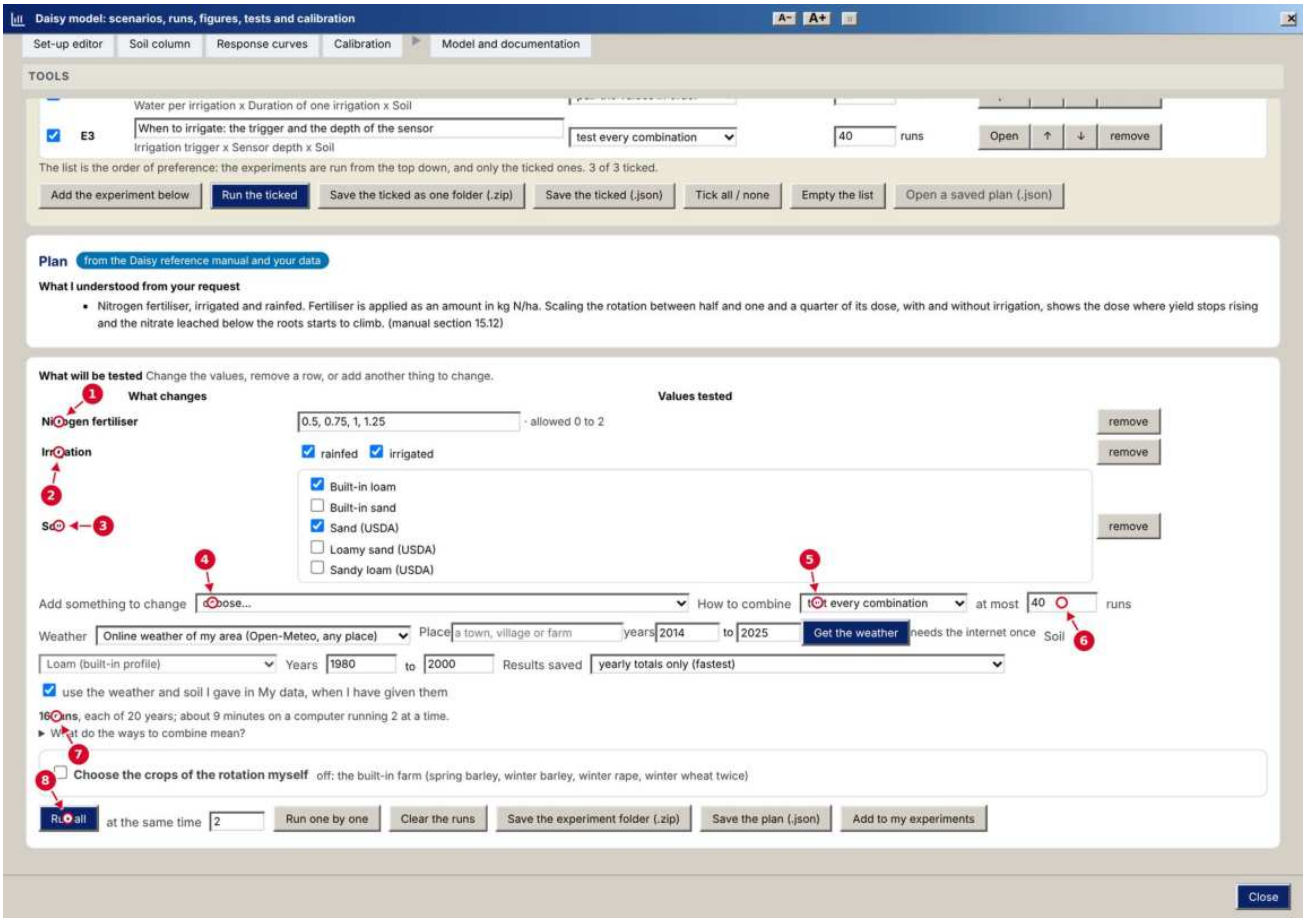



Figure 30.5: Nitrogen fertiliser, irrigated and rainfed. Screen 34 of the Daisy workbench.

**What the arrows point to.**

1. **Nitrogen fertiliser, factor on the rotation.** Varied over 0.5, 0.75, 1, 1.25.
2. **Irrigation.** Varied over 0, 1.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.6 A drier, wetter or warmer year

The weather file gives precipitation and air temperature to every model of the column. Scaling the rain and adding a warming to the same record shows how much the yield and the water use move with the weather, with the soil and the management unchanged.

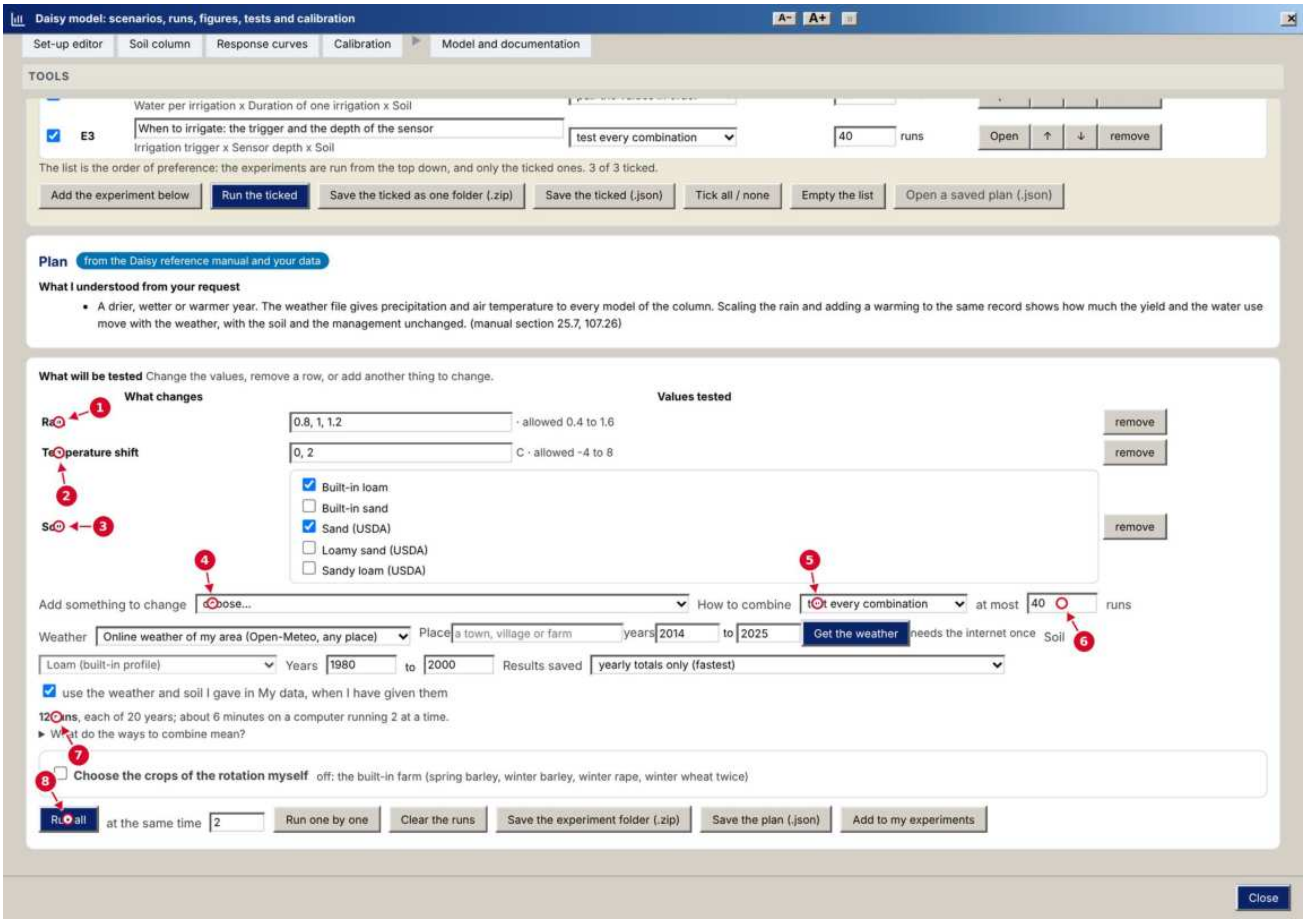



Figure 30.6: A drier, wetter or warmer year. Screen 35 of the Daisy workbench.

**What the arrows point to.**

1. **Rain, factor on the weather.** Varied over 0.8, 1, 1.2.
2. **Temperature shift.** Varied over 0, 2.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.7 The irrigation season and a limit on water

The irrigation of a crop is tested between a from date and a to date and can be given an amount. Start dates and a seasonal limit show how much of the yield is bought by the first and the last irrigations.

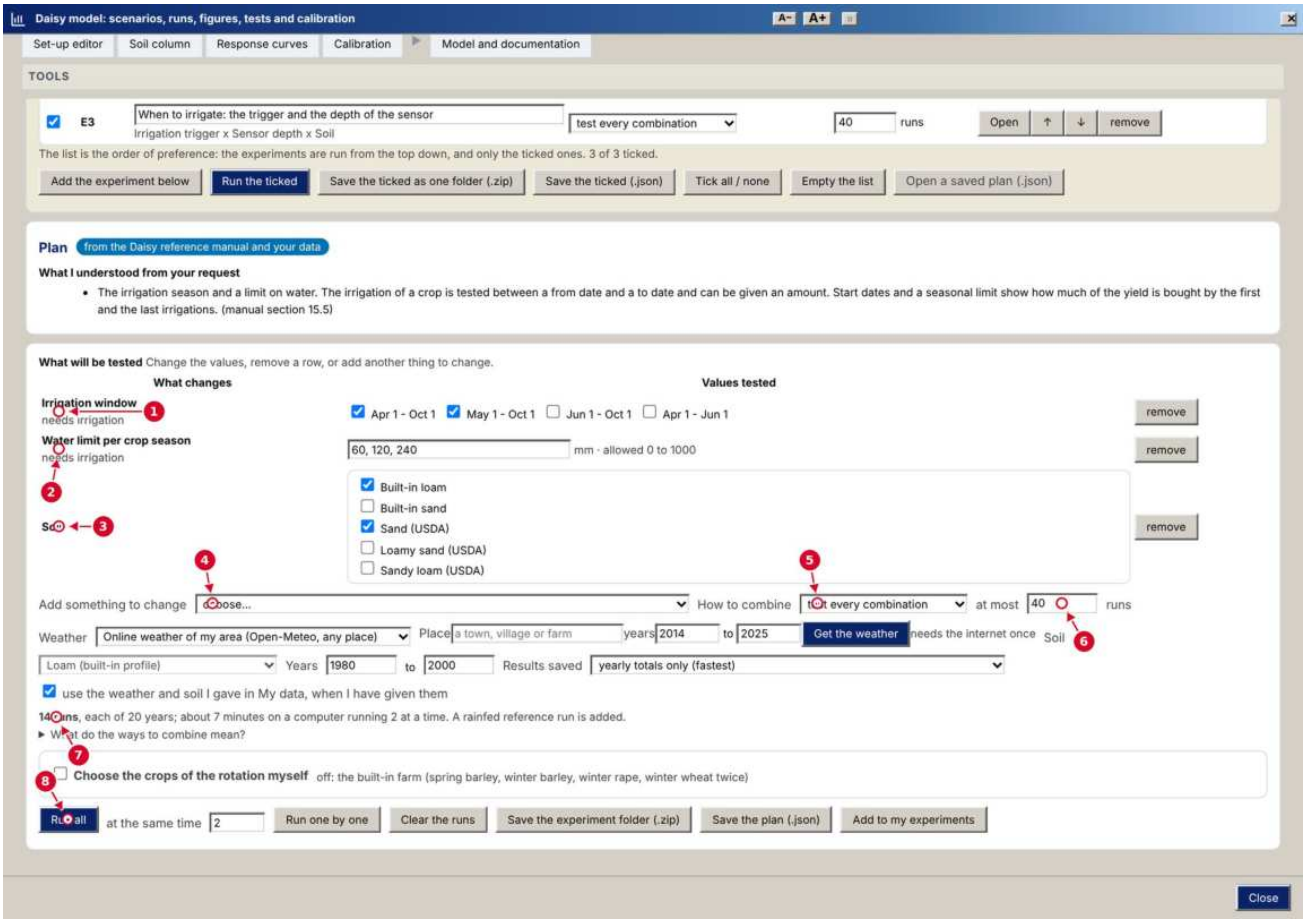



Figure 30.7: The irrigation season and a limit on water. Screen 36 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation window.** Varied over Apr 1 minus Oct 1, May 1 minus Oct 1, Jun 1 minus Sep 1.
2. **Water limit per crop season.** Varied over 60, 120, 240.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.8 Which soil needs the most water

A horizon gets its hydraulic properties from the hypres model by default, from the sand, silt, clay, humus and bulk density. Rainfed and irrigated runs on several textures show where irrigation pays.

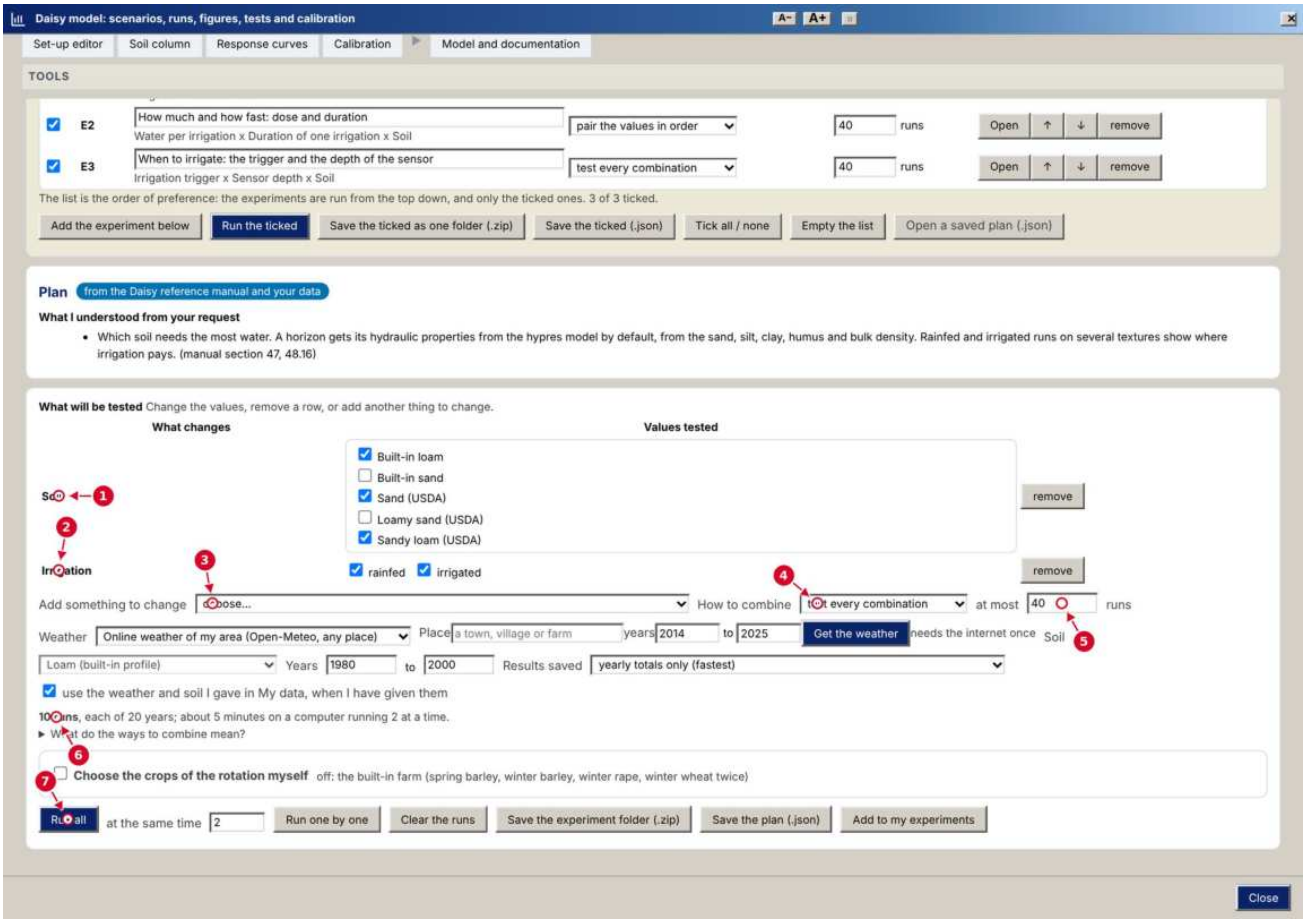



Figure 30.8: Which soil needs the most water. Screen 37 of the Daisy workbench.

**What the arrows point to.**

1. **Soil.** Varied over usda:sand, usda:sandyloam, loam, usda:siltloam, usda:clayloam.
2. **Irrigation.** Varied over 0, 1.
3. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
4. **How to combine.** Every combination, one thing at a time, or the values paired in order.
5. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
6. **Size of the experiment.** The number of runs and the time they take on this computer.
7. **Run all.** Runs every combination, several at a time.

### 30.0.9 Slow or fast irrigation: a constant water flux

Daisy applies irrigation at a rate in mm/h, so a slow rate keeps the water in the topsoil and a fast one can run off or percolate. The rates below keep the same dose per irrigation, so only the speed of application differs.

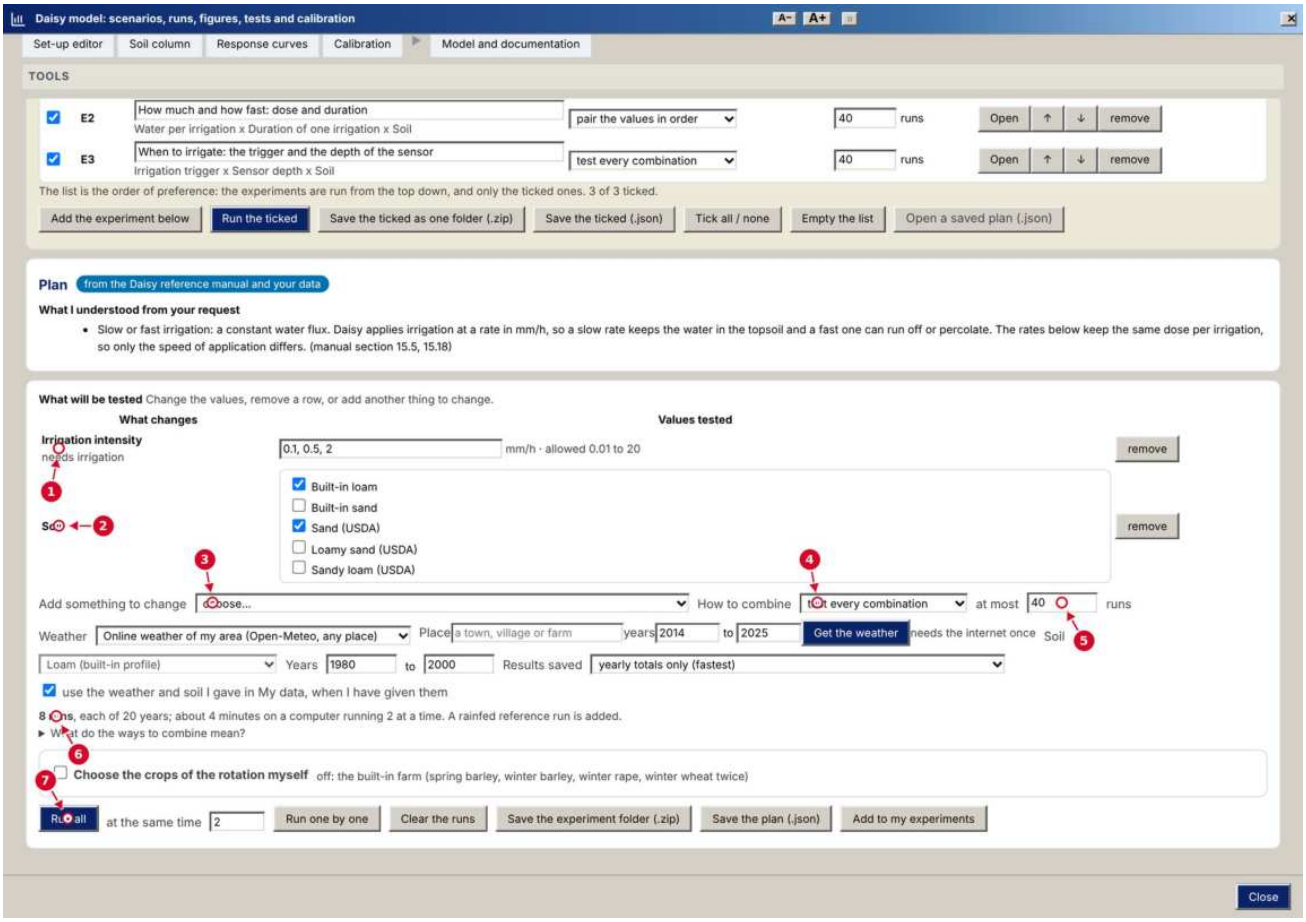



Figure 30.9: Slow or fast irrigation: a constant water flux. Screen 38 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation intensity (constant flux).** Varied over 0.1, 0.5, 2.
2. **Soil.** Varied over usda:sand, loam.
3. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
4. **How to combine.** Every combination, one thing at a time, or the values paired in order.
5. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
6. **Size of the experiment.** The number of runs and the time they take on this computer.
7. **Run all.** Runs every combination, several at a time.

### 30.0.10 Sensor depth against rooting depth

The tensiometer that triggers irrigation sits at one depth, while the roots take water from the whole root zone. A sensor above or below most of the roots triggers too early or too late; crossing the sensor depth with the rooting depth shows where it matters.

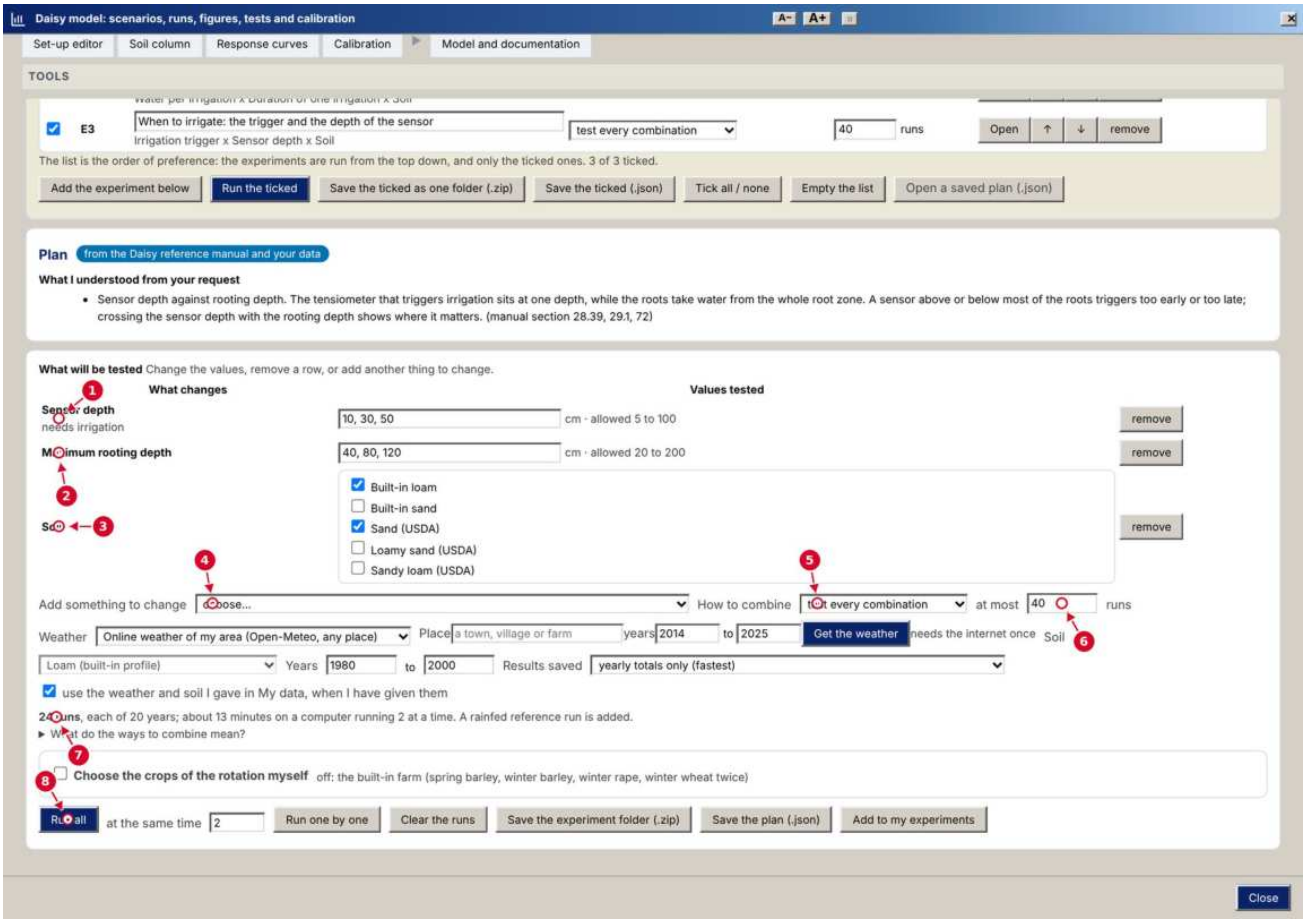



Figure 30.10: Sensor depth against rooting depth. Screen 39 of the Daisy workbench.

**What the arrows point to.**

1. **Sensor depth.** Varied over 10, 30, 50.
2. **Maximum rooting depth.** Varied over 40, 80, 120.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.11 When the fertiliser is applied and how much

The fertilize action is placed on a date. Moving it two weeks earlier or later, at three doses, shows how much nitrate is exposed to the rain before the crop takes it up.

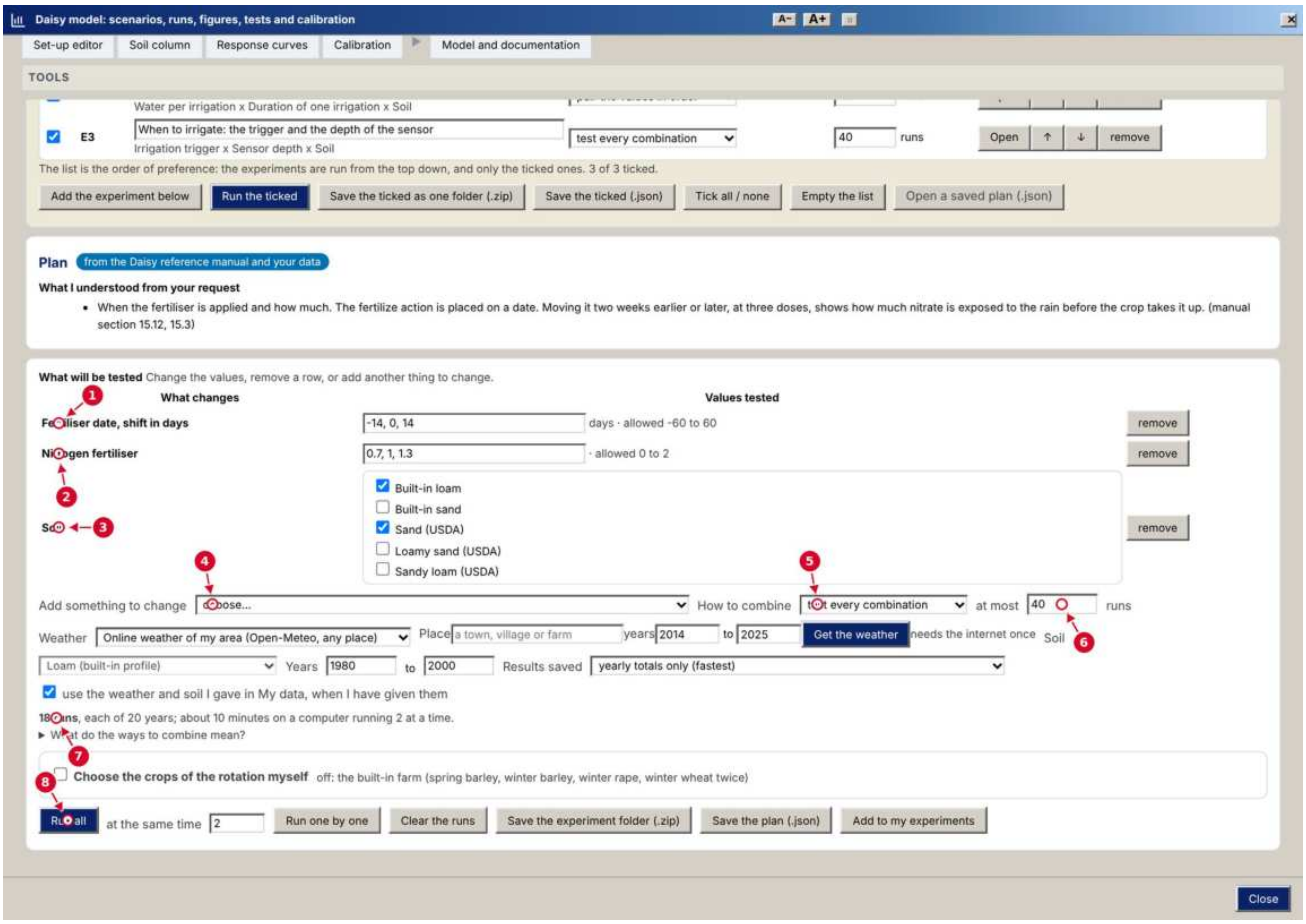



Figure 30.11: When the fertiliser is applied and how much. Screen 40 of the Daisy workbench.

**What the arrows point to.**

1. **Fertiliser date, shift in days.** Varied over -14, 0, 14.
2. **Nitrogen fertiliser, factor on the rotation.** Varied over 0.7, 1, 1.3.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.12 Warming with and without irrigation

A warmer year raises the evapotranspiration and shortens the crop cycle. Warming of 0 to 3 degrees, rainfed and irrigated, shows how much of the loss irrigation can give back.

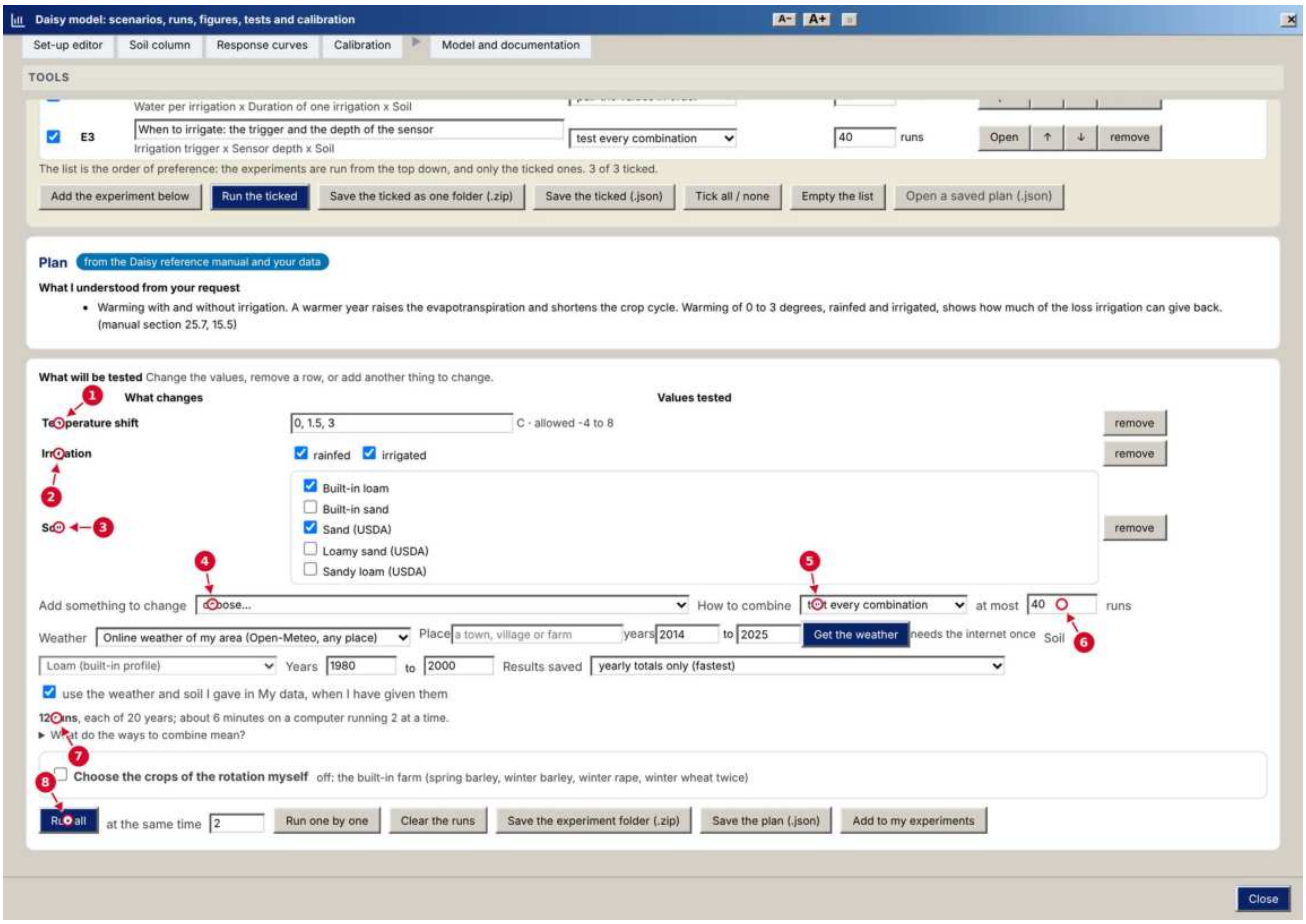



Figure 30.12: Warming with and without irrigation. Screen 41 of the Daisy workbench.

**What the arrows point to.**

1. **Temperature shift.** Varied over 0, 1.5, 3.
2. **Irrigation.** Varied over 0, 1.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.13 Tillage and the time and form of the fertiliser

The tillage action (plowing, rotavation, or none) mixes the residues and the topsoil; the fertiliser is taken from the fertilizer library by name, and the date can be moved. These three decide how much nitrate is exposed to leaching.

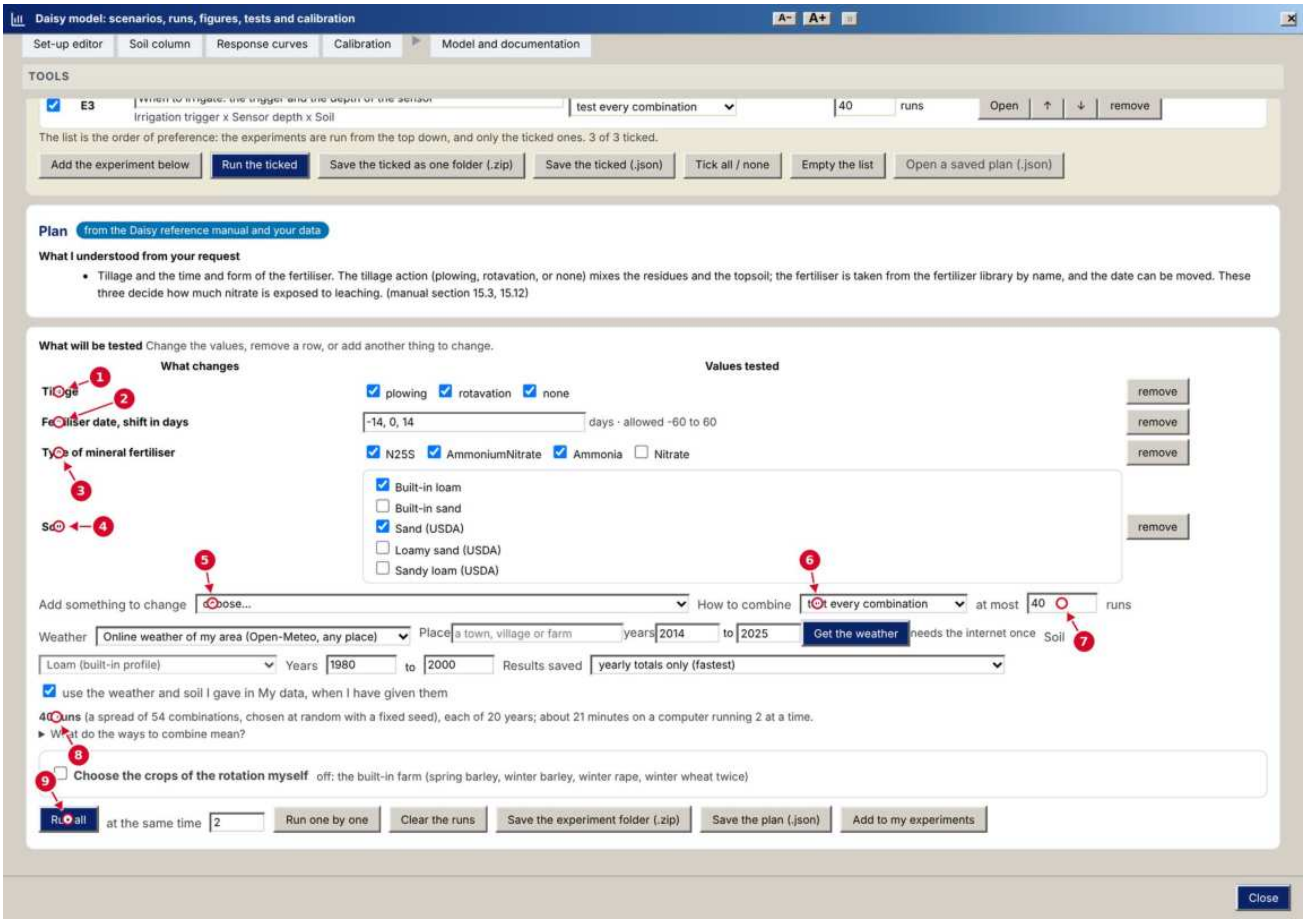



Figure 30.13: Tillage and the time and form of the fertiliser. Screen 42 of the Daisy workbench.

**What the arrows point to.**

1. **Tillage.** Varied over plowing, rotavation, none.
2. **Fertiliser date, shift in days.** Varied over -14, 0, 14.
3. **Type of mineral fertiliser.** Varied over N25S, AmmoniumNitrate, Ammonia.
4. **Soil.** Varied over usda:sand, loam.
5. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
6. **How to combine.** Every combination, one thing at a time, or the values paired in order.
7. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
8. **Size of the experiment.** The number of runs and the time they take on this computer.
9. **Run all.** Runs every combination, several at a time.

### 30.0.14 Evapotranspiration model and radiation

Daisy can take the reference evapotranspiration from the weather file (FAO56) or compute it from the energy balance, and the radiation can be scaled. The runs show how much of the answer depends on that choice.

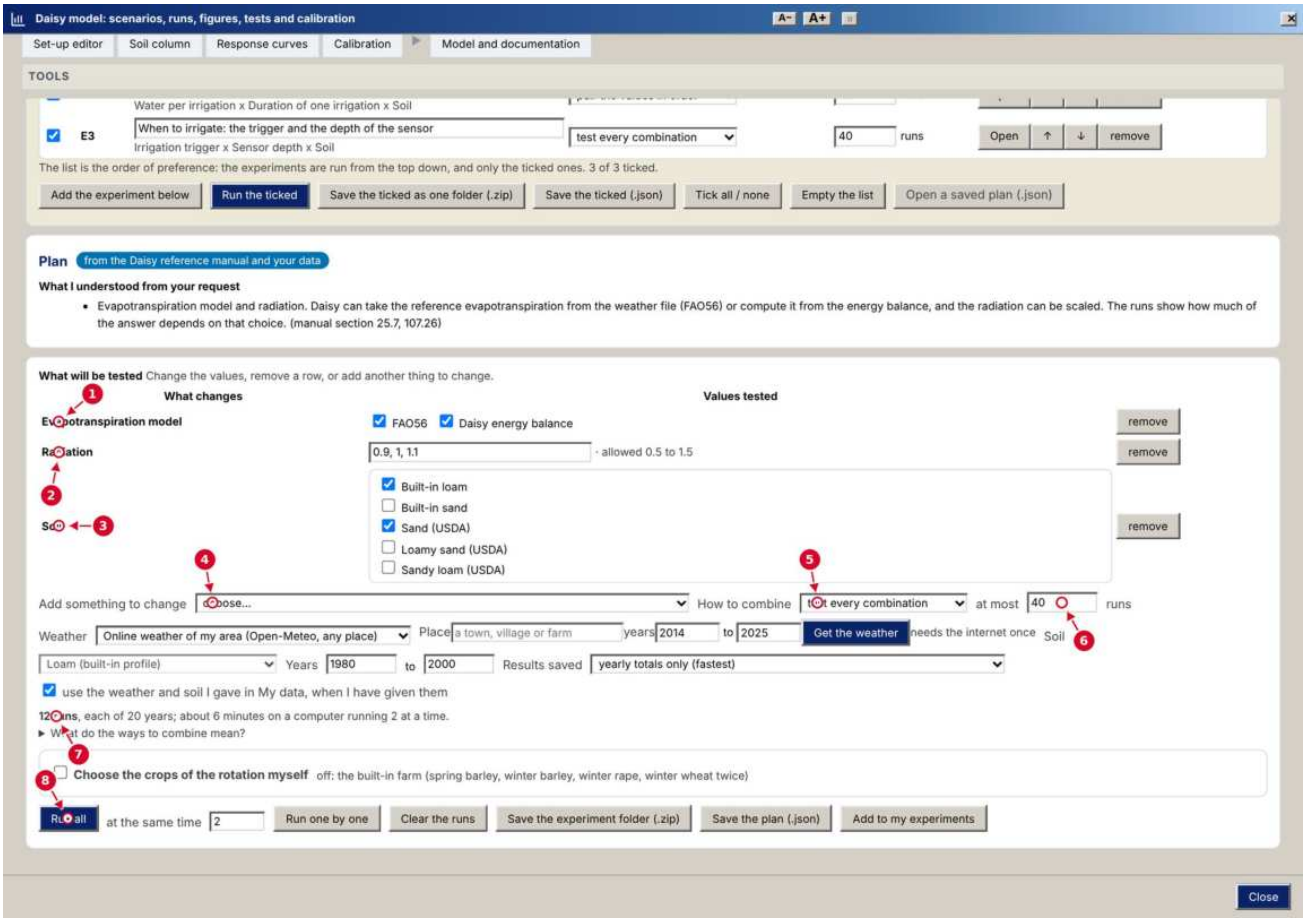



Figure 30.14: Evapotranspiration model and radiation. Screen 43 of the Daisy workbench.

**What the arrows point to.**

1. **Evapotranspiration model.** Varied over auto, default.
2. **Radiation, factor on the weather.** Varied over 0.9, 1, 1.1.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.15 A dry summer, a dry spring or a wet winter

The weather component can scale the precipitation month by month (twelve factors). Taking 30 percent of the rain from the summer or the spring, or adding 30 percent in the winter, shows which part of the year matters for yield and for leaching.

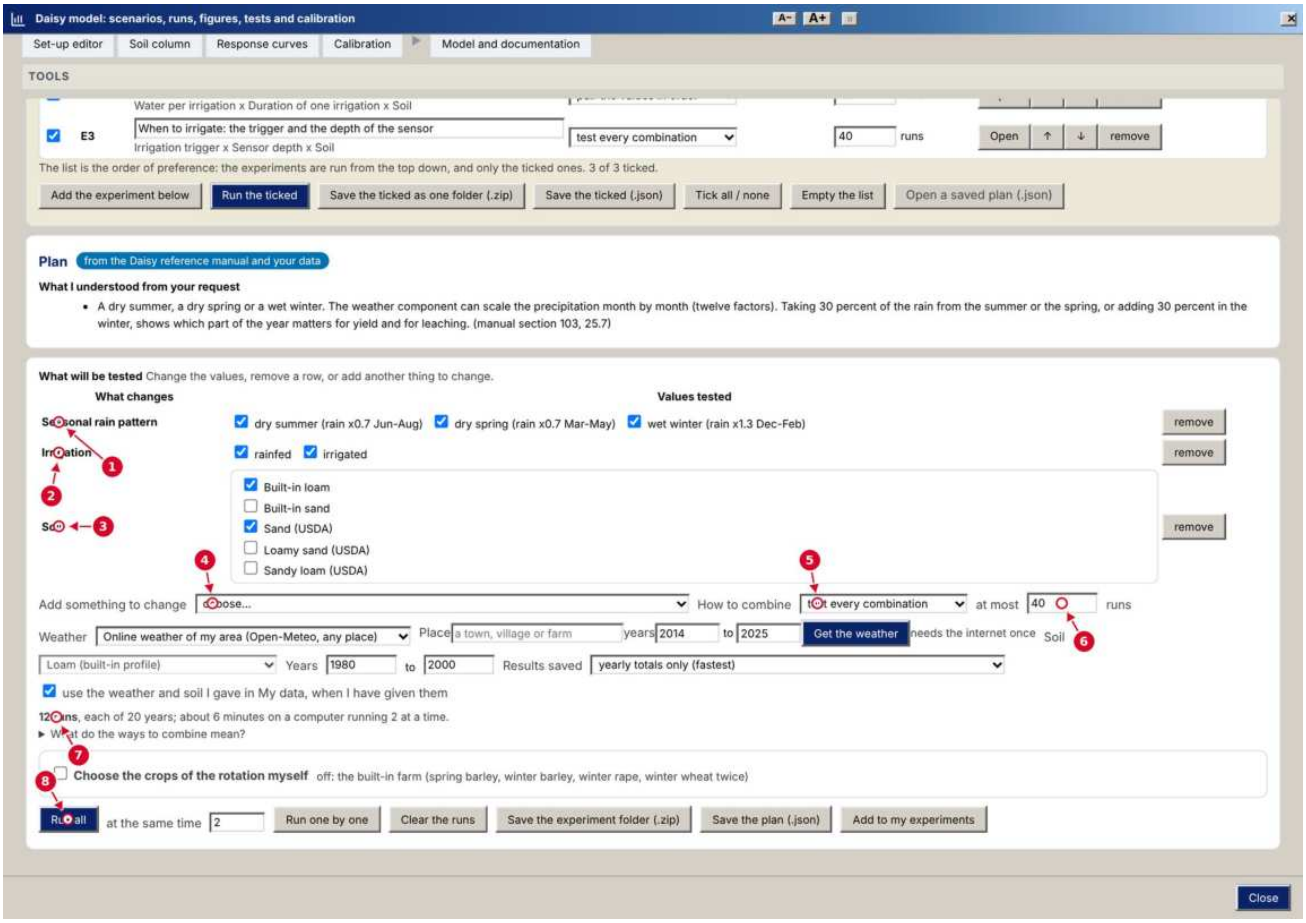



Figure 30.15: A dry summer, a dry spring or a wet winter. Screen 44 of the Daisy workbench.

**What the arrows point to.**

1. **Seasonal rain pattern.** Varied over dry summer (rain x0.7 Jun-Aug), dry spring (rain x0.7 Mar-May), wet winter (rain x1.3 Dec-Feb).
2. **Irrigation.** Varied over 0, 1.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.16 Sorption and half-life of a chemical

A chemical has a sorption coefficient (K_OC, L/kg) and a decomposition rate; the two decide whether it stays in the topsoil or reaches the depth where it is counted as leached.

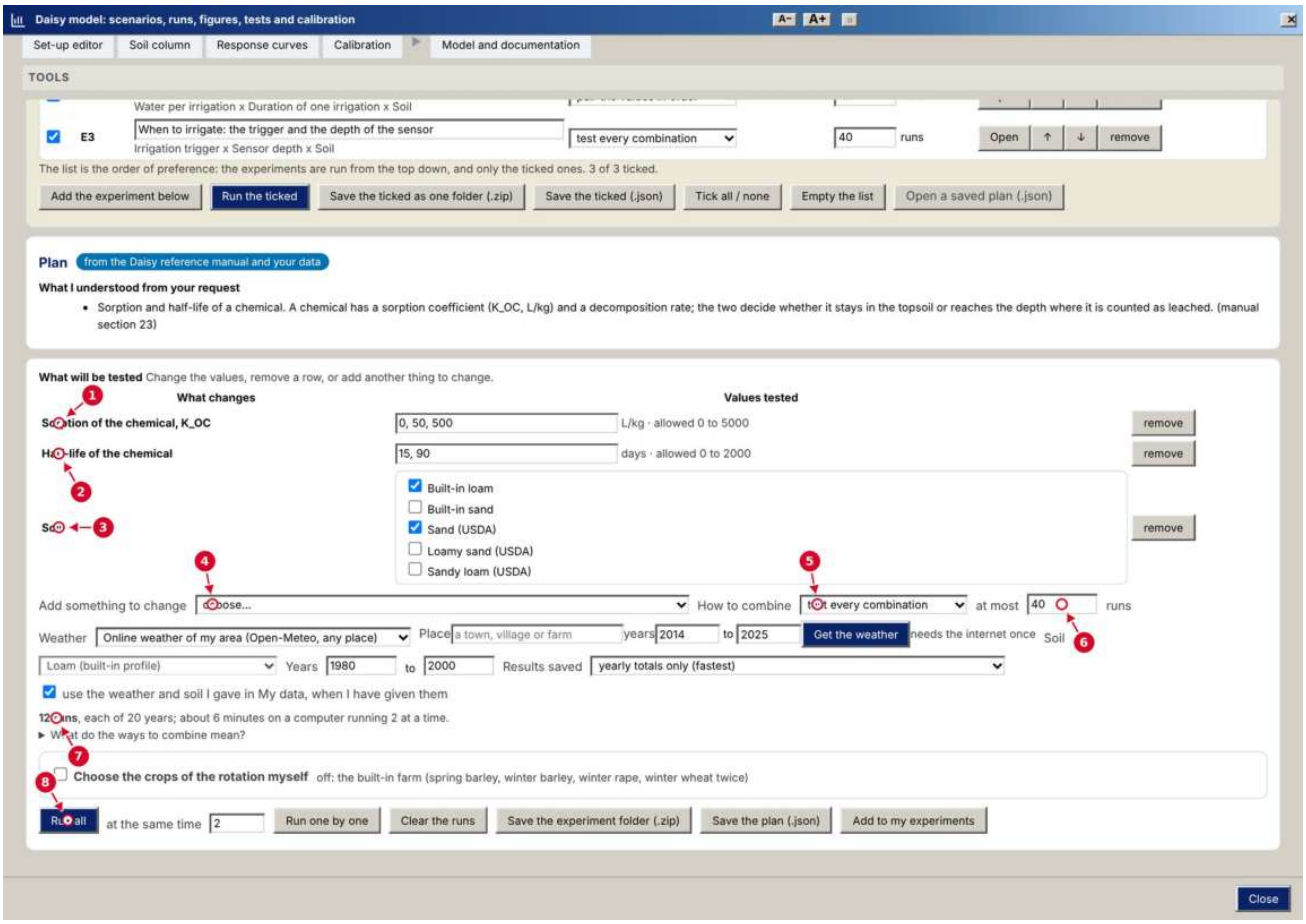



Figure 30.16: Sorption and half-life of a chemical. Screen 45 of the Daisy workbench.

**What the arrows point to.**

1. **Sorption of the chemical, K_OC.** Varied over 0, 50, 500.
2. **Half-life of the chemical.** Varied over 15, 90.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.17 Deficit irrigation: a water limit against the trigger

A crop can be given a limited amount of water in the season, and irrigation starts only when the soil water pressure is below the trigger. A strict trigger with a small limit, against a lax trigger with a large limit, shows whether the same water gives more yield when it is saved for the dry spell.

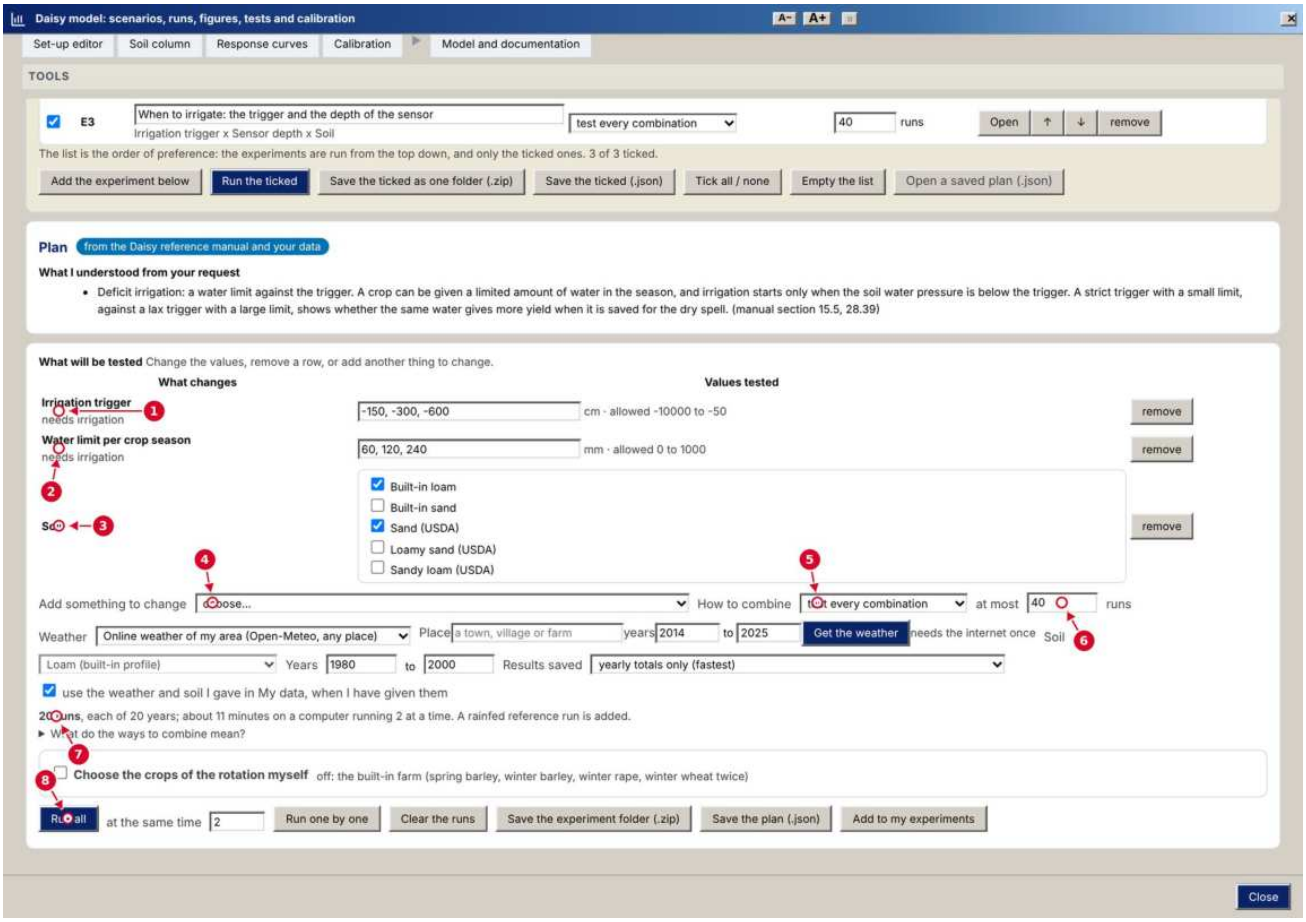



Figure 30.17: Deficit irrigation: a water limit against the trigger. Screen 46 of the Daisy workbench.

**What the arrows point to.**

1. **Irrigation trigger (soil suction).** Varied over -150, -300, -600.
2. **Water limit per crop season.** Varied over 60, 120, 240.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.18 Rooting depth of the crop

The crop model has a maximum rooting depth, which decides how much of the soil water and nitrate the roots can reach. Shallow to deep roots, rainfed and irrigated, show how much of a water shortage a deeper root zone absorbs.

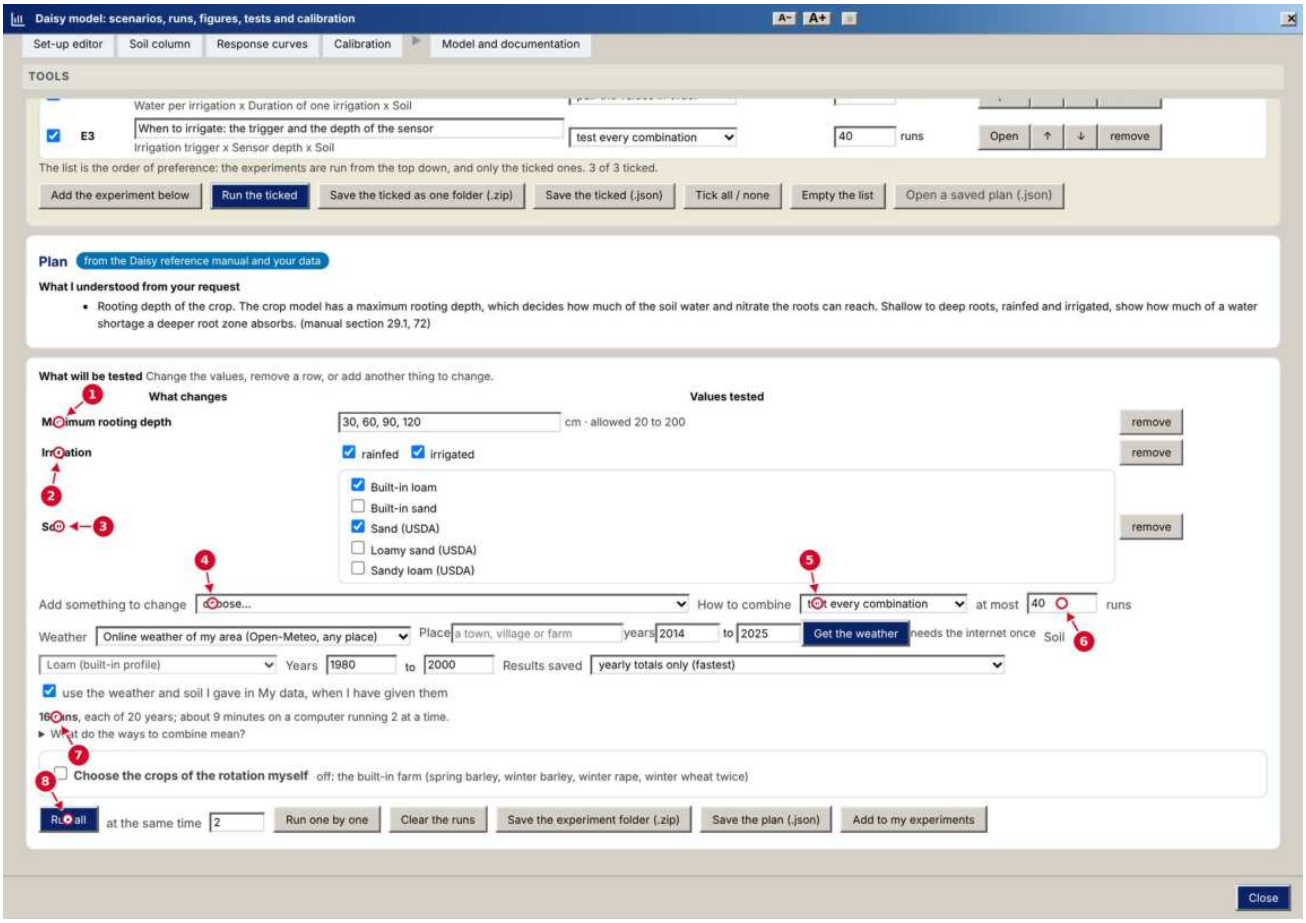



Figure 30.18: Rooting depth of the crop. Screen 47 of the Daisy workbench.

**What the arrows point to.**

1. **Maximum rooting depth.** Varied over 30, 60, 90, 120.
2. **Irrigation.** Varied over 0, 1.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.19 The same farm under other climates

The weather component reads a file for one place. Running the same soil, crops and management with the weather of several places shows how much of the result belongs to the climate.

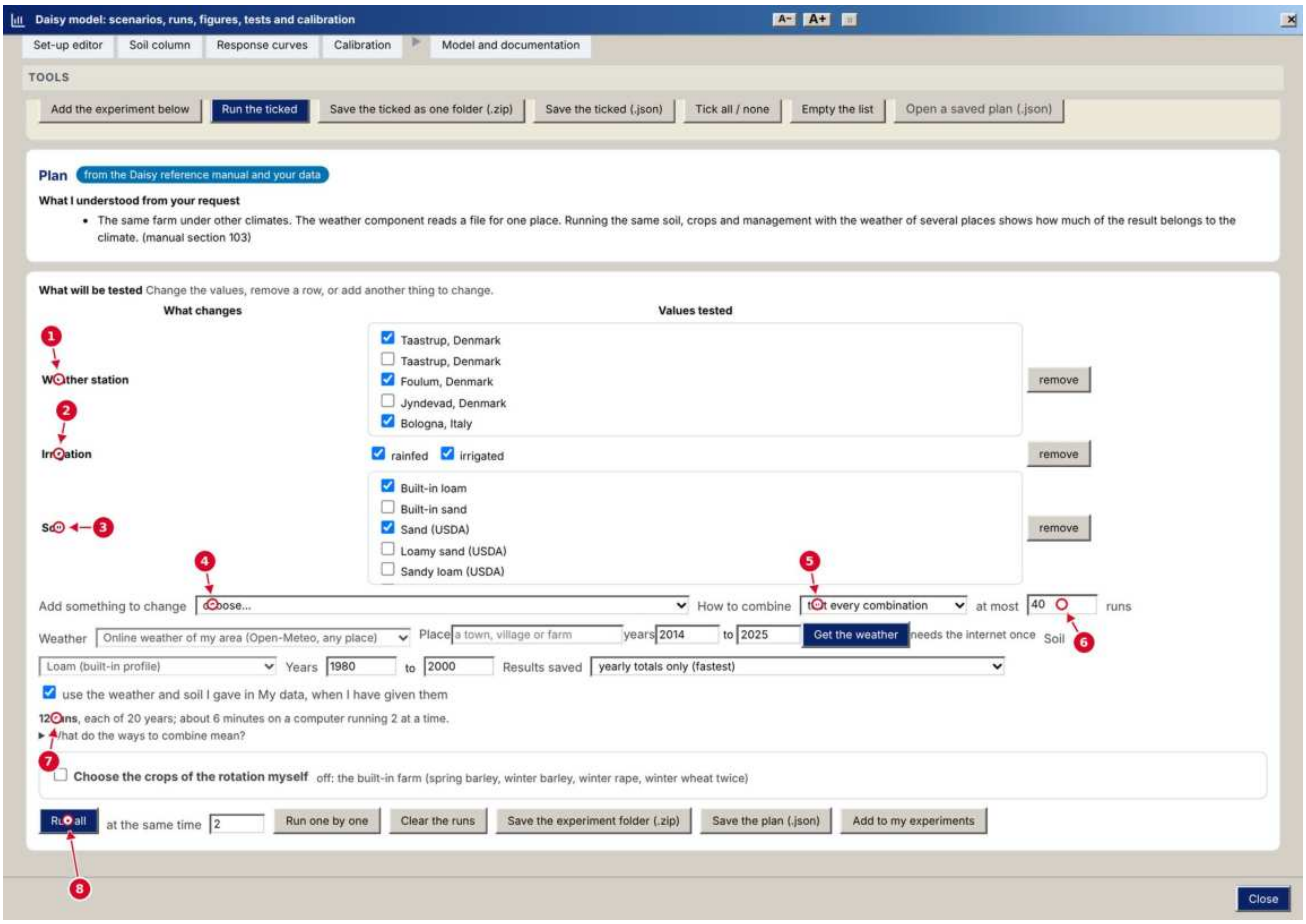



Figure 30.19: The same farm under other climates. Screen 48 of the Daisy workbench.

**What the arrows point to.**

1. **Weather station.** Varied over course, it-bologna, dk-foulum.
2. **Irrigation.** Varied over 0, 1.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.20 Tillage against fertiliser form

The tillage action mixes residues and topsoil; the fertilizer library holds several forms of mineral nitrogen. Crossing the two shows whether the form of the nitrogen matters more under one tillage than another.

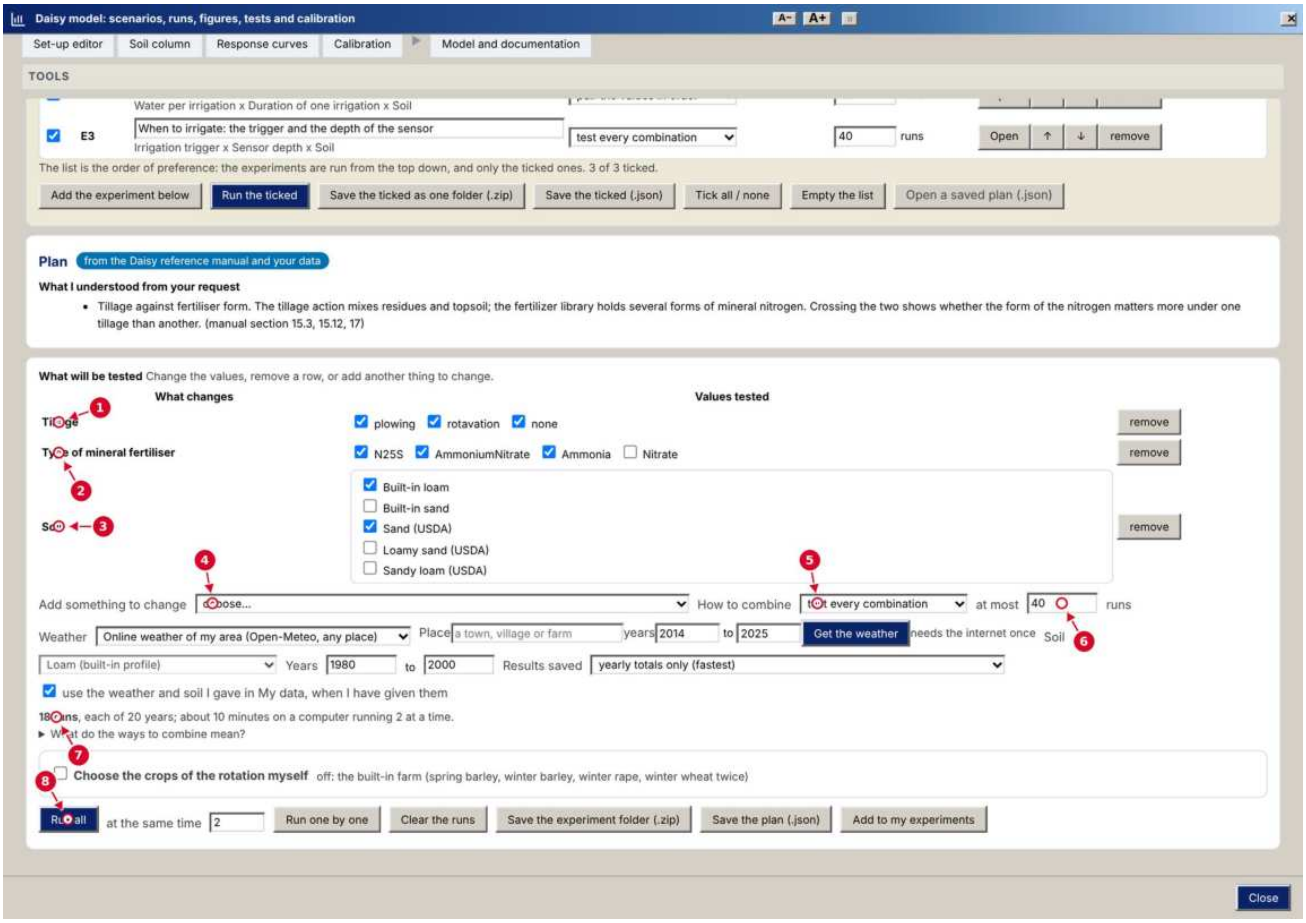



Figure 30.20: Tillage against fertiliser form. Screen 49 of the Daisy workbench.

**What the arrows point to.**

1. **Tillage.** Varied over plowing, rotavation, none.
2. **Type of mineral fertiliser.** Varied over N25S, AmmoniumNitrate, Ammonia.
3. **Soil.** Varied over usda:sand, loam.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.21 The numerical settings of the bioclimate model

NoOfIntervals: Number of vertical intervals in which we partition the canopy; max svat iterations: Max number of svat iterations before giving up on cobvergence; max svat absolute difference [mm/h]: Maximum absolute difference in svat ea values for convergence. Each is varied from half to double the value the farm uses; NoOfIntervals moved the yield, water or nitrogen in the test.

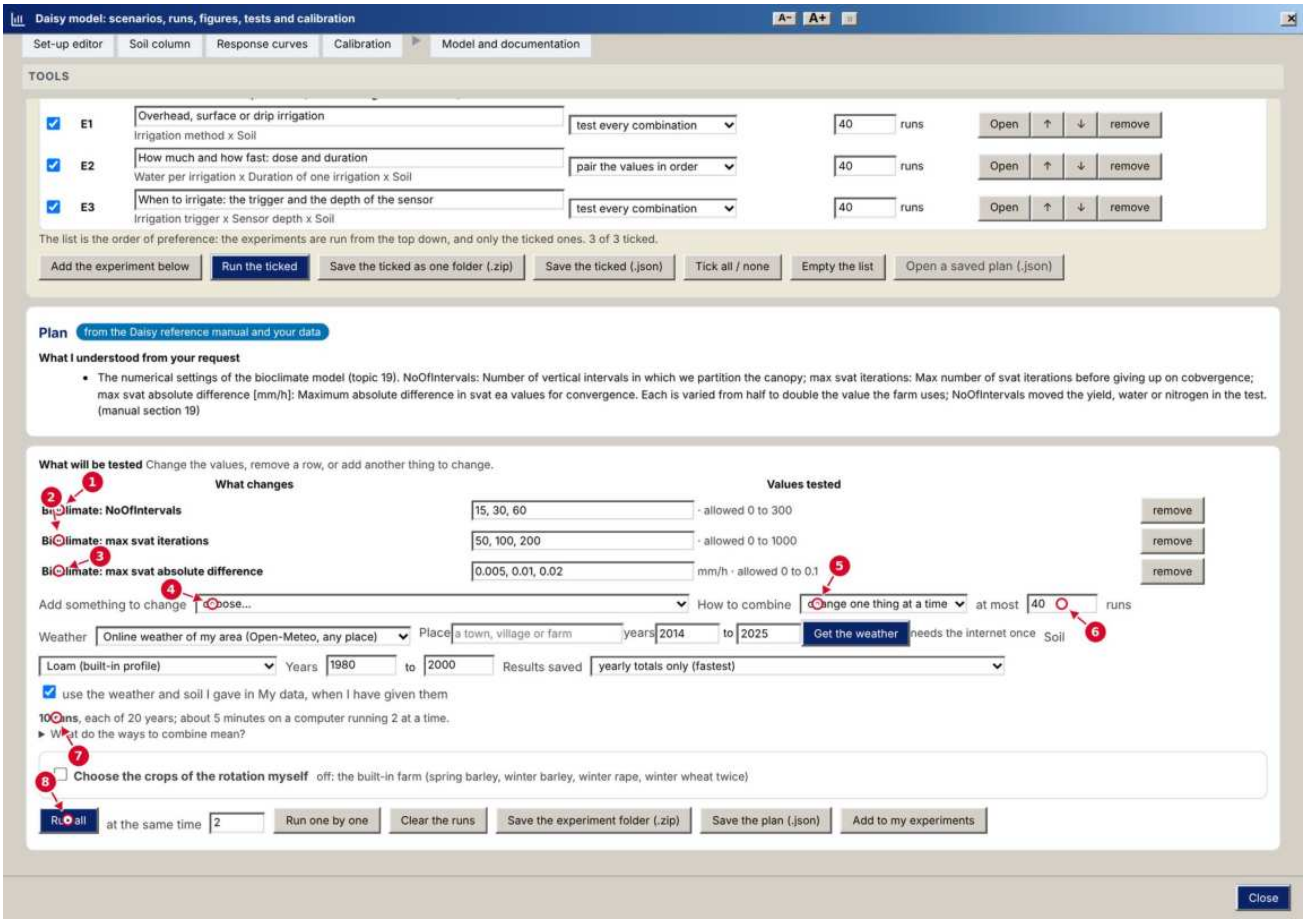



Figure 30.21: The numerical settings of the bioclimate model. Screen 50 of the Daisy workbench.

**What the arrows point to.**

1. **Bioclimate: NoOfIntervals (Daisy reference, chapter 19).** Varied over 15, 30, 60.
2. **Bioclimate: max svat iterations (Daisy reference, chapter 19).** Varied over 50, 100, 200.
3. **Bioclimate: max svat absolute difference (Daisy reference, chapter 19).** Varied over 0.005, 0.01, 0.02.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.22 The groundwater and the aquitard below the profile: aquitard

K aquitard [cm/h]: Conductivity of the aquitard; Z aquitard [cm]: Thickness of the aquitard. The aquitard begins below the bottommost soil horizon. Each is varied from half to double the value the farm uses; K aquitard, Z aquitard moved the yield, water or nitrogen in the test.

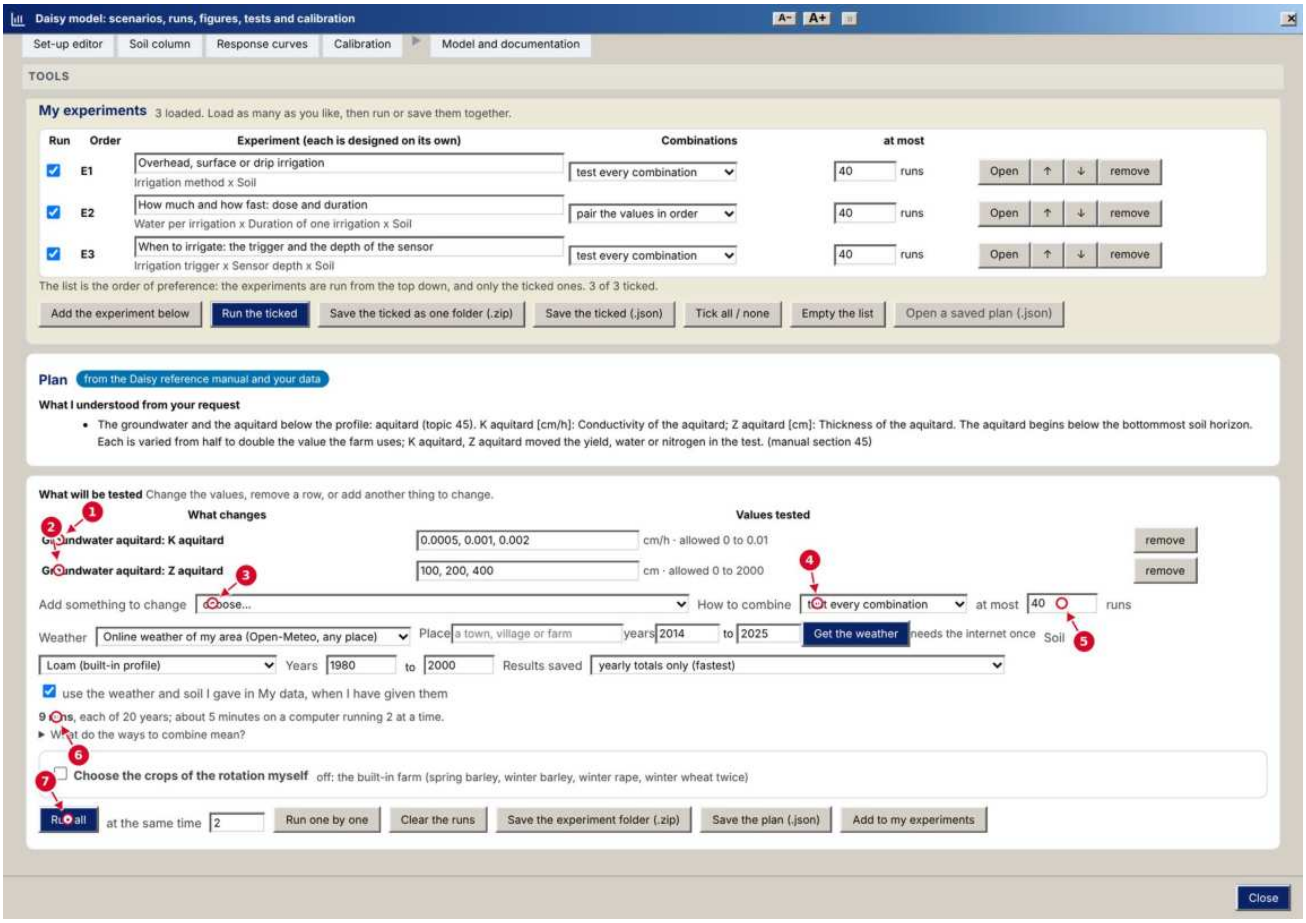



Figure 30.22: The groundwater and the aquitard below the profile: aquitard. Screen 51 of the Daisy workbench.

**What the arrows point to.**

1. **Groundwater aquitard: K aquitard (Daisy reference, chapter 45).** Varied over 0.0005, 0.001, 0.002.
2. **Groundwater aquitard: Z aquitard (Daisy reference, chapter 45).** Varied over 100, 200, 400.
3. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
4. **How to combine.** Every combination, one thing at a time, or the values paired in order.
5. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
6. **Size of the experiment.** The number of runs and the time they take on this computer.
7. **Run all.** Runs every combination, several at a time.

### 30.0.23 Turnover of organic matter and the mineral nitrogen it releases

K NH4 [h-1 ]: Maximal immobilization rate for ammonium; K NO3 [h-1 ]: Maximal immobilization rate for nitrate; CO2 threshold [h-1 ]: Turnover rate above which pools will contribute to 'CO2 fast'. Each is varied from half to double the value the farm uses; K NH4, K NO3, CO2 threshold moved the yield, water or nitrogen in the test.

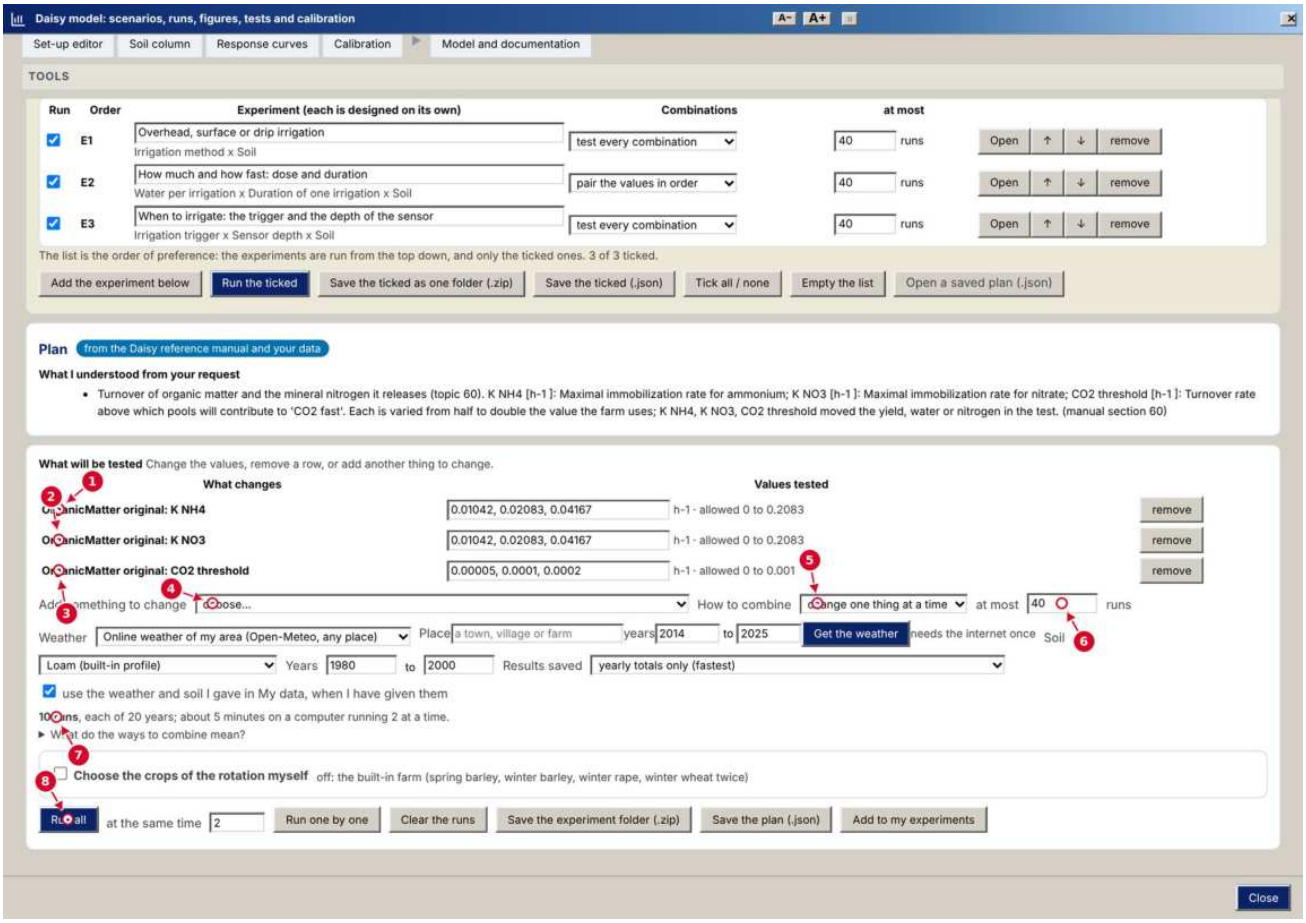



Figure 30.23: Turnover of organic matter and the mineral nitrogen it releases. Screen 52 of the Daisy workbench.

**What the arrows point to.**

1. **OrganicMatter original: K NH4 (Daisy reference, chapter 60).** Varied over 0.01042, 0.02083, 0.04167.
2. **OrganicMatter original: K NO3 (Daisy reference, chapter 60).** Varied over 0.01042, 0.02083, 0.04167.
3. **OrganicMatter original: CO2 threshold (Daisy reference, chapter 60).** Varied over 0.00005, 0.0001, 0.0002.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.24 Water at the soil surface: runoff and evaporation

temperature change rate [h-1 ]: Relative change of surface temperature. If you set this to a negative value, surface temperature will reflect temperature of incomming water or ponded; EpFactor [dimensionless]: The autumn value can be lower, due to muching. With a crop factor of 1.2 a combined Kc of 1.15 is reached at LAI=5. See also [Kjaersgaard et al., 2008]; ReservoirConstant [h-1 ]: Fraction of ponding above DetentionCapacity that runoffs each hour. Each is varied from half to double the value the farm uses; temperature change rate, EpFactor, ReservoirConstant moved the yield, water or nitrogen in the test.

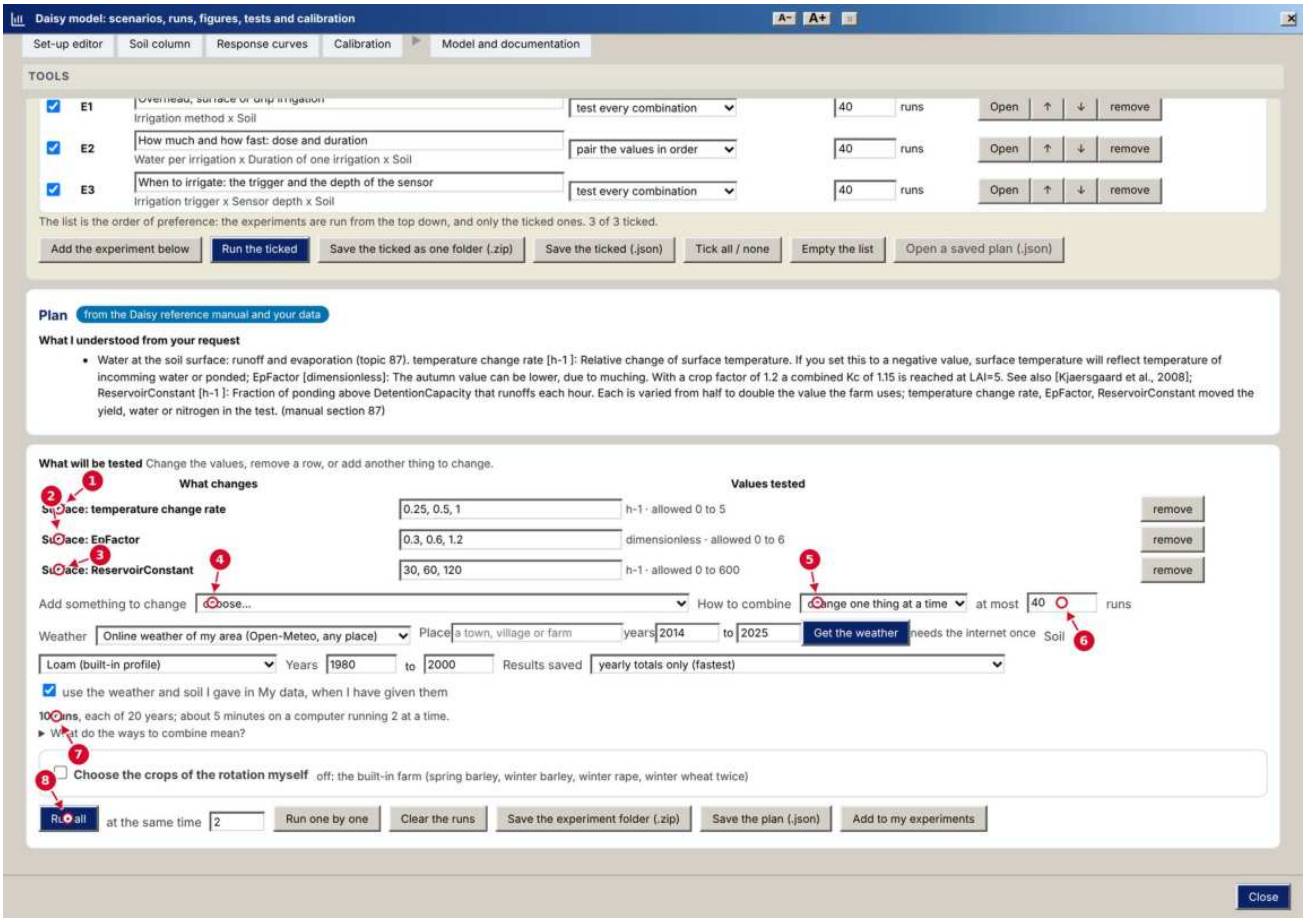



Figure 30.24: Water at the soil surface: runoff and evaporation. Screen 53 of the Daisy workbench.

**What the arrows point to.**

1. **Surface: temperature change rate (Daisy reference, chapter 87).** Varied over 0.25, 0.5, 1.
2. **Surface: EpFactor (Daisy reference, chapter 87).** Varied over 0.3, 0.6, 1.2.
3. **Surface: ReservoirConstant (Daisy reference, chapter 87).** Varied over 30, 60, 120.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.
8. **Run all.** Runs every combination, several at a time.

### 30.0.25 The limits on the chemistry sink terms

max sink total [dimensionless]: Maximum allowed sink term as a fraction of total content. If variable timesteps are enabled, Daisy will try to scale down the timestep in; max sink solute [dimensionless]: Maximum allowed sink term as a fraction of solute content. If variable timesteps are enabled, Daisy will try to scale down the timestep; max sink secondary [dimensionless]: Maximum allowed sink term as a fraction of secondary domain content. If variable timesteps are enabled, Daisy will try to scale down the timestep. Each is varied from half to double the value the farm uses; none of them moved yield, water or nitrogen in the test, so the effect shows in the process itself.

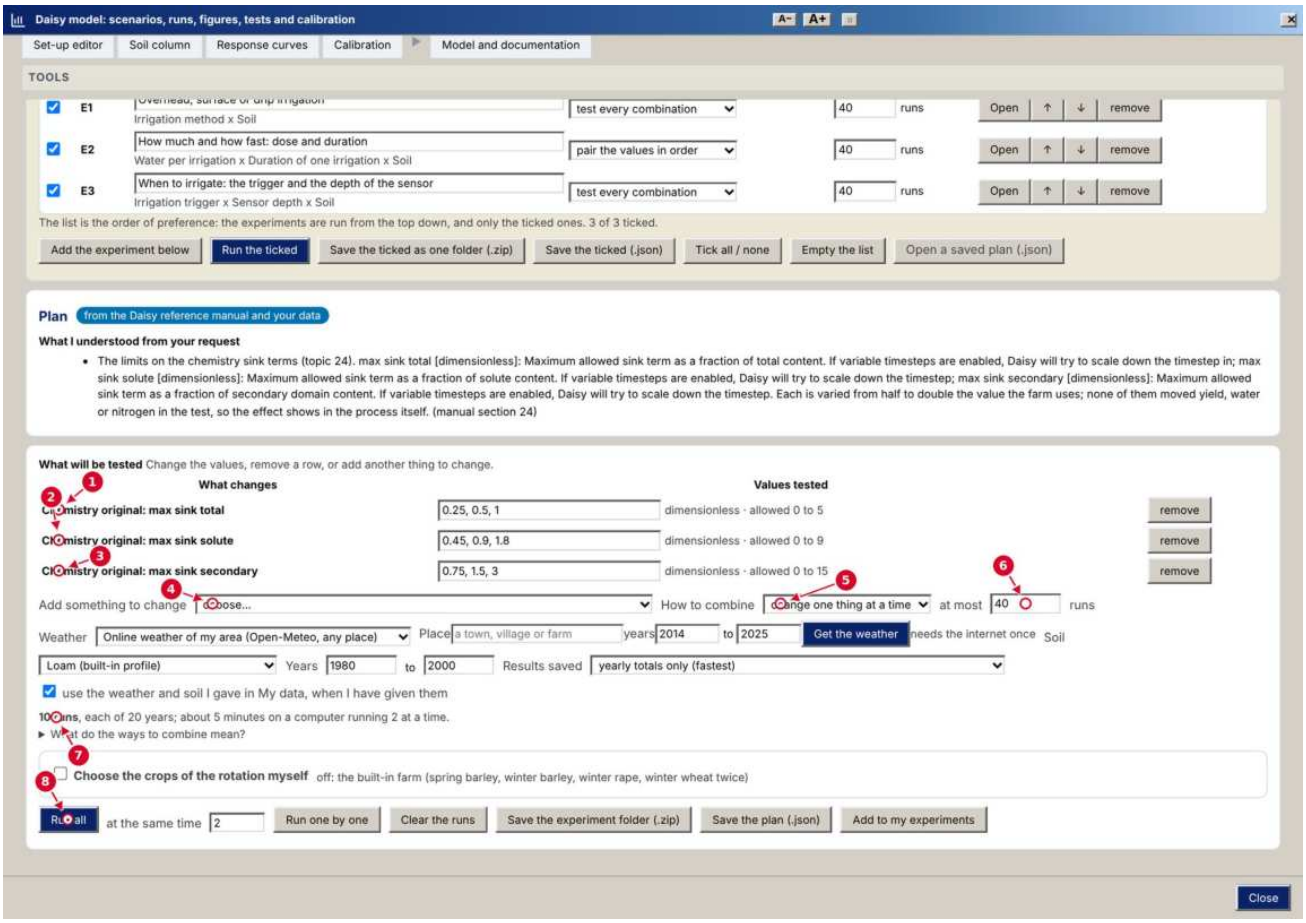



Figure 30.25: The limits on the chemistry sink terms. Screen 54 of the Daisy workbench.

**What the arrows point to.**

1. **Chemistry original: max sink total (Daisy reference, chapter 24).** Varied over 0.25, 0.5, 1.
2. **Chemistry original: max sink solute (Daisy reference, chapter 24).** Varied over 0.45, 0.9, 1.8.
3. **Chemistry original: max sink secondary (Daisy reference, chapter 24).** Varied over 0.75, 1.5, 3.
4. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
5. **How to combine.** Every combination, one thing at a time, or the values paired in order.
6. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
7. **Size of the experiment.** The number of runs and the time they take on this computer.

8. **Run all.** Runs every combination, several at a time.

### 30.0.26 Pipe drainage

L [cm]: Distance between pipes; rad [cm]: Inner radius of drain pipes. Each is varied from half to double the value the farm uses; none of them moved yield, water or nitrogen in the test, so the effect shows in the process itself.

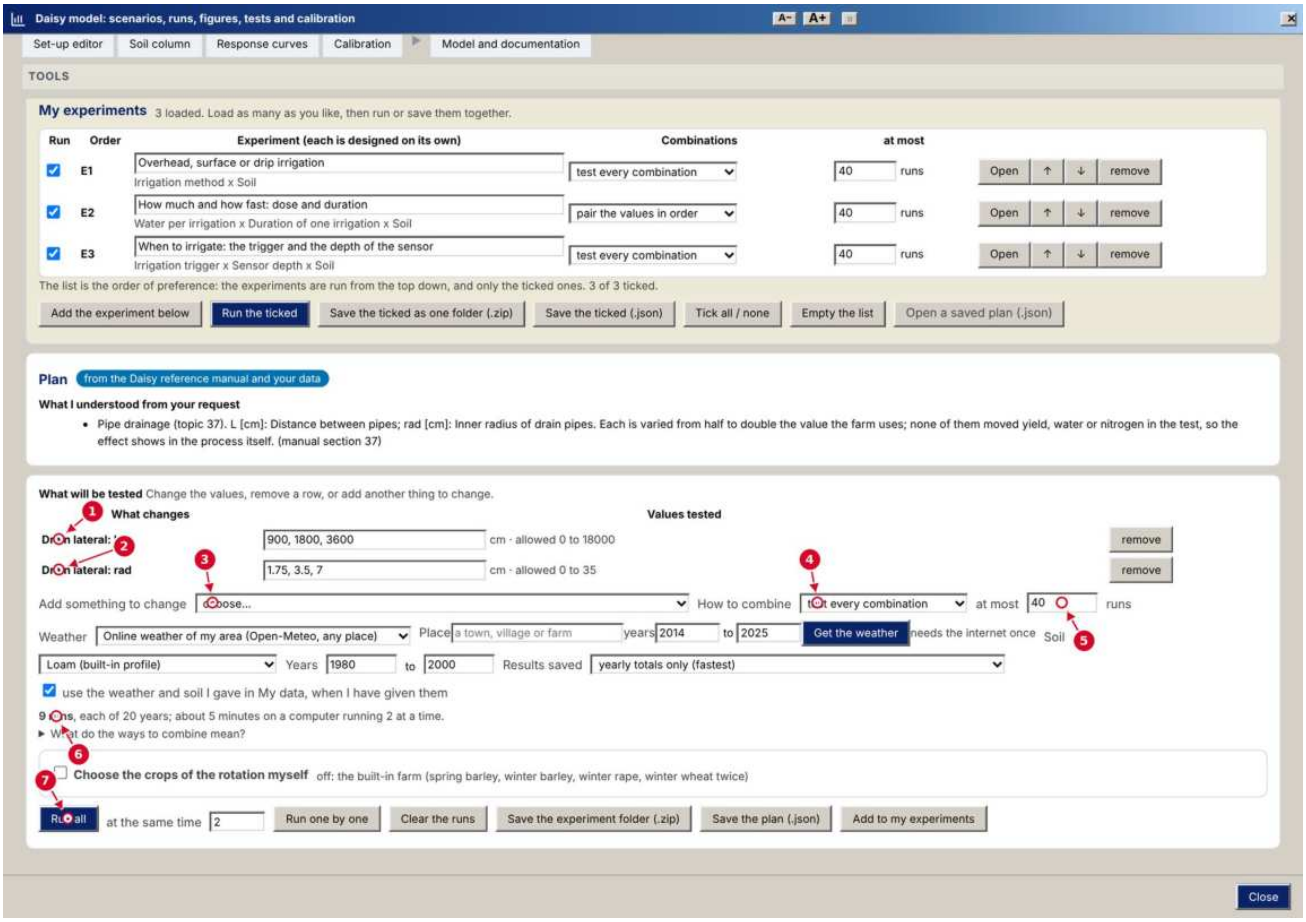



Figure 30.26: Pipe drainage. Screen 55 of the Daisy workbench.

**What the arrows point to.**

1. **Drain lateral: L (Daisy reference, chapter 37).** Varied over 900, 1800, 3600.
2. **Drain lateral: rad (Daisy reference, chapter 37).** Varied over 1.75, 3.5, 7.
3. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
4. **How to combine.** Every combination, one thing at a time, or the values paired in order.
5. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
6. **Size of the experiment.** The number of runs and the time they take on this computer.
7. **Run all.** Runs every combination, several at a time.

### 30.0.27 The groundwater and the aquitard below the profile: static

table [cm]: Groundwater level. Positive numbers indicate free drainage. Each is varied from half to double the value the farm uses; none of them moved yield, water or nitrogen in the test, so the effect shows in the process itself.

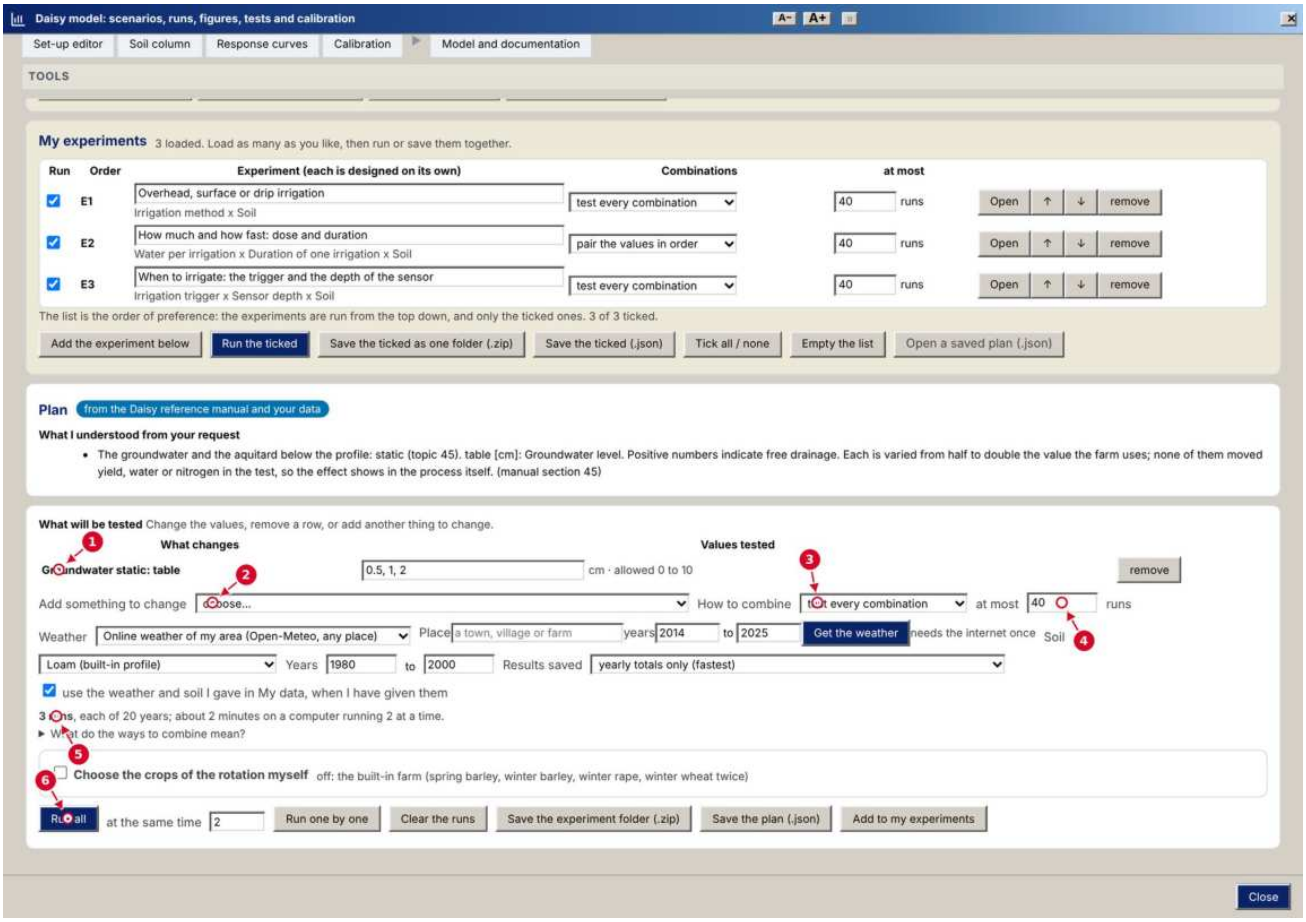



Figure 30.27: The groundwater and the aquitard below the profile: static. Screen 56 of the Daisy workbench.

**What the arrows point to.**

1. **Groundwater static: table (Daisy reference, chapter 45).** Varied over 0.5, 1, 2.
2. **Add something to change.** Adds a further factor, grouped by Irrigation, Nitrogen and tillage, Weather and climate, Soil and crop, and Chemicals.
3. **How to combine.** Every combination, one thing at a time, or the values paired in order.
4. **At most.** The largest number of runs; a spread of the combinations is chosen when there are more.
5. **Size of the experiment.** The number of runs and the time they take on this computer.
6. **Run all.** Runs every combination, several at a time.

## 30.1 All combinations in one table

| Group | Combination | Reference | Why |
|---|---|---|---|
| Irrigation | Overhead, surface or drip irrigation | 15.19, 15.20, 15.21 | The manual has three ways to apply water: over the canopy, on the soil surface, and into a soil volume. Run the three with the same trigger and dose, so only the way of applying water differs. |
| Nitrogen | Irrigation and nitrogen together | 15.5, 15.12 | Water moves nitrate and decides how much of the fertiliser the crop can use. Irrigation against rainfed, at three fertiliser doses, shows whether more water raises the yield only when the nitrogen is there, and what it costs in nitrate below the roots. |
| Irrigation | When to irrigate: the trigger and the depth of the sensor | 28.39, 15.5 | Irrigation starts when “soil water pressure above” is not true: the potential (cm) and the height (cm) are its two parameters. The potentials run from near field capacity to well before wilting for this soil texture; the depths stay inside the rooting depth of 100 cm. |
| Irrigation | How much and how fast: dose and duration | 15.5, 15.18 | The water application speed is a parameter in mm/h (default 2 mm/h) and the irrigation lasts a number of hours. The pairs below keep the speed of the manual (2 mm/h) while the dose grows, so the answer is the effect of the dose and not of the speed. |
| Nitrogen | Nitrogen fertiliser, irrigated and rainfed | 15.12 | Fertiliser is applied as an amount in kg N/ha. Scaling the rotation between half and one and a quarter of its dose, with and without irrigation, shows the dose where yield stops rising and the nitrate leached below the roots starts to climb. |
| Climate | A drier, wetter or warmer year | 25.7, 107.26 | The weather file gives precipitation and air temperature to every model of the column. Scaling the rain and adding a warming to the same record shows how much the yield and the water use move with the weather, with the soil and the management unchanged. |

| Group | Combination | Reference | Why |
|---|---|---|---|
| Irrigation | The irrigation season and a limit on water | 15.5 | The irrigation of a crop is tested between a from date and a to date and can be given an amount. Start dates and a seasonal limit show how much of the yield is bought by the first and the last irrigations. |
| Soil and crop | Which soil needs the most water | 47, 48.16 | A horizon gets its hydraulic properties from the hypres model by default, from the sand, silt, clay, humus and bulk density. Rainfed and irrigated runs on several textures show where irrigation pays. |
| Irrigation | Slow or fast irrigation: a constant water flux | 15.5, 15.18 | Daisy applies irrigation at a rate in mm/h, so a slow rate keeps the water in the topsoil and a fast one can run off or percolate. The rates below keep the same dose per irrigation, so only the speed of application differs. |
| Irrigation | Sensor depth against rooting depth | 28.39, 29.1, 72 | The tensiometer that triggers irrigation sits at one depth, while the roots take water from the whole root zone. A sensor above or below most of the roots triggers too early or too late; crossing the sensor depth with the rooting depth shows where it matters. |
| Nitrogen | When the fertiliser is applied and how much | 15.12, 15.3 | The fertilize action is placed on a date. Moving it two weeks earlier or later, at three doses, shows how much nitrate is exposed to the rain before the crop takes it up. |
| Climate | Warming with and without irrigation | 25.7, 15.5 | A warmer year raises the evapotranspiration and shortens the crop cycle. Warming of 0 to 3 degrees, rainfed and irrigated, shows how much of the loss irrigation can give back. |
| Nitrogen | Tillage and the time and form of the fertiliser | 15.3, 15.12 | The tillage action (plowing, rotavation, or none) mixes the residues and the topsoil; the fertiliser is taken from the fertilizer library by name, and the date can be moved. These three decide how much nitrate is exposed to leaching. |

| Group | Combination | Reference | Why |
|---|---|---|---|
| Climate | Evapotranspiration model and radiation | 25.7, 107.26 | Daisy can take the reference evapotranspiration from the weather file (FAO56) or compute it from the energy balance, and the radiation can be scaled. The runs show how much of the answer depends on that choice. |
| Climate | A dry summer, a dry spring or a wet winter | 103, 25.7 | The weather component can scale the precipitation month by month (twelve factors). Taking 30 percent of the rain from the summer or the spring, or adding 30 percent in the winter, shows which part of the year matters for yield and for leaching. |
| Chemicals | Sorption and half-life of a chemical | 23 | A chemical has a sorption coefficient (K_OC, L/kg) and a decomposition rate; the two decide whether it stays in the topsoil or reaches the depth where it is counted as leached. |
| Irrigation | Deficit irrigation: a water limit against the trigger | 15.5, 28.39 | A crop can be given a limited amount of water in the season, and irrigation starts only when the soil water pressure is below the trigger. A strict trigger with a small limit, against a lax trigger with a large limit, shows whether the same water gives more yield when it is saved for the dry spell. |
| Soil and crop | Rooting depth of the crop | 29.1, 72 | The crop model has a maximum rooting depth, which decides how much of the soil water and nitrate the roots can reach. Shallow to deep roots, rainfed and irrigated, show how much of a water shortage a deeper root zone absorbs. |
| Climate | The same farm under other climates | 103 | The weather component reads a file for one place. Running the same soil, crops and management with the weather of several places shows how much of the result belongs to the climate. |
| Nitrogen | Tillage against fertiliser form | 15.3, 15.12, 17 | The tillage action mixes residues and topsoil; the fertilizer library holds several forms of mineral nitrogen. Crossing the two shows whether the form of the nitrogen matters more under one tillage than another. |

| Group | Combination | Reference | Why |
|---|---|---|---|
| Parameters | The numerical settings of the bioclimate model (topic 19) | 19 | NoOfIntervals: Number of vertical intervals in which we partition the canopy; max svat iterations: Max number of svat iterations before giving up on cobvergence; max svat absolute difference [mm/h]: Maximum absolute difference in svat ea values for convergence. Each is varied from half to double the value the farm uses; NoOfIntervals moved the yield, water or nitrogen in the test. |
| Parameters | The groundwater and the aquitard below the profile: aquitard (topic 45) | 45 | K aquitard [cm/h]: Conductivity of the aquitard; Z aquitard [cm]: Thickness of the aquitard. The aquitard begins below the bottommost soil horizon. Each is varied from half to double the value the farm uses; K aquitard, Z aquitard moved the yield, water or nitrogen in the test. |
| Parameters | Turnover of organic matter and the mineral nitrogen it releases (topic 60) | 60 | K NH4 [h-1 ]: Maximal immobilization rate for ammonium; K NO3 [h-1 ]: Maximal immobilization rate for nitrate; CO2 threshold [h-1 ]: Turnover rate above which pools will contribute to ‘CO2 fast’. Each is varied from half to double the value the farm uses; K NH4, K NO3, CO2 threshold moved the yield, water or nitrogen in the test. |
| Parameters | Water at the soil surface: runoff and evaporation (topic 87) | 87 | temperature change rate [h-1 ]: Relative change of surface temperature. If you set this to a negative value, surface temperature will reflect temperature of incomming water or ponded; EpFactor [dimensionless]: The autumn value can be lower, due to muching. With a crop factor of 1.2 a combined Kc of 1.15 is reached at LAI=5. See also [Kjaersgaard et al., 2008]; ReservoirConstant [h-1 ]: Fraction of ponding above DetentionCapacity that runoffs each hour. Each is varied from half to double the value the farm uses; temperature change rate, EpFactor, ReservoirConstant moved the yield, water or nitrogen in the test. |

| Group | Combination | Reference | Why |
|---|---|---|---|
| Parameters | The limits on the chemistry sink terms (topic 24) | 24 | max sink total [dimensionless]: Maximum allowed sink term as a fraction of total content. If variable timesteps are enabled, Daisy will try to scale down the timestep in; max sink solute [dimensionless]: Maximum allowed sink term as a fraction of solute content. If variable timesteps are enabled, Daisy will try to scale down the timestep; max sink secondary [dimensionless]: Maximum allowed sink term as a fraction of secondary domain content. If variable timesteps are enabled, Daisy will try to scale down the timestep. Each is varied from half to double the value the farm uses; none of them moved yield, water or nitrogen in the test, so the effect shows in the process itself. |
| Parameters | Pipe drainage (topic 37) | 37 | L [cm]: Distance between pipes; rad [cm]: Inner radius of drain pipes. Each is varied from half to double the value the farm uses; none of them moved yield, water or nitrogen in the test, so the effect shows in the process itself. |
| Parameters | The groundwater and the aquitard below the profile: static (topic 45) | 45 | table [cm]: Groundwater level. Positive numbers indicate free drainage. Each is varied from half to double the value the farm uses; none of them moved yield, water or nitrogen in the test, so the effect shows in the process itself. |

## 30.2 The plan of each combination

### 30.2.1 Overhead, surface or drip irrigation

Design: factorial. Factors: method with overhead, surface, drip; soil with usda:sand, loam.

### 30.2.2 Irrigation and nitrogen together

Design: factorial. Factors: irrig with 0, 1; nfac with 0.5, 1, 1.25; soil with usda:sand, loam.

### 30.2.3 When to irrigate: the trigger and the depth of the sensor

Design: factorial. Factors: trig with -150, -300, -600; z with 10, 30, 60; soil with usda:sand, loam.

### 30.2.4 How much and how fast: dose and duration

Design: paired. Factors: mm with 10, 20, 30, 40; hours with 5, 10, 15, 20; soil with usda:sand, loam.

### 30.2.5 Nitrogen fertiliser, irrigated and rainfed

Design: factorial. Factors: nfac with 0.5, 0.75, 1, 1.25; irrig with 0, 1; soil with usda:sand, loam.

### 30.2.6 A drier, wetter or warmer year

Design: factorial. Factors: precip with 0.8, 1, 1.2; temp with 0, 2; soil with usda:sand, loam.

### 30.2.7 The irrigation season and a limit on water

Design: factorial. Factors: window with {‘label’: ‘Apr 1 minus Oct 1’, ‘set’: {‘w1’: ‘04-01’, ‘w2’: ‘10-01’}}, {‘label’: ‘May 1 minus Oct 1’, ‘set’: {‘w1’: ‘05-01’, ‘w2’: ‘10-01’}}, {‘label’: ‘Jun 1 minus Sep 1’, ‘set’: {‘w1’: ‘06-01’, ‘w2’: ‘09-01’}}; limit with 60, 120, 240; soil with usda:sand, loam.

### 30.2.8 Which soil needs the most water

Design: factorial. Factors: soil with usda:sand, usda:sandyloam, loam, usda:siltloam, usda:clayloam; irrig with 0, 1.

### 30.2.9 Slow or fast irrigation: a constant water flux

Design: factorial. Factors: flux with 0.1, 0.5, 2; soil with usda:sand, loam.

### 30.2.10 Sensor depth against rooting depth

Design: factorial. Factors: z with 10, 30, 50; root with 40, 80, 120; soil with usda:sand, loam.

### 30.2.11 When the fertiliser is applied and how much

Design: factorial. Factors: ntime with -14, 0, 14; nfac with 0.7, 1, 1.3; soil with usda:sand, loam.

### 30.2.12 Warming with and without irrigation

Design: factorial. Factors: temp with 0, 1.5, 3; irrig with 0, 1; soil with usda:sand, loam.

### 30.2.13 Tillage and the time and form of the fertiliser

Design: factorial. Factors: till with plowing, rotavation, none; ntime with -14, 0, 14; ntype with N25S, AmmoniumNitrate, Ammonia; soil with usda:sand, loam.

### 30.2.14 Evapotranspiration model and radiation

Design: factorial. Factors: bio with auto, default; radscale with 0.9, 1, 1.1; soil with usda:sand, loam.

### 30.2.15 A dry summer, a dry spring or a wet winter

Design: factorial. Factors: season with {'label': 'dry summer (rain x0.7 Jun-Aug)', 'set': {'pmonths': '1 1 1 1 1 0.7 0.7 0.7 1 1 1 1'}}, {'label': 'dry spring (rain x0.7 Mar-May)', 'set': {'pmonths': '1 1 0.7 0.7 0.7 1 1 1 1 1 1 1'}}, {'label': 'wet winter (rain x1.3 Dec-Feb)', 'set': {'pmonths': '1.3 1.3 1 1 1 1 1 1 1 1 1 1.3'}}; irrig with 0, 1; soil with usda:sand, loam.

### 30.2.16 Sorption and half-life of a chemical

Design: factorial. Factors: koc with 0, 50, 500; half with 15, 90; soil with usda:sand, loam.

### 30.2.17 Deficit irrigation: a water limit against the trigger

Design: factorial. Factors: trig with -150, -300, -600; limit with 60, 120, 240; soil with usda:sand, loam.

### 30.2.18 Rooting depth of the crop

Design: factorial. Factors: root with 30, 60, 90, 120; irrig with 0, 1; soil with usda:sand, loam.

### 30.2.19 The same farm under other climates

Design: factorial. Factors: station with course, it-bologna, dk-foulum; irrig with 0, 1; soil with usda:sand, loam.

### 30.2.20 Tillage against fertiliser form

Design: factorial. Factors: till with plowing, rotavation, none; ntype with N25S, AmmoniumNitrate, Ammonia; soil with usda:sand, loam.

### 30.2.21 The numerical settings of the bioclimate model (topic 19)

Design: oat. Factors: p_Bioclimate/NoOfIntervals with 15, 30, 60; p_Bioclimate/max_svat_iterations with 50, 100, 200; p_Bioclimate/max_svat_absolute_difference with 0.005, 0.01, 0.02.

### 30.2.22 The groundwater and the aquitard below the profile: aquitard (topic 45)

Design: factorial. Factors: p_Groundwater/aquitard/K_aquitard with 0.0005, 0.001, 0.002; p_Groundwater/aquitard/Z_aquitard with 100, 200, 400.

### 30.2.23 Turnover of organic matter and the mineral nitrogen it releases (topic 60)

Design: oat. Factors: p_OrganicMatter/original/K_NH4 with 0.01042, 0.02083, 0.04167; p_OrganicMatter/original/K_NO3 with 0.01042, 0.02083, 0.04167; p_OrganicMatter/original/CO2_threshold with 5e-05, 0.0001, 0.0002.

### 30.2.24 Water at the soil surface: runoff and evaporation (topic 87)

Design: oat. Factors: p_Surface/temperature_change_rate with 0.25, 0.5, 1; p_Surface/EpFactor with 0.3, 0.6, 1.2; p_Surface/ReservoirConstant with 30, 60, 120.

### 30.2.25 The limits on the chemistry sink terms (topic 24)

Design: oat. Factors: p_Chemistry/original/max_sink_total with 0.25, 0.5, 1; p_Chemistry/original/max_sink_so with 0.45, 0.9, 1.8; p_Chemistry/original/max_sink_secondary with 0.75, 1.5, 3.

### 30.2.26 Pipe drainage (topic 37)

Design: factorial. Factors: p_Drain/lateral/L with 900, 1800, 3600; p_Drain/lateral/rad with 1.75, 3.5, 7.

### 30.2.27 The groundwater and the aquitard below the profile: static (topic 45)

Design: factorial. Factors: p_Groundwater/static/table with 0.5, 1, 2.

# Chapter 31

# Results and analysis

The third group draws what a run produced.

## 31.1 The five scenarios, with their results

The screens that follow were taken after a real Daisy run of each scenario on the built-in farm with the built-in Taastrup weather for the years 1990 to 1993. The screens in this chapter show the figures as they were first drawn; the figures are now drawn in the style described at the end of this chapter (a title at the top left, light grid, colour blind safe colours, the legend below the figure, a colour bar for the maps of depth against time), and the numbers in them are the same.

## 31.2 Results of the irrigation scenario

### 31.2.1 Results of the irrigation question: the table of means

The irrigation question was run on the built-in farm for the years 1990 to 1993 with the built-in Taastrup weather. After the run the results appear under the options: a list that chooses the run to look at, three buttons, a table of the mean per season and the figures.

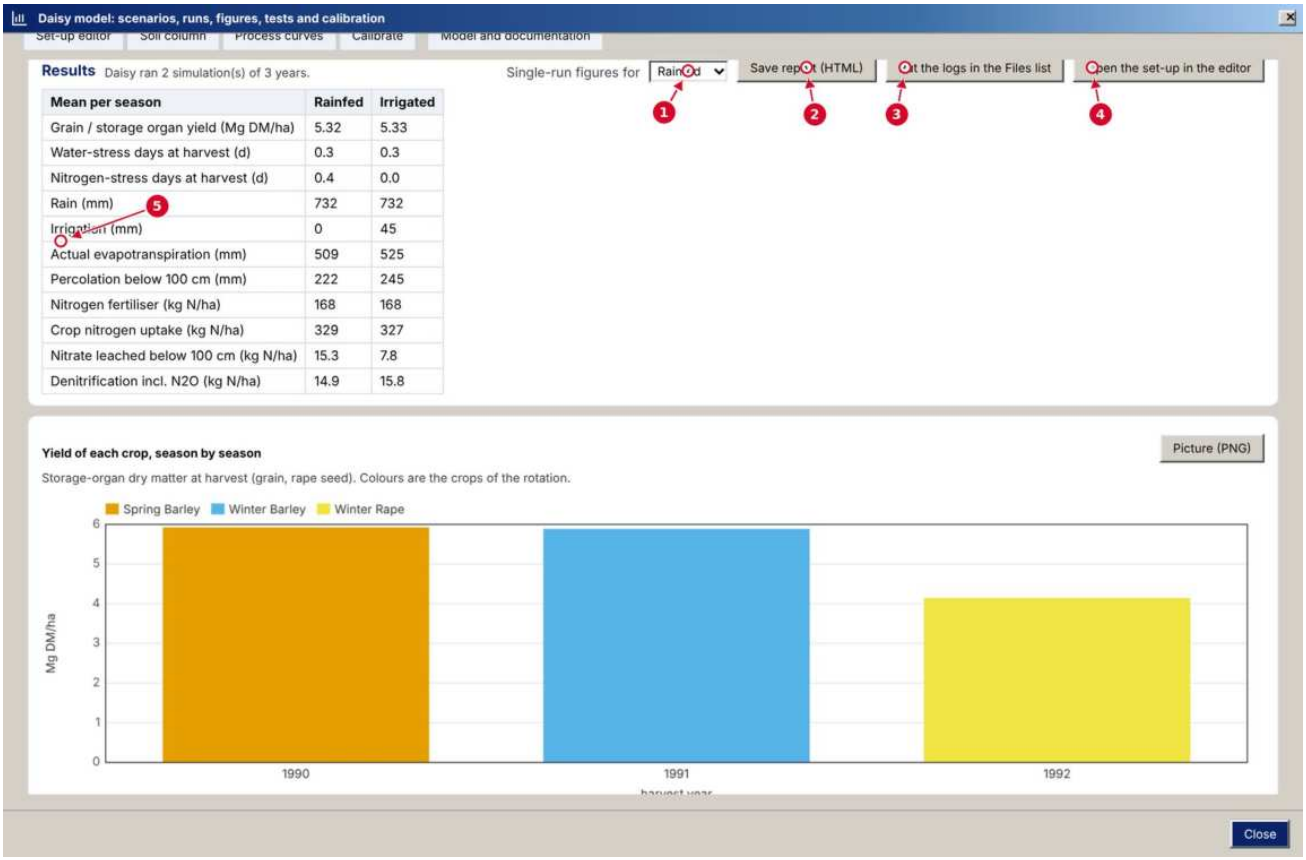



Figure 31.1: Results of the irrigation question: the table of means. Screen 57 of the Daisy workbench.

**What the arrows point to.**

1. **Run to look at.** Chooses the run whose single run figures are drawn.
2. **Save report.** Saves all results and figures as one HTML report.
3. **Put the logs in the list of loaded logs of the Log plots tab.** Keeps the Daisy logs of the runs for the Log plots and Mass balance tabs.
4. **Open the set-up in the editor.** Opens the Daisy set-up of the run in the Set-up editor.
5. **Mean per season.** The mean of each result over the seasons, one column for each run.

### 31.2.2 Yield of each crop, season by season (irrigation)

Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

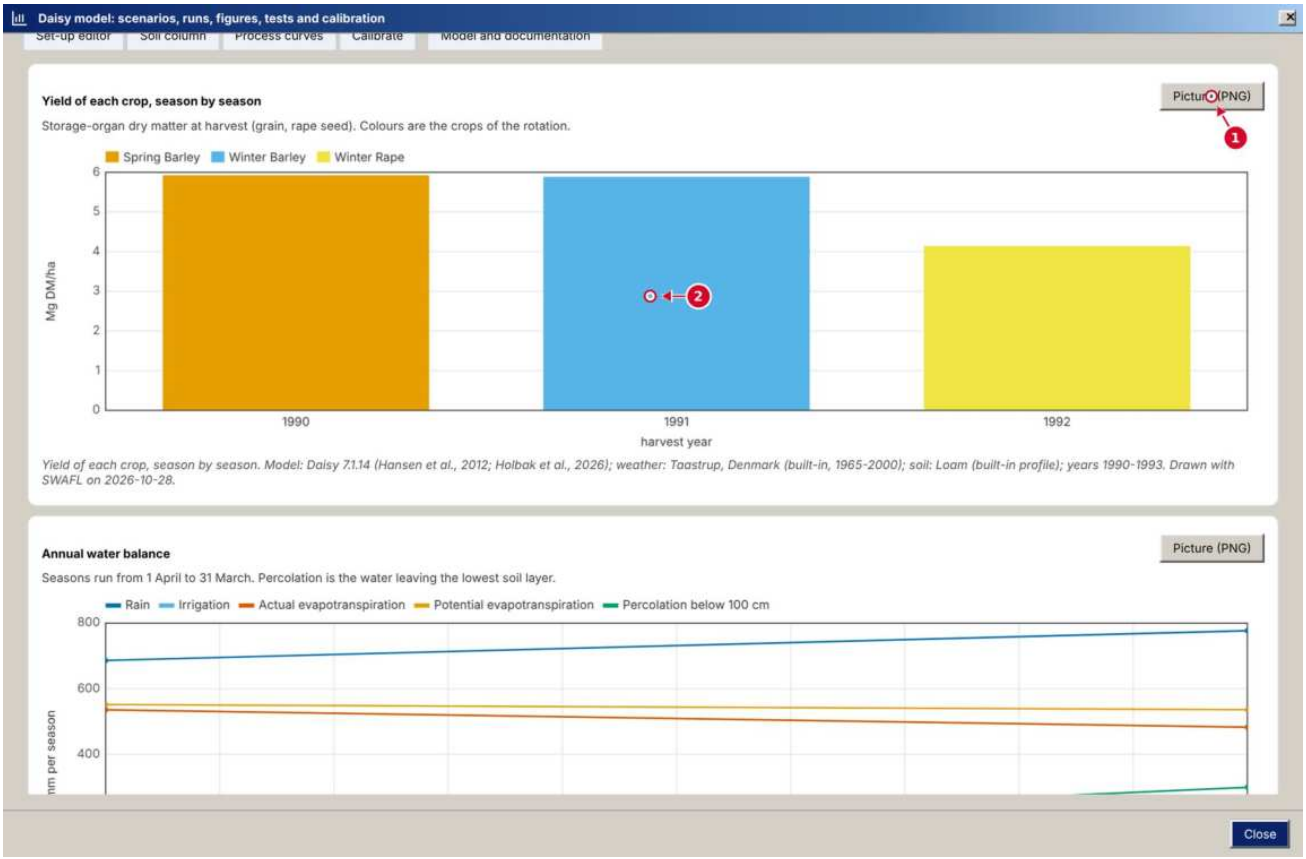



Figure 31.2: Yield of each crop, season by season (irrigation). Screen 58 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

### 31.2.3 Annual water balance (irrigation)

Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

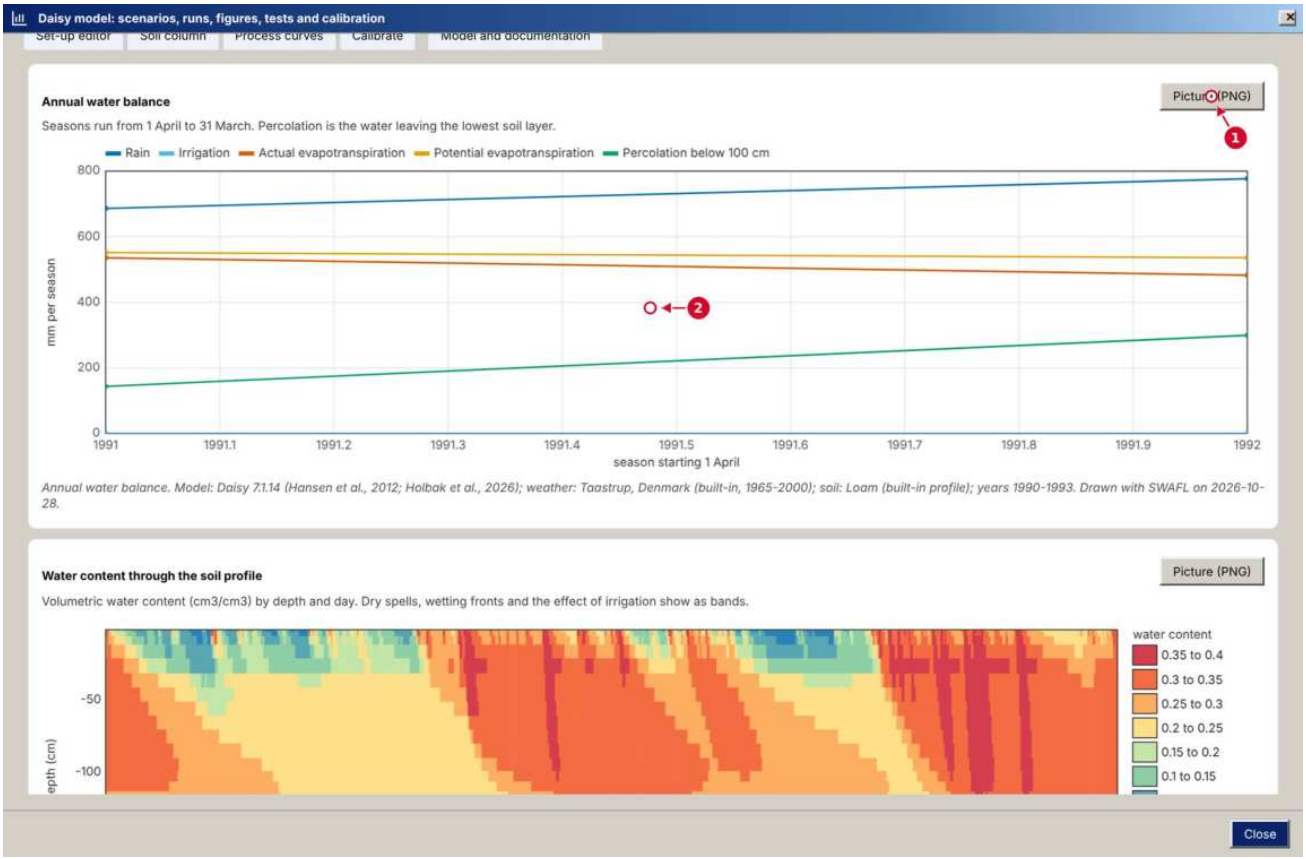



Figure 31.3: Annual water balance (irrigation). Screen 59 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

### 31.2.4 Water content through the soil profile (irrigation)

Soil water content against depth and time, from the daily log of Daisy.

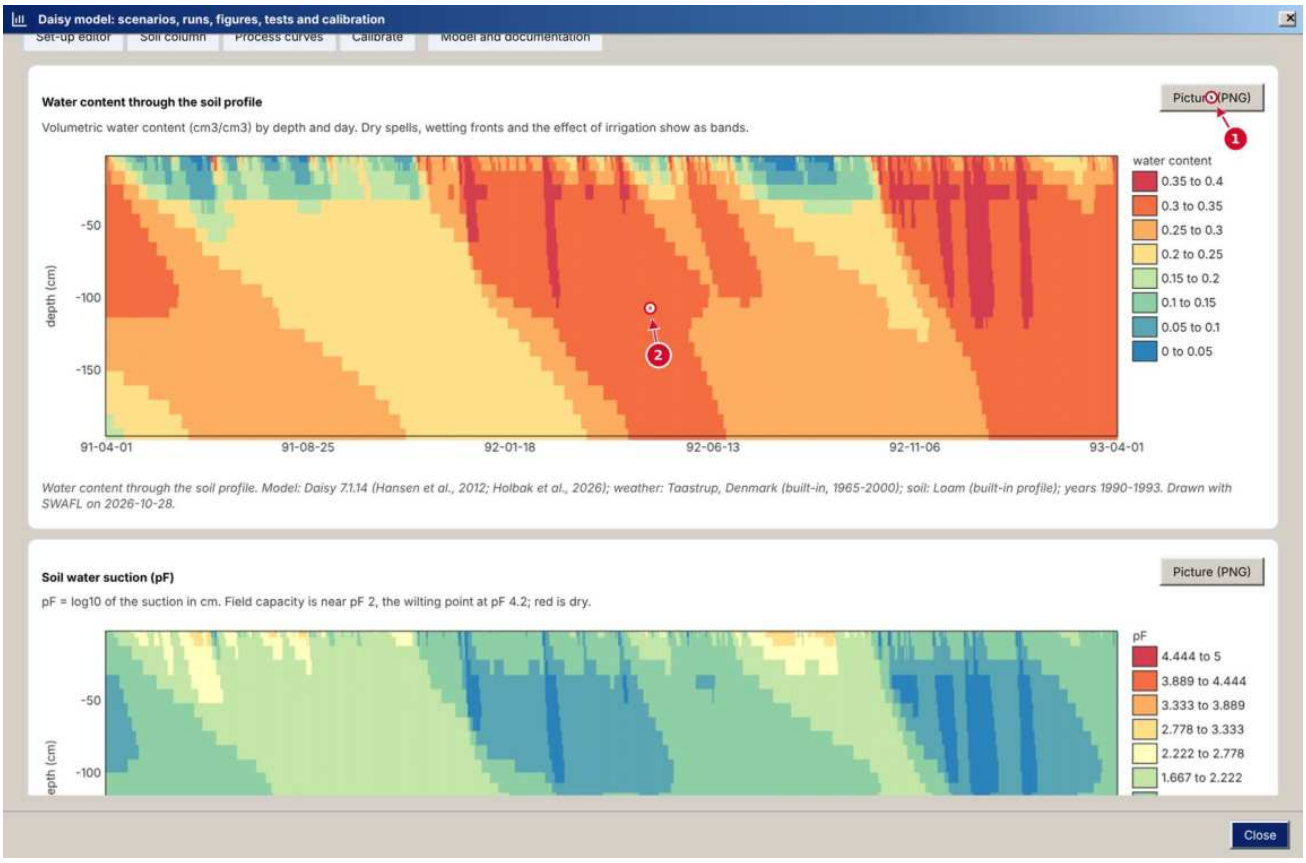



Figure 31.4: Water content through the soil profile (irrigation). Screen 60 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Soil water content against depth and time, from the daily log of Daisy.

### 31.2.5 Soil water suction (pF) (irrigation)

Soil water suction on the pF scale against depth and time; irrigation triggers act on this quantity.

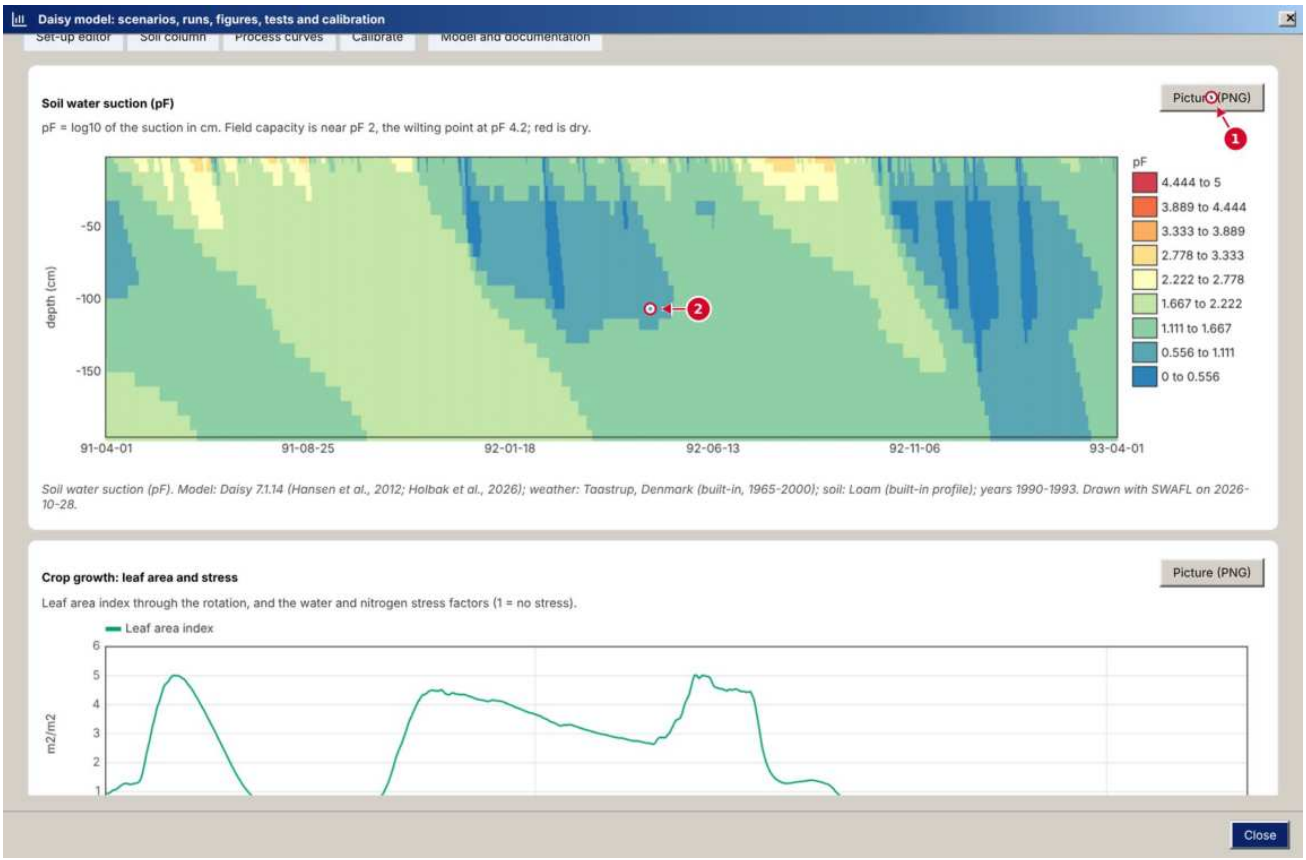



Figure 31.5: Soil water suction (pF) (irrigation). Screen 61 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Soil water suction on the pF scale against depth and time; irrigation triggers act on this quantity.

### 31.2.6 Crop growth: leaf area and stress (irrigation)

Leaf area index with the water and nitrogen stress of the crop through the seasons.

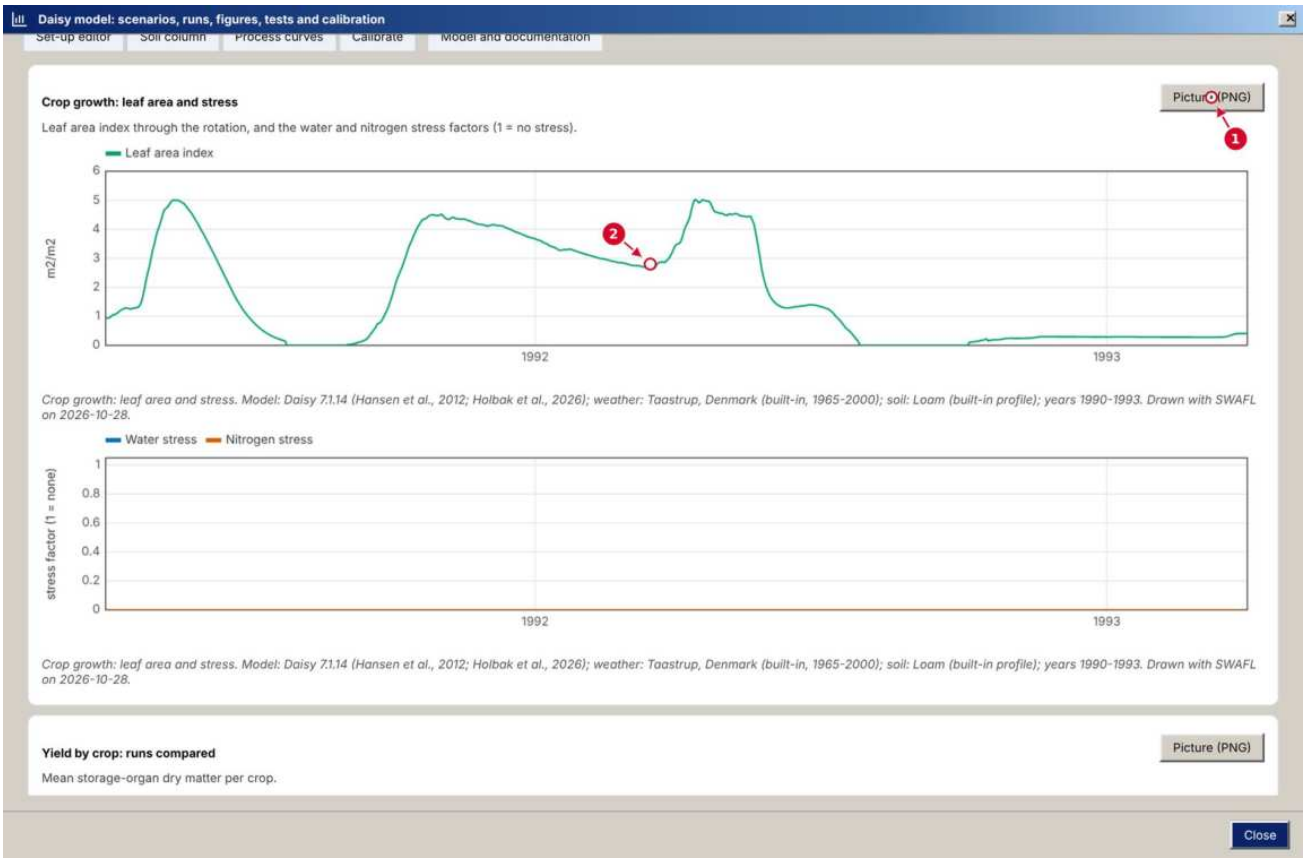



Figure 31.6: Crop growth: leaf area and stress (irrigation). Screen 62 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Leaf area index with the water and nitrogen stress of the crop through the seasons.

### 31.2.7 Yield by crop: runs compared (irrigation)

The mean yield of each crop in every ticked run, side by side.

Figure 31.7: Yield by crop: runs compared (irrigation). Screen 63 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The mean yield of each crop in every ticked run, side by side.

### 31.2.8 Water balance: runs compared (irrigation)

Irrigation, evapotranspiration and drainage of the runs side by side.

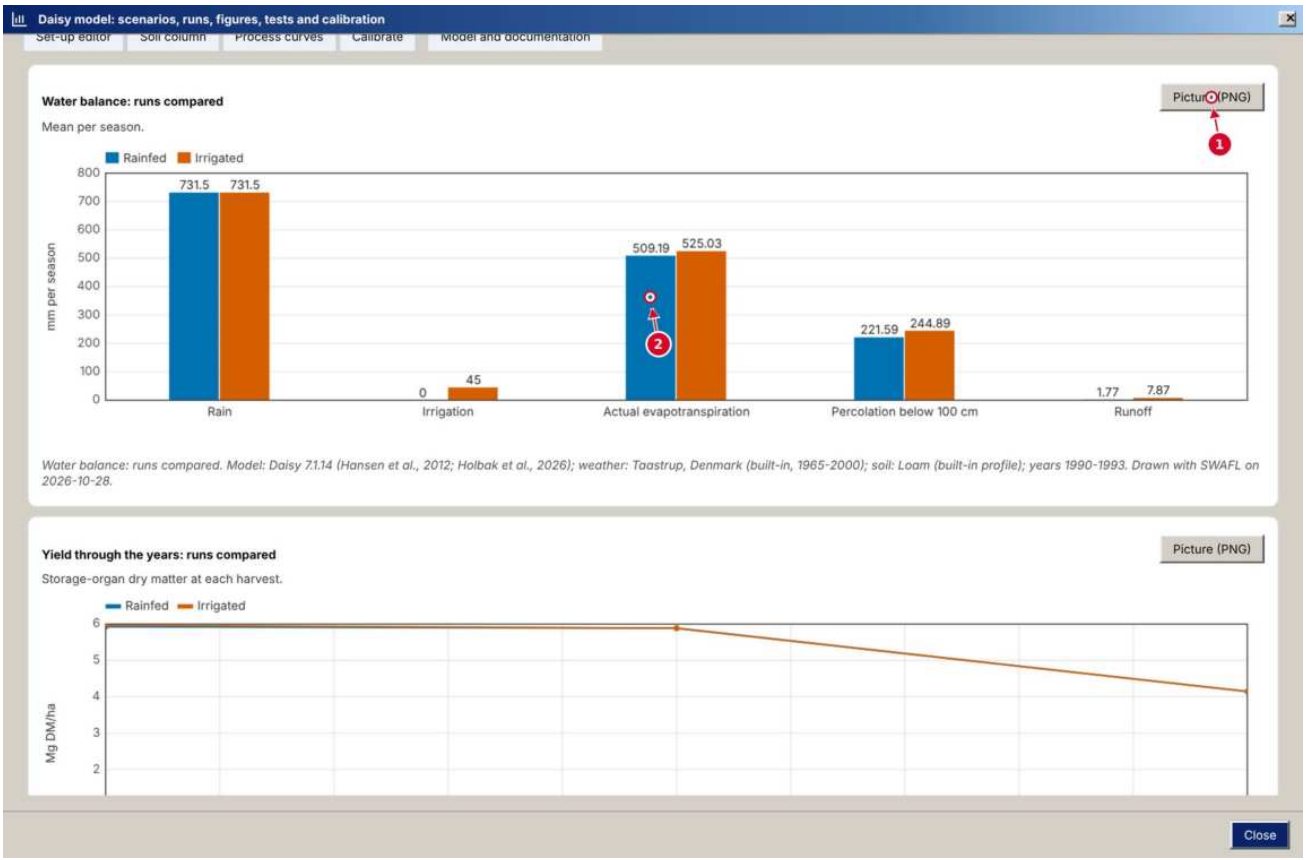



Figure 31.8: Water balance: runs compared (irrigation). Screen 64 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Irrigation, evapotranspiration and drainage of the runs side by side.

### 31.2.9 Yield through the years: runs compared (irrigation)

The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

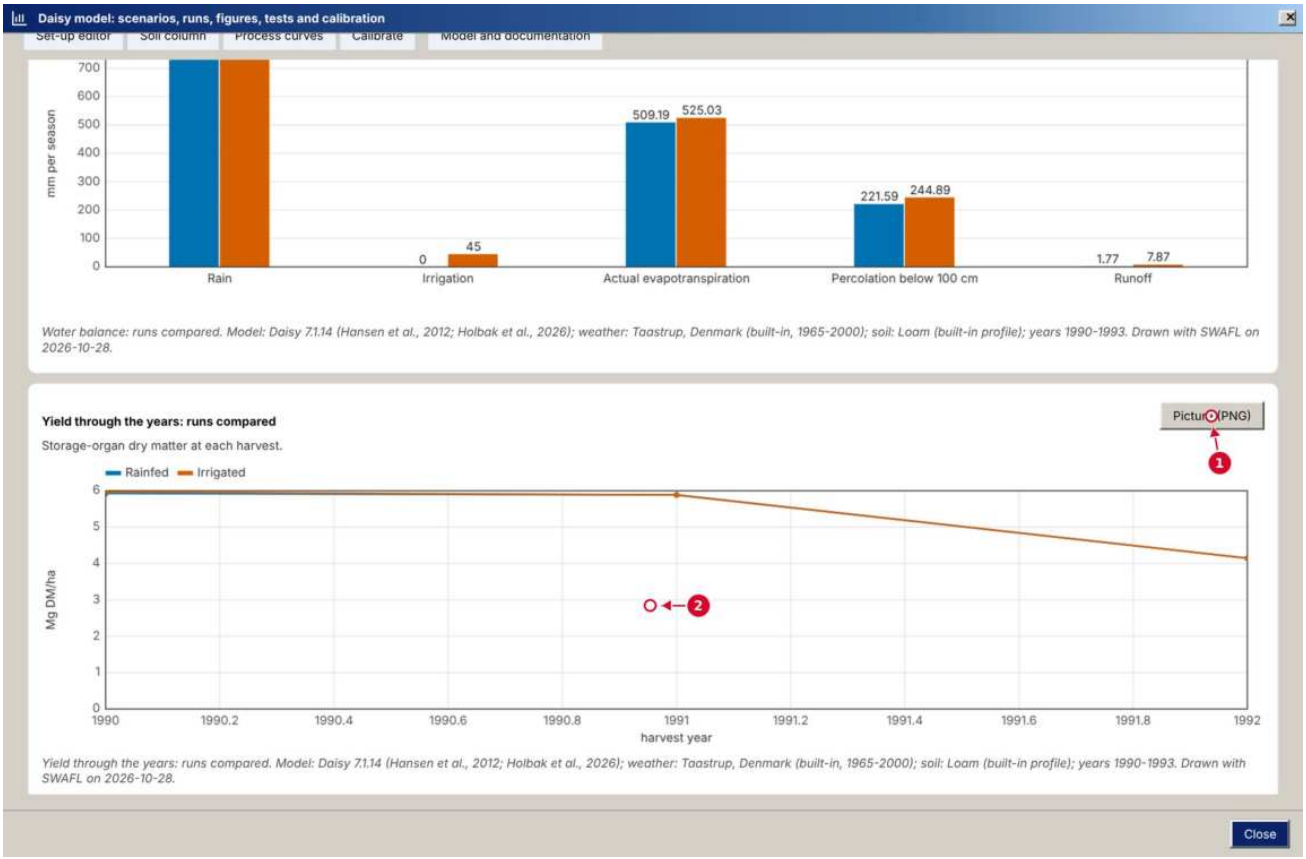



Figure 31.9: Yield through the years: runs compared (irrigation). Screen 65 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

## 31.3 Results of the nitrogen scenario

### 31.3.1 Results of the nitrogen question: the table of means

The nitrogen question was run on the built-in farm for the years 1990 to 1993 with the built-in Taastrup weather. After the run the results appear under the options: a list that chooses the run to look at, three buttons, a table of the mean per season and the figures.

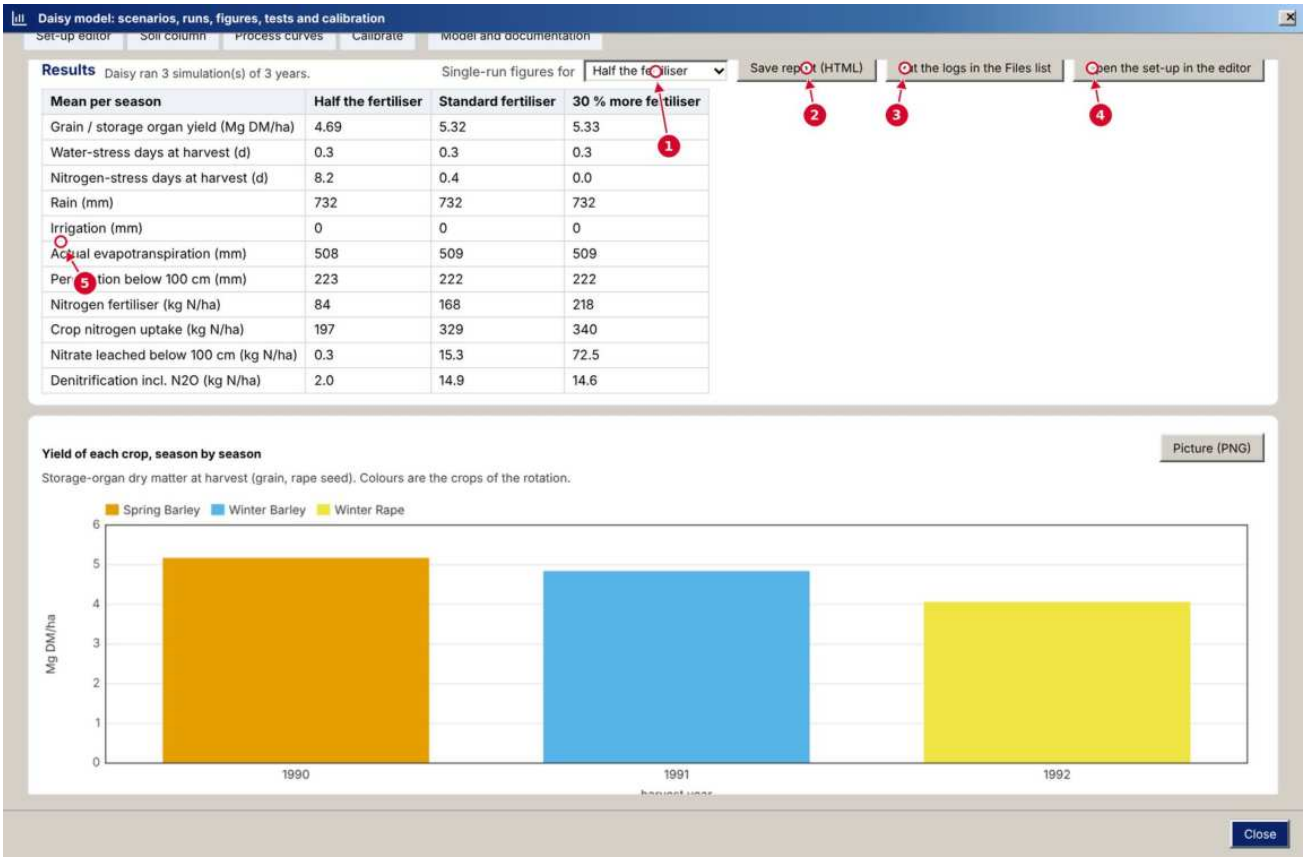



Figure 31.10: Results of the nitrogen question: the table of means. Screen 66 of the Daisy workbench.

**What the arrows point to.**

1. **Run to look at.** Chooses the run whose single run figures are drawn.
2. **Save report.** Saves all results and figures as one HTML report.
3. **Put the logs in the list of loaded logs of the Log plots tab.** Keeps the Daisy logs of the runs for the Log plots and Mass balance tabs.
4. **Open the set-up in the editor.** Opens the Daisy set-up of the run in the Set-up editor.
5. **Mean per season.** The mean of each result over the seasons, one column for each run.

### 31.3.2 Yield of each crop, season by season (nitrogen)

Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

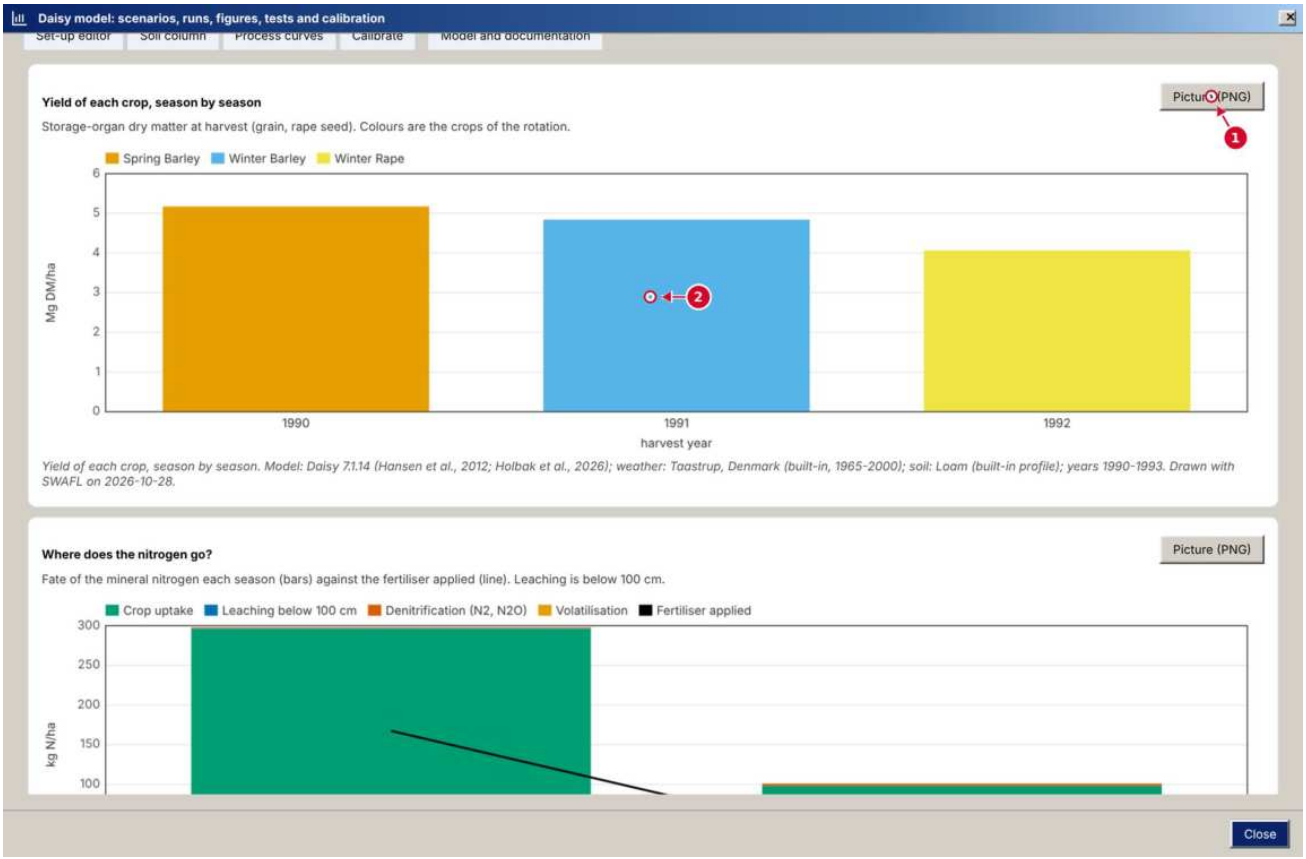



Figure 31.11: Yield of each crop, season by season (nitrogen). Screen 67 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

### 31.3.3 Where does the nitrogen go? (nitrogen)

The nitrogen balance of the run: what is applied, taken up by the crop, leached, denitrified and left in the soil.

Figure 31.12: Where does the nitrogen go? (nitrogen). Screen 68 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The nitrogen balance of the run: what is applied, taken up by the crop, leached, denitrified and left in the soil.

### 31.3.4 Nitrate in the soil solution (nitrogen)

Nitrate concentration of the soil solution against depth and time.

Figure 31.13: Nitrate in the soil solution (nitrogen). Screen 69 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Nitrate concentration of the soil solution against depth and time.

### 31.3.5 Annual water balance (nitrogen)

Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

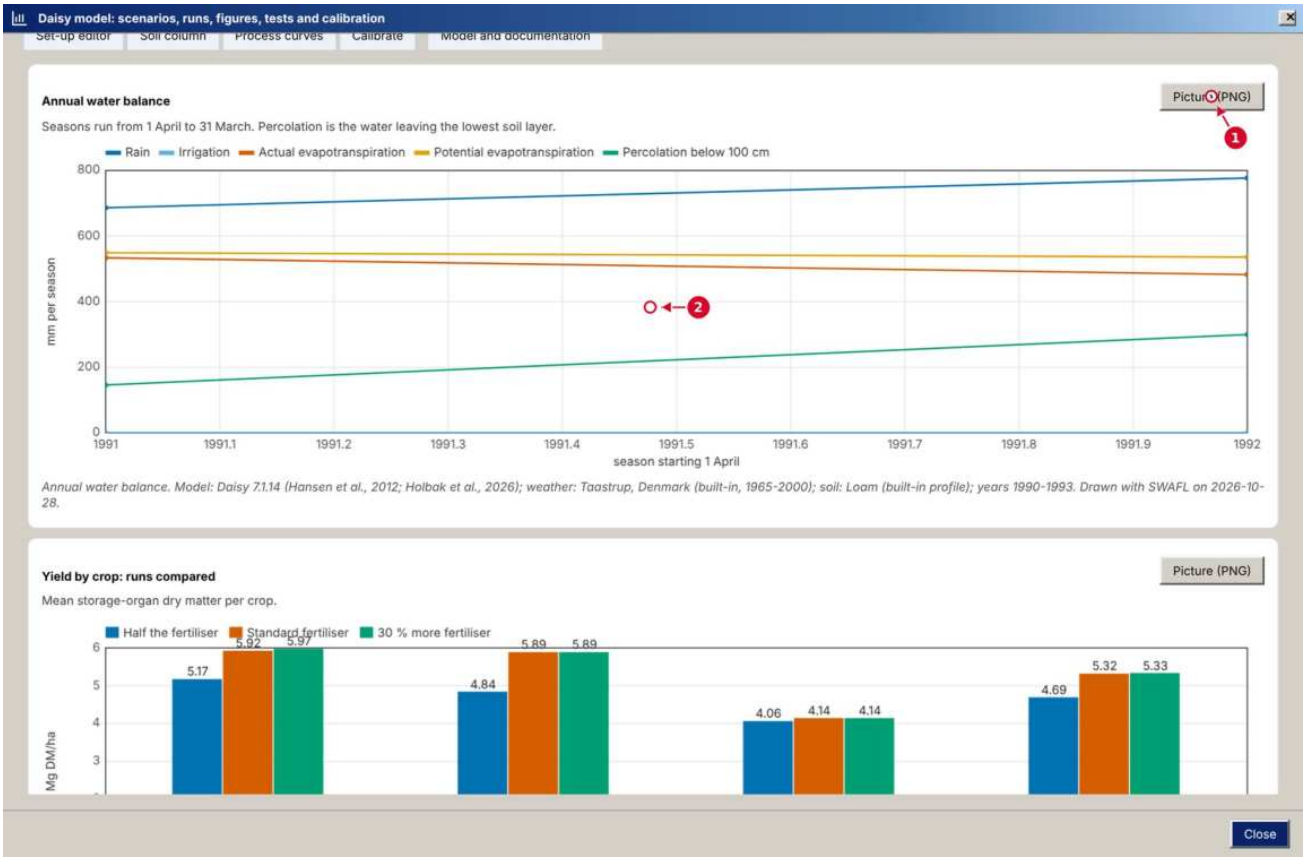



Figure 31.14: Annual water balance (nitrogen). Screen 70 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

### 31.3.6 Yield by crop: runs compared (nitrogen)

The mean yield of each crop in every ticked run, side by side.

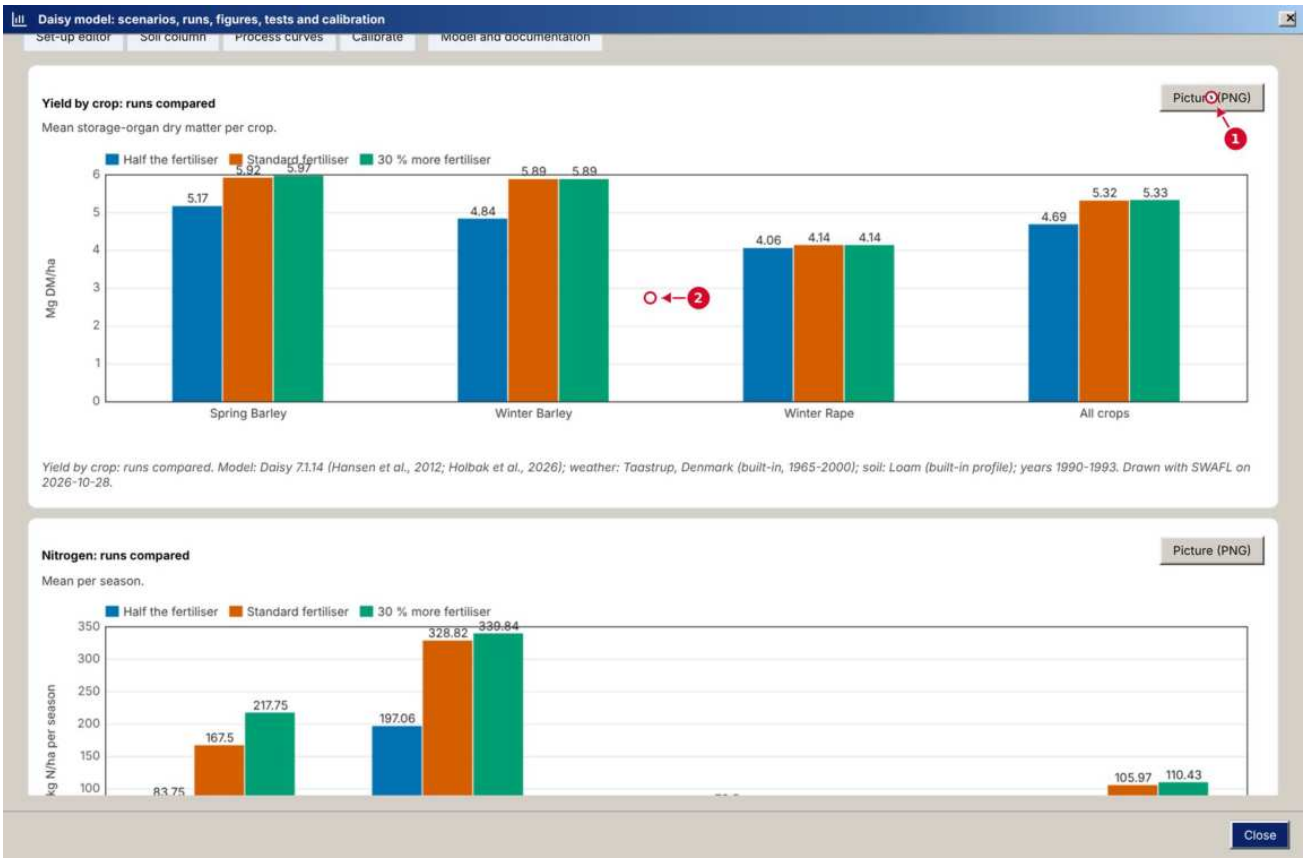



Figure 31.15: Yield by crop: runs compared (nitrogen). Screen 71 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The mean yield of each crop in every ticked run, side by side.

### 31.3.7 Nitrogen: runs compared (nitrogen)

Nitrate leached, denitrification and nitrogen uptake of the runs side by side.

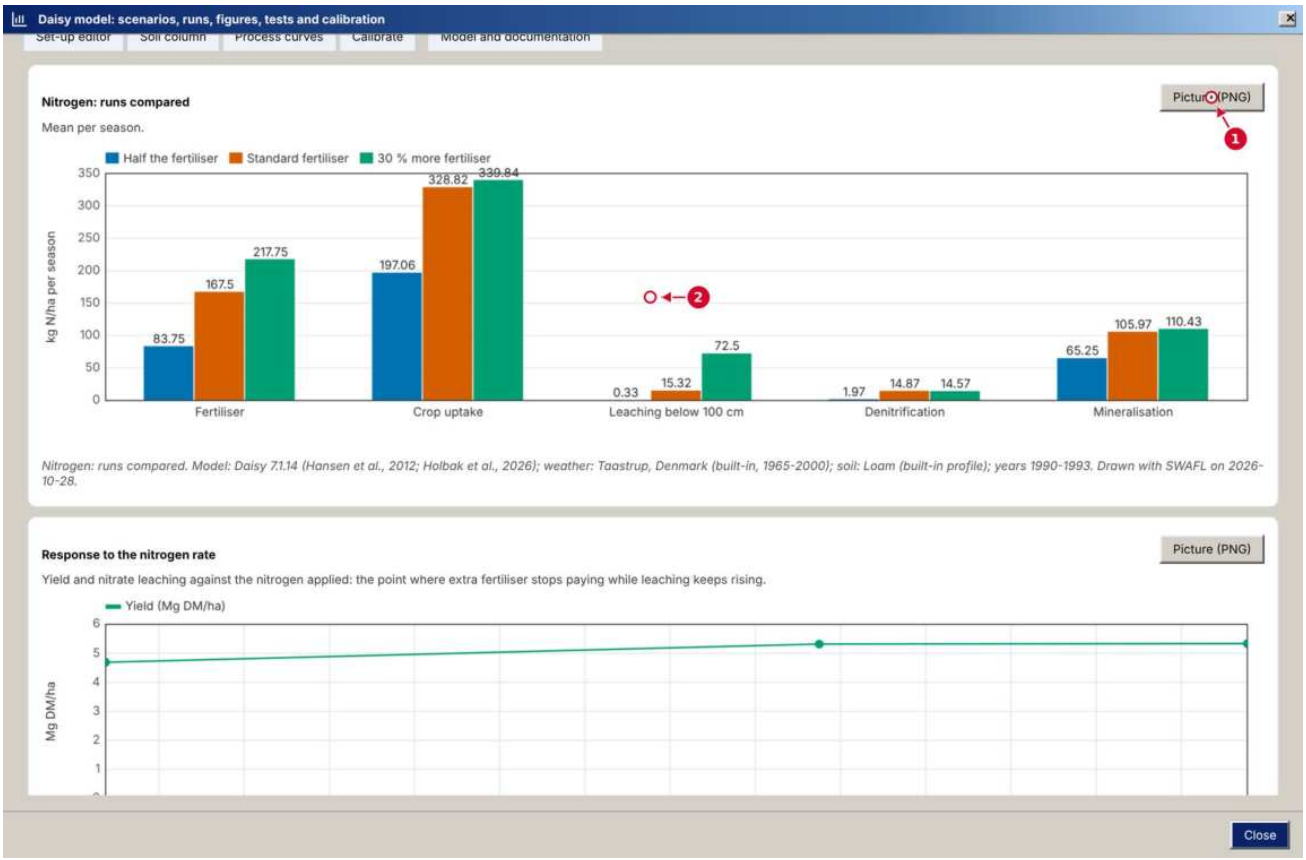



Figure 31.16: Nitrogen: runs compared (nitrogen). Screen 72 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Nitrate leached, denitrification and nitrogen uptake of the runs side by side.

### 31.3.8 Response to the nitrogen rate (nitrogen)

Yield and nitrate leaching against the fertiliser rate across the runs.

Figure 31.17: Response to the nitrogen rate (nitrogen). Screen 73 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Yield and nitrate leaching against the fertiliser rate across the runs.

### 31.3.9 Yield through the years: runs compared (nitrogen)

The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

Figure 31.18: Yield through the years: runs compared (nitrogen). Screen 74 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

## 31.4 Results of the climate scenario

### 31.4.1 Results of the climate question: the table of means

The climate question was run on the built-in farm for the years 1990 to 1993 with the built-in Taastrup weather. After the run the results appear under the options: a list that chooses the run to look at, three buttons, a table of the mean per season and the figures.

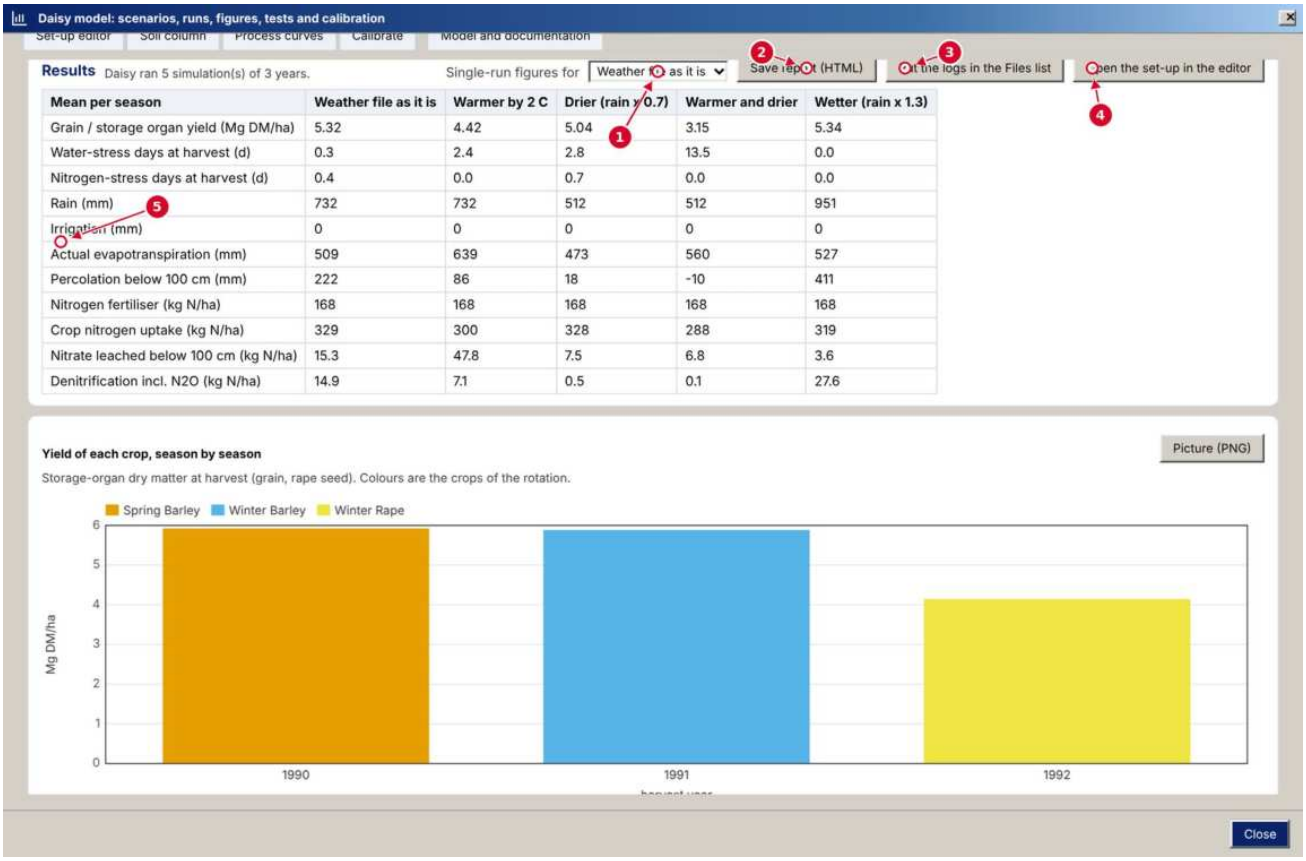



Figure 31.19: Results of the climate question: the table of means. Screen 75 of the Daisy workbench.

**What the arrows point to.**

1. **Run to look at.** Chooses the run whose single run figures are drawn.
2. **Save report.** Saves all results and figures as one HTML report.
3. **Put the logs in the list of loaded logs of the Log plots tab.** Keeps the Daisy logs of the runs for the Log plots and Mass balance tabs.
4. **Open the set-up in the editor.** Opens the Daisy set-up of the run in the Set-up editor.
5. **Mean per season.** The mean of each result over the seasons, one column for each run.

### 31.4.2 Yield of each crop, season by season (climate)

Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

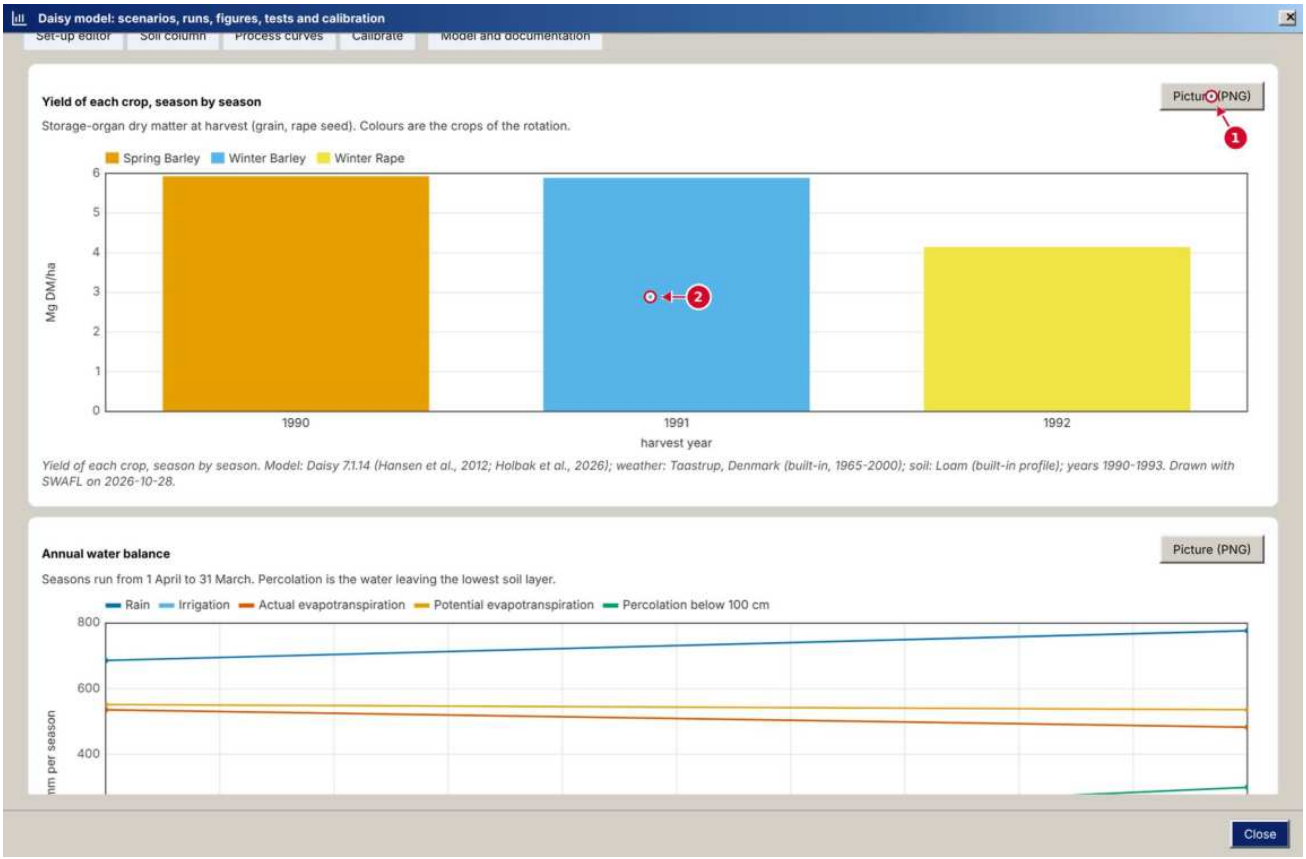



Figure 31.20: Yield of each crop, season by season (climate). Screen 76 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

### 31.4.3 Annual water balance (climate)

Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

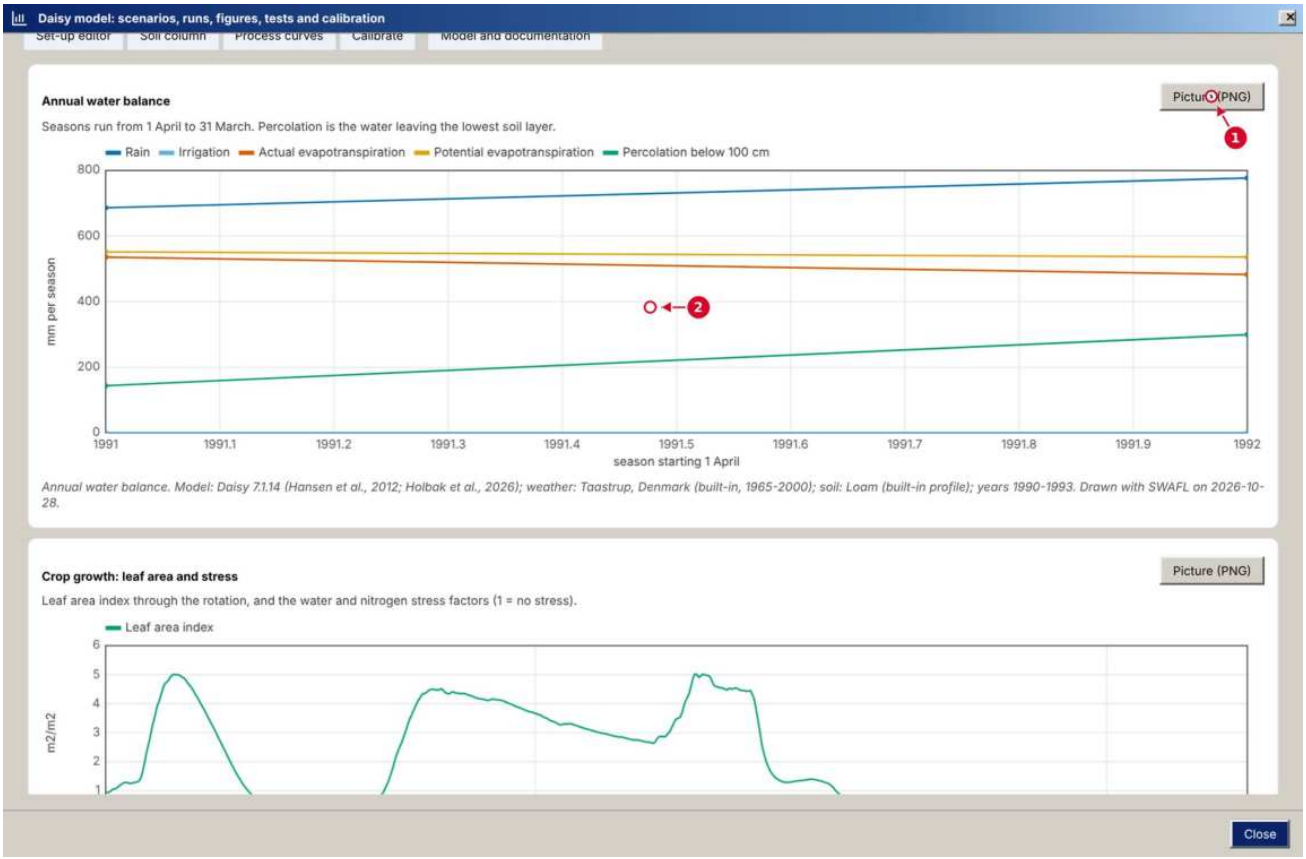



Figure 31.21: Annual water balance (climate). Screen 77 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

### 31.4.4 Crop growth: leaf area and stress (climate)

Leaf area index with the water and nitrogen stress of the crop through the seasons.

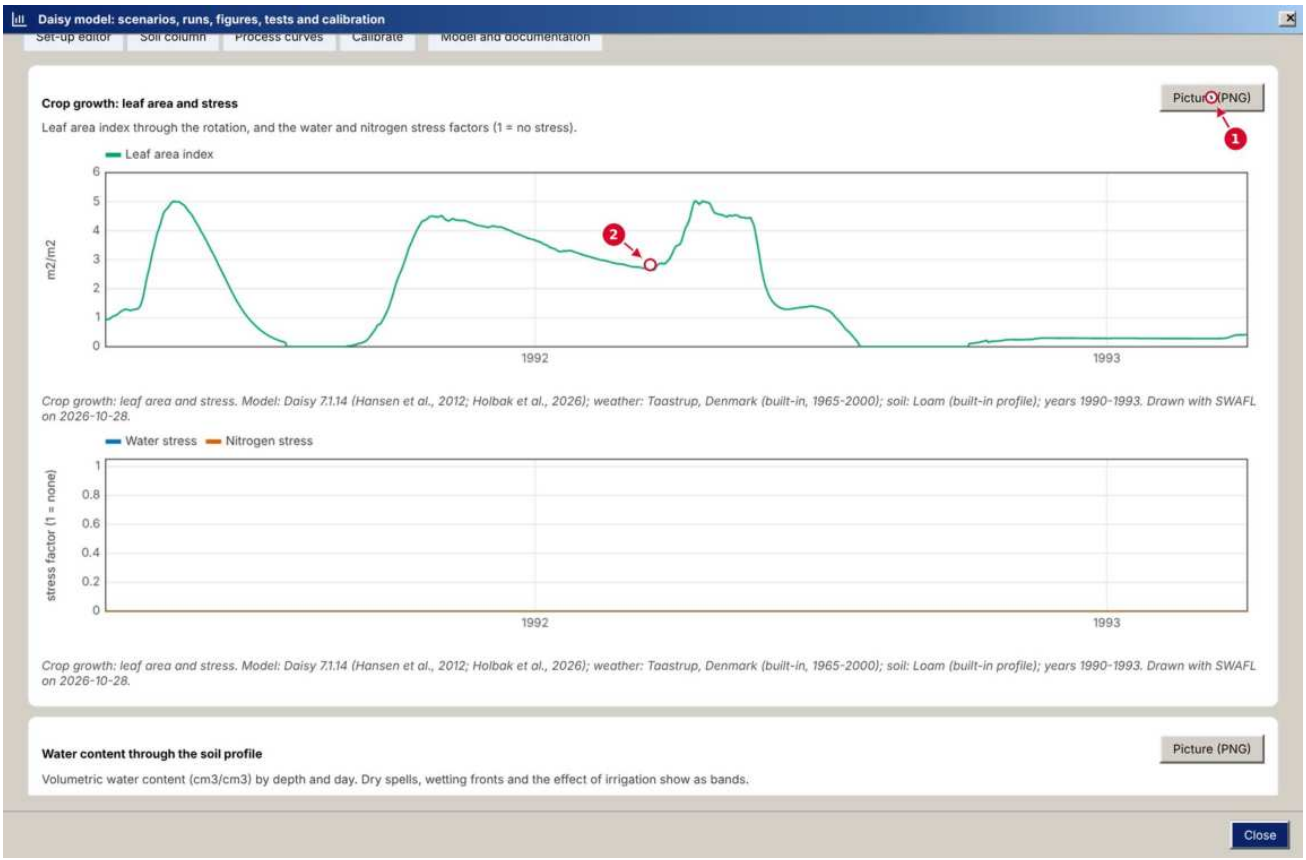



Figure 31.22: Crop growth: leaf area and stress (climate). Screen 78 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Leaf area index with the water and nitrogen stress of the crop through the seasons.

### 31.4.5 Water content through the soil profile (climate)

Soil water content against depth and time, from the daily log of Daisy.

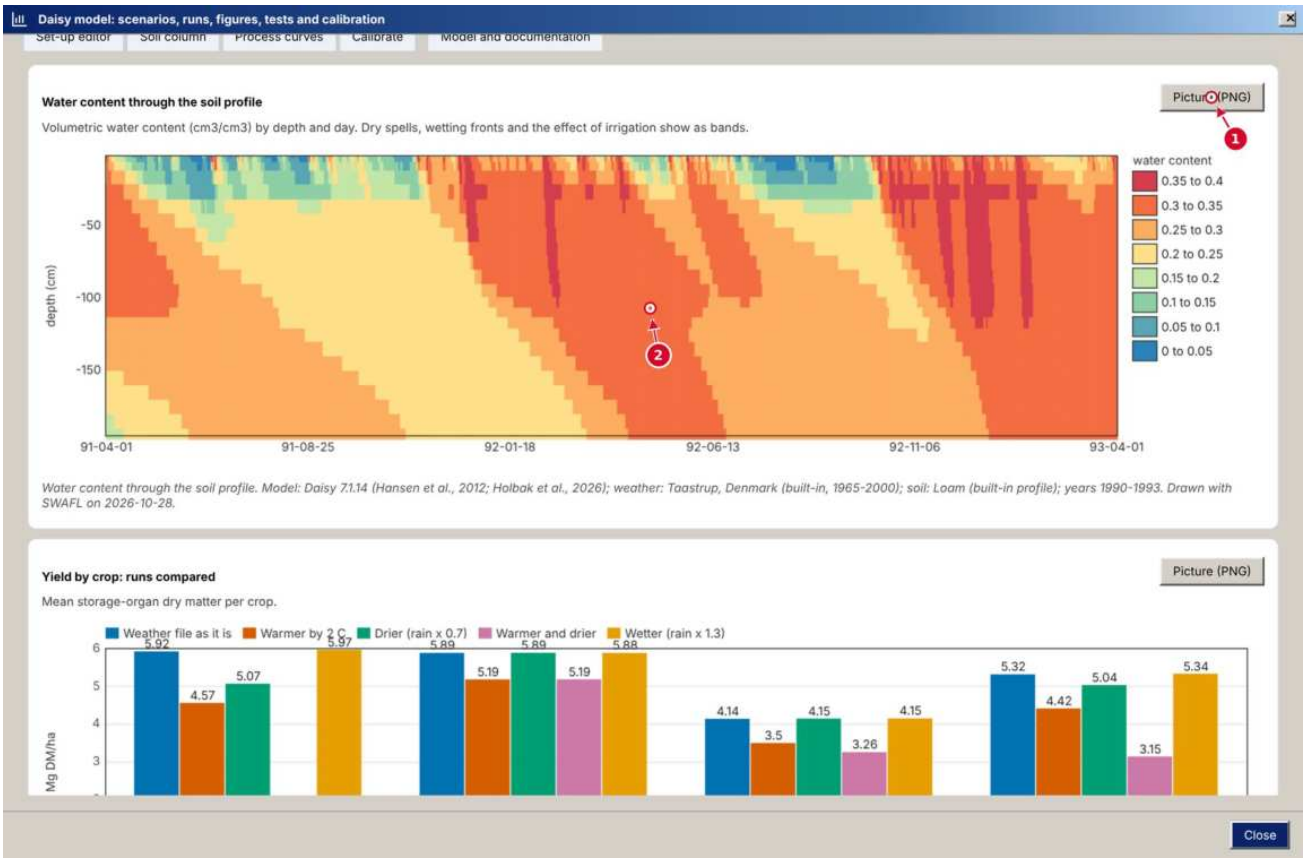



Figure 31.23: Water content through the soil profile (climate). Screen 79 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Soil water content against depth and time, from the daily log of Daisy.

### 31.4.6 Yield by crop: runs compared (climate)

The mean yield of each crop in every ticked run, side by side.

Figure 31.24: Yield by crop: runs compared (climate). Screen 80 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The mean yield of each crop in every ticked run, side by side.

### 31.4.7 Water balance: runs compared (climate)

Irrigation, evapotranspiration and drainage of the runs side by side.

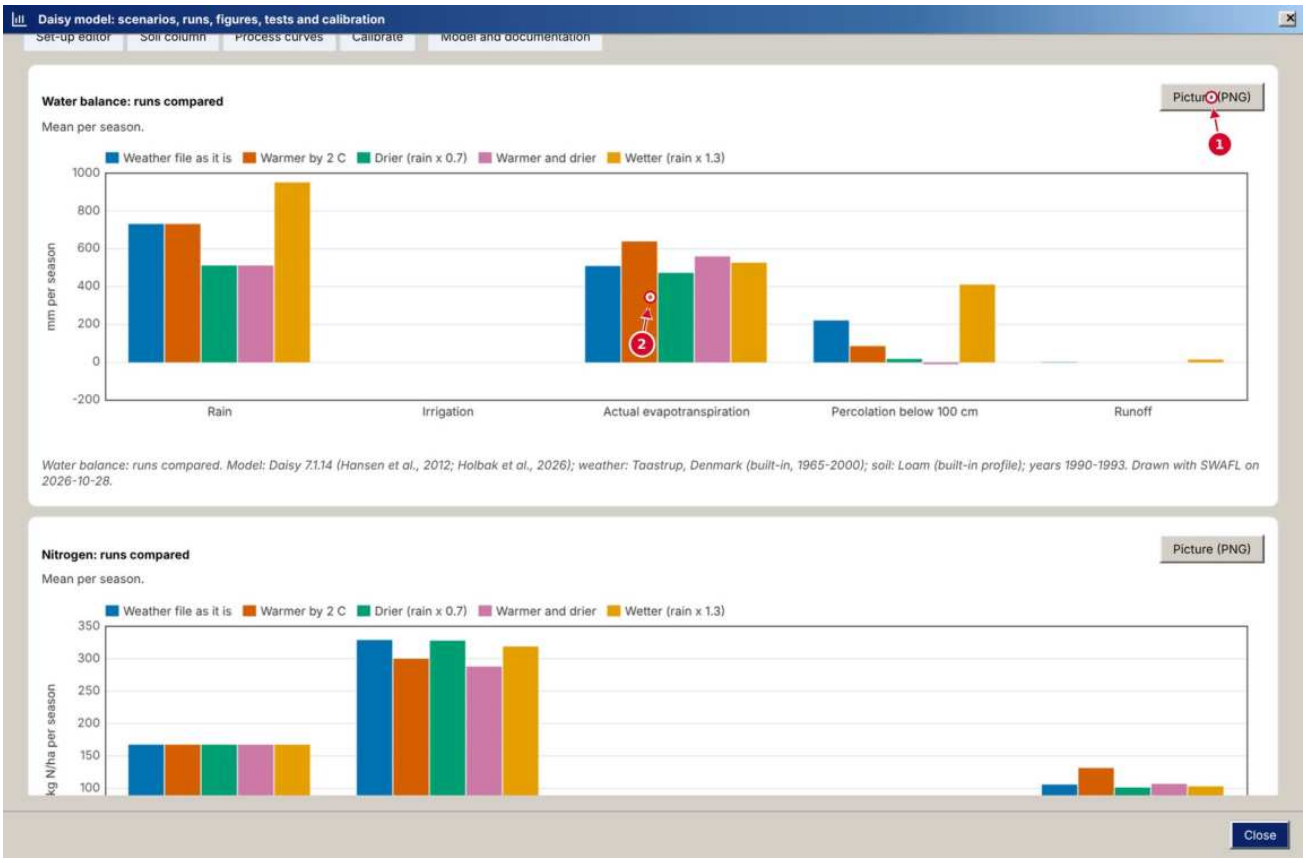



Figure 31.25: Water balance: runs compared (climate). Screen 81 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Irrigation, evapotranspiration and drainage of the runs side by side.

### 31.4.8 Nitrogen: runs compared (climate)

Nitrate leached, denitrification and nitrogen uptake of the runs side by side.

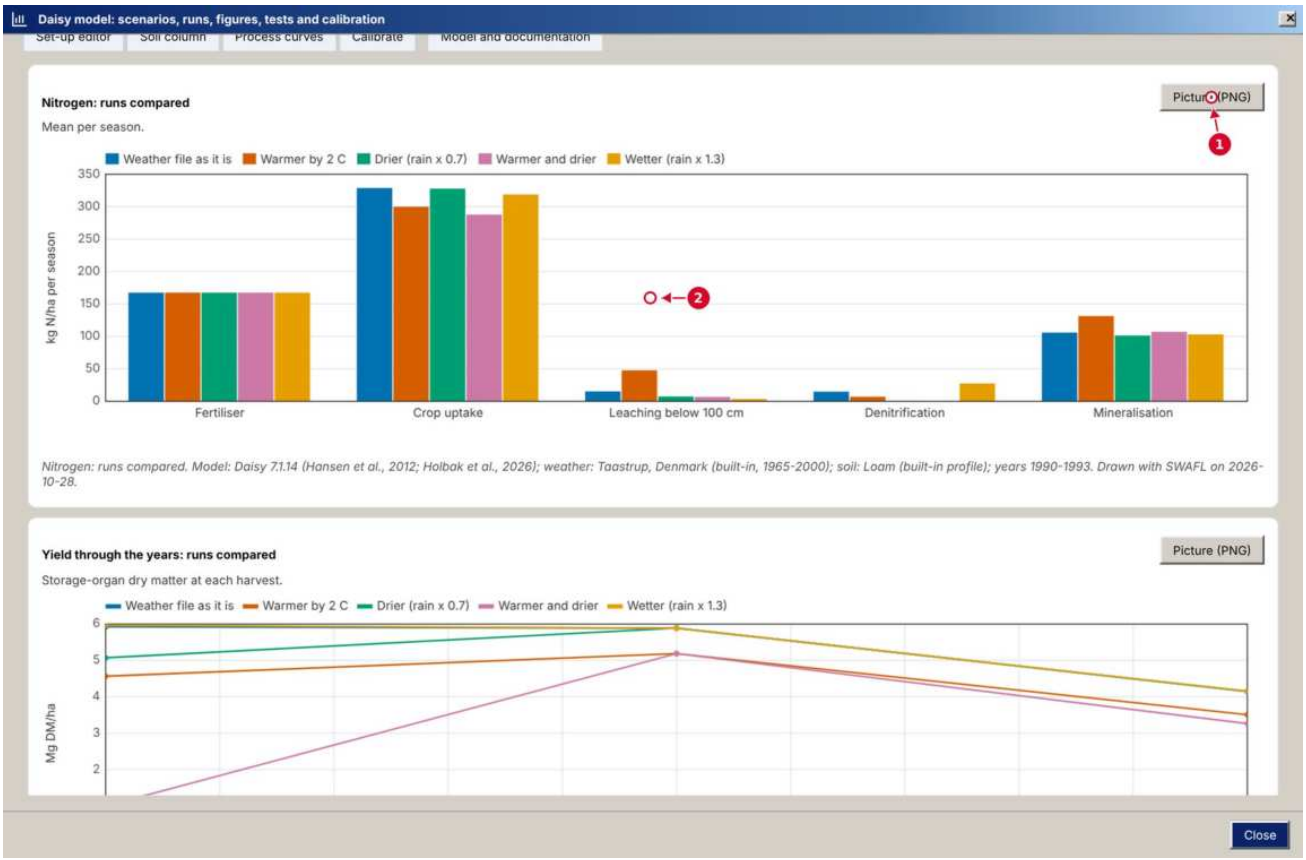



Figure 31.26: Nitrogen: runs compared (climate). Screen 82 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Nitrate leached, denitrification and nitrogen uptake of the runs side by side.

### 31.4.9 Yield through the years: runs compared (climate)

The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

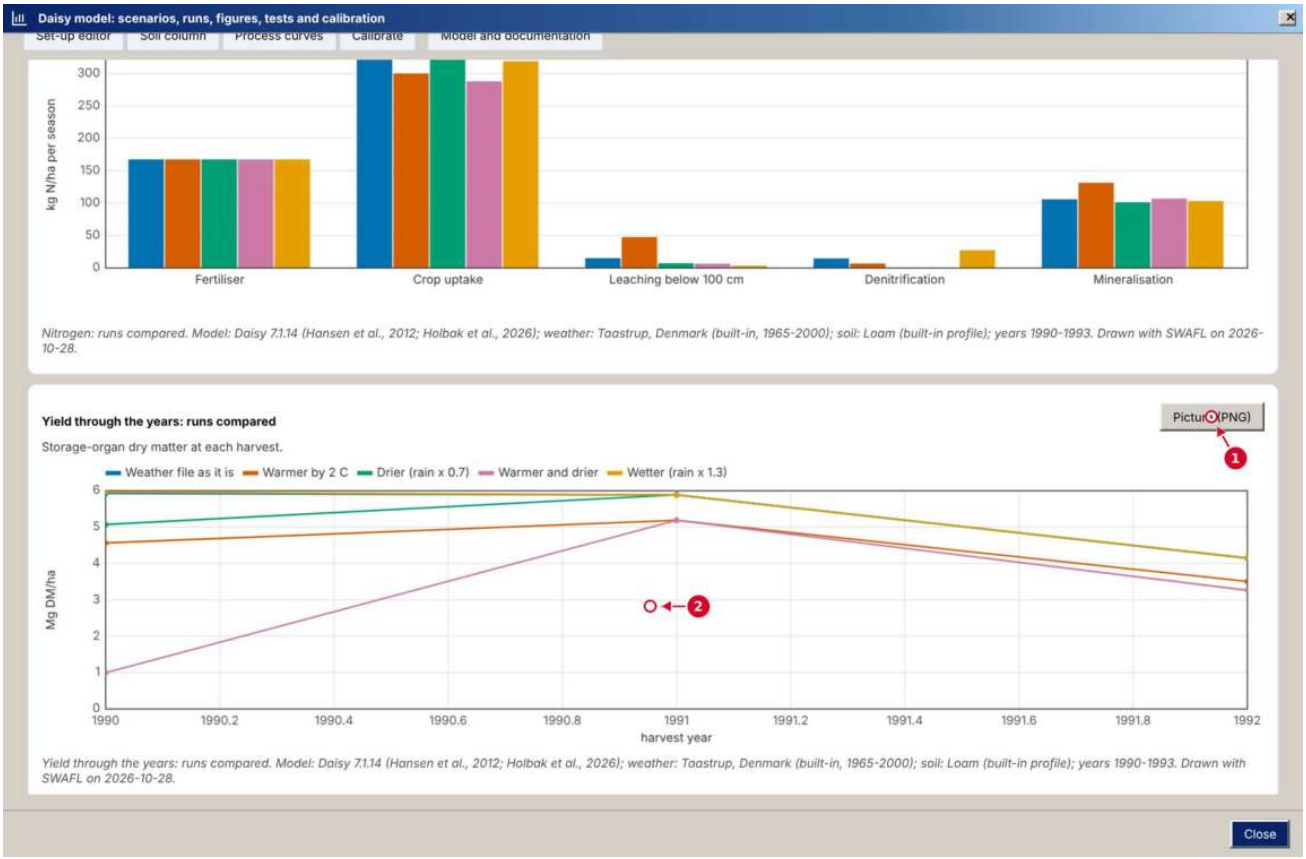



Figure 31.27: Yield through the years: runs compared (climate). Screen 83 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

## 31.5 Results of the chemical scenario

### 31.5.1 Results of the chemical question: the table of means

The chemical question was run on the built-in farm for the years 1990 to 1993 with the built-in Taastrup weather. After the run the results appear under the options: a list that chooses the run to look at, three buttons, a table of the mean per season and the figures.

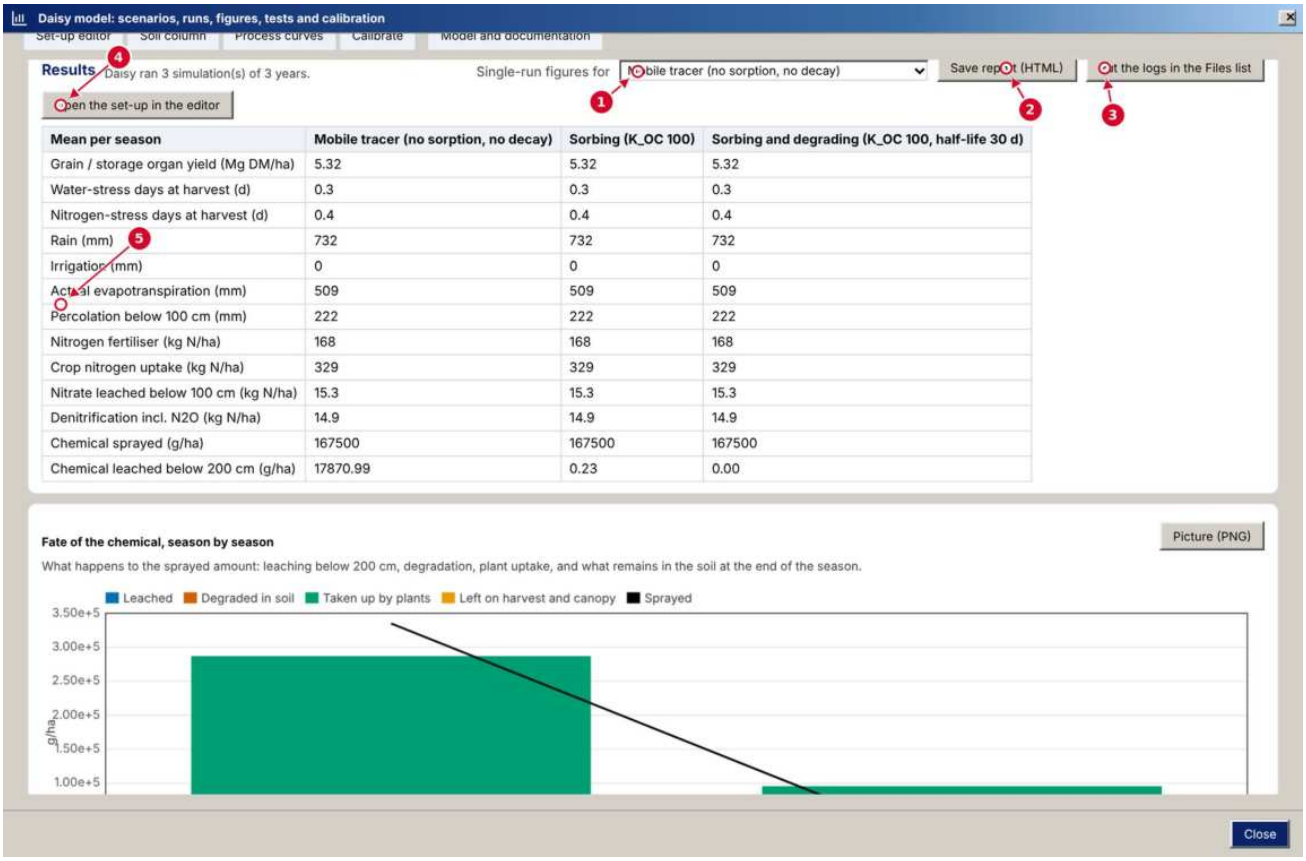



Figure 31.28: Results of the chemical question: the table of means. Screen 84 of the Daisy workbench.

**What the arrows point to.**

1. **Run to look at.** Chooses the run whose single run figures are drawn.
2. **Save report.** Saves all results and figures as one HTML report.
3. **Put the logs in the list of loaded logs of the Log plots tab.** Keeps the Daisy logs of the runs for the Log plots and Mass balance tabs.
4. **Open the set-up in the editor.** Opens the Daisy set-up of the run in the Set-up editor.
5. **Mean per season.** The mean of each result over the seasons, one column for each run.

### 31.5.2 Fate of the chemical, season by season (chemical)

How much of the chemical is in the soil, decayed, leached or taken up in each season.

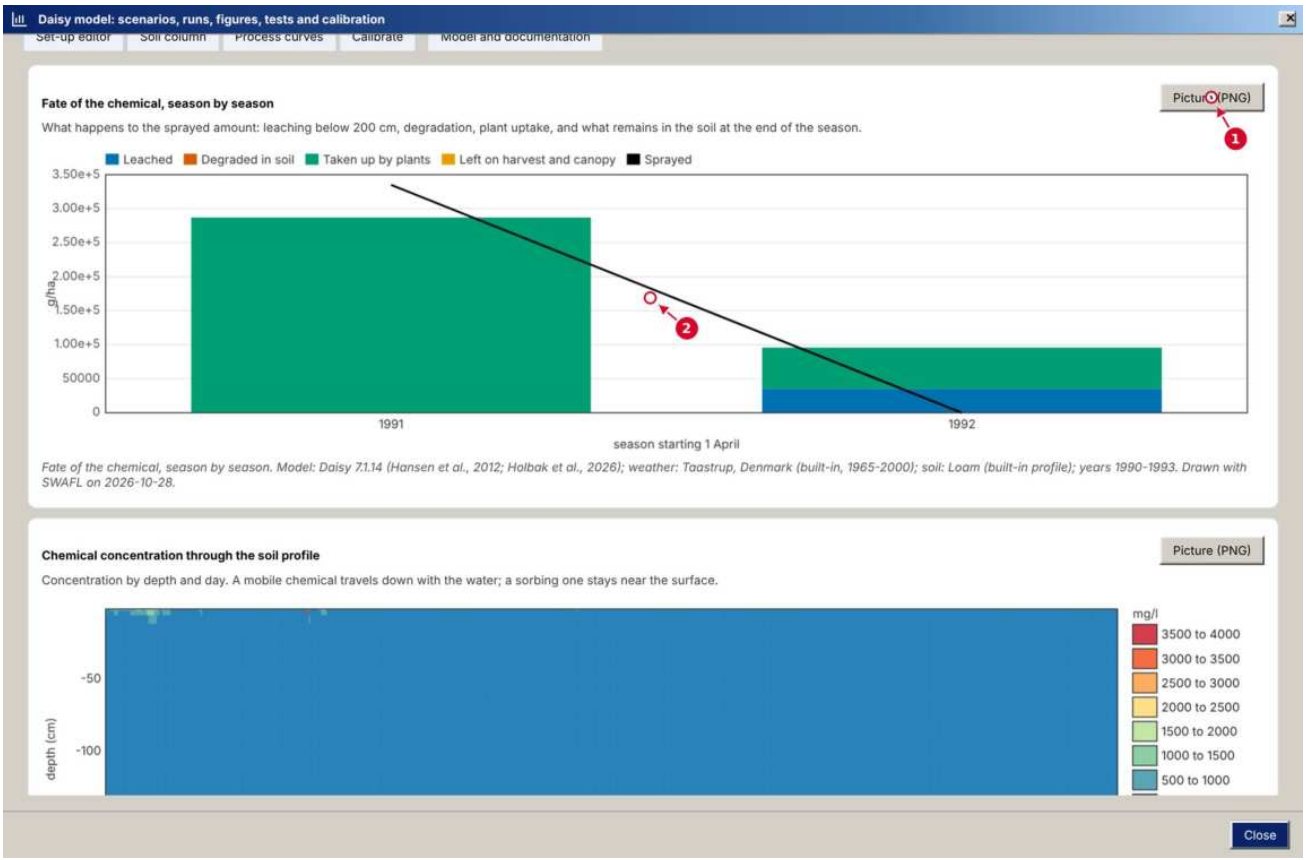



Figure 31.29: Fate of the chemical, season by season (chemical). Screen 85 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** How much of the chemical is in the soil, decayed, leached or taken up in each season.

### 31.5.3 Chemical concentration through the soil profile (chemical)

The concentration of the chemical against depth and time.

Figure 31.30: Chemical concentration through the soil profile (chemical). Screen 86 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The concentration of the chemical against depth and time.

### 31.5.4 Annual water balance (chemical)

Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

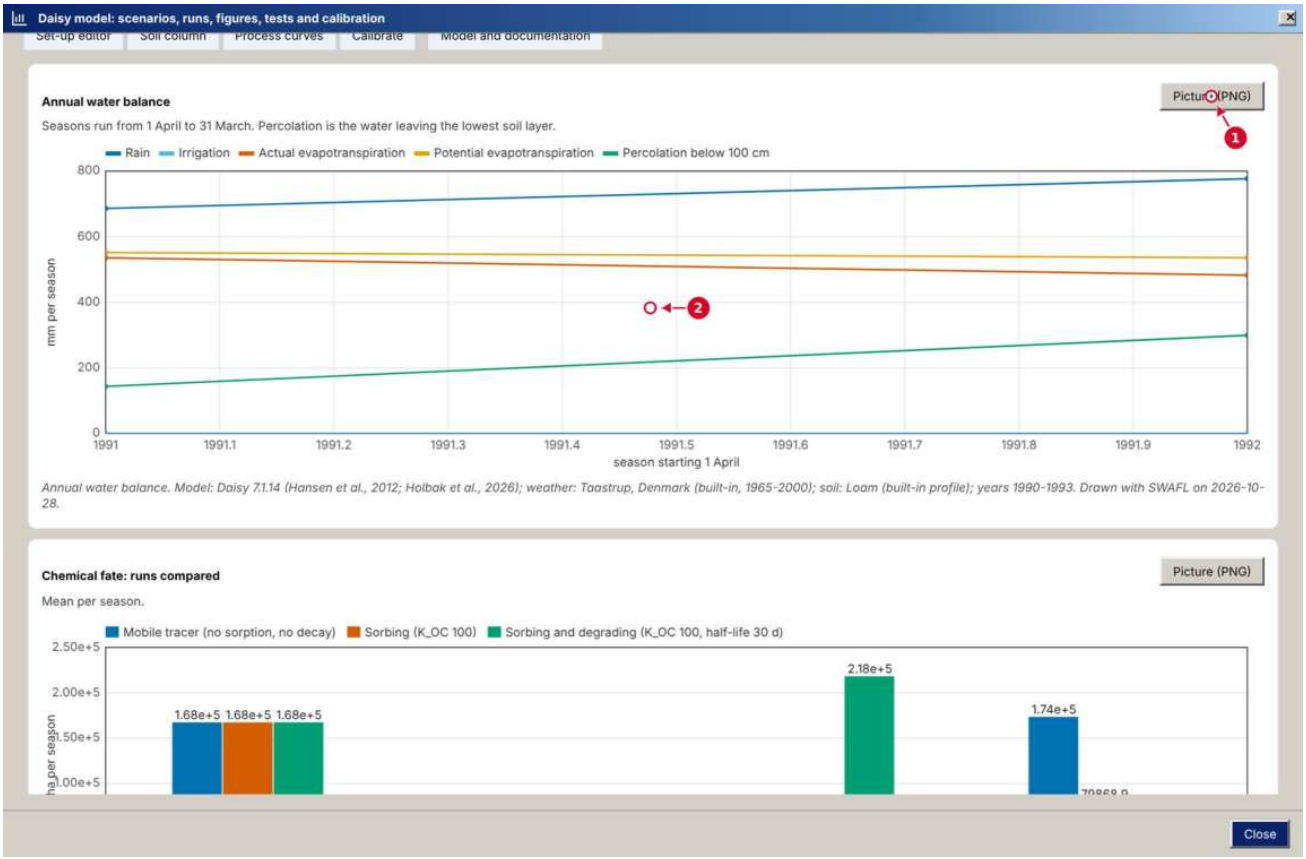



Figure 31.31: Annual water balance (chemical). Screen 87 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

### 31.5.5 Chemical fate: runs compared (chemical)

The fate of the chemical in the runs side by side: mobile tracer, sorbed and quickly decaying chemical.

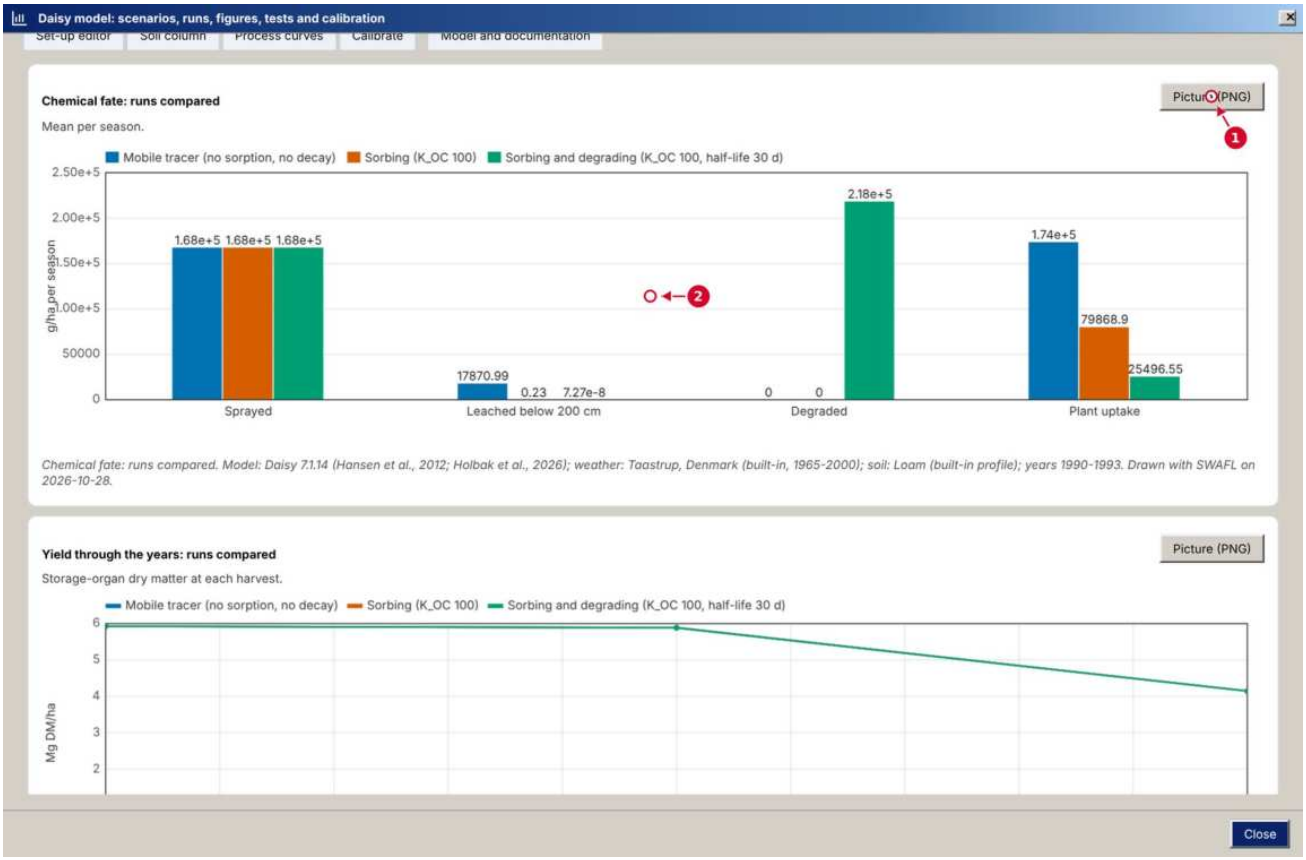



Figure 31.32: Chemical fate: runs compared (chemical). Screen 88 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The fate of the chemical in the runs side by side: mobile tracer, sorbed and quickly decaying chemical.

### 31.5.6 Yield through the years: runs compared (chemical)

The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

Figure 31.33: Yield through the years: runs compared (chemical). Screen 89 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

## 31.6 Results of the soil scenario

### 31.6.1 Results of the soil question: the table of means

The soil question was run on the built-in farm for the years 1990 to 1993 with the built-in Taastrup weather. After the run the results appear under the options: a list that chooses the run to look at, three buttons, a table of the mean per season and the figures.

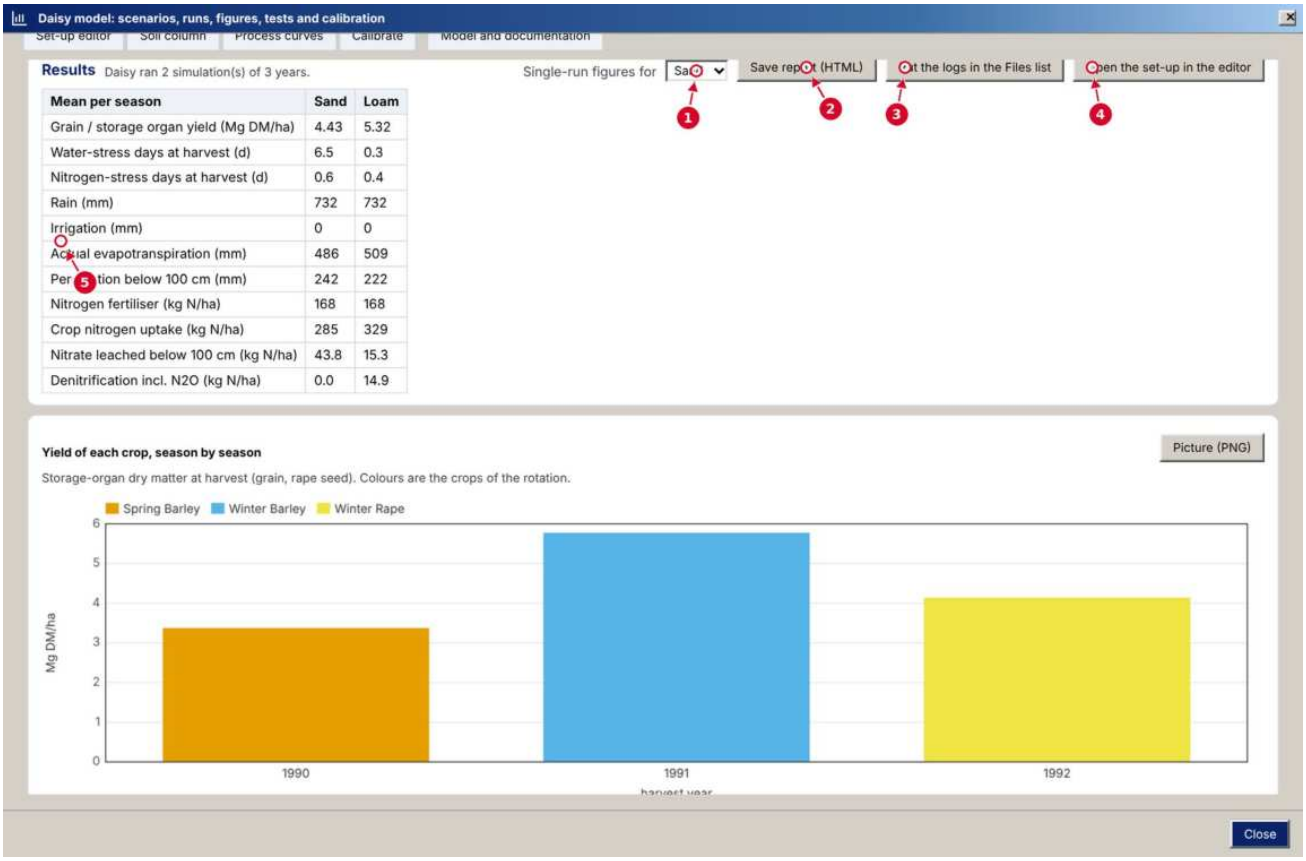



Figure 31.34: Results of the soil question: the table of means. Screen 90 of the Daisy workbench.

**What the arrows point to.**

1. **Run to look at.** Chooses the run whose single run figures are drawn.
2. **Save report.** Saves all results and figures as one HTML report.
3. **Put the logs in the list of loaded logs of the Log plots tab.** Keeps the Daisy logs of the runs for the Log plots and Mass balance tabs.
4. **Open the set-up in the editor.** Opens the Daisy set-up of the run in the Set-up editor.
5. **Mean per season.** The mean of each result over the seasons, one column for each run.

## 31.6.2 Yield of each crop, season by season (soil)

Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

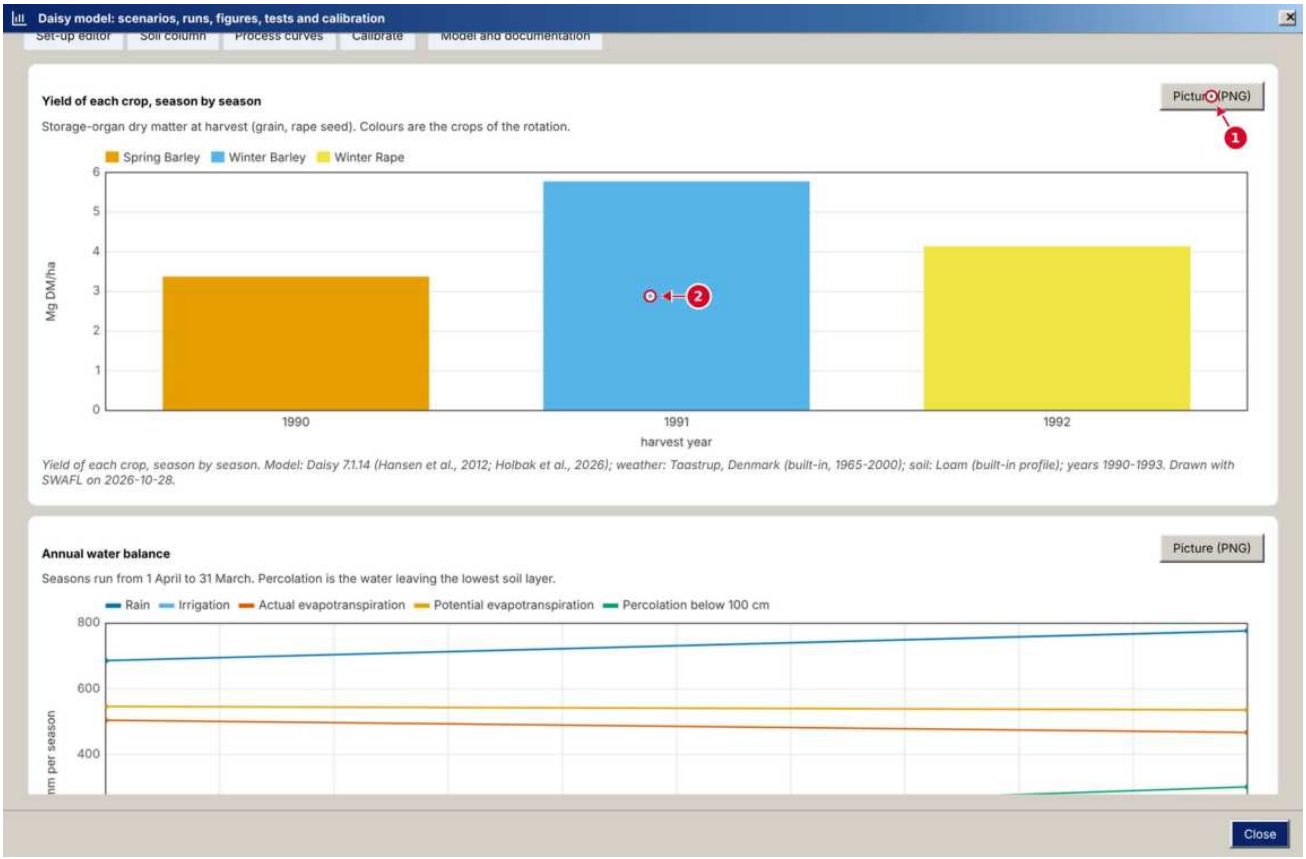



Figure 31.35: Yield of each crop, season by season (soil). Screen 91 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Grain or storage organ yield of each crop of the rotation in each year for the run chosen in the list above, in Mg dry matter per hectare.

### 31.6.3 Annual water balance (soil)

Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

Figure 31.36: Annual water balance (soil). Screen 92 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Rain, irrigation, evapotranspiration, drainage below the root zone and the change in soil water, summed for each year.

### 31.6.4 Water content through the soil profile (soil)

Soil water content against depth and time, from the daily log of Daisy.

Figure 31.37: Water content through the soil profile (soil). Screen 93 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Soil water content against depth and time, from the daily log of Daisy.

### 31.6.5 Soil water suction (pF) (soil)

Soil water suction on the pF scale against depth and time; irrigation triggers act on this quantity.

Figure 31.38: Soil water suction (pF) (soil). Screen 94 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Soil water suction on the pF scale against depth and time; irrigation triggers act on this quantity.

### 31.6.6 Nitrate in the soil solution (soil)

Nitrate concentration of the soil solution against depth and time.

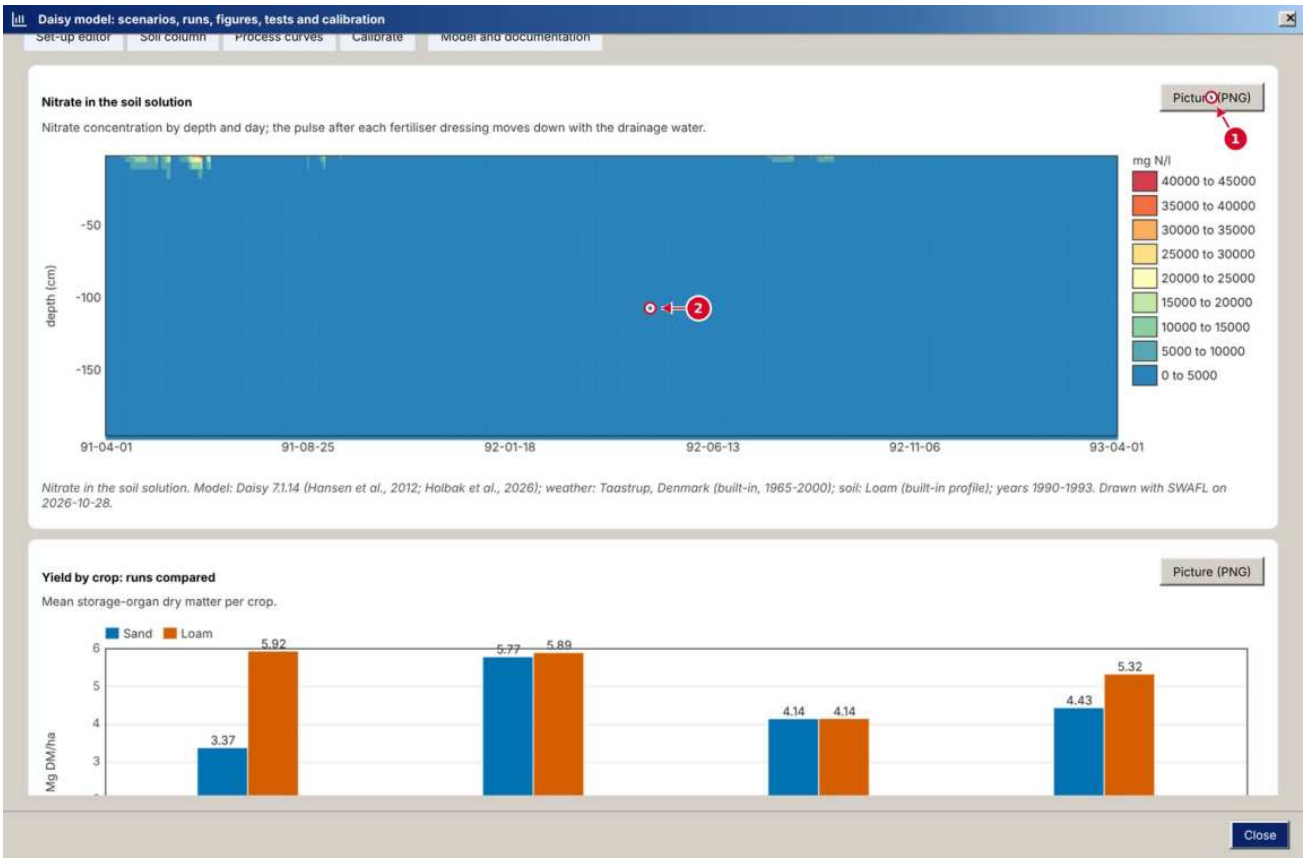



Figure 31.39: Nitrate in the soil solution (soil). Screen 95 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Nitrate concentration of the soil solution against depth and time.

### 31.6.7 Yield by crop: runs compared (soil)

The mean yield of each crop in every ticked run, side by side.

Figure 31.40: Yield by crop: runs compared (soil). Screen 96 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The mean yield of each crop in every ticked run, side by side.

### 31.6.8 Water balance: runs compared (soil)

Irrigation, evapotranspiration and drainage of the runs side by side.

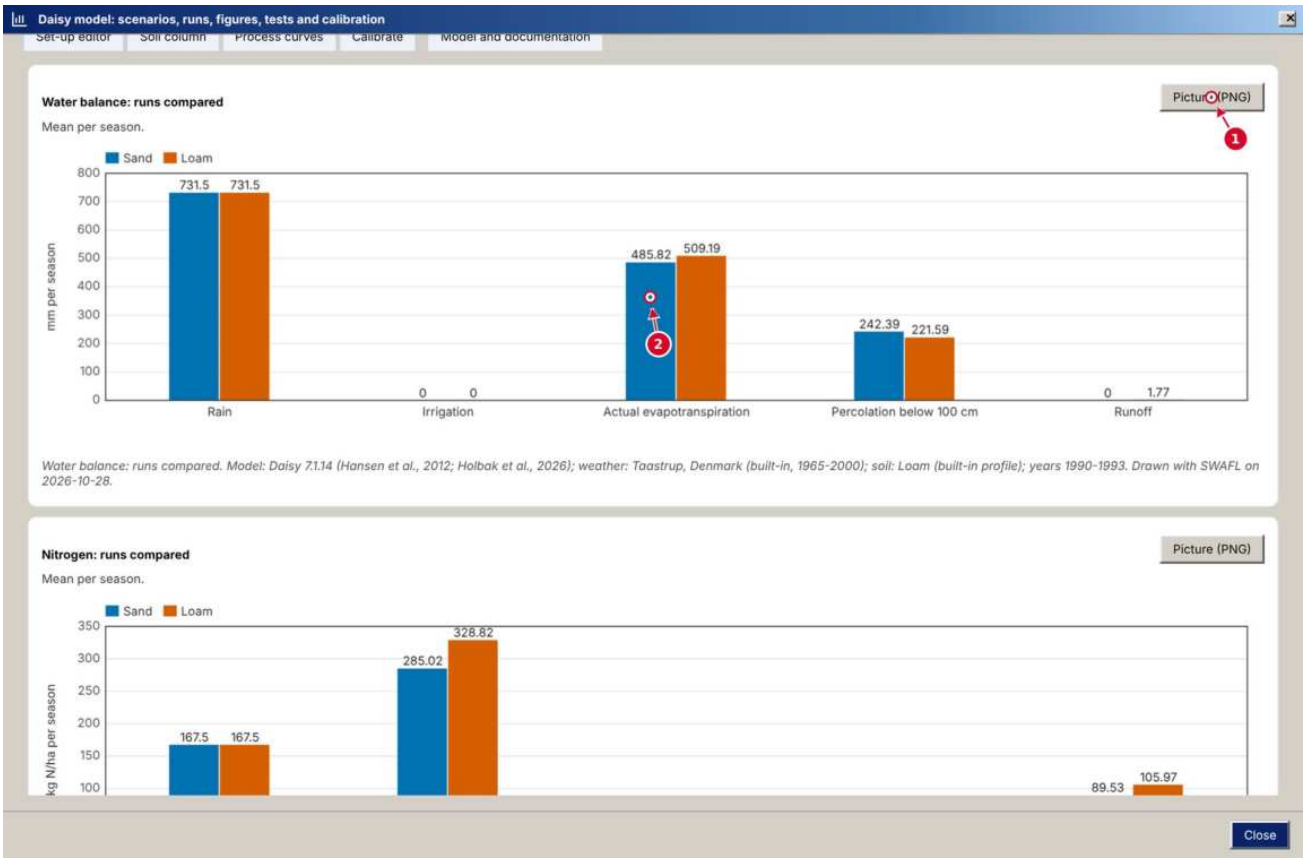



Figure 31.41: Water balance: runs compared (soil). Screen 97 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Irrigation, evapotranspiration and drainage of the runs side by side.

### 31.6.9 Nitrogen: runs compared (soil)

Nitrate leached, denitrification and nitrogen uptake of the runs side by side.

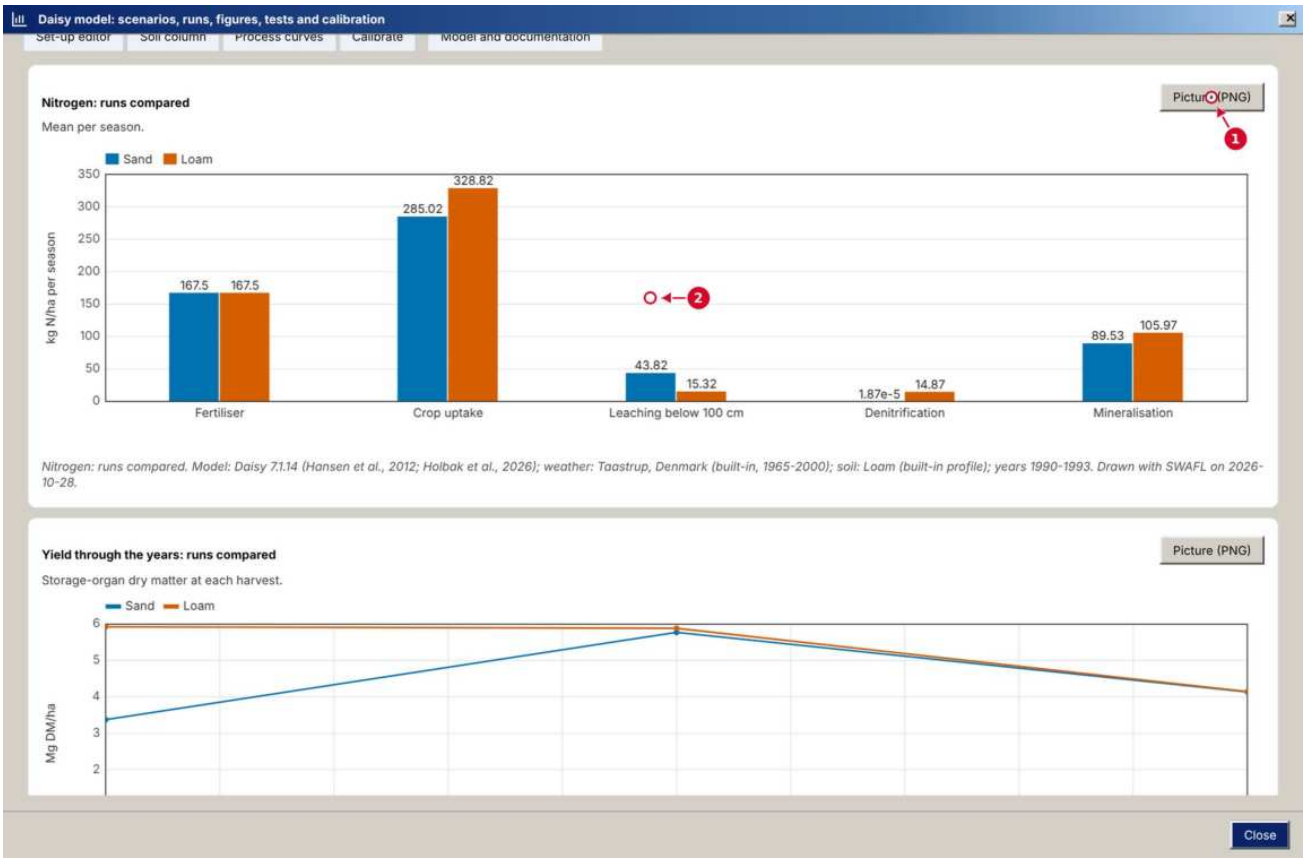



Figure 31.42: Nitrogen: runs compared (soil). Screen 98 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** Nitrate leached, denitrification and nitrogen uptake of the runs side by side.

## 31.6.10 Yield through the years: runs compared (soil)

The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

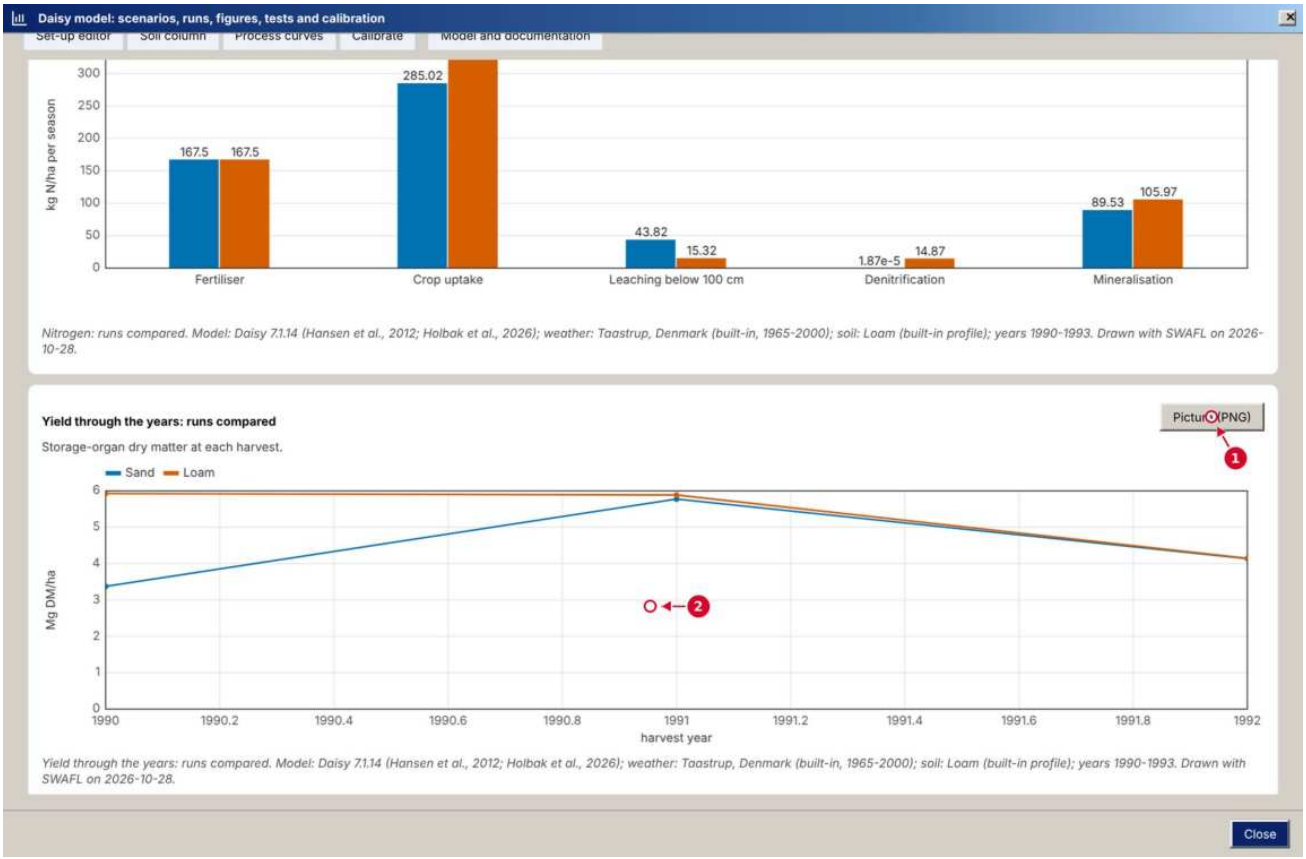



Figure 31.43: Yield through the years: runs compared (soil). Screen 99 of the Daisy workbench.

**What the arrows point to.**

1. **Picture (PNG).** Saves this figure as a picture, with its title and axes and nothing else: no citation and no caption is added.
2. **Figure.** The yield of each year in every run, to show how much of a difference is steady and how much depends on the weather of the year.

## 31.7 Processes, plots, balance and tests

### 31.7.1 Processes: the map

Processes is a map of the model after Fig. 1 of the Daisy paper. Each box is a process; clicking it explains it and draws its figures.

**The original figure, or every component in one figure.** The *Figure* box above the map chooses between two drawings. *Fig. 1 of the Daisy paper (original)*, the default, is the map as before, with the components that Fig. 1 has no box for as buttons under it. *Fig. 1 with every component of Daisy (standalone)* is the same figure, unchanged above the ground line, with every other component of Daisy drawn under the profile as a box of the same kind as the boxes of the map (the same rectangles, colours and serif labels), in one row group for each module: water and the surface (11), heat (3), nitrogen (1), organic matter and carbon (6), crop (10), management (7), chemicals and solutes (8), and soil and numerics (9). Ten of them come from the reference manual and had no box or button before: vernalization, rubisco nitrogen in the canopy, the water stress effect on production, vegetation without a crop model, reactions and chemical equilibria, maximal exfiltration, geometry and 2D movement, programs and batch runs, summaries and measured data, and units. Every box can be clicked like the boxes of the map. *Save as SVG*, *Save as PNG, 300 dpi* and *PNG, 600 dpi* save the figure shown as one picture.

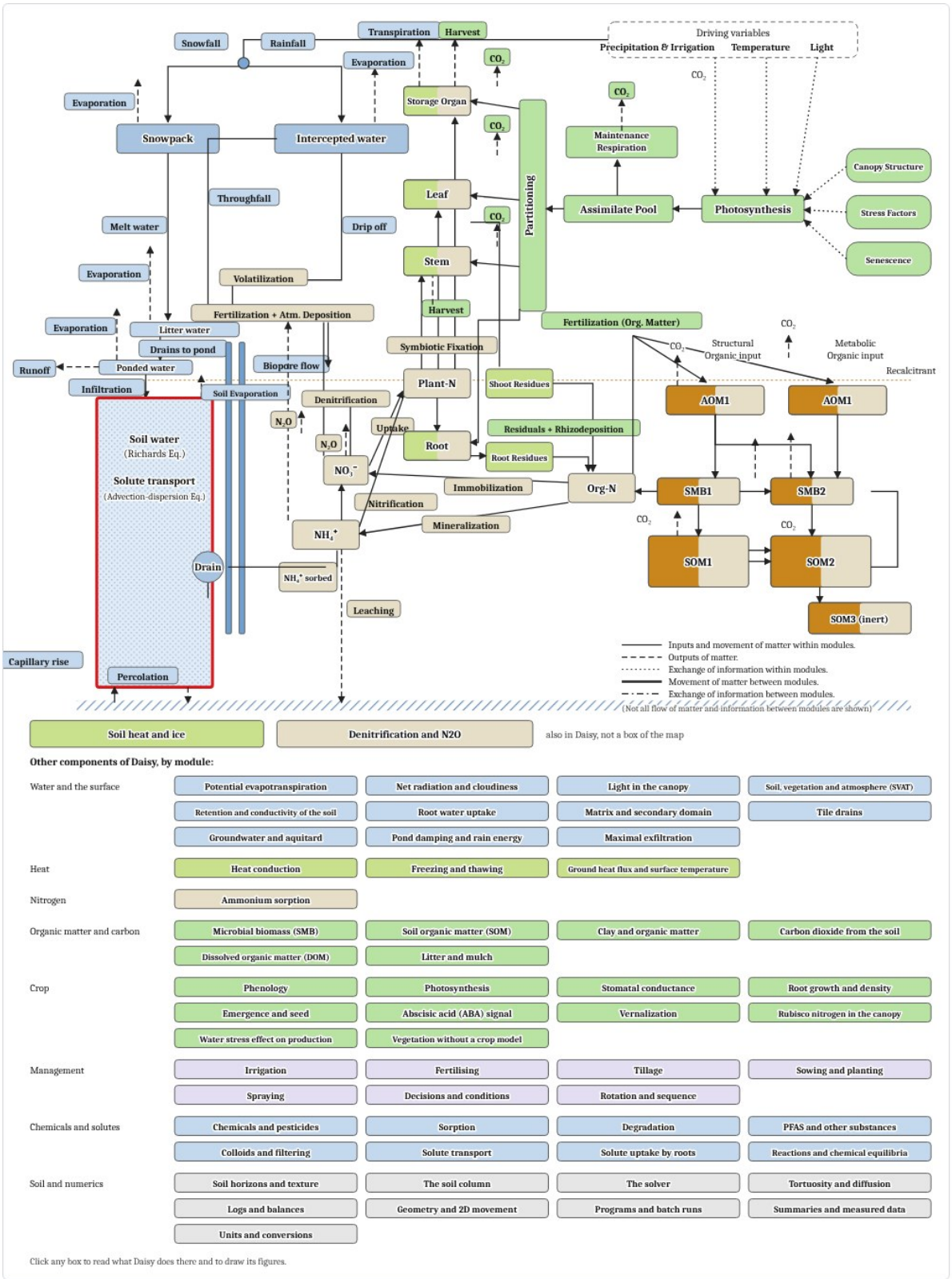



Figure 31.44: The standalone figure: Fig. 1 of the Daisy paper, unchanged, and under the profile every other component of Daisy in the same style, grouped by module (rendered from the program).

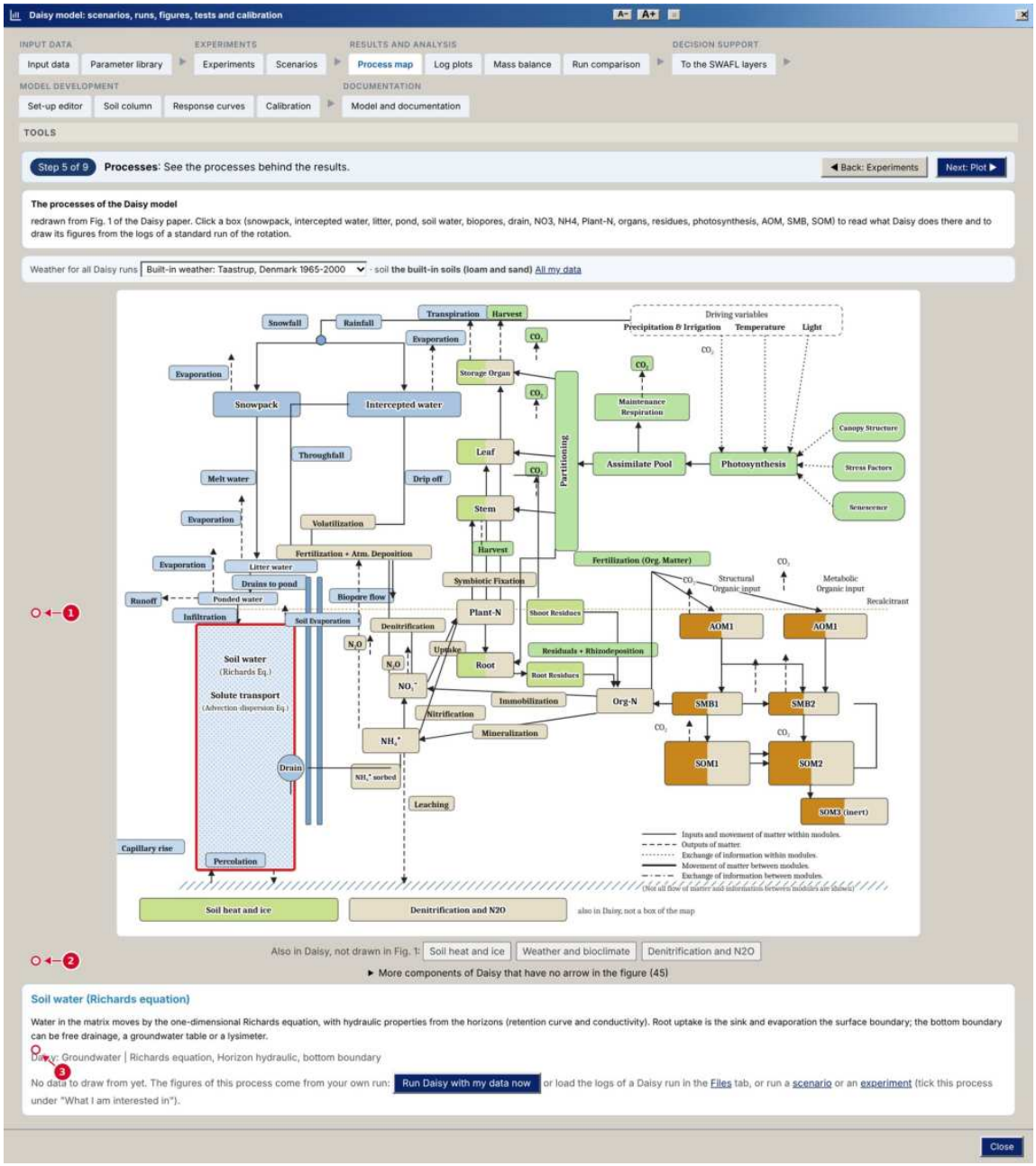



Figure 31.45: Processes: the map. Screen 100 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** One box for each process, coloured by group.
2. **Other processes.** Processes not drawn in the paper figure.
3. **Details.** Text and figures of the chosen process.

### 31.7.2 Processes: Weather and bioclimate

Daisy reads daily or hourly weather (air temperature, global radiation, precipitation, reference evapotranspiration and, if available, wind and vapour pressure) and the Bioclimate module splits the energy and water between canopy, litter, snow, pond and soil. The potential evapotranspiration is shared out by the cover of each layer.

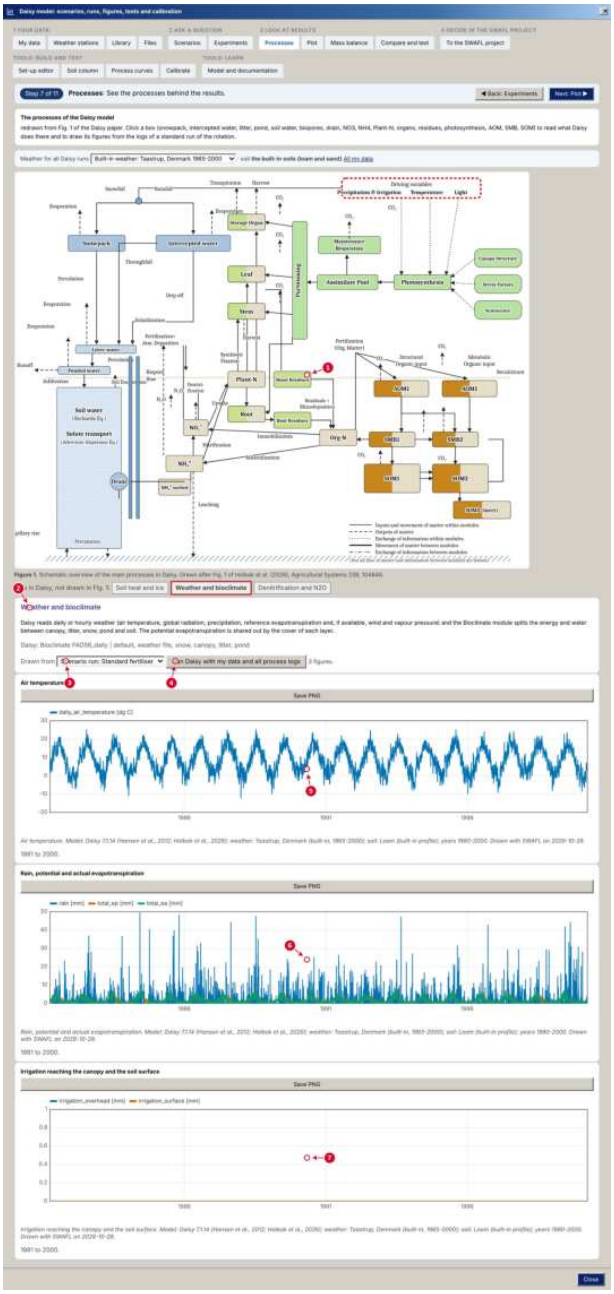


Figure 31.46: Processes: Weather and bioclimate. Screen 101 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.

5. **First figure.** The figure of the process.
6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.3 Processes: Snowpack

Precipitation falls as snow below the threshold temperature and builds a snowpack that holds liquid water, ages, compacts and melts; melt water percolates and, when the soil below is frozen, can run off or pond.

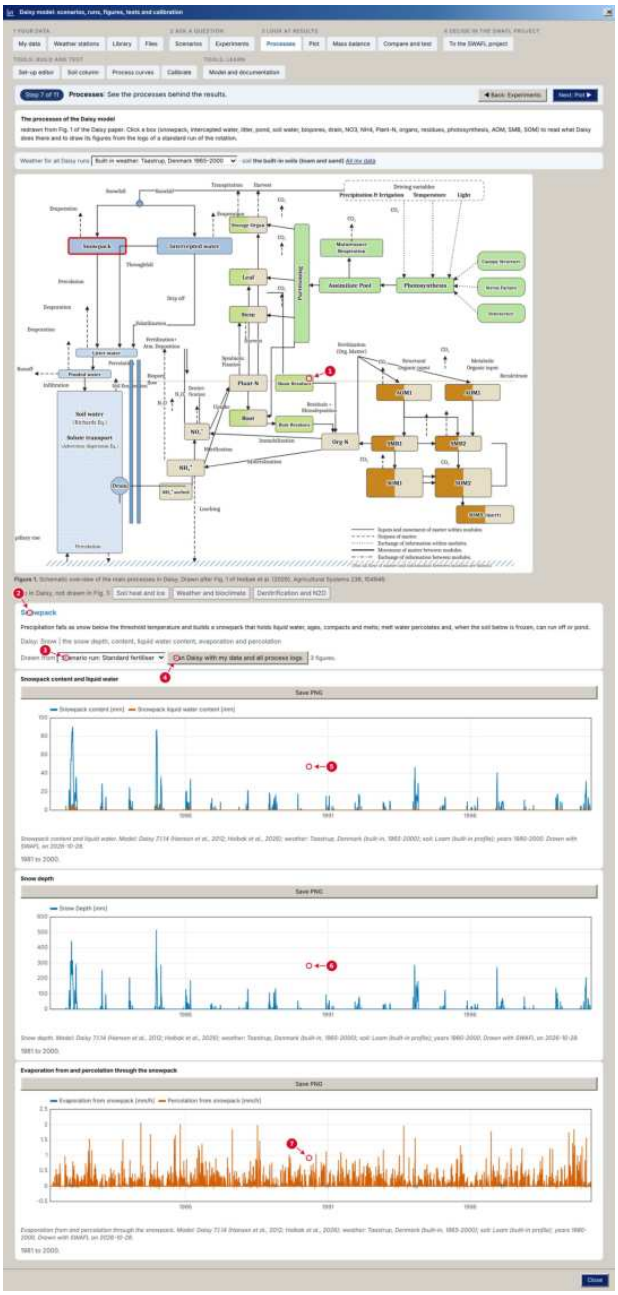


Figure 31.47: Processes: Snowpack. Screen 102 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.4 Processes: Canopy, litter and ponded water

Rain and irrigation are intercepted by the canopy and the litter layer and evaporate from them; what passes goes to the surface pond. The pond infiltrates into the matrix, the biopores (tertiary infiltration) or runs off. Tillage resets the surface.

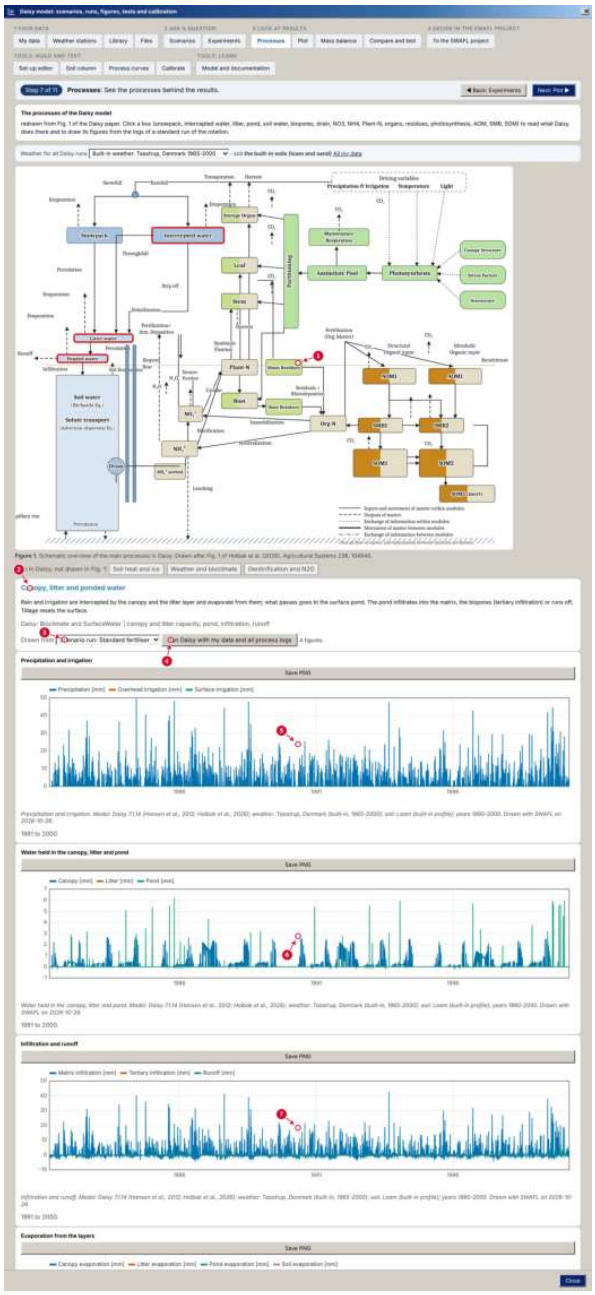


Figure 31.48: Processes: Canopy, litter and ponded water. Screen 103 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.5 Processes: Soil water (Richards equation)

Water in the matrix moves by the one-dimensional Richards equation, with hydraulic properties from the horizons (retention curve and conductivity). Root uptake is the sink and evaporation the surface boundary; the bottom boundary can be free drainage, a groundwater table or a lysimeter.

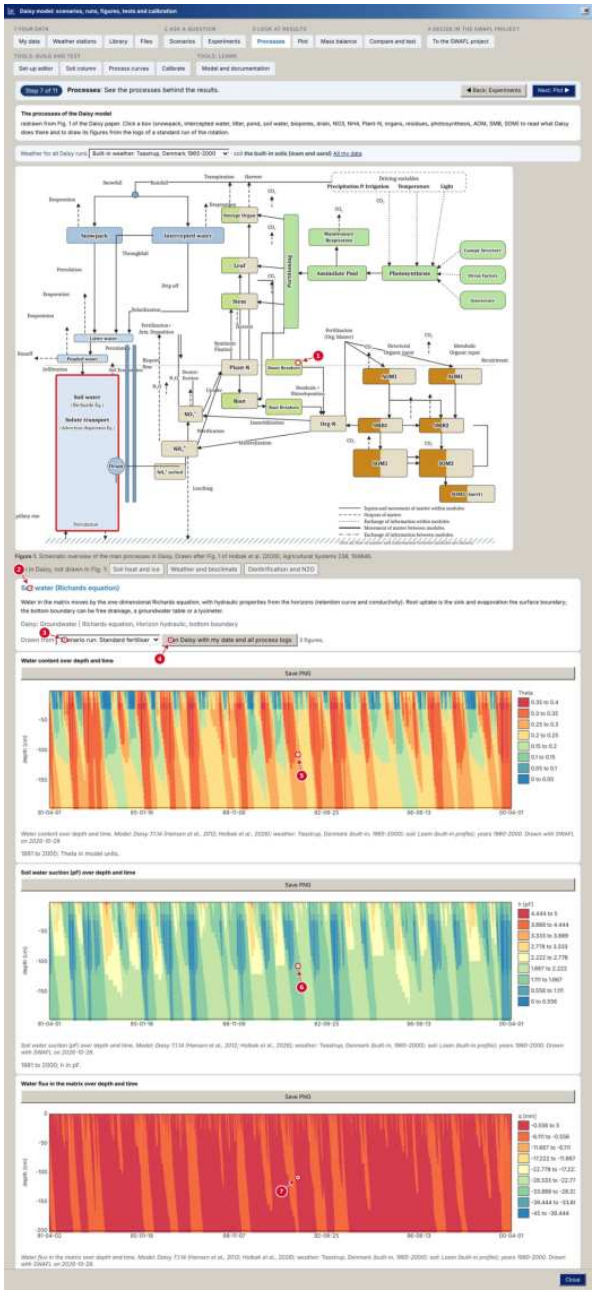


Figure 31.49: Processes: Soil water (Richards equation). Screen 104 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

## 31.7.6 Processes: Biopores (preferential flow)

Part of the water and solutes moves through biopores that bypass the matrix, filled from the surface pond and exchanging with the matrix along the way. This is the source of fast leaching of nitrate and pesticide.

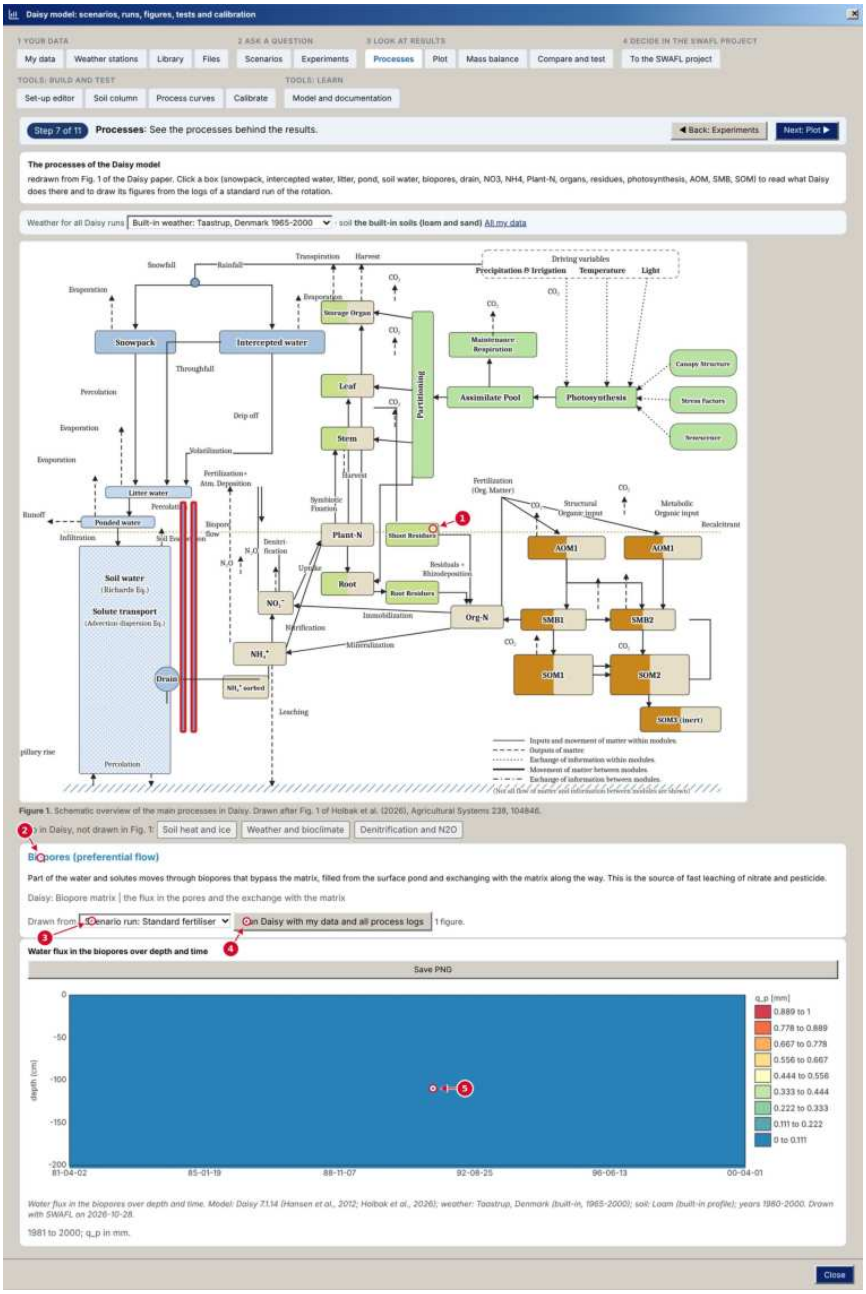



Figure 31.50: Processes: Biopores (preferential flow). Screen 105 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

## 31.7.7 Processes: Drains and groundwater

Tile drains are a sink in the layers at the drain depth, when the soil there is saturated; drained water takes nitrate with it. A groundwater table or a lysimeter bottom boundary closes the water balance at the bottom.

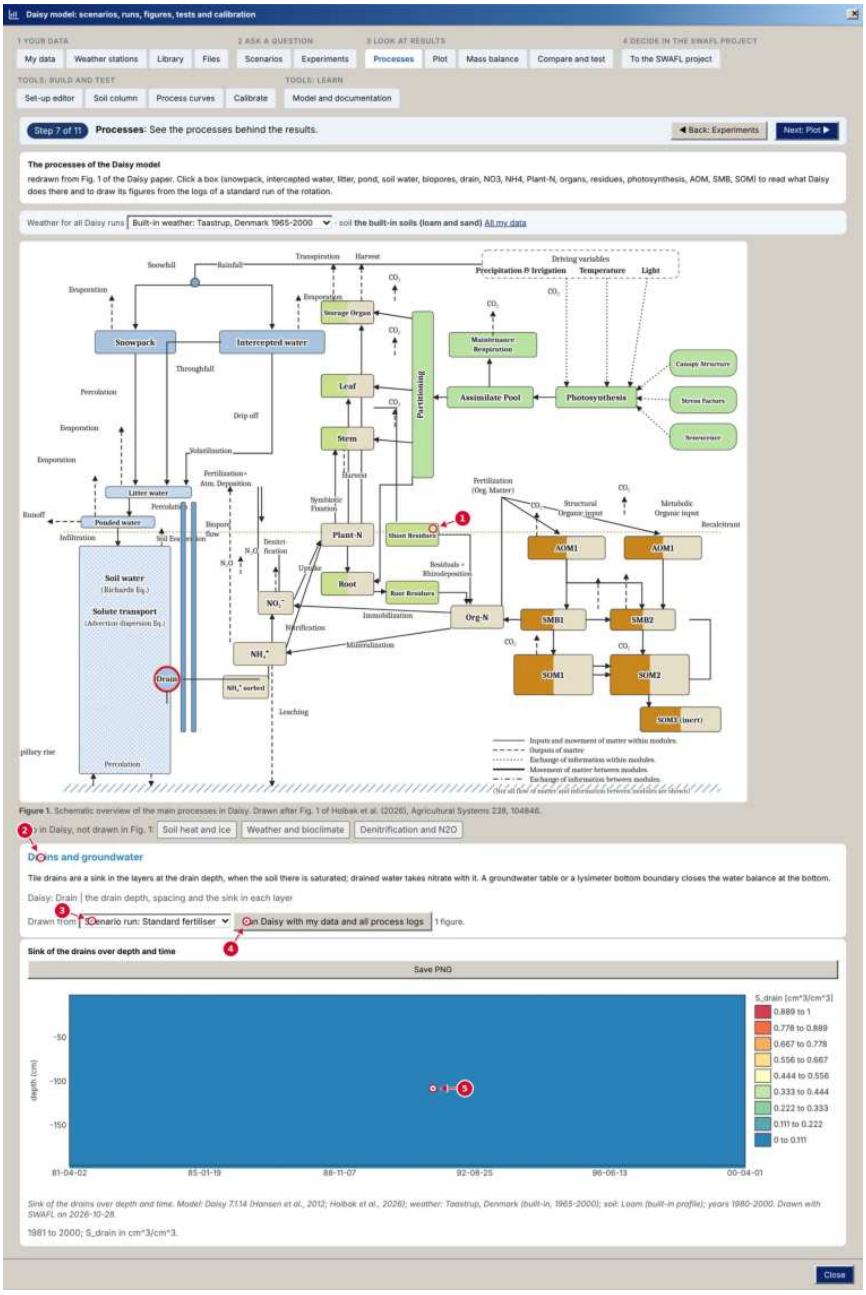



Figure 31.51: Processes: Drains and groundwater. Screen 106 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

### 31.7.8 Processes: Soil heat and ice

Heat moves by conduction and with the water flux; the surface boundary comes from the energy balance of the bioclimate. When the temperature falls below the freezing point part of the water turns to ice, which blocks the flow and releases latent heat.

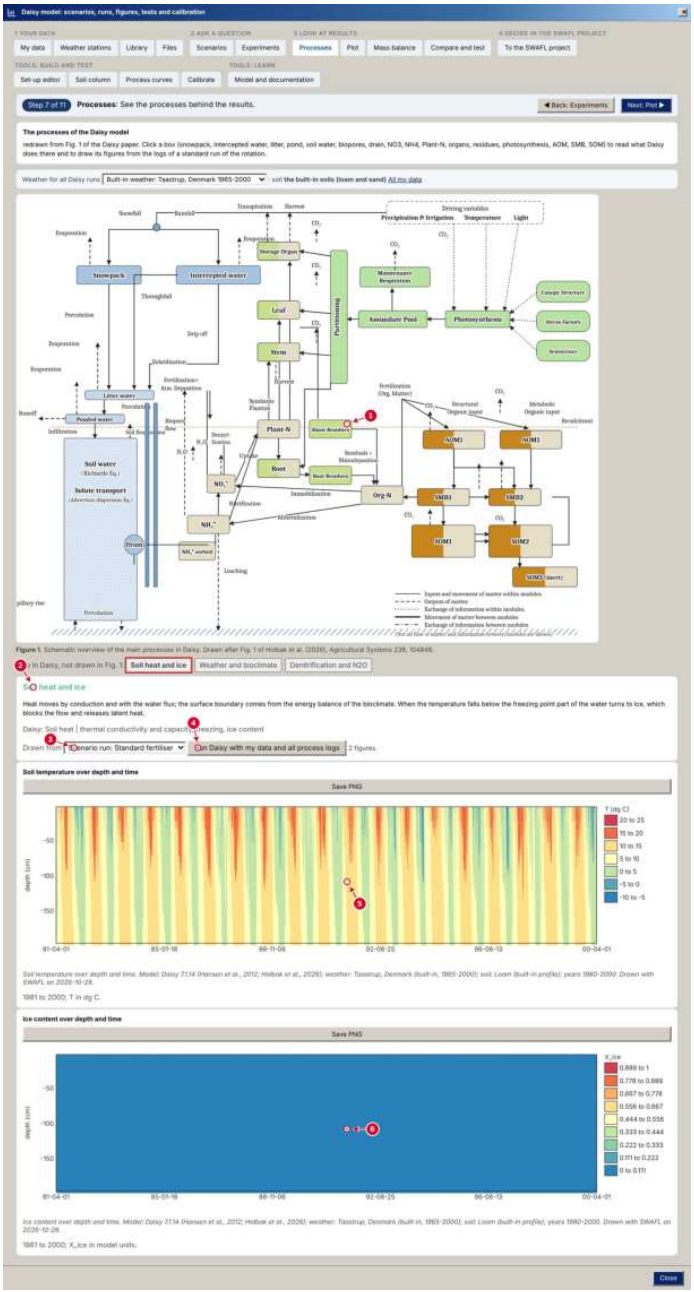


Figure 31.52: Processes: Soil heat and ice. Screen 107 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

6. **Second figure.** The next figure of the process.

### 31.7.9 Processes: Nitrate (solute transport)

Nitrate moves with the water by convection and dispersion and is taken up by roots; it is produced by nitrification and lost by denitrification, drains and leaching below the root zone.

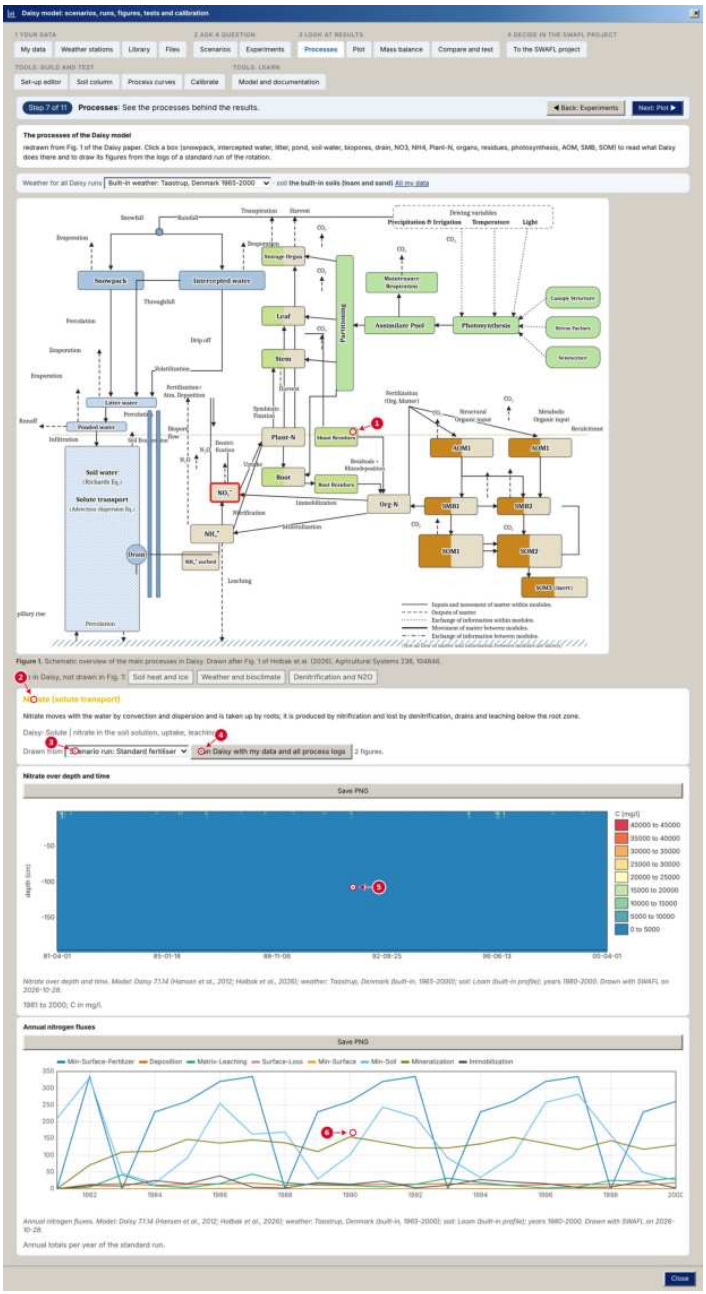


Figure 31.53: Processes: Nitrate (solute transport). Screen 108 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.
6. **Second figure.** The next figure of the process.

### 31.7.10 Processes: Ammonium (sorbed and dissolved)

Ammonium comes from mineralisation and fertiliser; it is sorbed on the soil, nitrified to nitrate (the rate depends on the temperature, the water content and the pH) and taken up by plants.

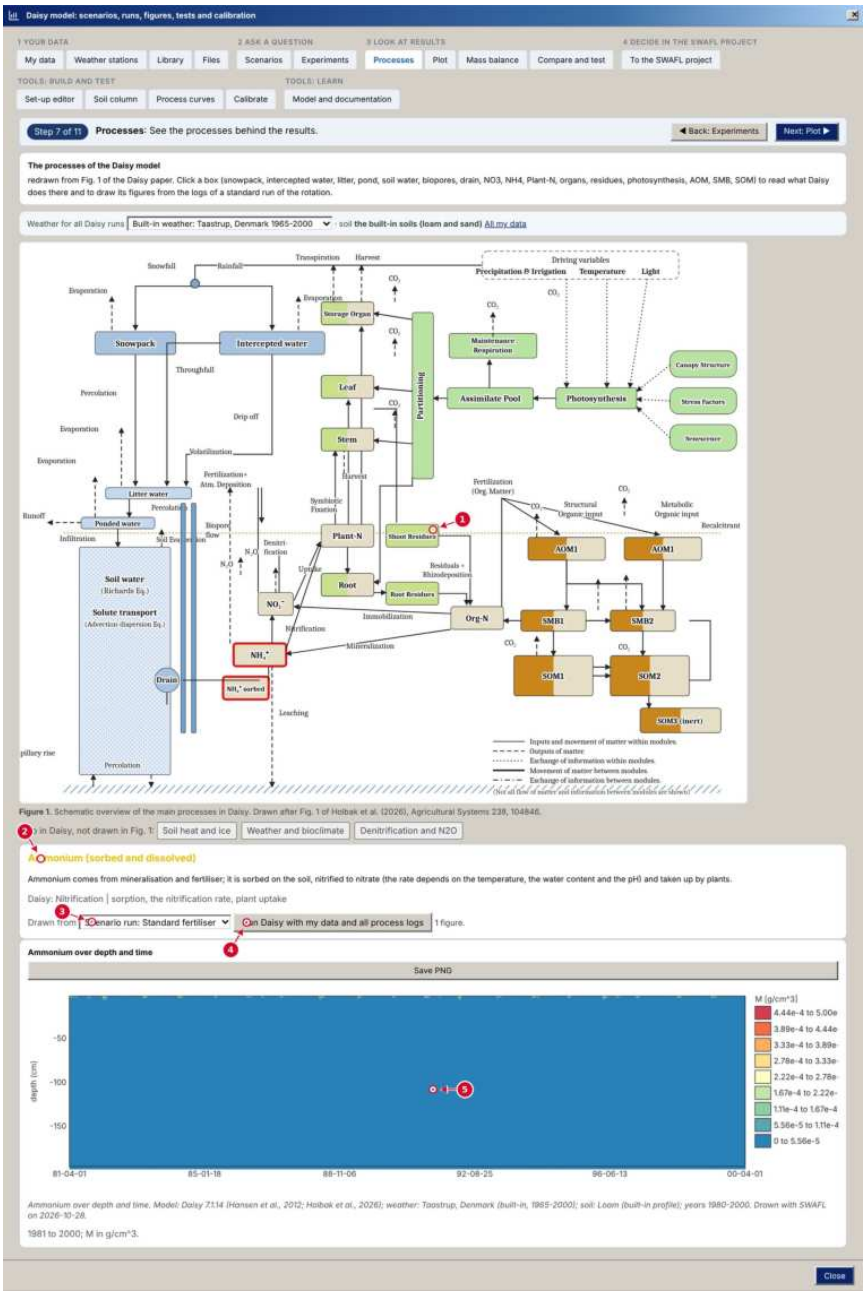



Figure 31.54: Processes: Ammonium (sorbed and dissolved). Screen 109 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

### 31.7.11 Processes: Organic matter pools (AOM, SMB, SOM)

Added organic matter (AOM1-3: crop residues, roots, manure) is decomposed by the soil microbial biomass (SMB1-2), which builds soil organic matter (SOM1-3) from slow to inert; each pool has a carbon and a nitrogen content with its own turnover and efficiency, and the C:N ratio decides whether nitrogen is mineralised or immobilised.

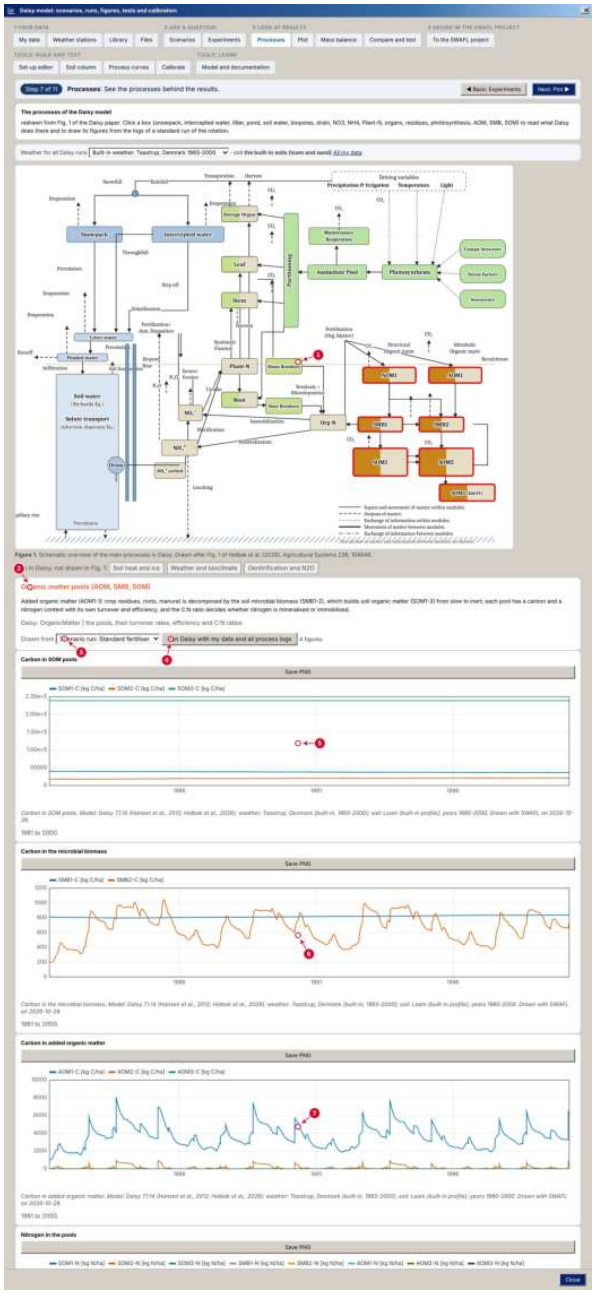


Figure 31.55: Processes: Organic matter pools (AOM, SMB, SOM). Screen 110 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.

4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.
6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.12 Processes: Carbon balance and residues

Carbon enters with the crop residues (shoot, root and stubble), fertiliser and manure and leaves as CO2 from the microbial pools and the plants; the balance shows how the soil carbon changes with the rotation and tillage.

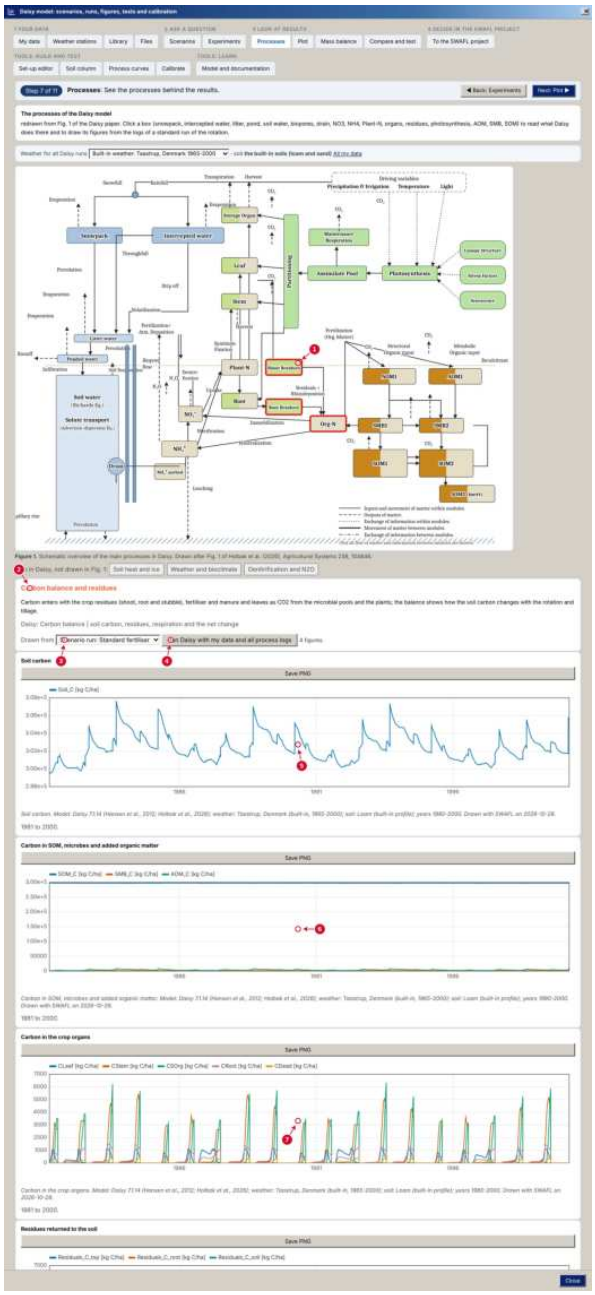


Figure 31.56: Processes: Carbon balance and residues. Screen 111 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.13 Processes: Denitrification and N2O

In wet, carbon-rich soil nitrate is reduced to N2 and N2O; the rate follows the CO2 production, the water content and the temperature, with a potential rate and a redox part, and N2O also comes from nitrification.

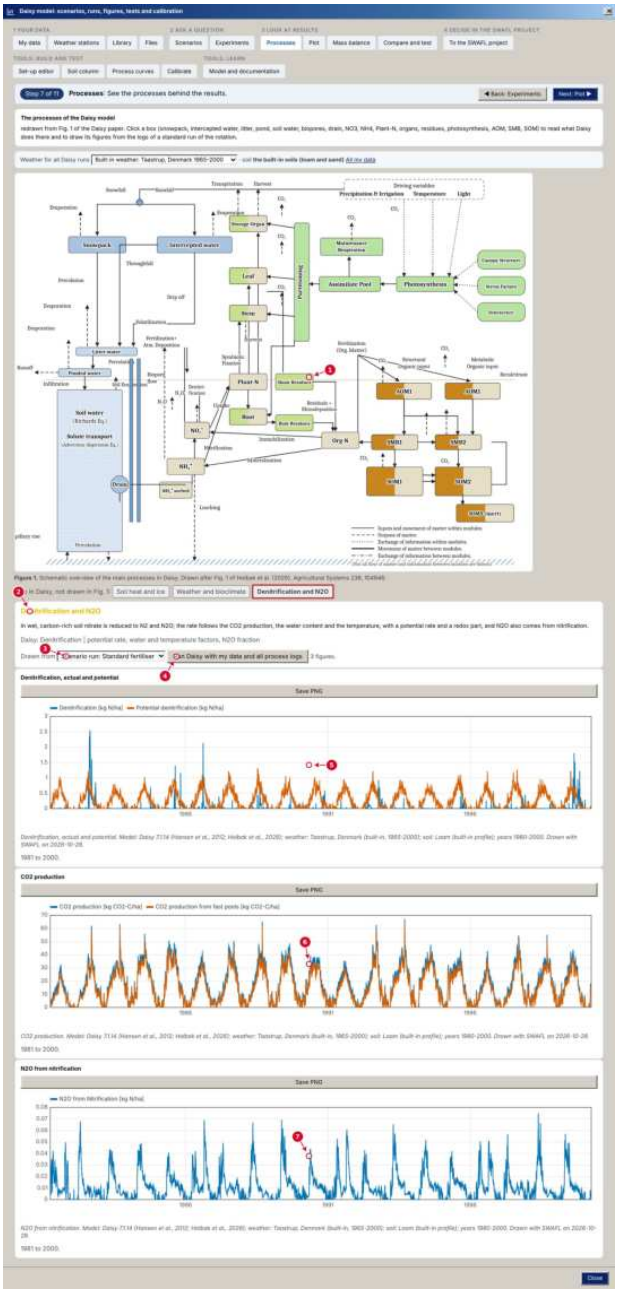


Figure 31.57: Processes: Denitrification and N2O. Screen 112 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.14 Processes: Photosynthesis and respiration

Canopy assimilation comes from the leaf area, the light, the temperature and the water and nitrogen stress; assimilate fills the pool, from which maintenance and growth respiration are paid and the rest is partitioned to the organs.

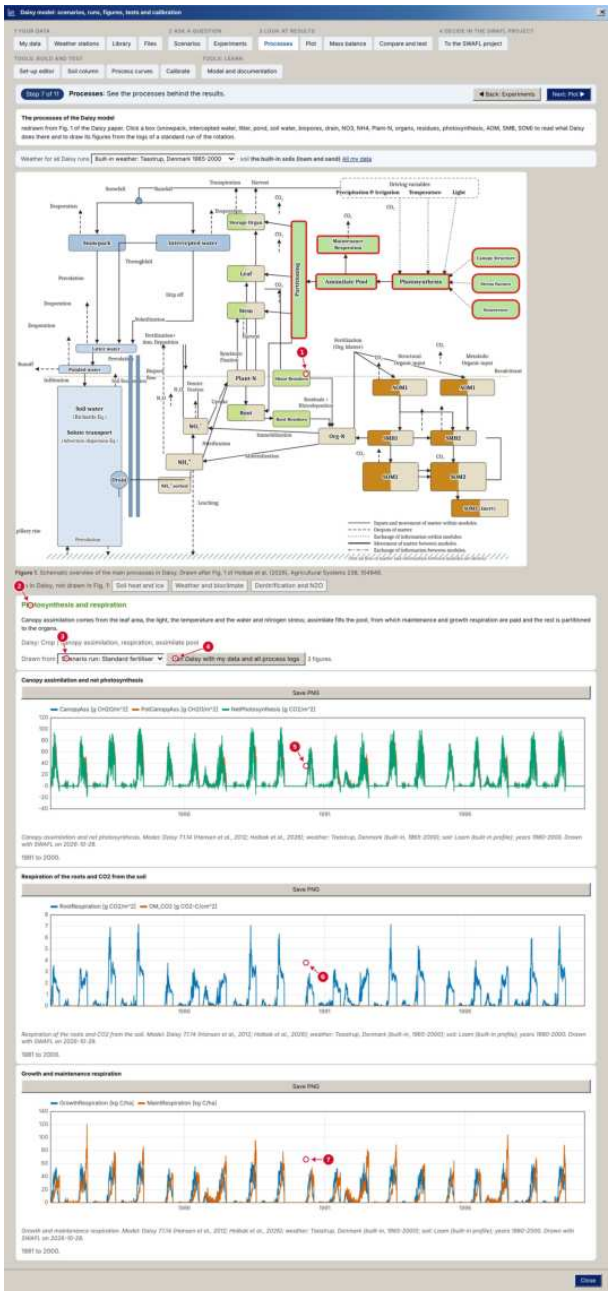


Figure 31.58: Processes: Photosynthesis and respiration. Screen 113 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.
5. **First figure.** The figure of the process.

6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.15 Processes: Crop growth: leaf, stem, root, storage organ

The crop develops with the development stage (DS), leaf area index (LAI), height and rooting depth; dry matter and nitrogen are partitioned to leaf, stem, storage organ and root, under water and nitrogen stress; at harvest the storage organ leaves the field and the rest becomes residue.

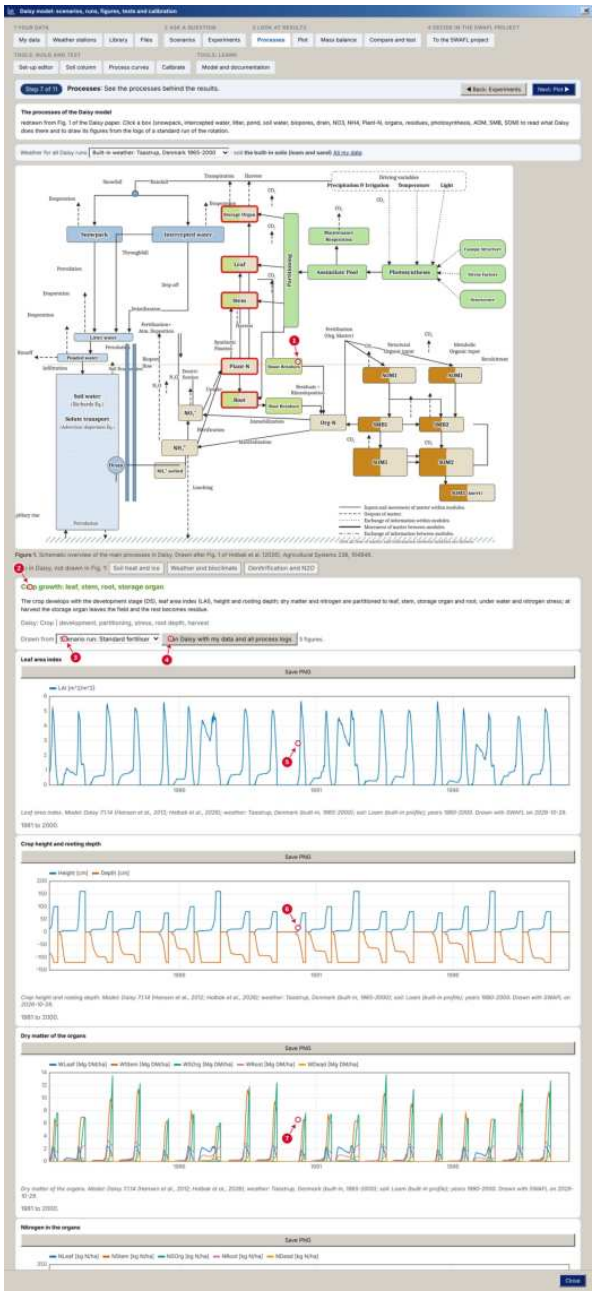


Figure 31.59: Processes: Crop growth: leaf, stem, root, storage organ. Screen 114 of the Daisy workbench.

**What the arrows point to.**

1. **Process map.** The process that is open has a red frame.
2. **Process.** Name of the process and what Daisy does there.
3. **Source of the figures.** The run whose logs are drawn.
4. **Run Daisy with all process logs.** Runs the model with every process log on.

5. **First figure.** The figure of the process.
6. **Second figure.** The next figure of the process.
7. **Third figure.** The next figure of the process.

### 31.7.16 Plot: a time series

Plot draws any column of a loaded log. The log, the view and the period are chosen in the first row, the columns in the list below it.

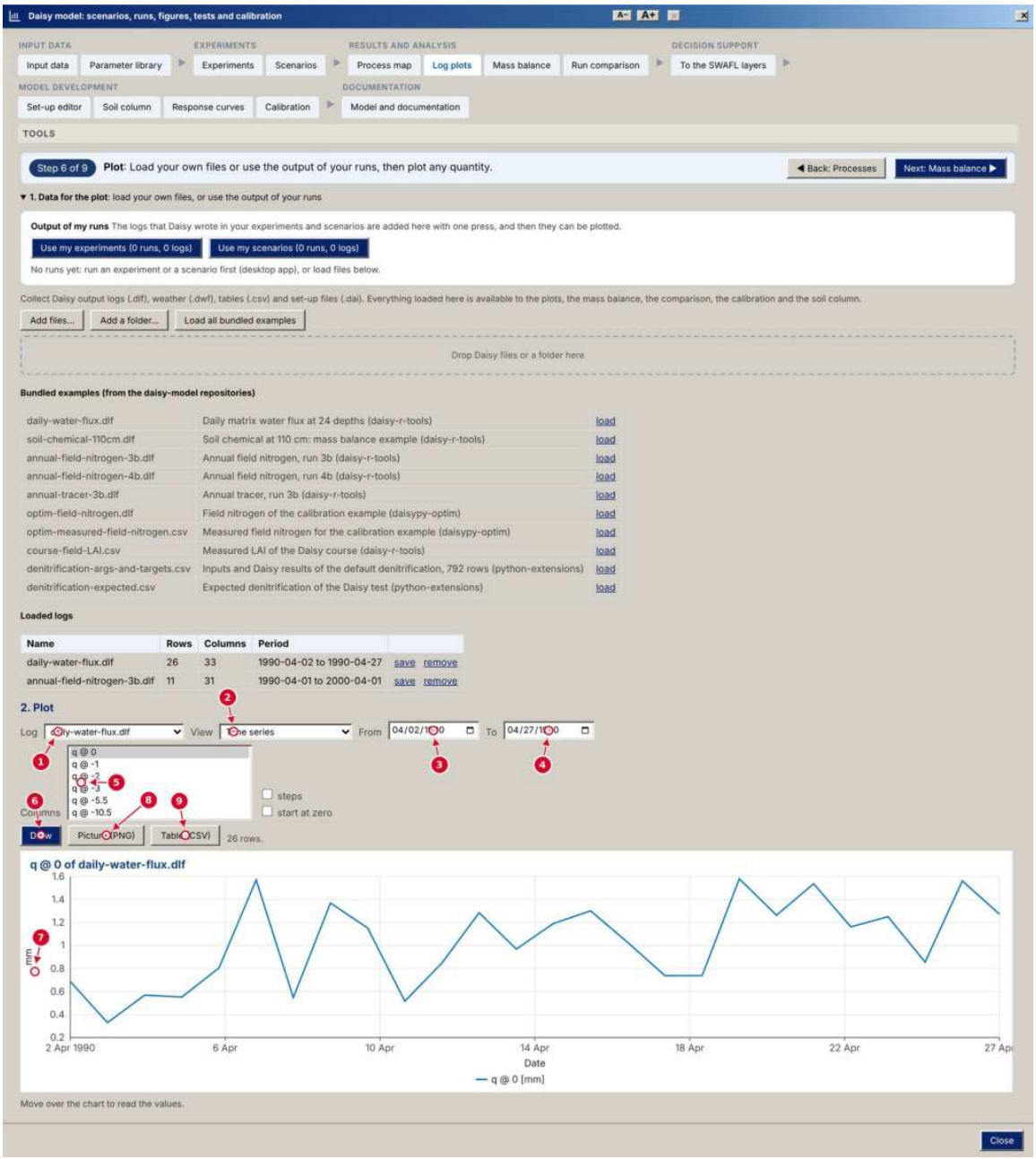



Figure 31.60: Plot: a time series. Screen 115 of the Daisy workbench.

**What the arrows point to.**

1. **Log.** The loaded log to draw.
2. **View.** Time series, annual values, depth profile through time, or profile at a date.
3. **From.** First date drawn.
4. **To.** Last date drawn.
5. **Columns.** Hold Ctrl or Shift to draw several columns together.
6. **Draw.** Draws the chosen columns.

7. **Chart.** Move over the chart to read the values.
8. **Picture (PNG).** Saves the figure as a picture with its source line.
9. **Table (CSV).** Saves the plotted numbers as a table.

### 31.7.17 Plot: twenty-one kinds of figure

*View* lists the figures in six groups; the line under it says what the chosen one shows, and the columns, the period, *Each period as its* (sum, mean, maximum or minimum) and *Smoothing over* appear only when the view uses them. The columns that fit the view are chosen by themselves; *Draw* draws it, *Picture (PNG)* and *Table (CSV)* save it.

| Group | View | What it shows |
|---|---|---|
| Over time | Full range over time | every day of the whole period: the mean of all runs, 10 to 90 % of them darker and the lowest to the highest lighter |
| | Time series | every chosen column against time |
| | Small multiples | one panel for each column, each with its own scale |
| | Raw and smoothed | the daily values as a thin line, the moving average (7, 15, 30 or 90 days) on top |
| | Cumulative over all the years | the running total from the start |
| Totals and means | Monthly values, Annual values | each month or year as its total, mean, maximum or minimum |
| | Anomaly | each year above (blue) or below (red) the mean of all years |
| | Stacked areas | parts of one total (for example evapotranspiration, percolation and runoff) on top of each other |
| The seasons | Through the year | the mean of all years for every day of the year, with 10 to 90 % of the years as the band |
| | Every year on one calendar | one line per year from January to December, coloured from the first year to the last, with the mean in black |
| | Monthly box plots | the spread of each calendar month (daily values, or the monthly totals of every year) |
| | Calendar heat map | one row per year, one column per day, the value as a colour |
| Distribution | Box plots, Violin plots | median, 25 to 75 %, 5 to 95 % and mean of each column; the violin adds the shape of the density |
| | Histogram and density | how often each value occurs, with the density curve and the mean |
| | Exceedance curve | the share of the time a value is reached or passed (a duration curve) |
| Relationship | Scatter with the fitted line | the second chosen column against the first, the least-squares line, its equation, $R^2$, RMSE, n and the 1:1 line when the units agree |
| Over depth | Depth profile through time | a variable over depth and time as colours |
| | Profiles of each month | the mean profile of every month, coloured through the seasons |
| | Profile at a date | the profile on one date, with Play |

**Runs: one log or several together.** Beside *View*, *Runs* chooses the data: *this log only*; *every run with this log, together* (the same log of every run of an experiment or a scenario, merged on their dates: each run becomes one line, box, violin, histogram or curve of the figure, for the first chosen column); or *two runs: this one and* another run, which in *Scatter* draws one run against the other with the 1:1 line, the fitted line and $R^2$. The runs of the session are loaded and the first daily log of an experiment is drawn when the tab opens, so the figures show the full period of the runs rather than the short example log.

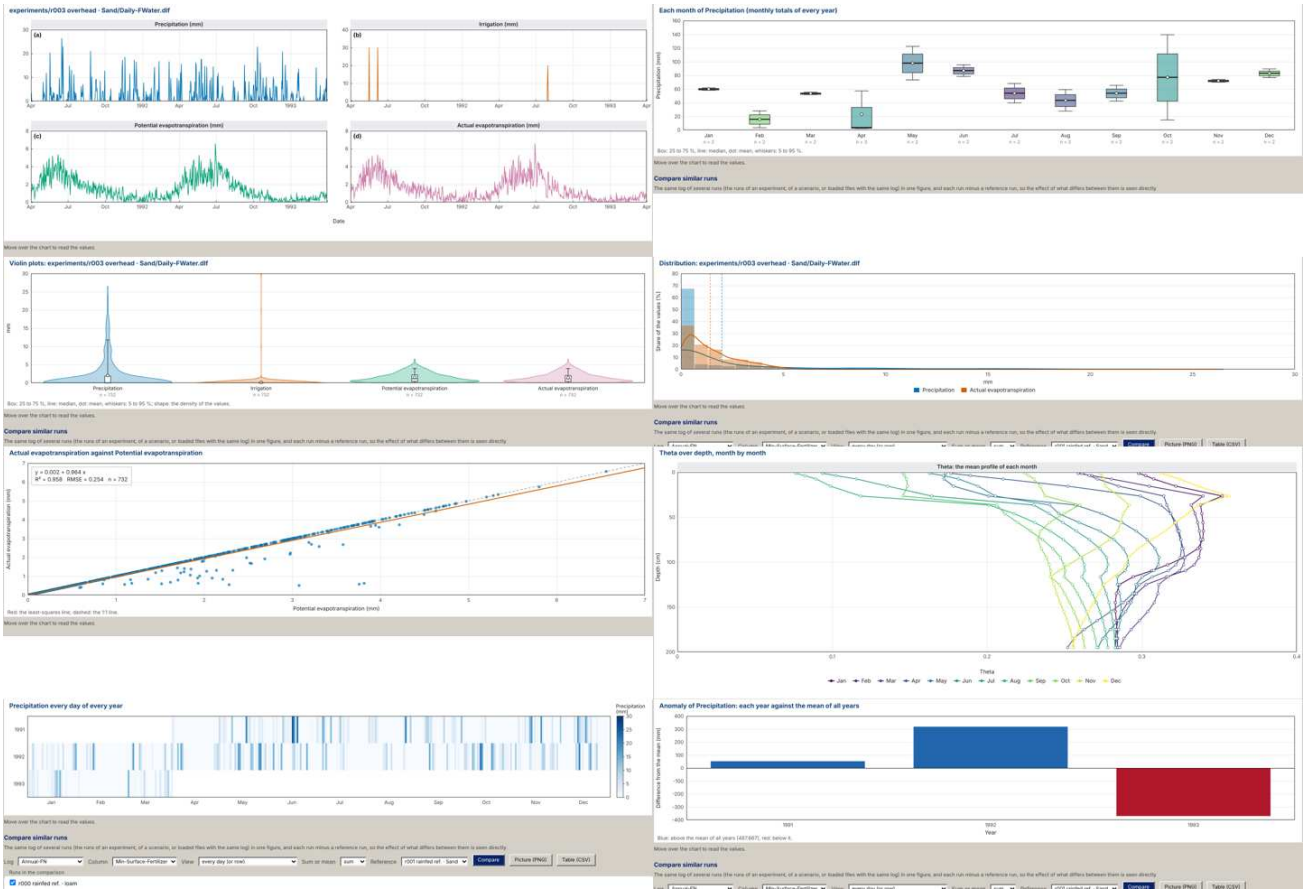


Figure 31.61: Eight of the views, drawn from one run of an experiment: small multiples, monthly box plots, violin plots, histogram and density, scatter with the fitted line, profiles of each month, calendar heat map and anomaly.

### 31.7.18 Plot: annual values

The annual view adds up, averages or takes the largest or smallest value of each year.

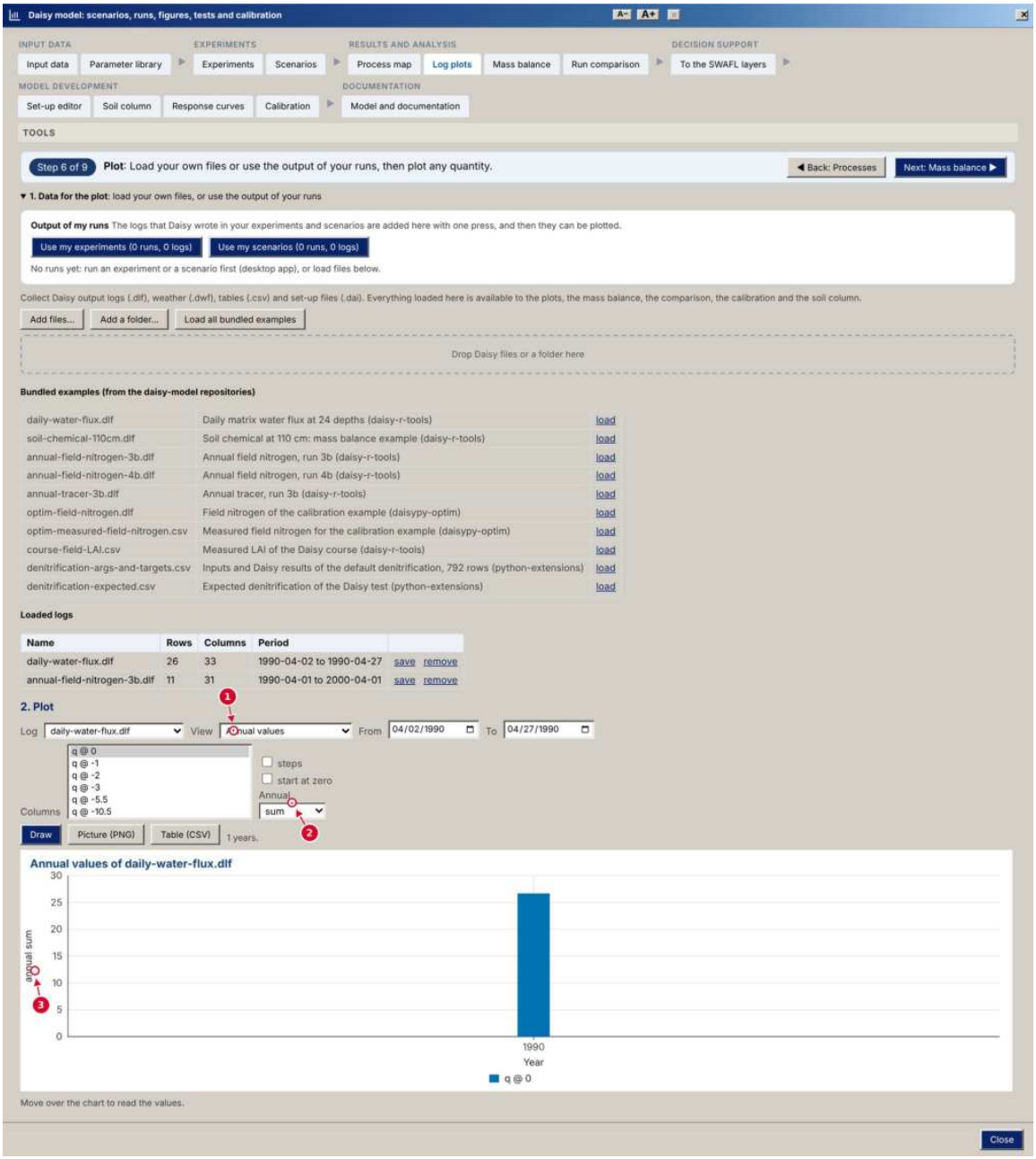



Figure 31.62: Plot: annual values. Screen 116 of the Daisy workbench.

**What the arrows point to.**

1. **Annual values.** The view that was chosen.
2. **Annual summary.** Sum, mean, maximum or minimum of each year.
3. **Chart.** One bar or point for each year.

## 31.7.19 Plot: depth profile through time

Logs with a depth column are drawn as a map of depth against time in spectral colours.

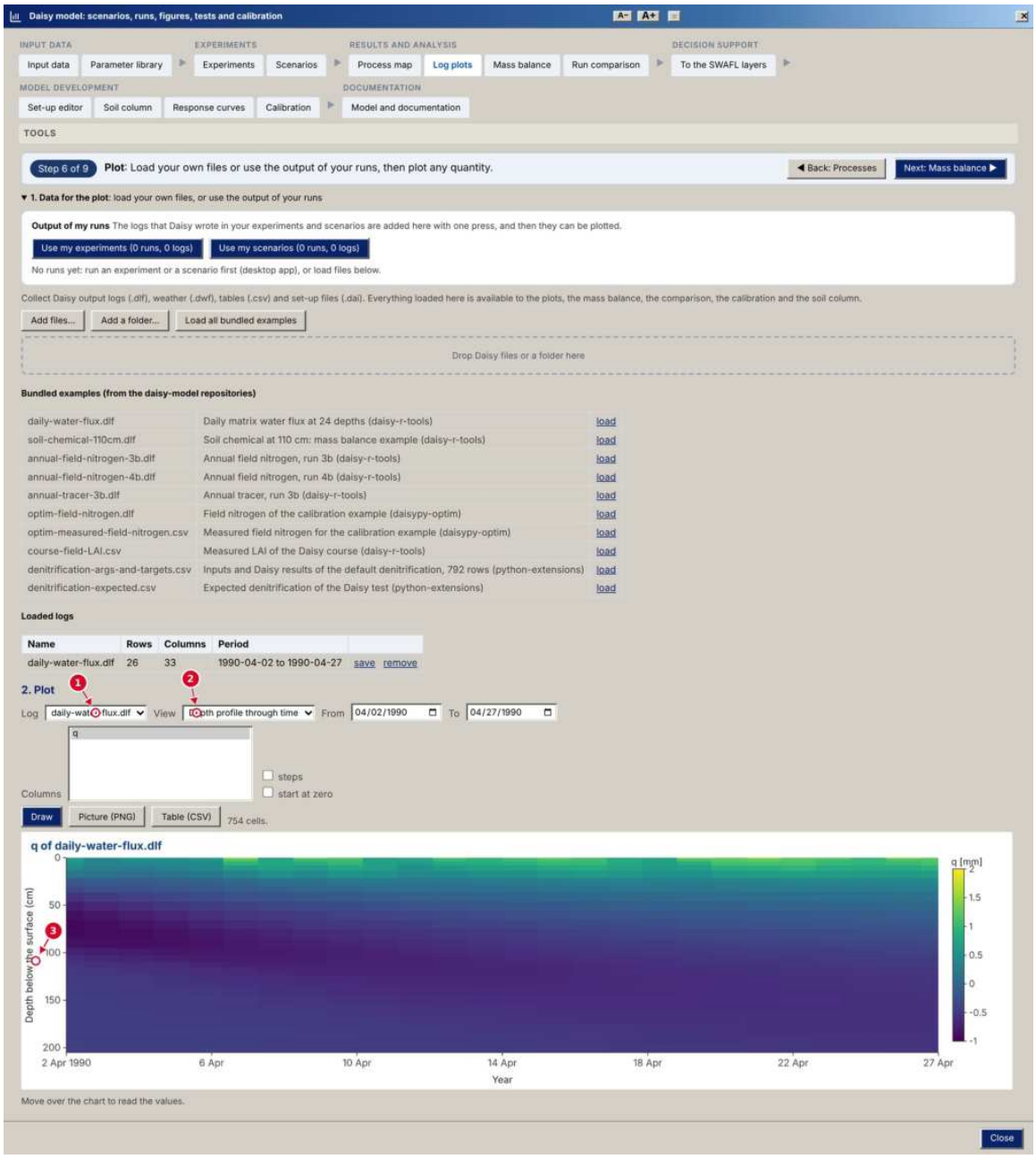



Figure 31.63: Plot: depth profile through time. Screen 117 of the Daisy workbench.

**What the arrows point to.**

1. **Log.** A log with depth, here the matrix water flux.
2. **View.** Depth profile through time.
3. **Map.** Colour shows the value; the scale is beside the map.

## 31.7.20 Plot: profile at a date, animated

The slider moves through the dates; Play runs the profile like a film.

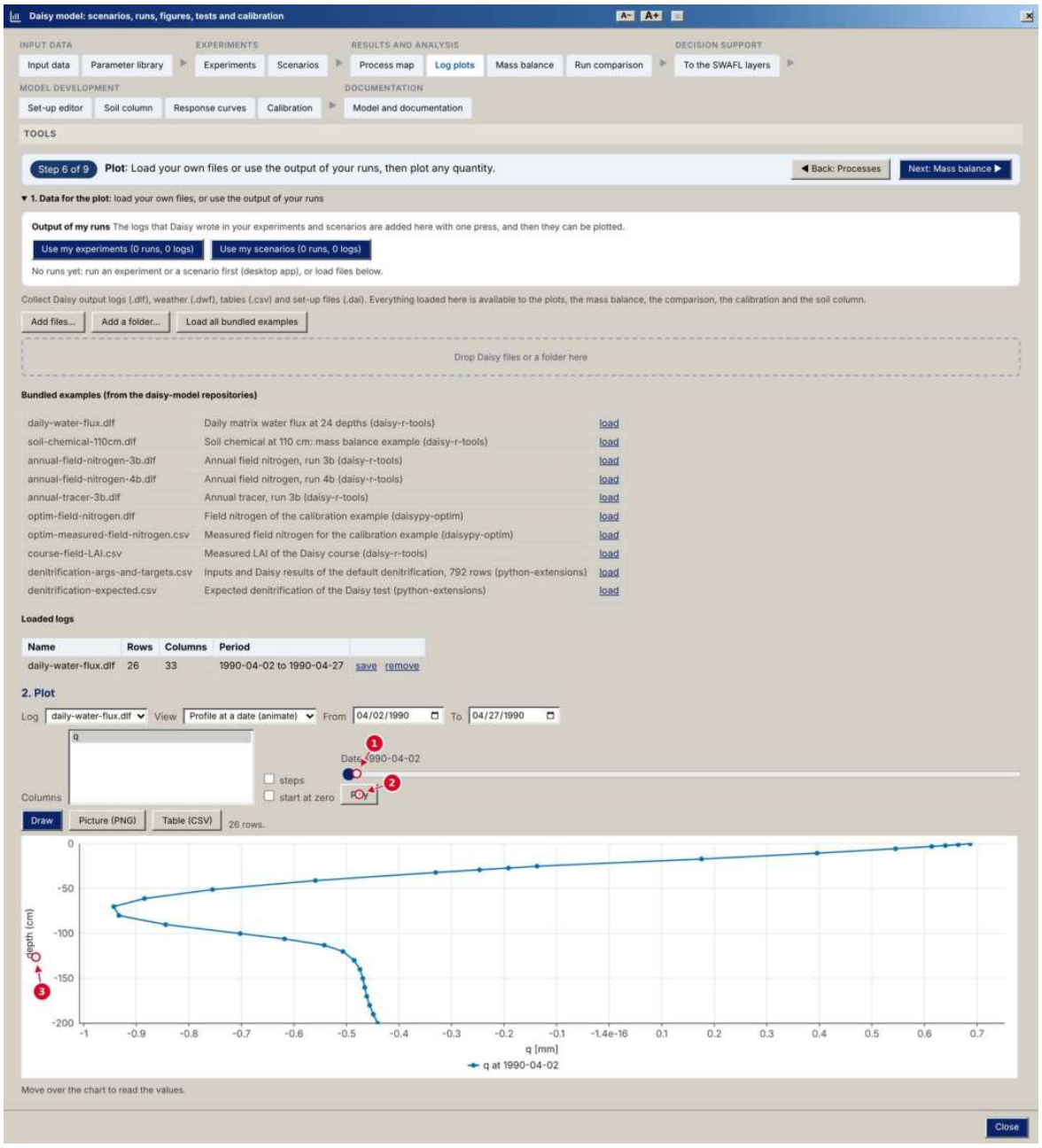



Figure 31.64: Plot: profile at a date, animated. Screen 118 of the Daisy workbench.

**What the arrows point to.**

1. **Date slider.** Chooses the date of the profile.
2. **Play.** Steps through the dates.
3. **Profile.** Value against depth on that date.

### 31.7.21 Plot: figures of every result

Under the plot, *3. Figures of every result* makes one figure of its own for every quantity of a log, or of every loaded log. Each column with values becomes a figure (logs with more than eight columns of one unit are drawn four columns to a figure); a log of several years also gets a figure of its annual totals, and every depth variable a depth-time map. Columns that are zero throughout are listed and left out.

Every figure stands alone: its number and title, the axis texts with units, the legend, a caption that says what it shows (the columns, the log and the period) and a table of its key numbers (mean, smallest and largest value with their dates, and the sum of the period). Each one has its own controls:

1. **Draw as.** Lines, steps, areas, cumulative sums (the end of each line is the total of the period) or points.
2. **Axis from zero.** Starts the value axis at zero.
3. **Zoom.** Drag across the figure to zoom in time; double-click to see the whole period again. The caption and the key numbers follow the zoom.
4. **Values under the cursor.** Shown under the figure as the cursor moves.
5. **Titles, Save, full screen.** Edit the title and the axis texts, save the figure as a picture (screen, 300 dpi or 600 dpi) or as a table, or show it on the whole screen.

The bar above the figures collapses or expands them all and saves every figure in one zip file.

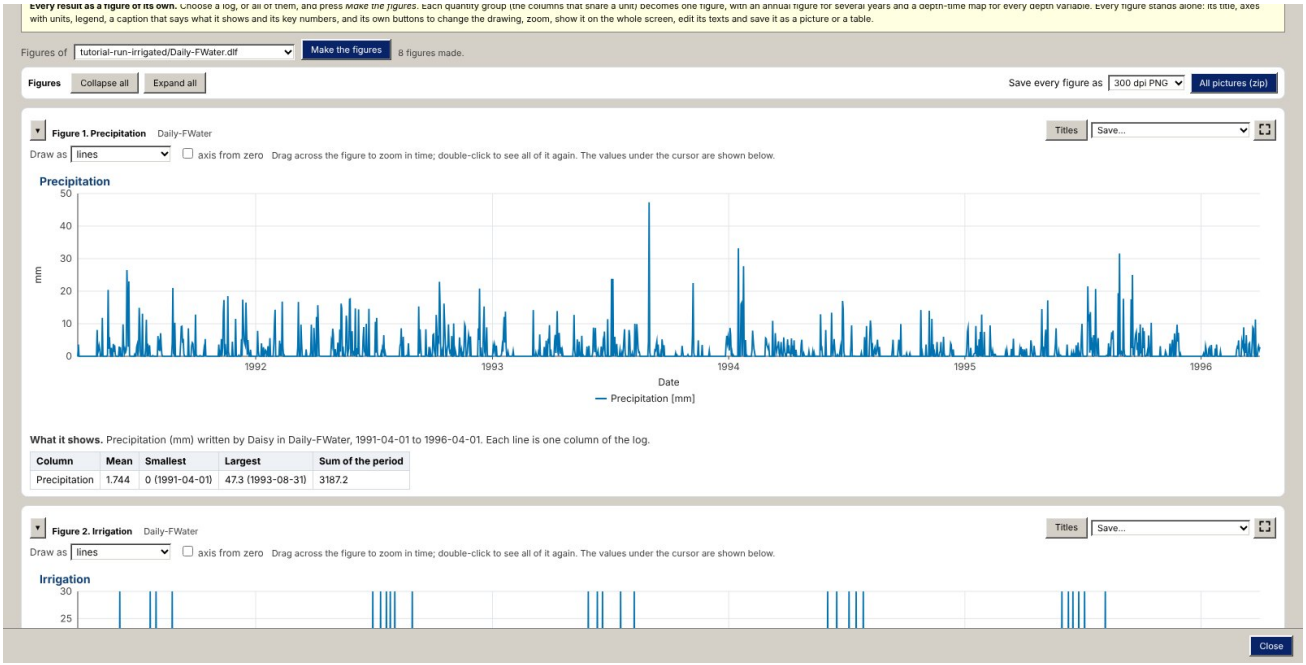



Figure 31.65: Figures of every result: one figure for each quantity of the daily water log of the irrigated run of the sample experiment (chapter Learn the Daisy files).

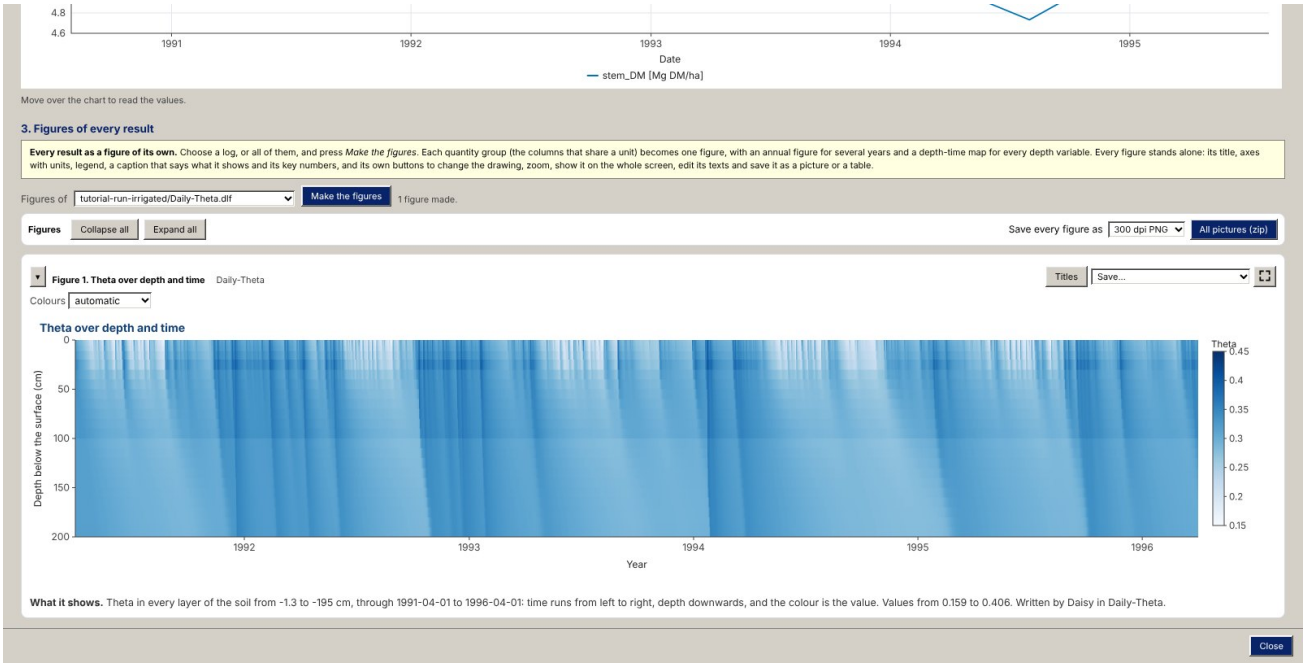



Figure 31.66: A depth-time map of the gallery: the water content of the irrigated run.

## 31.7.22 Mass balance: the form

The balance adds the inputs, the outputs and the content columns of a log. Choose columns proposes a set from the column names.

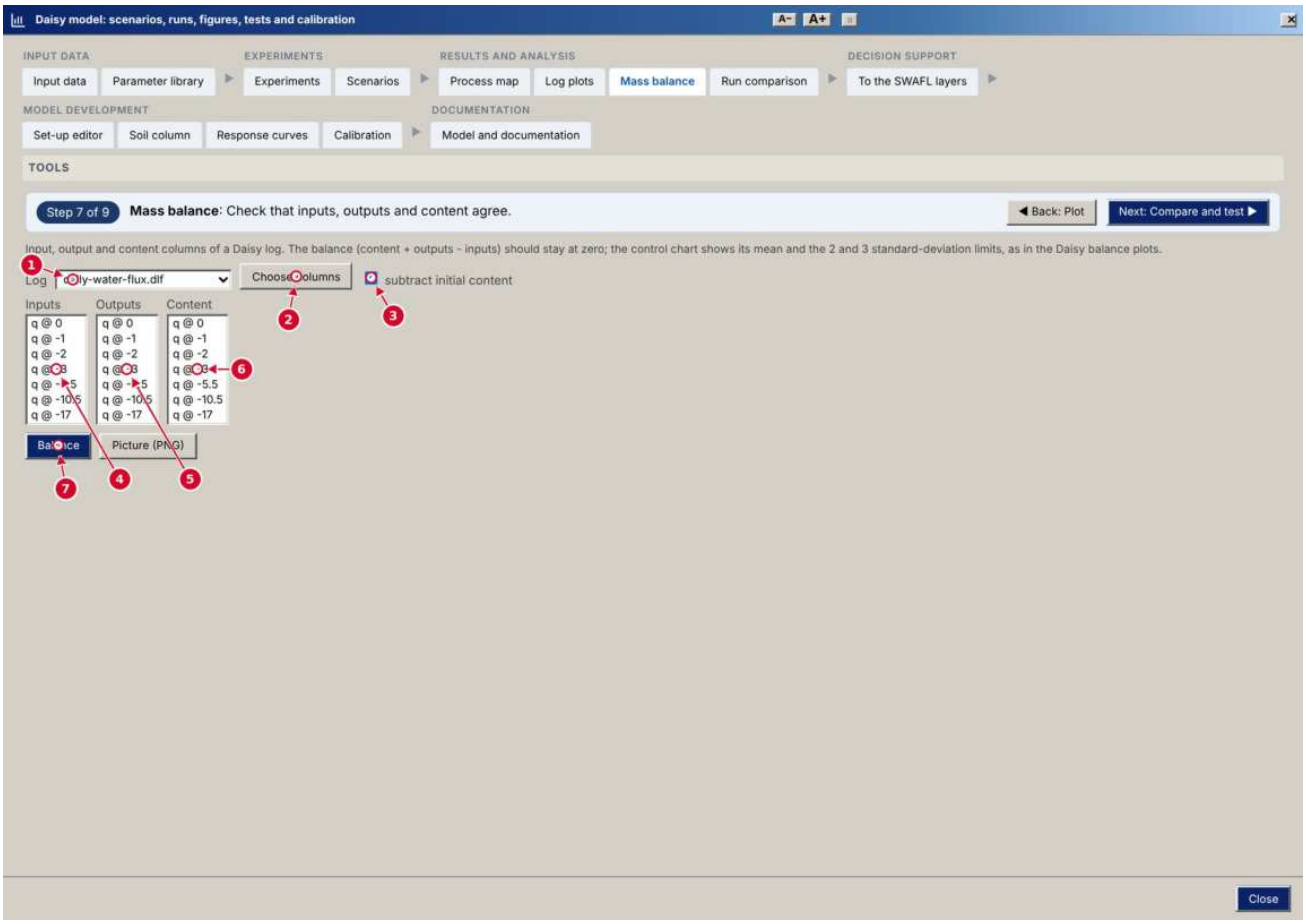



Figure 31.67: Mass balance: the form. Screen 119 of the Daisy workbench.

**What the arrows point to.**

1. **Log.** The log whose balance is checked.
2. **Choose columns.** Proposes inputs, outputs and content.
3. **Subtract initial content.** Counts only the change in content.
4. **Inputs.** Columns that add to the soil.
5. **Outputs.** Columns that leave the soil.
6. **Content.** Columns that hold the amount stored.
7. **Balance.** Calculates the balance.

### 31.7.23 Mass balance: the control chart

After Balance the error of each step is drawn with the mean and the limits of two and three standard deviations.

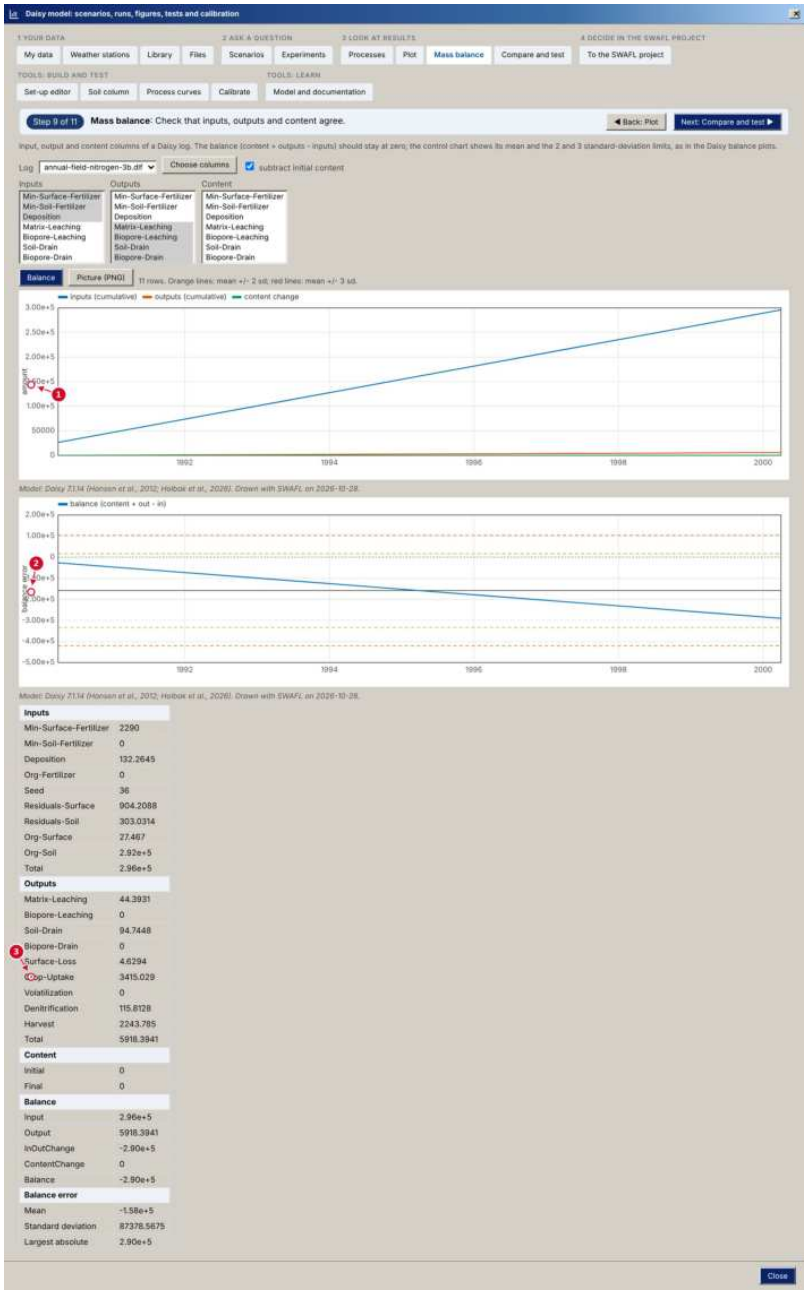



Figure 31.68: Mass balance: the control chart. Screen 120 of the Daisy workbench.

**What the arrows point to.**

1. **Balance through time.** Content plus outputs minus inputs; it should stay at zero.
2. **Control chart.** Mean and limits of two and three standard deviations.
3. **Summary.** Mean, spread and the number of steps outside the limits.

### 31.7.24 Run comparison: what to compare

Run comparison puts runs side by side and says, in plain words, how they differ. Three buttons choose what is compared, and the runs can come from three places: the logs in the Files list (a folder of a Daisy run counts as one run), the runs of the Experiments tab and the runs of the Scenarios tab.

| Button | What it compares | What it gives |
|---|---|---|
| Two logs | one log against another: two files, or a run against measurements | the verdict, the table of every quantity, three figures of a quantity |
| Two runs, every log | all the logs two runs have in common (A and B from any of the three places) | one row per log with its verdict and largest difference; click a log for its table and figures |
| Many runs, one quantity | any number of ticked runs on one quantity of one log, against a reference run | the ranking with mean, minimum, maximum, sum, difference to the reference (absolute and %), RMS difference, r, KGE and the verdict; all runs over time, the mean of each run with its range, the difference of each run to the reference |

**The run library.** The panel at the top of the tab gathers every run there is to compare, and every log of every run appears in the lists of Log A and Log B, grouped by run:

| Source | How it gets there |
|---|---|
| Runs of Experiments | by itself, as soon as the runs are made |
| Runs of Scenarios | by itself, as soon as the runs are made |
| Saved runs on this computer | every run ever made is kept in the folder *SWAFL runs* next to Daisy (older runs in Documents/SWAFL Daisy runs are listed too); *Show the runs saved on this computer* lists them, newest first, grouped by experiment, with a filter; tick them and press *Load the ticked*. When the session has no runs of its own, the newest experiment is loaded at once |
| Logs in Files | *Add logs from my computer*, *Add a run folder*, or drop files and folders on the tab |
| Samples that come with SWAFL | the sample experiment (rainfed and irrigated, seven logs each) and six logs of the daisy-model repositories; loaded by themselves when there are fewer than two runs, so there is always something to compare |

When Log A is chosen, Log B takes the same log of another run, so a comparison is always like with like. *Open the runs folder* opens the folder on the computer.

Two options apply to all three: *Years* (the first and last year compared) and *Values*: as written by Daisy, sum per year, mean per year, mean per month, or cumulative. Sums and means compare the totals of each year (for example the yearly drainage); the SML then no longer applies.

### 31.7.25 Run comparison: the verdict and the table

Run comparison puts two runs side by side and says, in plain words, how they differ. Choose run A and run B: the logs of two runs (two scenarios, the same set-up before and after a change, two variants of an experiment), or a run and a table of measurements. The comparison starts as soon as both are chosen; the arrows between them swap A and B.

- **Matching.** The rows are matched by date, so two runs of different lengths are compared on their common dates; logs without dates are matched row by row.
- **The verdict.** *Identical*, *The same within the meaningful level* or *Different*, with the number of quantities of each kind and the three largest relative differences. A quantity whose mean is zero in A (irrigation of a rainfed run) is reported with its two means.
- **The smallest meaningful level (SML).** The difference below which Daisy's own test suite (daisypy-test) counts two values as the same, for every column of every standard log (1 mm for the water balance). A difference within it is *below the meaningful level*; one under a thousandth of it is *identical* (the share is set under *Tolerances*). Quantities without an SML are compared with the absolute tolerance.

The table gives, for every quantity, its unit, the verdict, the SML, the means of A and B, the mean difference $\overline{B - A}$ and the same in per cent of A, the root mean square difference, the largest difference and its date, the mean absolute difference, the Kling-Gupta efficiency KGE, the correlation $r$ and the Nash-Sutcliffe efficiency of B on A,

$$\mathrm{NSE} = 1 - \frac{\sum_t (B_t - A_t)^2}{\sum_t (A_t - \overline{A})^2},$$

which is 1 when B equals A and below 0 when B is further from A than the mean of A is. *Show only the different ones* shortens the table. *Compare same-named logs of two folders* compares every log that two loaded folders have in common and gives one verdict line for each.

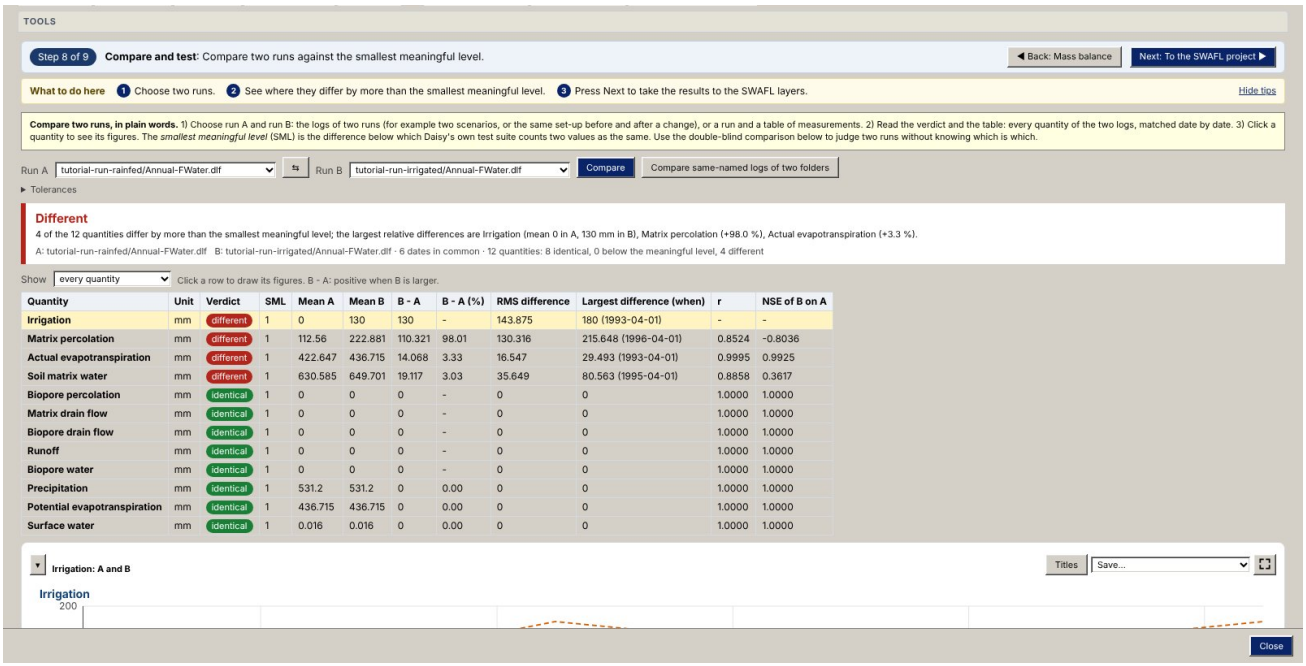
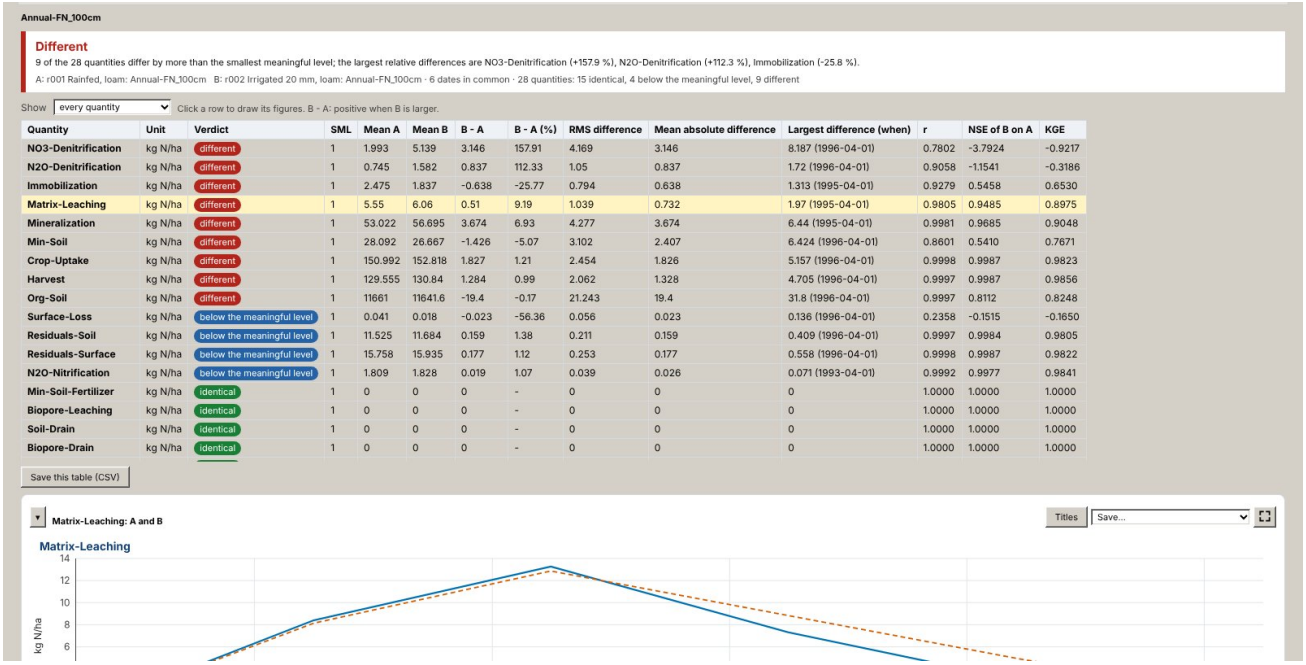



Figure 31.69: Run comparison: the two annual water balances of the sample experiment. The irrigated run (B) gets 130 mm of irrigation a season on average, drains 98 % more below the root zone and evaporates 3.3 % more.

$$\mathrm{KGE} = 1 - \sqrt{(r-1)^2 + \left(\frac{\sigma_B}{\sigma_A} - 1\right)^2 + \left(\frac{\overline{B}}{\overline{A}} - 1\right)^2}$$

is 1 when B has the same correlation, spread and mean as A (Gupta et al., 2009). *Save this table (CSV)* writes the table.



Annual-FN_100cm

**Different**

9 of the 28 quantities differ by more than the smallest meaningful level; the largest relative differences are NO3-Denitrification (+157.9 %), N2O-Denitrification (+112.3 %), Immobilization (-25.8 %).

A: r001 Rainfed, loam: Annual-FN_100cm B: r002 Irrigated 20 mm, loam: Annual-FN_100cm · 6 dates in common · 28 quantities: 15 identical, 4 below the meaningful level, 9 different

Show every quantity Click a row to draw its figures. B - A: positive when B is larger.

| Quantity | Unit | Verdict | SML | Mean A | Mean B | B - A | B - A (%) | RMS difference | Mean absolute difference | Largest difference (when) | r | NSE of B on A | KGE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NO3-Denitrification | kg N/ha | different | 1 | 1.993 | 5.139 | 3.146 | 157.91 | 4.169 | 3.146 | 8.187 (1996-04-01) | 0.7802 | -3.7924 | -0.9217 |
| N2O-Denitrification | kg N/ha | different | 1 | 0.745 | 1.582 | 0.837 | 112.33 | 1.05 | 0.837 | 1.72 (1996-04-01) | 0.9058 | -1.1541 | -0.3186 |
| Immobilization | kg N/ha | different | 1 | 2.475 | 1.837 | -0.638 | -25.77 | 0.794 | 0.638 | 1.313 (1995-04-01) | 0.9279 | 0.5458 | 0.6530 |
| Matrix-Leaching | kg N/ha | different | 1 | 5.55 | 6.06 | 0.51 | 9.19 | 1.039 | 0.732 | 1.97 (1995-04-01) | 0.9805 | 0.9485 | 0.8975 |
| Mineralization | kg N/ha | different | 1 | 53.022 | 56.695 | 3.674 | 6.93 | 4.277 | 3.674 | 6.44 (1995-04-01) | 0.9981 | 0.9685 | 0.9048 |
| Min-Soil | kg N/ha | different | 1 | 28.092 | 26.667 | -1.426 | -5.07 | 3.102 | 2.407 | 6.424 (1996-04-01) | 0.8601 | 0.5410 | 0.7671 |
| Crop-Uptake | kg N/ha | different | 1 | 150.992 | 152.818 | 1.827 | 1.21 | 2.454 | 1.826 | 5.157 (1996-04-01) | 0.9998 | 0.9987 | 0.9823 |
| Harvest | kg N/ha | different | 1 | 129.555 | 130.84 | 1.284 | 0.99 | 2.062 | 1.328 | 4.705 (1996-04-01) | 0.9997 | 0.9987 | 0.9856 |
| Org-Soil | kg N/ha | different | 1 | 11661 | 11641.6 | -19.4 | -0.17 | 21.243 | 19.4 | 31.8 (1996-04-01) | 0.9997 | 0.8112 | 0.8248 |
| Surface-Loss | kg N/ha | below the meaningful level | 1 | 0.041 | 0.018 | -0.023 | -56.36 | 0.056 | 0.023 | 0.136 (1996-04-01) | 0.2358 | -0.1515 | -0.1650 |
| Residuals-Soil | kg N/ha | below the meaningful level | 1 | 11.525 | 11.684 | 0.159 | 1.38 | 0.211 | 0.159 | 0.409 (1996-04-01) | 0.9997 | 0.9984 | 0.9805 |
| Residuals-Surface | kg N/ha | below the meaningful level | 1 | 15.758 | 15.935 | 0.177 | 1.12 | 0.253 | 0.177 | 0.558 (1996-04-01) | 0.9998 | 0.9987 | 0.9822 |
| N2O-Nitrification | kg N/ha | below the meaningful level | 1 | 1.809 | 1.828 | 0.019 | 1.07 | 0.039 | 0.026 | 0.071 (1993-04-01) | 0.9992 | 0.9977 | 0.9841 |
| Min-Soil-Fertilizer | kg N/ha | identical | 1 | 0 | 0 | 0 | - | 0 | 0 | 0 | 1.0000 | 1.0000 | 1.0000 |
| Biopore-Leaching | kg N/ha | identical | 1 | 0 | 0 | 0 | - | 0 | 0 | 0 | 1.0000 | 1.0000 | 1.0000 |
| Soil-Drain | kg N/ha | identical | 1 | 0 | 0 | 0 | - | 0 | 0 | 0 | 1.0000 | 1.0000 | 1.0000 |
| Biopore-Drain | kg N/ha | identical | 1 | 0 | 0 | 0 | - | 0 | 0 | 0 | 1.0000 | 1.0000 | 1.0000 |

Save this table (CSV)



Figure 31.70: Two runs, every log: run A (rainfed) against run B (irrigated), the verdict for the three logs and the table of the annual nitrogen log.

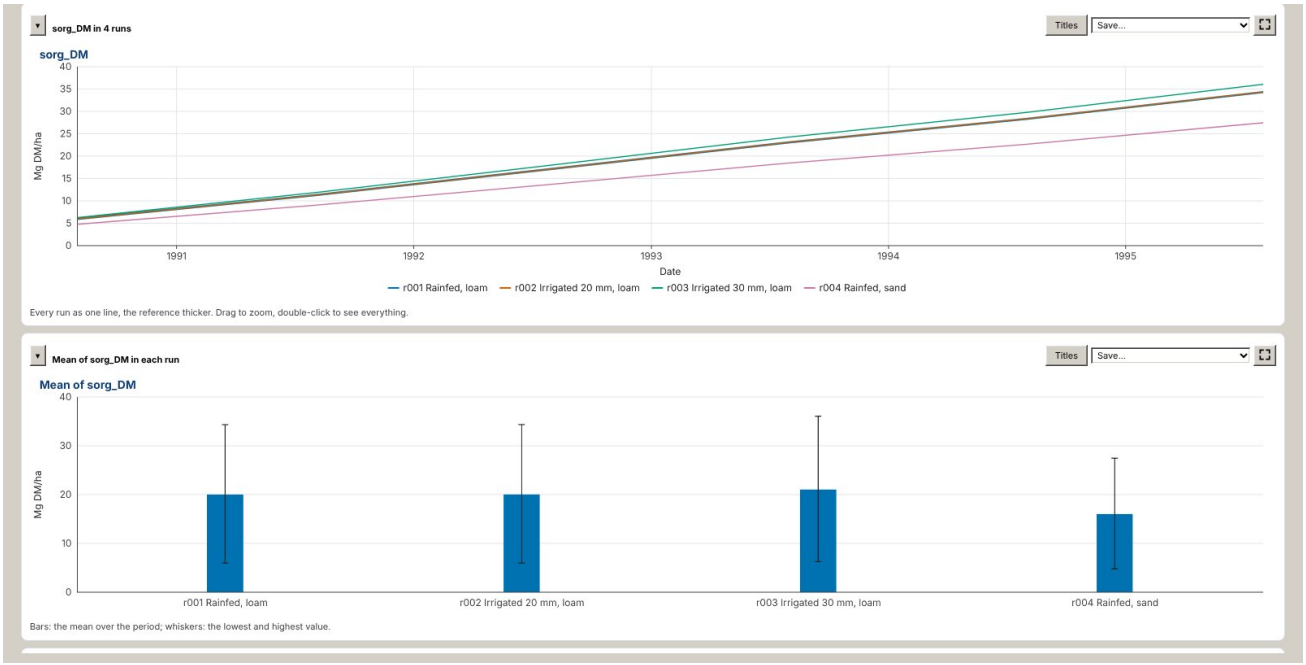




Figure 31.71: Many runs, one quantity: four runs of an experiment, cumulative yield, with the reference thicker and the mean of each run with its range.

## 31.7.26 Run comparison: the figures of a quantity

Click a quantity in the table to draw its three figures: A and B over time (B dashed), the difference B - A with the SML as a green band (a difference inside the band does not matter), and B against A with the 1:1 line (every date is one point). Each figure has the same controls as the other figures of the workbench.

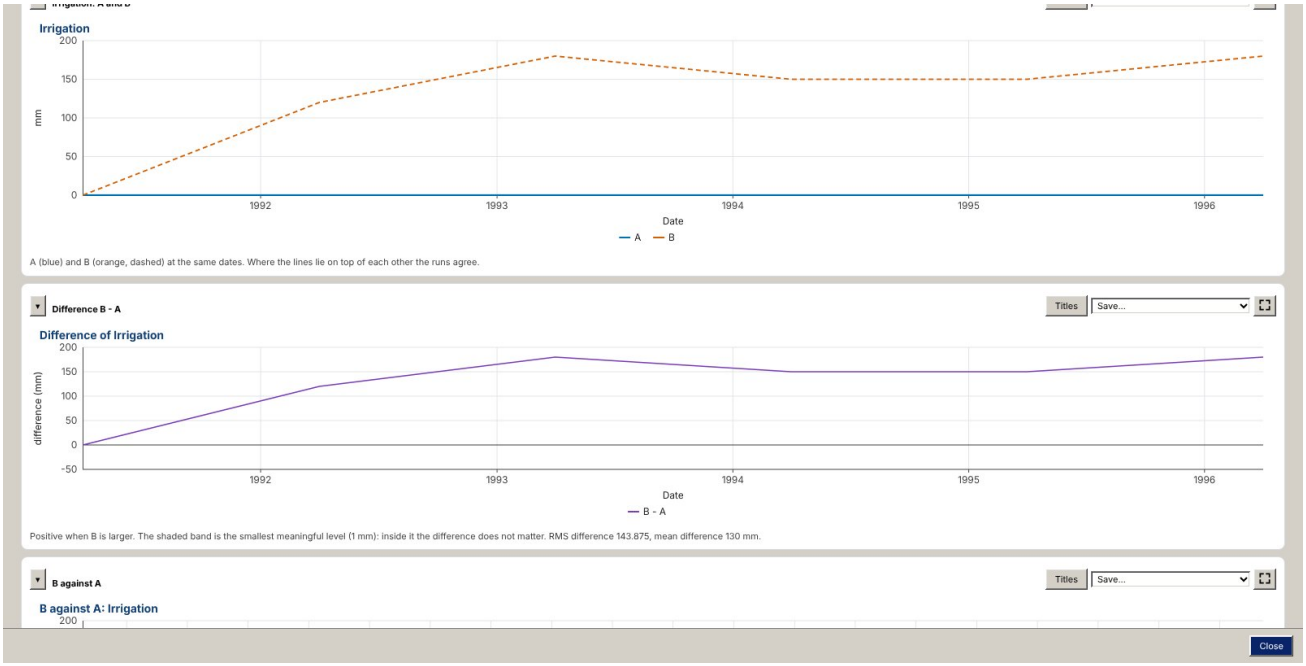



Figure 31.72: Run comparison: the figures of the chosen quantity.

### 31.7.27 Double-blind comparison

A judgement of two runs is easily coloured by knowing which is which (the new set-up, the expected answer). The double-blind comparison hides it from both judges, you and the program:

1. **Choose the two runs** (any log in Files, or any log of a run of Experiments or Scenarios, listed as *run : log*) and, if you have them, the **measurements** to judge against (a log or a table with the same column names).
2. **Start.** The program puts the runs behind the letters X and Y by a random draw and shows a fingerprint (the first 12 hexadecimal digits of the SHA-256 hash) of the hidden key, which holds the two names and a random salt. Neither the figures nor the numbers show the names, and the quantities come in a random order. Quantities that are identical in the two runs are left out.
3. **Judge.** For each quantity, read its three figures (X and Y, Y - X, Y against X) and press *X is better*, *Y is better* or *no real difference*. With measurements the program judges too, without the names: the run with the smaller root mean square error against the measurements is better, unless the two errors are within 2 % of each other.
4. **Reveal.** The names behind X and Y appear, the fingerprint is computed again from the key to show that it was not changed, and a table gives each quantity with your judgement, the program's and whether they agree, with the share of agreement. *Save the judgements (CSV)* keeps the record.

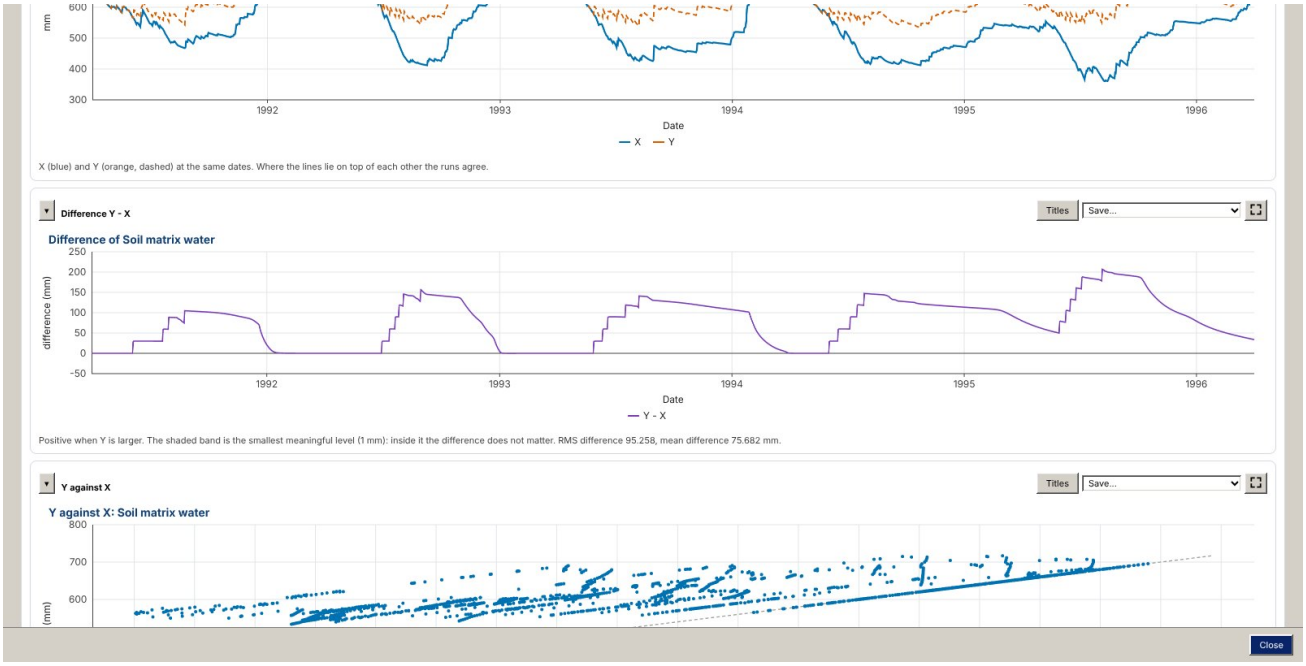



Figure 31.73: Double-blind comparison: the two daily water logs of the sample behind the letters X and Y, with the sealed key.

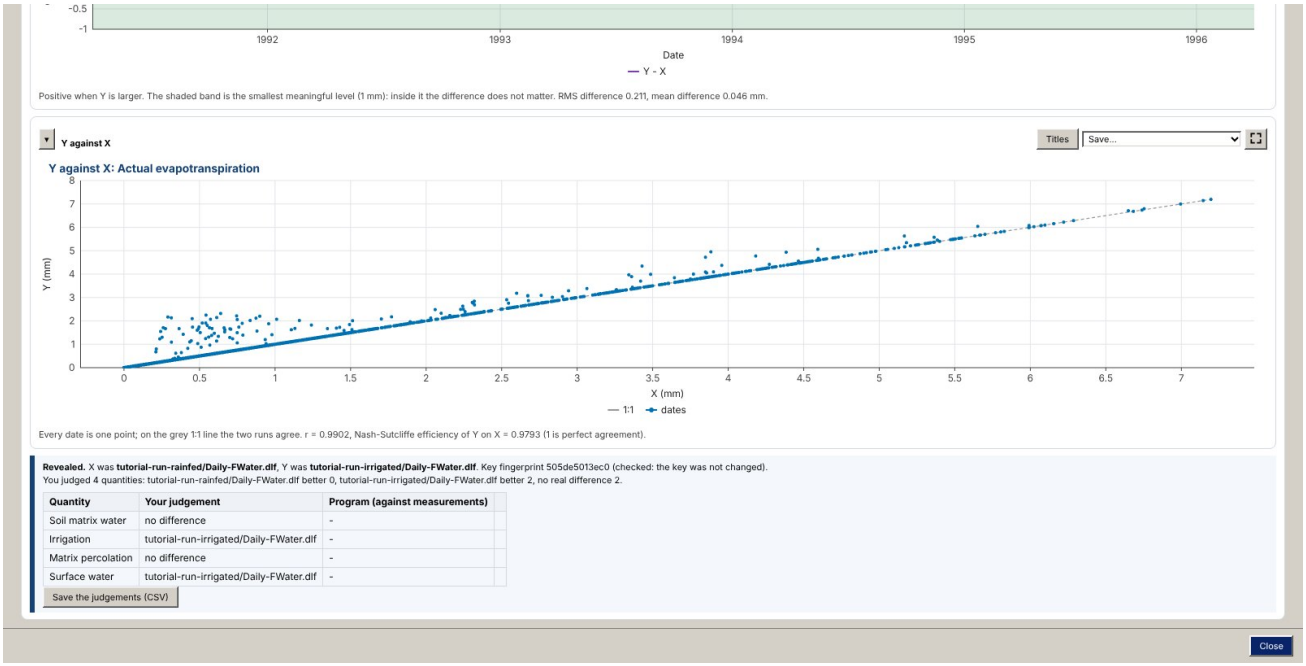



Figure 31.74: Double-blind comparison revealed: who was X and Y, the check of the key, and the judgements.

## 31.8 How the figures are drawn

Every figure of the workbench, in the scenarios, the processes, the experiments and the tabs Plot and Soil column, is drawn in the style of a journal figure and in the style of the figures of the SWAFL repository. The title is at the top left in bold. The panel is white with a light grid and the axes at the left and at the bottom only. A time axis has a tick for every year, and a run of less than a year has ticks for the months. The colours are the colour blind safe set (blue, orange, green, pink, yellow, sky blue), the legend is below the figure, and a map of depth against time has a colour bar with the unit written in the log file itself. Nothing but the title, the axes, the legend and the data is drawn: no citation and no caption is added to a figure, to a saved picture or to a saved report.

What each figure shows is written under its name, with the Daisy log it is drawn from, so that every figure can be traced to a file. A figure that has several panels is saved as one picture with all of its panels.

- The yield figure shows the storage organ at each harvest, coloured by crop, and below it the storage organ against the stem in the way of the daisyrVis exercise of the Daisy course.
- The water figures show the annual balance (rain and irrigation as bars, evapotranspiration and drainage as lines) and the running totals through the whole run.
- The water content and the suction are maps of depth against time. In the suction map the black marks FC and WP are field capacity (pF 2) and the wilting point (pF 4.2).
- The crop figure shows the leaf area index, the water and nitrogen stress and the dry matter of the plant parts. The stress of Daisy is 0 when there is none and 1 when the crop cannot grow.
- The nitrogen figure shows where the fertiliser goes (crop uptake, leaching below 100 cm, gaseous loss by denitrification and N2O, volatilisation) against the fertiliser applied and the mineralisation.

A column of a log that is zero throughout is left out of a process figure, and the note under the figure says which. A figure is never left empty without a sentence that says why.

## 31.9 The raw output of Daisy

Under the figures of a scenario, and under *One run in detail* of an experiment, the panel **The raw output of Daisy** gives the files exactly as Daisy wrote them, before any figure or summary of SWAFL.

1. **Kept on this computer.** The desktop app keeps the whole folder of every run in *SWAFL runs*, next to the Daisy that runs it: in the Daisy folder itself when it can be written, otherwise next to the Daisy that SWAFL keeps for itself (*%LOCALAPPDATA%/SWAFL/Daisy-7.1.14/SWAFL runs*). Runs of earlier versions in *Documents / SWAFL Daisy runs* are still listed and read. A scenario makes one folder named after the scenario and the time, with one sub-folder for each run. An experiment makes one folder with one sub-folder for each run, named by its number and its factors. The sub-folder *Run* holds the logs (.dlf), the .csv files and daisy.log, and the set-up that made them (Runfile.dai and the files it includes). The button *Open the folder* opens it in the file explorer. The folders stay until you delete them.
2. **Download all raw outputs (.zip).** One zip with the logs of all the runs of the scenario, untouched, and a README.
3. **Download outputs and set-up (.zip).** The same, with the files that were given to Daisy, so that the run can be repeated on any computer with Daisy 7.1.
4. **The list of files.** One row for each file with what it holds, the number of rows and columns, the size and a Download button.

A .dlf file is a text table: header lines (VERSION, LOGFILE, RUN, SIM…), a line of dashes, the column names, the units and then the numbers, separated by tabs. It opens in Excel or in any text editor, in R with the function read_dlf of the package daisyrVis, in Python with daisypy-io, and in SWAFL itself in the tab Files. The figures are drawn from these files only, so a figure can

always be checked against its file.

## 31.10 The efficiency indices

Every scenario that has what an index needs shows it in the summary table and in the figure “Efficiency indices”. Irrigation efficiency (IE) is the yield divided by the irrigation applied, in kg of yield per m3 of water (1 mm is 10 m3 per ha). Irrigation water use efficiency (IWUE) is the yield of the irrigated run minus the yield of the rainfed run, divided by the irrigation applied, in kg per m3; it needs a rainfed run, which the experiments add for every combination of the other factors. The deep percolation fraction (DPF) is the drainage below 100 cm divided by the irrigation plus the rainfall, in percent. The nitrogen efficiency is the partial factor productivity of nitrogen (PFPn), the yield divided by the nitrogen applied, in kg DM per kg N. The nitrogen leaching fraction (NLF) is the nitrogen leached below 100 cm divided by the nitrogen applied, in percent. An index that cannot be computed, for example IE without irrigation, is left out and not shown as zero.

Panel letters (a, b, c) mark the panels of the figures that have several.

## 31.11 The figures of one experiment

Every experiment of the study (E0 to E16) has its own question, its own design and its own figures, and the figures are drawn again from the runs each time results arrive, so they grow while the runs finish. Load an experiment with *Experiments of the study* at the top of the Experiments tab, then press *Run all*.

- The panel **Figures of this experiment** states the question and the design, and then asks which figures are wanted. All are ticked at first; untick the ones that are not needed, or open *Add a figure* and choose the kind (lines, bars, a surface over two factors, where the water goes, where the nitrogen goes, which factor matters most), the factor on the x axis, the factor that sets the colour, the factor that splits the figure into panels (a, b, c, …) and the response.
- Lines run with a colour bar when the colour factor is numeric (for example the duration of one irrigation). A surface is drawn smooth with white contour lines carrying their values, the tested combinations as dots and a colour bar. Panels of one figure share one scale, so they can be compared.
- An experiment of your own, or one that is changed, gets figures that follow its factors: the numeric factor with the most levels goes on the x axis, the next factor sets the colour and the third splits the panels.
- **Every run is one row.** The first figure, *The results of every run* (panel a), has one row for each run, named by its own values in short (for example *-14 d · N×0.7 · Sand · +2 °C · irrigated*), and one panel for each result (yield, irrigation, percolation, nitrate leached), each with its own scale and a dashed line at the mean of the runs. Runs are ordered by yield; rainfed references are open circles. Every combination therefore has its own result, side by side, and nothing is averaged away.
- **Main effects with their error.** The second figure, *Main effect of each factor* (panel b), has one panel for each factor that changes: the mean of the runs at each value, one standard error and the number of runs (n) under each value, on one y scale shared by the panels, with the mean of all runs as a dashed line. It is the figure of a screening or factorial design in a journal.
- **Complete or not.** The note above the figures says whether every combination of the values was run. When it was, the figures of the experiment follow (lines, bars, surfaces by factor). When only some combinations were run (the number of runs was smaller than the number of combinations), lines and surfaces by factor would join runs that differ in other values as well, so they are not drawn: the figures are then every run, the main effects, the effect ranking and the water and nitrogen balances run by run (one stacked bar for each run).

- **Style.** The figures follow the conventions of journal figures: grey facet strips over the panels, panel letters, the Okabe-Ito colours (readable by people with colour blindness), a light panel background with a fine grid, units in every axis title, and a one-line legend under the figure. Labels that are longer than their room are shortened with an ellipsis, never written over each other.
- **No empty figures.** A figure whose logs a run did not write is left out, without a message; a process opens only the figures that can be drawn.
- Below the figures, **the processes of Daisy in this experiment** are drawn as a diagram of three columns: on the left what the experiment changes (F1, F2, in orange frames), in the middle only the processes of Daisy it involves, in the fills of Fig. 1 of Daisy (blue for water, green for heat, the crop and residues, beige for nitrogen, brown for organic matter), and on the right what it measures (R1, R2, in blue frames). One arrow joins a factor to each process it acts on and a process to each result that comes from it; nothing else is drawn. A process that both receives a factor and gives a result has an orange frame inside a blue one. Click a process and the figures that depend on it light up; click it again to release. The scenarios draw the same diagram for their own factors and results.
- Every figure has the same toolbar: collapse, *Titles* (the title, the axis texts and the panel letter can be written as the paper needs them), and *Save* as a picture at screen resolution, 300 dpi or 600 dpi, copy, or the numbers behind the figure as a CSV table. A figure of several panels is saved as one picture.
- **Each experiment keeps its own outputs.** The runs of one experiment go to one folder named by the experiment, its title and the time (for example E5_Tillage-system-by-consolidation_2026-10-29-21-25), never mixed with another experiment. When the runs have finished, the program writes into that folder an *Outputs* folder with README.txt (the question and the design), plan.json (to load the experiment again), design.csv (every run and its factor levels), results.csv (the responses of every run) and a folder *figures* with every figure as a 300 dpi picture and its numbers. The button *Save the outputs of this experiment* writes them again, and *as a zip* downloads them.

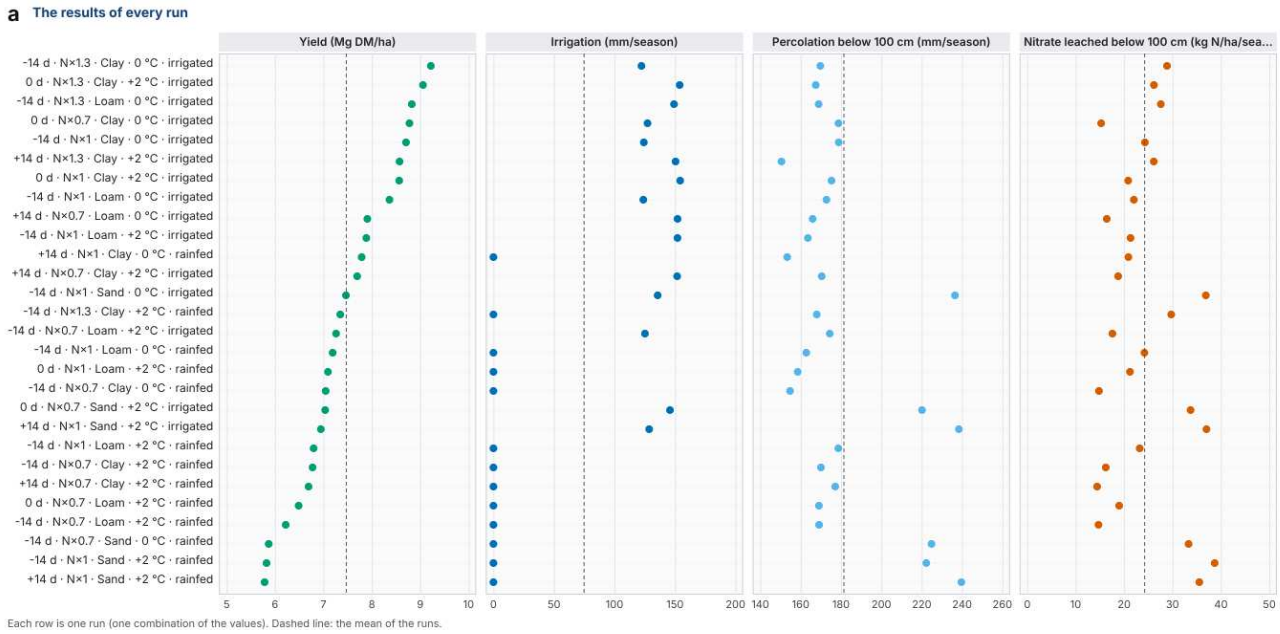



Figure 31.75: The results of every run: one row for each combination of a screening design (28 of 108 combinations), named by its values, with yield, irrigation, percolation and nitrate leached side by side; dashed lines are the means of the runs.

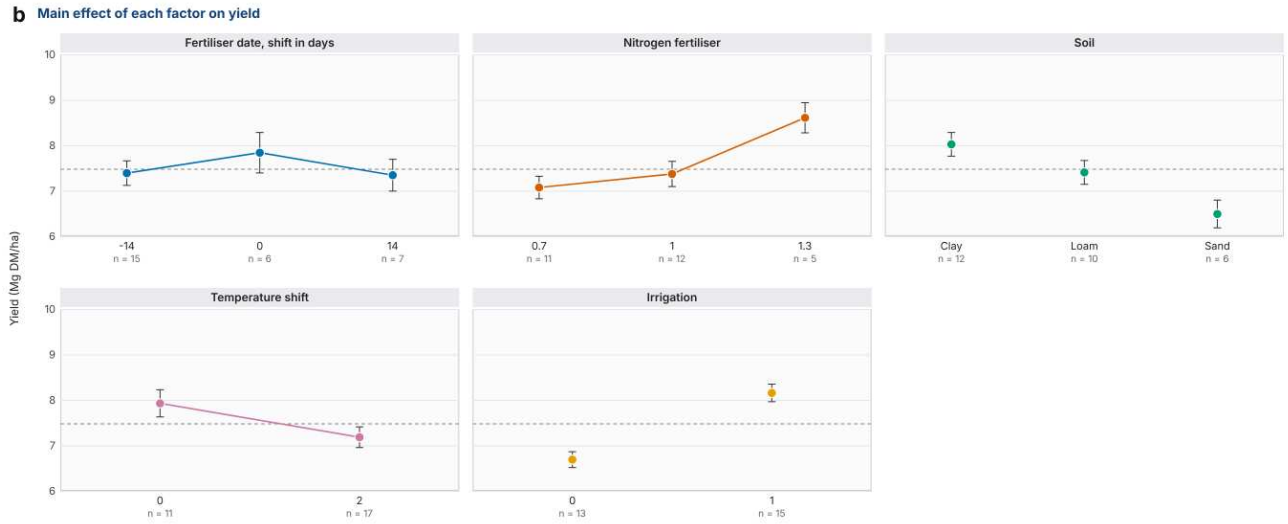



Figure 31.76: The main effect of each factor: the mean of the runs at each value with one standard error and the number of runs; one y scale for all panels, the dashed line is the mean of all runs.

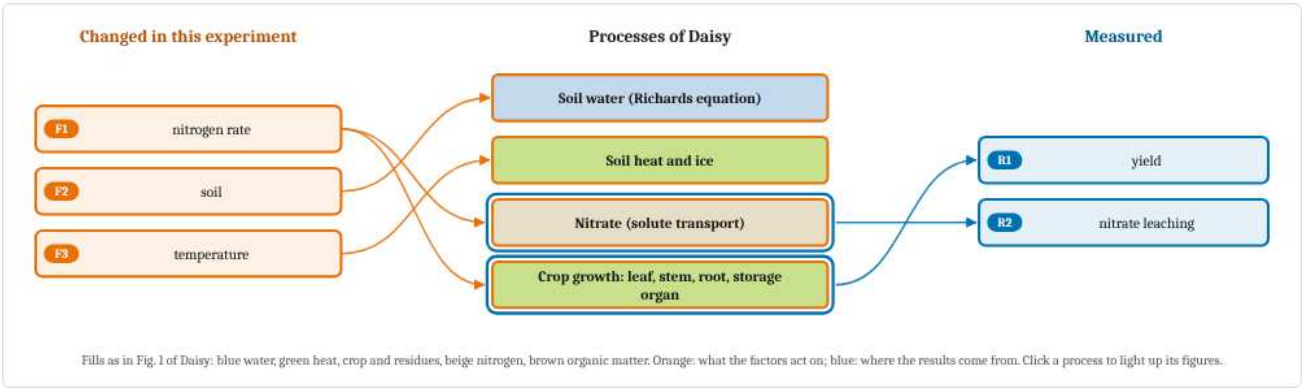



Figure 31.77: The processes of Daisy in one experiment: the factors (F, orange) act on the processes in the middle, drawn in the fills of Fig. 1 of Daisy, and the results (R, blue) come from them.

## 31.12 Efficiencies, budgets and trade-offs, from the literature

Under every experiment, and under every scenario with three or more runs, the group **Efficiencies, budgets and trade-offs, from the literature** draws 21 figures from the results measured in the runs. Nothing is typed in: every number comes from the logs of the runs, and every figure has the toolbar of every figure (*Titles*, *Save* as a picture at 300 or 600 dpi, copy, or the numbers behind it as a CSV table). The figures are drawn when they come into view, so the page opens at once. Point at a dot or a bar for its numbers. The box *Colour the runs by* at the top sets the quantity that gives the colour of the dots and lines in all figures, and *Second objective of the trade-off* sets the second axis of the Pareto figure. A figure whose quantities the runs did not measure is left out, without a message.

**The indices.** They are added to the results of every run, so they can also be chosen as the result of the whole experiment (Section 31.20).

| Result | Formula | Source |
|---|---|---|
| Nitrogen use efficiency, NUE (%) | 100× crop uptake of N ⁄ N input | EU Nitrogen Expert Panel 2015 |
| Nitrogen surplus (kg N per ha) | N input − crop uptake of N | EU Nitrogen Expert Panel 2015 |
| Reactive nitrogen lost | nitrate leached + denitrified + volatilised N | n/a |
| Nitrate leached per Mg of yield | nitrate leached ⁄ yield | van Groenigen et al. 2010 |
| Agronomic efficiency, AE (kg per kg N) | (yield − yield of the least fertilised run) ⁄ extra N applied | Dobermann 2005 |
| Apparent recovery, RE (%) | 100× extra uptake ⁄ extra N applied | Dobermann 2005 |
| Physiological efficiency, PE (kg per kg N) | yield gain ⁄ extra N taken up (AE = RE × PE ⁄ 100) | Dobermann 2005 |
| Water-use efficiency (kg per m3) | yield ⁄ evapotranspiration | French and Schultz 1984 |
| Yield per m3 of rain and irrigation | yield ⁄ (rain + irrigation) | Howell 2001 |
| Harvest index | storage organ ⁄ total dry matter | n/a |
| Relative evapotranspiration | actual ⁄ potential | Doorenbos and Kassam 1979 |
| Nitrate in the drainage (mg N per L) | 100× nitrate leached ⁄ drainage | EU Nitrates Directive 91/676/EEC (limit 11.3 mg N per L) |
| Chemical leached, degraded (%) | share of the amount sprayed | n/a |

**Nitrogen.**

1. *Nitrogen budget of every run*: a bar for every run, split into crop uptake, nitrate leached, denitrified, volatilised and the rest left in the soil, ranked by surplus.
2. *Nitrogen use efficiency against nitrogen surplus*: the diagram of the EU Nitrogen Expert Panel. The green box is the target (NUE 50 to 90 % and a surplus below 80 kg N per ha); left of it nitrogen is applied in excess, right of it the soil stock is mined.
3. *Yield against nitrogen use efficiency*, with the line of best fit.
4. *Nitrogen response curve*: yield against the nitrogen applied with a quadratic fit $y = a + bx + cx^2$, its R squared and, when the curve bends down, the rate of maximum yield.
5. *Agronomic efficiency, recovery and physiological efficiency*: three panels sharing the rows.
6. *Nitrate leached against nitrogen surplus*, with the line of best fit: the slope is the share of the surplus that leaves as nitrate.
7. *Nitrate in the drainage water*: nitrate leached against drainage with the lines of 5, 11.3 and 20 mg N per L; runs above the 11.3 line exceed the limit of the Nitrates Directive.
8. *Pathways of the reactive nitrogen lost*: the share of leaching, denitrification and volatilisation; a shift between them is pollution swapping.
9. *Nitrate leached per tonne of yield* against the nitrogen applied.

**Water.**

10. *Water balance of every run*: evapotranspiration, drainage, runoff and the change in storage as shares of rain plus irrigation.
11. *Water production function*: yield against evapotranspiration with a line; its slope is the water productivity per extra millimetre.
12. *Yield response factor Ky*: $1 - Y_a/Y_m = K_y\,(1 - ET_a/ET_m)$, the relative yield decrease against the relative evapotranspiration deficit with the dashed 1:1 line; the slope of the fitted line is Ky.
13. *Irrigation water productivity* against irrigation, when the experiment has a rainfed run to compare with.
14. *Water stress against nitrogen stress*: the days of each stress before harvest; runs above the dashed line lack nitrogen first, runs below it lack water first.
15. *Water and nitrogen efficiency together*: water-use efficiency against NUE with the front of the runs no other run beats on both.

**All results together.**

16. *Correlation of the results*: Pearson r between every pair of results (red: rise together, blue: one rises while the other falls).
17. *Trade-off and Pareto front*: yield against a second objective of your choice; the red line joins

the runs that no other run beats on both counts.

18. *All results of every run side by side*: parallel coordinates, each axis from the lowest to the highest value.
19. *Which factor moves the result most*: for each factor the mean result at its lowest and highest value against the mean of all runs, ranked by the length of the bar.
20. *Distribution of a result*: histogram and cumulative curve with the quartiles.
21. *Combined score of the runs*: every result scaled from the worst to the best run (inverted where less is better) and averaged with equal weights; stacked bars show what each result contributes.

Two more figures complete the group. *Dose-response of each factor on the main results* draws, for every factor, the mean of each main result at each value, scaled from 0 to 1 so that results in different units share one axis. For chemicals, *GUS leaching index* sets the simulated leaching against $GUS = \log_{10}(DT_{50})\,(4 - \log_{10} K_{OC})$ of Gustafson 1989: a chemical above 2.8 is a likely leacher and one below 1.8 a non-leacher.

## 31.13 Interactions and time series, for the paper

The group *Interactions and time series, for the paper* has five figures for the results of one experiment.

- **Interaction map of the factors.** Each cell is the share of the variance of the results explained by one factor alone (the diagonal, blue) or by a pair of factors beyond their separate effects (red, an interaction). Click a cell to open the interaction plot of that pair.
- **Interaction plot.** The mean result at each value of one factor, one line for each value of another, with the lowest and highest run as a whisker; the same numbers with the two factors swapped are in panel (b). Lines that are not parallel mean an interaction.
- **Effect of a factor through the year.** For matched pairs of runs that differ only in that factor, the difference of a daily quantity day by day, one line for each value of the other factor, with the band of 10 to 90 %.
- **Every run through the year.** One line for each run, coloured by its final result, with the lowest and highest run in black.
- **Runs against days.** One row for each run, ranked by the result, one column for each day of the year, coloured by the quantity.

The quantity is chosen in the box *Quantity* (the daily logs of the runs) and can be shown as the value or as the running total. The group is drawn from the logs, so it needs runs with daily logs, or at least two factors.

## 31.14 Where the results are saved: you choose the folder

SWAFL does not write the results of experiments to a hidden folder any more. While the runs are made, the files are kept in a temporary place that is shown only as a note; nothing is saved to the computer by itself. When the runs have finished, *Save the outputs of this experiment* and *Save both: outputs and raw Daisy files* ask for a folder with the usual window of the computer, and write into the folder chosen. *Open folder* shows what was saved. If the window is closed without a choice, nothing is written and the results stay on the screen for another try.

## 31.15 Hover texts and the details of the measured process map

The process map of an experiment or a scenario now draws every measured result with the range of its values in the runs (for example *yield 4.1 to 8.9 Mg per ha*) and every factor with its levels, and the result selected above the map is outlined. For a scenario the map shows only the factors

that the variants you ran changed and the results measured in the logs of those runs, not a generic list. Every button and every figure has a hover text that says what it does.

## 31.16 Every log of every run

Under the figures of an experiment, the panel **Every log of every run** turns everything Daisy wrote into figures: every log of the runs (the daily and the annual logs, and every log of the Daisy library that was added to the runs) and every quantity in it, one figure each. Nothing has to be chosen first; the quantities a paper usually shows come first (leaf area index, the storage organ, the development stage, evapotranspiration, transpiration, soil evaporation, percolation and leaching, crop nitrogen, nitrate, stress, water content), then all the others.

- **The layout of a journal figure.** One panel for each value of one factor (*Panels by*), one line for each value of a second factor (*Lines by*, or *every run* for one line per run), the other factors averaged over the runs that share these two values. All panels have one y scale and are lettered (a), (b), (c); the grey strip over a panel names its value; the legend is written once under the panels. Lines for numbers in three or more steps follow a colour ramp (viridis), other lines the Okabe-Ito colours.
- **Daily logs** are drawn *through the year* (the default): for every day of the year, the mean (or the median, *Line*) of all seasons and runs on that day, with 10 to 90 % of them as the shaded band, so the typical season and its spread are seen at once. *Over the whole period* draws every day of the run instead, with the lowest and the highest run as the band.
- **Annual logs** are drawn year by year, with open circles on the seasons and the lowest to the highest run as the band. The first record of an annual log (the start of the run, zero) is left out.
- **Depth logs** (water content, pressure, nitrate, temperature at each depth) give two figures: a map of depth (down) against time, one map for each value of the panel factor with one colour scale for all, and the profile, the mean at each depth with 10 to 90 % of the days and runs as the band.
- *Log* shows the figures of one log, and the search box finds a quantity by its name (LAI, nitrate, evaporation). The note above every figure says what it shows: the log, the statistic, the band, and the factors that were averaged.
- Every figure has the same toolbar as the others (titles, picture at screen resolution, 300 or 600 dpi, the numbers as a CSV table). *Save these figures (.zip)* saves all of them; they are also saved with the outputs of the experiment.

The daily figures need daily logs. **Results saved** under the plan is now *yearly totals and daily crop, water and soil water* by default, so every experiment writes them; *yearly totals only* is faster but gives the annual figures alone.

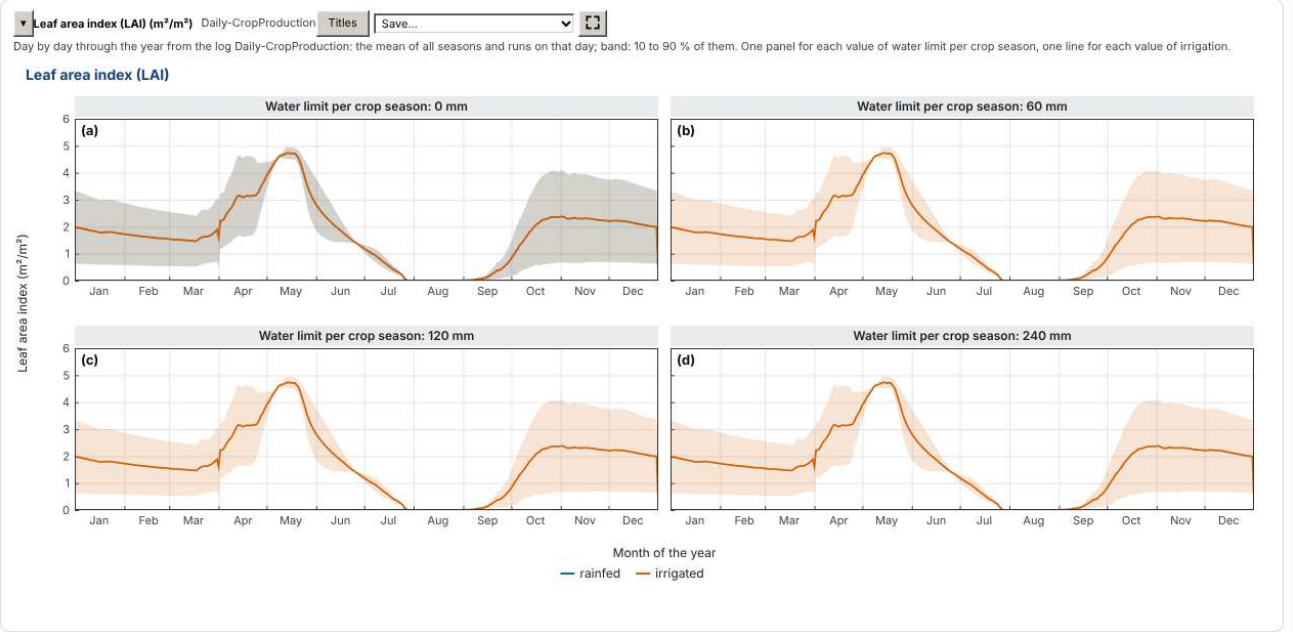



Figure 31.78: Leaf area index through the year: one panel for each water limit, one line for rainfed and irrigated, the mean of the seasons with 10 to 90 % as the band.

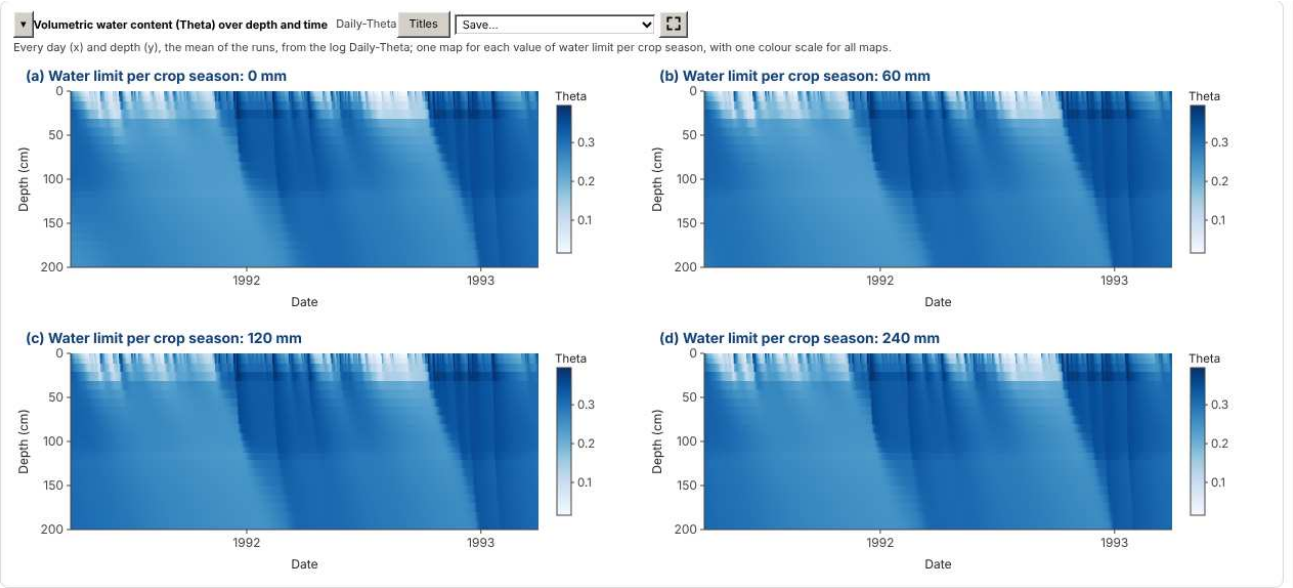



Figure 31.79: Water content over depth and time, one map for each water limit with one colour scale, followed by the profiles.

### 31.16.1 Monthly, yearly and cumulative

*Time* above the figures of every log has five choices: *through the year* (the mean of the seasons, day by day), *every day of the period*, *monthly*, *yearly* and *cumulative over all the years*. Monthly and yearly values are the totals of the month or the year for quantities that flow (water in mm, nitrogen in kg N/ha: rain, evapotranspiration, percolation, leaching) and the means for states (leaf area, water content, dry matter); the y axis says which ("per month", "per year"). *Cumulative* is the running total of every run from the start of the run, so the runs part as their totals grow

apart. Annual logs offer *yearly* and *cumulative*. The *Log plots* tab has the same choices for one log (*Monthly values*, *Annual values*, *Cumulative over all the years*).

The figures of each log are a group of their own that folds open and shut (the first is open), and the results of an experiment are in folding groups too: *Figures of this experiment*, *Which combination is best*, *Effects, interactions and trade-offs*, *Every log of every run*, *All the outputs of every run* and *One run in detail*.

## 31.17 Why a run gave no results

When Daisy gives no results for some runs, a panel **Why N runs gave no results** opens under the progress bar, and the list of runs names the cause of each one. The runs that failed for the same reason are put together, so every cause is explained once:

- **Daisy said**: the error line of Daisy, exactly.
- **Where**: the file, the line and the column, and the lines around it from the set-up of the run, with the place marked.
- **Root cause**, in plain words, sorted into one of these kinds: *a name Daisy does not know* (a crop, a log, a horizon used but not defined: often a file of the Daisy samples that still refers to its own run), *a numerical problem* (the solver of the Richards equation or of the solutes did not converge, or a value became not a number or negative: nearly always an extreme combination of soil hydraulic values, thin layers or a very large irrigation flux), *a value outside what Daisy accepts*, *a file that is missing*, *the weather* (years or gaps), *a name defined twice*, *a line Daisy could not read*, and *Daisy stopped without a message* (cancelled, memory, an antivirus).
- **What to do**: the change that removes the cause, and the runs that share it.

For example, *test-prd.dai:292:29: Unknown 'log' model 'crop'* came from the soil KarJyn_B, a horizon of the Daisy sample test-prd.dai used as it is: the sample file also carries the logs of its own run, which need a log model SWAFL runs do not load. SWAFL now takes only the definitions from such a sample file and leaves its logs out; a run with this soil, tested with Daisy 7.1.14, finishes and writes all its logs.

Figure 31.80: Why three runs gave no results: two causes, the line of the set-up and what to do.

## 31.18 Every simulated experiment

At the bottom of the Experiments tab, **See all the Daisy output files** opens every experiment that was run, in order, in the full-screen viewer: the folder of each experiment (its name and the time), the folder of each run in it (r000, r001, …), and in each run the files Daisy wrote, sorted (the annual logs, the daily logs, daisy.log), with their sizes. A click on a file opens it on the right in the colours of Notepad++ (line numbers, find, wrap); a file is read from the disk only when it is opened, so even hundreds of runs open at once. On the desktop the list is the folder *SWAFL runs* next to Daisy (and older runs in Documents, under *Older runs*); in the web version, the experiments of the session. *Save the folder (.zip)* saves them all.

The figures always follow the runs: when an experiment has new runs, the Log plots tab drops the runs of the previous one and loads the new ones, and the figures of every log, the summary and the ranking are drawn from the same runs.

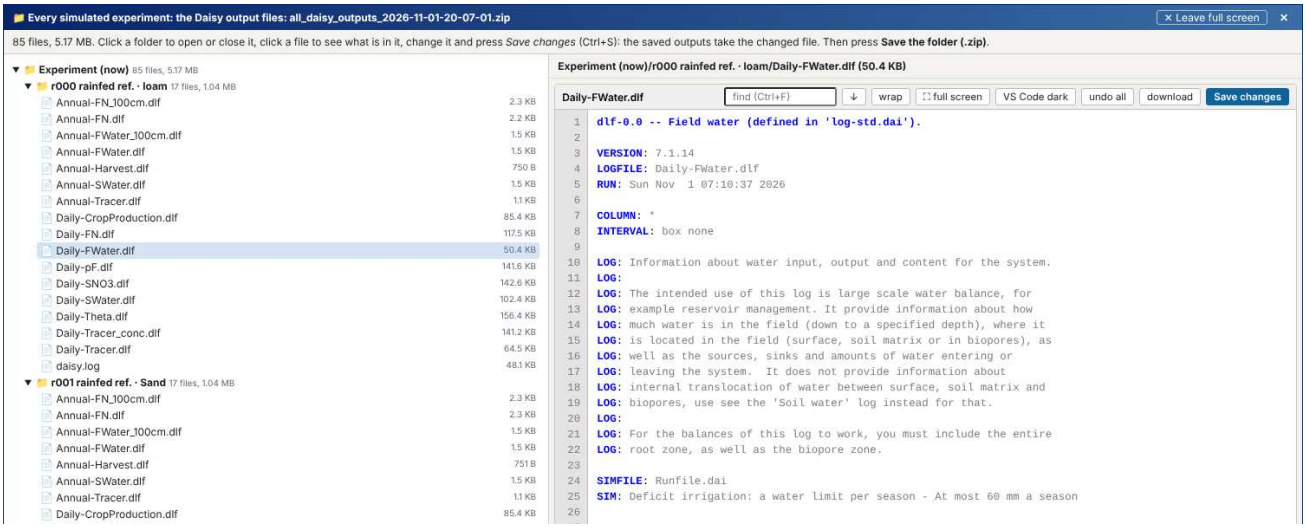
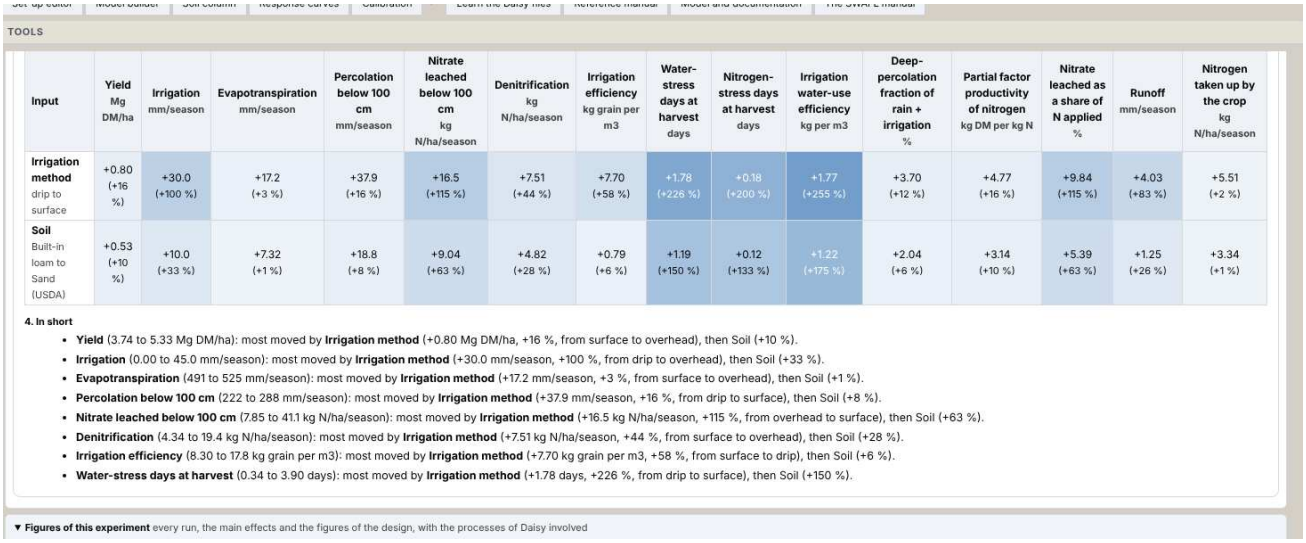



Figure 31.81: Every simulated experiment: experiment, run and files in order, a log open in the Notepad++ colours.

## 31.19 The summary of an experiment

The first group of the results, **Summary: every range and its effect**, gathers the whole experiment in four parts, and *Summary (CSV)* saves it (it is also *Outputs/summary.csv* with the outputs):

1. **What was varied**: every input with the values tested.
2. **The range of every result** over all runs: lowest, highest, mean, spread (coefficient of variation) and the best and the worst run by name.
3. **The effect of every input on every result**: a table with one row per input and one column per result. For a number the effect is the change of the mean from its lowest to its highest value, for a choice from the choice with the lowest result to the one with the highest; the share of the mean is in brackets. Blue cells rise, red cells fall, and the stronger the colour the larger the effect; pointing at a cell shows the mean at every value.
4. **In short**: for every result, the input that moves it most, by how much and between which values, then the next two.



TOOLS

| Input | Yield Mg DM/ha | Irrigation mm/season | Evapotranspiration mm/season | Percolation below 100 cm mm/season | Nitrate leached below 100 cm kg N/ha/season | Denitrification kg N/ha/season | Irrigation efficiency kg grain per m3 | Water-stress days at harvest days | Nitrogen-stress days at harvest days | Irrigation water-use efficiency kg per m3 | Deep-percolation fraction of rain + irrigation % | Partial factor productivity of nitrogen kg DM per kg N | Nitrate leached as a share of N applied % | Runoff mm/season | Nitrogen taken up by the crop kg N/ha/season |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Irrigation method** drip to surface | +0.80 (+16 %) | +30.0 (+100 %) | +17.2 (+3 %) | +37.9 (+16 %) | +16.5 (+115 %) | +7.51 (+44 %) | +7.70 (+58 %) | +1.78 (+226 %) | +0.18 (+200 %) | +1.77 (+255 %) | +3.70 (+12 %) | +4.77 (+16 %) | +9.84 (+115 %) | +4.03 (+83 %) | +5.51 (+2 %) |
| **Soil** Built-in loam to Sand (USDA) | +0.53 (+10 %) | +10.0 (+33 %) | +7.32 (+1 %) | +18.8 (+8 %) | +9.04 (+63 %) | +4.82 (+28 %) | +0.79 (+6 %) | +1.19 (+150 %) | +0.12 (+133 %) | +1.22 (+175 %) | +2.04 (+6 %) | +3.14 (+10 %) | +5.39 (+63 %) | +1.25 (+26 %) | +3.34 (+1 %) |

**4. In short**

- **Yield** (3.74 to 5.33 Mg DM/ha): most moved by **Irrigation method** (+0.80 Mg DM/ha, +16 %, from surface to overhead), then Soil (+10 %).
- **Irrigation** (0.00 to 45.0 mm/season): most moved by **Irrigation method** (+30.0 mm/season, +100 %, from drip to overhead), then Soil (+33 %).
- **Evapotranspiration** (491 to 525 mm/season): most moved by **Irrigation method** (+17.2 mm/season, +3 %, from surface to overhead), then Soil (+1 %).
- **Percolation below 100 cm** (222 to 288 mm/season): most moved by **Irrigation method** (+37.9 mm/season, +16 %, from drip to surface), then Soil (+8 %).
- **Nitrate leached below 100 cm** (7.85 to 41.1 kg N/ha/season): most moved by **Irrigation method** (+16.5 kg N/ha/season, +115 %, from overhead to surface), then Soil (+63 %).
- **Denitrification** (4.34 to 19.4 kg N/ha/season): most moved by **Irrigation method** (+7.51 kg N/ha/season, +44 %, from surface to overhead), then Soil (+28 %).
- **Irrigation efficiency** (8.30 to 17.8 kg grain per m3): most moved by **Irrigation method** (+7.70 kg grain per m3, +58 %, from surface to drip), then Soil (+6 %).
- **Water-stress days at harvest** (0.34 to 3.90 days): most moved by **Irrigation method** (+1.78 days, +226 %, from drip to surface), then Soil (+150 %).

▾ **Figures of this experiment** every run, the main effects and the figures of the design, with the processes of Daisy involved

Figure 31.82: The effect of every input on every result, and the summary in words.

## 31.20 Which combination is best

Every run of an experiment is one combination of the inputs. The group *Which combination is best* ranks all of them: each result is scaled from the worst run (0) to the best run (1), higher being better for yield, uptake and the efficiencies and lower being better for irrigation, leaching, percolation, denitrification, runoff and stress, and the score is the weighted mean. The weights are boxes over the table (yield 1, nitrate leached 1 and irrigation 0.5 at first; 0 leaves a result out), and the ranking follows every change. The three best combinations are marked, a bar shows each score, and under the table *the best value of each input* lists the values of every input in the order of the mean score of their runs, with the number of runs, followed by the best combination in words.

| Rank | Combination of the inputs | Yield Mg DM/ha | Irrigation mm/season | Nitrate leached below 100 cm kg N/ha/season | Score (0 to 1) |
|---|---|---|---|---|---|
| 1 | rainfed ref. · loam | 5.32 | 0.00 | 15.4 | 0.90 |
| 2 | drip · loam | 5.32 | 0.00 | 15.4 | 0.90 |
| 3 | rainfed ref. · Sand | 5.33 | 30.0 | 9.56 | 0.85 |
| 4 | drip · Sand | 5.33 | 30.0 | 9.56 | 0.85 |
| 5 | overhead · Sand | 5.33 | 45.0 | 7.85 | 0.80 |
| 6 | surface · loam | 5.33 | 45.0 | 7.85 | 0.80 |
| 7 | overhead · loam | 5.33 | 45.0 | 8.15 | 0.80 |
| 8 | surface · Sand | 3.74 | 45.0 | 41.1 | 0.00 |

**The best value of each input** (the mean score of the runs with that value, best first)

- **Irrigation method**: **drip (0.88, n = 2)** > overhead (0.80, n = 2) > surface (0.40, n = 2)
- **Soil**: **Built-in loam (0.85, n = 4)** > Sand (USDA) (0.62, n = 4)

**Best combination:** rainfed ref. · loam (score 0.90); next: drip · loam (0.90).

Figure 31.83: Every combination ranked by its score, with the best value of each input.

## 31.21 All the outputs of every run

The group *All the outputs of every run* lists every run of the experiment, not only one: each run folds open to its files (what each holds, rows, size, a download button). Tick files one by one, a whole run (the box in its title), or a kind of file in every run at once (*Tick this file in every run*,

for example Daily-FWater.dlf), then *Save the ticked files (.zip)*; *Save everything (.zip)* saves all of them. *with the set-up of each run* adds the files that made the run, and *with the outputs of the experiment* adds the figures (300 dpi, with the numbers as CSV), tables and README of the experiment (always with *Save everything*). The .zip has *Raw//outputs*, *Raw//setup* and *Outputs/*.

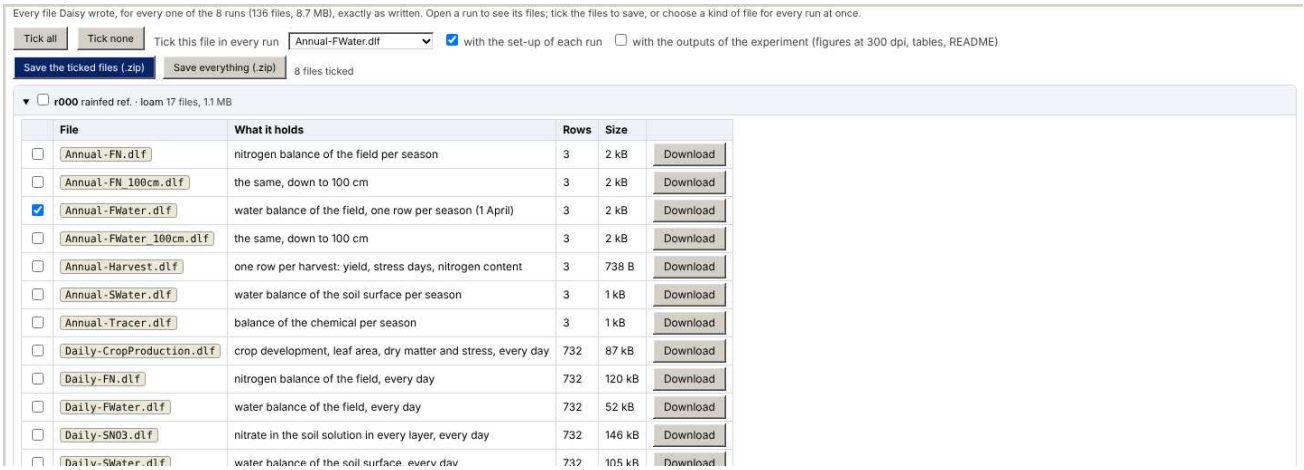



Figure 31.84: All the outputs of every run: one folding group per run, files ticked one by one or by kind.

## 31.22 Viewing and editing the outputs of Daisy before they are saved

The outputs here are the outputs of the Daisy simulation: the files Daisy wrote in every run (the .dlf logs such as Annual-Harvest, Daily-FWater or Daily-Theta, and daisy.log), not the figures. There are two ways to look at them and change them before they are saved.

- **One file at a time, in the style of Notepad++.** In *All the outputs of every run* (and in *One run in detail*), every file has **View / edit**. It opens the file in an editor that fills the screen: a tab for every file opened (click a tab to go back to it, × closes it), a grey margin with the line numbers, the header lines of the log in blue, the column names, the units in purple and the numbers in orange, find (Ctrl+F), wrap and *colours* (Notepad++, VS Code light or VS Code dark). Change the text and press *Save changes* (Ctrl+S): the file is marked *edited* in the list, and every save of the page (*Save the ticked files*, *Save everything*, *Download*) takes the edited text, while the text as Daisy wrote it is kept beside it in the .zip, in *outputs/_as_written_by_Daisy/. Back to the text of Daisy* drops the changes, *Download this file* saves the file alone, and on the desktop *Save into the run folder* writes it to *Edited/* in the folder of the run (the file Daisy wrote is left as it was). Esc or *Close* returns to the list.
- **All of them together. View and edit the Daisy outputs (full screen)**, in the results bar and in *All the outputs of every run*, opens every file Daisy wrote, one folder per run, in a viewer that fills the screen: the folders and files with their sizes on the left, the file in the Notepad++ editor on the right. *Save the folder (.zip)* saves all of them with the changes; on the desktop *Write them next to the runs* writes them into *Daisy outputs* in the folder of the experiment.

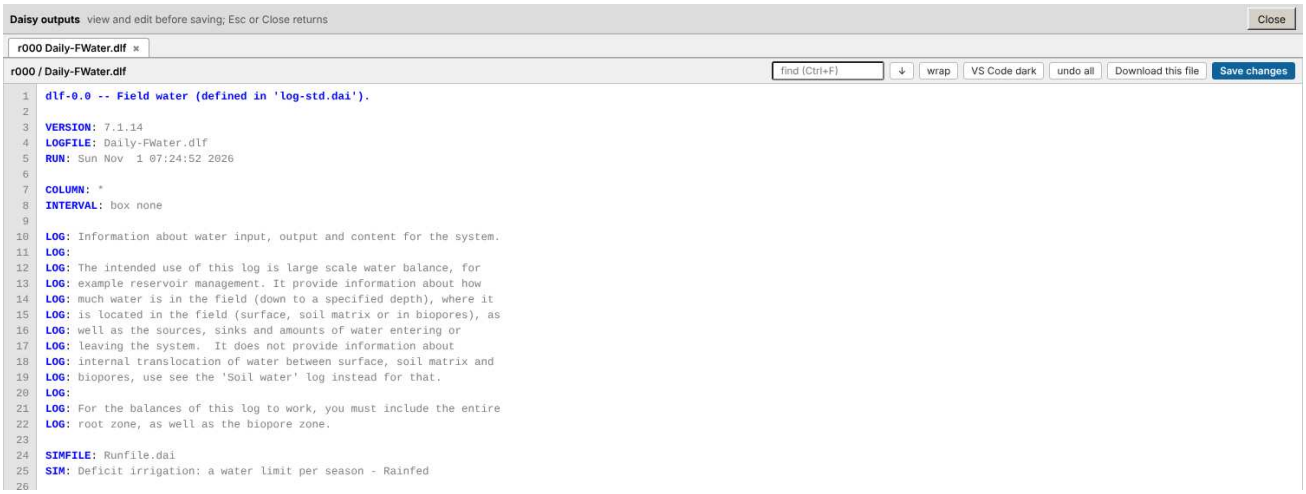



Figure 31.85: A Daisy output (Daily-FWater.dlf of one run) in the full-screen editor, in the colours of Notepad++.

## 31.23 Saving the outputs and the raw files of Daisy together

The results bar has three ways to save: *Save the outputs of this experiment* (README, plan, design, results table and every figure, written next to the runs on the desktop), *as a zip*, and **Save both: outputs and raw Daisy files**, one .zip with two folders:

- *Outputs/*: everything SWAFL made from the runs, the figures of every log included, each as a picture at 300 dpi with the numbers behind it as a CSV table.
- *Raw/*: one folder for every run, with *outputs/* (the .dlf logs and daisy.log exactly as Daisy wrote them) and *setup/* (Runfile.dai and the files it includes), so every run can be repeated and every figure checked against the numbers of Daisy.

On the desktop the outputs are written next to the runs as well. In the table of the raw output of a run, the name of the run is written once, beside all of its files.

## 31.24 Compare similar runs in Log plots

Under the plot of one log, *Compare similar runs* draws the same log of several runs in one figure: the runs of an experiment, of a scenario, or loaded files that hold the same log. The runs of the session are added to the loaded logs by themselves, so the comparison is ready when the tab opens. Choose the *Log* (every kind of log that two or more runs wrote), the *Column*, the *View* (every day, through the year, monthly, yearly or cumulative, with *Sum or mean* for monthly and yearly), tick the runs, and choose a *Reference*: a second figure then shows **each run minus the reference**, the effect of what differs between the runs (the rainfed reference of an experiment is chosen by itself). Each figure saves as a picture and the numbers of all runs as one CSV table.

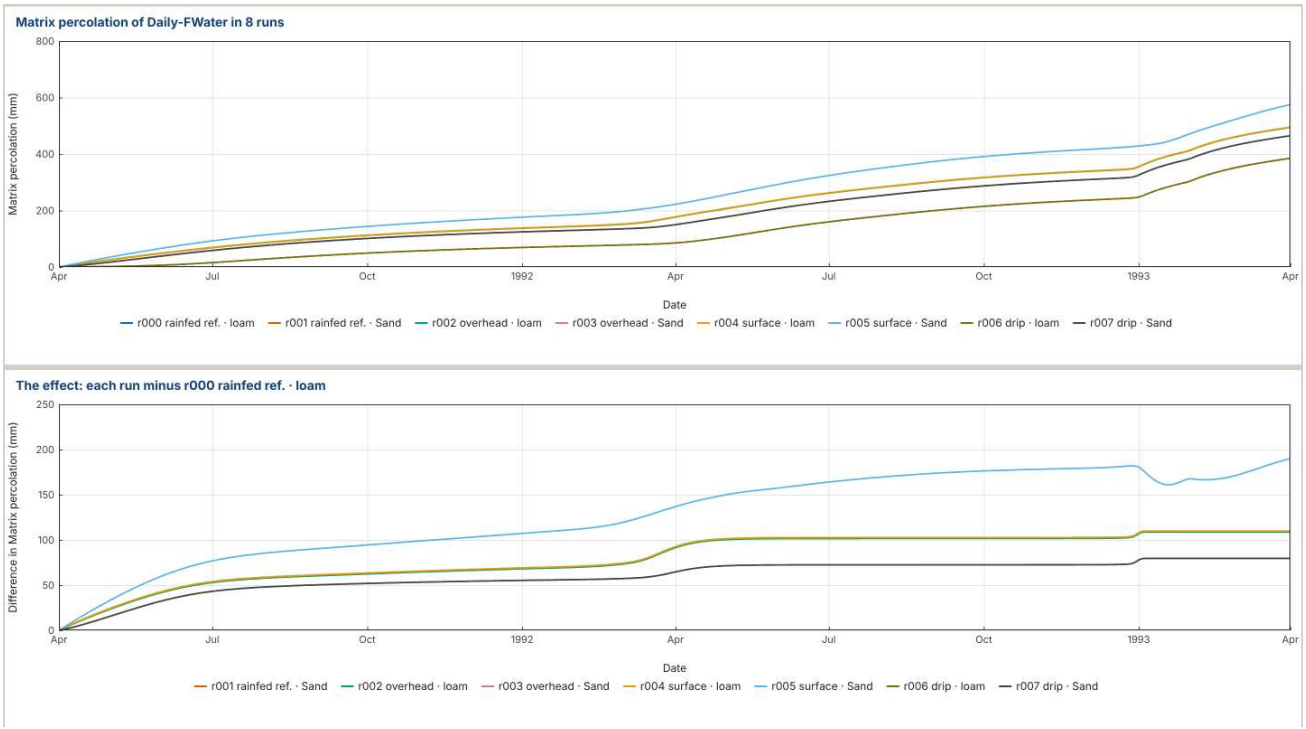



Figure 31.86: The cumulative matrix percolation of eight runs, and each run minus the rainfed reference.

## 31.25 The crops of the rotation

By default the farm grows spring barley, winter barley, winter rape and winter wheat (twice). In Scenarios and in Experiments the panel *Choose the crops of the rotation myself* replaces them: add any crop of the Daisy library, put the crops in the order of the rotation, and give each one its date of tillage, its date of sowing, its fertiliser (month and day with the kilograms of nitrogen per hectare, for example 04-05:160,05-05:70), the latest date of harvest and whether to harvest when the crop is ripe. The four crops of the built-in farm are tested; the others are templates (marked so in the list) whose dates and nitrogen should be checked against local practice. The nitrogen factor, the tillage, the fertiliser type and the irrigation of the scenario or the experiment act on the chosen crops in the same way.

# Chapter 32

# ⚑ From Daisy to the SWAFL project

Daisy calculates; the SWAFL project decides. The Daisy workbench is therefore arranged as one path that begins with the data of the site and ends in the layers of the SWAFL project, and every step of it is connected to the next. This chapter shows the path, the hand-over of the results, and the two layers of the SWAFL project that read real Daisy runs.

## 32.1 The guided path

The tabs are grouped as the work goes: **Input data** (Input data, Parameter library), **Experiments** (Experiments, Scenarios), **Results and analysis** (Process map, Log plots, Mass balance, Run comparison), **Decision support** (To the SWAFL layers), **Model development** (Set-up editor, Model builder, Soil column, Response curves, Calibration) and **Documentation** (Reference manual, Model and documentation). The nine tabs of the guided path carry their step number, 1 to 9 from left to right; holding the mouse over any tab tells what it is for.

A bar above each tab on the path shows the step (for example *Step 3 of 9*), what the step is for, and two buttons: **Back** goes to the step before and **Next** to the step after, each named on the button. Under it, a yellow line **What to do here** gives the three things to do on that tab, in order; **Hide tips** folds these lines on every tab, and **Show what to do here** brings them back. The tabs can still be opened in any order. Tabs outside the path show the bar *Tool*, with a button that returns to the last step of the path that was open.

| Step | Tab | What is done there |
|---|---|---|
| 1 | Input data | Set the weather and the soil of your site once, or look at a weather station or bring online weather of any place. Every simulation uses them. |
| 2 | Parameter library | Browse the soils, crops, fertilisers and chemicals of Daisy (this step can be skipped). |
| 3 | Experiments | Design your own set of runs from the factors of the model, or load a ready one. |
| 4 | Scenarios | Or choose a ready-made question and run it. |
| 5 | Process map | See the processes behind the results. |
| 6 | Log plots | Load your own files or use the output of your runs, then plot any quantity. |
| 7 | Mass balance | Check that inputs, outputs and content agree. |

| Step | Tab | What is done there |
|---|---|---|
| 8 | Run comparison | Compare two runs against the smallest meaningful level. |
| 9 | To the SWAFL layers | Take the results to the layers of the SWAFL project, which turn them into decisions. |

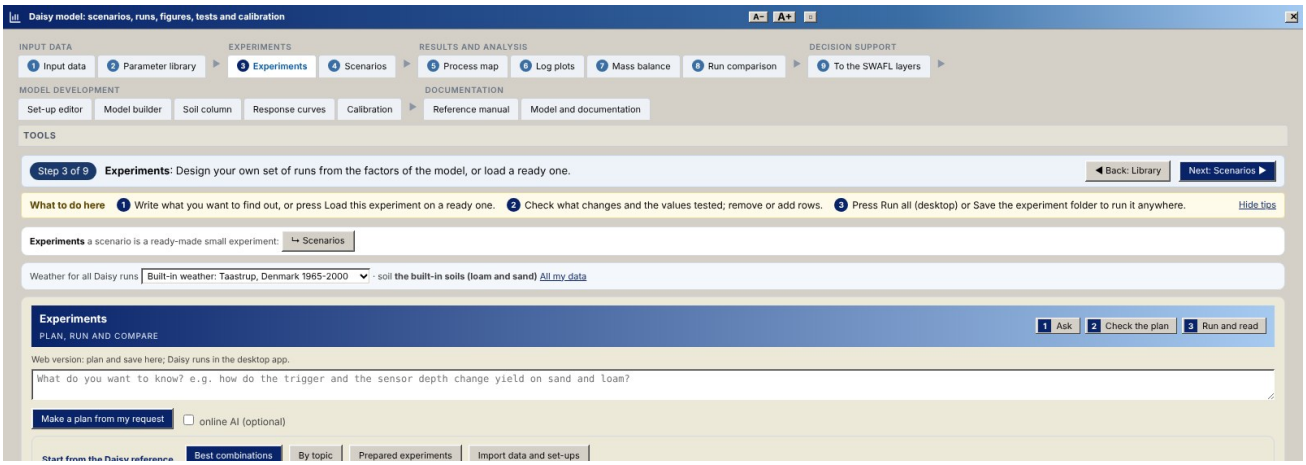



Figure 32.1: The tabs with the step numbers of the guided path, the bar of the step and the line What to do here.

### 32.1.1 The guided path: step 1, Input data

The path begins with the data of the site. Each tab shows a bar above its content with the step, its purpose, and the buttons Back and Next. The tabs can still be opened in any order from the row of tabs; the bar only shows where the tab lies on the path.

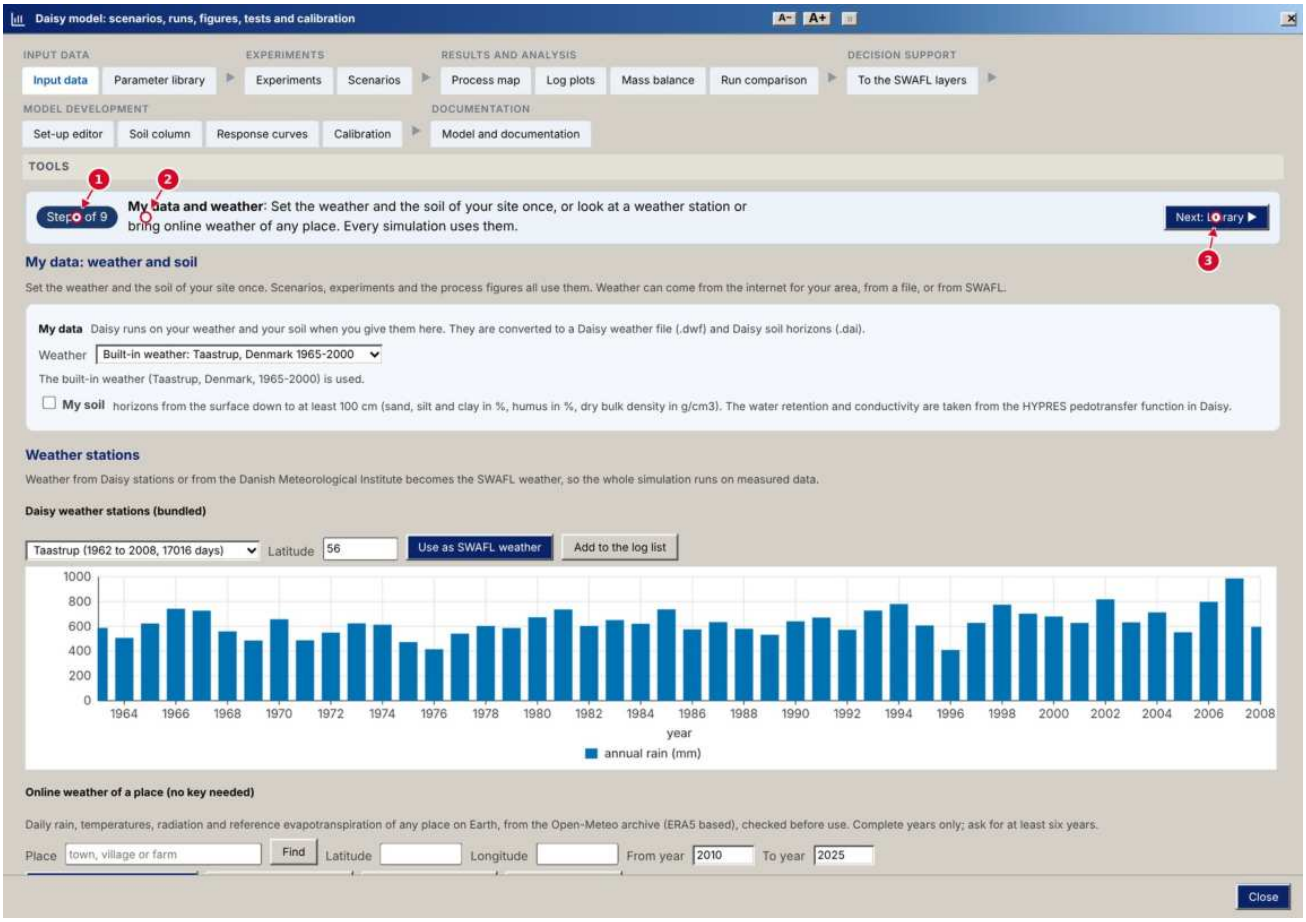



Figure 32.2: The guided path: step 1, Input data. Screen 123 of the Daisy workbench.

**What the arrows point to.**

1. **Where you are.** Step 1 of 9. The first step has no Back button.
2. **What to do here.** Set the weather and the soil of your site once.
3. **Next.** Goes to the Parameter library.

### 32.1.2 The guided path: step 4, Scenarios

Step 4 is where a ready-made question is chosen and run. After the run the results are drawn under the question, and Next leads on to the Process map and the other ways of looking at the results.

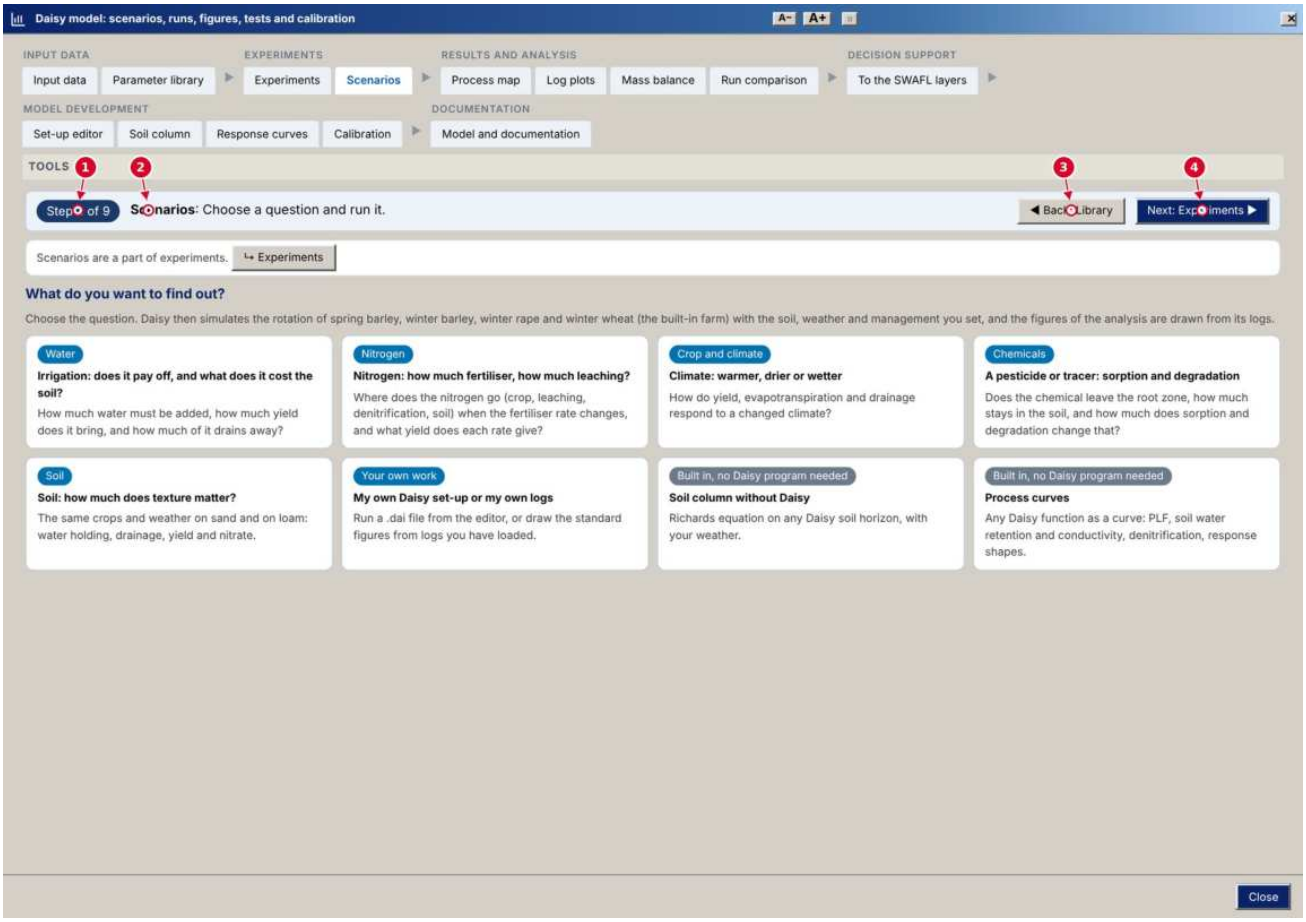



Figure 32.3: The guided path: step 4, Scenarios. Screen 124 of the Daisy workbench.

**What the arrows point to.**

1. **Where you are.** The number of the step on the guided path and the number of steps.
2. **What to do here.** The name of the step and one sentence on what it is for.
3. **Back.** Returns to the step before, with the name of that step.
4. **Next.** Goes on to the step after, with the name of that step.

### 32.1.3 The guided path: step 6, Log plots

Log plots draws any quantity of a run. The bar shows that three steps remain on the path: Mass balance, Run comparison, and the hand-over to the SWAFL project.

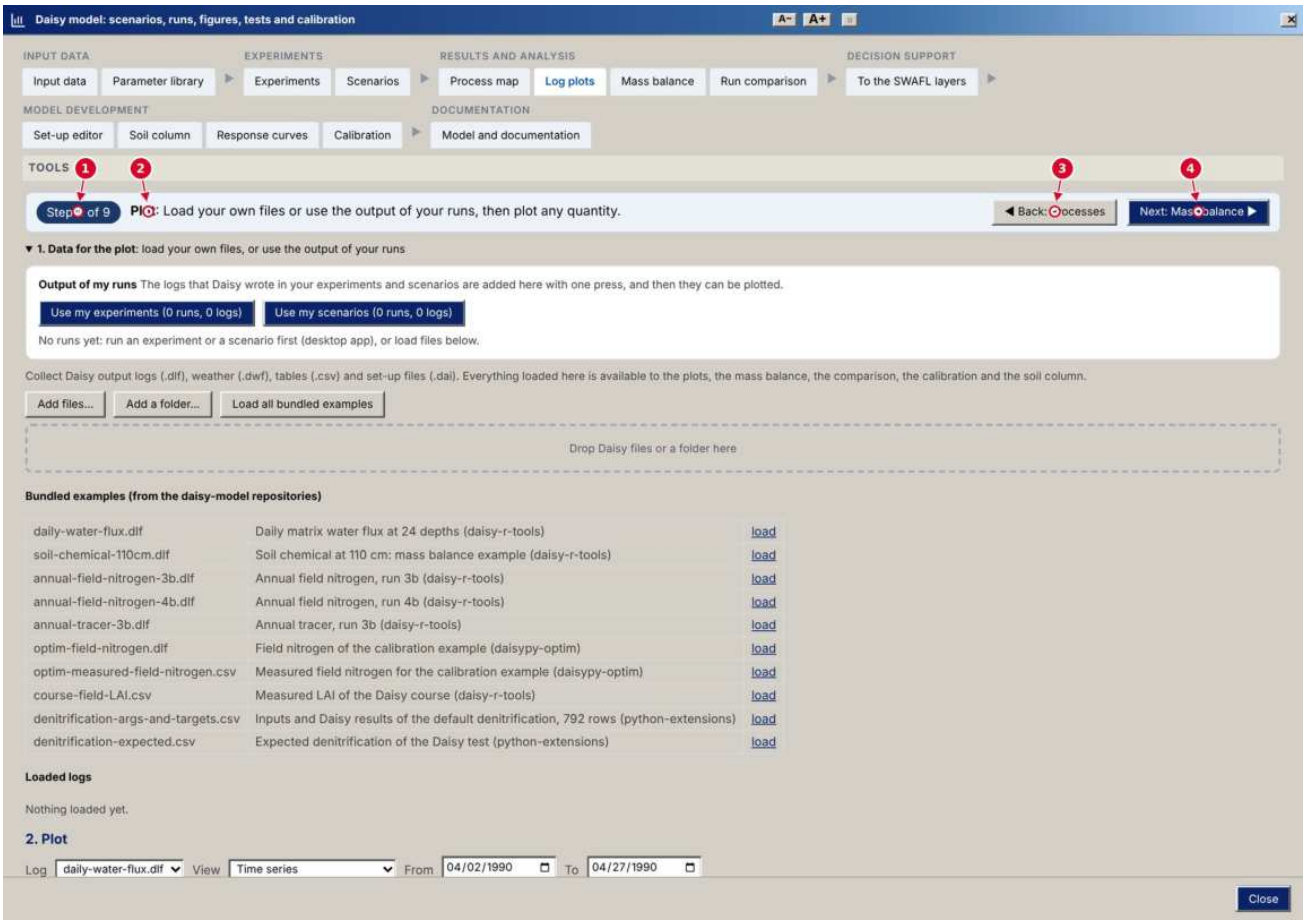



Figure 32.4: The guided path: step 6, Log plots. Screen 125 of the Daisy workbench.

**What the arrows point to.**

1. **Where you are.** The number of the step on the guided path and the number of steps.
2. **What to do here.** The name of the step and one sentence on what it is for.
3. **Back.** Returns to the step before, with the name of that step.
4. **Next.** Goes on to the step after, with the name of that step.

### 32.1.4 To the SWAFL layers: the last step of the path

The last step takes the results to the layers of the SWAFL project. Daisy calculates; the SWAFL project turns the results into decisions. The page answers three questions at a glance: have I run anything, which layer answers my question, and what does each layer still need.

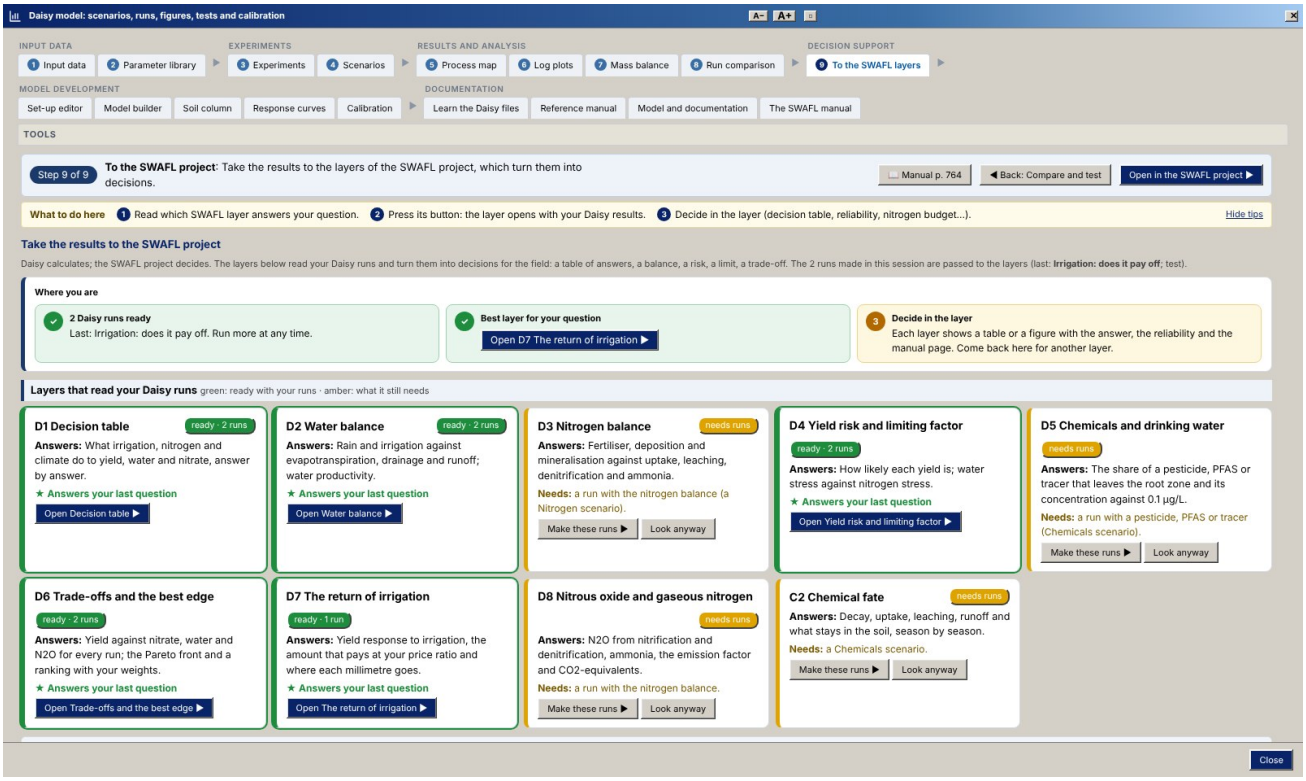



Figure 32.5: To the SWAFL layers after two runs of the irrigation question: the status panel at the top and every layer marked ready (green) or needing runs (amber). Screen 126 of the Daisy workbench.

**What you see, from the top.**

1. **Where you are.** Three steps with a tick when done: *Daisy runs ready* (how many, and the last question run), *Best layer for your question* (one button that opens the layer that answers the question you ran last) and *Decide in the layer*.
2. **Layers that read your Daisy runs.** One card for each layer (D1 Decision table to D8 Nitrous oxide, and C2 Chemical fate). Each card says what the layer *answers*. A green edge and the label *ready, n runs* mean the layer can show your runs: press *Open*. An amber edge and *needs runs* say in one line what the layer still needs (for example *a run with the nitrogen balance*), with *Make these runs*, which goes to Scenarios or Experiments, and *Look anyway*, which opens the layer with the bundled example runs.
3. **Answers your last question.** A star marks the layers that answer the theme of the question you ran last.
4. **Which layer answers which question** and **Every layer of the SWAFL project.** Folded at the bottom; open them to choose a layer by topic (Water, Nitrogen, Soil, Crop and climate, Chemicals, Experiment) or from all the layers of the project.

Before any run, every card is amber and the first step of the panel points to Scenarios and Experiments:

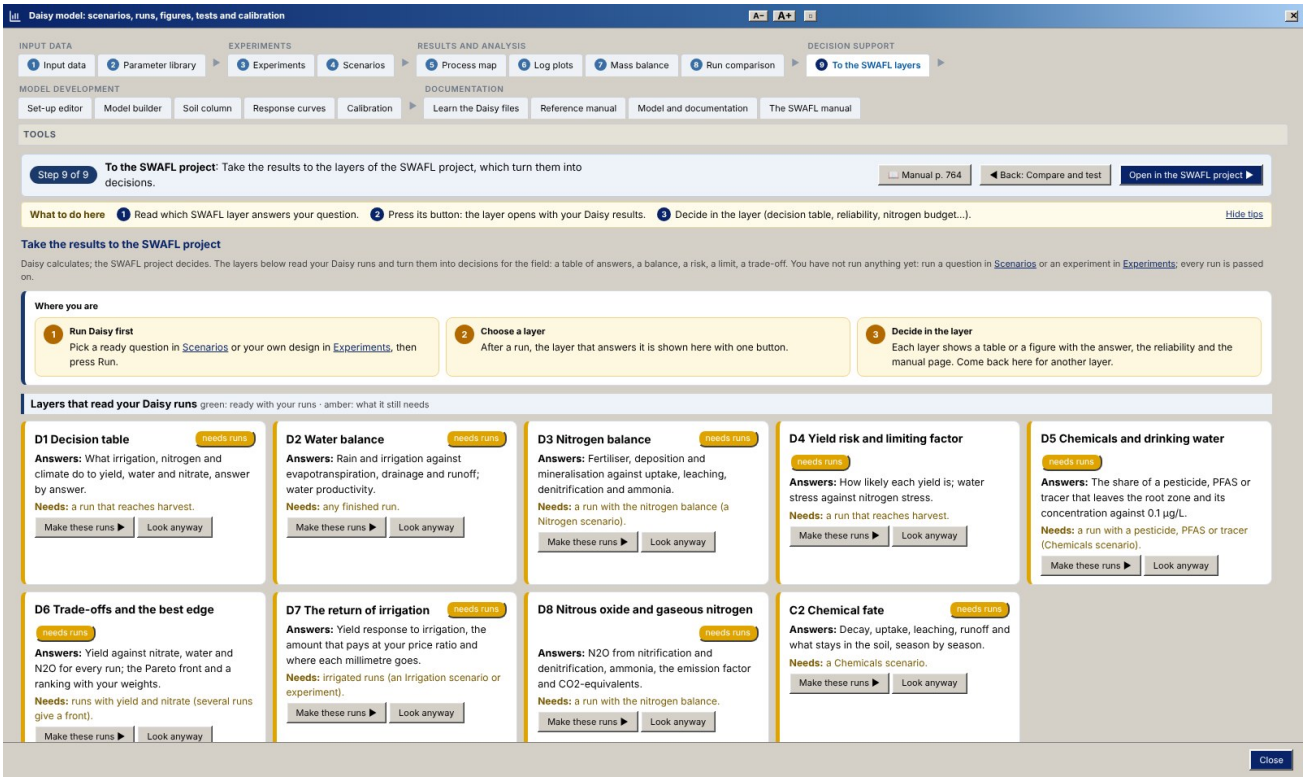



Figure 32.6: To the SWAFL layers before any run: every card says what it needs.

## 32.2 What is passed to the SWAFL project

When a scenario has been run in Scenarios, the workbench reads the logs of its runs (the annual harvest, the annual water and nitrogen fluxes at the surface and at 100 cm, and for the chemical scenario the annual and daily tracer logs) and passes a summary of each run to the layers of the SWAFL project: the yield, the water drained below the root zone, the nitrate leached, and for a chemical the decay, the crop uptake, the leaching, the runoff and the amount that stays in the soil, season by season. The layers then show **your** runs, with the weather and the soil you set, and say so in a line under the title. There are no ready-made figures: until the question has been run, a layer says *No Daisy run yet* and shows nothing else, and a figure is left out when the run it needs has not been made.

The soil of the run is matched to the soil of the project when the names agree (for example, the sand and the loam of the scenarios are the sand and the loam of the project), and the project is set to that soil when you press a button of the last step.

## 32.3 The layers that read Daisy runs

The group **Decision support (SWAFL layers)**, inside **Daisy model** in the Layers panel, has the layers *C2 Chemical fate*, *D1 Daisy decision table*, *D2 Daisy water balance*, *D3 Daisy nitrogen balance*, *D4 Yield risk and limiting factor*, *D5 Chemicals and drinking water*, *D6 Trade-offs and the best edge*, *D7 The return of irrigation* and *D8 Nitrous oxide and gaseous nitrogen*. They are drawn like every other layer, and they open from the last step of the path. Each shows in its first lines the weather, the soil and the years of the run it is drawn from.

**C2 Chemical fate.** Three buttons at the top choose the chemical: a mobile tracer (no sorption, no decay), a sorbing chemical, and a sorbing chemical that decays with a half-life of 30 days. For the soil in use the layer shows, from top to bottom:

1. *Where the chemical goes in each season*: stacked bars of the applied amount (decayed, taken up by the crop, leached below 2 m, lost by runoff, left in the soil), season by season, with a marker for what was applied plus what was already in the soil.
2. *What happened to the chemical over the seasons*: the totals and a plain statement of the risk (for example that more than 2 per cent of what was applied has passed below 2 m).
3. *A chemical goes where the water goes*: one dot per season, the drainage below 1 m against the amount leached.
4. *Where the chemical sits in the soil*: the concentration of the soil solution against depth, as the mean of all days and as the largest value.
5. *The three chemicals on this soil*: the leached, decayed and crop-taken shares of the applied amount and the amount left at the end, for the three runs side by side.

**D1 Daisy decision table.** For the questions that have been run the layer shows:

1. *Every question, one row for each answer*: rainfed and irrigated; half the fertiliser, the standard rate and 30 per cent more; the weather as it is, warmer by 2 degrees, drier (rain times 0.7), warmer and drier, and wetter (rain times 1.3). The columns are the yield, its change, the irrigation, the evapotranspiration, the drainage, the nitrate lost and the nitrogen taken up by the crop, as means of the seasons. The colour of a cell is green when the answer is better than the standard run and red when it is worse.
2. *What the table says*: three short statements, one for irrigation, one for nitrogen and one for climate, written from the numbers of the table.
3. *Yield season by season* and *Water season by season*: rainfed against irrigated, and rain, drainage and irrigation, for every season.

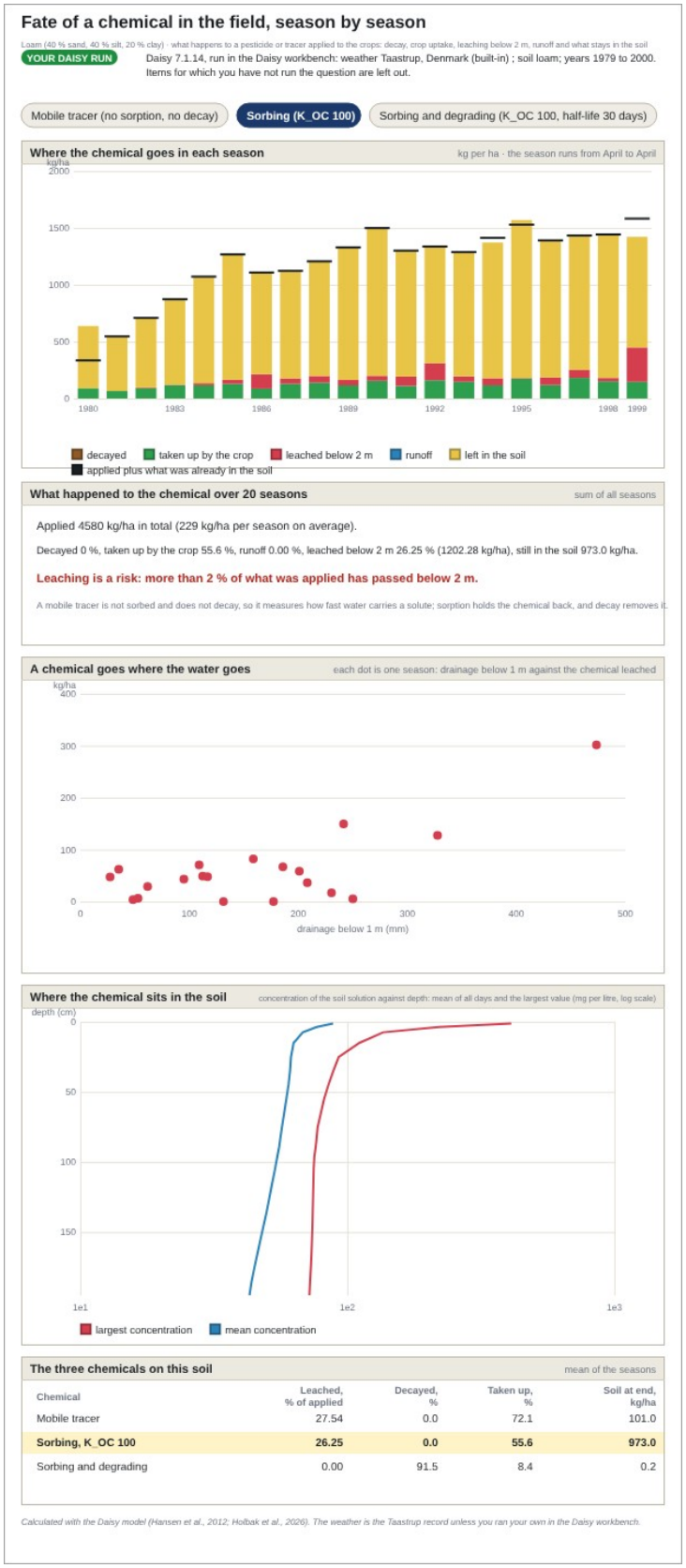



Figure 32.7: The layer C2 Chemical fate of the SWAFL project (from a run of the chemical question on loam).

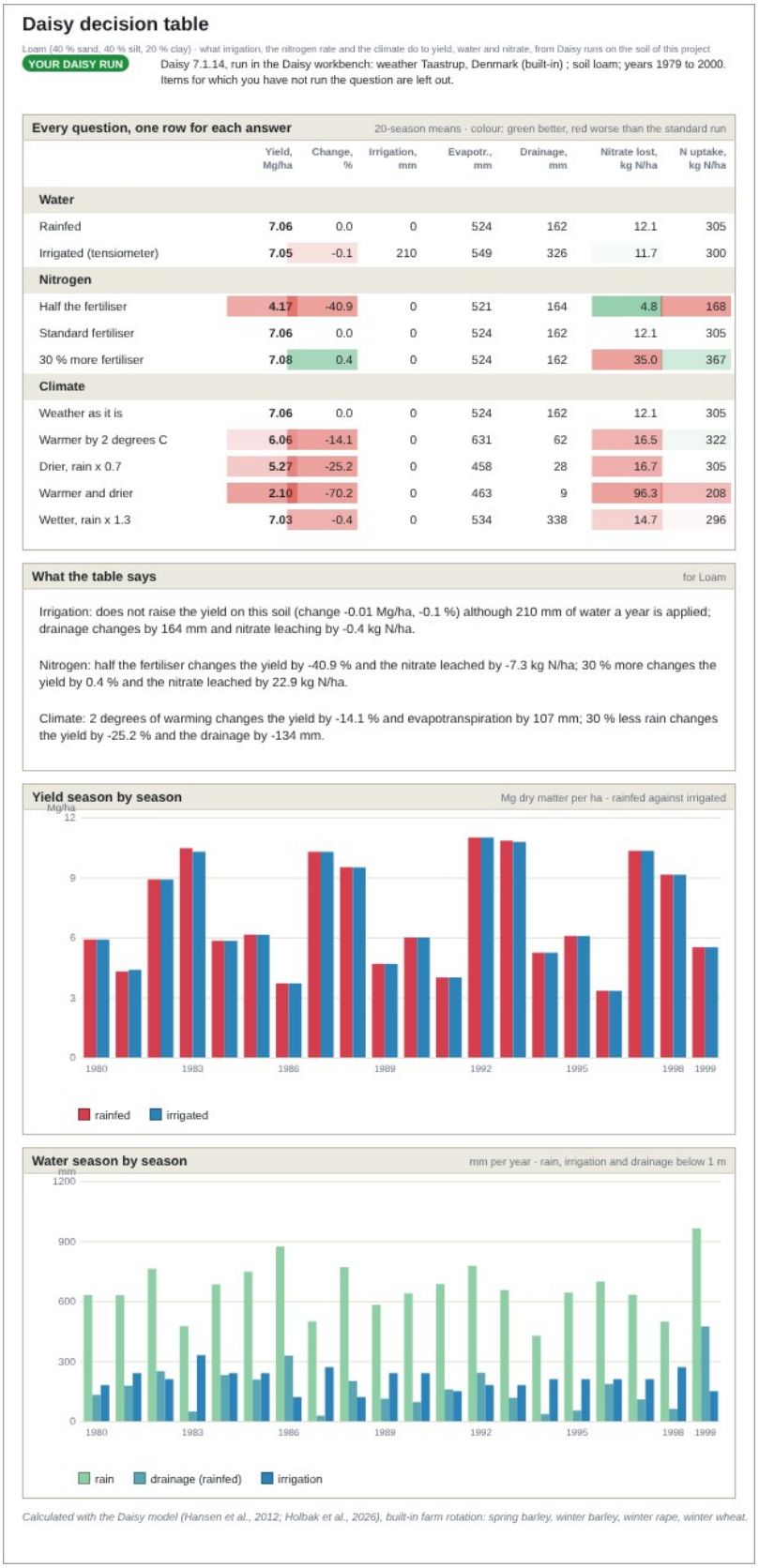



Figure 32.8: The layer D1 Daisy decision table of the SWAFL project (from the runs of the irrigation, nitrogen and climate questions on loam).

The three layers D2 to D4 read every run you made in this session. A row of buttons at the top chooses the run (rainfed, irrigated, half the fertiliser, warmer…), and a table at the bottom puts all the runs side by side; tapping a row of it chooses that run.

**D2 Daisy water balance.** Where the rain and the irrigation go.

1. *Water in and water out, season by season*: two bars for each season, the water that comes in (rain and irrigation) and the water that leaves (evapotranspiration, drainage below 1 m, runoff).
2. *The water balance in numbers*: the means of the seasons and the share of the input taken by each term; the change of the water in the soil, $\Delta S = P + I - ET_a - D - R$; the share of the potential evapotranspiration the crop used, $ET_a/ET_p$; and the water productivity, the yield per cubic metre evapotranspired, WP $= 1000\,Y/(10\,ET_a)$ in kg/m$^3$ with $Y$ in Mg/ha and $ET_a$ in mm (and per cubic metre of rain and irrigation when there is irrigation).
3. A plain statement: much of the water drains below the roots (drainage above 60 % of evapotranspiration), the crop is short of water ($ET_a/ET_p < 0.8$), or most of the water is used by the crop.

**D3 Daisy nitrogen balance.** What comes in (fertiliser, deposition, mineralisation of the soil organic matter) and where it goes.

1. *Nitrogen in, season by season*, with a black mark for the crop uptake, and *Nitrogen lost, season by season*: leached below 1 m, denitrified and volatilised.
2. *The nitrogen balance in numbers*: the means of each term; the nitrogen use efficiency, NUE $= U/(F + D_p)$, the uptake against the fertiliser and the deposition; the surplus $F + D_p - U$; and the nitrate leached for each kilogram of fertiliser.
3. A plain statement of the leaching: low (under 20 kg N/ha a year), moderate (20 to 40) or high (above 40).

**D4 Yield risk and limiting factor.** The mean yield hides the bad seasons.

1. *How likely is a yield?*: the exceedance curve of every run, the share of the seasons in which the yield reaches at least a value. A curve above another is better in every kind of season; crossing curves trade a better mean for more risk.
2. *Water stress and nitrogen stress, season by season*: the stress of the harvest log (the share of the potential production lost), water against nitrogen.
3. *The yield in numbers*: the mean; the yield reached in 9 seasons out of 10 (the 10th percentile), the median and the 90th percentile; the coefficient of variation; the share of seasons with a stress above 0.2; and which factor limits the crop more (one stress more than 1.2 times the other).

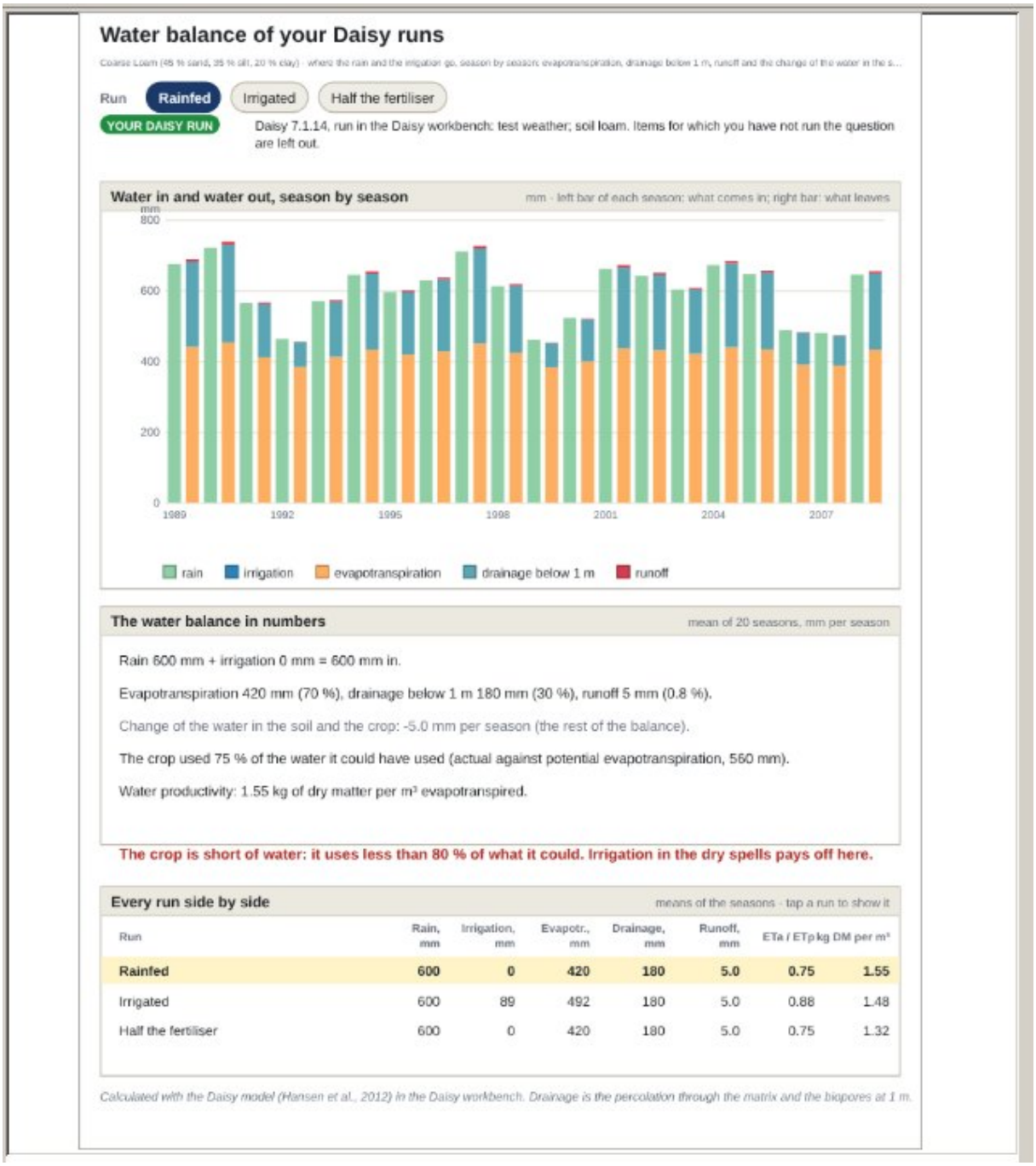



Figure 32.9: The layer D2 Daisy water balance (test data of three runs).

Figure 32.10: The layer D3 Daisy nitrogen balance (test data of three runs).

Figure 32.11: The layer D4 Yield risk and limiting factor (test data of three runs).

### 32.3.1 D5 to D8: chemicals, trade-offs, the return of irrigation and nitrous oxide

Four more layers read every run of the session, from Scenarios and from Experiments alike: a run of an experiment is passed on as soon as the experiment has run. Each layer lists all runs it can show as buttons at the top, and a tap on any row of its tables or on any dot of its figures shows that run.

**How the runs are named.** A run of an experiment is named in the layers by the short form of its values, factor after factor (for example *-14 d · N×0.7 · Sand · +2 °C · irrigated*: fertiliser 14 days

earlier, 70 % of the nitrogen, the sandy soil, 2 °C warmer, irrigated), so every combination can be told from the others at a glance; the full name is shown when the pointer rests on the run. A run of a scenario keeps its name. Names that are longer than the room of their column or button end with an ellipsis, buttons wrap to a new line by their real width, and the names next to the points of a scatter plot are placed where they touch neither a point nor another name; a name that finds no free place is left out (the run is still named in the buttons and the tables).

**D5 Chemicals and drinking water.** For every run with a chemical (the questions of the theme Chemicals, or an experiment with a chemical of the library):

1. *Dose of the product*: the results are scaled from the dose of the run to 0.1, 0.25, 0.5, 1 or 2 kg/ha a season. With linear sorption and first-order decay, as in these runs, every amount leached is proportional to the dose.
2. *Where each chemical goes*: one bar per chemical, the shares decayed, taken up, harvested or dissipated, lost by runoff, leached and left in the soil.
3. *Mean concentration in the water draining below the root zone*, the flow-weighted mean of all seasons on a log scale, against the limit of 0.1 µg/L for one pesticide and for the sum of 20 PFAS (Directive (EU) 2020/2184), with the number of seasons over the limit:

$$\bar{c} = \frac{\sum_t L_t}{\sum_t D_t} \cdot 10^5 \ \ \mu\mathrm{g\,L^{-1}}, \qquad L_t \text{ in kg ha}^{-1},\ D_t \text{ in mm}$$

(1 mm on 1 ha is 10 000 L). Drainage water is not yet groundwater: dilution and further decay below the root zone usually lower the concentration. 4. *Season by season* for the chosen chemical, with the line of 0.1 µg/L, and a plain statement of what the numbers say.

**D6 Trade-offs and the best edge.** Every run as one point: yield against nitrate leached (the size of a dot is the irrigation), yield against irrigation plus drainage, and yield against N2O. The green line is the best edge (the Pareto front): the runs that no other run beats on both yield and loss. *What matters most* (yield first, balanced, environment first, water saving) sets the weights of a score,

$$S = \frac{w_Y\,\hat{Y} + w_N\,(1-\hat{N}_L) + w_I\,(1-\hat{I}) + w_G\,(1-\hat{G})}{w_Y + w_N + w_I + w_G},$$

where each hat is the value scaled between the worst and the best run; the runs are ranked by $S$, and the runs on the edge are marked.

**D7 The return of irrigation.** For the irrigated runs against the rainfed run of the same question:

1. *Yield response to irrigation*: yield against the irrigation given, with a quadratic fit $Y = a + bI + cI^2$ when at least three amounts were run. *Price ratio* (the cost of 1 mm of irrigation on 1 ha in kg of crop, from free water to 20 kg) gives the amount that pays, where the slope of the curve equals the price ratio: $I^* = (p-b)/(2c)$ for $c < 0$.
2. *Where each millimetre of irrigation goes*: the extra evapotranspiration (used by the crop), the extra drainage, the extra runoff and the rest (stored in the soil), as shares of the irrigation, with the marginal water productivity $\Delta Y/I$ in kg per m³.
3. *Season by season*: the gain in yield and the irrigation given each season, and a plain statement.

**D8 Nitrous oxide and gaseous nitrogen.** N2O from denitrification and from nitrification and ammonia volatilised, season by season; the emission factor of every run, EF $= \mathrm{N_2O\text{-}N}/F$, against the IPCC default of 1 % (IPCC 2019 Refinement); the climate cost, $\mathrm{N_2O\text{-}N} \times 44/28 \times 273$ kg CO2-equivalent per ha (GWP100 of IPCC AR6), also per Mg of dry matter harvested; and the table of all runs.

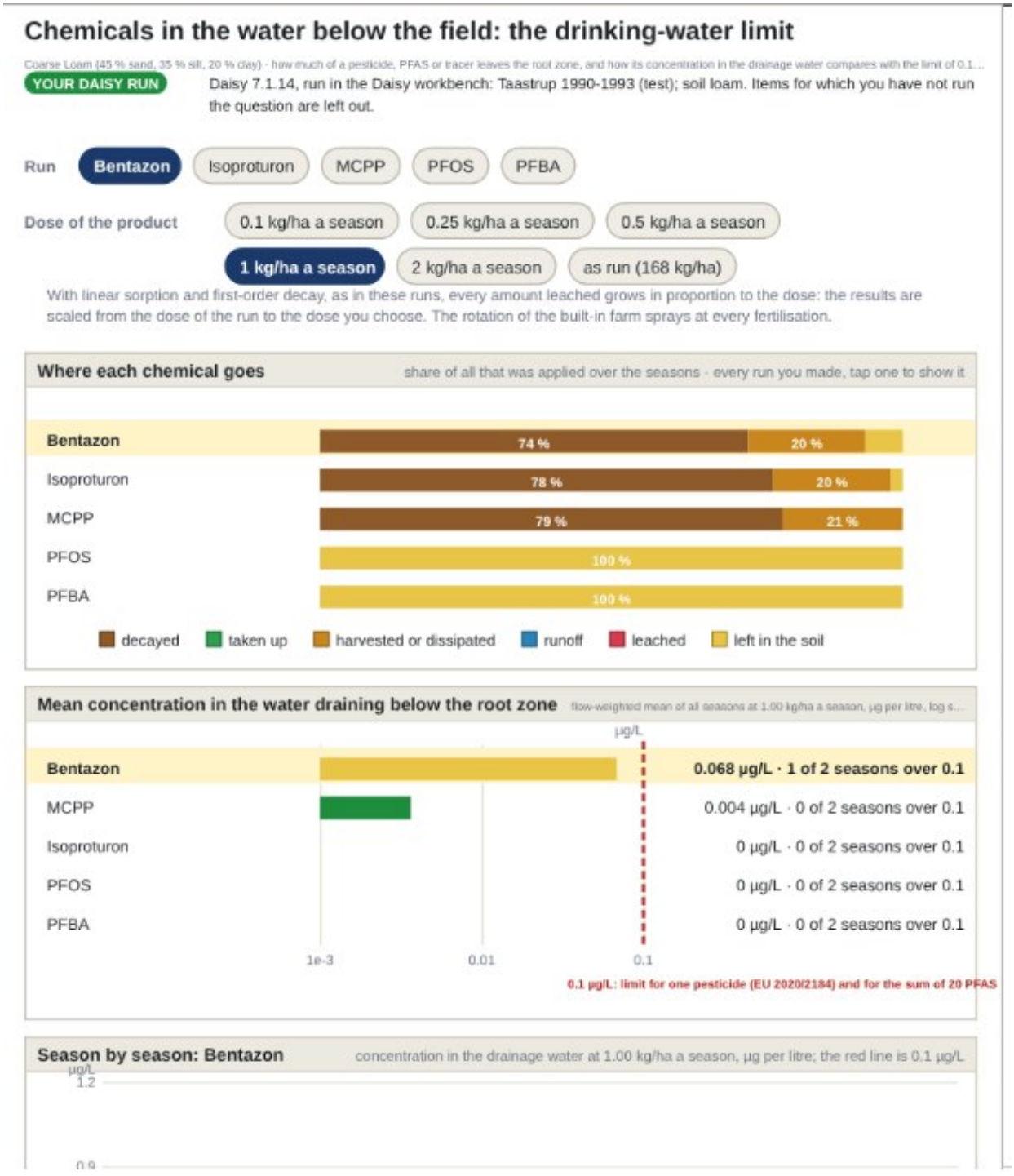



Figure 32.12: D5 Chemicals and drinking water, from five runs of the questions Pesticides of the Daisy library and PFAS (Daisy 7.1.14, loam, 1990 to 1993).

Figure 32.13: D6 Trade-offs and the best edge, from fourteen runs of the Water, Nitrogen, Soil and Chemicals questions.

Figure 32.14: D7 The return of irrigation, from the runs of the question Deficit irrigation.

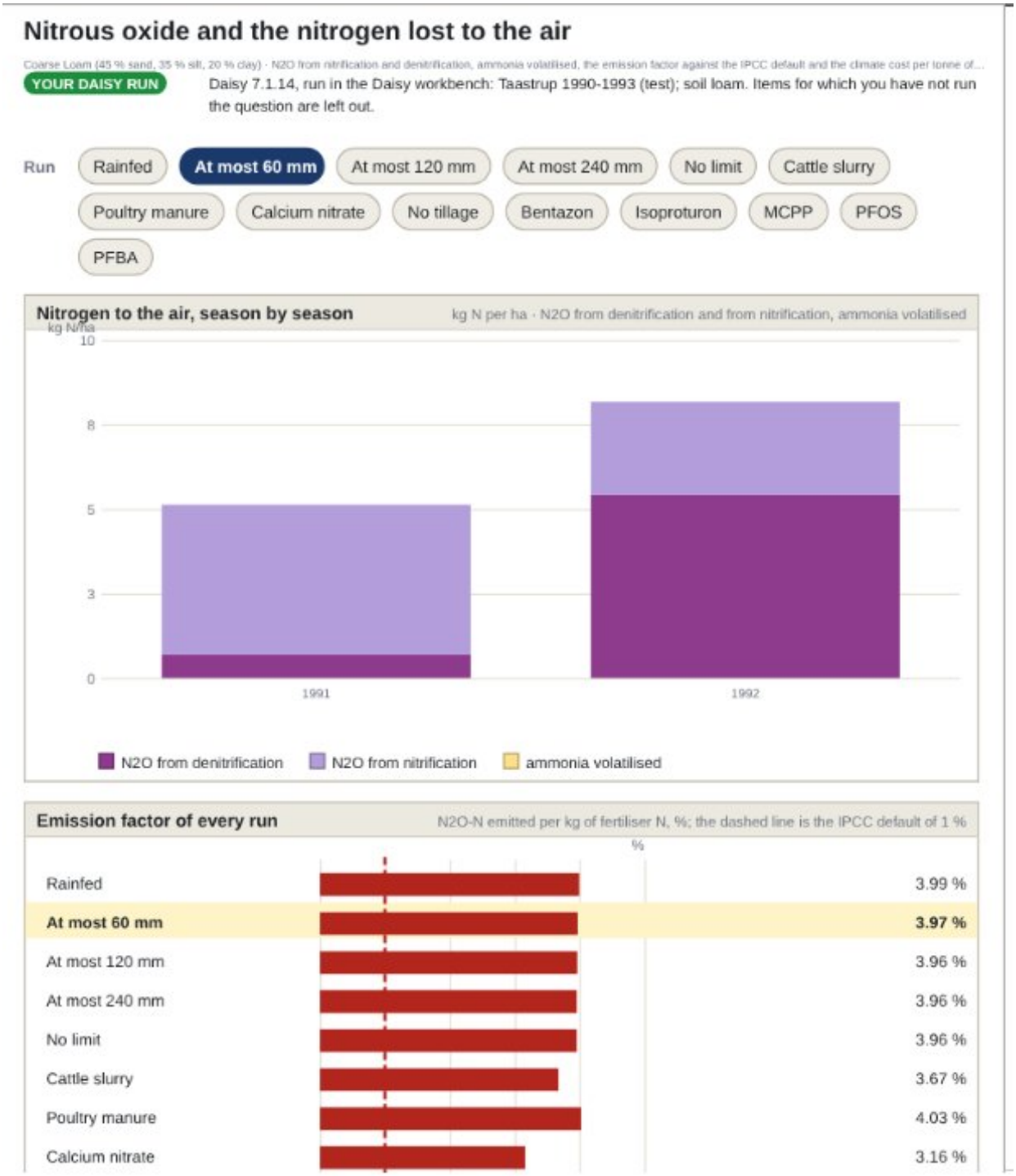



Figure 32.15: D8 Nitrous oxide and gaseous nitrogen.

The layer C2 Chemical fate now lists every chemical run as well, not only the three runs of the first chemical question.

### 32.3.2 To the SWAFL layers: every layer in one place

The last step of the path lists, in three parts: the nine layers that read your Daisy runs, each with the number of your runs it can show and a frame around those that answer the question

you ran last; which layer answers which kind of question (water, nitrogen, soil, crop and climate, chemicals, your own experiments), with the questions of each theme; and every layer of the SWAFL project, group by group, with a button to open it.

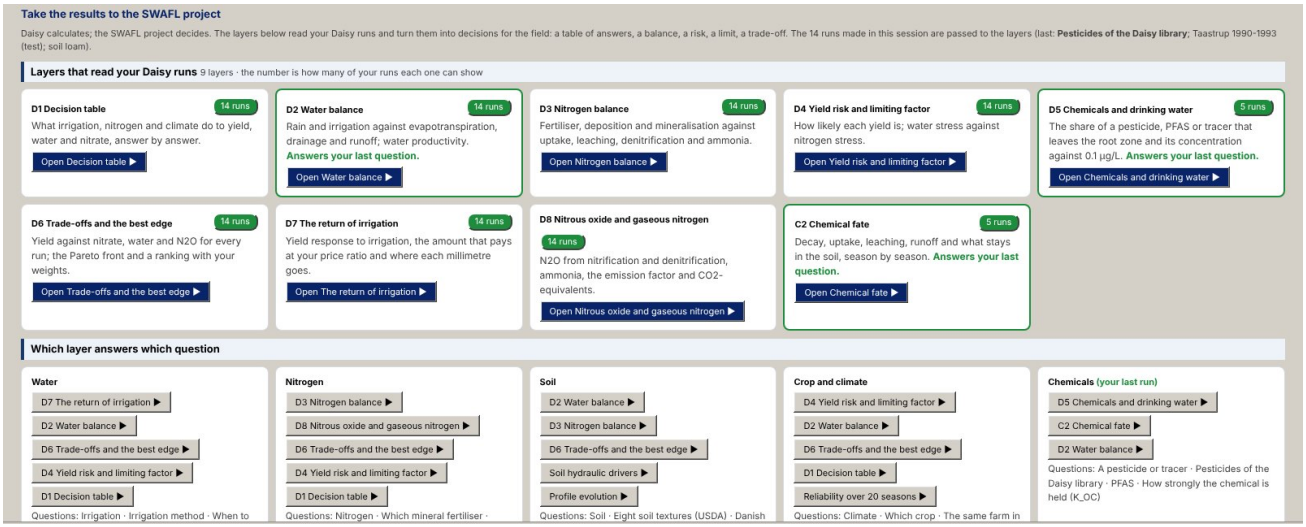



Figure 32.16: To the SWAFL layers: the layers that read your Daisy runs and which layer answers which question.

## 32.4 What to expect and what not

The layers are summaries of real Daisy runs, not new calculations: a number in a layer is a number of a Daisy log, or a sum or a mean of such numbers, and the line under the title says which runs it comes from. The weather and the soil are those of the run, so for another climate run the question again in Scenarios with the weather of the project or online weather of the place.

# Chapter 33

# ⚙ Build and test, and learn

The fourth group holds the tools that need no scenario, and the fifth explains the model.

### 33.0.1 Calibrate: the form

Parameters are searched with sequential, CMA-ES or Bayesian optimisation; each trial is one run.

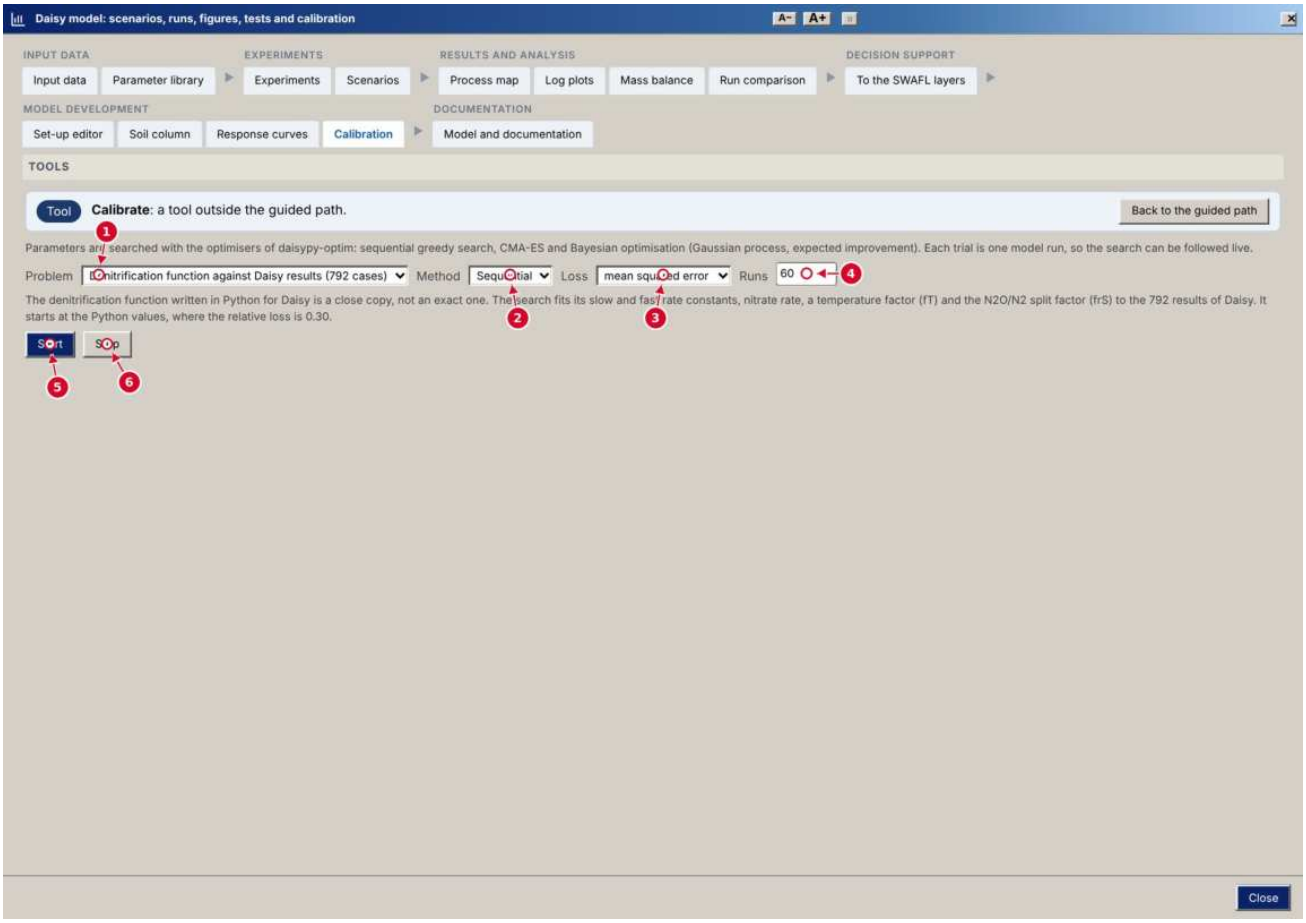



Figure 33.1: Calibrate: the form. Screen 127 of the Daisy workbench.

**What the arrows point to.**

1. **Problem.** What is calibrated.

2. **Method.** The search method.
3. **Loss.** The error to minimise.
4. **Runs.** Number of trials.
5. **Start.** Starts the search.
6. **Stop.** Stops it.

### 33.0.2 Calibrate: a search followed live

The loss and the best parameter set are drawn trial by trial.

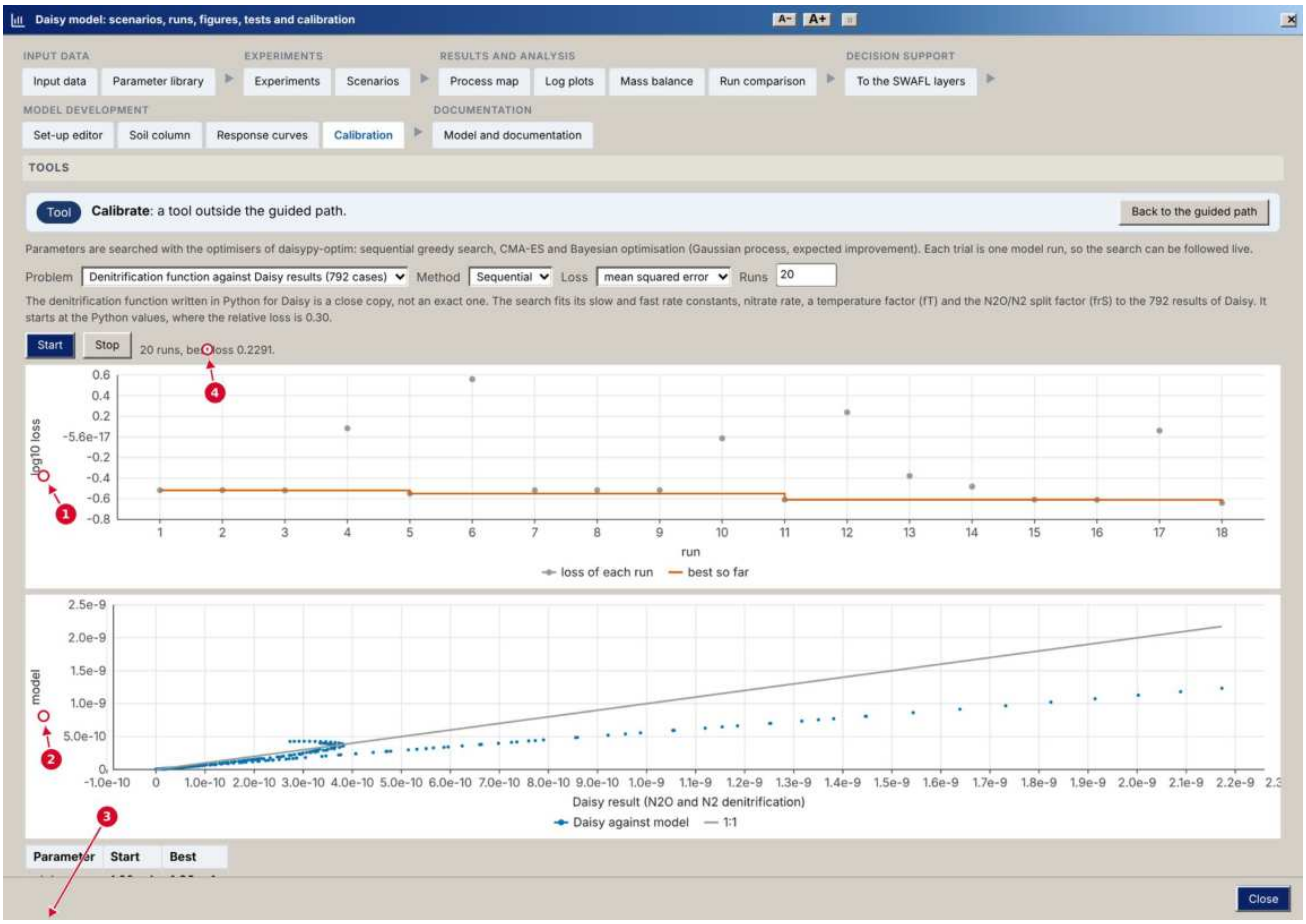



Figure 33.2: Calibrate: a search followed live. Screen 128 of the Daisy workbench.

**What the arrows point to.**

1. **Loss against trial.** Falls as the search improves.
2. **Parameters.** Parameter values tried.
3. **Best values.** Table of the best set.
4. **Status.** Progress.

### 33.0.3 Process curves: the start

Process curves draws any function of the Daisy library, such as a retention curve, a response to temperature or a stress function.

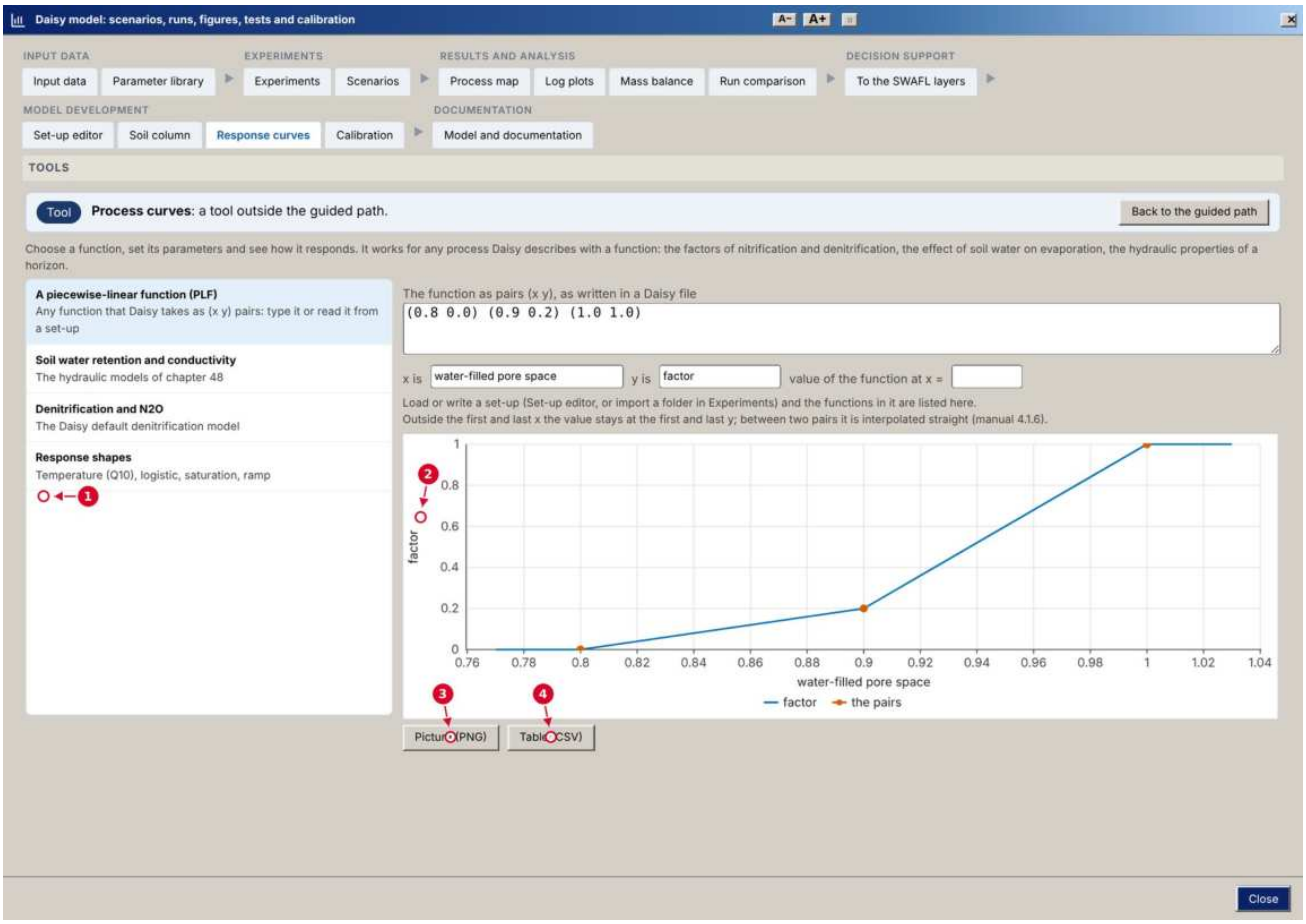



Figure 33.3: Process curves: the start. Screen 129 of the Daisy workbench.

**What the arrows point to.**

1. **Function list.** The functions that can be drawn.
2. **Drawing.** The chosen curve.
3. **Picture (PNG).** Saves the figure.
4. **Table (CSV).** Saves the numbers.

### 33.0.4 Set-up editor: the empty editor

The editor opens a Daisy set-up file (.dai), shows its outline and checks the brackets as you type.

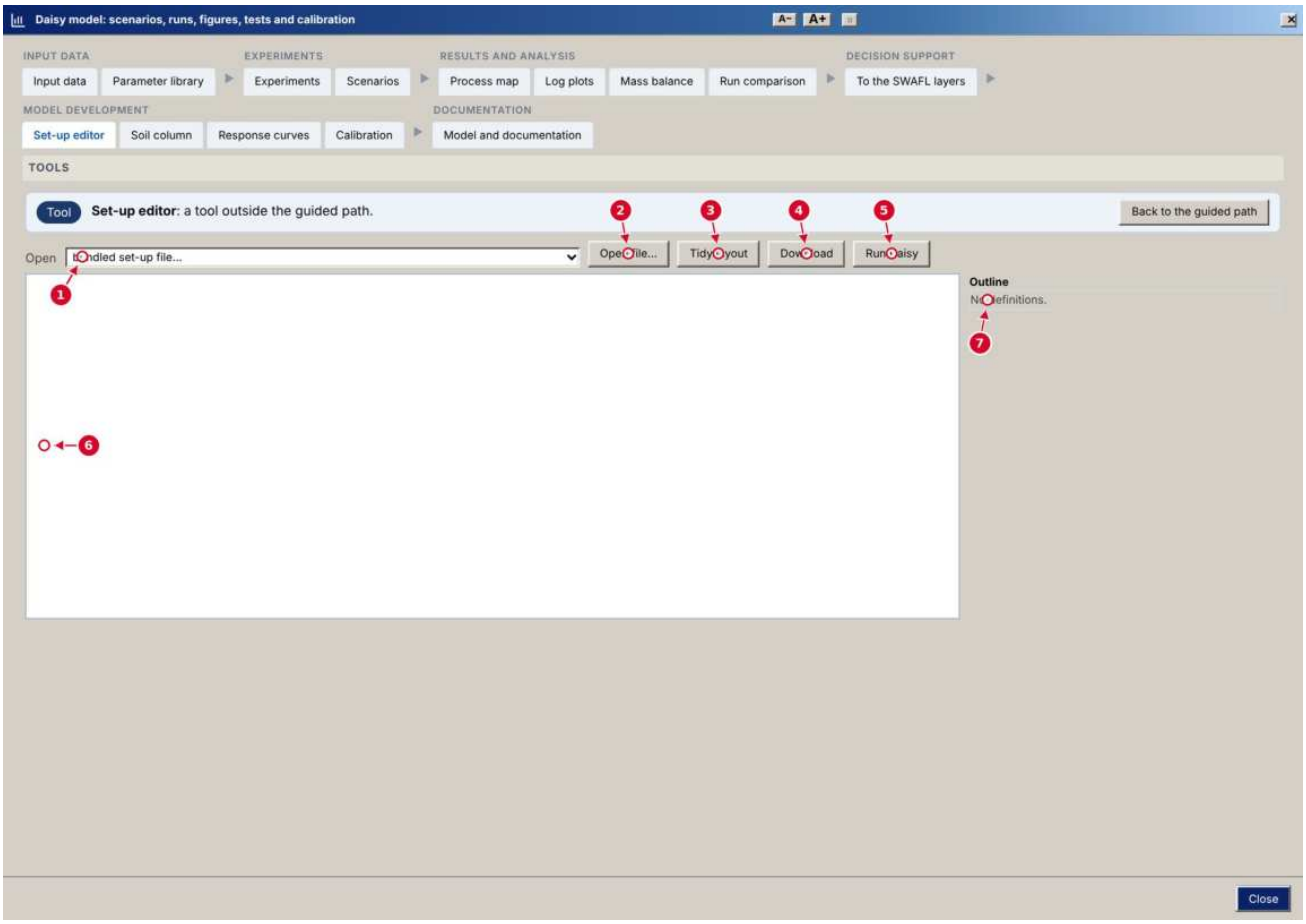



Figure 33.4: Set-up editor: the empty editor. Screen 130 of the Daisy workbench.

**What the arrows point to.**

1. **Bundled set-up.** Opens a set-up that comes with the program.
2. **Open file.** Opens a .dai file from your computer.
3. **Tidy layout.** Indents the file.
4. **Download.** Saves the file.
5. **Run Daisy.** Runs the set-up on the desktop app.
6. **Text.** The set-up text with colours.
7. **Outline.** The sections of the file.

### 33.0.5 Set-up editor: irrigation.dai open

With the irrigation set-up open the outline lists its sections and the definitions it uses.

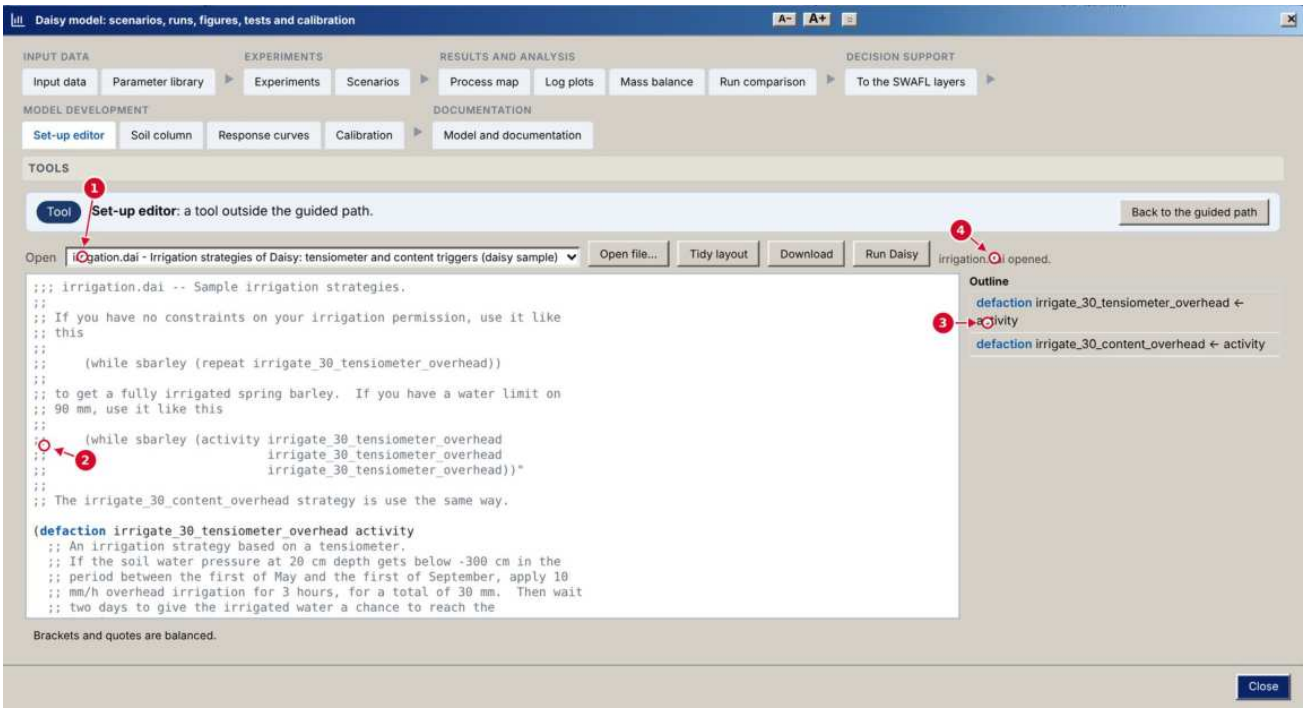



Figure 33.5: Set-up editor: irrigation.dai open. Screen 131 of the Daisy workbench.

**What the arrows point to.**

1. **Set-up chosen.** The file that is open.
2. **Text.** Keywords, values and comments in colours.
3. **Outline.** Click a section to jump to it.
4. **Status.** File name and number of lines.

### 33.0.6 Set-up editor: the placeholders explained

Many Daisy files contain names in braces. **${name}** is a parameter that Daisy fills by itself: from the default the file gives it, or from the value given where the log or the model is used (for example `(output ("Field nitrogen" (column "Askov") (unit "g N/m^2")))`). **{name}** is a value of a template that must be given before the file can run (the trial files of a calibration are written this way).

The panel **Placeholders** under *What the file defines* has a card for each one:

- what it means (from the declare line of the file, or from the reference manual when the name is an attribute of a model) and how many times the file uses it;
- how to fill it: *Leave it empty* when Daisy works it out by itself (the file name parts colfid, cropfid, chemfid and bioporefid, and the values the file computes; their box is grey and says *automatic*), *Leave it empty to keep the default* for a parameter, or *Give a value* for a template value, with the unit written after it in the file;
- the first line where it is used (a click shows that line in the editor);
- the button **▾ options**: a scrollable list of the values it can take, in groups, each with an explanation: *All* (*), the default of the file, the units Daisy knows (kg N/ha, g N/m^2, kg/ha, g/ha, mm and others, in words), the columns, crops, chemicals, biopores and organic matter defined in the file and in the Daisy library, and the values given to it in the bundled set-ups. A search box narrows the long lists. A click on a value puts it in the box.

**Download filled file** writes a copy of the file with the values given (empty boxes stay as they are); **Empty all** clears the boxes.

Figure 33.6: Placeholders of the standard logs: the card of column with its options open (every column, then the columns of the Daisy library), and the card of crop.

### 33.0.7 Model builder: any Daisy model as a form

**Model builder** (Model development) writes the definition of any model of the Daisy reference manual without typing in the Daisy language. Choose the **Component** (a chapter of the reference manual, for example *47 horizon* or *29 crop*) and the **Model** (a section of that chapter, for example *47.1 BSI3*); the box *find a component or model* narrows the lists. Give the model a **Name**.

The form has one row for every attribute of the model, as the manual gives it: the name, a box for the value, the unit, the type with its default, and the meaning. The rows are grouped as in the book:

- **What you set for this model**: its parameters, components and submodels. A row marked with a red star must be given and is always ticked; an optional row, or a row with a default, is ticked as soon as a value is typed in it (or by hand). Yes or no values are chosen from a list, a component from the names of the models of its chapter, a piecewise linear function (PLF) as pairs *x y; x y*.
- **Shared by every model of the component**: the parameters that the manual describes in the chapter itself (for a horizon: the hydraulic model, tortuosity, C/N ratio and the others).
- **Start values of the state variables** (optional).
- **Log variables you want to follow**: tick them (or *tick all*); they are listed under the definition, ready to be logged with a column or log definition (chapter 52 of the manual).

The definition is written under the form as you type, with Daisy names (the manual prints the underscores of the names as spaces) and the unit of each number as a comment. It is checked at once (*Brackets and quotes are balanced*), and the rows still to fill are named. **Open in the set-up editor** opens it as a new file, **Add to the file in the set-up editor** adds it to the file already there, **Copy** and **Save as .dai** keep it, and **Read this model in the reference manual** opens its section. In the Reference manual tab, the button **Fill in a form for this model** under every model opens this builder on that model. All the 683 models of the 95 components that a set-up file can define have a form.

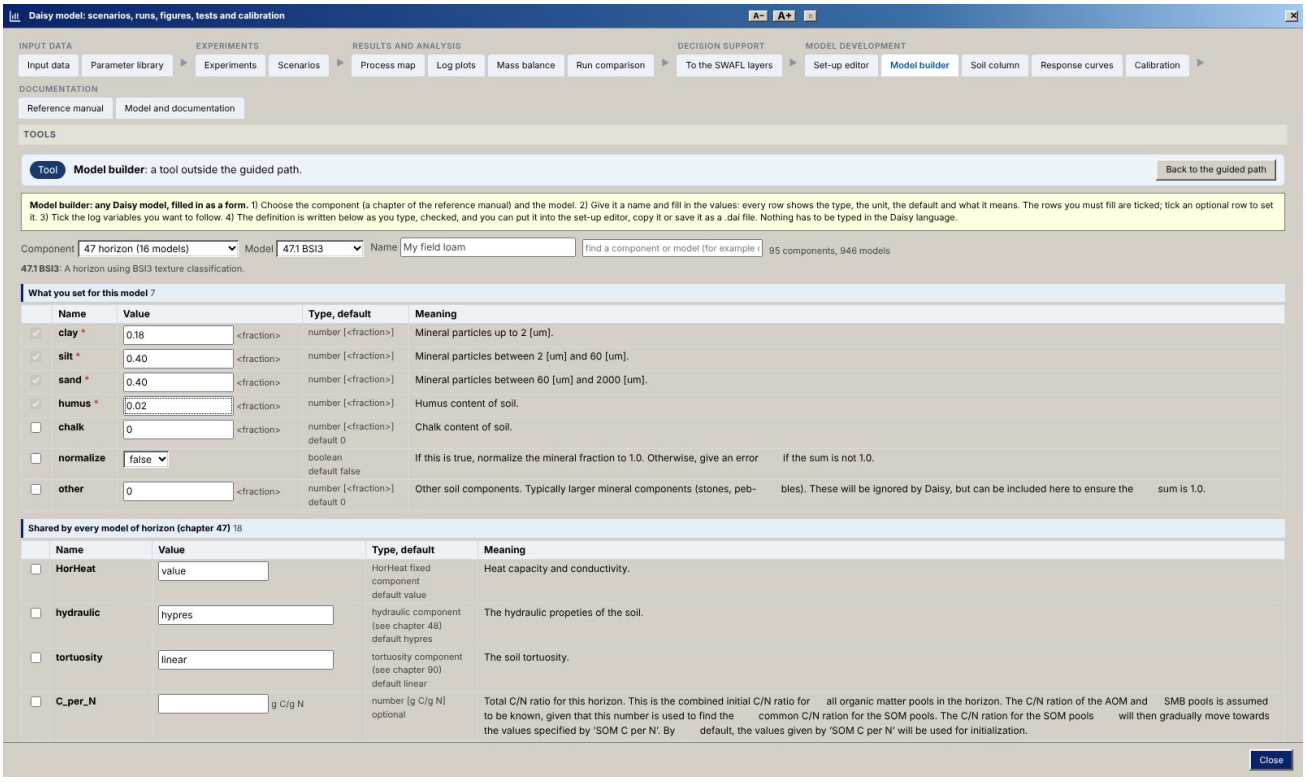



Figure 33.7: Model builder: the horizon model BSI3 filled in, with the parameters shared by every horizon.

### 33.0.8 Reference manual: the attributes in the groups of the book

The tab **Reference manual** holds the whole Daisy Program Reference Manual (P. Abrahamsen, 31 August 2026): 107 chapters, 1018 models and 3249 attributes, each with its type, unit, kind and meaning, and the 82 references. Under every model the attributes are grouped as in the book: **what you set** (parameters, components and submodels), **state variables**, and **log variables and log columns** (what the model can write to a log file, including the columns of the Field water, Soil water and Crop logs). Every attribute shows its kind; the variables listed after a *Log Variables* or *Table columns* heading of the book are log variables or log columns.

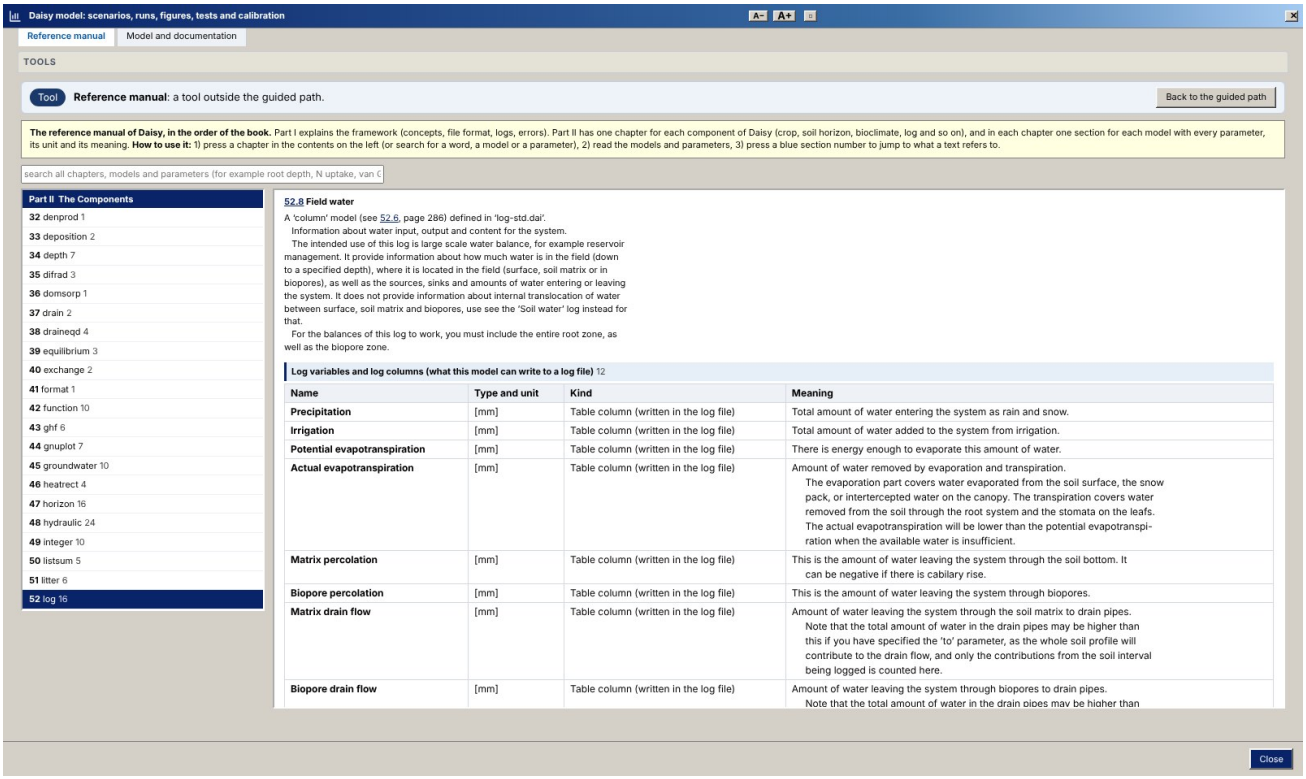



Figure 33.8: Reference manual: the Field water log (52.8) with its log columns.

### 33.0.9 Soil column: the form

A soil column of up to three layers is solved with the Richards equation on the weather chosen below.

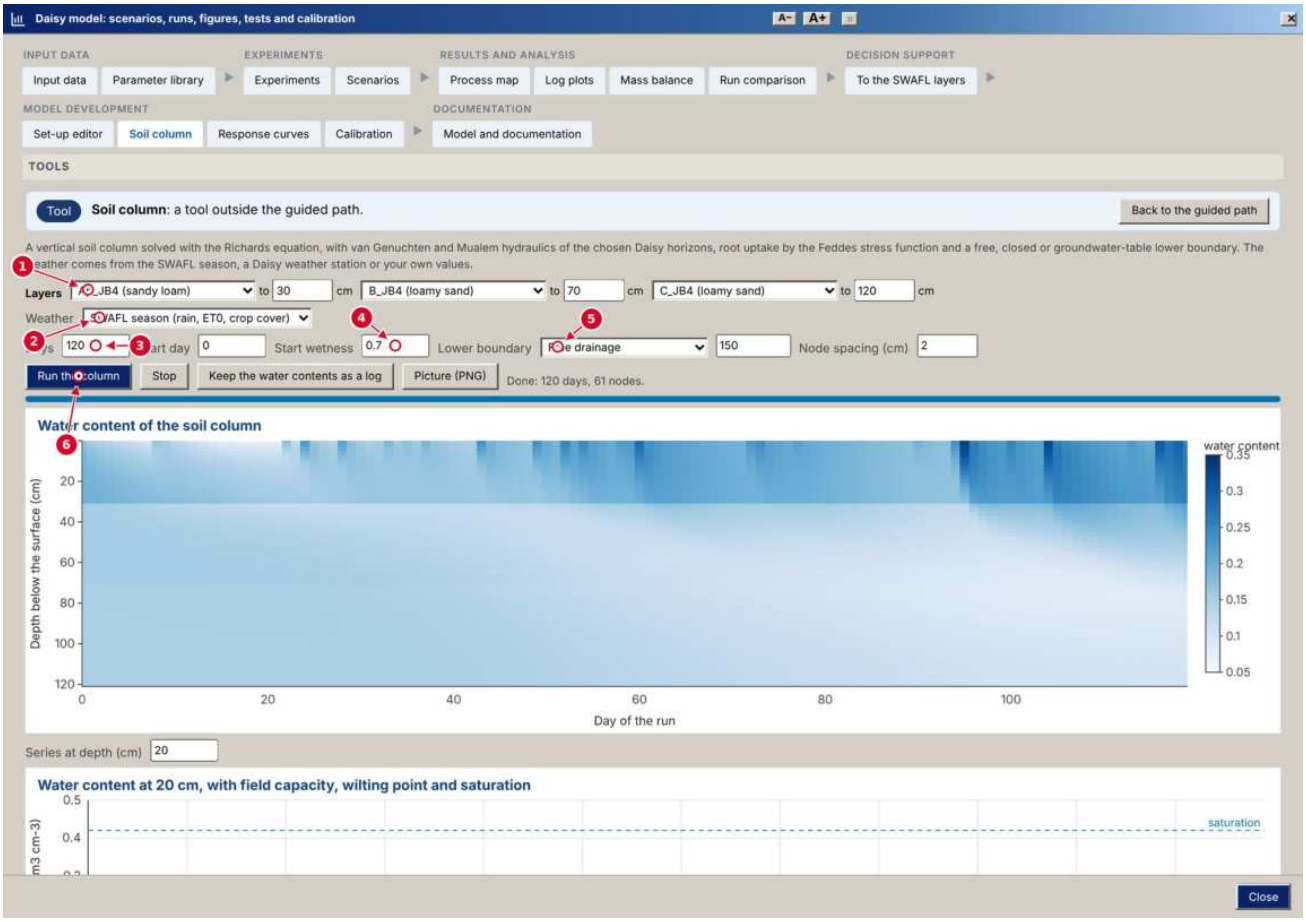



Figure 33.9: Soil column: the form. Screen 132 of the Daisy workbench.

**What the arrows point to.**

1. **Layer 1.** Horizon of the first layer and its lower depth.
2. **Weather.** SWAFL season, Daisy station, own values or file.
3. **Days.** Length of the simulation.
4. **Start wetness.** Initial relative saturation.
5. **Lower boundary.** Free drainage, closed or groundwater table.
6. **Run the column.** Starts the calculation.

### 33.0.10 Soil column: results

The figures show water content through depth and time, the series at a chosen depth and the water balance.

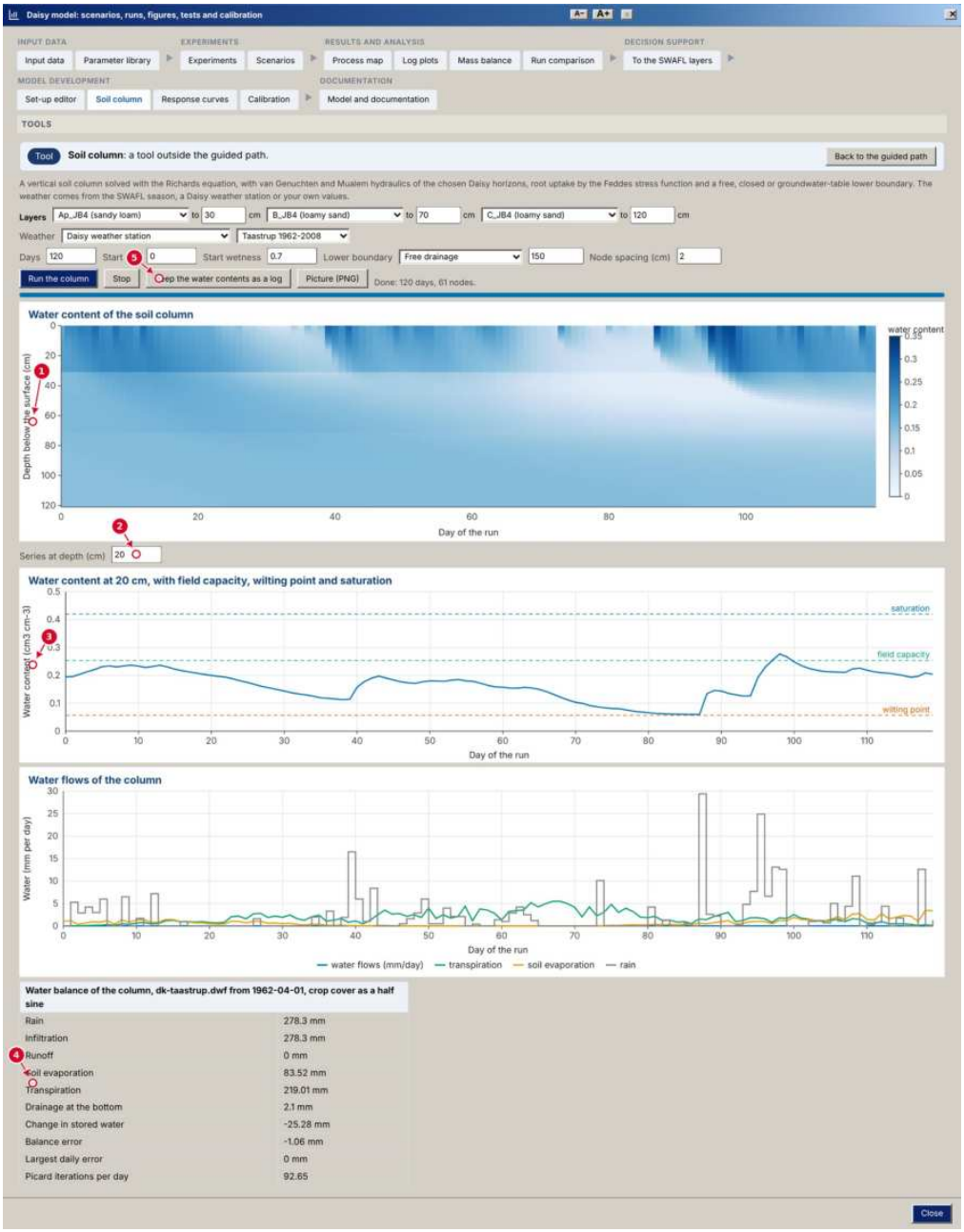



Figure 33.10: Soil column: results. Screen 133 of the Daisy workbench.

**What the arrows point to.**

1. **Water content map.** Depth against time.
2. **Depth of the series.** Depth in cm of the lower figure.
3. **Series at a depth.** Water content and suction at that depth.
4. **Summary.** Water balance of the run.
5. **Keep as a log.** Puts the water contents in the list of loaded logs of the Log plots tab.

### 33.0.11 Model and documentation

The Learn group describes the Daisy model, where each question is answered, how to get Daisy and how a time step is calculated.

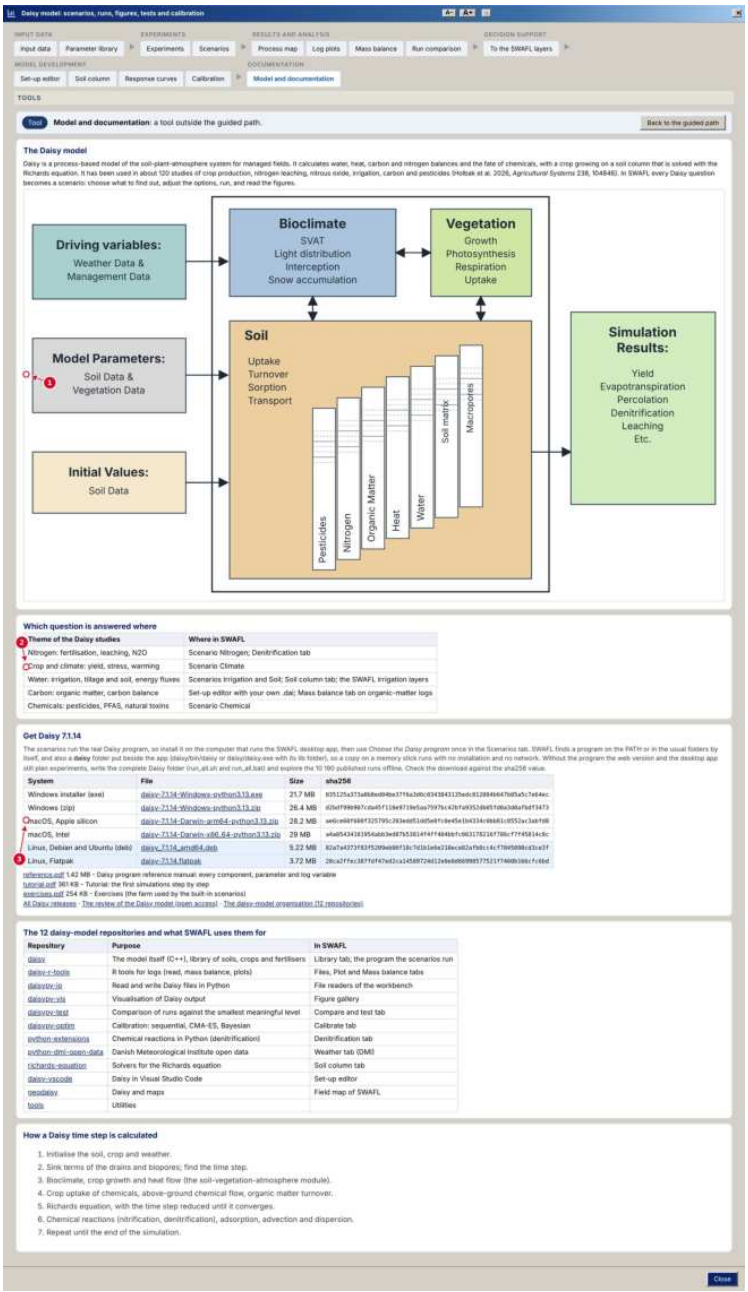



Figure 33.11: Model and documentation. Screen 134 of the Daisy workbench.

**What the arrows point to.**

1. **The Daisy model.** What the model is and what it simulates.
2. **Which question is answered where.** Themes of the Daisy studies and the tab for each.
3. **Get Daisy.** Downloads and checksums.

# Chapter 34

# ✎ Learn the Daisy files: Common, .dai files and a sample experiment

This chapter teaches how a Daisy experiment is made by hand, file by file, with a complete sample experiment that you can download from the program (*Daisy model, Documentation, Learn the Daisy files, Download the sample experiment*). Every file is printed in full. The labels ;<1>, ;<2> … at the ends of the lines are explained in the table under each file; in the program the same labels are drawn as orange numbers. The sample has two runs, spring barley without and with irrigation, on the Taastrup weather of 1990 to 1996. Both were run with Daisy 7.1.14 for this manual: each finishes with a water balance error of 0.0 mm, and their logs come with the sample.

The grammar follows chapters 2 to 6 of the Daisy Program Reference Manual (concepts, nomenclature, the file parser, log files and error messages). The program also has the whole reference manual (*Reference manual* tab) and the Model builder, which writes any definition from a form.

## 34.1 How a Daisy experiment is laid out

A Daisy experiment is a folder. The files that every run shares go into **Common**; each run has a folder of its own with one file, **run.dai**, which reads the common files and sets what is special about that run. Daisy writes the results of a run (the logs, **.dlf** files) into the folder of the run. This is how the Experiments tab of SWAFL writes its folders too.

```
MyFirstExperiment/                  <1> the experiment folder
  Common/                           <2> what every run reads
    my-soil.dai                     <3> soil horizons and the soil column
    my-management.dai               <4> crop, fertiliser, tillage and irrigation
    my-logs.dai                     <5> which results are written
    Daily-1965-2000.dwf             <6> the daily weather
  run-rainfed/                      <7> run 1
    run.dai                            reads Common, sets weather, column, management,
↪   period
    Annual-Harvest.dlf, Daily-Theta.dlf ...  <8> the logs Daisy writes here
  run-irrigated/                     run 2: the same, with irrigation (one line
↪   differs)
    run.dai
    *.dlf
```

Table 34.1: The parts of an experiment folder.

| Label | Part | What it is |
|---|---|---|
| <1> | MyFirstExperiment/ | One folder for the experiment. Give it a name without spaces. |
| <2> | Common/ | The files every run shares. A run reads them with `(path "../Common" &old)`. |
| <3> | my-soil.dai | Soil: `defhorizon` for each layer and `defcolumn` for the profile. |
| <4> | my-management.dai | Management: `defaction` for the season, the fertiliser and the irrigation. |
| <5> | my-logs.dai | The output list: which logs, when, into which .dlf file. |
| <6> | *.dwf | The weather, a Daisy weather file. SWAFL writes one from online weather or from your own file in Input data. |
| <7> | run-rainfed/run.dai | One run: the only file in its folder before it runs. |
| <8> | *.dlf | The logs: tables of results with a header, read by Log plots, Mass balance and Run comparison. |

### 34.1.1 Making an experiment, step by step

1. Make the folder `MyFirstExperiment` with a folder `Common` inside it.
2. Write the soil, `Common/my-soil.dai`: copy the file below, or build the horizons in the Model builder (component *horizon*) and the column (component *column*) and press *Save as .dai*.
3. Write the management, `Common/my-management.dai`, and the logs, `Common/my-logs.dai`.
4. Put a weather file in `Common`: the bundled one, one written by *Input data* from the online weather of your place, or your own.
5. Make a folder for each run with its `run.dai`. Copy the folder and change one line for each new run.
6. Run Daisy in each run folder (below), load the folders in *Log plots* and compare two runs in *Run comparison*.

The *Experiments* tab writes such a folder for you from a design (factors and their levels) and runs it; the next section shows what it writes.

## 34.2 The grammar of a .dai file

A set-up file is a list of brackets. Each bracket starts with a name and holds values. Text after a semicolon is a comment that Daisy ignores. Numbers carry their unit in square brackets, text is in double quotes, and a bracket can hold other brackets.

```
;; a comment: Daisy ignores everything after a semicolon   ;<1>
(dry_bulk_density 1.45 [g/cm^3])     ; an attribute: name, value, unit   ;<2>
(description "Ploughed topsoil")      ; a text value, in double quotes   ;<3>
(topsoil true)                        ; a boolean: true or false   ;<4>
(horizons (-30 [cm] "My Ap")           ; a list of pairs   ;<5>
          (-100 [cm] "My Bt"))
(water_factor (0.8 0.0) (0.9 0.2) (1.0 1.0))   ; a PLF: x y pairs of a piecewise
 ↪ linear function   ;<6>
(defhorizon "My C" "My Bt"              ; a definition: def + component, the new name,
 ↪ the model it builds on   ;<7>
  (humus 0.005 []))                     ; then only the attributes that differ
```

```
(input file "crop.dai")                ; read another file (the library of crops)   ;<8>
(path "../Common" &old)                  ; where to look for files; &old keeps the old
 ↪  search path   ;<9>
```

Table 34.2: The labels of the grammar example.

| Label | Where | What it does |
|---|---|---|
| <1> | `a comment` | A comment: everything after ; is ignored. ;; and ;;; are used for headings. |
| <2> | `(dry_bulk_density` | An attribute: (name value [unit]). The unit must be one Daisy knows; [] means no unit (a fraction). |
| <3> | `(description` | A string (text) value, always in double quotes. |
| <4> | `(topsoil true)` | A boolean value: true or false. |
| <5> | `(horizons` | A list: brackets inside a bracket. Here pairs of a depth and a horizon name. |
| <6> | `(water_factor` | A PLF (piecewise linear function): pairs of x and y. |
| <7> | `(defhorizon "My C"` | A definition: def followed by the component (horizon, column, crop, action, log...), the new name, and the model or earlier definition it builds on. Only the attributes that differ are written. |
| <8> | `(input file` | input file reads another set-up file, as if its text stood here. |
| <9> | `(path` | path sets the folders where Daisy looks for files. &old keeps the folders it already had, such as its own library. |

Table 34.3: The kinds of value (reference manual, chapter 4).

| Kind | Written as | Example |
|---|---|---|
| number | a number and its unit in square brackets; `[]` for a fraction | `(K_sat 0.5 [cm/h])`, `(clay 0.12 [])` |
| integer | a whole number | `(time 1990 1 1)` |
| boolean | `true` or `false` | `(topsoil true)` |
| string | text in double quotes | `(description "Ploughed topsoil")` |
| PLF | pairs of numbers: a piecewise linear function | `(water_factor (0.8 0.0) (0.9 0.2) (1.0 1.0))` |
| list | several values or brackets in one attribute | `(horizons (-30 [cm] "My Ap") (-100 [cm] "My Bt"))` |
| component | the name of a model, with its own attributes in brackets | `(hydraulic hypres (topsoil true))` |
| definition | `def` + component, a new name, the model it builds on, then the attributes | `(defhorizon "My C" "My Bt" (humus 0.005 []))` |

## 34.3 The soil: Common/my-soil.dai

The soil is built from horizons (the layers of the profile), and a column puts them together and adds the processes of the soil: water flow, the bottom boundary, organic matter and nitrogen.

```
;;; my-soil.dai -- the soil of the experiment: three horizons and the soil column
;;; A horizon is one layer of soil; the column stacks the horizons and adds water,
;;; heat, organic matter and nitrogen to them.
```

```
;; (1) A horizon: def + component name, its name in quotes, the model it is built on
 ↪  (USDA3),
;;     then its attributes, each as (name value [unit]).
(defhorizon "My Ap" USDA3   ;<1>
  (description "Ploughed topsoil, 0 to 30 cm")
  (clay 0.12 [])(silt 0.25 [])(sand 0.63 [])   ; fractions, they add up to 1   ;<2>
  (humus 0.025 [])                              ; 2.5 % organic matter   ;<3>
  (dry_bulk_density 1.45 [g/cm^3])   ;<4>
  (C_per_N 11.0 [g C/g N])
  (hydraulic hypres (topsoil true)))            ; water retention and conductivity
 ↪  from the HYPRES pedotransfer function   ;<5>

(defhorizon "My Bt" USDA3
  (description "Subsoil with more clay, 30 to 100 cm")
  (clay 0.22 [])(silt 0.20 [])(sand 0.58 [])
  (humus 0.010 [])
  (dry_bulk_density 1.60 [g/cm^3])
  (C_per_N 11.0 [g C/g N])
  (hydraulic hypres (topsoil false)))

;; (2) A horizon can also inherit from another one and change only what differs.
(defhorizon "My C" "My Bt"   ;<6>
  (description "Parent material, 100 to 200 cm")
  (humus 0.005 [])
  (dry_bulk_density 1.65 [g/cm^3]))

;; (3) The column: the whole soil profile of the field.
(defcolumn "My column" default   ;<7>
  (Soil (MaxRootingDepth 120 [cm])              ; roots cannot go deeper   ;<8>
        (border -100 [cm])                      ; fluxes are also logged at this depth
        (horizons (-30  [cm] "My Ap")           ; each pair: the lower boundary and the
 ↪  horizon above it   ;<9>
                  (-100 [cm] "My Bt")
                  (-200 [cm] "My C")))
  (Movement vertical (Tertiary none))           ; one-dimensional water flow, no
 ↪  biopores
  (Groundwater deep)                            ; free drainage at the bottom   ;<10>
  (OrganicMatter original (init (input 3000 [kg C/ha/y]) (end -25 [cm])))   ;<11>
  (Chemistry original   ;<12>
    (combine (N (reaction nitrification
                          (denitrification (water_factor (0.8 0.0) (0.9 0.2) (1.0
 ↪  1.0))))))))
```

Table 34.4: The labels of Common/my-soil.dai.

| Label | Where | What it does |
|---|---|---|
| <1> | `(defhorizon "My Ap" USDA3` | A new horizon called My Ap, built on the model USDA3 (texture in USDA classes). Names with spaces go in quotes. |
| <2> | `(clay 0.12` | The texture as fractions of the mineral soil: clay, silt and sand add up to 1. |
| <3> | `(humus 0.025` | Organic matter, as a fraction (0.025 = 2.5 %). |
| <4> | `(dry_bulk_density 1.45` | Dry bulk density in g/cm3. |

| Label | Where | What it does |
|---|---|---|
| <5> | `(hydraulic hypres` | The water retention and the conductivity come from the HYPRES pedotransfer function. A measured curve is written with (hydraulic M_vG (Theta_sat ...)(alpha ...)(n ...)(K_sat ...)). |
| <6> | `(defhorizon "My C" "My Bt"` | A horizon that inherits everything from My Bt and changes only humus and density. |
| <7> | `(defcolumn "My column" default` | The column, built on the default column model. |
| <8> | `(MaxRootingDepth` | The deepest the roots can reach in this soil. |
| <9> | `(horizons (-30` | The profile: each pair is the lower boundary of a horizon (negative, downwards) and its name. |
| <10> | `(Groundwater deep)` | The bottom boundary: free drainage. Other choices are a groundwater table, an aquitard or a lysimeter. |
| <11> | `(OrganicMatter original` | The soil organic matter pools, started in equilibrium with 3000 kg C/ha of input a year in the top 25 cm. |
| <12> | `(Chemistry original` | Nitrogen: nitrification and denitrification, with the water factor of denitrification as a PLF. |

## 34.4 The management: Common/my-management.dai

What the farmer does is written as actions. An action can be a list of operations that run one after the other (activity), and it can wait for a date or a condition between them.

```
;;; my-management.dai -- what the farmer does: the crop, the fertiliser, the tillage
 ↪  and the irrigation

;; (1) The library files of Daisy that define the crops, the tillage operations and
 ↪  the fertilisers.
(input file "tillage.dai")   ;<1>
(input file "crop.dai")
(input file "fertilizer.dai")

;; (2) One season of spring barley, as a list of operations done one after the other.
(defaction SBarley_rainfed activity   ;<2>
  (wait_mm_dd 3 20) (plowing)                                   ; wait until 20 March,
 ↪  then plough   ;<3>
  (wait_mm_dd 4 5)  (fertilize (N25S (weight 120 [kg N/ha])))  ; 5 April: 120 kg N/ha
 ↪  of mineral fertiliser   ;<4>
  (seed_bed_preparation) (sow "Spring Barley")                  ; prepare the seed bed
 ↪  and sow   ;<5>
  (wait (or (crop_ds_after "Spring Barley" 2.0) (mm_dd 9 1)))  ; wait for maturity (DS
 ↪  2) or 1 September   ;<6>
  (harvest "Spring Barley" (stub 8 [cm])))                     ; harvest, leaving 8 cm
 ↪  of stubble   ;<7>

;; (3) An irrigation rule: when the soil at 20 cm is drier than -300 cm (pF 2.5)
;;     between 1 May and 1 September, give 30 mm overhead (10 mm/h for 3 hours), then
 ↪  wait two days.
```

```
(defaction Irrigate_when_dry activity   ;<8>
  (wait (and (after_mm_dd 5 1)
             (before_mm_dd 9 1)
             (not (soil_water_pressure_above (height -20.0 [cm]) (potential -300
 ↪  [cm])))))
  (irrigate_overhead 10 [mm/h] (hours 3))   ;<9>
  (wait_days 2))

;; (4) The same season with irrigation: the rule repeats while the barley season runs.
(defaction SBarley_irrigated activity
  (while (SBarley_rainfed) (repeat Irrigate_when_dry)))   ;<10>
```

Table 34.5: The labels of Common/my-management.dai.

| Label | Where | What it does |
|---|---|---|
| <1> | `(input file "tillage.dai")` | The library files of Daisy that define the tillage operations, the crops and the fertilisers used below. |
| <2> | `(defaction SBarley_rainfed activity` | A new action called SBarley_rainfed: an activity, that is a list of operations done in turn. |
| <3> | `(wait_mm_dd 3 20) (plowing)` | wait_mm_dd waits until a month and day; plowing is a tillage operation of tillage.dai. |
| <4> | `(fertilize (N25S` | Fertilise with the mineral fertiliser N25S of fertilizer.dai, 120 kg N/ha. |
| <5> | `(sow "Spring Barley")` | Sow a crop of crop.dai. The name must be written exactly as there. |
| <6> | `(wait (or (crop_ds_after` | Wait until the crop reaches development stage 2 (maturity) or 1 September, whichever comes first. |
| <7> | `(harvest "Spring Barley"` | Harvest the crop, leaving 8 cm of stubble. |
| <8> | `(defaction Irrigate_when_dry activity` | An irrigation rule with a tensiometer at 20 cm: wait for a dry soil inside the irrigation window. |
| <9> | `(irrigate_overhead 10` | Sprinkle 10 mm/h for 3 hours (30 mm). |
| <10> | `(while (SBarley_rainfed) (repeat Irrigate_when_dry))` | Run the season and, at the same time, repeat the irrigation rule until the season ends. |

## 34.5 The results: Common/my-logs.dai

Daisy writes nothing unless it is told to. The output list names the logs (defined in Daisy's own log.dai), when to write a line, and the file to write it to.

```
;;; my-logs.dai -- which results Daisy writes, how often and into which file (.dlf)
;;; The log names ("harvest", "Field water", "Crop" ...) are defined in Daisy's own
 ↪  log.dai.   ;<3>

(input file "log.dai")   ;<1>

(output   ;<2>
  ;; once a year, on 1 April at midnight (the season runs from April to April)
  ("harvest"       (where "Annual-Harvest.dlf"))
```

```
 ("Field water"   (when (and (month 4) (mday 1) (hour 0))) (where
↪  "Annual-FWater.dlf"))   ;<4>
 ("Field water"   (when (and (month 4) (mday 1) (hour 0))) (to -100 [cm]) (where
↪  "Annual-FWater_100cm.dlf"))   ;<5>
 ("Field nitrogen" (when (and (month 4) (mday 1) (hour 0))) (to -100 [cm]) (where
↪  "Annual-FN_100cm.dlf"))
 ;; every day
 ("Crop"               (when daily) (where "Daily-CropProduction.dlf"))   ;<6>
 ("Field water"        (when daily) (where "Daily-FWater.dlf"))
 ("Soil Water Content" (when daily) (where "Daily-Theta.dlf")))
```

Table 34.6: The labels of Common/my-logs.dai.

| Label | Where | What it does |
|---|---|---|
| <1> | `(input file "log.dai")` | The library of standard logs: harvest, Field water, Field nitrogen, Crop, Soil Water Content and many more. |
| <2> | `(output` | The output list: one entry for each log file. |
| <3> | `("harvest"` | The harvest log writes one line at each harvest: the yield and the stress of the crop. |
| <4> | `(when (and (month 4)` | when: a condition for writing a line; here on 1 April at midnight, so each line is one season. |
| <5> | `(to -100 [cm])` | to: the balance of the soil down to 100 cm only, so the drainage is the flow at 1 m. |
| <6> | `(when daily)` | Write one line every day. |

## 34.6 The weather: a .dwf file

The weather file has a header of keywords and a table with one line per day (or hour). The header gives the station, its position, the period and fixed values such as the nitrogen deposition; the two lines over the table give the column names and their units.

```
dwf-0.0 -- Control_WG_daily.txt   ;<1>

# General information about the station itself.
Station: Taastrup   ;<2>
Elevation: 30 m
Longitude: 12 dgEast
Latitude: 56 dgNorth
TimeZone: 15 dgEast

# Information about the measurement conditions.
Surface: reference
ScreenHeight: 2.0 m

# General information about the available data.
Begin: 1965-01-01   ;<3>
End:   2000-12-31
Timestep: 24 hours

# Fixed deposit values.
NH4WetDep: 0.9 ppm   ;<4>
```

```
NH4DryDep: 2.2 kgN/ha/year
NO3WetDep: 0.6 ppm
NO3DryDep: 1.1 kgN/ha/year

TAverage: 8.6 dgC   ;<5>
TAmplitude: 8.6 dgC
MaxTDay: 209 yday

------------------------------------------------------------------------------
 ↪  ;<6>
Year    Month   Day Precip  T_min   T_max   VapPres Wind    DiffRad GlobRad RefEvap
 ↪  ;<7>
year    month   day mm/d    dgC dgC hPa m/s kWh/d/m^2   kWh/d/m^2   mm/d
1965    1   1   0   -2.36   5.84    6.53    5.16    0.33    0.75    0
... (one line per day follows)
```

Table 34.7: The labels of the weather file.

| Label | Where | What it does |
|---|---|---|
| <1> | `dwf-0.0` | The first line says that this is a Daisy weather file (dwf) and its version. |
| <2> | `Station:` | Station, elevation, longitude, latitude and time zone. |
| <3> | `Begin:` | The period and the time step of the data. |
| <4> | `NH4WetDep:` | Nitrogen in rain and dry deposition, added every day. |
| <5> | `TAverage:` | The mean and amplitude of the air temperature: they drive the soil temperature at the bottom. |
| <6> | `-----` | A line of dashes ends the header. |
| <7> | `Year` | The column names: date, precipitation, minimum and maximum temperature, vapour pressure, wind, radiation and reference evapotranspiration. |

Any daily weather can be written in this form. *Input data* writes a .dwf file from the online weather of a place (Open-Meteo, ERA5) or from a table of your own (CSV or Excel with date, rain, minimum and maximum temperature and radiation); the Daisy stations of the program are .dwf files too.

## 34.7 One run: run-rainfed/run.dai

The run file reads the common files and sets everything that belongs to this run: the weather, the column, the management and the period. A second run differs only in what it changes.

```
;;; run-rainfed/run.dai -- one run of the experiment: the files it reads and the
 ↪  settings of this run

(path "../Common" &old)                 ; also look for files in ../Common (&old keeps
 ↪  Daisy's own library)   ;<1>
(input file "my-soil.dai")              ; the soil horizons and the column   ;<2>
(input file "my-management.dai")        ; the crop and the field operations
(input file "my-logs.dai")              ; the results to write

;; the weather file gives radiation in kWh/d/m^2, a unit Daisy does not know: define
 ↪  it (1 kWh/d/m^2 = 41.67 W/m^2)
```

```
(defunit [kWh/d/m^2] SIfactor (mass 1) (time -3) (factor 41.67))   ;<3>
(weather default "../Common/Daily-1965-2000.dwf")   ; the daily weather of Taastrup,
 ↪ Denmark   ;<4>

(description "My first experiment: spring barley without irrigation")
(column "My column")                    ; the soil column to simulate   ;<5>
(manager activity (repeat SBarley_rainfed))   ; the management, repeated every year
 ↪ ;<6>

(time 1990 1 1)                         ; the start of the simulation   ;<7>
(stop 1996 4 1)                         ; the end
(activate_output (after 1991 3 31 23))  ; write the logs only after one year of
 ↪ warm-up   ;<8>
```

Table 34.8: The labels of run-rainfed/run.dai.

| Label | Where | What it does |
|---|---|---|
| <1> | `(path "../Common" &old)` | Look for files in ../Common as well as in Daisy's library. |
| <2> | `(input file "my-soil.dai")` | Read the soil, the management and the logs. |
| <3> | `(defunit [kWh/d/m^2]` | This weather file gives radiation in kWh/d/m2, which Daisy does not know: the unit is defined (41.67 W/m2). Without it Daisy stops with "No data for GlobRad". |
| <4> | `(weather default` | The weather file of the run. |
| <5> | `(column "My column")` | The soil column to simulate (defined in my-soil.dai). |
| <6> | `(manager activity (repeat SBarley_rainfed))` | The manager: the management action, repeated every year. The irrigated run has SBarley_irrigated here; that is its only difference. |
| <7> | `(time 1990 1 1)` | The start and the end of the simulation (year month day). |
| <8> | `(activate_output` | Start writing logs after one year of warm-up, so the start values of the soil do not count. |

The irrigated run is the same file with two lines changed: its description and the manager, `(manager activity (repeat SBarley_irrigated))`. Making a new run is copying a run folder and changing what the run is about.

## 34.8 Run it and read the results

Run each run.dai with Daisy (in its own folder), or open the file in the set-up editor and press *Run Daisy* in the desktop app. Each run writes its logs into its folder. The logs of the two runs of this sample are included, so you can read them now.

- Windows: open a command window in `run-rainfed` and type `"C:\Program Files\Daisy\bin\daisy.cmd" run.dai`, then the same in `run-irrigated`.
- SWAFL desktop: open `run.dai` in the set-up editor and press *Run Daisy*, or let the Experiments tab write and run the folders.
- Daisy prints its progress and, at the end, the water balance of the whole run, `Balance (= In - Out - Increase) = 0.0 [mm]`.

Table 34.9: What the two runs of the sample give (Daisy 7.1.14). Yield: dry matter of the grain at harvest; the water terms are the seasons from 1 April, from Annual-FWater.dlf.

| Season | Yield rainfed, Mg/ha | Yield irrigated, Mg/ha | Irrigation, mm | Evapotranspiration rainfed, mm | Evapotranspiration irrigated, mm | Percolation rainfed, mm | Percolation irrigated, mm |
|---|---|---|---|---|---|---|---|
| 1991 | 5.30 | 5.30 | 120 | 489 | 502 | 183 | 290 |
| 1992 | 5.87 | 5.88 | 180 | 522 | 551 | 280 | 430 |
| 1993 | 5.94 | 5.96 | 150 | 526 | 540 | 98 | 234 |
| 1994 | 5.24 | 5.24 | 150 | 454 | 471 | 106 | 159 |
| 1995 | 6.00 | 6.00 | 180 | 545 | 556 | 9 | 225 |

In this wet climate the irrigation (120 to 180 mm a season) raises evapotranspiration a little but mostly drains through the soil: the percolation below the root zone roughly doubles, while the yield hardly changes. This is the kind of answer an experiment gives. *Load the logs of both runs* in the Learn tab puts the fourteen logs in *Log plots*; *Load them and compare the two runs* opens *Run comparison* with the two water balances (section Run comparison).

## 34.9 When Daisy stops: messages and what to do

Daisy checks the file when it reads it (parser errors), before the run (completeness and consistency checks) and while it runs (runtime errors). The message gives the file, the line and the column. Each message below was written by Daisy 7.1.14 when the sample was run with one deliberate mistake. Each message gives the file, the line and the column (`run.dai:14:42`).

Table 34.10: Daisy messages, why they come and what to do. Every message was written by Daisy 7.1.14 on the sample with one deliberate mistake.

| Daisy says | Why | What to do |
|---|---|---|
| `run.dai:14:42: Unknown 'crop' model 'Spring Barly' (error)` | A name that is not defined: misspelt, written with other spaces or capitals, or its file is not read. The same message comes for any component (column, horizon, action...). | Write the name exactly as in the library file (crop.dai lists the crops), and read that file with (input file “crop.dai”). |
| `run.dai:13:19: Unknown 'column' model 'My colum' (error)` | The column named in (column “...”) is not defined. | Use the name of the defcolumn in my-soil.dai, letter for letter. |
| `run.dai:13:39: Unknown attribute 'dry_bulk_densty', did you miss a ')'? (error)` | An attribute name that the model does not have: misspelt, or a bracket closed too early so it lands in the wrong place. | Check the spelling in the reference manual (or build the model in the Model builder); count the brackets. |
| `run.dai:17:6: Unknown 'integer' model 'stop' (error) ... Unknown attribute 'activate_output', did you miss a ')'?` | A bracket is not closed: (time 1990 1 1 without its ). Daisy reads the next lines as part of it, so the first error is on the line after the mistake. | Look at the line above the first error. The set-up editor checks the brackets as you type. |
| `run.dai:13:52: expected [g/cm^3] got [g/cm3] (error)` | A unit written differently from the one Daisy knows. | Write the unit as the reference manual gives it ([g/cm^3], [cm/h], [kg N/ha]), or define a new one with defunit. |
| `Open 'my-log.dai' failed ... output is missing (error)` | A file named in (input file ...) does not exist in the folder or on the path, so what it defines is missing too. | Check the file name and the (path “../Common” &old) line. |
| `No data for 'GlobRad' (error) / Can't convert 'GlobRad' value of 0.39 [kWh/d/m^2] to [W/m^2] (warning)` | A column of the weather file is in a unit Daisy cannot convert. | Define the unit, as in run.dai: (defunit [kWh/d/m^2] SIfactor (mass 1) (time -3) (factor 41.67)). |
| `Value is -42 but should be >= 0 (error)` | A value outside its allowed range (a consistency check); here a follow-on error of the open bracket above. | The reference manual gives the range of every attribute; fix the first error first. |

| Daisy says | Why | What to do |
|---|---|---|
| `Balance (= In - Out - Increase) = 0.0 [mm]` | Not an error: the water balance of the whole run, printed at the end. It should be 0.0. | If it is not near zero, use a smaller irrigation rate (mm/h), check the hydraulic parameters, and look at the Mass balance tab. |

Daisy also writes every message into `daisy.log` in the run folder. Warnings (such as a value it cannot convert) do not stop the run but should be read.

## 34.10 Templates: one file, many runs

For calibration and for experiments with many runs, write the parameter that changes as a placeholder in braces. SWAFL (Calibration, Experiments) and daisypy-optim fill it with a value for each run.

```
;; template: the value of K_sat and of the fertiliser are filled in for each run
(defhorizon "My Ap" USDA3
  (clay 0.12 [])(silt 0.25 [])(sand 0.63 [])(humus 0.025 [])
  (dry_bulk_density 1.45 [g/cm^3])
  (hydraulic M_vG (Theta_res 0.0 []) (Theta_sat 0.40 []) (alpha 0.045 [cm^-1]) (n 1.5
 ↪  [])
                  (K_sat {Ksat} [cm/h]) (l -1.0 [])))   ;<1>
(defaction SBarley_N activity
  (wait_mm_dd 4 5) (fertilize (N25S (weight {Nrate} [kg N/ha]))))   ;<2>
```

Table 34.11: The labels of the template.

| Label | Where | What it does |
|---|---|---|
| <1> | `{Ksat}` | A placeholder: {name}. Each run replaces it with a number, for example 0.5. The Placeholders panel of the set-up editor lists them and lets you fill them in. |
| <2> | `{Nrate}` | Any value can be a placeholder; the unit stays outside the braces. |

The set-up editor lists the placeholders of a file in its *Placeholders* panel, says what each one means when it stands for an attribute of the reference manual, and fills them in; *Calibration* fills them for every trial of the search.

## 34.11 The Common folder that the Experiments tab writes

The Experiments and Scenarios tabs write the same kind of folder, so it can be run anywhere and read by hand. The folder of one experiment holds:

```
MyExperiment/
  Common/
    ArableFarmRotation.dai   the rotation: spring barley, winter barley, winter rape,
 ↪  winter wheat,
                            with the fertiliser and the tensiometer irrigation of the
 ↪  design
    ResFarm.dai              the soils of the project (horizons and columns)
    OutputLog.dai            the logs every run writes (annual and daily)
    UserSoil.dai             your own soil horizons, when you gave them in Input data
    <weather>.dwf            the weather of the runs
```

```
   <library files>          the files of the Daisy library a design uses (a crop, a
↪  fertiliser,
                           a horizon, a chemical, a management action or a program),
↪  copied
                           with the files they read in turn
  <run name>/run.dai       one folder for each run of the design
```

and the `run.dai` of each run has this form (the values are those of the run):

```
;; SWAFL scenario "Irrigation", variant "Irrigated"
(input file "../Common/ArableFarmRotation.dai")
(input file "../Common/ResFarm.dai")
(input file "../Common/OutputLog.dai")
;; the Daisy library used by this run
(path "../Common" &old)
(input file "winterwheat.dai")
(defunit [kWh/d/m^2] SIfactor (mass 1)(time -3)(factor 41.67))
(weather default "../Common/Daily-1965-2000.dwf"
  (PrecipCorrect 1)
  (TempOffset 0))
(description "Irrigation - Irrigated")
(log_prefix "irrigated-")
(manager activity
  (repeat "A_F_R"))
(defcolumn "SWAFL column" "ResFarm column"
  (Bioclimate FAO56_daily))
(column "SWAFL column")
(time 1990 01 01)
(stop 2010 04 1)
(activate_output (after 1991 03 31 23))
;; logs of the Daisy library added to this run
(output &old ("Soil Water Content"))
```

`PrecipCorrect` and `TempOffset` scale the rain and shift the temperature of the weather (the climate factors of a design); `(defcolumn "SWAFL column" "ResFarm column" ...)` builds the column of the run on a column of the project and changes only what the run is about; `(output &old (...))` adds a log to the output list of OutputLog.dai. The Experiments chapter lists every factor and the line it writes.

## 34.12 A checklist before you run

- Every bracket is closed: the set-up editor says *Brackets and quotes are balanced*.
- Every name you use is defined in a file that is read: crops in crop.dai, fertilisers in fertilizer.dai, tillage in tillage.dai, your own in Common.
- Units are written as the reference manual gives them, and the units of the weather file are known (or defined with `defunit`).
- Clay, silt and sand of each horizon add up to 1; the lowest horizon reaches the bottom of the column.
- `time` is before `stop`, and `activate_output` leaves a year of warm-up.
- The output list writes what you want to read: the annual balances for the water and nitrogen budgets, the daily logs for figures through time.

# Chapter 35

# ☰ The 107 reference topics

The Model and documentation tab and the references of the experiment combinations follow the 107 topics of the Daisy program reference manual. For each topic this chapter gives its purpose, its sections, and the questions of the workbench that touch it.

## 35.1 Topic 1: Introduction

What the manual is: a complete reference to all components and models Daisy provides, with their parameters, state variables and log variables, generated automatically from the code and listed alphabetically in part II. It is not an introduction to the model, a tutorial or a programmers manual.

## 35.2 Topic 2: Concepts

Components, models and parameterizations: a component is a group of exchangeable models (for example crop, bioclimate, soil horizon), a model is one implementation, and a parameterization is a model with chosen attribute values. Attributes can be optional, have defaults, and be log variables.

## 35.3 Topic 3: Nomenclature

Having covered the concepts, it is now possible to explain the nomenclature used in part II of the manual. As already explained, each library component has its own chapter, while all the fixed components are collected in chapter 107. All the library component chapters start with a short explanation of the purpose of that component. Each model (including those models that are part of a fixed component) has its own section. Here, we will explain how to read the content of these sections. This is best done with an example. Now, keep a finger here, and find section 15.12 (that's on page 80).

| Section | Name | What it covers | Components |
|---|---|---|---|
| 3.1 | Name and Description | The name of the section is 'fertilize', which is the name you have to specify in the setup file when you want to refer to the model. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 3.2 | Sample | The next element in the section is a sample, which shows the format recognized by the file parser model when it reads the setup files. | n/a |
| 3.3 | The Attribute List | After the sample comes a list with information about each attribute in the model. | n/a |
| 3.4 | Submodels | Sometimes it is useful to combine attributes in a model into groups. One example of this is the ‘cond’ management action model (section 15.4), where we use the ‘clauses’ group to emphasize the strong association between each condition, and the list of actions to perform when the condition is true. | n/a |

## 35.4 Topic 4: The file parser model

This section describes Daisy setup files. A Daisy setup file is read by the ‘file’ model of the ‘parser’ component, hence the strange name.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 4.1 | Types and Formatting | There are a few general rules for the formatting accepted by the ‘file’ parser. | n/a |
| 4.2 | Declaring new parameters | You can declare new parameters most places where you can specify parameter values. | n/a |
| 4.3 | Referring to parameters | Instead of a parameter value, you can specify the name of another parameter. | n/a |
| 4.4 | Toplevel | A Daisy setup file consist of definitions of parameterizations and a small number of commands. | n/a |

## 35.5 Topic 5: Log File Generation

The value of the ‘output’ attribute should be a sequence of log component parameterizations. Each parameterization should specify some state or log variables to be output during the simulation. The ‘log.dai’ file contains a number of standard log parameterizations, for example ‘Crop Production’, which will write information about the development and production of any crops in the field to a file named ‘crop prod.dlf’ located in the directory specified by the ‘directory’ attribute, or, by default, the directory where daisy was started.

## 35.6 Topic 6: Error Messages

Sometimes the simulation will fail to run for some reason or another. In these cases, Daisy will attempt to give an error message, which hopefully will give some indication of what went wrong. There are zillions of situations where Daisy might give some error message, in the following we describe some of them.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 6.1 | Parser errors | If the format of the setup files is wrong, for example if there are missing parentheses or unknown attributes, Daisy will give you information about which file the error was found, the line number, the column number, and what the problem was. | n/a |
| 6.2 | Completeness checks | After parsing the setup files, Daisy will check that all the required parameters and state variables are defined, and make some very simple consistency checks. | n/a |
| 6.3 | Consistency checks | After checking that all the required attributes are there, Daisy will create the simulation objects, and then perform a more throughout consistency check. | n/a |
| 6.4 | Runtime exceptions | Now Daisy is ready to run the simulation. However, some errors are too hard to find before the simulation has started. | Exception |
| 6.5 | Assertion failures | During debugging, a lot of ‘assert’ statements have been added to the code. Each of these statements will check if a certain condition is true, and if not, exit the program with a message like field.C:71: failed assertion ‘false’ A failed assertion message will always print the file name, the line number, and the assertion that failed, but the format may vary between different. | n/a |
| 6.6 | Program errors | Sometimes we can get errors that aren’t even caught by the assertions, they can manifest themselves as “Bus error” or “Segmentation violation” errors, or weird pop-up windows under win32. | n/a |

## 35.7 Topic 7: ABAproduction

The ‘ABAproduction’ component calculates the prod of ABA in soil.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 7.1 | none | No ABA production. | n/a |
| 7.2 | root | ABA production based on production in roots. | n/a |
| 7.3 | soil | ABA production based on soil location. | n/a |
| 7.4 | uptake | ABA production based on concentration in water uptake. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.8 Topic 8: AOM

A single Added Organic Matter pool. Used by Harvesting @ Root (see 107.7, page 595) , and Bioincorporation @ AOM (see 107.11, page 605) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 8.1 | AOM-DIRECT | Third AOM pool of already decomposed material. | n/a |
| 8.2 | AOM-FAST | Fast AOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003] Used by crop simple root am (see 29.47, page 178) , am root om (see 17.4, page 105) , vegetation permanent litter am (see 100.3, page 547) , vegetation afforestation root am (see 100.1, page 544) , Harvesting @ Root (see 107.7, page 595) , and Bioincorporation @ AOM (see 107.11, page 605) . | n/a |
| 8.3 | CROP-FAST | A 'AOM-FAST' model (see 8.2, page 54) build into Daisy. | n/a |
| 8.4 | AOM-SLOW | Slow AOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003] Used by crop simple root am (see 29.47, page 178) , am root om (see 17.4, page 105) , vegetation permanent litter am (see 100.3, page 547) , vegetation afforestation root am (see 100.1, page 544) , and Harvesting @ Root (see 107.7, page 595) . | n/a |
| 8.5 | AOM-SLOW-BIOINCORPORATION | A 'AOM-SLOW' model (see 8.4, page 55) build into Daisy. | n/a |
| 8.6 | AOM-SLOW-OLD | A 'AOM-SLOW' model (see 8.4, page 55) build into Daisy. | n/a |
| 8.7 | CROP-SLOW | A 'AOM-SLOW' model (see 8.4, page 55) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 8.8 | Ryegrass-SLOW | A ‘CROP-SLOW’ model (see 8.7, page 55) defined in ‘ryegrass.dai’. | n/a |
| 8.9 | Wclover-SLOW | A ‘CROP-SLOW’ model (see 8.7, page 55) defined in ‘wclover.dai’. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.9 Topic 9: AWI

Estimate area of air-water interface.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 9.1 | Brusseau2023 | Estimate AWI area from water and median particle size (Equation 5). See also [Brusseau, 2023] Used by column default AWI (see 26.1, page 153) . 9.2 Costanza2008x Estimate AWI area from water and median particle size (Equation 1 and 2). | n/a |
| 9.2 | Costanza2008x | Estimate AWI area from water and median particle size (Equation 5). See also [Brusseau, 2023] Used by column default AWI (see 26.1, page 153) . 9.2 Costanza2008x Estimate AWI area from water and median particle size (Equation 1 and 2). | n/a |

**Related theme.** water and heat movement in the soil.

## 35.10 Topic 10: ClayOM

Find the effect of clay on organic matter processing.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 10.1 | biomod | Clay influence on organic matter from BIOMOD project. All SMB pools are affected, but not the SOM pools. | a |
| 10.2 | old | Traditional clay influence on organic matter. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.11 Topic 11: MV Crop

Description of a crop for use by the MARKVAND model.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 11.1 | default | Standard MARKVAND crop model. | L ge, L gm, L ym, S Le, S Lm, z xA, z m |

**Related theme.** the crop: growth, roots and water use.

## 35.12 Topic 12: MV Soil

Description of a soil for use by the MARKVAND model.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 12.1 | default | Standard MARKVAND soil model. | z o, Theta fo, Theta wo, Theta fu, Theta wu, C e, c e, k qb |

**Related theme.** water and heat movement in the soil.

## 35.13 Topic 13: SMB

A single Soil MicroBiological pool.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 13.1 | SMB-FAST | Fast SMB pool parameterization. | n/a |
| 13.2 | SMB-FAST-2000 | A ‘SMB-FAST’ model (see 13.1, page 66) build into Daisy. | n/a |
| 13.3 | SMB-SLOW | Slow SMB pool parameterization. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.14 Topic 14: SOM

A single Soil Organic Matter pool.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 14.1 | SOM-FAST | Fast SOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003] Used by organic default som (see 60.1, page 327) . | n/a |
| 14.2 | SOM-FAST-OLD | A ‘SOM-FAST’ model (see 14.1, page 68) build into Daisy. | n/a |
| 14.3 | SOM-INERT | Inert SOM pool parameterization. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 14.4 | SOM-SLOW | Slow SOM pool parameterization by Sander Bruun. See also [Bruun et al., 2003] Used by organic default som (see 60.1, page 327) . | n/a |
| 14.5 | SOM-SLOW-OLD | A ‘SOM-SLOW’ model (see 14.4, page 68) build into Daisy. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.15 Topic 15: action

The ‘action’ component represents management on different abstraction levels, from a single tillage operation to strategies of how to manage a farm. Typically, but not necessarily, the high level management strategies are build by combining low level management operations.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 15.1 | BBCH | Management according to external BBCH scale. | crop, file, from, to, from start, to start, do, when |
| 15.2 | assert | Assert that condition is true, if not, stop the simulation. | condition, message |
| 15.3 | at | Do action at specific time. | month, day, do |
| 15.4 | cond | Perform the actions associated with the first true condition in the list. | clauses, condition, actions |
| 15.5 | crop | Manage a specific crop or multicrop. | spray, month, day, name, amount, secondary, crop, date |
| 15.6 | emerge | Force a crop to emerge. | crop |
| 15.7 | error | Write a error message to the user. | message |
| 15.8 | extern | Select an external scope, and perform action. | scope, action |
| 15.9 | extern fertigation | Continiues irrigation with mineral nitrogen mix. | volume, surface, scope, from, to |
| 15.10 | extern subsoil | Subsoil irrigation controlled externally. | volume, scope, from, SOLETE, to, constituents |
| 15.11 | fertilize base | Shared parameters for all fertilize actions. | am, equivalent weight, minimum weight, precision, target, from, to |
| 15.12 | fertilize | A ‘fertilize base’ model (see 15.11, page 80) build into Daisy. | am, from, SOLETE, to, LETE |
| 15.13 | incorporate fertilizer | A ‘fertilize base’ model (see 15.11, page 80) build into Daisy. | am, volume |
| 15.14 | harvest base | Common parameters for harvest operations. | crop, combine, stub, stem, sorg |
| 15.15 | cut | A ‘harvest base’ model (see 15.14, page 81) build into Daisy. | n/a |
| 15.16 | harvest | A ‘harvest base’ model (see 15.14, page 81) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 15.17 | if | If the condition is true, perform the first action, otherwise perform the second action. | then, else |
| 15.18 | irrigate base | Shared parameter for irrigate actions. | solute, name, value, hours, days, remaining time, temperature |
| 15.19 | irrigate overhead | A ‘irrigate base’ model (see 15.18, page 82) build into Daisy. | n/a |
| 15.20 | irrigate subsoil | A ‘irrigate base’ model (see 15.18, page 82) build into Daisy. | volume, from, SOLETE, to |
| 15.21 | irrigate surface | A ‘irrigate base’ model (see 15.18, page 82) build into Daisy. | n/a |
| 15.22 | markvand | Irrigate the field according to MARKVAND scheduling. | solute, name, value, dt, T sum, map, Daisy, MARKVAND |
| 15.23 | message | Write a message to the user. | message |
| 15.24 | mix | Mix soil content down to the specified depth. The effect is that nitrogen, water, temperature and such are averaged in the interval. | depth, random roughness, penetration, surface loose |
| 15.25 | disk harrowing | A ‘mix’ model (see 15.24, page 86) defined in ‘tillage.dai’. | n/a |
| 15.26 | rotavation | A ‘mix’ model (see 15.24, page 86) defined in ‘tillage.dai’. | n/a |
| 15.27 | seed bed preparation | A ‘mix’ model (see 15.24, page 86) defined in ‘tillage.dai’. | n/a |
| 15.28 | stubble cultivation | A ‘mix’ model (see 15.24, page 86) defined in ‘tillage.dai’. | n/a |
| 15.29 | nil | This action does nothing, always done. | n/a |
| 15.30 | panic | Write a error message to the user and stop the simulation. | message |
| 15.31 | pluck | Pluck a crop. Unlike the ‘harvest’ operation, this allows you to pluck selected parts of the above ground dry matter without killing the crop. | crop, stem, sorg |
| 15.32 | progn | Perform all the specified actions in the sequence listed. All the actions will be performed in the same time step. | actions |
| 15.33 | remove litter | Remove organic matter from soil surface. | n/a |
| 15.34 | remove solute | Remove a specific chemical from the field. | n/a |
| 15.35 | repeat | Perform all of the specified action. When done, repeat the action. The action may take several timesteps. | repeat, do |
| 15.36 | restore SOM | Restore the content of the soil organic matter pools. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 15.37 | sequence | Perform all the specified actions in the sequence listed. All the actions will be performed in the same time step. | do |
| 15.38 | activity | A ‘sequence’ model (see 15.37, page 88) build into Daisy. | n/a |
| 15.39 | set porosity | Set the porosity of the horizon at the specified depth. To get useful results, you need to use a hydraulic model that supports this. | depth, porosity |
| 15.40 | set surface detention capacity | Set amount of ponding the surface can retain. | height |
| 15.41 | sorption table | Print a sorption table for all chemicals. | Theta, start, intervals |
| 15.42 | sow base | Shared sowing parameters. | row position, row width, plant distance, plant position |
| 15.43 | sow | A ‘sow base’ model (see 15.42, page 89) build into Daisy. | crop |
| 15.44 | plant | A ‘sow’ model (see 15.43, page 90) build into Daisy. | n/a |
| 15.45 | sow name | A ‘sow base’ model (see 15.42, page 89) build into Daisy. | crop |
| 15.46 | sow object | A ‘sow base’ model (see 15.42, page 89) build into Daisy. | crop |
| 15.47 | spray | Spray a chemical (typically a pesticide) on the field. | amount |
| 15.48 | spray surface | Spray a chemical (typically a pesticide) on the field below the canopy. | amount |
| 15.49 | stop | Stop the simulation. | n/a |
| 15.50 | store SOM | Store the content of the soil organic matter pools. | n/a |
| 15.51 | swap | Swap two soil layers. The top layer start at the surface and goes down to ’middle’, and the second layer starts with ’middle’ and goes down to ’depth’. After the operation, the content (such as heat, water, and organic matter) will be averaged in each layer, and the bottom layer will be placed on top of what used to be the top layer. | depth, random roughness, middle |
| 15.52 | plowing | A ‘swap’ model (see 15.51, page 92) defined in ‘tillage.dai’. | n/a |
| 15.53 | t | This action does nothing, never done. | n/a |
| 15.54 | table | Read management actions from a Daisy data file. | file, emerge, harvest, volume, missing, tag, dim line, enable irrigation |
| 15.55 | wait | Wait until the specified condition is true. | condition |
| 15.56 | wait mm dd | Wait until a specific month and day in the year. | month, day |
| 15.57 | wait period | Waits the specified period. | hours, days, end time |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 15.58 | wait days | A ‘wait period’ model (see 15.57, page 95) build into Daisy. | n/a |
| 15.59 | wait hours | A ‘wait period’ model (see 15.57, page 95) build into Daisy. | n/a |
| 15.60 | warning | Write a warning to the user. | message |
| 15.61 | while | Perform all the specified actions in the sequence listed, but in the same timestep. | actions |
| 15.62 | with-column | Perform actions on a specific column. | column, actions |

**Related theme.** management.

**Combinations of the Experiments tab that use this topic.** Overhead, surface or drip irrigation; Overhead, surface or drip irrigation; Overhead, surface or drip irrigation; Irrigation and nitrogen together; Irrigation and nitrogen together; When to irrigate: the trigger and the depth of the sensor; How much and how fast: dose and duration; How much and how fast: dose and duration; Nitrogen fertiliser, irrigated and rainfed; The irrigation season and a limit on water; Slow or fast irrigation: a constant water flux; Slow or fast irrigation: a constant water flux; When the fertiliser is applied and how much; When the fertiliser is applied and how much; Warming with and without irrigation; Tillage and the time and form of the fertiliser; Tillage and the time and form of the fertiliser; Deficit irrigation: a water limit against the trigger; Tillage against fertiliser form; Tillage against fertiliser form.

## 35.16 Topic 16: adsorption

This component describes the adsorption of a chemical to the soil, which among other things affects how large a fraction can be transported with the water.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 16.1 | Freundlich | M = rho K C ref (C / C ref)m + Theta C. Formula from FOCUS page 93. See also [Freundlich and Heller, 1939, FOCUS, 2000] | m, K clay |
| 16.2 | Guo2020 | Linear sorption with K awi estimated from Equation 11. See also [Guo et al., 2020] Used by chemical PFOS Guo adsorption (see 23.25, page 142) , and chemical PFOA Guo adsorption (see 23.23, page 142) . 97 98 CHAPTER 16. | K clay, a, b |
| 16.3 | Langmuir | M = rho (my max C) / ((1/K) + C) + Theta C. | n/a |
| 16.4 | Python | Adsorption defined in user specified Python module. | module, extra |
| 16.5 | air | Henry’s law, fixed ratio between concentration in air and water. See also [?] | K wa |
| 16.6 | full | Full adsorption. Used for non-solutes, fully adsorped in the soil. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 16.7 | linear | M = rho K C + Theta C Used by chemical Atrazine adsorption (see 23.8, page 138) , chemical Bentazon adsorption (see 23.9, page 138) , chemical IPU adsorption (see 23.10, page 138) , chemical MCPP adsorption (see 23.12, page 138) , chemical Pendimethalin adsorption (see 23.13, page 138) , chemical Ioxynil adsorption (see 23.11, page 138) , chemical 2,4-D adsorption (see 23.7. | K clay |
| 16.8 | NH4 | A 'linear' model (see 16.7, page 100) build into Daisy. | n/a |
| 16.9 | NH4 Hansen | A 'linear' model (see 16.7, page 100) build into Daisy. | n/a |
| 16.10 | none | No adsorption. Used for solutes that are not adsorped to the soil. | n/a |
| 16.11 | vS S | Double langmuir description of NH4-N sorption to clay. See also [Van Schouwenburg and Schuffelen, 1963] | S edge, K edge |
| 16.12 | vS S Hansen | A 'vS S' model (see 16.11, page 101) build into Daisy. | n/a |
| 16.13 | vS S Styczen | A 'vS S' model (see 16.11, page 101) build into Daisy. | n/a |
| 16.14 | vS S old | Buggy implementation included as vS S before Daisy 6.47. Included here as reference. | n/a |

**Related theme.** chemicals and their sorption.

## 35.17 Topic 17: am

The 'am' component describes various kinds of fertilizer and other added matter such as crop residues. In particular, it describes how they decompose.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 17.1 | base | Common attributes for all added organic matter models. | name, creation, crop, part, om |
| 17.2 | initial | A 'base' model (see 17.1, page 103) build into Daisy. | layers, weight |
| 17.3 | organic | A 'base' model (see 17.1, page 103) build into Daisy. | weight, NH4 fraction, volatilization, NO3 fraction |
| 17.4 | root | A 'base' model (see 17.1, page 103) build into Daisy. | depth, dist, weight |
| 17.5 | state | A 'base' model (see 17.1, page 103) build into Daisy. | n/a |
| 17.6 | mineral | Mineral fertilizer. | weight, NH4 fraction, volatilization |

**Related theme.** nitrogen, carbon and organic matter.

## 35.18 Topic 18: average

Find the average of two numbers.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 18.1 | arithmetic | Arithmetic average ‘(a+b)/2’. | n/a |
| 18.2 | geometric | Geometric average ’sqrt(a*b)’. | n/a |
| 18.3 | harmonic | Harmonic average ‘2ab/(a+b)’. | n/a |

## 35.19 Topic 19: bioclimate

The ‘bioclimate’ component is responsible for distributing the water and energy provided by the weather component among the crops and soil for a given column.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 19.1 | base | Common parameters for all bioclimate parameterizations. | cloudiness, net radiation, pet, raddist, svat, deposition, NoOfIntervals, production stress |
| 19.2 | default | A ‘base’ model (see 19.1, page 109) build into Daisy. | n/a |
| 19.3 | FAO56 daily | A ‘default’ model (see 19.2, page 115) build into Daisy. | n/a |
| 19.4 | FAO56 hourly | A ‘default’ model (see 19.2, page 115) build into Daisy. | n/a |
| 19.5 | SSOC | A ‘default’ model (see 19.2, page 115) build into Daisy. | n/a |

**Related theme.** weather, radiation and evapotranspiration.

**Parameters that the workbench runs for this topic.** irrigation subsoil permanent, litter water storage, NoOfIntervals, canopy water storage, max svat iterations, max svat absolute difference, maxTdiff, maxEdiff, min sun angle.

## 35.20 Topic 20: biopore

A single class of biopores.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 20.1 | drain | Biopores that ends in the drain pipes. | pipe position, value, name |
| 20.2 | matrix | Biopores that ends in the matrix. | solute, name, value, debug, xplus, h bottom, max iterations, Theta |

**Related theme.** water and heat movement in the soil.

## 35.21 Topic 21: boolean

Generic representation of booleans. Used by IntegerCondClause @ condition (see 107.28, page 626) , and Stringer- CondClause @ condition (see 107.29, page 626) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 21.1 | false | Always false. | n/a |
| 21.2 | not | True if and only if the operand is not true. | operands |
| 21.3 | numbers | Base class for boolean expressions involving numbers. | operands |
| 21.4 | < | A ‘numbers’ model (see 21.3, page 121) build into Daisy. | n/a |
| 21.5 | <= | A ‘numbers’ model (see 21.3, page 121) build into Daisy. | n/a |
| 21.6 | > | A ‘numbers’ model (see 21.3, page 121) build into Daisy. | n/a |
| 21.7 | >= | A ‘numbers’ model (see 21.3, page 121) build into Daisy. | n/a |
| 21.8 | operands | Base class for boolean expressions involving multiple boolean operands. | operands |
| 21.9 | and | A ‘operands’ model (see 21.8, page 122) build into Daisy. | n/a |
| 21.10 | or | A ‘operands’ model (see 21.8, page 122) build into Daisy. | n/a |
| 21.11 | string-equal | True iff the supplied strings are identical. | values |
| 21.12 | true | Always true. | n/a |
| 21.13 | xor | True if and only if one operand is true, and one false. | operands |

## 35.22 Topic 22: bound

Specify one end of an interval boundary.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 22.1 | finite | Finite interval bound. | n/a |
| 22.2 | full | Maximum value for the interval boundary. | n/a |
| 22.3 | none | No boundary specified. | n/a |
| 22.4 | state | Bound used for checkpoints. | type |
| 22.5 | empty | A ‘state’ model (see 22.4, page 123) build into Daisy. | n/a |

## 35.23 Topic 23: chemical

This component should, for a specific chemical (typically a pesticide), provide a description of the properties of interest to Daisy.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 23.1 | base | Read chemical properties as normal Daisy parameters. | adsorption, value, molar mass, solubility, canopy dissipation, surface decompose, litter decompose, litter diffusion |
| 23.2 | FOCUS | A ‘base’ model (see 23.1, page 125) build into Daisy. | alpha, Value description |
| 23.3 | metabolite | A ‘FOCUS’ model (see 23.2, page 135) defined in ‘chemistry-base.dai’. | n/a |
| 23.4 | pesticide | A ‘FOCUS’ model (see 23.2, page 135) defined in ‘chemistry-base.dai’. | n/a |
| 23.5 | fungicide | A ‘pesticide’ model (see 23.4, page 137) defined in ‘chemistry-base.dai’. | n/a |
| 23.6 | herbicide | A ‘pesticide’ model (see 23.4, page 137) defined in ‘chemistry-base.dai’. | n/a |
| 23.7 | 2,4-D | A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’. | n/a |
| 23.8 | Atrazine | A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’. | n/a |
| 23.9 | Bentazon | A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’. | n/a |
| 23.10 | IPU | A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’. | n/a |
| 23.11 | Ioxynil | A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’. | n/a |
| 23.12 | MCPP | A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’. | n/a |
| 23.13 | Pendimethalin | A ‘herbicide’ model (see 23.6, page 137) defined in ‘chemistry.dai’. | n/a |
| 23.14 | insecticide | A ‘pesticide’ model (see 23.4, page 137) defined in ‘chemistry-base.dai’. | n/a |
| 23.15 | DDT | A ‘insecticide’ model (see 23.14, page 138) defined in ‘chemistry.dai’. | n/a |
| 23.16 | Heptachlor | A ‘insecticide’ model (see 23.14, page 138) defined in ‘chemistry.dai’. | n/a |
| 23.17 | default | A ‘base’ model (see 23.1, page 125) build into Daisy. | n/a |
| 23.18 | PFAS | A ‘default’ model (see 23.17, page 139) defined in ‘PFAS.dai’. | n/a |
| 23.19 | PFBA | A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’. | n/a |
| 23.20 | PFBS | A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’. | n/a |
| 23.21 | PFHxA | A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’. | n/a |
| 23.22 | PFOA | A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 23.23 | PFOA Guo | A ‘PFOA’ model (see 23.22, page 142) defined in ‘PFAS.dai’. | n/a |
| 23.24 | PFOS | A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’. | n/a |
| 23.25 | PFOS Guo | A ‘PFOS’ model (see 23.24, page 142) defined in ‘PFAS.dai’. | n/a |
| 23.26 | PFPeA | A ‘PFAS’ model (see 23.18, page 141) defined in ‘PFAS.dai’. | n/a |
| 23.27 | Tracer | A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’. | n/a |
| 23.28 | classic | A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’. | n/a |
| 23.29 | colloid | A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’. | n/a |
| 23.30 | solid | A ‘default’ model (see 23.17, page 139) defined in ‘chemistry-base.dai’. | n/a |
| 23.31 | nutrient | A ‘base’ model (see 23.1, page 125) build into Daisy. | n/a |
| 23.32 | C | A ‘nutrient’ model (see 23.31, page 143) build into Daisy. | n/a |
| 23.33 | DOC | A ‘C’ model (see 23.32, page 143) build into Daisy. | n/a |
| 23.34 | N | A ‘nutrient’ model (see 23.31, page 143) build into Daisy. | n/a |
| 23.35 | DON | A ‘N’ model (see 23.34, page 143) build into Daisy. | n/a |
| 23.36 | MIN | A ‘N’ model (see 23.34, page 143) build into Daisy. | n/a |
| 23.37 | NH4 | A ‘MIN’ model (see 23.36, page 143) build into Daisy. | n/a |
| 23.38 | NO3 | A ‘MIN’ model (see 23.36, page 143) build into Daisy. | n/a |

**Related theme.** chemicals and their sorption.

## 35.24 Topic 24: chemistry

Pesticides and other chemicals.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 24.1 | default | Handle chemicals and reactions. | reaction, trace |
| 24.2 | DOM | A ‘default’ model (see 24.1, page 145) build into Daisy. | n/a |
| 24.3 | N | A ‘default’ model (see 24.1, page 145) build into Daisy. | n/a |
| 24.4 | PFAS | A ‘default’ model (see 24.1, page 145) defined in ‘PFAS.dai’. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 24.5 | Tracer | A ‘default’ model (see 24.1, page 145) defined in ‘chemistry.dai’. | n/a |
| 24.6 | colloid | A ‘default’ model (see 24.1, page 145) defined in ‘chemistry.dai’. | n/a |
| 24.7 | pesticides | A ‘default’ model (see 24.1, page 145) defined in ‘chemistry.dai’. | n/a |
| 24.8 | multi | Handle multile chemistries. | combine, ignore, trace |
| 24.9 | none | A ‘multi’ model (see 24.8, page 146) build into Daisy. | n/a |
| 24.10 | nutrient | A ‘multi’ model (see 24.8, page 146) build into Daisy. | n/a |

**Related theme.** chemicals and their sorption.

**Parameters that the workbench runs for this topic.** max sink total, max sink solute, max sink secondary, min sink total.

## 35.25 Topic 25: cloudiness

Calculate cloudiness from meterological data.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 25.1 | clear sky | Calculate based on clear sky radiation: a * Si / Si0 + 1 minus a where Si is global radiation and Si0 is clear sky radiation. | cloudiness, a, Value description |
| 25.2 | ASCE | A ‘clear sky’ model (see 25.1, page 149) build into Daisy. | Kt |
| 25.3 | Taastrup | A ‘ASCE’ model (see 25.2, page 150) build into Daisy. | n/a |
| 25.4 | FAO56 | A ‘clear sky’ model (see 25.1, page 149) build into Daisy. | n/a |
| 25.5 | Kjaersgaard | A ‘FAO56’ model (see 25.4, page 151) build into Daisy. | n/a |
| 25.6 | const | The cloudiness never changes. | cloudiness |
| 25.7 | weather | Use value from weather file. | cloudiness |

**Related theme.** weather, radiation and evapotranspiration.

**Combinations of the Experiments tab that use this topic.** A drier, wetter or warmer year; Warming with and without irrigation; Evapotranspiration model and radiation; A dry summer, a dry spring or a wet winter.

## 35.26 Topic 26: column

A ‘column’ is an one-dimensional vertical description of the soil/crop/atmosphere system. The column component contains most of the other processes in Daisy as submodels.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 26.1 | default | Hansen et.al. 1990. with generic movement in soil. | Irrigation, SoilHeat, scope, Surface, Movement, Vegetation, Bioclimate, Drain |

## 35.27 Topic 27: condedge

Find the hydraulic conductivity between two cells.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 27.1 | arithmetic | Use the arithmetic average of the conductivity in the two cells. | n/a |
| 27.2 | geometric | Geometric average 'sqrt(a*b)'. | n/a |
| 27.3 | harmonic | Use harmonic average of the conductivity of the two cells. This corresponds to using the average hydraulic resistence. | n/a |
| 27.4 | pressure | Pressure dependent average of the two cells. Use harmonic average of the conductivity of unsaturated cells. | allow sideways, h lim |

**Related theme.** water and heat movement in the soil.

## 35.28 Topic 28: condition

A 'condition' component tests the state of the simulation, like whether the water pressure in a specific depth is above a given threshold. Logic conditions like 'and' and 'or' can be used for testing whether multiple conditions are fulfilled simultane- ously. Used by select component when (see 80, page 429) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 28.1 | BBCH after | True after specified BBCH. | crop, file, start, bbch |
| 28.2 | TSum above | Test if the temperature sum is above the specified value The temperature sum is the sum of the daily average air temperature since last reset. | TSum limit, reset mday, reset month |
| 28.3 | and | True iff all the listed conditions are true. The conditions are tested in the sequence listed, until a false is found, or the end of the list is reached. | operands |
| 28.4 | trafficable | A 'and' model (see 28.3, page 160) defined in 'tillage.dai'. | n/a |
| 28.5 | check | Test if a boolean expression is true. | n/a |
| 28.6 | crop dm over | True iff the crop has reached the specified amount of dry matter. | crop, weight, height |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 28.7 | crop dm sorg over | True iff the storage organ has reached the specified amount of dry matter. | crop, weight |
| 28.8 | crop ds after | True iff the crop has reached development stage ‘ds’. | crop, ds |
| 28.9 | crop stage after | True iff the crop has reached development stage ‘stage’. | crop, stage |
| 28.10 | daily | True at the end of each day. | n/a |
| 28.11 | daily air temperature above | Test if the daily air is warmer than the specified temperature. | temperature |
| 28.12 | daily precipitation above | Test if the daily precipitation is warmer than the specified value. | precipitation |
| 28.13 | every | Matches simulation with fixed time intervals. | seconds, minutes, hours, days, microseconds, next |
| 28.14 | extern | Test if a boolean expression is true,using extern log. | scope |
| 28.15 | false | Always false. | n/a |
| 28.16 | finished | True iff the simulation has finished. | n/a |
| 28.17 | hour | True, at the specified hour. | at |
| 28.18 | hourly | True at the end of each hour. | n/a |
| 28.19 | if | If the first condition is true, return the value of the second condition, else return the value of the third condition. | then, else |
| 28.20 | mday | True, at the specified day in the month. | at |
| 28.21 | microsecond | True, at the specified microsecond. | at |
| 28.22 | minute | True, at the specified minute. | at |
| 28.23 | minutely | True at the end of each minute. | n/a |
| 28.24 | mm dd base | Conditions based on month and day. | month, day, minute |
| 28.25 | after mm dd | A ‘mm dd base’ model (see 28.24, page 164) build into Daisy. | n/a |
| 28.26 | before mm dd | A ‘mm dd base’ model (see 28.24, page 164) build into Daisy. | n/a |
| 28.27 | mm dd | A ‘mm dd base’ model (see 28.24, page 164) build into Daisy. | n/a |
| 28.28 | month | True, at the specified month. | at |
| 28.29 | monthly | True at the end of each month. | n/a |
| 28.30 | not | True iff the operand is not true. | n/a |
| 28.31 | or | True iff any of the listed conditions are true. The conditions are tested in the sequence listed, until a true is found, or the end of the list is reached. | operands |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 28.32 | periodic | True if move than a specified walltime has passed since last time it was true. | n/a |
| 28.33 | running | True iff the simulation is still running. | n/a |
| 28.34 | second | True, at the specified second. | at |
| 28.35 | secondly | True at the end of each second. | n/a |
| 28.36 | soil inorganic N above | Test if the soil contains more mineral nitrogen than the specified amount. | amount, from, to |
| 28.37 | soil temperature above | Test if the soil is warmer than the specified temperature. | height, temperature |
| 28.38 | soil water content above | Test if the soil contains more water than the specified amount. | from, to |
| 28.39 | soil water pressure above | Test if the soil is wetter than the specified pressure potential. | height |
| 28.40 | time | Conditions based on a specific time. | time |
| 28.41 | after | A ‘time’ model (see 28.40, page 167) build into Daisy. | n/a |
| 28.42 | at | A ‘time’ model (see 28.40, page 167) build into Daisy. | n/a |
| 28.43 | before | A ‘time’ model (see 28.40, page 167) build into Daisy. | n/a |
| 28.44 | timestep | Add a timestep to a condition. It is true whenever ‘operand’ is true, but will let Daisy know what ‘timestep’ it represents. | timestep |
| 28.45 | true | Always true. | n/a |
| 28.46 | walltime | True if move than a specified walltime has passed since last time it was true. | seconds, minutes, hours, days, microseconds |
| 28.47 | weekly | True at the end of each week. | n/a |
| 28.48 | with | Test condition for specific column. | where, condition |
| 28.49 | yday | True, at the specified julian day. | at |
| 28.50 | year | True, at the specified year. | at |
| 28.51 | yearly | True at the end of each year. | n/a |

**Related theme.** management.

**Combinations of the Experiments tab that use this topic.** When to irrigate: the trigger and the depth of the sensor; Sensor depth against rooting depth; Deficit irrigation: a water limit against the trigger.

## 35.29 Topic 29: crop

The ‘crop’ component simulates a specific crop on the field, typically averaged over one square meter, not individual plants. Of particular interest is water and nitrogen uptake at different depths, and the vertical leaf area distribution, which are used for competition with other crops.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 29.1 | default | Standard Daisy crop model. Hansen, 1999. | Harvest, Root, Canopy, last time, sow time, emerge time, flowering time, ripe time |
| 29.2 | Beetroot | A ‘default’ model (see 29.1, page 171) defined in ‘beetroot.dai’. | n/a |
| 29.3 | Broccoli | A ‘default’ model (see 29.1, page 171) defined in ‘broccoli.dai’. | n/a |
| 29.4 | Broccoli minus transplanted | A ‘Broccoli’ model (see 29.3, page 173) defined in ‘broccoli.dai’. | n/a |
| 29.5 | Brussels sprouts | A ‘default’ model (see 29.1, page 171) defined in ‘brusselssprouts.dai’. | n/a |
| 29.6 | Brussels sprouts minus transplanted | A ‘Brussels sprouts’ model (see 29.5, page 173) defined in ‘brusselssprouts.dai’. | n/a |
| 29.7 | Celeriac | A ‘default’ model (see 29.1, page 171) defined in ‘celeriac.dai’. | n/a |
| 29.8 | Celeriac minus transplanted | A ‘Celeriac’ model (see 29.7, page 174) defined in ‘celeriac.dai’. | n/a |
| 29.9 | Fodder Beet | A ‘default’ model (see 29.1, page 171) defined in ‘fodderbeet.dai’. | n/a |
| 29.10 | Grass to grain | A ‘default’ model (see 29.1, page 171) defined in ‘grass.dai’. | n/a |
| 29.11 | Grass | A ‘Grass to grain’ model (see 29.10, page 174) defined in ‘grass.dai’. | n/a |
| 29.12 | Maize | A ‘default’ model (see 29.1, page 171) defined in ‘maize.dai’. | n/a |
| 29.13 | Ikuwala Maize | A ‘Maize’ model (see 29.12, page 174) defined in ‘maize.dai’. | n/a |
| 29.14 | Silage Maize | A ‘Maize’ model (see 29.12, page 174) defined in ‘maize.dai’. | n/a |
| 29.15 | Onion | A ‘default’ model (see 29.1, page 171) defined in ‘onion.dai’. | n/a |
| 29.16 | Onion minus planting of sets | A ‘Onion’ model (see 29.15, page 174) defined in ‘onion plantingofsets.dai’. | n/a |
| 29.17 | Pea | A ‘default’ model (see 29.1, page 171) defined in ‘pea.dai’. | n/a |
| 29.18 | Green Pea | A ‘Pea’ model (see 29.17, page 174) defined in ‘pea.dai’. | n/a |
| 29.19 | Pioneer Maize | A ‘default’ model (see 29.1, page 171) defined in ‘maize.dai’. | n/a |
| 29.20 | Potato; Koege | A ‘default’ model (see 29.1, page 171) defined in ‘potato.dai’. | n/a |
| 29.21 | Potato; SCRI | A ‘default’ model (see 29.1, page 171) defined in ‘potato.dai’. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 29.22 | Potato; FertOrgaNic | A ‘Potato; SCRI’ model (see 29.21, page 175) defined in ‘potato.dai’. | n/a |
| 29.23 | Potato; Agria | A ‘Potato; FertOrgaNic’ model (see 29.22, page 175) defined in ‘potato.dai’. | n/a |
| 29.24 | Potato; Folva | A ‘Potato; FertOrgaNic’ model (see 29.22, page 175) defined in ‘potato.dai’. | n/a |
| 29.25 | Potato | A ‘Potato; Folva’ model (see 29.24, page 175) defined in ‘potato.dai’. | n/a |
| 29.26 | Potato; Triada | A ‘Potato; FertOrgaNic’ model (see 29.22, page 175) defined in ‘potato.dai’. | n/a |
| 29.27 | Potato; SCRI minus AArhus | A ‘default’ model (see 29.1, page 171) defined in ‘earlypotato.dai’. | n/a |
| 29.28 | Early potato | A ‘Potato; SCRI minus AArhus’ model (see 29.27, page 176) defined in ‘earlypotato.dai’. | n/a |
| 29.29 | Rye | A ‘default’ model (see 29.1, page 171) defined in ‘rye.dai’. | n/a |
| 29.30 | Ryegrass | A ‘default’ model (see 29.1, page 171) defined in ‘ryegrass.dai’. | n/a |
| 29.31 | Spring Barley | A ‘default’ model (see 29.1, page 171) defined in ‘sbarley.dai’. | n/a |
| 29.32 | Spring Barley; Foulum | A ‘Spring Barley’ model (see 29.31, page 176) defined in ‘sbarley.dai’. | n/a |
| 29.33 | Spring Rape | A ‘default’ model (see 29.1, page 171) defined in ‘srape.dai’. | n/a |
| 29.34 | Spring Wheat | A ‘default’ model (see 29.1, page 171) defined in ‘swheat.dai’. | n/a |
| 29.35 | Sugar Beet | A ‘default’ model (see 29.1, page 171) defined in ‘sugarbeet.dai’. | n/a |
| 29.36 | Wclover | A ‘default’ model (see 29.1, page 171) defined in ‘wclover.dai’. | n/a |
| 29.37 | White cabbage | A ‘default’ model (see 29.1, page 171) defined in ‘whitecabbage.dai’. | n/a |
| 29.38 | White cabbage minus transplanted | A ‘White cabbage’ model (see 29.37, page 177) defined in ‘whitecabbage.dai’. | n/a |
| 29.39 | Early white cabbage minus transplanted | A ‘White cabbage minus transplanted’ model (see 29.38, page 177) defined in ‘earlywhitecabbage.dai’. | n/a |
| 29.40 | Winter Barley | A ‘default’ model (see 29.1, page 171) defined in ‘wbarley.dai’. | n/a |
| 29.41 | Winter Barley; Foulum | A ‘Winter Barley’ model (see 29.40, page 177) defined in ‘wbarley.dai’. | n/a |
| 29.42 | Winter Barley; Koge | A ‘default’ model (see 29.1, page 171) defined in ‘wbarley.dai’. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 29.43 | Winter Rape | A ‘default’ model (see 29.1, page 171) defined in ‘wrape.dai’. | n/a |
| 29.44 | Winter Wheat | A ‘default’ model (see 29.1, page 171) defined in ‘wwheat.dai’. | n/a |
| 29.45 | Winter Wheat JG | A ‘Winter Wheat’ model (see 29.44, page 177) defined in ‘wwheat.dai’. | n/a |
| 29.46 | Winter Wheat; Foulum | A ‘Winter Wheat’ model (see 29.44, page 177) defined in ‘wwheat.dai’. | n/a |
| 29.47 | simple | Forced growth crop model. | Root, Canopy, LAIvsDay, T sum, day, spring, root am, N b |

**Related theme.** the crop: growth, roots and water use.

**Combinations of the Experiments tab that use this topic.** Sensor depth against rooting depth; Rooting depth of the crop.

## 35.30 Topic 30: cstage

Crop phenological stage. The ‘default’ crop model uses a continius development stage [-1:2] in order to track the phenological age of a crop. The ‘cstage’ library allows translation of the internal model to other phenological models.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 30.1 | table | A table mapping Daisy DS to other model. | table, value, message |
| 30.2 | BBCH | A ‘table’ model (see 30.1, page 181) build into Daisy. | n/a |
| 30.3 | Daisy | A ‘table’ model (see 30.1, page 181) build into Daisy. | n/a |

**Related theme.** the crop: growth, roots and water use.

## 35.31 Topic 31: daisy/soil/soil.h

pH of soil.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 31.1 | neutral | A soil with a constant pH of 7. | n/a |
| 31.2 | year | A soil with a constant pH of 7. | pH depth |

**Related theme.** water and heat movement in the soil.

## 35.32 Topic 32: denprod

Find products (gasses) produced by denitrification.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 32.1 | Parton1996 | Find N2O from denitrification based on WFPS, NO3, and CO2. | N2O |

**Related theme.** nitrogen, carbon and organic matter.

## 35.33 Topic 33: deposition

Deposition of inorganic material from atmosphere.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 33.1 | const | Constant deposition in time interval. | before, wet, name, value, dry |
| 33.2 | weather | Rely solely on weather model for deposition. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.34 Topic 34: depth

Depth below soil surface.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 34.1 | PLF | Linear interpolation of depth. | table, time, value |
| 34.2 | const | Constant depth. | value |
| 34.3 | deep | A ‘const’ model (see 34.2, page 189) build into Daisy. | n/a |
| 34.4 | extern | Look up depth in an scope. | value, initial value |
| 34.5 | file | Linear interpolation of depth read from file. | file |
| 34.6 | season | Linear interpolation of depth within a year. | season |
| 34.7 | table | Linear interpolation of depth read from file. | file, offset, missing, tag, dim line |

## 35.35 Topic 35: difrad

The ‘difrad’ component should calculate the diffuse radiation from meterological data.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 35.1 | BRL | Diffuse radiation calculation based on the model of Ridley et al., 2010. Last predictor (persistence) is not taken into account. | n/a |
| 35.2 | DPF | Diffuse radiation calculated using the model of De Pury and Farquhar, 1997. See also [de Pury and Farquhar, 1997] | a, Value description, fa |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 35.3 | weather | Diffuse radiation using weather data. | n/a |

**Related theme.** weather, radiation and evapotranspiration.

## 35.36 Topic 36: domsorp

Sorption and desorption of DOM to SOM.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 36.1 | default | Transformation between two soil chemicals. | transform |

**Related theme.** nitrogen, carbon and organic matter.

## 35.37 Topic 37: drain

Lateral transport of water.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 37.1 | lateral | Pipe drainage. See also [Hooghoudt, 1940] | L, height, pipe position, eq depth, pipe outlet |
| 37.2 | none | No lateral transport. | n/a |

**Related theme.** water and heat movement in the soil.
**Parameters that the workbench runs for this topic.** L, rad.

## 35.38 Topic 38: draineqd

Find the equilibrium drain depth for the Hooghoudt drainage model.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 38.1 | MolenWesseling | Equivalent depth calculated with model by van der Molen and Wesseling. See also [der Molen and Wesseling, 1991] Used by drain lateral eq depth (see 37.1, page 197) . 38.2 Moody Equivalent depth calculated with model by Moody. | n/a |
| 38.2 | Moody | Equivalent depth calculated with model by Moody. See also [Moody, 1966] 38.3 Wesseling Equivalent depth calculated with model by Wesseling. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 38.3 | Wesseling | Equivalent depth calculated with model by Wesseling. See also [Wesseling, 1973] 38.4 none No modifications. | n/a |
| 38.4 | none | No modifications. See also [Hooghoudt, 1940] 199 200 CHAPTER 38. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.39 Topic 39: equilibrium

Find equilibrium between two soil chemicals.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 39.1 | Langmuir | A = (my max B) / (K + B) | n/a |
| 39.2 | goal A | Attempt to maintain A at at fixed level. | goal A, A solute, B solute |
| 39.3 | linear | A=KB. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.40 Topic 40: exchange

A named value to exchange with external models.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 40.1 | name | Exchange a string value. | value |
| 40.2 | number | Exchange a numeric value. | value, dimension |

## 35.41 Topic 41: format

Text formatting component.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 41.1 | LaTeX | Format text as LaTeX. | n/a |

## 35.42 Topic 42: function

Pure function of one parameter.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 42.1 | P96CO2 | A function [kg C/ha/d → ] build into Daisy. | n/a |
| 42.2 | P96NO3 | A function [ug N/g → ] build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 42.3 | P96WFPS | A function [ → ] build into Daisy. | n/a |
| 42.4 | Python | A function [? → ?] build into Daisy. | name, module |
| 42.5 | T scale | A base function [dg C → ] build into Daisy. | n/a |
| 42.6 | T min | A ‘T scale’ function [dg C → ] (see 42.5, page 209) build into Daisy. | n/a |
| 42.7 | T min 15 | A ‘T min’ function [dg C → ] (see 42.6, page 209) build into Daisy. | n/a |
| 42.8 | T nit | A ‘T scale’ function [dg C → ] (see 42.5, page 209) build into Daisy. | n/a |
| 42.9 | const | A function [? → ?] build into Daisy. | value |
| 42.10 | plf | A function [? → ?] build into Daisy. | n/a |

## 35.43 Topic 43: ghf

Ground heat flux.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 43.1 | FAO56 | Equation 45 and 46. Ground heat flux is 10 % of Rn during day, and 50 % of Rn during night. | n/a |
| 43.2 | const | The ground heat flux never changes. | G |
| 43.3 | none | A ‘const’ model (see 43.2, page 213) build into Daisy. | n/a |
| 43.4 | old | Based soil numeric cell 0 and 1. The model used in Daisy 6.08 and earlier, should not be used. | n/a |
| 43.5 | surface | Based on flux through soil surface. Calculated from previous timestep. | n/a |
| 43.6 | weather | Use value from weather file. | n/a |

**Related theme.** weather, radiation and evapotranspiration.

## 35.44 Topic 44: gnuplot

Plot a graph with gnuplot.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 44.1 | common | Common parameters. | extra, where, title, device, canvas, size, y |
| 44.2 | profile | A ‘common’ model (see 44.1, page 215) build into Daisy. | column |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 44.3 | soil | A ‘common’ model (see 44.1, page 215) build into Daisy. | file, top, min, dimension, bottom, type, left, right |
| 44.4 | time | A ‘common’ model (see 44.1, page 215) build into Daisy. | source, begin, ymin, y2min |
| 44.5 | vector | A ‘common’ model (see 44.1, page 215) build into Daisy. | file, top, dimension, begin, bottom, left, right, missing |
| 44.6 | xy | A ‘common’ model (see 44.1, page 215) build into Daisy. | source, ymin, y2min, xmin, x2min |
| 44.7 | multi | Generate multiple graphs for the gnuplot command file. | before, graph |

## 35.45 Topic 45: groundwater

The ‘groundwater’ component is responsible for specifying the groundwater table at each timestep.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 45.1 | common | All groundwater models can log height. | height |
| 45.2 | aquitard | A ‘common’ model (see 45.1, page 225) build into Daisy. | pressure table |
| 45.3 | deep | A ‘common’ model (see 45.1, page 225) build into Daisy. | n/a |
| 45.4 | extern | A ‘common’ model (see 45.1, page 225) build into Daisy. | table, initial table |
| 45.5 | file | A ‘common’ model (see 45.1, page 225) build into Daisy. | file, offset |
| 45.6 | fixed | A ‘common’ model (see 45.1, page 225) build into Daisy. | table |
| 45.7 | flux | A ‘common’ model (see 45.1, page 225) build into Daisy. | n/a |
| 45.8 | lysimeter | A ‘common’ model (see 45.1, page 225) build into Daisy. | n/a |
| 45.9 | source | A ‘common’ model (see 45.1, page 225) build into Daisy. | source, offset |
| 45.10 | static | A ‘common’ model (see 45.1, page 225) build into Daisy. | table, obsolete |

**Related theme.** water and heat movement in the soil.

**Parameters that the workbench runs for this topic.** K aquitard, Z aquitard, table.

## 35.46 Topic 46: heatrect

Heat transport in rectangular grid.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 46.1 | Mollerup | Finite volume solution to heat transfer by Mikkel Mollerup. | debug |
| 46.2 | linear | Linear temperature interpolation between top and bottom. | n/a |
| 46.3 | none | No heat transport. | n/a |
| 46.4 | top | Soil temperature is equal to upper boundary. This can be used for a poorly isolated soil column. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.47 Topic 47: horizon

A ‘horizon’ is a soil type with specific physical properties. It is the responsibility of the ‘horizon’ component to specify these properties. Used by SoilLayer @ horizon (see 107.19, page 612) , and SoilRegion @ horizon (see 107.20, page 613) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 47.1 | BSI3 | A horizon using BSI3 texture classification. | clay, silt, humus, normalize |
| 47.2 | DIN3 | A ‘BSI3’ model (see 47.1, page 233) build into Daisy. | n/a |
| 47.3 | MIT3 | A ‘BSI3’ model (see 47.1, page 233) build into Daisy. | n/a |
| 47.4 | BSI7 | A horizon using BSI7 texture classification. | clay, fine silt, medium silt, coarse silt, humus, normalize |
| 47.5 | MIT7 | A ‘BSI7’ model (see 47.4, page 235) build into Daisy. | n/a |
| 47.6 | DIN5 | A horizon using DIN5 texture classification. | clay, silt, humus, normalize |
| 47.7 | GEUS7 | A horizon using GEUS7 texture classification. | clay, fine silt, medium silt, coarse silt, humus, normalize |
| 47.8 | ISSS3 | A horizon using ISSS3 texture classification. | clay, silt, humus, normalize |
| 47.9 | ISSS4 | A horizon using ISSS4 texture classification. | clay, silt, humus, normalize |
| 47.10 | USDA3 | A horizon using USDA3 texture classification. | clay, silt, humus, normalize |
| 47.11 | FAO3 | A ‘USDA3’ model (see 47.10, page 242) build into Daisy. | n/a |
| 47.12 | USDA7 | A horizon using USDA7 texture classification. | clay, silt, humus, normalize |
| 47.13 | FAO7 | A ‘USDA7’ model (see 47.12, page 243) build into Daisy. | n/a |
| 47.14 | USPRA3 | A horizon using USPRA3 texture classification. | clay, silt, humus, normalize |
| 47.15 | USPRA4 | A horizon using USPRA4 texture classification. | clay, silt, humus, normalize |
| 47.16 | numeric | A horizon using explicit texture classification. | humus, fractions, limits, normalize |

**Related theme.** water and heat movement in the soil.

## 35.48 Topic 48: hydraulic

This component is responsible for specifying the soils hydraulic properties. Used by horizon component hydraulic (see 47, page 231) , and Soil @ Hydraulic (see 107.21, page 613) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 48.1 | B BaC | Brooks and Corey retention curve model with Burdine theory. | Theta sat, lambda, K sat, h, Theta res, h b, p |
| 48.2 | B BaC Bimodal | Brooks and Corey retention curve model with Burdine theory. | Theta sat, lambda, K sat, Theta res, h b, Theta b, K b |
| 48.3 | B C | Campbell retention curve model with Burdine theory. See also [Campbell, 1974, Burdine, 1953] | b, Theta sat, K sat, h, h b, p |
| 48.4 | B C inverse | Campbell retention curve model with Burdine theory. | Theta sat, K sat, h, Theta fc, Theta wp |
| 48.5 | B vG | van Genuchten retention curve model with Burdine theory. | alpha, Theta sat, K sat, h, Theta res, n |
| 48.6 | Cosby et al | Modified Campbell retention curve model with Burdine theory. Parameters estimated from soil texture as specified by Cosby et at. | n/a |
| 48.7 | MACRO | van Genuchten retention curve model with Mualem theory. The near saturated retention and hydraulic properties have been adjusted to take macropores into account. | alpha, Theta sat, K sat, Theta res, h b, Theta b, K b, n |
| 48.8 | M BaC | Brooks and Corey retention curve model with Mualem theory. | Theta sat, lambda, K sat, h, Theta res, h b, p |
| 48.9 | M BaC Bimodal | Brooks and Corey retention curve model with Mualem theory. | Theta sat, lambda, K sat, Theta res, h b, Theta b, K b |
| 48.10 | M BivG | Bimodal van Genuchten retention curve model with Mualem theory (Durner, 1994). | Theta sat, K sat, h, Theta res |
| 48.11 | M C | Campbell retention curve model with Mualem theory. See also [Campbell, 1974, Mualem, 1976] | b, Theta sat, K sat, h, h b, p |
| 48.12 | M vG | van Genuchten retention curve model with Mualem theory. | alpha, Theta sat, K sat, h, Theta res, n |
| 48.13 | M vG compact | van Genuchten retention curve model with Mualem theory and compaction. | Theta sat, Theta res, ref alpha, ref n, mod alpha, mod n |
| 48.14 | M vGip | Modified van Genuchten retention curve model with Mualem theory. | alpha, Theta sat, K sat, h, Theta res, n, he |
| 48.15 | M vGp | van Genuchten retention curve model with Mualem theory. The near saturated retention and hydraulic properties have been adjusted to take macropores into account. | alpha, Theta sat, K sat, Theta res, n, h m |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 48.16 | hypres | van Genuchten retention curve model with Mualem theory. The near saturated retention and hydraulic properties have been adjusted to take macropores into account. | K sat, h, Theta res |
| 48.17 | hyprop | HYPROP-FIT parameters for bimodal van Genuchten retention curve. It includes adsorptive water retention and scaling (code 1211). | alpha, m, ha, a, Value description, Theta sat, K sat, h |
| 48.18 | linear | Hysteresis using linear transformation. | a, b, debug, wet, dry, is wetting, c |
| 48.19 | mod C | Modified Campbell retention curve model with Burdine theory. | b, Theta sat, K sat, h, h b |
| 48.20 | old2 | Reads a file of lines in the format. | file, M intervals, inc |
| 48.21 | table | Reads a ddf file with column pF Theta Cw2 K. | file, Theta sat, missing, tag, dim line, K sat, Theta res, M intervals |
| 48.22 | wepp | van Genuchten retention curve model with Mualem theory. The near saturated retention and hydraulic properties have been adjusted to take macropores into account. | Theta res, time consolidation, average depth, p t, delta pc, crust thickness, day count, R c |
| 48.23 | Styczen | A ‘wepp’ model (see 48.22, page 267) build into Daisy. | n/a |
| 48.24 | yolo | Yolo soil. Haverkamp et.al., 1977. | M intervals |

**Related theme.** water and heat movement in the soil.

**Combinations of the Experiments tab that use this topic.** Which soil needs the most water.

## 35.49 Topic 49: integer

Generic representation of integers.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 49.1 | * | Use the product of its operands. | operands |
| 49.2 | + | Use the sum of its operands. | operands |
| 49.3 | minus | Negate integer or subtract integers. With one operand, negates it. With more than one operand, subtracts all but the first from the first. | operands |
| 49.4 | cond | Return the value of the first clause whose condition is true. | clauses |
| 49.5 | const | Always give the specified value. | value |
| 49.6 | div | Divide the first operand by the rest. | operands |
| 49.7 | max | Use the largest value of its operands. | operands |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 49.8 | min | Use the smallest value of its operands. | operands |
| 49.9 | mod | Modulo the first operand by the rest. | operands |
| 49.10 | sqr | Take the square of its argument. | n/a |

## 35.50 Topic 50: listsum

Condense a list of numbers to a single number.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 50.1 | arithmetic | Arithmetic average of all members of a list. | n/a |
| 50.2 | count | Count all members of a list. | n/a |
| 50.3 | max | The maximum of all members of a list. | n/a |
| 50.4 | min | The minimum of all members of a list. | n/a |
| 50.5 | sum | The sum of all members of a list. | n/a |

## 35.51 Topic 51: litter

Litter, surface residuals, or mulch below canopy.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 51.1 | none | The effect of surface residuals is ignored by the model. | n/a |
| 51.2 | permanent | A permanent litter layer cover the ground, as for example in a forest. | albedo, interception capacity |
| 51.3 | residue | A dynamic litter layer based on applied fertilizer and crop residuals. | albedo, water capacity, extinction coefficent, mass |
| 51.4 | Maize | A ‘residue’ model (see 51.3, page 275) build into Daisy. | n/a |
| 51.5 | Millet | A ‘residue’ model (see 51.3, page 275) build into Daisy. | n/a |
| 51.6 | mulch | A ‘residue’ model (see 51.3, page 275) build into Daisy. | alpha, retention, DOC name, DON name, density, Theta res, particle density, decompose height |

**Related theme.** nitrogen, carbon and organic matter.

## 35.52 Topic 52: log

Running a simulation is uninteresting, unless you can get access to the results in one way or another. The purpose of the ‘log’ component is to provide this access. Most ‘log’ models does this by writing a summary of the state to a log file.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 52.1 | checkpoint | Create a checkpoint of the entire simulation state, suitable for later hot start. | where, when |
| 52.2 | harvest | Create a log of all harvests. | where, print tags, print dimension, WStress, NStress |
| 52.3 | select | Select variables to log. | volume, summary, entries, from, SOLETE, to, OBSOLETE, when |
| 52.4 | extern | A ‘select’ model (see 52.3, page 283) build into Daisy. | numbers, value, name, where |
| 52.5 | table | A ‘select’ model (see 52.3, page 283) build into Daisy. | where, missing value, print tags, print dimension, flush |
| 52.6 | column | A ‘table’ model (see 52.5, page 285) defined in ‘log-std.dai’. | column |
| 52.7 | Field nitrogen | A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’. | unit, Deposition, Matrix-Leaching, Biopore-Leaching, Soil-Drain, Biopore-Drain, Surface-Loss, Min-Surface |
| 52.8 | Field water | A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’. | Precipitation, Irrigation, Potential evapotranspiration, Actual evapotranspiration, Matrix percolation, Biopore percolation |
| 52.9 | Soil nitrogen | A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’. | unit, NO3-Matrix-In, NO3-Biopore-In, NO3-Leak-Biopore, NO3-to-Biopores, NO3-from-Biopores, NO3-Soil-Drain, NO3-Indirect-Drain |
| 52.10 | Soil water | A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’. | Matrix infiltration, Matrix percolation, Biopore infiltration, Biopore percolation, Subsoil irrigation, Root extraction, Freezing, Tillage |
| 52.11 | biopore | A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’. | biopore |
| 52.12 | chemical | A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’. | n/a |
| 52.13 | Field chemical | A ‘chemical’ model (see 52.12, page 295) defined in ‘log-std.dai’. | unit, Spray, Deposit, Harvest, Dissipate, Litter Decompose, Litter Transform, Surface Decompose |
| 52.14 | Soil chemical | A ‘chemical’ model (see 52.12, page 295) defined in ‘log-std.dai’. | unit, In-Biopores, Leak-Biopores, Tillage, Drain-Biopores, Drain-Biopores-Indirect, Uptake, Decompose |
| 52.15 | crop | A ‘column’ model (see 52.6, page 286) defined in ‘log-std.dai’. | crop |
| 52.16 | Crop | A ‘crop’ model (see 52.15, page 298) defined in ‘log-std.dai’. | Height, Depth, WStem, WSOrg, WRoot, NStem, NSOrg, NRoot |

## 35.53 Topic 53: macro

Preferention flow in soil macro pores.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 53.1 | default | The area between ‘height start’ and ‘height end’ contains macropores, which are initiated when the water potential reach ‘pressure initiate’, and then immediately emptied down to ‘pressure end’. | distribution, height start, pressure initiate, S p |
| 53.2 | none | No macropores. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.54 Topic 54: mactrans

Macropore transportation of solutes.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 54.1 | default | Solute follows water. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.55 Topic 55: max exfiltration

Soil limited maximum exfiltration.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 55.1 | Hansen | Darcy formulated for water content with mimimum water at surface. | n/a |
| 55.2 | Theta | Darcy formulated for water content with mimimum water at surface. | h crit |
| 55.3 | gradient | Darcy formulated for water content with mimimum water at surface. | n/a |
| 55.4 | steady-state | Steady state flow from top cell to surface. | h crit |

**Related theme.** water and heat movement in the soil.

## 35.56 Topic 56: movement

This component handles the movement in the soil.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 56.1 | solute | Shared paramaters for handling solutes. | matrix solute |
| 56.2 | rectangle | A ‘solute’ model (see 56.1, page 307) build into Daisy. | Geometry, pipe outlet, drainpoints, heat |
| 56.3 | vertical | A ‘solute’ model (see 56.1, page 307) build into Daisy. | Geometry |

**Related theme.** water and heat movement in the soil.

## 35.57 Topic 57: net radiation

Calculate net radiation from meteorological data.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 57.1 | brunt | Brunt, 1932. Default parameterization by Jensen et.al., 1990. FAO recommendation. | A |
| 57.2 | brutsaert | Brutsaert, 1975 57.3 idso jackson Idso and Jackson, 1969 57.4 prata Prata, 1996 57.5 satterlund Satterlund, 1979 57.6 swinbank Swinbank, 1963 311 312 CHAPTER 57. | n/a |
| 57.3 | idso jackson | Idso and Jackson, 1969 57.4 prata Prata, 1996 57.5 satterlund Satterlund, 1979 57.6 swinbank Swinbank, 1963 311 312 CHAPTER 57. | n/a |
| 57.4 | prata | Prata, 1996 57.5 satterlund Satterlund, 1979 57.6 swinbank Swinbank, 1963 311 312 CHAPTER 57. | n/a |
| 57.5 | satterlund | Satterlund, 1979 57.6 swinbank Swinbank, 1963 311 312 CHAPTER 57. | n/a |
| 57.6 | swinbank | Swinbank, 1963 311 312 CHAPTER 57. | n/a |

**Related theme.** weather, radiation and evapotranspiration.

## 35.58 Topic 58: nitrification

The nitrification process, transforming ammonium into nitrate and nitrous oxide. Used by horizon component Nitrification (see 47, page 231) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 58.1 | soil | k 10 * M / (k + M). Michaelis-Menten kinetics, with nitrification based on total ammonium content. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 58.2 | solute | k 10 * C / (k + C). Michaelis-Menten kinetics, with nitrification based on ammonium solute. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.59 Topic 59: number

Generic representation of numbers. Used by select component expr (see 80, page 429) , and biopore component density (see 20, page 117) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 59.1 | * | Use the product of its operands. | operands |
| 59.2 | + | Use the sum of its operands. | operands |
| 59.3 | minus | Negate number or subtract numbers. With one operand, negates it. With more than one operand, subtracts all but the first from the first. | operands |
| 59.4 | / | Divide the first operand by the rest. | operands |
| 59.5 | apply | Apply function to operand. | function |
| 59.6 | child | Numbers based on another number. | value |
| 59.7 | convert | A ‘child’ model (see 59.6, page 316) build into Daisy. | value, dimension |
| 59.8 | dim | A ‘child’ model (see 59.6, page 316) build into Daisy. | value, dimension, warn known |
| 59.9 | identity | A ‘child’ model (see 59.6, page 316) build into Daisy. | dimension |
| 59.10 | const | Always give the specified value. | value |
| 59.11 | initial zero | A ‘const’ model (see 59.10, page 317) build into Daisy. | n/a |
| 59.12 | zero gradient | A ‘const’ model (see 59.10, page 317) build into Daisy. | n/a |
| 59.13 | daisy/soil/soil.h | Find pressure (h) for a given water content (Theta). | horizon, Theta |
| 59.14 | depth | Find soil value at specific depth. | h, column |
| 59.15 | depth K | A ‘depth’ model (see 59.14, page 318) build into Daisy. | n/a |
| 59.16 | depth Theta | A ‘depth’ model (see 59.14, page 318) build into Daisy. | n/a |
| 59.17 | exp | Take the exponential of its argument. | n/a |
| 59.18 | fetch | Fetch the value and dimension in the current scope. | name |
| 59.19 | get | Get the value of symbol in the current scope. | name, dimension |
| 59.20 | horizon | Find soil value at specific horizon. | h, horizon |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 59.21 | soil K | A ‘horizon’ model (see 59.20, page 319) build into Daisy. | n/a |
| 59.22 | soil Theta | A ‘horizon’ model (see 59.20, page 319) build into Daisy. | n/a |
| 59.23 | soil heat capacity | A ‘horizon’ model (see 59.20, page 319) build into Daisy. | n/a |
| 59.24 | soil heat conductivity | A ‘horizon’ model (see 59.20, page 319) build into Daisy. | n/a |
| 59.25 | if | Select between two numbers depending on a boolean expression. | then, else |
| 59.26 | initial C | Find initial content from concentration. | n/a |
| 59.27 | initial NH4 | A ‘initial C’ model (see 59.26, page 320) build into Daisy. | n/a |
| 59.28 | initial NO3 | A ‘initial C’ model (see 59.26, page 320) build into Daisy. | n/a |
| 59.29 | let | Bind symbols in ‘clauses’ in a new scope, and evaluate ‘expr’ in that scope. | clauses |
| 59.30 | ln | Take the natural logarithm of its argument. | n/a |
| 59.31 | log10 | Take the base 10 logarithm of its argument. | n/a |
| 59.32 | max | Use the largest value of its operands. | operands |
| 59.33 | min | Use the smallest value of its operands. | operands |
| 59.34 | plf | Look up argumen in a piecewise linear function. | domain, range, points, y |
| 59.35 | pow | Raise ‘base’ to the power of ‘exponent’. | base, exponent |
| 59.36 | solute uptake | Find root nitrogen uptake. | L, value, volume, Theta, Theta sat, PotNUpt |
| 59.37 | solute uptake2 | Find root nitrogen uptake. | L, value, volume, Theta, Theta sat |
| 59.38 | source | Extract information from a time series. | source, begin |
| 59.39 | source average | A ‘source’ model (see 59.38, page 324) build into Daisy. | n/a |
| 59.40 | source increase | A ‘source’ model (see 59.38, page 324) build into Daisy. | n/a |
| 59.41 | source sum | A ‘source’ model (see 59.38, page 324) build into Daisy. | n/a |
| 59.42 | source unique | A ‘source’ model (see 59.38, page 324) build into Daisy. | n/a |
| 59.43 | sqr | Take the square of its argument. | n/a |
| 59.44 | sqrt | Take the square root of its argument. | n/a |
| 59.45 | x | The value of the symbol ‘x’ in the current scope. | n/a |

## 35.60 Topic 60: organic

Turnover of organic matter in the soil.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 60.1 | default | Mineralization and immobilization in soil. See also [Hansen et al., 1991, Bruun et al., 2003] | am, Bioincorporation, domsorp, Value description, where, turnover rate, turnover halftime, smb |
| 60.2 | SOM2000 | A ‘default’ model (see 60.1, page 327) build into Daisy. | n/a |
| 60.3 | SOM2025 | A ‘default’ model (see 60.1, page 327) build into Daisy. | n/a |
| 60.4 | none | Ignore all soil organic matter dynamics. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

**Parameters that the workbench runs for this topic.** K NH4, K NO3, CO2 threshold, min AM C, min AM N.

## 35.61 Topic 61: parser

To start the simulation, many parameters must be specified and state variables must be given an initial value. It is the responsibility of the ‘parser’ component to read these data from an external source (typically a setup file), and convert them into the internal format. Used by Toplevel @ input (see 107.30, page 626) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 61.1 | file | Read a setup file containing lots of parentheses. | where |

## 35.62 Topic 62: pet

The ‘pet’ component should calculate the potential evapotranspiration from mete- orological data, as well as the crop and soil state.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 62.1 | FAO PM | Potential evopotranspiration using Penman-Monteith. See also [Allen et al., 1998] Used by bioclimate FAO56 daily pet (see 19.3, page 115) . | use wet, rb |
| 62.2 | FAO PM hourly | Potential evopotranspiration using Penman-Monteith. Modified to work with hourly data. | use wet, rb |
| 62.3 | Hargreaves | Potential evopotranspiration based on temperature. See also [Hargreaves and Samani, 1985] 62.4 Makkink Potential evopotranspiration using Makkink’s Equation. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 62.4 | Makkink | Potential evopotranspiration using Makkink's Equation. E M = beta M0 + beta M1 * (s * Si) / (lambda (s + gamma)) . E M is the reference evapotranspiration [mm/d]. beta M0 [mm] and beta M1 [] are empirical parameters. | n/a |
| 62.5 | AslyngHansen82 | A 'Makkink' model (see 62.4, page 338) build into Daisy. | n/a |
| 62.6 | Makkink57 | A 'Makkink' model (see 62.4, page 338) build into Daisy. | n/a |
| 62.7 | deBruin87 | A 'Makkink' model (see 62.4, page 338) build into Daisy. | n/a |
| 62.8 | PM | Potential evopotranspiration using Penman-Monteith. | rb |
| 62.9 | makkink | Potential evopotranspiration using Makkink's Equation (old version). | n/a |
| 62.10 | weather | Potential evopotranspiration using weather data. | n/a |

**Related theme.** weather, radiation and evapotranspiration.

## 35.63 Topic 63: phenology

The development process.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 63.1 | TSum | Crop phenology model purely based on temperature sums. The length of emergence, and the vegetative and reproductive fase are all based on the specified temperature sums. | EmrTSum, EmrThrs, VegTSum, VegThrs, RepTSum, RepThrs |
| 63.2 | default | Default crop phenology model. | EmrTSum, DSMature, DSRepeat |

**Related theme.** the crop: growth, roots and water use.

## 35.64 Topic 64: photosynthesis

Leaf photosynthesis.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 64.1 | Farquhar | Base parameterization for Farquhar derived photosynthesis models. | Xn, Value description, Ea Gamma, N-dist, Stomatacon, ABA effect, Gamma, gs |
| 64.2 | FC C3 | A ‘Farquhar’ model (see 64.1, page 345) build into Daisy. | beta, c Vm, Ea Vm, Eda Vm, Ea Jm, Ea ko, Ea kc, theta |
| 64.3 | FC C4 | A ‘Farquhar’ model (see 64.1, page 345) build into Daisy. | alpha, beta, theta, Q10vm, paab |
| 64.4 | GL | Goudriaan and Laar, 1978. | Fm |

**Related theme.** the crop: growth, roots and water use.

## 35.65 Topic 65: ponddamp

Dampening affect of ponding on soil erosion from rain.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 65.1 | EUROSEM | KH = exp (-b h) See also [Morgan et al., 1998] | b |
| 65.2 | Hairsine91 | KH = (h / dds)−0 .8 See also [Hairsine and Rose, 1991] 65.3 Park82 KH = 2.7183 * exp (-h / dds) See also [Park et al., 1982] 65.4 none KH = 1.0 Used by reaction colgen Jarvis99 ponddamp (see 70.6, page 387) . 351 352 CHAPTER 65. | n/a |
| 65.3 | Park82 | KH = 2.7183 * exp (-h / dds) See also [Park et al., 1982] 65.4 none KH = 1.0 Used by reaction colgen Jarvis99 ponddamp (see 70.6, page 387) . 351 352 CHAPTER 65. | n/a |
| 65.4 | none | KH = 1.0 Used by reaction colgen Jarvis99 ponddamp (see 70.6, page 387) . 351 352 CHAPTER 65. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.66 Topic 66: program

Run a program. Used by Toplevel @ run (see 107.30, page 626) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 66.1 | AM table | Generate a table of fertilizers. | n/a |
| 66.2 | Daisy | The Daisy crop/soil/atmosphere model. See also [Hansen et al., 1990, Hansen, 2002, Hansen et al., 1991] | harvest, stop, time, column, timestep, exchange, scope, output |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 66.3 | GP2D | Write root density table using 2D extension to Gerwitz and Page. | row position, WRoot, row width, Geometry, soil depth, crop depth, crop width |
| 66.4 | KM2 | Read KM2 precipitation data. | file |
| 66.5 | Osvaldo | Find the modelling error. This is done between one set of measured data and multiple sets of simulated data. | pars, par width, measured file |
| 66.6 | RS2WG | Prepare data from RainSim to the weather generator. | rshourly origin, wgcycle begin, wgcycle length, rshourly file, rsdaily file, dwfhourly file, hint file, hlim |
| 66.7 | WG2DWF | Prepare data from the weather generator for Daisy. | rshourly origin, wgcycle begin, wgcycle length, wg file, dwfdaily file, monthly file, yearly file, dint file |
| 66.8 | astdump | Generate a dump of the abstract syntax tree. | where |
| 66.9 | batch | Run a sequence of programs. | directory, run |
| 66.10 | cd | Change working directory. | directory |
| 66.11 | cpedata | Manipulate data from Agrovand. | file, value, begin, debug, day, missing, tag, dim line |
| 66.12 | docmodel | Document specific models. | component, format, where, models |
| 66.13 | document | Generate the components part of the reference manual. | format, where, print parameterizations |
| 66.14 | extract | Extract data from a data file. | file, missing, tag, dim line, handle, columns, rows |
| 66.15 | gnuplot | Generate a gnuplot command file. | extra, command file, graph |
| 66.16 | hmovie | Manipulate data from Agrovand. | file, dim, missing, tag, dim line, command file, x min, y min |
| 66.17 | hydraulic | Generate a table of the rentention curve and hydraulic conductivity. | horizon, intervals |
| 66.18 | minimize | Find local minimum for program. | scope, run, parameters, name, value, limit, epsilon |
| 66.19 | nwaps | Nwaps collects results from multiple directories. | file, fractiles, missing, directory, parent directory, scenario, scn sep, success file |
| 66.20 | post-process | Extract a subset of a soil profile log file. | file, top, before, dimension, where, bottom, left, right |
| 66.21 | rootmatch | Match root data with GP2D model. | file, gnuplot, x offset, x start, SpRtLength, DensRtTip, debug, row position |
| 66.22 | spawn | Spawn a number of programs in parallel. | file, program, directory, input directory, exe |
| 66.23 | weather | Weather data from a data file. | file, fractiles, begin |
| 66.24 | write | Write string to file. | where, what |
| 66.25 | wse | Generate a table of the water stress effect. | wse, intervals |

## 35.67 Topic 67: raddist

The ‘raddist’ component calculates the radiation distribution in the canopy.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 67.1 | default | Default model of radiation distribution in the canopy. | n/a |
| 67.2 | sun-shade | Sun-shade model of radiation distribution in the canopy. | sigma PAR, sigma NIR, Ps PAR, Ps NIR, Ph PAR, Pcb PAR, Pscb PAR, Pscd PAR |

**Related theme.** weather, radiation and evapotranspiration.

## 35.68 Topic 68: rainergy

Energy in rain.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 68.1 | Brown87 | Energy as a semi-empirical function of rain intensity. The energy content in the fraction that hits the canopy is ignored. | n/a |
| 68.2 | EUROSEM | Kinetic energy model taking vegetation into account. See also [Morgan et al., 1998] Used by reaction colgen Morgan98 rainergy (see 70.8, page 389) . 379 380 CHAPTER 68. | n/a |

**Related theme.** weather, radiation and evapotranspiration.

## 35.69 Topic 69: rate

Specify a rate or a halftime. Used by AOM component turnover (see 8, page 53) , SMB component turnover (see 13, page 65) , and SOM component turnover (see 14, page 67) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 69.1 | halftime | A rate specified through the equivalent halftime (rate = ln 2 / halftime). | halftime |
| 69.2 | rate | Specify rate directly. | rate |
| 69.3 | zero | A ‘rate’ model (see 69.2, page 381) build into Daisy. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.70 Topic 70: reaction

Generic transformations between soil chemicals.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 70.1 | DOM turnover | Turnover of dissolved organic matter. See also [Garnier et al., 2003] Used by chemistry DOM reaction (see 24.2, page 145) . | DOC name, DON name, where, efficiency, Value description |
| 70.2 | Python | Python interface for chemical reactions above and below ground. | top, module, extra, in, handle, tag, texture, size |
| 70.3 | adsorption | Maintain equilibrium between solute and sorbed from. | equilibrium, solute, adsorption rate, desorption rate, adsorption source |
| 70.4 | bound release | Release of chemicals bound to colloids from surface soil. | immobile, release |
| 70.5 | colgen | Shared parameter and log variable for colloid generation models. | ponddamp, surface release, dds, D |
| 70.6 | colgen Jarvis99 | A ‘colgen’ model (see 70.5, page 387) build into Daisy. | rainergy, Ms, As |
| 70.7 | colloid-generation | A ‘colgen Jarvis99’ model (see 70.6, page 387) defined in ‘chemistry-base.dai’. | n/a |
| 70.8 | colgen Morgan98 | A ‘colgen’ model (see 70.5, page 387) build into Daisy. | rainergy |
| 70.9 | colgen Styczen88 | A ‘colgen’ model (see 70.5, page 387) build into Daisy. | Ae, MA, MR |
| 70.10 | default | Transformation between two soil chemicals. | transform, A |
| 70.11 | denitrification | Denitrification in soil, (conversion of nitrate to atmospheric nitrogen). | alpha, K fast, alpha fast, redox height, converted fast, potential fast |
| 70.12 | equilibrium | Equilibrium between two soil chemicals. | equilibrium, secondary, surface, A, k BA, primary |
| 70.13 | filter velocity | Filtration of soil colloids. | immobile, mobile, fc primary, fc secondary, F primary, F secondary |
| 70.14 | colloid-filter | A ‘filter velocity’ model (see 70.13, page 393) defined in ‘chemistry-base.dai’. | n/a |
| 70.15 | nitrification | Nitrification. The actual nitrification specification is part of the horizon models, this reaction just applies the models and logs the result. | N2O |
| 70.16 | shoot | Pool based on shoot DM. Generate chemical (M) on canopy up to maximum (M max). | crop |
| 70.17 | shoot2 | Pool based on shoot DM. Generate chemical (M) on canopy up to maximum (M max). | crop, M pot |
| 70.18 | sorption | Kinetic linear sorption equilibrium. Faster than the ‘equilibrium’ reaction model, more flexible than the ‘adsorption’ reaction model. | solute, K clay, Equillibrium parameter, k sorption, k desorption, surface sorption, S sorption |

**Related theme.** nitrogen, carbon and organic matter.

## 35.71 Topic 71: retention

Specify a retention or a halftime.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 71.1 | PASTIS | Retention curve of Mulch used in PASTIS. h (Theta) = -(-h min)^(1-((Theta minus Theta res) / (2/3 Theta sat minus Theta res))) Used by litter mulch retention (see 51.6, page 276) . 71.2 exp Retention curve of Mulch used in Exponential. | n/a |
| 71.2 | exp | Retention curve of Mulch used in Exponential. Theta (h) = exp (k h) (Theta sat minus Theta res) + Theta res h (Theta) = ln ((Theta minus Theta res) / (Theta sat minus Theta res)) / k. | Theta sat, Theta res, h min |

**Related theme.** water and heat movement in the soil.

## 35.72 Topic 72: root

Root processes.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 72.1 | uptake | Roots that takes up water and nitrogen. | NH4 uptake, NO3 uptake, MaxPen, MaxWidth, h wp, MxNH4Up, MxNO3Up, Rxylem |
| 72.2 | zone | A ‘uptake’ base model (see 72.1, page 401) build into Daisy. | PenpFFac, PenClayFac, PenWaterFac, PenDSFac |
| 72.3 | classic | A ‘zone’ model (see 72.2, page 403) build into Daisy. | rootdens |

**Related theme.** the crop: growth, roots and water use.

## 35.73 Topic 73: rootdens

Root density calculations.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 73.1 | Anders Pedersen | Use exponential function for root density. In this variant of Gerwitz and Page, ’a’ is specified as a function of development stage. | q, a |
| 73.2 | DS Depth | Specify root density as a function of development stage. | entries, density |
| 73.3 | DS Rel | Specify root density as a function of development stage. | entries, density |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 73.4 | Depth Depth | Specify root density as a function of development stage. | entries, density |
| 73.5 | GP1D | Use exponential function for root density. See also [Gerwitz and Page, 1974] | DensRtTip, DensIgnore, a |
| 73.6 | GP2D | Use exponential function for root density in row crops. | DensRtTip, DensIgnore, debug, row position, row distance |
| 73.7 | Gerwitz+Page74 | Use exponential function for root density. See also [Gerwitz and Page, 1974] | DensRtTip, MinDens, a |
| 73.8 | growth | Dynamic root growth model. | Depth, DensRtTip, row position, row distance, Width, WRoot |
| 73.9 | local | Dynamic root growth model. | Depth, DensRtTip, row position, row distance, Width, neighbor effect, death rate, emerging |

**Related theme.** the crop: growth, roots and water use.

## 35.74 Topic 74: rubiscoN

Find total amount of Rubisco-N in leaves.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 74.1 | default | Standard estimation of RuBisCo N. | fraction, Value description, offset, subtract Pt |

**Related theme.** the crop: growth, roots and water use.

## 35.75 Topic 75: rubiscoNdist

The ‘rubiscoNdist’ component calculates the rubisco N distribution for photosyn- thesis in the canopy.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 75.1 | exp | Boegh et al.(2002) rubisco N-distribution model in the canopy for photosynthesis. | kn |
| 75.2 | expr | expr rubisco N-distribution model in the canopy. | value, obsolete, f photo |
| 75.3 | forced | Forced rubisco capacity distribution model in the canopy. | value, obsolete |
| 75.4 | uniform | Uniform rubisco N-distribution model in the canopy for photosynthesis. | f photo |

**Related theme.** the crop: growth, roots and water use.

## 35.76 Topic 76: scope

A scope maps names to values.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 76.1 | exchange | Exchange values with an external model. | entries |

## 35.77 Topic 77: scopesel

A method to choose a scope in a Daisy simulation.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 77.1 | multi | Combine multiple named scopes. | name |
| 77.2 | name | Select named scope. | name |
| 77.3 | null | Select the empty scope. | n/a |

## 35.78 Topic 78: secondary

Specify secondary domain. The secondary domain consist typically of soil fractures or other inter-aggregate pores small enough to be dominated by capillarity, yet so large that water moves fast enough that the solute equilibrium with the primary domain (typically intraaggregate pores) can not be maintained. This allows a pulse of water to be move through saturated or near saturated soil without solutes in the new water being mixed with solutes in the old water. The effects are twofold: It allows solutes applied to the surface to reach deeper soil layers much faster than it would otherwise, and it protects solutes in the soil matrix from being washed out with fast moving new water.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 78.1 | alpha | Shared base class for non-empty secondary domains. | alpha |
| 78.2 | cracks | A ‘alpha’ model (see 78.1, page 423) build into Daisy. | density, use secondary, aperture |
| 78.3 | pressure | A ‘alpha’ model (see 78.1, page 423) build into Daisy. | h lim, use secondary |
| 78.4 | none | No secondary domain. | n/a |

**Related theme.** nitrogen, carbon and organic matter.

## 35.79 Topic 79: seed

Initial growth after emergence. The initial growth process governs the growth of the crop until the point where there is enough leaf area for photosynthesis to take over.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 79.1 | LAI | Initial crop growth is governed by a forced LAI function. | SpLAIfac |
| 79.2 | release | Initial crop growth is governed by carbon released from seeds. | rate, initial weight, DM fraction, C fraction, N fraction |

**Related theme.** the crop: growth, roots and water use.

## 35.80 Topic 80: select

Select part of state.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 80.1 | array | Log all members of an array. | n/a |
| 80.2 | quiver | Convert edge flow to node based flow vectors. | n/a |
| 80.3 | value | Log a single numeric value. | n/a |
| 80.4 | content | A ‘value’ model (see 80.3, page 432) build into Daisy. | height, OBSOLETE, y |
| 80.5 | flow | A ‘value’ base model (see 80.3, page 432) build into Daisy. | volume, density |
| 80.6 | flow back | A ‘flow’ model (see 80.5, page 433) build into Daisy. | n/a |
| 80.7 | flow bottom | A ‘flow’ model (see 80.5, page 433) build into Daisy. | to |
| 80.8 | flux bottom | A ‘flow bottom’ model (see 80.7, page 433) build into Daisy. | n/a |
| 80.9 | flow front | A ‘flow’ model (see 80.5, page 433) build into Daisy. | n/a |
| 80.10 | flow left | A ‘flow’ model (see 80.5, page 433) build into Daisy. | n/a |
| 80.11 | flow right | A ‘flow’ model (see 80.5, page 433) build into Daisy. | n/a |
| 80.12 | flow top | A ‘flow’ model (see 80.5, page 433) build into Daisy. | from, OBSOLETE |
| 80.13 | flux top | A ‘flow top’ model (see 80.12, page 435) build into Daisy. | n/a |
| 80.14 | index | A ‘value’ model (see 80.3, page 432) build into Daisy. | n/a |
| 80.15 | number | A ‘value’ model (see 80.3, page 432) build into Daisy. | n/a |
| 80.16 | volume base | A ‘value’ model (see 80.3, page 432) build into Daisy. | volume, density, space, min, max, sum, density y |
| 80.17 | interval | A ‘volume base’ model (see 80.16, page 436) build into Daisy. | from, OBSOLETE, to |
| 80.18 | volume | A ‘volume base’ model (see 80.16, page 436) build into Daisy. | n/a |
| 80.19 | water | A ‘volume base’ model (see 80.16, page 436) build into Daisy. | h, h ice |
| 80.20 | water interval | A ‘water’ model (see 80.19, page 439) build into Daisy. | from, OBSOLETE, to |
| 80.21 | water volume | A ‘water’ model (see 80.19, page 439) build into Daisy. | n/a |

## 35.81 Topic 81: solute uptake

The ‘solute uptake’ component calculates uptake of nitrogen through roots.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 81.1 | diffusion | N is transported to the root surface based on diffusion and convection. | alpha, A, D, beta |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 81.2 | fixed sink | A ‘diffusion’ model (see 81.1, page 443) build into Daisy. | CRoot |
| 81.3 | variable sink | A ‘diffusion’ model (see 81.1, page 443) build into Daisy. | Value description, F relative, CRoot, CEst |
| 81.4 | none | No uptake. | n/a |

**Related theme.** the crop: growth, roots and water use.

## 35.82 Topic 82: solver

A way to solve the matrix equation ‘A x = b’.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 82.1 | cxsparse | Solve equation using CXSparse library described in: | n/a |
| 82.2 | none | Don’t solve the equation. | n/a |
| 82.3 | ublas | Solve equation using UBLAS lu functions. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.83 Topic 83: source

Time series, with possible error bars and formatting information.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 83.1 | arithmetic | Read a daisy log, weather or data file. Calculate a single value for each time step, based on the value in the various columns. | file, with, timestep, missing, tag, dim line, style, title |
| 83.2 | column | Read a a single column from a Daisy log, weather or data file. | file, with, timestep, dimension, offset, missing, tag, dim line |
| 83.3 | combine | Combine data from multiple sources with a single expression. | source, with, style, title, accumulate |
| 83.4 | merge | Merge multiple timeseries into one. Any errorbars on the original timeseries are ignored, but the merged timeseries may have errorbars if there are multiple values for the same time. | source, with, dimension, style, title, accumulate |

## 35.84 Topic 84: stomatacon

The ‘Stomatacon’ component calculates the stomata conductance of water vapour.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 84.1 | WSF | Common water stress effect parameters. | beta, ABA min, delta |
| 84.2 | BB base | A ‘WSF’ base model (see 84.1, page 455) build into Daisy. | m, b |
| 84.3 | BB | A ‘BB base’ model (see 84.2, page 455) build into Daisy. | n/a |
| 84.4 | Leuning | A ‘BB base’ model (see 84.2, page 455) build into Daisy. | Do |
| 84.5 | MNA | A ‘WSF’ model (see 84.1, page 455) build into Daisy. | alpha, m, b, obsolete, lambda |
| 84.6 | SHA12 | A ‘WSF’ model (see 84.1, page 455) build into Daisy. | alpha, m, min, lambda |
| 84.7 | SHA14 | A ‘WSF’ model (see 84.1, page 455) build into Daisy. | alpha, m, lambda |

**Related theme.** the crop: growth, roots and water use.

## 35.85 Topic 85: string

Generic representation of strings.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 85.1 | cond | Return the value of the first clause whose condition is true. | clauses |
| 85.2 | identity | Return the specified value. | value |
| 85.3 | number | Extract the value of a number. | n/a |
| 85.4 | dimension | A ‘number’ model (see 85.3, page 459) build into Daisy. | n/a |
| 85.5 | value | A ‘number’ model (see 85.3, page 459) build into Daisy. | precision |

## 35.86 Topic 86: summary

Summary reports for log parameterizations.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 86.1 | Rsqr | Calculate coefficient of determination. | measure, time, data, tag, value, print data |
| 86.2 | RsqrW | Calculate coefficient of determination. | measure, time, data, weight, tag, value, print data |
| 86.3 | balance | A summary model providing a balance for a log parameterization. | content, where, precision, input, output, require top |
| 86.4 | Biopore chemical balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.5 | Biopore water balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.6 | Field crop nitrogen balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 86.7 | Field mineral nitrogen balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.8 | Field nitrogen balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.9 | Field organic nitrogen balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.10 | Field water balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.11 | Matrix chemical balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.12 | Matrix water balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.13 | N balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.14 | Soil N balance | A ‘N balance’ model (see 86.13, page 464) defined in ‘log-std.dai’. | n/a |
| 86.15 | Soil NH4+ balance | A ‘N balance’ model (see 86.13, page 464) defined in ‘log-std.dai’. | n/a |
| 86.16 | Soil NO3- balance | A ‘N balance’ model (see 86.13, page 464) defined in ‘log-std.dai’. | n/a |
| 86.17 | Soil mineral-N balance | A ‘N balance’ model (see 86.13, page 464) defined in ‘log-std.dai’. | n/a |
| 86.18 | Soil organic-N balance | A ‘N balance’ model (see 86.13, page 464) defined in ‘log-std.dai’. | n/a |
| 86.19 | Soil chemical balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.20 | Soil water balance | A ‘balance’ model (see 86.3, page 462) defined in ‘log-std.dai’. | n/a |
| 86.21 | fractiles | Show fractiles for specified tags. | fractiles, tags, first |
| 86.22 | simple | A simple log file summary model. | fetch, where, precision, print sum, sum name |

## 35.87 Topic 87: surface

Surface processes.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 87.1 | default | Keep track of soil surface. | max exfiltration, Value description, albedo dry, albedo wet, pond section, DetentionCapacity, ReservoirConstant, LocalDetentionCapacity |
| 87.2 | simple | Don’t keep track of surface stuff. | n/a |
| 87.3 | const flux | A ‘simple’ model (see 87.2, page 469) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 87.4 | none | A ‘const flux’ model (see 87.3, page 469) build into Daisy. | n/a |
| 87.5 | const pressure | A ‘simple’ model (see 87.2, page 469) build into Daisy. | pressure |
| 87.6 | source flux | A ‘simple’ model (see 87.2, page 469) build into Daisy. | source, interpolate |

**Related theme.** water and heat movement in the soil.

**Parameters that the workbench runs for this topic.** temperature change rate, EpFactor, albedo dry, albedo wet, DetentionCapacity, ReservoirConstant, LocalDetentionCapacity, z mixing, R mixing.

## 35.88 Topic 88: svat

The task of the ‘svat’ component is to calculate the production stress, given the po- tential evapotranspiration, the actual evaporation from the surface, meteorological data, and the vegetation and soil state.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 88.1 | PMSW | Peter van der Keur’s SVAT model. See also [van der Keur et al., 2001] | albedo, z 0s, alpha u, arac, theta c, rcmin const, zeta, f3const |
| 88.2 | SSOC | Sun-Shade Open Canopy. See also [Plauborg et al., 2010] | hypostomatous, max iteration, z 0b, s, lambda, rho a, gamma, T s |
| 88.3 | none | Decoupling surface energy and water balance. Done using reference values and crop factors. | n/a |

**Related theme.** the crop: growth, roots and water use.

## 35.89 Topic 89: tertiary

Transport of water and solute outside the matrix. Used by movement component Tertiary (see 56, page 307) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 89.1 | biopores | Tertiary domain divided into biopore classes. | active, pressure initiate, classes, pressure limit, active msg, Possible values, none, old dt |
| 89.2 | none | No tertiary transport. | n/a |
| 89.3 | old | Tertiary water and solute movement based on the obsolete ’macro’ and ’mactrans’ components. | macro, mactrans |

**Related theme.** water and heat movement in the soil.

## 35.90 Topic 90: tortuosity

Solutes in the soil can't move the shortest way between two points. The tortuosity factor indicates how far the average solute have moved in absolute coordinates, when it has moved a given distance along the curved line. This component is responsible for calculating the soils tortuosity factor. Used by horizon component tortuosity (see 47, page 231) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 90.1 | M Q | Millington-Quirk. Thetaˆ(7/3) / Theta sat2 . | n/a |
| 90.2 | linear | Linear Impedance factor. a + b Theta. See also [Nye and Tinker, 1977] Used by horizon component tortuosity (see 47, page 231) . | a, b |

**Related theme.** water and heat movement in the soil.

## 35.91 Topic 91: transform

Generic transformations between soil components.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 91.1 | equilibrium | Two soil components reching for equilibrium. | equilibrium, k BA |

**Related theme.** nitrogen, carbon and organic matter.

## 35.92 Topic 92: transport

Solute transport in primary domain.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 92.1 | Hansen | Solute transport using convection-dispersion. | n/a |
| 92.2 | Mollerup | Coupled vertical and horizontal transport. See Mollerup 2007 for details. | debug, None, Timestep reduction, Streamline diffusion, upstream weight |
| 92.3 | convection | Pure forward calculation of flow except through upper boundary. J[edge] = q[edge] * C old[upstream] Used by movement vertical matrix solute (see 56.3, page 308) , and movement rectangle matrix solute (see 56.2, page 307) . 92.4 none Disable all transport except through boundaries. | n/a |
| 92.4 | none | Disable all transport except through boundaries. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.93 Topic 93: ui

Top level user interface. Used by Toplevel @ ui (see 107.30, page 626) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 93.1 | none | No user unterface. | n/a |
| 93.2 | progress | Write progress on standard output (or standard error). | n/a |

## 35.94 Topic 94: uifilter

Presentation of data in the user interface.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 94.1 | base | Base filter for the user interface. | n/a |
| 94.2 | raw | A 'base' model (see 94.1, page 493) build into Daisy. | n/a |
| 94.3 | simple | A 'base' model (see 94.1, page 493) build into Daisy. | default, name |

## 35.95 Topic 95: uiitem

User interface information about a item.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 95.1 | raw | Raw item for the user interface. | n/a |

## 35.96 Topic 96: unit

Specify units of physical quatities. Daisy will recognise both the build-in and user defined units, and convert be- tween different units representing the same physical dimension. This is done by converting to and from the base (usually defined by SI) unit for that dimension.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.1 | SI | Base parameterization for all SI based units. | time, length, mass, electric current, thermodynamic temperature, luminous intensity |
| 96.2 | SIfactor | A 'SI' model (see 96.1, page 497) build into Daisy. | n/a |
| 96.4 | % | A 'SIfactor' model (see 96.2, page 498) build into Daisy. | n/a |
| 96.5 | | A 'SIfactor' model (see 96.2, page 498) build into Daisy. | n/a |
| 96.6 | | A 'SIfactor' model (see 96.2, page 498) build into Daisy. | n/a |
| 96.7 | A | A 'SIfactor' model (see 96.2, page 498) build into Daisy. | n/a |
| 96.8 | J | A 'SIfactor' model (see 96.2, page 498) build into Daisy. | n/a |
| 96.9 | J/cm2 /h | A 'SIfactor' model (see 96.2, page 498) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.10 | J/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.11 | K | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.12 | K−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.13 | K−2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.14 | L/g | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.15 | L/kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.16 | L/mg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.17 | MJ/d/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.18 | MJ/h/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.19 | MJ/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.20 | MJ/m2 /d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.21 | MJ/m2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.22 | MPa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.23 | MPa−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.24 | Mg DM/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.25 | Mg DM/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.26 | Mg w.w./ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.27 | Mg/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.28 | Pa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.29 | Pa−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.30 | T w.w./ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.31 | W | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.32 | W m−2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.33 | W/cm2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.34 | W/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.35 | Wh/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.36 | ag/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.37 | cd | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.38 | cm | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.39 | cm R/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.40 | cm/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.41 | cm/dg C/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.42 | cm/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.43 | cm−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.44 | cm−2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.45 | cm2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.46 | cm2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.47 | cm2 /s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.48 | cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.49 | cm3 H2O/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.50 | cm3 /cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.51 | cm3 /g | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.52 | cm3 /ng | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.53 | d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.54 | d−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.55 | erg/cm2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.56 | fg/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.57 | fraction | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.58 | g | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.59 | g C/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.60 | g C/cm2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.61 | g C/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.62 | g C/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.63 | g DM/cm2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.64 | g DM/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.65 | g DM/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.66 | g N/cm2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.67 | g N/cm2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.68 | g N/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.69 | g N/l | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.70 | g N/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.71 | g N/m2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.72 | g w.w./m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.73 | g/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.74 | g/cm R/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.75 | g/cm/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.76 | g/cm2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.77 | g/cm2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.78 | g/cm2 /mm | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.79 | g/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.80 | g/cm3 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.81 | g/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.82 | g/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.83 | g/ha/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.84 | g/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.85 | g/ha/mm | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.86 | g/kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.87 | g/l | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.88 | g/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.89 | g/m2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.90 | g/mol | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.91 | h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.92 | hPa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.93 | h−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.94 | ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.95 | hours | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.96 | kJ/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.97 | kPa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.98 | kWh/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.99 | kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.100 | kg C/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.101 | kg C/ha/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.102 | kg C/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.103 | kg DM/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.104 | kg DM/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.105 | kg DM/m3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.106 | kg N/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.107 | kg N/ha/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.108 | kg N/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.109 | kg N/ha/mm | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.110 | kg N/ha/y | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.111 | kg m−2 s−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.112 | kg w.w./ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.113 | kg/Pa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.114 | kg/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.115 | kg/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.116 | kg/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.117 | kg/ha/mm | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.118 | kg/ha/y | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.119 | kg/m | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.120 | kg/m/s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.121 | kg/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.122 | kg/m2 /s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.123 | kg/m3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.124 | kg/mol | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.125 | kg/s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.126 | kgN/ha/year | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.127 | l/g | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.128 | l/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.129 | l/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.130 | l/kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.131 | l/mg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.132 | m | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.133 | m s−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.134 | m/s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.135 | mL/g | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.136 | mL/kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.137 | m−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.138 | m−2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.139 | m−2 kg s−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.140 | m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.141 | m2 /s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.142 | m3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.143 | m3 /cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.144 | m3 /kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.145 | mg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.146 | mg N/kg dry soil | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.147 | mg N/l | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.148 | mg/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.149 | mg/g | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.150 | mg/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.151 | mg/ha/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.152 | mg/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.153 | mg/l | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.154 | mg/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.155 | minutes | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.156 | ml/g | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.157 | ml/kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.158 | mm | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.159 | mm/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.160 | mm/dg C/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.161 | mm/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.162 | mm/s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.163 | mm−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.164 | mmol/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.165 | mmol/kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.166 | mmol/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.167 | mmol/m2 /s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.168 | mol | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.169 | mol/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.170 | mol/kg | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.171 | mol/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.172 | mol/m2 /s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.173 | mol/m3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.174 | ng/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.175 | ng/MPa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.176 | ng/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.177 | ng/kPa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.178 | ng/l | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.179 | ng/mm/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.180 | none | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.181 | pg/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.182 | ppm | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.183 | s | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.184 | s−1 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.185 | seconds | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.186 | t/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.187 | ug | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.188 | ug/L | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.189 | ug/L/MPa | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.190 | ug/ha | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.191 | ug/ha/d | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.192 | ug/ha/h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.193 | ug/l | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.194 | ug/m2 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.195 | ug/m2 /h | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.196 | ug/m3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.197 | um | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.198 | umol/cm3 | A ‘SIfactor’ model (see 96.2, page 498) build into Daisy. | n/a |
| 96.199 | base | A base unit. | n/a |
| 96.200 | | A ‘base’ model (see 96.199, page 529) build into Daisy. | n/a |
| 96.201 | | A ‘base’ model (see 96.199, page 529) build into Daisy. | n/a |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 96.202 | dgEast | A ‘base’ model (see 96.199, page 529) build into Daisy. | n/a |
| 96.203 | dgNorth | A ‘base’ model (see 96.199, page 529) build into Daisy. | n/a |
| 96.204 | dry soil fraction | A ‘base’ model (see 96.199, page 529) build into Daisy. | n/a |
| 96.205 | rad | A ‘base’ model (see 96.199, page 529) build into Daisy. | n/a |
| 96.206 | factor | Connvert to base units by multiplying with a factor. | base |
| 96.207 | dg | A ‘factor’ model (see 96.206, page 530) build into Daisy. | n/a |
| 96.208 | dgSouth | A ‘factor’ model (see 96.206, page 530) build into Daisy. | n/a |
| 96.209 | dgWest | A ‘factor’ model (see 96.206, page 530) build into Daisy. | n/a |
| 96.210 | new dg | A ‘factor’ model (see 96.206, page 530) build into Daisy. | n/a |
| 96.211 | ppm dry soil | A ‘factor’ model (see 96.206, page 530) build into Daisy. | n/a |
| 96.212 | offset | Connvert to base units by multiplying factor, then substracting offset. | base, offset |
| 96.213 | dg C | A ‘offset’ model (see 96.212, page 531) build into Daisy. | n/a |
| 96.214 | dg F | A ‘offset’ model (see 96.212, page 531) build into Daisy. | n/a |
| 96.215 | pF | log10 (- cmH2O). | n/a |

## 35.97 Topic 97: uz1d

The ‘uz1d’ component handles the horizontal water movement in the unsaturated zone soil matrix.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 97.1 | none | Disable transport Used by uzrect v+h horizontal (see 99.3, page 541) . 97.2 richards A numerical solution to Richard’s Equation. | n/a |
| 97.2 | richards | A numerical solution to Richard’s Equation. | max iterations, K average |

**Related theme.** water and heat movement in the soil.

## 35.98 Topic 98: uzmodel

The ‘uzmodel’ component handles the vertical water movement in the unsaturated zone soil matrix.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 98.1 | lr | Use gravitational water movement for wet soil, where h. | overflow warn, h fc, z top |
| 98.2 | none | No water movement, and no sink. | n/a |
| 98.3 | richards | A numerical solution to Richard's Equation. | debug, max iterations, K average |
| 98.4 | richards2 | A numerical solution to Richard's Equation. | debug, max iterations, K average, min timestep, low iterations, high iterations |

**Related theme.** water and heat movement in the soil.

## 35.99 Topic 99: uzrect

2D water movement in a rectangular grid.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 99.1 | Mollerup | A finite volume solution to matrix water transport. See Mollerup 2007 for details. | debug, max iterations, K average, Kedge |
| 99.2 | const | Steady-state water flow. | n/a |
| 99.3 | v+h | Transport water in the matrix in two phases, first vertical, then horizontal. | n/a |

**Related theme.** water and heat movement in the soil.

## 35.100 Topic 100: vegetation

That green stuff.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 100.1 | afforestation | A growing forest. | Root, Canopy, root am, litter am, Albedo, planting time, canopy height, root depth |
| 100.2 | crops | Keep track of all crops on the field. | crops, harvest |
| 100.3 | permanent | Permanent (non-crop) vegetation. | Root, Canopy, Height, litter am, Albedo, N uptake |
| 100.4 | Bush | A 'permanent' model (see 100.3, page 547) defined in 'vegetation.dai'. | n/a |
| 100.5 | Coniferous | A 'permanent' model (see 100.3, page 547) defined in 'vegetation.dai'. | n/a |
| 100.6 | Grass | A 'permanent' model (see 100.3, page 547) defined in 'vegetation.dai'. | n/a |
| 100.7 | Hardwood | A 'permanent' model (see 100.3, page 547) defined in 'vegetation.dai'. | n/a |

**Related theme.** the crop: growth, roots and water use.

## 35.101 Topic 101: vernalization

Requirement for a cold period before flowering.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 101.1 | default | Temperature sum dependent vernalization. | DSLim, TaLim, TaSum |
| 101.2 | none | No vernalization. | n/a |

**Related theme.** the crop: growth, roots and water use.

## 35.102 Topic 102: volume

A subset of 3D space. Used by Irrigation @ event volume (see 107.10, page 604) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 102.1 | box | A volume defined by intervals on each axis. By default, the intervals fill the entire axis. | top, bottom, left, right, front |

## 35.103 Topic 103: weather

A ‘wsource’ is a source of raw weatherdata.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 103.1 | weather | Weather interface implementation. | snow fraction, Value description, max rain, deposit, value, name, day length, reference evapotranspiration |
| 103.2 | base | A ‘weather’ base model (see 103.1, page 555) build into Daisy. | T min, AirTemp, Precip, RefEvap, VapPres, RelHum, Latitude, Longitude |
| 103.3 | const | A ‘base’ model (see 103.2, page 557) build into Daisy. | n/a |
| 103.4 | none | A ‘const’ model (see 103.3, page 561) build into Daisy. | n/a |
| 103.5 | null | A ‘const’ model (see 103.3, page 561) build into Daisy. | n/a |
| 103.6 | table | A ‘base’ model (see 103.2, page 557) build into Daisy. | file, missing, tag, dim line |
| 103.7 | default | A ‘table’ model (see 103.6, page 562) build into Daisy. | file, missing years, from, to |
| 103.8 | combine | A ‘weather’ model (see 103.1, page 555) build into Daisy. | entry, source, begin, use, reserve |
| 103.9 | indirect | A ‘weather’ model (see 103.1, page 555) build into Daisy. | source |
| 103.10 | time | A ‘indirect’ model (see 103.9, page 567) build into Daisy. | offset, from, to |

**Related theme.** weather, radiation and evapotranspiration.

## 35.104 Topic 104: wse

The water stress effect on crop growth.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 104.1 | full | Water stress has full effect on crop growth. This means that if there is 50% water stress, assimilate production will be cut into half. | n/a |
| 104.2 | none | Water stress has no effect on plant growth. | n/a |
| 104.3 | partial | Water stress has partial effect on crop growth. | n/a |

**Related theme.** the crop: growth, roots and water use.

## 35.105 Topic 105: xysource

XY data series.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 105.1 | arithmetic | Read a daisy log, weather or data file. Calculate an x and an y value for each time step, based on the value in the various columns. | file, with, missing, tag, dim line, style, title, y |
| 105.2 | combine | Combine data from multiple time series with a single expression. Data from times series are matched by date. | source, with, style, title, y |
| 105.3 | flux | Read a daisy 2d log file, extract flux through line. Values between 'begin' and 'end' will be accumulated, unless 'when' is set. | file, with, dimension, begin, missing, tag, dim line, style |
| 105.4 | frequency | Read a daisy log, weather or data file. Plot the frequency of a specific column. | file, with, missing, tag, dim line, style, title, sort |
| 105.5 | inline | A list of x, y pairs. | with, points, style, title, x dimension, y dimension |
| 105.6 | loop | Calculate x and y pairs based on a single variable. | with, begin, style, title, tag, y, step |
| 105.7 | merge | Merge multiple xy data series into one. | source, with, style, title, x dimension, y dimension |
| 105.8 | profile | Plot soil profile. | file, with, dimension, missing, tag, dim line, style, title |
| 105.9 | xycombine | Combine data from multiple time series with a single expression. Data from times series are matched by date. | source, with, style, title |

## 35.106 Topic 106: zone

A subset of 3D space. Used by SoilRegion @ volume (see 107.20, page 613) .

| Section | Name | What it covers | Components |
|---|---|---|---|
| 106.1 | box | A zone defined by intervals on each axis. By default, the intervals fill the entire axis. | top, bottom, left, right, front |
| 106.2 | polygon | A 2D region defined by points connected by lines. The object is assumed to be infinite in the third dimension. | n/a |

## 35.107 Topic 107: Fixed Components

Fixed components are similar to ordinary component, with the exceptions that there can only be one model, that is, only a single implementation of the component, and that it is not possible to define libraries of standard parameterizations for the model.

| Section | Name | What it covers | Components |
|---|---|---|---|
| 107.1 | DOM | A single Dissolved Organic Matter pool. | diffusion coefficient, efficiency, fractions, turnover rate, turnover halftime |
| 107.2 | SoluteUptakeResult | Result of the 'solute uptake' number model. | uptake, alpha, D, beta, C root |
| 107.3 | SoluteUptake2Result | Result of the 'solute uptake2' number model. | uptake, alpha, A, D, beta, C root |
| 107.4 | CanopySimple | Simple canopy model. | PARext, NIRext, EPext, IntcpCap, EpFac, Value description, EpFacWet, rs min |
| 107.5 | CanopyStandard | Standard canopy model. | PARext, NIRext, EPext, IntcpCap, EpFac, Value description, EpFacWet, rs min |
| 107.6 | CrpN | Default crop nitrogen parameters. | PtLeafCnc, CrLeafCnc, NfLeafCnc, PtStemCnc, CrStemCnc, NfStemCnc, PtSOrgCnc, CrSOrgCnc |
| 107.7 | Harvesting | Information about what happens to the crop at harvest and cut. | Root, Stem, SOrg, last cut, production delay, cut delay, sorg height, cut stress |
| 107.8 | Partition | Assimilate partitioning in the default crop model. The 'Root' parameter determine what fraction of the assimilate for growth goes to roots at a given development stage. | Root, Stem, RSR, max WRoot, NNI crit, NNI inc |
| 107.9 | Production | Crop production in the default crop model. | WRoot, ReMobilRt, StemRes, CH2OReleaseRate, E Root, E Stem, E SOrg, r Root |
| 107.10 | Irrigation | Keep track of active irrigation events. Usually not set explicitly, but may be found in a checkpint. | event, solute, name, value, volume, temperature, time left, target |
| 107.11 | Bioincorporation | Biological incorporation of organic matter in soil. | respiration, distribution |

| Section | Name | What it covers | Components |
|---|---|---|---|
| 107.12 | DOM-Element | A single element in a Dissolved Organic Matter pool. | J tertiary, S p, S drain |
| 107.13 | Fetch | A summary file line. | tag |
| 107.14 | FetchPretty | A summary file line. | tag, name |
| 107.15 | Harvest | Log of all harvests during the simulation. | crop, column, sow time, emerge time, flowering time, ripe time, harvest time, water productivity |
| 107.16 | Geometry1D | A one dimensional discretization of the soil. | zplus |
| 107.17 | GeometryRect | A rectangular discretization of the soil. | zplus, xplus |
| 107.18 | HorHeat | Heat capacity and conductivity per horizon. | intervals, K ice |
| 107.19 | SoilLayer | A location and content of a soil layer. The layers apply to the soil section not covered by the 'zones' parameter. | horizon |
| 107.20 | SoilRegion | A location and content of a soil zone. If several zones cover the same soil, the first one listed is used. | volume, horizon |
| 107.21 | Soil | The soil submodel provides the numeric and physical properties of the soil. | horizons, zones, Hydraulic, MaxRootingDepth, dispersivity |
| 107.22 | SoilHeat | Temperature and heat flux in soil. | q lim, value, h frozen, enable ice, ice dt, state, q, conductivity |
| 107.23 | SoilWater | Keep track of water and pressure in the soil matrix. | h, Theta, S permanent, default h, use last, initial h, value, initial Theta |
| 107.24 | Snow | Simulate snow pack on surface. Snow Hydrology , U.S. Corps of Engineers, 1956. | partial day, age, dZs, mtprime, mrprime, rho s, f c, Psa |
| 107.25 | Timestep | Relative time. | seconds, minutes, hours, days, microseconds |
| 107.26 | Weatherdata | Weather data. | T min, AirTemp, Precip, RefEvap, VapPres, RelHum, Latitude, Longitude |
| 107.27 | Time | Year, month, day and hour, minute, second and microsecond. | month, mday, minute, yday, wday |
| 107.28 | IntegerCondClause | If condition is true, return value. | condition, value |
| 107.29 | StringerCondClause | If condition is true, return value. | condition, value |
| 107.30 | Toplevel | The top level syntax for a Daisy setup file. | input, directory, run, path, OBSOLETE, install directory |

**Combinations of the Experiments tab that use this topic.** A drier, wetter or warmer year; Evapotranspiration model and radiation.

# Chapter 36

# The tested parameters

Thirty two parameters of the Daisy reference are run by the workbench. Each was tested with the Daisy program before it was offered, and its default is the value of the program.

## 36.1 Irrigation subsoil permanent of Bioclimate

Long term irrigation below soil surface.

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | irrigation subsoil permanent |
| Type | number |
| Unit | mm/h |
| Default | 0 |
| Sensitivity run | no |

## 36.2 Litter water storage of Bioclimate

Intercepted water on litter.

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | litter water storage |
| Type | number |
| Unit | mm |
| Default | 0 |
| Sensitivity run | no |

## 36.3 Noofintervals of Bioclimate

Number of vertical intervals in which we partition the canopy.

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |

| Item | Value |
|---|---|
| Key | NoOfIntervals |
| Type | integer |
| Unit | n/a |
| Default | 30 |
| Sensitivity run | yes |

## 36.4 Canopy water storage of Bioclimate

Intercepted water on canopy.

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | canopy water storage |
| Type | number |
| Unit | mm |
| Default | 0 |
| Sensitivity run | no |

## 36.5 Max svat iterations of Bioclimate

Max number of svat iterations before giving up on cobvergence.

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | max svat iterations |
| Type | integer |
| Unit | n/a |
| Default | 100 |
| Sensitivity run | no |

## 36.6 Max svat absolute difference of Bioclimate

Maximum absolute difference in svat ea values for convergence.

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | max svat absolute difference |
| Type | number |
| Unit | mm/h |
| Default | 0.01 |
| Sensitivity run | no |

## 36.7 Maxtdiff of Bioclimate

Largest temperature difference for convergence. 19.1. BASE 111

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | maxTdiff |
| Type | number |
| Unit | K |
| Default | 0.02 |
| Sensitivity run | no |

## 36.8 Maxediff of Bioclimate

Largest humidity difference for convergence.

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | maxEdiff |
| Type | number |
| Unit | Pa |
| Default | 1 |
| Sensitivity run | no |

## 36.9 Min sun angle of Bioclimate

Minimum sun angle above ground for some ‘raddist’ and ‘svat’ models. The ‘DPF’ raddist model will zero radiation if the angle is below this, and the

| Item | Value |
|---|---|
| Topic | 19 |
| Component | Bioclimate |
| Key | min sun angle |
| Type | number |
| Unit | rad |
| Default | 0.0628319 |
| Sensitivity run | no |

## 36.10 Max sink total of Chemistry

Maximum allowed sink term as a fraction of total content. If variable timesteps are enabled, Daisy will try to scale down the timestep in

| Item | Value |
|---|---|
| Topic | 24 |
| Component | Chemistry |
| Key | max sink total |
| Type | number |
| Unit | dimensionless |
| Default | 0.5 |
| Sensitivity run | no |

## 36.11 Max sink solute of Chemistry

Maximum allowed sink term as a fraction of solute content. If variable timesteps are enabled, Daisy will try to scale down the timestep

| Item | Value |
|---|---|
| Topic | 24 |
| Component | Chemistry |
| Key | max sink solute |
| Type | number |
| Unit | dimensionless |
| Default | 0.9 |
| Sensitivity run | no |

## 36.12 Max sink secondary of Chemistry

Maximum allowed sink term as a fraction of secondary domain content. If variable timesteps are enabled, Daisy will try to scale down the timestep

| Item | Value |
|---|---|
| Topic | 24 |
| Component | Chemistry |
| Key | max sink secondary |
| Type | number |
| Unit | dimensionless |
| Default | 1.5 |
| Sensitivity run | no |

## 36.13 Min sink total of Chemistry

24.9. NONE 147 Always allow this fraction of total content to be removed by sink term.

| Item | Value |
|---|---|
| Topic | 24 |
| Component | Chemistry |
| Key | min sink total |
| Type | number |
| Unit | dimensionless |
| Default | 0.01 |
| Sensitivity run | no |

## 36.14 L of Drain

Distance between pipes.

| Item | Value |
|---|---|
| Topic | 37 |
| Component | Drain |
| Key | L |
| Type | number |
| Unit | cm |

| Item | Value |
|---|---|
| Default | 1800 |
| Sensitivity run | no |

## 36.15 Rad of Drain

Inner radius of drain pipes.

| Item | Value |
|---|---|
| Topic | 37 |
| Component | Drain |
| Key | rad |
| Type | number |
| Unit | cm |
| Default | 3.5 |
| Sensitivity run | no |

## 36.16 K aquitard of Groundwater

Conductivity of the aquitard.

| Item | Value |
|---|---|
| Topic | 45 |
| Component | Groundwater |
| Key | K aquitard |
| Type | number |
| Unit | cm/h |
| Default | 0.001 |
| Sensitivity run | yes |

## 36.17 Z aquitard of Groundwater

Thickness of the aquitard. The aquitard begins below the bottommost soil horizon.

| Item | Value |
|---|---|
| Topic | 45 |
| Component | Groundwater |
| Key | Z aquitard |
| Type | number |
| Unit | cm |
| Default | 200 |
| Sensitivity run | yes |

## 36.18 Table of Groundwater

Groundwater level. Positive numbers indicate free drainage.

| Item | Value |
|---|---|
| Topic | 45 |

| Item | Value |
|---|---|
| Component | Groundwater |
| Key | table |
| Type | number |
| Unit | cm |
| Default | 1 |
| Sensitivity run | no |

## 36.19 K nh4 of OrganicMatter

Maximal immobilization rate for ammonium.

| Item | Value |
|---|---|
| Topic | 60 |
| Component | OrganicMatter |
| Key | K NH4 |
| Type | number |
| Unit | h-1 |
| Default | 0.020833 |
| Sensitivity run | yes |

## 36.20 K no3 of OrganicMatter

Maximal immobilization rate for nitrate.

| Item | Value |
|---|---|
| Topic | 60 |
| Component | OrganicMatter |
| Key | K NO3 |
| Type | number |
| Unit | h-1 |
| Default | 0.020833 |
| Sensitivity run | yes |

## 36.21 Co2 threshold of OrganicMatter

Turnover rate above which pools will contribute to ‘CO2 fast’.

| Item | Value |
|---|---|
| Topic | 60 |
| Component | OrganicMatter |
| Key | CO2 threshold |
| Type | number |
| Unit | h-1 |
| Default | 0.0001 |
| Sensitivity run | yes |

## 36.22 Min am c of OrganicMatter

Minimal amount of carbon in AOM ensuring it is not removed.

| Item | Value |
|---|---|
| Topic | 60 |
| Component | OrganicMatter |
| Key | min AM C |
| Type | number |
| Unit | g C/m2 |
| Default | 0.5 |
| Sensitivity run | yes |

## 36.23 Min am n of OrganicMatter

Minimal amount of nitrogen in AOM ensuring it is not removed.

| Item | Value |
|---|---|
| Topic | 60 |
| Component | OrganicMatter |
| Key | min AM N |
| Type | number |
| Unit | g N/m2 |
| Default | 0.05 |
| Sensitivity run | no |

## 36.24 Temperature change rate of Surface

Relative change of surface temperature. If you set this to a negative value, surface temperature will reflect temperature of incomming water or ponded

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | temperature change rate |
| Type | number |
| Unit | h-1 |
| Default | 0.5 |
| Sensitivity run | yes |

## 36.25 Epfactor of Surface

The autumn value can be lower, due to muching. With a crop factor of 1.2 a combined Kc of 1.15 is reached at LAI=5. See also [Kjaersgaard et al., 2008]

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | EpFactor |
| Type | number |
| Unit | dimensionless |
| Default | 0.6 |
| Sensitivity run | yes |

## 36.26 Albedo dry of Surface

Albedo of dry soil (pF >= 3)

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | albedo dry |
| Type | number |
| Unit | dimensionless |
| Default | 0.15 |
| Sensitivity run | no |

## 36.27 Albedo wet of Surface

Albedo of wet soil (pf <= 1.7)

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | albedo wet |
| Type | number |
| Unit | dimensionless |
| Default | 0.08 |
| Sensitivity run | no |

## 36.28 Detentioncapacity of Surface

Amount of ponding the surface can retain. If ponding in any part of the surface is above this, exceed will runoff.

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | DetentionCapacity |
| Type | number |
| Unit | mm |
| Default | 1000 |
| Sensitivity run | no |

## 36.29 Reservoirconstant of Surface

Fraction of ponding above DetentionCapacity that runoffs each hour.

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | ReservoirConstant |
| Type | number |
| Unit | h-1 |
| Default | 1 |
| Sensitivity run | yes |

| Item | Value |
|---|---|

## 36.30 Localdetentioncapacity of Surface

Amount of ponding the surface can retain locally. If ponding in any part of the surface is above this, exceed will be distributed to the rest of the surface.

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | LocalDetentionCapacity |
| Type | number |
| Unit | mm |
| Default | 10 |
| Sensitivity run | no |

## 36.31 Z mixing of Surface

Depth of mixing layer in the top of the soil. The mixing layer affect exchange between soil coloids, soil water and the surface, especially in connection with

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | z mixing |
| Type | number |
| Unit | cm |
| Default | 0.1 |
| Sensitivity run | yes |

## 36.32 R mixing of Surface

Resistance to mixing inorganic compounds between soil and ponding. Log Variables

| Item | Value |
|---|---|
| Topic | 87 |
| Component | Surface |
| Key | R mixing |
| Type | number |
| Unit | h/mm |
| Default | 1000000000 |
| Sensitivity run | yes |

# Chapter 37

# ⚖ The factors and responses of the Experiments tab

The Experiments tab offers 58 factors. Each has levels that are tried, a range that is accepted, and words that make the tab propose it when the question is typed in words. Fourteen responses are measured in every run.

## 37.1 Responses

| Response | Unit | Better when |
|---|---|---|
| Yield | Mg DM/ha | higher |
| Irrigation | mm/season | lower |
| Evapotranspiration | mm/season | not ranked |
| Percolation below 100 cm | mm/season | lower |
| Nitrate leached below 100 cm | kg N/ha/season | lower |
| Denitrification | kg N/ha/season | lower |
| Irrigation efficiency | kg grain per m3 | higher |
| Rain | mm/season | not ranked |
| Water-stress days at harvest | days | lower |
| Nitrogen-stress days at harvest | days | lower |
| Irrigation water-use efficiency (yield gain over the rainfed reference per mm) | kg DM per ha per mm | higher |
| Deep-percolation fraction of rain + irrigation | % | lower |
| Partial factor productivity of nitrogen | kg DM per kg N | higher |
| Nitrate leached as a share of N applied | % | lower |

## 37.2 Factors

Each row of *What will be tested* is one factor: one thing that changes between the runs. The text under the row says what the factor is; *more* opens where Daisy reads it, the typical values, what to type and what changes in the results. This chapter gives the same for all 63 factors.

### 37.2.1 How to fill in a row

A number factor has a box with its values, separated by commas, and a note after it: the unit and *allowed a to b*. Each value is one **level**: the runs combine the levels of every row. Three rows of

the Experiments tab as an example:

| Row as it appears | What to type | What it means |
|---|---|---|
| OrganicMatter original: min AM N, g N/m2 · allowed 0 to 0.5 | 0.025, 0.05, 0.1 | The same limit for nitrogen: an added organic matter pool with less nitrogen than this is merged into the soil. Typical: 0.025 to 0.1 g N per m² (default 0.05 g N/m², that is 0.5 kg N/ha). The workbench allows 0 to 0.5. |
| Surface: DetentionCapacity, mm · allowed 0 to 50 | 2.5, 5, 10 | The water depth that can stand (pond) on the field before the excess runs off. Typical: 2 to 10 mm for an arable field (rough, ploughed land holds more); the workbench allows 0 to 50. |
| Surface: LocalDetentionCapacity, mm · allowed 0 to 100 | 5, 10, 20 | The water that can pond in one part (patch) of the field before the excess spreads to the rest of the field. Typical: 5 to 20 mm (default 10); the workbench allows 0 to 100. |

*Allowed* is what the box accepts: from 0 to ten times the value the farm uses (or the Daisy default when the farm sets none). It is a guard against typing errors, not a limit of Daisy, and the sensible values are much narrower: they are given as *Typical* for every factor below. A value outside the box range is moved to its nearest end.

A choice factor (a soil, a fertiliser, a tillage, a crop of the library) has a list with a search box: tick the choices to compare. Every ticked choice is one level.

### 37.2.2 The number of runs

Under the rows: *How to combine* (every combination, one thing at a time, or the values in pairs), *number of runs*, *if there are more, choose*, *seed* and *rainfed references*. The number of runs is the number Daisy makes, the rainfed references included, and it is never exceeded:

1. The combinations are counted (for every combination: the product of the numbers of levels).
2. When they fit, every combination is run.
3. When there are more, as many as fit are chosen in the way set:
   - **balanced**: every value used about equally often.
   - **at random (with the seed)**.
   - **evenly spaced through the list of combinations**.
   - **the first ones in the list, in order**.
4. Irrigation factors need a rainfed reference for every soil and weather of the chosen runs, so the gain of irrigation is measured against the same soil and weather. They count as runs: with references the runs come in pairs, and an odd number gives one run fewer. Untick *rainfed references* to get exactly the number set.
5. The same plan and seed always give the same runs. *See the runs* lists them and counts how often each value is used.

The balanced choice is a Latin hypercube over the levels: each level of each factor is used in about the same number of runs, and the levels are mixed at random between the factors, so a small number of runs still covers every value. *My experiments* keeps the number, the way of choosing and the seed of every experiment, and shows how many runs each gives.

### 37.2.3 All factors at a glance

| Group | Factor | Unit | Typical values |
|---|---|---|---|
| Weather and climate | Rain, factor on the weather | n/a | 0.6 (very dry) to 1.4 (very wet); 1 is the weather as it is. |
| Weather and climate | Temperature shift | C | 0 (as it is) to +4 °C; negative values cool. |
| Weather and climate | Radiation, factor on the weather | n/a | 0.8 to 1.2. |

| Group | Factor | Unit | Typical values |
|---|---|---|---|
| Weather and climate | Seasonal rain pattern | n/a | The patterns offered. |
| Weather and climate | Weather station | n/a | Each station is a climate: Taastrup and Foulum (Denmark, humid), Bologna (Italy, dry summers). |
| Weather and climate | Evapotranspiration model | n/a | FAO-56 Penman-Monteith (daily) is the standard. |
| Nitrogen and tillage | Nitrogen fertiliser, factor on the rotation | n/a | 0 (no fertiliser) to 1.6. |
| Nitrogen and tillage | Type of mineral fertiliser | n/a | N25S (the farm), AmmoniumNitrate, Nitrate, Ammonia, CalciumNitrate; 72 slurries and manures. |
| Nitrogen and tillage | Fertiliser date, shift in days | days | -21 to +28 days. |
| Nitrogen and tillage | Tillage | n/a | plowing (the farm), rotavation, disk_harrowing, stubble_cultivation, none. |
| Nitrogen and tillage | Soil consolidation after tillage (time consolidation) | per day | 0.001 (slow) to 0.02 per day (fast). |
| Irrigation | Irrigation | n/a | rainfed, irrigated. |
| Irrigation | Water per irrigation | mm | 10 to 45 mm; light soils take small doses, heavy soils larger ones. |
| Irrigation | Duration of one irrigation | h | 1 to 24 h. |
| Irrigation | Irrigation intensity (constant flux) | mm/h | 0.05 to 2 mm/h. |
| Irrigation | Irrigation trigger (soil suction) | cm | -100 cm (very wet, pF 2) to -1000 cm (dry, pF 3); -300 cm is common. |
| Irrigation | Sensor depth | cm | 10 to 60 cm; inside the main root zone. |
| Irrigation | Irrigation method | n/a | overhead, surface, drip. |
| Irrigation | Drip depth | cm | 5-10, 10-20, 20-30 cm. |
| Irrigation | Water limit per crop season | mm | 60 to 240 mm (deficit irrigation); 0 for none. |
| Irrigation | Irrigation window | n/a | April to October in the north; shorter windows save water. |
| Soil and crop | Soil | n/a | The 12 USDA textures, the built-in loam and sand, and 63 horizons of the Daisy library. |
| Soil and crop | Maximum rooting depth | cm | 30 (shallow soil) to 150 cm. |
| Soil and crop | Crop of the Daisy library, grown every year | n/a | 84 crops of the library. |
| Soil and crop | Saturated hydraulic conductivity K_sat of every horizon (van Genuchten-Mualem), factor | times the soil value | 0.1 to 10 times; K_sat varies more than tenfold within one texture class. |
| Soil and crop | van Genuchten alpha (inverse air-entry pressure) of every horizon, factor | times the soil value | 0.5 to 2 times. |
| Soil and crop | van Genuchten n (pore-size distribution) of every horizon, factor on n - 1 | times the soil value | 0.7 to 1.3 times. |
| Soil and crop | Saturated water content Theta_sat (porosity) of every horizon, factor | times the soil value | 0.9 to 1.1 times. |
| Soil and crop | Mualem tortuosity l of every horizon, added | added to the soil value | -2 to +2 around the soil value (0.5 in Mualem 1976; fitted values are often negative). |
| Chemicals | Chemical of the Daisy library (pesticide, PFAS, tracer) | n/a | 35 chemicals of the library. |
| Chemicals | Sorption of the chemical, K_OC | L/kg | 0 (moves with water) to 1000 L/kg (strongly held). |
| Chemicals | Half-life of the chemical | days | 5 to 200 days. |
| The Daisy library | Management of the Daisy library (whole managements and single operations) | n/a | 82 actions of the library. |
| The Daisy library | Program of the Daisy library, run as it comes with Daisy | n/a | 55 programs. |
| Daisy reference, Bioclimate | NoOfIntervals | n/a | 10 to 60; 30 is the default. |

| Group | Factor | Unit | Typical values |
|---|---|---|---|
| Daisy reference, Bioclimate | max svat iterations | n/a | 50 to 200. |
| Daisy reference, Bioclimate | max svat absolute difference | mm/h | 0.005 to 0.02 mm/h. |
| Daisy reference, Bioclimate | maxTdiff | K | 0.01 to 0.05 K. |
| Daisy reference, Bioclimate | maxEdiff | Pa | 0.5 to 2 Pa. |
| Daisy reference, Bioclimate | min sun angle | rad | 0.03 to 0.13 rad (2 to 7 degrees). |
| Daisy reference, Chemistry original | max sink total | dimensionless | 0.25 to 1. |
| Daisy reference, Chemistry original | max sink solute | dimensionless | 0.5 to 1. |
| Daisy reference, Chemistry original | max sink secondary | dimensionless | 1 to 3. |
| Daisy reference, Chemistry original | min sink total | dimensionless | 0.005 to 0.02. |
| Daisy reference, Drain lateral | L | cm | 10 to 40 m (1000 to 4000 cm). |
| Daisy reference, Drain lateral | rad | cm | 2.5 to 7 cm. |
| Daisy reference, Groundwater aquitard | K aquitard | cm/h | 0.0001 (tight clay) to 0.01 cm/h. |
| Daisy reference, Groundwater aquitard | Z aquitard | cm | 50 to 500 cm. |
| Daisy reference, Groundwater static | table | cm | -100 to -300 cm for a real table; 1 for free drainage. |
| Daisy reference, OrganicMatter original | K NH4 | h-1 | 0.01 to 0.04 per hour (default 0.0208). |
| Daisy reference, OrganicMatter original | K NO3 | h-1 | 0.01 to 0.04 per hour (default 0.0208). |
| Daisy reference, OrganicMatter original | CO2 threshold | h-1 | 0.00005 to 0.0002 per hour. |
| Daisy reference, OrganicMatter original | min AM C | g C/m2 | 0.25 to 1 g C per m² (default 0.5). |
| Daisy reference, OrganicMatter original | min AM N | g N/m2 | 0.025 to 0.1 g N per m² (default 0.05 g N/m², that is 0.5 kg N/ha). The workbench allows 0 to 0.5. |
| Daisy reference, Surface | temperature change rate | h-1 | 0.25 to 1 per hour. |
| Daisy reference, Surface | EpFactor | dimensionless | 0.4 to 1 (default 0.6). |
| Daisy reference, Surface | albedo dry | dimensionless | 0.1 to 0.3 (light soils higher). |
| Daisy reference, Surface | albedo wet | dimensionless | 0.05 to 0.15. |
| Daisy reference, Surface | DetentionCapacity | mm | 2 to 10 mm for an arable field (rough, ploughed land holds more); the workbench allows 0 to 50. |
| Daisy reference, Surface | ReservoirConstant | h-1 | 1 (slow) to 60 per hour. |
| Daisy reference, Surface | LocalDetentionCapacity | mm | 5 to 20 mm (default 10); the workbench allows 0 to 100. |
| Daisy reference, Surface | z mixing | cm | 0.05 to 0.2 cm. |
| Daisy reference, Surface | R mixing | h/mm | Very large values (1e9 h/mm) mean almost no mixing. |

## 37.2.4 Weather and climate

### 37.2.4.1 Rain, factor on the weather

A factor on every rain of the weather file: 0.8 is 20 % less rain, 1.2 is 20 % more. The days with rain stay the same; each amount is multiplied.

| Item | Value |
|---|---|
| Where Daisy reads it | PrecipCorrect in the (weather …) of the run file. |
| Typical values | 0.6 (very dry) to 1.4 (very wet); 1 is the weather as it is. |
| Levels offered | 0.6, 0.8, 1, 1.2, 1.4 |
| Box accepts | 0.4 to 1.6 n/a |
| What to type | Several factors separated by commas, for example 0.7, 0.85, 1, 1.15. |
| What changes | Drainage and nitrate leaching follow rain most strongly; yield follows it on sandy soils without irrigation; irrigation need falls as rain rises. |
| Needs irrigation | no |
| Words that propose it | rain, rainfall, precipitation, wetter, drier, dry year, wet year, drought |

### 37.2.4.2 Temperature shift

Degrees added to every daily temperature of the weather file (a simple climate change).

| Item | Value |
|---|---|
| Where Daisy reads it | TempOffset in the (weather …) of the run file. |
| Typical values | 0 (as it is) to +4 °C; negative values cool. |
| Levels offered | 0, 1, 2, 3 |
| Box accepts | -4 to 8 C |
| What to type | Degrees, for example 0, 1, 2, 3. |
| What changes | Faster crop development (earlier ripening, often lower yield of cereals), more evapotranspiration and faster turnover of organic matter (more mineral nitrogen). |
| Needs irrigation | no |
| Words that propose it | temperature, warming, warmer, hotter, cooler, climate change, heat |

### 37.2.4.3 Radiation, factor on the weather

A factor on the global radiation of the weather file.

| Item | Value |
|---|---|
| Where Daisy reads it | GlobRadScale in the (weather …) of the run file. |
| Typical values | 0.8 to 1.2. |
| Levels offered | 0.9, 1, 1.1 |
| Box accepts | 0.5 to 1.5 n/a |
| What to type | Factors, for example 0.9, 1, 1.1. |
| What changes | Photosynthesis, potential evapotranspiration and the soil surface temperature. |
| Needs irrigation | no |
| Words that propose it | radiation, cloud, sunshine, solar |

### 37.2.4.4 Seasonal rain pattern

Changes the rain of chosen months only (for example a drier summer and a wetter winter).

| Item | Value |
|---|---|
| Where Daisy reads it | The weather file is rewritten month by month before the run. |
| Typical values | The patterns offered. |
| Levels offered | dry summer (rain x0.7 Jun-Aug), dry spring (rain x0.7 Mar-May), wet winter (rain x1.3 Dec-Feb) |

| Item | Value |
|---|---|
| What to type | Tick the patterns. |
| What changes | Summer patterns change yield and irrigation; winter patterns change drainage and leaching. |
| Needs irrigation | no |
| Words that propose it | seasonal, dry summer, dry spring, wet winter, monthly rain, drought in summer |

#### 37.2.4.5 Weather station

The weather file of the run: a built-in station, the weather of your SWAFL project or your own file.

| Item | Value |
|---|---|
| Where Daisy reads it | The .dwf in Common/ read by (weather default …). |
| Typical values | Each station is a climate: Taastrup and Foulum (Denmark, humid), Bologna (Italy, dry summers). |
| Levels offered | course, it-bologna, dk-foulum |
| What to type | Tick the stations to compare. |
| What changes | Everything: rain, temperature, radiation and their timing. |
| Needs irrigation | no |
| Words that propose it | station, site, location, climate of, bologna, foulum, jyndevad, valecov, jadwisin, bratislava |

#### 37.2.4.6 Evapotranspiration model

The model of evapotranspiration Daisy uses (the bioclimate).

| Item | Value |
|---|---|
| Where Daisy reads it | (Bioclimate …) of the column. |
| Typical values | FAO-56 Penman-Monteith (daily) is the standard. |
| Levels offered | auto, default |
| What to type | Tick the models to compare. |
| What changes | Potential and actual evapotranspiration, hence soil water, irrigation need and drainage. |
| Needs irrigation | no |
| Words that propose it | evapotranspiration model, penman, fao56, bioclimate, energy balance |

### 37.2.5 Nitrogen and tillage

#### 37.2.5.1 Nitrogen fertiliser, factor on the rotation

A factor on every fertiliser dose of the rotation: 1 is the farm, 0.5 half the nitrogen, 1.3 thirty per cent more.

| Item | Value |
|---|---|
| Where Daisy reads it | The (fertilize … (weight N [kg N/ha])) actions in the rotation file of Common/. |
| Typical values | 0 (no fertiliser) to 1.6. |
| Levels offered | 0.4, 0.7, 1, 1.3 |
| Box accepts | 0 to 2 n/a |
| What to type | Factors, for example 0, 0.5, 1, 1.5. |
| What changes | Yield rises then levels off; nitrate leaching rises steeply above the economic optimum. |
| Needs irrigation | no |
| Words that propose it | nitrogen, fertiliser, fertilizer, n rate, n dose, leaching, nitrate, pollution |

#### 37.2.5.2 Type of mineral fertiliser

The kind of mineral fertiliser (or slurry) given at every fertilisation of the rotation.

| Item | Value |
|---|---|
| Where Daisy reads it | The first word of (fertilize ( ...)) in the rotation file; slurries come from dk-fertilizer.dai. |
| Typical values | N25S (the farm), AmmoniumNitrate, Nitrate, Ammonia, CalciumNitrate; 72 slurries and manures. |
| Levels offered | N25S, AmmoniumNitrate, Ammonia, Nitrate |
| What to type | Tick the types. A slurry is given with the same total nitrogen as the mineral fertiliser. |
| What changes | Ammonia volatilisation, nitrification and leaching; slurries also add organic carbon and nitrogen to the soil. |
| Needs irrigation | no |
| Words that propose it | fertiliser type, fertilizer type, ammonium nitrate, ammonia, urea, form of nitrogen |

#### 37.2.5.3 Fertiliser date, shift in days

Moves every fertilisation of the rotation by this many days (negative: earlier).

| Item | Value |
|---|---|
| Where Daisy reads it | The (wait_mm_dd ...) before each (fertilize ...) in the rotation file. |
| Typical values | -21 to +28 days. |
| Levels offered | -14, 0, 14, 28 |
| Box accepts | -60 to 60 days |
| What to type | Days, for example -14, 0, 14. |
| What changes | Early: more leaching before the crop takes it up. Late: nitrogen stress in spring. |
| Needs irrigation | no |
| Words that propose it | fertiliser timing, fertilizer timing, when to fertili, split application, delay the fertili |

#### 37.2.5.4 Tillage

The tillage operation done before sowing.

| Item | Value |
|---|---|
| Where Daisy reads it | The tillage action of the rotation (plowing, rotavation, disk_harrowing, stubble_cultivation) or none. |
| Typical values | plowing (the farm), rotavation, disk_harrowing, stubble_cultivation, none. |
| Levels offered | plowing, rotavation, none |
| What to type | Tick the operations. |
| What changes | Mixing of residues and organic matter, the loosening of the topsoil (with consolidation) and so infiltration and evaporation. |
| Needs irrigation | no |
| Words that propose it | tillage, plough, plow, plowing, ploughing, rotavat, no-till, no till, cultivation |

#### 37.2.5.5 Soil consolidation after tillage (time consolidation)

How fast the topsoil loosened by tillage settles back (consolidation rate).

| Item | Value |
|---|---|
| Where Daisy reads it | time_consolidation in the tillage of the column (soil structure dynamics). |
| Typical values | 0.001 (slow) to 0.02 per day (fast). |
| Levels offered | 0.001, 0.01, 0.02 |
| Box accepts | 0.0005 to 0.05 per day |
| What to type | Rates, for example 0.001, 0.01, 0.02. |

| Item | Value |
|---|---|
| What changes | The water holding and conductivity of the tilled layer in the weeks after tillage. |
| Needs irrigation | no |
| Words that propose it | consolidation, time consolidation, soil structure, structure dynamics, wepp, shp, hydraulic properties in time, settling |

## 37.2.6 Irrigation

### 37.2.6.1 Irrigation

Whether the crop is irrigated at all.

| Item | Value |
|---|---|
| Where Daisy reads it | An (irrigate …) action in the management, or none. |
| Typical values | rainfed, irrigated. |
| Levels offered | 0, 1 |
| What to type | Tick one or both. |
| What changes | The difference between the two is the gain from irrigation. |
| Needs irrigation | no |
| Words that propose it | irrigation, irrigate, irrigated, rainfed |

### 37.2.6.2 Water per irrigation

The water given at each irrigation event.

| Item | Value |
|---|---|
| Where Daisy reads it | Written as a rate times a duration in (irrigate_overhead/surface/subsoil [mm/h] (hours )): 30 mm in 3 h is 10 mm/h. Above 20 mm/h the hours are lengthened, because Daisy solves the water flow better at moderate rates. |
| Typical values | 10 to 45 mm; light soils take small doses, heavy soils larger ones. |
| Levels offered | 10, 20, 30, 45 |
| Box accepts | 2 to 60 mm |
| What to type | Millimetres per event, for example 10, 20, 30. |
| What changes | Large doses on sand drain below the roots (deep percolation and leaching); small doses need more events. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | dose, amount, depth of water, mm per, water per irrigation |

### 37.2.6.3 Duration of one irrigation

How long one irrigation event lasts.

| Item | Value |
|---|---|
| Where Daisy reads it | (hours ) of the irrigate action; the rate is the dose divided by the hours. |
| Typical values | 1 to 24 h. |
| Levels offered | 3, 6, 12, 24 |
| Box accepts | 1 to 48 h |
| What to type | Hours, for example 3, 6, 12. |
| What changes | Long, slow events infiltrate better and run off less. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | duration, hours, how long, slow, intensity |

### 37.2.6.4 Irrigation intensity (constant flux)

A constant irrigation intensity instead of events (mm per hour, continuously while irrigating).

| Item | Value |
|---|---|
| Where Daisy reads it | The rate of the irrigate action. |
| Typical values | 0.05 to 2 mm/h. |
| Levels offered | 0.05, 0.1, 0.5, 2 |
| Box accepts | 0.01 to 20 mm/h |
| What to type | Rates, for example 0.1, 0.5, 1. |
| What changes | Low intensities imitate drip; high ones may pond. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | flux, intensity, mm per hour, mm/h, constant depth |

#### 37.2.6.5 Irrigation trigger (soil suction)

The soil water suction at the sensor that starts an irrigation (more negative is drier soil).

| Item | Value |
|---|---|
| Where Daisy reads it | The irrigation waits for (not (soil_water_pressure_above (height - [cm]) (potential [cm]))) in the irrigation action of the rotation file. |
| Typical values | -100 cm (very wet, pF 2) to -1000 cm (dry, pF 3); -300 cm is common. |
| Levels offered | -100, -200, -300, -600, -1000 |
| Box accepts | -10000 to -50 cm |
| What to type | Negative centimetres of water, for example -150, -300, -600. |
| What changes | Wetter triggers give more irrigation and drainage and less water stress. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | trigger, threshold, suction, tensiometer, pressure, when to irrigate |

#### 37.2.6.6 Sensor depth

The depth of the soil moisture sensor that triggers irrigation.

| Item | Value |
|---|---|
| Where Daisy reads it | (height - [cm]) of the soil_water_pressure_above condition. |
| Typical values | 10 to 60 cm; inside the main root zone. |
| Levels offered | 10, 20, 30, 50 |
| Box accepts | 5 to 100 cm |
| What to type | Depths in cm, for example 10, 20, 30. |
| What changes | Shallow sensors react fast and irrigate often; deep ones react late. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | sensor, sensor depth, probe, installation depth, where to put the sensor |

#### 37.2.6.7 Irrigation method

How the water is applied.

| Item | Value |
|---|---|
| Where Daisy reads it | irrigate_overhead (on the canopy), irrigate_surface (on the soil) or irrigate_subsoil (drip, in a depth range). |
| Typical values | overhead, surface, drip. |
| Levels offered | overhead, surface, drip |
| What to type | Tick the methods. |
| What changes | Overhead water is partly intercepted and evaporated; drip avoids this and wets the root zone directly. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | method, sprinkler, overhead, drip, surface irrigation, furrow, flood |

#### 37.2.6.8 Drip depth

The depth range where drip water is given.

| Item | Value |
|---|---|
| Where Daisy reads it | (from -a [cm]) (to -b [cm]) of irrigate_subsoil. |
| Typical values | 5-10, 10-20, 20-30 cm. |
| Levels offered | drip 5-10 cm, drip 10-20 cm, drip 15-25 cm, drip 20-30 cm |
| What to type | Tick the ranges. |
| What changes | Deeper drip loses less to evaporation but may wet below young roots. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | drip depth, buried drip, subsurface drip |

#### 37.2.6.9 Water limit per crop season

The most water that may be given in one crop season (0 is no limit).

| Item | Value |
|---|---|
| Where Daisy reads it | The irrigation action is allowed limit / dose times per season (an (activity …) of that many events instead of (repeat …)). |
| Typical values | 60 to 240 mm (deficit irrigation); 0 for none. |
| Levels offered | 0, 60, 120, 240 |
| Box accepts | 0 to 1000 mm |
| What to type | Millimetres per season, for example 0, 60, 120. |
| What changes | Water saved against yield lost: the basis of deficit irrigation. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | limit, deficit, water limit, quota, restricted, scarce water, allowance |

#### 37.2.6.10 Irrigation window

The dates between which irrigation is allowed.

| Item | Value |
|---|---|
| Where Daisy reads it | (after_mm_dd …) and (before_mm_dd …) in the wait of the irrigation action. |
| Typical values | April to October in the north; shorter windows save water. |
| Levels offered | Apr 1 - Oct 1, May 1 - Oct 1, Jun 1 - Oct 1, Apr 1 - Jun 1 |
| What to type | Tick the windows. |
| What changes | Early or late water mostly drains; the window should cover flowering and grain filling. |
| Needs irrigation | yes: the runs are irrigated, with a rainfed reference |
| Words that propose it | timing, window, season of irrigation, when in the season, start date |

### 37.2.7 Soil and crop

#### 37.2.7.1 Soil

The soil profile of the column.

| Item | Value |
|---|---|
| Where Daisy reads it | The horizons and the (defcolumn …) written in Common/ (UserSoil.dai) or the built-in loam of the farm. |
| Typical values | The 12 USDA textures, the built-in loam and sand, and 63 horizons of the Daisy library. |
| Levels offered | usda:sand, usda:sandyloam, loam, usda:siltloam, usda:clayloam |
| What to type | Tick the soils to compare. |

| Item | Value |
|---|---|
| What changes | Water holding (yield without irrigation), drainage and leaching, the right irrigation dose. |
| Needs irrigation | no |
| Words that propose it | soil, soils, texture, sand, loam, clay, silt |

#### 37.2.7.2 Maximum rooting depth

The deepest the roots may grow (0: the depth of the crop file).

| Item | Value |
|---|---|
| Where Daisy reads it | (MaxRootingDepth [cm]) of the soil in ResFarm.dai: the soil limits how deep any crop roots. |
| Typical values | 30 (shallow soil) to 150 cm. |
| Levels offered | 30, 60, 90, 120, 150 |
| Box accepts | 20 to 200 cm |
| What to type | Centimetres, for example 60, 90, 120. |
| What changes | Deeper roots reach more water and nitrate: higher yield without irrigation, less leaching. |
| Needs irrigation | no |
| Words that propose it | root, rooting depth, roots |

#### 37.2.7.3 Crop of the Daisy library, grown every year

One crop of the Daisy library grown every year instead of the rotation of the farm.

| Item | Value |
|---|---|
| Where Daisy reads it | A (defaction … (sow “”) … (harvest …)) written in the rotation file. |
| Typical values | 84 crops of the library. |
| Levels offered | n/a |
| What to type | Tick the crops. |
| What changes | Yield, water use and nitrogen uptake of each crop on the same soil and weather. |
| Needs irrigation | no |
| Words that propose it | crop of the library, any crop, grow every year, monoculture |

#### 37.2.7.4 Saturated hydraulic conductivity K_sat of every horizon (van Genuchten-Mualem), factor

Multiplies the saturated hydraulic conductivity K_sat of every horizon of the built-in soil: how fast the saturated soil conducts water.

| Item | Value |
|---|---|
| Where Daisy reads it | (hydraulic M_vG … (K_sat )) of every horizon in ResFarm.dai (built-in loam and sand; own soils use HYPRES). |
| Typical values | 0.1 to 10 times; K_sat varies more than tenfold within one texture class. |
| Levels offered | 0.25, 1, 4 |
| Box accepts | 0.01 to 100 times the soil value |
| What to type | Factors, for example 0.25, 1, 4. |
| What changes | Higher K_sat: faster infiltration, less runoff and ponding, more and earlier drainage and leaching; lower: wetter topsoil, more runoff and denitrification. |
| Needs irrigation | no |
| Words that propose it | ksat, k_sat, saturated conductivity, hydraulic conductivity, permeability, sensitivity |

#### 37.2.7.5 van Genuchten alpha (inverse air-entry pressure) of every horizon, factor

Multiplies alpha of the van Genuchten retention curve of every horizon (1/alpha is roughly the air-entry pressure).

| Item | Value |
|---|---|
| Where Daisy reads it | (hydraulic M_vG ... (alpha <cm^-1>)) of every horizon in ResFarm.dai. |
| Typical values | 0.5 to 2 times. |
| Levels offered | 0.5, 1, 2 |
| Box accepts | 0.1 to 10 times the soil value |
| What to type | Factors, for example 0.5, 1, 2. |
| What changes | Larger alpha: the soil drains at smaller suctions, holds less water at field capacity, less plant-available water and more drainage. |
| Needs irrigation | no |
| Words that propose it | alpha, van genuchten, air entry, retention curve, sensitivity |

#### 37.2.7.6 van Genuchten n (pore-size distribution) of every horizon, factor on n - 1

Scales n - 1 of the van Genuchten curve of every horizon (n stays above 1): the spread of pore sizes.

| Item | Value |
|---|---|
| Where Daisy reads it | (hydraulic M_vG ... (n <>)) of every horizon in ResFarm.dai; n becomes 1 + (n - 1) x factor. |
| Typical values | 0.7 to 1.3 times. |
| Levels offered | 0.7, 1, 1.3 |
| Box accepts | 0.3 to 2 times the soil value |
| What to type | Factors, for example 0.7, 1, 1.3. |
| What changes | Larger n: a steeper retention curve (sand-like), less water held at high suction, conductivity falling fast as the soil dries. |
| Needs irrigation | no |
| Words that propose it | van genuchten n, pore size distribution, retention curve, shape parameter, sensitivity |

#### 37.2.7.7 Saturated water content Theta_sat (porosity) of every horizon, factor

Multiplies the saturated water content Theta_sat (about the porosity) of every horizon.

| Item | Value |
|---|---|
| Where Daisy reads it | (hydraulic M_vG ... (Theta_sat <>)) of every horizon in ResFarm.dai; kept below 0.7. |
| Typical values | 0.9 to 1.1 times. |
| Levels offered | 0.9, 1, 1.1 |
| Box accepts | 0.6 to 1.5 times the soil value |
| What to type | Factors, for example 0.9, 1, 1.1. |
| What changes | More pore space stores more water: more water for the crop and slower drainage. |
| Needs irrigation | no |
| Words that propose it | theta_sat, porosity, saturated water content, sensitivity |

#### 37.2.7.8 Mualem tortuosity l of every horizon, added

Adds to the Mualem tortuosity-connectivity parameter l of every horizon.

| Item | Value |
|---|---|
| Where Daisy reads it | (hydraulic M_vG ... (l <>)) of every horizon in ResFarm.dai. |

| Item | Value |
|---|---|
| Typical values | -2 to +2 around the soil value (0.5 in Mualem 1976; fitted values are often negative). |
| Levels offered | -1, 0, 1 |
| Box accepts | -4 to 4 added to the soil value |
| What to type | Values added, for example -1, 0, 1. |
| What changes | Higher l: the unsaturated conductivity falls faster as the soil dries, less capillary rise and slower redistribution. |
| Needs irrigation | no |
| Words that propose it | tortuosity, mualem l, connectivity, sensitivity |

## 37.2.8 Chemicals

### 37.2.8.1 Chemical of the Daisy library (pesticide, PFAS, tracer)

A chemical of the Daisy library sprayed with every fertilisation (pesticides, PFAS, tracers).

| Item | Value |
|---|---|
| Where Daisy reads it | (defchemical ...) and (spray "" ...) in the rotation file. |
| Typical values | 35 chemicals of the library. |
| Levels offered | n/a |
| What to type | Tick the chemicals. |
| What changes | How much is leached, degraded and left in the soil. |
| Needs irrigation | no |
| Words that propose it | pesticide, pfas, atrazine, bentazon, which chemical, herbicide |

### 37.2.8.2 Sorption of the chemical, K_OC

The sorption of the tracer chemical to organic carbon (K_OC). Higher values hold the chemical in the topsoil.

| Item | Value |
|---|---|
| Where Daisy reads it | (adsorption linear (K_OC [L/kg])) of the chemical. |
| Typical values | 0 (moves with water) to 1000 L/kg (strongly held). |
| Levels offered | 0, 50, 100, 500 |
| Box accepts | 0 to 5000 L/kg |
| What to type | L/kg, for example 0, 50, 500. |
| What changes | Leaching falls steeply as K_OC rises. |
| Needs irrigation | no |
| Words that propose it | sorption, koc, pesticide, chemical, tracer, herbicide |

### 37.2.8.3 Half-life of the chemical

The half-life of the tracer chemical in the soil (0: it does not decay).

| Item | Value |
|---|---|
| Where Daisy reads it | decompose_rate of the chemical, as ln 2 / half-life. |
| Typical values | 5 to 200 days. |
| Levels offered | 0, 15, 30, 90 |
| Box accepts | 0 to 2000 days |
| What to type | Days, for example 10, 50, 150. |
| What changes | Short half-lives leave little to leach. |
| Needs irrigation | no |
| Words that propose it | half-life, half life, degradation, decay, persistence |

## 37.2.9 The Daisy library

### 37.2.9.1 Management of the Daisy library (whole managements and single operations)

A whole management or a single operation of the Daisy library.

| Item | Value |
|---|---|
| Where Daisy reads it | The (defaction ...) of the library read by the run. |
| Typical values | 82 actions of the library. |
| Levels offered | n/a |
| What to type | Tick the managements. |
| What changes | Depends on the management. |
| Needs irrigation | no |
| Words that propose it | management, managements, farm management, action |

### 37.2.9.2 Program of the Daisy library, run as it comes with Daisy

A program of the Daisy library, run as it comes with Daisy.

| Item | Value |
|---|---|
| Where Daisy reads it | The (defprogram ...) of the library; the run file only calls it. |
| Typical values | 55 programs. |
| Levels offered | n/a |
| What to type | Tick the programs. |
| What changes | Each program is its own simulation. |
| Needs irrigation | no |
| Words that propose it | program, sample program, example run |

## 37.2.10 Parameters of the Daisy reference

These 29 parameters were each written into a run and accepted by Daisy 7.1.14 at the default and at half and double that value. Their levels are fractions of the value the farm uses: 0.5, 0.75, 1, 1.5 and 2 times.

### 37.2.10.1 Bioclimate

**37.2.10.1.1 NoOfIntervals (n/a)** How many layers the canopy is split into for light and energy.

| Item | Value |
|---|---|
| Where Daisy reads it | Bioclimate (chapter 19). |
| Daisy default; the farm | 30; not set (the default is used) |
| Typical values | 10 to 60; 30 is the default. |
| Levels offered | 15, 23, 30, 45, 60 |
| Box accepts | 0 to 300 n/a |
| What to type | Whole numbers, for example 15, 30, 60. |
| What changes | A numerical setting: small changes in transpiration; more layers run slower. |
| Needs irrigation | no |
| Daisy reference | chapter 19: Number of vertical intervals in which we partition the canopy. |
| Words that propose it | bioclimate noofintervals, noofintervals of bioclimate, noofintervals |

**37.2.10.1.2 max svat iterations (n/a)** How many times the canopy energy balance (SVAT) may be repeated in a time step before Daisy accepts the result.

| Item | Value |
|---|---|
| Where Daisy reads it | Bioclimate (chapter 19). |
| Daisy default; the farm | 100; not set (the default is used) |
| Typical values | 50 to 200. |
| Levels offered | 50, 75, 100, 150, 200 |
| Box accepts | 0 to 1000 n/a |
| What to type | Whole numbers. |
| What changes | A numerical setting: no change to yield, water or nitrogen in the test. |
| Needs irrigation | no |
| Daisy reference | chapter 19: Max number of svat iterations before giving up on convergence. |
| Words that propose it | bioclimate max svat iterations, max_svat_iterations of bioclimate, max svat iterations, max_svat_iterations |

#### 37.2.10.1.3 max svat absolute difference (mm/h)

The accuracy at which the SVAT evaporation is accepted.

| Item | Value |
|---|---|
| Where Daisy reads it | Bioclimate (chapter 19). |
| Daisy default; the farm | 0.01; not set (the default is used) |
| Typical values | 0.005 to 0.02 mm/h. |
| Levels offered | 0.005, 0.0075, 0.01, 0.015, 0.02 |
| Box accepts | 0 to 0.1 mm/h |
| What to type | mm/h. |
| What changes | A numerical setting. |
| Needs irrigation | no |
| Daisy reference | chapter 19: Maximum absolute difference in svat ea values for convergence. |
| Words that propose it | bioclimate max svat absolute difference, max_svat_absolute_difference of bioclimate, max svat absolute difference, max_svat_absolute_difference |

#### 37.2.10.1.4 maxTdiff (K)

The accuracy of the canopy temperature in the SVAT solution.

| Item | Value |
|---|---|
| Where Daisy reads it | Bioclimate (chapter 19). |
| Daisy default; the farm | 0.02; not set (the default is used) |
| Typical values | 0.01 to 0.05 K. |
| Levels offered | 0.01, 0.015, 0.02, 0.03, 0.04 |
| Box accepts | 0 to 0.2 K |
| What to type | Kelvin. |
| What changes | A numerical setting. |
| Needs irrigation | no |
| Daisy reference | chapter 19: Largest temperature difference for convergence. |
| Words that propose it | bioclimate maxtdiff, maxtdiff of bioclimate, maxtdiff |

#### 37.2.10.1.5 maxEdiff (Pa)

The accuracy of the canopy vapour pressure in the SVAT solution.

| Item | Value |
|---|---|
| Where Daisy reads it | Bioclimate (chapter 19). |
| Daisy default; the farm | 1; not set (the default is used) |
| Typical values | 0.5 to 2 Pa. |
| Levels offered | 0.5, 0.75, 1, 1.5, 2 |
| Box accepts | 0 to 10 Pa |
| What to type | Pascal. |
| What changes | A numerical setting. |
| Needs irrigation | no |
| Daisy reference | chapter 19: Largest humidity difference for convergence. |

| Item | Value |
|---|---|
| Words that propose it | bioclimate maxediff, maxediff of bioclimate, maxediff |

#### 37.2.10.1.6 min sun angle (rad)

Below this sun angle the radiation models treat the sun as set.

| Item | Value |
|---|---|
| Where Daisy reads it | Bioclimate (chapter 19). |
| Daisy default; the farm | 0.0628319; not set (the default is used) |
| Typical values | 0.03 to 0.13 rad (2 to 7 degrees). |
| Levels offered | 0.03142, 0.04712, 0.06283, 0.09425, 0.1257 |
| Box accepts | 0 to 0.6283 rad |
| What to type | Radians. |
| What changes | Only early morning and late evening radiation. |
| Needs irrigation | no |
| Daisy reference | chapter 19: Minimum sun angle above ground for some ‘raddist’ and ‘svat’ models. The ‘DPF’ raddist model will zero radiation if the angle is below this, and the |
| Words that propose it | bioclimate min sun angle, min_sun_angle of bioclimate, min sun angle, min_sun_angle |

### 37.2.10.2 Chemistry original

#### 37.2.10.2.1 max sink total (dimensionless)

The largest share of a chemical that one time step may remove (reactions, uptake); above it Daisy shortens the step.

| Item | Value |
|---|---|
| Where Daisy reads it | Chemistry original (chapter 24). |
| Daisy default; the farm | 0.5; not set (the default is used) |
| Typical values | 0.25 to 1. |
| Levels offered | 0.25, 0.375, 0.5, 0.75, 1 |
| Box accepts | 0 to 5 dimensionless |
| What to type | Fractions. |
| What changes | A numerical setting for chemicals. |
| Needs irrigation | no |
| Daisy reference | chapter 24: Maximum allowed sink term as a fraction of total content. If variable timesteps are enabled, Daisy will try to scale down the timestep in |
| Words that propose it | chemistry max sink total, max_sink_total of chemistry, max sink total, max_sink_total |

#### 37.2.10.2.2 max sink solute (dimensionless)

The same for the dissolved part of a chemical.

| Item | Value |
|---|---|
| Where Daisy reads it | Chemistry original (chapter 24). |
| Daisy default; the farm | 0.9; not set (the default is used) |
| Typical values | 0.5 to 1. |
| Levels offered | 0.45, 0.675, 0.9, 1.35, 1.8 |
| Box accepts | 0 to 9 dimensionless |
| What to type | Fractions. |
| What changes | A numerical setting. |
| Needs irrigation | no |
| Daisy reference | chapter 24: Maximum allowed sink term as a fraction of solute content. If variable timesteps are enabled, Daisy will try to scale down the timestep |
| Words that propose it | chemistry max sink solute, max_sink_solute of chemistry, max sink solute, max_sink_solute |

#### 37.2.10.2.3 max sink secondary (dimensionless)

The same for the part in the secondary (slow) domain.

| Item | Value |
|---|---|
| Where Daisy reads it | Chemistry original (chapter 24). |
| Daisy default; the farm | 1.5; not set (the default is used) |
| Typical values | 1 to 3. |
| Levels offered | 0.75, 1.125, 1.5, 2.25, 3 |
| Box accepts | 0 to 15 dimensionless |
| What to type | Fractions. |
| What changes | A numerical setting. |
| Needs irrigation | no |
| Daisy reference | chapter 24: Maximum allowed sink term as a fraction of secondary domain content. If variable timesteps are enabled, Daisy will try to scale down the timestep |
| Words that propose it | chemistry max sink secondary, max_sink_secondary of chemistry, max sink secondary, max_sink_secondary |

#### 37.2.10.2.4 min sink total (dimensionless)

A share that may always be removed, however small the time step.

| Item | Value |
|---|---|
| Where Daisy reads it | Chemistry original (chapter 24). |
| Daisy default; the farm | 0.01; not set (the default is used) |
| Typical values | 0.005 to 0.02. |
| Levels offered | 0.005, 0.0075, 0.01, 0.015, 0.02 |
| Box accepts | 0 to 0.1 dimensionless |
| What to type | Fractions. |
| What changes | A numerical setting. |
| Needs irrigation | no |
| Daisy reference | chapter 24: 24.9. NONE 147 Always allow this fraction of total content to be removed by sink term. |
| Words that propose it | chemistry min sink total, min_sink_total of chemistry, min sink total, min_sink_total |

### 37.2.10.3 Drain lateral

#### 37.2.10.3.1 L (cm)

The distance between drain pipes.

| Item | Value |
|---|---|
| Where Daisy reads it | Drain lateral (chapter 37), used when the column has drains. |
| Daisy default; the farm | 1800; not set (the default is used) |
| Typical values | 10 to 40 m (1000 to 4000 cm). |
| Levels offered | 900, 1350, 1800, 2700, 3600 |
| Box accepts | 0 to 18000 cm |
| What to type | Centimetres, for example 1200, 1800, 2400. |
| What changes | Closer drains lower the water table faster (only with drains in the column). |
| Needs irrigation | no |
| Daisy reference | chapter 37: Distance between pipes. |
| Words that propose it | drain l, l of drain |

#### 37.2.10.3.2 rad (cm)

The inner radius of the drain pipes.

| Item | Value |
|---|---|
| Where Daisy reads it | Drain lateral (chapter 37). |

| Item | Value |
|---|---|
| Daisy default; the farm | 3.5; not set (the default is used) |
| Typical values | 2.5 to 7 cm. |
| Levels offered | 1.75, 2.625, 3.5, 5.25, 7 |
| Box accepts | 0 to 35 cm |
| What to type | Centimetres. |
| What changes | Only with drains in the column. |
| Needs irrigation | no |
| Daisy reference | chapter 37: Inner radius of drain pipes. |
| Words that propose it | drain rad, rad of drain |

#### 37.2.10.4 Groundwater aquitard

**37.2.10.4.1 K aquitard (cm/h)** The hydraulic conductivity of the low-permeability layer (aquitard) below the soil profile.

| Item | Value |
|---|---|
| Where Daisy reads it | Groundwater aquitard (chapter 45); the farm uses 0.001 cm/h. |
| Daisy default; the farm | 0.001; 0.001 |
| Typical values | 0.0001 (tight clay) to 0.01 cm/h. |
| Levels offered | 0.0005, 0.00075, 0.001, 0.0015, 0.002 |
| Box accepts | 0 to 0.01 cm/h |
| What to type | cm/h, for example 0.0005, 0.001, 0.002. |
| What changes | Lower values hold water above the aquitard: wetter subsoil, less drainage, more denitrification. |
| Needs irrigation | no |
| Daisy reference | chapter 45: Conductivity of the aquitard. |
| Words that propose it | groundwater k aquitard, k_aquitard of groundwater, k aquitard, k_aquitard |

**37.2.10.4.2 Z aquitard (cm)** The thickness of the aquitard below the deepest horizon.

| Item | Value |
|---|---|
| Where Daisy reads it | Groundwater aquitard (chapter 45); the farm uses 200 cm. |
| Daisy default; the farm | 200; 200 |
| Typical values | 50 to 500 cm. |
| Levels offered | 100, 150, 200, 300, 400 |
| Box accepts | 0 to 2000 cm |
| What to type | Centimetres, for example 100, 200, 400. |
| What changes | A thicker aquitard slows drainage like a lower conductivity. |
| Needs irrigation | no |
| Daisy reference | chapter 45: Thickness of the aquitard. The aquitard begins below the bottommost soil horizon. |
| Words that propose it | groundwater z aquitard, z_aquitard of groundwater, z aquitard, z_aquitard |

#### 37.2.10.5 Groundwater static

**37.2.10.5.1 table (cm)** The depth of a fixed groundwater table (positive numbers mean free drainage).

| Item | Value |
|---|---|
| Where Daisy reads it | Groundwater static (chapter 45). |
| Daisy default; the farm | 1; not set (the default is used) |
| Typical values | -100 to -300 cm for a real table; 1 for free drainage. |
| Levels offered | 0.5, 0.75, 1, 1.5, 2 |
| Box accepts | 0 to 10 cm |

| Item | Value |
|---|---|
| What to type | Centimetres (negative below the surface). |
| What changes | A shallow table feeds the roots by capillary rise and stops drainage. |
| Needs irrigation | no |
| Daisy reference | chapter 45: Groundwater level. Positive numbers indicate free drainage. |
| Words that propose it | groundwater table, table of groundwater, table |

#### 37.2.10.6 OrganicMatter original

**37.2.10.6.1 K NH4 (h-1)** The highest rate at which microbes take up (immobilise) ammonium from the soil.

| Item | Value |
|---|---|
| Where Daisy reads it | OrganicMatter original (chapter 60). |
| Daisy default; the farm | 0.020833; not set (the default is used) |
| Typical values | 0.01 to 0.04 per hour (default 0.0208). |
| Levels offered | 0.01042, 0.01562, 0.02083, 0.03125, 0.04167 |
| Box accepts | 0 to 0.2083 h-1 |
| What to type | Rates per hour. |
| What changes | More immobilisation: less mineral nitrogen for the crop and for leaching in the short term. |
| Needs irrigation | no |
| Daisy reference | chapter 60: Maximal immobilization rate for ammonium. |
| Words that propose it | organicmatter k nh4, k_nh4 of organicmatter, k nh4, k_nh4 |

**37.2.10.6.2 K NO3 (h-1)** The highest rate at which microbes take up (immobilise) nitrate.

| Item | Value |
|---|---|
| Where Daisy reads it | OrganicMatter original (chapter 60). |
| Daisy default; the farm | 0.020833; not set (the default is used) |
| Typical values | 0.01 to 0.04 per hour (default 0.0208). |
| Levels offered | 0.01042, 0.01562, 0.02083, 0.03125, 0.04167 |
| Box accepts | 0 to 0.2083 h-1 |
| What to type | Rates per hour. |
| What changes | As for ammonium, for nitrate. |
| Needs irrigation | no |
| Daisy reference | chapter 60: Maximal immobilization rate for nitrate. |
| Words that propose it | organicmatter k no3, k_no3 of organicmatter, k no3, k_no3 |

**37.2.10.6.3 CO2 threshold (h-1)** Pools of organic matter that turn over faster than this count as “fast” CO2.

| Item | Value |
|---|---|
| Where Daisy reads it | OrganicMatter original (chapter 60). |
| Daisy default; the farm | 0.0001; not set (the default is used) |
| Typical values | 0.00005 to 0.0002 per hour. |
| Levels offered | 0.00005, 0.000075, 0.0001, 0.00015, 0.0002 |
| Box accepts | 0 to 0.001 h-1 |
| What to type | Rates per hour. |
| What changes | How CO2 is reported (fast and slow), not the total. |
| Needs irrigation | no |
| Daisy reference | chapter 60: Turnover rate above which pools will contribute to ‘CO2 fast’. |
| Words that propose it | organicmatter co2 threshold, co2_threshold of organicmatter, co2 threshold, co2_threshold |

**37.2.10.6.4 min AM C (g C/m2)** Added organic matter (AOM: residues, manure) with less carbon than this is merged into the soil organic matter instead of being followed as its own pool.

| Item | Value |
|---|---|
| Where Daisy reads it | OrganicMatter original (chapter 60). |
| Daisy default; the farm | 0.5; not set (the default is used) |
| Typical values | 0.25 to 1 g C per m² (default 0.5). |
| Levels offered | 0.25, 0.375, 0.5, 0.75, 1 |
| Box accepts | 0 to 5 g C/m2 |
| What to type | g C/m² (1 g/m² = 10 kg/ha), for example 0.25, 0.5, 1. |
| What changes | A housekeeping limit of the model: small changes in mineralisation. |
| Needs irrigation | no |
| Daisy reference | chapter 60: Minimal amount of carbon in AOM ensuring it is not removed. |
| Words that propose it | organicmatter min am c, min_am_c of organicmatter, min am c, min_am_c |

**37.2.10.6.5 min AM N (g N/m2)** The same limit for nitrogen: an added organic matter pool with less nitrogen than this is merged into the soil.

| Item | Value |
|---|---|
| Where Daisy reads it | OrganicMatter original (chapter 60). |
| Daisy default; the farm | 0.05; not set (the default is used) |
| Typical values | 0.025 to 0.1 g N per m² (default 0.05 g N/m², that is 0.5 kg N/ha). The workbench allows 0 to 0.5. |
| Levels offered | 0.025, 0.0375, 0.05, 0.075, 0.1 |
| Box accepts | 0 to 0.5 g N/m2 |
| What to type | g N/m², for example 0.025, 0.05, 0.1. |
| What changes | A housekeeping limit: no change in yield, water or nitrogen in the test. |
| Needs irrigation | no |
| Daisy reference | chapter 60: Minimal amount of nitrogen in AOM ensuring it is not removed. |
| Words that propose it | organicmatter min am n, min_am_n of organicmatter, min am n, min_am_n |

### 37.2.10.7 Surface

**37.2.10.7.1 temperature change rate (h-1)** How fast the soil surface temperature follows the air temperature.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87). |
| Daisy default; the farm | 0.5; not set (the default is used) |
| Typical values | 0.25 to 1 per hour. |
| Levels offered | 0.25, 0.375, 0.5, 0.75, 1 |
| Box accepts | 0 to 5 h-1 |
| What to type | Rates per hour. |
| What changes | Soil temperature near the surface, hence early growth and turnover. |
| Needs irrigation | no |
| Daisy reference | chapter 87: Relative change of surface temperature. If you set this to a negative value, surface temperature will reflect temperature of incoming water or ponded |
| Words that propose it | surface temperature change rate, temperature_change_rate of surface, temperature change rate, temperature_change_rate |

**37.2.10.7.2 EpFactor (dimensionless)** The factor that turns potential evapotranspiration into potential evaporation from bare soil.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87). |
| Daisy default; the farm | 0.6; not set (the default is used) |
| Typical values | 0.4 to 1 (default 0.6). |
| Levels offered | 0.3, 0.45, 0.6, 0.9, 1.2 |
| Box accepts | 0 to 6 dimensionless |
| What to type | Factors, for example 0.4, 0.6, 0.8. |
| What changes | Higher values dry the topsoil faster between crops and before canopy closure. |
| Needs irrigation | no |
| Daisy reference | chapter 87: The autumn value can be lower, due to muching. With a crop factor of 1.2 a combined Kc of 1.15 is reached at LAI=5. See also [Kjaersgaard et al., 2008] |
| Words that propose it | surface epfactor, epfactor of surface, epfactor |

#### 37.2.10.7.3 albedo dry (dimensionless) The share of sunlight reflected by dry soil.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87). |
| Daisy default; the farm | 0.15; not set (the default is used) |
| Typical values | 0.1 to 0.3 (light soils higher). |
| Levels offered | 0.075, 0.1125, 0.15, 0.225, 0.3 |
| Box accepts | 0 to 1.5 dimensionless |
| What to type | Fractions. |
| What changes | Energy at the soil surface: small effect on evaporation. |
| Needs irrigation | no |
| Daisy reference | chapter 87: Albedo of dry soil (pF >= 3) |
| Words that propose it | surface albedo dry, albedo_dry of surface, albedo dry, albedo_dry |

#### 37.2.10.7.4 albedo wet (dimensionless) The share of sunlight reflected by wet soil.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87). |
| Daisy default; the farm | 0.08; not set (the default is used) |
| Typical values | 0.05 to 0.15. |
| Levels offered | 0.04, 0.06, 0.08, 0.12, 0.16 |
| Box accepts | 0 to 0.8 dimensionless |
| What to type | Fractions. |
| What changes | As for dry soil. |
| Needs irrigation | no |
| Daisy reference | chapter 87: Albedo of wet soil (pf <= 1.7) |
| Words that propose it | surface albedo wet, albedo_wet of surface, albedo wet, albedo_wet |

#### 37.2.10.7.5 DetentionCapacity (mm) The water depth that can stand (pond) on the field before the excess runs off.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87). The Daisy default of 1000 mm means no runoff at all; the farm uses 5 mm. |
| Daisy default; the farm | 1000; 5 |
| Typical values | 2 to 10 mm for an arable field (rough, ploughed land holds more); the workbench allows 0 to 50. |
| Levels offered | 2.5, 3.75, 5, 7.5, 10 |
| Box accepts | 0 to 50 mm |
| What to type | Millimetres, for example 2.5, 5, 10. |
| What changes | Smaller values give more surface runoff during heavy rain and less infiltration. |
| Needs irrigation | no |

| Item | Value |
|---|---|
| Daisy reference | chapter 87: Amount of ponding the surface can retain. If ponding in any part of the surface is above this, exceed will runoff. |
| Words that propose it | surface detentioncapacity, detentioncapacity of surface, detentioncapacity |

#### 37.2.10.7.6 ReservoirConstant (h-1)

The share of the ponded water above the detention capacity that runs off each hour.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87); the farm uses 60 per hour (all of it at once). |
| Daisy default; the farm | 1; 60 |
| Typical values | 1 (slow) to 60 per hour. |
| Levels offered | 30, 45, 60, 90, 120 |
| Box accepts | 0 to 600 h-1 |
| What to type | Per hour. |
| What changes | Higher values remove ponded water faster. |
| Needs irrigation | no |
| Daisy reference | chapter 87: Fraction of ponding above DetentionCapacity that runoffs each hour. |
| Words that propose it | surface reservoirconstant, reservoirconstant of surface, reservoirconstant |

#### 37.2.10.7.7 LocalDetentionCapacity (mm)

The water that can pond in one part (patch) of the field before the excess spreads to the rest of the field.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87), for fields split into patches. |
| Daisy default; the farm | 10; not set (the default is used) |
| Typical values | 5 to 20 mm (default 10); the workbench allows 0 to 100. |
| Levels offered | 5, 7.5, 10, 15, 20 |
| Box accepts | 0 to 100 mm |
| What to type | Millimetres, for example 5, 10, 20. |
| What changes | Only with several patches: how water moves between them. |
| Needs irrigation | no |
| Daisy reference | chapter 87: Amount of ponding the surface can retain locally. If ponding in any part of the surface is above this, exceed will be distributed to the rest of the surface. |
| Words that propose it | surface localdetentioncapacity, localdetentioncapacity of surface, localdetentioncapacity |

#### 37.2.10.7.8 z mixing (cm)

The depth of the thin top layer where soil, soil water and ponded water mix.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87). |
| Daisy default; the farm | 0.1; not set (the default is used) |
| Typical values | 0.05 to 0.2 cm. |
| Levels offered | 0.05, 0.075, 0.1, 0.15, 0.2 |
| Box accepts | 0 to 1 cm |
| What to type | Centimetres. |
| What changes | Loss of fertiliser and chemicals to runoff. |
| Needs irrigation | no |
| Daisy reference | chapter 87: Depth of mixing layer in the top of the soil. The mixing layer affect exchange between soil colloids, soil water and the surface, especially in connection with |
| Words that propose it | surface z mixing, z_mixing of surface, z mixing, z_mixing |

**37.2.10.7.9 R mixing (h/mm)** The resistance to the mixing of dissolved compounds between soil and ponded water.

| Item | Value |
|---|---|
| Where Daisy reads it | Surface (chapter 87). |
| Daisy default; the farm | 1000000000; not set (the default is used) |
| Typical values | Very large values (1e9 h/mm) mean almost no mixing. |
| Levels offered | 500000000, 750000000, 1000000000, 1500000000, 2000000000 |
| Box accepts | 0 to 10000000000 h/mm |
| What to type | h/mm. |
| What changes | As for z mixing. |
| Needs irrigation | no |
| Daisy reference | chapter 87: Resistance to mixing inorganic compounds between soil and ponding. |
| Words that propose it | surface r mixing, r_mixing of surface, r mixing, r_mixing |

## 37.3 The Daisy library, all of it, in the experiments

Everything of the Daisy library can be tested in an experiment. In *Add something to change*, the group **The Daisy library (all of it)** lists, with their numbers:

| What changes | Choices | What a run does with it |
|---|---|---|
| Soil | 63 soil horizons (and the USDA classes) | the texture and humus of the horizon down the profile with HYPRES, or the Daisy horizon itself when its file gives no texture |
| Crop of the Daisy library, grown every year | 84 crops | the crop grown every year with the dates of its template |
| Type of fertiliser | 79 fertilisers | mineral: in place of N25S; organic (slurry, manure): the same nitrogen, as fresh weight from its dry matter and nitrogen |
| Chemical of the Daisy library | 35 chemicals | the chemical followed under the name Tracer, with all its own properties (sorption, decay, uptake) |
| Management of the Daisy library | 82 actions | a whole management replaces the farm rotation; a single operation (harrowing, rolling, ridging…) is done each year in place of the ploughing |
| Program of the Daisy library | 55 programs | the program run as it comes with Daisy, with its own files and weather; the settings of SWAFL do not apply |
| Tillage | the tillage operations | in place of the ploughing |

The long lists have a search box. **Logs of the Daisy library added to every run** (176 logs, with a search box) adds any log of the library to the output of every run. The files of the Daisy samples that a choice needs (and the files they read in turn) are copied into the Common folder of the experiment and read through the path of the run; the files of the Daisy library itself are read from Daisy. In the tab **Parameter library**, every item has the button **Test it in an experiment** (for a log: **Write this log in every run of my experiment**), which puts it in the experiment.

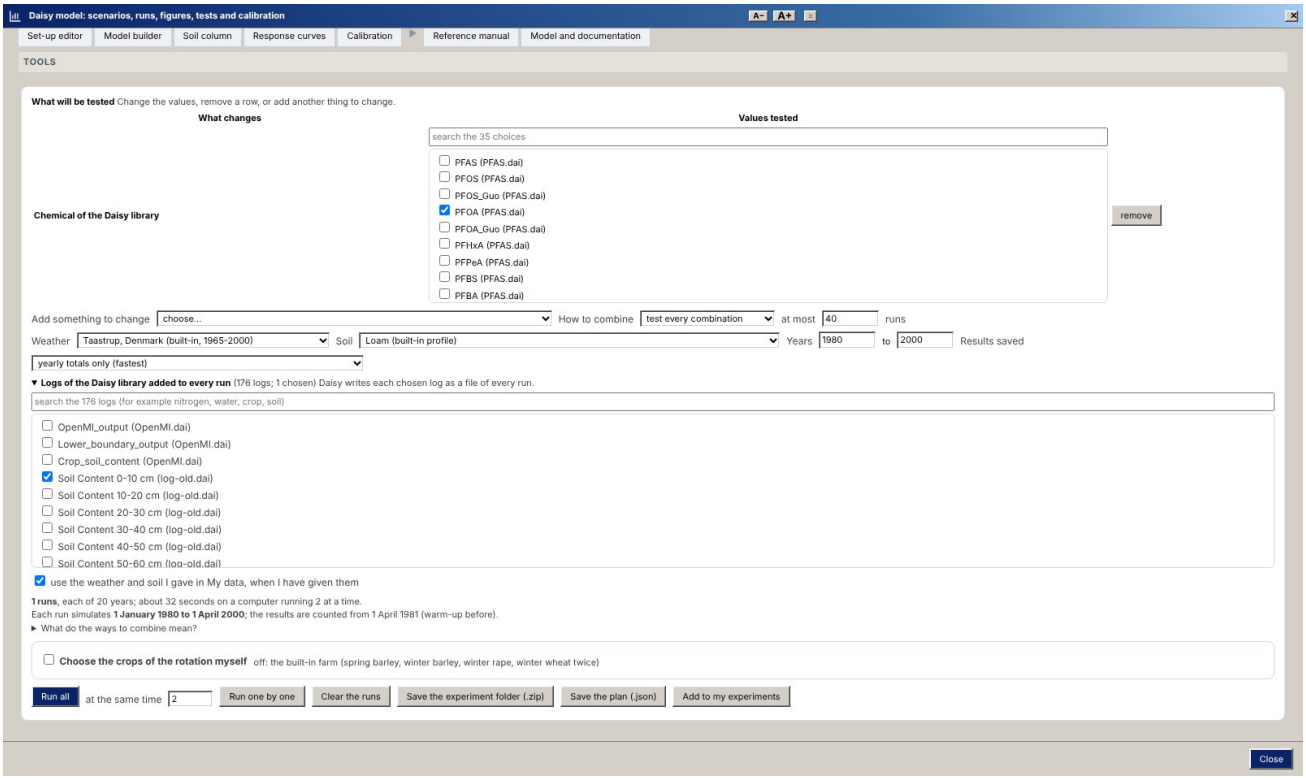



Figure 37.1: Experiments: a chemical of the Daisy library as the thing that changes, and a log of the library added to every run.

# Chapter 38

# The Daisy library

This chapter lists the entries of the Daisy library that come with the program and that the Library tab reads: 63 horizons, 84 crops, 79 fertilisers, 35 chemicals, 82 actions, 55 programs and 176 log types. The values are those of the library files; nothing is estimated.

## 38.1 The text of every item

In the Parameter library tab, every item shows under its values the text Daisy reads, in the colours of VS Code, with two comment lines that say where it comes from and how to use it in a set-up of your own ((input file “dk-fertilizer.dai”), or copy the block). The text can be changed in place; *open in set-up editor* opens it (as changed) in the Set-up editor, *open the whole file* opens the library file it belongs to, and *download* saves it as a .dai file. The table of values of a fertiliser or chemical has a third column, *What it means*: for example *volatilization*, the share of the ammonium lost as ammonia when the slurry is spread on the surface; *NH4_fraction*, the share of the nitrogen that is ammonium; *K_OC*, the sorption to organic carbon.

**A steady view.** A press on an item draws its values, its *Test it in an experiment* button and its text once; the box of the text keeps its size while the file is read, so nothing below it moves. The list and the values stay side by side (on a narrow screen the values go below the list), long values wrap inside the table, and the scroll bar of the tab is always shown, so the page never jumps from one layout to the other. Pressing the item that is already shown changes nothing.

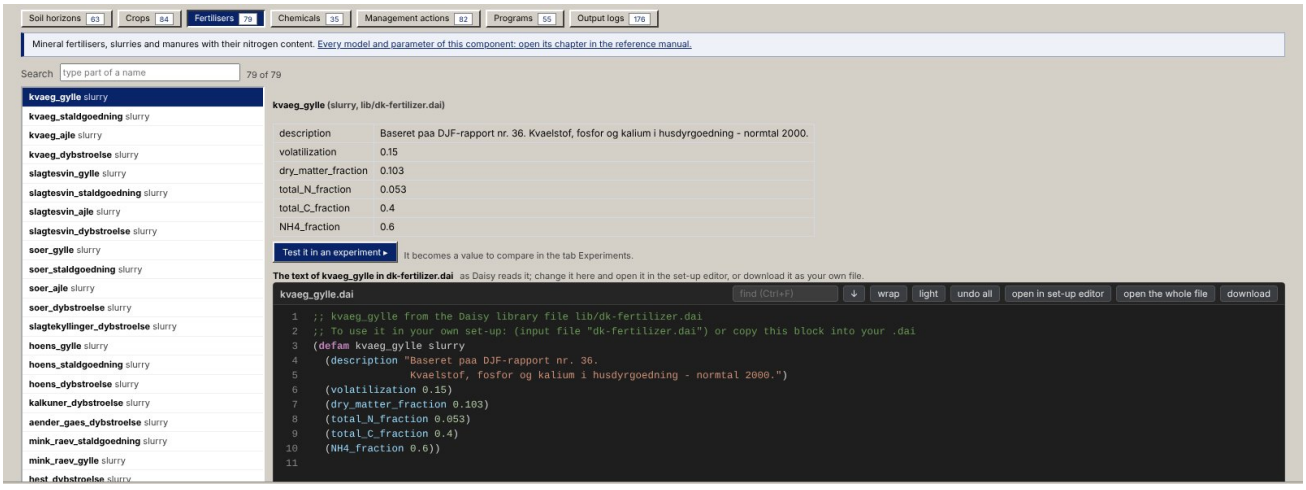



Figure 38.1: A slurry of the library: its values, what they mean, and its text in the colours of VS Code.

## 38.2 Soil horizons

Each horizon has its texture, its organic matter and its hydraulic parameters. Hydraulics follow the van Genuchten and Mualem model: the water content is the residual content plus the range between saturated and residual content times the effective saturation, and the conductivity follows from the same parameters. The water content and the conductivity at six suctions are calculated from the parameters of the library with these equations, to show the shape of each horizon.

| Horizon | Parent | Clay % | Silt % | Sand % | Humus % | C/N | Bulk density |
|---|---|---|---|---|---|---|---|
| Ap_JB1 | USDA3 | 3.9 | 6.4 | 87 | 2.8 | 14.2 | 1.45 |
| Ap_JB2 | USDA3 | 4.4 | 13 | 80 | 2.7 | 12.8 | 1.42 |
| Ap_JB3 | USDA3 | 6.8 | 13.5 | 76.9 | 2.8 | 13.3 | 1.44 |
| Ap_JB4 | USDA3 | 7.9 | 21.5 | 68 | 2.7 | 11.6 | 1.41 |
| Ap_JB5 | USDA3 | 12.3 | 22 | 63.1 | 2.5 | 12.8 | 1.51 |
| Ap_JB6 | USDA3 | 12.4 | 24.9 | 60.1 | 2.5 | 11 | 1.53 |
| Ap_JB7 | USDA3 | 17.5 | 25.8 | 54.2 | 2.4 | 10.4 | 1.54 |
| B_JB1 | USDA3 | 3.8 | 3.5 | 91.8 | 0.9 | 12.2 | 1.5 |
| B_JB2 | USDA3 | 4.1 | 10.1 | 84.9 | 0.96 | 12.5 | 1.45 |
| B_JB3 | USDA3 | 6.8 | 11.3 | 80.9 | 1.04 | 11 | 1.55 |
| B_JB4 | USDA3 | 7.1 | 19.4 | 72.7 | 0.84 | 12.2 | 1.51 |
| B_JB5 | USDA3 | 11.9 | 16.2 | 71.4 | 0.51 | 9.9 | 1.62 |
| B_JB6 | USDA3 | 12.5 | 21.8 | 65.2 | 0.52 | 8.1 | 1.64 |
| B_JB7 | USDA3 | 20 | 22 | 57.5 | 0.41 | 7.5 | 1.68 |
| C_JB1 | USDA3 | 2.9 | 2 | 94.9 | 0.2 | n/a | 1.56 |
| C_JB2 | USDA3 | 3.4 | 5.9 | 90.5 | 0.2 | n/a | 1.55 |
| C_JB3 | USDA3 | 7.5 | 11.5 | 80.7 | 0.21 | n/a | 1.63 |
| C_JB4 | USDA3 | 8 | 18.6 | 73.1 | 0.3 | n/a | 1.65 |
| C_JB5 | USDA3 | 14.9 | 22.1 | 62.9 | 0.24 | n/a | 1.79 |
| C_JB6 | USDA3 | 13.3 | 23.8 | 62.6 | 0.22 | n/a | 1.72 |
| C_JB7 | USDA3 | 19.8 | 23.8 | 56.2 | 0.21 | n/a | 1.73 |
| mini | default | n/a | n/a | n/a | n/a | n/a | n/a |
| Ap | default | 7.9 | 21.5 | 68 | 2.7 | 11.6 | 1.41 |
| B | default | 7.1 | 19.4 | 72.7 | 0.84 | 12.2 | 1.51 |
| C | default | 8 | 18.6 | 73.1 | 0.3 | n/a | 1.65 |
| Orh(g)_Valecov_PG | default | 0.141 | 0.407 | 0.452 | 0.022 | 8.6 | 1.514 |
| E(g)_Valecov_PG | default | 0.186 | 0.373 | 0.441 | 0.005 | 8.6 | 1.586 |
| E(g)/Ig_Valecov_PG | default | 0.218 | 0.31 | 0.472 | 0.003 | 8.6 | 1.593 |
| Ig_Valecov_PG | default | 0.214 | 0.272 | 0.515 | 0.002 | 8.6 | 1.649 |
| Pg_Valecov_PG | default | 0.199 | 0.192 | 0.609 | 0.001 | 8.6 | 1.64 |
| Askov Ap | ISSS4 | 0.113 | 0.277 | n/a | 0.026 | 11 | 1.55 |
| Askov B | ISSS4 | 0.235 | 0.253 | n/a | 0.005 | 11 | 1.48 |
| Askov C | ISSS4 | 0.244 | 0.283 | n/a | 0.005 | 11 | 1.57 |
| Jyndevad Ap | ISSS4 | 0.038 | 0.072 | n/a | 0.023 | 13 | 1.472 |
| Jyndevad C | ISSS4 | 0.028 | 0.023 | n/a | 0.004 | 13 | 1.469 |
| Foulum AP | ISSS4 | 0.115 | 0.11 | n/a | 0.023 | 11.2 | 1.45 |
| Foulum B | ISSS4 | 0.149 | 0.101 | n/a | 0.005 | 11.3 | 1.45 |
| Foulum C | ISSS4 | 0.164 | 0.106 | n/a | 0.002 | 9.1 | 1.55 |
| Taastrup Ap | USDA3 | 0.107 | 0.222 | 0.671 | 0.024 | n/a | 1.45 |
| Taastrup plow pan | USDA3 | 0.148 | 0.214 | 0.638 | 0.016 | n/a | 1.78 |
| Taastrup Bt | USDA3 | 0.222 | 0.195 | 0.583 | 0.016 | n/a | 1.66 |
| Taastrup C | USDA3 | 0.207 | 0.235 | 0.558 | 0.01 | n/a | 1.69 |
| Taastrup drain canyon | ISSS4 | 0.213 | 0.19 | n/a | 0.014 | n/a | 1.65 |
| WP1 IT CER 2005 L50 | default | 0.19 | 0.502 | 0.308 | 0.0121 | 10.2 | 1.2 |
| WP1 IT CER 2005 L100 | default | 0.16 | 0.357 | 0.483 | 0.0079 | 10.4 | 1.35 |
| Ap_JB1_Cosby | ISSS4 | 3.9 | 6.4 | n/a | 2.7 | 14.2 | n/a |

| Horizon | Parent | Clay % | Silt % | Sand % | Humus % | C/N | Bulk density |
|---|---|---|---|---|---|---|---|
| C_JB1_Cosby | ISSS4 | 2.9 | 2 | n/a | 0.2 | n/a | n/a |
| Ap_JB1_hypres | Ap_JB1_Cosby | n/a | n/a | n/a | n/a | n/a | 1.45 |
| C_JB1_hypres | Ap_JB1_Cosby | n/a | n/a | n/a | n/a | n/a | 1.56 |
| Jadorna 2004 | USDA3 | 0.03 | 0.3 | 0.67 | 0.028 | 11.65 | 1.5 |
| Jaddeep 2004 | USDA3 | 0.03 | 0.36 | 0.61 | 0.0047 | 8.87 | 1.48 |
| Jaddeeper 2004 | USDA3 | 0.06 | 0.17 | 0.77 | 0.003 | n/a | n/a |
| Jaddeepest 2004 | USDA3 | 0.06 | 0.06 | 0.88 | 0.001 | n/a | n/a |
| A | USDA3 | 18.9 | 42.7 | 38.4 | 4.4 | n/a | 1.6 |
| Aor Most7 | default | 0.2 | 0.65 | 0.15 | 0.0302 | 6 | 1.35 |
| B Most7 | default | 0.18 | 0.66 | 0.16 | 0.025 | 6 | 1.44 |
| C Most7 | default | 0.21 | 0.64 | 0.15 | 0.0121 | 8 | 1.35 |
| D Most7 | default | 0.21 | 0.65 | 0.14 | 0.0101 | 8 | 1.43 |
| S Most7 | default | 0.15 | 0.54 | 0.31 | 0.0081 | 8 | 1.43 |
| Ap_plowing | Ap | n/a | n/a | n/a | n/a | n/a | n/a |
| Jyn_15h | ISSS4 | 0.036 | 0.047 | n/a | 0.023 | 14.8 | 1.52 |
| KarJyn_B | ISSS4 | 0.044 | 0.013 | n/a | 0.011 | 16.9 | 1.51 |
| KarJyn_C | USDA3 | 0.043 | 0.008 | 0.946 | 0.003 | 13.3 | 1.58 |

### 38.2.1 Horizon Ap_JB1

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.062 |
| n | 1.445 |
| K_sat | 5.037 |
| Theta_sat | 0.395 |
| Theta_res | 0 |
| l | -0.973 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.393 | 2.56 |
| 1 | 10 | 0.349 | 0.468 |
| 2 | 100 | 0.172 | 0.00503 |
| 2.5 | 316.228 | 0.105 | 0.000315 |
| 3 | 1000 | 0.063 | 1.88e-05 |
| 4.2 | 15848.9 | 0.018 | 2.12e-08 |

### 38.2.2 Horizon Ap_JB2

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.054 |
| n | 1.382 |
| K_sat | 3.514 |
| Theta_sat | 0.412 |
| Theta_res | 0 |
| l | -0.922 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.410 | 1.6 |
| 1 | 10 | 0.373 | 0.31 |
| 2 | 100 | 0.211 | 0.00418 |
| 2.5 | 316.228 | 0.139 | 0.00028 |
| 3 | 1000 | 0.090 | 1.77e-05 |
| 4.2 | 15848.9 | 0.031 | 2.27e-08 |

### 38.2.3 Horizon Ap_JB3

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.057 |
| n | 1.337 |
| K_sat | 3.135 |
| Theta_sat | 0.406 |
| Theta_res | 0 |
| l | -1.461 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.404 | 1.22 |
| 1 | 10 | 0.368 | 0.225 |
| 2 | 100 | 0.221 | 0.00411 |
| 2.5 | 316.228 | 0.152 | 0.000356 |
| 3 | 1000 | 0.104 | 2.93e-05 |
| 4.2 | 15848.9 | 0.041 | 7.09e-08 |

### 38.2.4 Horizon Ap_JB4

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.048 |
| n | 1.304 |
| K_sat | 2.469 |
| Theta_sat | 0.42 |
| Theta_res | 0 |
| l | -1.492 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.418 | 0.908 |
| 1 | 10 | 0.389 | 0.185 |
| 2 | 100 | 0.253 | 0.0041 |
| 2.5 | 316.228 | 0.183 | 0.000373 |
| 3 | 1000 | 0.129 | 3.19e-05 |
| 4.2 | 15848.9 | 0.056 | 8.32e-08 |

### 38.2.5 Horizon Ap_JB5

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.05 |
| n | 1.251 |
| K_sat | 1.7699 |
| Theta_sat | 0.391 |
| Theta_res | 0 |
| l | -2.386 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.389 | 0.504 |
| 1 | 10 | 0.364 | 0.0984 |
| 2 | 100 | 0.255 | 0.00304 |
| 2.5 | 316.228 | 0.194 | 0.000364 |
| 3 | 1000 | 0.146 | 4.14e-05 |
| 4.2 | 15848.9 | 0.073 | 2.17e-07 |

### 38.2.6 Horizon Ap_JB6

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.044 |
| n | 1.246 |
| K_sat | 1.469 |
| Theta_sat | 0.386 |
| Theta_res | 0 |
| l | -2.365 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.384 | 0.429 |
| 1 | 10 | 0.363 | 0.0904 |
| 2 | 100 | 0.260 | 0.00303 |
| 2.5 | 316.228 | 0.201 | 0.000365 |
| 3 | 1000 | 0.152 | 4.13e-05 |
| 4.2 | 15848.9 | 0.077 | 2.12e-07 |

### 38.2.7 Horizon Ap_JB7

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.048 |
| n | 1.204 |
| K_sat | 1.351 |
| Theta_sat | 0.386 |
| Theta_res | 0 |
| l | -3.197 |

| Parameter | Value |
|---|---|

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.384 | 0.295 |
| 1 | 10 | 0.364 | 0.0576 |
| 2 | 100 | 0.274 | 0.00226 |
| 2.5 | 316.228 | 0.220 | 0.000319 |
| 3 | 1000 | 0.175 | 4.31e-05 |
| 4.2 | 15848.9 | 0.100 | 3.39e-07 |

### 38.2.8 Horizon B_JB1

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.082 |
| n | 1.58 |
| K_sat | 3.756 |
| Theta_sat | 0.377 |
| Theta_res | 0 |
| l | -0.335 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.374 | 2.22 |
| 1 | 10 | 0.308 | 0.296 |
| 2 | 100 | 0.110 | 0.000944 |
| 2.5 | 316.228 | 0.057 | 3.22e-05 |
| 3 | 1000 | 0.029 | 1.07e-06 |
| 4.2 | 15848.9 | 0.006 | 2.95e-10 |

### 38.2.9 Horizon B_JB2

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.069 |
| n | 1.458 |
| K_sat | 2.662 |
| Theta_sat | 0.405 |
| Theta_res | 0 |
| l | -0.3 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.402 | 1.34 |
| 1 | 10 | 0.351 | 0.202 |
| 2 | 100 | 0.164 | 0.00114 |
| 2.5 | 316.228 | 0.098 | 4.94e-05 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 3 | 1000 | 0.058 | 2.04e-06 |
| 4.2 | 15848.9 | 0.016 | 9.45e-10 |

### 38.2.10 Horizon B_JB3

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.071 |
| n | 1.388 |
| K_sat | 2.433 |
| Theta_sat | 0.409 |
| Theta_res | 0 |
| l | -0.937 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.406 | 1.02 |
| 1 | 10 | 0.357 | 0.153 |
| 2 | 100 | 0.188 | 0.00157 |
| 2.5 | 316.228 | 0.122 | 0.000103 |
| 3 | 1000 | 0.078 | 6.48e-06 |
| 4.2 | 15848.9 | 0.027 | 8.28e-09 |

### 38.2.11 Horizon B_JB4

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.058 |
| n | 1.349 |
| K_sat | 1.737 |
| Theta_sat | 0.395 |
| Theta_res | 0 |
| l | -0.717 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.393 | 0.696 |
| 1 | 10 | 0.357 | 0.119 |
| 2 | 100 | 0.209 | 0.00143 |
| 2.5 | 316.228 | 0.142 | 9.2e-05 |
| 3 | 1000 | 0.096 | 5.58e-06 |
| 4.2 | 15848.9 | 0.037 | 6.48e-09 |

### 38.2.12 Horizon B_JB5

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.066 |
| n | 1.274 |
| K_sat | 1.118 |
| Theta_sat | 0.362 |
| Theta_res | 0 |
| l | -1.504 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.360 | 0.315 |
| 1 | 10 | 0.328 | 0.048 |
| 2 | 100 | 0.212 | 0.00085 |
| 2.5 | 316.228 | 0.157 | 7.71e-05 |
| 3 | 1000 | 0.115 | 6.69e-06 |
| 4.2 | 15848.9 | 0.054 | 1.84e-08 |

### 38.2.13 Horizon B_JB6

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.054 |
| n | 1.249 |
| K_sat | 0.9584 |
| Theta_sat | 0.36 |
| Theta_res | 0 |
| l | -1.574 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.358 | 0.26 |
| 1 | 10 | 0.334 | 0.0454 |
| 2 | 100 | 0.231 | 0.000986 |
| 2.5 | 316.228 | 0.177 | 9.43e-05 |
| 3 | 1000 | 0.133 | 8.5e-06 |
| 4.2 | 15848.9 | 0.067 | 2.54e-08 |

### 38.2.14 Horizon B_JB7

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.053 |
| n | 1.169 |
| K_sat | 0.618 |
| Theta_sat | 0.352 |
| Theta_res | 0 |
| l | -2.809 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.350 | 0.0972 |
| 1 | 10 | 0.333 | 0.0165 |
| 2 | 100 | 0.260 | 0.000523 |
| 2.5 | 316.228 | 0.217 | 6.58e-05 |
| 3 | 1000 | 0.180 | 7.86e-06 |
| 4.2 | 15848.9 | 0.113 | 4.59e-08 |

### 38.2.15 Horizon C_JB1

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.085 |
| n | 1.721 |
| K_sat | 4.046 |
| Theta_sat | 0.355 |
| Theta_res | 0 |
| l | 1.18 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.353 | 2.78 |
| 1 | 10 | 0.280 | 0.271 |
| 2 | 100 | 0.075 | 6.93e-05 |
| 2.5 | 316.228 | 0.033 | 5.16e-07 |
| 3 | 1000 | 0.014 | 3.7e-09 |
| 4.2 | 15848.9 | 0.002 | 2.61e-14 |

### 38.2.16 Horizon C_JB2

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.08 |
| n | 1.557 |
| K_sat | 2.288 |
| Theta_sat | 0.37 |
| Theta_res | 0 |
| l | 1.188 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.367 | 1.3 |
| 1 | 10 | 0.306 | 0.133 |
| 2 | 100 | 0.115 | 0.000106 |
| 2.5 | 316.228 | 0.061 | 1.46e-06 |
| 3 | 1000 | 0.032 | 1.91e-08 |
| 4.2 | 15848.9 | 0.007 | 5.63e-13 |

### 38.2.17 Horizon C_JB3

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.075 |
| n | 1.37 |
| K_sat | 1.213 |
| Theta_sat | 0.354 |
| Theta_res | 0 |
| l | 0.24 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.351 | 0.465 |
| 1 | 10 | 0.308 | 0.0556 |
| 2 | 100 | 0.165 | 0.000273 |
| 2.5 | 316.228 | 0.109 | 1.12e-05 |
| 3 | 1000 | 0.072 | 4.38e-07 |
| 4.2 | 15848.9 | 0.026 | 1.77e-10 |

### 38.2.18 Horizon C_JB4

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.056 |
| n | 1.325 |
| K_sat | 1.0773 |
| Theta_sat | 0.354 |
| Theta_res | 0 |
| l | -0.191 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.352 | 0.402 |
| 1 | 10 | 0.322 | 0.0662 |
| 2 | 100 | 0.197 | 0.000668 |
| 2.5 | 316.228 | 0.138 | 3.72e-05 |
| 3 | 1000 | 0.096 | 1.93e-06 |
| 4.2 | 15848.9 | 0.039 | 1.52e-09 |

### 38.2.19 Horizon C_JB5

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.043 |
| n | 1.187 |
| K_sat | 0.472 |
| Theta_sat | 0.317 |
| Theta_res | 0 |
| l | -1.605 |

| Parameter | Value |
|---|---|

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.316 | 0.0948 |
| 1 | 10 | 0.302 | 0.0179 |
| 2 | 100 | 0.235 | 0.00049 |
| 2.5 | 316.228 | 0.193 | 5.02e-05 |
| 3 | 1000 | 0.157 | 4.75e-06 |
| 4.2 | 15848.9 | 0.094 | 1.56e-08 |

### 38.2.20 Horizon C_JB6

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.046 |
| n | 1.223 |
| K_sat | 0.6246 |
| Theta_sat | 0.3382 |
| Theta_res | 0 |
| l | -0.983 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.337 | 0.156 |
| 1 | 10 | 0.319 | 0.0286 |
| 2 | 100 | 0.234 | 0.0006 |
| 2.5 | 316.228 | 0.185 | 5.15e-05 |
| 3 | 1000 | 0.144 | 4.08e-06 |
| 4.2 | 15848.9 | 0.078 | 8.76e-09 |

### 38.2.21 Horizon C_JB7

Parent USDA3, defined in lib/dk-horizon.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.047 |
| n | 1.16 |
| K_sat | 0.4588 |
| Theta_sat | 0.3385 |
| Theta_res | 0 |
| l | -2.255 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.337 | 0.0701 |
| 1 | 10 | 0.323 | 0.0123 |
| 2 | 100 | 0.259 | 0.00037 |
| 2.5 | 316.228 | 0.219 | 4.26e-05 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 3 | 1000 | 0.183 | 4.58e-06 |
| 4.2 | 15848.9 | 0.117 | 2.06e-08 |

### 38.2.22 Horizon mini

Parent default, defined in lib/mini.dai. Hydraulic model n/a.

| Parameter | Value |
|---|---|

### 38.2.23 Horizon Ap

Parent default, defined in sample/DrainDrying.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.048 |
| n | 1.304 |
| Theta_sat | 0.42 |
| Theta_res | 0 |
| K_sat | 2.469 |
| l | -1.492 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.418 | 0.908 |
| 1 | 10 | 0.389 | 0.185 |
| 2 | 100 | 0.253 | 0.0041 |
| 2.5 | 316.228 | 0.183 | 0.000373 |
| 3 | 1000 | 0.129 | 3.19e-05 |
| 4.2 | 15848.9 | 0.056 | 8.32e-08 |

### 38.2.24 Horizon B

Parent default, defined in sample/DrainDrying.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.058 |
| n | 1.349 |
| Theta_sat | 0.395 |
| Theta_res | 0 |
| K_sat | 1.737 |
| l | -0.717 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.393 | 0.696 |
| 1 | 10 | 0.357 | 0.119 |
| 2 | 100 | 0.209 | 0.00143 |
| 2.5 | 316.228 | 0.142 | 9.2e-05 |
| 3 | 1000 | 0.096 | 5.58e-06 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 4.2 | 15848.9 | 0.037 | 6.48e-09 |

### 38.2.25 Horizon C

Parent default, defined in sample/DrainDrying.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| alpha | 0.056 |
| n | 1.325 |
| K_sat | 1.0773 |
| Theta_sat | 0.354 |
| Theta_res | 0 |
| l | -0.191 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.352 | 0.402 |
| 1 | 10 | 0.322 | 0.0662 |
| 2 | 100 | 0.197 | 0.000668 |
| 2.5 | 316.228 | 0.138 | 3.72e-05 |
| 3 | 1000 | 0.096 | 1.93e-06 |
| 4.2 | 15848.9 | 0.039 | 1.52e-09 |

### 38.2.26 Horizon Orh(g)_Valecov_PG

Parent default, defined in sample/cz-soil.dai. Hydraulic model M_BaC_Bimodal.

| Parameter | Value |
|---|---|
| K_sat | 1 |
| K_b | 0.2 |
| Theta_sat | 0.449 |
| Theta_b | 0.395 |
| Theta_res | 0.004 |
| h_b | -10 |
| lambda | 0.112 |

### 38.2.27 Horizon E(g)_Valecov_PG

Parent default, defined in sample/cz-soil.dai. Hydraulic model M_BaC_Bimodal.

| Parameter | Value |
|---|---|
| K_sat | 0.89 |
| K_b | 0.18 |
| Theta_sat | 0.423 |
| Theta_b | 0.386 |
| Theta_res | 0 |
| h_b | -10 |
| lambda | 0.117 |

### 38.2.28 Horizon E(g)/Ig_Valecov_PG

Parent default, defined in sample/cz-soil.dai. Hydraulic model M_BaC_Bimodal.

| Parameter | Value |
|---|---|
| K_sat | 0.75 |
| K_b | 0.15 |
| Theta_sat | 0.417 |
| Theta_b | 0.385 |
| Theta_res | 0 |
| h_b | -10 |
| lambda | 0.107 |

### 38.2.29 Horizon Ig_Valecov_PG

Parent default, defined in sample/cz-soil.dai. Hydraulic model M_BaC_Bimodal.

| Parameter | Value |
|---|---|
| K_sat | 0.58 |
| K_b | 0.12 |
| Theta_sat | 0.384 |
| Theta_b | 0.366 |
| Theta_res | 0 |
| h_b | -10 |
| lambda | 0.117 |

### 38.2.30 Horizon Pg_Valecov_PG

Parent default, defined in sample/cz-soil.dai. Hydraulic model M_BaC_Bimodal.

| Parameter | Value |
|---|---|
| K_sat | 0.48 |
| K_b | 0.1 |
| Theta_sat | 0.383 |
| Theta_b | 0.364 |
| Theta_res | 0 |
| h_b | -10 |
| lambda | 0.17 |

### 38.2.31 Horizon Askov Ap

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0 |
| Theta_sat | 0.392 |
| alpha | 0.0385 |
| n | 1.211 |
| K_sat | 7.52 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.391 | 1.87 |
| 1 | 10 | 0.374 | 0.357 |
| 2 | 100 | 0.286 | 0.00602 |
| 2.5 | 316.228 | 0.229 | 0.000388 |
| 3 | 1000 | 0.181 | 2.21e-05 |
| 4.2 | 15848.9 | 0.101 | 2.08e-08 |

### 38.2.32 Horizon Askov B

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0 |
| Theta_sat | 0.445 |
| alpha | 0.2605 |
| n | 1.135 |
| K_sat | 14.5 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.435 | 0.492 |
| 1 | 10 | 0.378 | 0.0154 |
| 2 | 100 | 0.286 | 9.77e-05 |
| 2.5 | 316.228 | 0.245 | 6.77e-06 |
| 3 | 1000 | 0.210 | 4.61e-07 |
| 4.2 | 15848.9 | 0.145 | 7.25e-10 |

### 38.2.33 Horizon Askov C

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0 |
| Theta_sat | 0.401 |
| alpha | 0.057 |
| n | 1.131 |
| K_sat | 5 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.399 | 0.498 |
| 1 | 10 | 0.382 | 0.0652 |
| 2 | 100 | 0.314 | 0.001 |
| 2.5 | 316.228 | 0.273 | 7.66e-05 |
| 3 | 1000 | 0.236 | 5.43e-06 |
| 4.2 | 15848.9 | 0.164 | 8.84e-09 |

### 38.2.34 Horizon Jyndevad Ap

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model “B_BaC”.

| Parameter | Value |
|---|---|
| Theta_res | 0.036 |
| Theta_sat | 0.435 |
| lambda | 0.446 |
| h_b | -8.4 |
| K_sat | 6.76e-05 |

### 38.2.35 Horizon Jyndevad C

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model “B_BaC”.

| Parameter | Value |
|---|---|
| Theta_res | 0.017 |
| Theta_sat | 0.443 |
| lambda | 0.687 |
| h_b | -8.7 |
| K_sat | 0.000178 |

### 38.2.36 Horizon Foulum AP

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| K_sat | 2.995 |
| Theta_sat | 0.46 |
| Theta_res | 0 |
| alpha | 0.05334 |
| n | 1.287 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.458 | 0.974 |
| 1 | 10 | 0.424 | 0.153 |
| 2 | 100 | 0.278 | 0.00136 |
| 2.5 | 316.228 | 0.203 | 6.66e-05 |
| 3 | 1000 | 0.147 | 2.99e-06 |
| 4.2 | 15848.9 | 0.066 | 1.65e-09 |

### 38.2.37 Horizon Foulum B

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| K_sat | 1.796 |
| Theta_sat | 0.386 |
| Theta_res | 0 |
| alpha | 0.0398 |
| n | 1.2585 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.385 | 0.576 |
| 1 | 10 | 0.365 | 0.113 |
| 2 | 100 | 0.261 | 0.00158 |
| 2.5 | 316.228 | 0.199 | 8.83e-05 |
| 3 | 1000 | 0.149 | 4.37e-06 |
| 4.2 | 15848.9 | 0.073 | 2.95e-09 |

### 38.2.38 Horizon Foulum C

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| K_sat | 1.038 |
| Theta_sat | 0.345 |
| Theta_res | 0 |
| alpha | 0.0283 |
| n | 1.22 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.344 | 0.308 |
| 1 | 10 | 0.333 | 0.0736 |
| 2 | 100 | 0.262 | 0.00173 |
| 2.5 | 316.228 | 0.210 | 0.000116 |
| 3 | 1000 | 0.165 | 6.56e-06 |
| 4.2 | 15848.9 | 0.090 | 5.83e-09 |

### 38.2.39 Horizon Taastrup Ap

Parent USDA3, defined in sample/dk-soil.dai. Hydraulic model n/a.

| Parameter | Value |
|---|---|

### 38.2.40 Horizon Taastrup plow pan

Parent USDA3, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0 |
| Theta_sat | 0.339353 |
| alpha | 0.0465371 |
| n | 1.19395 |
| K_sat | * 0.5, 0.0925 |
| l | -2.92948 |

### 38.2.41 Horizon Taastrup Bt

Parent USDA3, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0 |
| l | -4.0323 |
| n | 1.14768 |
| alpha | 0.0565146 |
| K_sat | * 0.5, 0.537042 |
| Theta_sat | 0.356969 |

### 38.2.42 Horizon Taastrup C

Parent USDA3, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0 |
| Theta_sat | 0.348314 |
| alpha | 0.0476257 |
| n | 1.15336 |
| K_sat | 1.5 |
| l | -3.60322 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.347 | 0.214 |
| 1 | 10 | 0.332 | 0.0392 |
| 2 | 100 | 0.269 | 0.00155 |
| 2.5 | 316.228 | 0.228 | 0.000221 |
| 3 | 1000 | 0.192 | 3e-05 |
| 4.2 | 15848.9 | 0.126 | 2.37e-07 |

### 38.2.43 Horizon Taastrup drain canyon

Parent ISSS4, defined in sample/dk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0 |
| Theta_sat | 0.36384 |
| alpha | 0.0351307 |
| n | 1.14609 |
| K_sat | 0.526775 |
| l | -3.62743 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.363 | 0.0803 |
| 1 | 10 | 0.352 | 0.0172 |
| 2 | 100 | 0.295 | 0.000809 |
| 2.5 | 316.228 | 0.254 | 0.000118 |
| 3 | 1000 | 0.216 | 1.6e-05 |
| 4.2 | 15848.9 | 0.144 | 1.24e-07 |

### 38.2.44 Horizon WP1 IT CER 2005 L50

Parent default, defined in sample/it-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| K_sat | 1.96 |
| Theta_res | 0.01 |
| Theta_sat | 0.5 |
| alpha | 0.0623 |
| n | 1.306 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.497 | 0.646 |
| 1 | 10 | 0.453 | 0.0885 |
| 2 | 100 | 0.284 | 0.000607 |
| 2.5 | 316.228 | 0.206 | 2.76e-05 |
| 3 | 1000 | 0.148 | 1.17e-06 |
| 4.2 | 15848.9 | 0.069 | 5.65e-10 |

### 38.2.45 Horizon WP1 IT CER 2005 L100

Parent default, defined in sample/it-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| K_sat | 1.239 |
| Theta_res | 0.01 |
| Theta_sat | 0.45 |
| alpha | 0.0636 |
| n | 1.327 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.447 | 0.439 |
| 1 | 10 | 0.405 | 0.0598 |
| 2 | 100 | 0.245 | 0.000366 |
| 2.5 | 316.228 | 0.174 | 1.56e-05 |
| 3 | 1000 | 0.123 | 6.21e-07 |
| 4.2 | 15848.9 | 0.056 | 2.59e-10 |

### 38.2.46 Horizon Ap_JB1_Cosby

Parent ISSS4, defined in sample/pedo-soil.dai. Hydraulic model Cosby_et_al.

| Parameter | Value |
|---|---|

### 38.2.47 Horizon C_JB1_Cosby

Parent ISSS4, defined in sample/pedo-soil.dai. Hydraulic model Cosby_et_al.

| Parameter | Value |
|---|---|

### 38.2.48 Horizon Ap_JB1_hypres

Parent Ap_JB1_Cosby, defined in sample/pedo-soil.dai. Hydraulic model hypres.

| Parameter | Value |
|---|---|

### 38.2.49 Horizon C_JB1_hypres

Parent Ap_JB1_Cosby, defined in sample/pedo-soil.dai. Hydraulic model hypres.

| Parameter | Value |
|---|---|

### 38.2.50 Horizon Jadorna 2004

Parent USDA3, defined in sample/pl-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| l | 0.5 |
| n | 1.36188 |
| alpha | 0.0308718 |
| K_sat | 0.6 |
| Theta_sat | 0.397147 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.396 | 0.308 |
| 1 | 10 | 0.378 | 0.0835 |
| 2 | 100 | 0.251 | 0.00122 |
| 2.5 | 316.228 | 0.172 | 5.32e-05 |
| 3 | 1000 | 0.115 | 1.97e-06 |
| 4.2 | 15848.9 | 0.042 | 6.52e-10 |

### 38.2.51 Horizon Jaddeep 2004

Parent USDA3, defined in sample/pl-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| l | 1.1338 |
| n | 1.43343 |
| alpha | 0.0319665 |
| K_sat | 0.19 |
| Theta_sat | 0.41415 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.413 | 0.114 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 1 | 10 | 0.392 | 0.0318 |
| 2 | 100 | 0.238 | 0.000263 |
| 2.5 | 316.228 | 0.150 | 6.92e-06 |
| 3 | 1000 | 0.092 | 1.52e-07 |
| 4.2 | 15848.9 | 0.028 | 1.43e-11 |

### 38.2.52 Horizon Jaddeeper 2004

Parent USDA3, defined in sample/pl-soil.dai. Hydraulic model Cosby_et_al.

| Parameter | Value |
|---|---|

### 38.2.53 Horizon Jaddeepest 2004

Parent USDA3, defined in sample/pl-soil.dai. Hydraulic model Cosby_et_al.

| Parameter | Value |
|---|---|

### 38.2.54 Horizon A

Parent USDA3, defined in sample/safir-site.dai. Hydraulic model n/a.

| Parameter | Value |
|---|---|

### 38.2.55 Horizon Aor Most7

Parent default, defined in sample/sk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0.08 |
| Theta_sat | 0.4237 |
| alpha | 0.04611 |
| n | 1.17693 |
| K_sat | 1 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.422 | 0.178 |
| 1 | 10 | 0.407 | 0.0286 |
| 2 | 100 | 0.336 | 0.000447 |
| 2.5 | 316.228 | 0.293 | 3.09e-05 |
| 3 | 1000 | 0.254 | 1.93e-06 |
| 4.2 | 15848.9 | 0.187 | 2.29e-09 |

### 38.2.56 Horizon B Most7

Parent default, defined in sample/sk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0.0709 |
| Theta_sat | 0.3766 |
| alpha | 0.0157 |
| n | 1.1258 |
| K_sat | 1.02 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.376 | 0.169 |
| 1 | 10 | 0.373 | 0.0482 |
| 2 | 100 | 0.345 | 0.00254 |
| 2.5 | 316.228 | 0.317 | 0.000261 |
| 3 | 1000 | 0.286 | 2.06e-05 |
| 4.2 | 15848.9 | 0.224 | 3.62e-08 |

### 38.2.57 Horizon C Most7

Parent default, defined in sample/sk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0.0324 |
| Theta_sat | 0.4089 |
| alpha | 0.0172 |
| n | 1.1514 |
| K_sat | 0.95 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.408 | 0.201 |
| 1 | 10 | 0.403 | 0.0572 |
| 2 | 100 | 0.360 | 0.00267 |
| 2.5 | 316.228 | 0.319 | 0.000249 |
| 3 | 1000 | 0.276 | 1.81e-05 |
| 4.2 | 15848.9 | 0.193 | 2.64e-08 |

### 38.2.58 Horizon D Most7

Parent default, defined in sample/sk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0.0406 |
| Theta_sat | 0.4196 |
| alpha | 0.0359 |
| n | 1.1447 |
| K_sat | 0.75 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.419 | 0.11 |
| 1 | 10 | 0.407 | 0.0205 |
| 2 | 100 | 0.347 | 0.000455 |
| 2.5 | 316.228 | 0.305 | 3.59e-05 |
| 3 | 1000 | 0.266 | 2.5e-06 |
| 4.2 | 15848.9 | 0.192 | 3.73e-09 |

### 38.2.59 Horizon S Most7

Parent default, defined in sample/sk-soil.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| Theta_res | 0.0158 |
| Theta_sat | 0.3569 |
| alpha | 0.0203 |
| n | 1.1579 |
| K_sat | 0.5 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.356 | 0.106 |
| 1 | 10 | 0.350 | 0.028 |
| 2 | 100 | 0.306 | 0.00109 |
| 2.5 | 316.228 | 0.266 | 9.48e-05 |
| 3 | 1000 | 0.227 | 6.63e-06 |
| 4.2 | 15848.9 | 0.153 | 9.18e-09 |

### 38.2.60 Horizon Ap_plowing

Parent Ap, defined in sample/test-colloids.dai. Hydraulic model original.

| Parameter | Value |
|---|---|
| K_sat | 0.0001 |

### 38.2.61 Horizon Jyn_15h

Parent ISSS4, defined in sample/test-prd.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| K_sat | 125 |
| Theta_sat | 0.438 |
| Theta_res | 0.061 |
| alpha | 0.056 |
| n | 1.76 |
| l | 0.5 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.437 | 98.5 |
| 1 | 10 | 0.391 | 22.3 |
| 2 | 100 | 0.161 | 0.0261 |
| 2.5 | 316.228 | 0.103 | 0.000313 |
| 3 | 1000 | 0.079 | 3.54e-06 |
| 4.2 | 15848.9 | 0.063 | 7.41e-11 |

### 38.2.62 Horizon KarJyn_B

Parent ISSS4, defined in sample/test-prd.dai. Hydraulic model M_vG.

| Parameter | Value |
|---|---|
| K_sat | 110.8 |
| Theta_sat | 0.43 |
| Theta_res | 0.043 |
| alpha | 0.069 |
| n | 1.87 |
| l | 0.5 |

| pF | Suction, cm | Water content, cm3 per cm3 | Conductivity, cm per h |
|---|---|---|---|
| 0 | 1 | 0.429 | 90.1 |
| 1 | 10 | 0.364 | 16.2 |
| 2 | 100 | 0.114 | 0.00721 |
| 2.5 | 316.228 | 0.069 | 6.14e-05 |
| 3 | 1000 | 0.053 | 5.04e-07 |
| 4.2 | 15848.9 | 0.044 | 4.93e-12 |

### 38.2.63 Horizon KarJyn_C

Parent USDA3, defined in sample/test-prd.dai. Hydraulic model n/a.

| Parameter | Value |
|---|---|

## 38.3 Fertilisers

| Fertiliser | Parent | Dry matter | Total N | Total C | NH4 share | Volatilisation |
|---|---|---|---|---|---|---|
| kvaeg_gylle | slurry | 0.103 | 0.053 | 0.4 | 0.6 | 0.15 |
| kvaeg_staldgoedning | slurry | 0.196 | 0.027 | 0.4 | 0.251 | 0.15 |
| kvaeg_ajle | slurry | 0.033 | 0.162 | 0.4 | 0.9 | 0.15 |
| kvaeg_dybstroelse | slurry | 0.289 | 0.025 | 0.4 | 0.267 | 0.15 |
| slagtesvin_gylle | slurry | 0.054 | 0.09 | 0.4 | 0.75 | 0.15 |
| slagtesvin_staldgoedning | slurry | 0.23 | 0.039 | 0.4 | 0.349 | 0.15 |
| slagtesvin_ajle | slurry | 0.02 | 0.18 | 0.4 | 0.899 | 0.15 |
| slagtesvin_dybstroelse | slurry | 0.33 | 0.035 | 0.4 | 0.25 | 0.15 |
| soer_gylle | slurry | 0.05 | 0.101 | 0.4 | 0.75 | 0.15 |
| soer_staldgoedning | slurry | 0.23 | 0.037 | 0.4 | 0.35 | 0.15 |
| soer_ajle | slurry | 0.025 | 0.237 | 0.4 | 0.9 | 0.15 |
| soer_dybstroelse | slurry | 0.33 | 0.022 | 0.4 | 0.25 | 0.15 |
| slagtekyllinger_dybstroelse | slurry | 0.484 | 0.077 | 0.4 | 0.3 | 0.15 |
| hoens_gylle | slurry | 0.12 | 0.053 | 0.4 | 0.651 | 0.15 |
| hoens_staldgoedning | slurry | 0.4 | 0.053 | 0.4 | 0.35 | 0.15 |
| hoens_dybstroelse | slurry | 0.633 | 0.122 | 0.4 | 0.3 | 0.15 |
| kalkuner_dybstroelse | slurry | 0.48 | 0.126 | 0.4 | 0.3 | 0.15 |
| aender_gaes_dybstroelse | slurry | 0.35 | 0.079 | 0.4 | 0.3 | 0.15 |
| mink_raev_staldgoedning | slurry | 0.25 | 0.044 | 0.4 | 0.45 | 0.15 |
| mink_raev_gylle | slurry | 0.065 | 0.066 | 0.4 | 0.699 | 0.15 |
| hest_dybstroelse | slurry | 0.26 | 0.032 | 0.4 | 0.25 | 0.15 |
| faar_geder_dybstroelse | slurry | 0.346 | 0.034 | 0.4 | 0.25 | 0.15 |
| spildevandsslam | slurry | 0.141 | 0.304 | 0.4 | 0.292 | 0.15 |
| komposteret_husholdningsaffald | slurry | 0.5 | 0.019 | 0.4 | 0.107 | 0.15 |
| kartoffelfrugtsaft | slurry | 0.02 | 0.055 | 0.4 | 0.5 | 0.15 |
| novogro_slam | slurry | 0.035 | 0.031 | 0.4 | 0.1 | 0.15 |
| novogro-30_slam | slurry | 0.3 | 0.02 | 0.4 | 0.1 | 0.15 |
| pressesaft_groentilfabrik | slurry | 0.012 | 0.031 | 0.4 | 0.1 | 0.15 |
| svinegylle_useparret | slurry | 0.061 | 0.12 | 0.4 | 0.78 | 0.15 |
| svinegylle_separeret_tyk | slurry | 0.245 | 0.048 | 0.4 | 0.322 | 0.15 |
| svinegylle_separeret_tynd | slurry | 0.023 | 0.278 | 0.4 | 0.864 | 0.15 |
| kvaeggylle_useparret | slurry | 0.059 | 0.056 | 0.4 | 0.628 | 0.15 |
| kvaeggylle_separeret_tyk | slurry | 0.185 | 0.03 | 0.4 | 0.291 | 0.15 |
| kvaeggylle_separeret_tynd | slurry | 0.03 | 0.097 | 0.4 | 0.672 | 0.15 |
| Ammonia | mineral | n/a | n/a | n/a | 1 | n/a |
| AmmoniumNitrate | mineral | n/a | n/a | n/a | 0.5 | n/a |
| Nitrate | mineral | n/a | n/a | n/a | 0 | n/a |
| N25S | mineral | n/a | n/a | n/a | 0.508 | n/a |
| NPK01 | mineral | n/a | n/a | n/a | 0.53 | n/a |
| NPK02 | mineral | n/a | n/a | n/a | 0.57 | n/a |
| NP | mineral | n/a | n/a | n/a | 0.65 | n/a |
| CalciumNitrate | mineral | n/a | n/a | n/a | 0.08 | n/a |
| slurry | organic | n/a | n/a | n/a | n/a | 0 |
| cattle_slurry | slurry | 0.059 | 0.1 | 0.375 | 0.7 | n/a |
| pig_slurry | slurry | 0.059 | 0.1 | 0.375 | 0.7 | n/a |
| manure | slurry | n/a | n/a | n/a | n/a | n/a |
| cattle_manure | manure | 0.059 | 0.1 | 0.375 | 0.7 | n/a |
| pig_manure | manure | 0.059 | 0.1 | 0.375 | 0.7 | n/a |
| horse_manure | manure | 0.297 | 0.014 | 0.375 | 0.25 | 0 |
| Foulum_slurry | slurry | 0.0341 | 0.0615 | 0.323 | 0.57 | n/a |
| pigcz01_slurry | pig_slurry | 0.0249 | 0.1242 | 0.41017 | 0.88106 | 0.017 |
| pigcz02_slurry | pig_slurry | 0.0565 | 0.0853 | 0.41017 | 0.615475 | 0.017 |
| pigcz03_slurry | pig_slurry | 0.0584 | 0.08621 | 0.418 | 0.554 | 0.017 |
| pigcz04_slurry | pig_slurry | 0.05335 | 0.0552 | 0.4025 | 0.35688 | 0.017 |
| pigcz05_slurry | pig_slurry | 0.0729 | 0.056775 | 0.41 | 0.51387 | 0.017 |
| pig_farmyard_manure_cz01 | pig_manure | 0.2401 | 0.0355 | 0.316 | 0.33651 | 0.065 |
| pig_farmyard_manure_cz02 | pig_manure | 0.211 | 0.0419 | 0.365787 | 0.26084 | 0.065 |

| Fertiliser | Parent | Dry matter | Total N | Total C | NH4 share | Volatilisation |
|---|---|---|---|---|---|---|
| pig_farmyard_manure_cz03 | pigcz03_manure | 0.3215 | 0.03471 | 0.316 | 0.15552 | 0.065 |
| pig_farmyard_manure_cz04 | pigcz04_manure | 0.2332 | 0.0542 | 0.3317 | 0.58191 | 0.065 |
| pig_farmyard_manure_cz05 | pigcz05_manure | 0.19625 | 0.057 | 0.35875 | 0.55 | 0.065 |
| cattle_manurecz | pig_manure | 0.24 | n/a | n/a | n/a | n/a |
| cattle_dungwatercz | pigcz03_slurry | 0.02 | n/a | n/a | n/a | n/a |
| calcium_nitratecz | mineral | n/a | n/a | n/a | 0 | n/a |
| amonium_sulphatecz | mineral | n/a | n/a | n/a | 1 | n/a |
| ureacz | mineral | n/a | n/a | n/a | 1 | n/a |
| DAM390cz | mineral | n/a | n/a | n/a | 0.75 | n/a |
| LAV27cz | mineral | n/a | n/a | n/a | 0.5 | n/a |
| LA33cz | mineral | n/a | n/a | n/a | 0.5 | n/a |
| NPK11-12-12cz | mineral | n/a | n/a | n/a | 0.5 | n/a |
| NP26-14cz | mineral | n/a | n/a | n/a | 0.5 | n/a |
| NP20-20cz | mineral | n/a | n/a | n/a | 0.5 | n/a |
| KAS | mineral | n/a | n/a | n/a | 0.5 | n/a |
| N15B | mineral | n/a | n/a | n/a | 0.06 | n/a |
| N46 | mineral | n/a | n/a | n/a | 1 | n/a |
| NP 12-23-0 | mineral | n/a | n/a | n/a | 0.57 | n/a |
| NPK 14-3-18 | mineral | n/a | n/a | n/a | 0.54 | n/a |
| NS 24-7 | mineral | n/a | n/a | n/a | 0.5 | n/a |
| cattleJad_manure04 | organic | 0.243 | 0.01895 | 0.235 | 0.077 | 0.15 |
| jynd2005_slurry | slurry | 0.025 | 0.196 | 0.4 | 0.77 | n/a |

## 38.4 Chemicals

### 38.4.1 Chemical PFAS

Parent default, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| decompose_rate | 0 |
| canopy_dissipation_rate | 0 |
| canopy_washoff_coefficient | 1 |
| crop_uptake_reflection_factor | 1 |

### 38.4.2 Chemical PFOS

Parent PFAS, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| cite | Gauthier2024 |
| description | Aqueous diffusion coefficient from Table 1 |
| molar_mass | 550.13 |
| diffusion_coefficient | 5.27e-06 |
| adsorption | cite Brusseau2021x, description KOC and Kclay estimates from Jakobsen et al. (2025). KAWI calculated from equation in Figure 5 from Brusseau & Van Glubt (2021), K_AWI 0.0151, K_OC 604, K_clay 63 |

### 38.4.3 Chemical PFOS_Guo

Parent PFOS, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| adsorption | cite guo2020mathematical, description a and b from section 4 ‘Data and Parameters’ in Guo et al. (2020). KOC and Kclay estimated from Jakobsen et al. (2025), a 0.0034, b 0.107, K_OC 604, K_clay 63 |

### 38.4.4 Chemical PFOA

Parent PFAS, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| cite | Gauthier2024 |
| description | Aqueous diffusion coefficient from Table 1 |
| molar_mass | 414.07 |
| diffusion_coefficient | 5.46e-06 |
| adsorption | cite Brusseau2021x, description KOC and Kclay estimates from Jakobsen et al., (2025). KAWI calculated from equation in Figure 5 from Brusseau & Van Glubt (2021), K_AWI 0.00337, K_OC 91, K_clay 4 |

### 38.4.5 Chemical PFOA_Guo

Parent PFOA, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| adsorption | cite Constanza2019x, description a and b from Constanza et al. (2019) Table S1. KOC and Kclay estimated from Jakobsen et al. (2025), a 0.0116, b 0.033, K_OC 91, K_clay 4 |

### 38.4.6 Chemical PFHxA

Parent PFAS, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| cite | Gauthier2024 |
| description | Aqueous diffusion coefficient from Table 1 |
| molar_mass | 314.05 |
| diffusion_coefficient | 5.93e-06 |
| adsorption | cite Brusseau2021x, description KOC and Kclay estimates from Jakobsen et al., (2025). KAWI calculated from equation in Figure 5 from Brusseau & Van Glubt (2021), K_AWI 0.000196, K_OC 14, K_clay 5 |

### 38.4.7 Chemical PFPeA

Parent PFAS, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| cite | Gauthier2024 |
| description | Aqueous diffusion coefficient from Table 1 |
| molar_mass | 264.05 |
| diffusion_coefficient | 6.33e-06 |
| adsorption | cite Brusseau2021x, description KOC and Kclay estimates from Jakobsen et al., (2025). KAWI calculated from equation in Figure 5 from Brusseau & Van Glubt (2021), K_AWI 7.09e-05, K_OC 55, K_clay 8 |

### 38.4.8 Chemical PFBS

Parent PFAS, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| cite | Gauthier2024 |
| description | Aqueous diffusion coefficient from Table 1 |
| molar_mass | 338.19 |
| diffusion_coefficient | 6.79e-06 |
| adsorption | cite Brusseau2021x, description KOC and Kclay estimates from Jakobsen et al., (2025). KAWI calculated from equation in Figure 5 from Brusseau & Van Glubt (2021), K_AWI 0.000294, K_OC 9, K_clay 11 |

### 38.4.9 Chemical PFBA

Parent PFAS, from lib/PFAS.dai.

| Parameter | Value |
|---|---|
| cite | Gauthier2024 |

| Parameter | Value |
|---|---|
| description | Aqueous diffusion coefficient from Table 1 |
| molar_mass | 214.04 |
| diffusion_coefficient | 6.73e-06 |
| adsorption | cite Brusseau2021x, description KOC and Kclay estimates from Jakobsen et al., (2025). KAWI calculated from equation in Figure 5 from Brusseau & Van Glubt (2021), K_AWI 2.36e-05, K_OC 3, K_clay 9 |

### 38.4.10 Chemical solid

Parent default, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|
| crop_uptake_reflection_factor | 1 |
| canopy_dissipation | 0 |
| canopy_washoff_coefficient | 1 |
| diffusion_coefficient | 1 |
| adsorption | full |
| decompose | 0 |

### 38.4.11 Chemical classic

Parent default, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|
| cite | focusgw2002 |
| decompose_depth_factor | (-105, 0); (-95, 0.3); (-65, 0.3); (-55, 0.5); (-35, 0.5); (-25, 1); (0, 1) |
| canopy_dissipation | 0.0083 |
| canopy_washoff_coefficient | 1 |
| crop_uptake_reflection_factor | 1 |
| diffusion_coefficient | 4.6e-06 |

### 38.4.12 Chemical pesticide

Parent FOCUS, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|

### 38.4.13 Chemical fungicide

Parent pesticide, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|

### 38.4.14 Chemical herbicide

Parent pesticide, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|

### 38.4.15 Chemical insecticide

Parent pesticide, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|

### 38.4.16 Chemical metabolite

Parent FOCUS, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|

### 38.4.17 Chemical Tracer

Parent default, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|
| canopy_dissipation | 0 |
| canopy_washoff_coefficient | 1 |
| crop_uptake_reflection_factor | 1 |
| decompose | 0 |
| diffusion_coefficient | 1.8e-05 |
| adsorption | none |

### 38.4.18 Chemical colloid

Parent default, from lib/chemistry-base.dai.

| Parameter | Value |
|---|---|
| canopy_dissipation | 0 |
| canopy_washoff_coefficient | 1 |
| diffusion_coefficient | 1e-12 |
| decompose | 0 |

### 38.4.19 Chemical Atrazine

Parent herbicide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| crop_uptake_reflection_factor | 0.8 |
| decompose | 7e-05 |
| diffusion_coefficient | 8e-06 |
| adsorption | K_clay 7 |

### 38.4.20 Chemical Bentazon

Parent herbicide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 40 |
| adsorption | K_OC 34 |

### 38.4.21 Chemical IPU

Parent herbicide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 29 |
| adsorption | K_OC 100 |

### 38.4.22 Chemical MCPP

Parent herbicide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 7 |
| adsorption | K_OC 20 |

### 38.4.23 Chemical Pendimethalin

Parent herbicide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 0.0004 |
| adsorption | K_OC 16000 |

### 38.4.24 Chemical Ioxynil

Parent herbicide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 10 |
| adsorption | K_OC 350 |

### 38.4.25 Chemical 2,4-D

Parent herbicide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 10 |
| adsorption | K_OC 20 |

### 38.4.26 Chemical Heptachlor

Parent insecticide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 300 |
| adsorption | K_OC 24000 |

### 38.4.27 Chemical DDT

Parent insecticide, from lib/chemistry.dai.

| Parameter | Value |
|---|---|
| decompose | 2000 |
| adsorption | K_OC 2000000 |

### 38.4.28 Chemical common

Parent default, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|
| canopy_dissipation_rate | 0.0083 |
| canopy_washoff_coefficient | 1 |
| crop_uptake_reflection_factor | 1 |
| diffusion_coefficient | 4.6e-06 |

### 38.4.29 Chemical pesticide

Parent common, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|

### 38.4.30 Chemical herbicide

Parent pesticide, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|

### 38.4.31 Chemical colloid

Parent common, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|
| diffusion_coefficient | 4.6e-06 |
| decompose_rate | 0 |

### 38.4.32 Chemical Pendimethalin-base

Parent herbicide, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|
| decompose_rate | 0.0004 |

### 38.4.33 Chemical Pendimethalin

Parent Pendimethalin-base, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|
| adsorption | none |

### 38.4.34 Chemical Pendimethalin-immobile

Parent Pendimethalin-base, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|
| adsorption | full |

### 38.4.35 Chemical Pendimethalin-colloid

Parent Pendimethalin-base, from sample/test-colloids.dai.

| Parameter | Value |
|---|---|
| adsorption | none |
| diffusion_coefficient | 4.6e-06 |

## 38.5 Crops

### 38.5.1 Crop Beet; V1

Parent sold, defined in lib/beet-old.dai. Beet; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 385 |
| DSRate1 | 0.00087 |
| DSRate2 | 0.00087 |
| DS_Emr | 0.01 |
| DS_reset | false |
| TempEff1 | (0, 0); (100, 100) |
| TempEff2 | (0, 0); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| TLim1 | 3 |
| TLim2 | 8 |

**Canopy**

| Parameter | Value |
|---|---|
| InitGrowth | 4.6 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 1.39 |
| alpha | 1 |
| beta | 1.3043 |
| SpLAI | 0.011 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.6 |

**Root**

| Parameter | Value |
|---|---|
| DptEmr | 10 |

| Parameter | Value |
|---|---|
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 150 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.52, 0.2); (2, 0.2) |
| SOrg | (0, 0); (0.78, 0.6); (2, 0.6) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_SOrg | (0, 0.85); (2, 0.85) |
| E_Leaf | (0, 0.83); (2, 0.83) |
| r_Root | (0, 0.065); (2, 0.065) |
| r_SOrg | (0, 0.005); (2, 0.005) |
| r_Leaf | (0, 0.03); (2, 0.03) |

### CrpN

| Parameter | Value |
|---|---|
| SeedN | 0.02 |
| PtLeafCnc | (0, 0.055); (0.026, 0.053); (0.052, 0.052); (0.078, 0.05); (0.104, 0.048); (0.13, 0.047); (0.157, 0.045); (0.183, 0.044); (0.209, 0.043); (0.235, 0.041); (0.261, 0.04) |
| PtRootCnc | (0, 0.02); (0.026, 0.019); (0.052, 0.019); (0.078, 0.018); (0.104, 0.018); (0.13, 0.017); (0.157, 0.017); (0.183, 0.016); (0.209, 0.016); (0.235, 0.015); (0.261, 0.015) |
| PtSOrgCnc | (0, 0.03); (0.026, 0.027); (0.052, 0.024); (0.078, 0.022); (0.104, 0.019); (0.13, 0.017); (0.157, 0.015); (0.183, 0.013); (0.209, 0.012); (0.235, 0.011); (0.261, 0.01) |
| CrLeafCnc | (0, 0.035); (0.026, 0.033); (0.052, 0.03); (0.078, 0.028); (0.104, 0.026); (0.13, 0.024); (0.157, 0.023); (0.183, 0.021); (0.209, 0.02); (0.235, 0.018); (0.261, 0.017) |
| CrRootCnc | (0, 0.012); (0.026, 0.012); (0.052, 0.011); (0.078, 0.011); (0.104, 0.01); (0.13, 0.01); (0.157, 0.009); (0.183, 0.009); (0.209, 0.009); (0.235, 0.008); (0.261, 0.008) |
| CrSOrgCnc | (0, 0.012); (0.026, 0.012); (0.052, 0.011); (0.078, 0.011); (0.104, 0.01); (0.13, 0.01); (0.157, 0.009); (0.183, 0.009); (0.209, 0.009); (0.235, 0.008); (0.261, 0.008) |
| NfLeafCnc | (0, 0.017); (0.026, 0.016); (0.052, 0.015); (0.078, 0.014); (0.104, 0.014); (0.13, 0.013); (0.157, 0.012); (0.183, 0.012); (0.209, 0.011); (0.235, 0.011); (0.261, 0.01) |
| NfRootCnc | (0, 0.01); (0.026, 0.009); (0.052, 0.009); (0.078, 0.008); (0.104, 0.008); (0.13, 0.007); (0.157, 0.007); (0.183, 0.006); (0.209, 0.006); (0.235, 0.005); (0.261, 0.005) |

| Parameter | Value |
|---|---|
| NfSOrgCnc | (0, 0.01); (0.026, 0.009); (0.052, 0.009); (0.078, 0.008); (0.104, 0.008); (0.13, 0.007); (0.157, 0.007); (0.183, 0.006); (0.209, 0.006); (0.235, 0.005); (0.261, 0.005) |

**Harvest**

| Parameter | Value |
|---|---|
| At | -0.0072 |
| Bt | 0.69 |
| beta | 0.075 |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| C_Leaf | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.
IntcpCap: 0.5.

### 38.5.2 Crop Beetroot

Parent default, defined in lib/beetroot.dai. Beetroot

**Devel**

| Parameter | Value |
|---|---|
| defined_until_ds | 1 |
| EmrTSum | 330 |
| DSRate1 | 0.0085 |
| TempEff1 | (0, 0.01); (5, 0.5); (20, 1); (25, 0.5) |
| PhotEff1 | (1, 0); (8, 1); (12, 1); (24, 1) |
| DSRate2 | 0.007 |
| TempEff2 | (0, 1); (30, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4 |
| Qeff | 0.055 |
| TempEff | (0, 0.01); (5, 0.01); (10, 0.75); (20, 1); (35, 1); (40, 0.9); (45, 0.01) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.2 |
| SpLAIfac | (0, 1); (1, 1) |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.025 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| LeafAIMod | (0, 1); (0.12, 1); (0.33, 0.74); (0.95, 0.47) |
| PARref | 0.06 |
| PARext | 0.68 |
| EPext | 0.5 |
| HvsWStem | (0, 1); (1, 2) |
| HvsDS | (0, 0); (0.5, 35); (2, 35) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.16 |
| PenPar2 | 6 |
| MaxPen | 180 |
| Rad | 0.05 |
| h_wp | -15000 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.3); (0.2, 0.2); (0.4, 0.2); (0.6, 0.2); (0.85, 0.15); (1.1, 0.1) |
| Leaf | (0, 0.6); (0.2, 0.4); (0.4, 0.35); (0.6, 0.3); (0.85, 0.125); (1.1, 0) |
| Stem | (0, 0); (1.1, 0) |
| RSR | (0, 0.5); (1, 0.5); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.75 |
| r_SOrg | 0.005 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0.2, 0); (0.4, 0.01); (0.6, 0.01); (0.85, 0.015); (1.1, 0.09); (2, 0.1) |
| RtDR | (0.2, 0); (0.6, 0); (0.95, 0); (1, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.007) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.005) |
| PtLeafCnc | (0, 0.035); (1, 0.03); (2, 0.03) |
| CrLeafCnc | (0, 0.02); (1, 0.015); (2, 0.01) |
| NfLeafCnc | (0, 0.0075); (1, 0.005); (2, 0.005) |
| PtStemCnc | (0, 0.035); (1, 0.03); (2, 0.03) |
| CrStemCnc | (0, 0.02); (1, 0.01); (2, 0.01) |
| NfStemCnc | (0, 0.0075); (1, 0.005); (2, 0.005) |
| PtSOrgCnc | (0, 0.025); (1, 0.02); (2, 0.02) |
| CrSOrgCnc | (0, 0.01); (1, 0.01); (2, 0.01) |
| NfSOrgCnc | (0, 0.005); (1, 0.005); (2, 0.005) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |
| sorg_height | -15 |

### 38.5.3 Crop Broccoli

Parent default, defined in lib/broccoli.dai. Broccoli

**Devel**

| Parameter | Value |
|---|---|
| defined_until_ds | 2 |
| EmrTSum | 175 |
| DSRate1 | 0.027 |
| TempEff1 | (0, 0); (7, 0); (16, 1); (18, 1); (30, 0) |
| PhotEff1 | (0, 0); (8, 1); (16, 1); (24, 0.8) |
| DSRate2 | 0.05 |
| TempEff2 | (0, 0); (7, 0); (16, 1); (18, 1); (30, 0) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 3.6 |
| Qeff | 0.045 |
| TempEff | (0, 0); (15, 1); (25, 1); (40, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |
| SpLAIfac | (0, 3); (0.2, 1.5); (0.4, 1.25); (0.6, 1); (2, 1) |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.0165 |

| Parameter | Value |
|---|---|
| SpSOrgAI | 0.00165 |
| SpStemAI | 0 |
| LAIDist0 | 0, 0.1, 1 |
| LAIDist1 | 0.1, 0.2, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| PARrel | 0.05 |
| HvsDS | (0, 0); (1, 50); (2, 50) |
| HvsWStem | (0, 0.1); (100, 1) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 5 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.05 |
| h_wp | -15000 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.1, 0.4); (0.15, 0.1); (0.4, 0.1); (0.8, 0.1); (1, 0.1); (1.5, 0.1); (2, 0.1) |
| Leaf | (0, 0.85); (0.1, 0.7); (0.15, 0.8); (0.4, 0.75); (0.8, 0.65); (1, 0.45); (1.5, 0.2); (2, 0.05) |
| Stem | (0, 0.15); (0.1, 0.3); (0.15, 0.2); (0.4, 0.25); (0.8, 0.35); (1, 0.5); (1.5, 0.5); (2, 0.15) |
| RSR | (0, 1); (1, 0.5); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_SOrg | 0.025 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 1 |
| LfDR | (0, 0); (0.5, 0); (0.6, 0); (1, 0); (1.5, 0.005); (2, 0.01) |
| RtDR | (0, 0); (0.3, 0); (0.6, 0); (1, 0.01); (1.5, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.025); (1, 0.0225); (2, 0.0225) |
| CrRootCnc | (0, 0.02); (1, 0.0175); (2, 0.0175) |
| NfRootCnc | (0, 0.0125); (1, 0.01); (2, 0.01) |
| PtLeafCnc | (0, 0.065); (1, 0.06); (2, 0.06) |
| CrLeafCnc | (0, 0.045); (1, 0.0425); (2, 0.0425) |
| NfLeafCnc | (0, 0.02); (1, 0.0175); (2, 0.0175) |
| PtStemCnc | (0, 0.045); (1, 0.0425); (2, 0.0425) |
| CrStemCnc | (0, 0.025); (1, 0.0225); (2, 0.0225) |
| NfStemCnc | (0, 0.0125); (1, 0.01); (2, 0.01) |
| PtSOrgCnc | (0, 0.06); (1, 0.0575); (2, 0.0575) |
| CrSOrgCnc | (0, 0.0525); (1, 0.05); (2, 0.05) |
| NfSOrgCnc | (0, 0.0225); (1, 0.02); (2, 0.02) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |

### 38.5.4 Crop Broccoli minus transplanted

Parent Broccoli, defined in lib/broccoli.dai. Broccoli minus transplanted

**Devel**

| Parameter | Value |
|---|---|
| DS | 0.1 |

**Prod**

| Parameter | Value |
|---|---|
| CH2OPool | 0.05 |
| NCrop | 0.02 |
| WLeaf | 0.175 |
| WRoot | 0.25 |
| WStem | 0.075 |

**Root**

| Parameter | Value |
|---|---|
| Depth | 5 |

**Harvest**

| Parameter | Value |
|---|---|
| production_delay | 14 |

### 38.5.5 Crop Brussels sprouts

Parent default, defined in lib/brusselssprouts.dai. Brussels sprouts

**Devel**

| Parameter | Value |
|---|---|
| defined_until_ds | 2 |
| EmrTSum | 250 |
| DSRate1 | 0.0044 |
| TempEff1 | (0, 0.01); (11, 1); (19, 1); (32, 0.01) |
| PhotEff1 | (0, 0); (8, 1); (16, 1); (24, 0.8) |
| DSRate2 | 0.007 |
| TempEff2 | (0, 1); (30, 1) |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 3.8 |
| Qeff | 0.045 |
| TempEff | (0, 0.01); (11, 1); (19, 1); (30, 0.7); (35, 0.01) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |
| SpLAIfac | (0, 3); (0.15, 1.5); (0.3, 1.25); (0.45, 1); (1, 1) |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.008 |
| SpSOrgAI | 0.0008 |
| SpStemAI | 0 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| PARrel | 0.05 |
| HvsDS | (0, 0); (0.7, 70); (2, 70) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 5 |
| PenPar1 | 0.15 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.05 |
| h_wp | -15000 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.1, 0.35); (0.15, 0.15); (0.25, 0.15); (0.35, 0.15); (0.4, 0.15); (0.5, 0.15); (0.75, 0.12); (1, 0.1) |
| Leaf | (0, 0.95); (0.1, 0.9); (0.25, 0.85); (0.3, 0.75); (0.35, 0.7); (0.4, 0.6); (0.5, 0.6); (0.75, 0.1); (1, 0.05) |
| Stem | (0, 0.05); (0.1, 0.1); (0.25, 0.15); (0.3, 0.25); (0.35, 0.3); (0.4, 0.4); (0.5, 0.4); (0.75, 0.1); (1, 0.05) |
| RSR | (0, 0.5); (1, 0.25) |

**Prod**

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_SOrg | 0.02 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 1 |
| LfDR | (0, 0); (0.5, 0); (0.6, 0); (1, 0.007) |
| RtDR | (0, 0); (0.3, 0); (0.6, 0); (1, 0.005) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.0175); (2, 0.0175) |
| CrRootCnc | (0, 0.01); (1, 0.0075); (2, 0.0075) |
| NfRootCnc | (0, 0.0075); (1, 0.005); (2, 0.005) |
| PtLeafCnc | (0, 0.065); (1, 0.05); (2, 0.05) |
| CrLeafCnc | (0, 0.03); (1, 0.0275); (2, 0.0275) |
| NfLeafCnc | (0, 0.015); (1, 0.012); (2, 0.012) |
| PtStemCnc | (0, 0.065); (1, 0.05); (2, 0.05) |
| CrStemCnc | (0, 0.03); (1, 0.0275); (2, 0.0275) |
| NfStemCnc | (0, 0.015); (1, 0.012); (2, 0.012) |
| PtSOrgCnc | (0, 0.065); (1, 0.05); (2, 0.05) |
| CrSOrgCnc | (0, 0.03); (1, 0.0275); (2, 0.0275) |
| NfSOrgCnc | (0, 0.015); (1, 0.012); (2, 0.012) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |

### 38.5.6 Crop Brussels sprouts minus transplanted

Parent Brussels sprouts, defined in lib/brusselssprouts.dai. Brussels sprouts minus transplanted

**Devel**

| Parameter | Value |
|---|---|
| DS | 0.1 |

**Prod**

| Parameter | Value |
|---|---|
| CH2OPool | 0.05 |
| NCrop | 0.02 |
| WLeaf | 0.225 |
| WRoot | 0.25 |
| WStem | 0.025 |

**Root**

| Parameter | Value |
|---|---|
| Depth | 5 |

**Harvest**

| Parameter | Value |
|---|---|
| production_delay | 14 |

### 38.5.7 Crop Celeriac

Parent default, defined in lib/celeriac.dai. Celeriac

**Devel**

| Parameter | Value |
|---|---|
| defined_until_ds | 1 |
| EmrTSum | 350 |
| DSRate1 | 0.005 |
| TempEff1 | (0, 0.01); (5, 0.5); (15, 1); (22, 1); (28, 0.5) |
| PhotEff1 | (0, 0.4); (8, 1); (16, 1); (24, 0.8) |
| DSRate2 | 0.007 |
| TempEff2 | (0, 1); (30, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4.3 |
| Qeff | 0.045 |
| TempEff | (0, 0.01); (5, 0.01); (15, 1); (22, 1); (35, 0.8); (40, 0.01) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |
| SpLAIfac | (0, 1); (1, 1) |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |
| LeafAIMod | (0, 1); (0.2, 1); (0.4, 0.44); (1, 0.44) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.68 |
| EPext | 0.5 |
| HvsWStem | (0, 0.1); (50, 1) |
| HvsDS | (0, 0); (0.5, 45); (2, 45) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 5 |
| PenPar1 | 0.038 |
| PenPar2 | 5 |
| MaxPen | 48 |
| Rad | 0.05 |
| h_wp | -15000 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.35, 0.25); (0.55, 0.1); (0.7, 0.06); (1.1, 0) |
| Leaf | (0, 0.75); (0.35, 0.55); (0.55, 0.35); (0.7, 0.25); (0.85, 0.15); (1.1, 0) |
| Stem | (0, 0.25); (0.35, 0.2); (0.55, 0.15); (0.7, 0.1); (0.85, 0.05); (1.1, 0) |
| RSR | (0, 0.5); (1, 0.5); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_SOrg | 0.01 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0.19, 0); (0.2, 0.025); (0.25, 0.005); (0.55, 0.005); (0.85, 0.011); (1.1, 0.01); (2, 0.1) |
| RtDR | (0.19, 0); (0.2, 0.01); (0.25, 0); (0.35, 0); (0.55, 0.005); (1.1, 0.005); (2, 0.01) |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (0.5, 0.02); (1, 0.02); (2, 0.02) |
| CrRootCnc | (0, 0.01); (0.5, 0.01); (1, 0.01); (2, 0.01) |
| NfRootCnc | (0, 0.007); (0.5, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.045); (0.5, 0.04); (1, 0.03); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (0.5, 0.025); (1, 0.015); (2, 0.015) |
| NfLeafCnc | (0, 0.01); (0.5, 0.0075); (1, 0.005); (2, 0.005) |
| PtStemCnc | (0, 0.045); (0.5, 0.04); (1, 0.03); (2, 0.03) |
| CrStemCnc | (0, 0.03); (0.5, 0.025); (1, 0.015); (2, 0.015) |
| NfStemCnc | (0, 0.01); (0.5, 0.0075); (1, 0.005); (2, 0.005) |
| PtSOrgCnc | (0, 0.0375); (0.5, 0.035); (1, 0.03); (2, 0.03) |
| CrSOrgCnc | (0, 0.0225); (0.5, 0.02); (1, 0.015); (2, 0.015) |
| NfSOrgCnc | (0, 0.0075); (0.5, 0.005); (1, 0.005); (2, 0.005) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |
| sorg_height | -10 |

### 38.5.8 Crop Celeriac minus transplanted

Parent Celeriac, defined in lib/celeriac.dai. Celeriac minus transplanted

**Devel**

| Parameter | Value |
|---|---|
| DS | 0.2 |

**Prod**

| Parameter | Value |
|---|---|
| CH2OPool | 0.5 |
| NCrop | 0.25 |
| WLeaf | 1.75 |
| WRoot | 2.5 |
| WStem | 0.75 |

**Root**

| Parameter | Value |
|---|---|
| Depth | 7 |

**Harvest**

| Parameter | Value |
|---|---|
| production_delay | 21 |

### 38.5.9 Crop Froegraes

Parent default, defined in lib/dk-grass.dai. Froegraes

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DS_Emr | 0.01 |
| DSRate1 | 0.022 |
| TempEff1 | (-10, 0.01); (0, 0.01); (20, 0.9); (25, 1); (35, 1.2) |
| PhotEff1 | (0, 0.1); (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.028 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |
| defined_until_ds | 2. |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.9 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Qeff | 0.05 |
| DSEff | (0, 1); (0.3, 1); (1, 1); (2, 1) |
| Fm | 2.5 |
| TempEff | (-20, 0.0001); (5, 0.0001); (10, 0.9); (25, 1); (35, 0.01); (50, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.2 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.025 |
| HvsDS | (0, 1); (1, 30); (2, 30) |
| LeafAIMod | (0); (2, 1) |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.6, 1); (2, 0) |
| StemPhotEff | 0.2 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| IntcpCap | 0.5 |

**Root**

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 60 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |
| Root | (0, 0.5); (0.33, 0.25); (0.53, 0.2); (1, 0); (2, 0) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.43, 0.83); (0.53, 0.56); (0.95, 0.36); (1.14, 0) |
| Stem | (0, 0.1); (0.33, 0.15); (0.43, 0.17); (0.53, 0.44); (0.95, 0.64); (1.14, 0.62); (1.45, 0.3); (2, 0.05) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.57 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0 |
| ReMobilDS | 1.38 |
| ReMobilRt | 0.1 |
| LfDR | (0.09, 0); (0.1, 0); (0.35, 0); (1, 0.02); (2, 0.1) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |
| Large_RtDR | 0.05 |
| IntDSRelRtRes | 0.2 |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (0.6, 0.06); (1, 0.04); (2, 0.02) |
| CrLeafCnc | (0, 0.04); (0.6, 0.04); (1, 0.03); (2, 0.015) |
| NfLeafCnc | (0, 0.004); (0.6, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.025); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.02); (1, 0.012); (2, 0.006) |
| NfStemCnc | (0, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (2, 0.014) |

| Parameter | Value |
|---|---|
| TLLeafEff | (0, 0.9); (2, 0.9) |
| TLRootEff | (0, 0.1); (2, 0.1) |

**Harvest**

| Parameter | Value |
|---|---|
| Dead | CROP-SLOW |
| Stem | CROP-SLOW |
| Leaf | CROP-SLOW |
| SOrg | CROP-SLOW initial_fraction 0 |
| Root | CROP-SLOW |
| DSmax | 3 |
| DSnew | 0.2 |

## 38.5.10 Crop Graes

Parent Froegraes, defined in lib/dk-grass.dai. Graes

**Devel**

| Parameter | Value |
|---|---|
| PhotEff1 | (0, 0.1); (10, 0); (11, 0); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |

**LeafPhot**

| Parameter | Value |
|---|---|
| DSEff | (0, 0.68); (0.3, 0.68); (1, 0.68); (2, 0.68) |
| Fm | 2 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |

**Root**

| Parameter | Value |
|---|---|
| MxNH4Up | 1.2e-07 |
| MxNO3Up | 1.2e-08 |

## 38.5.11 Crop Red fescue

Parent Froegraes, defined in lib/dk-grass.dai. Red fescue

cite: redfescue.

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DS_Emr | 0.01 |
| DSRate1 | 0.022 |
| TempEff1 | (-10, 0.01); (0, 0.01); (20, 0.9); (25, 1); (35, 1.2) |
| PhotEff1 | (0, 0.1); (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.028 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |
| defined_until_ds | 2. |

**LeafPhot**

| Parameter | Value |
|---|---|
| Qeff | 0.05 |
| Fm | 2.16 |
| DSEff | (0, 0.5); (0.3, 0.6); (1, 1); (2, 1) |
| TempEff | (-20, 0.0001); (5, 0.0001); (10, 0.9); (25, 1); (35, 0.01); (50, 0) |

**CrpN**

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.05); (0.6, 0.05); (1, 0.04); (2, 0.02) |
| CrLeafCnc | (0, 0.04); (0.6, 0.03); (1, 0.02); (2, 0.015) |
| NfLeafCnc | (0, 0.004); (0.6, 0.004); (2, 0.004) |

## 38.5.12 Crop Silomajs

Parent default, defined in lib/dk-maize.dai. Silomajs

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 300 |
| DS_Emr | 0.01 |
| DSRate1 | 0.016 |
| TempEff1 | 0. 0.3, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |
| PhotEff1 | 0. 0.9, 12. 1, 14. 0.95, 16. 0.9, 24. 0.9 |
| DSRate2 | 0.021 |
| TempEff2 | 0. 0.3, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 5 |
| DSEff | (0, 1); (0.3, 1); (1, 1); (2, 1) |
| Qeff | 0.04 |
| TempEff | (-20, 0.001); (6, 0.1); (10, 1); (15, 1); (20, 1) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.25 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.01 |
| LeafAIMod | (0, 1); (2, 1) |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.6, 0); (2, 0) |
| StemPhotEff | 0.2 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.6, 0); (2, 0) |
| SOrgPhotEff | 0.2 |
| LAIDist0 | 0, 1, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.8 |
| EPext | 0.5 |
| HvsDS | (0, 1); (0.2, 50); (0.5, 150); (1, 200); (2, 200) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 100 |
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 9e-09 |
| MxNO3Up | 9e-09 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.67, 0.35); (1.37, 0.1); (1.6, 0.1); (2, 0.05) |
| Leaf | (0.35, 0.7); (0.67, 0.5); (1, 0.15); (1.18, 0.05); (1.85, 0.05); (2, 0.05) |
| Stem | (0.35, 0.3); (0.67, 0.5); (1, 0.85); (1.18, 0.05); (1.8, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.68 |
| r_Root | 0.015 |
| r_Leaf | 0.025 |

| Parameter | Value |
|---|---|
| r_Stem | 0.01 |
| r_SOrg | 0.03 |
| ShldResC | 0 |
| ReMobilDS | 1.4 |
| ReMobilRt | 0.02 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.5 |
| LfDR | (0, 0); (0.1, 0); (1.5, 0.005); (2, 0.005) |
| RtDR | (0, 0); (0.2, 0); (1.2, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.005) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.004) |
| PtLeafCnc | (0, 0.044); (0.367, 0.044); (0.5, 0.04); (1, 0.03); (1.5, 0.02); (1.8, 0.018) |
| CrLeafCnc | (0, 0.01); (0.367, 0.01); (0.5, 0.009); (1, 0.008); (1.5, 0.008); (1.8, 0.008) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0.838, 0.02); (1.057, 0.015); (1.244, 0.014) |
| CrStemCnc | (0.838, 0.01); (1.057, 0.01); (1.244, 0.008) |
| NfStemCnc | (0.838, 0.004); (1.057, 0.004); (1.244, 0.004) |
| PtSOrgCnc | (1.057, 0.02); (1.244, 0.018); (2, 0.018) |
| CrSOrgCnc | (1.057, 0.016); (1.244, 0.012); (2, 0.012) |
| NfSOrgCnc | (1.057, 0.014); (1.244, 0.01); (2, 0.01) |

### 38.5.13 Crop Silomajs; Foulum

Parent Silomajs, defined in lib/dk-maize.dai. Silomajs; Foulum
cite: redfescue.

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 180 |
| DSRate1 | 0.02 |
| DSRate2 | 0.021 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 6.5 |
| DSEff | (0, 1); (0.3, 1); (1, 1); (2, 0.6) |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.02 |

| Parameter | Value |
|---|---|
| LeafAIMod | (0, 0.35); (0.4, 0.45); (0.5, 0.5); (0.6, 0.55); (0.7, 0.58); (0.8, 0.62); (0.9, 0.75); (1.05, 0.9); (1.08, 1); (1.11, 1.05); (1.2, 1.08); (1.3, 1.1); (1.4, 1.1); (1.5, 1.1); (1.6, 1); (1.7, 0.9); (1.8, 0.5); (2, 0.4) |

## Root

| Parameter | Value |
|---|---|
| MxNH4Up | 8e-08 |
| MxNO3Up | 8e-08 |

## Partit

| Parameter | Value |
|---|---|
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |
| Root | (0, 0.6); (0.67, 0.3); (1.37, 0.1); (1.6, 0.1); (2, 0.05) |
| Leaf | (0.35, 0.63); (0.49, 0.54); (0.67, 0.34); (0.9, 0.32); (1, 0.3); (1.11, 0.3); (1.2, 0.1); (1.3, 0); (1.8, 0); (2, 0) |
| Stem | (0.35, 0.37); (0.49, 0.46); (0.67, 0.66); (0.9, 0.68); (1, 0.7); (1.11, 0); (1.2, 0); (1.3, 0); (1.8, 0); (2, 0) |

## Prod

| Parameter | Value |
|---|---|
| E_Leaf | 0.75 |
| E_Stem | 0.78 |
| E_SOrg | 0.8 |
| r_Root | 0.01 |
| r_Leaf | 0.01 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.007) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.005) |
| PtLeafCnc | (0, 0.027); (0.367, 0.028); (0.5, 0.029); (0.8, 0.03); (0.9, 0.031); (1, 0.034); (1.1, 0.034); (1.2, 0.034); (1.3, 0.034); (1.4, 0.03); (1.5, 0.025); (1.8, 0.015) |
| CrLeafCnc | (0, 0.009); (0.367, 0.009); (0.5, 0.009); (0.8, 0.009); (0.9, 0.009); (1, 0.01); (1.1, 0.015); (1.2, 0.017); (1.3, 0.01); (1.4, 0.009); (1.5, 0.009); (1.8, 0.009) |
| NfLeafCnc | (0, 0.004); (0.8, 0.004); (0.9, 0.004); (1, 0.006); (1.1, 0.008); (1.2, 0.009); (1.3, 0.006); (1.4, 0.004); (1.5, 0.004); (1.8, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.022); (0.367, 0.022); (0.5, 0.022); (0.8, 0.022); (0.9, 0.022); (1, 0.018); (1.1, 0.018); (1.2, 0.016); (1.3, 0.012); (1.4, 0.011); (1.5, 0.01) |
| CrStemCnc | (0, 0.01); (0.367, 0.01); (0.5, 0.01); (0.8, 0.01); (0.9, 0.01); (1, 0.008); (1.1, 0.008); (1.2, 0.008); (1.3, 0.008); (1.4, 0.008); (1.5, 0.008) |

| Parameter | Value |
|---|---|
| NfStemCnc | (0, 0.004); (0.367, 0.004); (0.5, 0.004); (0.8, 0.004); (0.9, 0.004); (1, 0.004); (1.1, 0.004); (1.2, 0.004); (1.3, 0.004); (1.4, 0.004); (1.5, 0.004) |
| PtSOrgCnc | (1.057, 0.02); (1.244, 0.015); (2, 0.012) |
| CrSOrgCnc | (1.057, 0.016); (1.244, 0.012); (2, 0.011) |
| NfSOrgCnc | (1.057, 0.014); (1.244, 0.01); (2, 0.01) |

### 38.5.14 Crop Aert

Parent default, defined in lib/dk-pea.dai. Aert

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.027 |
| TempEff1 | (0, 0); (4, 0.01); (19, 1); (24, 1); (3000, 0.01) |
| PhotEff1 | (0, 0.1); (11, 0.1); (16, 1); (20, 1.2); (240, 1.2) |
| DSRate2 | 0.027 |
| TempEff2 | (0, 0); (16, 1); (30, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 3.5 |
| DSEff | (0, 1); (0.3, 0.88); (1, 0.85); (2, 0.85) |
| Qeff | 0.04 |
| TempEff | (-20, 0.001); (0, 0.01); (5, 0.4); (10, 0.7); (15, 0.9); (20, 1); (25, 1); (30, 0.9); (35, 0.8); (40, 0.5); (45, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.033 |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.2, 0); (2, 0) |
| StemPhotEff | 0.2 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.8, 0); (2, 0) |
| SOrgPhotEff | 0.2 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARrel | 0.02 |
| PARref | 0.06 |
| PARext | 0.55 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| IntcpCap | 0.5 |

| Parameter | Value |
|---|---|

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 85 |
| rootdens | SpRtLength 30, DensRtTip 0.1 |
| Rad | 0.035 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.75, 0.6); (0.91, 0.05); (2, 0.005) |
| Leaf | (0, 0.82); (0.25, 0.82); (0.51, 0.7); (1, 0.3); (1.2, 0); (2, 0) |
| Stem | (0, 0.18); (0.25, 0.18); (0.51, 0.3); (1, 0.2); (1.2, 0.1); (1.5, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.57 |
| r_Root | 0.025 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.1 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.5 |
| LfDR | (0, 0); (1.5, 0.01); (2, 0.03) |
| RtDR | (0, 0); (0.8, 0.01); (2, 0.01) |

## CrpN

| Parameter | Value |
|---|---|
| DS_fixate | 0.2 |
| fixate_factor | 0.8 |
| TLLeafEff | (1.09, 1); (2, 1) |
| PtRootCnc | (0, 0.02); (0.27, 0.02); (1.93, 0.02) |
| CrRootCnc | (0, 0.015); (0.27, 0.015); (1.93, 0.015) |
| NfRootCnc | (0, 0.01); (0.27, 0.01); (1.93, 0.01) |

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.06); (1, 0.06); (1.5, 0.025); (1.8, 0.02) |
| CrLeafCnc | (0, 0.02); (1, 0.02); (1.5, 0.016); (1.8, 0.012) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (1.5, 0.004); (1.8, 0.004) |
| PtStemCnc | (0, 0.03); (1, 0.025); (1.5, 0.015); (1.8, 0.011) |
| CrStemCnc | (0, 0.01); (1, 0.01); (1.5, 0.01); (1.8, 0.01) |
| NfStemCnc | (0, 0.004); (1, 0.004); (1.5, 0.004); (1.8, 0.004) |
| PtSOrgCnc | (0, 0.034); (1, 0.034); (2, 0.034) |
| CrSOrgCnc | (0, 0.02); (1, 0.02); (2, 0.02) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.81 |
| EconomicYield_N | 0.92 |

## 38.5.15 Crop Rug

Parent default, defined in lib/dk-rye.dai. Rug

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.02 |
| TempEff1 | -10. 0.01 |
| PhotEff1 | (0, 0.1); (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.025 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.33 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4 |
| DSEff | (0, 1); (0.3, 1); (1, 0.92); (2, 0.9) |
| Qeff | 0.05 |
| TempEff | (-20, 0); (4, 0); (10, 1); (25, 1); (35, 0.01); (50, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.025 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| PARrel | 0.05 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 100); (2, 100) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 100 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 7.3e-08 |
| MxNO3Up | 7.3e-09 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.5); (0.53, 0.25); (1, 0.05); (2, 0) |
| Leaf | (0, 0.7); (0.33, 0.65); (0.43, 0.6); (0.53, 0.55); (0.62, 0.4); (0.77, 0.18); (0.95, 0.05); (1.14, 0); (1.38, 0); (2, 0) |
| Stem | (0, 0.3); (0.33, 0.35); (0.43, 0.4); (0.53, 0.45); (0.62, 0.6); (0.77, 0.75); (0.95, 0.07); (1.14, 0.04); (1.38, 0.02); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.2 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0 |
| LfDR | (0, 0); (0.05, 0.01); (0.2, 0); (0.25, 0.01); (0.9, 0.03); (2, 0.1) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.011); (1, 0.011); (2, 0.01) |
| CrRootCnc | (0, 0.009); (1, 0.009); (2, 0.008) |
| NfRootCnc | (0, 0.006); (1, 0.006); (2, 0.006) |
| PtLeafCnc | (0.34, 0.055); (0.5, 0.032); (1.28, 0.025); (1.52, 0.021); (1.93, 0.017) |
| CrLeafCnc | (0.34, 0.02); (0.5, 0.019); (1.28, 0.018); (1.52, 0.016); (1.93, 0.013) |
| NfLeafCnc | (0.34, 0.006); (0.5, 0.006); (1.28, 0.0055); (1.52, 0.005); (1.93, 0.005) |
| PtStemCnc | (0.5, 0.018); (1.12, 0.01); (1.28, 0.006); (1.53, 0.0055); (1.93, 0.0055) |
| CrStemCnc | (0.5, 0.0054); (1.12, 0.0039); (1.28, 0.0033); (1.53, 0.0031); (1.93, 0.0025) |
| NfStemCnc | (0.5, 0.002); (1.12, 0.002); (1.28, 0.002); (1.53, 0.002); (1.93, 0.002) |
| PtSOrgCnc | (1.12, 0.014); (2, 0.014) |
| CrSOrgCnc | (1.12, 0.012); (2, 0.012) |
| NfSOrgCnc | (1.12, 0.01); (2, 0.01) |

### Harvest

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.16 Crop Vaarbyg

Parent default, defined in lib/dk-sbarley.dai. Vaarbyg

### Devel

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.032 |
| TempEff1 | (0, 0); (4, 0.01); (19, 1); (24, 1); (30, 0.01) |
| PhotEff1 | (0, 0.1); (11, 0.1); (16, 1); (20, 1.2); (24, 1.2) |
| DSRate2 | 0.021 |
| TempEff2 | (0, 0); (16, 1); (30, 1) |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 4.4 |
| DSEff | (0, 1); (0.3, 1); (1, 1); (2, 1) |
| Qeff | 0.04 |
| TempEff | -20. 0.001 |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.2 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.038 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 75); (2, 75) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 90 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 1.3e-07 |
| MxNO3Up | 1.3e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.65); (0.51, 0.55); (0.72, 0.15); (0.95, 0.05); (2, 0) |
| Leaf | (0, 1); (0.25, 0.7); (0.51, 0.55); (0.6, 0.5); (0.72, 0.23); (0.83, 0.01); (0.95, 0); (2, 0) |
| Stem | (0, 0); (0.25, 0.3); (0.51, 0.45); (0.6, 0.5); (0.72, 0.77); (0.83, 0.99); (0.95, 0.65); (1.51, 0); (2, 0) |
| RSR | (0, 0.5); (0.75, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.7 |
| LfDR | (0, 0); (0.5, 0); (1.2, 0.01); (2, 0.08) |
| RtDR | (0, 0); (0.2, 0); (1.2, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.02) |
| CrLeafCnc | (0, 0.03); (1, 0.02); (2, 0.01) |
| NfLeafCnc | (0, 0.006); (1, 0.006); (2, 0.006) |
| PtStemCnc | (0, 0.02); (1, 0.012); (2, 0.006) |
| CrStemCnc | (0, 0.01); (1, 0.008); (2, 0.005) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.017); (1, 0.017); (2, 0.017) |
| CrSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |
| NfSOrgCnc | (0, 0.011); (1, 0.011); (2, 0.011) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.17 Crop Vinterbyg

Parent default, defined in lib/dk-wbarley.dai. Vinterbyg

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 200 |
| DS_Emr | 0.01 |
| DSRate1 | 0.025 |
| TempEff1 | -10. 0.01 |
| PhotEff1 | (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.03 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.25 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4 |
| DSEff | (0, 1); (0.3, 1); (1, 0.95); (2, 0.94) |
| Qeff | 0.05 |
| TempEff | (-20, 0); (2, 0); (10, 1); (25, 1); (35, 0.01); (50, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 100); (2, 100) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 100 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 1e-07 |
| MxNO3Up | 1e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.5); (0.53, 0.25); (1, 0.05); (2, 0) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.43, 0.83); (0.53, 0.75); (0.62, 0.56); (0.77, 0.2); (0.95, 0.09); (1.14, 0.05); (1.38, 0); (2, 0) |
| Stem | (0, 0.1); (0.33, 0.15); (0.43, 0.17); (0.53, 0.25); (0.62, 0.44); (0.77, 0.8); (0.95, 0.55); (1.14, 0.4); (1.38, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.2 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |

| Parameter | Value |
|---|---|
| LfDR | (0, 0); (0.05, 0.01); (0.2, 0); (0.25, 0.01); (0.9, 0.03); (2, 0.1) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.02) |
| CrLeafCnc | (0, 0.03); (1, 0.02); (2, 0.01) |
| NfLeafCnc | (0, 0.006); (1, 0.006); (2, 0.006) |
| PtStemCnc | (0, 0.02); (1, 0.014); (2, 0.007) |
| CrStemCnc | (0, 0.01); (1, 0.008); (2, 0.005) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.017); (1, 0.017); (2, 0.017) |
| CrSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |
| NfSOrgCnc | (0, 0.012); (1, 0.012); (2, 0.012) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.18 Crop Vinterraps

Parent default, defined in lib/dk-wrape.dai. Vinterraps

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 180 |
| DS_Emr | 0.01 |
| DSRate1 | 0.03 |
| TempEff1 | -10. 0.01, 20. 0.9, 25. 1, 35. 1.2 |
| PhotEff1 | 10. 0.29, 11. 0.55, 12. 0.75, 13. 0.89, 14. 1, 15. 1.08, 16. 1.14, 17. 1.18, 24. 1.18 |
| DSRate2 | 0.025 |
| TempEff2 | 10. 0.14, 15. 0.66, 20. 1, 25. 1.23 |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.33 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 5 |
| DSEff | (0, 1); (0.3, 1); (1, 0.91); (2, 0.87) |
| Qeff | 0.05 |
| TempEff | -20. 0, 2. 0, 10. 1, 25. 1, 35. 0.01, 50. 0 |

## Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.25 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.02 |
| LeafAIMod | (0); (2, 1) |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.6, 1); (2, 0) |
| StemPhotEff | 0.2 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.4, 1); (2, 0) |
| SOrgPhotEff | 0.25 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.7 |
| EPext | 0.5 |
| HvsDS | (0, 1); (0.34, 5); (0.45, 10); (0.69, 50); (1, 160); (2, 160) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 1e-08 |
| MxNO3Up | 1e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.25); (0.66, 0.15); (1, 0.07); (2, 0.07) |
| Leaf | (0, 0.95); (0.33, 0.95); (0.43, 0.9); (0.53, 0.75); (0.62, 0.56); (0.77, 0.14); (0.95, 0.06); (1.14, 0.04); (1.18, 0); (2, 0) |
| Stem | (0, 0.05); (0.33, 0.05); (0.43, 0.1); (0.53, 0.25); (0.62, 0.44); (0.77, 0.6); (0.95, 0.48); (1.14, 0.18); (1.18, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.5 |
| r_Root | 0.015 |
| r_Leaf | 0.06 |
| r_Stem | 0.01 |
| r_SOrg | 0.03 |
| ShldResC | 0 |
| ReMobilDS | 1.18 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0.09, 0); (0.1, 0.01); (0.15, 0.005); (1, 0.005); (1.05, 0.2); (1.2, 0.1); (1.25, 0.05); (2, 0.02) |
| RtDR | (0.6, 0); (1.7, 0); (2, 0.005) |
| ExfoliationFac | 0.9 |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0.33, 0.033); (0.41, 0.02); (0.58, 0.023); (1.4, 0.01); (2, 0.01) |
| CrRootCnc | (0.33, 0.008); (0.41, 0.008); (0.58, 0.008); (1.4, 0.008); (2, 0.008) |
| NfRootCnc | (0.33, 0.007); (0.41, 0.007); (0.58, 0.007); (1.4, 0.007); (2, 0.007) |
| PtLeafCnc | (0.33, 0.06); (0.41, 0.06); (0.65, 0.06); (0.8, 0.06); (1.09, 0.05); (1.4, 0.03); (2, 0.025) |
| CrLeafCnc | (0.33, 0.044); (0.41, 0.044); (0.65, 0.044); (0.8, 0.044); (1.09, 0.04); (1.4, 0.022); (2, 0.02) |
| NfLeafCnc | (0.33, 0.004); (0.41, 0.004); (0.65, 0.004); (0.8, 0.004); (1.09, 0.004); (1.4, 0.004); (2, 0.004) |
| PtStemCnc | (0.65, 0.05); (0.8, 0.035); (1.09, 0.016); (1.6, 0.008); (2, 0.008) |
| CrStemCnc | (0.65, 0.03); (0.8, 0.02); (1.09, 0.008); (1.6, 0.006); (2, 0.006) |
| NfStemCnc | (0.65, 0.004); (0.8, 0.004); (1.09, 0.004); (1.6, 0.004); (2, 0.004) |
| PtSOrgCnc | (1.09, 0.045); (1.4, 0.024); (2, 0.022) |
| CrSOrgCnc | (1.09, 0.03); (1.4, 0.019); (2, 0.018) |
| NfSOrgCnc | (1.09, 0.004); (1.4, 0.004); (2, 0.004) |
| TLLeafEff | (1.09, 1); (1.4, 1); (2, 1) |

### Harvest

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.58 |
| EconomicYield_N | 0.82 |

## 38.5.19 Crop Vinterhvede

Parent default, defined in lib/dk-wwheat.dai. Vinterhvede

### Devel

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.026 |
| TempEff1 | (-10, 0.01); (0, 0.01); (20, 0.9); (25, 1); (35, 1.2) |
| PhotEff1 | (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.025 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

### Vernal

| Parameter | Value |
|---|---|
| DSLim | 0.25 |
| TaLim | 5 |
| TaSum | -50 |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 5 |
| DSEff | (0, 1); (0.3, 1); (1, 0.86); (2, 0.81) |
| Qeff | 0.05 |
| TempEff | (-20, 0); (4, 0); (10, 1); (25, 1); (35, 0.01); (50, 0) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| PARrel | 0.05 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 80); (2, 100) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 100 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |

| Parameter | Value |
|---|---|
| MxNH4Up | 7.4e-09 |
| MxNO3Up | 7.4e-09 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.5); (0.53, 0.5); (1, 0.05); (2, 0) |
| Leaf | (0, 0.7); (0.4, 0.7); (0.55, 0.7); (0.62, 0.5); (0.77, 0.2); (0.95, 0.1); (1.38, 0.05); (2, 0) |
| Stem | (0, 0.3); (0.4, 0.3); (0.55, 0.3); (0.62, 0.5); (0.77, 0.8); (0.95, 0.9); (1.2, 0.25); (1.38, 0.08); (2, 0) |
| RSR | (0, 0.5); (1, 0.5); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.016 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.4 |
| ReMobilDS | 1.5 |
| ReMobilRt | 0.03 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.7 |
| LfDR | (0, 0); (0.1, 0); (0.25, 0); (0.5, 0.01); (0.9, 0.01); (1.2, 0.02); (1.8, 0.05); (2, 0.05) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.011); (1, 0.011); (2, 0.01) |
| CrRootCnc | (0, 0.009); (1, 0.009); (2, 0.008) |
| NfRootCnc | (0, 0.006); (1, 0.006); (2, 0.006) |
| PtLeafCnc | (0.34, 0.06); (0.5, 0.034); (1.28, 0.031); (1.52, 0.0268); (1.93, 0.0184) |
| CrLeafCnc | (0.34, 0.02); (0.5, 0.02); (1.28, 0.02); (1.52, 0.018); (1.93, 0.0153) |
| NfLeafCnc | (0.34, 0.006); (0.5, 0.006); (1.28, 0.006); (1.52, 0.006); (1.93, 0.006) |
| PtStemCnc | (0.5, 0.018); (1.12, 0.0115); (1.28, 0.0083); (1.53, 0.0075); (1.93, 0.006) |
| CrStemCnc | (0.5, 0.0054); (1.12, 0.0039); (1.28, 0.0033); (1.53, 0.0031); (1.93, 0.0025) |
| NfStemCnc | (0.5, 0.002); (1.12, 0.002); (1.28, 0.002); (1.53, 0.002); (1.93, 0.002) |
| PtSOrgCnc | (1.12, 0.019); (2, 0.019) |
| CrSOrgCnc | (1.12, 0.014); (2, 0.014) |
| NfSOrgCnc | (1.12, 0.011); (2, 0.011) |

### Harvest

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.20 Crop Potato; SCRI minus AArhus

Parent default, defined in lib/earlypotato.dai. Potato; SCRI minus AArhus

### Devel

| Parameter | Value |
|---|---|
| EmrTSum | 425 |
| DS_Emr | 0.01 |
| DSRate1 | 0.07 |
| TempEff1 | (2, 0); (13, 1); (19, 1); (29, 0) |
| PhotEff1 | (12, 1); (16, 1) |
| DSRate2 | 0.009 |
| TempEff2 | (2, 0); (13, 1); (19, 1); (29, 0) |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 3 |
| DSEff | (0, 1.2); (0.3, 1.2); (1, 1); (1.6, 0.75); (2, 0.75) |
| Qeff | 0.045 |
| TempEff | (3, 0.01); (10, 0.75); (15, 1); (20, 1); (26, 0.75); (33, 0.01) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.033 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.5 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 20); (1.25, 35) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 20 |
| PenPar1 | 0.3 |
| PenPar2 | 4 |
| MaxPen | 50 |

| Parameter | Value |
|---|---|
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |
| rootdens | SpRtLength 200, DensRtTip 0.2 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.2); (1, 0.2); (1.36, 0) |
| Leaf | (0, 0.95); (1, 0.75); (1.05, 0.2); (1.25, 0.18); (1.6, 0) |
| Stem | (0, 0.05); (1, 0.25); (1.05, 0.5); (1.25, 0.2); (1.6, 0.03); (2, 0) |
| RSR | (0, 0.5); (0.75, 0.5); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.72 |
| E_Leaf | 0.66 |
| E_Stem | 0.72 |
| E_SOrg | 0.85 |
| r_Root | 0.01 |
| r_Leaf | 0.03 |
| r_Stem | 0.015 |
| r_SOrg | 0.0045 |
| ShldResC | 0.1 |
| ReMobilDS | 1.5 |
| ReMobilRt | 0.01 |
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (1.6, 0); (1.75, 0.02); (2, 0.1) |
| RtDR | (0, 0); (1.6, 0); (2, 0) |
| ExfoliationFac | 0 |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.06); (1.25, 0.05); (2, 0.035) |
| CrLeafCnc | (0, 0.045); (1, 0.045); (1.25, 0.04); (2, 0.03) |
| NfLeafCnc | (0, 0.01); (1, 0.01); (2, 0.01) |
| PtStemCnc | (0, 0.04); (1, 0.04); (1.6, 0.016); (2, 0.013) |
| CrStemCnc | (0, 0.012); (1.25, 0.012); (2, 0.01) |
| NfStemCnc | (0, 0.004); (2, 0.004) |
| PtSOrgCnc | (1, 0.02); (1.25, 0.015); (2, 0.012) |
| CrSOrgCnc | (1, 0.008); (1.25, 0.008); (2, 0.0075) |
| NfSOrgCnc | (1, 0.004); (2, 0.004) |
| TLLeafEff | (1.09, 1); (2, 1) |

## Harvest

| Parameter | Value |
|---|---|
| DSmax | 0 |
| sorg_height | -10 |

### 38.5.21 Crop Early potato

Parent Potato; SCRI minus AArhus, defined in lib/earlypotato.dai. Early potato

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 275 |
| DS_Emr | 0.01 |
| DSRate1 | 0.059 |
| TempEff1 | (7, 0.01); (18, 1); (29, 0.01) |
| PhotEff1 | (0, 0); (8, 1); (16, 1); (24, 0.9) |
| DSRate2 | 0.0225 |
| TempEff2 | (7, 0); (18, 1); (29, 0) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 3.2 |
| DSEff | (0, 1); (0.3, 1); (1, 1); (1.6, 1); (2, 1) |
| Qeff | 0.045 |
| TempEff | (3, 0.01); (12, 0.75); (15, 1); (20, 1); (26, 0.75); (33, 0.01) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.024 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.65 |
| EPext | 0.5 |
| HvsDS | (0, 0); (1, 60); (2.1, 60) |

**Partit**

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.37, 0.32); (0.83, 0.09); (1, 0); (1.14, 0); (1.35, 0); (1.57, 0); (2.1, 0) |
| Leaf | (0, 0.77); (0.37, 0.67); (0.83, 0.39); (1, 0.23); (1.14, 0.1); (1.35, 0); (1.57, 0); (2.1, 0) |
| Stem | (0, 0.16); (0.37, 0.15); (0.83, 0.16); (1, 0.16); (1.14, 0.15); (1.35, 0.02); (1.57, 0); (2.1, 0) |
| RSR | (0, 0.5); (0.75, 0.5); (2, 0.25) |

## 38.5.22 Crop Early white cabbage minus transplanted

Parent White cabbage minus transplanted, defined in lib/earlywhitecabbage.dai. Early white cabbage minus transplanted

**Devel**

| Parameter | Value |
|---|---|
| defined_until_ds | 2 |
| DSRate1 | 0.013 |
| TempEff1 | (0, 0); (3, 0); (13, 1); (18, 1); (26, 0) |
| PhotEff1 | (0, 0); (8, 1); (16, 1); (24, 0.8) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 6.5 |
| Qeff | 0.048 |
| TempEff | (0, 0.01); (10, 1); (20, 1); (25, 0.8); (35, 0.01) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.018 |
| SpSOrgAI | 0.0018 |

**Partit**

| Parameter | Value |
|---|---|
| Root | (0, 0.3); (0.15, 0.15); (0.3, 0.15); (0.5, 0.1); (0.75, 0.1); (1, 0.1); (1.1, 0.1) |
| Leaf | (0, 0.8); (0.15, 0.65); (0.3, 0.5); (0.5, 0.3); (0.75, 0.2); (1, 0.1); (1.1, 0.1) |
| Stem | (0, 0.2); (0.15, 0.35); (0.3, 0.25); (0.5, 0.25); (0.75, 0.15); (1, 0.1); (1.1, 0.1) |
| RSR | (0, 0.5); (1, 0.25) |

## 38.5.23 Crop Fodder Beet

Parent default, defined in lib/fodderbeet.dai. Fodder Beet

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 250 |
| DS_Emr | 0.01 |
| DSRate1 | 0.005 |
| TempEff1 | -1. 0, 20. 1, 25. 0.5 |

| Parameter | Value |
|---|---|
| PhotEff1 | 1. 0, 12. 1, 16. 1, 24. 1 |
| DSRate2 | 0.007 |
| TempEff2 | 30. 1 |

## LeafPhot

| Parameter | Value |
|---|---|
| Fm | 3.5 |
| Qeff | 0.056 |
| TempEff | 5. 0, 10. 0.75, 20. 1, 30. 1, 40. 0.01, 50. 0 |

## Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.17 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.011 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.67 |
| EPext | 0.5 |
| HvsDS | (0, 1); (0.02, 4); (0.08, 10); (0.14, 30); (0.21, 40); (0.27, 55) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 200 |
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.25); (0.08, 0.2); (0.14, 0.1); (0.27, 0.07); (0.34, 0.06) |
| Leaf | (0, 0.8); (0.18, 0.8); (0.2, 0.2); (0.25, 0.2); (0.35, 0.15) |
| Stem | (0, 0); (1.1, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.66 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0.2, 0); (0.25, 0.01); (0.8, 0.01); (1, 0.02); (2, 0.1) |
| RtDR | (0.2, 0); (0.5, 0.01); (2, 0.01) |
| ExfoliationFac | 0 |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.005) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.004) |
| PtLeafCnc | (0, 0.06); (0.1, 0.05); (0.125, 0.045); (0.2, 0.04); (0.3, 0.04); (0.4, 0.04); (0.5, 0.04) |
| CrLeafCnc | (0, 0.03); (0.1, 0.03); (0.2, 0.03); (0.3, 0.02); (0.8, 0.02) |
| NfLeafCnc | (0, 0.01); (0.1, 0.01); (0.2, 0.01); (0.3, 0.01); (0.8, 0.01) |
| PtStemCnc | (0, 0.02); (2, 0.02) |
| CrStemCnc | (0, 0.01); (2, 0.01) |
| NfStemCnc | (0, 0.004); (2, 0.004) |
| PtSOrgCnc | (0.1, 0.025); (0.2, 0.018); (0.3, 0.013); (0.4, 0.011); (0.5, 0.011) |
| CrSOrgCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| NfSOrgCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| TLLeafEff | (1.09, 1); (2, 1) |

### Harvest

| Parameter | Value |
|---|---|
| DSmax | 0 |
| sorg_height | -15 |

## 38.5.24 Crop Grass; V1

Parent old, defined in lib/grass-old.dai. Grass; V1

### Devel

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DSRate1 | 0.00145 |
| DSRate2 | 0.00145 |
| DS_Emr | 0.01 |
| DS_reset | true |
| TempEff1 | (0, 1); (100, 100) |

| Parameter | Value |
|---|---|
| TempEff2 | (0, 1); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

### Vernal

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

### LeafPhot

| Parameter | Value |
|---|---|
| TLim1 | 4 |
| TLim2 | 10 |

### Canopy

| Parameter | Value |
|---|---|
| InitGrowth | 4.7 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 1000 |
| alpha | 3 |
| beta | 1.87 |
| SpLAI | 0.017 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.6 |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 70 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.2); (2, 0.2) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_Leaf | (0, 0.7); (2, 0.7) |
| r_Root | (0, 0.065); (2, 0.065) |
| r_Leaf | (0, 0.04); (2, 0.04) |

### CrpN

| Parameter | Value |
|---|---|
| SeedN | 0.02 |
| PtLeafCnc | (0, 0.055); (0.044, 0.052); (0.087, 0.049); (0.131, 0.046); (0.175, 0.043); (0.218, 0.041); (0.262, 0.038); (0.305, 0.036); (0.349, 0.034); (0.393, 0.032); (0.436, 0.03) |
| PtRootCnc | (0, 0.02); (0.044, 0.019); (0.087, 0.019); (0.131, 0.018); (0.175, 0.018); (0.218, 0.017); (0.262, 0.017); (0.305, 0.016); (0.349, 0.016); (0.393, 0.015); (0.436, 0.015) |
| CrLeafCnc | (0, 0.035); (0.044, 0.034); (0.087, 0.033); (0.131, 0.032); (0.175, 0.031); (0.218, 0.03); (0.262, 0.029); (0.305, 0.028); (0.349, 0.027); (0.393, 0.026); (0.436, 0.025) |
| CrRootCnc | (0, 0.008); (0.044, 0.008); (0.087, 0.008); (0.131, 0.008); (0.175, 0.008); (0.218, 0.008); (0.262, 0.008); (0.305, 0.008); (0.349, 0.008); (0.393, 0.008); (0.436, 0.008) |
| NfLeafCnc | (0, 0.015); (0.044, 0.014); (0.087, 0.013); (0.131, 0.012); (0.175, 0.012); (0.218, 0.011); (0.262, 0.01); (0.305, 0.01); (0.349, 0.009); (0.393, 0.009); (0.436, 0.008) |
| NfRootCnc | (0, 0.008); (0.044, 0.007); (0.087, 0.007); (0.131, 0.006); (0.175, 0.006); (0.218, 0.006); (0.262, 0.005); (0.305, 0.005); (0.349, 0.005); (0.393, 0.004); (0.436, 0.004) |

### Harvest

| Parameter | Value |
|---|---|
| index | 0 |
| CStem | -4.242e+43 |
| CSOrg | -4.242e+43 |
| alpha | -4.242e+43 |
| beta | 0.075 |
| Stem | (); () |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| DSmax | 3 |
| DSnew | 0.07 |
| C_Stem | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.

IntcpCap: 0.5.

### 38.5.25 Crop Grass to grain

Parent default, defined in lib/grass.dai. Grass to grain

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DS_Emr | 0.01 |
| DSRate1 | 0.022 |
| TempEff1 | (-10, 0.01); (0, 0.01); (20, 0.9); (25, 1); (35, 1.2) |
| PhotEff1 | (0, 0.1); (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.028 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |
| defined_until_ds | 2. |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.9 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Qeff | 0.05 |
| Fm | 2.16 |
| TempEff | (-20, 0.0001); (5, 0.0001); (10, 0.9); (25, 1); (35, 0.01); (50, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.2 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.025 |
| HvsDS | (0, 1); (1, 30); (2, 30) |
| LeafAIMod | (0); (2, 1) |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.6, 1); (2, 0) |
| StemPhotEff | 0.2 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| IntcpCap | 0.5 |

**Root**

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 70 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |
| Root | (0, 0.5); (0.33, 0.25); (0.53, 0.2); (1, 0); (2, 0) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.43, 0.83); (0.53, 0.56); (0.95, 0.36); (1.14, 0) |
| Stem | (0, 0.1); (0.33, 0.15); (0.43, 0.17); (0.53, 0.44); (0.95, 0.64); (1.14, 0.62); (1.45, 0.3); (2, 0.05) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.57 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0 |
| ReMobilDS | 1.38 |
| ReMobilRt | 0.1 |
| LfDR | (0.09, 0); (0.1, 0); (0.35, 0); (1, 0.02); (2, 0.1) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |
| Large_RtDR | 0.05 |
| IntDSRelRtRes | 0.2 |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.03); (2, 0.02) |
| CrLeafCnc | (0, 0.03); (1, 0.02); (2, 0.015) |
| NfLeafCnc | (0, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (2, 0.014) |

| Parameter | Value |
|---|---|
| TLLeafEff | (0, 0.9); (2, 0.9) |
| TLRootEff | (0, 0.1); (2, 0.1) |

**Harvest**

| Parameter | Value |
|---|---|
| Dead | CROP-SLOW |
| Stem | CROP-SLOW |
| Leaf | CROP-SLOW |
| SOrg | CROP-SLOW initial_fraction 0 |
| Root | CROP-SLOW |
| DSmax | 3 |
| DSnew | 0.2 |

## 38.5.26 Crop Grass

Parent Grass to grain, defined in lib/grass.dai. Grass to grain

**Devel**

| Parameter | Value |
|---|---|
| PhotEff1 | (0, 0.1); (10, 0); (11, 0); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |

## 38.5.27 Crop Grass

Parent default, defined in lib/grass2.dai. Grass

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DS_Emr | 0.01 |
| DSRate1 | 0.022 |
| TempEff1 | (-10, 0.01); (0, 0.01); (20, 0.9); (25, 1); (35, 1.2) |
| PhotEff1 | (0, 0.1); (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.028 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.99 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 2.16 |
| Qeff | 0.036 |

| Parameter | Value |
|---|---|
| TempEff | (-20, 0); (5, 0); (10, 0.9); (25, 1); (35, 0.01); (50, 0) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.2 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.02 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 40); (2, 40) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 70 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.55); (0.33, 0.5); (0.53, 0.5); (1, 0.3); (1.5, 0.1); (2, 0) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.43, 0.83); (0.53, 0.75); (0.62, 0.65); (0.77, 0.3); (0.95, 0.5); (1.14, 0.2); (1.38, 0.2); (2, 0.1) |
| Stem | (0, 0.1); (0.33, 0.15); (0.43, 0.17); (0.53, 0.25); (0.62, 0.35); (0.77, 0.7); (0.95, 0.5); (1.14, 0.8); (1.38, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.005 |
| r_Leaf | 0.02 |
| r_Stem | 0.005 |
| r_SOrg | 0.005 |
| ShldResC | 0.1 |
| ReMobilDS | 1.2 |
| ReMobilRt | 0.1 |

| Parameter | Value |
|---|---|
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (0.1, 0); (0.35, 0); (1, 0.02); (2, 0.2) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.007); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (1, 0.03); (2, 0.02) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

## 38.5.28 Crop Maize

Parent default, defined in lib/maize.dai. Maize

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 300 |
| DS_Emr | 0.01 |
| DSRate1 | 0.024 |
| TempEff1 | 0. 0, 6. 0, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |
| PhotEff1 | 0. 0.9, 12. 1, 14. 0.95, 16. 0.9, 24. 0.9 |
| DSRate2 | 0.015 |
| TempEff2 | 0. 0.3, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 6 |
| Qeff | 0.04 |
| TempEff | (-20, 0.001); (6, 0.1); (10, 1); (15, 1); (20, 1) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.25 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.01 |

| Parameter | Value |
|---|---|
| LeafAIMod | (0, 1); (2, 1) |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.6, 0); (2, 0) |
| StemPhotEff | 0.2 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.6, 0); (2, 0) |
| SOrgPhotEff | 0.2 |
| LAIDist0 | 0, 1, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.8 |
| EPext | 0.5 |
| HvsDS | (0, 1); (0.2, 50); (0.5, 150); (1, 200); (2, 200) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.67, 0.35); (1.37, 0.1); (1.6, 0.1); (2, 0.05) |
| Leaf | (0.35, 0.7); (0.67, 0.5); (1, 0.15); (1.18, 0.05); (1.85, 0.05); (2, 0.05) |
| Stem | (0.35, 0.3); (0.67, 0.5); (1, 0.85); (1.18, 0.05); (1.8, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.68 |
| r_Root | 0.015 |
| r_Leaf | 0.025 |
| r_Stem | 0.01 |
| r_SOrg | 0.03 |
| ShldResC | 0.35 |
| ReMobilDS | 1.4 |
| ReMobilRt | 0.02 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.5 |
| LfDR | (0, 0); (0.1, 0); (1.5, 0.005); (2, 0.005) |

| Parameter | Value |
|---|---|
| RtDR | (0, 0); (0.2, 0); (1.2, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.005) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.004) |
| PtLeafCnc | (0, 0.044); (0.367, 0.044); (0.5, 0.04); (1, 0.03); (1.5, 0.018); (1.8, 0.015) |
| CrLeafCnc | (0, 0.01); (0.367, 0.01); (0.5, 0.009); (1, 0.008); (1.5, 0.008); (1.8, 0.008) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0.838, 0.02); (1.057, 0.014); (1.244, 0.011) |
| CrStemCnc | (0.838, 0.01); (1.057, 0.01); (1.244, 0.008) |
| NfStemCnc | (0.838, 0.004); (1.057, 0.004); (1.244, 0.004) |
| PtSOrgCnc | (1.057, 0.02); (1.244, 0.015); (2, 0.015) |
| CrSOrgCnc | (1.057, 0.016); (1.244, 0.012); (2, 0.012) |
| NfSOrgCnc | (1.057, 0.014); (1.244, 0.01); (2, 0.01) |

### 38.5.29 Crop Pioneer Maize

Parent default, defined in lib/maize.dai. Pioneer Maize
cite: vries-maize.

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.0265 |
| TempEff1 | 0. 0.3, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |
| PhotEff1 | 0. 0.9, 12. 1, 14. 0.95, 16. 0.9, 24. 0.9 |
| DSRate2 | 0.017 |
| TempEff2 | 0. 0.3, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 6 |
| Qeff | 0.04 |
| TempEff | -20. 0.001, 0. 0.01, 5. 0.01, 10. 0.1, 15. 0.5, 20. 0.8, 25. 1, 35. 1, 40. 0.9, 45. 0.75, 50. 0.07 |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.13 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.03 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (0.2, 50); (0.5, 150); (1, 200); (2, 200) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.67, 0.25); (1.37, 0); (1.6, 0); (2, 0) |
| Leaf | (0, 0.95); (0.35, 0.59); (0.67, 0.2); (1, 0.12); (1.18, 0.09); (1.37, 0); (2, 0) |
| Stem | (0, 0.05); (0.35, 0.41); (0.67, 0.66); (1, 0.64); (1.18, 0.31); (1.37, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.68 |
| r_Root | 0.015 |
| r_Leaf | 0.016 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.35 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (0.1, 0); (1.5, 0.005); (2, 0.1) |
| RtDR | (0, 0); (0.2, 0); (1.2, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.005) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.004) |
| PtLeafCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrLeafCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrStemCnc | (0, 0.03); (1, 0.02); (2, 0.02) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

**Harvest**

| Parameter | Value |
|---|---|
| Dead | CROP-SLOW |
| Stem | CROP-SLOW |
| Leaf | CROP-SLOW |
| SOrg | CROP-SLOW |
| Root | CROP-SLOW |
| DSmax | 0.8 |
| DSnew | 0.2 |

## 38.5.30 Crop Ikuwala Maize

Parent Maize, defined in lib/maize.dai. Ikuwala Maize

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 300 |
| DSRate1 | 0.02 |
| DSRate2 | 0.02 |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.3 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.03 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARext | 0.56 |

**Prod**

| Parameter | Value |
|---|---|
| ShldResC | 0.3 |
| LfDR | (0, 0); (0.1, 0); (1.4, 0.01); (1.8, 0.1); (2, 0.2) |
| ExfoliationFac | 0 |

**Partit**

| Parameter | Value |
|---|---|
| Stem | (0, 0.3); (0.35, 0.3); (0.67, 0.5); (1, 0.64); (1.18, 0.31); (1.37, 0.3); (2, 0.1) |

## 38.5.31 Crop Silage Maize

Parent Maize, defined in lib/maize.dai. Silage Maize

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 300 |
| DS_Emr | 0.01 |
| DSRate1 | 0.016 |
| TempEff1 | 0. 0.3, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |
| PhotEff1 | 0. 0.9, 12. 1, 14. 0.95, 16. 0.9, 24. 0.9 |
| DSRate2 | 0.021 |
| TempEff2 | 0. 0.3, 10. 0.3, 15. 0.75, 25. 1, 35. 1.2 |

**Prod**

| Parameter | Value |
|---|---|
| ShldResC | 0 |

## 38.5.32 Crop Onion

Parent default, defined in lib/onion.dai. Onion

**Devel**

| Parameter | Value |
|---|---|
| defined_until_ds | 1 |
| EmrTSum | 270 |
| DSRate1 | 0.01625 |
| TempEff1 | (0, 0); (5, 0); (25, 1) |
| PhotEff1 | (0, 0); (12, 0); (15, 1); (24, 1) |
| DSRate2 | 0.01 |
| TempEff2 | (0, 1); (30, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 3.3 |
| Qeff | 0.045 |
| TempEff | (0, 0.01); (6, 0.01); (20, 1); (25, 1); (31, 0.01) |

## Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.25 |
| SpLAIfac | (0, 1); (1, 1) |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.1 |
| LeafAIMod | (0, 1); (0.07, 1); (0.12, 0.405); (0.25, 0.2775); (0.9, 0.15) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0, 1 |
| PARref | 0.06 |
| PARext | 0.5 |
| EPext | 0.5 |
| HvsWStem | (0, 0.1); (1, 1) |
| HvsDS | (0, 0); (0.4, 50); (1, 50) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 5 |
| PenPar1 | 0.02 |
| PenPar2 | 0 |
| MaxPen | 30 |
| Rad | 0.05 |
| h_wp | -15000 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.2, 0.3); (0.4, 0.1); (0.6, 0.05); (0.8, 0.05); (1, 0) |
| Leaf | (0, 0.75); (0.2, 0.55); (0.4, 0.4); (0.6, 0); (0.8, 0); (1, 0) |
| Stem | (0, 0.2); (0.2, 0.15); (0.4, 0.05); (0.6, 0); (0.8, 0); (1, 0) |
| RSR | (0, 0.5); (1, 0.5); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_SOrg | 0.0025 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |

| Parameter | Value |
|---|---|
| Large_RtDR | 0.05 |
| LfDR | (0.2, 0); (0.6, 0); (0.8, 0.03); (1, 0.1) |
| RtDR | (0.2, 0); (0.6, 0); (0.8, 0.01); (1, 0.02) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.024); (0.25, 0.022); (1, 0.02) |
| CrRootCnc | (0, 0.012); (0.25, 0.011); (1, 0.01) |
| NfRootCnc | (0, 0.009); (0.25, 0.008); (1, 0.007) |
| PtLeafCnc | (0, 0.05); (0.25, 0.0375); (1, 0.025) |
| CrLeafCnc | (0, 0.03); (0.25, 0.021); (1, 0.012) |
| NfLeafCnc | (0, 0.02); (0.25, 0.0135); (1, 0.007) |
| PtStemCnc | (0, 0.05); (0.25, 0.0375); (1, 0.025) |
| CrStemCnc | (0, 0.03); (0.25, 0.021); (1, 0.012) |
| NfStemCnc | (0, 0.02); (0.25, 0.0135); (1, 0.007) |
| PtSOrgCnc | (0, 0.045); (0.25, 0.0325); (1, 0.02) |
| CrSOrgCnc | (0, 0.03); (0.25, 0.0225); (1, 0.015) |
| NfSOrgCnc | (0, 0.015); (0.25, 0.011); (1, 0.007) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |
| sorg_height | -5 |

### 38.5.33 Crop Onion minus planting of sets

Parent Onion, defined in lib/onion_plantingofsets.dai. Onion minus planting of sets

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DSRate1 | 0.017 |

**Prod**

| Parameter | Value |
|---|---|
| NCrop | 0.6 |
| WSOrg | 20 |
| CH2OPool | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4.1 |
| Qeff | 0.046 |

### 38.5.34 Crop Pea; V1

Parent old, defined in lib/pea-old.dai. Pea; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DSRate1 | 0.00137 |
| DSRate2 | 0.00137 |
| DS_Emr | 0.01 |
| DS_reset | false |
| TempEff1 | (0, 1); (100, 100) |
| TempEff2 | (0, 1); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| TLim1 | 4 |
| TLim2 | 10 |

**Canopy**

| Parameter | Value |
|---|---|
| InitGrowth | 5.1 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 0.724 |
| alpha | 1.4 |
| beta | 0.551 |
| SpLAI | 0.018 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |

**Root**

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 60 |

| Parameter | Value |
|---|---|
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.92, 0.1); (2, 0.1) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_Leaf | (0, 0.6); (2, 0.6) |
| r_Root | (0.92, 0.1); (0.9201, 0.05) |
| r_Leaf | (0, 0.02); (2, 0.02) |

### CrpN

| Parameter | Value |
|---|---|
| SeedN | 1 |
| DS_fixate | 0.207 |
| PtLeafCnc | (0.276, 0.034); (0.512, 0.032); (0.749, 0.03); (0.985, 0.029); (1.222, 0.027); (1.458, 0.026); (1.695, 0.024); (1.931, 0.023) |
| PtRootCnc | (0.276, 0.045); (0.512, 0.04); (0.749, 0.036); (0.985, 0.032); (1.222, 0.029); (1.458, 0.026); (1.695, 0.023); (1.931, 0.021) |
| CrLeafCnc | (0.276, 0.034); (0.512, 0.032); (0.749, 0.03); (0.985, 0.029); (1.222, 0.027); (1.458, 0.026); (1.695, 0.024); (1.931, 0.023) |
| CrRootCnc | (0.276, 0.045); (0.512, 0.04); (0.749, 0.036); (0.985, 0.032); (1.222, 0.029); (1.458, 0.026); (1.695, 0.023); (1.931, 0.021) |
| NfLeafCnc | (0.276, 0.01); (0.512, 0.009); (0.749, 0.009); (0.985, 0.008); (1.222, 0.007); (1.458, 0.007); (1.695, 0.006); (1.931, 0.006) |
| NfRootCnc | (0.276, 0.011); (0.512, 0.01); (0.749, 0.009); (0.985, 0.008); (1.222, 0.008); (1.458, 0.007); (1.695, 0.007); (1.931, 0.006) |

### Harvest

| Parameter | Value |
|---|---|
| index | 0.55 |
| CStem | 0.0085 |
| CSOrg | 0.0038 |
| alpha | 0.25 |
| beta | 0.125 |
| Stem | (); () |

| Parameter | Value |
|---|---|
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| C_Stem | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.
IntcpCap: 0.5.

### 38.5.35 Crop Pea

Parent default, defined in lib/pea.dai. Pea

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.027 |
| TempEff1 | (0, 0); (4, 0.01); (19, 1); (24, 1); (30, 0.01) |
| PhotEff1 | (0, 0.1); (11, 0.1); (16, 1); (20, 1.2); (24, 1.2) |
| DSRate2 | 0.027 |
| TempEff2 | (0, 0); (16, 1); (30, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4 |
| Qeff | 0.04 |
| TempEff | (-20, 0.001); (0, 0.01); (5, 0.4); (10, 0.7); (15, 0.9); (20, 1); (25, 1); (30, 0.9); (35, 0.8); (40, 0.5); (45, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.033 |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.2, 0); (2, 0) |
| StemPhotEff | 0.2 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.8, 0); (2, 0) |
| SOrgPhotEff | 0.2 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARrel | 0.02 |
| PARref | 0.06 |
| PARext | 0.55 |

| Parameter | Value |
|---|---|
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| IntcpCap | 0.5 |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 85 |
| rootdens | SpRtLength 30, DensRtTip 0.1 |
| Rad | 0.035 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.75, 0.6); (0.91, 0.05); (2, 0.005) |
| Leaf | (0, 0.82); (0.25, 0.82); (0.51, 0.7); (1, 0.3); (1.2, 0); (2, 0) |
| Stem | (0, 0.18); (0.25, 0.18); (0.51, 0.3); (1, 0.2); (1.2, 0.1); (1.5, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.57 |
| r_Root | 0.025 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.1 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.5 |
| LfDR | (0, 0); (1.5, 0.01); (2, 0.03) |
| RtDR | (0, 0); (0.8, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| DS_fixate | 0.2 |
| fixate_factor | 0.8 |
| TLLeafEff | (1.09, 1); (2, 1) |

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (0.27, 0.02); (1.93, 0.02) |
| CrRootCnc | (0, 0.015); (0.27, 0.015); (1.93, 0.015) |
| NfRootCnc | (0, 0.01); (0.27, 0.01); (1.93, 0.01) |
| PtLeafCnc | (0, 0.06); (1, 0.06); (1.5, 0.025); (1.8, 0.02) |
| CrLeafCnc | (0, 0.02); (1, 0.02); (1.5, 0.016); (1.8, 0.012) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (1.5, 0.004); (1.8, 0.004) |
| PtStemCnc | (0, 0.03); (1, 0.025); (1.5, 0.015); (1.8, 0.011) |
| CrStemCnc | (0, 0.01); (1, 0.01); (1.5, 0.01); (1.8, 0.01) |
| NfStemCnc | (0, 0.004); (1, 0.004); (1.5, 0.004); (1.8, 0.004) |
| PtSOrgCnc | (0, 0.034); (1, 0.034); (2, 0.034) |
| CrSOrgCnc | (0, 0.02); (1, 0.02); (2, 0.02) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.81 |
| EconomicYield_N | 0.92 |

## 38.5.36 Crop Green Pea

Parent Pea, defined in lib/pea.dai. Green Pea

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 1 |
| EconomicYield_N | 1 |

## 38.5.37 Crop Potato; V1

Parent sold, defined in lib/potato-old.dai. Potato; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 485 |
| DSRate1 | 0.000909 |
| DSRate2 | 0.000909 |
| DS_Emr | 0.01 |
| DS_reset | false |
| TempEff1 | (0, 1); (100, 100) |
| TempEff2 | (0, 1); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

### LeafPhot

| Parameter | Value |
|---|---|
| TLim1 | 4 |
| TLim2 | 10 |

### Canopy

| Parameter | Value |
|---|---|
| InitGrowth | 5.1 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 1.545 |
| alpha | 0.3 |
| beta | 0.454 |
| SpLAI | 0.017 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.6 |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 150 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.57, 0.05); (2, 0.05) |
| SOrg | (0, 0); (0.636, 0.85); (2, 0.85) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_SOrg | (0, 0.85); (2, 0.85) |
| E_Leaf | (0, 0.72); (2, 0.72) |
| r_Root | (0, 0.065); (2, 0.065) |
| r_SOrg | (0, 0.003); (2, 0.003) |
| r_Leaf | (0, 0.03); (2, 0.03) |

**CrpN**

| Parameter | Value |
|---|---|
| SeedN | 1 |
| PtLeafCnc | (0.364, 0.055); (0.491, 0.053); (0.618, 0.052); (0.745, 0.05); (0.873, 0.048); (1, 0.047); (1.127, 0.045); (1.255, 0.044); (1.382, 0.043); (1.509, 0.041); (1.636, 0.04) |
| PtRootCnc | (0.364, 0.02); (0.491, 0.019); (0.618, 0.019); (0.745, 0.018); (0.873, 0.018); (1, 0.017); (1.127, 0.017); (1.255, 0.016); (1.382, 0.016); (1.509, 0.015); (1.636, 0.015) |
| PtSOrgCnc | (0.364, 0.035); (0.491, 0.033); (0.618, 0.03); (0.745, 0.028); (0.873, 0.026); (1, 0.024); (1.127, 0.023); (1.255, 0.021); (1.382, 0.02); (1.509, 0.018); (1.636, 0.017) |
| CrLeafCnc | (0.364, 0.035); (0.491, 0.033); (0.618, 0.03); (0.745, 0.028); (0.873, 0.026); (1, 0.024); (1.127, 0.023); (1.255, 0.021); (1.382, 0.02); (1.509, 0.018); (1.636, 0.017) |
| CrRootCnc | (0.364, 0.012); (0.491, 0.012); (0.618, 0.011); (0.745, 0.011); (0.873, 0.01); (1, 0.01); (1.127, 0.009); (1.255, 0.009); (1.382, 0.009); (1.509, 0.008); (1.636, 0.008) |
| CrSOrgCnc | (0.364, 0.015); (0.491, 0.014); (0.618, 0.013); (0.745, 0.012); (0.873, 0.012); (1, 0.011); (1.127, 0.01); (1.255, 0.01); (1.382, 0.009); (1.509, 0.009); (1.636, 0.008) |
| NfLeafCnc | (0.364, 0.017); (0.491, 0.016); (0.618, 0.015); (0.745, 0.014); (0.873, 0.014); (1, 0.013); (1.127, 0.012); (1.255, 0.012); (1.382, 0.011); (1.509, 0.011); (1.636, 0.01) |
| NfRootCnc | (0.364, 0.01); (0.491, 0.009); (0.618, 0.009); (0.745, 0.008); (0.873, 0.008); (1, 0.007); (1.127, 0.007); (1.255, 0.006); (1.382, 0.006); (1.509, 0.005); (1.636, 0.005) |
| NfSOrgCnc | (0.364, 0.01); (0.491, 0.009); (0.618, 0.009); (0.745, 0.008); (0.873, 0.008); (1, 0.007); (1.127, 0.007); (1.255, 0.006); (1.382, 0.006); (1.509, 0.005); (1.636, 0.005) |

**Harvest**

| Parameter | Value |
|---|---|
| At | -0.0072 |
| Bt | 0.69 |
| beta | 0.075 |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| C_Leaf | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.
IntcpCap: 0.5.

## 38.5.38 Crop Potato; Koege

Parent default, defined in lib/potato.dai. Potato; Koege

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DS_Emr | 0.01 |
| DSRate1 | 0.007 |
| TempEff1 | -1. 0, 20. 1, 25. 0.5 |

| Parameter | Value |
|---|---|
| PhotEff1 | 1. 0, 12. 1, 16. 1, 24. 1 |
| DSRate2 | 0.007 |
| TempEff2 | 30. 1 |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 3.8 |
| Qeff | 0.056 |
| TempEff | 5. 0, 10. 0.75, 20. 1, 30. 1, 40. 0.01, 50. 0 |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.17 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.015 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsWStem | (0, 9999); (1, 9999) |
| HvsDS | (0, 1); (1, 40); (2, 40) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 70 |
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.4); (0.35, 0.1); (0.52, 0.1); (1.1, 0) |
| Leaf | (0, 0.7); (0.35, 0.36); (0.52, 0.08); (0.7, 0.07); (0.86, 0.05); (1.1, 0) |
| Stem | (0, 0.3); (0.35, 0.13); (0.52, 0.13); (0.7, 0.1); (0.86, 0.05); (1.1, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0.2, 0); (0.25, 0.01); (0.8, 0.01); (1, 0.02); (2, 0.1) |
| RtDR | (0.2, 0); (0.5, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.005) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.004) |
| PtLeafCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrLeafCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrStemCnc | (0, 0.03); (1, 0.02); (2, 0.02) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.01); (1, 0.01); (2, 0.01) |
| CrSOrgCnc | (0, 0.008); (1, 0.008); (2, 0.008) |
| NfSOrgCnc | (0, 0.005); (1, 0.005); (2, 0.005) |

**Harvest**

| Parameter | Value |
|---|---|
| Dead | CROP-SLOW |
| Stem | CROP-SLOW |
| Leaf | CROP-SLOW |
| SOrg | CROP-SLOW |
| Root | CROP-SLOW |
| DSmax | 0.8 |
| DSnew | 0.2 |
| sorg_height | -10 |

### 38.5.39 Crop Potato; SCRI

Parent default, defined in lib/potato.dai. Potato; SCRI

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 425 |
| DS_Emr | 0.01 |
| DSRate1 | 0.07 |
| TempEff1 | (2, 0); (13, 1); (19, 1); (29, 0) |

| Parameter | Value |
|---|---|
| PhotEff1 | (12, 1); (16, 1) |
| DSRate2 | 0.009 |
| TempEff2 | (2, 0); (13, 1); (19, 1); (29, 0) |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 3 |
| DSEff | (0, 1.2); (0.3, 1.2); (1, 1); (1.6, 0.75); (2, 0.75) |
| Qeff | 0.045 |
| TempEff | (3, 0.01); (10, 0.75); (15, 1); (20, 1); (26, 0.75); (33, 0.01) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.033 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.5 |
| EPext | 0.5 |
| HvsWStem | (0, 9999); (1, 9999) |
| HvsDS | (0, 1); (1, 20); (1.25, 35) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 20 |
| PenPar1 | 0.3 |
| PenPar2 | 4 |
| MaxPen | 50 |
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |
| rootdens | SpRtLength 200, DensRtTip 0.2 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.2); (1, 0.2); (1.36, 0) |
| Leaf | (0, 0.95); (1, 0.75); (1.05, 0.2); (1.25, 0.18); (1.6, 0) |
| Stem | (0, 0.05); (1, 0.25); (1.05, 0.5); (1.25, 0.2); (1.6, 0.03); (2, 0) |
| RSR | (0, 0.5); (0.75, 0.5); (2, 0.25) |

**Prod**

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.72 |
| E_Leaf | 0.66 |
| E_Stem | 0.72 |
| E_SOrg | 0.85 |
| r_Root | 0.01 |
| r_Leaf | 0.03 |
| r_Stem | 0.015 |
| r_SOrg | 0.0045 |
| ShldResC | 0.1 |
| ReMobilDS | 1.5 |
| ReMobilRt | 0.01 |
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (1.6, 0); (1.75, 0.02); (2, 0.1) |
| RtDR | (0, 0); (1.6, 0); (2, 0) |
| ExfoliationFac | 0 |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.06); (1.25, 0.05); (2, 0.035) |
| CrLeafCnc | (0, 0.045); (1, 0.045); (1.25, 0.04); (2, 0.03) |
| NfLeafCnc | (0, 0.01); (1, 0.01); (2, 0.01) |
| PtStemCnc | (0, 0.04); (1, 0.04); (1.6, 0.016); (2, 0.013) |
| CrStemCnc | (0, 0.012); (1.25, 0.012); (2, 0.01) |
| NfStemCnc | (0, 0.004); (2, 0.004) |
| PtSOrgCnc | (1, 0.02); (1.25, 0.015); (2, 0.012) |
| CrSOrgCnc | (1, 0.008); (1.25, 0.008); (2, 0.0075) |
| NfSOrgCnc | (1, 0.004); (2, 0.004) |
| TLLeafEff | (1.09, 1); (2, 1) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |
| sorg_height | -10 |

## 38.5.40 Crop Potato; FertOrgaNic

Parent Potato; SCRI, defined in lib/potato.dai. Potato; FertOrgaNic

**Devel**

| Parameter | Value |
|---|---|
| DSRate1 | 0.06 |
| DSRate2 | 0.009 |

**Root**

| Parameter | Value |
|---|---|
| PenPar1 | 0.06 |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 4 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.2); (1, 0.2); (1.36, 0) |
| Leaf | (0, 0.95); (1, 0.8); (1.05, 0.3); (1.25, 0.25); (1.4, 0) |
| Stem | (0, 0.05); (1, 0.2); (1.05, 0.45); (1.25, 0.05); (1.4, 0) |
| RSR | (0, 0.5); (0.75, 0.5); (2, 0.25) |
| nitrogen_stress_limit | 0.01 |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.2 |
| SpLAIfac | (0, 100); (1, 100) |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.02 |
| LeafAIMod | (1.5, 1); (2, 0.3) |

### Prod

| Parameter | Value |
|---|---|
| LfDR | (0, 0); (1.5, 0); (1.6, 0.02); (2, 0.1) |
| ExfoliationFac | 0 |

### CrpN

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.06); (1, 0.05); (1.3, 0.04); (1.5, 0.025); (2, 0.02) |
| CrLeafCnc | (0, 0.045); (1, 0.04); (1.3, 0.03); (1.5, 0.02); (2, 0.01) |
| NfLeafCnc | (0, 0.01); (1, 0.01); (2, 0.01) |
| PtStemCnc | (0, 0.04); (1, 0.035); (1.3, 0.02); (1.5, 0.01); (2, 0.01) |
| CrStemCnc | (0, 0.02); (1, 0.015); (1.3, 0.01); (1.5, 0.005); (2, 0.005) |
| PtSOrgCnc | (0, 0.02); (1.3, 0.015); (2, 0.01) |
| CrSOrgCnc | (0, 0.008); (1.3, 0.008); (2, 0.007) |
| NH4_root_min | 1e-08 |
| NO3_root_min | 1e-08 |

## 38.5.41 Crop Potato; Agria

Parent Potato; FertOrgaNic, defined in lib/potato.dai. Potato; Agria

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 260 |
| TempEff1 | (6.4, 0); (12, 1.9); (18, 0.4); (33, 0) |
| TempEff2 | (2, 0); (13, 1); (19, 1); (29, 1) |

**Partit**

| Parameter | Value |
|---|---|
| Root | (0, 0.2); (1, 0.2); (1.36, 0) |
| Leaf | (0, 0.95); (1, 0.8); (1.05, 0.3); (1.25, 0.25); (1.4, 0) |
| Stem | (0, 0.05); (1, 0.2); (1.05, 0.15); (1.25, 0.02); (1.4, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.5 |

## 38.5.42 Crop Potato; Folva

Parent Potato; FertOrgaNic, defined in lib/potato.dai. Potato; Folva

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 330 |
| DSRate1 | 0.055 |
| DSRate2 | 0.011 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4.8 |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.5 |
| SpLAIfac | (0, 100); (1, 100) |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.02 |
| LeafAIMod | (1.5, 1.1); (2, 0.3) |

**Partit**

| Parameter | Value |
|---|---|
| Root | (0, 0.2); (1, 0.2); (1.36, 0) |
| Leaf | (0, 0.95); (1, 0.8); (1.05, 0.2); (1.25, 0.15); (1.35, 0) |
| Stem | (0, 0.05); (1, 0.2); (1.05, 0.35); (1.25, 0); (1.35, 0) |
| nitrogen_stress_limit | 0.01 |

**Prod**

| Parameter | Value |
|---|---|
| LfDR | (0, 0); (1.48, 0); (2, 0.1) |
| ExfoliationFac | 0 |

**Root**

| Parameter | Value |
|---|---|
| PenPar1 | 0.06 |

**CrpN**

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.06); (1, 0.05); (1.3, 0.04); (1.5, 0.025); (2, 0.02) |
| CrLeafCnc | (0, 0.045); (1, 0.04); (1.3, 0.03); (1.5, 0.02); (2, 0.01) |
| NfLeafCnc | (0, 0.01); (1, 0.01); (2, 0.01) |
| PtStemCnc | (0, 0.04); (1, 0.035); (1.3, 0.02); (1.5, 0.01); (2, 0.01) |
| CrStemCnc | (0, 0.02); (1, 0.015); (1.3, 0.01); (1.5, 0.005); (2, 0.005) |
| PtSOrgCnc | (0, 0.02); (1.3, 0.015); (2, 0.01) |
| CrSOrgCnc | (0, 0.008); (1.3, 0.008); (2, 0.007) |
| NH4_root_min | 1e-08 |
| NO3_root_min | 1e-08 |

## 38.5.43 Crop Potato; Triada

Parent Potato; FertOrgaNic, defined in lib/potato.dai. Potato; Triada

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 330 |
| TempEff2 | (15, 0.7); (20, 1.6); (25, 0.7) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 5 |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.4 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.015 |
| LeafAIMod | (1, 1); (2, 0.62) |

**CrpN**

| Parameter | Value |
|---|---|
| NH4_root_min | 3e-05 |
| NO3_root_min | 3e-05 |

**Partit**

| Parameter | Value |
|---|---|
| Leaf | (0, 0.95); (1, 0.9); (1.05, 0.4); (1.25, 0.28); (1.4, 0) |
| Stem | (0, 0.05); (1, 0.1); (1.05, 0.35); (1.25, 0.02); (1.4, 0) |

**water_stress_effect**

| Parameter | Value |
|---|---|
| y_half | 0.75 |

## 38.5.44 Crop Potato

Parent Potato; Folva, defined in lib/potato.dai. Potato; Folva

## 38.5.45 Crop Rye

Parent default, defined in lib/rye.dai. Rye

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.02 |
| TempEff1 | -10. 0.01 |
| PhotEff1 | (0, 0.1); (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.025 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.33 |
| TaLim | 5 |
| TaSum | -50 |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 4 |
| Qeff | 0.05 |
| TempEff | (-20, 0); (4, 0); (10, 1); (25, 1); (35, 0.01); (50, 0) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| PARrel | 0.05 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 100); (2, 100) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.5); (0.53, 0.25); (1, 0.05); (2, 0) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.43, 0.83); (0.53, 0.75); (0.62, 0.56); (0.77, 0.2); (0.95, 0.09); (1.14, 0); (1.38, 0); (2, 0) |
| Stem | (0, 0.1); (0.33, 0.15); (0.43, 0.17); (0.53, 0.25); (0.62, 0.44); (0.77, 0.8); (0.95, 0.64); (1.14, 0.52); (1.38, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.2 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0 |
| LfDR | (0, 0); (0.05, 0.01); (0.2, 0); (0.25, 0.01); (0.9, 0.03); (2, 0.1) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.012); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.009) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0.34, 0.06); (0.5, 0.036); (1.28, 0.033); (1.52, 0.0288); (1.93, 0.0194) |
| CrLeafCnc | (0.34, 0.021); (0.5, 0.021); (1.28, 0.021); (1.52, 0.019); (1.93, 0.0183) |
| NfLeafCnc | (0.34, 0.008); (0.5, 0.008); (1.28, 0.008); (1.52, 0.008); (1.93, 0.008) |
| PtStemCnc | (0.5, 0.02); (1.12, 0.0125); (1.28, 0.0093); (1.53, 0.0085); (1.93, 0.0057) |
| CrStemCnc | (0.5, 0.0064); (1.12, 0.0049); (1.28, 0.0043); (1.53, 0.0041); (1.93, 0.0035) |
| NfStemCnc | (0.5, 0.003); (1.12, 0.003); (1.28, 0.003); (1.53, 0.003); (1.93, 0.003) |
| PtSOrgCnc | (1.12, 0.025); (2, 0.03) |
| CrSOrgCnc | (1.12, 0.015); (2, 0.015) |
| NfSOrgCnc | (1.12, 0.014); (2, 0.014) |

### Harvest

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.46 Crop Ryegrass

Parent default, defined in lib/ryegrass.dai. Ryegrass

### Devel

| Parameter | Value |
|---|---|
| EmrTSum | 50 |
| DS_Emr | 0.01 |
| DSRate1 | 0.022 |

| Parameter | Value |
|---|---|
| TempEff1 | (-4, 0.02); (6, 0.2); (13, 0.7); (20, 1); (29, 1.2); (35, 1.2) |
| PhotEff1 | (0, 0.005); (11, 0.29); (12, 0.55); (13, 0.75); (15, 1); (16, 1); (17, 1.1); (18, 1.14); (20, 1.18); (28, 1.18) |
| DSRate2 | 0.02 |
| TempEff2 | (-4, 0.02); (5, 0.2); (14, 0.7); (20, 1); (29, 1.2); (35, 1.2) |
| defined_until_ds | 2. |

### Vernal

| Parameter | Value |
|---|---|
| DSLim | 0.33 |
| TaLim | 5 |
| TaSum | -50 |

### LeafPhot

| Parameter | Value |
|---|---|
| Qeff | 0.065 |
| Fm | 3.9 |
| TempEff | (-20, 0.0001); (1, 0.01); (4.5, 0.5); (7.5, 0.875); (8, 0.92); (10, 0.97); (12, 1); (16, 1); (20, 1); (23, 0.97); (29, 0.9); (33, 0.8); (35, 0.1); (37, 0.01); (50, 0) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.3 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.014 |
| HvsDS | (0, 1); (0.2, 5); (0.4, 8.5); (0.6, 20); (1, 30); (1.3, 35); (2, 35) |
| LeafAIMod | (0, 1); (1, 1); (2, 0.7) |
| SpStemAI | 0.0035 |
| StemAIMod | (0, 1); (1, 1); (2, 0.7) |
| StemPhotEff | 0.7 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.3, 0.8 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| EpFac | 1 |
| IntcpCap | 0.5 |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 2 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |

| Parameter | Value |
|---|---|
| MaxPen | 100 |
| rootdens | SpRtLength 108, DensRtTip 0.1 |
| Rad | 0.01 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| RSR | (0, 0.7); (1, 0.5); (2, 0.4) |
| Root | (0, 0.8); (0.33, 0.6); (0.53, 0.45); (1, 0.45); (2, 0.35) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.53, 0.83); (0.73, 0.56); (0.95, 0.35); (1.14, 0.28); (1.45, 0.23); (2, 0.2) |
| Stem | (0, 0.1); (0.33, 0.15); (0.53, 0.17); (0.73, 0.44); (0.95, 0.65); (1.14, 0.68); (1.45, 0.77); (2, 0.8) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.04 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.57 |
| r_Root | 0.01 |
| r_Leaf | 0.03 |
| r_Stem | 0.005 |
| r_SOrg | 0.005 |
| ShldResC | 0.1 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |
| LfDR | (0, 0); (0.5, 0.025); (2, 0.05) |
| RtDR | (0, 0); (0.5, 0.025); (2, 0.03) |
| Large_RtDR | 0.05 |
| RtDR_T_factor | (-5, 0); (5, 1) |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.02) |
| CrRootCnc | (0, 0.015); (1, 0.012); (2, 0.0075) |
| NfRootCnc | (0, 0.012); (2, 0.001) |
| PtLeafCnc | (0, 0.04); (1, 0.03); (2, 0.025) |
| CrLeafCnc | (0, 0.02); (1, 0.015); (2, 0.012) |
| NfLeafCnc | (0, 0.015); (2, 0.01) |
| PtStemCnc | (0, 0.02); (1, 0.018); (2, 0.015) |
| CrStemCnc | (0, 0.015); (1, 0.01); (2, 0.01) |
| NfStemCnc | (0, 0.01); (2, 0.008) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (2, 0.014) |
| TLLeafEff | (0, 0.9); (2, 0.9) |
| TLRootEff | (0, 0.6); (2, 0.6) |

**Harvest**

| Parameter | Value |
|---|---|
| Dead | Ryegrass-SLOW |
| Stem | Ryegrass-SLOW |
| Leaf | Ryegrass-SLOW |
| SOrg | () |
| Root | () |
| DSmax | 2 |
| cut_delay | (1500, 1); (2000, 20); (7000, 45) |

## 38.5.47 Crop Spring Barley; V1

Parent old, defined in lib/sbarley-old.dai. Spring Barley; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 160 |
| DSRate1 | 0.00129 |
| DSRate2 | 0.00129 |
| DS_Emr | 0.01 |
| DS_reset | false |
| TempEff1 | (0, 0); (100, 100) |
| TempEff2 | (0, 0); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| TLim1 | 4 |
| TLim2 | 10 |

**Canopy**

| Parameter | Value |
|---|---|
| InitGrowth | 4.65 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 0.58 |
| alpha | 3 |
| beta | 1.87 |
| SpLAI | 0.02 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |

| Parameter | Value |
|---|---|
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (1.067, 0.15); (2, 0.15) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_Leaf | (0, 0.72); (2, 0.72) |
| r_Root | (1.55, 0.065); (1.5501, 0.015) |
| r_Leaf | (1.55, 0.015); (1.5501, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| SeedN | 0.4 |
| PtRootCnc | (0.258, 0.0179); (0.387, 0.0164); (0.516, 0.015); (0.645, 0.0138); (0.774, 0.0127); (0.903, 0.0116); (1.032, 0.0106); (1.161, 0.0098) |
| CrRootCnc | (0.258, 0.0129); (0.387, 0.012); (0.516, 0.0112); (0.645, 0.0104); (0.774, 0.0097); (0.903, 0.009); (1.032, 0.0084); (1.161, 0.0078) |
| NfRootCnc | (0.258, 0.01); (0.387, 0.0095); (0.516, 0.009); (0.645, 0.0085); (0.774, 0.0081); (0.903, 0.0077); (1.032, 0.0073); (1.161, 0.0069) |
| PtLeafCnc | (0.258, 0.0493); (0.387, 0.0402); (0.516, 0.0328); (0.645, 0.0267); (0.774, 0.0218); (0.903, 0.0177); (1.032, 0.0145); (1.161, 0.0118) |
| CrLeafCnc | (0.258, 0.0297); (0.387, 0.0252); (0.516, 0.0215); (0.645, 0.0183); (0.774, 0.0155); (0.903, 0.0132); (1.032, 0.0112); (1.161, 0.0095) |

| Parameter | Value |
|---|---|
| NfLeafCnc | (0.258, 0.0198); (0.387, 0.017); (0.516, 0.0145); (0.645, 0.0124); (0.774, 0.0107); (0.903, 0.0091); (1.032, 0.0078); (1.161, 0.0067) |

**Harvest**

| Parameter | Value |
|---|---|
| index | 0.5 |
| CStem | 0.0065 |
| CSOrg | 0.00215 |
| alpha | 0.8 |
| beta | 0.09 |
| Stem | (); () |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| C_Stem | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.
IntcpCap: 0.5.

### 38.5.48 Crop Spring Barley

Parent default, defined in lib/sbarley.dai. Spring Barley

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.032 |
| TempEff1 | (0, 0); (4, 0.01); (19, 1); (24, 1); (30, 0.01) |
| PhotEff1 | (0, 0.1); (11, 0.1); (16, 1); (20, 1.2); (24, 1.2) |
| DSRate2 | 0.021 |
| TempEff2 | (0, 0); (16, 1); (30, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 3.5 |
| Qeff | 0.04 |
| TempEff | -20. 0.001 |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.031 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 75); (2, 75) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.65); (0.51, 0.55); (0.72, 0.15); (0.95, 0.05); (2, 0) |
| Leaf | (0, 1); (0.25, 0.7); (0.51, 0.55); (0.6, 0.5); (0.72, 0.23); (0.83, 0.01); (0.95, 0); (2, 0) |
| Stem | (0, 0); (0.25, 0.3); (0.51, 0.45); (0.6, 0.5); (0.72, 0.77); (0.83, 0.99); (0.95, 1); (1.51, 0); (2, 0) |
| RSR | (0, 0.5); (0.75, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.7 |
| LfDR | (0, 0); (0.5, 0); (1.2, 0.01); (2, 0.08) |
| RtDR | (0, 0); (0.2, 0); (1.2, 0.01); (2, 0.01) |

## CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (1, 0.03); (2, 0.02) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.49 Crop Spring Barley; Foulum

Parent Spring Barley, defined in lib/sbarley.dai. Spring Barley; Foulum

**Devel**

| Parameter | Value |
|---|---|
| DSRate2 | 0.025 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.023 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.7, 1); (2, 0) |
| SOrgPhotEff | 0.5 |

**Prod**

| Parameter | Value |
|---|---|
| CH2OReleaseRate | 0.03 |
| ShldResC | 0.4 |
| ReMobilDS | 1 |
| ReMobilRt | 0.05 |
| ExfoliationFac | 0.7 |
| LfDR | (0, 0); (1, 0); (1.2, 0.02); (1.3, 0.08); (2, 0.1) |

**Partit**

| Parameter | Value |
|---|---|
| Root | (0, 0.65); (0.51, 0.55); (0.72, 0.15); (0.95, 0.02); (2, 0) |
| Leaf | (0, 1); (0.25, 0.7); (0.51, 0.55); (0.6, 0.5); (0.72, 0.23); (0.83, 0.01); (0.95, 0); (2, 0) |

| Parameter | Value |
|---|---|
| Stem | (0, 0); (0.25, 0.3); (0.51, 0.45); (0.6, 0.5); (0.72, 0.77); (0.83, 0.99); (0.95, 1); (1.21, 0); (2, 0) |
| RSR | (0, 0.5); (0.75, 0.25); (2, 0.25) |

**Root**

| Parameter | Value |
|---|---|
| MxNH4Up | 5e-07 |
| MxNO3Up | 2.5e-07 |

**CrpN**

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.06); (0.8, 0.04); (2, 0.02) |
| CrLeafCnc | (0, 0.03); (0.8, 0.03); (2, 0.015) |
| NfLeafCnc | (0, 0.004); (0.8, 0.004); (2, 0.004) |
| PtStemCnc | (0.5, 0.06); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0.5, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0.5, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

## 38.5.50 Crop Sugar Beet

Parent default, defined in lib/sbeet.dai. Sugar Beet

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DS_Emr | 0.01 |
| DSRate1 | 0.007 |
| TempEff1 | (-1, 0); (0, 0.01); (5, 0.5); (20, 1); (25, 0.5) |
| PhotEff1 | (1, 0); (12, 1); (16, 1); (24, 1) |
| DSRate2 | 0 |
| TempEff2 | (0, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 3.8 |
| Qeff | 0.056 |
| TempEff | (-20, 0); (5, 0); (10, 0.75); (20, 1); (30, 1); (40, 0.01); (50, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.17 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.015 |
| LAIDist0 | 0, 1, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 40); (2, 40) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 200 |
| SpRtLength | 100 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.4); (0.35, 0.1); (0.52, 0.1); (1.1, 0) |
| Leaf | (0, 0.7); (0.35, 0.36); (0.52, 0.08); (0.7, 0.07); (0.86, 0.05); (1.1, 0) |
| Stem | (0, 0.3); (0.35, 0.13); (0.52, 0.13); (0.7, 0.1); (0.86, 0.05); (1.1, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (0.09, 0); (0.1, 0.01); (0.15, 0.01); (1, 0.02); (2, 0.1) |
| RtDR | (0, 0); (0.2, 0); (0.5, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (1, 0.03); (2, 0.02) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

### 38.5.51 Crop Spring Rape; V1

Parent old, defined in lib/srape-old.dai. Spring Rape; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 110 |
| DSRate1 | 0.0011 |
| DSRate2 | 0.0011 |
| DS_Emr | 0.01 |
| DS_reset | false |
| TempEff1 | (0, 1); (100, 100) |
| TempEff2 | (0, 1); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| TLim1 | 6 |
| TLim2 | 12 |

**Canopy**

| Parameter | Value |
|---|---|
| InitGrowth | 4.32 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 0.5 |
| alpha | 1.2 |
| beta | 0.44 |

| Parameter | Value |
|---|---|
| SpLAI | 0.018 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (1, 0.15); (2, 0.15) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_Leaf | (0, 0.65); (2, 0.65) |
| r_Root | (1, 0.065); (1.0001, 0.015) |
| r_Leaf | (1, 0.025); (1.0001, 0.015) |

### CrpN

| Parameter | Value |
|---|---|
| SeedN | 0.02 |
| PtLeafCnc | (0.306, 0.07); (0.419, 0.061); (0.533, 0.054); (0.647, 0.047); (0.761, 0.042); (0.875, 0.036); (0.989, 0.032); (1.103, 0.028); (1.217, 0.025); (1.331, 0.022); (1.444, 0.019) |
| PtRootCnc | (0.306, 0.02); (0.419, 0.019); (0.533, 0.017); (0.647, 0.016); (0.761, 0.015); (0.875, 0.014); (0.989, 0.013); (1.103, 0.012); (1.217, 0.011); (1.331, 0.011); (1.444, 0.01) |
| CrLeafCnc | (0.306, 0.03); (0.419, 0.027); (0.533, 0.024); (0.647, 0.022); (0.761, 0.019); (0.875, 0.017); (0.989, 0.016); (1.103, 0.014); (1.217, 0.012); (1.331, 0.011); (1.444, 0.01) |
| CrRootCnc | (0.306, 0.013); (0.419, 0.012); (0.533, 0.012); (0.647, 0.011); (0.761, 0.011); (0.875, 0.01); (0.989, 0.01); (1.103, 0.009); (1.217, 0.009); (1.331, 0.008); (1.444, 0.008) |

| Parameter | Value |
|---|---|
| NfLeafCnc | (0.306, 0.02); (0.419, 0.018); (0.533, 0.016); (0.647, 0.015); (0.761, 0.013); (0.875, 0.012); (0.989, 0.011); (1.103, 0.01); (1.217, 0.009); (1.331, 0.008); (1.444, 0.007) |
| NfRootCnc | (0.306, 0.01); (0.419, 0.01); (0.533, 0.009); (0.647, 0.009); (0.761, 0.009); (0.875, 0.008); (0.989, 0.008); (1.103, 0.008); (1.217, 0.008); (1.331, 0.007); (1.444, 0.007) |

**Harvest**

| Parameter | Value |
|---|---|
| index | 0.3 |
| CStem | 0.0045 |
| CSOrg | 0.00375 |
| alpha | 1 |
| beta | 0.1 |
| Stem | (); () |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| DSmax | 0 |
| DSnew | 0 |
| C_Stem | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.
IntcpCap: 0.5.

## 38.5.52 Crop Spring Rape

Parent default, defined in lib/srape.dai. Spring Rape

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 125 |
| DS_Emr | 0.01 |
| DSRate1 | 0.035 |
| TempEff1 | (0, 0); (4, 0.01); (19, 1); (24, 1); (30, 0.01) |
| PhotEff1 | 11. 0.1 |
| DSRate2 | 0.0225 |
| TempEff2 | 16. 1 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4 |
| Qeff | 0.05 |
| TempEff | -20. 0, 6. 0.001, 12. 1, 25. 1, 35. 0.01, 50. 0 |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.025 |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.5, 1); (2, 0) |
| StemPhotEff | 0.2 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.4, 1); (2, 0) |
| SOrgPhotEff | 0.25 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.7 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 50); (2, 80) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.65); (0.51, 0.55); (0.72, 0.15); (0.95, 0.05); (2, 0) |
| Leaf | (0, 0.9); (0.25, 0.8); (0.51, 0.7); (0.6, 0.6); (0.72, 0.4); (0.83, 0.2); (0.95, 0.1); (1.51, 0.1); (2, 0) |
| Stem | (0, 0.1); (0.25, 0.2); (0.51, 0.3); (0.6, 0.4); (0.72, 0.6); (0.83, 0.8); (0.95, 0.9); (1.1, 0.5); (1.2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.5 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |

| Parameter | Value |
|---|---|
| r_SOrg | 0.03 |
| ShldResC | 0.3 |
| ReMobilDS | 1.18 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (0.25, 0); (0.5, 0.005); (1, 0.01); (1.05, 0.03); (1.1, 0.03) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.005); (2, 0.01) |
| ExfoliationFac | 0.9 |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.06); (2, 0.06) |
| CrLeafCnc | (0, 0.044); (1, 0.044); (2, 0.044) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0.65, 0.05); (0.8, 0.035); (1, 0.02); (1.6, 0.01); (2, 0.008) |
| CrStemCnc | (0.65, 0.03); (0.8, 0.02); (1.09, 0.01); (1.6, 0.008); (2, 0.006) |
| NfStemCnc | (0.65, 0.004); (0.8, 0.004); (1.09, 0.004); (1.6, 0.004); (2, 0.004) |
| PtSOrgCnc | (1.09, 0.05); (1.4, 0.025); (2, 0.025) |
| CrSOrgCnc | (1.09, 0.037); (1.4, 0.02); (2, 0.02) |
| NfSOrgCnc | (1.09, 0.004); (1.4, 0.004); (2, 0.004) |
| TLLeafEff | (1.09, 1); (1.4, 1); (2, 1) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.58 |
| EconomicYield_N | 0.82 |

## 38.5.53 Crop Sugar Beet

Parent default, defined in lib/sugarbeet.dai. Sugar Beet

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 325 |
| DS_Emr | 0.01 |
| DSRate1 | 0.005 |
| TempEff1 | -1. 0, 20. 1, 25. 0.5 |
| PhotEff1 | 3. 0, 8. 1, 16. 1, 24. 1 |
| DSRate2 | 0.007 |
| TempEff2 | 30. 1 |
| defined_until_ds | 0.8 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 7 |
| Qeff | 0.056 |
| TempEff | 5. 0, 10. 0.75, 20. 1, 30. 1, 40. 0.01, 50. 0 |

## Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.17 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.012 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0.1, 0.8, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (0.02, 4); (0.08, 10); (0.14, 30); (0.21, 40); (0.27, 55) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.2 |
| PenPar2 | 4 |
| MaxPen | 200 |
| Rad | 0.05 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.25); (0.08, 0.2); (0.14, 0.1); (0.27, 0.07); (0.34, 0.06) |
| Leaf | (0, 0.7); (0.35, 0.3); (0.52, 0.08); (0.7, 0.07); (0.86, 0.05); (1.1, 0) |
| Stem | (0, 0); (1, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.66 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |

| Parameter | Value |
|---|---|
| r_Leaf | 0.05 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0.2, 0); (0.25, 0.01); (0.8, 0.01); (1, 0); (2, 0) |
| RtDR | (0.2, 0); (0.5, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.005) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.004) |
| PtLeafCnc | (0, 0.04); (1, 0.035); (2, 0.02) |
| CrLeafCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrStemCnc | (0, 0.03); (1, 0.02); (2, 0.02) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.01); (1, 0.01); (2, 0.01) |
| CrSOrgCnc | (0, 0.008); (1, 0.008); (2, 0.008) |
| NfSOrgCnc | (0, 0.005); (1, 0.005); (2, 0.005) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |
| sorg_height | -15 |

## 38.5.54 Crop Spring Wheat

Parent default, defined in lib/swheat.dai. Spring Wheat

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.04 |
| TempEff1 | (0, 0); (4, 0.01); (19, 1); (24, 1); (30, 0.01) |
| PhotEff1 | (0, 0.1); (11, 0.1); (16, 1); (20, 1.2); (24, 1.2) |
| DSRate2 | 0.025 |
| TempEff2 | (0, 0); (16, 1); (30, 1) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4 |
| Qeff | 0.05 |
| TempEff | (-20, 0.001); (0, 0.01); (5, 0.4); (10, 0.7); (15, 0.9); (20, 1); (25, 1); (30, 0.9); (35, 0.8); (40, 0.5); (45, 0) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.031 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 100); (2, 100) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.65); (0.51, 0.55); (0.72, 0.15); (0.95, 0.05); (2, 0) |
| Leaf | (0, 0.82); (0.25, 0.7); (0.51, 0.55); (0.6, 0.5); (0.72, 0.23); (0.83, 0.01); (0.95, 0); (2, 0) |
| Stem | (0, 0.18); (0.25, 0.3); (0.51, 0.45); (0.6, 0.5); (0.72, 0.77); (0.83, 0.99); (0.95, 1); (1.51, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.016 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |

| Parameter | Value |
|---|---|
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (0.5, 0.02); (1, 0.01); (2, 0.2) |
| RtDR | (0, 0); (0.2, 0); (1.2, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (1, 0.03); (2, 0.02) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

### 38.5.55 Crop Winter Barley; V1

Parent old, defined in lib/wbarley-old.dai. Winter Barley; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 80 |
| DSRate1 | 0.00163 |
| DSRate2 | 0.00163 |
| DS_Emr | 0.01 |
| DS_reset | true |
| TempEff1 | (0, 1); (100, 100) |
| TempEff2 | (0, 1); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| TLim1 | 4 |
| TLim2 | 10 |

**Canopy**

| Parameter | Value |
|---|---|
| InitGrowth | 3.67 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 0.53 |
| alpha | 1.8 |
| beta | 1.1755 |
| SpLAI | 0.014 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.608, 0.1); (2, 0.1) |

## Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_Leaf | (0, 0.72); (2, 0.72) |
| r_Root | (0.898, 0.065); (0.8981, 0.015) |
| r_Leaf | (0.898, 0.015); (0.8981, 0.01) |

## CrpN

| Parameter | Value |
|---|---|
| SeedN | 0.4 |
| PtLeafCnc | (0.163, 0.05); (0.347, 0.04); (0.531, 0.031); (0.714, 0.025); (0.898, 0.02); (1.082, 0.016); (1.265, 0.013) |
| PtRootCnc | (0.163, 0.018); (0.347, 0.016); (0.531, 0.015); (0.714, 0.013); (0.898, 0.012); (1.082, 0.011); (1.265, 0.01) |
| CrLeafCnc | (0.163, 0.03); (0.347, 0.025); (0.531, 0.021); (0.714, 0.017); (0.898, 0.014); (1.082, 0.012); (1.265, 0.01) |

| Parameter | Value |
|---|---|
| CrRootCnc | (0.163, 0.013); (0.347, 0.012); (0.531, 0.011); (0.714, 0.01); (0.898, 0.009); (1.082, 0.009); (1.265, 0.008) |
| NfLeafCnc | (0.163, 0.02); (0.347, 0.017); (0.531, 0.014); (0.714, 0.012); (0.898, 0.01); (1.082, 0.008); (1.265, 0.007) |
| NfRootCnc | (0.163, 0.01); (0.347, 0.009); (0.531, 0.009); (0.714, 0.008); (0.898, 0.008); (1.082, 0.007); (1.265, 0.007) |

**Harvest**

| Parameter | Value |
|---|---|
| index | 0.45 |
| CStem | 0.0065 |
| CSOrg | 0.00215 |
| alpha | 0.8 |
| beta | 0.09 |
| Stem | (); () |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| DSmax | 0 |
| DSnew | 0 |
| C_Stem | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.

IntcpCap: 0.5.

### 38.5.56 Crop Winter Barley

Parent default, defined in lib/wbarley.dai. Winter Barley

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 200 |
| DS_Emr | 0.01 |
| DSRate1 | 0.025 |
| TempEff1 | -10. 0.01 |
| PhotEff1 | (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.03 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.25 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 4 |
| Qeff | 0.05 |
| TempEff | (-20, 0); (2, 0); (10, 1); (25, 1); (35, 0.01); (50, 0) |

## Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 100); (2, 100) |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

## Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.5); (0.53, 0.25); (1, 0.05); (2, 0) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.43, 0.83); (0.53, 0.75); (0.62, 0.56); (0.77, 0.2); (0.95, 0.09); (1.14, 0.05); (1.38, 0); (2, 0) |
| Stem | (0, 0.1); (0.33, 0.15); (0.43, 0.17); (0.53, 0.25); (0.62, 0.44); (0.77, 0.8); (0.95, 0.64); (1.14, 0.62); (1.38, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |

| Parameter | Value |
|---|---|
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.2 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (0.05, 0.01); (0.2, 0); (0.25, 0.01); (0.9, 0.03); (2, 0.1) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (1, 0.03); (2, 0.02) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.57 Crop Winter Barley; Koge

Parent default, defined in lib/wbarley.dai. Winter Barley; Koge

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 200 |
| DS_Emr | 0.01 |
| DSRate1 | 0.022 |
| TempEff1 | -10. 0.01 |
| PhotEff1 | (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.028 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.85 |
| TaLim | 5 |
| TaSum | -50 |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 4 |
| Qeff | 0.05 |
| TempEff | (-20, 0); (2, 0); (10, 1); (25, 1); (35, 0.01); (50, 0) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| PARrel | 0.07 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 90); (2, 90) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.32, 0.5); (0.6, 0.25); (1, 0); (2, 0) |
| Leaf | (0, 0.9); (0.19, 0.83); (0.26, 0.85); (0.45, 0.82); (0.6, 0.32); (0.86, 0.15); (1, 0.26); (1.26, 0); (1.5, 0); (2, 0) |
| Stem | (0, 0.1); (0.19, 0.17); (0.26, 0.15); (0.45, 0.18); (0.6, 0.68); (0.86, 0.85); (1, 0.34); (1.26, 0.27); (1.5, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.2 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.75 |
| LfDR | (0.9, 0); (1.2, 0.03); (2, 0.15) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.02); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.008); (2, 0.008) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0, 0.06); (1, 0.04); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (1, 0.03); (2, 0.02) |
| NfLeafCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.01) |
| CrStemCnc | (0, 0.01); (1, 0.01); (2, 0.006) |
| NfStemCnc | (0, 0.004); (1, 0.004); (2, 0.004) |
| PtSOrgCnc | (0, 0.024); (1, 0.024); (2, 0.024) |
| CrSOrgCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfSOrgCnc | (0, 0.014); (1, 0.014); (2, 0.014) |

## 38.5.58 Crop Winter Barley; Foulum

Parent Winter Barley, defined in lib/wbarley.dai. Winter Barley; Foulum

**Prod**

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| ShldResC | 0.3 |
| ReMobilDS | 1.2 |
| ReMobilRt | 0.05 |
| Large_RtDR | 0.05 |
| LfDR | (0, 0); (0.05, 0.01); (0.2, 0); (0.25, 0.01); (0.9, 0.03); (2, 0.1) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.012); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.009) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |

| Parameter | Value |
|---|---|
| PtLeafCnc | (0.34, 0.06); (0.5, 0.036); (1.28, 0.032); (1.52, 0.0288); (1.93, 0.0194) |
| CrLeafCnc | (0.34, 0.021); (0.5, 0.021); (1.28, 0.021); (1.52, 0.019); (1.93, 0.0183) |
| NfLeafCnc | (0.34, 0.008); (0.5, 0.008); (1.28, 0.008); (1.52, 0.008); (1.93, 0.008) |
| PtStemCnc | (0.5, 0.02); (1.12, 0.0125); (1.28, 0.0093); (1.53, 0.0085); (1.93, 0.0085) |
| CrStemCnc | (0.5, 0.0064); (1.12, 0.006); (1.28, 0.006); (1.53, 0.006); (1.93, 0.006) |
| NfStemCnc | (0.5, 0.004); (1.12, 0.004); (1.28, 0.004); (1.53, 0.004); (1.93, 0.004) |
| PtSOrgCnc | (1.12, 0.015); (2, 0.023) |
| CrSOrgCnc | (1.12, 0.014); (2, 0.017) |
| NfSOrgCnc | (1.12, 0.013); (2, 0.014) |

### Harvest

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.59 Crop Wclover

Parent default, defined in lib/wclover.dai. v.30 Developed by Henning Hoegh Jensen
min_light_fraction: 0.01.

### Devel

| Parameter | Value |
|---|---|
| EmrTSum | 50 |
| DS_Emr | 0.05 |
| DSRate1 | 0.025 |
| TempEff1 | (0, 0.001); (4, 0.005); (11, 0.35); (17, 0.7); (20, 0.8); (32, 1); (39, 1.2) |
| PhotEff1 | (0, 0.1); (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.02 |
| TempEff2 | (0, 0.001); (5, 0.015); (14, 0.4); (20, 0.8); (24, 0.9); (29, 1); (39, 1.2) |
| defined_until_ds | 2 |

### Vernal

| Parameter | Value |
|---|---|
| DSLim | 0.33 |
| TaLim | 5 |
| TaSum | -50 |

### LeafPhot

| Parameter | Value |
|---|---|
| Qeff | 0.06 |
| Fm | 1.8 |

| Parameter | Value |
|---|---|
| TempEff | (-20, 0.0001); (6.5, 0.1); (8, 0.3); (12, 0.8); (16, 1); (20, 1); (33, 1); (37, 0.03); (50, 0) |

## Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.3 |
| SpLAIfac | (0, 100); (1, 100) |

## Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.0175 |
| HvsDS | (0, 1); (0.25, 5); (0.4, 8); (0.7, 16); (1, 22); (2, 30) |
| LeafAIMod | (0, 1); (1, 1); (2, 0.75) |
| SpStemAI | 0.003 |
| StemAIMod | (0, 1); (1, 1); (2, 1) |
| StemPhotEff | 1 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.6, 0.9 |
| PARref | 0.06 |
| PARext | 0.7 |
| EPext | 0.8 |
| EpFac | 1 |
| IntcpCap | 0.5 |

## Root

| Parameter | Value |
|---|---|
| DptEmr | 2 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 100 |
| rootdens | SpRtLength 90, DensRtTip 0.1 |
| Rad | 0.014 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 1e-11 |
| MxNO3Up | 1e-08 |

## Partit

| Parameter | Value |
|---|---|
| RSR | (0, 0.5); (1, 0.3); (2, 0.2) |
| Root | (0, 0.95); (0.33, 0.95); (0.53, 0.85); (1, 0.75); (2, 0.5) |
| Leaf | (0, 0.65); (0.33, 0.55); (0.43, 0.5); (0.63, 0.4); (0.95, 0.25); (1.2, 0.2); (1.45, 0.12); (2, 0) |
| Stem | (0, 0.35); (0.33, 0.45); (0.43, 0.5); (0.63, 0.6); (0.95, 0.75); (1.2, 0.7); (1.45, 0.6); (2, 0) |

## Prod

| Parameter | Value |
|---|---|
| NCrop | 0.04 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.57 |
| r_Root | 0.025 |
| r_Leaf | 0.03 |
| r_Stem | 0.005 |
| r_SOrg | 0.005 |
| ShldResC | 0.3 |
| ReMobilDS | 1 |
| ReMobilRt | 0.1 |
| LfDR | (0, 0); (0.8, 0.004); (1, 0.01); (1.4, 0.1); (2, 0.25) |
| RtDR | (0, 0); (0.5, 0); (0.7, 0.01); (2, 0.15) |
| Large_RtDR | 0.03 |
| RtDR_T_factor | (0, 0); (5, 1) |

**CrpN**

| Parameter | Value |
|---|---|
| DS_fixate | 0.1 |
| DS_cut_fixate | 0.15 |
| fixate_factor | 0.02 |
| PtRootCnc | (0, 0.025); (1, 0.025); (2, 0.025) |
| CrRootCnc | (0, 0.015); (1, 0.013); (2, 0.011) |
| NfRootCnc | (0, 0.008); (2, 0.008) |
| PtLeafCnc | (0, 0.048); (1, 0.046); (2, 0.043) |
| CrLeafCnc | (0, 0.035); (1, 0.035); (2, 0.035) |
| NfLeafCnc | (0, 0.03); (2, 0.03) |
| PtStemCnc | (0, 0.02); (1, 0.02); (2, 0.02) |
| CrStemCnc | (0, 0.016); (1, 0.016); (2, 0.016) |
| NfStemCnc | (0, 0.015); (1, 0.015); (2, 0.015) |
| PtSOrgCnc | (0, 0.03); (1, 0.03); (2, 0.03) |
| CrSOrgCnc | (0, 0.028); (1, 0.028); (2, 0.028) |
| NfSOrgCnc | (0, 0.022); (2, 0.022) |
| TLLeafEff | (0, 0.9); (2, 0.9) |
| TLRootEff | (0, 0.1); (2, 0.1) |

**Harvest**

| Parameter | Value |
|---|---|
| Dead | Wclover-SLOW |
| Stem | Wclover-SLOW |
| Leaf | Wclover-SLOW |
| SOrg | Wclover-SLOW initial_fraction 0.9 |
| Root | Wclover-SLOW |
| DSmax | 2 |
| DSnew | 0.27 |
| cut_delay | (500, 10); (1000, 35); (2000, 45); (7000, 75) |

## 38.5.60 Crop White cabbage

Parent default, defined in lib/whitecabbage.dai. White cabbage

**Devel**

| Parameter | Value |
|---|---|
| defined_until_ds | 1.1 |
| EmrTSum | 200 |
| DSRate1 | 0.0061 |
| TempEff1 | (0, 0); (5, 0); (15, 1); (20, 1); (28, 0) |
| PhotEff1 | (0, 0); (8, 1); (16, 1); (24, 0.8) |
| DSRate2 | 0.007 |
| TempEff2 | (0, 1); (30, 1) |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 4.4 |
| Qeff | 0.045 |
| TempEff | (0, 0.01); (5, 0.01); (15, 1); (20, 1); (25, 0.8); (35, 0.01) |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |
| SpLAIfac | (0, 3); (0.15, 1.5); (0.3, 1.25); (0.45, 1); (1, 1) |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.012 |
| SpSOrgAI | 0.0012 |
| SpStemAI | 0 |
| LAIDist0 | 0, 0.1, 1 |
| LAIDist1 | 0, 0.1, 1 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |
| PARrel | 0.05 |
| HvsDS | (0, 0); (0.6, 50); (1, 50) |
| HvsWStem | (0, 0.1); (100, 1) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 5 |
| PenPar1 | 0.21 |
| PenPar2 | 4 |
| MaxPen | 240 |
| Rad | 0.05 |
| h_wp | -15000 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 3.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.3); (0.15, 0.15); (0.3, 0.15); (0.5, 0.1); (0.75, 0.1); (1, 0.1); (1.1, 0.1) |
| Leaf | (0, 0.8); (0.15, 0.65); (0.3, 0.45); (0.5, 0.25); (0.75, 0.15); (1, 0.05); (1.1, 0.05) |
| Stem | (0, 0.2); (0.15, 0.35); (0.3, 0.2); (0.5, 0.2); (0.75, 0.1); (1, 0.05); (1.1, 0.05) |
| RSR | (0, 0.5); (1, 0.25) |

**Prod**

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_SOrg | 0.025 |
| r_Root | 0.015 |
| r_Leaf | 0.03 |
| r_Stem | 0.01 |
| ShldResC | 0.3 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 1 |
| LfDR | (0, 0); (0.5, 0); (0.75, 0); (1, 0.01) |
| RtDR | (0, 0); (0.5, 0); (0.75, 0); (1, 0.005) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.03); (1, 0.02); (2, 0.02) |
| CrRootCnc | (0, 0.015); (1, 0.01); (2, 0.01) |
| NfRootCnc | (0, 0.01); (1, 0.0075); (2, 0.0075) |
| PtLeafCnc | (0, 0.04); (1, 0.03); (2, 0.03) |
| CrLeafCnc | (0, 0.03); (1, 0.015); (2, 0.015) |
| NfLeafCnc | (0, 0.0125); (1, 0.01); (2, 0.01) |
| PtStemCnc | (0, 0.04); (1, 0.03); (2, 0.03) |
| CrStemCnc | (0, 0.03); (1, 0.015); (2, 0.015) |
| NfStemCnc | (0, 0.0125); (1, 0.01); (2, 0.01) |
| PtSOrgCnc | (0, 0.03); (1, 0.025); (2, 0.025) |
| CrSOrgCnc | (0, 0.0175); (1, 0.015); (2, 0.015) |
| NfSOrgCnc | (0, 0.01); (1, 0.0075); (2, 0.0075) |

**Harvest**

| Parameter | Value |
|---|---|
| DSmax | 0 |

## 38.5.61 Crop White cabbage minus transplanted

Parent White cabbage, defined in lib/whitecabbage.dai. White cabbage minus transplanted

**Devel**

| Parameter | Value |
|---|---|
| DS | 0.1 |

**Prod**

| Parameter | Value |
|---|---|
| CH2OPool | 0.05 |
| NCrop | 0.02 |
| WLeaf | 0.28 |
| WRoot | 0.1 |
| WStem | 0.12 |

**Root**

| Parameter | Value |
|---|---|
| Depth | 5 |

**Harvest**

| Parameter | Value |
|---|---|
| production_delay | 14 |

## 38.5.62 Crop Winter Rape; V1

Parent old, defined in lib/wrape-old.dai. Winter Rape; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 175 |
| DSRate1 | 0.00129 |
| DSRate2 | 0.00129 |
| DS_Emr | 0.01 |
| DS_reset | true |
| TempEff1 | (0, 1); (100, 100) |
| TempEff2 | (0, 1); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| TLim1 | 2 |
| TLim2 | 6 |

### Canopy

| Parameter | Value |
|---|---|
| InitGrowth | 3.22 |
| DSinit | 0.07 |
| WLfInit | 20 |
| DS1 | 0.516 |
| alpha | 1.8 |
| beta | 1.096 |
| SpLAI | 0.0092 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 4 |
| PenPar1 | 0.2 |
| PenPar2 | 2 |
| MaxPen | 130 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.011 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.4); (0.338, 0.25); (2, 0.25) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_Leaf | (1.032, 0.65); (1.033, 0.55) |
| r_Root | (1.032, 0.065); (1.033, 0.03) |
| r_Leaf | (1.032, 0.025); (1.033, 0.015) |

### CrpN

| Parameter | Value |
|---|---|
| SeedN | 0.02 |
| PtLeafCnc | (0.065, 0.055); (0.227, 0.047); (0.389, 0.04); (0.551, 0.034); (0.713, 0.029); (0.876, 0.025); (1.038, 0.021); (1.2, 0.018) |
| PtRootCnc | (0.065, 0.03); (0.227, 0.028); (0.389, 0.026); (0.551, 0.024); (0.713, 0.022); (0.876, 0.021); (1.038, 0.019); (1.2, 0.018) |
| CrLeafCnc | (0.065, 0.032); (0.227, 0.029); (0.389, 0.027); (0.551, 0.024); (0.713, 0.022); (0.876, 0.02); (1.038, 0.019); (1.2, 0.017) |
| CrRootCnc | (0.065, 0.015); (0.227, 0.015); (0.389, 0.014); (0.551, 0.014); (0.713, 0.013); (0.876, 0.013); (1.038, 0.012); (1.2, 0.012) |
| NfLeafCnc | (0.065, 0.01); (0.227, 0.009); (0.389, 0.009); (0.551, 0.008); (0.713, 0.007); (0.876, 0.007); (1.038, 0.006); (1.2, 0.006) |
| NfRootCnc | (0.065, 0.01); (0.227, 0.009); (0.389, 0.009); (0.551, 0.008); (0.713, 0.007); (0.876, 0.007); (1.038, 0.006); (1.2, 0.006) |

**Harvest**

| Parameter | Value |
|---|---|
| index | 0.35 |
| CStem | 0.0092 |
| CSOrg | 0.00365 |
| alpha | 1 |
| beta | 0.1 |
| Stem | (); () |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| DSmax | 0 |
| DSnew | 0 |
| C_Stem | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.

IntcpCap: 0.5.

### 38.5.63 Crop Winter Rape

Parent default, defined in lib/wrape.dai. Winter Rape

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 180 |
| DS_Emr | 0.01 |
| DSRate1 | 0.03 |
| TempEff1 | -10. 0.01, 20. 0.9, 25. 1, 35. 1.2 |
| PhotEff1 | 10. 0.29, 11. 0.55, 12. 0.75, 13. 0.89, 14. 1, 15. 1.08, 16. 1.14, 17. 1.18, 24. 1.18 |
| DSRate2 | 0.025 |
| TempEff2 | 10. 0.14, 15. 0.66, 20. 1, 25. 1.23 |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.33 |
| TaLim | 5 |
| TaSum | -50 |

### LeafPhot

| Parameter | Value |
|---|---|
| Fm | 6 |
| Qeff | 0.05 |
| TempEff | -20. 0, 2. 0, 10. 1, 25. 1, 35. 0.01, 50. 0 |

### Seed

| Parameter | Value |
|---|---|
| DSLAI05 | 0.25 |

### Canopy

| Parameter | Value |
|---|---|
| SpLAI | 0.02 |
| LeafAIMod | (0); (2, 1) |
| SpStemAI | 0.005 |
| StemAIMod | (0, 1); (1.6, 1); (2, 0) |
| StemPhotEff | 0.2 |
| SpSOrgAI | 0.005 |
| SOrgAIMod | (0, 1); (1.4, 1); (2, 0) |
| SOrgPhotEff | 0.25 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.7 |
| EPext | 0.5 |
| HvsDS | (0, 1); (0.34, 5); (0.45, 10); (0.69, 50); (1, 160); (2, 160) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.25); (0.66, 0.15); (1, 0.07); (2, 0.07) |

| Parameter | Value |
|---|---|
| Leaf | (0, 0.95); (0.33, 0.95); (0.43, 0.9); (0.53, 0.75); (0.62, 0.56); (0.77, 0.2); (0.95, 0.09); (1.14, 0.05); (1.18, 0); (2, 0) |
| Stem | (0, 0.05); (0.33, 0.05); (0.43, 0.1); (0.53, 0.25); (0.62, 0.44); (0.77, 0.8); (0.95, 0.64); (1.14, 0.22); (1.18, 0); (2, 0) |
| RSR | (0, 0.5); (1, 0.25); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.2 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.5 |
| r_Root | 0.015 |
| r_Leaf | 0.06 |
| r_Stem | 0.01 |
| r_SOrg | 0.03 |
| ShldResC | 0 |
| ReMobilDS | 1.18 |
| ReMobilRt | 0.1 |
| Large_RtDR | 0.05 |
| LfDR | (0.09, 0); (0.1, 0.01); (0.15, 0.005); (1, 0.005); (1.05, 0.2); (1.2, 0.1); (1.25, 0.05); (2, 0.02) |
| RtDR | (0.6, 0); (1.7, 0); (2, 0.005) |
| ExfoliationFac | 0.9 |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0.33, 0.033); (0.41, 0.02); (0.58, 0.023); (1.4, 0.01); (2, 0.01) |
| CrRootCnc | (0.33, 0.008); (0.41, 0.008); (0.58, 0.008); (1.4, 0.008); (2, 0.008) |
| NfRootCnc | (0.33, 0.007); (0.41, 0.007); (0.58, 0.007); (1.4, 0.007); (2, 0.007) |
| PtLeafCnc | (0.33, 0.06); (0.41, 0.06); (0.65, 0.06); (0.8, 0.06); (1.09, 0.06); (1.4, 0.035); (2, 0.03) |
| CrLeafCnc | (0.33, 0.044); (0.41, 0.044); (0.65, 0.044); (0.8, 0.044); (1.09, 0.044); (1.4, 0.026); (2, 0.023) |
| NfLeafCnc | (0.33, 0.004); (0.41, 0.004); (0.65, 0.004); (0.8, 0.004); (1.09, 0.004); (1.4, 0.004); (2, 0.004) |
| PtStemCnc | (0.65, 0.05); (0.8, 0.035); (1.09, 0.02); (1.6, 0.011); (2, 0.008) |
| CrStemCnc | (0.65, 0.03); (0.8, 0.02); (1.09, 0.01); (1.6, 0.008); (2, 0.006) |
| NfStemCnc | (0.65, 0.004); (0.8, 0.004); (1.09, 0.004); (1.6, 0.004); (2, 0.004) |
| PtSOrgCnc | (1.09, 0.05); (1.4, 0.025); (2, 0.025) |
| CrSOrgCnc | (1.09, 0.037); (1.4, 0.02); (2, 0.02) |
| NfSOrgCnc | (1.09, 0.004); (1.4, 0.004); (2, 0.004) |
| TLLeafEff | (1.09, 1); (1.4, 1); (2, 1) |

### Harvest

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.58 |
| EconomicYield_N | 0.82 |

### 38.5.64 Crop Winter Wheat; V1

Parent old, defined in lib/wwheat-old.dai. Winter Wheat; V1

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DSRate1 | 0.00114 |
| DSRate2 | 0.00114 |
| DS_Emr | 0.01 |
| DS_reset | true |
| TempEff1 | (0, 1); (100, 100) |
| TempEff2 | (0, 1); (100, 100) |
| PhotEff1 | (0, 1); (1, 1) |

**Vernal**

| Parameter | Value |
|---|---|
| required | false |
| DSLim1 | 0 |
| DSLim2 | 0 |
| TaLim | 0 |
| TaSum | 0 |

**LeafPhot**

| Parameter | Value |
|---|---|
| TLim1 | 4 |
| TLim2 | 10 |

**Canopy**

| Parameter | Value |
|---|---|
| InitGrowth | 4.66 |
| DSinit | 0.07 |
| WLfInit | 30 |
| DS1 | 0.514 |
| alpha | 1.8 |
| beta | 1.1428 |
| SpLAI | 0.014 |
| HvsDS | (0, 1); (1, 100); (2, 100) |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| EPext | 0.5 |

**Root**

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| SpRtLength | 0.1 |
| DensRtTip | 0.1 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 0 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-08 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.6); (0.8, 0.1); (2, 0.1) |

### Resp

| Parameter | Value |
|---|---|
| E_Root | (0, 0.54); (2, 0.54) |
| E_Leaf | (0, 0.72); (2, 0.72) |
| r_Root | (0.914, 0.065); (0.9141, 0.015) |
| r_Leaf | (0.914, 0.015); (0.9141, 0.008) |

### CrpN

| Parameter | Value |
|---|---|
| SeedN | 0.4 |
| PtLeafCnc | (0.114, 0.05); (0.229, 0.043); (0.343, 0.038); (0.457, 0.033); (0.571, 0.028); (0.686, 0.024); (0.8, 0.021); (0.914, 0.018); (1.029, 0.016); (1.143, 0.014); (1.257, 0.012) |
| PtRootCnc | (0.114, 0.018); (0.229, 0.017); (0.343, 0.016); (0.457, 0.015); (0.571, 0.014); (0.686, 0.013); (0.8, 0.013); (0.914, 0.012); (1.029, 0.011); (1.143, 0.011); (1.257, 0.01) |
| CrLeafCnc | (0.114, 0.03); (0.229, 0.027); (0.343, 0.024); (0.457, 0.022); (0.571, 0.019); (0.686, 0.017); (0.8, 0.016); (0.914, 0.014); (1.029, 0.012); (1.143, 0.011); (1.257, 0.01) |
| CrRootCnc | (0.114, 0.013); (0.229, 0.012); (0.343, 0.012); (0.457, 0.011); (0.571, 0.011); (0.686, 0.01); (0.8, 0.01); (0.914, 0.009); (1.029, 0.009); (1.143, 0.008); (1.257, 0.008) |
| NfLeafCnc | (0.114, 0.02); (0.229, 0.018); (0.343, 0.016); (0.457, 0.015); (0.571, 0.013); (0.686, 0.012); (0.8, 0.011); (0.914, 0.01); (1.029, 0.009); (1.143, 0.008); (1.257, 0.007) |
| NfRootCnc | (0.114, 0.01); (0.229, 0.01); (0.343, 0.009); (0.457, 0.009); (0.571, 0.009); (0.686, 0.008); (0.8, 0.008); (0.914, 0.008); (1.029, 0.008); (1.143, 0.007); (1.257, 0.007) |

### Harvest

| Parameter | Value |
|---|---|
| index | 0.45 |
| CStem | 0.0065 |
| CSOrg | 0.00215 |

| Parameter | Value |
|---|---|
| alpha | 0.8 |
| beta | 0.09 |
| Stem | (); () |
| Leaf | (); () |
| SOrg | (); () |
| Root | (); () |
| DSmax | 0 |
| DSnew | 0 |
| C_Stem | 0.42 |
| C_SOrg | 0.42 |
| C_Root | 0.42 |

EpFac: 1.
IntcpCap: 0.5.

### 38.5.65 Crop Winter Wheat

Parent default, defined in lib/wwheat.dai. Winter Wheat

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 100 |
| DS_Emr | 0.01 |
| DSRate1 | 0.026 |
| TempEff1 | (-10, 0.01); (0, 0.01); (20, 0.9); (25, 1); (35, 1.2) |
| PhotEff1 | (10, 0.29); (11, 0.55); (12, 0.75); (13, 0.89); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.025 |
| TempEff2 | (0, 0); (10, 0.14); (15, 0.66); (20, 1); (25, 1.23) |

**Vernal**

| Parameter | Value |
|---|---|
| DSLim | 0.25 |
| TaLim | 5 |
| TaSum | -50 |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 5 |
| Qeff | 0.05 |
| TempEff | (-20, 0); (4, 0); (10, 1); (25, 1); (35, 0.01); (50, 0) |

**Seed**

| Parameter | Value |
|---|---|
| DSLAI05 | 0.15 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.022 |
| LAIDist0 | 0, 0, 1 |
| LAIDist1 | 0, 0.2, 0.9 |
| PARref | 0.06 |
| PARext | 0.6 |
| PARrel | 0.05 |
| EPext | 0.5 |
| HvsDS | (0, 1); (1, 80); (2, 100) |

### Root

| Parameter | Value |
|---|---|
| DptEmr | 10 |
| PenPar1 | 0.25 |
| PenPar2 | 4 |
| MaxPen | 120 |
| Rad | 0.005 |
| h_wp | -15000 |
| Rxylem | 10 |
| MxNH4Up | 2.5e-07 |
| MxNO3Up | 2.5e-07 |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.5); (0.53, 0.5); (1, 0.05); (2, 0) |
| Leaf | (0, 0.7); (0.4, 0.7); (0.55, 0.7); (0.62, 0.5); (0.77, 0.2); (0.95, 0.1); (1.38, 0.05); (2, 0) |
| Stem | (0, 0.3); (0.4, 0.3); (0.55, 0.3); (0.62, 0.5); (0.77, 0.8); (0.95, 0.9); (1.2, 0.5); (1.38, 0.1); (2, 0) |
| RSR | (0, 0.5); (1, 0.5); (2, 0.25) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| E_Root | 0.69 |
| E_Leaf | 0.68 |
| E_Stem | 0.66 |
| E_SOrg | 0.7 |
| r_Root | 0.015 |
| r_Leaf | 0.016 |
| r_Stem | 0.01 |
| r_SOrg | 0.01 |
| ShldResC | 0.4 |
| ReMobilDS | 1.5 |
| ReMobilRt | 0.03 |
| Large_RtDR | 0.05 |
| ExfoliationFac | 0.7 |
| LfDR | (0, 0); (0.1, 0); (0.25, 0); (0.5, 0.01); (0.9, 0.01); (1.2, 0.02); (1.8, 0.05); (2, 0.05) |
| RtDR | (0, 0); (0.6, 0); (1.2, 0.01); (2, 0.01) |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.012); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.009) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0.34, 0.06); (0.5, 0.036); (1.28, 0.033); (1.52, 0.0288); (1.93, 0.0194) |
| CrLeafCnc | (0.34, 0.021); (0.5, 0.021); (1.28, 0.021); (1.52, 0.019); (1.93, 0.0183) |
| NfLeafCnc | (0.34, 0.008); (0.5, 0.008); (1.28, 0.008); (1.52, 0.008); (1.93, 0.008) |
| PtStemCnc | (0.5, 0.02); (1.12, 0.0125); (1.28, 0.0093); (1.53, 0.0085); (1.93, 0.0057) |
| CrStemCnc | (0.5, 0.0064); (1.12, 0.0049); (1.28, 0.0043); (1.53, 0.0041); (1.93, 0.0035) |
| NfStemCnc | (0.5, 0.003); (1.12, 0.003); (1.28, 0.003); (1.53, 0.003); (1.93, 0.003) |
| PtSOrgCnc | (1.12, 0.025); (2, 0.03) |
| CrSOrgCnc | (1.12, 0.015); (2, 0.015) |
| NfSOrgCnc | (1.12, 0.014); (2, 0.014) |

**Harvest**

| Parameter | Value |
|---|---|
| EconomicYield_W | 0.8 |
| EconomicYield_N | 0.94 |

## 38.5.66 Crop Winter Wheat; Foulum

Parent Winter Wheat, defined in lib/wwheat.dai. Winter Wheat; Foulum

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 200 |
| DSRate1 | 0.022 |

**Canopy**

| Parameter | Value |
|---|---|
| SpLAI | 0.021 |
| LeafAIMod | (0.63, 1); (1.13, 0.85); (1.25, 0.76) |
| PARrel | 0.07 |
| HvsDS | (0, 1); (1, 80); (2, 80) |

**Partit**

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.33, 0.5); (0.53, 0.25); (1, 0.05); (2, 0) |
| Leaf | (0, 0.9); (0.33, 0.85); (0.43, 0.83); (0.53, 0.75); (0.62, 0.56); (0.77, 0.2); (0.95, 0.09); (1.14, 0.05); (1.38, 0); (2, 0) |
| Stem | (0, 0.1); (0.33, 0.15); (0.43, 0.17); (0.53, 0.25); (0.62, 0.44); (0.77, 0.8); (0.95, 0.64); (1.14, 0.32); (1.38, 0); (2, 0) |

**Prod**

| Parameter | Value |
|---|---|
| NCrop | 0.4 |
| ReMobilDS | 1.1 |
| ReMobilRt | 0.15 |
| ExfoliationFac | 0.9 |
| LfDR | (0.9, 0); (1.2, 0.05); (1.8, 0.05); (2, 0.1) |

**CrpN**

| Parameter | Value |
|---|---|
| PtRootCnc | (0, 0.012); (1, 0.012); (2, 0.01) |
| CrRootCnc | (0, 0.01); (1, 0.01); (2, 0.009) |
| NfRootCnc | (0, 0.007); (1, 0.007); (2, 0.007) |
| PtLeafCnc | (0.34, 0.06); (0.5, 0.036); (1.28, 0.032); (1.52, 0.0288); (1.93, 0.0194) |
| CrLeafCnc | (0.34, 0.021); (0.5, 0.021); (1.28, 0.021); (1.52, 0.019); (1.93, 0.0183) |
| NfLeafCnc | (0.34, 0.008); (0.5, 0.008); (1.28, 0.008); (1.52, 0.008); (1.93, 0.008) |
| PtStemCnc | (0.5, 0.02); (1.12, 0.0125); (1.28, 0.0093); (1.53, 0.0085); (1.93, 0.0085) |
| CrStemCnc | (0.5, 0.0064); (1.12, 0.006); (1.28, 0.006); (1.53, 0.006); (1.93, 0.006) |
| NfStemCnc | (0.5, 0.004); (1.12, 0.004); (1.28, 0.004); (1.53, 0.004); (1.93, 0.004) |
| PtSOrgCnc | (1.12, 0.015); (2, 0.023) |
| CrSOrgCnc | (1.12, 0.014); (2, 0.017) |
| NfSOrgCnc | (1.12, 0.013); (2, 0.014) |

## 38.5.67 Crop Winter Wheat JG

Parent Winter Wheat, defined in lib/wwheat.dai. Winter Wheat JG

cite: gyldengren2020effects.

**Devel**

| Parameter | Value |
|---|---|
| DSRate1 | 0.032 |
| TempEff1 | (0, 0); (2, 0.06); (7, 0.21); (12, 0.4); (16, 0.59); (20, 0.8); (24, 0.95); (28, 1); (32, 0.91); (34, 0.79); (35, 0.71); (40, 0.05) |
| PhotEff1 | (0, 0); (6, 0.05); (10, 0.1); (11, 0.2); (12, 0.35); (13, 0.8); (14, 1); (15, 1.08); (16, 1.14); (17, 1.18); (24, 1.18) |
| DSRate2 | 0.048 |
| TempEff2 | (0, 0); (4, 0.03); (8, 0.1); (12, 0.21); (15, 0.31); (18, 0.46); (22, 0.79); (24, 0.89); (27, 0.97); (32, 1); (34, 0.98); (35, 0.95); (40, 0.65) |

**LeafPhot**

| Parameter | Value |
|---|---|
| Fm | 6 |
| TempEff | (0, 0); (4, 0.25); (8, 0.55); (12, 0.95); (16, 1.3); (20, 1.6); (24, 1.9); (28, 2); (32, 1.82); (34, 1.58); (35, 1.42); (40, 0.1) |
| DSEff | (0, 1); (0.3, 1); (0.4, 1); (0.8, 1); (1, 1); (1.3, 0.85); (1.7, 0.75); (2, 0.7) |

### Seed

| Parameter | Value |
|---|---|
| DM_fraction | 87 |
| C_fraction | 45 |
| N_fraction | 2 |
| T_factor | (20, 100) |
| rate | 0.4 |

### Root

| Parameter | Value |
|---|---|
| PenPar1 | 1 |
| PenPar2 | 0 |
| MaxPen | 175 |
| NO3_uptake | F_relative 10 |
| PenWaterFac | (0, 1); (0.989, 1); (0.99, 0); (1, 0) |

### Prod

| Parameter | Value |
|---|---|
| NCrop | -1 |
| LfDR | (0, 0); (0.1, 0); (0.25, 0); (0.5, 0.015); (0.9, 0.025); (1.3, 0.05); (1.7, 0.06); (2, 0.08) |
| RtDR | (0, 0); (0.6, 0.005); (1, 0.02); (2, 0.03) |

### Canopy

| Parameter | Value |
|---|---|
| LeafAIMod | (0, 1.3); (0.5, 1); (0.9, 0.8); (2, 0.8) |
| HvsDS | (0, 1); (1, 80); (2, 90) |

### Partit

| Parameter | Value |
|---|---|
| Root | (0, 0.5); (0.28, 0.45); (0.5, 0.4); (1, 0); (2, 0) |
| Leaf | (0.1, 0.72); (0.28, 0.58); (0.5, 0.53); (0.65, 0.4); (0.85, 0.25); (0.95, 0.25); (1.1, 0); (1.35, 0); (2, 0) |
| Stem | (0.1, 0.28); (0.28, 0.42); (0.5, 0.47); (0.65, 0.6); (0.85, 0.75); (0.95, 0.62); (1.1, 0.38); (1.35, 0.18); (1.5, 0); (2, 0) |
| NNI_crit | 1.4 |
| NNI_inc | 0.65 |

### CrpN

| Parameter | Value |
|---|---|
| PtRootCnc | (0.19, 0.025); (0.47, 0.026); (0.77, 0.019); (1.2, 0.015); (2, 0.014) |
| CrRootCnc | (0.19, 0.017); (0.47, 0.019); (0.77, 0.016); (1.2, 0.012); (2, 0.009) |
| NfRootCnc | (0.19, 0.009); (1.2, 0.003); (2, 0.003) |
| PtLeafCnc | (0.19, 0.051); (0.47, 0.049); (0.77, 0.045); (1, 0.038); (2, 0.021) |

| Parameter | Value |
|---|---|
| CrLeafCnc | (0.19, 0.039); (0.47, 0.037); (0.77, 0.03); (1, 0.025); (2, 0.014) |
| NfLeafCnc | (0.19, 0.01); (0.47, 0.01); (0.77, 0.007); (1, 0.006); (2, 0.005) |
| PtStemCnc | (0.19, 0.038); (0.47, 0.036); (0.77, 0.021); (1, 0.015); (2, 0.011) |
| CrStemCnc | (0.19, 0.023); (0.47, 0.014); (0.77, 0.009); (1, 0.007); (2, 0.006) |
| NfStemCnc | (0.19, 0.01); (0.47, 0.007); (0.77, 0.004); (1, 0.002); (2, 0.002) |
| PtSOrgCnc | (1, 0.017); (2, 0.0175) |
| CrSOrgCnc | (1, 0.0125); (2, 0.0125) |
| NfSOrgCnc | (1, 0.0005); (2, 0.0005) |

### 38.5.68 Crop no stress Barley

Parent Spring Barley, defined in sample/DrainDrying.dai. no stress Barley
enable_N_stress: false.
water_stress_effect: none.

### 38.5.69 Crop Clovercz

Parent Pea, defined in sample/cz-management.dai. Clovercz

### 38.5.70 Crop Oatcz

Parent Rye, defined in sample/cz-management.dai. Oatcz

### 38.5.71 Crop Triticale

Parent Rye, defined in sample/cz-management.dai. Triticale

### 38.5.72 Crop Rye-grasscz

Parent Grass to grain, defined in sample/cz-management.dai. Rye-grasscz

### 38.5.73 Crop Mustardcz

Parent Winter Rape, defined in sample/cz-management.dai. Mustardcz

### 38.5.74 Crop Potato; CER

Parent Potato; Agria, defined in sample/it-management.dai. Potato; CER

### 38.5.75 Crop Potato; Folva; SAFIR

Parent Potato; Folva, defined in sample/safir-common.dai. Potato; Folva; SAFIR
enable_N_stress: false.
water_stress_effect: none.
**Root**

| Parameter | Value |
|---|---|
| ABAprod | Fulai Liu uptake |
| MaxWidth | 1.5 |

LeafPhot: Potato Farquhar.

**Prod**

| Parameter | Value |
|---|---|
| NCrop | -1 |

**Seed**

| Parameter | Value |
|---|---|
| rate | 0.3 |
| DM_fraction | 0.2 |
| C_fraction | 0.4676 |
| N_fraction | 2 |

**Canopy**

| Parameter | Value |
|---|---|
| LeafAIMod | (0, 2.2); (0.4, 1.1); (1.5, 1.1); (2, 0.3) |

### 38.5.76 Crop SiteCrop

Parent Potato; Folva; SAFIR, defined in sample/safir-site.dai. SiteCrop
Devel: original.
Canopy: True.
Root: True.
LeafPhot: original.
Partit: True.
CrpN: True.

### 38.5.77 Crop Lettuce

Parent Potato, defined in sample/safir-site.dai.

### 38.5.78 Crop Maize_SK

Parent Maize, defined in sample/sk-management.dai. Maize_SK

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 200 |
| DSRate1 | 0.019 |
| DSRate2 | 0.017 |

**CrpN**

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.038); (0.5, 0.038); (1, 0.03); (1.5, 0.018); (1.8, 0.015); (2, 0.013) |
| CrLeafCnc | (0, 0.01); (0.367, 0.01); (0.5, 0.009); (1, 0.008); (2, 0.008) |

## 38.5.79 Crop Spring Barley_SK

Parent Spring Barley, defined in sample/sk-management.dai. Spring Barley_SK

**Devel**

| Parameter | Value |
|---|---|
| DSRate1 | 0.035 |
| DSRate2 | 0.025 |

**CrpN**

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.04); (1, 0.04); (2, 0.03) |

## 38.5.80 Crop Winter Wheat_SK

Parent Winter Wheat, defined in sample/sk-management.dai. Winter Wheat_SK

**Devel**

| Parameter | Value |
|---|---|
| EmrTSum | 150 |
| DSRate1 | 0.017 |
| DSRate2 | 0.025 |

**CrpN**

| Parameter | Value |
|---|---|
| PtLeafCnc | (0, 0.05); (0.5, 0.035); (1.28, 0.033); (1.52, 0.025); (2, 0.012) |
| CrLeafCnc | (0, 0.021); (0.5, 0.021); (1.28, 0.021); (1.52, 0.015); (2, 0.01) |

## 38.5.81 Crop NewSB

Parent Spring Barley, defined in sample/test-organic.dai. Spring Barley

**Harvest**

| Parameter | Value |
|---|---|
| Dead | NewSB-SLOW |
| Stem | NewSB-SLOW |
| Leaf | NewSB-SLOW |
| SOrg | NewSB-SLOW |
| Root | NewSB-SLOW |

### 38.5.82 Crop Potato; Folva; ABA

Parent Potato; Folva, defined in sample/test-prd.dai. Potato; Folva; ABA
enable_N_stress: false.
water_stress_effect: none.
**Root**

| Parameter | Value |
|---|---|
| ABAprod | Fulai Liu uptake |

LeafPhot: Potato.
**Prod**

| Parameter | Value |
|---|---|
| NCrop | -1 |

**Seed**

| Parameter | Value |
|---|---|
| rate | 0.3 |
| DM_fraction | 0.2 |
| C_fraction | 0.4676 |
| N_fraction | 2 |

**Canopy**

| Parameter | Value |
|---|---|
| LeafAIMod | (0, 2.2); (0.4, 1.1); (1.5, 1.1); (2, 0.3) |

### 38.5.83 Crop Potato; Folva; PRD

Parent Potato; Folva; ABA, defined in sample/test-prd.dai. Potato; Folva; PRD
**Root**

| Parameter | Value |
|---|---|
| rootdens | row_position 50, row_distance 1 |
| MaxWidth | 1.5 |

### 38.5.84 Crop Potato; Exp

Parent Potato; Folva; PRD, defined in sample/test-prd.dai. Potato; Exp

## 38.6 Actions

An action is a management operation of a set-up: sowing, tillage, fertilising, irrigation and so on. The text of each action is shown as it is defined in the library.

### 38.6.1 Action OpenMI_fertigation

Parent extern_fertigation, from lib/OpenMI.dai.

```
(defaction
  OpenMI_fertigation
  extern_fertigation
  "Control fertigation through OpenMI."
  (surface "Surface Irrigation")
  (overhead "Overhead Irrigation")
  (subsoil "Subsoil Irrigation")
  (NO3 "NO3 input")
  (NH4 "NH4 input"))
```

### 38.6.2 Action MARKVAND

Parent markvand, from lib/markvand.dai.

```
(defaction
  MARKVAND
  markvand ;;"Markvand map for standard crops."
  (map
    ("Beet; V1" "roer")
    ("Silomajs" "majs")
    ("Aert" "aerter")
    ("Vaarbyg" "vaarbyg")
    ("Fodder Beet" "roer")
    ("Maize" "majs")
    ("Pioneer Maize" "majs")
    ("Silage Maize" "majs")
    ("Pea" "aerter")
    ("Green Pea" "aerter")
    ("Pea; V1" "aerter")
    ("Potato; Koege" "kartofler; sildige")
    ("Potato; SCRI" "kartofler; sildige")
    ("Potato" "kartofler; sildige")
    ("Potato; V1" "kartofler; sildige")
    ("Spring Barley" "vaarbyg")
    ("Spring Barley; Foulum" "vaarbyg")
    ("Spring Barley; V1" "vaarbyg")
    ("Spring Rape" "vaarraps")
    ("Spring Rape; V1" "vaarraps")
    ("Sugar Beet" "roer")
    ("Winter Barley" "vinterbyg")
    ("Winter Wheat" "vinterhvede")
    ("Winter Rape" "vinterraps")
    ("Winter Rye" "vinterrug")))
```

### 38.6.3 Action plowing

Parent swap, from lib/tillage.dai.

```
(defaction plowing swap (middle -9.0) (depth -18.0))
```

### 38.6.4 Action rotavation

Parent mix, from lib/tillage.dai.

```
(defaction rotavation mix -15.0)
```

### 38.6.5 Action disk_harrowing

Parent mix, from lib/tillage.dai.

```
(defaction disk_harrowing mix -10.0 (penetration 0.8))
```

### 38.6.6 Action stubble_cultivation

Parent mix, from lib/tillage.dai.

```
(defaction stubble_cultivation mix -8.0 (penetration 0.6))
```

### 38.6.7 Action seed_bed_preparation

Parent mix, from lib/tillage.dai.

```
(defaction seed_bed_preparation mix -8.0)
```

### 38.6.8 Action Barley no stress

Parent activity, from sample/DrainDrying.dai.

```
(defaction
  "Barley no stress"
  activity
  (wait_mm_dd 4 02)
  (plowing)
  (wait_mm_dd 4 05)
  (seed_bed_preparation)
  (sow "no stress Barley")
  (wait_mm_dd 08 25)
  (harvest "no stress Barley" (stub 8 [cm]) (stem 0.70)))
```

### 38.6.9 Action MyMan

Parent activity, from sample/OpenMI_management.dai.

```
(defaction
  MyMan
  activity
  (wait (at 1987 3 20 1))
  (plowing)
  (wait (at 1987 4 5 1))
  (progn (sow "Grass") (sow "Spring Barley"))
  (wait (at 1987 9 5 1))
  (harvest "Spring Barley")
  (wait (at 1987 10 10 1))
  (harvest
    "Grass"
    (stub 8.0 [cm]) ;Leave 8 cm stub.
    (stem 1.00 [])) ;Harvest everything above stub.)
```

### 38.6.10 Action WW with PFAS cont slurry

Parent activity, from sample/PFAS-sample.dai.

```
(defaction
  "WW with PFAS cont slurry"
  activity
  (wait_mm_dd 09 10)
  (seed_bed_preparation)
  (sow "Winter Wheat")
  (wait_mm_dd 03 20)
 (fertilize ("pig_slurry" (volatilization 10 [%]) (weight 40 [T w.w./ha])) (to -1 [cm]))
 (irrigate_subsoil 0.1 [mm/h] (hours 1) (to -1 [cm]) (solute (PFBA 0.0001 [ppm]))) ;; PFAS added toge
  (wait (or (crop_ds_after "Winter Wheat" 2.0) (mm_dd 9 1)))
  (harvest "Winter Wheat" (stub 8 [cm]) (stem 0.0) (leaf 0.0))
  (wait_mm_dd 9 02)
  (plowing))
```

### 38.6.11 Action WW with PFAS cont irrigation

Parent activity, from sample/PFAS-sample.dai.

```
(defaction
  "WW with PFAS cont irrigation"
  activity
  (wait
    (and
      (after_mm_dd 05 01)
      (before_mm_dd 09 01)
      (not (soil_water_pressure_above (height -25 [cm]) (potential -300 [cm])))))
   (irrigate_overhead 10 [mm/h] (hours 3) (solute (PFOS 5.1e-6 [ppm]) (PFOA 7e-
6 [ppm]))) ;; Wastewater effluent contaminated with PFAS. Estimate from 'Simulating PFAS transport in
irrigated farmland usingPRZM5, LEACHM, and HYDRUS-1D models' https://doi.org/10.1002/jeq2.20639
  (wait_days 3))
```

### 38.6.12 Action PFAS cont WW

Parent activity, from sample/PFAS-sample.dai.

```
(defaction
  "PFAS cont WW"
  activity
  (while "WW with PFAS cont slurry" (repeat "WW with PFAS cont irrigation")))
```

### 38.6.13 Action harrowing

Parent mix, from sample/cz-management.dai.

```
(defaction harrowing mix -7.0)
```

### 38.6.14 Action roller

Parent mix, from sample/cz-management.dai.

```
(defaction roller mix -2.0)
```

### 38.6.15 Action sweep

Parent mix, from sample/cz-management.dai.

```
(defaction sweep mix -10.0)
```

### 38.6.16 Action ridging

Parent mix, from sample/cz-management.dai.

```
(defaction ridging mix -5.0)
```

### 38.6.17 Action haulm_destruction

Parent mix, from sample/cz-management.dai.

```
(defaction haulm_destruction mix -2.0)
```

### 38.6.18 Action surface_levelling

Parent mix, from sample/cz-management.dai.

```
(defaction surface_levelling mix -10.0)
```

### 38.6.19 Action manual_burrying_drip_lines

Parent mix, from sample/cz-management.dai.

```
(defaction manual_burrying_drip_lines mix -5.0)
```

### 38.6.20 Action manual_removal_of_drip_lines

Parent mix, from sample/cz-management.dai.

```
(defaction manual_removal_of_drip_lines mix -5.0)
```

### 38.6.21 Action WP1 CZ RISWC 2004 WARMUP

Parent activity, from sample/cz-management.dai.

```
(defaction
  "WP1 CZ RISWC 2004 WARMUP"
 activity ;;1991 Spring Barley Jarek with Clover Chlumecky (as pea) ; FD 16/11/04: Spring barley is no
  (wait (mm_dd 4 3)) ; fertilize  200 kg SA 21%/ha (=amonia)
  (fertilize (amonium_sulphatecz (weight 48.0 [kg N/ha])))
  (wait (mm_dd 4 4))
  (seed_bed_preparation)
  (wait (mm_dd 4 6)) ;(sow "Spring Barley") operation made inactive by FD 16/11/04
  (sow "Clovercz") ;(wait (mm_dd 4 7)) ;(harrowing) operation made inactive FD 16/11/04
 (wait (mm_dd 6 6)) ;(spray MCPA 640 [g/ha])  ;; = in Herbicide Aminex Pur (2,0 l/ha)(320g MCPA/l) ;fe
mineral)
 (fertilize (ureacz (weight 17.0 [kg N/ha]))) ;(wait (mm_dd 6 29)) ;(spray Propiconazole 125 [g/ha])
 (wait (mm_dd 8 18)) ;(harvest "Spring Barley" (stem 0)(leaf 0)(sorg 0.95)) operation made inactive b
 (harvest "Clovercz" (stub 10.0 [cm]) (stem 1.0)) ;;1992 Clover Chlumecky (as pea) + rye-
grass 20 kg/ha (as grass)+ oats 20 kg/ha (as rye) ;FD 16/11/04: rye-grass is not modelled, oats only re
  (wait (mm_dd 4 7)) ;(sow "Rye-grasscz") operation made inactive by FD 16/11/04
  (sow "Oatcz") ;(fertilize (mocuvka=stale? (weight xx.x [kg N/ha]))) !!!!!
 (fertilize (cattle_dungwatercz (weight 20.0 [Mg w.w./ha]))) ;(plowing) operation made inactive FD 1
  (wait (mm_dd 4 8)) ; fertilize  100 kg LAV 27%/ha (=amonium nitrate)
  (fertilize (LAV27cz (weight 27.0 [kg N/ha])))
 (wait (mm_dd 6 10)) ;(harvest "Clovercz" (stub 10.0 [cm])(stem 1.0)) operation made inactive by FD 1
  (wait (mm_dd 6 11)) ; fertilize  100 kg LAV 27%/ha (=amonium nitrate)
  (fertilize (LAV27cz (weight 27.0 [kg N/ha])))
```

```
  (wait (mm_dd 8 20)) ;(harvest "Clovercz" (stub 10.0 [cm])(stem 1.0)) operation made inactive by FD 1
  (harvest "Oatcz" (stub 10.0 [cm]) (stem 1.0) (sorg 0.95)) ;operation added by FD 16/11/04
   (wait (mm_dd 9 1))
   (plowing)
   (wait (mm_dd 9 14)) ; fertilize  300 kg SA 21%/ha (=ammonia)
   (fertilize (amonium_sulphatecz (weight 63.0 [kg N/ha])))
  (sow "Winter Barley") ;(wait (mm_dd 9 16)) ;(spray Chlortoluron 2000 [g/ha]) ;; = in Herbicide Dicur
   (wait (mm_dd 5 17)) ; fertilize  xx kg LA 33%/ha - the dose estimated by FD
   (fertilize (LA33cz (weight 50.0 [kg N/ha])))
   (wait (mm_dd 7 25))
   (harvest "Winter Barley" (stem 0.95) (leaf 0.95) (sorg 0.95))
   (wait (mm_dd 7 28))
   (fertilize (cattle_manurecz (weight 35.0 [T w.w./ha])))
   (wait (mm_dd 7 29))
   (plowing)
   (wait (mm_dd 7 30))
   (disk_harrowing)
   (seed_bed_preparation)
   (wait (mm_dd 8 1)) ; ferttilize  300 kg LA 33%/ha
   (fertilize (LA33cz (weight 99.0 [kg N/ha])))
   (wait (mm_dd 8 2)) ; sow mustard (=winter rape)
  (sow "Mustardcz") ;(roller) operation made inactive by FD 16/10/04 ;(wait (mm_dd 10 22)) ;(harvest "
   (wait (mm_dd 11 5))
   (plowing) ;;1994 Potato; RISWC Karin 2500 kg/ha
   (wait (mm_dd 4 19))
   (sweep)
  (wait (mm_dd 4 22)) ;fertilize 730kg Simferta (N:P:K=11:12:12)=80,3kgN+88kgP2O5+88kgK2O
   (fertilize (NPK11-12-12cz (weight 80.3 [kg N/ha])))
   (wait (mm_dd 4 26))
   (seed_bed_preparation)
   (wait (mm_dd 4 27))
  (sow "Potato; Agria") ;(wait (mm_dd 5 13)) ;(harrowing) ;vlaceni sitovymi branami operation made in
   (wait (mm_dd 10 4))
   (harvest "Potato; Agria" (stem 0.00) (leaf 0.00) (sorg 0.95))
   (wait (mm_dd 11 12))
  (plowing) ;;1995 Spring Barley Akcent 180 kg/ha with Clover Start 20kg/ha (=spring rape) ;FD 16/11/0
  (wait (mm_dd 4 8)) ;fertilize 500kg Simferta NPK (N:P:K=11:12,4:12,4)=55kgN+62kgP2O5+62kgK2O
   (fertilize (NPK11-12-12cz (weight 55.0 [kg N/ha])))
   (seed_bed_preparation)
   (wait (mm_dd 4 9)) ;(sow "Spring Barley") operation made inactive by FD 16/11/04
  (sow "Clovercz") ;(wait (mm_dd 4 10)) ;(harrowing) operation made inactive by FD 16/10/04
  (wait (mm_dd 6 4)) ;(spray MCPA 640 [g/ha]) ;; = in Herbicide Aminex Pur (2,0 l/ha)(320g MCPA/l) ;fe
   (fertilize (DAM390cz (weight 18.0 [kg N/ha])))
  (wait (mm_dd 8 11)) ;(harvest "Spring Barley" (stem 0)(leaf 0)(sorg 0.95)) made inactive FD 16/11/04
   (wait (mm_dd 9 30))
  (harvest "Clovercz" (stub 10.0 [cm]) (stem 1.0)) ;;1996 Clover Start (=pea)+ pea 20 kg/ha + rye-
grass - jílek (=grass) 30 kg/ha ; rygrass and clover not modelled, pea only remains)
   (wait (mm_dd 4 10))
  (sow "Pea") ;(sow "Rye-grasscz") operation made inactive by FD 16/11/04 ;(wait (mm_dd 4 11)) ;(rollc
grasscz" (stub 10.0 [cm])(stem 1.0)) made inactive by FD 16/11/04 ;(wait (mm_dd 7 23)) ; (harvest "Clo
grasscz" (stub 10.0 [cm])(stem 1.0)) made inactive by FD 16/11/04
  (wait (mm_dd 9 24)) ; (harvest "Clovercz" (stub 10.0 [cm])(stem 1.0)) made inactive by FD 16/11/04
  (harvest "Pea" (stub 10.0 [cm]) (stem 1.0) (sorg 0.80)) ; sorg added by FD 16/11/04 ; (harvest "Rye-
grasscz" (stub 10.0 [cm])(stem 1.0)) made inactive by FD 16/11/04
   (wait (mm_dd 9 25))
```

```
   (plowing)
  (wait (mm_dd 9 28)) ;fertilize 600kg Simferta NPK (N:P:K=11:12,4:12,4)=66kgN+74,4kgP2O5+74,4kgK2O
   (fertilize (NPK11-12-12cz (weight 66.0 [kg N/ha])))
   (seed_bed_preparation)
   (wait (mm_dd 9 29))
  (sow "Winter Wheat") ; WW Hana 220 kg/ha ;(wait (mm_dd 9 30)) ;(roller) operation made inactive by FD
   (wait (mm_dd 3 15)) ;fertilize 300kg LAV 27% = 81 kg N/ha
  (fertilize (LAV27cz (weight 81.0 [kg N/ha]))) ;(wait (mm_dd 4 23)) ;(spray Prochloraz 675 [g/ha]) ;
chlorin 720 [g/ha]) ;; = in RR Retacel extra R68 (1,0 l/ha)(720g Chl./l)  ;(wait (mm_dd 4 28)) ;(spr
  (wait (mm_dd 5 31)) ;fertilize 300kg LAV 27% = 81 kg N/ha
  (fertilize (LAV27cz (weight 81.0 [kg N/ha]))) ;(wait (mm_dd 6 11)) ;(spray Epoxiconazole 50.4 [g/ha
   (wait (mm_dd 8 19))
   (harvest "Winter Wheat" (stem 0) (leaf 0) (sorg 0.95))
   (wait (mm_dd 8 28))
   (stubble_cultivation) ;podmitka
   (wait (mm_dd 9 2))
   (sweep)
   (harrowing)
   (disk_harrowing) ;smykování a vláèení
   (wait (mm_dd 9 12))
   (fertilize (cattle_manurecz (weight 35.0 [T w.w./ha])))
   (wait (mm_dd 9 13))
   (plowing) ;;1998 Potato; RISWC Rosela 3000 kg/ha
   (wait (mm_dd 4 12))
   (disk_harrowing)
  (wait (mm_dd 4 25)) ;fertilize 800kg Simferta (N:P:K=11:12,4:12,4)=88kgN+99,2kgP2O5+99,2kgK2O
   (fertilize (NPK11-12-12cz (weight 88.0 [kg N/ha])))
   (wait (mm_dd 4 27))
   (seed_bed_preparation)
   (wait (mm_dd 4 28))
  (sow "Potato; Agria") ;(wait (mm_dd 5 14)) ;(spray Terbuthylazine 210 [g/ha]) ;(spray Terbutryn 490
   (wait (mm_dd 9 10))
  (harvest "Potato; Agria" (stem 0.00) (leaf 0.00) (sorg 0.95)) ;;1999 Spring Barley Primus 200 kg/ha
  (wait (mm_dd 4 29)) ;fertilize 250kg NP (Norsk Hydro) (N:P=26:14)=65kgN+35kgP2O5 (fertilize (Ammoni
   (fertilize (NP26-14cz (weight 65.0 [kg N/ha])))
   (seed_bed_preparation) ;(stop)
   (wait (mm_dd 4 30))
     (sow  "Spring  Barley")  ;(wait  (mm_dd  4  31))  corrected  by  FD  16/11/04  -
see the following line: ;(wait (mm_dd 5 1)) ;(roller) operation made inactive by FD 16/11/04 ;(wait (m
  (wait (mm_dd 6 4)) ;fertilize 111,11 kg LAV 27% = 30 kg N/ha
  (fertilize (LAV27cz (weight 30.0 [kg N/ha]))) ;(wait (mm_dd 6 8)) ;(spray Propiconazole 125 [g/ha])
   (wait (mm_dd 6 28)) ;fertilize 111,11 kg LAV 27% = 30 kg N/ha
   (fertilize (LAV27cz (weight 30.0 [kg N/ha])))
   (wait (mm_dd 8 3))
   (harvest "Spring Barley" (stem 0) (leaf 0) (sorg 0.95))
   (plowing)
   (wait (mm_dd 8 23))
   (seed_bed_preparation) ;this line was added by FD 16/11/04
  (sow "Winter Rape") ;Pronto 4,8 kg/ha (hybrid) ;(wait (mm_dd 8 24)) ;(spray Metazachlor 400 [g/ha])
   (wait (mm_dd 4 6)) ;fertilize 180,0 kg LAV 27% = 48,6 kg N/ha
   (fertilize (LAV27cz (weight 48.6 [kg N/ha])))
  (wait (mm_dd 4 14)) ;(spray Chlorpyrifos 300 [g/ha]) ;(spray Cypermethrin 30 [g/ha]) ;; = in Insect
   (wait (mm_dd 4 16)) ;fertilize 210,0 kg LAV 27% = 56,5 kg N/ha
   (fertilize (LAV27cz (weight 56.5 [kg N/ha])))
  (wait (mm_dd 4 26)) ;(spray Deltamethrin 2.5 [g/ha]) ;; = in Insecticide Decis Flow 2,5 EW (0,1 l/ha
```

```
  (fertilize (DAM390cz (weight 36.0 [kg N/ha])))
  (wait (mm_dd 7 23))
  (harvest "Winter Rape" (stem 0) (leaf 0) (sorg 0.95))
  (wait (mm_dd 7 27))
  (stubble_cultivation) ;podmítka
  (wait (mm_dd 9
```

### 38.6.22 Action SBarley w. MF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "SBarley w. MF"
  activity
  (wait_mm_dd 3 05)
  (fertilize (N25S (weight 115 [kg N/ha])))
  (plowing)
  (wait_mm_dd 4 05)
  (seed_bed_preparation)
  (sow "Spring Barley")
  (wait (or (crop_ds_after "Spring Barley" 2.0) (mm_dd 08 20)))
  (harvest "Spring Barley" (stub 8 [cm]) (stem 0.70)))
```

### 38.6.23 Action SBarley w. OF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "SBarley w. OF"
  activity
  (wait_mm_dd 3 05)
  (fertilize ("pig_slurry" (volatilization 5 [%]) (weight 28 [T w.w./ha])) (to -1 [cm]))
  (plowing)
  (wait_mm_dd 4 05)
  (seed_bed_preparation)
  (sow "Spring Barley")
  (wait (or (crop_ds_after "Spring Barley" 2.0) (mm_dd 08 20)))
  (harvest "Spring Barley" (stub 8 [cm]) (stem 0.70)))
```

### 38.6.24 Action WWheat w. MF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "WWheat w. MF"
  activity
  (wait_mm_dd 09 10)
  (seed_bed_preparation)
  (sow "Winter Wheat")
  (wait_mm_dd 04 05)
  (fertilize (N25S (weight 060 [kg N/ha])))
  (wait_mm_dd 05 05)
  (fertilize (N25S (weight 100 [kg N/ha])))
  (wait (or (crop_ds_after "Winter Wheat" 2.0) (mm_dd 9 1)))
  (harvest "Winter Wheat" (stub 8 [cm]) (stem 0.0) (leaf 0.0))
  (wait_mm_dd 9 02)
  (plowing))
```

### 38.6.25 Action WWheat w. OF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "WWheat w. OF"
  activity
  (wait_mm_dd 09 10)
  (seed_bed_preparation)
  (sow "Winter Wheat")
  (wait_mm_dd 03 20)
 (fertilize ("pig_slurry" (volatilization 10 [%]) (weight 40 [T w.w./ha])) (to -1 [cm]))
  (wait (or (crop_ds_after "Winter Wheat" 2.0) (mm_dd 9 1)))
  (harvest "Winter Wheat" (stub 8 [cm]) (stem 0.0) (leaf 0.0))
  (wait_mm_dd 9 02)
  (plowing))
```

### 38.6.26 Action WBarley w. MF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "WBarley w. MF"
  activity
  (wait_mm_dd 9 01)
  (plowing)
  (wait_mm_dd 9 10)
  (seed_bed_preparation)
  (sow "Winter Barley")
  (wait_mm_dd 3 25)
  (fertilize (N25S (weight 60 [kg N/ha])))
  (wait_mm_dd 4 25)
  (fertilize (N25S (weight 81 [kg N/ha])))
  (wait (or (crop_ds_after "Winter Barley" 2.0) (mm_dd 8 9)))
  (harvest "Winter Barley" (stub 8 [cm]) (stem 0.0) (leaf 0.0)))
```

### 38.6.27 Action WBarley w. MF & OF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "WBarley w. MF & OF"
  activity
  (wait_mm_dd 9 01)
  (plowing)
  (wait_mm_dd 9 10)
  (seed_bed_preparation)
  (sow "Winter Barley")
  (wait_mm_dd 3 25)
  (fertilize (N25S (weight 68 [kg N/ha])))
  (wait_mm_dd 4 20)
 (fertilize ("pig_slurry" (volatilization 10 [%]) (weight 20 [T w.w./ha])) (to -1 [cm]))
  (wait (or (crop_ds_after "Winter Barley" 2.0) (mm_dd 8 9)))
  (harvest "Winter Barley" (stub 8 [cm])))
```

### 38.6.28 Action WBarley w. OF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "WBarley w. OF"
  activity
  (wait_mm_dd 9 01)
  (plowing)
  (wait_mm_dd 9 10)
  (seed_bed_preparation)
  (sow "Winter Barley")
  (wait_mm_dd 3 25)
 (fertilize ("pig_slurry" (volatilization 10 [%]) (weight 37 [T w.w./ha])) (to -1 [cm]))
  (wait (or (crop_ds_after "Winter Barley" 2.0) (mm_dd 8 9)))
  (harvest "Winter Barley" (stub 8 [cm])))
```

### 38.6.29 Action WRape w. MF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "WRape w. MF"
  activity
  (wait_mm_dd 8 15)
  (plowing)
  (wait_mm_dd 8 20)
  (fertilize (N25S (weight 25 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Winter Rape")
  (wait_mm_dd 3 05)
  (fertilize (N25S (weight 130 [kg N/ha])))
  (wait (or (crop_ds_after "Winter Rape" 2.0) (mm_dd 8 9)))
  (harvest "Winter Rape" (stub 8 [cm]) (stem 0.0) (leaf 0.0)))
```

### 38.6.30 Action WRape w. OF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "WRape w. OF"
  activity
  (wait_mm_dd 8 15)
 (fertilize ("pig_slurry" (volatilization 5 [%]) (weight 20 [T w.w./ha])) (to -1 [cm]))
  (plowing)
  (wait_mm_dd 8 20)
  (seed_bed_preparation)
  (sow "Winter Rape")
  (wait_mm_dd 3 05)
 (fertilize ("pig_slurry" (volatilization 10 [%]) (weight 26 [T w.w./ha])) (to -1 [cm]))
  (wait (or (crop_ds_after "Winter Rape" 2.0) (mm_dd 8 9)))
  (harvest "Winter Rape" (stub 8 [cm]) (stem 0.0) (leaf 0.0)))
```

### 38.6.31 Action Maize w. MF & OF

Parent activity, from sample/dk-management.dai.

```
(defaction
  "Maize w. MF & OF"
  activity
  (wait_mm_dd 4 05)
```

```
  (fertilize
    ("cattle_slurry" (volatilization 5 [%]) (weight 50 [T w.w./ha]))
    (to -1 [cm]))
  (plowing)
  (wait_mm_dd 4 20)
  (fertilize (Ammonia (weight 100 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Silage Maize")
  (wait_mm_dd 10 1)
  (harvest "Silage Maize" (stub 8 [cm])))
```

### 38.6.32 Action Pea w. no F

Parent activity, from sample/dk-management.dai.

```
(defaction
  "Pea w. no F"
  activity
  (wait_mm_dd 3 31)
  (plowing)
  (wait_mm_dd 4 05)
  (seed_bed_preparation)
  (sow "Pea")
  (wait (or (crop_ds_after "Pea" 2.0) (mm_dd 8 15)))
  (harvest "Pea" (sorg 0.99) (leaf 0.01) (stem 0.01) (stub 10 [cm])))
```

### 38.6.33 Action SBarley & Pea

Parent activity, from sample/dk-management.dai.

```
(defaction
  "SBarley & Pea"
  activity
  (wait_mm_dd 3 05)
  (fertilize (N25S (weight 60 [kg N/ha])))
  (plowing)
  (wait_mm_dd 4 05)
  (seed_bed_preparation)
  (progn (sow "Spring Barley") (sow "Pea"))
  (wait_mm_dd 7 05)
  (progn (harvest "Spring Barley" (stub 8 [cm])) (harvest "Pea" (stub 8 [cm]))))
```

### 38.6.34 Action Sowing Grass

Parent activity, from sample/dk-management.dai.

```
(defaction
  "Sowing Grass"
  activity
  (wait_mm_dd 8 20)
  (fertilize
    ("cattle_slurry" (volatilization 5 [%]) (weight 20 [T w.w./ha]))
    (to -1 [cm]))
  (plowing)
  (wait_mm_dd 09 01)
  (seed_bed_preparation)
  (sow "Grass")
```

```
  (wait_mm_dd 10 10)
  (harvest "Grass" (stub 8 [cm]) (stem 0.00) (leaf 0.00)))
```

### 38.6.35 Action Grass minus 4 cuts

Parent activity, from sample/dk-management.dai.

```
(defaction
  "Grass - 4 cuts"
  activity
  (wait_mm_dd 03 15)
  (fertilize (AmmoniumNitrate (weight 100 [kg N/ha])))
  (wait_mm_dd 05 25)
  (harvest "Grass" (stub 8 [cm]) (stem 0.90) (leaf 0.90))
  (wait_days 1)
  (fertilize (AmmoniumNitrate (weight 100 [kg N/ha])))
  (wait_mm_dd 07 01)
  (harvest "Grass" (stub 8 [cm]) (stem 0.80) (leaf 0.80))
  (wait_days 1)
  (fertilize (AmmoniumNitrate (weight 75 [kg N/ha])))
  (wait_mm_dd 08 10)
  (harvest "Grass" (stub 8 [cm]) (stem 0.80) (leaf 0.80))
  (wait_days 1)
  (fertilize (AmmoniumNitrate (weight 50 [kg N/ha])))
  (wait_mm_dd 10 10)
  (harvest "Grass" (stub 8 [cm]) (stem 0.80) (leaf 0.80)))
```

### 38.6.36 Action Grass for cutting minus 2y

Parent activity, from sample/dk-management.dai.

```
(defaction
  "Grass for cutting - 2y"
  activity
  "Sowing Grass"
  "Grass - 4 cuts"
  "Grass - 4 cuts")
```

### 38.6.37 Action SBarley w. catch crop

Parent activity, from sample/dk-management.dai.

```
(defaction
  "SBarley w. catch crop"
  activity
  (wait_mm_dd 3 05)
  (fertilize (N25S (weight 115 [kg N/ha])))
  (plowing)
  (wait_mm_dd 4 05)
  (seed_bed_preparation)
  (sow "Spring Barley")
  (wait_mm_dd 7 05)
  (sow "Grass")
  (wait (or (crop_ds_after "Spring Barley" 2.0) (mm_dd 08 20)))
  (harvest "Spring Barley" (stub 8 [cm]) (stem 0.70)))
```

### 38.6.38 Action Potato Management

Parent activity, from sample/dk-management.dai.

```
(defaction
  "Potato Management"
  activity
  (wait_mm_dd 04 24)
  (fertilize (N25S (weight 160 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Potato; SCRI")
  (wait_mm_dd 10 22)
  (harvest "Potato; SCRI" (stem 0.0) (leaf 0.0)))
```

### 38.6.39 Action harrowing

Parent mix, from sample/dk-veg-man.dai.

```
(defaction "harrowing" mix -15 (penetration 0.8))
```

### 38.6.40 Action irrigation

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "irrigation"
  activity
  (wait (not (soil_water_pressure_above (height -20.0 [cm]) (potential -300 [cm]))))
  (irrigate_overhead 10 [mm/h] (hours 3))
  (wait_days 2))
```

### 38.6.41 Action irrigation onion

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "irrigation onion"
  activity
  (wait (not (soil_water_pressure_above (height -10.0 [cm]) (potential -250 [cm]))))
  (irrigate_overhead 10 [mm/h] (hours 2))
  (wait_days 1))
```

### 38.6.42 Action Beetroot management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Beetroot management"
  activity
  (wait_mm_dd 4 1)
  (plowing)
  (wait_mm_dd 5 20)
  (fertilize (KAS (weight 40 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Beetroot")
  (wait_mm_dd 6 10)
  (while (wait_mm_dd 6 23) (repeat "irrigation"))
  (fertilize (KAS (weight 95 [kg N/ha])))
```

```
  (while (wait_mm_dd 8 15) (repeat "irrigation"))
  (fertilize (N15B (weight 30 [kg N/ha])))
  (while (wait_mm_dd 10 8) (repeat "irrigation"))
  (harvest "Beetroot" (stub 0.0 [cm]) (sorg 1.0) (stem 0.0) (leaf 0.0)))
```

### 38.6.43 Action Broccoli minus transplanted management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Broccoli - transplanted management"
  activity
  (wait_mm_dd 4 1)
  (plowing)
  (wait_mm_dd 06 05)
  (fertilize (KAS (weight 85 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Broccoli - transplanted")
  (while (wait_mm_dd 07 01) (repeat "irrigation"))
  (fertilize ("NPK 14-3-18" (weight 85 [kg N/ha])))
  (while (wait_mm_dd 07 21) (repeat "irrigation"))
  (fertilize (N15B (weight 40 [kg N/ha])))
  (while
    (wait (or (crop_ds_after "Broccoli - transplanted" 1.8 []) (mm_dd 10 15)))
    (repeat "irrigation"))
  (harvest "Broccoli - transplanted" (stub 0.0 [cm]) (sorg 1.0) (stem 0.0) (leaf 0.0)))
```

### 38.6.44 Action Brussels sprouts management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Brussels sprouts management"
  activity
  (wait_mm_dd 04 25)
  (fertilize (KAS (weight 105 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Brussels sprouts")
  (while (wait_mm_dd 6 21) (repeat "irrigation"))
  (fertilize (KAS (weight 50 [kg N/ha])))
  (while (wait_mm_dd 7 31) (repeat "irrigation"))
  (fertilize (KAS (weight 50 [kg N/ha])))
 (while (wait_mm_dd 10 1) (repeat "irrigation")) ;(wait_mm_dd 10 15) ;(fertilize (N46 (weight 25 [kg
  (wait_mm_dd 12 15)
  (harvest "Brussels sprouts" (stub 0.0 [cm]) (sorg 1.0) (stem 0.0) (leaf 0.0)))
```

### 38.6.45 Action Brussels sprouts minus transplanted management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Brussels sprouts - transplanted management"
  activity
  (wait_mm_dd 05 07)
  (fertilize (KAS (weight 130 [kg N/ha])))
```

```
  (harrowing)
  (seed_bed_preparation)
  (sow "Brussels sprouts - transplanted")
  (while (wait_mm_dd 6 21) (repeat "irrigation"))
  (fertilize (KAS (weight 50 [kg N/ha])))
  (while (wait_mm_dd 7 31) (repeat "irrigation"))
  (fertilize (KAS (weight 50 [kg N/ha])))
  (while (wait_mm_dd 10 1) (repeat "irrigation"))
  (wait_mm_dd 12 01)
  (harvest
    "Brussels sprouts - transplanted"
    (stub 0.0 [cm])
    (sorg 1.0)
    (stem 0.0)
    (leaf 0.0)))
```

### 38.6.46 Action Celeriac minus transplanted management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Celeriac - transplanted management"
  activity
  (wait_mm_dd 4 1)
  (plowing)
  (wait_mm_dd 5 5)
  (fertilize (KAS (weight 100 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Celeriac - transplanted")
  (while (wait_mm_dd 6 30) (repeat "irrigation"))
  (fertilize (KAS (weight 70 [kg N/ha])))
  (while (wait_mm_dd 8 15) (repeat "irrigation"))
  (fertilize (N15B (weight 50 [kg N/ha])))
  (while (wait_mm_dd 10 15) (repeat "irrigation"))
  (harvest "Celeriac - transplanted" (stub 0.0 [cm]) (sorg 1.0) (stem 0.0) (leaf 0.0)))
```

### 38.6.47 Action Early potato management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Early potato management"
  activity
  (wait_mm_dd 04 15)
  (fertilize ("NPK 14-3-18" (weight 130 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Early potato")
  (wait_mm_dd 06 01)
  (while
    (wait
      (or
        (and
          (crop_ds_after "Early potato" 1.1 [])
          (crop_dm_over "Early potato" 4650 [kg DM/ha] (height 0 [cm])))
        (mm_dd 08 01)))
```

```
    (repeat "irrigation"))
  (if
    (crop_ds_after "Early potato" 0.1 [])
    (harvest "Early potato" (stub 0.0 [cm]) (sorg 1.0) (stem 0.0) (leaf 0.0))))
```

### 38.6.48 Action Onion management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Onion management"
  activity
  (wait_mm_dd 04 12)
  (fertilize ("NP 12-23-0" (weight 45 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Onion")
  (wait_mm_dd 05 15)
  (fertilize (KAS (weight 45 [kg N/ha])))
  (while (wait_mm_dd 06 15) (repeat "irrigation onion"))
  (fertilize ("NPK 14-3-18" (weight 30 [kg N/ha])))
  (while (wait_mm_dd 08 15) (repeat "irrigation onion"))
  (while
    (wait (or (crop_ds_after "Onion" 0.85 []) (mm_dd 09 01)))
    (repeat "irrigation onion"))
  (wait (or (crop_ds_after "Onion" 0.85 []) (mm_dd 09 15)))
  (harvest "Onion" (stub 0.0 [cm]) (sorg 1.0) (stem 0.0) (leaf 0.0)))
```

### 38.6.49 Action Onion minus planting of sets management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Onion - planting of sets management"
  activity
  (wait_mm_dd 04 01)
  (fertilize ("NP 12-23-0" (weight 45 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Onion - planting of sets")
  (wait_mm_dd 04 15) ;;04 15
  (while (wait_mm_dd 05 05) (repeat "irrigation onion"))
  (fertilize ("KAS" (weight 55 [kg N/ha])))
  (while (wait_mm_dd 06 05) (repeat "irrigation onion"))
  (fertilize ("NPK 14-3-18" (weight 30 [kg N/ha])))
  (while (wait_mm_dd 07 20) (repeat "irrigation onion"))
  (while
    (wait (or (crop_ds_after "Onion - planting of sets" 0.85 []) (mm_dd 08 15)))
    (repeat "irrigation onion"))
  (wait (or (crop_ds_after "Onion - planting of sets" 0.85 []) (mm_dd 09 01)))
  (harvest "Onion - planting of sets" (stub 0.0 [cm]) (sorg 1.0) (stem 0.0) (leaf 0.0)))
```

### 38.6.50 Action White cabbage minus transplanted management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
```

```
  "White cabbage - transplanted management"
  activity
  (wait_mm_dd 05 01)
  (fertilize (KAS (weight 125 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "White cabbage - transplanted")
  (while (wait_mm_dd 06 21) (repeat "irrigation"))
  (fertilize ("NPK 14-3-18" (weight 75 [kg N/ha])))
  (while (wait_mm_dd 7 31) (repeat "irrigation"))
  (fertilize ("NS 24-7" (weight 50 [kg N/ha])))
  (while (wait_mm_dd 10 01) (repeat "irrigation"))
  (wait (or (crop_ds_after "White cabbage - transplanted" 0.85 []) (mm_dd 10 25)))
  (harvest
    "White cabbage - transplanted"
    (stub 0.0 [cm])
    (sorg 1.0)
    (stem 0.0)
    (leaf 0.0)))
```

### 38.6.51 Action Early white cabbage minus transplanted management

Parent activity, from sample/dk-veg-man.dai.

```
(defaction
  "Early white cabbage - transplanted management"
  activity
  (wait_mm_dd 04 01)
  (fertilize (KAS (weight 150 [kg N/ha])))
  (harrowing)
  (seed_bed_preparation)
  (sow "Early white cabbage - transplanted")
  (while (wait_mm_dd 04 25) (repeat "irrigation"))
  (fertilize ("NPK 14-3-18" (weight 50 [kg N/ha])))
  (while (wait_mm_dd 05 20) (repeat "irrigation"))
  (while
   (wait (or (crop_ds_after "Early white cabbage - transplanted" 0.8 []) (mm_dd 07 01)))
    (repeat "irrigation"))
  (harvest
    "Early white cabbage - transplanted"
    (stub 0.0 [cm])
    (sorg 1.0)
    (stem 0.0)
    (leaf 0.0)))
```

### 38.6.52 Action none

Parent activity, from sample/genweather.dai.

```
(defaction none activity)
```

### 38.6.53 Action irrigate_30_tensiometer_overhead

Parent activity, from sample/irrigation.dai.

```
(defaction
  irrigate_30_tensiometer_overhead
```

```
 activity ;; An irrigation strategy based on a tensiometer. ;; If the soil water pressure at 20 cm dep
300 cm in the ;; period between the first of May and the first of September, apply 10 ;; mm/h overhead i
  (wait
    (and
      (after_mm_dd 5 1) ;After May the first.
           (before_mm_dd  8  1)  ;Before  August  the  first.  ;;  Potential  below  -
300 cm at 20 cm depth.
    (not (soil_water_pressure_above (height -20.0 [cm]) (potential -300 [cm])))))) ;; Overhead irriga
 (irrigate_overhead 10 [mm/h] (hours 3)) ;; Wait two days to allow water to reach the tensiometer.
  (wait_days 2))
```

### 38.6.54 Action irrigate_30_content_overhead

Parent activity, from sample/irrigation.dai.

```
(defaction
  irrigate_30_content_overhead
 activity ;; An irrigation strategy based on a soil water content. ;; If the soil contain less than 60
  (wait
    (and
      (after_mm_dd 5 1) ;After May the first.
    (before_mm_dd 8 1) ;Before August the first. ;; Less than 60 mm water in the top half meter of the
    (not (soil_water_content_above 60 [mm] (from 0 [cm]) (to -50 [cm])))))) ;; Overhead irrigation of
 (irrigate_overhead 10 [mm/h] (hours 3)) ;; Wait one day to allow water to reach the tensiometer.
  (wait_days 1))
```

### 38.6.55 Action it_irrigate_crop

Parent activity, from sample/it-management.dai.

```
(defaction
  it_irrigate_crop
  activity
  (wait (and (after_mm_dd 5 10) (before_mm_dd 7 15)))
  (irrigate_overhead 25)
  (wait_days 15))
```

### 38.6.56 Action WP1 IT 2005 warmup

Parent activity, from sample/it-management.dai.

```
(defaction
  "WP1 IT 2005 warmup"
  activity ;;StartTime 96
  (wait (mm_dd 10 25))
  (sow "Winter Wheat") ;;Grano Standard 97
  (wait (mm_dd 3 31))
  (fertilize (AmmoniumNitrate (weight 75.0 [kg N/ha])))
  (wait (mm_dd 7 10))
  (harvest "Winter Wheat" (stem 0.00) (leaf 0.00) (sorg 0.95))
  (wait (mm_dd 8 15))
  (plowing)
  (wait (mm_dd 10 15))
  (disk_harrowing)
  (wait (mm_dd 10 25))
  (sow "Winter Wheat") ;;Grano Standard 98
  (wait (mm_dd 3 31))
```

```
(fertilize (AmmoniumNitrate (weight 75.0 [kg N/ha])))
(wait (mm_dd 7 10))
(harvest "Winter Wheat" (stem 0.00) (leaf 0.00) (sorg 0.95))
(wait (mm_dd 8 15))
(plowing)
(wait (mm_dd 10 15))
(disk_harrowing)
(wait (mm_dd 11 15))
(disk_harrowing) ;;Patata Modified (Tomato 99)
(wait (mm_dd 4 15))
(seed_bed_preparation)
(wait (mm_dd 5 15))
(sow "Potato; CER")
(fertilize (Nitrate (weight 40.0 [kg N/ha])))
(wait (mm_dd 5 30))
(fertilize (Nitrate (weight 40.0 [kg N/ha])))
(wait (mm_dd 6 15))
(fertilize (Nitrate (weight 40.0 [kg N/ha])))
(while
  (wait (mm_dd 7 15))
  (activity
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop))
(wait (mm_dd 8 25))
(harvest "Potato; CER" (stem 0.00) (leaf 0.00) (sorg 0.95))
(wait (mm_dd 8 30))
(plowing)
(wait (mm_dd 10 15))
(disk_harrowing)
(wait (mm_dd 10 20))
(seed_bed_preparation) ;;Bietola Standard (00)
(wait (mm_dd 2 5))
(seed_bed_preparation)
(wait (mm_dd 2 20))
(sow "Sugar Beet")
(wait (mm_dd 2 21))
(fertilize (AmmoniumNitrate (weight 75.0 [kg N/ha])))
(wait (mm_dd 4 15))
(fertilize (AmmoniumNitrate (weight 75.0 [kg N/ha])))
(wait (mm_dd 7 10))
(irrigate_overhead 50)
(wait (mm_dd 9 25))
(harvest "Sugar Beet" (stem 0) (leaf 0) (sorg 0.95))
(wait (mm_dd 9 30))
(wait trafficable)
(plowing)
(wait (mm_dd 10 15))
(disk_harrowing)
(wait (mm_dd 10 20))
(seed_bed_preparation)
(wait (mm_dd 10 25))
```

```
(sow "Winter Wheat") ;;Grano Standard 01
(wait (mm_dd 3 31))
(fertilize (AmmoniumNitrate (weight 75.0 [kg N/ha])))
(wait (mm_dd 7 10))
(harvest "Winter Wheat" (stem 0.00) (leaf 0.00) (sorg 0.95))
(wait (mm_dd 8 15))
(plowing)
(wait (mm_dd 10 15))
(disk_harrowing)
(wait (mm_dd 11 15))
(disk_harrowing) ;;Pea Modified (Sojabeen) 02
(wait (mm_dd 5 1))
(sow "Pea")
(fertilize (Nitrate (weight 60.0 [kg N/ha])))
(wait (mm_dd 9 20))
(harvest "Pea" (stem 0.00) (leaf 0.00) (sorg 0.95))
(wait (mm_dd 9 30))
(plowing)
(wait (mm_dd 10 15))
(disk_harrowing)
(wait (mm_dd 11 15))
(disk_harrowing) ;;Patata Standard 03
(wait (mm_dd 2 15))
(seed_bed_preparation)
(wait (mm_dd 3 1))
(sow "Potato; CER") ;;(wait (mm_dd 3 1)) ;;(ridge (R_crust 0 [h]) ;;(switch 0.333) ;;(z (30) [cm]))
(wait (mm_dd 3 15))
(fertilize (AmmoniumNitrate (weight 100.0 [kg N/ha])))
(wait (mm_dd 4 15))
(fertilize (AmmoniumNitrate (weight 100.0 [kg N/ha])))
(while
  (wait (mm_dd 7 15))
  (activity
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop
    it_irrigate_crop))
(wait (mm_dd 7 25))
(harvest "Potato; CER" (stem 0.00) (leaf 0.00) (sorg 0.95))
(wait (mm_dd 8 30))
(plowing)
(wait (mm_dd 10 15))
(disk_harrowing)
(wait (mm_dd 10 20))
(seed_bed_preparation)
(wait (mm_dd 10 25))
(sow "Winter Wheat") ;;Grano Standard 04
(wait (mm_dd 3 31))
(fertilize (AmmoniumNitrate (weight 75.0)))
(wait (mm_dd 7 10))
(harvest "Winter Wheat" (stem 0.00) (leaf 0.00) (sorg 0.95))
(wait (mm_dd 8 15))
(plowing)
```

```
  (wait (mm_dd 10 15))
  (disk_harrowing))
```

### 38.6.57 Action Maize_99

Parent activity, from sample/pl-management.dai.

```
(defaction
  Maize_99
  activity
  (wait_mm_dd 4 20)
  (fertilize (AmmoniumNitrate (weight 70 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Maize")
  (wait (or (crop_ds_after "Maize" 2.0) (mm_dd 9 25)))
  (harvest "Maize" (stub 8 [cm]) (stem 1.00 []) (leaf 1.00 []))
  (wait_mm_dd 9 27)
  (stubble_cultivation))
```

### 38.6.58 Action Potato_00

Parent activity, from sample/pl-management.dai.

```
(defaction
  Potato_00
  activity
  (wait_mm_dd 10 23)
  (fertilize ("cattleJad_manure04" (weight 25 [Mg w.w./ha])))
  (plowing)
  (wait_mm_dd 4 15)
  (fertilize (AmmoniumNitrate (weight 90 [kg N/ha])))
  (wait_mm_dd 4 19)
  (seed_bed_preparation)
  (sow "Potato; Triada")
  (wait (or (crop_ds_after "Potato; IHAR 2004" 2.0 []) (mm_dd 9 10)))
  (harvest "Potato; IHAR 2004" (stem 0.00 []) (leaf 0.00 []))
  (wait_mm_dd 9 23))
```

### 38.6.59 Action SWheat_01

Parent activity, from sample/pl-management.dai.

```
(defaction
  SWheat_01
  activity
  (wait_mm_dd 11 12)
  (plowing)
  (wait_mm_dd 4 03)
  (fertilize (AmmoniumNitrate (weight 20 [kg N/ha])))
  (wait_mm_dd 4 06)
  (seed_bed_preparation)
  (sow "Spring Wheat")
  (wait (or (crop_ds_after "Spring Wheat" 2.0) (mm_dd 8 12)))
  (harvest "Spring Wheat" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation))
```

### 38.6.60 Action WWheat_02

Parent activity, from sample/pl-management.dai.

```
(defaction
  WWheat_02
  activity
  (wait_mm_dd 9 20)
  (fertilize (AmmoniumNitrate (weight 24 [kg N/ha])))
  (wait_mm_dd 9 28)
  (seed_bed_preparation)
  (sow "Winter Wheat")
  (wait_mm_dd 5 21)
  (fertilize (AmmoniumNitrate (weight 46 [kg N/ha])))
  (wait (or (crop_ds_after "Winter Wheat" 2.0) (mm_dd 8 03)))
  (harvest "Winter Wheat" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation))
```

### 38.6.61 Action SWheat_03

Parent activity, from sample/pl-management.dai.

```
(defaction
  SWheat_03
  activity
  (wait_mm_dd 11 06)
  (plowing)
  (wait_mm_dd 4 10)
  (fertilize (AmmoniumNitrate (weight 20 [kg N/ha])))
  (wait_mm_dd 4 12)
  (seed_bed_preparation) ;7/2-06 cto; Because Spring wheat else don't give any yield
  (sow ("Spring Wheat" (Canopy (SpLAIfac (0.0 100) (1.0 100))))) ;;(sow "Spring Wheat")
  (wait_mm_dd 4 29)
  (fertilize (AmmoniumNitrate (weight 68 [kg N/ha])))
  (wait (or (crop_ds_after "Spring Wheat" 2.0) (mm_dd 8 10)))
  (harvest "Spring Wheat" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation))
```

### 38.6.62 Action WP1 PL 2004 warmup

Parent activity, from sample/pl-management.dai.

```
(defaction
  "WP1 PL 2004 warmup"
  activity
  Maize_99
  Potato_00
  SWheat_01
  WWheat_02
  SWheat_03)
```

### 38.6.63 Action Lettuce_management

Parent activity, from sample/safir-site.dai.

```
(defaction
  Lettuce_management
  activity
```

```
  (wait_mm_dd 10 18)
  (plowing)
  (wait_mm_dd 4 18)
  (seed_bed_preparation)
  (wait_mm_dd 5 5)
  (sow ("Lettuce"))
  (wait_mm_dd 7 21)
  (fertilize (mineral (weight 89.68 [kg N/ha]) (NH4_fraction 0.5 [])))
  (wait_mm_dd 8 1)
  (fertilize (mineral (weight 89.68 [kg N/ha]) (NH4_fraction 0.5 [])))
  (wait_mm_dd 8 18)
  (fertilize (mineral (weight 89.68 [kg N/ha]) (NH4_fraction 0.5 [])))
  (wait_mm_dd 9 25)
  (harvest "Lettuce"))
```

### 38.6.64 Action warmup

Parent n/a, from sample/safir-site.dai.

```
(defaction warmup (Lettuce_management))
```

### 38.6.65 Action Crop_management

Parent activity, from sample/safir-site.dai.

```
(defaction
  Crop_management
  activity
  (wait_mm_dd 10 18)
  (plowing)
  (wait_mm_dd 4 18)
  (seed_bed_preparation)
  (wait_mm_dd 5 5)
  (sow ("SiteCrop") (seed 2765 [kg w.w./ha]) (plant_distance 0.30 [m]))
  (wait_mm_dd 9 25)
  (harvest "SiteCrop"))
```

### 38.6.66 Action irrigation_subsoil_full

Parent activity, from sample/safir-site.dai.

```
(defaction
  irrigation_subsoil_full
  activity
  "Fully irrigated soil."
  (wait_mm_dd 7 21 (hour 10))
  (progn
    (irrigate_subsoil
      2.6
      [mm/h]
      (remaining_time 5.0 [h])
      (solute (NO3 1.725 [kg N/ha/mm]) (NH4 1.725 [kg N/ha/mm]))
      (volume left-drip))
    (irrigate_subsoil
      2.6
      [mm/h]
      (remaining_time 5.0 [h])
```

```
    (solute (NO3 1.725 [kg N/ha/mm]) (NH4 1.725 [kg N/ha/mm]))
    (volume right-drip)))
  (wait_mm_dd 7 26 (hour 10))
  (progn
    (irrigate_subsoil 2.6 [mm/h] (remaining_time 5.0 [h]) (volume left-drip))
    (irrigate_subsoil 2.6 [mm/h] (remaining_time 5.0 [h]) (volume right-drip)))
  (wait_mm_dd 7 28 (hour 10))
  (progn
    (irrigate_subsoil 2.75 [mm/h] (remaining_time 5.0 [h]) (volume left-drip))
    (irrigate_subsoil 2.75 [mm/h] (remaining_time 5.0 [h]) (volume right-drip)))
  (wait_mm_dd 8 1 (hour 10))
  (progn
    (irrigate_subsoil
      2.6
      [mm/h]
      (remaining_time 5.0 [h])
      (solute (NO3 1.725 [kg N/ha/mm]) (NH4 1.725 [kg N/ha/mm]))
      (volume left-drip))
    (irrigate_subsoil
      2.6
      [mm/h]
      (remaining_time 5.0 [h])
      (solute (NO3 1.725 [kg N/ha/mm]) (NH4 1.725 [kg N/ha/mm]))
      (volume right-drip)))
  (wait_mm_dd 8 18 (hour 10))
  (progn
    (irrigate_subsoil
      2.6
      [mm/h]
      (remaining_time 5.0 [h])
      (solute (NO3 1.725 [kg N/ha/mm]) (NH4 1.725 [kg N/ha/mm]))
      (volume left-drip))
    (irrigate_subsoil
      2.6
      [mm/h]
      (remaining_time 5.0 [h])
      (solute (NO3 1.725 [kg N/ha/mm]) (NH4 1.725 [kg N/ha/mm]))
      (volume right-drip))))
```

### 38.6.67 Action irrigation_subsoil_PRD_left

Parent activity, from sample/safir-site.dai.

```
(defaction
  irrigation_subsoil_PRD_left
  activity
  "PRD irrigated soil, left side."
  (wait_mm_dd 7 21 (hour 10))
  (irrigate_subsoil
    3
    [mm/h]
    (remaining_time 6.0 [h])
    (solute (NO3 1.25 [kg N/ha/mm]) (NH4 1.25 [kg N/ha/mm]))
    (volume left-drip))
  (wait_mm_dd 7 28 (hour 10))
  (irrigate_subsoil 3.5 [mm/h] (remaining_time 2.0 [h]) (volume left-drip))
```

```
  (wait_mm_dd 8 1 (hour 10))
  (irrigate_subsoil
    2.7
    [mm/h]
    (remaining_time 4.0 [h])
    (solute (NO3 2.076 [kg N/ha/mm]) (NH4 2.076 [kg N/ha/mm]))
    (volume left-drip))
  (wait_mm_dd 8 11 (hour 10))
  (irrigate_subsoil
    2.7
    [mm/h]
    (remaining_time 4.0 [h])
    (solute (NO3 2.076 [kg N/ha/mm]) (NH4 2.076 [kg N/ha/mm]))
    (volume left-drip)))
```

### 38.6.68 Action irrigation_subsoil_PRD_right

Parent activity, from sample/safir-site.dai.

```
(defaction
  irrigation_subsoil_PRD_right
  activity
  "PRD irrigated soil, right side."
  (wait_mm_dd 7 21 (hour 10))
  (irrigate_subsoil
    3
    [mm/h]
    (remaining_time 6.0 [h])
    (solute (NO3 1.25 [kg N/ha/mm]) (NH4 1.25 [kg N/ha/mm]))
    (volume right-drip))
  (wait_mm_dd 7 26 (hour 10))
  (irrigate_subsoil 3 [mm/h] (remaining_time 6.0 [h]) (volume right-drip))
  (wait_mm_dd 8 1 (hour 10))
  (irrigate_subsoil
    2.7
    [mm/h]
    (remaining_time 4.0 [h])
    (solute (NO3 2.076 [kg N/ha/mm]) (NH4 2.076 [kg N/ha/mm]))
    (volume right-drip))
  (wait_mm_dd 8 11 (hour 10))
  (irrigate_subsoil
    2.7
    [mm/h]
    (remaining_time 4.0 [h])
    (solute (NO3 2.076 [kg N/ha/mm]) (NH4 2.076 [kg N/ha/mm]))
    (volume right-drip)))
```

### 38.6.69 Action SBeet_management

Parent activity, from sample/sk-management.dai.

```
(defaction
  SBeet_management
  activity
  (wait_mm_dd 3 18)
  (fertilize ("cattle_manure" (weight 40 [T w.w./ha])) (to -5 [cm]))
  (wait_mm_dd 3 21)
```

```
  (seed_bed_preparation)
  (sow "Sugar Beet")
  (wait_mm_dd 5 12)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 5 20)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 6 1)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 7 3)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 8 10)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 8 12)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 8 15)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 8 18)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Sugar Beet" 2.0) (mm_dd 10 31)))
  (harvest "Sugar Beet" (leaf 0.00 []))
  (disk_harrowing)
  (wait_days 2)
  (plowing))
```

### 38.6.70 Action SBarley_94

Parent activity, from sample/sk-management.dai.

```
(defaction
  SBarley_94
  activity
  (wait_mm_dd 3 29)
  (fertilize (AmmoniumNitrate (weight 70 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Spring Barley_SK")
  (wait_mm_dd 5 31)
  (irrigate_overhead 2 [mm/h] (hours 15))
  (wait_mm_dd 6 8)
  (irrigate_overhead 2 [mm/h] (hours 15))
  (wait_mm_dd 6 15)
  (irrigate_overhead 2 [mm/h] (hours 15))
  (wait (or (crop_ds_after "Spring Barley_SK" 2.0) (mm_dd 8 15)))
  (harvest "Spring Barley_SK" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation))
```

### 38.6.71 Action WWheat_95

Parent activity, from sample/sk-management.dai.

```
(defaction
  WWheat_95
  activity
  (wait_mm_dd 8 21)
  (rotavation)
  (wait_mm_dd 10 1)
  (fertilize (AmmoniumNitrate (weight 40 [kg N/ha])))
  (seed_bed_preparation)
```

```
  (sow "Winter Wheat_SK")
  (wait_mm_dd 3 1)
  (wait trafficable)
  (fertilize (AmmoniumNitrate (weight 50 [kg N/ha])))
  (wait_mm_dd 5 2)
  (fertilize (AmmoniumNitrate (weight 30 [kg N/ha])))
  (wait_mm_dd 5 22)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Winter Wheat_SK" 2.0) (mm_dd 8 31)))
  (harvest "Winter Wheat_SK" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation)
  (wait_mm_dd 10 1)
  (plowing))
```

### 38.6.72 Action Maize_1996

Parent activity, from sample/sk-management.dai.

```
(defaction
  Maize_1996
  activity
  (wait_mm_dd 4 20)
  (fertilize (AmmoniumNitrate (weight 120 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Maize_SK")
  (wait_mm_dd 6 2)
  (irrigate_overhead 2 [mm/h] (hours 15))
  (wait_mm_dd 6 10)
  (irrigate_overhead 2 [mm/h] (hours 20))
  (wait_mm_dd 7 17)
  (irrigate_overhead 2 [mm/h] (hours 15))
  (wait_mm_dd 7 21)
  (irrigate_overhead 2 [mm/h] (hours 15))
  (wait (or (crop_ds_after "Maize_SK" 2.0) (mm_dd 10 10)))
  (harvest "Maize_SK" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation)
  (wait_mm_dd 10 15)
  (fertilize ("cattle_manure" (weight 40 [T w.w./ha])) (to -5 [cm])))
```

### 38.6.73 Action Maize_1997

Parent activity, from sample/sk-management.dai.

```
(defaction
  Maize_1997
  activity
  (wait_mm_dd 4 21)
  (fertilize (AmmoniumNitrate (weight 80 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Maize_SK")
  (wait_mm_dd 5 12)
  (fertilize (AmmoniumNitrate (weight 40 [kg N/ha])))
  (wait_mm_dd 6 1)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 10)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 25)
```

```
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 8 7)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 8 8)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 8 14)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 8 16)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Maize_SK" 2.0) (mm_dd 10 5)))
  (harvest "Maize_SK" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (wait_mm_dd 10 6)
  (fertilize (AmmoniumNitrate (weight 40 [kg N/ha])))
  (stubble_cultivation)
  (wait_days 2)
  (rotavation))
```

### 38.6.74 Action SBarley_98

Parent activity, from sample/sk-management.dai.

```
(defaction
  SBarley_98
  activity
  (wait_mm_dd 3 31)
  (fertilize (AmmoniumNitrate (weight 60 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Spring Barley_SK")
  (wait_mm_dd 5 5)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 5 11)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 5 21)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 5 28)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 5)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 20)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Spring Barley_SK" 2.0) (mm_dd 8 15)))
  (harvest "Spring Barley_SK" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation)
  (wait_mm_dd 10 1)
  (rotavation))
```

### 38.6.75 Action Pea_99

Parent activity, from sample/sk-management.dai.

```
(defaction
  Pea_99
  activity
  (wait_mm_dd 3 15)
  (fertilize (AmmoniumNitrate (weight 30 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Pea")
```

```
  (wait_mm_dd 5 6)
  (irrigate_overhead 2 [mm/h] (hours 12))
  (wait_mm_dd 5 24)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait_mm_dd 5 30)
  (irrigate_overhead 2 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Pea" 2.0) (mm_dd 8 15)))
  (harvest "Pea" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation)
  (wait_mm_dd 10 15)
  (fertilize ("cattle_manure" (weight 40 [T w.w./ha])) (to -5 [cm])))
```

### 38.6.76 Action Maize_2000

Parent activity, from sample/sk-management.dai.

```
(defaction
  Maize_2000
  activity
  (wait_mm_dd 4 21)
  (fertilize (AmmoniumNitrate (weight 120 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Maize_SK")
  (wait_mm_dd 5 10)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 2)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 12)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 20)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 7 1)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 7 9)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 7 20)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 8 10)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 8 18)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 8 25)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Maize_SK" 2.0) (mm_dd 10 01)))
  (harvest "Maize_SK" (stub 8 [cm]) (stem 0.00 []) (leaf 0.00 []))
  (stubble_cultivation))
```

### 38.6.77 Action S_Barley_01

Parent activity, from sample/sk-management.dai.

```
(defaction
  S_Barley_01
  activity
  (wait_mm_dd 3 15)
  (fertilize (AmmoniumNitrate (weight 50 [kg N/ha])))
  (seed_bed_preparation)
```

```
  (sow "Spring Barley_SK")
  (wait_mm_dd 4 30)
  (fertilize (AmmoniumNitrate (weight 30 [kg N/ha])))
  (wait_mm_dd 5 10)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 5 21)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 3)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 12)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Spring Barley_SK" 2.0) (mm_dd 8 31)))
  (harvest "Spring Barley_SK" (stub 8 [cm]) (stem 1.00 []) (leaf 1.00 []))
  (stubble_cultivation))
```

### 38.6.78 Action W_Wheat_02

Parent activity, from sample/sk-management.dai.

```
(defaction
  W_Wheat_02
  activity
  (wait_mm_dd 8 21)
  (rotavation)
  (wait_mm_dd 10 1)
  (fertilize (AmmoniumNitrate (weight 30 [kg N/ha])))
  (seed_bed_preparation)
  (sow "Winter Wheat_SK")
  (wait_mm_dd 3 15)
  (wait trafficable)
  (fertilize (AmmoniumNitrate (weight 60 [kg N/ha])))
  (wait_mm_dd 5 10)
  (fertilize (AmmoniumNitrate (weight 30 [kg N/ha])))
  (wait_mm_dd 5 12)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 5 22)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 5 31)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait_mm_dd 6 20)
  (irrigate_overhead 3 [mm/h] (hours 10))
  (wait (or (crop_ds_after "Winter Wheat_SK" 2.0) (mm_dd 8 31)))
  (harvest "Winter Wheat_SK" (stub 8 [cm]) (stem 1.00 []) (leaf 1.00 []))
  (stubble_cultivation)
  (wait_mm_dd 10 1)
  (plowing))
```

### 38.6.79 Action WP1 SK 2003 warmup

Parent activity, from sample/sk-management.dai.

```
(defaction
  "WP1 SK 2003 warmup"
  activity
  SBeet_management
  SBarley_94
  WWheat_95
```

```
  Maize_1996
  Maize_1997
  SBarley_98
  Pea_99
  Maize_2000
  S_Barley_01
  W_Wheat_02)
```

### 38.6.80 Action sow_Spring_Barley

Parent n/a, from sample/test-aba.dai.

```
(defaction
  sow_Spring_Barley
  (sow
    ("Spring Barley"
    (enable_N_stress false) ;; ABA production in roots is affected by soil water potential.
    (Root (ABAprod "Fulai Liu uptake")) ;; Stomata conductivity is affected by ABA. ;; Doesn't work ri
      (LeafPhot
        FC_C3
        (Xn 6e-3 [mol/mol/s])
       (Stomatacon Leuning (m 11 []) (b 0.01 [mol/m^2/s]) (beta 1555311.743 [cm^3/g]))
        (TempEff (0 1) (100 1))))))
```

### 38.6.81 Action MyMan

Parent activity, from sample/test-organic.dai.

```
(defaction
  MyMan
  activity
  (wait (at 1987 3 20 1))
  (plowing)
  (wait (at 1987 4 4 1))
  (fertilize (mineral (weight 100.0 [kg N/ha]) (NH4_fraction 0.5 [])))
  (wait (at 1987 4 5 1))
  (sow "NewSB")
  (wait (or (crop_ds_after "NewSB" 2.0) (at 1987 9 5 1)))
  (harvest "NewSB"))
```

### 38.6.82 Action DK 2005 PRD

Parent activity, from sample/test-prd.dai.

```
(defaction
  "DK 2005 PRD"
  activity ;; Virtual experiment with PRD based on FertOrgaNic treatment 3.
  (wait (mm_dd 12 2))
  (plowing)
  (wait (mm_dd 4 7))
  (fertilize (jynd2005_slurry (weight 12.73)) (from 0) (to -10))
  (wait (mm_dd 4 8))
  (seed_bed_preparation)
  (wait (mm_dd 4 11))
 (sow ("Potato; Exp") (seed 2765 [kg w.w./ha]) (row_width 1 [m])) ;; Watering left side.
  (wait (mm_dd 5 28))
  (irrigate_subsoil
```

```
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 5 29))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 5 30))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 1))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 3))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 9))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 10))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 11))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l]))) ;; Watering right side.
(wait (mm_dd 6 17))
```

```
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 21))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 22))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 24))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 25))
(irrigate_subsoil
  2
  [mm/h]
  (hours 2)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l]))) ;; Watering left side.
(wait (mm_dd 6 27))
(irrigate_subsoil
  2.133
  [mm/h]
  (hours 3)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 28))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 6 29))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
```

```
(wait (mm_dd 6 30))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 1))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 3))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 5))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 7))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 9))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l]))) ;; Watering right side.
(wait (mm_dd 7 11))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 12))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume right-drip)
```

```
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 13))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 14))
(irrigate_subsoil
  2
  [mm/h]
  (hours 3)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 15))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 18))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 20))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 22))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume right-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l]))) ;; Watering left side.
(wait (mm_dd 7 25))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 7 28))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
```

```
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 8 1))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 8 5))
(irrigate_subsoil
  2
  [mm/h]
  (hours 1)
  (volume left-drip)
  (solute (NH4 0 [mg N/l]) (NO3 0 [mg N/l])))
(wait (mm_dd 8 15))
(harvest "Potato; Exp" (stem 0.00) (leaf 0.00) (sorg 1.00)))
```

## 38.7 Programs

| Name | Parent | File |
|---|---|---|
| Common | Daisy | sample/DrainDrying.dai |
| Drainage test | Common | sample/DrainDrying.dai |
| Drying test | Common | sample/DrainDrying.dai |
| main | batch | sample/DrainDrying.dai |
| JB1_PFAS | Daisy | sample/PFAS-sample.dai |
| Oak | Daisy | sample/afforestation.dai |
| Andeby | Daisy | sample/batch.dai |
| Andeby Cosby | Andeby | sample/batch.dai |
| Andeby HYPRES | Andeby | sample/batch.dai |
| Cosby et.al. vs. HYPRES | batch | sample/batch.dai |
| RefSim | Daisy | sample/checkpoint-1988-4-1+1.dai |
| RefSim Checkpoint | RefSim | sample/checkpoint-1988-4-1+1.dai |
| WP1 CZ RISWC 2004 | Daisy | sample/cz-sample.dai |
| GenWeather | Daisy | sample/genweather.dai |
| main | write | sample/heat-properties.dai |
| WP1 IT CER 2005 | Daisy | sample/it-sample.dai |
| WP1 PL IHAR 2004 | Daisy | sample/pl-sample.dai |
| MyMatch | rootmatch | sample/rootmatch.dai |
| Safir | Daisy | sample/safir-common.dai |
| Field-2006 | Safir | sample/safir-site.dai |
| Crop-Subsoil-Full | Field-2006 | sample/safir-site.dai |
| Crop-Subsoil-PRD | Field-2006 | sample/safir-site.dai |
| main | write | sample/sample-heat.dai |
| WP1 SK RIIDLE 2003 | Daisy | sample/sk-sample.dai |
| RefSim | Daisy | sample/test-cp.dai |
| MySim | Daisy | sample/test-minimize.dai |
| MyOpt | minimize | sample/test-minimize.dai |
| MySim | Daisy | sample/test-organic.dai |
| convert | post-process | sample/test-post.dai |
| print_it | write | sample/test-python.dai |
| RefSim | Daisy | sample/test-refman.dai |
| SCN1 | RefSim Checkpoint | sample/test-scn.dai |
| SCN2 | RefSim Checkpoint | sample/test-scn.dai |
| scenarios | batch | sample/test-scn.dai |
| Base | Daisy | sample/test-spawn.dai |

| Name | Parent | File |
|---|---|---|
| N-A-S | Base | sample/test-spawn.dai |
| N-A-W | Base | sample/test-spawn.dai |
| N-A-P | Base | sample/test-spawn.dai |
| N-J-S | Base | sample/test-spawn.dai |
| N-J-W | Base | sample/test-spawn.dai |
| N-J-P | Base | sample/test-spawn.dai |
| N-F-S | Base | sample/test-spawn.dai |
| N-F-W | Base | sample/test-spawn.dai |
| N-F-P | Base | sample/test-spawn.dai |
| W-A-S | Base | sample/test-spawn.dai |
| W-A-W | Base | sample/test-spawn.dai |
| W-A-P | Base | sample/test-spawn.dai |
| W-J-S | Base | sample/test-spawn.dai |
| W-J-W | Base | sample/test-spawn.dai |
| W-J-P | Base | sample/test-spawn.dai |
| W-F-S | Base | sample/test-spawn.dai |
| W-F-W | Base | sample/test-spawn.dai |
| W-F-P | Base | sample/test-spawn.dai |
| all | spawn | sample/test-spawn.dai |
| AndebyFarm | Daisy | sample/test.dai |

## 38.8 Log types

| Name | Parent | File |
|---|---|---|
| OpenMI_output | extern | lib/OpenMI.dai |
| Lower_boundary_output | OpenMI_output | lib/OpenMI.dai |
| Crop_soil_content | OpenMI_output | lib/OpenMI.dai |
| Soil Content 0-10 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 10-20 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 20-30 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 30-40 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 40-50 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 50-60 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 60-70 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 70-80 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 80-90 cm | Total Soil Content | lib/log-old.dai |
| Soil Content 90-100 cm | Total Soil Content | lib/log-old.dai |
| Hourly Water Flux | Matrix Water Flux | lib/log-old.dai |
| Daily Water Flux | Hourly Water Flux | lib/log-old.dai |
| Soil Nitrate Concentration | column | lib/log-old.dai |
| Soil Water NO3 Concentration | column | lib/log-old.dai |
| Root Zone Water Balance 0-100 cm | Root Zone Water Balance | lib/log-old.dai |
| N Balance 0-100 cm | N Balance | lib/log-old.dai |
| Monthly N Balance 100 cm | N Balance 0-100 cm | lib/log-old.dai |
| Soil Chemical | Soil Chemicals | lib/log-old.dai |
| Soil Chemicals 0-100 cm | Soil Chemicals | lib/log-old.dai |
| Soil Chemical 0-100 cm | Soil Chemical | lib/log-old.dai |
| Harvest | harvest | lib/log-old.dai |
| Evapotranspiration | simple | lib/log-old.dai |
| Surface Water Balance | column | lib/log-old.dai |
| column | table | lib/log-std.dai |
| crop | column | lib/log-std.dai |
| chemical | column | lib/log-std.dai |
| biopore | column | lib/log-std.dai |
| Crop | crop | lib/log-std.dai |
| Field water balance | balance | lib/log-std.dai |
| Field water | column | lib/log-std.dai |

| Name | Parent | File |
|---|---|---|
| Soil water balance | balance | lib/log-std.dai |
| Biopore water balance | balance | lib/log-std.dai |
| Matrix water balance | balance | lib/log-std.dai |
| Soil water | column | lib/log-std.dai |
| Field nitrogen balance | balance | lib/log-std.dai |
| Field mineral nitrogen balance | balance | lib/log-std.dai |
| Field crop nitrogen balance | balance | lib/log-std.dai |
| Field organic nitrogen balance | balance | lib/log-std.dai |
| Field nitrogen | column | lib/log-std.dai |
| N_balance | balance | lib/log-std.dai |
| Soil NO3- balance | N_balance | lib/log-std.dai |
| Soil NH4+ balance | N_balance | lib/log-std.dai |
| Soil mineral-N balance | N_balance | lib/log-std.dai |
| Soil organic-N balance | N_balance | lib/log-std.dai |
| Soil N balance | N_balance | lib/log-std.dai |
| Soil nitrogen | column | lib/log-std.dai |
| Field chemical | chemical | lib/log-std.dai |
| Soil chemical balance | balance | lib/log-std.dai |
| Biopore chemical balance | balance | lib/log-std.dai |
| Matrix chemical balance | balance | lib/log-std.dai |
| Soil chemical | chemical | lib/log-std.dai |
| Total Soil Content | column | lib/log.dai |
| Soil N Residuals | column | lib/log.dai |
| Soil C Residuals | column | lib/log.dai |
| NO3 Root Uptake | column | lib/log.dai |
| NH4 Root Uptake | column | lib/log.dai |
| Soil NO3 Concentration | column | lib/log.dai |
| Soil NO3-N Concentration | column | lib/log.dai |
| Soil NH4-N Concentration | column | lib/log.dai |
| Soil NH4-N Content | column | lib/log.dai |
| Soil Temperature | column | lib/log.dai |
| Soil Water Potential | column | lib/log.dai |
| Soil Water Potential (pF) | column | lib/log.dai |
| Soil Water Sink | column | lib/log.dai |
| Soil Hydraulic Conductivity (log10) | column | lib/log.dai |
| Soil Heat Conductivity | column | lib/log.dai |
| Soil Heat Capacity | column | lib/log.dai |
| Soil Ice | column | lib/log.dai |
| Soil Ice Buffer | column | lib/log.dai |
| Preferential Water Flux | column | lib/log.dai |
| Matrix Water Flux | column | lib/log.dai |
| Preferential Chemical Transport | chemical | lib/log.dai |
| Matrix Chemical Transport | chemical | lib/log.dai |
| Preferential NO3 Transport | column | lib/log.dai |
| Preferential NH4 Transport | column | lib/log.dai |
| Evapotranspiration | simple | lib/log.dai |
| Surface water balance2 | n/a | lib/log.dai |
| Surface Water Balance | column | lib/log.dai |
| Denitrification | column | lib/log.dai |
| Soil Heat Balance | column | lib/log.dai |
| Snow | column | lib/log.dai |
| Surface Chemicals | chemical | lib/log.dai |
| Chemical Content | chemical | lib/log.dai |
| Chemical Concentration | chemical | lib/log.dai |
| Chemical Content Secondary | Chemical Content | lib/log.dai |
| Chemical Concentration Secondary | Chemical Concentration | lib/log.dai |
| reaction | column | lib/log.dai |
| Chemistry | reaction | lib/log.dai |
| Weather | column | lib/log.dai |
| Crop Photosynthesis | crop | lib/log.dai |
| Crop Production | crop | lib/log.dai |

| Name | Parent | File |
|---|---|---|
| Forced LAI | crop | lib/log.dai |
| Simple Crop Production | crop | lib/log.dai |
| Old Crop Production | column | lib/log.dai |
| Vegetation | column | lib/log.dai |
| Permanent Vegetation | column | lib/log.dai |
| Root Density | crop | lib/log.dai |
| Root form parameter | crop | lib/log.dai |
| Bioclimate | column | lib/log.dai |
| PMSW | column | lib/log.dai |
| SSOC | column | lib/log.dai |
| Groundwater | column | lib/log.dai |
| Bioincorporation | column | lib/log.dai |
| aom | column | lib/log.dai |
| AOM | aom | lib/log.dai |
| AOM Pools | aom | lib/log.dai |
| Organic Matter | column | lib/log.dai |
| DOM Pools | column | lib/log.dai |
| pool | column | lib/log.dai |
| DOM Pool N Balance | balance | lib/log.dai |
| DOM Pool C Balance | balance | lib/log.dai |
| DOM Pool | pool | lib/log.dai |
| DOM-N | pool | lib/log.dai |
| DOM-C | pool | lib/log.dai |
| Drain Sink Water | column | lib/log.dai |
| Biopore Drain Sink Water | column | lib/log.dai |
| Drain Sink NO3 | column | lib/log.dai |
| Drain Sink NH4 | column | lib/log.dai |
| Soil Water Content | column | lib/log.dai |
| Fertilizer | column | lib/log.dai |
| Fertilizer Yearly | Fertilizer | lib/log.dai |
| Crop Carbon Balance | simple | lib/log.dai |
| Litter Carbon Balance | simple | lib/log.dai |
| Soil Carbon Balance | simple | lib/log.dai |
| Total Carbon Balance | simple | lib/log.dai |
| C Input | simple | lib/log.dai |
| Carbon Balance | crop | lib/log.dai |
| Tertiary biopores water | biopore | lib/log.dai |
| Matrix to biopores | column | lib/log.dai |
| Biopores to matrix | column | lib/log.dai |
| Surface water | balance | lib/log.dai |
| Snow water | balance | lib/log.dai |
| Canopy water | balance | lib/log.dai |
| Litter water | balance | lib/log.dai |
| Ponded water | balance | lib/log.dai |
| Surface water | column | lib/log.dai |
| Colloids balance | balance | lib/log.dai |
| Colloids pool balance | balance | lib/log.dai |
| Colloids | column | lib/log.dai |
| Failure | column | lib/log.dai |
| LeafPhot | crop | lib/log.dai |
| Stomatacon | LeafPhot | lib/log.dai |
| Time | column | lib/log.dai |
| Drain lateral | column | lib/log.dai |
| biopore-chemical | chemical | lib/log.dai |
| Biopore content | balance | lib/log.dai |
| Biopore chemical | biopore-chemical | lib/log.dai |
| Biopore water | biopore | lib/log.dai |
| M2B | biopore | lib/log.dai |
| B2M | biopore | lib/log.dai |
| Crop uptake | crop | lib/log.dai |
| QtTime | Qt | lib/ui-Qt.dai |

| Name | Parent | File |
|---|---|---|
| AWI | column | sample/PFAS-sample.dai |
| Afforestation | column | sample/afforestation.dai |
| SoilpH | column | sample/afforestation.dai |
| DailyWeather | table | sample/genweather.dai |
| MinMaxWeather | table | sample/genweather.dai |
| HourlyWeather | table | sample/genweather.dai |
| Sample water content | column | sample/log-sample.dai |
| Root zone water | crop | sample/safir-common.dai |
| TDR | crop | sample/safir-site.dai |
| ABAExtraction | crop | sample/test-aba.dai |
| H2OExtraction | crop | sample/test-aba.dai |
| ABACrop | crop | sample/test-aba.dai |
| Colloid bound sorption | balance | sample/test-colloids.dai |
| Colloid bound | balance | sample/test-colloids.dai |
| Colloid bound | chemical | sample/test-colloids.dai |
| compare | Rsqr | sample/test-minimize.dai |
| ABAExtraction | crop | sample/test-prd.dai |
| H2OExtraction | crop | sample/test-prd.dai |
| ABACrop | crop | sample/test-prd.dai |
| Root form GP2D | crop | sample/test-prd.dai |
| Root zone water | crop | sample/test-prd.dai |

# Chapter 39

# The weather of the bundled stations

The stations of the weather stations on the Input data page are described here from their files. Every number below is calculated from the daily values of the file: the sums of rain and reference evapotranspiration, the means of temperature and radiation, and the number of days.

## 39.1 CZRISWC03

| Item | Value |
|---|---|
| File | cz-valecov.dwf |
| Note | The file was compiled by F. Dolezal on 14/1/2006. |
| Elevation | 461 m |
| Longitude | 14.5 dgEast |
| Latitude | 49.64 dgNorth |
| TimeZone | 15 dgEast |
| Surface | reference |
| ScreenHeight | 2 m |
| Begin | 1991-1-1 |
| End | 2005-12-19 |
| NH4WetDep | 0.57 ppm |
| NH4DryDep | 8.5 kgN/ha/year |
| NO3WetDep | 0.5 ppm |
| NO3DryDep | 8.8 kgN/ha/year |
| TAverage | 7.8 dgC |
| TAmplitude | 10.1 dgC |
| MaxTDay | 201 yday |
| Timestep | 24 hours |
| PrecipCorrect | 1.21 1.19 1.23 1.13 1.06 1.07 1.08 1.08 1.08 1.12 1.19 1.17 |
| Columns | Year, Month, Day, GlobRad, AirTemp, Precip, RefEvap, RelHum, Wind |
| Days | 5467 |

### 39.1.1 Years of CZRISWC03

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1991 | 365 | 581 | 7.1 | 128.5 | 633 |
| 1992 | 366 | 582 | 8.6 | 135.1 | 718 |

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1993 | 365 | 618 | 8.1 | 134.1 | 716 |
| 1994 | 365 | 697 | 8.9 | 134.6 | 714 |
| 1995 | 365 | 718 | 7.7 | 128.7 | 626 |
| 1996 | 366 | 588 | 6.2 | 127.4 | 599 |
| 1997 | 365 | 665 | 8.0 | 137.7 | 675 |
| 1998 | 365 | 783 | 8.0 | 131.4 | 679 |
| 1999 | 365 | 583 | 7.9 | 126.1 | 653 |
| 2000 | 366 | 683 | 8.4 | 130.3 | 682 |
| 2001 | 365 | 872 | 7.2 | 118.9 | 601 |
| 2002 | 365 | 816 | 8.2 | 117.2 | 666 |
| 2003 | 365 | 539 | 7.4 | 129.5 | 745 |
| 2004 | 366 | 720 | 6.8 | 117.2 | 629 |
| 2005 | 353 | 730 | 7.4 | 128.6 | 624 |

### 39.1.2 Months of CZRISWC03

The mean of all years of the file.

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 1 | 37.0 | -1.9 | 39.0 | 9.9 |
| 2 | 32.6 | -0.9 | 70.8 | 16.7 |
| 3 | 46.7 | 2.8 | 109.1 | 34.8 |
| 4 | 38.1 | 7.2 | 167.3 | 63.3 |
| 5 | 74.2 | 12.6 | 221.0 | 99.1 |
| 6 | 71.0 | 15.4 | 226.7 | 107.3 |
| 7 | 112.4 | 17.3 | 220.1 | 114.8 |
| 8 | 84.0 | 17.7 | 198.7 | 104.7 |
| 9 | 54.6 | 12.7 | 133.1 | 60.8 |
| 10 | 42.6 | 8.0 | 79.9 | 32.4 |
| 11 | 41.1 | 2.7 | 39.3 | 12.9 |
| 12 | 44.0 | -1.6 | 29.4 | 7.6 |

## 39.2 Foulum

| Item | Value |
|---|---|
| File | dk-foulum.dwf |
| Elevation | 15 m |
| Longitude | 9 dgEast |
| Latitude | 56 dgNorth |
| TimeZone | 10 dgEast |
| Surface | reference |
| ScreenHeight | 2.0 m |
| Begin | 1990-01-01 |
| End | 2003-12-31 |
| Timestep | 24 hours |
| NH4WetDep | 0.9 ppm |
| NH4DryDep | 2.2 kgN/ha/year |
| NO3WetDep | 0.6 ppm |
| NO3DryDep | 1.1 kgN/ha/year |
| TAverage | 7.3 dgC |
| TAmplitude | 15.6 dgC |
| MaxTDay | 209 yday |

| Item | Value |
|---|---|
| Columns | Year, Month, Day, GlobRad, AirTemp, Precip, RefEvap |
| Days | 5113 |

### 39.2.1 Years of Foulum

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1990 | 365 | 859 | 8.8 | 113.8 | 586 |
| 1991 | 365 | 741 | 7.7 | 110.3 | 553 |
| 1992 | 366 | 768 | 8.4 | 114.8 | 598 |
| 1993 | 365 | 788 | 7.0 | 109.7 | 540 |
| 1994 | 365 | 914 | 8.1 | 114.0 | 581 |
| 1995 | 365 | 686 | 7.6 | 118.2 | 602 |
| 1996 | 366 | 495 | 6.4 | 111.7 | 555 |
| 1997 | 365 | 772 | 8.0 | 119.7 | 613 |
| 1998 | 365 | 889 | 7.8 | 99.4 | 499 |
| 1999 | 365 | 945 | 8.4 | 110.3 | 568 |
| 2000 | 366 | 883 | 8.6 | 102.0 | 519 |
| 2001 | 365 | 753 | 7.7 | 109.5 | 550 |
| 2002 | 365 | 911 | 8.5 | 113.6 | 593 |
| 2003 | 365 | 551 | 8.2 | 117.0 | 599 |

### 39.2.2 Months of Foulum

The mean of all years of the file.

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 1 | 72.0 | 1.4 | 19.8 | 6.1 |
| 2 | 63.9 | 1.5 | 43.7 | 12.3 |
| 3 | 57.5 | 3.1 | 90.9 | 30.7 |
| 4 | 46.6 | 6.4 | 148.4 | 54.9 |
| 5 | 42.9 | 10.8 | 211.0 | 91.2 |
| 6 | 58.0 | 13.4 | 222.5 | 98.7 |
| 7 | 63.4 | 16.0 | 218.8 | 106.0 |
| 8 | 74.4 | 16.3 | 177.4 | 86.5 |
| 9 | 78.8 | 12.4 | 107.6 | 46.6 |
| 10 | 82.3 | 8.1 | 57.9 | 22.8 |
| 11 | 69.8 | 4.1 | 24.1 | 8.1 |
| 12 | 72.8 | 1.3 | 14.1 | 4.3 |

## 39.3 Jyndevad

| Item | Value |
|---|---|
| File | dk-jyndevad.dwf |
| Elevation | 15 m |
| Longitude | 9 dgEast |
| Latitude | 54 dgNorth |
| TimeZone | 15 dgEast |
| Surface | reference |
| ScreenHeight | 2.0 m |

| Item | Value |
|---|---|
| Begin | 1997-01-01 |
| End | 2005-12-21 |
| Timestep | 24 hours |
| NH4WetDep | 0.9 ppm |
| NH4DryDep | 2.2 kgN/ha/year |
| NO3WetDep | 0.6 ppm |
| NO3DryDep | 1.1 kgN/ha/year |
| TAverage | 7.9 dgC |
| TAmplitude | 7.8 dgC |
| MaxTDay | 209 yday |
| PrecipCorrect | 1.41 1.42 1.35 1.24 1.13 1.11 1.10 1.10 1.11 1.14 1.23 1.37 |
| Columns | Year, Month, Day, GlobRad, AirTemp, Precip, Wind, RelHum |
| Days | 3277 |

### 39.3.1 Years of Jyndevad

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1997 | 365 | 812 | 8.7 | 116.7 | n/a |
| 1998 | 365 | 1085 | 8.7 | 104.0 | n/a |
| 1999 | 365 | 1083 | 9.2 | 115.5 | n/a |
| 2000 | 366 | 808 | 9.6 | 107.9 | n/a |
| 2001 | 365 | 840 | 8.5 | 111.6 | n/a |
| 2002 | 365 | 1002 | 9.5 | 109.4 | n/a |
| 2003 | 365 | 669 | 8.8 | 121.8 | n/a |
| 2004 | 366 | 976 | 8.9 | 110.6 | n/a |
| 2005 | 355 | 781 | 9.3 | 120.4 | n/a |

### 39.3.2 Months of Jyndevad

The mean of all years of the file.

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 1 | 71.2 | 1.8 | 22.0 | n/a |
| 2 | 71.9 | 2.4 | 48.2 | n/a |
| 3 | 61.5 | 3.9 | 97.1 | n/a |
| 4 | 50.3 | 7.6 | 151.2 | n/a |
| 5 | 56.7 | 11.9 | 209.3 | n/a |
| 6 | 79.8 | 14.6 | 219.1 | n/a |
| 7 | 95.7 | 16.8 | 201.7 | n/a |
| 8 | 78.6 | 17.3 | 175.5 | n/a |
| 9 | 66.9 | 14.1 | 118.5 | n/a |
| 10 | 107.9 | 9.4 | 63.2 | n/a |
| 11 | 71.5 | 5.2 | 26.9 | n/a |
| 12 | 83.2 | 2.5 | 17.0 | n/a |

## 39.4 Taastrup

| Item | Value |
|---|---|
| File | dk-taastrup.dwf |
| # weather data. The general information have form of keyword | value |
| Elevation | 30 m |
| Longitude | 12 dgEast |
| Latitude | 56 dgNorth |
| TimeZone | 15 dgEast |
| Surface | reference |
| ScreenHeight | 2.0 m |
| Begin | 1962-04-01 |
| End | 2008-10-31 |
| Timestep | 24 hours |
| Note | Precipitation from DMI Copenhagen used for 2002-06-02 to 2002-07-28. |
| NH4WetDep | 0.9 ppm |
| NH4DryDep | 2.2 kgN/ha/year |
| NO3WetDep | 0.6 ppm |
| NO3DryDep | 1.1 kgN/ha/year |
| TAverage | 7.8 dgC |
| TAmplitude | 8.5 dgC |
| MaxTDay | 209 yday |
| Columns | Year, Month, Day, GlobRad, AirTemp, Precip, RefEvap |
| Days | 17016 |

### 39.4.1 Years of Taastrup

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1962 | 275 | 507 | 9.3 | 135.0 | 519 |
| 1963 | 365 | 587 | 6.9 | 115.5 | 585 |
| 1964 | 366 | 506 | 7.3 | 117.0 | 584 |
| 1965 | 365 | 623 | 7.1 | 116.2 | 568 |
| 1966 | 365 | 742 | 7.2 | 106.9 | 486 |
| 1967 | 365 | 726 | 8.2 | 110.0 | 502 |
| 1968 | 366 | 559 | 7.6 | 116.6 | 552 |
| 1969 | 365 | 485 | 7.2 | 118.2 | 604 |
| 1970 | 365 | 658 | 6.5 | 113.4 | 546 |
| 1971 | 365 | 487 | 8.1 | 116.0 | 604 |
| 1972 | 366 | 549 | 7.7 | 110.3 | 526 |
| 1973 | 365 | 624 | 7.9 | 123.5 | 613 |
| 1974 | 365 | 613 | 8.2 | 117.5 | 620 |
| 1975 | 365 | 473 | 8.9 | 122.5 | 622 |
| 1976 | 366 | 416 | 7.6 | 125.2 | 615 |
| 1977 | 365 | 541 | 7.8 | 111.6 | 568 |
| 1978 | 365 | 603 | 7.3 | 117.7 | 618 |
| 1979 | 365 | 586 | 6.5 | 111.3 | 558 |
| 1980 | 366 | 673 | 6.8 | 111.6 | 547 |
| 1981 | 365 | 736 | 7.0 | 113.2 | 551 |
| 1982 | 365 | 603 | 8.0 | 123.6 | 612 |
| 1983 | 365 | 650 | 8.6 | 116.4 | 633 |
| 1984 | 366 | 621 | 8.1 | 106.1 | 583 |
| 1985 | 365 | 737 | 6.6 | 106.7 | 575 |
| 1986 | 365 | 576 | 7.2 | 112.1 | 604 |
| 1987 | 365 | 634 | 6.4 | 102.8 | 488 |
| 1988 | 366 | 580 | 8.4 | 109.1 | 580 |
| 1989 | 365 | 532 | 8.8 | 119.9 | 649 |
| 1990 | 365 | 641 | 9.0 | 116.5 | 617 |

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1991 | 365 | 671 | 7.7 | 108.5 | 519 |
| 1992 | 366 | 572 | 8.5 | 117.4 | 632 |
| 1993 | 365 | 727 | 7.2 | 110.0 | 563 |
| 1994 | 365 | 780 | 8.6 | 115.0 | 611 |
| 1995 | 365 | 607 | 8.5 | 117.2 | 632 |
| 1996 | 366 | 409 | 7.0 | 110.6 | 572 |
| 1997 | 365 | 628 | 8.4 | 118.4 | 618 |
| 1998 | 365 | 774 | 8.1 | 101.7 | 534 |
| 1999 | 365 | 702 | 8.8 | 113.5 | 601 |
| 2000 | 366 | 680 | 9.3 | 106.9 | 574 |
| 2001 | 365 | 627 | 8.3 | 109.7 | 579 |
| 2002 | 365 | 818 | 9.2 | 112.7 | 685 |
| 2003 | 365 | 632 | 8.7 | 118.5 | 695 |
| 2004 | 366 | 712 | 8.5 | 111.8 | 616 |
| 2005 | 365 | 552 | 8.7 | 115.6 | 608 |
| 2006 | 365 | 797 | 9.4 | 112.6 | 647 |
| 2007 | 365 | 986 | 9.6 | 107.6 | 761 |
| 2008 | 305 | 596 | 10.6 | 134.7 | 726 |

### 39.4.2 Months of Taastrup

The mean of all years of the file.

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 1 | 46.9 | 0.2 | 19.9 | 7.6 |
| 2 | 30.6 | 0.1 | 45.0 | 11.2 |
| 3 | 38.4 | 2.2 | 92.3 | 29.5 |
| 4 | 40.3 | 6.1 | 156.0 | 59.9 |
| 5 | 46.4 | 11.0 | 212.2 | 94.6 |
| 6 | 58.8 | 14.5 | 230.1 | 105.1 |
| 7 | 68.9 | 16.3 | 214.8 | 106.2 |
| 8 | 66.2 | 16.3 | 176.9 | 88.3 |
| 9 | 62.0 | 13.0 | 115.1 | 51.3 |
| 10 | 58.8 | 9.0 | 59.2 | 24.4 |
| 11 | 59.1 | 4.8 | 25.5 | 9.7 |
| 12 | 52.1 | 1.6 | 14.1 | 6.0 |

## 39.5 ITCER04 (Bologna)

| Item | Value |
|---|---|
| File | it-bologna.dwf |
| Elevation | 18 m |
| Longitude | 11 dgEast |
| Latitude | 41 dgNorth |
| TimeZone | 15 dgEast |
| Surface | reference |
| ScreenHeight | 2 m |
| Begin | 1994-01-01 |
| # PA | Fixed date format below |
| End | 2005-09-12 |
| NH4WetDep | 0 ppm |

| Item | Value |
|---|---|
| NH4DryDep | 0 kgN/ha/year |
| NO3WetDep | 0 ppm |
| NO3DryDep | 0 kgN/ha/year |
| TAverage | 14.03 dgC |
| TAmplitude | 11.04 dgC |
| MaxTDay | 208 yday |
| Timestep | 24 hours |
| PrecipCorrect | 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 |
| Columns | Year, Month, Day, GlobRad, AirTemp, Precip, RefEvap |
| Days | 4273 |

### 39.5.1 Years of ITCER04 (Bologna)

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1994 | 365 | 694 | 15.7 | 122.5 | 999 |
| 1995 | 365 | 888 | 13.9 | 124.6 | 813 |
| 1996 | 366 | 820 | 13.8 | 323.7 | 1519 |
| 1997 | 365 | 1053 | 13.9 | 193.9 | 1137 |
| 1998 | 365 | 514 | 13.6 | 209.7 | 966 |
| 1999 | 365 | 754 | 13.7 | 206.5 | 1139 |
| 2000 | 366 | 576 | 14.3 | 163.8 | 1052 |
| 2001 | 365 | 797 | 13.9 | 162.0 | 1040 |
| 2002 | 365 | 900 | 14.0 | 129.7 | 884 |
| 2003 | 365 | 654 | 14.5 | 145.1 | 1017 |
| 2004 | 366 | 919 | 13.4 | 138.0 | 853 |
| 2005 | 255 | 460 | 14.3 | 162.7 | 727 |

### 39.5.2 Months of ITCER04 (Bologna)

The mean of all years of the file.

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 1 | 36.9 | 3.2 | 89.3 | 34.6 |
| 2 | 43.1 | 5.4 | 126.1 | 49.2 |
| 3 | 47.8 | 9.3 | 157.8 | 71.7 |
| 4 | 82.7 | 13.0 | 180.7 | 82.3 |
| 5 | 44.2 | 18.2 | 241.7 | 117.4 |
| 6 | 65.3 | 23.3 | 280.8 | 145.3 |
| 7 | 48.5 | 24.8 | 280.2 | 154.3 |
| 8 | 76.5 | 24.5 | 228.6 | 138.7 |
| 9 | 76.2 | 19.1 | 180.2 | 93.2 |
| 10 | 68.7 | 14.5 | 137.8 | 62.2 |
| 11 | 87.1 | 8.4 | 95.4 | 36.4 |
| 12 | 75.6 | 3.8 | 66.3 | 26.8 |

## 39.6 Jadwisin

| Item | Value |
|---|---|
| File | pl-jadwisin.dwf |
| # Contact | Plant Breeding and Acclimatisation Institute, Department of |
| # Agronomy, Jadwisin, 05-140 Serock, Poland; e-mail | n/a |
| # weather data. The general information have form of keyword | value |
| #Comments | n/a |
| #Ra was estimated for that period according to Blacks formula \| Ra = Ro (0.209+0.565 S/So) \| \| Elevation \| 105 m \| \| Longitude \| 23 dgEast \| \| Latitude \| 52 dgNorth \| \| TimeZone \| 15 dgEast \| \| Surface \| reference \| \| ScreenHeight \| 2.0 m \| \| Begin \| 1992-1-01 \| \| End \| 2005-12-31 \| \| Timestep \| 24 hours \| \| Note \| lines, all starting withNote:’. | |
| Deposition | 16 kgN/ha/year |
| PAverage | 493 mm |
| TAverage | 7.9 dgC |
| TAmplitude | 11.75 dgC |
| MaxTDay | 211 yday |
| PrecipCorrect | 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 1.1 |
| Columns | Year, Month, Day, GlobRad, AirTemp, RelHum, Wind, Precip |
| Days | 5114 |

### 39.6.1 Years of Jadwisin

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1992 | 366 | 518 | 8.8 | 97.0 | n/a |
| 1993 | 365 | 499 | 7.8 | 97.9 | n/a |
| 1994 | 365 | 666 | 8.8 | 102.7 | n/a |
| 1995 | 365 | 575 | 8.3 | 100.6 | n/a |
| 1996 | 366 | 424 | 6.8 | 114.2 | n/a |
| 1997 | 365 | 578 | 7.9 | 104.6 | n/a |
| 1998 | 365 | 618 | 8.1 | 137.1 | n/a |
| 1999 | 365 | 586 | 8.7 | 113.2 | n/a |
| 2000 | 366 | 447 | 9.1 | 116.0 | n/a |
| 2001 | 365 | 540 | 7.8 | 109.8 | n/a |
| 2002 | 365 | 417 | 8.7 | 117.5 | n/a |
| 2003 | 365 | 475 | 7.6 | 121.3 | n/a |
| 2004 | 366 | 466 | 8.1 | 123.8 | n/a |
| 2005 | 365 | 444 | 8.0 | 130.0 | n/a |

### 39.6.2 Months of Jadwisin

The mean of all years of the file.

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 1 | 25.1 | -1.7 | 27.3 | n/a |
| 2 | 27.9 | -0.7 | 51.5 | n/a |
| 3 | 30.0 | 2.1 | 100.6 | n/a |
| 4 | 40.1 | 8.3 | 141.1 | n/a |
| 5 | 47.0 | 13.9 | 200.4 | n/a |
| 6 | 67.8 | 16.6 | 210.9 | n/a |

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 7 | 75.9 | 18.8 | 204.6 | n/a |
| 8 | 39.4 | 18.3 | 179.5 | n/a |
| 9 | 54.1 | 13.2 | 118.4 | n/a |
| 10 | 35.9 | 8.0 | 67.7 | n/a |
| 11 | 38.2 | 2.3 | 32.9 | n/a |
| 12 | 36.8 | -1.6 | 20.7 | n/a |

## 39.7 Bratislava

| Item | Value |
|---|---|
| File | sk-bratislava.dwf |
| # weather data. The general information have form of keyword | value |
| Elevation | 131 m |
| Longitude | 17 dgEast |
| Latitude | 48 dgNorth |
| TimeZone | 15 dgEast |
| Surface | reference |
| ScreenHeight | 2.0 m |
| Begin | 1990-01-01 |
| End | 2006-05-31 |
| Timestep | 24 hours |
| NH4WetDep | 1.3 ppm |
| NH4DryDep | 6.7 kgN/ha/year |
| NO3WetDep | 0.8 ppm |
| NO3DryDep | 4.3 kgN/ha/year |
| TAverage | 9.8 dgC |
| TAmplitude | 11.5 dgC |
| MaxTDay | 213 yday |
| PrecipCorrect | 1.29 1.29 1.31 1.17 1.11 1.10 1.12 1.11 1.10 1.16 1.25 1.25 |
| Columns | Year, Month, Day, GlobRad, AirTemp, Precip, VapPres, Wind |
| Days | 5995 |

### 39.7.1 Years of Bratislava

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 1990 | 365 | 507 | 10.9 | 159.3 | n/a |
| 1991 | 365 | 596 | 9.8 | 134.8 | n/a |
| 1992 | 366 | 499 | 11.3 | 143.6 | n/a |
| 1993 | 365 | 527 | 10.0 | 143.5 | n/a |
| 1994 | 365 | 582 | 11.6 | 139.5 | n/a |
| 1995 | 365 | 820 | 10.3 | 135.0 | n/a |
| 1996 | 366 | 605 | 9.0 | 134.6 | n/a |
| 1997 | 365 | 518 | 10.0 | 144.8 | n/a |
| 1998 | 365 | 556 | 11.0 | 137.6 | n/a |
| 1999 | 365 | 624 | 10.9 | 135.2 | n/a |
| 2000 | 366 | 529 | 11.8 | 148.2 | n/a |
| 2001 | 365 | 506 | 10.7 | 138.1 | n/a |
| 2002 | 365 | 689 | 11.6 | 131.5 | n/a |
| 2003 | 365 | 466 | 11.3 | 154.1 | n/a |

| Year | Days | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapo-transpiration, mm |
|---|---|---|---|---|---|
| 2004 | 366 | 611 | 10.6 | 133.6 | n/a |
| 2005 | 365 | 640 | 10.4 | 139.3 | n/a |
| 2006 | 151 | 296 | 5.4 | 125.9 | n/a |

### 39.7.2 Months of Bratislava

The mean of all years of the file.

| Month | Rain, mm | Mean temperature, C | Mean radiation, W per m2 | Reference evapotranspiration, mm |
|---|---|---|---|---|
| 1 | 30.3 | -0.2 | 40.7 | n/a |
| 2 | 32.4 | 1.6 | 82.9 | n/a |
| 3 | 37.3 | 5.7 | 123.6 | n/a |
| 4 | 46.0 | 10.9 | 179.4 | n/a |
| 5 | 54.9 | 16.2 | 236.9 | n/a |
| 6 | 55.8 | 19.5 | 250.0 | n/a |
| 7 | 61.1 | 21.2 | 240.5 | n/a |
| 8 | 55.8 | 21.3 | 215.3 | n/a |
| 9 | 57.5 | 15.6 | 146.9 | n/a |
| 10 | 41.6 | 10.4 | 91.1 | n/a |
| 11 | 46.8 | 5.1 | 44.8 | n/a |
| 12 | 43.7 | 0.3 | 31.7 | n/a |

# Chapter 40

# ☰ Index of the components of the Daisy reference

Every component named in the sections of the 107 topics, in alphabetical order, with the topic and the section where the reference describes it. There are 1498 entries.

| Component | Topic | Section | Section name |
|---|---|---|---|
| a | 10 | 10.1 | biomod |
| a | 16 | 16.2 | Guo2020 |
| a | 25 | 25.1 | clear sky |
| a | 35 | 35.2 | DPF |
| a | 48 | 48.17 | hyprop |
| a | 48 | 48.18 | linear |
| A | 57 | 57.1 | brunt |
| A | 70 | 70.10 | default |
| A | 70 | 70.12 | equilibrium |
| a | 73 | 73.1 | Anders Pedersen |
| a | 73 | 73.5 | GP1D |
| a | 73 | 73.7 | Gerwitz+Page74 |
| A | 81 | 81.1 | diffusion |
| a | 90 | 90.2 | linear |
| A | 107 | 107.3 | SoluteUptake2Result |
| A solute | 39 | 39.2 | goal A |
| ABA effect | 64 | 64.1 | Farquhar |
| ABA min | 84 | 84.1 | WSF |
| accumulate | 83 | 83.3 | combine |
| accumulate | 83 | 83.4 | merge |
| action | 15 | 15.8 | extern |
| actions | 15 | 15.4 | cond |
| actions | 15 | 15.32 | progn |
| actions | 15 | 15.61 | while |
| actions | 15 | 15.62 | with-column |
| active | 89 | 89.1 | biopores |
| active msg | 89 | 89.1 | biopores |
| Actual evapotranspiration | 52 | 52.8 | Field water |
| adsorption | 23 | 23.1 | base |
| adsorption rate | 70 | 70.3 | adsorption |
| adsorption source | 70 | 70.3 | adsorption |
| Ae | 70 | 70.9 | colgen Styczen88 |
| age | 107 | 107.24 | Snow |
| AirTemp | 103 | 103.2 | base |
| AirTemp | 107 | 107.26 | Weatherdata |
| albedo | 51 | 51.2 | permanent |
| albedo | 51 | 51.3 | residue |

| Component | Topic | Section | Section name |
|---|---|---|---|
| albedo | 88 | 88.1 | PMSW |
| Albedo | 100 | 100.1 | afforestation |
| Albedo | 100 | 100.3 | permanent |
| albedo dry | 87 | 87.1 | default |
| albedo wet | 87 | 87.1 | default |
| allow sideways | 27 | 27.4 | pressure |
| alpha | 23 | 23.2 | FOCUS |
| alpha | 48 | 48.5 | B vG |
| alpha | 48 | 48.7 | MACRO |
| alpha | 48 | 48.12 | M vG |
| alpha | 48 | 48.14 | M vGip |
| alpha | 48 | 48.15 | M vGp |
| alpha | 48 | 48.17 | hyprop |
| alpha | 51 | 51.6 | mulch |
| alpha | 64 | 64.3 | FC C4 |
| alpha | 70 | 70.11 | denitrification |
| alpha | 78 | 78.1 | alpha |
| alpha | 81 | 81.1 | diffusion |
| alpha | 84 | 84.5 | MNA |
| alpha | 84 | 84.6 | SHA12 |
| alpha | 84 | 84.7 | SHA14 |
| alpha | 107 | 107.2 | SoluteUptakeResult |
| alpha | 107 | 107.3 | SoluteUptake2Result |
| alpha fast | 70 | 70.11 | denitrification |
| alpha u | 88 | 88.1 | PMSW |
| am | 15 | 15.11 | fertilize base |
| am | 15 | 15.12 | fertilize |
| am | 15 | 15.13 | incorporate fertilizer |
| am | 60 | 60.1 | default |
| amount | 15 | 15.5 | crop |
| amount | 15 | 15.47 | spray |
| amount | 15 | 15.48 | spray surface |
| amount | 28 | 28.36 | soil inorganic N above |
| aperture | 78 | 78.2 | cracks |
| arac | 88 | 88.1 | PMSW |
| As | 70 | 70.6 | colgen Jarvis99 |
| at | 28 | 28.17 | hour |
| at | 28 | 28.20 | mday |
| at | 28 | 28.21 | microsecond |
| at | 28 | 28.22 | minute |
| at | 28 | 28.28 | month |
| at | 28 | 28.34 | second |
| at | 28 | 28.49 | yday |
| at | 28 | 28.50 | year |
| average depth | 48 | 48.22 | wepp |
| b | 16 | 16.2 | Guo2020 |
| b | 48 | 48.3 | B C |
| b | 48 | 48.11 | M C |
| b | 48 | 48.18 | linear |
| b | 48 | 48.19 | mod C |
| b | 65 | 65.1 | EUROSEM |
| b | 84 | 84.2 | BB base |
| b | 84 | 84.5 | MNA |
| b | 90 | 90.2 | linear |
| B solute | 39 | 39.2 | goal A |
| base | 59 | 59.35 | pow |
| base | 96 | 96.206 | factor |
| base | 96 | 96.212 | offset |
| bbch | 28 | 28.1 | BBCH after |
| before | 33 | 33.1 | const |
| before | 44 | 44.7 | multi |

| Component | Topic | Section | Section name |
|---|---|---|---|
| before | 66 | 66.20 | post-process |
| begin | 44 | 44.4 | time |
| begin | 44 | 44.5 | vector |
| begin | 59 | 59.38 | source |
| begin | 66 | 66.11 | cpedata |
| begin | 66 | 66.23 | weather |
| begin | 103 | 103.8 | combine |
| begin | 105 | 105.3 | flux |
| begin | 105 | 105.6 | loop |
| beta | 64 | 64.2 | FC C3 |
| beta | 64 | 64.3 | FC C4 |
| beta | 81 | 81.1 | diffusion |
| beta | 84 | 84.1 | WSF |
| beta | 107 | 107.2 | SoluteUptakeResult |
| beta | 107 | 107.3 | SoluteUptake2Result |
| Bioclimate | 26 | 26.1 | default |
| Bioincorporation | 60 | 60.1 | default |
| biopore | 52 | 52.11 | biopore |
| Biopore infiltration | 52 | 52.10 | Soil water |
| Biopore percolation | 52 | 52.8 | Field water |
| Biopore percolation | 52 | 52.10 | Soil water |
| Biopore-Drain | 52 | 52.7 | Field nitrogen |
| Biopore-Leaching | 52 | 52.7 | Field nitrogen |
| bottom | 44 | 44.3 | soil |
| bottom | 44 | 44.5 | vector |
| bottom | 66 | 66.20 | post-process |
| bottom | 102 | 102.1 | box |
| bottom | 106 | 106.1 | box |
| c | 48 | 48.18 | linear |
| C e | 12 | 12.1 | default |
| c e | 12 | 12.1 | default |
| C fraction | 79 | 79.2 | release |
| C root | 107 | 107.2 | SoluteUptakeResult |
| C root | 107 | 107.3 | SoluteUptake2Result |
| c Vm | 64 | 64.2 | FC C3 |
| Canopy | 29 | 29.1 | default |
| Canopy | 29 | 29.47 | simple |
| Canopy | 100 | 100.1 | afforestation |
| Canopy | 100 | 100.3 | permanent |
| canopy dissipation | 23 | 23.1 | base |
| canopy height | 100 | 100.1 | afforestation |
| canvas | 44 | 44.1 | common |
| CEst | 81 | 81.3 | variable sink |
| CH2OReleaseRate | 107 | 107.9 | Production |
| classes | 89 | 89.1 | biopores |
| clauses | 15 | 15.4 | cond |
| clauses | 49 | 49.4 | cond |
| clauses | 59 | 59.29 | let |
| clauses | 85 | 85.1 | cond |
| clay | 47 | 47.1 | BSI3 |
| clay | 47 | 47.4 | BSI7 |
| clay | 47 | 47.6 | DIN5 |
| clay | 47 | 47.7 | GEUS7 |
| clay | 47 | 47.8 | ISSS3 |
| clay | 47 | 47.9 | ISSS4 |
| clay | 47 | 47.10 | USDA3 |
| clay | 47 | 47.12 | USDA7 |
| clay | 47 | 47.14 | USPRA3 |
| clay | 47 | 47.15 | USPRA4 |
| cloudiness | 19 | 19.1 | base |
| cloudiness | 25 | 25.1 | clear sky |

| Component | Topic | Section | Section name |
|---|---|---|---|
| cloudiness | 25 | 25.6 | const |
| cloudiness | 25 | 25.7 | weather |
| coarse silt | 47 | 47.4 | BSI7 |
| coarse silt | 47 | 47.7 | GEUS7 |
| column | 15 | 15.62 | with-column |
| column | 44 | 44.2 | profile |
| column | 52 | 52.6 | column |
| column | 59 | 59.14 | depth |
| column | 66 | 66.2 | Daisy |
| column | 107 | 107.15 | Harvest |
| columns | 66 | 66.14 | extract |
| combine | 15 | 15.14 | harvest base |
| combine | 24 | 24.8 | multi |
| command file | 66 | 66.15 | gnuplot |
| command file | 66 | 66.16 | hmovie |
| component | 66 | 66.12 | docmodel |
| condition | 15 | 15.2 | assert |
| condition | 15 | 15.4 | cond |
| condition | 15 | 15.55 | wait |
| condition | 28 | 28.48 | with |
| condition | 107 | 107.28 | IntegerCondClause |
| condition | 107 | 107.29 | StringerCondClause |
| conductivity | 107 | 107.22 | SoilHeat |
| constituents | 15 | 15.10 | extern subsoil |
| content | 86 | 86.3 | balance |
| converted fast | 70 | 70.11 | denitrification |
| creation | 17 | 17.1 | base |
| CrLeafCnc | 107 | 107.6 | CrpN |
| CRoot | 81 | 81.2 | fixed sink |
| CRoot | 81 | 81.3 | variable sink |
| crop | 15 | 15.1 | BBCH |
| crop | 15 | 15.5 | crop |
| crop | 15 | 15.6 | emerge |
| crop | 15 | 15.14 | harvest base |
| crop | 15 | 15.31 | pluck |
| crop | 15 | 15.43 | sow |
| crop | 15 | 15.45 | sow name |
| crop | 15 | 15.46 | sow object |
| crop | 17 | 17.1 | base |
| crop | 28 | 28.1 | BBCH after |
| crop | 28 | 28.6 | crop dm over |
| crop | 28 | 28.7 | crop dm sorg over |
| crop | 28 | 28.8 | crop ds after |
| crop | 28 | 28.9 | crop stage after |
| crop | 52 | 52.15 | crop |
| crop | 70 | 70.16 | shoot |
| crop | 70 | 70.17 | shoot2 |
| crop | 107 | 107.15 | Harvest |
| crop depth | 66 | 66.3 | GP2D |
| crop width | 66 | 66.3 | GP2D |
| crops | 100 | 100.2 | crops |
| CrSOrgCnc | 107 | 107.6 | CrpN |
| CrStemCnc | 107 | 107.6 | CrpN |
| crust thickness | 48 | 48.22 | wepp |
| cut delay | 107 | 107.7 | Harvesting |
| cut stress | 107 | 107.7 | Harvesting |
| D | 70 | 70.5 | colgen |
| D | 81 | 81.1 | diffusion |
| D | 107 | 107.2 | SoluteUptakeResult |
| D | 107 | 107.3 | SoluteUptake2Result |
| Daisy | 15 | 15.22 | markvand |

| Component | Topic | Section | Section name |
|---|---|---|---|
| data | 86 | 86.1 | Rsqr |
| data | 86 | 86.2 | RsqrW |
| date | 15 | 15.5 | crop |
| day | 15 | 15.3 | at |
| day | 15 | 15.5 | crop |
| day | 15 | 15.56 | wait mm dd |
| day | 28 | 28.24 | mm dd base |
| day | 29 | 29.47 | simple |
| day | 66 | 66.11 | cpedata |
| day count | 48 | 48.22 | wepp |
| day length | 103 | 103.1 | weather |
| days | 15 | 15.18 | irrigate base |
| days | 15 | 15.57 | wait period |
| days | 28 | 28.13 | every |
| days | 28 | 28.46 | walltime |
| days | 107 | 107.25 | Timestep |
| dds | 70 | 70.5 | colgen |
| death rate | 73 | 73.9 | local |
| debug | 20 | 20.2 | matrix |
| debug | 46 | 46.1 | Mollerup |
| debug | 48 | 48.18 | linear |
| debug | 66 | 66.11 | cpedata |
| debug | 66 | 66.21 | rootmatch |
| debug | 73 | 73.6 | GP2D |
| debug | 92 | 92.2 | Mollerup |
| debug | 98 | 98.3 | richards |
| debug | 98 | 98.4 | richards2 |
| debug | 99 | 99.1 | Mollerup |
| Decompose | 52 | 52.14 | Soil chemical |
| decompose height | 51 | 51.6 | mulch |
| default | 94 | 94.3 | simple |
| default h | 107 | 107.23 | SoilWater |
| delta | 84 | 84.1 | WSF |
| delta pc | 48 | 48.22 | wepp |
| DensIgnore | 73 | 73.5 | GP1D |
| DensIgnore | 73 | 73.6 | GP2D |
| density | 51 | 51.6 | mulch |
| density | 73 | 73.2 | DS Depth |
| density | 73 | 73.3 | DS Rel |
| density | 73 | 73.4 | Depth Depth |
| density | 78 | 78.2 | cracks |
| density | 80 | 80.5 | flow |
| density | 80 | 80.16 | volume base |
| density y | 80 | 80.16 | volume base |
| DensRtTip | 66 | 66.21 | rootmatch |
| DensRtTip | 73 | 73.5 | GP1D |
| DensRtTip | 73 | 73.6 | GP2D |
| DensRtTip | 73 | 73.7 | Gerwitz+Page74 |
| DensRtTip | 73 | 73.8 | growth |
| DensRtTip | 73 | 73.9 | local |
| Deposit | 52 | 52.13 | Field chemical |
| deposit | 103 | 103.1 | weather |
| deposition | 19 | 19.1 | base |
| Deposition | 52 | 52.7 | Field nitrogen |
| depth | 15 | 15.24 | mix |
| depth | 15 | 15.39 | set porosity |
| depth | 15 | 15.51 | swap |
| depth | 17 | 17.4 | root |
| Depth | 52 | 52.16 | Crop |
| Depth | 73 | 73.8 | growth |
| Depth | 73 | 73.9 | local |

| Component | Topic | Section | Section name |
|---|---|---|---|
| desorption rate | 70 | 70.3 | adsorption |
| DetentionCapacity | 87 | 87.1 | default |
| device | 44 | 44.1 | common |
| diffusion coefficient | 107 | 107.1 | DOM |
| dim | 66 | 66.16 | hmovie |
| dim line | 15 | 15.54 | table |
| dim line | 34 | 34.7 | table |
| dim line | 48 | 48.21 | table |
| dim line | 66 | 66.11 | cpedata |
| dim line | 66 | 66.14 | extract |
| dim line | 66 | 66.16 | hmovie |
| dim line | 83 | 83.1 | arithmetic |
| dim line | 83 | 83.2 | column |
| dim line | 103 | 103.6 | table |
| dim line | 105 | 105.1 | arithmetic |
| dim line | 105 | 105.3 | flux |
| dim line | 105 | 105.4 | frequency |
| dim line | 105 | 105.8 | profile |
| dimension | 40 | 40.2 | number |
| dimension | 44 | 44.3 | soil |
| dimension | 44 | 44.5 | vector |
| dimension | 59 | 59.7 | convert |
| dimension | 59 | 59.8 | dim |
| dimension | 59 | 59.9 | identity |
| dimension | 59 | 59.19 | get |
| dimension | 66 | 66.20 | post-process |
| dimension | 83 | 83.2 | column |
| dimension | 83 | 83.4 | merge |
| dimension | 105 | 105.3 | flux |
| dimension | 105 | 105.8 | profile |
| dint file | 66 | 66.7 | WG2DWF |
| directory | 66 | 66.9 | batch |
| directory | 66 | 66.10 | cd |
| directory | 66 | 66.19 | nwaps |
| directory | 66 | 66.22 | spawn |
| directory | 107 | 107.30 | Toplevel |
| dispersivity | 107 | 107.21 | Soil |
| Dissipate | 52 | 52.13 | Field chemical |
| dist | 17 | 17.4 | root |
| distribution | 53 | 53.1 | default |
| distribution | 107 | 107.11 | Bioincorporation |
| DM fraction | 79 | 79.2 | release |
| do | 15 | 15.1 | BBCH |
| do | 15 | 15.3 | at |
| do | 15 | 15.35 | repeat |
| do | 15 | 15.37 | sequence |
| Do | 84 | 84.4 | Leuning |
| DOC name | 51 | 51.6 | mulch |
| DOC name | 70 | 70.1 | DOM turnover |
| domain | 59 | 59.34 | plf |
| domsorp | 60 | 60.1 | default |
| DON name | 51 | 51.6 | mulch |
| DON name | 70 | 70.1 | DOM turnover |
| Drain | 26 | 26.1 | default |
| Drain-Biopores | 52 | 52.14 | Soil chemical |
| Drain-Biopores-Indirect | 52 | 52.14 | Soil chemical |
| drainpoints | 56 | 56.2 | rectangle |
| dry | 33 | 33.1 | const |
| dry | 48 | 48.18 | linear |
| ds | 28 | 28.8 | crop ds after |
| DSLim | 101 | 101.1 | default |

| Component | Topic | Section | Section name |
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| DSMature | 63 | 63.2 | default |
| DSRepeat | 63 | 63.2 | default |
| dt | 15 | 15.22 | markvand |
| dwfdaily file | 66 | 66.7 | WG2DWF |
| dwfhourly file | 66 | 66.6 | RS2WG |
| dZs | 107 | 107.24 | Snow |
| E Root | 107 | 107.9 | Production |
| E SOrg | 107 | 107.9 | Production |
| E Stem | 107 | 107.9 | Production |
| Ea Gamma | 64 | 64.1 | Farquhar |
| Ea Jm | 64 | 64.2 | FC C3 |
| Ea kc | 64 | 64.2 | FC C3 |
| Ea ko | 64 | 64.2 | FC C3 |
| Ea Vm | 64 | 64.2 | FC C3 |
| Eda Vm | 64 | 64.2 | FC C3 |
| efficiency | 70 | 70.1 | DOM turnover |
| efficiency | 107 | 107.1 | DOM |
| electric current | 96 | 96.1 | SI |
| else | 15 | 15.17 | if |
| else | 28 | 28.19 | if |
| else | 59 | 59.25 | if |
| emerge | 15 | 15.54 | table |
| emerge time | 29 | 29.1 | default |
| emerge time | 107 | 107.15 | Harvest |
| emerging | 73 | 73.9 | local |
| EmrThrs | 63 | 63.1 | TSum |
| EmrTSum | 63 | 63.1 | TSum |
| EmrTSum | 63 | 63.2 | default |
| enable ice | 107 | 107.22 | SoilHeat |
| enable irrigation | 15 | 15.54 | table |
| end time | 15 | 15.57 | wait period |
| entries | 52 | 52.3 | select |
| entries | 73 | 73.2 | DS Depth |
| entries | 73 | 73.3 | DS Rel |
| entries | 73 | 73.4 | Depth Depth |
| entries | 76 | 76.1 | exchange |
| entry | 103 | 103.8 | combine |
| EPext | 107 | 107.4 | CanopySimple |
| EPext | 107 | 107.5 | CanopyStandard |
| EpFac | 107 | 107.4 | CanopySimple |
| EpFac | 107 | 107.5 | CanopyStandard |
| EpFacWet | 107 | 107.4 | CanopySimple |
| EpFacWet | 107 | 107.5 | CanopyStandard |
| epsilon | 66 | 66.18 | minimize |
| eq depth | 37 | 37.1 | lateral |
| equilibrium | 70 | 70.3 | adsorption |
| equilibrium | 70 | 70.12 | equilibrium |
| equilibrium | 91 | 91.1 | equilibrium |
| Equillibrium parameter | 70 | 70.18 | sorption |
| equivalent weight | 15 | 15.11 | fertilize base |
| event | 107 | 107.10 | Irrigation |
| Exception | 6 | 6.4 | Runtime exceptions |
| exchange | 66 | 66.2 | Daisy |
| exe | 66 | 66.22 | spawn |
| exponent | 59 | 59.35 | pow |
| extinction coefficent | 51 | 51.3 | residue |
| extra | 16 | 16.4 | Python |
| extra | 44 | 44.1 | common |
| extra | 66 | 66.15 | gnuplot |
| extra | 70 | 70.2 | Python |
| f c | 107 | 107.24 | Snow |

| Component | Topic | Section | Section name |
|---|---|---|---|
| f photo | 75 | 75.2 | expr |
| f photo | 75 | 75.4 | uniform |
| F primary | 70 | 70.13 | filter velocity |
| F relative | 81 | 81.3 | variable sink |
| F secondary | 70 | 70.13 | filter velocity |
| f3const | 88 | 88.1 | PMSW |
| fa | 35 | 35.2 | DPF |
| fc primary | 70 | 70.13 | filter velocity |
| fc secondary | 70 | 70.13 | filter velocity |
| fetch | 86 | 86.22 | simple |
| file | 15 | 15.1 | BBCH |
| file | 15 | 15.54 | table |
| file | 28 | 28.1 | BBCH after |
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| gnuplot | 66 | 66.21 | rootmatch |
| goal A | 39 | 39.2 | goal A |
| graph | 44 | 44.7 | multi |
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| h ice | 80 | 80.19 | water |
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| K sat | 48 | 48.9 | M BaC Bimodal |
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| Pscb PAR | 67 | 67.2 | sun-shade |
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| Q10vm | 64 | 64.3 | FC C4 |
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| ref n | 48 | 48.13 | M vG compact |
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| release | 70 | 70.4 | bound release |
| RelHum | 103 | 103.2 | base |
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| retention | 51 | 51.6 | mulch |
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| rho s | 107 | 107.24 | Snow |
| right | 44 | 44.3 | soil |
| right | 44 | 44.5 | vector |
| right | 66 | 66.20 | post-process |
| right | 102 | 102.1 | box |
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| ripe time | 107 | 107.15 | Harvest |
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| solute | 107 | 107.10 | Irrigation |
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| z o | 12 | 12.1 | default |
| z top | 98 | 98.1 | lr |
| z xA | 11 | 11.1 | default |
| zeta | 88 | 88.1 | PMSW |
| zones | 107 | 107.21 | Soil |
| zplus | 107 | 107.16 | Geometry1D |
| zplus | 107 | 107.17 | GeometryRect |

# Chapter 41

# ⚠ Limits, messages and glossary of the Daisy workbench

## 41.1 What the workbench can and cannot do

| Question | Answer |
|---|---|
| Which runs need the Daisy program? | The scenarios, the experiments, the Processes figures that are drawn from a new run, the set-up editor run and the calibration against the program. They need the desktop app. Daisy 7.1.14 comes with SWAFL (the folder `daisy`), so nothing has to be installed. |
| The run says *spawn ...daisy-7.1.xx-Windows-python3.13.exe EACCES*, or *Daisy gave no results (code -1)* | The file chosen as the Daisy program was the Daisy **installer** (the file that is downloaded from the Daisy site). An installer asks for administrator rights, so Windows refuses to run it as the program (EACCES). Choose nothing: SWAFL uses its own Daisy. If you want your own installation, run the installer once, then in *Choose the Daisy program* pick `bin\daisy-bin.exe` in the folder where it installed Daisy (for example `C:\Program Files\Daisy\bin\daisy-bin.exe`). This version of SWAFL ignores an installer that was chosen earlier, and says so when the installer is chosen. |
| Which tools work without it? | The soil column, the process curves, the denitrification calibration, the Plot, Mass balance and Compare tabs for logs that are loaded, the Library and the Weather stations tabs. |
| Does the web version run Daisy? | No. A web page cannot start a program on the computer. The web version draws the tools above and shows the screens; the runs are made in the desktop app. |
| How many parameters can be varied? | Thirty two parameters of the Daisy reference are tested and can be varied in an experiment, together with the factors of the farm such as rain, temperature, irrigation trigger and fertiliser. The other parameters of the 107 topics are described but not varied. |
| Which years can be simulated? | Daisy starts one year after the first year of the weather. A weather record of at least six years is therefore needed. |
| Is online weather complete? | Only complete years up to last year are asked for. A request that ends in the current year is refused by the weather service. |
| What are the sources of online weather? | Daily data of the Open-Meteo archive, which is based on the ERA5 reanalysis. They are checked before use and the checks are shown. |

## 41.2 Messages and what to do

| Message | Meaning | What to do |
|---|---|---|
| No data to draw from yet | The Process map tab has no run to read. | Press Run Daisy with my data and all process logs, run a scenario, or load logs in Files. |
| Not used: radiation is not in the data | The weather has no radiation column, and the latitude is not known. | Give the place in Input data, so that the latitude is found, or type the latitude. |
| Ask for at least six years | The weather record is too short for Daisy. | Choose a longer period of years. |
| No connection to the weather service | The internet is not reachable. | Use a weather file in Input data, or try again when online. |
| The weather service answered HTTP 400 | The request was refused, for example because it ended after the last complete year. | Choose a last year before the current year. |
| Daisy is not available here | The program is not found or the web version is open. | Open the desktop app and choose the Daisy program once in Scenarios. |

## 41.3 Glossary

**Action.** A management operation of a set-up, such as sowing, tillage, fertilising or irrigation.

**Biopore.** A channel in the soil, made by roots or worms, through which water and solutes move faster than through the soil matrix.

**Bioclimate.** The part of Daisy that shares energy and water between canopy, litter, snow, pond and soil.

**Calibration.** The search for parameter values that bring the results of the model close to measurements or to results of another model.

**Combination.** A set of factors that the Start from the Daisy reference list offers to compare, with the reason it is offered.

**Component.** A named part of the Daisy model with its own parameters, such as a crop, a horizon or a log.

**Design.** The plan of an experiment: one factor at a time, factorial, Latin hypercube or another plan.

**Experiment.** A set of runs that differ in chosen factors, designed and run together.

**Factor.** A quantity that is changed from run to run, with the levels that are tried.

**Field capacity.** The water content that a soil holds against drainage, near pF 2.

**Horizon.** A layer of the soil profile with its own texture and hydraulic properties.

**Log.** A file in which Daisy writes the values of chosen variables through time; the extension is dlf.

**Mass balance.** The check that what enters a system, what leaves it and what it holds agree.

**pF.** The common logarithm of the suction of soil water in cm. Field capacity is near pF 2 and the wilting point at pF 4.2.

**Process.** A part of the model drawn in the map of the Process map tab, such as soil water or nitrate.

**Response.** A result measured in every run, such as yield, irrigation or nitrate leached.

**Richards equation.** The equation of water movement in unsaturated soil, solved by the soil column and by Daisy.

**Raw output.** The files Daisy wrote in a run, exactly as written: the logs (.dlf), the .csv files and daisy.log. SWAFL keeps them in the folder SWAFL runs next to Daisy and gives them for download under the figures.

**Scenario.** A ready question about the built-in farm with the runs that answer it.

**Set-up.** The file, with the extension dai, that describes a whole simulation.

**Trigger.** The soil water potential at which irrigation starts.

**van Genuchten and Mualem model.** The description of the retention and conductivity curves of a soil used by the horizons of the library.

**Weather station.** A weather file in the Daisy format, with the extension dwf.

**Wilting point.** The water content at which plants cannot take up water, near pF 4.2.

# Symbols and units

Table A.1. Symbols used in this manual, grouped by chapter of first use. Units are SI-derived as shown; $[-]$ is dimensionless.

| Symbol | Meaning | Unit |
|---|---|---|
| $\theta$, $\theta_r$, $\theta_s$ | Volumetric water content; residual; saturated | $\mathrm{m^3\ m^{-3}}$ |
| $\theta_{fc}$, $\theta_{wp}$ | Water content at field capacity; at wilting point | $\mathrm{m^3\ m^{-3}}$ |
| $h$, $\psi$ | Soil-water suction (positive); matric potential | cm; kPa |
| $\alpha$, $n$, $m$ | van Genuchten parameters ($m = 1 - 1/n$) | $\mathrm{cm^{-1}}$, [1], [1] |
| $K_s$, $K(h)$ | Saturated and unsaturated hydraulic conductivity | $\mathrm{mm\ h^{-1}}$ |
| AWC | Plant-available water capacity | $\mathrm{mm\ cm^{-1}}$ |
| $f$, $f_T$ | Relative depletion of the sensor layer; trigger depletion | [1] |
| $R_{\max}$, $R_d$ | Maximum and present rooting depth | cm |
| $P$, $I$, $E$, $T$, $D$ | Rain, irrigation, soil evaporation, transpiration, drainage | $\mathrm{mm\ d^{-1}}$ or $\mathrm{mm\ a^{-1}}$ |
| $E_0$ | Atmospheric evaporative demand | $\mathrm{mm\ d^{-1}}$ |
| $\mathrm{ET_{pk}}$ | Peak 5-day mean of $E + T$ | $\mathrm{mm\ d^{-1}}$ |
| $k_t$ | Calibrated transpiration coefficient of Engine 2 | [1] |
| $Z_n$ | Required (net) depth per irrigation | mm |
| $Z(\tau)$, $F(t)$ | Cumulative infiltration (Kostiakov; Green and Ampt) | mm |
| $k$, $A$ | Kostiakov intake constants | $\mathrm{mm\ min^{-A}}$; [1] |
| $\psi_f$ | Wetting-front suction | mm |
| $L$, $S$, $w$, $n$ | Field length; slope; spacing or width; Manning roughness | m; %; m; $\mathrm{s\ m^{-1/3}}$ |
| $q$, $Q$ | Unit and total inflow | $\mathrm{L\ s^{-1}\ m^{-1}}$, $\mathrm{L\ s^{-1}}$ |
| $t_x$, $t_L$, $t_{co}$, $\tau$ | Advance time; time to end; cut-off time; opportunity time | min |
| $\sigma_y$, $\sigma_z$ | Surface and subsurface shape factors | [1] |
| $E_a$, $E_s$ | Application and storage efficiency | [1] or % |
| DP, RO, WD | Deep percolation, runoff, wind drift and interception fractions | [1] |
| CU, DU, $E_u$ | Christiansen uniformity; distribution uniformity; emission uniformity | [1] or % |
| $S_e$, $S_l$, $L_L$, $H_0$, $D$ | Emitter spacing; lateral spacing; lateral length; inlet head; inside diameter | m, m, m, m, mm |
| $c_v$, $x$ | Emitter coefficient of variation; flow exponent | [1] |
| $r$, $a$, $u$ | Bulb radius; aspect ratio; upper semi-axis ratio | mm, [1], [1] |
| $d$, $P$, $h_s$, $R$ | Nozzle diameter; pressure; pressure head; wetted radius | mm, kPa, m, m |

| Symbol | Meaning | Unit |
|---|---|---|
| $S_s$, $S_l$ | Sprinkler spacing along and between laterals | m |
| $u$ | Wind speed | m s$^{-1}$ |
| AR | Mean gross application rate | mm h$^{-1}$ |
| $C$ | Hazen and Williams roughness coefficient | [1] |
| $J$ | Hydraulic gradient | m m$^{-1}$ |
| TDH, $H_{\text{man}}$ | Total dynamic head; manometric head | m |
| $A_f$, $Q_s$ | Irrigated area; farm supply | ha; L s$^{-1}$ |

# Control value lists

Table B.1. Index lists behind the Properties controls (defaults in bold). All other controls are described with their ranges in Chapter 3.

| Control | Values |
|---|---|
| Trigger (sensor suction) | −50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm |
| Sensor depth | 5, 10, 15, …, 100 cm (20 values), default **20** cm |
| Nitrogen rate | 10, 25, 40, 55, 70, 85, 100, 115, 130, 150 |
| Mulch storage capacity (CAP) | 0.5, 1, 2, 4 mm |
| Rainfall factor | 0.5 to 1.4 |
| Temperature offset | −1 to +3.5 °C |
| Mulch strength (vapour-flux factor) | 0.1 to 1.0 (1.0 = bare soil) |
| Structural consolidation rate | 0.001, 0.0025, 0.005, 0.01, 0.02 $\mathrm{d}^{-1}$ |
| Crop | the simulated cereal/oilseed rotation and the 48 library crops of Table 21.2 |

Table B.2. Mulch materials (thermal weight / storage capacity in mm): husk 0.5 / 2, straw 0.4 / 4, chips 0.3 / 2, sawdust 0.6 / 1, grass 0.7 / 4, plastic 0.1 / 0.5.

# Simulation data sets

The annual meta-model (Engine 1) is built on the following Daisy 7.1 run sets of the study (Chapter 5, Table 5.1): E2, E3, E4, E5 (300 runs), E7, E11, E13 and E14. The study as a whole comprises 10,190 simulation runs in 17 experiments, available from the project repository (`Results/` and `Software/` folders). The 20-year weather record (1980 to 1999) is the baseline of the study; the projections to 2100 are analogue-year scenarios (Chapter 4).

# Provenance ledger of the numbered equations

All 134 numbered equations are labelled in the text and listed here: 38 **[Literature]**, 18 **[Data-driven]** and 78 **[Engineered]**. A **[Literature]** label means the relation is the published one; it does not mean that SWAFL has been validated in the field. The surface, drip and sprinkler design relations marked in Paper 3 were tested against published worked examples; nothing has been tested against field measurements.

| Eq. | Label | Basis |
|---|---|---|
| 4.1 | Engineered | calendar bookkeeping and choice of analogue year; no physical law |
| 4.2 | Engineered | calendar bookkeeping and choice of analogue year; no physical law |
| 5.1 | Literature | van Genuchten (1980) |
| 5.2 | Literature | Mualem (1976) with van Genuchten (1980) |
| 5.3 | Literature | standard definition of plant-available water between field capacity and wilting point |
| 5.4 | Data-driven | ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits |
| 5.5 | Data-driven | ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits |
| 5.6 | Data-driven | ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits |
| 5.7 | Data-driven | ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits |
| 5.8 | Data-driven | ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits |
| 5.9 | Data-driven | ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits |
| 5.10 | Data-driven | ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits |
| 5.11 | Data-driven | set-specific corrections from the Daisy run sets; clip bounds engineered |
| 5.12 | Engineered | closure constraint and bounds chosen for the workbench |
| 5.13 | Engineered | canopy curves, parameters chosen for the workbench; not a measured leaf-area record |
| 5.14 | Engineered | canopy curves, parameters chosen for the workbench; not a measured leaf-area record |

| Eq. | Label | Basis |
|---|---|---|
| 5.15 | Engineered | stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen |
| 5.16 | Engineered | stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen |
| 5.17 | Engineered | tipping-bucket redistribution with a chosen coefficient |
| 5.18 | Engineered | rooting rules with constants chosen for the workbench |
| 5.19 | Engineered | rooting rules with constants chosen for the workbench |
| 5.20 | Engineered | rooting rules with constants chosen for the workbench |
| 5.21 | Engineered | demand partitioning with chosen weights and a calibration factor |
| 5.22 | Engineered | demand partitioning with chosen weights and a calibration factor |
| 5.23 | Engineered | demand partitioning with chosen weights and a calibration factor |
| 5.24 | Data-driven | scaling of the daily series to the Engine 1 twenty-year means |
| 5.25 | Literature | water-balance identity |
| 5.26 | Engineered | atmosphere factor with chosen constants |
| 5.27 | Literature | readily available water as in FAO-56 practice, with an engineered clip |
| 6.1 | Data-driven | counts over the simulated daily series |
| 6.2 | Data-driven | counts over the simulated daily series |
| 6.3 | Literature | inverse of the van Genuchten curve, solved by bisection |
| 6.4 | Literature | same relation as Eq. (5.27) without the clip |
| 6.5 | Data-driven | ratio of Daisy run-set E2 yields |
| 7.1 | Literature | inverse of Eq. (5.3); the 30 cm floor is engineered |
| 7.2 | Engineered | series-resistance chain after van den Honert (1948); resistance values chosen for the workbench |
| 7.3 | Engineered | series-resistance chain after van den Honert (1948); resistance values chosen for the workbench |
| 7.4 | Engineered | series-resistance chain after van den Honert (1948); resistance values chosen for the workbench |
| 7.5 | Literature | Nobel (2009) |
| 7.6 | Engineered | indicator defined for the workbench |
| 8.1 | Engineered | exponential relaxation with chosen constants |
| 8.2 | Engineered | porosity relation with chosen coefficients |
| 8.3 | Literature | van Genuchten (1980) |
| 8.4 | Literature | Young-Laplace (Jurin) capillary law |
| 8.5 | Data-driven | differences of Daisy run-set E5 results |
| 8.6 | Data-driven | ordinary least squares applied to Daisy results |
| 8.7 | Engineered | diagnostic defined for the workbench |
| 9.1 | Data-driven | differences and ratios of Daisy run-set E14 and E11 results |

| Eq. | Label | Basis |
|---|---|---|
| 9.2 | Engineered | two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench |
| 9.3 | Engineered | two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench |
| 9.4 | Engineered | drying-front geometry chosen for drawing |
| 9.5 | Engineered | drying-front geometry chosen for drawing |
| 9.6 | Data-driven | differences and ratios of Daisy run-set E14 and E11 results |
| 9.7 | Data-driven | differences and ratios of Daisy run-set E14 and E11 results |
| 9.8 | Engineered | index defined for the workbench |
| 10.1 | Engineered | drawing rule |
| 10.2 | Data-driven | one-step differences of Daisy results |
| 11.1 | Literature | Saxton and Rawls (2006), implemented as written |
| 11.2 | Literature | Saxton and Rawls (2006), implemented as written |
| 11.3 | Engineered | rational-polynomial approximation of a Rawls-type suction |
| 11.4 | Engineered | initial and wetted water contents chosen for the workbench |
| 11.5 | Literature | Green and Ampt (1911) |
| 11.6 | Literature | Kostiakov (1932) |
| 11.7 | Literature | ordinary least squares on the log-log plot |
| 11.8 | Literature | Manning (1891) |
| 11.9 | Literature | volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3) |
| 11.10 | Literature | volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3) |
| 11.11 | Literature | volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3) |
| 11.12 | Engineered | alternative advance models set by worked examples and course notes |
| 11.13 | Engineered | alternative advance models set by worked examples and course notes |
| 11.15 | Literature | inversion of the Kostiakov law |
| 11.16 | Engineered | cut-off rules chosen as design options |
| 11.17 | Literature | standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997) |
| 11.18 | Literature | standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997) |
| 11.19 | Literature | standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997) |
| 11.20 | Engineered | hydrograph shapes chosen for drawing; volume conserving |

| Eq. | Label | Basis |
|---|---|---|
| 11.21 | Engineered | hydrograph shapes chosen for drawing; volume conserving |
| 11.22 | Engineered | search limit; the furrow stream limit follows Criddle et al. (1956) |
| 11.23 | Engineered | optimisation score and seasonal bookkeeping chosen for the workbench |
| 11.24 | Engineered | optimisation score and seasonal bookkeeping chosen for the workbench |
| 12.1 | Engineered | depth and demand rules chosen for the workbench |
| 12.2 | Engineered | depth and demand rules chosen for the workbench |
| 12.3 | Literature | Hazen and Williams (1905) |
| 12.4 | Literature | Keller and Karmeli (1974) emitter law |
| 12.5 | Engineered | pressure-compensating emitter model with chosen constants |
| 12.6 | Literature | definitions of mean and minimum discharge |
| 12.7 | Literature | Keller and Karmeli (1974) |
| 12.8 | Engineered | combined spread chosen to drive the bulb quantiles |
| 12.9 | Engineered | wetted-bulb geometry and loss partition; constants chosen for the workbench |
| 12.10 | Engineered | wetted-bulb geometry and loss partition; constants chosen for the workbench |
| 12.11 | Engineered | wetted-bulb geometry and loss partition; constants chosen for the workbench |
| 12.12 | Engineered | wetted-bulb geometry and loss partition; constants chosen for the workbench |
| 12.13 | Engineered | wetted-bulb geometry and loss partition; constants chosen for the workbench |
| 12.14 | Engineered | wetted-bulb geometry and loss partition; constants chosen for the workbench |
| 12.15 | Engineered | wetted-bulb geometry and loss partition; constants chosen for the workbench |
| 12.16 | Engineered | five-quantile delivery and run-time rules chosen for the workbench |
| 12.17 | Literature | definition of the requirement over the area served by one emitter |
| 12.18 | Engineered | five-quantile delivery and run-time rules chosen for the workbench |
| 12.19 | Engineered | five-quantile delivery and run-time rules chosen for the workbench |
| 12.20 | Engineered | layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice |
| 12.21 | Engineered | layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice |
| 12.22 | Engineered | Christiansen-type reduction with a chosen factor 0.4 |
| 12.23 | Engineered | layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice |

| Eq. | Label | Basis |
|---|---|---|
| 12.24 | Engineered | layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice |
| 12.25 | Engineered | layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice |
| 12.26 | Engineered | layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice |
| 13.1 | Engineered | depression storage reduction with slope |
| 13.2 | Literature | Torricelli orifice equation |
| 13.3 | Engineered | range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen |
| 13.4 | Engineered | pattern and wind shapes chosen to reproduce published qualitative patterns |
| 13.5 | Engineered | pattern and wind shapes chosen to reproduce published qualitative patterns |
| 13.6 | Literature | overlap by superposition of single-sprinkler patterns |
| 13.7 | Literature | Christiansen (1942) |
| 13.8 | Engineered | loss and ponding rules chosen for the workbench |
| 13.9 | Engineered | loss and ponding rules chosen for the workbench |
| 13.10 | Engineered | loss and ponding rules chosen for the workbench |
| 13.11 | Literature | efficiency and distribution uniformity of Merriam and Keller (1978) |
| 13.12 | Engineered | run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified |
| 13.13 | Engineered | run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified |
| 13.14 | Engineered | run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified |
| 13.15 | Engineered | run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified |
| 13.16 | Engineered | run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified |
| 13.17 | Literature | Blasius friction law for smooth pipe |
| 13.18 | Literature | 20 % rule of Keller and Bliesner (1990) |
| 13.19 | Engineered | head and sets bookkeeping |
| 13.20 | Literature | hydraulic power definition |
| 13.21 | Engineered | head and sets bookkeeping |
| 13.22 | Literature | Christiansen reduction factor with Hazen-Williams exponent |
| 13.23 | Engineered | head and sets bookkeeping |

| Eq. | Label | Basis |
|---|---|---|
| 14.1 | Engineered | bookkeeping of Engine 2 series |
| 14.2 | Engineered | bookkeeping of Engine 2 series |
| 14.3 | Literature | water-balance identity |
| 14.4 | Engineered | bookkeeping of Engine 2 series |
| 14.5 | Engineered | bookkeeping of Engine 2 series |

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# Equation reference

This appendix prints every numbered equation of the program in full, in the order of the chapters, with its provenance and the layers of the program that draw on it. The equations added in version 1.1 (Eqs. 5.12a, 6.4a and those of Chapters 14, 18 to 23) are printed in their chapters.

**Eq. 4.1**: dates

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

*Provenance (Eq. 4.1):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

**Eq. 4.2**: dates

$$a = \begin{cases} \text{argsort}_y\,(P_{\text{thr},y})_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

*Provenance (Eq. 4.2):* **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

**Eq. 5.1**: engine

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\,S_e(h) \tag{5.1}$$

*Provenance (Eq. 5.1):* **[Literature]**: van Genuchten (1980).

**Eq. 5.2**: engine

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

*Provenance (Eq. 5.2):* **[Literature]**: Mualem (1976) with van Genuchten (1980).

**Eq. 5.3**: engine

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

*Provenance (Eq. 5.3):* **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

**Eq. 5.4**: engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

**Eq. 5.5**: engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

**Eq. 5.6**: engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

**Eq. 5.7**: engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

**Eq. 5.8**: engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

**Eq. 5.9**: engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

**Eq. 5.10**: engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

*Provenance (Eq. 5.4 to 5.10):* **[Data-driven]**: ratios of Daisy ensemble results (run sets E2 to E14); the clip bounds are engineered limits.

**Eq. 5.11**: engine

$$Q \leftarrow Q\ \text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ \ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N). \tag{5.11}$$

*Provenance (Eq. 5.11):* **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

**Eq. 5.12**: engine

$$ET \in [\,P + I - D - 30,\ P + I - D + 15\,] \text{ (if not, } ET \text{ is clipped and } E, T \text{ are scaled by the same factor)},$$
$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

*Provenance (Eq. 5.12):* **[Engineered]**: closure constraint and bounds chosen for the workbench.

**Eq. 5.13**: engine

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

*Provenance (Eq. 5.13):* **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

**Eq. 5.14**: engine

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

*Provenance (Eq. 5.14):* **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

**Eq. 5.15**: engine

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

*Provenance (Eq. 5.15):* **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

**Eq. 5.16**: engine

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

*Provenance (Eq. 5.16):* **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

**Eq. 5.17**: engine

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\;\; w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

*Provenance (Eq. 5.17):* **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

**Eq. 5.18**: engine

$$R_{\max} = \operatorname{clip}\Big(R_c\,(0.85 + 0.3\,\text{sand})[1 + 0.12\log_{10}(|h_T|/300)][1 - 0.18\,\tfrac{N-100}{100}];\; 25,\; 100\Big)\ \text{cm}, \tag{5.18}$$

*Provenance (Eq. 5.18):* **[Engineered]**: rooting rules with constants chosen for the workbench.

**Eq. 5.19**: engine

$$R_d \leftarrow \max\Big\{R_d,\; \min\big\{R_{\max}\operatorname{clip}(\tfrac{c-0.04}{0.96};0,1)^{0.8},\; R_{\max},\; R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

*Provenance (Eq. 5.19):* **[Engineered]**: rooting rules with constants chosen for the workbench.

**Eq. 5.20**: engine

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

*Provenance (Eq. 5.20):* **[Engineered]**: rooting rules with constants chosen for the workbench.

**Eq. 5.21**: engine

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65\cdot\tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

*Provenance (Eq. 5.21):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

**Eq. 5.22**: engine

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

*Provenance (Eq. 5.22):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

**Eq. 5.23**: engine

$$q_i^{T} = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \operatorname{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

*Provenance (Eq. 5.23):* **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

**Eq. 5.24**: engine

$$S_P = \frac{P}{\overline{P_{\text{sim}}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T_{f,\text{sim}}}}, \quad S_E = \frac{E}{\overline{E_{\text{sim}}}}, \quad S_T = \frac{T}{\overline{T_{\text{sim}}}}, \quad S_D = \frac{D}{\overline{D_{\text{sim}}}}, \tag{5.24}$$

*Provenance (Eq. 5.24):* **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

**Eq. 5.25**: engine

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

*Provenance (Eq. 5.25):* **[Literature]**: water-balance identity.

**Eq. 5.26**: engine

$$E_{0,\text{bare}} = 4.2\,(1 + 0.07\,\Delta T)\,\frac{0.35 + 0.65 \cdot \frac{1}{2}[1 + \sin(2\pi(\text{doy} - 100)/365)]}{0.675}\,\text{clip}(1 - 0.12(\rho - 1);\,0.85,\,1.12)\sqrt{\gamma_y}\ \text{mm d}^{-1}, \quad (5.26)$$

*Provenance (Eq. 5.26):* **[Engineered]**: atmosphere factor with chosen constants.

**Eq. 5.27**: engine

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\text{max}};\,25,\,150\big)\ \text{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \quad (5.27)$$

*Provenance (Eq. 5.27):* **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

**Eq. 6.1**: control

$$N_y(h_T) = \#\big\{\, d \in \text{growing season of year } y : h_d > h_T \,\big\} \quad (183 \text{ days from 1 April}), \quad (6.1)$$

*Provenance (Eq. 6.1):* **[Data-driven]**: counts over the simulated daily series.

**Eq. 6.2**: control

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_{y} N_y. \quad (6.2)$$

*Provenance (Eq. 6.2):* **[Data-driven]**: counts over the simulated daily series.

**Eq. 6.3**: control

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \quad (6.3)$$

*Provenance (Eq. 6.3):* **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

**Eq. 6.4**: control

$$Z_n = f \cdot \text{AWC} \cdot R_{\text{max}} \quad [\text{mm}] \quad (6.4)$$

*Provenance (Eq. 6.4):* **[Literature]**: same relation as Eq. (5.27) without the clip.

**Eq. 6.5**: control

$$Y_{\text{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \quad (6.5)$$

*Provenance (Eq. 6.5):* **[Data-driven]**: ratio of Daisy run-set E2 yields.

**Eq. 7.1**: field

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \quad (7.1)$$

*Provenance (Eq. 7.1):* **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

**Eq. 7.2**: field

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \quad (7.2)$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \quad (7.3)$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \quad (7.4)$$

*Provenance (Eq. 7.2 to 7.4):* **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

**Eq. 7.3**: field

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55},\quad R_{\text{root}} = 0.10,\quad R_{\text{xyl}} = 0.05,\quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}},\qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

*Provenance (Eq. 7.2 to 7.4):* **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

**Eq. 7.4**: field

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55},\quad R_{\text{root}} = 0.10,\quad R_{\text{xyl}} = 0.05,\quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}},\qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

*Provenance (Eq. 7.2 to 7.4):* **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

**Eq. 7.5**: field

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH}\ \approx\ 135.5\,\ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

*Provenance (Eq. 7.5):* **[Literature]**: Nobel (2009).

**Eq. 7.6**: field

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

*Provenance (Eq. 7.6):* **[Engineered]**: indicator defined for the workbench.

**Eq. 8.1**: structure

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t},\qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

*Provenance (Eq. 8.1):* **[Engineered]**: exponential relaxation with chosen constants.

**Eq. 8.2**: structure

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big),\qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

*Provenance (Eq. 8.2):* **[Engineered]**: porosity relation with chosen coefficients.

**Eq. 8.3**: structure

$$\theta(h) = \theta_r + (\phi - \theta_r)\,S_e(h). \tag{8.3}$$

*Provenance (Eq. 8.3):* **[Literature]**: van Genuchten (1980).

**Eq. 8.4**: structure

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h}\ \approx\ \frac{0.149\ \text{cm}^2}{h}\quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

*Provenance (Eq. 8.4):* **[Literature]**: Young-Laplace (Jurin) capillary law.

**Eq. 8.5**: structure

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*),\qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

*Provenance (Eq. 8.5):* **[Data-driven]**: differences of Daisy run-set E5 results.

**Eq. 8.6**: structure

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

*Provenance (Eq. 8.6):* **[Data-driven]**: ordinary least squares applied to Daisy results.

**Eq. 8.7**: structure

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k : \frac{\sum_{i \le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d \in \text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

*Provenance (Eq. 8.7):* **[Engineered]**: diagnostic defined for the workbench.

**Eq. 9.1**: mulch

$$\Delta E = E_{1.0} - E_{\varphi}, \qquad \Delta I = I_{1.0} - I_{\varphi}, \qquad \Delta D = D_{\varphi} - D_{1.0}, \tag{9.1}$$

*Provenance (Eq. 9.1):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

**Eq. 9.2**: mulch

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

*Provenance (Eq. 9.2):* **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

**Eq. 9.3**: mulch

$$Q(t) = q_0 + \begin{cases} e_0\,t, & t \le t_1 \\ (Q_1 - q_0) + 2\,e_0\,\tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

*Provenance (Eq. 9.3):* **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

**Eq. 9.4**: mulch

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

*Provenance (Eq. 9.4):* **[Engineered]**: drying-front geometry chosen for drawing.

**Eq. 9.5**: mulch

$$\log_{10} h(z) = \log_{10} h_s + (\log_{10} h_i - \log_{10} h_s)(1 - e^{-(z-\ell)/\lambda}) \quad (z \ge \ell), \qquad h = 10^5\ \text{cm for } z < \ell. \tag{9.5}$$

*Provenance (Eq. 9.5):* **[Engineered]**: drying-front geometry chosen for drawing.

**Eq. 9.6**: mulch

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

*Provenance (Eq. 9.6):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

**Eq. 9.7**: mulch

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

*Provenance (Eq. 9.7):* **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

**Eq. 9.8**: mulch

$$\mathcal{I} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{I} > +0.05 \\ \text{drainage-dominated} & \mathcal{I} < -0.05 \\ \text{on the boundary} & |\mathcal{I}| \le 0.05 \end{cases} \tag{9.8}$$

*Provenance (Eq. 9.8):* **[Engineered]**: index defined for the workbench.

**Eq. 10.1**: loops

$$w_k = 5 + \text{clip}(\mu_k;\, 0,\, 4). \tag{10.1}$$

*Provenance (Eq. 10.1):* **[Engineered]**: drawing rule.

**Eq. 10.2**: loops

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D} \quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

*Provenance (Eq. 10.2):* **[Data-driven]**: one-step differences of Daisy results.

**Eq. 11.1**: surface

$$\begin{aligned} \theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\text{OM} + 0.005\,S\,\text{OM} - 0.013\,C\,\text{OM} + 0.068\,S\,C + 0.031, \\ \theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02), \\ \theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\text{OM} + 0.006\,S\,\text{OM} - 0.027\,C\,\text{OM} + 0.452\,S\,C + 0.299, \\ \theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015), \end{aligned} \tag{11.1}$$

*Provenance (Eq. 11.1):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

**Eq. 11.2**: surface

$$\begin{aligned} \theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\text{OM} - 0.018\,S\,\text{OM} - 0.027\,C\,\text{OM} - 0.584\,S\,C + 0.078, \\ \theta_{(S-33)} &= \theta_{(S-33),t} + (0.636\,\theta_{(S-33),t} - 0.107), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043, \\ B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}. \end{aligned} \tag{11.2}$$

*Provenance (Eq. 11.2):* **[Literature]**: Saxton and Rawls (2006), implemented as written.

**Eq. 11.3**: surface

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

*Provenance (Eq. 11.3):* **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

**Eq. 11.4**: surface

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

*Provenance (Eq. 11.4):* **[Engineered]**: initial and wetted water contents chosen for the workbench.

**Eq. 11.5**: surface

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

*Provenance (Eq. 11.5):* **[Literature]**: Green and Ampt (1911).

**Eq. 11.6**: surface

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

*Provenance (Eq. 11.6):* **[Literature]**: Kostiakov (1932).

**Eq. 11.7**: surface

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

*Provenance (Eq. 11.7):* **[Literature]**: ordinary least squares on the log-log plot.

**Eq. 11.8**: surface

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

*Provenance (Eq. 11.8):* **[Literature]**: Manning (1891).

**Eq. 11.9**: surface

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

*Provenance (Eq. 11.9):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

**Eq. 11.10**: surface

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

*Provenance (Eq. 11.10):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

**Eq. 11.11**: surface

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

*Provenance (Eq. 11.11):* **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

**Eq. 11.12**: surface

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

*Provenance (Eq. 11.12):* **[Engineered]**: alternative advance models set by worked examples and course notes.

**Eq. 11.13**: surface

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

*Provenance (Eq. 11.13):* **[Engineered]**: alternative advance models set by worked examples and course notes.

**Eq. 11.15**: surface

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

*Provenance (Eq. 11.15):* **[Literature]**: inversion of the Kostiakov law.

**Eq. 11.16**: surface

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\,\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

*Provenance (Eq. 11.16):* **[Engineered]**: cut-off rules chosen as design options.

**Eq. 11.17**: surface

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

*Provenance (Eq. 11.17):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

**Eq. 11.18**: surface

$$\begin{aligned} &V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)), \quad V_{\text{req}} = Z_n\,L\,w, \\ &V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx, \\ &V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx, \\ &V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \end{aligned} \tag{11.18}$$

*Provenance (Eq. 11.18):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

**Eq. 11.19**: surface

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

*Provenance (Eq. 11.19):* **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

**Eq. 11.20**: surface

$$V_c(t) = w\sum_i k\left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x\ \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

*Provenance (Eq. 11.20):* **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

**Eq. 11.21**: surface

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t - t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

*Provenance (Eq. 11.21):* **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

**Eq. 11.22**: surface

$$q_{\max} = \min\Big\{q_{\max}^{\text{sys}},\ \frac{0.04}{S}\cdot\frac{1000}{60}\Big\}\ \text{L s}^{-1} \quad (S \text{ in \%}), \tag{11.22}$$

*Provenance (Eq. 11.22):* **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

**Eq. 11.23**: surface

$$\text{score} = \min(E_a, 1) - 3\max(0,\, 0.8 - E_s) - 3\,[c_b < 1]\,\max(0,\, 0.05 - \text{RO}) - 5\max(0,\, E_a - 1). \tag{11.23}$$

*Provenance (Eq. 11.23):* **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

**Eq. 11.24**: surface

$$\begin{aligned} n_{\text{ev}} &= \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g, \\ N_{0M} &= \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \end{aligned} \tag{11.24}$$

*Provenance (Eq. 11.24):* **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

**Eq. 12.1**: drip

$$Z_n = \min\!\big\{ Z_n^{\text{DSS}},\; \max(4,\; \text{ET}_{\text{pk}}\,\Delta t) \big\} \;\; \text{mm}, \tag{12.1}$$

*Provenance (Eq. 12.1):* **[Engineered]**: depth and demand rules chosen for the workbench.

**Eq. 12.2**: drip

$$\text{ET}_{\text{pk}} = \text{clip}\Big\{ \max_{d} \tfrac{1}{5} \sum_{j=0}^{4} (E_{d+j} + T_{d+j});\; 2.5,\; 9 \Big\}. \tag{12.2}$$

*Provenance (Eq. 12.2):* **[Engineered]**: depth and demand rules chosen for the workbench.

**Eq. 12.3**: drip

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\text{m}^3\,\text{s}^{-1}), \quad D \text{ in m}. \tag{12.3}$$

*Provenance (Eq. 12.3):* **[Literature]**: Hazen and Williams (1905).

**Eq. 12.4**: drip

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\; h_j/H_0 \geq 0.01), \tag{12.4}$$

*Provenance (Eq. 12.4):* **[Literature]**: Keller and Karmeli (1974) emitter law.

**Eq. 12.5**: drip

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

*Provenance (Eq. 12.5):* **[Engineered]**: pressure-compensating emitter model with chosen constants.

**Eq. 12.6**: drip

$$\bar{q} = \frac{1}{N+1} \sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600} \;\; (\text{L}\,\text{s}^{-1}), \tag{12.6}$$

*Provenance (Eq. 12.6):* **[Literature]**: definitions of mean and minimum discharge.

**Eq. 12.7**: drip

$$E_u = \operatorname{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

*Provenance (Eq. 12.7):* **[Literature]**: Keller and Karmeli (1974).

**Eq. 12.8**: drip

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

*Provenance (Eq. 12.8):* **[Engineered]**: combined spread chosen to drive the bulb quantiles.

**Eq. 12.9**: drip

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

*Provenance (Eq. 12.9):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Eq. 12.10**: drip

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

*Provenance (Eq. 12.10):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Eq. 12.11**: drip

$$a = \operatorname{clip}\Big\{\Big[0.7 + 1.6\Big(\frac{K_s}{K_s + 15}\Big)^{1.1}\Big]\sqrt{\operatorname{clip}(2/q;\ 0.7,\ 1.4)};\ 0.75,\ 2.4\Big\} \tag{12.11}$$

*Provenance (Eq. 12.11):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Eq. 12.12**: drip

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

*Provenance (Eq. 12.12):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Eq. 12.13**: drip

$$P_w = \operatorname{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

*Provenance (Eq. 12.13):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Eq. 12.14**: drip

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z > R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

*Provenance (Eq. 12.14):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Eq. 12.15**: drip

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\mathrm{L},\ \text{area in } \mathrm{m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

*Provenance (Eq. 12.15):* **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

**Eq. 12.16**: drip

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1+\sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

*Provenance (Eq. 12.16):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

**Eq. 12.17**: drip

$$V_{\text{req}} = Z_n\,S_e\,S_l\ \ (\text{L}), \qquad V_{\text{st},i} = \min\{V_{\text{root},i},\ V_{\text{req}}\}, \qquad V_{\text{ex},i} = V_{\text{root},i} - V_{\text{st},i}. \tag{12.17}$$

*Provenance (Eq. 12.17):* **[Literature]**: definition of the requirement over the area served by one emitter.

**Eq. 12.18**: drip

$$t = \text{clip}\Big\{\frac{V_{\text{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \text{h}. \tag{12.18}$$

*Provenance (Eq. 12.18):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

**Eq. 12.19**: drip

$$E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

*Provenance (Eq. 12.19):* **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

**Eq. 12.20**: drip

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \Big\lfloor \frac{W_f}{S_l} \Big\rfloor, \quad n_{\text{zone}} = \Big\lceil \frac{N_L\,Q_{\text{lat}}}{Q_s} \Big\rceil, \quad n_l = \Big\lceil \frac{N_L}{n_{\text{zone}}} \Big\rceil, \quad Q_z = n_l\,Q_{\text{lat}}, \tag{12.20}$$

*Provenance (Eq. 12.20):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

**Eq. 12.21**: drip

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

*Provenance (Eq. 12.21):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

**Eq. 12.22**: drip

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

*Provenance (Eq. 12.22):* **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

**Eq. 12.23**: drip

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\,Q_z\,\text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

*Provenance (Eq. 12.23):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

**Eq. 12.24**: drip

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\mathrm{L\,s^{-1}\,ha^{-1}}), \tag{12.24}$$

*Provenance (Eq. 12.24):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

**Eq. 12.25**: drip

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

*Provenance (Eq. 12.25):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

**Eq. 12.26**: drip

$$\text{score} = \min(E_a, 1) - 2.5\,\max(0,\ 0.9 - E_s), \tag{12.26}$$

*Provenance (Eq. 12.26):* **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

**Eq. 13.1**: sprinkler

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

*Provenance (Eq. 13.1):* **[Engineered]**: depression storage reduction with slope.

**Eq. 13.2**: sprinkler

$$q = 0.95\,A_n \sqrt{2g\,h} \times 6 \times 10^4\ \ (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2\ \mathrm{m^2}, \tag{13.2}$$

*Provenance (Eq. 13.2):* **[Literature]**: Torricelli orifice equation.

**Eq. 13.3**: sprinkler

$$R = c_R \sqrt{d\,h}\ f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

*Provenance (Eq. 13.3):* **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

**Eq. 13.4**: sprinkler

$$\widehat{p}(x) = \begin{cases} 1 - x^m, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^6), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^5), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

*Provenance (Eq. 13.4):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

**Eq. 13.5**: sprinkler

$$R_e(\theta) = R\,\text{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

*Provenance (Eq. 13.5):* **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

**Eq. 13.6**: sprinkler

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_k a_k, \tag{13.6}$$

*Provenance (Eq. 13.6):* **[Literature]**: overlap by superposition of single-sprinkler patterns.

**Eq. 13.7**: sprinkler

$$\mathrm{CU} = 1 - \frac{\sum_{k} |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

*Provenance (Eq. 13.7):* **[Literature]**: Christiansen (1942).

**Eq. 13.8**: sprinkler

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2 + 0.5u + 0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}\big[1 + 0.025\,(T_a - 20)\big]\Big[0.6 + 0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

*Provenance (Eq. 13.8):* **[Engineered]**: loss and ponding rules chosen for the workbench.

**Eq. 13.9**: sprinkler

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \mathrm{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \mathrm{sup} - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \mathrm{sup} - \mathrm{RO}. \tag{13.9}$$

*Provenance (Eq. 13.9):* **[Engineered]**: loss and ponding rules chosen for the workbench.

**Eq. 13.10**: sprinkler

$$I_c = \min\{I,\ I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

*Provenance (Eq. 13.10):* **[Engineered]**: loss and ponding rules chosen for the workbench.

**Eq. 13.11**: sprinkler

$$\begin{aligned} E_a &= \frac{\overline{\min(Z_k, Z_n)}}{\mathrm{gross}}, & E_s &= \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, & \mathrm{DU} &= \frac{\overline{Z}_{25}}{\overline{Z}}, \\ \mathrm{DP} &= \frac{\overline{\max(Z_k - Z_n, 0)}}{\mathrm{gross}}, & \mathrm{RO} &= \frac{\overline{\mathrm{RO}_k}}{\mathrm{gross}}, & \mathrm{WD} &= w_d + \frac{\overline{I_c}}{\mathrm{gross}}, \end{aligned} \tag{13.11}$$

*Provenance (Eq. 13.11):* **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

**Eq. 13.12**: sprinkler

$$T = \min\{T :\ \overline{Z}_{25}(T) \ge Z_n\}, \qquad T \in [1,\ 2400]\ \mathrm{min}, \tag{13.12}$$

*Provenance (Eq. 13.12):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

**Eq. 13.13**: sprinkler

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

*Provenance (Eq. 13.13):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

**Eq. 13.14**: sprinkler

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x / S_s \rfloor, \quad N_m = \lfloor L_y / S_l \rfloor, \tag{13.14}$$

*Provenance (Eq. 13.14):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

**Eq. 13.15**: sprinkler

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}}\ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right)\ \text{days}, \tag{13.15}$$

*Provenance (Eq. 13.15):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

**Eq. 13.16**: sprinkler

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\, N\, q_h, \tag{13.16}$$

*Provenance (Eq. 13.16):* **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

**Eq. 13.17**: sprinkler

$$J = \frac{0.472\, Q^{1.75}}{D^{4.75}}\ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

*Provenance (Eq. 13.17):* **[Literature]**: Blasius friction law for smooth pipe.

**Eq. 13.18**: sprinkler

$$h_{f,\text{lat}} = 1.1\, F\, J(Q_L, D)\,(L_x - S_s/2) \le 0.2\, h_s, \qquad Q_L = 1000\, N\, q_h, \tag{13.18}$$

*Provenance (Eq. 13.18):* **[Literature]**: 20 % rule of Keller and Bliesner (1990).

**Eq. 13.19**: sprinkler

$$H_{\text{man}} = z_{\text{lift}} + J\, L_{\text{conv}} + J\, L_{\text{main}} + h_s + 0.75\, h_{f,\text{lat}}, \tag{13.19}$$

*Provenance (Eq. 13.19):* **[Engineered]**: head and sets bookkeeping.

**Eq. 13.20**: sprinkler

$$P_{\text{CV}} = \frac{1000\,(Q_m/3.6\times 10^{6})\, H_{\text{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\text{kW}} = 0.7355\, P_{\text{CV}}, \tag{13.20}$$

*Provenance (Eq. 13.20):* **[Literature]**: hydraulic power definition.

**Eq. 13.21**: sprinkler

$$N_L = \left\lfloor \frac{10^4 A_f / L_L}{S_l} \right\rfloor, \quad n_l = \min\!\left(N_L,\ \left\lfloor \frac{Q_s}{Q_{\text{lat}}} \right\rfloor\right), \quad n_{\text{set}} = \left\lceil \frac{N_L}{n_l} \right\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

*Provenance (Eq. 13.21):* **[Engineered]**: head and sets bookkeeping.

**Eq. 13.22**: sprinkler

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m\, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

*Provenance (Eq. 13.22):* **[Literature]**: Christiansen reduction factor with Hazen-Williams exponent.

**Eq. 13.23**: sprinkler

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\!\left(\frac{10^4 A_f}{S_s S_l}\right), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \left\lceil \frac{Q_{\text{tot}}}{Q_s} \right\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60}\ \text{h}. \tag{13.23}$$

*Provenance (Eq. 13.23):* **[Engineered]**: head and sets bookkeeping.

**Eq. 14.1**: dashboard

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm}\,\text{d}^{-1}). \tag{14.1}$$

*Provenance (Eq. 14.1):* **[Engineered]**: bookkeeping of Engine 2 series.
**Eq. 14.2**: dashboard

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

*Provenance (Eq. 14.2):* **[Engineered]**: bookkeeping of Engine 2 series.
**Eq. 14.3**: dashboard

$$\Delta S = P + I - E_{\text{int}} - E - T - D, \tag{14.3}$$

*Provenance (Eq. 14.3):* **[Literature]**: water-balance identity.
**Eq. 14.4**: dashboard

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

*Provenance (Eq. 14.4):* **[Engineered]**: bookkeeping of Engine 2 series.
**Eq. 14.5**: dashboard

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

*Provenance (Eq. 14.5):* **[Engineered]**: bookkeeping of Engine 2 series.